



**5 Series MSO (MSO54, MSO56, MSO58, MSO58LP)
6 Series MSO (MSO64)
Programmer Manual**





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Preface

This programmer guide provides you with the information required to use Programmable Interface commands to remotely control your instrument.

The programmer manual is divided into the following major topics:

- **Getting started.** This topic introduces you to the programming information and provides basic information about setting up your instrument for remote control.
- **Command syntax.** This topic provides an overview of the command syntax that you use to communicate with the instrument and other general information about commands, such as how commands and queries are constructed, how to enter commands, constructed mnemonics, and argument types.
- **Command groups.** This topic contains all the commands listed in functional groups. Each group consists of an overview of the commands in that group and a table that lists all the commands and queries for that group. You can click a command in the listing to display a detailed description of the command.
- **Commands listed in alphabetical order.** This topic contains all commands listed in alphabetical order. Command details, syntax, and examples are provided.
- **Status and events.** This topic discusses the status and event reporting system for the programming interfaces. This system informs you of certain significant events that occur within the instrument. Topics that are discussed include registers, queues, event handling sequences, synchronization methods, and messages that the instrument might return, including error messages.
- **Appendices.** These topics contain miscellaneous information, such as a list of reserved words, a table of the factory initialization (default) settings, and interface specifications that can be helpful when using commands to remotely control the instrument.

Getting Started

This manual explains the use of commands to remotely control your oscilloscope. With this information, you can write computer programs to perform functions, such as setting the front-panel controls, taking measurements, performing statistical calculations, and exporting data for use in other programs.

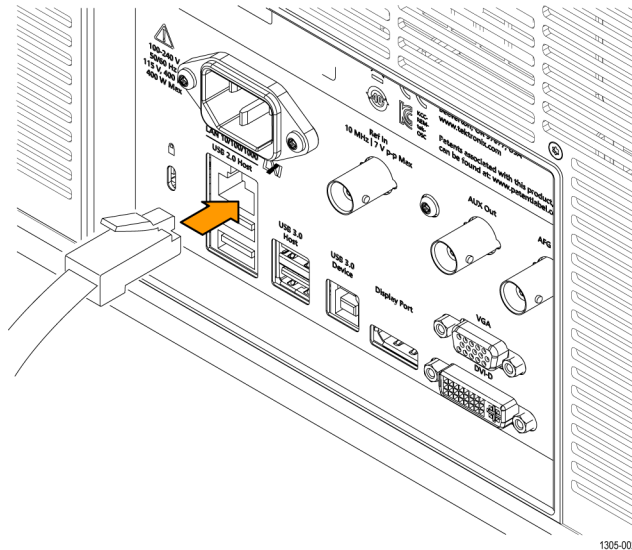
Familiarity with the User Manual for your oscilloscope is assumed. You can download the User Manual from the Tektronix website at www.tek.com.

NOTE. Most examples in this document assume that both *HEADer* and *VERBoSe* are set to ON.

Setting Up Remote Communications Hardware

You can remotely control communications between your oscilloscope and a PC via Ethernet or USB cables.

Ethernet If you are using Ethernet, start by connecting an appropriate Ethernet cable to the Ethernet port (RJ-45 connector) on the rear panel of your oscilloscope. This connects the oscilloscope to a 10BASE-T/100BASE-T/1000BASE-T local area network.

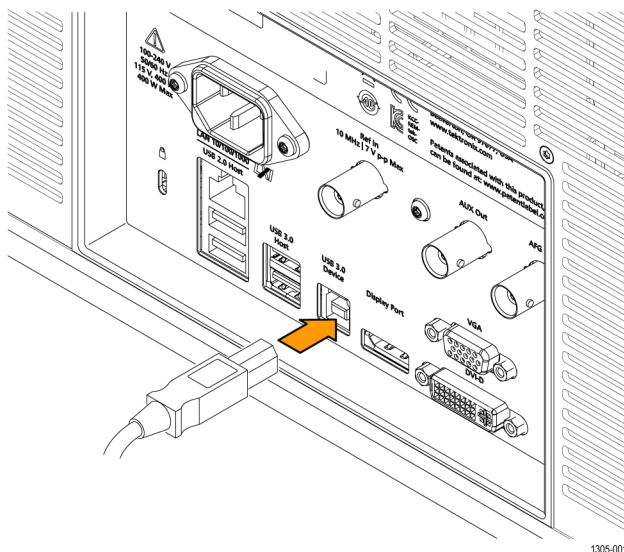


To change the Ethernet settings on your oscilloscope, do the following:

1. Select the **Utility** drop-down menu.
2. Select the **I/O** menu.
3. Select the **LAN** panel.
4. In the menu, if you are on a DHCP Ethernet network that supplies the IP address automatically by a DHCP, tap **Auto**.
5. In the menu, if you want to supply your own network settings, tap **Manual** to set a hard coded TCP/IP address.

USB

If you are using USB, start by connecting the appropriate USB cable to the USB 3.0 super-speed (SS) Device port on the rear panel of your oscilloscope. This port requires that the cable connected from the port to the host computer meets the USB 3.0 specification for super speed connections. Typically, such cables should be 3 feet or shorter in length, but this is determined by the quality of the cable and, with higher quality cables, this length can be extended. (It is also dependent upon the drive capability of the host USB port to which the instrument is connected.) The use of high quality short cables is recommended to avoid USB connection problems.



With USB, the system automatically configures itself. To verify that the USB is enabled:

1. Select the **Utility** drop-down menu.
2. Select the **I/O** menu.
3. Touch **USB Device Port** to open the USB Device Port configuration menu.
4. If USB is disabled, tap **USB Device Port** to enable the USB Device port.

After connection, the host, with appropriate software, can list the oscilloscope as a USB device with the following parameters: (See Table 1-1.)

Table 1-1: USB Device Parameters

Parameter	Value
Manufacturer ID	0x0699 (decimal 1689)
Product ID	0x0522 (You can send the USBTMC:PRODUCTID:HEXadecimal? query to read the value)
Serial number	Serial number
Manufacturer description	"Tektronix"
Interface description	"USBTMC-USB488"

Setting Up Remote Communications Software

Connect your oscilloscope directly to a computer to let the PC analyze your data, collect screen images, or to control the oscilloscope using a program of your own creation. Three ways to connect your oscilloscope to a computer are through the VISA drivers, the e*Scope Web-enabled tools, or via a socket server.

Using VISA VISA lets you use your MS-Windows computer to acquire data from your oscilloscope for use in an analysis package that runs on your PC, such as Microsoft Excel, National Instruments LabVIEW, Tektronix OpenChoice Desktop software, or your own custom software. You can use a common communications connection, such as USB or Ethernet, to connect the computer to the oscilloscope.

To set up VISA communications between your oscilloscope and a computer:

1. Load the VISA drivers on your computer. Also, load your application, such as OpenChoice Desktop. You will find the drivers and OpenChoice Desktop software on the appropriate CD that comes with your oscilloscope or at the Tektronix software finder Web page (www.tektronix.com/downloads).
2. Connect the oscilloscope to your computer with the appropriate USB or Ethernet cable. Cycle the power on the oscilloscope.
3. Select the **Utility** drop-down menu
4. Select **I/O** menu.
5. If you are using USB, the system sets itself up automatically for you, if USB is enabled. Check **USB DEVICE PORT** panel to be sure that USB is enabled. If it is not enabled, toggle the On/Off button to On.
6. To use Ethernet, select the **LAN** panel. Use the controls to adjust your network settings, as needed. For more information, see the e*Scope setup information below.
7. If you want to change socket server parameters, select the **Socket Server** panel and enter new values through the resulting panel controls.
8. Run your application software on your computer.

Quick Tips

- The USB 3.0 super-speed (SS) device port is the correct USB port for computer connectivity. Use the rear- and front-panel USB 2.0 or 3.0 host ports to connect your oscilloscope to USB flash drives, hard drives, keyboards or mice. Use the USB Device port to connect your oscilloscope to a PC.

There are both USB 2.0 and 3.0 host ports on the instrument. The device port is USB 3.0. Printers are not supported on host ports. PictBridge printers are not supported on the device port.

Using the LXI Web Page and e*Scope

With e*Scope, you can access your Internet-connected instrument from a web browser.

NOTE. *This procedure presumes that you have connected the oscilloscope to a network and configures the oscilloscope to work on your network. See the oscilloscope Installation and Safety instructions, or the embedded Help on the oscilloscope, for instructions on how to configure the oscilloscope for network use.*

To set up e*Scope communications between your oscilloscope and a web browser running on a remote computer:

1. Connect the oscilloscope to your computer network with an appropriate Ethernet cable.
2. Select the **Utility** drop-down menu.
3. Select the **I/O** menu.
4. Select the **LAN** panel.
5. At the top left of the panel, there is an indicator light which turns red if the device detects a fault.
6. Tap **Test Connection** to check if your oscilloscope can find an attached network.
7. Record the IP address shown on your oscilloscope.
8. Start a Web browser on a PC that has access to the same network to which the oscilloscope is attached.
9. Enter the IP address of the oscilloscope in the browser address line. You should then see the LXI Welcome page on your Web browser on your computer screen.
10. For e*Scope, click the Instrument Control (e*Scope) link on the left side of the LXI Welcome page. You should then see a new tab (or window) open in your browser, showing the oscilloscope screen. You can use the PC mouse to interact with the e*Scope screen and controls.

Using a Socket Server

A socket server provides two-way communication over an Internet Protocol-based computer network. You can use your oscilloscope's socket server feature to let your oscilloscope talk to a remote-terminal device or computer.

To set up and use a socket server between your oscilloscope and a remote terminal or computer:

1. Connect the oscilloscope to your computer network with an appropriate Ethernet cable.
2. Select the **Utility** drop-down menu.
3. Select the **I/O** menu.
4. Tap **Socket Server**.
5. On the resulting Socket Server panel, tap the top entry to toggle the Socket Server On.
6. Choose whether the protocol should be **None** or **Terminal**. A communication session run by a human at a keyboard typically uses a terminal protocol. An automated session might handle its own communications without using such a protocol.
7. If required, change the port number by rotating multipurpose knob **a**.

8. If required, press **OK** to set the new port number.
9. After setting up the socket server parameters, you can now have the computer “talk” to the oscilloscope. If you are running an MS Windows PC, you could run its default client with its command-like interface. One way to do this is by typing “**Telnet**” in the Run window. The Telnet window will open on the PC.

NOTE. *On MS Windows 7, you must first enable Telnet in order for it to work.*

10. Start a terminal session between your computer and your oscilloscope by typing in an open command with the oscilloscope LAN address and port #. You can obtain the LAN address by pushing the **LAN** panel to view the resulting LAN setting panel. You can obtain the port # by tapping the **Socket Server** panel and viewing the **Port** item.

For example, if the oscilloscope IP address was **123.45.67.89** and the port # was the default of **4000**, you could open a session by writing into the MS Windows Telnet screen:

```
open 123.45.67.89 4000
```

The oscilloscope will send a help screen to the computer when it is done connecting.

11. You can now type in a standard query, as found in the programmer manual, such as ***idn?**

The Telnet session window will respond by displaying a character string describing your instrument. You can type in more queries and view more results on this Telnet session window. You can find the syntax for relevant queries and related status codes in other sections of this manual.

NOTE. *Do not use the computer's backspace key during an MS Windows' Telnet session with the oscilloscope.*

Socket Server Terminal Protocol Mode Commands. Following are Tektronix Instrument Control Terminal Session Control commands:

!t <timeout> : set the response timeout in milliseconds.

!d : send device clear to the instrument.

!r : read response from instrument.

!h : print this usage info.

NOTE. *Commands containing a ? are treated as queries, and the responses are read automatically.*

Documentation Documentation for your instrument is available for download at www.tek.com.

Dynamic programmatic interface

This programmatic interface is dynamic. This means the instrument will not recognize certain commands until the objects referenced by those commands actually exist. For example, commands related to measurements are not recognized until measurements are added. Therefore, the response to a *LRN? query will not normally include the instrument's complete command set.

The following command groups are not available when the instrument is in its default state:

- Measurement
- Math
- Bus
- Search and Mark
- Plot

Adding an instance of one of those components will cause all commands related to that component to be recognized. For example, sending the [MEASUREMENT:ADDNew](#) command adds a measurement at which point the measurement commands will be recognized. Conversely, once all instances of a component have been deleted, the commands related to that component will no longer be recognized.

Implicit activation

When you send a command or query related to a dynamic object (such as Math1, Bus3, or Meas2) to the instrument and that instance does not yet exist, the instrument:

- creates a default instance with the name you specified
- adds all relevant commands and queries to the set of recognized commands
- responds to the command or query

Example 1 *RST followed by a *LRN? will not return any MATH<x> commands because in the default state, the instrument does not have any math waveforms. However querying :MATH:MATH1:DEFinition? will add MATH1 with the default math

expression Ch1 - Ch2. Then the query will return the expected result. Note that if Ch1 or Ch2 is not active, they will be activated as part of this action. A *LRN issued after this will return MATH commands in addition to other available commands.

Example 2 *RST followed by :MEASUREMENT:MEAS3:TYPE? creates a measurement named MEAS3 and return its type. Since the default type is Period, you will get Period as the response. A *LRN issued after this will return all MEASUREMENT:MEAS3 commands in addition to other available commands.

Not all commands in these groups start implicit activation. ADDNew, DELEte, and LIST commands do not result in implicit activation.

Creating, deleting, and listing dynamic instances

You can create a new default instance of a dynamic feature by using the ADDNew command. For example, :MEASUREMENT:ADDNew "Meas1" will create a new measurement named Meas1. Meas1 will be a Period measurement since the default type for measurements is Period. Note that you can change Meas1 type to any other supported type using :MEASUREMENT:MEAS1:TYPE command. The DELEte command deletes the named dynamic instance. For example :BUS:DELEte "B2" will delete a bus named B2 if it exists. You can delete only one instance at a time. The LIST command returns a list of all dynamic instances currently in existence. For example, if you have added three measurements named Meas1, Meas2 and Meas3, :MEASUREMENT:LIST? returns MEAS1, MEAS2, MEAS3.

Command Syntax

You can control the operations and functions of the oscilloscope through the Ethernet port or the USB 3.0 super speed (SS) device port using commands and queries. The related topics listed below describe the syntax of these commands and queries. The topics also describe the conventions that the oscilloscope uses to process them. See the *Command Groups* topic in the table of contents for a listing of the commands by command group, or use the index to locate a specific command.

Backus-Naur Form Notation

This documentation describes the commands and queries using Backus-Naur Form (BNF) notation. Refer to the following table for the symbols that are used.

Table 2-1: Symbols for Backus-Naur Form

Symbol	Meaning
< >	Defined element
=	Is defined as
	Exclusive OR
{ }	Group; one element is required
[]	Optional; can be omitted
. . .	Previous element(s) may be repeated

Command and Query Structure

Commands consist of set commands and query commands (usually called commands and queries). Commands modify oscilloscope settings or tell the oscilloscope to perform a specific action. Queries cause the oscilloscope to return data and status information.

Most commands have both a set form and a query form. The query form of the command differs from the set form by its question mark at the end. For example, the set command `ACQuire:MODE` has a query form `ACQuire:MODE?`. Not all commands have both a set and a query form. Some commands have set only and some have query only.

Messages

A command message is a command or query name followed by any information the oscilloscope needs to execute the command or query. Command messages may contain five element types, defined in the following table.

Table 2-2: Command Message Elements

Symbol	Meaning
<Header>	This is the basic command name. If the header ends with a question mark, the command is a query. The header may begin with a colon (:) character. If the command is concatenated with other commands, the beginning colon is required. Never use the beginning colon with command headers beginning with a star (*).
<Mnemonic>	This is a header subfunction. Some command headers have only one mnemonic. If a command header has multiple mnemonics, a colon (:) character always separates them from each other.
<Argument>	This is a quantity, quality, restriction, or limit associated with the header. Some commands have no arguments while others have multiple arguments. A <space> separates arguments from the header. A <comma> separates arguments from each other.
<Comma>	A single comma is used between arguments of multiple-argument commands. Optionally, there may be white space characters before and after the comma.
<Space>	A white space character is used between a command header and the related argument. Optionally, a white space may consist of multiple white space characters.

Commands Commands cause the oscilloscope to perform a specific function or change one of the settings. Commands have the structure:

```
[ : ] <Header> [ <Space> <Argument> [ <Comma> <Argument> ] . . . ]
```

A command header consists of one or more mnemonics arranged in a hierarchical or tree structure. The first mnemonic is the base or root of the tree and each subsequent mnemonic is a level or branch off the previous one. Commands at a higher level in the tree may affect those at a lower level. The leading colon (:) always returns you to the base of the command tree.

Queries Queries cause the oscilloscope to return status or setting information. Queries have the structure:

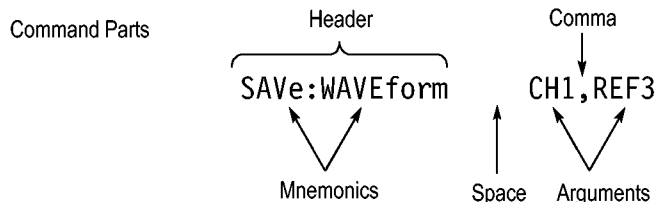
- `[<:><Header>`
- `[<:><Header>[<Space><Argument> [<Comma><Argument>]...]`

You can specify a query command at any level within the command tree unless otherwise noted. These branch queries return information about all the mnemonics below the specified branch or level.

Headers Use the `HEADER` command to control whether the oscilloscope returns headers as part of the query response. If header is on, the query response returns command headers, then formats itself as a valid set command. When header is off, the response includes only the values. This may make it easier to parse and extract the information from the response. The table below shows the difference in responses.

Table 2-3: Comparison of Header Off and Header On Responses

Query	Header Off	Header On
TIME?	"14:30:00"	:TIME "14:30:00"
ACQUIRE:NUMAVG?	100	:ACQUIRE:NUMAVG 100



Clearing the oscilloscope

You can clear the Output Queue and reset the oscilloscope to accept a new command or query by using the selected Device Clear (DCL) function.

Command Entry

The following rules apply when entering commands:

- You can enter commands in upper or lower case.
- You can precede any command with white space characters. White space characters include any combination of the ASCII control characters 00 through 09 and 0B through 20 hexadecimal (0 through 9 and 11 through 32 decimal).
- The oscilloscope ignores commands consisting of any combination of white space characters and line feeds.

Abbreviating

You can abbreviate many oscilloscope commands. Each command in this documentation shows the minimum acceptable abbreviations in capitals. For example, you can enter the command `ACQuire:NUMAVg` simply as `ACQ:NUMA` or `acq:numa`.

Abbreviation rules may change over time as new oscilloscope models are introduced. Thus, for the most robust code, use the full spelling.

If you use the `HEADer` command to have command headers included as part of query responses, you can further control whether the returned headers are abbreviated or are full-length with the `VERBoSe` command.

Concatenating

You can concatenate any combination of set commands and queries using a semicolon (;). The oscilloscope executes concatenated commands in the order received.

When concatenating commands and queries, you must follow these rules:

1. Separate completely different headers by a semicolon and by the beginning colon on all commands except the first one. For example, the commands `TRIGger:MODE NORMa1` and `ACQuire:NUMAVg 8`, can be concatenated into the following single command:

```
TRIGger:MODE NORMa1;:ACQuire:NUMAVg 8
```

2. If concatenated commands have headers that differ by only the last mnemonic, you can abbreviate the second command and eliminate the beginning colon. For example, you can concatenate the commands `ACQuire:MODE ENVe1ope` and `ACQuire:NUMAVg 8` into a single command:

```
ACQuire:MODE ENVe1ope; NUMAVg 8
```

The longer version works equally well:

```
ACQuire:MODE ENVe1ope;:ACQuire:NUMAVg 8
```

3. Never precede a star (*) command with a colon:

```
ACQuire:STATE 1;*OPC
```

Any commands that follow will be processed as if the star command was not there so the commands, `ACQuire:MODE ENVe1ope;*OPC;NUMAVg 8` will set the acquisition mode to envelope and set the number of acquisitions for averaging to 8.

4. When you concatenate queries, the responses to all the queries are concatenated into a single response message. For example, if the display graticule is set to Full and the display style is set to dotsonly, the concatenated query `DISPlay:GRATicule?;STYle?` will return the following.

If the header is on:

```
DISPLAY:GRATICULE FULL;:DISPLAY:STYLE DOTSONLY
```

If the header is off:

FULL;DOTSONLY

1. Set commands and queries may be concatenated in the same message. For example,

ACQuire:MODE SAMple;NUMAVg?;STATE?

is a valid message that sets the acquisition mode to sample. The message then queries the number of acquisitions for averaging and the acquisition state. Concatenated commands and queries are executed in the order received.

Here are some invalid concatenations:

DISPlay:STYle DOTsonly OFF;ACQuire:NUMAVg 8 (no colon before ACQuire)

DISPlay:GRAticule FULL;;STYle DOTSONLY OFF (extra colon before STYle.

DISPlay:GRAticule FULL;:*TRG (colon before a star (*) command)

ACQUIRE:FASTACQ:PALETTE TEMPerature;FASTAcq:STATE ON (levels of the mnemonics are different; either remove the second use of FASTACQ: or place :ACQUIRE in front of FASTAcq:STATE)

Terminating

This documentation uses <EOM> (End of Message) to represent a message terminator.

Table 2-4: End of Message Terminator

Symbol	Meaning
<EOM>	Message terminator

The end-of-message terminator must be the END message (EOI asserted concurrently with the last data byte). The last data byte may be an ASCII line feed (LF) character.

This oscilloscope does not support ASCII LF only message termination. The oscilloscope always terminates outgoing messages with LF and EOI.

Constructed Mnemonics

Some header mnemonics specify one of a range of mnemonics. For example, a channel mnemonic can be CH1, CH2, CH3, CH4, CH5, CH6, CH7, or CH8 depending on the number of FlexChannels in your instrument. You use these mnemonics in the command just as you do any other mnemonic. For example, there is a CH1:POSition command, and there is also a CH2:POSition command. In the command descriptions, this list of choices is abbreviated as CH<x>.

Bus Mnemonics Commands specify the bus to use as a mnemonic in the header.

Table 2-5: Bus Mnemonics

Symbol	Meaning
B<x>	A bus specifier; <x> is ≥1.

Channel Mnemonics Commands specify the channel to use as a mnemonic in the header.

Table 2-6: Channel Mnemonics

Symbol	Meaning
CH<x>	A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.
CH<x>_D<x>	A digital channel specifier; <x> in CH<x> is 1 through 8 and is limited by the number of FlexChannels in your instrument. <x> in D<x> is 0 through 7. Together they define a FlexChannel digital input.

Cursor Position Mnemonics When cursors are displayed, commands may specify which cursor of the pair to use.

Table 2-7: Cursor Mnemonics

Symbol	Meaning
CURSOR<x>	A cursor selector; <x> is must be 1 or 2.

Math Specifier Mnemonics Commands can specify the mathematical waveform to use as a mnemonic in the header.

Table 2-8: Math Specifier Mnemonics

Symbol	Meaning
MATH<x>	A math waveform specifier; <x> is ≥1.

Measurement Specifier Mnemonics Commands can specify which measurement to set or query as a mnemonic in the header.

Table 2-9: Measurement Specifier Mnemonics

Symbol	Meaning
MEAS<x>	A measurement specifier; <x> is ≥1.

Reference Waveform Mnemonics

Commands can specify the reference waveform to use as a mnemonic in the header.

Table 2-10: Reference Waveform Mnemonics

Symbol	Meaning
REF<x>	A reference waveform specifier; <x> is ≥ 1 .
REF<x>_D<x>	A digital reference waveform specifier; <x> is ≥ 1 .

View Mnemonics

Commands can specify the view to use as a mnemonic in the header.

Table 2-11: Waveview Mnemonics

Symbol	Meaning
WAVEView<x>	A waveview specifier; <x> must be equal to 1.
PLOTView<x>	A plotview specifier; <x> must be equal to 1.
MATHFFTView<x>	A mathfftview specifier; <x> must be equal to 1.

Search Mnemonics

Commands can specify a search to use as a mnemonic in the header.

Table 2-12: Search Mnemonics

Symbol	Meaning
SEARCH<x>	A Search specifier; <x> is ≥ 1 .

Zoom Mnemonics

Commands can specify a zoom to use as a mnemonic in the header.

Table 2-13: Zoom Mnemonics

Symbol	Meaning
ZOOM<x>	A zoom specifier; <x> must be equal to 1.

Argument Types

Commands use arguments such as enumeration, numeric, quoted string and block. Each of these arguments are listed in detail below.

Enumeration

Enter these arguments as unquoted text words. Like key words, enumeration arguments follow the same convention where the portion indicated in uppercase is required and that in lowercase is optional.

For example: ACQuire:MODE SAMPlE

Numeric Many oscilloscope commands require numeric arguments. The syntax shows the format that the oscilloscope returns in response to a query. This is also the preferred format when sending the command to the oscilloscope, though any of the formats will be accepted. This documentation represents these arguments as described below.

Table 2-14: Numeric Arguments

Symbol	Meaning
<NR1>	Signed integer value
<NR2>	Floating point value without an exponent
<NR3>	Floating point value with an exponent
<bin>	Signed or unsigned integer in binary format

Most numeric arguments will be automatically forced to a valid setting, by either rounding or truncating, when an invalid number is input, unless otherwise noted in the command description.

Quoted String Some commands accept or return data in the form of a quoted string, which is simply a group of ASCII characters enclosed by a single quote (') or double quote ("). The following is an example of a quoted string: "This is a quoted string". This documentation represents these arguments as follows:

Table 2-15: Quoted String Argument

Symbol	Meaning
<QString>	Quoted string of ASCII text

A quoted string can include any character defined in the 7-bit ASCII character set. Follow these rules when you use quoted strings:

1. Use the same type of quote character to open and close the string. For example: "this is a valid string".
2. You can mix quotation marks within a string as long as you follow the previous rule. For example: "this is an 'acceptable' string".
3. You can include a quote character within a string by repeating the quote. For example: "here is a "" mark".
4. Strings can have upper or lower case characters.
5. A carriage return or line feed embedded in a quoted string does not terminate the string. The return is treated as another character in the string.
6. The maximum length of a quoted string returned from a query is 1000 characters.

Here are some invalid strings:

- "Invalid string argument" (quotes are not of the same type)
- "test<EOI>" (termination character is embedded in the string)

Block Some commands use a block argument form to define a range or type of value, as defined in the table below.

Table 2-16: Block Argument

Symbol	Meaning
<NZDig>	A nonzero digit character in the range of 1–9
<Dig>	A digit character, in the range of 0–9
<DChar>	A character with the hexadecimal equivalent of 00 through FF (0 through 255 decimal)
<Block>	A block of data bytes defined as: <Block> ::= {#<NZDig><Dig>[<Dig>...][<DChar>...][#0[<DChar>...]<terminator>]}

<NZDig> specifies the number of <Dig> elements that follow. Taken together, the <NZDig> and <Dig> elements form a decimal integer that specifies how many <DChar> elements follow.

Command groups

The programmable interface conforms to Tektronix standard codes and formats except where noted. The interface also conforms to IEEE Std 488.2-1987 except where noted.

Acquisition command group

Acquisition commands set up the modes and functions that control how the instrument acquires signals and processes them into waveforms. Using these commands for acquiring waveforms, you can do the following:

- Start and stop acquisitions.
- Control whether each waveform is simply acquired, averaged, or enveloped over successive acquisitions of that waveform.
- Set the controls or conditions that start and stop acquisitions.
- Control acquisition of acquired channel waveforms.
- Set acquisition parameters.

Table 2-17: Acquisition commands

Command	Description
ACquire?	Queries the current acquisition state.
ACquire:FASTAcq:PALEtte	Sets or queries the waveform grading for fast acquisition mode.
ACquire:FASTAcq:STATE	Sets or queries the state of fast acquisition mode.
ACquire:MAXSamplerate?	Returns the maximum real-time sample rate.
ACquire:MODE	Sets or queries the acquisition mode.
ACquire:NUMAcq?	Returns the number of waveform acquisitions that have occurred since starting acquisition with the <code>ACquire:STATE RUN</code> command.
ACquire:NUMAVg	Sets or queries number of acquisitions for an averaged waveform.
ACquire:SEquence:CURrent?	In single sequence acquisition mode, this query returns the number of acquisitions or measurements in the sequence completed so far.
ACquire:SEquence:MODE	In single sequence acquisition, the single sequence stop after count is based on number of acquisitions or measurements. Number of acquisitions is the only mode supported for this product.
ACquire:SEquence:NUMSEquence	In single sequence acquisition mode, specify the number of acquisitions or measurements that comprise the sequence.
ACquire:STATE	Starts, stops, or returns acquisition state.
ACquire:STOPAfter	Sets or queries whether the acquisition is continuous or single sequence.

AFG Command Group

Use the AFG commands for Arbitrary Function Generator functionality. Requires option AFG.

Table 2-18: AFG commands

Command	Description
AFG:AMPLitude	Sets (or queries) the AFG amplitude in volts, peak to peak.
AFG:ARBitrary:SOUrce	Sets or queries the source name for the Arbitrary Waveform.
AFG:FREQuency	Sets (or queries) the AFG frequency, in Hz.
AFG:FUNcTion	Sets (or queries) which AFG function to execute.
AFG:HIGHLevel	Sets (or queries) the high level value of the output waveform, in volts.
AFG:LOWLevel	Sets (or queries) the low level value of the output waveform, in volts.
AFG:NOISEAdd:PERCent	Sets (or queries) the AFG additive noise level as a percentage.
AFG:NOISEAdd:STATE	Sets (or queries) the AFG additive noise state.
AFG:OFFSet	Sets (or queries) the AFG offset value, in volts.
AFG:OUTPut:LOAd:IMPEDance	Sets (or queries) the AFG output load impedance.
AFG:OUTPut:STATE	Sets (or queries) the AFG output state.
AFG:PERIod	Sets (or queries) the period of the AFG waveform, in seconds.
AFG:PULse:WIDth	Sets (or queries) the AFG pulse width, in seconds.
AFG:RAMP:SYMmetry	Sets (or queries) the AFG ramp symmetry as a percentage.
AFG:SQUare:DUty	Sets (or queries) the AFG duty cycle, as a percentage.

Alias command group

Alias commands allow you to define new commands as a sequence of standard commands. You might find this useful when repeatedly using the same commands to perform certain tasks like setting up measurements.

Aliases are similar to macros but do not include the capability to substitute parameters into alias bodies. The alias mechanism obeys the following rules:

- The alias name must consist of a valid IEEE 488.2 message unit, which may not appear in a message preceded by a colon, comma, or a command or query program header.
- The alias name may not appear in a message followed by program data, a colon, comma, or question mark.
- An alias name must be distinct from any keyword or keyword short form.
- An alias name cannot be redefined without first being deleted using one of the alias deletion functions.
- Alias names do not appear in response messages.
- The Alias commands are defined in Tektronix Standard Codes and Formats. Deviations between that standard and what is specified here will be considered errors unless specifically noted in the command description in this document.

Table 2-19: Alias commands

Command	Description
ALias	Sets or queries the alias state.
ALias:CATalog?	Returns a list of the currently defined alias labels.
ALias:DEFine	Assigns a sequence of program messages to an alias label.
ALias:DELEte	Removes a specified alias.
ALias:DELEte:ALL	Deletes all existing aliases.
ALias:DELEte:NAME	Removes a specified alias.
ALias:STATE	Sets or queries the alias state.

Bus command group

Use the commands in the Bus Command Group to configure a bus. These commands let you:

- Specify the bus type.
- Specify the signals to be used in the bus.
- Specify its display style.

NOTE. Bus commands are present once a bus has been added.

Bus Mnemonics Commands specify the bus to use as a mnemonic in the header.

Table 2-20: Bus mnemonics

Symbol	Meaning
B<x>	A bus specifier; <x> is ≥ 1 .

Table 2-21: Bus commands

Command	Description
BUS:ADDNew	Adds the specified bus.
BUS:B<x>:ARINC429A:SOUrce	Sets or queries the source for the specified ARINC429 bus.
BUS:B<x>:ARINC429A:THRESHold	Sets or queries the ARINC429 upper threshold for the specified bus.
BUS:B<x>:ARINC429A:POLARITY	Sets or queries the source polarity for the specified ARINC429 bus.
BUS:B<x>:ARINC429A:DATAFORmat	Sets or queries the format of the DATA field for the specified ARINC429 bus.
BUS:B<x>:ARINC429A:BITRate	Sets or queries the ARINC429 bit rate for the specified bus.
BUS:B<x>:ARINC429A:BITRate:CUSTom	Sets or queries the ARINC429 custom bit rate for the specified bus.
BUS:B<x>:AUDio:BITDelay	Sets or queries the number of delay bits for the specified AUDIO bus.
BUS:B<x>:AUDio:BITOrder	Specifies the bit order for the specified AUDIO bus.
BUS:B<x>:AUDio:CLOCK:POLarity	Sets or queries the clock polarity for the specified AUDIO bus.
BUS:B<x>:AUDio:CLOCK:SOUrce	Sets or queries the clock source waveform for the specified AUDIO bus.
BUS:B<x>:AUDio:CLOCK:THReshold	Sets or queries the Audio Clock source threshold for the specified bus.
BUS:B<x>:AUDio:DATA:POLarity	Specifies the data polarity for the specified AUDIO bus.
BUS:B<x>:AUDio:DATA:SIZE	Specifies the number of bits per word for the specified AUDIO bus.
BUS:B<x>:AUDio:DATA:SOUrce	Specifies the audio data source waveform for the specified AUDIO bus.
BUS:B<x>:AUDio:DATA:THReshold	Sets or queries the Audio Data source threshold for the specified bus.
BUS:B<x>:AUDio:DATA:WORDSize	Sets or queries the Audio bits per word for the specified bus.
BUS:B<x>:AUDio:FRAME:CLOCKBITSPERCHANNEL	Sets or queries the Audio bits of sync width for the specified bus.
BUS:B<x>:AUDio:FRAME:SIZE	Specifies the number of channels in each frame for the specified AUDIO bus.
BUS:B<x>:AUDio:TYPE	Specifies the audio format (type) for the specified AUDIO bus.

Table 2-21: Bus commands (cont.)

Command	Description
BUS:B<x>:AUDio:WORDSel:POLarity	Sets or queries the word select polarity for the specified AUDIO bus.
BUS:B<x>:AUDio:WORDSel:SOUrce	Specifies the word select source waveform for the AUDIO bus.
BUS:B<x>:AUDio:WORDSel:THReshold	Sets or queries the Audio Word Select source threshold for the specified bus.
BUS:B<x>:CAN:BITRate	Sets or queries the CAN bit rate.
BUS:B<x>:CAN:BITRate:VALue	Sets or queries CAN bit rate.
BUS:B<x>:CAN:FD:BITRate	Sets or queries the increased data phase bit rate used by CAN FD packets on the specified CAN bus.
BUS:B<x>:CAN:FD:BITRate:CUSTom	Sets or queries the custom bit rate for the increased data phase of CAN FD packets on the specified CAN bus.
BUS:B<x>:CAN:SAMPLEpoint	Sets or queries the sample point for the specified CAN bus.
BUS:B<x>:CAN:SIGNAL	Sets or queries the signal type for the specified CAN bus.
BUS:B<x>:CAN:SOUrce	Sets or queries the CAN source channel.
BUS:B<x>:CAN:STANDard	Sets or queries which CAN standard specification to analyze the specified CAN bus with.
BUS:B<x>:CAN:THReshold	Sets or queries the source channel threshold for the specified CAN bus.
BUS:B<x>:DISplay:FORMat	Sets or queries how the data is represented in the busform for the specified bus.
BUS:B<x>:DISplay:LAYout	This command sets or queries the format a bus layer should use.
BUS:B<x>:ETHERnet:DATAMINUSTHRESHold	Sets or queries the Ethernet DATA Minus source threshold for the specified bus.
BUS:B<x>:ETHERnet:DATAPLUSTHRESHold	Sets or queries the Ethernet DATA Plus source threshold for the specified bus.
BUS:B<x>:ETHERnet:IPVFOUR	Sets or queries whether IPV4 packets are available for triggering on Ethernet.
BUS:B<x>:ETHERnet:LOWTHRESHold	Sets or queries the Ethernet DATA source Low threshold for the specified bus.
BUS:B<x>:ETHERnet:QTAGGING	Sets or queries whether Q-Tagging packets are available for triggering on Ethernet.
BUS:B<x>:ETHERnet:SIGNALTYpe	Sets or queries the Ethernet signal type for the specified bus.
BUS:B<x>:ETHERnet:SOUrce	Specifies the Ethernet data source for differential input.
BUS:B<x>:ETHERNET:SOUrce:DMINus	Sets or queries the Ethernet DMINus source.
BUS:B<x>:ETHERNET:SOUrce:DPLUs	Sets or queries the Ethernet DPLUs source.
BUS:B<x>:ETHERnet:THRESHold	Sets or queries the Ethernet DATA source High threshold for the specified bus.
BUS:B<x>:ETHERnet:TYPe	Specifies the Ethernet standard type: 10Base-T or 100Base-T.
BUS:B<x>:FLEXray:BITRate	Sets or queries the FlexRay bus bit rate.
BUS:B<x>:FLEXray:BITRate:CUSTom	Sets or queries the FlexRay custom bit rate for the specified bus.
BUS:B<x>:FLEXray:CHannel	Sets or queries the FlexRay bus channel.
BUS:B<x>:FLEXray:LOWTHRESHold	Sets or queries the FlexRay data source low threshold for the specified bus.
BUS:B<x>:FLEXray:SIGNAL	Sets or queries the FlexRay probe.
BUS:B<x>:FLEXray:SOUrce	Sets or queries the FlexRay bus source.
BUS:B<x>:FLEXray:SOUrce:TXRX	Sets or queries the FlexRay TxRx data source for the specified bus.
BUS:B<x>:FLEXray:THRESHold	Sets or queries the FlexRay data source threshold for the specified bus.
BUS:B<x>:FLEXray:TXRXTHRESHold	Sets or queries the FlexRay data source TxRx threshold for the specified bus.

Table 2-21: Bus commands (cont.)

Command	Description
BUS:B<x>:I2C:CLOCK:SOUrce	Sets or queries the I2C clock (SCLK) source for the specified bus.
BUS:B<x>:I2C:CLOCK:THReshold	Sets or queries the I2C Clock (SCLK) source threshold for the specified bus.
BUS:B<x>:I2C:DATa:SOUrce	Sets or queries the I2C data (SDA) source for the specified bus.
BUS:B<x>:I2C:DATa:THReshold	Sets or queries the I2C Data (SDA) source threshold for the specified bus.
BUS:B<x>:I2C:RWINADDR	Determines whether decoded I2C slave addresses are pure seven-bit values, or have the R/W* combined with them.
BUS:B<x>:LABel:COLor	Sets or queries the color of the specified bus label.
BUS:B<x>:LABel:FONT:BOLD	Sets or queries the bold state of the specified bus label.
BUS:B<x>:LABel:FONT:ITALic	Sets or queries the italic state of the specified bus label.
BUS:B<x>:LABel:FONT:SIZE	Sets or queries the font size of the specified bus label.
BUS:B<x>:LABel:FONT:TYPE	Sets or queries the font type of the specified bus label, such as Arial or Times New Roman.
BUS:B<x>:LABel:FONT:UNDERline	Sets or queries the underline state of the specified bus label.
BUS:B<x>:LABel:name	Sets or queries the waveform label for the specified bus.
BUS:B<x>:LABel:XPOS	Sets or queries the x-position of the specified bus' label.
BUS:B<x>:LABel:YPOS	Sets or queries the y-position of the specified bus' label.
BUS:B<x>:LIN:BITRate	Sets or queries the LIN bus bit rate.
BUS:B<x>:LIN:IDFORmat	Sets or queries LIN bus id format.
BUS:B<x>:LIN:POLarity	Sets or queries the LIN bus polarity.
BUS:B<x>:LIN:SAMPLEpoint	Specifies the point to sample during each bit period, as a percent, for the specified LIN bus.
BUS:B<x>:LIN:SOUrce	Sets or queries sets the LIN bus source.
BUS:B<x>:LIN:SOUrce:THReshold	Sets or queries the LIN source threshold for the specified bus.
BUS:B<x>:LIN:STANDard	Sets or queries the LIN bus standard.
BUS:B<x>:LIN:BITRate:CUSTom	Sets or queries LIN custom bit rate for the specified bus.
BUS:B<x>:MIL1553B:SOUrce	Sets or queries the source for the specified MIL-STD-1553 bus.
BUS:B<x>:MIL1553B:THRESHold	Sets or queries the MIL-STD-1553 upper threshold for the specified bus.
BUS:B<x>:MIL1553B:LOWTHRESHold	Sets or queries the MIL-STD-1553 lower threshold for the specified bus.
BUS:B<x>:MIL1553B:POLarity	Sets or queries the source polarity for the specified MIL-STD-1553 bus.
BUS:B<x>:MIL1553B:RESPonsetime:MINimum	Sets or queries the minimum response time to a valid command issued for the specified MIL-STD-1553 bus.
BUS:B<x>:MIL1553B:RESPonsetime:MAXimum	Sets or queries the maximum response time to a valid command issued for the specified MIL-STD-1553 bus.
BUS:B<x>:PARAllel:ALLTHResholds	Sets or queries the threshold for all sources for the parallel bus.
BUS:B<x>:PARAllel:ALLTHResholds:APPLY	Sets all of the data source thresholds to the value of the allMRefs parameter for the parallel bus.
BUS:B<x>:PARAllel:BIT<n>SOUrce	Sets or queries the specified bit source for specified parallel bus.
BUS:B<x>:PARAllel:BIT<n>SOUrce:THReshold	Sets or queries the specified bit source threshold for the specified parallel bus.

Table 2-21: Bus commands (cont.)

Command	Description
BUS:B<x>:PARAllel:CLOCK:EDGE	Determines which edges of its clock signal cause a clocked parallel bus to sample new states.
BUS:B<x>:PARAllel:CLOCK:ISCLOCKED	Determines whether the bus operates in a clocked or asynchronous fashion.
BUS:B<x>:PARAllel:CLOCK:SOURce	Sets or queries the Parallel clock source for the specified bus.
BUS:B<x>:PARAllel:CLOCK:SOURce:THReshold	Sets or queries the bit source threshold for the parallel bus.
BUS:B<x>:RS232C:BITRate	Sets or queries the RS-232 bit rate for the specified bus.
BUS:B<x>:RS232C:BITRate:CUSTom	Sets or queries the RS232 custom bit rate for the specified bus.
BUS:B<x>:RS232C:DATABits	Sets or queries the number of RS-232 data bits for the specified bus.
BUS:B<x>:RS232C:DELMiter	Sets or queries the RS-232 delimiting value for a packet on the specified bus.
BUS:B<x>:RS232C:DISplaymode	Sets or queries the display mode for the specified bus.
BUS:B<x>:RS232C:PARity	Sets or queries the RS-232 parity for the specified bus.
BUS:B<x>:RS232C:POLarity	Sets or queries the RS-232 polarity for the specified bus.
BUS:B<x>:RS232C:SOURce	Sets or queries the RS-232 polarity for the specified bus.
BUS:B<x>:RS232C:SOURce:THReshold	Sets or queries the RS232 source threshold for the specified bus.
BUS:B<x>:SENT:CHANWidth	Sets or queries SENT fast channel bit widths for the specified bus.
BUS:B<x>:SENT:NIBBLECount	Sets or queries SENT data nibbles for the specified bus.
BUS:B<x>:SENT:NUMCHANnel	Sets or queries SENT fast data channels for the specified bus.
BUS:B<x>:SENT:PAUSEPULSe	Sets or queries SENT pause pulse for the specified bus.
BUS:B<x>:SENT:POLARITY	Sets or queries SENT Idle State signal polarity for the specified bus.
BUS:B<x>:SENT:SLOW	Sets or queries the SENT slow channel configuration for the specified bus.
BUS:B<x>:SENT:SOURce	Sets or queries the SENT DATA source for the specified bus.
BUS:B<x>:SENT:THRESHold	Sets or queries the SENT DATA source threshold for the specified bus.
BUS:B<x>:SENT:TICKTIME	Sets or queries the SENT bus Clock Tick parameter for the specified bus.
BUS:B<x>:SENT:TICKTOLerance	Sets or queries the SENT bus Tick Tolerance percent parameter for the specified bus.
BUS:B<x>:SPI:BITOrder	Sets or queries the shift direction used to de-serialize data for the SPI mode of the bus.
BUS:B<x>:SPI:CLOCK:POLarity	Sets or queries the SPI clock (SCLK) polarity for the specified bus.
BUS:B<x>:SPI:CLOCK:SOURce	Sets or queries the SPI clock (SCLK) source for the specified bus.
BUS:B<x>:SPI:CLOCK:THReshold	Sets or queries the SPI Clock (SCLK) source threshold for the specified bus.
BUS:B<x>:SPI:DATA:POLarity	Sets or queries the SPI data (DATA) polarity for the specified bus.
BUS:B<x>:SPI:DATA:SIZE	Sets or queries the number of bits per word for the specified bus.
BUS:B<x>:SPI:DATA:SOURce	Sets or queries the SPI data (DATA) source for the specified bus.
BUS:B<x>:SPI:DATA:THReshold	Sets or queries the SPI Data (DATA) source threshold for the specified bus.
BUS:B<x>:SPI:FRAMING	Sets or queries the SPI bus framing.
BUS:B<x>:SPI:IDLETime	Sets or queries the SPI bus idle time.
BUS:B<x>:SPI:SElect:POLarity	Sets or queries the SPI Slave Select (SS) polarity for the specified bus.

Table 2-21: Bus commands (cont.)

Command	Description
BUS:B<x>:SPI:SElect:SOUrce	Sets or queries the SPI Slave Select (SS) source for the specified bus.
BUS:B<x>:SPI:SElect:THReshold	Sets or queries the SPI Select (SS) source threshold for the specified bus.
BUS:B<x>:SPMI:SCLk:SOUrce	This command sets or queries the SPMI Clock (SCLK) source for the specified bus.
BUS:B<x>:SPMI:SCLk:THReshold	This command sets or queries the SPMI Clock (SCLK) source threshold for the specified bus.
BUS:B<x>:SPMI:SDATa:SOUrce	This command sets or queries the SPMI Data (SDATA) source for the specified bus.
BUS:B<x>:SPMI:SDATa:THReshold	This command sets or queries the SPMI Data (SDATA) source threshold for the specified bus.
BUS:B<x>:TYPE	Sets or queries the bus type specified.
BUS:B<x>:USB:BITRate	Sets or queries the USB bit rate for the specified bus.
BUS:B<x>:USB:SOUrce	Sets or queries the USB Data Source for the specified bus.
BUS:B<x>:USB:SOUrce:DMINus	Sets or queries the USB Data Source for D- input for the specified bus.
BUS:B<x>:USB:SOUrce:DPLUS	Sets or queries the USB Data Source for D+ input for the specified bus.
BUS:B<x>:USB:DATAMINUSTHRESHold	Sets or queries the USB DATA Minus source threshold for the specified bus.
BUS:B<x>:USB:DATAPLUSTHRESHold	Sets or queries the USB DATA Plus source threshold for the specified bus.
BUS:B<x>:USB:LOWTHRESHold	Sets or queries the USB DATA source Low threshold for the specified bus.
BUS:B<x>:USB:SIGNALTYpe	Sets or queries the USB signal type for the specified bus.
BUS:B<x>:USB:THRESHold	Sets or queries the USB DATA source High threshold for the specified bus.
BUS:DELeTe	Deletes the specified bus.
BUS:LIST?	Lists all currently defined bus.
BUSTABle:ADDNew	Adds the specified bus table.
BUSTABle:DELeTe	Deletes the specified bus table.
BUSTABle:LIST?	Lists all currently defined bus tables.
DISPlay:WAVEView<x>:BUS:B<x>:STATE	Sets or queries the state of the specified bus in the specified Waveview.
DISPlay:WAVEView<x>:BUS:B<x>:VERTical:POSition	Sets or queries the vertical position of the specified bus in the specified Waveview.

Calibration command group

The Calibration commands provide information about the current state of instrument calibration and allow you to initiate signal path calibration (SPC).

NOTE. When running SPC through the remote interface, calibration status cannot be obtained until after the SPC completes, which can take several minutes. Any remote command that performs an action on the oscilloscope is also disabled until the SPC is complete.

Table 2-22: Calibration commands

Command	Description
*CAL?	Instructs the instrument to perform signal path calibration and returns the calibration status when complete. Takes several minutes to run.
CALibrate?	Returns the calibration status.
CALibrate:INTERNAL	Starts the signal path calibration. Takes several minutes to run.
CALibrate:INTERNAL:START	Starts the signal path calibration.
CALibrate:INTERNAL:STATUS?	Returns the status of the signal path calibration.
CALibrate:PWRUpstatus?	Returns the current status of the power-up calibration.
TOUCHSCreen:CALibrate	Launches the touchscreen calibration.
TOUCHSCreen:STATe	Sets or queries the enabled state of the touch screen.

Cursor command group

Use the commands in the Cursor Command Group to control the cursor display and readout. You can use these commands to control the setups for each cursor, such as waveform source, and cursor position.

You can also use the commands to select one of the following cursor functions:

- **Off.** Shuts off the display of all cursors.
- **Vertical bars.** Displays vertical bar cursors, which provide traditional horizontal unit readouts for Cursor 1 (bar1), Cursor 2 (bar2), the delta between them, and 1/delta (results in frequency when the horizontal unit is time). Vertical bars are another name for vertical screen cursors.
- **Horizontal bars.** Displays horizontal bar cursors, which provide traditional vertical unit readouts for Cursor 1 (bar1), Cursor 2 (bar2), and the delta between them. Horizontal bars are another name for horizontal screen cursors.
- **Waveform cursors.** Consists of two cursors you can independently assign to a waveform. Waveform cursors enable you to conveniently measure waveform amplitude and time at specific points on the waveform. In XY or XYZ format, waveform cursors indicate the amplitude position of an XY pair (Ch1 vs Ch2 voltage, where Ch1 is the X axis and Ch2 is the Y axis) relative to the trigger.
- **Screen cursors.** Consist of two pairs of horizontal and vertical bar cursors. You can use these cursors to indicate an arbitrary position within the waveform display area. Screen cursors are basically just turning on horizontal bars and vertical bars at the same time. These cursors have no association with any waveform, except that they inherit the color of the waveform they are assigned to.

NOTE. *Cursor commands are available once a view has been added.*

Table 2-23: Cursor commands

Command	Description
DISplay:MATHFFTView<x>:CURSor:ASOURce?	Queries the cursor source for cursor A in the specified Math-FFT waveform.
DISplay:MATHFFTView<x>:CURSor:BSOURce?	Queries the cursor source for cursor B in the specified Math-FFT waveform.
DISplay:MATHFFTView<x>:CURSor:DDT?	Queries the delta V over delta T cursor readout value in the specified Math-FFT waveform.
DISplay:MATHFFTView<x>:CURSor:FUNCTION	Sets or queries the cursor type in the specified Math-FFT waveform.
DISplay:MATHFFTView<x>:CURSor:HBArs:APOSition	Sets or returns the horizontal cursor A position in the specified Math-FFT waveform.

Table 2-23: Cursor commands (cont.)

Command	Description
DISPlay:MATHFFTView<x>:CURSor:HBArs:AUNIts?	Queries cursor A vertical units of the specified cursor in the specified view.
DISPlay:MATHFFTView<x>:CURSor:HBArs:BPOsition	Sets or returns the vertical cursor B position in the specified Math-FFT waveform.
DISPlay:MATHFFTView<x>:CURSor:HBArs:BUNIts?	Queries the cursor B vertical units in the specified Math-FFT waveform.
DISPlay:MATHFFTView<x>:CURSor:HBArs:DELTA?	Queries the delta V cursor readout value of the specified cursor in the specified view.
DISPlay:MATHFFTView<x>:CURSor:MODE	Sets or returns the cursor tracking mode in the specified Math-FFT waveform.
DISPlay:MATHFFTView<x>:CURSor:ONEOVERDELTAVALUE?	Queries the one over delta T cursor readout value in the specified Math-FFT waveform.
DISPlay:MATHFFTView<x>:CURSor:SCREEN:AXPOsition	Sets or returns the horizontal cursor A position in the specified Math-FFT waveform.
DISPlay:MATHFFTView<x>:CURSor:SCREEN:AYPOsition	Sets or returns the vertical cursor A position in the specified Math-FFT waveform.
DISPlay:MATHFFTView<x>:CURSor:SCREEN:BXPOsition	Sets or returns the horizontal cursor B position of the specified cursor in the specified view.
DISPlay:MATHFFTView<x>:CURSor:SCREEN:BYPOsition	Sets or returns the vertical cursor B position in the specified Math-FFT waveform.
DISPlay:MATHFFTView<x>:CURSor:STATE	Sets or queries the visible state of cursors in the specified Math-FFT waveform.
DISPlay:MATHFFTView<x>:CURSor:VBArS:APOsition	Sets or queries the horizontal cursor A position in the specified Math-FFT waveform.
DISPlay:MATHFFTView<x>:CURSor:VBArS:AUNIts?	Queries the vertical cursor A measurement units for the specified Math-FFT view.
DISPlay:MATHFFTView<x>:CURSor:VBArS:BPOsition	Sets or queries the horizontal cursor B position in the specified Math-FFT waveform.
DISPlay:MATHFFTView<x>:CURSor:VBArS:BUNIts?	Queries the vertical cursor B measurement units for the specified Math-FFT view.
DISPlay:MATHFFTView<x>:CURSor:VBArS:DELTA?	Queries the delta T cursor readout value of the specified cursor in the specified Math-FFT waveform.
DISPlay:MATHFFTView<x>:CURSor:WAVEform:APOsition	Sets or queries the waveform cursor A position in the specified plot view.
DISPlay:MATHFFTView<x>:CURSor:WAVEform:BPOsition	Sets or queries the waveform cursor B position in the specified plot view.
DISPlay:PLOTView<x>:CURSor:ASOUrce?	Queries the cursor source for plot cursor A.
DISPlay:PLOTView<x>:CURSor:BSOUrce?	Queries the cursor source for plot cursor B.
DISPlay:PLOTView<x>:CURSor:DDT?	Queries the delta V over delta T cursor readout value of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:FUNCTION	Sets or queries the cursor mode of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:HBArs:APOsition	Sets or queries the vertical cursor A position of the specified cursor in the specified view.

Table 2-23: Cursor commands (cont.)

Command	Description
DISPlay:PLOTView<x>:CURSor:HBArs:AUNIts?	Sets or queries the cursor A vertical units of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:HBArs:BPOStion	Sets or queries the vertical cursor B position of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:HBArs:BUNIts?	Sets or queries the cursor B vertical units of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:HBArs:DELTA?	Sets or queries the delta V cursor readout value of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:MODE	Sets or queries the cursor tracking mode of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:ONEOVERDELTAVALUE?	Sets or queries the one over delta T cursor readout value of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:SCREEN:AXPOStion	Sets or queries the horizontal cursor A position of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:SCREEN:AYPOStion	Sets or queries the vertical cursor A position of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:SCREEN:BXPOStion	Sets or queries the horizontal cursor B position of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:SCREEN:BYPOStion	Sets or queries the vertical cursor B position of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:SPLITMODE	Sets or queries the cursor source mode in the specified view.
DISPlay:PLOTView<x>:CURSor:STATE	Sets or queries the visible state of the cursor of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:VBArS:APOStion	Sets or queries the horizontal cursor A position of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:VBArS:BPOStion	Sets or queries the horizontal cursor B position of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:VBArS:DELTA?	Queries the delta T cursor readout value of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:VBArS:UNIts?	Queries the VBArS cursor readout units of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:WAVEform:APOStion	Sets or queries the horizontal cursor A position of the specified cursor in the specified view.
DISPlay:PLOTView<x>:CURSor:WAVEform:BPOStion	Sets or queries the horizontal cursor B position of the specified cursor in the specified view.
DISPlay:REFFFTView<x>:CURSor:ASOURce?	Queries the cursor source for plot cursor A
DISPlay:REFFFTView<x>:CURSor:BSOURce?	Queries the cursor source for plot cursor B.
DISPlay:REFFFTView<x>:CURSor:DDT?	Queries the delta V over delta T cursor readout value of the specified cursor in the specified view.
DISPlay:REFFFTView<x>:CURSor:FUNCTion	Sets or queries the cursor type of the specified cursor in the specified view.

Table 2-23: Cursor commands (cont.)

Command	Description
DISplay:REFFFTView<x>:CURSor:HBArs:APOSition	Sets or queries the vertical cursor A position of the specified cursor in the specified view.
DISplay:REFFFTView<x>:CURSor:HBArs:AUNIts?	Queries cursor A vertical units of the specified cursor in the specified view.
DISplay:REFFFTView<x>:CURSor:HBArs:BPOSition	Sets or queries the vertical cursor B position of the specified cursor in the specified view.
DISplay:REFFFTView<x>:CURSor:HBArs:BUNIts?	Queries the cursor B vertical units of the specified cursor in the specified view.
DISplay:REFFFTView<x>:CURSor:HBArs:DELta?	Queries the delta V cursor readout value of the specified cursor in the specified view.
DISplay:REFFFTView<x>:CURSor:MODE	Sets or queries the cursor tracking mode of the specified cursor in the specified view.
DISplay:REFFFTView<x>:CURSor:ONEOVERDELTAVALUE?	Queries the one over delta T cursor readout value of the specified cursor in the specified view.
DISplay:REFFFTView<x>:CURSor:SCREEN:AXPOSition	Sets or queries the horizontal cursor A position of the specified cursor in the specified view.
DISplay:REFFFTView<x>:CURSor:SCREEN:AYPOSition	Sets or queries the vertical cursor A position of the specified cursor in the specified view.
DISplay:REFFFTView<x>:CURSor:SCREEN:BXPOSition	Sets or queries the horizontal cursor B position of the specified cursor in the specified view.
DISplay:REFFFTView<x>:CURSor:SCREEN:BYPOSition	Sets or queries the vertical cursor B position of the specified cursor in the specified view.
DISplay:REFFFTView<x>:CURSor:SPLITMODE	Sets or queries whether both cursors have same or different source.
DISplay:REFFFTView<x>:CURSor:STATE	Sets or queries the visible state of the cursor of the specified cursor n the specified view.
DISplay:REFFFTView<x>:CURSor:VBArS:APOSition	Sets or queries the horizontal cursor A position of the specified cursor in the specified view.
DISplay:REFFFTView<x>:CURSor:VBArS:BPOSition	Sets or queries the horizontal cursor B position of the specified cursor in the specified view.
DISplay:REFFFTView<x>:CURSor:VBArS:DELta?	Queries the delta T cursor readout value of the specified cursor in the specified view.
DISplay:REFFFTView<x>:CURSor:VBArS:UNIts?	Queries cursor A vertical units of the specified cursor in the specified view.
DISplay:REFFFTView<x>:CURSor:WAVEform:AHPOSition?	Queries the value of the cursor A horizontal position.
DISplay:REFFFTView<x>:CURSor:WAVEform:APOSition	Sets or returns the waveform cursor A position in the specified plot view.
DISplay:REFFFTView<x>:CURSor:WAVEform:AVPOSition?	Queries the value of the cursor A vertical position.
DISplay:REFFFTView<x>:CURSor:WAVEform:BHPOSition?	Queries the value of the cursor B horizontal position.

Table 2-23: Cursor commands (cont.)

Command	Description
DISPlay:REFFFTView<x>:CURSor:WAVEform:BPOStion	Sets or returns the waveform cursor B position in the specified plot view.
DISPlay:REFFFTView<x>:CURSor:WAVEform:BVPOStion?	Queries the value of the cursor B vertical position.
DISPlay:WAVEView<x>:CURSor?	Queries the cursor parameters for the specified waveview.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>?	Queries the cursor parameters for the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>:ASOURce	Sets or queries the cursor A source of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>:BSOURce	Sets or queries the cursor B source of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>:DDT?	Returns the delta V over delta T cursor readout value of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>:FUNCTion	Sets or queries the cursor mode (SCREEN or DATA) of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>:HBArS:APOStion	Sets or queries the vertical cursor A position of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>:HBArS:AUNIts?	Queries the cursor A vertical units of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>:HBArS:BPOStion	Sets or queries the vertical cursor B position of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>:HBArS:BUNIts?	Queries the cursor B vertical units of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>:HBArS:DELTA?	Sets or queries the delta V cursor readout value over the history period.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>:MODE	Sets or queries the cursor tracking mode of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>:ONEOVERDELTAVALUE?	Sets or queries the one over delta T cursor readout value of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>:SCREEN:AXPOStion	Sets or queries the horizontal cursor A position of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>:SCREEN:AYPOStion	Sets or queries the vertical cursor A position of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>:SCREEN:BXPOStion	Sets or queries the horizontal cursor B position of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>:SCREEN:BYPOStion	Sets or queries the vertical cursor B position of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>:SPLITMODE	Sets or queries the cursor source mode in the specified view.
DISPlay:WAVEView<x>:CURSor:CURSOR<x>:STATE	Sets or queries the visible state of the cursor of the specified cursor in the specified waveview.

Table 2-23: Cursor commands (cont.)

Command	Description
DISPlay:WAVEView<x>:CURSor: CURSOR<x>:VBArS:APOSition	Sets or queries the horizontal cursor A position of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor: CURSOR<x>:VBArS:BPOSition	Sets or queries the horizontal cursor B position of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor: CURSOR<x>:VBArS:DELTA?	Sets or queries the delta T cursor readout value of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor: CURSOR<x>:VBArS:UNIts?	Queries cursor A vertical units of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor: CURSOR<x>:WAVEform:APOSition	Sets or queries the horizontal cursor A position of the specified cursor in the specified waveview.
DISPlay:WAVEView<x>:CURSor: CURSOR<x>:WAVEform:BPOSition	Sets or queries the horizontal cursor B position of the specified cursor in the specified waveview.

Digital command group

Use the commands in the Digital Command Group to acquire up to 64 digital signals and analyze them. Digital channels are only available when a digital probe is attached to the super channel.

Table 2-24: Digital commands

Command	Description
CH<x>_DALL:LABel:COLor	Sets or queries the color of the specified digital group label.
CH<x>_DALL:LABel:FONT:BOLD	Sets or queries the bold state of the specified digital group.
CH<x>_DALL:LABel:FONT:ITALic	Sets or queries the italic state of the specified digital group.
CH<x>_DALL:LABel:FONT:SIZE	Sets or queries the font size of the specified digital group.
CH<x>_DALL:LABel:FONT:TYPE	Sets or queries the font type of the specified digital group, such as Arial or Times New Roman.
CH<x>_DALL:LABel:FONT:UNDERline	Sets or queries the underline state of the specified digital group.
CH<x>_DALL:LABel:NAME	Sets or queries the label of the specified digital group.
CH<x>_D<x>:LABel:COLor	Sets or queries the color of the label of the specified digital bit.
CH<x>_D<x>:LABel:FONT:BOLD	Sets or queries the bold state of the label of the specified digital bit.
CH<x>_D<x>:LABel:FONT:ITALic	Sets or queries the italic state of the label of the specified digital bit.
CH<x>_D<x>:LABel:FONT:SIZE	Sets or queries the font size of the label of the specified digital bit.
CH<x>_D<x>:LABel:FONT:TYPE	Sets or queries the font type of the label of the specified digital bit.
CH<x>_D<x>:LABel:FONT:UNDERline	Sets or queries the underline state of the label of the specified digital bit.
CH<x>_D<x>:LABel:NAME	Sets or queries the label of the specified digital bit.
REF<x>_DALL:LABel:COLor	Sets or queries the color of the specified digital group.
REF<x>_DALL:LABel:FONT:BOLD	Sets or queries the bold state of the specified digital group.
REF<x>_DALL:LABel:FONT:ITALic	Sets or queries the italic state of the specified digital group.
REF<x>_DALL:LABel:FONT:SIZE	Sets or queries the font size of the specified digital group.
REF<x>_DALL:LABel:FONT:TYPE	Sets or queries the font type of the specified digital group.
REF<x>_DALL:LABel:FONT:UNDERline	Sets or queries the underline state of the specified digital group.
REF<x>_DALL:LABel:NAME	Sets or queries the label of the specified digital group.
REF<x>_DALL:LABel:XPOS	Sets or queries the x-position of the label of the specified digital group.
REF<x>_DALL:LABel:YPOS	Sets or queries the y-position of the label of the specified digital group.
REF<x>_D<x>:LABel:COLor	Sets or queries the color of the label of the specified digital channel.
REF<x>_D<x>:LABel:FONT:BOLD	Sets or queries the bold state of the label of the specified digital bit.
REF<x>_D<x>:LABel:FONT:ITALic	Sets or queries the italic state of the label of the specified digital bit.
REF<x>_D<x>:LABel:FONT:SIZE	Sets or queries the font size of the label of the specified digital bit.
REF<x>_D<x>:LABel:FONT:TYPE	Sets or queries the font type of the label of the specified digital bit.
REF<x>_D<x>:LABel:FONT:UNDERline	Sets or queries the underline state of the label of the specified digital bit.
REF<x>_D<x>:LABel:NAME	Sets or queries the label of the specified digital bit.
REF<x>_D<x>:LABel:XPOS	Sets or queries the x-position of the label of the specified digital bit.

Table 2-24: Digital commands (cont.)

Command	Description
REF<x>_D<x>:LABel:YPOS	Sets or queries the y-position of the label of the specified digital channel.
DIGGRP<x>:D<x>:THReshold	Sets or queries the threshold level in volts for specified digital channel.

Display control command group

Display commands can be found in this section as well as the sections of related components. These commands control general oscilloscope settings, such as the intensity of the graticule, stacked or overlay display mode, and the fastacq color palette. Display commands also control how and where waveforms are shown, their position on screen, and zoom settings applied to the view. For example, display commands can turn on or off the display of channels or set the selected source.

Some actions can create a new view which can have its own settings. For example, adding a histogram will create a new view where the Histogram plot is displayed. Each view acts as a separate window within the oscilloscope application and can be rearranged or annotated as desired.

A WaveView is the primary view used for viewing inputs and time-domain signals. Buses, non-FFT maths, refs, analog and digital channels, and time-trends are displayed in the WaveView.

A PlotView is used for viewing measurement results and other plotted data. Histograms, eye diagrams, XY/XYZ plots, FFTs, and other plots are shown in individual PlotViews.

Each of these views can have separate settings, zoom, cursors, and annotations. Display commands which are view specific have a view parameter in the programmable interface, such as `DISPLAY:WAVEView<x>: ...` or `DISPLAY:PLOTView<x>: ...`. Selected source can also be specified on a per-view basis. The overall selected source is determined by the selected view and the selected source within that view.

PlotViews and WaveViews have some differences in command syntax due to differences in view functionality. For example, WaveViews and PlotViews have a different zoom model. The commands for specifying the zoom reflects these differences, and different command syntax is available depending on the view.

Table 2-25: Display control commands

Command	Description
<code>DISplay?</code>	Returns current display settings.
<code>DISplay:COLors</code>	Sets or queries the color mode for the graticule and waveform display.
<code>DISplay:GLObal:B<x>:STATE</code>	Sets or queries the display mode (on or off) of the specified bus.
<code>DISplay:GLObal:CH<x>:STATE</code>	Sets or queries the display mode (on or off) of the specified channel (both analog and digital).
<code>DISplay:GLObal:MATH<x>:STATE</code>	Sets or queries the display mode (on or off) of the specified math.
<code>DISplay:GLObal:PLOT<x>:STATE</code>	Sets or queries the display mode (on or off) of the specified time trend plot.
<code>DISplay:GLObal:REF<x>:STATE</code>	Sets or queries the display mode (on or off) of the specified reference.
<code>DISplay:INTENSITY?</code>	Returns the waveform and graticule saturation levels.
<code>DISplay:INTENSITY:BACKLight</code>	Sets or queries the waveform backlight intensity settings.

Table 2-25: Display control commands (cont.)

Command	Description
DISPlay:INTENSITy:BACKLight:AUTODim:ENable	Sets or queries the state of the display auto-dim feature.
DISPlay:INTENSITy:BACKLight:AUTODim:TIME	Sets or queries the amount of time, in minutes, to wait for no user interface activity before automatically dimming the display.
DISPlay:MATHFFTView<x>:AUTOScale	Sets or returns the enabled state of autoscale for plots.
DISPlay:MATHFFTView<x>:GRIDlines	Sets or queries the grid lines setting of the plot.
DISPlay:MATHFFTView<x>:MATH:MATH<x>:STATE	Sets or queries the state of the specified math waveform in the specified waveview.
DISPlay:MATHFFTView<x>:XAXIS:SCALE	Sets or queries the x-axis scale setting for FFT Math waveforms.
DISPlay:MATHFFTView<x>:YAXIS:SCALE	Sets or queries the vertical scale setting for FFT Maths.
DISPlay:PERSistence	Sets or queries display persistence setting.
DISPlay:PERSistence:RESET	Clears the persistence data.
DISPlay:PLOTView<x>:AUTOScale	Sets or queries the enabled state of autoscale for plots.
DISPlay:PLOTView<x>:GRIDlines	Sets or queries the Grid lines setting of the specified plot.
DISPLAY:PLOTVIEW<x>:XAXIS:SCALE	Sets or queries the horizontal scale setting for applicable plot.
DISPLAY:PLOTVIEW<x>:YAXIS:SCALE	Sets or queries the vertical scale setting for applicable plots.
DISPlay:REFFFTView<x>:AUTOScale	Sets or queries the enabled state of auto-scale for plots.
DISPlay:REFFFTView<x>:GRIDlines	Sets or returns the grid lines setting of the plot.
DISPlay:REFFFTView<x>:REF:REF<x>:STATE	Sets or queries the state of the specified reference waveform in the specified waveview.
DISPlay:REFFFTView<x>:XAXIS:SCALE	Sets or queries the x-axis scale setting for REFFFT.
DISPlay:SElect:BUS	Sets or queries the overall selected bus.
DISPlay:SElect:MATH	Sets or queries the overall selected math.
DISPlay:SElect:REFerence	Sets or queries the overall selected reference waveform.
DISPlay:SElect:SOUrce	Sets or queries the overall selected source.
DISPlay:SElect:VIEW	Sets or queries the selected view.
DISPlay:SElect:WAVEView<x>:SOUrce	Sets or queries the selected source in the given waveview.
DISPlay:VARpersist	Sets or queries the persistence decay time.
DISPlay:WAVEView<x>:BUS:B<x>:STATE	Sets or queries the state of the specified bus in the specified waveview.
DISPlay:WAVEView<x>:BUS:B<x>:VERTical:POSition	Sets or queries the vertical position of the specified bus in the specified waveview.
DISPlay:WAVEView<x>:CH<x>:STATE	Sets or queries the state of the specified channel in the specified waveview.
DISPlay:WAVEView<x>:CH<x>:VERTical:POSition	Sets or queries the vertical position of the specified channel in the specified waveview in divisions.
DISPlay:WAVEView<x>:CH<x>:VERTical:SCALE	Sets or queries the vertical scale of the specified channel in volts per division within the specified waveview.
DISPlay:WAVEView<x>:CH<x>_DALL:STATE	Sets or queries the display state of the specified digital channels in the specified waveview.

Table 2-25: Display control commands (cont.)

Command	Description
DISPlay:WAVEView<x>:CH<x>_DALL:VERTical:POSition	Sets or queries the vertical position of the specified digital channel in the specified waveview in divisions.
DISPlay:WAVEView<x>:CH<x>_D<x>:STATE	Sets or queries the display state of the specified digital channel in the specified waveview.
DISPlay:WAVEView<x>:FILTer	Sets or queries the type of interpolation filter for the display.
DISPlay:WAVEView<x>:GRAticule	Selects or queries the type of graticule that is displayed.
DISPlay:WAVEView<x>:INTENSITy:GRAticule	Sets or queries the graticule saturation level.
DISPlay:WAVEView<x>:INTENSITy:WAVEform	Sets or queries the waveform saturation level.
DISPlay:WAVEView<x>:MATH:MATH<x>:AUTOScale	Sets or queries whether the specified math gets auto-scaled when the math equation changes within the specified waveview.
DISPlay:WAVEView<x>:MATH:MATH<x>:STATE	Sets or queries the state of the specified math waveform in the specified waveview.
DISPlay:WAVEView<x>:MATH:MATH<x>:VERTical:POSition	Sets or queries the vertical position in divisions of the specified math waveform.
DISPlay:WAVEView<x>:MATH:MATH<x>:VERTical:SCALE	Sets or queries the vertical scale of the specified math waveform.
DISPlay:WAVEView<x>:PLOT:PLOT<x>:AUTOScale	Sets or queries whether the specified math gets auto-scaled when the math equation changes within the specified waveview.
DISPlay:WAVEView<x>:PLOT:PLOT<x>:STATE	Sets or queries the state of the specified trend waveform in the specified waveview.
DISPlay:WAVEView<x>:PLOT:PLOT<x>:VERTical:POSition	Sets or queries the vertical position of the specified time trend in the specified waveview in divisions.
DISPlay:WAVEView<x>:PLOT:PLOT<x>:VERTical:SCALE	Sets or queries the vertical scale of the specified time trend in volts per division in the specified waveview.
DISPlay:WAVEView<y>:REF<x>_DALL:FRAMe	Sets or returns the selected frame of the specified digital ref.
DISPlay:WAVEView<y>:REF:REF<x>:FRAMe	sets or returns the selected frame of the specified analog ref.
DISPlay:WAVEView<x>:REF:REF<x>:STATE	Sets or queries the state of the specified reference waveform in the specified waveview.
DISPlay:WAVEView<x>:REF:REF<x>:VERTical:POSition	Sets or queries the vertical position in divisions of the specified reference waveform.
DISPlay:WAVEView<x>:REF:REF<x>:VERTical:SCALE	Sets or queries the vertical scale of the specified reference waveform.
DISPlay:WAVEView<x>:STYle	Sets or queries the waveforms are displayed for analysis mode.
DISPlay:WAVEView<x>:VIEWStyle	Sets or queries the waveform layout style used by the display.
DISPlay:PLOTVIEW<x>:XAXIS:SCALE	Sets or queries the horizontal scale setting for applicable plots, either Linear or Log.
DISPlay:PLOTVIEW<x>:YAXIS:SCALE	Sets or queries the vertical scale setting for applicable plots, either Linear or Log.

DVM Command Group

Use the commands in the DVM command group for Digital Voltmeter functionality. Requires DVM option (free with product registration).

Table 2-26: DVM Command Group

Command	Description
DVM	Resets the Digital Voltmeter measurements and history.
DVM:AUTORange	Sets or queries the autorange state for the Digital Voltmeter.
DVM:MEASUrement:FREQuency?	Returns the current frequency value for the Digital Voltmeter.
DVM:MEASUrement:HIStory:AVErage?	Returns the average readout value for the Digital Voltmeter function over the history period.
DVM:MEASUrement:HIStory:MAXimum?	Returns the maximum readout value for the DVM function over the history period.
DVM:MEASUrement:HIStory:MINimum?	Returns the minimum readout value for the DVM function over the history period.
DVM:MEASUrement:INFMAXimum?	Returns the maximum DVM readout value over the entire time that the DVM has been on since the last change using the DVM:MODE or DVM:SOURce commands or DVM RESET.
DVM:MEASUrement:INFMINimum?	Returns the minimum readout value of the DVM function over the entire time that the DVM has been on since the last change using the DVM:MODE or DVM:SOURce commands or DVM RESET.
DVM:MEASUrement:VALue?	Returns the DVM readout value.
DVM:MODE	Specifies or queries the mode to use for the Digital Voltmeter.
DVM:SOURce	Sets or queries the source for the Digital Voltmeter.
DVM:TRIGger:FREQuency:COUNTer	Sets or queries the state of the trigger frequency counter.

Ethernet Command Group

Use the commands in the Ethernet Command Group to set up the 10BASE-T, 100BASE-TX, 1000BASE-TX or 100BASE-T Ethernet remote interface.

Table 2-27: Ethernet Commands

Command	Description
ETHERnet:DHCPbootp	Specifies the network initialization search for a DHCP/BOOTP server.
ETHERnet:DNS:IPADdress	Specifies the network Domain Name Server (DNS) IP address.
ETHERnet:DOMAINname	Specifies the network domain name.
ETHERnet:ENET:ADDress?	Returns the Ethernet address (MAC address) value assigned to the oscilloscope.
ETHERnet:GATEWay:IPADdress	Specifies the network gateway IP address.
ETHERnet:IPADdress	Specifies the IP address assigned to the oscilloscope.
ETHERnet:LXI:LAN:RESET	Resets the LXI local area network.
ETHERnet:LXI:LAN:SERVICENAME	Specifies the mDNS service name used for the LXI interface.
ETHERnet:LXI:LAN:STATus?	Returns the LXI network status.

Table 2-27: Ethernet Commands (cont.)

Command	Description
ETHERnet:NAME	Sets or queries the instrument Ethernet hostname assigned to the oscilloscope.
ETHERnet:NETWORKCONFig	Specifies the Ethernet network configuration setting.
ETHERnet:PING	Causes the oscilloscope to ping the gateway IP address.
ETHERnet:PING:STATus?	Returns the results from sending the ETHERnet:PING command to ping the gateway IP address.
ETHERnet:SUBNETMask	Specifies the network subnet mask value.

File system command group

Use the commands in the File System Command Group to help you use the built-in hard disk drive. You can use the commands to do the following:

- List the contents of the current directory
- Create and delete directories
- Create, copy, read, rename, or delete a file

When using these commands, keep the following points in mind:

- File arguments are always enclosed within double quotes: "C:\MYDIR\TEK00001.SET"
- File names follow the MS-DOS format: [DRIVE:][\PATH\]filename
- Path separators can be either forward slashes (/) or back slashes (\)

NOTE. Using a back slash as a path separator can produce some unexpected results, depending on how your controller application treats escaped characters. Many applications recognize the sequence of a back slash followed by an alphabetic character as an escaped character, and, as such, interpret that alphabetic character as a control character. For example, the sequence "\n" might be interpreted as a newline character; "\t" might be interpreted as a tab character. To ensure that this interpretation does not occur, you can use double back slashes. For example, "C:\\testfile.txt".

- Some FILESystem commands can fail because a file has read-only attributes. You will not be able to delete or replace such files until this attribute is removed. Refer to the operating system help on file properties for further information.

Table 2-28: File system commands

Command	Description
FILESystem?	Returns the file system state.
FILESystem:COpy	Copies one or more files to a new file.
FILESystem:CWD	Sets or queries the current working directory for FILESystem commands.
FILESystem:DELEte	Deletes a named file or directory.
FILESystem:DIR?	Returns a list of directory contents.
FILESystem:HOMEDir?	Returns the current user's home directory.
FILESystem:LDIR?	Returns a list of directory contents.
FILESystem:MKDir	Makes a new directory.
FILESystem:READFile	Copies the named file to the interface.
FILESystem:REName	Assigns a new name to an existing file.

Table 2-28: File system commands (cont.)

Command	Description
FILESystem:RMDir	Deletes the named directory.
FILESystem:UNMOUNT:DRive	Unmount the USB drive.
FILESystem:WRITEFile	Copies the block data to a named file.

Horizontal command group

Horizontal commands control the time base of the instrument. You can set the time per division (or time per point) of the main time base. You can use the Horizontal commands to do the following:

- Set the scale, horizontal position and reference, and units of the time base
- Get the screen resolution, time of first point and time of last point, or get all the horizontal settings
- Enable or disable the display of the time base

Table 2-29: Horizontal commands

Command	Description
ACquire:NUMFRAMESACquired?	Returns the number of FastFrame frames which have been acquired.
CH<x>:SCALERATio	Sets or returns the scale ration for the specified analog channel.
HORizontal?	Queries the current horizontal settings.
HORizontal:ACQDURATION?	Returns the time base duration.
HORizontal:DELAy:MODE	Sets or queries the horizontal delay mode.
HORizontal:DELAy:TIME	Sets or queries the horizontal delay time (position) that is used when delay is on.
HORizontal:DIVisions?	Returns the number of graticule divisions over which the waveform is displayed.
HORizontal:FASTframe?	Returns all information under horizontal:fastframe.
HORizontal:FASTframe:COUNt	Sets or returns the number of frames.
HORizontal:FASTframe:MAXFRames?	Returns the maximum number of frames.
HORizontal:FASTframe:MULtipleframes:MODE	Sets or returns the overlay display type.
HORizontal:FASTframe:REF:FRame	Sets or returns the reference frame number.
HORizontal:FASTframe:REF:INCLUde	Sets or returns whether the reference frame delta information is shown in the display.
HORizontal:FASTframe:SELECTED	Sets or returns the selected frame number for acquired frames.
HORizontal:FASTframe:STATE	Sets or returns the state of FastFrame.
HORizontal:FASTframe:SUMFrame?	Sets or returns the summary frame type.
HORizontal:FASTframe:SUMFrame:STATE	Sets or returns the state of FastFrame summary frame.
HORizontal:FASTframe:TIMEStamp:REFERENCE?	Returns the time-stamp of the FastFrame Reference frame.
HORizontal:FASTframe:TIMEStamp:SELECTED?	Returns the time-stamp of the FastFrame Selected acquired frame.
HORizontal:FASTframe:TIMEStamp:DELTA?	Returns the time difference between the Selected and Reference time-stamps.
HORizontal:FASTframe:TIMEStamp:ALL?	Returns the time stamp of all frames.
HORizontal:FASTframe:XZEro:ALL?	Returns the sub-sample time between the trigger sample (designated by PT_OFF) and the occurrence of the actual trigger for the waveform specified by the DATA:SOURce command for all frames.

Table 2-29: Horizontal commands (cont.)

Command	Description
<code>HORizontal:FASTframe:XZErO:REF?</code>	Returns the sub-sample time between the trigger sample (designated by PT_OFF) and the occurrence of the actual trigger for the waveform specified by the DATA:SOURce command for the reference frame.
<code>HORizontal:FASTframe:XZErO:SELECTED?</code>	Returns the sub-sample time between the trigger sample (designated by PT_OFF) and the occurrence of the actual trigger for the waveform specified by the DATA:SOURce command for the selected frame.
<code>HORizontal:MAIn:INTERPRatio?</code>	Returns the main horizontal time base interpolation ratio.
<code>HORizontal:MODE</code>	Sets or queries the horizontal mode.
<code>HORizontal:MODE:AUTOMATIC:FASTAcq:RECOrdlength:MAXimum:VALue</code>	Sets or queries the horizontal FastAcq maximum record length.
<code>HORizontal:MODE:AUTOMATIC:FASTAcq:RECOrdlength:MAXimum:ZOOMOVERride</code>	Sets or queries the flag which allows override of the horizontal FastAcq maximum record length.
<code>HORizontal:MODE:MANual:CONFIgure</code>	Sets or queries which horizontal control (scale or record length) will also change when the sample rate is adjusted.
<code>HORizontal:MODE:RECOrdlength</code>	Sets or queries the record length.
<code>HORizontal:MODE:SAMPLERate</code>	Sets or queries the sample rate.
<code>HORizontal:MODE:SCALE</code>	Sets or queries the horizontal scale.
<code>HORizontal:POSition</code>	Sets or queries the waveform horizontal position, in percent, that is used when delay is off.
<code>HORizontal:PREViewstate?</code>	Returns the display system preview state.
<code>HORizontal:RECOrdlength</code>	Sets or queries the horizontal record length.
<code>HORizontal:ROLL?</code>	Queries the horizontal roll mode status.
<code>HORizontal:SAMPLERate</code>	Sets or queries the horizontal sample rate.
<code>HORizontal:SAMPLERate:ANALYZemode:MINimum:OVERRide</code>	Sets or queries the flag which allows override of the horizontal analyze minimum sample rate.
<code>HORizontal:SAMPLERate:ANALYZemode:MINimum:VALue</code>	Sets or queries the minimum sample rate used by Analysis Automatic horizontal mode.
<code>HORizontal:SCALE</code>	Sets or queries the horizontal scale.

Mask command group

Mask commands compare incoming waveforms to standard or user-defined masks. A mask is a set of polygonal regions on the screen. Unlike limit testing, the inside of a mask is the region where waveform data would not normally fall. Standards with eye patterns usually have three masks, but some have four.

Table 2-30: Mask commands

Command	Description
EYEMASK:MASK<x>:COUNT:HITS?	Returns the total number of hit violations for all segments in the specified eye diagram mask test.
EYEMASK:MASK<x>:COUNT:SEG<y>:HITS?	Returns the number of hit violations for the specified segment (area) in the specified eye diagram mask test.
EYEMASK:MASK<x>:CREATor?	Returns the name of the eye diagram plot that created the mask.
EYEMASK:MASK<x>:ENabled	Enables or disables eye mask testing in the specified plot.
EYEMASK:MASK<x>:MASKfile	Sets or queries the current mask definition file name for the specified mask test.
EYEMASK:MASK<x>:TEST:SAMple:THReshold	Sets or queries the total number of hit violations that will cause a mask test failure.
EYEMASK:MASK<x>:TEST:STATUS?	Returns the mask hit test status for the specified mask test.

Math command group

Use the commands in the Math Command Group to create and define math waveforms. Use the available math functions to define your math waveform.

The math waveform you create depends on sources listed in the math expression. If you change these sources, the math waveforms you previously defined will be affected.

Math expressions can be simple, containing no mathematical computation, such as CH1, which specifies that a waveform shows the signal source of channel 1. Math expressions can also be complex, consisting of 100 plus characters and comprising many sources, functions, and operands.

The acquisition of a live waveform can stop for several reasons: You can turn off the channel, stop the waveform (via Run/Stop from the Horiz/Acq menu), or stop the trigger (via Run/Stop from the Trig menu). When you turn off the channel, math continues and data is acquired but is not displayed. When you stop either the waveform or the trigger, the math calculation stops, and the last math calculation performed is displayed.

When a live waveform updates or a reference waveform is altered, math waveforms containing those waveforms as sources are also updated to reflect the changes. Also, sources must exist but do not need to be displayed to be used in and to update math waveforms.

NOTE. Math commands are present once a math has been added.

Table 2-31: Math commands

Command	Description
DISplay:GLObal:MATH<x>:STATE	Sets or queries the global state of the specified math.
DISplay:SElect:MATH	Sets or queries the overall selected math.
DISplay:WAVEView<x>:MATH:MATH<x>:STATE	Sets or queries the state of the specified math waveform in the specified waveview.
DISplay:WAVEView<x>:MATH:MATH<x>:VERTical:POSition	Sets or queries the vertical position in divisions of the specified math waveform.
DISplay:WAVEView<x>:MATH:MATH<x>:VERTical:SCAle	Sets or queries the vertical scale of the specified math waveform.
MATH:ADDNew	Adds the specified math.
MATHArbflt<x>:FILEpath	Sets the file path for a file of filter coefficients and reads the file.
MATH:DELeTe	Deletes the specified math.
MATH:LIST?	Lists all currently defined math waveforms.
MATH:MATH<x>:AVG:MODE	Sets or queries the math average mode flag.
MATH:MATH<x>:AVG:WEIGHt	Sets or queries the number of acquisitions at which the averaging algorithm will begin exponential averaging.

Table 2-31: Math commands (cont.)

Command	Description
MATH:MATH<x>:FUNCTION	Sets or queries the basic math arithmetic function.
MATH:MATH<x>:LABel:COLor	Sets or queries color of the specified math's label.
MATH:MATH<x>:LABel:FONT:BOLD	Sets or queries bold state of the specified math label.
MATH:MATH<x>:LABel:FONT:ITALic	Sets or queries italic state of the specified math label.
MATH:MATH<x>:LABel:FONT:SIZE	Sets or queries font size of the specified math label.
MATH:MATH<x>:LABel:FONT:TYPE	Sets or queries font type of the specified math label.
MATH:MATH<x>:LABel:FONT:UNDERline	Sets or queries the underline state of the specified math label.
MATH:MATH<x>:LABel:XPOS	Sets or queries the X screen offset where the math waveform label is displayed.
MATH:MATH<x>:LABel:YPOS	Sets or queries the Y screen offset where the math waveform label is displayed.
MATH:MATH<x>:DEFine	Defines new waveforms using mathematical expressions.
MATH:MATH<x>:GATing	Specifies or returns the gating setting.
MATH:MATH<x>:LABel:NAME	Sets or queries the label string.
MATH:MATH<x>:SOUrce<x>	Sets or queries the specified math source.
MATH:MATH<x>:SPECTral:HORZ	Sets or queries the horizontal display scale of the spectral math waveform.
MATH:MATH<x>:SPECTral:MAG	Sets or queries the units of the SpectralMag function in the specified math definition string.
MATH:MATH<x>:SPECTral:PHASE	Sets or queries the units of a SpectralPhase function in the specified math definition string.
MATH:MATH<x>:SPECTral:SOUrce	Sets or queries the specified spectral math source.
MATH:MATH<x>:SPECTral:SUPPress	Sets or queries whether suppression threshold for the specified math waveform is enabled.
MATH:MATH<x>:SPECTral:SUPPress:VALue	Sets or queries in volts the value of suppression threshold of the specified math waveform.
MATH:MATH<x>:SPECTral:TYPE	Sets or queries the FFT type selected for spectral analysis.
MATH:MATH<x>:SPECTral:UNWRap	Sets or queries whether phase unwrap of the spectral analyzer output data is enabled.
MATH:MATH<x>:SPECTral:UNWRap:DEGrees	Sets or queries in degrees the value of unwrap phase.
MATH:MATH<x>:SPECTral:WINDow	Sets or queries the window function used to multiply the spectral analyzer input data for the specified math waveform.
MATH:MATH<x>:TYPE	Sets or queries the math type.
MATH:MATH<x>:VUNIT	Sets or queries the math custom vertical units.

Measurement command group

Use the commands in the Measurement Command Group to control the automated measurement system.

Measurement commands can set and query measurement parameters. You can assign parameters, such as waveform sources and reference levels, differently for each measurement.

Clock recovery, edge, filter, gating, population and range measurement commands can be either global or per-measurement.

Global clock recovery commands are of the form

`:MEASUREMENT:CLOCKRecovery:XXXX`

Global edge commands are of the form `:MEASUREMENT:XXXX`

Global filter commands are of the form `:MEASUREMENT:FILTERs:XXXX`

Global gating commands are of the form `:MEASUREMENT:GATING:XXXX`

Global population commands are of the form

`:MEASUREMENT:POPulation:XXXX`

Global range commands are of the form `:MEASUREMENT:MEASRange:XXXX`

Reference levels for measurements can be global, per-measurement or per-source. The default is global. Per-measurement settings are used when local reference levels are enabled for the measurement with the command `:MEASUREMENT:MEAS?:GLOBalref 0`. Per-source settings are used when per-source settings are enabled with the command `:MEASUREMENT:REFLeveLs:TYPE PerSource` and per-source settings are selected for the measurement with the command `:MEASUREMENT:MEAS?:GLOBalref 0`.

Global reference level commands are of the form

`:MEASUREMENT:REFLeveLs:XXXX`

Per-source reference level commands are of the form

`:MEASUREMENT:CH1:REFLeveLs:XXXX`

`:MEASUREMENT:MATH1:REFLeveLs:XXXX`

Table 2-32: Measurement commands

Command	Description
MEASUREMENT?	This command returns all measurement parameters.
MEASUREMENT:ADDMEAS	This command adds a measurement.
MEASUREMENT:ADDNew	Adds the specified measurement.
MEASUREMENT:ANNOtate	This command sets or queries the annotation state for measurements.
MEASUREMENT:AUTOset	Performs an analysis jitter autoset.
MEASUREMENT:CH<x>:REFLeveLs:ABSolute:FALLHigh	This command sets or queries the value used as the high reference level of the falling edge when the source ref level method is set to absolute.
MEASUREMENT:CH<x>:REFLeveLs:ABSolute:FALLLow	This command sets or queries the value used as the low reference level of the falling edge when the source ref level method is set to absolute.

Table 2-32: Measurement commands (cont.)

Command	Description
MEASUrement:CH<x>:REFLevels:ABSolute:FALLMid	This command sets or queries the value used as the mid reference level of the falling edge when the source ref level method is set to absolute.
MEASUrement:CH<x>:REFLevels:ABSolute:HYSTeresis	This command sets or queries the value of the hysteresis of the reference level when the source ref level method is set to absolute.
MEASUrement:CH<x>:REFLevels:ABSolute:RISEHigh	This command sets or queries the value used as the high reference level of the rising edge when the source ref level method is set to absolute.
MEASUrement:CH<x>:REFLevels:ABSolute:RISELow	This command sets or queries the value used as the low reference level of the rising edge when the source ref level method is set to absolute.
MEASUrement:CH<x>:REFLevels:ABSolute:RISEMid	This command sets or queries the value used as the mid reference level of the rising edge when the source ref level method is set to absolute.
MEASUrement:CH<x>:REFLevels:ABSolute:TYPE	This command sets or queries the reference level type for the source.
MEASUrement:CH<x>:REFLevels:BASETop	This command sets or queries the method used to calculate the TOP and BASE, used to calculate reference levels for the measurement.
MEASUrement:CH<x>:REFLevels:METHod	This command sets or queries the method used to calculate reference levels for the source.
MEASUrement:CH<x>:REFLevels:PERCent:FALLHigh	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the high reference level of the falling edge when the source ref level method is set to percent.
MEASUrement:CH<x>:REFLevels:PERCent:FALLLow	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the falling edge when the source ref level method is set to percent.
MEASUrement:CH<x>:REFLevels:PERCent:FALLMid	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the falling edge when the source ref level method is set to percent.
MEASUrement:CH<x>:REFLevels:PERCent:HYSTeresis	This command sets or queries the percentage (where 100% is equal to MAX and 0% is equal to MIN) used to calculate the hysteresis of the reference level when the source ref level method is set to percent.
MEASUrement:CH<x>:REFLevels:PERCent:RISEHigh	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the high reference level of the rising edge when the source ref level method is set to percent.
MEASUrement:CH<x>:REFLevels:PERCent:RISELow	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the low reference level of the rising edge when the source ref level method is set to percent.
MEASUrement:CH<x>:REFLevels:PERCent:RISEMid	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the rising edge when the source ref level method is set to percent.
MEASUrement:CH<x>:REFLevels:PERCent:TYPE	This command specifies or queries the reference level percent type for the source.

Table 2-32: Measurement commands (cont.)

Command	Description
MEASUrement:CLOCKRecovery:ADVanced:METHod	This command sets or queries the global advanced clock recovery method.
MEASUrement:CLOCKRecovery:CLOCKFrequency	This command sets or queries the global clock frequency used for fixed constant clock recovery.
MEASUrement:CLOCKRecovery:CLOCKMultiplier	This command sets or queries the global clock multiplier used for explicit clock recovery.
MEASUrement:CLOCKRecovery:CONSTCLOCKMODE	This command sets or queries the global constant clock mode used for constant clock recovery.
MEASUrement:CLOCKRecovery:DAMPing	This command sets or queries the global damping value used for PLL clock recovery.
MEASUrement:CLOCKRecovery:DATAPath	This command sets or queries the global file containing the data pattern used for known data pattern clock recovery.
MEASUrement:CLOCKRecovery:DATARate	This command sets or queries the global nominal data bit rate used for nominal data rate clock recovery.
MEASUrement:CLOCKRecovery:EXPLICITCLOCKMODE	This command sets or queries the global explicit clock mode used for explicit clock recovery.
MEASUrement:CLOCKRecovery:JTFBandwidth	This command sets or queries the global JTF bandwidth used for PLL clock recovery.
MEASUrement:CLOCKRecovery:LOOPBandwidth	This command sets or queries the global loop bandwidth used for PLL clock recovery.
MEASUrement:CLOCKRecovery:MEANAUTOCalculate	This command sets or queries how often the clock is calculated for constant clock recovery.
MEASUrement:CLOCKRecovery:METHod	This command sets or queries the global clock recovery method.
MEASUrement:CLOCKRecovery:MODEl	This command sets or queries the global PLL clock recovery model used for PLL clock recovery.
MEASUrement:CLOCKRecovery:NOMINALOFFset	This command sets or queries the global offset value used for explicit clock recovery.
MEASUrement:CLOCKRecovery:NOMINALOFFset:SELECTIONtype	This command sets or queries the global offset type used for explicit clock recovery.
MEASUrement:CLOCKRecovery:STANDARD	This command sets or queries the global communications standard used for PLL clock recovery.
MEASUrement:DELeTe	The command deletes the specified measurement.
MEASUrement:DIRacmodel	This command sets or queries the dirac model used to separate random from deterministic jitter for jitter measurements.
MEASUrement:DISPLAYUnits	This command sets or queries the display units used for jitter summary measurements.
MEASUrement:EDGE<x>	Sets or queries the type of the edge for the measurement.
MEASUrement:EYERENDER	This command sets or queries the state of high-performance eye rendering for an eye diagram.
MEASUrement:FILTers:BLANKingtime	This command sets or queries the global filter blanking time.

Table 2-32: Measurement commands (cont.)

Command	Description
MEASUrement:FILTerS:HIGHPass:FREQ	This command sets or queries the global high pass filter frequency.
MEASUrement:FILTerS:HIGHPass:SPEC	This command sets or queries the global high pass filter order.
MEASUrement:FILTerS:LOWPass:FREQ	This command sets or queries the global low pass filter cutoff frequency.
MEASUrement:FILTerS:LOWPass:SPEC	This command sets or queries the global low pass filter order.
MEASUrement:FILTerS:RAMpTime	This command sets or queries the global filter ramp time.
MEASUrement:GATing	This command sets or queries the global gating type.
MEASUrement:GATing:ACTive	This command sets or queries the global gating active level used for logic gating.
MEASUrement:GATing:HYSTeresis	This command sets or queries the global gating hysteresis value used for logic gating.
MEASUrement:GATing:LOGICSource	This command sets or queries the gating data source used for logic gating.
MEASUrement:GATing:MIDRef	This command sets or queries the global gating mid ref value used for logic gating.
MEASUrement:GATing:SEARChSource	This command sets or queries the global gating search source used for logic gating.
MEASUrement:INTERp	This command sets or queries the interpolation mode used to locate edge crossings.
MEASUrement:JITTermodel	This command sets or queries the model used to separate random from deterministic jitter for jitter measurements.
MEASUrement:LIST?	Lists all currently defined measurements.
MEASUrement:LOCKRJ	Sets or queries the state of RJ locking.
MEASUrement:LOCKRJValue	Sets or queries the RJ lock value.
MEASUrement:MATH<x>:REFLevels:ABSolute:FALLHigh	This command sets or queries the value used as the high reference level of the falling edge when the source ref level method is set to absolute.
MEASUrement:MATH<x>:REFLevels:ABSolute:FALLLow	This command sets or queries the value used as the low reference level of the falling edge when the source ref level method is set to absolute.
MEASUrement:MATH<x>:REFLevels:ABSolute:FALLMid	This command sets or queries the value used as the mid reference level of the falling edge when the source ref level method is set to absolute.
MEASUrement:MATH<x>:REFLevels:ABSolute:HYSTeresis	This command sets or queries the value of the hysteresis of the reference level when the source ref level method is set to absolute.
MEASUrement:MATH<x>:REFLevels:ABSolute:RISEHigh	This command sets or queries the value used as the high reference level of the rising edge when the source ref level method is set to absolute.
MEASUrement:MATH<x>:REFLevels:ABSolute:RISELow	This command sets or queries the value used as the low reference level of the rising edge when the source ref level method is set to absolute.
MEASUrement:MATH<x>:REFLevels:ABSolute:RISEMid	This command sets or queries the value used as the mid reference level of the rising edge when the source ref level method is set to absolute.
MEASUrement:MATH<x>:REFLevels:ABSolute:TYPE	This command sets or queries the reference level type for the source.
MEASUrement:MATH<x>:REFLevels:BASETop	This command sets or queries the method used to calculate the TOP and BASE used to calculate reference levels for the source.

Table 2-32: Measurement commands (cont.)

Command	Description
MEASUrement:MATH<x>:REFLevels:METHod	This command sets or queries the method used to calculate reference levels for the source.
MEASUrement:MATH<x>:REFLevels:PERCent: FALLHigh	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the high reference level of the falling edge when the source ref level method is set to percent.
MEASUrement:MATH<x>:REFLevels:PERCent: FALLLow	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the falling edge when the source ref level method is set to percent.
MEASUrement:MATH<x>:REFLevels:PERCent: FALLMid	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the falling edge when the source ref level method is set to percent.
MEASUrement:MATH<x>:REFLevels:PERCent: HYSTeresis	This command sets or queries the percentage (where 100% is equal to MAX and 0% is equal to MIN) used to calculate the hysteresis of the reference level when the source ref level method is set to percent.
MEASUrement:MATH<x>:REFLevels:PERCent: RISEHigh	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the high reference level of the rising edge when the source ref level method is set to percent.
MEASUrement:MATH<x>:REFLevels:PERCent: RISELow	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the low reference level of the rising edge when the source ref level method is set to percent.
MEASUrement:MATH<x>:REFLevels:PERCent: RISEMid	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the rising edge when the source ref level method is set to percent.
MEASUrement:MATH<x>:REFLevels:PERCent: TYPE	This command specifies or queries the reference level percent type for the source.
MEASUrement:MEAS<x>:BER	This command sets or queries the BER value for the measurement.
MEASUrement:MEAS<x>:BER:TARGETBER	This command sets or queries the target BER value for the measurement.
MEASUrement:MEAS<x>:BIN	This command sets or queries the bin count for the measurement.
MEASUrement:MEAS<x>:BITCfgmode	This command sets or queries whether the measurement returns the mean or mode statistic result when the measurement type is bit amplitude/high/low.
MEASUrement:MEAS<x>:BITEnd	This command sets or queries the bit end as a percentage of the unit interval.
MEASUrement:MEAS<x>:BITPcnt	This command sets or queries the bit center as a percentage of the unit interval.
MEASUrement:MEAS<x>:BITStart	This command sets or queries the bit start as a percentage of the unit interval.
MEASUrement:MEAS<x>:BITType	This command sets or queries the bit type for the measurement.
MEASUrement:MEAS<x>:BURSTEDGTYPE	This command sets or queries the burst edge type for the measurement.
MEASUrement:MEAS<x>:CCRESULTS: ALLAcqs:MAXimum?	This query-only command returns the maximum cycle-cycle value for the specified measurement for all acquisitions.
MEASUrement:MEAS<x>:CCRESULTS: ALLAcqs:MEAN?	This query-only command returns the mean cycle-cycle value for the specified measurement for all acquisitions.

Table 2-32: Measurement commands (cont.)

Command	Description
MEASUrement:MEAS<x>:CCRESUlts:ALLAcqs:MINimum?	This query-only command returns the minimum cycle-cycle value for the specified measurement for all acquisitions.
MEASUrement:MEAS<x>:CCRESUlts:ALLAcqs:PK2PK?	This query-only command returns the peak to peak cycle-cycle statistic for the specified measurement for all acquisitions.
MEASUrement:MEAS<x>:CCRESUlts:ALLAcqs:POPulation?	This query-only command returns the population of all cycle-cycle statistics for the specified measurement for all acquisitions accumulated since statistics were last reset.
MEASUrement:MEAS<x>:CCRESUlts:ALLAcqs:STDDev?	This query-only command returns the standard deviation cycle-cycle for the specified measurement for all acquisitions.
MEASUrement:MEAS<x>:CCRESUlts:CURRentacq:MAXimum?	This query-only command returns the maximum cycle-cycle value for the specified measurement for the current acquisition.
MEASUrement:MEAS<x>:CCRESUlts:CURRentacq:MEAN?	This query-only command returns the mean cycle-cycle value for the specified measurement for the current acquisition.
MEASUrement:MEAS<x>:CCRESUlts:CURRentacq:MINimum?	This query-only command returns the minimum cycle-cycle value for the specified measurement for the current acquisition.
MEASUrement:MEAS<x>:CCRESUlts:CURRentacq:PK2PK?	This query-only command returns the peak to peak cycle-cycle statistic for the specified measurement for the current acquisition.
MEASUrement:MEAS<x>:CCRESUlts:CURRentacq:POPulation?	This query-only command returns the population of the cycle-cycle statistics for the specified measurement for the current acquisition.
MEASUrement:MEAS<x>:CCRESUlts:CURRentacq:STDDev?	This query-only command returns the standard deviation cycle-cycle for the specified measurement for the current acquisition.
MEASUrement:MEAS<x>:CLOCKRecovery:ADVanced:METHOD	This command sets or queries the advanced clock recovery method when advanced clock recovery is used for the measurement.
MEASUrement:MEAS<x>:CLOCKRecovery:CLOCKFrequency	This command sets or queries the clock frequency used when fixed constant clock recovery is used for the measurement.
MEASUrement:MEAS<x>:CLOCKRecovery:CLOCKMultiplier	This command sets or queries the clock multiplier used when explicit clock recovery is used for the measurement.
MEASUrement:MEAS<x>:CLOCKRecovery:CONSTCLOCKMODE	This command sets or queries the constant clock mode used when constant clock recovery is used for the measurement.
MEASUrement:MEAS<x>:CLOCKRecovery:DAMPing	This command sets or queries the damping value used when PLL clock recovery is used for the measurement.
MEASUrement:MEAS<x>:CLOCKRecovery:DATAPath	This command sets or queries the file containing the data pattern used when known data pattern clock recovery is used for the measurement.
MEASUrement:MEAS<x>:CLOCKRecovery:DATARate	This command sets or queries the nominal data bit rate when nominal data rate clock recovery is used for the measurement.
MEASUrement:MEAS<x>:CLOCKRecovery:EXPLICITCLOCKMODE	This command sets or queries the explicit clock mode used when explicit clock recovery is used for the measurement.
MEASUrement:MEAS<x>:CLOCKRecovery:GLOBal	This command sets or queries the clock recovery settings global flag for the measurement.

Table 2-32: Measurement commands (cont.)

Command	Description
MEASUrement:MEAS<x>:CLOCKRecovery:JTFBandwidth	This command sets or queries the JTF bandwidth used when PLL clock recovery is used for the measurement.
MEASUrement:MEAS<x>:CLOCKRecovery:LOOPBandwidth	This command sets or queries the loop bandwidth used when PLL clock recovery is used for the measurement.
MEASUrement:MEAS<x>:CLOCKRecovery:MEANAUTOCalculate	This command sets or queries how often the clock is calculated when constant clock recovery is used for the measurement.
MEASUrement:MEAS<x>:CLOCKRecovery:METHod	This command sets or queries the clock recovery method for the measurement.
MEASUrement:MEAS<x>:CLOCKRecovery:MODEl	This command sets or queries the PLL clock recovery model used when PLL clock recovery is used for the measurement.
MEASUrement:MEAS<x>:CLOCKRecovery:NOMINALOFFset	This command sets or queries the offset value used when explicit clock recovery is used for the measurement.
MEASUrement:MEAS<x>:CLOCKRecovery:NOMINALOFFset:SELECTIONtype	This command sets or queries the offset type used when explicit clock recovery is used for the measurement.
MEASUrement:MEAS<x>:CLOCKRecovery:STAndard	This command sets or queries the communications standard when PLL clock recovery is used for the measurement.
MEASUrement:MEAS<x>:COMMONMode:FILTers:STATE	This command sets or queries whether a filter is used for the measurement when the measurement type is AC common mode.
MEASUrement:MEAS<x>:COMMONMode:SOURCEs	This command sets or queries the number of sources for the measurement when the measurement type is AC common mode.
MEASUrement:MEAS<x>:CYCLEmode	This command sets or queries the cycle mode for the measurement.
MEASUrement:MEAS<x>:DElay:EDGE<x>	This command sets or queries the 'to edge' type when EDGE? is EDGE1 and the 'from edge' type when EDGE? is EDG2, for the measurement when the measurement type is DELAY.
MEASUrement:MEAS<x>:DISPlaystat:ENABLE	Turns on and off the display of statistics in measurement badges.
MEASUrement:MEAS<x>:EDGE<x>	This command sets or queries the type of the specified edge, rise or fall, for the measurement.
MEASUrement:MEAS<x>:EDGEIncre	This command sets or queries the edge increment value for the measurement.
MEASUrement:MEAS<x>:EDGES:FROMLevel	This command sets or queries the 'from level' edge for the measurement.
MEASUrement:MEAS<x>:EDGES:LEVel	This sets or queries the level type for the 'time outside level' measurement.
MEASUrement:MEAS<x>:EDGES:LOWERFREQuency	This command sets or queries the lower frequency for the measurement when the measurement type is phase noise. Lower frequencies are ignored.
MEASUrement:MEAS<x>:EDGES:N	The command sets or queries the number of accumulation cycles for the measurement when the measurement type is nperiod.
MEASUrement:MEAS<x>:EDGES:SLEWRATEMethod	This command sets or queries the slew rate method for the measurement.
MEASUrement:MEAS<x>:EDGES:TOLevel	This command sets or queries the 'to level' edge for the measurement.

Table 2-32: Measurement commands (cont.)

Command	Description
MEASUrement:MEAS<x>:EDGES:UPPERFREQuency	This command sets or queries the upper frequency for the measurement when the measurement type is phase noise. Higher frequencies are ignored.
MEASUrement:MEAS<x>:FILTErs:BLANKingtime	This command sets or queries the filter blanking time for the measurement.
MEASUrement:MEAS<x>:FILTErs:GLOBal	This command sets or queries the global flag for filter settings for the measurement.
MEASUrement:MEAS<x>:FILTErs:HIGHPass:FREQ	This command sets or queries the high pass filter frequency for the measurement.
MEASUrement:MEAS<x>:FILTErs:HIGHPass:SPEC	This command sets or queries the high pass filter order for the measurement.
MEASUrement:MEAS<x>:FILTErs:LOWPass:FREQ	This command sets or queries the low pass filter cutoff frequency for the measurement.
MEASUrement:MEAS<x>:FILTErs:LOWPass:SPEC	This command sets or queries the low pass filter order for the measurement.
MEASUrement:MEAS<x>:FILTErs:RAMPTime	This command sets or queries the filter ramp time for the measurement.
MEASUrement:MEAS<x>:FROMedge	This command sets or queries the 'from edge' type for the measurement.
MEASUrement:MEAS<x>:FROMEDGESEARCHDIRect	This command sets or queries the 'from edge' search direction for the measurement.
MEASUrement:MEAS<x>:GATing	This command sets or queries the gating type for the measurement.
MEASUrement:MEAS<x>:GATing:ACTive	This command sets or queries the gating active level when the gating type is logic.
MEASUrement:MEAS<x>:GATing:GLOBal	This command sets or queries the gating settings global flag.
MEASUrement:MEAS<x>:GATing:HYSTeresis	This command sets or queries the gating hysteresis value when the gating type is logic.
MEASUrement:MEAS<x>:GATing:LOGICSource	This command sets or queries the gating data source when the gating type is logic.
MEASUrement:MEAS<x>:GATing:MIDRef	This command sets or queries the gating mid ref value when the gating type is logic.
MEASUrement:MEAS<x>:GATing:SEARCHSource	This command sets or queries the gating search source when the gating type is search.
MEASUrement:MEAS<x>:GLOBalref	This command sets or queries the reference levels global flag for the measurement.
MEASUrement:MEAS<x>:HIGHREFVoltage	This command sets or queries the high reference voltage value for the 'time outside level' measurement.
MEASUrement:MEAS<x>:IDLETime	This command sets or queries the idle time for the measurement when the measurement type is burst width.
MEASUrement:MEAS<x>:JITTERSummary:DCD	This command sets or queries whether DCD is included in the jitter summary for the measurement.
MEASUrement:MEAS<x>:JITTERSummary:DDJ	This command sets or queries whether DDJ is included in the jitter summary for the measurement.

Table 2-32: Measurement commands (cont.)

Command	Description
MEASUrement:MEAS<x>:JITTERSummary:DJDD	This command sets or queries whether DJ-dd is included in the jitter summary for the measurement.
MEASUrement:MEAS<x>:JITTERSummary:EYEWIDTHBER	This command sets or queries whether EyeWidth@BER is included in the jitter summary for the measurement.
MEASUrement:MEAS<x>:JITTERSummary:NPJ	This command sets or queries whether NPJ is included in the jitter summary for the measurement.
MEASUrement:MEAS<x>:JITTERSummary:PJ	This command sets or queries whether PJ is included in the jitter summary for the measurement.
MEASUrement:MEAS<x>:JITTERSummary:RJDD	This command sets or queries whether RJ-dd is included in the jitter summary for the measurement.
MEASUrement:MEAS<x>:JITTERSummary:TIE	This command sets or queries whether TIE is included in the jitter summary for the measurement.
MEASUrement:MEAS<x>:JITTERSummary:TJBER	This command sets or queries whether TJ@BER is included in the jitter summary for the measurement.
MEASUrement:MEAS<x>:LABel	This command sets or queries the label for the measurement.
MEASUrement:MEAS<x>:LOWREFVoltage	This command sets or queries the low reference voltage value for the 'time outside level' measurement.
MEASUrement:MEAS<x>:MAXCycle	Sets or queries the maximum cycle value for the DDRTERRN and DDRTERRMN measurements.
MEASUrement:MEAS<x>:MEASRange:GLOBal	This command sets or queries the range settings global flag for the measurement.
MEASUrement:MEAS<x>:MEASRange:MAX	This command sets or queries the range maximum value for the measurement.
MEASUrement:MEAS<x>:MEASRange:MIN	This command sets or queries the range minimum value for the measurement.
MEASUrement:MEAS<x>:MEASRange:STATE	This command sets or queries the range state for the measurement.
MEASUrement:MEAS<x>:MINCycle	Sets or queries the minimum cycle value for the DDRTERRN and DDRTERRMN measurements.
MEASUrement:MEAS<x>:PATTERNDETECTION	This command sets or queries the pattern detection type for the measurement.
MEASUrement:MEAS<x>:PATTERNLENgth	This command sets or queries the pattern length for the measurement.
MEASUrement:MEAS<x>:PATTERNType	This command sets or queries the pattern type for the measurement.
MEASUrement:MEAS<x>:PERFREQ:EDGE	This command sets or queries the edge type of a Period/Frequency measurement.
MEASUrement:MEAS<x>:POLarity	This command sets or queries the polarity for the measurement when the measurement type is burst width.
MEASUrement:MEAS<x>:POPulation:GLOBal	This command sets or queries the population settings global flag.
MEASUrement:MEAS<x>:POPulation:LIMIT:STATE	This command sets or queries the population limit state for the measurement.

Table 2-32: Measurement commands (cont.)

Command	Description
MEASUrement:MEAS<x>:POPulation:LIMIT:VALue	This command sets or queries the population limit value for the measurement.
MEASUrement:MEAS<x>:REFLevels:ABSolute:FALLHigh	This command sets or queries the value used as the high reference level of the falling edge when the measurement's ref level method is set to absolute.
MEASUrement:MEAS<x>:REFLevels<x>:ABSolute:FALLLow	This command sets or queries the value used as the low reference level of the falling edge when the measurement's ref level method is set to absolute.
MEASUrement:MEAS<x>:REFLevels<x>:ABSolute:FALLMid	This command sets or queries the value used as the mid reference level of the falling edge when the measurement's ref level method is set to absolute.
MEASUrement:MEAS<x>:REFLevels<x>:ABSolute:HYSTeresis	This command sets or queries the value of the hysteresis of the reference level when the measurement's ref level method is set to absolute.
MEASUrement:MEAS<x>:REFLevels<x>:ABSolute:RISEHigh	This command sets or queries the value used as the high reference level of the rising edge when the measurement's ref level method is set to absolute.
MEASUrement:MEAS<x>:REFLevels<x>:ABSolute:RISELow	This command sets or queries the value used as the low reference level of the rising edge when the measurement's ref level method is set to absolute.
MEASUrement:MEAS<x>:REFLevels<x>:ABSolute:RISEMid	This command sets or queries the value used as the mid reference level of the rising edge when the measurement's ref level method is set to absolute.
MEASUrement:MEAS<x>:REFLevels<x>:ABSolute:TYPE	This command sets or queries the reference level type for the measurement.
MEASUrement:MEAS<x>:REFLevels<x>:BASETop	This command sets or queries the method used to calculate the TOP and BASE used to calculate reference levels for the measurement.
MEASUrement:MEAS<x>:REFLevels<x>:METHod	This command sets or queries the method used to calculate reference levels for the measurement.
MEASUrement:MEAS<x>:REFLevels<x>:PERCent:FALLHigh	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the high reference level of the falling edge when the measurement's ref level method is set to percent.
MEASUrement:MEAS<x>:REFLevels<x>:PERCent:FALLLow	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the low reference level of the falling edge when the measurement's ref level method is set to percent.
MEASUrement:MEAS<x>:REFLevels<x>:PERCent:FALLMid	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the falling edge when the measurement's ref level method is set to percent.
MEASUrement:MEAS<x>:REFLevels<x>:PERCent:HYSTeresis	This command sets or queries the percentage (where 100% is equal to MAX and 0% is equal to MIN) used to calculate the hysteresis of the reference level when the measurement's ref level method is set to percent.
MEASUrement:MEAS<x>:REFLevels<x>:PERCent:RISEHigh	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the high reference level of the rising edge when the measurement's ref level method is set to percent.
MEASUrement:MEAS<x>:REFLevels<x>:PERCent:RISELow	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the low reference level of the rising edge when the measurement's ref level method is set to percent.

Table 2-32: Measurement commands (cont.)

Command	Description
MEASUrement:MEAS<x>:REFLevels<x>:PERCent:RISEMid	This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the rising edge when the measurement's ref level method is set to percent.
MEASUrement:MEAS<x>:REFLevels<x>:PERCent:TYPE	This command specifies or queries the reference level percent type for the measurement.
MEASUrement:MEAS<x>:REFMode	This command sets or queries the reference level mode for the measurement.
MEASUrement:MEAS<x>:REFVoltage	This command sets or queries the reference voltage value for the measurement.
MEASUrement:MEAS<x>:RESUlts:ALLAcqs:MAXimum?	This command is identical to that described in the DPOJet programmer manual.
MEASUrement:MEAS<x>:RESUlts:ALLAcqs:MEAN?	This command is identical to that described in the DPOJet programmer manual.
MEASUrement:MEAS<x>:RESUlts:ALLAcqs:MINimum?	This command is identical to that described in the DPOJet programmer manual.
MEASUrement:MEAS<x>:RESUlts:ALLAcqs:PK2PK?	This command is identical to that described in the DPOJet programmer manual.
MEASUrement:MEAS<x>:RESUlts:ALLAcqs:POPulation?	This command is identical to that described in the DPOJet programmer manual.
MEASUrement:MEAS<x>:RESUlts:ALLAcqs:STDDev?	This command is identical to that described in the DPOJet programmer manual.
MEASUrement:MEAS<x>:RESUlts:CURRentacq:MAXimum?	This query-only command returns the maximum value found for the specified measurement since the last statistical reset.
MEASUrement:MEAS<x>:RESUlts:CURRentacq:MEAN?	This query-only command returns the mean value for the measurement for the current acquisition.
MEASUrement:MEAS<x>:RESUlts:CURRentacq:MINimum?	This query-only command returns the minimum value found for the specified measurement since the last statistical reset.
MEASUrement:MEAS<x>:RESUlts:CURRentacq:PK2PK?	This query-only command returns the peak-to-peak value for the specified measurement for the current acquisition.
MEASUrement:MEAS<x>:RESUlts:CURRentacq:POPulation?	This query-only command returns the population for the specified measurement for the current acquisition.
MEASUrement:MEAS<x>:RESUlts:CURRentacq:STDDev?	This query-only command returns the standard deviation for the specified measurement for all acquisitions accumulated since statistics were last reset.
MEASUrement:MEAS<x>:SIGNALType	This command sets or queries the signal type of source 1 for the measurement.
MEASUrement:MEAS<x>:SOURce<x>	This command sets or queries the measurement source.
MEASUrement:MEAS<x>:SSC:NOMinalfreq	This command sets or queries the user-defined frequency for the measurement when the measurement type is SSC.
MEASUrement:MEAS<x>:SSC:NOMinalfreq:SELECTIONtype	This command sets or queries the frequency detection type for the measurement when the measurement type is SSC.

Table 2-32: Measurement commands (cont.)

Command	Description
MEASUrement:MEAS<x>:TCKAVG	This command sets or queries the average clock period value used in DDR measurements.
MEASUrement:MEAS<x>:TIMINGMode	This command sets or queries the Timing mode for the specified DDR measurement.
MEASUrement:MEAS<x>:TOEdge	This command sets or queries the 'to edge' type for the measurement.
MEASUrement:MEAS<x>:TOEDGESEARCHDIRect	This command sets or queries the 'to edge' search direction for the measurement.
MEASUrement:MEAS<x>:TRANSition	This command sets or queries the transition edges flag for the measurement.
MEASUrement:MEAS<x>:TYPE	This command sets or queries the measurement type.
MEASUrement:MEAS<x>:WINDOWLENgth	This command sets or queries the window length for the measurement.
MEASUrement:MEASRange:MAX	This command sets or queries the global range maximum value.
MEASUrement:MEASRange:MIN	This command sets or queries the global range minimum value.
MEASUrement:MEASRange:STATE	Sets or queries the global range state.
MEASUrement:MINUI	This command sets or queries the minimum number of unit intervals required for BUJ analysis.
MEASUrement:POPulation:LIMIT:STATE	This command sets or queries the global population limit state.
MEASUrement:POPulation:LIMIT:VALue	This command sets or queries the global population limit value.
MEASUrement:REFLevels:ABSolute:FALLHigh	Sets or queries the value used as the high reference level of the falling edge.
MEASUrement:REFLevels:ABSolute:FALLLow	Sets or queries the value used as the low reference level of the falling edge.
MEASUrement:REFLevels:ABSolute:FALLMid	Sets or queries the value used as the mid reference level of the falling edge.
MEASUrement:REFLevels:ABSolute:HYSTeresis	Sets or queries the value of the hysteresis of the reference level.
MEASUrement:REFLevels:ABSolute:RISEHigh	Sets or queries the value used as the high reference level of the rising edge.
MEASUrement:REFLevels:ABSolute:RISELow	Sets or queries the value used as the low reference level of the rising edge.
MEASUrement:REFLevels:ABSolute:RISEMid	Sets or queries the value used as the mid reference level of the rising edge.
MEASUrement:REFLevels:ABSolute:TYPE	Sets or queries the reference level type.
MEASUrement:REFLevels:BASETop	Sets or queries the method used to calculate the TOP and BASE, used to calculate reference levels.
MEASUrement:REFLevels:JITTERMODE	Sets or queries how often reference levels are calculated on Jitter measurements.
MEASUrement:REFLevels:METHod	Sets or queries the method used to calculate reference levels.
MEASUrement:REFLevels:MODE	This command sets or queries how often reference levels are calculated.
MEASUrement:REFLevels:PERCent:FALLHigh	Sets or queries the percentage used to calculate the high reference level of the falling edge.
MEASUrement:REFLevels:PERCent:FALLLow	Sets or queries the percentage used to calculate the mid reference level of the falling edge.

Table 2-32: Measurement commands (cont.)

Command	Description
MEASUrement:REFLevels:PERCent:FALLMid	Sets or queries the percentage used to calculate the mid reference level of the falling edge.
MEASUrement:REFLevels:PERCent:HYSTeresis	Sets or queries the percentage used to calculate the hysteresis of the reference level.
MEASUrement:REFLevels:PERCent:RISEHigh	Sets or queries the percentage used to calculate the high reference level of the rising edge.
MEASUrement:REFLevels:PERCent:RISELow	Sets or queries the percentage used to calculate the low reference level of the rising edge.
MEASUrement:REFLevels:PERCent:RISEMid	Sets or queries the percentage used to calculate the mid reference level of the rising edge.
MEASUrement:REFLevels:PERCent:TYPE	Sets or queries the reference level percent type.
MEASUrement:REFLevels:TYPE	This command sets or queries the shared reference level method used for sources of measurement calculations.
MEASUrement:REF<x>:REFLevels:ABSolute:FALLHigh	Sets or queries the value used as the high reference level of the falling edge.
MEASUrement:REF<x>:REFLevels:ABSolute:FALLLow	Sets or queries the value used as the low reference level of the falling edge.
MEASUrement:REF<x>:REFLevels:ABSolute:FALLMid	Sets or queries the value used as the mid reference level of the falling edge.
MEASUrement:REF<x>:REFLevels:ABSolute:HYSTeresis	Sets or queries the value of the hysteresis of the reference level.
MEASUrement:REF<x>:REFLevels:ABSolute:RISEHigh	Sets or queries the value used as the high reference level of the rising edge.
MEASUrement:REF<x>:REFLevels:ABSolute:RISELow	Sets or queries the value used as the low reference level of the rising edge.
MEASUrement:REF<x>:REFLevels:ABSolute:RISEMid	Sets or queries the value used as the mid reference level of the rising edge.
MEASUrement:REF<x>:REFLevels:ABSolute:TYPE	Sets or queries the reference level type.
MEASUrement:REF<x>:REFLevels:BASETop	Sets or queries the method used to calculate the TOP and BASE, used to calculate reference levels.
MEASUrement:REF<x>:REFLevels:METHod	Sets or queries the method used to calculate reference levels.
MEASUrement:REF<x>:REFLevels:PERCent:FALLHigh	Sets or queries the percentage used to calculate the high reference level of the falling edge.
MEASUrement:REF<x>:REFLevels:PERCent:FALLLow	Sets or queries the percentage used to calculate the low reference level of the falling edge.
MEASUrement:REF<x>:REFLevels:PERCent:FALLMid	Sets or queries the percentage used to calculate the mid reference level of the falling edge.

Table 2-32: Measurement commands (cont.)

Command	Description
MEASUrement:REF<x>:REFLevels:PERCent:HYSTeresis	Sets or queries the percentage used to calculate the hysteresis of the reference level.
MEASUrement:REF<x>:REFLevels:PERCent:RISEHigh	Sets or queries the percentage used to calculate the high reference level of the rising edge.
MEASUrement:REF<x>:REFLevels:PERCent:RISELow	Sets or queries the percentage used to calculate the low reference level of the rising edge.
MEASUrement:REF<x>:REFLevels:PERCent:RISEMid	Sets or queries the percentage used to calculate the mid reference level of the rising edge.
MEASUrement:REF<x>:REFLevels:PERCent:TYPE	Sets or queries the reference level percent type.
MEASUrement:STATistics:CYCLEMode	This command sets or queries whether cycle-cycle statistics are calculated for all measurements.

Miscellaneous command group

Miscellaneous commands do not fit into other categories.

Several commands and queries are common to all devices. The 488.2-1987 standard defines these commands. The common commands begin with an asterisk (*) character.

Table 2-33: Miscellaneous commands

Command	Description
AUTOSAVEPITIMEOUT	Sets or queries the idle time from the programmable interface before auto-save occurs.
AUTOSAVEUITIMEOUT	Sets or queries the idle time from the user interface before auto-save occurs.
AUTOSet	Sets or queries the vertical, horizontal, and trigger controls of the instrument to automatically acquire and display the selected waveform.
AUTOSet:ACQuisition:ENABle	Sets or queries the Autoset acquisition setting adjustment.
AUTOSet:ENABle	Sets or queries the Autoset enable/disable mode.
AUTOSet:HORizontal:ENABle	Sets or queries Autoset's adjustment of horizontal settings.
AUTOSet:TRIGger:ENABle	Sets or queries Autoset's adjustment of trigger settings.
AUTOSet:VERTical:ENABle	sets or queries Autoset's adjustment of vertical settings.
AUTOSet:VERTical:OPTIMize	Sets or queries which vertical settings Autoset will optimize when the display mode is set to Overlay mode.
AUXout:EDGE	Sets or queries the direction in which the trigger output signal will transition when a trigger occurs.
AUXout:SOUrce	Sets or queries the trigger source at the BNC connection.
CLEAR	Clears acquisitions, measurements, and waveforms.
DATE?	Queries the date that the instrument displays.
*DDT	Sets or queries the commands that will be executed by the group execute trigger.
FPAnel:PRESS	Turns off the displayed menu.
FPAnel:TURN	Emulates a knob turn.
HEADer	Sets or queries the Response Header Enable State.
ID?	Returns identifying information about the instrument and its firmware.
*IDN?	Returns the instrument identification code.
LICense?	Queries all license parameters.
LICENSE:APPID?	Returns a comma-separated list of the active application IDs.
LICense:COUNT?	Returns a count of the number of active licenses installed.
LICense:ERRor?	This query-only command prompts the instrument to return all events and their messages (delimited by commas), and removes the returned events from the Event Queue (alias for ALLEV?).
LICense:GMT?	Returns the GMT time in ISO 8601 format, the local date, 24 hour time and time-zone offset.
LICense:HID?	Returns the instrument HostID unique identifier.

Table 2-33: Miscellaneous commands (cont.)

Command	Description
LICense:INSTall	Accepts a <block data> license and installs it on the instrument.
LICense:ITEM?	Returns the details pertaining to a specific license.
LICense:LIST?	Returns the active license nomenclatures as a comma-separated list of strings.
LICense:VALidate?	Accepts a license nomenclature as an argument and returns the status of the license.
LIC:UNINSTALL?	Returns the exit license information for the user to return to their TekAMS account.
LOCK	Sets or queries the front panel lock state.
*LRN?	Returns a listing of instrument settings.
MAINWindow:RRBDisplaystate	Sets the display state of the Results readout bar to ON (displayed) or OFF (not displayed).
NEWpass	Changes the password for user protected data.
PASSWord	Provides access for changing user protected data.
PAUSE	Causes the interface to pause the specified number of seconds before processing any other commands.
REM	Specifies a comment which is ignored by the instrument.
ROSc:SOUrce	Selects or queries the selected source for the time base reference oscillator.
ROSc:STATE?	Returns whether the time base reference oscillator is locked.
SET?	Returns a listing of instrument settings.
SOCKETServer:ENable	Enables or disables the socket server which supports a telnet or other TCPIP socket connection to send commands and queries to the instrument.
SOCKETServer:PORT	Sets the TCPIP port for the socket server connection.
SOCKETServer:PROTOCol	Sets or queries the protocol for the socket server.
TEKSecure	Initializes both waveform and setup memories.
TIME?	Queries the time displayed by the instrument.
TIME:ZONE	Sets the time zone to the one specified.
TIME:ZONE:UTCDELTA	Sets or queries the time zone using the difference between the desired time zone and UTC.
TOTaluptime?	Returns the total number of hours the oscilloscope has been turned on since the NV memory was last programmed.
TOUCHSCReen:STATe	Sets or queries the enabled state of the touch screen.
*TRG	Performs the group execute trigger (GET).
*TST?	Tests the interface and returns status.
UNDO	Reverts the scope settings to a state before the previous command or user interface action.
UNLock	Unlocks front panel.
USBDevice:CONFIgure	Used to configure the rear USB port to be off or enabled as a USBTMC device.
VERBoSe	Sets or queries the verbose state.

Plot command group

Plot commands let you select the type and control the appearance of your plots.

Table 2-34: Plot commands

Command	Description
PLOT:ADDNew	Adds the specified plot.
PLOT:DELeTe	Deletes the specified plot.
PLOT:LIST?	Lists all currently defined plots.
PLOT:PLOT<x>:BAThtub:BER	Sets or queries the bathtub BER value.
PLOT:PLOT<x>:BAThtub:XAXISUnits	Sets or queries the X-Axis unit, either unit intervals or seconds.
PLOT:PLOT<x>:BITType	Sets or queries the bit type to display for the specified eye diagram plot.
PLOT:PLOT<x>:LABel:COLor	Sets or queries the color of the specified trend label.
PLOT:PLOT<x>:LABel:FONT:BOLD	Sets or queries the bold state of the specified trend label.
PLOT:PLOT<x>:LABel:FONT:ITALic	Sets or queries the italic state of the specified trend label.
PLOT:PLOT<x>:LABel:FONT:SIZE	Sets or queries the font size of the specified trend label.
PLOT:PLOT<x>:LABel:FONT:TYPE	Sets or queries the font type of the specified trend label.
PLOT:PLOT<x>:LABel:FONT:UNDERline	Sets or queries the underline state of the specified trend label.
PLOT:PLOT<x>:LABel:NAMe	Sets or queries the specified trend's label.
PLOT:PLOT<x>:LABel:XPOS	Sets or queries the x-position of the specified trend label.
PLOT:PLOT<x>:LABel:YPOS	Sets or queries the y-position of the specified trend label.
PLOT:PLOT<x>:MASK?	Returns the name of the mask test associated with the specified eye diagram plot.
PLOT:PLOT<x>:NUMBins	Sets or queries the current histogram resolution.
PLOT:PLOT<x>:SOUrce<x>	Sets or queries the measurement source.
PLOT:PLOT<x>:SPECTrum:BASE	Sets or queries the spectrum base. Undefined for non-spectrum plots.
PLOT:PLOT<x>:SPECTrum:DYNRange	Sets or queries the dynamic range value.
PLOT:PLOT<x>:TYPE	Sets or queries the current plot type for the selected plot.

Power command group

Table 2-35: Power commands

Command	Description
POWer:ADDNew	Adds the specified power measurement badge.
POWer:DELeTe	Deletes the specified power measurement badge.
POWer:POWer<x>:AUTOSet	Executes power autoset for the specified power measurement badge.
POWer:POWer<x>:CLRESPONSE:AMP[1-10]Val	Sets or queries the generator amplitude value of the specified configuration step for the Control Loop Response power measurement.
POWer:POWer<x>:CLRESPONSE:AMPMode	Sets or queries the amplitude mode for the Control Loop Response power measurement.

Table 2-35: Power commands (cont.)

Command	Description
POWER:POWER<x>:CLRESPONSE:CONSTAMPLitude	Sets or queries the constant amplitude voltage for the Control Loop Response power measurement.
POWER:POWER<x>:CLRESPONSE:FREQ[1-11]Val	Sets or queries the generator frequency value of the specified configuration step for the Control Loop Response power measurement.
POWER:POWER<x>:CLRESPONSE:GENERator	Sets or queries the generator source used to send stimulus signals to the DUT, for the Control Loop Response power measurement.
POWER:POWER<x>:CLRESPONSE:IMPEDance	Sets or queries the vertical termination impedance for the Control Loop Response power measurement.
POWER:POWER<x>:CLRESPONSE:INPUTSource	Sets or queries the input source for the Control Loop Response power measurement.
POWER:POWER<x>:CLRESPONSE:OUTPUTSource	Sets or queries the output source for the Control Loop Response power measurement.
POWER:POWER<x>:CLRESPONSE:PPD	Sets or queries the points per decade (PPD) value for the Control Loop Response power measurement.
POWER:POWER<x>:CLRESPONSE:STARTFREquency	Sets or queries the start frequency value for the Control Loop Response power measurement.
POWER:POWER<x>:CLRESPONSE:STOPFREquency	Sets or queries the stop frequency value for the Control Loop Response power measurement.
POWER:POWER<x>:CYCLEAmp:INPUTSource	Sets or queries the input source for cycle amplitude measurement of the specified power measurement number.
POWER:POWER<x>:CYCLEBase:INPUTSource	Sets or queries the input source for cycle base measurement of the specified power measurement number.
POWER:POWER<x>:CYCLEMAX:INPUTSource	Sets or queries the input source for cycle maximum measurement of the specified power measurement number.
POWER:POWER<x>:CYCLEMin:INPUTSource	Sets or queries the input source for cycle minimum measurement of the specified power measurement number.
POWER:POWER<x>:CYCLEPKPK:INPUTSource	Sets or queries the input source for cycle peak-to-peak measurement of the specified power measurement number.
POWER:POWER<x>:CYCLETop:INPUTSource	Sets or queries the input source for cycle top measurement of the specified power measurement number.
POWER:POWER<x>:DIDT:INPUTSource	Sets or queries the input source for di/dt measurement of the specified power measurement number.
POWER:POWER<x>:DIDT:SOURCEEDGEType	Sets or queries the edge type for di/dt measurement of the specified power measurement number.
POWER:POWER<x>:DVDT:INPUTSource	Sets or queries the input source for dv/dt measurement of the specified power measurement number.
POWER:POWER<x>:DVDT:SOURCEEDGEType	Sets or queries the edge type for dv/dt measurement of the specified power measurement number.

Table 2-35: Power commands (cont.)

Command	Description
POWER:POWer<x>:EFFICIENCY:INPUTType	Sets or queries the input type (AC or DC) for efficiency measurement of the specified power measurement number.
POWER:POWer<x>:EFFICIENCY:IOUT1SOUrce	Sets or queries the output 1 current source for efficiency measurement of the specified power measurement number.
POWER:POWer<x>:EFFICIENCY:IOUT2SOUrce	Sets or queries the output 2 current source for efficiency measurement of the specified power measurement number.
POWER:POWer<x>:EFFICIENCY:IOUT3SOUrce	Sets or queries the output 3 current source for efficiency measurement of the specified power measurement number.
POWER:POWer<x>:EFFICIENCY:ISOUrce	Sets or queries the current source for efficiency measurement of the specified power measurement number.
POWER:POWer<x>:EFFICIENCY:NUMOFOutputs	Sets or queries the number of outputs for efficiency measurement of the specified power measurement number.
POWER:POWer<x>:EFFICIENCY:OUTPUT1Type	Sets or queries the Output1 type for efficiency measurement of the specified power measurement number.
POWER:POWer<x>:EFFICIENCY:OUTPUT2Type	Sets or queries the Output2 type for efficiency measurement of the specified power measurement number.
POWER:POWer<x>:EFFICIENCY:OUTPUT3Type	Sets or queries the Output3 type for efficiency measurement of the specified power measurement number.
POWER:POWer<x>:EFFICIENCY:OUTPUTType	Sets or queries the Output type for efficiency measurement of the specified power measurement number.
POWER:POWer<x>:EFFICIENCY:VOUT1SOUrce	Sets or queries the output 1 voltage source for efficiency measurement of the specified power measurement number.
POWER:POWer<x>:EFFICIENCY:VOUT2SOUrce	Sets or queries the output 2 voltage source for efficiency measurement of the specified power measurement number.
POWER:POWer<x>:EFFICIENCY:VOUT3SOUrce	Sets or queries the output 3 voltage source for efficiency measurement of the specified power measurement number.
POWER:POWer<x>:EFFICIENCY:VSOUrce	Sets or queries the voltage source for efficiency measurement of the specified power measurement number.
POWER:POWer<x>:FREQUENCY:EDGE	Sets or queries the edge type for frequency measurement of the specified power measurement number.
POWER:POWer<x>:FREQUENCY:INPUTSource	Sets or queries the input source for frequency measurement of the specified power measurement number.
POWER:POWer<x>:GATing	Sets or queries the gating type for the specified power measurement badge.
POWER:POWer<x>:GATing:GLOBal	Sets or queries the gating settings for the specified power measurement badge.
POWER:POWer<x>:HARMONICS:CLASS	Sets or queries the class type for the harmonics measurement of the specified power measurement number.
POWER:POWer<x>:HARMONICS:CMETHod	Sets or queries the fundamental current method for the harmonics measurement of the specified power measurement number.

Table 2-35: Power commands (cont.)

Command	Description
POWER:POWER<x>:HARMONICS:FUNDCURRENT	Sets or queries the fundamental current value for the harmonics measurement of the specified power measurement number.
POWER:POWER<x>:HARMONICS:HORDER	Sets or queries the order value for the harmonics measurement of the specified power measurement number.
POWER:POWER<x>:HARMONICS:HSOURCE	Sets or queries the source type for the harmonics measurement of the specified power measurement number.
POWER:POWER<x>:HARMONICS:IPOWER	Sets or queries the input power value for the harmonics measurement of the specified power measurement number.
POWER:POWER<x>:HARMONICS:ISOURCE	Sets or queries the current source for SOA measurement of the specified power measurement number.
POWER:POWER<x>:HARMONICS:LINEFREQUENCY	Sets or queries the value for the line frequency for the power Harmonics measurement.
POWER:POWER<x>:HARMONICS:ODDEVEN	Sets or queries the harmonics value analysis format of the specified power measurement number.
POWER:POWER<x>:HARMONICS:PFACtor	Sets or queries the value of power factor for the harmonics measurement of the specified power measurement number.
POWER:POWER<x>:HARMONICS:POWERRating	Sets or queries the power level for the harmonics measurement of the specified power measurement number.
POWER:POWER<x>:HARMONICS:RCURRENT	Sets or queries the rated current for the harmonics measurement of the specified power measurement number.
POWER:POWER<x>:HARMONICS:STANDARD	Sets or queries the test mode for harmonics measurement of the specified power measurement number.
POWER:POWER<x>:HARMONICS:STARTFREQUENCY	Sets or queries the value for the start frequency for the power Harmonics measurement. in the range of 1 Hz to 1 GHz.
POWER:POWER<x>:HARMONICS:UNITS	Sets or queries the harmonics results units of the specified power measurement number.
POWER:POWER<x>:HARMONICS:VSOURCE	Sets or queries the voltage source for SOA measurement of the specified power measurement number.
POWER:POWER<x>:INDUCTANCE:EDGESource	Sets or queries the edge source for inductance measurement of the specified power measurement number.
POWER:POWER<x>:INDUCTANCE:ISOURCE	Sets or queries the current source for inductance measurement of the specified power measurement number.
POWER:POWER<x>:INDUCTANCE:VSOURCE	Sets or queries the voltage source for inductance measurement of the specified power measurement number.
POWER:POWER<x>:INPUTCAP:ISOURCE	Sets or queries the inrush current input source of the specified Input Capacitance measurement.
POWER:POWER<x>:INPUTCAP:PEAKCURRENT	Sets or queries the peak current value of the specified Input Capacitance measurement.

Table 2-35: Power commands (cont.)

Command	Description
POWER:POWer<x>:INPUTCAP:PEAKVOLTage	Sets or queries the peak voltage value of the specified Input Capacitance measurement.
POWER:POWer<x>:INPUTCAP:VSOURce	Sets or queries the input voltage source of the specified Input Capacitance measurement.
POWER:POWer<x>:INRUSHcurrent:INPUTSOurce	Sets or queries the input source of the specified Inrush Current measurement.
POWER:POWer<x>:INRUSHcurrent:PEAKCURREnt	Sets or queries the peak current value of the specified Inrush Current measurement.
POWER:POWer<x>:IVSINTEGRALV:ISOURce	Sets or queries the current source for I vs Integral V measurement of the specified power measurement number.
POWER:POWer<x>:IVSINTEGRALV:VSOURce	Sets or queries the voltage source for I vs Integral V measurement of the specified power measurement number.
POWER:POWer<x>:LABel	Sets or queries the custom name for the specified power measurement badge.
POWER:POWer<x>:LINERIPPLE:INPUTSOurce	Sets or queries the input source for line ripple measurement of the specified power measurement number.
POWER:POWer<x>:LINERIPPLE:LFREQuency	Sets or queries the frequency present for line ripple measurement of the specified power measurement number.
POWER:POWer<x>:MAGNETICLOSS:ISOURce	Sets or queries the current source for magnetic measurement of the specified power measurement number.
POWER:POWer<x>:MAGNETICLOSS:VSOURce	Sets or queries the voltage source for magnetic measurement of the specified power measurement number.
POWER:POWer<x>:MAGPROPERTY:AREAofcrosssection	Sets or queries the coil cross section area for magnetic measurement of the specified power measurement number.
POWER:POWer<x>:MAGPROPERTY:EDGESOURce	Sets or queries the edge signal source for magnetic measurement of the specified power measurement number.
POWER:POWer<x>:MAGPROPERTY:ISOURce	Sets or queries the current source for magnetic measurement of the specified power measurement number.
POWER:POWer<x>:MAGPROPERTY:LENgth	Sets or queries the conductor length for magnetic measurement of the specified power measurement number.
POWER:POWer<x>:MAGPROPERTY:PRIMARYTURNs	Sets or queries the number of primary turns for magnetic measurement of the specified power measurement number.
POWER:POWer<x>:MAGPROPERTY:SEC1SOURce	Sets or queries the current source for secondary winding 1 for magnetic measurement of the specified power measurement number.
POWER:POWer<x>:MAGPROPERTY:SEC1TURNs	Sets or queries the number of turns of secondary winding 1 for magnetic measurement in the specified power measurement badge
POWER:POWer<x>:MAGPROPERTY:SEC2SOURce	Sets or queries the current source for secondary winding 2 for magnetic measurement of the specified power measurement number.

Table 2-35: Power commands (cont.)

Command	Description
POWer:POWer<x>: MAGPROPERTY:SEC2TURNs	Sets or queries the number of turns of secondary winding 2 for magnetic measurement of the specified power measurement number.
POWer:POWer<x>: MAGPROPERTY:SEC3SOURce	Sets or queries the current source for secondary winding 3 for magnetic measurement of the specified power measurement number.
POWer:POWer<x>: MAGPROPERTY:SEC3TURNs	Sets or queries the number of turns of secondary winding 3 for magnetic measurement of the specified power measurement number.
POWer:POWer<x>: MAGPROPERTY:SEC4SOURce	Sets or queries the current source for secondary winding 4 for magnetic measurement of the specified power measurement number.
POWer:POWer<x>: MAGPROPERTY:SEC4TURNs	Sets or queries the number of turns of secondary winding 4 for magnetic measurement of the specified power measurement number.
POWer:POWer<x>: MAGPROPERTY:SEC5SOURce	Sets or queries the current source for secondary winding 5 for magnetic measurement of the specified power measurement number.
POWer:POWer<x>: MAGPROPERTY:SEC5TURNs	Sets or queries the number of turns of secondary winding 5 for magnetic measurement of the specified power measurement number.
POWer:POWer<x>: MAGPROPERTY:SEC6SOURce	Sets or queries the current source for secondary winding 6 for magnetic measurement of the specified power measurement number.
POWer:POWer<x>: MAGPROPERTY:SEC6TURNs	Sets or queries the number of turns of secondary winding 6 for magnetic measurement of the specified power measurement number.
POWer:POWer<x>: MAGPROPERTY:SECWINDings	Sets or queries the number of secondary windings for magnetic measurement of the specified power measurement number.
POWer:POWer<x>: MAGPROPERTY:UNITs	Sets or queries the units for magnetic measurement of the specified power measurement number.
POWer:POWer<x>: MAGPROPERTY:VSOURce	Sets or queries the voltage source for magnetic measurement of the specified power measurement number.
POWer:POWer<x>:NDUTYCYCLE: EDGEType	Sets or queries the clock edge type for negative duty cycle measurement of the specified power measurement number.
POWer:POWer<x>:NDUTYCYCLE: INPUTSource	Sets or queries the input source for negative duty cycle measurement of the specified power measurement number.
POWer:POWer<x>:NPULSEWIDTH: INPUTSource	Sets or queries the input source for negative pulse width measurement of the specified power measurement number.
POWer:POWer<x>:PDUTYCYCLE: EDGEType	Sets or queries the clock edge type for positive duty cycle measurement of the specified power measurement number.
POWer:POWer<x>:PDUTYCYCLE: INPUTSource	Sets or queries the input source for positive duty cycle measurement of the specified power measurement number.
POWer:POWer<x>:PERIOD:EDGe	Sets or queries the edge type for period measurement of the specified power measurement number.
POWer:POWer<x>:PERIOD: INPUTSource	Sets or queries the input source for period measurement of the specified power measurement number.

Table 2-35: Power commands (cont.)

Command	Description
POWer:POWer<x>: POWERQUALITY:CCYCles	Sets or queries the calculate cycles over full cycles settings for the specified power quality measurement badge.
POWer:POWer<x>: POWERQUALITY:FReference	Sets or queries the frequency reference type for power quality measurement of the specified power measurement number.
POWer:POWer<x>: POWERQUALITY:ISOURce	Sets or queries the current source for power quality measurement of the specified power measurement number.
POWer:POWer<x>: POWERQUALITY:VSOURce	Sets or queries the voltage source for power quality measurement of the specified power measurement number.
POWer:POWer<x>:PRESET	Sets or queries the input source for positive pulse width measurement of the specified power measurement number.
POWer:POWer<x>:PSRR: AMP[1-10]Val	Sets or queries the generator amplitude value of the specified configuration step for the Power Supply Rejection Ratio (PSRR) power measurement.
POWer:POWer<x>:PSRR: AMPMode	Sets or queries the amplitude mode for the Power Supply Rejection Ratio (PSRR) power measurement.
POWer:POWer<x>:PSRR: CONSTAmplitude	Sets or queries the constant amplitude voltage for the Power Supply Rejection Ratio (PSRR) power measurement.
POWer:POWer<x>:PSRR: FREQ[1-11]Val	Sets or queries the generator frequency value of the specified configuration step for the Power Supply Rejection Ratio (PSRR) power measurement.
POWer:POWer<x>:PSRR: GENerator	Sets or queries the generator source used to send stimulus signals to the DUT for the Power Supply Rejection Ratio (PSRR) power measurement.
POWer:POWer<x>:PSRR: IMPEDance	Sets or queries the vertical termination impedance for the Power Supply Rejection Ratio (PSRR) power measurement.
POWer:POWer<x>:PSRR: INPUTSource	Sets or queries the input source for the Power Supply Rejection Ratio (PSRR) power measurement.
POWer:POWer<x>:PSRR: OUTPUTSource	Sets or queries the output source for the Power Supply Rejection Ratio (PSRR) power measurement.
POWer:POWer<x>:PSRR:PPD	Sets or queries the points per decade (PPD) value for the Power Supply Rejection Ratio (PSRR) power measurement.
POWer:POWer<x>:PSRR: STARTFREquency	Sets or queries the start frequency value for the Power Supply Rejection Ratio (PSRR) power measurement.
POWer:POWer<x>:PSRR: STOPFREquency	Sets or queries the stop frequency value for the Power Supply Rejection Ratio (PSRR) power measurement.
POWer:POWer<x>:RDSON: DEVICeType	Sets or queries the device type for the power drain source on resistance measurement for RDSON measurement of the specified power measurement number.
POWer:POWer<x>:RDSON: ISOURce	Sets or queries the current source for RDSON measurement of the specified power measurement number.
POWer:POWer<x>:RDSON: VSOURce	Sets or queries the voltage source for RDSON measurement of the specified power measurement number.

Table 2-35: Power commands (cont.)

Command	Description
POWER:POWER<x>:REFLevels:ABSolute:FALLHigh	Sets or queries the falling edge for high reference level in absolute units for the specified power measurement badge.
POWER:POWER<x>:REFLevels:ABSolute:FALLLow	Sets or queries the falling edge for low reference level in absolute units for the specified power measurement badge.
POWER:POWER<x>:REFLevels:ABSolute:FALLMid	Sets or queries the falling edge for mid reference level in absolute units for the specified power measurement badge.
POWER:POWER<x>:REFLevels:ABSolute:HYSteresis	Sets or queries the absolute hysteresis value for the specified power measurement badge.
POWER:POWER<x>:REFLevels:ABSolute:RISEHigh	Sets or queries the rising edge for high reference level in absolute units for the specified power measurement badge.
POWER:POWER<x>:REFLevels:ABSolute:RISELow	Sets or queries the rising edge for low reference level in absolute units for the specified power measurement badge.
POWER:POWER<x>:REFLevels:ABSolute:RISEMid	Sets or queries the rising edge for mid reference level in absolute units for the specified power measurement badge.
POWER:POWER<x>:REFLevels:ABSolute:TYPE	Sets or queries the type of measurement levels when reference level is set to absolute for the specified power measurement badge.
POWER:POWER<x>:REFLevels:BASETop	Sets or queries the reference level base top method for the specified power measurement badge.
POWER:POWER<x>:REFLevels:METHod	Sets or queries the method to configure reference level values for the specified power measurement badge.
POWER:POWER<x>:REFLevels:PERCent:FALLHigh	Sets or queries the falling edge for high reference level in percentage for the specified power measurement badge.
POWER:POWER<x>:REFLevels:PERCent:FALLLow	Sets or queries the falling edge for low reference level in percentage for the specified power measurement badge.
POWER:POWER<x>:REFLevels:PERCent:FALLMid	Sets or queries the falling edge for mid reference level in percentage for the specified power measurement badge.
POWER:POWER<x>:REFLevels:PERCent:HYSteresis	Sets or queries the hysteresis in percentage for the specified power measurement badge.
POWER:POWER<x>:REFLevels:PERCent:RISEHigh	Sets or queries the rising edge for high reference level in percentage for the specified power measurement badge.
POWER:POWER<x>:REFLevels:PERCent:RISELow	Sets or queries the rising edge for low reference level in percentage for the specified power measurement badge.
POWER:POWER<x>:REFLevels:PERCent:RISEMid	Sets or queries the rising edge for mid reference level in percentage for the specified power measurement badge.
POWER:POWER<x>:REFLevels:PERCent:TYPE	Sets or queries the reference levels for the specified power measurement badge.
POWER:POWER<x>:RESUltS:ALLAcqs:MAXimum?	Queries the maximum value of all acquisitions for the measurement parameter of the specified power measurement number.

Table 2-35: Power commands (cont.)

Command	Description
POWER:POWER<x>:RESULTS:ALLAcqs:MEAN?	Queries the mean value of all acquisitions for the measurement parameter of the specified power measurement number.
POWER:POWER<x>:RESULTS:ALLAcqs:MINimum?	Queries the minimum value of all acquisitions for the measurement parameter of the specified power measurement number.
POWER:POWER<x>:RESULTS:ALLAcqs:PK2PK?	Queries the peak-to-peak value of all acquisitions for the measurement parameter of the specified power measurement number.
POWER:POWER<x>:RESULTS:ALLAcqs:POPulation?	Queries the population (number of complete cycles) of all acquisitions for the measurement parameter of the specified power measurement number.
POWER:POWER<x>:RESULTS:ALLAcqs:STDDev?	Queries the standard deviation value of all acquisitions for the measurement parameter of the specified power measurement number.
POWER:POWER<x>:RESULTS:CURRentacq:F1MAG?	Queries the first harmonics magnitude value for the specified power measurement badge.
POWER:POWER<x>:RESULTS:CURRentacq:F3MAG?	Queries the third harmonics magnitude value for the specified power measurement badge.
POWER:POWER<x>:RESULTS:CURRentacq:FREQUENCY?	Queries the fundamental frequency for the specified power measurement badge.
POWER:POWER<x>:RESULTS:CURRentacq:IRMS?	Queries the RMS current value for the specified power measurement badge.
POWER:POWER<x>:RESULTS:CURRentacq:MAXimum?	Queries the maximum value of the current acquisition for the measurement parameter of the specified power measurement number.
POWER:POWER<x>:RESULTS:CURRentacq:MEAN?	Queries the mean value of the current acquisition for the measurement parameter of the specified power measurement number.
POWER:POWER<x>:RESULTS:CURRentacq:MINimum?	Queries the minimum value of the current acquisition for the measurement parameter of the specified power measurement number.
POWER:POWER<x>:RESULTS:CURRentacq:PK2PK?	Queries the peak-to-peak value of the current acquisition for the measurement parameter of the specified power measurement number.
POWER:POWER<x>:RESULTS:CURRentacq:POHCL?	Queries the limit of partial odd harmonic current for the specified power measurement badge.
POWER:POWER<x>:RESULTS:CURRentacq:POHCM?	Queries the measured value of partial odd harmonic current for the specified power measurement badge.
POWER:POWER<x>:RESULTS:CURRentacq:POHCS?	Queries the status of partial odd harmonic current for the specified power measurement badge.
POWER:POWER<x>:RESULTS:CURRentacq:POPulation?	Queries the population (number of complete cycles) of the current acquisition for the measurement parameter of the specified power measurement number.
POWER:POWER<x>:RESULTS:CURRentacq:RMS?	Queries the RMS value of the source selected for the specified power measurement badge.
POWER:POWER<x>:RESULTS:CURRentacq:STATUS?	Queries the status of the measurement for the specified power measurement badge.

Table 2-35: Power commands (cont.)

Command	Description
POWER:POWER<x>:RESULTS:CURRentacq:STDDev?	Queries the standard deviation value of the current acquisition for the measurement parameter of the specified power measurement number.
POWER:POWER<x>:RESULTS:CURRentacq:THDF?	Queries the total harmonic distortion (fundamental) value for the specified power measurement badge.
POWER:POWER<x>:RESULTS:CURRentacq:THDR?	Queries the total harmonic distortion (RMS) value for the specified power measurement badge.
POWER:POWER<x>:RESULTS:CURRentacq:TRPWR?	Queries the true power value for the specified power measurement badge.
POWER:POWER<x>:RESULTS:CURRentacq:VRMS?	Queries the RMS voltage value for the specified power measurement badge.
POWER:POWER<x>:SEQSETup	Runs the power measurement sequence setup.
POWER:POWER<x>:SEQuence	Sets or queries the run state of a single sequence power measurement.
POWER:POWER<x>:SOA:ISOURce	Sets or queries the current source for SOA measurement of the specified power measurement number.
POWER:POWER<x>:SOA:POINT	Sets or queries the X or Y coordinate value for a specified SOA mask point.
POWER:POWER<x>:SOA:RECAAllmask	Recalls or queries the recall mask file name of the specified power measurement number.
POWER:POWER<x>:SOA:RECAAllmask:FILENAME	Sets or queries the file name for saving SOA mask file name of the specified power measurement number.
POWER:POWER<x>:SOA:SAVemask	Saves the mask file as per the name configured and at the configured path or queries the mask file name, path, and file type for the SOA measurement of the specified power measurement number.
POWER:POWER<x>:SOA:SAVemask:AUTOINCrement	Sets or queries the state of auto-increment for saved SOA mask file names of the specified power measurement number.
POWER:POWER<x>:SOA:SAVemask:FILENAME	Sets or queries the mask file name for SOA measurement of the specified power measurement number.
POWER:POWER<x>:SOA:SAVemask:FOLDer	Sets or queries the mask file folder path for SOA measurement of the specified power measurement number.
POWER:POWER<x>:SOA:VSOURce	Sets or queries the voltage source for SOA measurement of the specified power measurement number.
POWER:POWER<x>:SWITCHINGLOSS:DEVICEType	Sets or queries the conduction calculation method for switching loss measurement of the specified power measurement number.
POWER:POWER<x>:SWITCHINGLOSS:GATESource	Sets or queries the gate voltage (V_g) for the switching loss measurement of the specified power measurement number.
POWER:POWER<x>:SWITCHINGLOSS:ILEVELAbs	Sets or queries the current level (Ton-Start & Stop) in absolute units for switching loss measurement of the specified power measurement number.
POWER:POWER<x>:SWITCHINGLOSS:ILEVELPct	Sets or queries the current level (Ton-Start & Stop) in percentage for switching loss measurement of the specified power measurement number.

Table 2-35: Power commands (cont.)

Command	Description
POWer:POWer<x>: SWITCHINGLOSS:ISOURce	Sets or queries the current source for the switching loss measurement of the specified power measurement number.
POWer:POWer<x>: SWITCHINGLOSS:LEVELUNIts	Sets or queries the level units for switching loss measurement of the specified power measurement number.
POWer:POWer<x>: SWITCHINGLOSS:RDSOn	Sets or queries the RDS(on) value for switching loss measurement of the specified power measurement number.
POWer:POWer<x>: SWITCHINGLOSS: SWLCONFIgType	Sets or queries the configuration type for the switching loss measurement of the specified power measurement number.
POWer:POWer<x>: SWITCHINGLOSS:VCESat	Sets or queries the value for the VCE(sat) value for switching loss measurement of the specified power measurement number.
POWer:POWer<x>: SWITCHINGLOSS:VGLevel	Sets or queries the gate voltage value (V_g Level Ton-Start) for the switching loss measurement of the specified power measurement number.
POWer:POWer<x>: SWITCHINGLOSS:VLEVELAbs	Sets or queries the voltage level (Ton-Start & Stop) in absolute units for switching loss measurement of the specified power measurement number.
POWer:POWer<x>: SWITCHINGLOSS:VLEVELPct	Sets or queries the voltage level (Ton-Start & Stop) in percentage for switching loss measurement of the specified power measurement number.
POWer:POWer<x>: SWITCHINGLOSS:VSOURce	Sets or queries the voltage source for the switching loss measurement of the specified power measurement number.
POWer:POWer<x>: SWITCHINGRIPPLE:INPUTSOurce	Sets or queries the input source for switching ripple measurement of the specified power measurement number.
POWer:POWer<x>: SWITCHINGRIPPLE:LFREQuency	Sets or queries the switching frequency for switching ripple measurement of the specified power measurement number.
POWer:POWer<x>:TURNOFFtime: FREQuency	Sets or queries the input frequency used by the AC or DC converter of the specified Turn Off Time measurement.
POWer:POWer<x>:TURNOFFtime: INPUTLEVel	Sets or returns the input frequency used by the AC or DC converter of the specified Turn Off Time measurement.
POWer:POWer<x>:TURNOFFtime: INPUTSOurce	Sets or queries the input source of the specified Turn Off Time measurement.
POWer:POWer<x>:TURNOFFtime: MAXTime	Sets or queries the maximum turn off time of the specified Turn Off Time measurement.
POWer:POWer<x>:TURNOFFtime: MAXVoltage	Sets or queries the maximum voltage of the specified Turn OffTime measurement.
POWer:POWer<x>:TURNOFFtime: NUMOUTputs	Sets or queries the number of outputs of the specified Turn Off Time power measurement.
POWer:POWer<x>:TURNOFFtime: OUTPUT1SOURce	Sets or queries the output 1 source of the specified Turn Off Time measurement.
POWer:POWer<x>:TURNOFFtime: OUTPUT1VOLTage	Sets or queries the output 1 voltage level of the of the specified Turn Off Time power measurement.

Table 2-35: Power commands (cont.)

Command	Description
POWER:POWER<x>:TURNOFFtime:OUTPUT2SOURce	Sets or queries the output 2 source of the specified Turn Off Time measurement.
POWER:POWER<x>:TURNOFFtime:OUTPUT2VOLTage	Sets or queries the output 2 voltage level of the specified Turn Off Time power measurement.
POWER:POWER<x>:TURNOFFtime:OUTPUT3SOURce	Sets or queries the output 3 source of the specified Turn Off Time measurement.
POWER:POWER<x>:TURNOFFtime:OUTPUT3VOLTage	Sets or queries the output 3 voltage level of the specified Turn Off Time power measurement.
POWER:POWER<x>:TURNOFFtime:OUTPUT4SOURce	Sets or queries the output 4 source of the specified Turn Off Time measurement.
POWER:POWER<x>:TURNOFFtime:OUTPUT4VOLTage	Sets or queries the output 4 voltage level of the specified Turn Off Time power measurement.
POWER:POWER<x>:TURNOFFtime:OUTPUT5SOURce	Sets or queries the output 5 source of the specified Turn Off Time measurement.
POWER:POWER<x>:TURNOFFtime:OUTPUT5VOLTage	Sets or queries the output 5 voltage level of the specified Turn Off Time power measurement.
POWER:POWER<x>:TURNOFFtime:OUTPUT6SOURce	Sets or queries the output 6 source of the specified Turn Off Time measurement.
POWER:POWER<x>:TURNOFFtime:OUTPUT6VOLTage	Sets or queries the output 6 voltage level of the specified Turn Off Time power measurement.
POWER:POWER<x>:TURNOFFtime:OUTPUT7SOURce	Sets or queries the output 7 source of the specified Turn Off Time measurement.
POWER:POWER<x>:TURNOFFtime:OUTPUT7VOLTage	Sets or queries the output 7 voltage level of the specified Turn Off Time power measurement.
POWER:POWER<x>:TURNOFFtime:TYPE	Sets or queries the type of AC/DC converter used in the specified Turn Off Time power measurement.
POWER:POWER<x>:TURNONtime:FREQuency	Sets or queries the input frequency used by the AC or DC converter of the specified Turn On Time measurement.
POWER:POWER<x>:TURNONtime:INPUTLEVel	Sets or queries the input voltage level of the specified Turn On Time measurement.
POWER:POWER<x>:TURNONtime:INPUTSOurce	Sets or queries the input source of the specified Turn On Time measurement.
POWER:POWER<x>:TURNONtime:MAXTime	Sets or queries the maximum turn on time of the specified Turn On Time measurement.
POWER:POWER<x>:TURNONtime:MAXVoltage	Sets or queries the maximum voltage setting of the specified Turn On Time measurement.
POWER:POWER<x>:TURNONtime:NUMOUTputs	Sets or queries the number of outputs for the specified Turn On Time power measurement.

Table 2-35: Power commands (cont.)

Command	Description
POWer:POWer<x>:TURNONtime:OUTPUT1SOURce	Sets or queries the output 1 source of the specified Turn On Time measurement.
POWer:POWer<x>:TURNONtime:OUTPUT1VOLTage	Sets or queries the output 1 voltage level of the specified Turn On Time power measurement.
POWer:POWer<x>:TURNONtime:OUTPUT2SOURce	Sets or queries the output 2 source of the specified Turn On Time measurement.
POWer:POWer<x>:TURNONtime:OUTPUT2VOLTage	Sets or queries the output 2 voltage level of the specified Turn On Time power measurement.
POWer:POWer<x>:TURNONtime:OUTPUT3SOURce	Sets or queries the output 3 source of the specified Turn On Time measurement.
POWer:POWer<x>:TURNONtime:OUTPUT3VOLTage	Sets or queries the output 3 voltage level of the specified Turn On Time power measurement.
POWer:POWer<x>:TURNONtime:OUTPUT4SOURce	Sets or queries the output 4 source of the specified Turn On Time measurement.
POWer:POWer<x>:TURNONtime:OUTPUT4VOLTage	Sets or queries the output 4 voltage level of the specified Turn On Time power measurement.
POWer:POWer<x>:TURNONtime:OUTPUT5SOURce	Sets or queries the output 5 source of the specified Turn On Time measurement.
POWer:POWer<x>:TURNONtime:OUTPUT5VOLTage	Sets or queries the output 5 voltage level of the specified Turn On Time power measurement.
POWer:POWer<x>:TURNONtime:OUTPUT6SOURce	Sets or queries the output 6 source of the specified Turn On Time measurement.
POWer:POWer<x>:TURNONtime:OUTPUT6VOLTage	Sets or queries the output 6 voltage level of the specified Turn On Time power measurement.
POWer:POWer<x>:TURNONtime:OUTPUT7SOURce	Sets or queries the output 7 source of the specified Turn On Time measurement.
POWer:POWer<x>:TURNONtime:OUTPUT7VOLTage	Sets or queries the output 7 voltage level of the specified Turn On Time power measurement.
POWer:POWer<x>:TURNONtime:TYPE	Sets or queries the type of AC/DC converter used in the specified Turn On Time power measurement.
POWer:POWer<x>:TYPE	Sets or queries the measurement type for the specified power measurement badge.
POWERTABLE:ADDNew	Adds the power harmonics table.
POWERTABLE:DELeTe	Deletes the power harmonics table.
POWERTABLE:LIST?	Lists all defined power harmonics tables.

Save and Recall command group

Use the commands in the Save and Recall Command Group to store and retrieve internal waveforms and settings. When you save a setup, you save all the settings of the instrument. When you recall a setup, the instrument restores itself to the state that it was in when you originally saved that setting.

Table 2-36: Save and Recall commands

Command	Description
FACtory	Resets the instrument to factory default settings.
RECALL:SESSion	Restores the state of the instrument from a saved session file.
RECALL:SETUp	Recalls saved instrument settings.
RECALL:WAVEform	Recalls a stored waveform to a reference memory location.
SAVe:EVENTable:BUS	Saves bus results table to the specified file.
SAVe:EVENTable:CUSTom	Saves the results table to the specified file path and name.
SAVe:EVENTable:CUSTom:COMMENTS	Sets or queries comments to be included in saved results table files.
SAVe:EVENTable:CUSTom:DATAFormat	Sets or queries the data format to use for saving results table data.
SAVe:EVENTable:CUSTom:INCLUDEREFs	Sets or queries whether to include displayed reference waveforms with saved results table files.
SAVe:EVENTable:MEASUrement	Saves data (measurement) results to the specified file.
SAVe:IMAGe	Saves a capture of the screen contents to the specified image file.
SAVe:REPOrt	Saves a report to the specified file or, if no argument is specified, uses the folder and file name specified by the related commands.
SAVe:REPOrt:COMMENTS	Sets or queries the comments to be included in saved report files.
SAVe:SESSion	Saves the state of the instrument, including reference waveforms, to a saved session file.
SAVe:SETUp	Saves the current instrument state to the specified file.
SAVe:SETUp:INCLUDEREFs	Sets or queries whether displayed reference waveforms are to be included in saved setups.
SAVe:WAVEform	Saves the specified waveform(s) to the specified destination file(s).
SAVe:WAVEform:SOURCEList?	Returns a list of the available waveforms that can be specified as the source for the SAvE:WAVEform command.

Save On command Group

Use this group of commands to program the oscilloscope to save images, measurements, waveforms, or the instrument setup, on triggers that you select.

Table 2-37: Save On commands

Command	Description
SAVEON:FILE:DEST	Sets or queries the location where files are saved.
SAVEON:FILE:NAME	Sets or queries the file name to use when SAVEON:TRIGer is ON.
SAVEON:IMAGe:FILEFormat	Sets or queries the file format to be used for saved image files.
SAVEON:IMAGe	Sets or queries whether to save a screen capture when a trigger occurs.
SAVEON:TRIGger	Sets or queries whether to save a file when a trigger occurs.
SAVEON:WAVEform	Sets or queries whether to save a waveform when a limit test failure, mask failure, or trigger occurs.
SAVEON:WAVEform:FILEFormat	Sets or queries the file format for saving waveform.
SAVEON:WAVEform:SOURce	Sets or queries the sources for saving waveforms.

Search and Mark command group

Use search and mark commands to seek out and identify information in waveform records that warrant further investigation.

Table 2-38: Search and Mark commands

Command	Description
SEARCH:ADDNew	Adds the specified search.
SEARCH:DELeTe	Deletes the specified search.
SEARCH:LIST?	Lists all currently defined searches.
SEARCH:SEARCH<x>:COPy	Copies the search criteria to or from the trigger.
SEARCH:SEARCH<x>:NAVigate	Sets the navigation action for search marks.
SEARCH:SEARCH<x>:TOTAL?	Queries the total number of found search marks for this search.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:CONDition	Specifies a field or condition for an ARINC429 bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:ERRTYPe	Sets or queries the error type when searching on an ARINC429 bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:LABel:QUALifier	Sets or queries the qualifier to be used when searching on label data for an ARINC429 bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:LABel:VALue	Sets or queries the low value when searching on an ARINC429 label field.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:LABel:HIVALue	Sets or queries the high value when searching on an ARINC429 label field.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:SDI:VALue	Sets or queries the when searching on an ARINC429 SDI field.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:SSM:VALue	Sets or queries the when searching on an ARINC429 SSM field.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:DATa:QUALifier	Sets or queries the qualifier to be used when searching on data in the DATA field for an ARINC429 bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:DATa:VALue	Sets or queries the low value when searching on an ARINC429 data field.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:DATa:HIVALue	Sets or queries the high value when searching on an ARINC429 data field.
SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:CONDition	Sets or queries the condition (word select / frame sync, or matching data) to be used when searching on an audio bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATa:HITDMVALue	Sets or queries the binary data string for the high data word to be used when searching on an TDM audio bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATa:HIVALue	Sets or queries the binary data string for the high data word to be used when searching on an audio bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATa:OFFSet	Sets or queries the data offset value (TDM channel) to be used when searching on a TDM type audio bus signal.

Table 2-38: Search and Mark commands (cont.)

Command	Description
SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATA:QUALifier	Sets or queries the qualifier to be used when searching on an audio bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATA:TDMVALue	Sets or queries the binary data string for the single or low data word to be used when searching on an TDM audio bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATA:VALue	Sets or queries the binary data string for the single or low data word to be used when searching on an audio bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATA:WORD	Sets or queries the alignment of the data (left, right or either) to be used when searching on a non-TDM type audio bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:CONDition	Sets or queries the CAN bus trigger condition.
SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:DIRection	Sets the CAN search type.
SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:OFFSet	Sets or queries the data offset value, in bytes, to use when searching on the CAN data field.
SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:QUALifier	Sets or queries the CAN bus trigger data qualifier.
SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:SIZE	Sets or queries the CAN bus trigger data size.
SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:VALue	Sets or queries the binary data value to be used when searching on a CAN bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:ERRType	Sets or queries the type of error condition for a CAN bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:FD:BRSBit	Sets or queries the value of the bit rate switch bit (BRS bit) for a CAN bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:FD:ESIBit	Sets or queries the value of the error state indicator bit (ESI bit) for a CAN bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:FRAMEType	Sets or queries CAN bus trigger frame type.
SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:IDentifier:MODE	Sets or queries the CAN bus trigger identifier mode.
SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:IDentifier:VALue	Sets or queries CAN bus trigger identifier value.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition	Specifies a field or condition within an Ethernet frame to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:HIVALue	Sets or queries the binary data value to be used when searching on an Ethernet bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:OFFSet	Specifies the data offset value, in bytes, to use when searching on the Ethernet data field.

Table 2-38: Search and Mark commands (cont.)

Command	Description
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:QUALifier	Sets the qualifier to be used when searching on an Ethernet bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:SIZE	Specifies the number of contiguous TCP/IPV4/MAC client bytes to use when searching on the Ethernet data field.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:VALue	Specifies the binary value to use when searching on the Ethernet data field.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:IPHeader:DESTinationaddr:VALue	Specifies the 32-bit value to use when searching on the Ethernet IPv4 header address destination field.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:IPHeader:PROTOcol:VALue	Specifies the 8-bit value to use when searching on the Ethernet IPv4 header protocol field.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:IPHeader:SOUrceaddr:VALue	Specifies the 32-bit value to use when searching on the Ethernet IPv4 header address source field.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:MAC:ADDRess:DESTination:VALue	Specifies the 48-bit value to use when searching on the Ethernet MAC address destination field.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:MAC:ADDRess:SOUrce:VALue	Specifies the 48-bit value to use when searching on the Ethernet MAC address source field.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:MAC:LENGth:HIVALue	Specifies the 16-bit high value to use when searching on the Ethernet MAC length/type.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:MAC:LENGth:VALue	Specifies the 16-bit value to use when searching on the Ethernet MAC length/type.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:QTAG:VALue	Specifies the 32-bit value to use when searching on the Ethernet Q-Tag field.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:TCPHeader:ACKnum:VALue	Specifies the 32-bit value to use when searching on the Ethernet TCP header acknowledgement number.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:TCPHeader:DESTinationport:VALue	Specifies the 16-bit value to use when searching on the Ethernet TCP header destination port number.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:TCPHeader:SEQnum:VALue	Specifies the 32-bit value to use when searching on the Ethernet TCP header sequence number.
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:TCPHeader:SOUrceport:VALue	Specifies the 16-bit value to use when searching on the Ethernet TCP header source port number.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:CONDition	Sets or queries FlexRay bus search trigger condition.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:CYCLEcount:VALue	Sets or queries the FlexRay bus search cycle count value.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATA:VALue	Sets or queries the FlexRay bus search cycle count value.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:CYCLEcount:HIVALue	Sets or queries the FlexRay bus search cycle count value.

Table 2-38: Search and Mark commands (cont.)

Command	Description
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:CYCLEcount:QUALifier	Sets or queries the FlexRay bus search trigger cycle count qualifier.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATa:HIVALue	Sets or queries the FlexRay bus search trigger data value.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATa:OFFSet	Sets or queries the FlexRay bus search trigger data offset.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATa:QUALifier	Sets or queries the FlexRay bus search trigger data qualifier.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATa:SIZE	Sets or queries the FlexRay bus search trigger data size.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:EOFTYPE	Sets or queries the FlexRay bus search trigger end of file type.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:ERRTYPE	Sets or queries the FlexRay bus search trigger error type.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXray:FRAMEID:HIVALue	Sets or queries the high value when searching on a FlexRay bus frame id field.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXray:FRAMEID:QUALifier	Sets the qualifier to be used when searching on a FlexRay bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXray:FRAMEID:VALue	Sets the low value when searching on a FlexRay bus id field.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXray:FRAMEType	Sets or queries the FlexRay bus search trigger frame type.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:CRC	Sets or queries the FlexRay bus search trigger header CRC.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:CYCLECount	Sets or queries the FlexRay bus search trigger header cycle count.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:FRAMEID	Sets or queries the FlexRay bus search trigger header frame id.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:INDBits	Sets or queries the FlexRay bus search header Indicator Bits.
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:PAYLENgth	Sets or queries the FlexRay bus search trigger header payload length.
SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:ADDRess:MODE	Sets or queries the address mode for the specified I2C bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:ADDRess:VALue	Sets or queries the address string when the search condition for the specified search is Address or AddressData, to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:CONDition	Sets or queries the trigger condition for the specified I2C bus trigger search to determine where to place a mark.

Table 2-38: Search and Mark commands (cont.)

Command	Description
SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:DATa:DIRection	Sets or queries the direction of the data for the I2C bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:DATa:SIZE	Sets or queries the length of the data string in bytes used for the specified I2C bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:DATa:VALue	Sets or queries the data value of the data token for the specified I2C bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:DATa:HIVALue	Specifies the high data value to be used in a LIN search.
SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:DATa:QUALifier	Sets or queries the LIN bus search trigger data qualifier.
SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:DATa:SIZE	Sets or queries the LIN bus search trigger data size.
SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:DATa:VALue	Sets or queries the LIN bus search trigger data value.
SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:ERRTYPE	Sets or queries the LIN bus search trigger error type.
SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:IDentifier:VALue	Sets or queries the LIN bus search trigger identifier value.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:CONDition	Sets or queries the field or condition for a MIL-STD-1553 bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:ERRTYPE	Sets or queries the type of error condition for a MIL-STD-1553 bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:TRBit	Sets or queries the value of the command word Transmit / Receive bit for a MIL-STD-1553 bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:PARity	Sets or queries the value of the command word parity bit for a MIL-STD-1553 bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:COUNT	Sets or queries the value of the command word "word count" field for a MIL-STD-1553 bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:SUBADdress	Sets or queries the value of the command word subaddress field for a MIL-STD-1553 bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:ADDRess:QUALifier	Sets or queries the qualifier to be used when searching on command word addresses for a MIL-STD-1553 bus.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:ADDRess:VALue	Sets or queries the low value when searching on command word addresses for a MIL-STD-1553 bus.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:ADDRess:HIVALue	Sets or queries the high value when searching on command word addresses for a MIL-STD-1553 bus.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:DATA:PARity	Sets or queries the value of the command word parity bit for a MIL-STD-1553 bus to search on.

Table 2-38: Search and Mark commands (cont.)

Command	Description
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:DATa:VALue	Sets or queries the value when searching on data words for a MIL-STD-1553 bus.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:PARity	Sets or queries the value of the status word parity bit for a MIL-STD-1553 bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:ADDRess:QUALifier	Sets or queries the qualifier to be used when searching on status word addresses for a MIL-STD-1553 bus.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:ADDRess:VALue	Sets or queries the low value when searching on status word addresses for a MIL-STD-1553 bus.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:ADDRess:HIVALue	Sets or queries the high value when searching on status word addresses for a MIL-STD-1553 bus.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:ME	Sets or queries the value of the message error bit (ME bit, bit 9) in a status word for a MIL-STD-1553 bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:INSTR	Sets or queries the value of the instrumentation bit (INSTR bit, bit 10) in a status word for a MIL-STD-1553 bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:SRQ	Sets or queries the value of the status word service request bit (SRQ bit, bit 11) in a status word for a MIL-STD-1553 bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:BCR	Sets or queries the value of the broadcast command received bit (BCR bit, bit 15) in a status word for a MIL-STD-1553 bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:BUSY	Sets or queries the value of the busy bit (BUSY bit, bit 16) in a status word for a MIL-STD-1553 bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:SUBSF	Ssets or queries the value of the subsystem flag bit (SUBSF bit, bit 17) in a status word for a MIL-STD-1553 bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:DBCA	Sets or queries the value of the dynamic bus control acceptance bit (DBCA bit, bit 18) in a status word for a MIL-STD-1553 bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:TF	Sets or queries the value of the terminal flag bit (TF bit, bit 19) in a status word for a MIL-STD-1553 bus to search on.
SEARCH:SEARCH<x>:TRIGger:A:BUS:PARallel:DATa:VALue	Sets or queries the data value for a parallel bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:RS232C:DATa:VALue	Sets or queries the data string value for the specified RS232c bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:RS232C:CONDition	Sets or queries the condition for an RS232C bus search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:RS232C:DATa:SIZE	Sets or queries the length of the data string in bytes to be used for an RS232 bus search to determine where to place a mark when the search condition is Data.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition	Sets or queries the search condition for a SENT bus.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:ERRType	Sets or queries the error type to be used when searching on SENT data.

Table 2-38: Search and Mark commands (cont.)

Command	Description
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:ERRType:CRC	Sets or queries the CRC error type to be used when searching on SENT data.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN1A:HIVALue	Sets or queries the high binary fast channel 1 value to use when searching on a SENT bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN1A:QUALifier	Sets or queries the qualifier to be used when searching on SENT fast packet bus data for device channel 1.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN1A:VALue	Sets or queries the binary fast channel 1 value to be used when searching on a SENT bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN2B:HIVALue	Sets or queries the high binary fast channel 2 value to use when searching on a SENT bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN2B:QUALifier	Sets or queries the qualifier to be used when searching on SENT fast packet bus data for device channel 2.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN2B:VALue	Sets or queries the binary fast channel 2 value to be used when searching on a SENT bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:COUNTER:HIVALue	Sets or queries the high binary fast message counter value to use when searching on a SENT bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:COUNTER:QUALifier	Sets or queries the qualifier to be used when searching on SENT fast packet bus data for the secure format counter.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:COUNTER:VALue	Sets or queries the binary fast message counter value to be used when searching on a SENT bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:INVERTNIBble:VALue	Sets or queries the binary fast message inverted nibble value to be used when searching on a SENT bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:STATUS:VALue	Sets or queries the binary status value to be used when searching on a SENT bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:PAUSE:QUALifier	Sets or queries the qualifier to be used when searching on SENT pause pulses.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:PAUSE:TICKs:HIVALue	Sets or queries the maximum number of pause clock ticks to be used when searching on a SENT bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:PAUSE:TICKs:VALue	Sets or queries the minimum number of pause clock ticks to be used when searching on a SENT bus signal.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:SLOW:DATA:HIVALue	Sets or queries the high binary Slow channel data value to use when searching on SENT Slow packet bus data.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:SLOW:DATA:QUALifier	Sets or queries the qualifier to be used when searching on SENT Slow packet bus data.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:SLOW:DATA:VALue	Sets or queries the binary slow channel data value to be used when searching on SENT Slow packet bus data.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:SLOW:IDentifier:VALue	Sets or queries the binary Slow identifier value to be used when searching on SENT Slow packet bus data.

Table 2-38: Search and Mark commands (cont.)

Command	Description
SEARCH:SEARCH<x>:TRIGger:A:BUS:SOURce	Sets or queries the bus source for the specified bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SPI:CONDition	Sets or queries the trigger condition for the specified SPI bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SPI:DATA:SIZE	Sets or queries the length of the data string in bytes used for the specified SPI bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SPI:DATA:VALue	Sets or queries the data value of the data token for the specified SPI bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:CONDition	This command sets or queries the search condition for an SPMI bus.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:DATA:VALue	This command specifies the binary data string used for SPMI when the search condition is MASTERREAd, MASTERWRItE, REGREAd, REGWRItE, EXTREGREAd, EXTREGWRItE, LONGEXTREGREAd, LONGEXTREGWRItE, or REG0WRItE.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:MASTERADDRess:VALue	This command sets or queries the binary data string that identifies the master address used in SPMI search.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:NOReSPonse	This command sets or queries whether or not to search for No Response frames.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:REGISTERADDRess:VALue	This command sets or queries the binary data string that identifies the register address used in SPMI triggering.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:SLAVEADDRess:VALue	This command sets or queries the binary data string that identifies the slave address used in SPMI.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ADDRESS:HIVALue	Sets or queries the address value for normal token to be used with In Range and Out of Range qualifiers for the specified USB bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ADDRESS:VALue	Sets or queries the address value for normal token for the specified USB bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:CONDition	Sets or queries the search condition for the specified USB bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:HIVALue	Sets or queries the data value for data token used with In Range and Out of Range qualifiers for the specified USB bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:OFFSet	Sets or queries data offset for the specified USB bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:SIZE	Sets or queries the length of the data string in bytes for the specified USB bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:TYPE	Sets or queries the data packet type for the specified USB bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:VALue	Sets or queries the data value for data token for the specified bus trigger search to determine where to place a mark.

Table 2-38: Search and Mark commands (cont.)

Command	Description
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ENDPoint:VALue	Sets or queries the endpoint value for normal token for the specified USB bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ERRType	Sets or queries the error type for the specified USB bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:HANDSHAKType	Sets or queries the handshake type for the specified USB bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SOFFRAMENUMber	Sets or queries the frame number string to use for the Start of Frame for the specified USB bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPECIALType	Sets or queries the PID value for the USB bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:ET:VALue	Sets or queries the Endpoint Type value for the specified USB bus trigger split token field search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:HUB:VALue	Sets or queries the hub address of the specified USB bus trigger on split token field search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:PORT:VALue	Sets or queries the port address for the specified USB bus trigger on split token field search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:SC:VALue	Sets or queries the Start/Complete value for the specified USB bus trigger on split token field search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:SE:VALue	Sets or queries the Start/End value for the specified USB bus trigger on split token field search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:TOKENType	Sets or queries the token type when the specified USB bus trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:EDGE:SLOpe	Sets or queries the slope for an edge trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:DATARate	Sets or queries the DDR read search data rate for DDR3 and LPDDR3 standards of the specified search.
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:DATASource	Sets or queries the DDR read data source when the search type is DDR READ.
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:HYSteresis	Sets or queries the DDR read hysteresis reference level value, when the search type is DDR READ.
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:MARgin	Sets or queries the DDR read margin reference level value, when the search type is DDR READ.
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:POSTAMBLE:LENGth	Sets or queries the DDR read postamble length when the search type is DDR READ.
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:PREAMBLE:TYPE	Sets or queries the DDR read preamble type when the search type is DDR READ.
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:HIGH	Sets or queries the DDR read data high reference level value, when the search type is DDR READ.

Table 2-38: Search and Mark commands (cont.)

Command	Description
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:LOW	Sets or queries the DDR read data low reference level value, when the search type is DDR READ.
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:MID	Sets or queries the DDR read data mid reference level value, when the search type is DDR READ.
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:STROBE:HIGh	Sets or queries the DDR read reference level strobe high value, when the search type is DDR READ.
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:STROBE:LOW	Sets or queries the DDR read reference level strobe low value, when the search type is DDR READ.
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:STROBE:MID	Sets or queries the DDR read reference level strobe mid value, when the search type is DDR READ.
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLEVELMode	Sets or queries the DDR read reference level mode to auto or manual, when the search type is DDR READ.
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:STANdard	Sets or queries the DDR read search standard as DDR3 or LPDDR3.
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:STROBESource	Sets or queries the DDR read strobe source when the search type is DDR READ.
SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:DATARate	Sets or queries the DDR read/write data rate for DDR3 and LPDDR3 standards of the specified search.
SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:DATASource	Sets or queries the DDR read/write data source when the search type is DDR READWRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:HYSteresis	Sets or queries the DDR read/write hysteresis reference level value, when the search type is DDR READWRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:MARgin	Sets or queries the DDR read/write margin reference level value, when the search type is DDR READWRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:POSTAMBLE:LENGth	Sets or queries the DDR read/write postamble length when the search type is DDR READWRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:PREAMBLE:TYPE	Sets or queries the DDR read/write preamble type when the search type is DDR READWRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLevel:DATA:HIGh	Sets or queries the DDR read/write data high reference level value, when the search type is DDR READWRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLevel:DATA:LOW	Sets or queries the DDR read/write data low reference level value, when the search type is DDR READWRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLevel:DATA:MID	Sets or queries the DDR read/write data mid reference level value, when the search type is DDR READWRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLevel:STROBE:HIGh	Sets or queries the DDR read/write reference level strobe high value, when the search type is DDR READWRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLevel:STROBE:LOW	Sets or queries the DDR read/write reference level strobe low value, when the search type is DDR READWRITE.

Table 2-38: Search and Mark commands (cont.)

Command	Description
SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLevel:STROBE:MID	Sets or queries the DDR read/write reference level strobe mid value, when the search type is DDR READWRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLEVELMode	Sets or queries the DDR read/write reference level mode to auto or manual, when the search type is DDR READWRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:STANdard	Sets or queries the ddr readwrite search standard as DDR3 or LPDDR3.
SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:STROBESource	Sets or queries the DDR readwrite strobe source when the search type is DDR READWRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:DATARate	Sets or queries the DDR write search data rate for DDR3 and LPDDR3 standards of the specified search.
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:DATASource	Sets or queries the DDR write data source when the search type is DDR Write.
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:HYSteresis	Sets or queries the DDR write hysteresis reference level value when the search type is DDR WRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:MARgin	Sets or queries the DDR write margin reference level value, when the search type is DDR WRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:POSTAMBLE:LENGth	Sets or queries the DDR write postamble length when the search type is DDR WRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:PREAMBLE:TYPE	Sets or queries the DDR write preamble type when the search type is DDR WRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLevel:DATA:HIGh	Sets or queries the DDR write data high reference level value, when the search type is DDR WRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLevel:DATA:LOW	Sets or queries the DDR write data low reference level value, when the search type is DDR WRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLevel:DATA:MID	Sets or queries the DDR write data mid reference level value, when the search type is DDR WRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLevel:STROBE:HIGh	Sets or queries the DDR write reference level strobe high value, when the search type is DDR WRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLevel:STROBE:LOW	Sets or queries the DDR write reference level strobe low value, when the search type is DDR WRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLevel:STROBE:MID	Sets or queries the DDR write reference level strobe mid value, when the search type is DDR WRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLEVELMode	Sets or queries the DDR write reference level mode to auto or manual, when the search type is DDR WRITE.
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:STANdard	Sets or queries the DDR write search standard as DDR3 or LPDDR3.
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:STROBESource	Sets or queries the DDR write strobe source when the search type is DDR Write.

Table 2-38: Search and Mark commands (cont.)

Command	Description
SEARCH:SEARCH<x>:TRIGger:A:EDGE:SLOpe	Sets or queries the slope for an edge trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:EDGE:SOUrce	Sets or queries the source waveform for an edge trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:LOGIc:FUNctIon	Sets or queries the logic operator for a pattern or state trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:LOGIc:USEClockedge	Sets whether or not Logic search uses a clock source.
SEARCH:SEARCH<x>:TRIGger:A:RUNT:WHEn	Sets or queries the condition setting for a runt trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:RUNT:WIDth	Sets or queries the width setting for a runt trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:SETHold:CLOCK:EDGE	Sets or queries the clock slope setting for a setup/hold trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:SETHold:CLOCK:SOUrce	Sets or queries the clock source setting for a setup/hold trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:SETHold:HOLDTime	Sets or queries the hold time setting for a setup/hold trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:SETHold:SETTime	Sets or queries the setup time setting for a setup/hold trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:TIME	Sets or queries the time setting for a timeout trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:TRANSition:DELTATime	Sets or queries the transition time setting for a transition trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:TRANSition:WHEn	Sets or queries the condition setting for a transition trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:TYPE	Sets or queries the trigger type setting for a search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:WINDow:WIDth	Sets or queries the width setting for a window search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:CONDition	Sets or queries the LIN bus search trigger condition.
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATa:QUALifier	Sets or queries the qualifier to be used when searching on a USB bus signal.
SEARCH:SEARCH<x>:TRIGger:A:EDGE:THReshold	Sets or queries the source threshold level for an edge trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:LOGIc:CLOCK:THReshold	Sets or queries the logic clock threshold for a logic trigger search to determine where to place a mark.

Table 2-38: Search and Mark commands (cont.)

Command	Description
SEARCH:SEARCH<x>:TRIGger:A:LOGic:DELTime	Specifies the Logic search delta time value.
SEARCH:SEARCH<x>:TRIGger:A:LOGic:INPUT:CLOCK:SOUrce	Sets or queries the channel to use as the clock source for logic trigger.
SEARCH:SEARCH<x>:TRIGger:A:LOGic:LEVel:CH<x>	Sets or queries the voltage level to use for logic trigger search.
SEARCH:SEARCH<x>:TRIGger:A:LOGic:LEVel:MATH<x>	Sets the voltage level to use for logic trigger search.
SEARCH:SEARCH<x>:TRIGger:A:LOGic:LEVel:REF<x>	Sets the voltage level to use for logic trigger search.
SEARCH:SEARCH<x>:TRIGger:A:LOGic:LOGICPattern:CH<x>	Sets or queries the conditions used for generating an A logic pattern.
SEARCH:SEARCH<x>:TRIGger:A:LOGic:LOGICPattern:CH<x>_D<x>	Sets or queries the conditions used for generating an A logic pattern.
SEARCH:SEARCH<x>:TRIGger:A:LOGic:LOGICPattern:MATH<x>	Sets or queries the conditions used for generating an A logic pattern.
SEARCH:SEARCH<x>:TRIGger:A:LOGic:LOGICPattern:REF<x>	Sets or queries the conditions used for generating an A logic pattern.
SEARCH:SEARCH<x>:TRIGger:A:LOGic:POLarity	Sets or queries the polarity for the clock channel when Use Clock Edge is set to Yes for Logic search type.
SEARCH:SEARCH<x>:TRIGger:A:LOGic:WHEn	Sets or queries the condition for generating an A or B logic search with respect to the defined input pattern.
SEARCH:SEARCH<x>:TRIGger:A:PULSEWidth:HIGHLimit	Specifies the upper limit to use, in seconds, when searching for a pulse whose duration is inside or outside a range of two values.
SEARCH:SEARCH<x>:TRIGger:A:PULSEWidth:LOGICQUALification	Specifies whether or not to use logic qualification for a pulse width search.
SEARCH:SEARCH<x>:TRIGger:A:PULSEWidth:LOWLimit	Specifies the lower limit to use, in seconds, when searching for a pulse whose duration is inside or outside a range of two values.
SEARCH:SEARCH<x>:TRIGger:A:PULSEWidth:POLarity	Specifies the polarity for a pulse width search.
SEARCH:SEARCH<x>:TRIGger:A:PULSEWidth:SOUrce	Sets and queries the source for the pulsewidth search input.
SEARCH:SEARCH<x>:TRIGger:A:PULSEWidth:THReshold	Sets or queries the source threshold level for a pulse width trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:PULSEWidth:WHEn	Specifies to search for a pulse with a specified width.
SEARCH:SEARCH<x>:TRIGger:A:RUNT:LOGICQUALification	Specifies whether or not to use logic qualification for a runt search.

Table 2-38: Search and Mark commands (cont.)

Command	Description
SEARCH:SEARCH<x>:TRIGger:A:RUNT:POLarity	Specifies the polarity for the runt search.
SEARCH:SEARCH<x>:TRIGger:A:RUNT:SOUrce	Sets and queries the source for the Runt search input.
SEARCH:SEARCH<x>:TRIGger:A:RUNT:THReshold:HIGH	Sets or queries the source threshold HIGH level for a runt trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:RUNT:THReshold:LOW	Sets or queries the source threshold LOW level for a runt trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:SETHold:CLOCK:THReshold	Sets or queries the clock threshold setting for a setup/hold trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:SETHold:LEVel:CH<x>	Sets or queries the voltage level to use for setup & hold trigger search.
SEARCH:SEARCH<x>:TRIGger:A:SETHold:LEVel:MATH<x>	Sets or queries the voltage level to use for setup & hold trigger search.
SEARCH:SEARCH<x>:TRIGger:A:SETHold:LEVel:REF<x>	Sets or queries the voltage level to use for setup & hold trigger search.
SEARCH:SEARCH<x>:TRIGger:A:SETHold:LOGICPattern:CH<x>	Sets or queries the conditions used for generating an A logic pattern.
SEARCH:SEARCH<x>:TRIGger:A:SETHold:LOGICPattern:CH<x>_D<x>	Sets or queries the conditions used for generating an A logic pattern.
SEARCH:SEARCH<x>:TRIGger:A:SETHold:LOGICPattern:MATH<x>	Sets or queries the conditions used for generating an A logic pattern.
SEARCH:SEARCH<x>:TRIGger:A:SETHold:LOGICPattern:REF<x>	Sets and returns the conditions used for generating an A logic pattern.
SEARCH:SEARCH<x>:TRIGger:A:STATE	Sets or queries the enabled state of the search.
SEARCH:SEARCH<x>:TRIGger:A:STOPAcq	Sets or queries whether acquisitions are stopped when a search hit is found.
SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:LOGICQUALification	Sets whether or not to use logic qualification for a timeout search.
SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:POLarity	Sets or queries the polarity to be used for a Timeout search.
SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:SOUrce	Sets and queries the source for timeout search input.
SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:THReshold	Sets or queries the source threshold level for a timeout trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:TRANSition:LOGICQUALification	Specifies whether or not to use logic qualification for a transition search.
SEARCH:SEARCH<x>:TRIGger:A:TRANSition:POLarity	Specifies the polarity for the transition search.

Table 2-38: Search and Mark commands (cont.)

Command	Description
SEARCH:SEARCH<x>:TRIGger:A:TRANSition:SOUrce	Sets and queries the source for the transition search input.
SEARCH:SEARCH<x>:TRIGger:A:TRANSition:THReshold:HIGH	Sets or queries the source threshold HIGH level for a transition trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:TRANSition:THReshold:LOW	Sets or queries the source threshold LOW level for a transition trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:WINDow:CROSSIng	Sets or queries the window trigger threshold crossing of the selected trigger Source.
SEARCH:SEARCH<x>:TRIGger:A:WINDow:LOGICQUALification	Specifies or queries whether or not to use logic qualification for a window search.
SEARCH:SEARCH<x>:TRIGger:A:WINDow:POLarity	Sets or queries the window trigger threshold crossing of the selected trigger Source.
SEARCH:SEARCH<x>:TRIGger:A:WINDow:SOUrce	Sets and queries the source for the window search input.
SEARCH:SEARCH<x>:TRIGger:A:WINDow:THReshold:HIGH	Sets or queries the source threshold HIGH level for a transition trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:WINDow:THReshold:LOW	Sets or queries the source threshold LOW level for a transition trigger search to determine where to place a mark.
SEARCH:SEARCH<x>:TRIGger:A:WINDow:WHEn	Sets or queries the window search event.
SEARCH:SELEcted	Sets or queries the selected search.
SEARCHTABLE	This command adds or deletes a new search event table in an Option 5-WIN (Microsoft Windows 10 OS) TekExpress compliance testing application.

Self Test command group

The Self Test commands control the selection and execution of diagnostic tests.

Table 2-39: Self Test commands

Command	Description
DIAG:LOOP:OPTion	Sets or queries the type of looping desired.
DIAG:LOOP:OPTion:NTIMes	Sets or queries how many loops to run.
DIAG:LOOP:STOP	Stops diagnostics looping.
DIAG:MODE	Sets or queries the diagnostics mode.
DIAG:RESUlt?	Returns the diagnostics results.
DIAG:RESUlt:FLAg?	Returns the status of the diagnostics (single area).
DIAG:RESUlt:LOG?	Returns the status of the diagnostic area.
DIAG:SElect	Selects or queries an available diagnostic area.
DIAG:STATE	Sets the instrument operating state.
TOUCHSCReen:CALibrate	Starts the touchscreen calibration procedure.

Status and Error command group

Use the commands in the Status and Error command Group to determine the status of the instrument and control events.

Several commands and queries used with the instrument are common to all devices. The IEEE Std 488.2-1987 defines these commands and queries. The common commands begin with an asterisk (*) character.

Table 2-40: Status and Error commands

Command	Description
ALLev?	Returns all events and their messages.
BUSY?	Returns instrument status.
*CLS	Clears status.
DESE	Sets or queries the bits in the Device Event Status Enable Register.
*ESE	Sets or queries the bits in the Event Status Enable Register.
*ESR?	Returns the contents of the Standard Event Status Register.
EVENT?	Returns event code from the event queue.
EVMsg?	Returns event code, message from the event queue.
EVQty?	Returns number of events that are enabled in the queue.
*OPC	Generates the operation complete message in the standard event status register when all pending operations are finished Or returns "1" when all current operations are finished.
*OPT?	Returns a comma separated list of installed options as an arbitrary ASCII string (no quotes).
*PSC	Sets or queries the power on status flag.
*PUD	Sets or queries a string of protected user data.
*RST	Resets the instrument to factory default settings.
*SRE	Sets or queries the bits in the Service Request Enable Register.
*STB?	Returns the contents of the Status Byte Register.
*WAI	Prevents the instrument from executing further commands until all pending operations finish.

Trigger command group

Use the commands in the Trigger Command Group to control all aspects of triggering for the instrument.

There are two triggers: A and B. Where appropriate, the command set has parallel constructions for each trigger.

You can set the A or B triggers to edge mode. Edge triggering lets you display a waveform at or near the point where the signal passes through a voltage level of your choosing.

You can also set A or B triggers to pulse or logic modes. With pulse triggering, the instrument triggers whenever it detects a pulse of a certain width or height. Logic triggering lets you logically combine the signals on one or more channels. The instrument then triggers when it detects a certain combination of signal levels. The trigger types of Pulse Width, Timeout, Runt, Window, and Rise/Fall Time can be further qualified by a logic pattern. This is referred to as logic qualification.

Table 2-41: Trigger commands

Command	Description
TRIGger	Forces a trigger event to occur or returns current trigger parameters for the instrument.
TRIGger:{A B}:BUS:B<x>:ARINC429A:CONDition	Specifies a field or condition for an ARINC429 bus to trigger on.
TRIGger:{A B}:BUS:B<x>:ARINC429A:ERRType	Sets or queries the error type when triggering on an ARINC429 bus signal.
TRIGger:{A B}:BUS:B<x>:ARINC429A:LABel:QUALifier	Sets or queries the qualifier to be used when triggering on label data for an ARINC429 bus signal.
TRIGger:{A B}:BUS:B<x>:ARINC429A:LABel:VALue	Sets or queries the low value when triggering on an ARINC429 label field.
TRIGger:{A B}:BUS:B<x>:ARINC429A:LABel:HIVALue	Sets or queries the high value when triggering on an ARINC429 label field.
TRIGger:{A B}:BUS:B<x>:ARINC429A:SDI:VALue	sSets or queries the when triggering on an ARINC429 SDI field.
TRIGger:{A B}:BUS:B<x>:ARINC429A:SSM:VALue	Sets or queries the when triggering on an ARINC429 SSM field.
TRIGger:{A B}:BUS:B<x>:ARINC429A:DATA:QUALifier	Sets or queries the qualifier to be used when triggering on data in the DATA field for an ARINC429 bus signal.
TRIGger:{A B}:BUS:B<x>:ARINC429A:DATA:VALue	Sets or queries the low value when triggering on an ARINC429 data field.
TRIGger:{A B}:BUS:B<x>:ARINC429A:DATA:HIVALue	Sets or queries the high value when trigger on an ARINC429 data field.
TRIGger:{A B}:BUS:B<x>:AUDio:CONDition	Sets the condition (word select, start of frame, or matching data) to be used when triggering on an audio bus signal.
TRIGger:{A B}:BUS:B<x>:AUDio:DATA:HIVALue	Sets the upper word value to be used when triggering on an audio bus signal.

Table 2-41: Trigger commands (cont.)

Command	Description
TRIGger:{A B}:BUS:B<x>:AUDio:DATa:HITDMVALue	Sets or queries the binary data string for the high data word to be used when triggering on an TDM audio bus signal.
TRIGger:{A B}:BUS:B<x>:AUDio:DATa:OFFSet	Sets the data offset value to be used when triggering on an audio bus signal.
TRIGger:{A B}:BUS:B<x>:AUDio:DATa:QUALifier	Sets the qualifier (<, >, =, <=, >=, not =, in range, out of range) to be used when triggering on an audio bus signal.
TRIGger:{A B}:BUS:B<x>:AUDio:DATa:TDMVALue	Sets or queries the binary data string for the single or low data word to be used when triggering on an TDM audio bus signal.
TRIGger:{A B}:BUS:B<x>:AUDio:DATa:VALue	Sets the lower word value to be used when triggering on an audio bus signal.
TRIGger:{A B}:BUS:B<x>:AUDio:DATa:WORD	Sets the alignment of the data (left, right or either) to be used to trigger on an audio bus signal.
TRIGger:{A B}:BUS:B<x>:CAN:CONDition	Sets the condition (start of frame, frame type, identifier, matching data, EOF, missing ACK field, bit-stuffing error) to be used when triggering on a CAN bus signal.
TRIGger:{A B}:BUS:B<x>:CAN:DATa:DIRection	Sets the data direction (read, write or “nocare”) to be used to search on a CAN bus signal.
TRIGger:{A B}:BUS:B<x>:CAN:DATa:OFFSet	Sets or queries the data offset value, in bytes, to use when triggering on the CAN data field.
TRIGger:{A B}:BUS:B<x>:CAN:DATa:QUALifier	Sets the qualifier (<, >, =, not =, <=, >=) to be used when triggering on a CAN bus signal.
TRIGger:{A B}:BUS:B<x>:CAN:DATa:SIZE	Sets the length of the data string, in bytes, to be used when triggering on a CAN bus signal.
TRIGger:{A B}:BUS:B<x>:CAN:DATa:VALue	Sets the binary data value to be used when triggering on a CAN bus signal.
TRIGger:{A B}:BUS:B<x>:CAN:ERRType	Sets or queries the type of error condition for a CAN bus to triggering on.
TRIGger:{A B}:BUS:B<x>:CAN:FD:BRSBit	Sets or queries the value of the bit rate switch bit (BRS bit) for a CAN bus to triggering on.
TRIGger:{A B}:BUS:B<x>:CAN:FD:ESIBit	Sets or queries the value of the error state indicator bit (ESI bit) for a CAN bus to triggering on.
TRIGger:{A B}:BUS:B<x>:CAN:FRAMetype	Sets the frame type (data, remote, error or overload) to be used when triggering on a CAN bus signal.
TRIGger:{A B}:BUS:B<x>:CAN:IDentifier:MODE	Sets the addressing mode (standard or extended format) to be used when triggering on a CAN bus signal.
TRIGger:{A B}:BUS:B<x>:CAN:IDentifier:VALue	Sets the binary address value to be used when triggering on a CAN bus signal.
TRIGger:{A B}:BUS:B<x>:ETHERnet:CONDition	Specifies a field or condition within an Ethernet frame to trigger on.
TRIGger:{A B}:BUS:B<x>:ETHERnet:DATa:HIVALue	When the Ethernet trigger condition is set to DATa, and the qualifier is set to either INrange orOUTrange, this command specifies the upper data value of the range.
TRIGger:{A B}:BUS:B<x>:ETHERnet:DATa:OFFSet	When the Ethernet trigger condition is set to DATa, this command specifies where in the data field to look for the data trigger value.
TRIGger:{A B}:BUS:B<x>:ETHERnet:DATa:QUALifier	Sets or queries the qualifier to be used when triggering on an Ethernet bus signal.

Table 2-41: Trigger commands (cont.)

Command	Description
TRIGger:{A B}:BUS:B<x>:ETHERnet:DATA:SIZE	When the Ethernet trigger condition is set to DATA, this command specifies the number of contiguous TCP/IPv4/MAC client data bytes to trigger on.
TRIGger:{A B}:BUS:B<x>:ETHERnet:DATA:VALue	When the Ethernet trigger condition is set to DATA, and the qualifier is set to LESSthan, MOREthan, EQUAL, UNEQUAL, LESSEQUAL or MOREEQUAL, this command specifies the value to trigger on.
TRIGger:{A B}:BUS:B<x>:ETHERnet:IPHeader:DESTinationaddr:VALue	When the Ethernet trigger condition is set to IPHeader, this command specifies the value of the 32-bit destination address that is to be used in the trigger.
TRIGger:{A B}:BUS:B<x>:ETHERnet:IPHeader:PROTOcol:VALue	When the Ethernet trigger condition is set to IPHeader, this command specifies the value of the 8-bit protocol field that is to be used in the trigger.
TRIGger:{A B}:BUS:B<x>:ETHERnet:IPHeader:SOURceaddr:VALue	When the Ethernet trigger condition is set to IPHeader, this command specifies the value of the 32-bit source address that is to be used in the trigger.
TRIGger:{A B}:BUS:B<x>:ETHERnet:MAC:ADDRess:DESTination:VALue	When the Ethernet trigger condition is set to MACADDRESS, this command specifies the 48-bit MAC destination address that is to be used in the trigger.
TRIGger:{A B}:BUS:B<x>:ETHERnet:MAC:ADDRess:SOURce:VALue	When the Ethernet trigger condition is set to MACADDRESS, this command specifies the 48-bit MAC source address value that is to be used in the trigger.
TRIGger:{A B}:BUS:B<x>:ETHERnet:MAC:LENGth:HIVALue	When the Ethernet trigger condition is set to MACLENGTH, and the qualifier is set to INrange or OUTrange, this command specifies the upper data value of the range.
TRIGger:{A B}:BUS:B<x>:ETHERnet:MAC:LENGth:VALue	When the Ethernet trigger condition is set to MACLENGTH, and the qualifier is set to LESSthan, MOREthan, EQUAL, UNEQUAL, LESSEQUAL or MOREEQUAL, this command specifies the 16-bit value to trigger on.
TRIGger:{A B}:BUS:B<x>:ETHERnet:QTAG:VALue	When the Ethernet trigger condition is set to QTAG, this command specifies the 32-bit Q-Tag value to trigger on.
TRIGger:{A B}:BUS:B<x>:ETHERnet:TCPHeader:ACKnum:VALue	When the Ethernet trigger condition is set to TCPHeader, this command specifies the 32-bit acknowledgement number that is to be used in the trigger.
TRIGger:{A B}:BUS:B<x>:ETHERnet:TCPHeader:DESTinationport:VALue	When the Ethernet trigger condition is set to TCPHeader, this command specifies the 16-bit destination port address value that is to be used in the trigger.
TRIGger:{A B}:BUS:B<x>:ETHERnet:TCPHeader:SEQnum:VALue	When the Ethernet trigger condition is set to TCPHeader, this command specifies the 32-bit sequence number that is to be used in the trigger.
TRIGger:{A B}:BUS:B<x>:ETHERnet:TCPHeader:SOURceport:VALue	When the Ethernet trigger condition is set to TCPHeader, this command specifies the 16-bit source port address that is to be used in the trigger.
TRIGger:{A B}:BUS:B<x>:FLEXray:CONDition	Specifies the condition to use when triggering on a FlexRay bus signal (start of frame, frame type, ID, cycle count, header, data, ID and data, EOF, error).
TRIGger:{A B}:BUS:B<x>:FLEXray:CYCLEcount:HIVALue	Specifies the high value when triggering on a FlexRay bus cycle count field.
TRIGger:{A B}:BUS:B<x>:FLEXray:CYCLEcount:QUALifier	Specifies the qualifier (<, >, =, <=, >=, ≠, in range, out of range) to use when triggering on the FlexRay bus cycle count field.
TRIGger:{A B}:BUS:B<x>:FLEXray:CYCLEcount:VALue	Specifies the low value when triggering on the FlexRay bus cycle count field.
TRIGger:{A B}:BUS:B<x>:FLEXray:DATA:HIVALue	Specifies the high value when triggering on the FlexRay bus data field.
TRIGger:{A B}:BUS:B<x>:FLEXray:DATA:OFFSet	Specifies the offset of the data string, in bytes, when triggering on the FlexRay bus data field.

Table 2-41: Trigger commands (cont.)

Command	Description
TRIGger:{A B}:BUS:B<x>:FLEXray:DATA:QUALifier	Specifies the qualifier (<, >, =, <=, >=, ≠, in range, out of range) to use when triggering on the FlexRay bus data field.
TRIGger:{A B}:BUS:B<x>:FLEXray:DATA:SIZE	Specifies the length of the data string, in bytes, when triggering on the FlexRay bus data field.
TRIGger:{A B}:BUS:B<x>:FLEXray:DATA:VALue	Specifies the low value when triggering on the FlexRay bus data field.
TRIGger:{A B}:BUS:B<x>:FLEXray:EOFTYPE	Specifies the end of file type (static, dynamic or any) when triggering on the FlexRay bus EOF field.
TRIGger:{A B}:BUS:B<x>:FLEXray:ERRTYPE	Specifies the error type when triggering on the FlexRay bus signal.
TRIGger:{A B}:BUS:B<x>:FLEXray:FRAMEID:HIVALue	Specifies the high value when triggering on the FlexRay bus frame ID field.
TRIGger:{A B}:BUS:B<x>:FLEXray:FRAMEID:QUALifier	Specifies the qualifier to use when triggering on the FlexRay bus frame ID field.
TRIGger:{A B}:BUS:B<x>:FLEXray:FRAMEID:VALue	Specifies the low value when triggering on the FlexRay bus frame ID field.
TRIGger:{A B}:BUS:B<x>:FLEXray:FRAMEType	Specifies the frame type (normal, payload, null, sync or startup) when triggering on the FlexRay bus signal.
TRIGger:{A B}:BUS:B<x>:FLEXray:HEADER:CRC	Specifies the CRC portion of the binary header string when triggering on the FlexRay bus signal.
TRIGger:{A B}:BUS:B<x>:FLEXray:HEADER:CYCLEcount	Specifies the cycle count portion of the binary header string when triggering on the FlexRay bus header.
TRIGger:{A B}:BUS:B<x>:FLEXray:HEADER:FRAMEID	Specifies the frame ID portion of the binary header string when triggering on the FlexRay bus header.
TRIGger:{A B}:BUS:B<x>:FLEXray:HEADER:INDBits	Specifies the indicator bits portion of the binary header string when triggering on the FlexRay bus header.
TRIGger:{A B}:BUS:B<x>:FLEXray:HEADER:PAYLength	Specifies the payload length portion of the binary header string when triggering on the FlexRay bus header.
TRIGger:{A B}:BUS:B<x>:I2C:ADDRess:MODE	Specifies the I ² C address mode to 7 or 10-bit.
TRIGger:{A B}:BUS:B<x>:I2C:ADDRess:VALue	Specifies the binary address string used for the I ² C trigger if the trigger condition is ADDRESS or ADDRANDDATA.
TRIGger:{A B}:BUS:B<x>:I2C:CONDition	Specifies the trigger condition for an I ² C trigger.
TRIGger:{A B}:BUS:B<x>:I2C:DATA:DIRection	Specifies the I ² C trigger type to be valid on a Read, Write, or No Care condition.
TRIGger:{A B}:BUS:B<x>:I2C:DATA:SIZE	Specifies the length of the data string in bytes to be used for an I ² C trigger if the trigger condition is DATA or ADDRANDDATA.
TRIGger:{A B}:BUS:B<x>:I2C:DATA:VALue	Specifies the binary data string used for I ² C triggering if the trigger condition is DATA or ADDRANDDATA.
TRIGger:{A B}:BUS:B<x>:LIN:CONDition	Specifies the trigger condition for LIN.
TRIGger:{A B}:BUS:B<x>:LIN:DATA:HIVALue	Specifies the high data value string used for a LIN bus trigger when the trigger condition is DATA or IDANDDATA and the data qualifier is INRANGE or OUTRANGE.

Table 2-41: Trigger commands (cont.)

Command	Description
TRIGger:{A B}:BUS:B<x>:LIN:DATa:QUALifier	Specifies the LIN data qualifier. This only applies if the trigger condition is IDANDDATA or DATA.
TRIGger:{A B}:BUS:B<x>:LIN:DATa:SIZE	Specifies the length of the data string in bytes to be used for LIN trigger.
TRIGger:{A B}:BUS:B<x>:LIN:DATa:VALue	Specifies the binary data string to be used for LIN trigger condition if trigger condition is ID or IDANDDATA.
TRIGger:{A B}:BUS:B<x>:LIN:ERRTYPE	Specifies the error type be used for LIN trigger.
TRIGger:{A B}:BUS:B<x>:LIN:IDentifier:VALue	Specifies the binary address string used for LIN trigger if the trigger condition is ID or IDANDDATA.
TRIGger:{A B}:BUS:B<x>:MIL1553B:CONDition	Sets or queries the field or condition for a MIL-STD-1553 bus to trigger on.
TRIGger:{A B}:BUS:B<x>:MIL1553B:ERRType	Sets or queries the type of error condition for a MIL-STD-1553 bus to trigger on.
TRIGger:{A B}:BUS:B<x>:MIL1553B:COMMAND:TRBit	Sets or queries the value of the command word Transmit / Receive bit for a MIL-STD-1553 bus to trigger on.
TRIGger:{A B}:BUS:B<x>:MIL1553B:COMMAND:PARity	Sets or queries the value of the command word parity bit for a MIL-STD-1553 bus to triggering on.
TRIGger:{A B}:BUS:B<x>:MIL1553B:COMMAND:COUNT	Sets or queries the value of the command word "word count" field for a MIL-STD-1553 bus to triggering on.
TRIGger:{A B}:BUS:B<x>:MIL1553B:COMMAND:SUBAddress	Sets or queries the value of the command word "subaddress" field for a MIL-STD-1553 bus to triggering on.
TRIGger:{A B}:BUS:B<x>:MIL1553B:COMMAND:ADDRess:QUALifier	Sets or queries the qualifier to be used when triggering on command word addresses for a MIL-STD-1553 bus.
TRIGger:{A B}:BUS:B<x>:MIL1553B:COMMAND:ADDRess:VALue	Sets or queries the low value when triggering on command word addresses for a MIL-STD-1553 bus.
TRIGger:{A B}:BUS:B<x>:MIL1553B:COMMAND:ADDRess:HIVALue	Sets or queries the high value when triggering on command word addresses for a MIL-STD-1553 bus.
TRIGger:{A B}:BUS:B<x>:MIL1553B:DATa:PARity	Sets or queries the value of the command word parity bit for a MIL-STD-1553 bus to triggering on.
TRIGger:{A B}:BUS:B<x>:MIL1553B:DATa:VALue	Sets or queries the value when triggering on data words for a MIL-STD-1553 bus.
TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:PARity	Sets or queries the value of the status word parity bit for a MIL-STD-1553 bus to triggering on.
TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:ADDRess:QUALifier	Sets or queries the qualifier to be used when triggering on status word addresses for a MIL-STD-1553 bus.
TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:ADDRess:VALue	Sets or queries the low value when triggering on status word addresses for a MIL-STD-1553 bus.
TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:ADDRess:HIVALue	Sets or queries the high value when triggering on status word addresses for a MIL-STD-1553 bus.
TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:BIT:ME	Sets or queries the value of the message error bit (ME bit, bit 9) in a status word for a MIL-STD-1553 bus to triggering on.
TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:BIT:INSTR	Sets or queries the value of the instrumentation bit (INSTR bit, bit 10) in a status word for a MIL-STD-1553 bus to triggering on.

Table 2-41: Trigger commands (cont.)

Command	Description
TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:BIT:SRQ	Sets or queries the value of the status word service request bit (SRQ bit, bit 11) in a status word for a MIL-STD-1553 bus to triggering on.
TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:BIT:BCR	Sets or queries the value of the broadcast command received bit (BCR bit, bit 15) in a status word for a MIL-STD-1553 bus to triggering on.
TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:BIT:BUSY	Sets or queries the value of the busy bit (BUSY bit, bit 16) in a status word for a MIL-STD-1553 bus to triggering on.
TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:BIT:SBSF	Sets or queries the value of the subsystem flag bit (SBSF bit, bit 17) in a status word for a MIL-STD-1553 bus to triggering on.
TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:BIT:DBCA	Sets or queries the value of the dynamic bus control acceptance bit (DBCA bit, bit 18) in a status word for a MIL-STD-1553 bus to triggering on.
TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:BIT:TF	Sets or queries the value of the terminal flag bit (TF bit, bit 19) in a status word for a MIL-STD-1553 bus to triggering on.
TRIGger:{A B}:BUS:B<x>:MIL1553B:TIME:QUALifier	Sets or queries the qualifier to be used when triggering on response time / inter message gap time for a MIL-STD-1553 bus.
TRIGger:{A B}:BUS:B<x>:MIL1553B:TIME:LESSLimit	Sets or queries the lower limit to be used when triggering on response time / inter message gap time for a MIL-STD-1553 bus.
TRIGger:{A B}:BUS:B<x>:MIL1553B:TIME:MORELimit	Sets or queries the upper limit to be used when triggering on response time / inter message gap time for a MIL-STD-1553 bus.
TRIGger:{A B}:BUS:B<x>:PARallel:DATA:VALue	Specifies the binary data string used for a Parallel Bus trigger.
TRIGger:{A B}:BUS:B<x>:RS232C:CONDition	Specifies the condition for an RS-232C trigger.
TRIGger:{A B}:BUS:B<x>:RS232C:DATA:SIZE	Sets or queries the length of the data string in bytes to be used for an RS232 trigger.
TRIGger:{A B}:BUS:B<x>:RS232C:DATA:VALue	Sets or queries the data value of the data token for an RS232 trigger.
TRIGger:{A B}:BUS:B<x>:SENT:CONDition	Sets or queries the trigger condition for a SENT bus.
TRIGger:{A B}:BUS:B<x>:SENT:ERRType	Sets or queries the error type to be used when triggering on SENT data.
TRIGger:{A B}:BUS:B<x>:SENT:ERRType:CRC	Sets or queries the CRC error type to be used when triggering on SENT data.
TRIGger:{A B}:BUS:B<x>:SENT:FAST:CHAN1A:HIVALue	Sets or queries the high binary Fast channel 1 value to use when triggering on a SENT bus signal.
TRIGger:{A B}:BUS:B<x>:SENT:FAST:CHAN1A:QUALifier	Sets or queries the qualifier to be used when triggering on SENT fast packet bus data for device channel 1.
TRIGger:{A B}:BUS:B<x>:SENT:FAST:CHAN1A:VALue	Sets or queries the binary fast channel 1 value to be used when triggering on a SENT bus signal.
TRIGger:{A B}:BUS:B<x>:SENT:FAST:CHAN2B:HIVALue	Sets or queries the high binary fast channel 2 value to use when triggering on a SENT Fast bus signal.
TRIGger:{A B}:BUS:B<x>:SENT:FAST:CHAN2B:QUALifier	Sets or queries the qualifier to be used when triggering on SENT Fast packet bus data for device channel 2.
TRIGger:{A B}:BUS:B<x>:SENT:FAST:CHAN2B:VALue	Sets or queries the binary Fast channel 2 value to be used when triggering on a SENT bus signal. The trigger condition must be set to FAST.

Table 2-41: Trigger commands (cont.)

Command	Description
TRIGger:{A B}:BUS:B<x>:SENT:FAST:COUNTER:HIVALue	Sets or queries the high binary Fast message counter value to be used when triggering on a SENT bus signal.
TRIGger:{A B}:BUS:B<x>:SENT:FAST:COUNTER:QUALifier	Sets or queries the qualifier to be used when triggering on SENT Fast packet bus data for the secure format counter.
TRIGger:{A B}:BUS:B<x>:SENT:FAST:COUNTER:VALue	Sets or queries the binary fast message counter value to be used when triggering on a SENT bus signal.
TRIGger:{A B}:BUS:B<x>:SENT:FAST:INVERTNibble:VALue	Sets or queries the binary fast message inverted nibble value to be used when triggering on a SENT bus signal.
TRIGger:{A B}:BUS:B<x>:SENT:FAST:STATus:VALue	Sets or queries the binary status value to be used when triggering on a SENT bus signal.
TRIGger:{A B}:BUS:B<x>:SENT:PAUSE:QUALifier	Sets or queries the qualifier to be used when triggering on SENT pause pulses.
TRIGger:{A B}:BUS:B<x>:SENT:SLOW:DATA:HIVALue	Sets or queries the high binary Slow channel data value to use when triggering on SENT Slow packet bus data.
TRIGger:{A B}:BUS:B<x>:SENT:SLOW:DATA:QUALifier	This command sets or queries the binary identifier value to use when triggering on Slow packet bus data.
TRIGger:{A B}:BUS:B<x>:SENT:SLOW:DATA:VALue	This command sets or queries the binary slow channel value to use when triggering on Slow packet bus data.
TRIGger:{A B}:BUS:B<x>:SENT:SLOW:Identifier:VALue	This command sets or queries the qualifier to use when triggering on SENT Slow packet bus data.
TRIGger:{A B}:BUS:B<x>:SPI:CONDition	Specifies the trigger condition for a SPI trigger.
TRIGger:{A B}:BUS:B<x>:SPI:DATa:SIZE	Specifies the length of the data string to be used for a SPI trigger if the trigger condition is DATA.
TRIGger:{A B}:BUS:B<x>:SPI:DATa:VALue	Specifies the binary data string used for SPI triggering if the trigger condition is DATA.
TRIGger:{A B}:BUS:B<x>:SPMI:CONDition	This command sets or queries the trigger condition for the specified SPMI bus.
SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:DATa:SIZE	This command sets or queries the length of the data string, in bytes, to be used when triggering on an SPMI bus signal.
TRIGger:{A B}:BUS:B<x>:SPMI:DATa:VALue	This command sets or queries the binary data string used for SPMI triggering.
TRIGger:{A B}:BUS:B<x>:SPMI:MASTERADDRESS:VALue	This command sets or queries the binary data string that identifies the master address used in SPMI triggering if the trigger condition is MASTERREAD, MASTERWRITE, or DEVIDESCMASMASTERREAD.
TRIGger:{A B}:BUS:B<x>:SPMI:NORESPonse	This command sets or queries whether or not to trigger on No Response frames.
TRIGger:{A B}:BUS:B<x>:SPMI:REGISTERADDRESS:VALue	This command sets or queries the binary data string that identifies the register address used in SPMI triggering.
TRIGger:{A B}:BUS:B<x>:SPMI:SLAVEADDRESS:VALue	This command sets or queries the binary data string that identifies the slave address used in SPMI triggering.
TRIGger:{A B}:BUS:B<x>:USB:ADDRESS:HIVALue	Specifies the binary address string for the upper limit for inside-of-range and outside-of-range qualifiers for the USB trigger.
TRIGger:{A B}:BUS:B<x>:USB:ADDRESS:VALue	Specifies the binary address string to be used for USB trigger.

Table 2-41: Trigger commands (cont.)

Command	Description
TRIGger:{A B}:BUS:B<x>:USB:CONDition	Specifies the trigger condition for the USB trigger.
TRIGger:{A B}:BUS:B<x>:USB:DATA:HIVALue	Specifies the binary data string for the upper limit for inside-of-range and outside-of-range qualifiers for the USB trigger when the trigger condition is DATAPacket.
TRIGger:{A B}:BUS:B<x>:USB:DATA:OFFSet	Specifies the data offset in bytes to trigger on.
TRIGger:{A B}:BUS:B<x>:USB:DATA:QUALifier	Sets the qualifier to be used when triggering on a USB bus signal.
TRIGger:{A B}:BUS:B<x>:USB:DATA:SIZE	Specifies the number of contiguous data bytes to trigger on.
TRIGger:{A B}:BUS:B<x>:USB:DATA:TYPE	Specifies the data type for when the trigger condition is set to DATAPacket.
TRIGger:{A B}:BUS:B<x>:USB:DATA:VALue	Specifies the binary data string to be used for the USB trigger when the trigger condition is DATAPacket.
TRIGger:{A B}:BUS:B<x>:USB:ENDPoint:VALue	Specifies the binary endpoint string to be used for the USB trigger.
TRIGger:{A B}:BUS:B<x>:USB:ERRType	Specifies the error type to be used when the trigger condition is set to ERROR.
TRIGger:{A B}:BUS:B<x>:USB:HANDSHAKEType	Specifies the handshake type for the USB trigger.
TRIGger:{A B}:BUS:B<x>:USB:SOFFRAMENUMber	Specifies the binary data string to be used for start of frame number, when the trigger condition is Token Packet and the token type is Start of Frame.
TRIGger:{A B}:BUS:B<x>:USB:SPECIALType	Specifies the packet ID (PID) for the special packet.
TRIGger:{A B}:BUS:B<x>:USB:SPLit:ET:VALue	When triggering on a high-speed USB split transaction, this command specifies the split transaction endpoint type value to trigger on.
TRIGger:{A B}:BUS:B<x>:USB:SPLit:HUB:VALue	When triggering on a high-speed USB split transaction, this command specifies the split transaction hub address value to trigger on.
TRIGger:{A B}:BUS:B<x>:USB:SPLit:PORT:VALue	When triggering on a high-speed USB split transaction, this command specifies the split transaction port address value to trigger on.
TRIGger:{A B}:BUS:B<x>:USB:SPLit:SC:VALue	When triggering on a high-speed USB split transaction, this command specifies whether to trigger on the start or complete phase of the split transaction, based on the Start/Complete bit field value.
TRIGger:{A B}:BUS:B<x>:USB:SPLit:SE:VALue	When triggering on a high-speed USB split transaction, this command specifies the split transaction start/end bit value to trigger on.
TRIGger:{A B}:BUS:B<x>:USB:TOKENType	Specifies the token type for the USB trigger.
TRIGger:{A B}:BUS:SOURce	Sets or queries the source for a bus trigger.
TRIGger:{A B}:EDGE:COUPling	Sets or queries the type of coupling for the edge trigger.
TRIGger:{A B}:EDGE:SLOpe	Sets or queries the slope for the edge trigger.
TRIGger:{A B}:EDGE:SOURce	Sets or queries the source for the edge trigger.
TRIGger:{A B}:LEVel:CH<x>	Sets or queries the level for the trigger for the channel.
TRIGger:{A B}:LOGic:DELTatime	Specifies or queries the Logic trigger delta time value.
TRIGger:{A B}:LOGic:FUNCTion	Sets or queries the logical combination of the input channels for the logic trigger.

Table 2-41: Trigger commands (cont.)

Command	Description
TRIGger:{A B}:LOGic:INPut:CLOCK:SOUrce	Specifies the channel to use as the clock source.
TRIGger:{A B}:LOGic:POLarity	Sets or queries the polarity for the clock channel when Use Clock Edge is set to Yes for Logic trigger type.
TRIGger:{A B}:LOGic:USECLockedge	Sets whether or not Logic trigger type uses clock source.
TRIGger:{A B}:LOGic:WHEn	Sets or queries the condition for generating a logic trigger with respect to the defined input pattern.
TRIGger:{A B}:LOGICPattern: {CH<x> CH<x>_D<x>}	Sets or queries the Logic Pattern that is used along with the Define Logic choice to determine when the logic trigger occurs.
TRIGger:{A B}:LOWerthreshold:CH<x>	Sets or queries CH<x> lower trigger level.
TRIGger:{A B}:PULSEWidth: LOGICQUALification	Sets whether or not to use logic qualification for a pulse width trigger.
TRIGger:{A B}:PULSEWidth:HIGHLimit	Specifies the upper limit to use, in seconds, when triggering on detection of a pulse whose duration is inside or outside a range of two values.
TRIGger:{A B}:PULSEWidth:LOWLimit	Specifies the lower limit to use, in seconds, when triggering on detection of a pulse whose duration is inside or outside a range of two values.
TRIGger:{A B}:PULSEWidth:POLarity	Specifies the polarity for a pulse width trigger.
TRIGger:{A B}:PULSEWidth:SOUrce	Specifies the source waveform for a pulse width trigger.
TRIGger:{A B}:PULSEWidth:WHEn	Specifies to trigger when a pulse is detected with a width (duration) that is less than, greater than, equal to, or unequal to a specified value.
TRIGger:{A B}:RUNT:LOGICQUALification	Specifies whether or not to use logic qualification for a runt trigger.
TRIGger:{A B}:RUNT:POLarity	Specifies the polarity for the runt trigger.
TRIGger:{A B}:RUNT:SOUrce	Specifies the source waveform for the runt trigger.
TRIGger:{A B}:RUNT:WHEn	Specifies the type of pulse width the trigger checks for when it detects a runt.
TRIGger:{A B}:RUNT:WIDth	Specifies the width, in seconds, for a runt trigger.
TRIGger:{A B}:SETHold:CLOCK:EDGE	Specifies the clock edge polarity for setup and hold triggering.
TRIGger:{A B}:SETHold:CLOCK:SOUrce	Specifies the clock source for the setup and hold triggering.
TRIGger:{A B}:SETHold:HOLDTime	Specifies the hold time for setup and hold violation triggering.
TRIGger:{A B}:SETHold:SETTime	Specifies the setup time for setup and hold violation triggering.
TRIGger:{A B}:SETHOLDLogicval: {CH<x> CH<x>_D<x>}	Sets or queries whether the specified channel is included (INCLude) or not included (DONTInclude) in the Setup & Hold trigger input configuration.
TRIGger:{A B}:TIMEOut: LOGICQUALification	Specifies whether or not to use logic qualification for a timeout trigger.
TRIGger:{A B}:TIMEOut:POLarity	When triggering using the TIMEOut trigger type, this commands specifies the polarity to be used.
TRIGger:{A B}:TIMEOut:SOUrce	When triggering using the TIMEOut trigger type, this command specifies the source.
TRIGger:{A B}:TIMEOut:TIME	When triggering using the TIMEOut trigger type, this command specifies the timeout time, in seconds.
TRIGger:{A B}:TRANSition:DELTatime	Specifies the delta time used in calculating the transition value for the transition trigger.
TRIGger:{A B}:TRANSition:POLarity	Specifies the polarity for the transition trigger.

Table 2-41: Trigger commands (cont.)

Command	Description
TRIGger:{A B}:TRANsition:SOUrce	Specifies the source for a transition trigger.
TRIGger:{A B}:TRANsition:WHEn	Specifies whether to check for a transitioning signal that is faster or slower than the specified delta time.
TRIGger:{A B}:TRANsition:LOGICQUALification	Specifies whether or not to use logic qualification for a transition trigger.
TRIGger:{A B}:TYPE	Sets or queries the type of A or B trigger.
TRIGger:{A B}:UPPerthreshold:CH<x>	Sets or queries the CH<x> upper trigger level.
TRIGger:{A B}:WINdow:CROSSIng	Sets or queries the window trigger threshold crossing of the selected trigger Source.
TRIGger:{A B}:WINdow:LOGICQUALification	Specifies whether or not to use logic qualification for a window trigger.
TRIGger:{A B}:WINdow:SOUrce	Sets or queries the source for a window trigger.
TRIGger:{A B}:WINdow:WHEn	Sets or queries the window trigger event.
TRIGger:{A B}:WINdow:WIDTh	Sets or queries the minimum width for a window violation.
TRIGger:{A B B:RESET}	Sets the A, B, or B Reset trigger level automatically to 50% of the range of the minimum and maximum values of the trigger input signal.
TRIGger:A:HOLDoff:BY	Sets or queries the type of holdoff for the A trigger.
TRIGger:A:HOLDoff:TIME	Sets or queries the A trigger holdoff time.
TRIGger:A:LOGICQUALification	Sets or queries the type of logic qualification to perform.
TRIGger:A:MODE	Sets or queries the A trigger mode.
TRIGger:AUXLevel	Sets or queries the Auxiliary Input voltage level to use for an edge trigger.
TRIGger:B:BY	Sets or queries B trigger time or event qualifiers.
TRIGger:B:EVENTS:COUNT	Sets or queries the number of events that must occur before the B trigger occurs.
TRIGger:B:RESET	Sets the B reset trigger level to 50%.
TRIGger:B:RESET:EDGE:COUPling	Sets or queries the trigger coupling for a sequential edge trigger reset when the Source is set to an analog channel.
TRIGger:B:RESET:EDGE:LEVel	Sets the voltage level to use for an Edge Reset trigger when triggering on an analog channel waveform.
TRIGger:B:RESET:EDGE:SLOPe	Sets or queries the trigger slope for a sequential edge trigger reset.
TRIGger:B:RESET:EDGE:SOUrce	Sets or queries the trigger source for the A→B sequential edge trigger reset feature.
TRIGger:B:RESET:TIMEOut:TIME	Sets or queries the reset timer for a sequential time out trigger reset.
TRIGger:B:RESET:TYPE	Sets or queries the type of A→B sequential trigger reset.
TRIGger:B:STATE	Returns the current state of the triggering system.
TRIGger:B:Time	Sets or queries the B trigger delay time.
TRIGger:STATE?	Returns the current state of the triggering system.
VISual:AREA<x>:ASPEctratio	Sets or queries whether the aspect ratio of the specified Visual Trigger area is locked.
VISual:AREA<x>:FLIP:HORizontal	Flips the specified Visual Trigger area horizontally around its center point.
VISual:AREA<x>:FLIP:VERTical	Flips the specified Visual Trigger area vertically around its center point.
VISual:AREA<x>:HEIGht	Sets or queries the height of the specified Visual Trigger area.
VISual:AREA<x>:HITType	Sets or queries the area hit logic true condition for the specified Visual Trigger area.

Table 2-41: Trigger commands (cont.)

Command	Description
VISual:AREA<x>:RESET	Sets the specified Visual Trigger area shape to a default-sized triangle.
VISual:AREA<x>:ROTation	Sets or queries the rotation angle of the specified Visual Trigger area.
VISual:AREA<x>:SHAPE	Sets or queries the current shape of the area.
VISual:AREA<x>:SOUrce	Sets or queries the signal source for the specified Visual Trigger area.
VISual:AREA<x>:VERTICES	Sets or queries the X and Y vertex coordinate values for all vertices of the specified Visual Trigger area.
VISual:AREA<x>:WIDTH	Sets or queries the width of the specified Visual Trigger area.
VISual:AREA<x>:XPOSition	Sets or queries the horizontal (X-axis) center position of the specified Visual Trigger area.
VISual:AREA<x>:YPOSition	Sets or queries the vertical (Y-axis) center position of the specified Visual Trigger area.
VISual:DELETEALL	Deletes all Visual Trigger areas.
VISual:ENable	Sets or queries the status (on or off) of the Visual Trigger mode.
VISual:EQuation	Sets or queries the Visual Trigger area combination logic equation.
VISual:SHOWAReas	Shows or hides all Visual Trigger areas.
VISual:SHOWCRiteria	Sets or queries display of the area names and hit criteria for all visual trigger areas.
VISual:SHOWEQuation	Shows or hides the Visual Trigger area combination logic equation.

Vertical command group

Use the commands in the Vertical Command Group to control the vertical setup of all live (channel) waveforms for acquisition and to control the display of channel, reference, and math waveforms. Analog channels are available when analog probes are attached to FlexChannel inputs.

You can replace VOLts with SCAle in the vertical commands. This provides program compatibility with earlier models of Tektronix instruments.

Table 2-42: Vertical commands

Command	Description
CH<x>?	Returns vertical parameters for the specified channel.
CH<x>:BANdwidth	Sets or queries the bandwidth of the specified channel.
CH<x>:CLIPping?	Queries whether the specified channel's input signal is clipping (exceeding) the channel vertical scale setting.
CH<x>:COUPling	Sets or queries the coupling setting for the specified channel.
CH<x>:DESKew	Sets or queries the deskew time for the specified channel.
CH<x>:LABel:COLor	Sets or queries the color of the specified channel label.
CH<x>:LABel:FONT:BOLD	Sets or queries the bold state of the specified channel label.
CH<x>:LABel:FONT:ITALic	Sets or queries the italic state of the specified channel label.
CH<x>:LABel:FONT:SIZE	Sets or queries the font size of the specified channel label.
CH<x>:LABel:FONT:TYPE	Sets or queries the font type of the specified channel label.
CH<x>:LABel:FONT:UNDERline	Sets or queries the underline state of the specified channel label.
CH<x>:LABel:NAME	Defines or queries the label for the channel waveform.
CH<x>:LABel:XPOS	Sets or queries the X display coordinate for the channel waveform label.
CH<x>:LABel:YPOS	Sets or queries the Y display coordinate for the channel waveform label.
CH<x>:OFFSet	Sets or queries the channel offset.
CH<x>:POSition	Sets or queries the vertical position for the specified analog channel.
CH<x>:PRObe?	Returns all information concerning the probe that is attached to the specified channel.
CH<x>:PRObe:AUTOZero	Autozeros the probe attached to the specified channel.
CH<x>:PRObe:DEGAUSS	Starts a degauss cycle of the probe attached to the specified channel.
CH<x>:PRObe:DEGAUSS:STATE?	Queries whether the probe attached to the specified channel requires a degauss operation.
CH<x>:PRObe:FORCEDRange	Sets the attached probe to its default range, or it queries the default range of the probe.
CH<x>:PRObe:GAIN?	Returns the gain of the probe that is attached to the specified channel.
CH<x>:PRObe:ID?	Returns the type and serial number of the probe that is attached to the specified channel.
CH<x>:PRObe:ID:SERnumber?	Returns the serial number of the probe that is attached to the specified channel.
CH<x>:PRObe:ID:TYPE?	Returns the type of probe that is attached to the specified channel.

Table 2-42: Vertical commands (cont.)

Command	Description
CH<x>:PRObe:INPUTMode	Sets or queries the input mode of the probe that is attached to the specified channel.
CH<x>:PRObe:INPUTMode:AOffset	Sets or queries the A mode offset value of the probe that is attached to the specified channel.
CH<x>:PRObe:INPUTMode:BOffset	Sets or queries the B mode offset value of the probe that is attached to the specified channel.
CH<x>:PRObe:INPUTMode:COFFSet	Sets or queries the common mode offset value of the probe that is attached to the specified channel.
CH<x>:PRObe:INPUTMode:DOFFSet	Sets or queries the differential mode offset value of the probe that is attached to the specified channel.
CH<x>:PRObe:RESistance?	Queries the resistance of the probe that is attached to the specified channel.
CH<x>:PRObe:SET	Sets or queries aspects of probe accessory user interfaces, for example probe attenuation factors.
CH<x>:PRObe:UNIts?	Returns the units of the probe that is attached to the specified channel.
CH<x>:PROBECal?	Returns the probe calibration status.
CH<x>:PROBEControl	Sets or queries the probe range.
CH<x>:PROBEFunc:EXTAtten	Sets the attenuation value for the specified channel to the specified scale factor. Or queries the user-specified attenuation.
CH<x>:PROBEFunc:EXTDBatten	Sets the attenuation value for the specified channel to the specified value in decibels. Or queries the user-specified attenuation in decibels.
CH<x>:PROBEFunc:EXTUnits	Sets or queries the alternate unit for the external attenuator of the specified channel.
CH<x>:PROBEFunc:EXTUnits:STATE	Sets or queries the custom units enable state for the specified channel.
CH<x>:SCALE	Sets or returns the vertical scale for the specified analog channel.
CH<x>:TERmination	Sets channel input termination.
CH<x>:VTERm:BIAS	Sets or queries the termination voltage for the specified channel (if control is available).
CONFIguration:ANALog:BANDWidth?	Queries the maximum licensed bandwidth of the instrument.
REF:ADDNew	Adds the specified reference. Argument is of the form "REF<NR1> ", where $NR1 \geq 1$.
REF:DELeTe	Deletes the specified reference.
REF:REF<x>:DESKew	Sets or queries the deskew time for the specified reference.
REF:REF<x>:LABel:COLor	Sets or queries the color of the specified ref label.
REF:REF<x>:LABel:FONT:BOLD	Sets or queries the bold state of the specified reference label.
REF:REF<x>:LABel:FONT:ITALic	Sets or queries the italic state of the specified reference label.
REF:REF<x>:LABel:FONT:SIZE	Sets or queries the font size of the specified reference label.
REF:REF<x>:LABel:FONT:TYPE	Sets or queries the font type of the specified reference label.
REF:REF<x>:LABel:FONT:UNDERline	Sets or queries the underline state of the specified reference label.

Table 2-42: Vertical commands (cont.)

Command	Description
REF:REF<x>:LABel:NAME	Sets or queries the label of the designated waveform.
REF:REF<x>:LABel:XPOS	Sets or queries the position of the reference waveform label on the X axis.
REF:REF<x>:LABel:YPOS	Sets or queries the position of the reference waveform label on the Y axis.
REF:LIST?	Lists all currently defined references.
REF:REF<x>:SOUrce	Sets or queries the filename used by the given reference.
VERTical:DESKew:FROM: CUSTOMPROPAGation	Sets or queries a target (FROM) delay that you can specify when the propagation delay of the target (FROM) probe used for deskew cannot be detected automatically.
VERTical:DESKEW:STATIC	Sets or queries the target channel for performing channel-to-channel deskew adjustment.
VERTical:DESKew:TO:CUSTOMPROPAGation	Sets or queries a target (TO) delay that can be specified by the user when the propagation delay of the target (TO) probe used for deskew cannot be detected automatically.
VERTical:DESKEW:FROM:SOUrce	Sets or queries the source channel for performing channel-to-channel deskew adjustment.
VERTical:DESKEW:TO:SOUrce	Sets or queries the target channel for performing channel-to-channel deskew adjustment.

Waveform Transfer command group

Use the commands in the Waveform Transfer Command Group to transfer waveform data points from the instrument. Waveform data points are a collection of values that define a waveform. One data value usually represents one data point in the waveform record. When working with envelope waveforms, each data value is either the minimum or maximum of a min/max pair.

Before you transfer waveform data, you must specify the data format, record length, and waveform source.

Data formats. Acquired waveform data uses eight or more bits to represent each data point. The number of bits used depends on the acquisition mode specified when you acquired the data. Data acquired in SAMple or ENvelope mode uses eight bits per waveform data point. Data acquired in AVERage mode uses up to 14 bits per point.

The instrument can transfer waveform data in either ASCII or binary format. You specify the format with the DATA:ENCdg command. The instrument uses signed, 4 byte integers and floating point values; it does not support unsigned floating point values.

ASCII data is represented by signed integer or floating point values. An example ASCII waveform data string can look like this:

```
CURVE<space>-110,-109,-110,-110,-109,-107,  
-109,-107,-106,-105,-103,-100,-97,-90,-84,-80
```

Use ASCII to obtain more readable and easier to format output than binary. However, ASCII can require more bytes to send the same values than it does with binary. This can reduce transmission speeds.

Binary data can be represented by signed integer or floating point values. The range of the values depends on the byte width specified. When the byte width is one, signed integer data ranges from -128 to 127, and positive integer values range from 0 to 255. When the byte width is two, the values range from -32768 to 32767. When a MATH (or REF that came from a MATH) is used, 32-bit floating point values are used that are four bytes in width.

The defined binary formats specify the order in which the bytes are transferred. The following are the four binary formats:

- RIBinary specifies signed integer data-point representation with the most significant byte transferred first.
- SRIBinary is the same as RIBinary except that the byte order is swapped, meaning that the least significant byte is transferred first. This format is useful when transferring data to PCs.
- RFBinary specifies floating point data-point representation with the most significant byte transferred first.
- SRFBinary is the same as RFBinary except that the byte order is swapped, meaning that the least significant byte is transferred first. This format is useful when transferring data to PCs.

Waveform data and record lengths. You can transfer multiple points for each waveform record. You can transfer a portion of the waveform or you can transfer the entire record. You can use the `DATA:START` and `DATA:STOP` commands to specify the first and last data points of the waveform record.

When transferring data from the instrument, you must specify the first and last data points in the waveform record. Setting `DATA:START` to 1 and `DATA:STOP` to the record length will always return the entire waveform.

Waveform data locations and memory allocation. The `DATA:SOURce` command specifies the waveform source when transferring a waveform from the instrument.

Waveform preamble. Each waveform that you transfer has an associated waveform preamble that contains information such as the horizontal scale, the vertical scale, and other settings in effect when the waveform was created. Refer to the individual `WFMOutpre?` commands for more information.

Scaling waveform data. Once you transfer the waveform data to the controller, you can convert the data points into voltage values for analysis using information from the waveform preamble.

Transferring waveform data from the instrument.

You can transfer waveforms from the instrument to an external controller using the following sequence:

1. Select the waveform source(s) using `DATA:SOURce`.
2. Specify the waveform data format using `DATA:ENCdg`.
3. Specify the number of bytes per data point using `WFMOutpre:BYT_Nr`.

NOTE. *MATH waveforms (and REF waveforms that came from a MATH) are always set to four bytes.*

4. Specify the portion of the waveform that you want to transfer using `DATA:START` and `DATA:STOP`.
5. Transfer waveform preamble information using `WFMOutpre`.
6. Transfer waveform data from the instrument using `CURVe?`.

Table 2-43: Waveform Transfer commands

Command	Description
<code>CURVe</code>	The query format transfers waveform data from instrument specified by the <code>DATA:SOURce</code> command.
<code>DATA</code>	Sets the format and location of the waveform data that is transferred with the <code>CURVe</code> Command. Or queries the format and location of the waveform data that is transferred with the <code>CURVe?</code> command.
<code>DATA:ENCdg</code>	Sets or queries the format of outgoing waveform data.
<code>DATA:SOURce</code>	Sets or queries the location of waveform data transferred from the instrument.
<code>DATA:SOURce:AVAILable?</code>	Returns a list of enumerations representing the source waveforms that are currently available for <code>:CURVe?</code> queries.
<code>DATA:START</code>	Sets or queries the starting point in waveform transfer.
<code>DATA:STOP</code>	Sets or queries the ending data point in waveform transfer.
<code>DATA:WIDTH</code>	Specifies the width, in bytes per point, for waveform data transferred from the oscilloscope via the <code>CURVe?</code> query.
<code>WAVFrm?</code>	Returns <code>WFMOutpre?</code> and <code>CURVe?</code> data for the waveform as specified by the <code>DATA:SOURce</code> command.
<code>WFMOutpre?</code>	Returns the waveform formatting data for the waveform specified by the <code>DATA:SOURce</code> command.
<code>WFMOutpre:ASC_Fmt?</code>	Returns the format for ASCII data transferred from the instrument.
<code>WFMOutpre:BIT_Nr</code>	Sets or queries the number of bits per waveform point that outgoing waveforms contain.
<code>WFMOutpre:BN_Fmt</code>	Sets or queries the format of binary data for the waveform.

Table 2-43: Waveform Transfer commands (cont.)

Command	Description
WFMOutpre:BYT_Nr	Sets or queries the data width for the waveform.
WFMOutpre:BYT_Or	Sets or queries the byte order of waveform points.
WFMOutpre:DOMain?	Returns the domain of the outgoing waveform.
WFMOutpre:ENCdg	Sets or queries the type of encoding for outgoing waveforms.
WFMOutpre:NR_Pt?	Returns the number of points for the waveform transmitted in response to a CURVe? query.
WFMOutpre:PT_Fmt?	Returns the point format for the waveform.
WFMOutpre:PT_Off?	Returns the trigger point relative to DATA:START for the waveform.
WFMOutpre:WFIId?	Returns a string describing the acquisition parameters for the waveform.
WFMOutpre:XINcr?	Returns the horizontal sampling interval.
WFMOutpre:XUNit?	Returns the horizontal units for the waveform.
WFMOutpre:XZEro?	Returns the (sub-sample) time between the trigger sample and the occurrence of the actual waveform trigger.
WFMOutpre:YMUlt?	Returns the vertical scale factor per digitizing level for the waveform.
WFMOutpre:YOFf?	Returns the vertical offset in digitizing levels for the waveform.
WFMOutpre:YUNit?	Returns the vertical units for the waveform.
WFMOutpre:YZEro?	Returns the vertical offset for the waveform.

Zoom command group

Zoom commands let you expand and position the waveform display horizontally and vertically, without changing the time base or vertical settings.

NOTE. Zoom commands are available once a view has been added.

Table 2-44: Zoom commands

Command	Description
DISplay:MATHFFTView<x>:ZOOM:XAXIS:FROM	Sets or queries the start of the zoom x axis in the specified plot view.
DISplay:MATHFFTView<x>:ZOOM:XAXIS:TO	Sets or queries the end of the zoom x axis in the specified plot view.
DISplay:MATHFFTView<x>:ZOOM:YAXIS:FROM	Sets or queries the start of the zoom y axis in the specified plot view.
DISplay:MATHFFTView<x>:ZOOM:YAXIS:TO	Sets or queries the end of the zoom y axis in the specified plot view.
DISplay:PLOTView<x>:ZOOM:XAXIS:FROM	Sets or queries the start of the zoom x-axis in the specified plot view.
DISplay:PLOTView<x>:ZOOM:XAXIS:TO	Sets or queries the end of the zoom x-axis in the specified plot view.
DISplay:PLOTView<x>:ZOOM:YAXIS:FROM	Sets or queries the start of the zoom y-axis in the specified plot view.
DISplay:PLOTView<x>:ZOOM:YAXIS:TO	Sets or queries the end of the zoom y-axis in the specified plot view.
DISplay:REFFFTView<x>:ZOOM:XAXIS:FROM	Sets or returns the left edge of the zoom x-axis in the specified plot view.
DISplay:REFFFTView<x>:ZOOM:XAXIS:TO	Sets or queries the right edge of the zoom x-axis in the specified plot view.
DISplay:REFFFTView<x>:ZOOM:YAXIS:FROM	Sets or queries the bottom value of the zoom y-axis in the specified plot view.
DISplay:REFFFTView<x>:ZOOM:YAXIS:TO	Sets or queries the top value of the zoom y-axis in the specified plot view.
DISplay:WAVEView<x>:ZOOM:ZOOM<x>:HORizontal:POSition	Sets or queries the horizontal zoom position (of the specified zoom in the specified waveview) of the zoomed waveform or zoom waveform in the display, around which the zoom waveform displays.
DISplay:WAVEView<x>:ZOOM:ZOOM<x>:HORizontal:SCALe	Sets or queries the zoom horizontal scale factor, of the specified zoom in the specified waveview, in which the zoom waveform is displayed.
DISplay:WAVEView<x>:ZOOM:ZOOM<x>:HORizontal:WINSCLe	Sets or queries the zoom window horizontal scale in the specified waveview.
DISplay:WAVEView<x>:ZOOM:ZOOM<x>:STATe	Sets or queries the specified zoom in the specified waveview on or off.
DISplay:WAVEView<x>:ZOOM:ZOOM<x>:VERTical:POSition	Sets or queries the vertical position of the specified zoom in the specified waveview, in which the zoom waveform is displayed.
DISplay:WAVEView<x>:ZOOM:ZOOM<x>:VERTical:SCALe	Sets or queries the zoom vertical scale factor of the specified waveform for the specified zoom in the specified waveview.
DISplay:WAVEView<x>:Zoom?	Queries the zoom parameters of the specified waveview.
DISplay:WAVEView<x>:ZOOM:ZOOM<x>?	Queries the zoom parameters of the specified zoom in the specified waveview.

Commands listed in alphabetical order

NOTE. Some of the following commands may not be available on your instrument model. Also, some of the following commands are only available if your instrument has the associated option installed.

ACQuire? (Query Only)

Queries the current acquisition state.

Group	Acquisition
Syntax	ACQuire?
Examples	ACQUIRE? might return :ACQUIRE:STATE 1;NUMENV INFINITE;STOPAFTER RUNSTOP;SEQUENCE:MODE NUMACQS;NUMSEQUENCE 1;:ACQUIRE:NUMAVG 16;FASTACQ:STATE 0;PALETTE TEMPERATURE;:ACQUIRE:MODE SAMPLE;SAMPLINGMODE IT.

ACQuire:FASTAcq:PALEtte

Sets or queries the waveform grading for fast acquisition mode.

Group	Acquisition
Syntax	ACQuire:FASTAcq:PALEtte {NORMal TEMPerature SPECTral INVERTed} ACQuire:FASTAcq:PALEtte?
Arguments	NORMal colors traces according to their channel. TEMPerature colors all traces using a multicolored palette, where “intensity” is represented by hue; blue for least frequently hit, red for most frequently hit. All traces share this palette. This is the default color palette. SPECTral colors all traces using a multicolored palette, where “intensity” is represented by hue; red for least frequently hit, blue for most frequently hit. All traces share this palette. INVERTed Inverts the normal display hues and lightness levels based on sample intensity. The areas of lowest sample density appear the brightest, while the areas with the highest sample density appear the darkest.

Examples `ACQUIRE:FASTACQ:PALETTE TEMPERATURE` sets the acquisition fast acquisition palette to Temperature.

`ACQUIRE:FASTACQ:PALETTE?` might return `:ACQUIRE:FASTACQ:PALETTE TEMPERATURE`.

ACQuire:FASTAcq:STATE

Sets or queries the state of fast acquisition mode.

Group Acquisition

Syntax `ACQuire:FASTAcq:STATE {<NR1>|OFF|ON}`
`ACQuire:FASTAcq:STATE?`

Arguments `<NR1> = 0` disables FASTAcq; any other value turns this feature on.
`OFF` disables the FASTAcq feature.
`ON` enables the FASTAcq feature.

Examples `ACQUIRE:FASTACQ:STATE ON` turns on fast acquisition mode.

`ACQUIRE:FASTACQ:STATE?` might return `:ACQUIRE:FASTACQ:STATE 0` indicating that fast acquisition mode is off.

ACQuire:MAXSamplerate? (Query Only)

This query returns the maximum real-time sample rate.

Group Acquisition

Syntax `ACQuire:MAXSamplerate?`

Examples `ACQUIRE:MAXSAMPLERATE?` might return `:ACQUIRE:MAXSAMPLERATE 6.25E9`.

ACQuire:MODe

This command sets or queries the selected acquisition mode of the instrument.

Group	Acquisition
Syntax	ACQuire:MODE {SAmple PEAKdetect HIRes AVERage ENVELOpe} ACQuire:MODE?
Related Commands	ACQuire:NUMAVg CURVe
Arguments	SAmple specifies that the displayed data point value is the first sampled value that is taken during the acquisition interval. In sample mode, all waveform data has 8 bits of precision. You can request 16 bit data with a CURVe query but the lower-order 8 bits of data will be zero. SAmple is the default mode. PEAKdetect specifies the display of high-low range of the samples taken from a single waveform acquisition. The high-low range is displayed as a vertical column that extends from the highest to the lowest value sampled during the acquisition interval. PEAKdetect mode can reveal the presence of aliasing or narrow spikes. HIRes specifies Hi Res mode where the displayed data point value is the average of all the samples taken during the acquisition interval. This is a form of averaging, where the average comes from a single waveform acquisition. The number of samples taken during the acquisition interval determines the number of data values that compose the average. AVERage specifies averaging mode, in which the resulting waveform shows an average of SAmple data points from several separate waveform acquisitions. The instrument processes the number of waveforms you specify into the acquired waveform, creating a running exponential average of the input signal. The number of waveform acquisitions that go into making up the average waveform is set or queried using the ACQuire:NUMAVg command. ENVELOpe specifies envelope mode, where the resulting waveform shows the PEAKdetect range of data points from several separate waveform acquisitions. The number of waveform acquisitions that go into making up the envelope waveform is set or queried using the ACQuire:NUMAVg command. The instrument acquires data after each trigger event using Sample mode; it then determines the pix map location of each sample point and accumulates it with stored data from previous acquisitions. A Pix map is a two dimensional array. The value at each point in the array is a counter that reflects the hit intensity. Infinite and noninfinite persist display modes affect how pix maps are accumulated. Zoom, Math, FastAcq, FastFrame, XY, Roll, and Interpolated Time (IT) Sampling Mode are conflicting features to WFMDB acqMode. Turning on one of them generally turns the other one off. Selection of some standard masks (for example, eye masks, which require option MTM) changes the acquisition mode to WFMDB.

Examples `ACQUIRE:MODE ENVELOPE` sets the acquisition mode to display a waveform that is an envelope of many individual waveform acquisitions.

`ACQUIRE:MODE?` might return `:ACQUIRE:MODE AVERAGE`, indicating that the displayed waveform is the average of the specified number of waveform acquisitions.

ACQUIRE:NUMACQ? (Query Only)

This query-only command returns the number of waveform acquisitions that have occurred since the last time acquisitions were stopped.

Group Acquisition

Syntax `ACQUIRE:NUMACQ?`

Related Commands [ACQUIRE:STATE](#)

Examples `ACQUIRE:NUMACQ?` might return `:ACQUIRE:NUMACQ 350`, indicating that 350 acquisitions have occurred since executing an `ACQUIRE:STATE RUN` command.

ACQUIRE:NUMAVG

This command sets or queries the number of waveform acquisitions that make up an averaged waveform. Ranges from 2 to 10240.

Group Acquisition

Syntax `ACQUIRE:NUMAVG <NR1>`
`ACQUIRE:NUMAVG?`

Related Commands [ACQUIRE:MODE](#)

Arguments `<NR1>` is the number of waveform acquisitions to average.

Examples `ACQUIRE:NUMAVG 10` specifies that 10 waveform averages will be performed before exponential averaging starts.

`ACQUIRE:NUMAVG?` might return `:ACQUIRE:NUMAVG 75`, indicating that there are 75 acquisitions specified for averaging.

ACQuire:NUMFRAMESACQuired? (Query Only)

This query returns the number of FastFrame frames which have been acquired.

Group Horizontal

Syntax ACQuire:NUMFRAMESACQuired?

Examples ACQUIRE:NUMFRAMESACQUIRED? might return
:ACQUIRE:NUMFRAMESACQUIRED 4 indicating 4 frames have been acquired.

ACQuire:SEQuence:CURrent? (Query Only)

In single sequence acquisition mode, this query returns the number of acquisitions or measurements in the sequence completed so far.

Group Acquisition

Syntax ACQuire:SEQuence:CURrent?

Examples ACQUIRE:SEQUENCE:CURRENT? might return :ACQUIRE:SEQUENCE:CURRENT
5.

ACQuire:SEQuence:MODe

In single sequence acquisition, the single sequence stop after count is based on the number of acquisitions.

Group Acquisition

Syntax ACQuire:SEQuence:MODe NUMACQS

Arguments NUMACQS is the number of acquisitions.

Examples ACQUIRE:SEQUENCE:MODE? might return :ACQUIRE:SEQUENCE:MODE
NUMACQS indicating the acquisition sequence mode is set to NUMACQS.

ACquire:SEquence:NUMSEquence

In single sequence acquisition mode, specify the number of acquisitions or measurements that comprise the sequence. The default is 1.

Group	Acquisition
Syntax	ACquire:SEquence:NUMSEquence <NR1> ACquire:SEquence:NUMSEquence?
Arguments	<NR1> is the number of acquisitions or measurements that comprise the sequence.
Examples	ACQUIRE:SEQUENCE:NUMSEQUENCE 2 sets the number of acquisition in a sequence is set to 2. ACQUIRE:SEQUENCE:NUMSEQUENCE? might return ACQUIRE:SEQUENCE:NUMSEQUENCE 1, indicating that the number of acquisition in a sequence is set to 1.

ACquire:STATE

This command starts or stops acquisitions. When state is set to ON or RUN, a new acquisition will be started. If the last acquisition was a single acquisition sequence, a new single sequence acquisition will be started. If the last acquisition was continuous, a new continuous acquisition will be started.

If RUN is issued in the middle of completing a single sequence acquisition (for example, averaging or enveloping), the acquisition sequence is restarted, and any accumulated data is discarded. Also, the instrument resets the number of acquisitions. If the RUN argument is issued while in continuous mode, a reset occurs and acquired data continues to acquire.

If acquire:stopafter is SEQUENCE, this command leaves the instrument in single sequence, unlike the run/stop button which takes the instrument out of single sequence.

Group	Acquisition
Syntax	ACquire:STATE {<NR1> OFF ON RUN STOP} ACquire:STATE?
Related Commands	ACquire:STOPAfter

Arguments	<NR1> = 0 stops acquisitions; any other value starts acquisitions. OFF stops acquisitions. ON starts acquisitions. RUN starts acquisitions. STOP stops acquisitions.
Examples	ACQUIRE:STATE RUN starts the acquisition of waveform data and resets the count of the number of acquisitions. ACQUIRE:STATE? might return :ACQUIRE:STATE 0, indicating that the acquisition is stopped.

ACQuire:STOPAfter

This command sets or queries whether the instrument continually acquires acquisitions or acquires a single sequence. Pressing SINGLE on the front panel button is equivalent to sending these commands: ACQUIRE:STOPAFTER SEQUENCE and ACQUIRE:STATE 1.

Group	Acquisition
Syntax	ACQuire:STOPAfter {RUNStop SEQUENCE} ACQuire:STOPAfter?
Related Commands	ACQuire:STATE
Arguments	RUNStop specifies that the instrument will continually acquire data, if ACQuire:STATE is turned on. SEQUENCE specifies that the next acquisition will be a single-sequence acquisition.
Examples	ACQUIRE:STOPAFTER RUNSTOP sets the instrument to continually acquire data. ACQUIRE:STOPAFTER? might return :ACQUIRE:STOPAFTER SEQUENCE, indicating that the next acquisition the instrument makes will be of the single-sequence type.

AFG:AMPLitude

Sets (or queries) the AFG amplitude in volts, peak to peak.

Conditions	Requires option AFG.
Group	AFG
Syntax	AFG:AMPLitude <NR3> AFG:AMPLitude?
Arguments	<NR3> is a floating point number that represents the AFG amplitude, peak to peak, in volts.
Examples	AFG:AMPLITUDE 1.0 sets the AFG amplitude to 1.0 volts, peak to peak. AFG:AMPLITUDE? might return :AFG:AMPLITUDE 3.0000 indicating the amplitude is set to 3.0 Volts.

AFG:ARBitrary:SOURce

This command sets or queries the source name for the Arbitrary Waveform. Currently supported sources are either waveform file (.wfm) or text file (.csv).

Conditions	Requires option AFG.
Group	AFG
Syntax	AFG:ARBitrary:SOURce <QString> AFG:ARBitrary:SOURce?
Arguments	<QString> is the source name.
Examples	AFG:ARBITRARY:SOURCE "E:/Waveforms/Square.wfm" sets the source waveform to E:/Waveforms/Square.wfm. AFG:ARBITRARY:SOURCE? might return "E:/Waveforms/Square.wfm" indicating the source is set to E:/Waveforms/Square.wfm.

AFG:FREQuency

Sets (or queries) the AFG frequency, in Hz.

Conditions	Requires option AFG.
Group	AFG
Syntax	AFG:FREQUENCY <NR3> AFG:FREQUENCY?
Arguments	<NR3> is the floating point number that represents the AFG frequency, in Hz.
Examples	AFG:FREQUENCY 100.0E3 sets the AFG frequency to 100 kHz. AFG:FREQUENCY? might return :AFG:FREQUENCY 312.5000E+3 indicating the frequency is set to 312.5 kHz.

AFG:FUNCTION

Sets (or queries) which AFG function to execute.

Conditions	Requires option AFG.
Group	AFG
Syntax	AFG:FUNCTION {SINE SQUare PULSe RAMP NOISe DC SINC GAUSSian LOREntz ERISe EDEcAY HAVERSINE CARDIac ARBiTrary} AFG:FUNCTION?

Arguments	SINE
	SQUare
	PULSe
	RAMP
	NOISe
	DC – The DC level is controlled by AFG:OFFSet .
	SINC (Sin(x)/x)
	GAUSSian
	LOREntz

ERISe
EDECAy
HAVERSINe
CARDIac
ARBitrary

Examples **AFG:FUNC** LOREN specifies to generate the Lorentz function.

AFG:FUNCTION? might return **:AFG:FUNCTION SINE** indicating the AFG function is set to sine.

AFG:HIGhLevel

This command sets (or queries) the high level value of the output waveform, in volts, when using the arbitrary function generator feature.

Conditions Requires option AFG.

Group AFG

Syntax **AFG:HIGhLevel** <NR3>
AFG:HIGhLevel?

Arguments <NR3> is a floating point number that represents the AFG high level value, in volts.

Examples **AFG:HIGhLEVEL 1.0** sets the AFG high level value to 1.0 volts.

AFG:HIGhLEVEL? might return **:AFG:HIGhLEVEL 1.5000** indicating the high level is set to 1.5 Volts.

AFG:LOWLevel

This command sets (or queries) the low level value of the output waveform, in volts, when using the arbitrary function generator feature.

Conditions Requires option AFG.

Group AFG

Syntax	<code>AFG:LOWLevel <NR3></code> <code>AFG:LOWLevel?</code>
Arguments	NR3 is the floating point number that represents the AFG low level value, in volts.
Examples	<code>AFG:LOWLEVEL 1.0</code> sets the AFG low level value to 1.00 volts. <code>AFG:LOWLEVEL?</code> might return <code>:AFG:LOWLEVEL -1.5000</code> indicating the low level is set to —1.5 Volts.

AFG:NOISEAdd:PERCent

Sets (or queries) the AFG additive noise level as a percentage. Minimum is 0.0%, maximum is 100.0% and increment is 1.0%.

Conditions	Requires option AFG.
Group	AFG
Syntax	<code>AFG:NOISEAdd:PERCent <NR3></code> <code>AFG:NOISEAdd:PERCent?</code>
Arguments	Floating point number that represents the AFG additive noise level, as a percentage.
Examples	<code>AFG:NOISEADD:PERCENT 50</code> sets the AFG additive noise level to 50 percent. <code>AFG:NOISEADD:PERCENT?</code> might return <code>:AFG:NOISEADD:PERCENT 0</code> indicating that no noise is added.

AFG:NOISEAdd:STATE

Sets (or queries) the AFG additive noise state.

Conditions	Requires option AFG.
Group	AFG

Syntax AFG:NOISEAdd:STATE {0|1|OFF|ON}
AFG:NOISEAdd:STATE?

Arguments 1 or ON turns on the AFG additive noise state.
0 or OFF turns it off.

Examples AFG:NOISEADD:STATE ON turns on the additive noise state.
AFG:NOISEADD:STATE? might return :AFG:NOISEADD:STATE 0 indicating the noise additive state is off.

AFG:OFFSet

Sets (or queries) the AFG offset value, in volts.

Conditions Requires option AFG.

Group AFG

Syntax AFG:OFFSet <NR3>
AFG:OFFSet?

Arguments <NR3> is a floating point number that represents the AFG offset, in volts.

Examples AFG:OFFSET 1.0 sets the AFG offset to 1.0 volts.
AFG:OFFSET? might return AFG:OFFSET 0.0E+0 indicating there is no offset.

AFG:OUTPut:LOAd:IMPEDance

Sets (or queries) the AFG output load impedance.

Conditions Requires option AFG.

Group AFG

Syntax AFG:OUTPut:LOAd:IMPEDance {FIFTy|HIGHZ}
AFG:OUTPut:LOAd:IMPEDance?

Arguments	FIFtY sets the output load impedance to 50 Ohms. HIGHZ sets the output load impedance to the high-impedance state.
Examples	AFG:OUTP:LOA:IMPED FIF sets the AFG output load impedance to 50 Ohms. AFG:OUTPUT:LOAD:IMPEDANCE? might return AFG:OUTPUT:LOAD:IMPEDANCE HIGHZ indicating the load impedance is set to the high impedance state.

AFG:OUTPut:STATE

Sets (or queries) the AFG output state.

Conditions	Requires option AFG.
Group	AFG
Syntax	AFG:OUTPut:STATE {0 1 OFF ON} AFG:OUTPut:STATE?
Arguments	1 or ON turns on the AFG output state. 0 or OFF turns it off.
Examples	AFG:OUTPUT:STATE ON turns on the AFG output state. AFG:OUTPUT:STATE? might return AFG:OUTPUT:STATE 1 indicating the AFG output is on.

AFG:PERIod

Sets (or queries) the period of the AFG waveform, in seconds.

Conditions	Requires option AFG.
Group	AFG
Syntax	AFG:PERIod <NR3> AFG:PERIod?

Arguments	NR3 is the floating point number that represents the AFG period value, in seconds.
Returns	The query response is returned in high precision NR3 format (up to 12 digits with more than 4 trailing 0 digits after the decimal point is omitted).
Examples	AFG:PERIOD 1 sets the AFG period value to 1 second. AFG:PERIOD? might return :AFG:PERIOD 3.2000E-6 indicating the AFG period is set to 3.2 μs.

AFG:PULse:WIDth

Sets (or queries) the AFG pulse width, in seconds.

Conditions	Requires option AFG.
Group	AFG
Syntax	AFG:PULse:WIDth <NR3> AFG:PULse:WIDth?
Arguments	NR3 is the floating point number that represents the pulse width, in seconds.
Examples	AFG:PULSE:WIDTH 100.0E-6 sets the AFG pulse width to 100 microseconds. AFG:PULSE:WIDTH? might return :AFG:PULSE:WIDTH 1.0000E-6 indicating the pulse width is set to 1 μs.

AFG:RAMP:SYMmetry

Sets (or queries) the AFG ramp symmetry in percent. Minimum is 0.0%, maximum is 100.0% and increment is 0.10%.

Conditions	Requires option AFG.
Group	AFG
Syntax	AFG:RAMP:SYMmetry <NR3> AFG:RAMP:SYMmetry?

Arguments Floating point number that represents the AFG ramp symmetry, as a percentage.

Examples AFG:RAMP:SYMMETRY 50.0 sets the AFG ramp symmetry to 50 percent.
AFG:RAMP:SYMMETRY? might return :AFG:RAMP:SYMMETRY 50.0000 indicating the symmetry is set to 50%.

AFG:SQUare:DUTy

Sets (or queries) the AFG duty cycle in percent. The minimum is 10.0%, maximum is 90.0% and increment is 0.10%.

Conditions Requires option AFG.

Group AFG

Syntax AFG:SQUare:DUTy <NR3>
AFG:SQUare:DUTy?

Arguments Floating point number that represents the AFG duty cycle, as a percentage.

Examples AFG:SQUARE:DUTY 50.0 sets the AFG duty cycle to 50 percent.
AFG:SQUARE:DUTY? might return :AFG:SQUARE:DUTY 50.0000 indicating the duty cycle is set to 50%.

ALias

This command sets or queries the state of alias functionality, and it is identical to the ALIAS:STATE command.

Group Alias

Syntax ALias {OFF|ON|<NR1>}
ALias?

Related Commands [ALias:STATE](#)

Arguments	OFF turns Alias expansion off. ON turns Alias expansion on. When a defined alias is received, the specified command sequence is substituted for the alias and executed. <NR1> = 0 disables Alias mode; any other value enables Alias mode.
Examples	ALIAS ON turns the alias feature on. When a defined alias is received, the specified command sequence is substituted for the alias and executed. ALIAS? might return :ALIAS:STATE 1, indicating that the alias feature is on.

ALias:CATalog? (Query Only)

This query-only command returns a list of the currently defined alias labels, separated by commas. If no aliases are defined, the query returns the string "".

Group	Alias
Syntax	ALias:CATalog?
Examples	ALIAS:CATALOG? might return the string :ALIAS:CATALOG "SETUP1", "TESTMENU1", "DEFAULT" showing that there are three aliases named SETUP1, TESTMENU1, and DEFAULT.

ALias:DEFine

This command assigns a sequence of program messages to an alias label. These messages are then substituted for the alias whenever it is received as a command or query, provided that ALIAS:STATE is turned on. The query form of this command returns the definitions of a selected alias.

NOTE. Attempting to give two aliases the same name causes an error. To give a new alias the name of an existing alias, the existing alias must first be deleted.

Group	Alias
Syntax	ALias:DEFine <QString><,>{<QString> <Block>} ALias:DEFine?
Related Commands	ALias:STATE

Arguments	The first <QString> is the alias label. This label cannot be a command name. Labels must start with a letter and can contain only letters, numbers, and underscores; other characters are not allowed. The label must be less than or equal to 12 characters. The second <QString> or <Block> is a complete sequence of program messages. The messages can contain only valid commands that must be separated by semicolons and must follow all rules for concatenating commands. The sequence must be less than or equal to 256 characters.
Examples	ALIAS:DEFINE "ST1",":RECALL:SETUP 'C:/mySetup.set';:AUTOSET EXECUTE" defines an alias named "ST1" that recalls a previously saved setup and performs an autoset. ALIAS:DEFINE? "ST1" returns :ALIAS:DEFINE "ST1",#247 :RECALL:SETUP 'C:/mySetup.set';:AUTOSET EXECUTE.

ALias:DELEte (No Query Form)

This command removes a specified alias and is identical to ALias:DELEte:NAME. An error message is generated if the named alias does not exist.

Group	Alias
Syntax	ALias:DELEte <QString>
Related Commands	*ESR? ALias:DELEte:ALL
Arguments	<QString> is the name of the alias to be removed. Using ALias:DELEte without specifying an alias causes an execution error. <QString> must be a previously defined value.
Examples	ALIAS:DELETE "SETUP1" deletes the alias named SETUP1.

ALias:DELEte:ALL (No Query Form)

This command deletes all existing aliases.

Group	Alias
Syntax	ALIAS:DELETE:ALL
Related Commands	ALIAS:DELETE ALIAS:DELETE:NAME
Examples	ALIAS:DELETE:ALL deletes all existing aliases.

Alias:DELETE:NAME (No Query Form)

This command removes a specified alias. An error message is generated if the named alias does not exist. This command is identical to `ALIAS:DELETE`.

Group	Alias
Syntax	ALIAS:DELETE:NAME <QString>
Arguments	<QString> is the name of the alias to remove. Using <code>ALIAS:DELETE:NAME</code> without specifying an alias causes an exception error. <QString> must be a previously defined alias.
Examples	ALIAS:DELETE:NAME "STARTUP" deletes the alias named STARTUP.

Alias:STATE

This command turns aliases on or off and is identical to the [ALIAS](#) command. The query form of this command returns the state of the aliases.

Group	Alias
Syntax	ALIAS:STATE {<NR1> OFF ON} ALIAS:STATE?
Arguments	<NR1> = 0 turns off aliases; any other value turns on aliases. OFF turns alias expansion off.

ON turns alias expansion on. When a defined alias is received, the specified command sequence is substituted for the alias and executed.

Examples `ALIAS:STATE OFF` turns off the alias feature.

`ALIAS:STATE?` might return `:ALIAS:STATE ON`, indicating that alias expansion is currently turned on.

ALLEV? (Query Only)

This query-only command prompts the instrument to return all events and their messages (delimited by commas), and removes the returned events from the Event Queue. Use the `*ESR?` query to enable the events to be returned. This command is similar to repeatedly sending `*EVMsg?` queries to the instrument.

Group Status and Error

Syntax `ALLEV?`

Related Commands [*ESR?](#)
[EVMsg?](#)

Examples `ALLEV?` might return `:ALLEV 2225,"Measurement error, No waveform to measure; "420,"Query UNTERMINATED;"`.

AUTOSAVEPITIMEOUT

This command sets or queries the idle time from the programmable interface before auto-save occurs.

Group Miscellaneous

Syntax `AUTOSAVEPITIMEOUT <NR1>`
`AUTOSAVEPITIMEOUT?`

Arguments `<NR1>`

Examples `AUTOSAVEPITIMEOUT 100000` sets the timeout to 100 seconds.
`AUTOSAVEPITIMEOUT?` might return `:AUTOSAVEPITIMEOUT 180000`
indicating the time out value is 180 seconds.

AUTOSAVEUITIMEOUT

This command sets or queries the idle time from the user interface before auto-save occurs.

Group Miscellaneous

Syntax `AUTOSAVEUITIMEOUT <NR1>`
`AUTOSAVEUITIMEOUT?`

Arguments `<NR1>`

Examples `AUTOSAVEUITIMEOUT 4000` sets the timeout to 4 seconds.
`AUTOSAVEUITIMEOUT?` might return `:AUTOSAVEUITIMEOUT 3000` indicating
the time out value is 3 seconds.

AUTOSet (No Query Form)

This command (no query format) sets the vertical, horizontal, and trigger controls of the instrument to automatically acquire and display the selected waveform.

Group Miscellaneous

Syntax `AUTOSet EXECute`

Arguments `EXECute` autosets the displayed waveform; this is equivalent to pressing the front panel Autoset button.

Examples `AUTOSET EXECute` autosets the displayed waveform.

AUTOSet:ACQuisition:ENABle

This command sets or queries the Autoset acquisition setting adjustment. Settings affected may include, but not be limited to, acquisition mode, and FastAcq mode.

Group Miscellaneous

Syntax `AUTOSet:ACQuisition:ENABle {ON|1|OFF|0}`
`AUTOSet:ACQuisition:ENABle?`

Related Commands [AUTOSet:HORizontal:ENABle](#)
[AUTOSet:TRIGger:ENABle](#)
[AUTOSet:VERTical:ENABle](#)

Arguments ON or 1 enables Autoset to change acquisition settings.
OFF or 0 disables Autoset from changing acquisition settings.

Examples `AUTOSet:ACQuisition:ENABle OFF` turns off Autoset's adjustment of the acquisition settings.
`AUTOSet:ACQuisition:ENABle?` might return 0, indicating Autoset's adjustment of the acquisition settings is turned off.

AUTOSet:ENABle

This command sets or queries the Autoset enable/disable feature. This is useful for classroom purposes where the instructor wants the students to achieve the desired instrument settings without the benefit of the Autoset feature.

Group Miscellaneous

Syntax `AUTOSet:ENABle {ON|1|OFF|0}`
`AUTOSet:ENABle?`

Related Commands [AUTOSet:ACQuisition:ENABle](#)
[AUTOSet:HORizontal:ENABle](#)
[AUTOSet:TRIGger:ENABle](#)
[AUTOSet:VERTical:ENABle](#)

Arguments	ON or 1 enables Autoset. OFF or 0 disables Autoset.
Examples	AUTOSet:ENABle ON turns on the Autoset feature. AUTOSet:ENABle? might return 1, indicating that Autoset is enabled.

AUTOSet:HORizontal:ENABle

This command sets or queries Autoset's adjustment of horizontal settings. Settings affected may include, but not be limited to, horizontal scale, horizontal position, and horizontal delay mode.

Group	Miscellaneous
Syntax	AUTOSet:HORizontal:ENABle {ON 1 OFF 0} AUTOSet:HORizontal:ENABle?
Related Commands	AUTOSet:ACQuisition:ENABle AUTOSet:TRIGger:ENABle AUTOSet:VERTical:ENABle
Arguments	ON or 1 enables Autoset to change horizontal settings. OFF or 0 disables Autoset from changing horizontal settings.
Examples	AUTOSet:HORizontal:ENABle OFF turns off Autoset's adjustment of the horizontal settings. AUTOSet:HORizontal:ENABle? might return 0, indicating Autoset's adjustment of the horizontal settings is turned off.

AUTOSet:TRIGger:ENABle

This command sets or queries Autoset's adjustment of trigger settings. Settings affected may include, but not be limited to, trigger level, trigger source, and trigger coupling.

Group	Miscellaneous
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Syntax	<code>AUTOSet:TRIGger:ENABle {ON 1 OFF 0}</code> <code>AUTOSet:TRIGger:ENABle?</code>
Related Commands	AUTOSet:ACQuisition:ENABle AUTOSet:HORizontal:ENABle AUTOSet:VERTical:ENABle
Arguments	ON or 1 enables Autoset to change trigger settings. OFF or 0 disables Autoset from changing trigger settings.
Examples	<code>AUTOSet:TRIGger:ENABle OFF</code> turns off Autoset's adjustment of the trigger settings. <code>AUTOSet:TRIGger:ENABle?</code> might return 0, indicating Autoset's adjustment of the trigger settings is turned off.

AUTOSet:VERTical:ENABle

This command sets or queries Autoset's adjustment of vertical settings. Settings affected may include, but not be limited to, vertical scale, vertical position, and vertical offset.

Group	Miscellaneous
Syntax	<code>AUTOSet:VERTical:ENABle {ON 1 OFF 0}</code> <code>AUTOSet:VERTical:ENABle?</code>
Related Commands	AUTOSet:ACQuisition:ENABle AUTOSet:HORizontal:ENABle AUTOSet:TRIGger:ENABle
Arguments	ON or 1 enables Autoset to change vertical settings. OFF or 0 disables Autoset from changing vertical settings.
Examples	<code>AUTOSet:VERTical:ENABle OFF</code> turns off Autoset's adjustment of the vertical settings.

AUTOSet:VERTical:ENABle? might return 0, indicating Autoset's adjustment of the vertical settings is turned off.

AUTOSet:VERTical:OPTIMize

This command sets or queries which vertical settings Autoset will optimize when the display mode is set to Overlay mode (all waveforms are in one common graticule in the Waveform View).

Group Miscellaneous

Syntax AUTOSet:VERTical:OPTIMize {RESOLution|VISIBility}
AUTOSet:VERTical:OPTIMize?

Related Commands [DISplay:WAVEView<x>:VIEWStyle](#)

Arguments RESOLution uses as much of the ADC's (Analog to Digital Converter) range as possible to provide the best vertical resolution and measurement accuracy, but waveforms will overlap each other.

VISIBility vertically scales and positions waveforms so they are visually separated from each other at the expense of vertical resolution and measurement accuracy.

Examples AUTOSet:VERTical:OPTIMize VISIBility will vertically scale and position all active waveforms so that they are visually separated from each other when autoset is executed.

AUTOSet:VERTical:OPTIMize? might return
AUTOSet:VERTical:OPTIMize RESOLUTION, which indicates that, when executed, autoset will vertically scale/position the active waveforms to maximize the ADC range while in Overlay display mode.

AUXout:EDGE

This command sets or queries the direction in which the Auxiliary Output signal will transition when a trigger occurs.

Group Miscellaneous

Syntax	<code>AUXout:EDGE {RISing FALling}</code> <code>AUXout:EDGE?</code>
Arguments	<code>RISing</code> sets the polarity to the rising edge. <code>FALling</code> sets the polarity to the falling edge.
Examples	<code>AUXOUT:EDGE RISING</code> sets the polarity to rising edge. <code>AUXOUT:EDGE?</code> might return <code>:AUXOUT:EDGE FALLING</code> , indicating that the polarity is set to falling edge.

AUXout:SOUrce

This command sets or queries the source at the Auxiliary Output BNC connection.

Group	Miscellaneous
Syntax	<code>AUXout:SOUrce {ATRIGger REFOUT AFG}</code> <code>AUXout:SOUrce?</code>
Arguments	<code>ATRIGger</code> sets the source at the BNC connector to the main trigger. <code>REFOUT</code> sets the source at the BNC connector to the reference output. <code>AFG</code> sets the source at the BNC connector to the AFG output.
Examples	<code>AUXOUT:SOURCE?</code> might return <code>:AUXOUT:SOURCE ATRIGGER</code> , indicating that the source at the BNC connector is set to the A trigger.

BUS:ADDNew (No Query Form)

This command adds the specified bus. This command creates/adds the bus but does not display it (turn it on). In order to enable bus decoding and see the bus display on screen, send the [DISplay:WAVEView<x>:BUS:B<x>:STATE ON](#) command.

Group	Bus
Syntax	<code>BUS:ADDNew <QString></code>

Related Commands	BUS:B<x>:TYPE DISplay:WAVEView<x>:BUS:B<x>:STATE
Arguments	<QString> is a quoted string of the form "B<NR1>" where NR1 is ≥ 1 .
Examples	<code>BUS:ADDNEW "B5";:display:waveview1:bus:b5:state on</code> creates bus 3 with the default type of Parallel, and then turns it on.

BUS:B<x>:ARINC429A:BITRate

This command sets or queries the ARINC429 bit rate for the specified bus. The bus number is specified by x. If you select Custom, use `BUS:B<x>:ARINC429A:BITRate:CUSTom` to set the bit rate.

Conditions	Requires option SR-AERO.
Group	Bus
Syntax	<code>BUS:B<x>:ARINC429A:BITRate {LOW HI CUSTom}</code> <code>BUS:B<x>:ARINC429A:BITRate?</code>
Related Commands	<code>B<x></code> is the number of the bus. BUS:B<x>:ARINC429A:BITRate:CUSTom
Arguments	Arguments specify the bit rate.
Examples	<code>BUS:B1:ARINC429A:BITRATE LOW</code> sets the bit rate to handle low speed signals. <code>BUS:B1:ARINC429A:BITRATE?</code> might return <code>:BUS:B1:ARINC429A:BITRATE HI</code> , indicating the bit rate is set to handle high speed signals.

BUS:B<x>:ARINC429A:BITRate:CUSTom

This command sets or queries the ARINC429 custom bit rate for the specified bus. The bus is specified by x.

Conditions	Requires option SR-AERO.
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Group	Bus
Syntax	<code>BUS:B<x>:ARINC429A:BITRate:CUSTOm <NR1></code> <code>BUS:B<x>:ARINC429A:BITRate:CUSTOm?</code>
Related Commands	BUS:B<x>:ARINC429A:BITRate
Arguments	<code>B<x></code> is the number of the bus. <code><NR1></code> is the ARINC429 custom bit rate for the specified bus.
Examples	<code>BUS:B1:ARINC429A:BITRATE:CUSTOM 12500</code> sets the bit rate to 12,500. <code>BUS:B1:ARINC429A:BITRATE:CUSTOM?</code> might return <code>:BUS:B1:ARINC429A:BITRATE:CUSTOM 100000</code> , indicating the bit rate is set to 100,000.

BUS:B<x>:ARINC429A:DATAFormat

This command sets or queries the format of the DATA field for the specified ARINC429 bus. The bus is specified by x.

Conditions	Requires option SR-AERO.
Group	Bus
Syntax	<code>BUS:B<x>:ARINC429A:DATAFormat {DATA SDIDATA SDIDATASSM}</code> <code>BUS:B<x>:ARINC429A:DATAFormat?</code>
Arguments	<code>B<x></code> is the number of the bus. <code>DATA</code> specifies a DATA field width of 19 bits (covering bits 11 through 29 of the 32 bit packet) <code>SDIDATA</code> specifies a DATA field width of 21 bits (covering bits 9 through 29 of the 32 bit packet) <code>SDIDATASSM</code> specifies a DATA field width of 23 bits (covering bits 9 through 31 of the 32 bit packet)
Examples	<code>BUS:B1:ARINC429A:DATAFORMAT SDIDATASSM</code> sets Bus 1 ARINC429 DATA field width to 23 bits.

`BUS:B1:ARINC429A:DATAFORMAT?` might return
`:BUS:B1:ARINC429A:DATAFORMAT DATA`, indicating a DATA field width
of 19 bits.

BUS:B<x>:ARINC429A:POLARITY

This command sets or queries the source polarity for the specified ARINC429 bus. The bus is specified by x.

Conditions	Requires option SR-AERO.
Group	Bus
Syntax	<code>BUS:B<x>:ARINC429A:POLARITY {NORMa1 INVERTed}</code> <code>BUS:B<x>:ARINC429A:POLARITY?</code>
Arguments	<code>B<x></code> is the number of the bus. <code>NORMa1</code> specifies normal polarity. <code>INVERTed</code> specifies inverted polarity.
Examples	<code>BUS:B3:ARINC429A:POLARITY INVERTed</code> sets Bus 3 ARINC429 polarity to inverted. <code>BUS:B2:ARINC429A:POLARITY?</code> might return <code>:BUS:B2:ARINC429A:POLARITY NORMAL</code> , indicating that the Bus 2 ARINC429 polarity is set to normal.

BUS:B<x>:ARINC429A:SOUrce

This command sets or queries the source for the specified ARINC429 bus. The bus is specified by x.

Conditions	Requires option SR-AERO.
Group	Bus
Syntax	<code>BUS:B<x>:ARINC429A:SOUrce {CH<x> Math<x> REF<x>}</code> <code>BUS:B<x>:ARINC429A:SOUrce?</code>

Arguments **B<x>** is the number of the bus.

CH<x> specifies an analog channel as the source waveform for the ARINC429 bus.

Math<x> specifies a math waveform as the source waveform for the ARINC429 bus.

REF<x> specifies a reference waveform as the source waveform for the ARINC429 bus.

Examples **BUS:B1:ARINC429A:SOURCE CH1** sets channel 1 as the source for the ARINC429 bus.

BUS:B1:ARINC429A:SOURCE? might return **:BUS:B1:ARINC429A:SOURCE MATH2** indicating that the source is set to MATH2.

BUS:B<x>:ARINC429A:THRESHold

This command sets or queries the ARINC429 upper threshold for the specified bus. The bus is specified by x.

Conditions Requires option SR-AERO.

Group Bus

Syntax **BUS:B<x>:ARINC429A:THRESHold <NR3>**
BUS:B<x>:ARINC429A:THRESHold?

Related Commands [BUS:B<x>:ARINC429A:SOUrce](#)

Arguments **B<x>** is the number of the bus.

<NR3> is the ARINC429 lower threshold for the specified bus.

Examples **BUS:B3:ARINC429A:THRESHOLD -200e-3** sets the Bus 3 ARINC429 upper threshold to -200 mV.

BUS:B2:ARINC429A:THRESHOLD? might return **:BUS:B2:ARINC429A:THRESHOLD -500.0000E-3** indicating the Bus 2 ARINC429 upper threshold is set to -500 mV.

BUS:B<x>:AUDIO:BITDelay

This command sets or queries the number of delay bits for the specified AUDIO bus. The bus is specified by x.

NOTE. *This command only applies to the TDM Audio type.*

Conditions	Requires option SR-AUDIO.
Group	Bus
Syntax	BUS:B<x>:AUDIO:BITDelay <NR1> BUS:B<x>:AUDIO:BITDelay?
Arguments	B<x> is the number of the bus. <NR1> specifies the number of delay bits.
Examples	BUS:B1:AUDIO:BITDELAY 2 sets the bit delay to 2. BUS:B1:AUDIO:BITDELAY? might return :BUS:B1:AUDIO:BITDELAY 1 indicating that the number of delay bits is 1.

BUS:B<x>:AUDIO:BITOrder

Specifies the bit order for the specified AUDIO bus. The bus is specified by x.

Conditions	Requires option SR-AUDIO.
Group	Bus
Syntax	BUS:B<x>:AUDIO:BITOrder {MSB LSB} BUS:B<x>:AUDIO:BITOrder?
Arguments	B<x> is the number of the bus. MSB specifies that the most significant bit will be expected first in the order. LSB specifies that the least significant bit will be expected first in the order.

Examples `BUS:B1:AUDIO:BITORDER LSB` sets the bit order to LSB.

`BUS:B1:AUDIO:BITORDER?` might return `:BUS:B1:AUDIO:BITORDER MSB` indicating that the MSB is first in the bit order.

BUS:B<x>:AUDio:CLOCK:POLarity

This command sets or queries the clock source polarity for the specified AUDIO bus. The bus is specified by x.

Conditions Requires option SR-AUDIO.

Group Bus

Syntax `BUS:B<x>:AUDio:CLOCK:POLarity {FALL|RISE}`
`BUS:B<x>:AUDio:CLOCK:POLarity?`

Arguments B<x> is the number of the bus.

FALL sets falling edge as the clock polarity.

RISE sets rising edge as the clock polarity.

Examples `BUS:B1:AUDIO:CLOCK:POLARITY FALL` sets the clock polarity to Fall.

`BUS:B1:AUDIO:CLOCK:POLARITY?` might return `:BUS:B1:AUDIO:CLOCK:POLARITY RISE` indicating that the clock polarity is set to Rise.

BUS:B<x>:AUDio:CLOCK:SOUrce

This command sets or queries the clock source waveform for the specified AUDIO bus. The bus is specified by x.

Conditions Requires option SR-AUDIO.

Group Bus

Syntax `BUS:B<x>:AUDio:CLOCK:SOUrce`
`{CH<x>|CH<x>_D<x>|Math<x>|REF<x>|REF<x>_D<x>}`
`BUS:B<x>:AUDio:CLOCK:SOUrce?`

Arguments	B<x> is the number of the bus. CH<x> specifies an analog channel as the clock source waveform for the audio bus. CH<x>_D<x> specifies a digital channel as the clock source waveform for the specified audio bus. Math<x> specifies a math waveform as the clock source waveform for the audio bus. REF<x> specifies a reference waveform as the clock source waveform for the audio bus. REF<x>_D<x> specifies a digital reference waveform as the clock source waveform for the specified audio bus.
Examples	BUS:B1:AUDIO:CLOCK:SOURCE CH1_D1 sets D1 of channel 1 as the clock source for the audio bus. BUS:B1:AUDIO:CLOCK:SOURCE? might return :BUS:B1:AUDIO:CLOCK:SOURCE CH1 indicating that the clock source is set to channel 1.

BUS:B<x>:AUDio:CLOCK:THReshold

This command sets or queries the audio clock source threshold for the specified bus. The bus is specified by x.

Conditions	Requires option SR-AUDIO.
Group	Bus
Syntax	BUS:B<x>:AUDio:CLOCK:THReshold <NR3> BUS:B<x>:AUDio:CLOCK:THReshold?
Related Commands	BUS:B<x>:AUDio:CLOCK:SOUrce
Arguments	B<x> is the number of the bus. <NR3> is the audio clock source threshold for the specified bus.
Examples	BUS:B3:AUDIO:CLOCK:THRESHOLD 500.0E-3 sets the Bus 3 audio Cclock source threshold to 500.0 mV.

`BUS:B2:AUDIO:CLOCK:THRESHOLD?` might return
`:BUS:B2:AUDIO:CLOCK:THRESHOLD 1.0` indicates the Bus 2 audio clock
source threshold is set to 1.0 V.

BUS:B<x>:AUDio:DATa:POLarity

This command sets or queries the audio data source polarity for the specified audio bus. The bus is specified by x.

Conditions Requires option SR-AUDIO.

Group Bus

Syntax `BUS:B<x>:AUDio:DATa:POLarity {HIGH|LOW}`
`BUS:B<x>:AUDio:DATa:POLarity?`

Arguments B<x> is the number of the bus.

HIGH specifies positive data polarity for the audio bus.

LOW specifies negative data polarity for the audio bus.

Examples `BUS:B1:AUDIO:DATA:POLARITY LOW` sets the data polarity to LOW.

`BUS:B1:AUDIO:DATA:POLARITY?` might return
`:BUS:B1:AUDIO:DATA:POLARITY HIGH` indicating that the data polarity
is set to HIGH.

BUS:B<x>:AUDio:DATa:SIZE

This command sets or queries the number of bits per channel for the specified audio bus. The bus is specified by x.

NOTE. *This command only applies to the TDM Audio type.*

Conditions Requires option SR-AUDIO.

Group Bus

Syntax	<code>BUS:B<x>:AUDio:DATA:SIZE <NR1></code> <code>BUS:B<x>:AUDio:DATA:SIZE?</code>
Arguments	<code>B<x></code> is the number of the bus. <code>NR1</code> specifies the number of bits per word.
Examples	<code>BUS:B1:AUDIO:DATA:SIZE 8</code> sets the number of bits per word to 8. <code>BUS:B1:AUDIO:DATA:SIZE?</code> might return <code>:BUS:B1:AUDIO:DATA:SIZE 24</code> indicating that the number of bits per word is set to 24.

BUS:B<x>:AUDio:DATA:SOUrce

This command sets or queries the audio data source for the specified audio bus. The bus is specified by `x`.

Conditions	Requires option SR-AUDIO.
Group	Bus
Syntax	<code>BUS:B<x>:AUDio:DATA:SOUrce</code> <code>{CH<x> CH<x>_D<x> MATH<x> REF<x> REF<x>_D<x>}</code> <code>BUS:B<x>:AUDio:DATA:SOUrce?</code>
Arguments	<code>B<x></code> is the number of the bus. <code>CH<x></code> specifies an analog channel as the data source waveform for the audio bus. <code>CH<x>_D<x></code> specifies a digital channel as the data source waveform for the audio bus. <code>MATH<x></code> specifies an math waveform as the data source waveform for the audio bus. <code>REF<x></code> specifies an reference waveform as the data source waveform for the audio bus. <code>REF<x>_D<x></code> specifies a digital reference waveform as the data source waveform for the specified audio bus.
Examples	<code>BUS:B1:AUDIO:DATA:SOURCE CH1_D1</code> sets the data source to D1 of FlexChannel 1.

BUS:B1:AUDIO:DATA:SOURCE? might return :BUS:B1:AUDIO:DATA:SOURCE CH3 indicating that the data source is set to CH3.

BUS:B<x>:AUDio:DATa:THReshold

This command sets or queries the audio data source threshold for the specified bus. The bus is specified by x.

Conditions Requires option SR-AUDIO.

Group Bus

Syntax BUS:B<x>:AUDio:DATa:THReshold <NR3>
BUS:B<x>:AUDio:DATa:THReshold?

Related Commands [BUS:B<x>:AUDio:DATa:SOUrce](#)

Arguments B<x> is the number of the bus.
<NR3> is the audio data source threshold for the specified bus.

Examples BUS:B3:AUDIO:DATA:THRESHOLD 1.5 sets the Bus 3 audio clock source threshold to 500.0 mV.
BUS:B2:AUDIO:DATA:THRESHOLD? might return
:BUS:B2:AUDIO:DATA:THRESHOLD 500.0000E-3 indicates the Bus 2 audio data source threshold is set to 500 mV.

BUS:B<x>:AUDio:DATa:WORDSize

This command sets or queries the audio bits per word for the specified bus. The bus is specified by x.

Group Bus

Syntax BUS:B<x>:AUDio:DATa:WORDSize <NR1>
BUS:B<x>:AUDio:DATa:WORDSize?

Arguments	B<x> is the number of the bus. <NR1> is the audio bits per word for the specified bus.
Examples	BUS:B1:Audio:DATA:WORDSize 24 sets the audio bits per word to 24 bits. BUS:B1:Audio:DATA:WORDSize? might return :BUS:B1:AUDIO:DATA:WORDSIZE 24 indicating the bits per word is 24.

BUS:B<x>:AUDIO:FRAME:CLOCKBITSPERCHANNEL

This command sets or queries the audio bits of sync width for the specified bus. The bus is specified by x.

NOTE. *This command only applies to the TDM Audio type.*

Conditions	Requires option SR-AUDIO.
Group	Bus
Syntax	BUS:B<x>:AUDIO:FRAME:CLOCKBITSPERCHANNEL <NR1> BUS:B<x>:AUDIO:FRAME:CLOCKBITSPERCHANNEL?
Arguments	B<x> is the number of the bus. <NR1> is the audio bits of sync width for the specified bus.
Examples	BUS:B1:Audio:FRAME:CLOCKBITSPERCHANNEL 32 sets the number of bits to 32. BUS:B1:Audio:FRAME:CLOCKBITSPERCHANNEL? might return :BUS:B1:AUDIO:FRAME:CLOCKBITSPERCHANNEL 24 indicating there are 24 bits of sync width for the bus.

BUS:B<x>:AUDIO:FRAME:SIZE

This command sets or queries the number of audio channels in each frame for the specified AUDIO bus. The bus is specified by x.

NOTE. *This command only applies to the TDM Audio type.*

Conditions	Requires option SR-AUDIO.
Group	Bus
Syntax	<code>BUS:B<x>:AUDIO:FRAME:SIZE <NR1></code> <code>BUS:B<x>:AUDIO:FRAME:SIZE?</code>
Arguments	<code>B<x></code> is the number of the bus. <code><NR1></code> specifies the number of channels in each frame.
Examples	<code>BUS:B1:AUDIO:FRAME:SIZE 2</code> sets the frame size to 2. <code>BUS:B1:AUDIO:FRAME:SIZE?</code> might return <code>:BUS:B1:AUDIO:FRAME:SIZE 8</code> indicating that the number of channels in each frame is set to 8.

BUS:B<x>:AUDIO:TYPE

This command sets or queries the audio format (type) for the specified audio bus. The bus is specified by x.

Conditions	Requires option SR-AUDIO.
Group	Bus
Syntax	<code>BUS:B<x>:AUDIO:TYPE {I2S LJ RJ TDM}</code> <code>BUS:B<x>:AUDIO:TYPE?</code>
Arguments	<code>B<x></code> is the number of the bus. <code>I2S</code> specifies the I2S audio format. <code>LJ</code> specifies the left-justified audio format. <code>RJ</code> specifies the right-justified audio format. <code>TDM</code> specifies the time-division multiplexing audio format.
Examples	<code>BUS:B1:AUDIO:TYPE RJ</code> sets right-justified as the audio format. <code>BUS:B1:AUDIO:TYPE?</code> might return <code>:BUS:B1:AUDIO:TYPE I2S</code> indicating that the audio format is set to I2S.

BUS:B<x>:AUDIO:WORDSel:POLarity

This command sets or queries the word select source polarity for the specified audio bus. The bus is specified by x.

Conditions Requires option SR-AUDIO.

Group Bus

Syntax BUS:B<x>:AUDIO:WORDSel:POLarity {NORMal|INVERTed}
BUS:B<x>:AUDIO:WORDSel:POLarity?

Arguments B<x> is the number of the bus.

NORMal specifies positive polarity.

INVERTed specifies negative polarity.

Examples BUS:B1:AUDIO:WORDSEL:POLARITY NORMal sets normal as the word select polarity.

BUS:B1:AUDIO:WORDSEL:POLARITY? might return
:BUS:B1:AUDIO:WORDSEL:POLARITY NORMAL indicating that the word select polarity is set to normal.

BUS:B<x>:AUDIO:WORDSel:SOURce

This command sets or queries the audio word select source waveform for the specified audio bus. The bus is specified by x.

Conditions Requires option SR-AUDIO.

Group Bus

Syntax BUS:B<x>:AUDIO:WORDSel:SOURce
{CH<x>|CH<x>_D<x>|MATH<x>|REF<x>|REF<x>_D<x>}
BUS:B<x>:AUDIO:WORDSel:SOURce?

Arguments B<x> is the number of the bus.

CH<x> specifies an analog channel as the word select source waveform.

CH<x>_D<x> specifies a digital channel as the word select source waveform.
 MATH<x> specifies an math waveform as the word select source waveform.
 REF<x> specifies an reference waveform as the word select source waveform.
 REF<x>_D<x> specifies a digital reference waveform as the word select source waveform for the specified audio bus.

Examples `BUS:B1:AUDIO:WORDSEL:SOURCE CH1` sets CH1 as the word select source.
 `BUS:B1:AUDIO:WORDSEL:SOURCE CH2_D3` sets D3 of FlexChannel 2 as the word select source.
 `BUS:B1:AUDIO:WORDSEL:SOURCE?` might return
 `:BUS:B1:AUDIO:WORDSEL:SOURCE CH2` indicating that the word select source is set to CH2.

BUS:B<x>:AUDio:WORDSel:THReshold

This command sets or queries the audio word select source threshold for the specified bus. The bus is specified by x.

Conditions Requires option SR-AUDIO.

Group Bus

Syntax `BUS:B<x>:AUDio:WORDSel:THReshold <NR3>`
 `BUS:B<x>:AUDio:WORDSel:THReshold?`

Arguments B<x> is the number of the bus.
 <NR3> is the audio word select source threshold for the specified bus.

Examples `BUS:B4:AUDIO:WORDSEL:THRESHOLD 300.0E-3` sets the Bus 4 audio word select source threshold to 300 mV.
 `BUS:B1:AUDIO:WORDSEL:THRESHOLD?` might return
 `:BUS:B1:AUDIO:WORDSEL:THRESHOLD 1.25` indicates the Bus 1 audio word select source threshold is set to 1.25 V.

BUS:B<x>:CAN:BITRate

This command sets or queries the CAN bit rate. The bus number is specified by x. If you select Custom, use [BUS:B<x>:CAN:BITRate:VALue](#) to set the bit rate.

Conditions Requires option SR-AUTO.

Group Bus

Syntax `BUS:B<x>:CAN:BITRate`
`{RATE10K|RATE100K|RATE1M|RATE125K|RATE153K|`
`RATE20K|RATE25K|RATE250K|RATE31K|RATE33K|RATE400K|RATE50K|`
`RATE500K|RATE62K| RATE68K|RATE800K|RATE83K|RATE92K|CUSTom}`
`BUS:B<x>:CAN:BITRate?`

Related Commands [BUS:B<x>:CAN:BITRate:VALue](#)

Arguments B<x> is the number of the bus.
 Arguments specify the bit rate.

Examples `BUS:B1:CAN:BITRate RATE1M` sets the CAN bit rate to 1 Mb.
`BUS:B1:CAN:BITRate?` might return `:BUS:B1:CAN:BITRATE RATE500K`, indicating the CAN bit rate is set to 500 k.

BUS:B<x>:CAN:BITRate:VALue

This command sets or queries CAN bit rate when Custom is selected by [BUS:B<x>:CAN:BITRate](#). The bus number is specified by x.

Conditions Requires option SR-AUTO.

Group Bus

Syntax `BUS:B<x>:CAN:BITRate:VALue <NR3>`
`BUS:B<x>:CAN:BITRate:VALue?`

Related Commands [BUS:B<x>:CAN:BITRate](#)

Arguments	B<x> is the number of the bus. <NR3> specifies the CAN bit rate.
Returns	<NR1> is the bit rate.
Examples	BUS:B1:CAN:BITRate:VALue 400.0E+3 sets the bit rate to 400 k. BUS:B1:CAN:BITRate:VALue? might return :BUS:B1:CAN:BITRATE:VALUE 500000, indicating that the bit rate is set to 500,000.

BUS:B<x>:CAN:FD:BITRate

This command sets or queries the increased data phase bit rate used by CAN FD packets on the specified CAN bus. The bus is specified by x. If you select Custom, use BUS:B<x>:CAN:FD:BITRate:CUSTom to set the bit rate.

Conditions	Requires option SR-AUTO.
Group	Bus
Syntax	BUS:B<x>:CAN:FD:BITRate {RATE1M RATE2M RATE3M RATE4M RATE5M RATE6M RATE7M RATE8M RATE9M RATE10M RATE11M RATE12M RATE13M RATE14M RATE15M RATE16M CUSTom} BUS:B<x>:CAN:FD:BITRate?

Related Commands [BUS:B<x>:CAN:FD:BITRate:CUSTom](#)

Arguments	B<x> is the number of the bus. Arguments specify the bit rate.
Examples	BUS:B1:CAN:BITRATE RATE1M sets the CAN FD bit rate to 1 Mbps. BUS:B1:CAN:BITRATE? might return :BUS:B1:CAN:BITRATE RATE4M, indicating that the CAN FD bit rate is 4 Mbps.

BUS:B<x>:CAN:FD:BITRate:CUSTom

This command sets or queries the custom bit rate for the increased data phase of CAN FD packets on the specified CAN bus. The bus is specified by x.

Conditions	Requires option SR-AUTO.
Group	Bus
Syntax	BUS:B<x>:CAN:FD:BITRate:CUSTom <NR1> BUS:B<x>:CAN:FD:BITRate:CUSTom?
Related Commands	BUS:B<x>:CAN:BITRate
Arguments	B<x> is the number of the bus. <NR1> is the custom FD bit rate for the specified bus.
Examples	BUS:B1:CAN:FD:BITRATE:CUSTOM 1000000 sets the bit rate to 1,000,000. BUS:B1:CAN:FD:BITRATE:CUSTOM? might return :BUS:B1:CAN:FD:BITRATE:CUSTOM 4000000, indicating the bit rate is set to 4,000,000.

BUS:B<x>:CAN:SAMPLEpoint

This command sets or queries the sample point for the specified CAN bus. The bus is specified by x.

Conditions	Requires option SR-AUTO.
Group	Bus
Syntax	BUS:B<x>:CAN:SAMPLEpoint <NR1> BUS:B<x>:CAN:SAMPLEpoint?
Arguments	B<x> is the number of the bus. <NR1> is the sample point, in percent, for the specified CAN bus.

Examples `BUS:B1:CAN:SAMPLEpoint 40` sets the sample point to 40%.

`BUS:B1:CAN:SAMPLEpoint?` might return `:BUS:B1:CAN:SAMPLEPOINT 50` indicating the sample point is at 50%.

BUS:B<x>:CAN:SIGNa1

This command sets or queries the signal type for the specified CAN bus. The bus is specified by x.

Conditions Requires option SR-AUTO.

Group Bus

Syntax `BUS:B<x>:CAN:SIGNa1 {DIFFerential|CANH|CANL|RX|TX}`
`BUS:B<x>:CAN:SIGNa1?`

Arguments B<x> is the number of the bus.
 Arguments are the CAN bus signal types.

Examples `BUS:B1:CAN:SINa1 CANL` sets the signal type to CANL.

`BUS:B1:CAN:SINa1?` might return `:BUS:B1:CAN:SIGNAL CANH` indicating the signal type is CANH.

BUS:B<x>:CAN:SOUrce

This command sets or queries the CAN source channel for the specified CAN bus. The bus number is specified by x.

Conditions Requires option SR-AUTO.

Group Bus

Syntax `BUS:B<x>:CAN:SOUrce`
`{CH<x>|CH<x>_D<x>|MATH<x>|REF<x>|REF<x>_D<x>}`
`BUS:B<x>:CAN:SOUrce?`

Arguments	B<x> is the number of the bus. Arguments specify a channel.
Examples	BUS:B1:CAN:SOURce MATH6 sets the source channel to MATH6. BUS:B1:CAN:SOURce? might return :BUS:B1:CAN:SOURCE CH1 , indicating that the CAN source channel is CH1.

BUS:B<x>:CAN:STANDARD

This command sets or queries which CAN standard specification to analyze the specified CAN bus with. The bus is specified by x.

Conditions	Requires option SR-AERO.
Group	Bus
Syntax	BUS:B<x>:CAN:STANDARD {CAN2X FDISO FDNONISO} BUS:B<x>:CAN:STANDARD?
Arguments	B<x> is the number of the bus. CAN2X sets the CAN bus standard to CAN 2.0. FDISO sets the CAN bus standard to ISO CAN FD (11898-1:2015). FDNONISO sets the CAN bus standard to non-ISO CAN FD (Bosch:2012).
Examples	BUS:B1:CAN:STANDARD FDISO sets the CAN standard to ISO CAN FD. BUS:B1:CAN:STANDARD? might return :BUS:B1:CAN:STANDARD CAN2X , indicating that the CAN standard is CAN 2.0.

BUS:B<x>:CAN:THRESHOLD

This command sets or queries the source channel threshold for the specified CAN bus. The bus is specified by x.

Conditions	Requires option SR-AUTO.
Group	Bus

Syntax	<code>BUS:B<x>:CAN:THReshold <NR3></code> <code>BUS:B<x>:CAN:THReshold?</code>
Arguments	<code>B<x></code> is the number of the bus. <code><NR3></code> is the source channel threshold for the specified CAN bus.
Examples	<code>BUS:B1:CAN:THReshold 5</code> sets the threshold to 5 V. <code>BUS:B1:CAN:THReshold?</code> might return <code>:BUS:B1:CAN:THRESHOLD 0.0E+0</code> indicating the threshold is set to 0.0 V.

BUS:B<x>:DISplay:FORMat

This command sets or queries how the data is represented in the bus form for the specified bus. The bus is specified by x.

Group	Bus
Syntax	<code>BUS:B<x>:DISplay:FORMat</code> <code>{HEX BINARY MIXEDASCII MIXEDHEX ASCII DECIMAL MIXED}</code> <code>BUS:B<x>:DISplay:FORMat?</code>
Arguments	<code>B<x></code> is the number of the bus.
NOTE. <i>Different bus types support only a subset of these arguments.</i>	
HEX specifies hexadecimal format. BINARY specifies binary format. MIXEDASCII specifies mixed ASCII format. MIXEDHEX specifies mixed hexadecimal format. ASCII specifies ASCII format. DECIMAL specifies decimal format. MIXED specifies mixed format.	
Examples	<code>BUS:B1:DISplay:FORMat BINARY</code> sets the format to binary. <code>BUS:B1:DISplay:FORMat?</code> might return <code>:BUS:B1:DISPLAY:FORMAT HEX</code> indicating the format is set to hexadecimal.

BUS:B<x>:DISPlay:LAYout

This command sets or queries what to display for the specified bus. The bus is specified by x.

Group	Bus
Syntax	<code>BUS:B<x>:DISPlay:LAYout {BUS BUSANDWAVEFORM}</code> <code>BUS:B<x>:DISPlay:LAYout?</code>
Arguments	<code>B<x></code> is the number of the bus. BUS specifies displaying the bus form only. BUSANDWAVEFORM specifies displaying the bus form and the constituent source waveform(s). This argument is not available for some bus types and some bus configurations
Examples	<code>BUS:B1:DISPlay:LAYout</code> BUS specifies displaying the bus form only. <code>BUS:B1:DISPlay:LAYout?</code> might return <code>:BUS:B1:DISPLAY:LAYOUT BUS</code> indicating the bus will be displayed without displaying its constituent source waveform(s).

BUS:B<x>:ETHERnet:DATAMINUSTHRESHold

This command sets or queries the Ethernet D- source threshold for the specified bus. This threshold only applies when the Ethernet signal type is single ended. The bus is specified by x.

Conditions	Requires option SR-ENET.
Group	Bus
Syntax	<code>BUS:B<x>:ETHERnet:DATAMINUSTHRESHold <NR3></code> <code>BUS:B<x>:ETHERnet:DATAMINUSTHRESHold?</code>
Related Commands	BUS:B<x>:ETHERNET:SOURce:DMINus BUS:B<x>:ETHERNET:SOURce:DPLUs BUS:B<x>:ETHERnet:SIGNALTYpe

Arguments	B<x> is the number of the bus. <NR3> is the Ethernet D- source threshold for the specified bus.
Examples	BUS:B3:ETHERNET:DATAMINUSTHRESHOLD 250.0E-3 sets the Bus 3 Ethernet DATA Minus source threshold to 250.0000 mV. BUS:B2:ETHERNET:DATAMINUSTHRESHOLD? might return BUS:B2:ETHERNET:DATAMINUSTHRESHOLD 1.0 indicates the Bus 2 Ethernet D- source threshold is set to 1.0 V.

BUS:B<x>:ETHERnet:DATAPLUSTHRESHold

This command sets or queries the Ethernet D+ source threshold for the specified bus. This threshold only applies when the Ethernet signal type is single ended. The bus is specified by x.

Conditions Requires option SR-ENET.

Group Bus

Syntax **BUS:B<x>:ETHERnet:DATAPLUSTHRESHold <NR3>**
BUS:B<x>:ETHERnet:DATAPLUSTHRESHold?

Related Commands B<x> is the number of the bus.

[BUS:B<x>:ETHERNET:SOUrce:DMINus](#)

[BUS:B<x>:ETHERNET:SOUrce:DPLUs](#)

[BUS:B<x>:ETHERnet:SIGNALTYpe](#)

Arguments <NR3> is the Ethernet D+ source threshold for the specified bus.

Examples **BUS:B3:ETHERNET:DATAPLUSTHRESHOLD 250.0E-3** sets the Bus 3 Ethernet D+ source threshold to 250.0000 mV.

BUS:B2:ETHERNET:DATAMINUSTHRESHOLD? might return
:BUS:B2:ETHERNET:DATAPLUSTHRESHOLD 1.0 indicating the Bus 2 Ethernet D+ source threshold is set to 1.0 V.

BUS:B<x>:ETHERnet:IPVFOUR

This command sets or queries whether IPV4 packets are available for triggering on Ethernet. The bus is specified by x.

Conditions Requires option SR-ENET.

Group Bus

Syntax BUS:B<x>:ETHERnet:IPVFOUR {YES|NO}
BUS:B<x>:ETHERnet:IPVFOUR?

Arguments B<x> is the number of the bus.

YES specifies that IPV4 packets are available.

NO specifies that IPV4 packets are not available.

Examples BUS:B1:ETHERNET:IPVFOUR NO will specify that Bus 1 does not have IPV4 packets available.

BUS:B3:ETHERNET:IPVFOUR? might return :BUS:B3:ETHERNET:IPVFOUR YES indicating that Bus 3 has IPV4 packets available.

BUS:B<x>:ETHERnet:LOWTHRESHold

This command sets or queries the Ethernet source Low threshold for the specified bus. This threshold only applies when the Ethernet signal type is differential. The bus is specified by x.

Conditions Requires option SR-ENET.

Group Bus

Syntax BUS:B<x>:ETHERnet:LOWTHRESHold <NR3>
BUS:B<x>:ETHERnet:LOWTHRESHold?

Related Commands [BUS:B<x>:ETHERnet:SIGNALTYpe](#)

Arguments	B<x> is the number of the bus. <NR3> is the Ethernet source Low threshold for the specified bus.
Examples	BUS:B1:ETHERnet:LOWTHRESHold -200e-3 sets the threshold to -200 mV. BUS:B1:ETHERnet:LOWTHRESHold? might return :BUS:B1:ETHERNET:LOWTHRESHOLD -500.0000E-3 indicating the threshold is set to -500 mV.

BUS:B<x>:ETHERnet:QTAGGING

This command sets or queries whether Q-Tagging packets are available for triggering on Ethernet. The bus is specified by x.

Conditions	Requires option SR-ENET.
Group	Bus
Syntax	BUS:B<x>:ETHERnet:QTAGGING {YES NO} BUS:B<x>:ETHERnet:QTAGGING?
Arguments	B<x> is the number of the bus. YES specifies that Q-Tagging packets are available. NO specifies that Q-Tagging packets are not available.
Examples	BUS:B1:ETHERnet:QTAGGING YES turns on Q-Tagging. BUS:B1:ETHERnet:QTAGGING? might return :BUS:B1:ETHERNET:QTAGGING YES indicating that Q-tagging packets are available.

BUS:B<x>:ETHERnet:SIGNALType

This command sets or queries the Ethernet signal type for the specified bus. The bus is specified by x.

Conditions	Requires option SR-ENET.
Group	Bus

Syntax `BUS:B<x>:ETHERnet:SIGNALType {SINGLE|DIFF}`
`BUS:B<x>:ETHERnet:SIGNALType?`

Arguments `B<x>` is the number of the bus.

`SINGLE` specifies single-ended signals.

`DIFF` specifies differential signals.

Examples `BUS:B1:ETHERnet:SIGNALType SINGLE` specifies single-ended signals.

`BUS:B1:ETHERnet:SIGNALType?` might return
`:BUS:B1:ETHERnet:SIGNALType DIFF` indicating differential signals are specified.

BUS:B<x>:ETHERnet:SOURce

This command sets or queries the Ethernet data (SDATA) source for the specified bus. This command controls the source channel when the signal type is differential. The bus number is specified by `<x>`.

Conditions Requires option SR-ENET.

Group Bus

Syntax `BUS:B<x>:ETHERnet:SOURce {CH<x>|MATH<x>|REF<x>}`

`BUS:B<x>:ETHERnet:SOURce?`

Related Commands [BUS:B<x>:ETHERnet:THRESHold](#)

Arguments `B<x>` is the number of the bus.

`CH<x>` specifies to use one of the analog channels as the Ethernet data source for differential input.

`MATH<x>` specifies to use a math waveform as the source for Ethernet data differential input

`REF<x>` specifies to use one of the reference waveforms as the Ethernet data source for differential input.

Examples `BUS:B1:ETHERNET:SOURCE CH4` specifies to use the channel 4 waveform as the source for Ethernet data.

`BUS:B1:ETHERNET:SOURCE?` might return CH2, indicating that channel 2 is the currently specified source for Ethernet data.

BUS:B<x>:ETHERNET:SOURce:DMINus

This command sets or queries the Ethernet D- source for the specified bus. this command specifies the source channel to use when the signal type is single ended. The bus is specified by x.

Group Bus

Syntax `BUS:B<x>:ETHERNET:SOURce:DMINus {CH<x>|MATH<x>|REF<x>}`
`BUS:B<x>:ETHERNET:SOURce:DMINus?`

Arguments B<x> is the number of the bus.

CH<x>, MATH<x> or REF<x> set the D- source to the specified signal source.

Examples `BUS:B1:ETHERNET:SOURce:DMINus CH1` sets the D- source to channel 1.

`BUS:B1:ETHERNET:SOURce:DMINus?` might return
`:BUS:B1:ETHERNET:SOURce:DMINus CH2` indicating the D- source is set to channel 2.

BUS:B<x>:ETHERNET:SOURce:DPLUs

This command sets or queries the Ethernet D+ source for the specified bus. this command specifies the source channel to use when the signal type is single ended. The bus is specified by x.

Group Bus

Syntax `BUS:B<x>:ETHERNET:SOURce:DPLUs {CH<x>|MATH<x>|REF<x>}`
`BUS:B<x>:ETHERNET:SOURce:DPLUs?`

Arguments B<x> is the number of the bus.

CH<x>, MATH<x> or REF<x> set the D+ source to the specified signal source.

Examples `BUS:B1:ETHERNET:SOURce:DPLUS Ch5` sets the D+ source to channel 5.

`BUS:B1:ETHERNET:SOURce:DPLUS?` might return
`:BUS:B1:ETHERNET:SOURCE:DPLUS CH5` indicating the D+ source is set to channel 5.

BUS:B<x>:ETHERnet:THRESHold

This command sets or queries the Ethernet DATA source High threshold for the specified bus. The bus is specified by x.

Conditions Requires option SR-ENET.

Group Bus

Syntax `BUS:B<x>:ETHERnet:THRESHold <NR3>`
`BUS:B<x>:ETHERnet:THRESHold?`

Related Commands [BUS:B<x>:ETHERnet:SOURce](#)

Arguments `B<x>` is the number of the bus.
`<NR3>` is the Ethernet DATA source High threshold for the specified bus.

Examples `BUS:B4:ETHERNET:THRESHOLD 1.0` sets the Bus 4 Ethernet DATA source High threshold to 1.0 V.

`BUS:B3:ETHERNET:THRESHOLD?` might return
`BUS:B3:ETHERNET:THRESHOLD 225.00000E-3` indicates the Bus 3 Ethernet DATA source High threshold is set to 225.0 mV.

BUS:B<x>:ETHERnet:TYPE

This command specifies the Ethernet standard speed. The bus number is specified by <x>.

Conditions Requires option SR-ENET.

Group Bus

Syntax	<code>BUS:B<x>:ETHERnet:TYPE {TENBASET HUNDREDBASETX}</code> <code>BUS:B<x>:ETHERnet:TYPE?</code>
Arguments	<code>B<x></code> is the number of the bus. TENBASET specifies the Ethernet speed as 10Base-T. HUNDREDBASETX specifies the Ethernet speed as 100Base-T.
Examples	<code>BUS:B1:ETHERNET:TYPE HUNDREDBASETX</code> specifies the Ethernet speed as 100Base-T. <code>BUS:B1:ETHERNET:TYPE?</code> might return TENBASET, indicating that 10Base-T is the currently specified Ethernet speed.

BUS:B<x>:FLEXray:BITRate

This command sets or queries the FlexRay bus bit rate. The bus is specified by x. If you select Custom, use [BUS:B<x>:FLEXray:BITRate:CUSTom](#) to set the bit rate.

Conditions	Requires option SR-AUTO.
Group	Bus
Syntax	<code>BUS:B<x>:FLEXray:BITRate {CUSTOM RATE2M RATE5M RATE10M}</code> <code>BUS:B<x>:FLEXray:BITRate?</code>
Arguments	<code>B<x></code> is the number of the bus. Arguments specify the bit rate.
Examples	<code>BUS:B1:FLEXRAY:BITRate RATE2M</code> sets the bit rate to 2 Mb. <code>BUS:B1:FLEXRAY:BITRate?</code> might return <code>:BUS:B1:FLEXRAY:BITRATE RATE10M</code> , indicating the bit rate is 10 Mb.

BUS:B<x>:FLEXray:BITRate:CUSTom

This command sets or queries the FlexRay custom bit rate for the specified bus. The bus is specified by x.

Conditions	Requires option SR-AUTO.
Group	Bus
Syntax	<code>BUS:B<x>:FLEXray:BITRate:CUSTom <NR1></code> <code>BUS:B<x>:FLEXray:BITRate:CUSTom?</code>
Arguments	<code>B<x></code> is the number of the bus. <code><NR1></code> is the FlexRay custom bit rate for the specified bus.
Examples	<code>BUS:B1:FLEXray:BITRate:CUSTom 10000000</code> sets the bit rate to 10,000,000. <code>BUS:B1:FLEXray:BITRate:CUSTom?</code> might return <code>:BUS:B1:FLEXRAY:BITRATE:CUSTOM 10000000</code> indicating the bit rate is set to 10,000,000.

BUS:B<x>:FLEXray:CHannel

This command sets or queries the FlexRay channel type for the specified bus. The bus number is specified by x.

Conditions	Requires option SR-AUTO.
Group	Bus
Syntax	<code>BUS:B<x>:FLEXray:CHanne1 {A B}</code> <code>BUS:B<x>:FLEXray:CHanne1?</code>
Arguments	<code>B<x></code> is the number of the bus. <code>A</code> specifies the A channel. <code>B</code> specifies the B channel.
Examples	<code>BUS:B1:FLEXRAY:CHANNEL B</code> sets the FlexRay channel to B. <code>BUS:B1:FLEXRAY:CHANNEL?</code> might return <code>:BUS:B1:FLEXRAY:CHANNEL A</code> , indicating the channel is set to A.

BUS:B<x>:FLEXray:LOWTHRESHold

This command sets or queries the FlexRay data source low threshold for the specified bus. The bus is specified by x.

Conditions Requires option SR-AUTO.

Group Bus

Syntax BUS:B<x>:FLEXray:LOWTHRESHold <NR3>
BUS:B<x>:FLEXray:LOWTHRESHold?

Arguments <NR3> is the FlexRay data source low threshold for the specified bus.
B<x> is the number of the bus.

Examples BUS:B1:FLEXray:LOWTHRESHold 50.0e-3 sets the threshold to 50 mV.
BUS:B1:FLEXray:LOWTHRESHold? might
:BUS:B1:FLEXray:LOWTHRESHold 0.0E+0 indicating the threshold is
set to 0.0 V.

BUS:B<x>:FLEXray:SIGnal

This command sets or queries the FlexRay signal type for the specified bus. The bus number is specified by x.

Conditions Requires option SR-AUTO.

Group Bus

Syntax BUS:B<x>:FLEXray:SIGnal {BDIFFBP|BM|TXRX}
BUS:B<x>:FLEXray:SIGnal?

Arguments B<x> is the number of the bus.
BDIFFBP sets the FlexRay signal type to BDIFFBP.
BM sets the FlexRay signal type to BM.
TXRX sets the FlexRay signal type to TXRX.

Examples `BUS:B1:FLEXRAY:SIGNAL BM` sets the FlexRay channel type to BM.
 `BUS:B1:FLEXRAY:SIGNAL?` might return `:BUS:B1:FLEXRAY:SIGNAL BDIFFBP`, indicating the FlexRay channel type is set to BDIFFBP.

BUS:B<x>:FLEXray:SOUrce

This command sets or queries the Flexray bus data source for the specified bus when the signal type is BDIFFBP or BM. The bus number is specified by x.

Conditions Requires option SR-AUTO.

Group Bus

Syntax `BUS:B<x>:FLEXray:SOUrce {CH<x>|MATH<x>|REF<x>}`
 `BUS:B<x>:FLEXray:SOUrce?`

Arguments B<x> is the number of the bus.
 Arguments are the available FlexRay sources.

Examples `BUS:B1:FLEXRAY:SOUrce MATH4` sets the FlexRay source to Math 4.
 `BUS:B1:FLEXRAY:SOUrce?` might return `:BUS:B1:FLEXRAY:SOUrce CH1`, indicating the source is channel 1.

BUS:B<x>:FLEXray:SOUrce:TXRX

This command sets or queries the FlexRay TxRx data source for the specified bus when the signal type is TXRX. The bus is specified by x.

Conditions Requires option SR-AUTO.

Group Bus

Syntax `BUS:B<x>:FLEXray:SOUrce:TXRX`
 `{CH<x>|CH<x>_D<x>|MATH<x>|REF<x>|REF<x>_D<x>}`

Arguments Arguments are the available sources.
 B<x> is the number of the bus.

Examples `BUS:B1:FLEXray:SOURce:TXRX CH1` sets the TXRX source the channel 1.
`BUS:B1:FLEXray:SOURce:TXRX?` might return
`:BUS:B1:FLEXRAY:SOURCE:TXRX CH1_D0` indicating the TXRX source is set to CH1_D0.

BUS:B<x>:FLEXray:THRESHold

This command sets or queries the FlexRay data source high threshold for the specified bus. The bus is specified by x.

Conditions Requires option SR-AUTO.

Group Bus

Syntax `BUS:B<x>:FLEXray:THRESHold <NR3>`
`BUS:B<x>:FLEXray:THRESHold?`

Arguments `B<x>` is the number of the bus.
`<NR3>` is the FlexRay data source high threshold for the specified bus.

Examples `BUS:B1:FLEXray:THRESHold 50.0-3` sets the high threshold to 50 mV.
`BUS:B1:FLEXray:THRESHold?` might return `:BUS:B1:FLEXRAY:THRESHOLD 0.0E+0` indicating the high threshold is set to 0.0 V.

BUS:B<x>:FLEXray:TXRXTHRESHold

This command sets or queries the FlexRay data source TxRx threshold for the specified bus. The bus is specified by x.

Conditions Requires option 5-SREMBD or SUP5-SREMBD.

Group Bus

Syntax `BUS:B<x>:FLEXray:TXRXTHRESHold <NR3>`
`BUS:B<x>:FLEXray:TXRXTHRESHold?`

Arguments	B<x> is the number of the bus. <NR3> is the TxRx threshold.
Examples	BUS:B1:FLEXray:TXRXTHRESHo1d 50.0e-3 sets the threshold to 50 mV. BUS:B1:FLEXray:TXRXTHRESHo1d? might return :BUS:B1:FLEXRAY:TXRXTHRESHOLD 0.0E+0 indicating the threshold is set to 0.0 V.

BUS:B<x>:I2C:CLOCK:SOURce

This command sets or queries the I2C clock (SCLK) source for the specified bus. The bus is specified by x.

Conditions	Requires option 5-SREMBD or SUP5-SREMBD.
Group	Bus
Syntax	BUS:B<x>:I2C:CLOCK:SOURce {CH<x> CH<x>_D<x> MATH<x> REF<x> REF<x>_D<x>} BUS:B<x>:I2C:CLOCK:Source?
Related Commands	BUS:B<x>:I2C:DATA:SOURce BUS:B<x>:I2C:RWINADDR
Arguments	B<x> is the number of the bus. CH<x> specifies an analog channel to use as the I2C SCLK source. CH<x>_D<x> specifies a digital channel to use as the I2C SCLK source. MATH<x> specifies a math waveform to use as the I2C SCLK source. REF<x> specifies a reference waveform to use as the I2C SCLK source. REF<x>_D<x> specifies a digital reference waveform as the clock source waveform for the specified I2C bus.
Examples	BUS:B1:I2C:CLOCK:SOURCE CH1 sets the I2C SCLK source to CH1. BUS:B1:I2C:CLOCK:SOURCE? might return :BUS:B1:I2C:CLOCK:SOURCE CH4_D5, indicating that the I2C SCLK source is set to D5 of FlexChannel 4.

BUS:B<x>:I2C:CLOCK:THReshold

This command sets or queries the I2C Clock (SCLK) source threshold for the specified bus. The bus is specified by x.

Conditions Requires option 5-SREMBD or SUP5-SREMBD.

Group Bus

Syntax BUS:B<x>:I2C:CLOCK:THReshold <NR3>
BUS:B<x>:I2C:CLOCK:THReshold?

Arguments B<x> is the number of the bus.
<NR3> is the I2C Clock (SCLK) source threshold for the specified bus.

Examples BUS:B1:I2C:CLOCK:THReshold 50.0e-3 sets the threshold to 50 mV.
BUS:B1:I2C:CLOCK:THReshold? might return
:BUS:B1:I2C:CLOCK:THRESHOLD 0.0E+0 indicating the threshold is set to 0 V.

BUS:B<x>:I2C:DATa:SOUrce

This command sets or queries the I2C data (SDA) source for the specified I2C bus. The bus is specified by x.

Conditions Requires option 5-SREMBD or SUP5-SREMBD.

Group Bus

Syntax BUS:B<x>:I2C:DATa:SOUrce
{CH<x> | CH<x>_D<x> | MATH<x> | REF<x> | REF<x>_D<x>}
BUS:B<x>:I2C:DATa:SOUrce?

Related Commands [BUS:B<x>:I2C:CLOCK:SOUrce](#)
[BUS:B<x>:I2C:RWINADDR](#)

Arguments CH<x> specifies an analog channel to use as the I2C SDA source.
 CH<x>_D<x> specifies a digital channel to use as the I2C SDA source.
 MATH<x> specifies a math waveform to use as the I2C SDA source.
 REF<x> specifies a reference waveform to use as the I2C SDA source.
 REF<x>_D<x> specifies a digital reference waveform as the data source waveform for the specified I2C bus.

Examples BUS:B1:I2C:DATA:SOURCE CH1_D5 sets the I2C SDA source to CH1_D5.
 BUS:B1:I2C:DATA:SOURCE? might return :BUS:B1:I2C:DATA:SOURCE MATH1, indicating that the I2C SDA source is set to MATH1.

BUS:B<x>:I2C:DATA:THReshold

This command sets or queries the I2C Data (SDA) source threshold for the specified bus. The bus is specified by x.

Conditions Requires option 5-SREMBD or SUP5-SREMBD.

Group Bus

Syntax BUS:B<x>:I2C:DATA:THReshold <NR3>
 BUS:B<x>:I2C:DATA:THReshold?

Arguments B<x> is the number of the bus.
 <NR3> is the I2C Data (SDA) source threshold for the specified bus.

Examples BUS:B1:I2C:DATA:THReshold 50.0e-3 sets the threshold to 50 mV.
 BUS:B1:I2C:DATA:THReshold? might return :BUS:B1:I2C:DATA:THRESHOLD 0.0E+0 indicating the threshold is set to 0 V.

BUS:B<x>:I2C:RWINADDR

This command sets or queries the manner in which seven-bit I2C addresses are represented in the busform display of the specified bus. The bus is specified by x.

Conditions Requires option 5-SREMBD or SUP5-SREMBD.

Group	Bus
Syntax	<code>BUS:B<x>:I2C:RWINADDR {0 1}</code> <code>BUS:B<x>:I2C:RWINADDR?</code>
Related Commands	BUS:B<x>:I2C:CLOCK:SOURce BUS:B<x>:I2C:DATA:SOURce
Arguments	<code>B<x></code> is the number of the bus. 0 displays seven-bit slave addresses as integers in the range of 0 to 127, with the state of the R/W* bit from the LSB of the slave address byte. For example, the slave address byte of 0b10100101 is displayed as the value 0x52 R. 1 displays the entire slave address byte as a number, with the R/W* signal as its LSB (bit 0) and the slave address in bits 7..1. For example, the slave address byte of 0b10100101 is displayed as the value 0xA5 R.
Examples	<code>BUS:B1:I2C:RWINADDR 0</code> displays seven-bit slave addresses as integers in the range of 0 to 127. <code>BUS:B1:I2C:RWINADDR?</code> might return <code>:BUS:B1:I2C:RWINADDR 1</code>, indicating that the entire slave address byte is displayed as a number, with the R/W* signal as its LSB (bit 0) and the slave address in bits 7..1.

BUS:B<x>:LABEL:COLor

This command sets or queries the color of the specified bus label. The bus is specified by x.

Group	Bus
Syntax	<code>BUS:B<x>:LABEL:COLor <QString></code> <code>BUS:B<x>:LABEL:COLor?</code>
Arguments	<code>B<x></code> is the number of the bus. <code><QString></code> is the bus label color. To return the color to the default color, send an empty string as in this example: <code>:BUS:B1:LABEL:COLOR ""</code>.

Examples `BUS:B1:LABe1:COLoR "#FF0000"` sets the label color to red.
`BUS:B1:LABe1:COLoR?` might return `:BUS:B1:LABEL:COLOR "#FF0000"` indicating the color is red.

BUS:B<x>:LABe1:FONT:BOLD

This command sets or queries the bold state of the specified bus label. The bus is specified by x.

Group Bus

Syntax `BUS:B<x>:LABe1:FONT:BOLD {ON|OFF|1|0}`
`BUS:B<x>:LABe1:FONT:BOLD?`

Arguments `B<x>` is the number of the bus.
`ON` displays the label in bold font.
`OFF` does not display the label in bold font.
`1` displays the label in bold font.
`0` does not display the label in bold font.

Examples `BUS:B1:LABe1:FONT:BOLD OFF` turns off bold font.
`BUS:B1:LABe1:FONT:BOLD?` might return `:BUS:B1:LABEL:FONT:BOLD 1` indicating a bold font.

BUS:B<x>:LABe1:FONT:ITALic

This command sets or queries the italic state of the specified bus label. The bus is specified by x.

Group Bus

Syntax `BUS:B<x>:LABe1:FONT:ITALic {ON|OFF|1|0}`
`BUS:B<x>:LABe1:FONT:ITALic?`

Arguments `B<x>` is the number of the bus.
`ON` displays the label in italic font.

OFF does not display the label in italic font.

1 displays the label in italic font.

0 does not display the label in italic font.

Examples `BUS:B1LABe1:FONT:ITALiC OFF` turns off italic font.
`BUS:B1LABe1:FONT:ITALiC?` might return `:BUS:B1:LABEL:FONT:ITALIC 1` indicating the font is italic.

BUS:B<x>:LABe1:FONT:SIZE

This command sets or queries the font size of the specified bus label. The bus is specified by x.

Group Bus

Syntax `BUS:B<x>:LABe1:FONT:SIZE <NR1>`
`BUS:B<x>:LABe1:FONT:SIZE?`

Arguments `B<x>` is the number of the bus.
`<NR1>` is the font size.

Examples `BUS:B1:LABe1:FONT:SIZE 10` sets the font size to 10 points.
`BUS:B1:LABe1:FONT:SIZE?` might return `:BUS:B1:LABEL:FONT:SIZE 20` indicating the font size is 20 points.

BUS:B<x>:LABe1:FONT:TYPE

This command sets or queries the font type of the specified bus label, such as Arial or Times New Roman. The bus is specified by x.

Group Bus

Syntax `BUS:B<x>:LABe1:FONT:TYPE <QString>`
`BUS:B<x>:LABe1:FONT:TYPE?`

Arguments	B<x> is the number of the bus. <QString> is the specified font type. Available fonts include: DejaVu Sans, DejaVu Sans Mono, DejaVu Serif, Frutiger LT Std, Monospace, Sans Serif, Serif, Ubuntu, Ubuntu Condensed, and Ubuntu Mono.
Examples	BUS:B1:LABe1:FONT:TYPE Monospace selects a monospace font. BUS:B1:LABe1:FONT:TYPE? might return :BUS:B1:LABEL:FONT:TYPE "Frutiger LT Std 55 Roman" indicating the font type is Frutiger LT Std.

BUS:B<x>:LABe1:FONT:UNDERline

This command sets or queries the underline state of the specified bus label. The bus is specified by x.

Group	Bus
Syntax	BUS:B<x>:LABe1:FONT:UNDERline {ON OFF 1 0} BUS:B<x>:LABe1:FONT:UNDERline?
Arguments	B<x> is the number of the bus. ON displays the label in underlined font. OFF does not display the label in underlined font. 1 displays the label in underlined font. 0 does not display the label in underlined font.
Examples	BUS:B:LABe1:FONT:UNDERline ON turns on underline font. BUS:B:LABe1:FONT:UNDERline? might return :BUS:B1:LABEL:FONT:UNDERLINE 0 indicating underline is off.

BUS:B<x>:LABe1:name

This command sets or queries the label for the specified bus. The bus is specified by x.

Group	Bus
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Syntax `BUS:B<x>:LABel:name <QString>`
 `BUS:B<x>:LABel:name?`

Related Commands [BUS:B<x>:TYPe](#)

Arguments `B<x>` is the number of the bus.

 `<QString>` is an alphanumeric string of text enclosed in quotes. The text string is limited to 30 characters. It contains the text label information for the bus.

Examples `BUS:B1:LABEL:NAME "TEST"` sets the waveform label for the bus B1 to Test.

 `BUS:B1:LABEL:NAME?` might return `:BUS:B1:LABEL:NAME "BUS 1"`, indicating that the waveform label for the bus B1 is set to "Bus 1".

BUS:B<x>:LABel:XPOS

This command sets or queries the x-position of the specified bus label. The bus is specified by x.

Group Bus

Syntax `BUS:B<x>:LABel:XPOS <NR3>`
 `BUS:B<x>:LABel:XPOS?`

Arguments `B<x>` is the number of the bus.

 `<NR3>` is the x-position, in pixels relative to the left edge of the screen of the specified bus label.

Examples `BUS:B1:LABel:XPOS 90` set the x position to 90.

 `BUS:B1:LABel:XPOS?` might return `:BUS:B1:LABEL:XPOS 45.0000` indicating the x position is 45.0 pixels to the right of the left edge of the display.

BUS:B<x>:LABel:YPOS

This command sets or queries the y-position of the specified bus label. The bus is specified by x.

Group Bus

Syntax	<code>BUS:B<x>:LABe1:YPOS <NR3></code> <code>BUS:B<x>:LABe1:YPOS?</code>
Arguments	<code>B<x></code> is the number of the bus. <code><NR3></code> is the y-position, in pixels relative to the baseline of the waveform, of the specified bus label.
Examples	<code>BUS:B1:LABe1:YPOS 1.0e1</code> set the y position to 10.0. <code>BUS:B1:LABe1:YPOS?</code> might return <code>:BUS:B1:LABEL:YPOS 0.0E+0</code> indicating the y position is 0.0 pixels from the baseline of the waveform.

BUS:B<x>:LIN:BITRate

This command sets or queries the LIN bus bit rate. The bus number is specified by x. If you select Custom, use [BUS:B<x>:LIN:BITRate:CUSTom](#) to set the bit rate.

Conditions	Requires option SR-AUTO.
Group	Bus
Syntax	<code>BUS:B<x>:LIN:BITRate</code> <code>{RATE10K RATE1K RATE19K RATE2K RATE4K RATE9K CUSTom}</code> <code>BUS:B<x>:LIN:BITRate?</code>
Related Commands	BUS:B<x>:LIN:BITRate:CUSTom
Arguments	<code>B<x></code> is the number of the bus. Arguments are the available bit rates.
Examples	<code>BUS:B1:LIN:BITRate Rate4k</code> sets the bit rate to 4 kb. <code>BUS:B1:LIN:BITRate?</code> might return <code>:BUS:B1:LIN:BITRATE RATE19K</code> , indicating that the bit rate is set to 19 kb.

BUS:B<x>:LIN:BITRate:CUSTom

This command sets or queries LIN custom bit rate for the specified bus. The bus is specified by x.

Conditions	Requires option SR-AUTO.
Group	Bus
Syntax	<code>BUS:B<x>:LIN:BITRate:CUSTom <NR1></code> <code>BUS:B<x>:LIN:BITRate:CUSTom?</code>
Related Commands	BUS:B<x>:LIN:BITRate
Arguments	<code>B<x></code> is the number of the bus. <code><NR1></code> is the LIN custom bit rate for the specified bus.
Examples	<code>BUS:B1:LIN:BITRate:CUSTom 9000</code> sets the bit rate to 9,000. <code>BUS:B1:LIN:BITRate:CUSTom?</code> might return <code>:BUS:B1:LIN:BITRate:CUSTom 10000</code> indicating the bit rate is set to 10,000.

BUS:B<x>:LIN:IDFORmat

This command sets or queries LIN bus identifier format for the specified bus. The bus number is specified by x.

Conditions	Requires option SR-AUTO.
Group	Bus
Syntax	<code>BUS:B<x>:LIN:IDFORmat {NOPARity PARity}</code> <code>BUS:B<x>:LIN:IDFORmat?</code>
Arguments	<code>B<x></code> is the number of the bus. <code>NOPARity</code> specifies an id format that includes parity. <code>PARity</code> specifies an id format that separates parity.
Examples	<code>BUS:B1:LIN:IDFORmat PARITY</code> set the id format to parity. <code>BUS:B1:LIN:IDFORmat?</code> might return <code>:BUS:B1:LIN:IDFORMAT NOPARITY</code> , indicating the id format is set to no parity.

BUS:B<x>:LIN:POLarity

This command sets or queries the LIN source polarity for the specified bus. The bus number is specified by x.

Conditions Requires option SR-AUTO.

Group Bus

Syntax BUS:B<x>:LIN:POLarity {INVerted|NORmal}
BUS:B<x>:LIN:POLarity?

Arguments B<x> is the number of the bus.

INVerted specifies inverted polarity.

NORmal specifies normal polarity.

Examples BUS:B1:LIN:POLarity Inverted sets the polarity to inverted.

BUS:B1:LIN:POLarity? might return :BUS:B1:LIN:POLARITY NORMAL,
indicating the bus polarity is set to normal.

BUS:B<x>:LIN:SAMPLEpoint

Specifies the LIN sample point, for the specified LIN bus. The bus is specified by x.

Conditions Requires option SR-AUTO.

Group Bus

Syntax BUS:B<x>:LIN:SAMPLEpoint <NR1>
BUS:B<x>:LIN:SAMPLEpoint?

Arguments B<x> is the number of the bus.

<NR1> is a percentage that represents the point at which to sample during each bit period.

Examples `BUS:B1:LIN:SAMPLEPOINT 10` sets the sample point to 10% of the bit period
`BUS:B1:LIN:SAMPLEPOINT?` might return `BUS:B1:LIN:SAMPLEPOINT 50` indicating that the sample point is set to 50% of the bit period

BUS:B<x>:LIN:SOURce

This command sets or queries the LIN data source for the specified bus. The bus number is specified by x.

Conditions Requires option SR-AUTO.

Group Bus

Syntax `BUS:B<x>:LIN:SOURce`
`{CH<x>|CH<x>_D<x>|MATH<x>|REF<x>|REF<x>_D<x>}`
`BUS:B<x>:LIN:SOURce?`

Arguments B<x> is the number of the bus.
Arguments specify the available sources.

Examples `BUS:B1:LIN:SOURce MATH4` sets the source to MATH 4
`BUS:B1:LIN:SOURce?` might return `:BUS:B1:LIN:SOURce CH1`, indicating the source is set to channel 1.

BUS:B<x>:LIN:SOURce:THReshold

This command sets or queries the LIN source threshold for the specified bus. The bus is specified by x.

Conditions Requires option SR-AUTO.

Group Bus

Syntax `BUS:B<x>:LIN:SOURce:THReshold <NR3>`
`BUS:B<x>:LIN:SOURce:THReshold?`

Arguments	B<x> is the number of the bus. <NR3> the LIN source threshold for the specified bus.
Examples	BUS:B1:LIN:SOURce:THResho1d 50.0e-3 sets the threshold to 50 mV. BUS:B1:LIN:SOURce:THResho1d? might return :BUS:B1:LIN:SOURCE:THRESHOLD 0.0E+0 indicating the threshold is 0.0 V.

BUS:B<x>:LIN:STANDard

This command sets or queries the LIN bus standard for the specified bus. The bus number is specified by x.

Conditions	Requires option SR-AUTO.
Group	Bus
Syntax	BUS:B<x>:LIN:STANDard {MIXed V1X V2X} BUS:B<x>:LIN:STANDard?
Arguments	B<x> is the number of the bus. MIXed specifies both versions 1.x and 2.x of the LIN standard. V1X specifies version 1.x of the LIN standard. V2X specifies version 2.x of the LIN standard.
Examples	BUS:B1:LIN:STANDard V1X sets the standard to version 1.x. BUS:B1:LIN:STANDard? might return :BUS:B1:LIN:STANDARD V2X, indicating the standard is set to version 2.x for bus B1.

BUS:B<x>:MIL1553B:LOWTHRESHold

This command sets or queries the MIL-STD-1553 lower threshold for the specified bus. The bus is specified by <x>.

Conditions	Requires option SR-AERO.
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Group	Bus
Syntax	<code>BUS:B<x>:MIL1553B:LOWTHRESHo1d <NR3></code>
Related Commands	BUS:B<x>:MIL1553B:SOUrce
Arguments	<code>B<x></code> is the number of the bus. <code><NR3></code> is the MIL-STD-1553 lower threshold for the specified bus.
Examples	<code>BUS:B3:MIL1553B:LOWTHRESHOLD -200e-3</code> sets the Bus 3 MIL-STD-1553 lower threshold to -200 mV. <code>BUS:B2:MIL1553B:LOWTHRESHOLD?</code> might return <code>:BUS:B2:MIL1553B:LOWTHRESHOLD -500.0000E-3</code> indicating the Bus 2 MIL-STD-1553 lower threshold is set to -500 mV.

BUS:B<x>:MIL1553B:POLarity

This command sets or queries the source polarity for the specified MIL-STD-1553 bus. The bus is specified by x.

Conditions	Requires option SR-AERO.
Group	Bus
Syntax	<code>BUS:B<x>:MIL1553B:POLarity {NORMa1 INVERTed}</code> <code>BUS:B<x>:MIL1553B:POLarity?</code>
Arguments	<code>B<x></code> is the number of the bus. <code>NORMa1</code> specifies normal polarity. <code>INVERTed</code> specifies inverted polarity.
Examples	<code>BUS:B3:MIL1553B:POLARITY INVERTED</code> sets Bus 3 MIL-STD-1553 polarity to inverted. <code>BUS:B2:MIL1553B:POLARITY?</code> might return <code>:BUS:B2:MIL1553B:POLARITY NORMAL</code>, indicating that the Bus 2 MIL-STD-1553 polarity is set to normal.

BUS:B<x>:MIL1553B:RESPonsetime:MAXimum

This command sets or queries the maximum response time to a valid command issued for the specified MIL-STD-1553 bus. The bus is specified by x.

Conditions	Requires option SR-AERO.
Group	Bus
Syntax	<code>BUS:B<x>:MIL1553B:RESPonsetime:MAXimum <NR3></code> <code>BUS:B<x>:MIL1553B:RESPonsetime:MAXimum?</code>
Related Commands	BUS:B<x>:MIL1553B:RESPonsetime:MINimum
Arguments	<code>B<x></code> is the number of the bus. <code><NR3></code> is a floating point number that specifies the maximum response time, in seconds.
Examples	<code>BUS:B1:MIL1553B:RESPONSETIME:MAXIMUM 15.0E-6</code> specifies the maximum response time to a valid command received to be 15.0 microseconds. <code>BUS:B1:MIL1553B:RESPONSETIME:MAXIMUM?</code> might return <code>:BUS:B1:MIL1553B:RESPONSETIME:MAXIMUM 12.000E-6</code> , indicating a maximum response time of 12 microseconds.

BUS:B<x>:MIL1553B:RESPonsetime:MINimum

This command sets or queries the minimum response time to a valid command issued for the specified MIL-STD-1553 bus. The bus is specified by x.

Conditions	Requires option SR-AERO.
Group	Bus
Syntax	<code>BUS:B<x>:MIL1553B:RESPonsetime:MINimum <NR3></code> <code>BUS:B<x>:MIL1553B:RESPonsetime:MINimum?</code>
Related Commands	BUS:B<x>:MIL1553B:RESPonsetime:MAXimum

Arguments	<NR3> is a floating point number that specifies the minimum response time, in seconds.
Examples	<code>BUS:B1:MIL1553B:RESPONSETIME:MINIMUM 5.0E-6</code> specifies the minimum response time to a valid command received to be 5.0 microseconds. <code>BUS:B1:MIL1553B:RESPONSETIME:MINIMUM?</code> might return <code>:BUS:B1:MIL1553B:RESPONSETIME:MINIMUM 4.000E-6</code>, indicating a minimum response time of 4 microseconds.

BUS:B<x>:MIL1553B:SOUrce

This command sets or queries the source for the specified MIL-STD-1553 bus. The bus is specified by x.

Conditions	Requires option SR-AERO.
Group	Bus
Syntax	<code>BUS:B<x>:MIL1553B:SOUrce {CH<x> Math<x> REF<x>}</code> <code>BUS:B<x>:MIL1553B:SOUrce?</code>
Arguments	<code>B<x></code> is the number of the bus. <code>CH<x></code> specifies an analog channel as the source waveform for the MIL-STD-1553 bus. <code>Math<x></code> specifies a math waveform as the source waveform for the MIL-STD-1553 bus. <code>REF<x></code> specifies a reference waveform as the source waveform for the MIL-STD-1553 bus.
Examples	<code>BUS:B1:MIL1553B:SOUrce CH1</code> sets channel 1 as the source for the MIL-STD-1553 bus. <code>BUS:B1:MIL1553B:SOUrce?</code> might return <code>:BUS:B1:MIL1553B:SOUrce MATH2</code> indicating that the source is set to MATH2.

BUS:B<x>:MIL1553B:THRESHold

This command sets or queries the MIL-STD-1553 upper threshold for the specified bus. The bus is specified by x.

Conditions	Requires option SR-AERO.
Group	Bus
Syntax	<code>BUS:B<x>:MIL1553B:THRESHo1d <NR3></code> <code>BUS:B<x>:MIL1553B:THRESHo1d</code>
Related Commands	BUS:B<x>:MIL1553B:SOUrce
Arguments	<code>B<x></code> is the number of the bus. <code><NR3></code> is the MIL-STD-1553 upper threshold for the specified bus.
Examples	<code>BUS:B3:MIL1553B:THRESHOLD 2.5</code> sets the Bus 3 MIL-STD-1553 upper threshold to 2.5 V. <code>BUS:B2:MIL1553B:THRESHOLD?</code> might return <code>:BUS:B2:MIL1553B:THRESHOLD 500.0000E-3</code> indicating the Bus 2 MIL-STD-1553 upper threshold is set to 500 mV.

BUS:B<x>:PARAllel:ALLTHResholds

This command sets or queries a threshold value for sources for the parallel bus. Use the [BUS:B<x>:PARAllel:ALLTHResholds:APPLY](#) command to set the thresholds to this value. The bus is specified by x.

Group	Bus
Syntax	<code>BUS:B<x>:PARAllel:ALLTHResholds <NR3></code>
Related Commands	BUS:B<x>:PARAllel:ALLTHResholds:APPLY
Arguments	<code>B<x></code> is the number of the bus. <code><NR3></code> is the source threshold.
Examples	<code>BUS:B4:PARALLEL:ALLTHRESHOLDS 1.0</code> sets the threshold of all the sources in parallel Bus 4 to 1.0 V.

BUS:B3:PARALLEL:ALLTHRESHOLDS? might return
 BUS:B3:PARALLEL:ALLTHRESHOLDS 500.00000E-3 indicates the threshold for
 all sources in parallel Bus 3 is currently set to 500.0 mV.

BUS:B<x>:PARAllel:ALLTHResholds:APPLY (No Query Form)

This command sets all of the data source thresholds to the value set by
[BUS:B<x>:PARAllel:ALLTHResholds](#) for the parallel bus. The bus is specified
 by x.

Group Bus

Syntax BUS:B<x>:PARAllel:ALLTHResholds:APPLY

Related Commands [BUS:B<x>:PARAllel:ALLTHResholds](#)

Examples BUS:B1:PARAllel:ALLTHResholds:APPLY sets all data source thresholds to
 the value set by [BUS:B<x>:PARAllel:ALLTHResholds](#).

BUS:B<x>:PARAllel:BIT<n>SOURce

This command sets or queries the specified bit source for specified parallel bus.
 The bus is specified by x. The bit is specified by n and is an integer in the range of
 1 to 64.

Group Bus

Syntax BUS:B<x>:PARAllel:BIT<n>SOURce
 {CH<x> | CH<x>_D<x> | MATH<x> | REF<x> | REF<x>_D<x> | NONE}

Related Commands [BUS:B<x>:PARAllel:BIT<n>SOURce:THReshold](#)

Arguments B<x> is the number of the bus.
 CH<x> is the specified bit source.
 CH<x>_D<x> is the specified bit source.
 MATH<x> is the specified bit source.
 REF<x> is the specified bit source.

REF<x>_D<x> specifies a digital reference waveform as the bit<x> source waveform for the specified parallel bus.

NONE disables the bit source.

Examples `BUS:B1:PARAllel:BIT1Source CH1` sets the bit 1 source to channel 1.
 `BUS:B1:PARAllel:BIT1Source?` might return
 `:BUS:B1:PARALLEL:BIT1SOURCE CH1_D0` indicating the bit1 source is CH1_D0.

BUS:B<x>:PARAllel:BIT<n>SOURce:THReshold

This command sets or queries the specified bit source threshold for the specified parallel bus. The bus is specified by x. The bit is specified by n and is an integer in the range of 1 to 64.

Group Bus

Syntax `BUS:B<x>:PARAllel:BIT<n>SOURce:THReshold <NR3>`

Related Commands [BUS:B<x>:PARAllel:BIT<n>SOURce](#)

Arguments `B<x>` is the number of the bus.
 `<NR3>` is the specified bit source threshold for the specified parallel bus.

Examples `BUS:B3:PARALLEL:BIT2SOURCE:THRESHOLD 1.0` sets the threshold of bit source 2 of parallel Bus 3 to 1.0 V.
 `BUS:B1:PARALLEL:BIT4SOURCE:THRESHOLD?` might return
 `:BUS:B1:PARALLEL:BIT4SOURCE:THRESHOLD 250.00000E-3` indicates the threshold of bit source 4 of parallel Bus 1 is currently set to 250.0 mV.

BUS:B<x>:PARAllel:CLOCK:EDGE

This command sets or queries the clock edge for the parallel bus. The bus is specified by x.

Group Bus

Syntax	<code>BUS:B<x>:PARAllel:CLOCK:EDGE {FALLING RISING EITHER}</code> <code>BUS:B<x>:PARAllel:CLOCK:EDGE?</code>
Related Commands	BUS:B<x>:PARAllel:CLOCKSOURCE
Arguments	<code>B<x></code> is the number of the bus. <code>FALLING</code> decodes on the falling edge of the clocked parallel bus signal. <code>RISING</code> decodes on the rising edge of the clocked parallel bus signal. <code>EITHER</code> decodes on the rising or falling edge of the clocked parallel bus signal.
Examples	<code>BUS:B1:PARALLEL:CLOCK:EDGE FALLING</code> sets the decoding to happen on the falling edge of its clocked parallel bus B1. <code>BUS:B1:PARALLEL:CLOCK:EDGE?</code> might return <code>:BUS:B1:PARALLEL:CLOCK:EDGE RISING</code>, indicating that when bus B1 operates in Clocked Parallel mode, it samples new data values on the rising edge of its clock source signal.

BUS:B<x>:PARAllel:CLOCK:ISCLOCKED

This command determines whether the bus operates in a clocked or asynchronous fashion. The bus is specified by `x`.

Group	Bus
Syntax	<code>BUS:B<x>:PARAllel:CLOCK:ISCLOCKED {OFF ON NR1>}</code> <code>BUS:B<x>:PARAllel:CLOCK:ISCLOCKED?</code>
Arguments	<code>B<x></code> is the number of the bus. <code>OFF</code> argument specifies an asynchronous bus. <code>ON</code> argument specifies a clocked bus. <code><NR1> = 0</code> specifies an asynchronous bus; any other value specifies a clocked bus.
Examples	<code>BUS:B1:PARAllel:CLOCK:ISCLOCKED 0</code> sets the bus to operate asynchronously. <code>BUS:B1:PARAllel:CLOCK:ISCLOCKED?</code> might return <code>:BUS:B1:PARALLEL:CLOCK:ISCLOCKED 1</code> indicating the bus is clocked.

BUS:B<x>:PARAllel:CLOCKSOURCE

This command sets or queries the Parallel clock bit source for the specified bus. The bus is specified by x.

Group	Bus
Syntax	<pre>BUS:B<x>:PARAllel:CLOCKSOURCE {CH<x> CH<x>_D<x> MATH<x> REF<x> REF<x>_D<x> NONE} BUS:B<x>:PARAllel:CLOCKSOURCE?</pre>
Related Commands	BUS:B<x>:PARAllel:CLOCK:ISCLOCKED
Arguments	B<x> is the number of the bus. CH<x> specifies an analog FlexChannel to use as the bus clock source. CH<x>_D<x> specifies a digital channel on a specified FlexChannel to use as the bus clock source. MATH<x> specifies the math channel to use as the bus clock source. REF<x> specifies the reference channel to use as the bus clock source. REF<x>_D<x> specifies a digital reference waveform as the clock source waveform for the specified parallel bus. NONE specifies the reference channel to use as the bus clock source.
Examples	BUS:B1:PARALLEL:CLOCK:SOURCE CH3_D6 sets the Parallel clock source for the bus B1 to D6 of FlexChannel 3. BUS:B1:PARALLEL:CLOCK:SOURCE? might return :BUS:B1:PARALLEL:CLOCK:SOURCE CH8_D9, indicating that the Parallel clock source for the bus B1 is set to D9 of FlexChannel 8.

BUS:B<x>:PARAllel:CLOCKSOURCE:THReshold

This command sets or queries the clock source threshold for the parallel bus. The bus is specified by x.

Group Bus

Syntax `BUS:B<x>:PARAllel:CLOCKSource:THReshoId <NR3>`
`BUS:B<x>:PARAllel:CLOCKSource:THReshoId?`

Related Commands [BUS:B<x>:PARAllel:CLOCK:ISCLOCKED](#)

Arguments <NR3> is the clock bit source threshold for the parallel bus.

Examples `BUS:B4:PARALLEL:CLOCKSource:THRESHOLD 1.5` sets the threshold of the clock source of parallel Bus 4 to 1.5 V.

`BUS:B3:PARALLEL:CLOCKSource:THRESHOLD?` might return
`:BUS:B3:PARALLEL:CLOCKSource:THRESHOLD 750.00000E-3` indicates the threshold of the clock source of parallel Bus 3 is currently set to 750.0 mV.

BUS:B<x>:RS232C:BITRate

This command sets or queries the RS-232C bit rate for bus<x>, where the bus number is specified by <x>. If you select Custom, use [BUS:B<x>:RS232C:BITRate:CUSTom](#) to set the bit rate.

Conditions Requires option SR-COMP.

Group Bus

Syntax `BUS:B<x>:RS232C:BITRate`
`{CUSTOM|RATE300|RATE1K|RATE2K|RATE9K|`
`RATE19K|RATE38K|RATE115K|RATE921K}`
`BUS:B<x>:RS232C:BITRate?`

Arguments B<x> is the number of the bus.

Arguments are the available bit rates.

Examples `BUS:B1:RS232C:BITRATE RATE9K` sets the bit rate for the RS-232C bus B1 to 9000 bits-per-second.

`BUS:B1:RS232C:BITRATE?` might return `:BUS:B1:RS232C:BITRATE 9000`, indicating that the bit rate for the RS-232C bus B1 is set to 9000 bits-per-second.

BUS:B<x>:RS232C:BITRate:CUSTom

This command sets or queries the RS-232C custom bit rate for the specified bus. The bus is specified by x.

Conditions	Requires option SR-COMP.
Group	Bus
Syntax	<code>BUS:B<x>:RS232C:BITRate:CUSTom <NR1></code> <code>BUS:B<x>:RS232C:BITRate:CUSTom?</code>
Related Commands	B<x> is the number of the bus. BUS:B<x>:RS232C:BITRate
Arguments	<NR1> is the custom bit rate.
Examples	<code>BUS:B1:RS232C:BITRate:CUSTOM 9500</code> sets the bit rate for the RS-232C bus B1 to 9500 bits-per-second. <code>BUS:B1:RS232C:BITRate:CUSTOM?</code> might return <code>:BUS:B1:RS232C:BITRate:CUSTOM 9500</code> , indicating that the bit rate for the RS-232C bus B1 is set to 9500 bits-per-second.

BUS:B<x>:RS232C:DATABits

This command sets or queries the RS-232C data width for bus<x>, where the bus number is specified by <x>.

Conditions	Requires option SR-COMP.
Group	Bus
Syntax	<code>BUS:B<x>:RS232C:DATABits {7 8 9}</code> <code>BUS:B<x>:RS232C:DATABits?</code>
Arguments	B<x> is the number of the bus. <NR1> specifies the number of bits in the RS-232C data frame.

Examples `BUS:B1:RS232C:DATABITS 8` sets the data bits to 8 for the RS-232C bus B1.

`BUS:B5:RS232C:DATABITS?` might return `:BUS:B1:RS232C:DATABITS 7`, indicating that the data bits is set to 7 for the RS-232C bus B5.

BUS:B<x>:RS232C:DELIMiter

This command sets or queries the RS-232C string delimiter on bus <x>, where the bus number is specified by <x>. This command only applies when Packet view is turned On.

Conditions Requires option SR-COMP.

Group Bus

Syntax `BUS:B<x>:RS232C:DELIMiter {NUL|CR|LF|SPace|XFF}`
`BUS:B<x>:RS232C:DELIMiter?`

Related Commands [BUS:B<x>:RS232C:DISplaymode](#)

Arguments B<x> is the number of the bus.

NUL specifies NULL (0x00) delimiting value for a packet.

CR specifies CR (0x0D) delimiting value for a packet.

LF specifies LF (0x0A) delimiting value for a packet.

XFF specifies XFF (0xFF) delimiting value for a packet.

SPace specifies SPace delimiting value for a packet.

Examples `BUS:B1:RS232C:DELIMITER LF` sets the delimiter to 0x0A for the RS-232C bus B1.

`BUS:B1:RS232C:DELIMITER?` might return `:BUS:B1:RS232C:DELIMITER NULL`, indicating that the delimiter to 0x00 for the RS-232C bus B1.

BUS:B<x>:RS232C:DISplaymode

This command sets or queries the RS-232C display mode for the specified bus. The bus is specified by x.

Conditions	Requires option SR-COMP.
Group	Bus
Syntax	<code>BUS:B<x>:RS232C:DISPlaymode {FRame PACKET}</code> <code>BUS:B<x>:RS232C:DISPlaymode?</code>
Related Commands	BUS:B<x>:RS232C:DELIMiter
Arguments	B<x> is the number of the bus. FRame displays each frame as a single entity. PACKET displays a group of frames terminated with a single frame defined by the <code>BUS:B<x>:RS232C:DELIMiter</code> command.
Examples	<code>BUS:B1:RS232C:DISPLAYMODE FRAME</code> sets the display mode for the RS-232C bus B1 to Frame. <code>BUS:B1:RS232C:DISPLAYMODE?</code> might return <code>:BUS:B1:RS232C:DISPLAYMODE PACKET</code>, indicating that the display mode for the RS-232C bus B1 is set to Packet.

BUS:B<x>:RS232C:PARity

This command sets or queries the RS-232C parity for bus <x>, where the bus number is specified by <x>.

Conditions	Requires option SR-COMP.
Group	Bus
Syntax	<code>BUS:B<x>:RS232C:PARity {NONE EVEN ODD}</code> <code>BUS:B<x>:RS232C:PARity?</code>
Arguments	B<x> is the number of the bus. NONE specifies no parity. EVEN specifies even parity. ODD specifies odd parity.

Examples `BUS:B1:RS232C:PARITY ODD` sets the parity for the RS-232C bus B1 to odd.

`BUS:B1:RS232C:PARITY?` might return `:BUS:B1:RS232C:PARITY NONE`, indicating that the display mode for the RS-232C bus B1 is set to none.

BUS:B<x>:RS232C:POLarity

This command sets or queries the RS-232C source polarity for bus <x>, where the bus number is specified by <x>.

Conditions Requires option SR-COMP.

Group Bus

Syntax `BUS:B<x>:RS232C:POLarity {NORma1|INVERTed}`
`BUS:B<x>:RS232C:POLarity?`

Arguments B<x> is the number of the bus.

NORma1 sets the RS-232C bus polarity to positive.

INVERTed sets the RS-232C bus polarity to negative.

Examples `BUS:B1:RS232C:POLARITY INVERTed` sets the polarity for the RS-232C bus B1 to Inverted.

`BUS:B1:RS232C:POLARITY?` might return `:BUS:B1:RS232C:POLARITY NORMAL`, indicating that the polarity for the RS-232C bus B1 is set to Normal.

BUS:B<x>:RS232C:SOUrce

This command sets or queries the RS-232C source for bus <x>, where the bus number is specified by <x>.

Conditions Requires option SR-COMP.

Group Bus

Syntax `BUS:B<x>:RS232C:SOUrce`
`{CH<x>|CH<x>_D<x>|REF<x>|MATH<x>|REF<x>_D<x>}`
`BUS:B<x>:RS232C:SOUrce?`

Arguments **B<x>** is the number of the bus.

CH<x> specifies an analog channel to use as the RS-232C source.

CH<x>_D<x> specifies a digital channel of a specified FlexChannel to use for the RS-232C source.

MATH<x> specifies a math channel to use for the RS-232C source.

REF<x> specifies a reference channel to use for the RS-232C source.

REF<x>_D<x> specifies a digital reference waveform as the source waveform for the specified RS-232C bus.

Examples **BUS:B1:RS232C:SOURCE CH1_D0** sets the source for the RS-232C bus B1 to D0 of FlexChannel 1.

BUS:B1:RS232C:SOURCE? might return **:BUS:B1:RS232C:SOURCE CH1**, indicating that the source for the RS-232C bus B1 is CH1.

BUS:B<x>:RS232C:SOURce:THReshold

This command sets or queries the RS-232C source threshold for the specified bus. The bus is specified by x.

Conditions Requires option SR-COMP.

Group Bus

Syntax **BUS:B<x>:RS232C:SOURce:THResho1d <NR3>**
BUS:B<x>:RS232C:SOURce:THResho1d?

Arguments **B<x>** is the number of the bus.

<NR3> is the RS-232C source threshold for the specified bus.

Examples **BUS:B1:RS232C:SOURce:THRESHOLD 50.0e-3** sets the threshold to 50 mV.

BUS:B1:RS232C:SOURce:THRESHOLD? might return **:BUS:B1:RS232C:SOURce:THRESHOLD 0.0E+0** indicating the threshold is set to 0.0 V.

BUS:B<x>:SENT:CHANWidth

This command sets or queries SENT fast channel bit widths for the specified bus.

Conditions Requires option SRAUTOSEN.

Group Bus

Syntax BUS:B<x>:SENT:CHANWidth
{TWELVETwelve|FOURTEENTen|SIXTEENeight}
BUS:B<x>:SENT:CHANWidth?

Related Commands [BUS:B<x>:SENT:NUMCHANnel](#)

Arguments B<x> is the number of the bus waveform.

TWELVETwelve sets both Fast Channel 1 and Fast Channel 2 to 12 bits wide.
FOURTEENTen sets Fast Channel 1 to 14 bits and Fast Channel 2 to 10 bits.
SIXTEENeight sets Fast Channel 1 to 16 bits and Fast Channel 2 to eight bits.

Examples BUS:B1:SENT:CHANWIDTH TWELVETWELVE sets the channel width of both Fast Channel 1 and Fast Channel 2 to twelve bits.

BUS:B1:SENT:CHANWIDTH? might return :BUS:B1:SENT:CHANWIDTH SIXTEENEIGHT, which indicates that the Fast Channel 1 width is 16 and the Fast Channel 2 width is eight.

BUS:B<x>:SENT:NIBBLECount

This command sets or queries SENT data nibbles for the specified bus.

Conditions Requires option SRAUTOSEN.

Group Bus

Syntax BUS:B<x>:SENT:NIBBLECount {THREE|FOUR|SIX}
BUS:B<x>:SENT:NIBBLECount?

Arguments	B<x> is the number of the bus waveform. THREE specifies regular data with 3 nibbles FOUR specifies regular data with 4 nibbles SIX specifies regular data with 6 nibbles
Examples	BUS:B1:SENT:NIBBLECOUNT THREE sets the data nibble count to three. BUS:B1:SENT:NIBBLECOUNT? might return :BUS:B1:SENT:NIBBLECOUNT SIX, which indicates the nibble count is set to six.

BUS:B<x>:SENT:NUMCHANnel

This command sets or queries SENT fast data channels for the specified bus.

Conditions	Requires option SRAUTOSEN.
Group	Bus
Syntax	BUS:B<x>:SENT:NUMCHANnel {ONE TWO} BUS:B<x>:SENT:NUMCHANnel?
Arguments	B<x> is the number of the bus waveform. ONE specifies a SENT bus with one fast channel. TWO specifies a SENT bus with two fast channels.
Examples	BUS:B1:SENT:NUMCHANnel TWO sets the number of SENT channels to two. BUS:B1:SENT:NUMCHANNEL? might return :BUS:B1:SENT:NUMCHANNEL ONE to indicate that the number of channels is set to one.

BUS:B<x>:SENT:PAUSEPULSe

This command sets or queries SENT pause pulse for the specified bus.

Conditions	Requires option SRAUTOSEN.
Group	Bus

Syntax	<code>BUS:B<x>:SENT:PAUSEPULSe {NO YES}</code> <code>BUS:B<x>:SENT:PAUSEPULSe?</code>
Arguments	<code>B<x></code> is the number of the bus waveform. <code>NO</code> specifies no pause pulse. <code>YES</code> specifies a pause pulse is used.
Examples	<code>BUS:B1:SENT:PAUSEPULSE NO</code> sets the pause pulse value to indicate no pause pulse. <code>BUS:B1:SENT:PAUSEPULSE?</code> might return <code>:BUS:B1:SENT:PAUSEPULSE YES</code>, which indicates the SENT bus is set to use pause pulses.

BUS:B<x>:SENT:POLARITY

This command sets or queries SENT Idle State signal polarity for the specified bus.

Conditions	Requires option SRAUTOSEN.
Group	Bus
Syntax	<code>BUS:B<x>:SENT:POLARITY {INverted NORma1}</code> <code>BUS:B<x>:SENT:POLARITY?</code>
Arguments	<code>B<x></code> is the number of the bus waveform. <code>INverted</code> specifies inverted polarity. <code>NORma1</code> specifies normal polarity.
Examples	<code>BUS:B1:SENT:POLARITY INVERTED</code> sets the idle state of the bus to inverted polarity. <code>BUS:B1:SENT:POLARITY?</code> might return <code>:BUS:B1:SENT:POLARITY NORMAL</code> to indicate the bus idle state polarity is set to normal

BUS:B<x>:SENT:SLOW

This command sets or queries the SENT slow channel configuration for the specified bus.

Conditions	Requires option SRAUTOSEN.
Group	Bus
Syntax	<code>BUS:B<x>:SENT:SLOW {None ENHANCED4 ENHANCED8 SHORT}</code> <code>BUS:B<x>:SENT:SLOW?</code>
Arguments	<code>B<x></code> is the number of the bus waveform. <code>None</code> specifies no slow channel configured. <code>ENHANCED4</code> specifies Enhanced 4 slow channel configuration. <code>ENHANCED8</code> specifies Enhanced 8 slow channel configuration. <code>SHORT</code> specifies short slow channel configuration.
Examples	<code>BUS:B1:SENT:SLOW SHORT</code> sets the SENT bus to use the short slow channel configuration. <code>BUS:B1:SENT:SLOW?</code> might return <code>:BUS:B1:SENT:SLOW NONE</code> to indicate that the SENT bus is set to have no slow channel.

BUS:B<x>:SENT:SOURce

This command sets or queries the SENT DATA source for the specified bus.

Conditions	Requires option SRAUTOSEN.
Group	Bus
Syntax	<code>BUS:B<x>:SENT:SOURce</code> <code>{CH<x> CH<x>_D<x> Math<x> REF<x> REF<x>_D<x>}</code> <code>BUS:B<x>:SENT:SOURce?</code>
Arguments	<code>B<x></code> is the number of the bus waveform. <code>CH<x></code> specifies an analog channel as the clock source waveform for the audio bus. <code>CH<x>_D<x></code> specifies a digital channel as the clock source waveform for the specified audio bus. <code>Math<x></code> specifies a math waveform as the clock source waveform for the audio bus.

REF<x> specifies a reference waveform as the clock source waveform for the audio bus

REF<x>_D<x> specifies a digital reference waveform as the clock source waveform for the specified audio bus.

Examples **BUS:B1:SENT:SOURCE CH1_D1** sets D1 of channel 1 as the source for the SENT bus.

BUS:B1:SENT:SOURCE? might return **:BUS:B1:SENT:SOURCE CH1** indicating that the source is set to channel 1

BUS:B<x>:SENT:THRESHold

This command sets or queries the SENT DATA source threshold for the specified bus.

Conditions Requires option SRAUTOSEN.

Group Bus

Syntax **BUS:B<x>:SENT:THRESHold <NR3>**
BUS:B<x>:SENT:THRESHold?

Arguments **B<x>** is the number of the bus waveform.

<NR3> sets the data source threshold value in volts.

Examples **BUS:B1:SENT:THRESHOLD 500.0E-3** sets the Bus 1 source threshold to 500.0 mV.

BUS:B1:SENT:THRESHOLD? might return **:BUS:B1:SENT:THRESHOLD 2.5000** to indicate the Bus 1 source threshold is 2.5 V.

BUS:B<x>:SENT:TICKTIME

This command sets or queries the SENT bus Clock Tick parameter for the specified bus.

Conditions Requires option SRAUTOSEN.

Group	Bus
Syntax	<code>BUS:B<x>:SENT:TICKTIME <NR3></code> <code>BUS:B<x>:SENT:TICKTIME?</code>
Arguments	<code>B<x></code> is the number of the bus waveform. <code><NR3></code> is the SENT clock tick time, in seconds.
Examples	<code>BUS:B1:SENT:TICKTIME 3.0E-6</code> sets the SENT bus to clock tick to 3 microseconds. <code>BUS:B1:SENT:TICKTIME?</code> might return <code>:BUS:B1:SENT:TICKTIME 10.0000E-6</code> to indicate the clock tick is set to 10 microseconds.

BUS:B<x>:SENT:TICKTOLerance

This command sets or queries the SENT bus Tick Tolerance percent parameter for the specified bus.

Conditions	Requires option SRAUTOSEN.
Group	Bus
Syntax	<code>BUS:B<x>:SENT:TICKTOLerance <NR3></code> <code>BUS:B<x>:SENT:TICKTOLerance?</code>
Arguments	<code>B<x></code> is the number of the bus waveform. <code><NR3></code> is the tick tolerance percentage.
Examples	<code>BUS:B1:SENT:TICKTOLERANCE 20.0</code> sets the tick tolerance to 20%. <code>US:B1:SENT:TICKTOLERANCE?</code> might return <code>:BUS:B1:SENT:TICKTOLERANCE 22.5</code> to indicate the tolerance is set to 22.5%.

BUS:B<x>:SPI:BITOrder

This command sets or queries the SPI bit order for the specified bus. The bus is specified by x.

Conditions	Requires option 5-SREMBD or SUP5-SREMBD.
Group	Bus
Syntax	<code>BUS:B<x>:SPI:BITOrder {LSB MSB}</code> <code>BUS:B<x>:SPI:BITOrder?</code>
Related Commands	BUS:B<x>:SPI:DATA:SIZE
Arguments	<code>B<x></code> is the number of the bus waveform. <code>LSB</code> specifies that each bit becomes the recovered value's new LSB, after shifting previously recovered bits one place to the left. The decoding happens right to left. <code>MSB</code> specifies that each successive bit from the bus's data line becomes the new MSB of the recovered value, shifting any previously recovered bits one place to the right. The decoding happens left to right.
Examples	<code>BUS:B1:SPI:BITORDER LSB</code> sets each bit order for the SPI bus B1 to LSB. <code>BUS:B1:SPI:BITORDER?</code> might return <code>:BUS:B1:SPI:BITORDER MSB</code>, indicating that the bit order for the SPI bus B1 is set to MSB.

BUS:B<x>:SPI:CLOCK:POLarity

This command sets or queries the SPI clock (SCLK) source polarity for the specified bus. The bus is specified by `x`.

Conditions	Requires option 5-SREMBD or SUP5-SREMBD.
Group	Bus
Syntax	<code>BUS:B<x>:SPI:CLOCK:POLarity {FALL RISE}</code> <code>BUS:B<x>:SPI:CLOCK:POLarity?</code>
Related Commands	BUS:B<x>:SPI:CLOCK:SOURce
Arguments	<code>B<x></code> is the number of the bus waveform. <code>FALL</code> sets the clock to the falling edge of the signal.

RISE sets the clock to the rising edge of the signal.

Examples `BUS:B1:SPI:CLOCK:POLARITY FALL` sets the SPI clock polarity to the falling edge for the bus B1.

`BUS:B1:SPI:CLOCK:POLARITY?` might return
`:BUS:B1:SPI:CLOCK:POLARITY RISE`, indicating that the SPI clock polarity for the bus B1 is set to the rising edge.

BUS:B<x>:SPI:CLOCK:SOURce

This command sets or queries the SPI clock (SCLK) source for the specified bus. The bus is specified by x.

Conditions Requires option 5-SREMBD or SUP5-SREMBD.

Group Bus

Syntax `BUS:B<x>:SPI:CLOCK:SOURce`
`{CH<x>|CH<x>_D<x>|MATH<x>|REF<x>|REF<x>_D<x>}`
`BUS:B<x>:SPI:CLOCK:SOURce?`

Related Commands [BUS:B<x>:SPI:CLOCK:POLarity](#)

Arguments `B<x>` is the number of the bus waveform.

`CH<x>` designates an analog channel as the bus SPI clock source.

`CH<x>_D<x>` designates an digital channel as the bus SPI clock source.

`MATH<x>` designates a math waveform as the clock source.

`REF<x>` designates a reference waveform as the clock source.

`REF<x>_D<x>` specifies a digital reference waveform as the clock source waveform for the specified SPI bus.

Examples `BUS:B1:SPI:CLOCK:SOURCE CH5` sets the SPI clock source for the bus B1 to CH5.

`BUS:B1:SPI:CLOCK:SOURCE?` might return `:BUS:B1:SPI:CLOCK:SOURCE MATH1`, indicating that the SPI clock source for the bus B1 is set to MATH1.

BUS:B<x>:SPI:CLOCK:THReshold

This command sets or queries the SPI Clock (SCLK) source threshold for the specified bus. The bus is specified by x.

Conditions Requires option 5-SREMBD or SUP5-SREMBD.

Group Bus

Syntax BUS:B<x>:SPI:CLOCK:THReshold <NR3>
BUS:B<x>:SPI:CLOCK:THReshold?

Arguments B<x> is the number of the bus waveform.
<NR3> is the SPI Clock (SCLK) source threshold for the specified bus.

Examples BUS:B1:SPI:CLOCK:THReshold 50.0e-3 sets the threshold to 50.0 mV.
BUS:B1:SPI:CLOCK:THReshold? might return
:BUS:B1:SPI:CLOCK:THRESHOLD 0.0E+0, indicating the threshold is set to 0.0 V.

BUS:B<x>:SPI:DATa:POLarity

This command sets or queries the SPI Data (Data) source polarity for the bus number specified by x.

Conditions Requires option 5-SREMBD or SUP5-SREMBD.

Group Bus

Syntax BUS:B<x>:SPI:DATa:POLarity {HIGH|LOW}
BUS:B<x>:SPI:DATa:POLarity?

Arguments B<x> is the number of the bus waveform.
HIGH sets the SPI data polarity to active high.
LOW sets the SPI data polarity to active low.

- Examples** `BUS:B4:SPI:DATA:POLARITY HIGH` sets the SPI data polarity for the bus B4 to active high.
- `BUS:B1:SPI:DATA:POLARITY?` might return `:BUS:B1:SPI:DATA:POLARITY LOW`, indicating that the SPI SDA polarity for the bus B1 is set to active low.

BUS:B<x>:SPI:DATA:SIZE

This command sets or queries the number of bits per word for the specified SPI bus. The bus is specified by x.

- Conditions** Requires option 5-SREMBD or SUP5-SREMBD.
- Group** Bus
- Syntax** `BUS:B<x>:SPI:DATA:SIZE <NR1>`
`BUS:B<x>:SPI:DATA:SIZE?`
- Arguments** `B<x>` is the number of the bus waveform.
- `<NR1>` is the data size for the specified bus. The minimum value is 2 and maximum is 32.
- Examples** `BUS:B1:SPI:DATA:SIZE 7` sets the data size for SPI bus B1 to seven bits per word.
- `BUS:B1:SPI:DATA:SIZE?` might return `:BUS:B1:SPI:DATA:SIZE 8`, indicating that the data size for SPI bus B1 is set to eight bits per word.

BUS:B<x>:SPI:DATA:SOURce

This command sets or queries the SPI Data (Data) source for the bus number specified by x.

- Conditions** Requires option 5-SREMBD or SUP5-SREMBD.
- Group** Bus
- Syntax** `BUS:B<x>:SPI:DATA:SOURce`
`{CH<x>|CH<x>_D<x>|MATH<x>|REF<x>|REF<x>_D<x>}`
`BUS:B<x>:SPI:DATA:SOURce?`

Arguments **B<x>** is the number of the bus waveform.

CH<x> designates an analog channel as the data source for the specified SPI bus.

CH<x>_D<x> designates an digital channel as the bus SPI clock source.

MATH<x> designates a math waveform as the data source.

REF<x> designates a reference waveform as the data source.

REF<x>_D<x> specifies a digital reference waveform as the data source waveform for the specified SPI bus.

Examples **BUS:B1:SPI:DATA:SOURCE CH2** sets the SPI data source for the bus B1 to CH2.

BUS:B1:SPI:DATA:SOURCE? might return **:BUS:B1:SPI:DATA:SOURCE CH3**, indicating that the SPI data source for the bus B1 is set to CH3.

BUS:B<x>:SPI:DATA:THReshold

This command sets or queries the SPI Data (Data) source threshold for the specified bus. The bus is specified by x.

Conditions Requires option 5-SREMBD or SUP5-SREMBD.

Group Bus

Syntax **BUS:B<x>:SPI:DATA:THReshold <NR3>**
BUS:B<x>:SPI:DATA:THReshold?

Arguments **B<x>** is the number of the bus waveform.

<NR3> is the SPI Data (SDA) source threshold for the specified bus.

Examples **BUS:B1:SPI:DATA:THReshold 50.0e-3** sets the threshold to 50.0 mV.

BUS:B1:SPI:DATA:THReshold? might return
:BUS:B1:SPI:DATA:THRESHOLD 0.0E+0 indicating the threshold is set to 0.0 V.

BUS:B<x>:SPI:FRAMING

This command sets or queries the SPI framing setting for the specified bus. The bus number is specified by x.

Conditions	Requires option 5-SREMBD or SUP5-SREMBD.
Group	Bus
Syntax	<code>BUS:B<x>:SPI:FRAMING {IDLE SS}</code> <code>BUS:B<x>:SPI:FRAMING?</code>
Arguments	<code>B<x></code> is the number of the bus waveform. <code>IDLE</code> specifies IDLE SPI framing. <code>SS</code> specifies SS SPI framing.
Examples	<code>BUS:B1:SPI:FRAMING IDLE</code> sets the SPI framing to IDLE. <code>BUS:B1:SPI:FRAMING?</code> might return <code>:BUS:B1:SPI:FRAMING SS</code> , indicating that the SPI framing is set to SS.

BUS:B<x>:SPI:IDLETime

This command sets or queries the SPI idle time for the specified bus. The bus is specified by x.

Conditions	Requires option 5-SREMBD or SUP5-SREMBD.
Group	Bus
Syntax	<code>BUS:B<x>:SPI:IDLETime <NR3></code> <code>BUS:B<x>:SPI:IDLETime?</code>
Arguments	<code>B<x></code> is the number of the bus waveform. <code><NR3></code> specifies the SPI idle time.
Examples	<code>BUS:B1:SPI:IDLETime 0.000004</code> sets the idle time to 4 μ s. <code>BUS:B1:SPI:IDLETime?</code> might return <code>:BUS:B1:SPI:IDLETIME 5.0000E-6</code> , indicating that the idle time is set to 5 μ s.

BUS:B<x>:SPI:SElect:POLarity

This command sets or queries the SPI Slave Select (SS) polarity for the specified bus. The bus is specified by x.

Conditions Requires option 5-SREMBD or SUP5-SREMBD.

Group Bus

Syntax BUS:B<x>:SPI:SElect:POLarity {LOW|HIGH}
BUS:B<x>:SPI:SElect:POLarity?

Related Commands [BUS:B<x>:SPI:SElect:SOUrce](#)

Arguments B<x> is the number of the bus waveform.
LOW sets an active low polarity.
HIGH sets an active high polarity.

Examples BUS:B1:SPI:SELECT:POLARITY HIGH sets the SPI Slave Select polarity for the bus B1 to active high.
BUS:B1:SPI:SELECT:POLARITY? might return
:BUS:B1:SPI:SELECT:POLARITY HIGH, indicating that the SPI Slave Select polarity for the bus B1 is set to active high.

BUS:B<x>:SPI:SElect:SOUrce

This command sets or queries the SPI Slave Select (SS) source for the specified bus. The bus is specified by x.

Conditions Requires option 5-SREMBD or SUP5-SREMBD.

Group Bus

Syntax BUS:B<x>:SPI:SElect:SOUrce
{CH<x> | CH<x>_D<x> | MATH<x> | REF<x> | REF<x>_D<x>}
BUS:B<x>:SPI:SElect:SOUrce?

Related Commands [BUS:B<x>:SPI:SElect:POLarity](#)

Arguments	B<x> is the number of the bus waveform. CH<x> designates an analog channel as the buses' SPI Slave Select source. CH<x>_D<x> designates a digital channel as the buses' SPI Slave Select source. MATH<x> designates a math waveform as the Slave Select source. REF<x> designates a reference waveform as the Slave Select source. REF<x> _D<x> specifies a digital reference waveform as the clock source waveform for the specified SPI bus.
Examples	BUS:B1:SPI:SELECT:SOURCE MATH3 sets the SPI Slave Select source for the bus B1 to MATH3. BUS:B1:SPI:SELECT:SOURCE? might return :BUS:B1:SPI:SELECT:SOURCE CH3, indicating that the SPI Slave Select source for the bus B1 is set to CH3.

BUS:B<x>:SPI:SElect:THReshold

This command sets or queries the SPI Select (SS) source threshold for the specified bus. The bus is specified by x.

Conditions	Requires option 5-SREMBD or SUP5-SREMBD.
Group	Bus
Syntax	BUS:B<x>:SPI:SElect:THReshold <NR3> BUS:B<x>:SPI:SElect:THReshold?
Arguments	B<x> is the number of the bus waveform. <NR3> is the SPI Select (SS) source threshold for the specified bus.
Examples	BUS:B1:SPI:SElect:THReshold 50.0e-3 sets the threshold to 50.0 mV. BUS:B1:SPI:SElect:THReshold? might return :BUS:B1:SPI:SELECT:THRESHOLD 0.0E+0 indicating the threshold is set to 0.0 V.

BUS:B<x>:SPMI:SCLk:SOUrce

This command sets or queries the SPMI Clock (SCLK) source for the specified bus.

Conditions Requires option SRPM

Group Bus

Syntax `BUS:B<x>:SPMI:SCLk:SOUrce {CH<x>|CH<x>_Dx|
Math<x>|REF<x>|REF<x>_D<x>}`
`BUS:B<x>:SPMI:SCLk:SOUrce`

Related Commands [BUS:B<x>:SPMI:SDATa:SOUrce](#)

Arguments B<x> is the number of the bus waveform.

CH<x> specifies an analog channel as the clock source waveform for the SPMI bus.

CH<x>_Dx specifies a digital channel and bit as the clock source waveform for the specified SPMI bus.

Math<x> specifies a math waveform as the clock source waveform for the specified SPMI bus.

REF<x> specifies a reference waveform as the clock source waveform for the specified SPMI bus.

REF<x>_D<x> specifies a digital reference waveform and bit as the clock source waveform for the specified SPMI bus.

Examples `BUS:B1:SPMI:SCLk:SOUrce CH1_D1` sets D1 of channel 1 as the clock source for the SPMI bus B1.

`BUS:B1:SPMI:SCLk:SOUrce?` might return `:BUS:B3:SPMI:SCLk:SOUrce CH3`, indicating that the data source for bus B3 is set to channel 3.

BUS:B<x>:SPMI:SCLk:THReshold

This command sets or queries the SPMI Clock (SCLK) source threshold for the specified bus.

Conditions Requires option SRPM

Group	Bus
Syntax	<code>BUS:B<x>:SPMI:SCLk:THResho1d <NR3></code> <code>BUS:B<x>:SPMI:SCLk:THResho1d?</code>
Related Commands	TRIGger:{A B}:BUS:B<x>:SPMI:CONDition
Arguments	<code>B<x></code> is the number of the bus waveform. <code><NR3></code> is the clock (SCLK) source threshold value for the specified SPMI bus.
Examples	<code>BUS:B3:SPMI:SCLK:THRESHOLD 500.0E-3</code> sets the Bus3 clock source threshold to 500.0 mV. <code>BUS:B1:SPMI:SCLK:THRESHOLD?</code> might return <code>:BUS:B1:SPMI:SCLK:THRESHOLD 250.0E-3</code> to indicate Bus1 clock threshold is set to 250.0 mV.

BUS:B<x>:SPMI:SDATa:SOUrce

This command sets or queries the SPMI Data (SDATA) source for the specified bus.

Conditions	Requires option SRPM
Group	Bus
Syntax	<code>BUS:B<x>:SPMI:SDATa:SOUrce {CH<x> CH<x>_Dx} </code> <code>Math<x> REF<x> REF<x>_D<x>}</code> <code>BUS:B<x>:SPMI:SDATa:SOUrce</code>
Related Commands	BUS:B<x>:SPMI:SCLk:SOUrce
Arguments	<code>B<x></code> is the number of the bus waveform. <code>CH<x></code> specifies an analog channel as the data source waveform for the SPMI bus. <code>CH<x>_Dx</code> specifies a digital channel and bit as the data source waveform for the specified SPMI bus. <code>Math<x></code> specifies a math waveform as the data source waveform for the specified SPMI bus.

REF<x> specifies a reference waveform as the data source waveform for the specified SPMI bus.

REF<x>_D<x> specifies a digital reference waveform and bit as the data source waveform for the specified SPMI bus.

Examples `BUS:B4:SPMI:SDATa:Source CH1_D2` sets bit D2 of digital channel 1 as the data source for the SPMI bus B4.

`BUS:B1:SPMI:SCLk:Source?` might return `:BUS:B3:SPMI:SCLk:Source CH5` indicating that the data source is set to channel 5 for bus B3.

BUS:B<x>:SPMI:SDATa:THReshold

This command sets or queries the SPMI Data (SDATA) source threshold for the specified bus.

Conditions Requires option SRPM

Group Bus

Syntax `BUS:B<x>:SPMI:SDATa:THReshold <NR3>`
`BUS:B<x>:SPMI:SDATa:THReshold?`

Related Commands [TRIGger:{A|B}:BUS:B<x>:SPMI:CONDition](#)

Arguments B<x> is the number of the bus waveform.

<NR3> is the data (SDATA) source threshold value for the specified SPMI bus.

Examples `BUS:B2:SPMI:SDATA:THRESHOLD 400.0E-3` sets Bus2 data source threshold to 400.0 mV.

`BUS:B1:SPMI:SDATA:THRESHOLD?` might return `:BUS:B1:SPMI:SDATA:THRESHOLD 220.0E-3` to indicate that Bus1 data source threshold is set to 220.0 mV

BUS:B<x>:TYPE

This command sets or queries the bus type or standard for the specified bus. The bus is specified by x.

Group	Bus
Syntax	<pre>BUS:B<x>:TYPE {ARINC429 AUDio CAN ETHeRnet FLEXRAY I2C LIN MIL1553B PARa11e1 RS232C SENT SPI SPMI USB} BUS:B<x>:TYPE?</pre>
Arguments	B<x> is the number of the bus waveform. ARINC429 specifies the ARINC 429 avionics serial bus. AUDio specifies an audio bus. CAN specifies a Controller Area Network bus. ETHeRnet specifies the Ethernet bus. FLEXRAY specifies a FlexRay bus. I2C specifies the Inter-IC bus. LIN specifies a Local Interconnect Network bus. MIL1553B specifies the MIL-STD-1553 avionics serial bus. PARa11e1 specifies a parallel bus. RS232C specifies the RS-232 Serial bus. SENT specifies the Single Edge Nibble Transmission (SENT) automotive serial bus. SPI specifies the Serial Peripheral Interface bus. SPMI Specifies a System Power Management Interface bus USB specifies the Universal Serial Bus.
Examples	BUS:B1:TYPE I2C sets the bus B1 type to I2C. BUS:B1:TYPE? might return :BUS:B1:TYPE SPI, indicating that the bus B1 type is set to SPI and that the bus behavior is governed by the :BUS:B1:SPI:xxxx commands.

BUS:B<x>:USB:BITRate

This command sets or queries the USB data rate for bus <x>, where the bus number is specified by <x>.

Conditions	Requires option SR-USB2.
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Group	Bus
Syntax	<code>BUS:B<x>:USB:BITRate {FULL HIGH LOW}</code> <code>BUS:B<x>:USB:BITRate?</code>
Arguments	<code>B<x></code> is the number of the bus waveform. <code>FULL</code> indicates the bit rate is 12 Mbps. <code>HIGH</code> indicates the bit rate is 480 Mbps. <code>LOW</code> indicates the bit rate is 1.5 Mbps.
Examples	<code>BUS:B1:USB:BITRATE FULL</code> sets the bit rate to 12 Mbps. <code>BUS:B1:USB:BITRATE?</code> might return <code>:BUS:B1:USB:BITRATE LOW</code>, indicating that the bit rate is 1.5 Mbps.

BUS:B<x>:USB:DATAMINUSTHRESHold

This command sets or queries the USB D- source threshold for the specified bus. The bus is specified by `x`.

Conditions	Requires option SR-USB2.
Group	Bus
Syntax	<code>BUS:B<x>:USB:DATAMINUSTHRESHold <NR3></code> <code>BUS:B<x>:USB:DATAMINUSTHRESHold?</code>
Arguments	<code>B<x></code> is the number of the bus waveform. <code><NR3></code> is the Minus threshold.
Examples	<code>BUS:B1:USB:DATAMINUSTHRESHold 50.0e-3</code> sets the threshold to 50.0 mV. <code>BUS:B1:USB:DATAMINUSTHRESHold?</code> might return <code>:BUS:B1:USB:DATAMINUSTHRESHOLD 0.0E+0</code> indicating the threshold is set to 0.0 V.

BUS:B<x>:USB:DATAPLUSTHRESHold

This command sets or queries the USB D+ source threshold for the specified bus. The bus is specified by x.

Conditions Requires option SR-USB2.

Group Bus

Syntax BUS:B<x>:USB:DATAPLUSTHRESHold <NR3>
BUS:B<x>:USB:DATAPLUSTHRESHold?

Arguments B<x> is the number of the bus waveform.
<NR3> is the Plus threshold.

Examples BUS:B1:USB:DATAPLUSTHRESHold 50.0e-3 sets the threshold to 50.0 mV.
BUS:B1:USB:DATAPLUSTHRESHold? might return
:BUS:B1:USB:DATAPLUSTHRESHOLD 0.0E+0 indicating the threshold is set to 0.0 V.

BUS:B<x>:USB:LOWTHRESHold

This command sets or queries the USB data source threshold for the specified bus when the signal type is differential. The bus is specified by x.

Conditions Requires option SR-USB2.

Group Bus

Syntax BUS:B<x>:USB:LOWTHRESHold <NR3>
BUS:B<x>:USB:LOWTHRESHold?

Arguments B<x> is the number of the bus waveform.
<NR3> is the Low threshold.

Examples `BUS:B1:USB:LOWTHRESHold 50.0e-3` sets the threshold to 50.0 mV.

`BUS:B1:USB:LOWTHRESHold?` might return `:BUS:B1:USB:LOWTHRESHOLD 0.0E+0` indicating the threshold is set to 0.0 V.

BUS:B<x>:USB:SIGNALTYpe

This command sets or queries the USB signal type for the specified bus. The bus is specified by x.

Conditions Requires option SR-USB2.

Group Bus

Syntax `BUS:B<x>:USB:SIGNALTYpe {SINGLE|DIFF}`
`BUS:B<x>:USB:SIGNALTYpe?`

Arguments B<x> is the number of the bus waveform.

SINGLE specifies single-ended signals.

DIFF specifies differential signals.

Examples `BUS:B1:USB:SIGNALTYpe SINGLE` specifies single-ended signals.

`BUS:B1:USB:SIGNALTYpe?` might return `:BUS:B1:USB:SIGNALTYpe DIFF` indicating the signal type is differential.

BUS:B<x>:USB:SOUrce

This command sets or queries the USB data source when the signal type is differential for bus <x>. The bus number is specified by <x>.

Conditions Requires option SR-USB2.

Group Bus

Syntax `BUS:B<x>:USB:SOUrce {CH<x>|MATH<x>|REF<x>}`
`BUS:B<x>:USB:SOUrce?`

Related Commands [BUS:B<x>:USB:SOUrce:DMINus](#)
[BUS:B<x>:USB:SOUrce:DPLUs](#)

Arguments B<x> is the number of the bus waveform.
CH<x> specifies an analog channel as the data source for the specified USB bus.
MATH<x> specifies a math channel as the data source for the specified USB bus.
REF<x> specifies a reference waveform as the data source.

Examples BUS:B1:USB:SOURCE D2 sets the USB data source for bus B1 to D2.
BUS:B1:USB:SOURCE? might return :BUS:B1:USB:SOURCE D3, indicating that the USB data source for bus B1 is D3.

BUS:B<x>:USB:SOUrce:DMINus

This command sets or queries the USB D- (SDATAMINUS) source for bus <x> when the signal type is single ended. The bus number is specified by <x>.

Conditions Requires option SR-USB2.

Group Bus

Syntax BUS:B<x>:USB:SOUrce:DMINus
{CH<x>|CH<x>_D<x>|MATH<x>|REF<x>|REF<x>_D<x>}
BUS:B<x>:USB:SOUrce:DMINus?

Related Commands [BUS:B<x>:USB:SOUrce](#)
[BUS:B<x>:USB:SOUrce:DPLUs](#)
[BUS:B<x>:USB:SIGNALTYpe](#)

Arguments B<x> is the number of the bus waveform.
CH<x> specifies an analog channel as the D- source for the specified USB bus.
CH<x>_D<x> specifies a digital channel as the D- source for the specified USB bus.
MATH<x> specifies a math channel as the D- source for the specified USB bus.
REF<x> specifies a reference waveform as the source

REF<x>_D<x> specifies a digital reference waveform as the clock source waveform for the specified USB bus.

Examples `BUS:B1:USB:SOURCE:DMINUS CH2` sets the USB Data Source for D- input to CH2.

`BUS:B1:USB:SOURCE:DMINUS?` might return `:BUS:B1:USB:SOURCE:DMINUS CH4`, indicating that CH4 is set to be the D- input for USB data.

BUS:B<x>:USB:SOURce:DPLUs

This command sets or queries the USB dataPlus (SDATAPLUS) source for the specified bus when the signal type is single ended. The bus is specified by x.

Conditions Requires option SR-USB2.

Group Bus

Syntax `BUS:B<x>:USB:SOURce:DPLUS`
 `{CH<x> | CH<x>_D<x> | MATH<x> | REF<x> | REF<x>_D<x>}`
 `BUS:B<x>:USB:SOURce:DPLUS?`

Related Commands [BUS:B<x>:USB:SOURce](#)
 [BUS:B<x>:USB:SOURce:DMINus](#)
 [BUS:B<x>:USB:SIGNALTYpe](#)

Arguments B<x> is the number of the bus waveform.

CH<x> specifies an analog channel as the D+ source for the specified USB bus

CH<x>_D<x> specifies a digital channel as the D+ source for the specified USB bus

MATH<x> specifies a math channel as the D+ source for the specified USB bus.

REF<x> specifies a reference waveform as the source.

REF<x>_D<x> specifies a digital reference waveform as the clock source waveform for the specified USB bus.

Examples `BUS:B1:USB:SOURCE:DPLUS CH2` sets the USB Data Source for D+ input to CH2.

`BUS:B1:USB:SOURCE:DPLUS?` might return `:BUS:B1:USB:SOURCE:DPLUS CH3`, indicating that CH3 is set to be the D+ input for USB data.

BUS:B<x>:USB:THRESHold

This command sets or queries the USB DATA source High threshold for the specified bus when the signal source is differential. The bus is specified by x.

Conditions Requires option SR-USB2.

Group Bus

Syntax `BUS:B<x>:USB:THRESHold <NR3>`
`BUS:B<x>:USB:THRESHold?`

Arguments `B<x>` is the number of the bus waveform.
`<NR3>` is the USB DATA source High threshold for the specified bus.

Examples `BUS:B1:USB:THRESHold 50.0e-3` sets the threshold to 50.0 mV.
`BUS:B1:USB:THRESHold?` might return `:BUS:B1:USB:THRESHOLD 0.0E+0` indicating the threshold is set to 0.0 V.

BUS:DELeTe (No Query Form)

This command deletes the specified bus.

Group Bus

Syntax `BUS:DELeTe <QString>`

Arguments `<QString>` specifies the bus to delete and is of the form "B<NR1>", where `<NR1>` is ≥ 1 .

Examples `BUS:DELETE B1` deletes bus B1.

BUS:LIST? (Query Only)

This query returns a comma separated list of all currently defined buses.

Group	Bus
Syntax	BUS:LIST?
Returns	Returns all currently defined buses.
Examples	BUS:LIST? might return :BUS:LIST B1,B4 indicating the bus 1 and bus 4 are defined.

BUSTABLE:ADDNew (No Query Form)

Adds the specified bus table. Argument is of the form "TABLE<NR1>", where <NR1> is ≥ 1).

Group	Bus
Syntax	BUSTABle:ADDNew <QString>
Arguments	<QString> is a quoted string that is the name of the new bus table.
Examples	BUSTABLE:ADDNEW "Table1" adds bus table Table1.

BUSTABLE:DELeTe (No Query Form)

Deletes the specified bus table. Argument is of the form "TABLE<NR1>", where <NR1> is ≥ 1).

Group	Bus
Syntax	BUSTABle:DELeTe <QString>
Arguments	<QString> is a quoted string that is the name of the bus table to delete.

Examples `BUSTABLE:DELETE "Table1"` deletes bus table Table1.

BUSTABLE:LIST? (Query Only)

This query lists all currently defined bus tables.

Group Bus

Syntax `BUSTABLE:LIST?`

Returns Returns a list of all currently defined bus tables.

Examples `BUSTABLE:LIST?` might return `:BUSTABLE:LIST TABLE1` indicating TABLE1 is currently the only defined bus table.

BUSY? (Query Only)

This query-only command returns the status of the instrument. This command allows you to synchronize the operation of the instrument with your application program.

Group Status and Error

Syntax `BUSY?`

Related Commands [*OPC](#)
 [*WAI](#)

Returns `<NR1> = 0` means that the instrument is not busy processing a command whose execution time is extensive.

 `<NR1> = 1` means that the instrument is busy processing Commands that Generate an OPC Message (See Table 2-46.).

Examples `BUSY?` might return `:BUSY 1`, indicating that the instrument is currently busy.

*CAL? (Query Only)

This query-only command starts signal path calibration (SPC) and returns the status upon completion.

NOTE. When running SPC through the remote interface, calibration status cannot be obtained until after the SPC completes. SPC takes approximately 5 minutes per channel which means a total of 40 minutes on an 8-channel model. Any remote command that performs an action on the oscilloscope is also disabled until the SPC is complete.

Group	Calibration
Syntax	*CAL?
Returns	0 indicates SPC passed. -1 indicates SPC failed or did not complete.
Examples	*CAL? starts the signal path calibration and returns the status upon completion.

CALibrate? (Query Only)

This query returns the status of signal path calibration.

NOTE. When running SPC through the remote interface, calibration status cannot be obtained until after the SPC completes, which can take several minutes.

Group	Calibration
Syntax	CALibrate?
Examples	CALIBRATE? might return :CALIBRATE:INTERNAL:STATUS PASS, indicating the calibration status.

CALibrate:INTERNAL (No Query Form)

This command (no query form) starts the signal path calibration (SPC) of the instrument. You can use the [CALibrate:INTERNAL:STATUS?](#) query to return the current status of the signal path calibration of the instrument.

NOTE. When running SPC through the remote interface, calibration status cannot be obtained until after the SPC completes. SPC takes approximately 5 minutes per channel which means a total of 40 minutes on an 8-channel model. Any remote command that performs an action on the oscilloscope is also disabled until the SPC is complete.

Group Calibration

Syntax CALibrate:INTERNAL

Related Commands [CALibrate:INTERNAL:STATUS?](#)

Examples CALIBRATE:INTERNAL starts the signal path calibration of the instrument.

CALibrate:INTERNAL:START (No Query Form)

This command (no query form) starts the signal path calibration (SPC) of the analog channels. This command is the same as the [CALibrate:INTERNAL](#) command. You can use the [CALibrate:INTERNAL:STATUS?](#) query to return the current status of the signal path calibration of the instrument.

NOTE. When running SPC through the remote interface, calibration status cannot be obtained until after the SPC completes. SPC takes approximately 5 minutes per channel which means a total of 40 minutes on an 8-channel model. Any remote command that performs an action on the oscilloscope is also disabled until the SPC is complete.

Group Calibration

Syntax CALibrate:INTERNAL:START

Related Commands [CALibrate:INTERNAL:STATUS?](#)

Examples `CALIBRATE:INTERNAL:START` starts the signal path calibration.

CALibrate:INTERNAL:STATus? (Query Only)

This query-only command returns the current status of the signal path calibration.

NOTE. When running SPC through the remote interface, calibration status cannot be obtained until after the SPC completes. SPC takes approximately 5 minutes per channel which means a total of 40 minutes on an 8-channel model. Any remote command that performs an action on the oscilloscope is also disabled until the SPC is complete.

Group Calibration

Syntax `CALibrate:INTERNAL:STATus?`

Related Commands [*CAL?](#)

Returns This query will return one of the following:

- `INIT` indicates the instrument has not had signal path calibration run. The instrument may need to be readjusted at the Tektronix service center.
- `PASS` indicates that the signal path calibration completed successfully.
- `FAIL` indicates that the signal path calibration did not complete successfully.

Examples `CALIBRATE:INTERNAL:STATUS?` might return `PASS`, indicating that the current status of the signal path calibration is that the signal path calibration completed successfully.

CALibrate:PWRUpstatus? (Query Only)

This query-only command returns the current status of the power-up calibration.

Group Calibration

Syntax `CALibrate:PWRUpstatus?`

- Returns** This query will return one of the following:
- 0 Indicating the power-up calibration failed.
 - 1 Indicating the power-up calibration passed.

Examples CALIBRATE:PWRUPSTATUS? might return 0, indicating that the power-up calibration failed.

CH<n>:PRObe:STATUs? (Query Only)

Reads the probe unsigned integer error value.

Conditions Requires a probe that supports the relevant error messages.

Group Vertical

Syntax CH<n>:PRObe:STATUs?

Returns Returns an integer number that represents the sum total of binary error bits B0 – B15. The error bits are not displayed; they are concatenated into the integer value. The following is a list of the error for each bit.

- B0 – Probe disabled
- B1 – Jaws open
- B2 – Over range
- B3 – Probe temperature out of limits
- B4 – Degauss needed
- B5 – Probe tip missing
- B6 – Probe tip failed
- B7 – Probe tip not supported
- B8 through B15 – Reserved

Examples CH4:PROBE:STATUs? might return CH4:PROBE:STATUs 0, indicating that the probe is not reporting any errors.

CH4:PROBE:STATUs? might return CH4:PROBE:STATUs 20, indicating that the probe is reporting an over range error and a Degauss needed error.

CH4:PROBE:STATUs? might return CH4:PROBE:STATUs 2, indicating that the probe is reporting an open jaws error.

CH<x>? (Query Only)

This query-only command returns the vertical parameters for the specified channel. The channel is specified by x.

Group Vertical

Syntax CH<x>?

Examples CH1? might return the following vertical parameters for channel 1: :CH1:BANDWIDTH 1.0000E+09;COUPLING DC;DESKEW0.0000E+00;OFFSET 0.0000E+00;POSITION 0.0000E+00;SCALE 5.0000E-01;PROBCAL INIT;PROBE:GAIN 1.0000E+00;RESISTANCE 1.0000E+06;UNITS "V";ID:TYPE "1X";SERNUMBER "N/A";:CH1:PROBEFUNC:EXTATTEN 1.0000E+00;EXTUNITS "None";:CH1:LABEL:NAME "";XPOS 5;YPOS 5.

CH<x>:BANDwidth

This command sets or queries the selectable low-pass bandwidth limit filter of the specified channel. The channel is specified by x.

The query form of this command always returns the approximate realized bandwidth of the channel.

Available arguments depend upon the instrument and the attached accessories.

Group Vertical

Syntax CH<x>:BANDwidth {<NR3>|FUL1}
CH<x>:BANDwidth?

Arguments CH<x> is the channel number.

<NR3> is the desired bandwidth. The instrument rounds this value to an available bandwidth using geometric rounding and then uses this value to set the upper bandwidth.

FULL disables any optional bandwidth limiting. The specified channel operates at its maximum bandwidth.

Examples CH1:BANDWIDTH 20 sets the bandwidth of Channel 1 to 20 MHz.
CH2:BANDWIDTH?, might return :CH2:BANDWIDTH 500.0000E+06, indicating that there is bandwidth limiting on Channel 2.

CH<x>:CLIPping? (Query Only)

Queries whether the specified channel's input signal is clipping (exceeding) the channel A/D converter range. The channel is specified by x.

Group Vertical

Syntax CH<x>:CLIPping?

Related Commands [CH<x>:SCAla](#)
[CH<x>:POSition](#)

Returns This query will return one of the following:
0 indicates the channel is not clipping.
1 indicates the channel is clipping.

Examples CH4:CLIPping? might return CH4:CLIPping? 1, indicating that the input signal range is exceeding (clipping) the channel 4 A/D converter range.

CH<x>:COUPLing

This command sets or queries the input coupling setting for the specified analog channel. The channel is specified by x.

NOTE. The available arguments depend on the attached accessories.

Group Vertical

Syntax	CH<x>:COUPling {AC DC DCREJ} CH<x>:COUPling?
Arguments	CH<x> is the channel number. AC sets the specified channel to AC coupling. DC sets the specified channel to DC coupling. DCREJect sets DC Reject coupling when probes are attached that support that feature.
Examples	CH2:COUPLING AC sets Channel 2 coupling to AC. CH3:COUPLING? might return :CH3:COUPLING DC, indicating that Channel 3 is set to DC coupling.

CH<x>:DESKew

This command sets or queries the horizontal deskew time for the specified channel. The channel is specified by x.

Group	Vertical
Syntax	CH<x>:DESKew <NR3> CH<x>:DESKew?
Arguments	CH<x> is the channel number. <NR3> is the deskew time for this channel, ranging from -125 ns to +125 ns with a resolution of 40 ps. Out-of-range values are clipped.
Examples	CH4:DESKEW 5.0E-9 sets the deskew time for Channel 4 to 5 ns. CH2:DESKEW? might return :CH2:DESKEW 2.0000E-09, indicating that the deskew time for Channel 2 is set to 2 ns.

CH<x>:LABel:COLor

This command sets or queries the color of the specified channel label. The channel is specified by x.

Group	Vertical
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Syntax CH<x>:LABel:COLor <QString>

Arguments CH<x> is the channel number.
<QString> is the label color. To return the color to the default color, send an empty string as in this example: :CH5:LABEL:COLOR "".

Examples CH2:LABel:COLor "#FFFF00" sets the label color to yellow.
CH2:LABel:COLor? might return :CH2:LABEL:COLOR "#FF0000" indicating the color is red.

CH<x>:LABel:FONT:BOLD

This command sets or queries the bold state of the specified channel label. The channel is specified by x.

Group Vertical

Syntax CH<x>:LABel:FONT:BOLD {ON|OFF|<NR1>}

Arguments CH<x> is the channel number.
OFF argument turns off bold font.
ON argument turns on bold font.
<NR1> = 0 turns off bold font; any other value turns on bold font.

Examples CH2:LABel:FONT:BOLD OFF turns off the bold font.
CH2:LABel:FONT:BOLD? might return :CH2:LABEL:FONT:BOLD 1 indicating a bold font.

CH<x>:LABel:FONT:ITALic

This command sets or queries the italic state of the specified channel label. The channel is specified by x.

Group Vertical

Syntax CH<x>:LABel:FONT:ITALic {ON|OFF|<NR1>}

Arguments	CH<x> is the channel number. OFF argument turns off italic font. ON argument turns on italic font. <NR1> = 0 turns off italic font; any other value turns on italic font.
Examples	CH2:LABel:FONT:ITALic ON set the font to italic. CH2:LABel:FONT:ITALic? might return :CH2:LABEL:FONT:ITALIC 0 indicating the font is not italic.

CH<x>:LABel:FONT:SIZE

This command sets or queries the font size of the specified channel label. The channel is specified by x.

Group	Vertical
Syntax	CH<x>:LABel:FONT:SIZE <NR1>
Arguments	CH<x> is the channel number. <NR1> is the font size.
Examples	CH2:LABel:FONT:SIZE 16 sets the font size to 16 point size. CH2:LABel:FONT:SIZE? might return :CH2:LABEL:FONT:SIZE 20 indicating a 20 point font size.

CH<x>:LABel:FONT:TYPE

This command sets or queries the font type of the specified channel label, such as Arial or Times New Roman. The channel is specified by x.

Group	Vertical
Syntax	CH<x>:LABel:FONT:TYPE <QString>

Arguments	CH<x> is the channel number. <QString> is the specified font type.
Examples	CH2:LABel:FONT:TYPE "Monospace" sets the font to a mono space font. CH2:LABel:FONT:TYPE? might return :CH2:LABEL:FONT:TYPE "Frutiger LT Std 55 Roman".

CH<x>:LABel:FONT:UNDERline

This command sets or queries the underline state of the specified channel label. The channel is specified by x.

Group	Vertical
Syntax	CH<x>:LABel:FONT:UNDERline {ON OFF <NR1>}
Arguments	CH<x> is the channel number. OFF argument turns off underlined font. ON argument turns on underlined font. <NR1> = 0 turns off underlined font; any other value turns on underlined font.
Examples	CH2:LABel:FONT:UNDERline ON sets the font to underlined. CH2:LABel:FONT:UNDERline? might return :CH2:LABEL:FONT:UNDERLINE 0 indicating the font is not underlined.

CH<x>:LABel:NAME

This command sets or queries the label attached to the displayed waveform for the specified channel. The channel is specified by x.

Group	Vertical
Syntax	CH<x>:LABel:NAME <QString> CH<x>:LABel:NAME?

Arguments	CH<x> is the channel number. <QString> is an alphanumeric character string, ranging from 1 through 32 characters in length.
Examples	CH2:LABEL:NAME "Pressure" changes the waveform label for the Channel 2 waveform to "Pressure". CH3:LABEL:NAME? might return :CH3:LABEL:NAME "Force", indicating that the waveform label for the Channel 3 waveform is "Force".

CH<x>:LABel:XPOS

This command sets or queries the X-position of the specified channel label. The channel is specified by x.

Group	Vertical
Syntax	CH<x>:LABel:XPOS <NR3> CH<x>:LABel:XPOS?
Arguments	CH<x> is the channel number. <NR3> is the location (in pixels) where the waveform label for the selected channel is displayed, relative to the left edge of the screen.
Examples	CH3:LABEL:XPOS 5 moves the waveform label for Channel 3 so that it begins 5 pixels to the right of the left edge of the screen. CH2:LABEL:XPOS? might return :CH2:LABEL:XPOS 5, indicating that the waveform label for the Channel 2 currently 5 pixels to the right of the left edge of the screen.

CH<x>:LABel:YPOS

This command sets or queries the Y-position of the specified channel label. The channel is specified by x.

Group	Vertical
Syntax	CH<x>:LABel:YPOS <NR3> CH<x>:LABel:YPOS?

Arguments	CH<x> is the channel number. <NR3> is the location (in pixels) where the waveform label for the selected channel is displayed, relative to the baseline of the waveform. Positive values are above the baseline and negative values are below.
Examples	CH3:LABEL:YPOS -20 moves the waveform label for the Channel 3 20 pixels below the baseline of the waveform. CH2:LABEL:YPOS? might return :CH2:LABEL:YPOS 0, indicating that the waveform label for the Channel 2 is currently located at the baseline of the waveform.

CH<x>:OFFSet

This command sets or queries the vertical offset for the specified analog channel.

Group	Vertical
Syntax	CH<x>:OFFSet <NR3> CH<x>:OFFSet?
Arguments	CH<x> is the channel number. <NR3> is the offset value for the specified channel.
Examples	CH3:OFFSET 2.0E-3 sets the offset for Channel 3 to 2 mV. CH4:OFFSET? might return :CH4:OFFSET 1.0000E-03, indicating that the offset for Channel 4 is set to 1 mV.

CH<x>:POSition

This command sets or queries the vertical position for the specified analog channel.

Group	Vertical
Syntax	CH<x>:POSition <NR1>

Arguments CH<x> is the channel number.
<NR1> is the vertical position for the specified analog channel.

Examples CH2:POSition -2.0 sets the position to -2 divisions.
CH2:POSition? might return :CH2:POSITION -2.2400 indicating the position is -2.24 divisions.

CH<x>:PRObe? (Query Only)

This query-only command returns all information concerning the probe that is attached to the specified channel. The channel is specified by x.

Group Vertical

Syntax CH<x>:PRObe?

Related Commands [CH<x>:PROBECal?](#)

Examples CH2:PROBE? might return :CH2:PROBE:GAIN 1.0000E-01; RESISTANCE 1.0000E+07; UNITS "V"; ID:TYPE "10X" 'SERNUMBER "N/A" for a 10X probe, indicating that (among other parameters) the attenuation factor for the probe attached to Channel 2 is 100.0 mV (assuming that probe units are set to volts).

CH<x>:PRObe:AUTOZero (No Query Form)

This command executes the attached probe's Auto Zero function, for probes that support this feature. See your probe documentation for more details. The channel is specified by x.

Group Vertical

Syntax CH<x>:PRObe:AUTOZero EXECute

Arguments CH<x> is the channel number.
EXECute sets the probe attached to the specified channel to autozero.

Examples CH1:PROBE:AUTOZERO EXECUTE sets the probe attached to the Channel 1 to autozero.

CH<x>:PRObe:DEGAUSS (No Query Form)

This command starts a degauss cycle of the TekVPI probe attached to the specified channel. The channel is specified by x.

Group Vertical

Syntax CH<x>:PRObe:DEGAUSS EXECute

Arguments CH<x> is the channel number.
EXECute starts the degauss cycle.

Examples CH1:PROBE:DEGAUSS EXECUTE causes the probe attached to Channel 1 to degauss.

CH<x>:PRObe:DEGAUSS:STATE? (Query Only)

This command queries whether the probe attached to the specified channel requires a degauss operation. The channel is specified by x.

Group Vertical

Syntax CH<x>:PRObe:DEGAUSS:STATE?

Returns Required indicates the probe should be degaussed before taking measurements.
Recommended indicates the measurement accuracy might be improved by degaussing the probe.
Passed indicates the probe is degaussed.
Failed indicates the degauss operation failed.
Inprocess indicates the probe degauss operation is currently in progress.

Examples CH2:PROBE:DEGAUSS:STATE? might return :CH2:PROBE:DEGAUSS:STATE PASSED, indicating that the probe attached to the Channel 2 is degaussed.

CH<x>:PRObe:FORCEDRange

This command sets the attached TekVPI probe to the specified range, or it queries the range of the probe attached to the specified channel. The channel is specified by x.

Group	Vertical
Syntax	CH<x>:PRObe:FORCEDRange <NR3> CH<x>:PRObe:FORCEDRange?
Arguments	CH<x> is the channel number. <NR3> specifies the probe dynamic range.
Examples	If a TCP0030 current probe is attached to the Channel 1 input, CH1:PROBE:FORCEDRANGE 5.0 sets the attached probe to its 5 Ampere range. CH3:PROBE:FORCEDRANGE? might return :CH1:PROBE:FORCEDRANGE 30.0000, indicating that the range of the probe attached to the Channel 3 is set to 30 Amperes.

CH<x>:PRObe:GAIN? (Query Only)

This query-only command returns the gain factor of the probe that is attached to the specified channel. The channel is specified by x. The gain of a probe is the output divided by the input transfer ratio. For example, a common 10x probe has a gain of 0.1.

Group	Vertical
Syntax	CH<x>:PRObe:GAIN?
Examples	CH2:PROBE:GAIN? might return :CH2:PROBE:GAIN 100.0000E-3, indicating that the attached 10X probe delivers 0.1 V to the Channel 2 BNC for every 1.0 V applied to the probe input.

CH<x>:PRObe:ID? (Query Only)

This query-only command returns the type and serial number of the probe that is attached to the specified channel. The channel is specified by x.

Group	Vertical
Syntax	CH<x>:PRObe:ID?
Examples	CH2:PROBE:ID? might return :CH2:PROBE:ID:TYPE "10X";SERNUMBER "N/A", indicating that a passive 10X probe of unknown serial number is attached to Channel 2.

CH<x>:PRObe:ID:SERnumber? (Query Only)

This query-only command returns the serial number of the probe that is attached to the specified channel. The channel is specified by x.

NOTE. For Level 0 and 1 probes, the serial number will be "N/A".

Group	Vertical
Syntax	CH<x>:PRObe:ID:SERnumber?
Examples	CH1:PROBE:ID:SERNUMBER? might return :CH1:PROBE:ID:SERNUMBER "B010289", indicating that the serial number of the probe attached to Channel 1 is B010289.

CH<x>:PRObe:ID:TYPE? (Query Only)

This query-only command returns the type of probe that is attached to the specified channel. The channel is specified by x.

Group	Vertical
Syntax	CH<x>:PRObe:ID:TYPE?
Examples	CH1:PROBE:ID:TYPE? might return :CH1:PROBE:ID:TYPE "TCP0030", indicating that a TCP0030 current probe is attached to Channel 1.

CH<x>:PRObe:INPUTMode

Sets or queries the input mode of the probe that is attached to the specified channel.

Conditions Requires a probe with dual inputs.

Group Vertical

Syntax CH<x>:PRObe:INPUTMode {A|B|C|D}
CH<x>:PRObe:INPUTMode?

Arguments CH<x> is the channel number.

A sets the probe to send single-ended A signals to the instrument.

B sets the probe to send single-ended B signals to the instrument.

C sets the probe to send common-mode signals to the instrument.

D sets the probe to send differential signals to the instrument.

Examples CH8:PROBE:INPUTMODE C sets the probe input type to send common mode signals to channel 8.

CH6:PROBE:INPUTMODE? might return :CH6:PROBE:INPUTMODE A, indicating that the input mode of the probe that is attached to the Channel 6 is set to single ended A signals.

CH<x>:PRObe:INPUTMode:AOffset

Sets or queries the A mode offset control of the probe that is attached to the specified channel.

Conditions Requires a probe with dual inputs.

Group Vertical

Syntax CH<x>:PRObe:INPUTMode:AOffset <NR3>
CH<x>:PRObe:INPUTMode:AOffset?

Arguments	CH<x> is the channel number. <NR3> sets the A mode offset value, in vertical units (V or A).
Examples	CH2:PROBE:INPUTMODE:AOFFSET 5.0 sets the A mode offset control value of the probe that is attached to the Channel 2 to 5. CH1:PROBE:INPUTMODE:AOFFSET? might return CH1:PROBE:INPUTMODE:AOFFSET? 0.0000, indicating that the A mode offset value of the probe that is attached to the channel 1 is set to 0.0.

CH<x>:PROBe:INPUTMode:BOFFSet

Sets or queries the B mode offset control of the probe that is attached to the specified channel.

Conditions	Requires a probe with dual inputs.
Group	Vertical
Syntax	CH<x>:PROBe:INPUTMode:BOFFSet <NR3> CH<x>:PROBe:INPUTMode:BOFFSet?
Arguments	CH<x> is the channel number. <NR3> sets the B mode offset value, in vertical units (V or A).
Examples	CH2:PROBE:INPUTMODE:BOFFSET 5.0 sets the B mode offset value of the probe that is attached to the Channel 2 to 5. CH5:PROBE:INPUTMODE:BOFFSET? might return CH5:PROBE:INPUTMODE:BOFFSET? 1.2500, indicating that the B mode offset value of the probe that is attached to the channel 5 is set to 1.25.

CH<x>:PROBe:INPUTMode:COFFSet

Sets or queries the common mode offset value of the probe that is attached to the specified channel.

Conditions	Requires a probe with dual inputs.
-------------------	------------------------------------

Group	Vertical
Syntax	CH<x>:PRObe:INPUTMode:COFFSet <NR3> CH<x>:PRObe:INPUTMode:COFFSet?
Arguments	CH<x> is the channel number. <NR3> sets the C (common) mode offset value, in vertical units (V or A).
Examples	CH7:PROBE:INPUTMODE:COFFSET 3.2 sets the C mode offset value of the probe that is attached to the Channel 7 to 3.2. CH5:PROBE:INPUTMODE:COFFSET? might return CH5:PROBE:INPUTMODE:COFFSET? 1.0250, indicating that the C mode offset value of the probe that is attached to the channel 5 is set to 1.025.

CH<x>:PRObe:INPUTMode:DOFFSet

Sets or queries the differential offset value of the probe that is attached to the specified channel.

Conditions	Requires a probe with dual inputs.
Group	Vertical
Syntax	CH<x>:PRObe:INPUTMode:DOFFSet <NR3> CH<x>:PRObe:INPUTMode:DOFFSet?
Arguments	CH<x> is the channel number. <NR3> sets the D (differential) mode offset value, in vertical units (V or A).
Examples	CH7:PROBE:INPUTMODE:DOFFSET 3.2 sets the D mode offset value of the probe that is attached to the Channel 7 to 3.2. CH5:PROBE:INPUTMODE:DOFFSET? might return CH5:PROBE:INPUTMODE:DOFFSET? 1.0250, indicating that the D mode offset value of the probe that is attached to the channel 5 is set to 1.025.

CH<x>:PRObe:RESistance? (Query Only)

This query-only command returns the resistance of the probe that is attached to the specified channel. The channel is specified by x.

Group	Vertical
Syntax	CH<x>:PRObe:RESistance?
Examples	CH2:PROBE:RESISTANCE? might return :CH2:PROBE:RESISTANCE 1.0000E+06, indicating that the input resistance of the probe attached to Channel 2 is 1 MΩ.

CH<x>:PRObe:SET

This command sets or queries aspects of probe accessory user interfaces, for example probe attenuation factors or probe audible over range. The available arguments for this command will vary depending on the accessory you attach to the instrument. The channel is specified by x.

Group	Vertical
Syntax	CH<x>:PRObe:SET <QString> CH<x>:PRObe:SET?
Arguments	CH<x> is the channel number. <QString> is a quoted string representing a settable aspect of the attached accessory.
Examples	CH6:PRObe:SET "ATTENUATION 10X" sets the probe to 10X attenuation. CH6:PRObe:SET? might return CH6:PROBE:SET "ATTENUATION 25X" indicating that the probe is set to the selection for 25X attenuation.

CH<x>:PRObe:UNIts? (Query Only)

This query-only command returns a string describing the units of measure for the probe attached to the specified channel. The channel is specified by x.

Group Vertical

Syntax CH<x>:PRObe:UNITs?

Related Commands [CH<x>:PROBEFunc:EXTUnits](#)

Examples CH4:PROBE:UNITs? might return :CH4:PROBE:UNITs "V", indicating that the units of measure for the probe attached to Channel 4 are volts.

CH<x>:PROBECal? (Query Only)

This query-only command returns the probe calibration state for the specified channel. The channel is specified by x.

Group Vertical

Syntax CH<x>:PROBECal?

Returns This query will return one of the following:

- **Failed** signifies that the probe calibration has failed for the selected channel.
- **Default** signifies that the probe calibration has not yet been run for the selected channel.
- **Passed** signifies the probe calibration has passed for the selected channel.
- **Running** signifies the probe calibration is running.

Examples CH2:PROBECAL? might return :CH2:PROBECAL PASSED indicating that the probe calibration has passed for Channel 2.

CH<x>:PROBECOntrol

This command sets or queries multirange probe range-control policy preference of the probe that is attached to CH<x>. The channel number is specified by <x>.

Group Vertical

Syntax CH<x>:PROBEControl {AUTO|MANual}
CH<x>:PROBEControl?

Arguments CH<x> is the channel number.

AUTO sets the values. The probe range is automatically calculated.

MANual allows you to select various valid values for the probe connected to a particular channel.

Examples CH2:PROBECONTROL AUTO sets the values and the probe range is automatically calculated.

CH2:PROBECONTROL? might return :CH2:PROBECONTROL MANUAL indicating that you can select various valid values for the probe connected to channel 2.

CH<x>:PROBEFunc:EXTAtten

This command is used to specify the attenuation value as a multiplier to the given scale factor on the specified channel. The channel is specified by x.

The query form of this command returns the user-specified attenuation.

Group Vertical

Syntax CH<x>:PROBEFunc:EXTAtten <NR3>
CH<x>:PROBEFunc:EXTAtten?

Related Commands [CH<x>:PROBEFunc:EXTDBatten](#)

Arguments CH<x> is the channel number.

<NR3> is the attenuation value, which is specified as a multiplier in the range from 1.00E-10 to 1.00E+10.

Examples CH1:PROBEFUNC:EXTATTEN 167.00E-3 specifies an external attenuation, which is connected between the your input signal and the input of the probe attached to Channel 1.

CH2:PROBEFUNC:EXTATTEN? might return :CH2:PROBEFUNC:EXTATTEN 1.0000E+00, indicating that the probe attached to Channel 2 is connected directly to the user's signal.

CH<x>:PROBEFunc:EXTDBatten

This command sets or queries the input-output ratio (expressed in decibel units) of external attenuation or gain between the signal and the instrument input channels. The channel is specified by x.

The query form of this command returns the user-specified attenuation in decibels.

Group Vertical

Syntax CH<x>:PROBEFunc:EXTDBatten <NR3>
CH<x>:PROBEFunc:EXTDBatten?

Related Commands [CH<x>:PROBEFunc:EXTAtten](#)

Arguments CH<x> is the channel number.

<NR3> is the attenuation value, which is specified in the range from -200.00 dB to 200.00 dB.

Examples CH3:PROBEFunc:EXTDBATTEN 2.5 specifies an external 2.5 dB attenuator on Channel 3.

CH1:PROBEFunc:EXTDBATTEN? might return
:CH1:PROBEFunc:EXTDBATTEN 2.5000E+00, indicating that the attenuation for Channel 1 is 2.5 dB.

CH<x>:PROBEFunc:EXTUnits

This command sets the unit of measurement for the external attenuator of the specified channel. The channel is specified by x. The alternate units are used if they are enabled. Use the [CH<x>:PROBEFunc:EXTUnits:STATE](#) command to enable or disable the alternate units.

Group Vertical

Syntax CH<x>:PROBEFunc:EXTUnits <QString>
CH<x>:PROBEFunc:EXTUnits?

Related Commands [CH<x>:PRObe:UNIts?](#)

Arguments	CH<x> is the channel number. <QString> indicates the attenuation unit of measurement for the specified channel.
Examples	CH4:PROBEFunc:EXTUNITS "Pascals" sets the unit of measurement for the Channel 4 external attenuator. CH2:PROBEFunc:EXTUNITS? might return :CH2:PROBEFunc:EXTUNITS "Pascals", indicating that the Channel 2 external attenuator units of measurement are Pascals.

CH<x>:PROBEFunc:EXTUnits:STATE

This command sets or queries the custom units enable state for the specified channel. The channel is specified by x.

Group	Vertical
Syntax	CH<x>:PROBEFunc:EXTUnits:STATE {ON OFF <NR1>}
Arguments	CH<x> is the channel number. OFF argument turns off external units. ON argument turns on external units. <NR1> = 0 turns off external units; any other value turns on external units.
Examples	CH2:PROBEFunc:EXTUnits:STATE ON turns on external units. CH2:PROBEFunc:EXTUnits:STATE? might return :CH2:PROBEFunc:EXTUNITS:STATE 0 indicating that external units are off for the specified channel.

CH<x>:SCAle

This command sets or returns the vertical scale for the specified analog channel. The channel is specified by x.

Group	Vertical
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Syntax	<code>CH<x>:SCALE <NR3></code>
Arguments	<code>CH<x></code> is the channel number. <code><NR3></code> is the vertical scale for the specified analog channel.
Examples	<code>CH2:SCALE 200E-3</code> sets the scale to 200 mV per division. <code>CH2:SCALE?</code> might return <code>:CH2:SCALE 500.0000E-3</code> indicating the vertical scale for the specified channel is 500 mV per division.

CH<x>:SCALERATio

This command sets or returns the scale ration for the specified analog channel.

Group	Horizontal
Syntax	<code>CH<x>:SCALERATio <NR2></code> <code>CH<x>:SCALERATio?</code>
Arguments	<code>CH<x></code> is the channel number. <code><NR2></code> is the scale ration for the specified analog channel.
Examples	<code>CH2:SCALERATio 2.0</code> sets the scale ratio to 2.0. <code>CH2:SCALERATio?</code> might return <code>:CH2:SCALERATIO 1.0000</code> indicating the scale ratio is 1.0.

CH<x>:TERmination

This command sets or queries the vertical termination for the specified analog channel. The channel is specified by x.

NOTE. The available arguments depend on the instrument model and the attached accessories.

Group	Vertical
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Syntax CH<x>:TERmination <NR3>
CH<x>:TERmination?

Arguments CH<x> is the channel number.

<NR3> specifies the channel input resistance, which can be specified as 50 Ω or 1,000,000 Ω .

Examples CH4:TERMINATION 50.0E+0 establishes 50 Ω impedance on Channel 1.

CH2:TERMINATION? might return :CH2:TERMINATION 50.0E+0, indicating that Channel 2 is set to 50 Ω impedance.

CH<x>:VTERm:BIAS

Sets or queries the termination bias voltage for the specified channel (if control is available).

Conditions Requires a probe with dual inputs.

Group Vertical

Syntax CH<x>:VTERm:BIAS <NR3>
CH<x>:VTERm:BIAS?

Arguments CH<x> is the channel number.

<NR> is the termination voltage.

Examples CH1:VTERm:BIAS 1.5 sets the termination bias voltage on channel 1 to 1.5 Volts

CH2:VTERm:BIAS? might return CH2:VTERm:BIAS 0.0000, indicating that the termination bias voltage for channel 2 is set to 0 volts.

CH<x>_DALL:LABel:COLor

This command sets or queries the color of the specified digital group label. The channel is specified by x.

Group Digital

Syntax	<code>CH<x>_DALL:LABe1:COLor <QString></code>
Arguments	<code>CH<x></code> is the channel number. <code><QString></code> is the color of the digital group label. To return the color to the default color, send an empty string as in this example: <code>:CH5_DALL:LABEL:COLOR ""</code>.
Examples	<code>CH1_DALL:LABe1:COLor "#FF0000"</code> sets the font color to red. <code>CH1_DALL:LABe1:COLor?</code> might return <code>:CH1_DALL:LABEL:COLOR "#FFFF00"</code> indicating the font color is yellow.

CH<x>_DALL:LABe1:FONT:BOLD

This command sets or queries the bold state of the specified digital group. The channel is specified by x.

Group	Digital
Syntax	<code>CH<x>_DALL:LABe1:FONT:BOLD {ON OFF <NR1>}</code>
Arguments	<code>CH<x></code> is the channel number. OFF argument turns off bold font. ON argument turns on bold font. <code><NR1> = 0</code> turns off bold font; any other value turns on bold font.
Examples	<code>CH1_DALL:LABe1:FONT:BOLD ON</code> sets the font to bold. <code>CH1_DALL:LABe1:FONT:BOLD?</code> might return <code>:CH1_DALL:LABEL:FONT:BOLD 0</code> indicating the font is not bold.

CH<x>_DALL:LABe1:FONT:ITALic

This command sets or queries the italic state of the specified digital group. The channel is specified by x.

Group	Digital
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Syntax CH<x>_DALL:LABeI:FONT:ITALiC {ON|OFF|<NR1>}

Arguments CH<x> is the channel number.
 OFF argument turns off italic font.
 ON argument turns on italic font.
 <NR1> = 0 turns off italic font; any other value turns on italic font.

Examples CH1_DALL:LABeI:FONT:ITALiC 1 turns on italic font.
 CH1_DALL:LABeI:FONT:ITALiC? might return
 :CH1_DALL:LABEL:FONT:ITALIC 0 indicating the font is not italic.

CH<x>_DALL:LABeI:FONT:SIZE

This command sets or queries the font size of the specified digital group. The channel is specified by x.

Group Digital

Syntax CH<x>_DALL:LABeI:FONT:SIZE <NR1>

Arguments CH<x> is the channel number.
 <NR1> is the font size.

Examples CH1_DALL:LABeI:FONT:SIZE 16 sets the font size to 16 points.
 CH1_DALL:LABeI:FONT:SIZE? might return :CH1_DALL:LABEL:FONT:SIZE
 20 indicating the font size is 20 points.

CH<x>_DALL:LABeI:FONT:TYPE

This command sets or queries the font type of the specified digital group, such as Arial or Times New Roman. The channel is specified by x.

Group Digital

Syntax CH<x>_DALL:LABeI:FONT:TYPE <QString>

Arguments	CH<x> is the channel number. <QString> is the font type.
Examples	CH1_DALL:LABel:FONT:TYPE "Monospace" sets the font to a monospace font. CH1_DALL:LABel:FONT:TYPE? might return :CH1_DALL:LABEL:FONT:TYPE "Frutiger LT Std 55 Roman".

CH<x>_DALL:LABel:FONT:UNDERline

This command sets or queries the underline state of the specified digital group. The channel is specified by x.

Group	Digital
Syntax	CH<x>_DALL:LABel:FONT:UNDERline {ON OFF <NR1>}
Arguments	CH<x> is the channel number. OFF argument turns off underline font. ON argument turns on underline font. <NR1> = 0 turns off underline font; any other value turns on underline font.
Examples	CH1_DALL:LABel:FONT:UNDERline ON specifies an underlined font. CH1_DALL:LABel:FONT:UNDERline? might return :CH1_DALL:LABEL:FONT:UNDERLINE 0 indicating underline is off.

CH<x>_DALL:LABel:NAME

This command sets or queries the label of the specified digital group. The channel is specified by x.

Group	Digital
Syntax	CH<x>_DALL:LABel:NAME <QString>

Arguments	CH<x> is the channel number. <QString> is the name of the group.
Examples	CH1_DALL:LABe1:NAME "Clock Out" sets the label name to Clock Out. CH1_DALL:LABe1:NAME? might return :CH1_DALL:LABEL:NAME "This is the digital name".

CH<x>_D<x>:LABe1:COLor

This command sets or queries the color of the label of the specified digital bit.
The channel is specified by x.

Group	Digital
Syntax	CH<x>_D<x>:LABe1:COLor <QString>
Arguments	CH<x> is the channel number. <QString> is the label color. To return the color to the default color, send an empty string as in this example: :CH5_D1:LABEL:COLOR "".
Examples	CH1_D1:LABe1:COLor "#FF0000" sets the color to red. CH1_D1:LABe1:COLor? might return :CH1_D1:LABEL:COLOR "#FFFF00" indicating the color is yellow.

CH<x>_D<x>:LABe1:FONT:BOLD

This command sets or queries the bold state of the label of the specified digital bit.
The channel is specified by x.

Group	Digital
Syntax	CH<x>_D<x>:LABe1:FONT:BOLD {ON OFF <NR1>}
Arguments	CH<x> is the channel number. OFF argument turns off bold font.

ON argument turns on bold font.

<NR1> = 0 turns off bold font; any other value turns on bold font.

Examples CH1_D1:LABe1:FONT:BOLD ON sets the font to bold.

CH1_D1:LABe1:FONT:BOLD? might return :CH1_D1:LABEL:FONT:BOLD 0 indicating the font is not bold.

CH<x>_D<x>:LABe1:FONT:ITALic

This command sets or queries the italic state of the label of the specified digital bit. The channel is specified by x.

Group Digital

Syntax CH<x>_D<x>:LABe1:FONT:ITALic {ON|OFF|<NR1>}

Arguments CH<x> is the channel number.

OFF argument turns off italic font.

ON argument turns on italic font.

<NR1> = 0 turns off italic font; any other value turns on italic font.

Examples CH1_D1:LABe1:FONT:ITALic OFF turns off italic font.

CH1_D1:LABe1:FONT:ITALic? might return :CH1_D1:LABEL:FONT:ITALIC 1 indicating the font is italic.

CH<x>_D<x>:LABe1:FONT:SIZE

This command sets or queries the font size of the label of the specified digital bit. The channel is specified by x.

Group Digital

Syntax CH<x>_D<x>:LABe1:FONT:SIZE <NR1>

Arguments	CH<x> is the channel number. <NR1> is the font size.
Examples	CH1_D1:LABEl:FONT:SIZE 16 sets the font size to 16 points. CH1_D1:LABEl:FONT:SIZE? might return :CH1_D1:LABEL:FONT:SIZE 20 indicating the font size is 20 points.

CH<x>_D<x>:LABEl:FONT:TYPE

This command sets or queries the font type of the label of the specified digital bit, such as Arial or Times New Roman. The channel is specified by x.

Group	Digital
Syntax	CH<x>_D<x>:LABEl:FONT:TYPE <QString>
Arguments	CH<x> is the channel number. <QString> is the font type of the label.
Examples	CH1_D1:LABEl:FONT:TYPE "Monospace" sets the font to Monospace. CH1_D1:LABEl:FONT:TYPE? might return :CH1_D1:LABEL:FONT:TYPE "Frutiger LT Std 55 Roman".

CH<x>_D<x>:LABEl:FONT:UNDERline

This command sets or queries the underline state of the label of the specified digital bit. The channel is specified by x.

Group	Digital
Syntax	CH<x>_D<x>:LABEl:FONT:UNDERline {ON OFF <NR1>}
Arguments	CH<x> is the channel number. OFF argument turns off underline font. ON argument turns on underline font.

<NR1> = 0 turns off underline font; any other value turns on underline font.

Examples CH1_D1:LABel:FONT:UNDERline ON turns on underline font.
CH1_D1:LABel:FONT:UNDERline? might return
:CH1_D1:LABEL:FONT:UNDERLINE 0 indicating the underline font is off.

CH<x>_D<x>:LABel:NAME

Sets or queries the label of the specified digital bit. The channel is specified by x.

Group Digital

Syntax CH<x>_D<x>:LABel:NAME <QString>

Arguments CH<x> is the channel number.
<QString> is the label.

Examples CH1_D1:LABel:NAME "Clock in" sets the name to Clock in.
CH1_D1:LABel:NAME? might return :CH1_D1:LABEL:NAME "Digital 1".

CLEAR (No Query Form)

This command clears acquisitions, measurements, and waveforms.

Group Miscellaneous

Syntax CLEAR

Examples CLEAR clears all acquisitions, measurements, and waveforms.

*CLS (No Query Form)

This command (no query form) clears the following:

- Event Queue
- Standard Event Status Register
- Status Byte Register (except the MAV bit)

If the *CLS command immediately follows an <EOI>, the Output Queue and MAV bit (Status Byte Register bit 4) are also cleared. MAV indicates that information is in the output queue. The device clear (DCL) control message will clear the output queue and thus MAV. *CLS does not clear the output queue or MAV.

*CLS can suppress a Service Request that is to be generated by an *OPC. This will happen if a single sequence acquisition operation is still being processed when the *CLS command is executed.

Group Status and Error

Syntax *CLS

Related Commands [DESE](#)
[*ESE](#)
[*ESR?](#)
[EVENT?](#)
[EVMsg?](#)
[*SRE](#)
[*STB?](#)

Examples *CLS clears the instrument status data structures.

CONFIGuration:ANALog:BANDWidth? (Query Only)

This command queries the maximum licensed bandwidth of the instrument.

Group Vertical

Syntax CONFIGuration:ANALog:BANDwidth?

Returns	The maximum licensed bandwidth of the instrument is returned.
Examples	<code>CONFIGuration:ANALog:BANDwidth?</code> might return <code>:CONFIGURATION:ANALOG:BANDWIDTH 2.0000E+9</code> indicating the bandwidth is 2.0 GHz.

CURVe

This command transfers waveform data to and from the instrument. Each waveform that is transferred has an associated waveform preamble that contains information such as data format and scale.

The `CURVe?` query transfers data from the instrument. The data source is specified by the `DATA:SOURce` command. The first and last data points are specified by the `DATA:START` and `DATA:STOP` commands.

The `CURVe` command transfers waveform data to the instrument. Only one waveform can be transferred at a time. The waveform will only be displayed if the reference is displayed.

For digital sources, `CH<x>_D<n>` or `CH<x>_DALL`, when the `:DATA:WIDTH` is 1, the returned data is state only. When the `:DATA:WIDTH` is 2, the returned data is transition data with 2 bits per digital channel representing the transition information as follows:

- 0 0 low
- 0 1 high
- 1 1 multiple transitions in interval ending with high
- 1 0 multiple transitions in interval ending with low

For individual digital channels (such as `CH<x>_D<n>`), `:DATA:WIDTH 2` provides the 2-bit transition data with the upper 14 bits zero. `:DATA:WIDTH 1` provides only the state in the LSB with the upper 7 bits all zero.

For `CH<x>_DALL` sources, `:DATA:WIDTH 2` provides the 2-bit transition data for each of the 8 constituent channels with the D7 bit represented in the 2 most significant bits, D6 in the next 2, and so on. `:DATA:WIDTH 1` provides the states of each of the 8 constituent channels with D7 represented in the most significant bit and D0 in the least significant bit.

Depending on the sample rate, multi-transition data may not be available and `:CURVe?` queries for digital channels with `:DATA:WIDTH 2` may result in a warning event "Execution warning: Multi-transition data not available". In this case, the transition data returned will be 0 0 or 0 1.

For MATH sources, only 8-byte double precision floating point data is returned in `:CURVe?` queries.

Group	Waveform Transfer
Syntax	CURVe {<Block> <asc curve>} CURVe?
Related Commands	DATA:SOURce DATA:START DATA:STOP SAVE:WAVEform SAVEON:WAVEform:FILEFormat WFMOutpre?
Arguments	<Block> is the waveform data in binary format. The waveform is formatted as: #<x><yyy><data><newline>, where: <x> is the number of y bytes. For example, if <yyy>=500, then <x>=3) NOTE. <x> is hexadecimal format. The letters A-F denote several y bytes between 10 and 15 digits. <yyy> is the number of bytes to transfer. If width is 1, then all bytes on the bus are single data points. If width is 2, then all bytes on the bus are 2-byte pairs. If width is 4, then all bytes on the bus are 4-byte pairs. <data> is the curve data. <newline> is a single byte new line character at the end of the data. <asc curve> is the waveform data in ASCII format. The format for ASCII data is <NR1>[,<NR1>...], where each <NR1> represents a data point.
Examples	CURVe <Block> sets the format of the waveform data, transferred to and from the instrument, to binary format. CURVe? with ASCII encoding, start and stop of 1 and 10 respectively, and a width set to 1 might return :CURVE 61,62,61,60,60,-59,-59,-58,-58,-59

NOTE. Curve data is transferred from the instrument asynchronously and, depending upon the length of the curve record, such transfers can require several seconds to complete. During this time, the instrument will not respond to user controls. You can interrupt these asynchronous data transfers by sending a device clear message to the instrument or by interrupting the query with another command or query. Verify that curve data is completely transferred.

It is recommended that you follow such queries with an **ESR?* query and verify that the error bit returned and, if set, check the event queue to ascertain the reason for the error. If the error was caused by an interrupted query, then the asynchronous data transfer had not completed when the **ESR?* query was sent. In this case, you might need increase your program's time-out value to ensure that all data is transferred and read.

DATa

This command sets or queries the format and location of the waveform data that is transferred with the [CURVe](#) command.

Group Waveform Transfer

Syntax DATa {INIT|SNAP}
DATa?

Related Commands [CURVe](#)
[DATa:STARt](#)
[DATa:STOP](#)
[DATa:ENCdg](#)
[SAVEON:WAVEform:FILEFormat](#)
[WFMOutpre:NR_Pt?](#)

Arguments INIT initializes the waveform data parameters to their factory defaults except for DATa:STOP, which is set to the current acquisition record length.

SNAP Sets DATa:STARt and DATa:STOP to match the current waveform cursor positions of WAVEVIEW1 CURSOR1 if these waveform cursors are currently on. If these waveform cursors are not on when the DATa SNAP command is sent, it is silently ignored and DATa:STARt and :STOP remain unchanged.

If either of the cursors is outside the record, DATa:STARt and :STOP are set to nearest values within the record. This will cause DATa:STARt and/or :STOP to snap to the beginning or end of the record whichever is nearest, and the following

event is set: 500, "Execution warning", "One or both cursors outside of waveform record".

If there is no acquired waveform for the cursor source, then DATA:START and :STOP remain unchanged and the following event is set: 500, "Execution warning", "No acquired waveform record".

If DATA:START is greater than :STOP, the values of DATA:START and STOP are swapped and the following event is set: 530, "Data start > stop, Values swapped internally".

Examples DATA INIT initializes the waveform data parameters to their factory defaults.

DATA? might return DATA:ENCDG RIBINARY;SOURCE CH1;START 1;STOP 1000;FRAMESTART 1;FRAMESTOP 1000.

DATA:ENCdg

This command sets or queries the format of outgoing waveform data. This command is equivalent to setting [WFMOutpre:ENCdg](#), [WFMOutpre:BN_Fmt](#), and [WFMOutpre:BYT_Or](#). Setting the DATA:ENGdg value causes the corresponding WFMOutpre values to be updated and vice versa.

NOTE. *Values are constrained (for outbound data) to the format of the data specified by DATA:SOURce.*

Group Waveform Transfer

Syntax DATA:ENCdg {ASCII|RIBinary|RPBinary|FPBinary|SRIBinary|SRPbinary|SFPbinary}
DATA:ENCdg?

Related Commands [WFMOutpre:ENCdg](#)
[WFMOutpre:BN_Fmt](#)
[WFMOutpre:BYT_Or](#)

Arguments

- **ASCIi** specifies the ASCII representation of signed INT, FLOAT. If ASCII is the value, then :BN_Fmt and :BYT_Or are ignored.

- **RIBinary** specifies signed integer data point representation with the most significant byte transferred first.

When :BYT_Nr is 1, the range is from -128 through 127. When :BYT_Nr is 2, the range is from -32,768 through 32,767. When :BYT_Nr is 8, then the waveform being queried is set to Fast Acquisition mode. Center screen is 0 (zero). The upper limit is the top of the screen and the lower limit is the bottom of the screen. This is the default argument.

- **RPBinary** specifies the positive integer data-point representation, with the most significant byte transferred first.

When :BYT_Nr is 1, the range from 0 through 255. When :BYT_Nr is 2, the range is from 0 to 65,535. When :BYT_Nr is 8, then the waveform being queried is set to Fast Acquisition mode. The center of the screen is 127. The upper limit is the top of the screen and the lower limit is the bottom of the screen.

- **FPBinary** specifies the floating point (width = 4) data.

The range is from -3.4×10^{38} to 3.4×10^{38} . The center of the screen is 0. The upper limit is the top of the screen and the lower limit is the bottom of the screen.

The **FPBinary** argument is only applicable to math waveforms or ref waveforms saved from math waveforms.

- **SRIbinary** is the same as **RIBinary** except that the byte order is swapped, meaning that the least significant byte is transferred first. This format is useful when transferring data to IBM compatible PCs.
- **SRPbinary** is the same as **RPBinary** except that the byte order is swapped, meaning that the least significant byte is transferred first. This format is useful when transferring data to PCs.
- **SFPbinary** specifies floating point data in IBM PC format. The **SFPbinary** argument only works on math waveforms or ref waveforms saved from math waveforms.

Table 2-45: DATa and WFMOutpre Parameter Settings

DATa:ENCdg Setting	WFMOutpre Settings			
	:ENCdg	:BN_Fmt	:BYT_Or	:BYT_NR
ASCIi	ASC	N/A	N/A	1,2,4,8
RIBinary	BIN	RI	MSB	1,2,8
RPBinary	BIN	RP	MSB	1,2,8
FPBinary	BIN	FP	MSB	4
SRIbinary	BIN	RI	LSB	1,2,8

Table 2-45: DATA and WFMOutpre Parameter Settings (cont.)

DATA:ENCdg Setting	WFMOutpre Settings			
	:ENCdg	:BN_Fmt	:BYT_Or	:BYT_NR
SRPbinary	BIN	RP	LSB	1,2,8
SFPbinary	BIN	FP	LSB	4

Examples DATA:ENCDG RPBinary sets the data encoding format to be a positive integer where the most significant byte is transferred first.

DATA:ENCDG? might return :DATA:ENCDG SRPBINARY for the format of the outgoing waveform data.

DATA:SOURce

This command sets or queries the location of waveform data that is transferred from the instrument by the CURVe? Query.

Group Waveform Transfer

Syntax DATA:SOURce <wfm> [<, ><wfm>]
DATA:SOURce?

Related Commands [CURVe](#)
[DATA](#)

Arguments <wfm> is the location of the waveform data that will be transferred from the instrument to the controller. It can consist of CH<x>, MATH<x>, REF<x>, DIGITALALL. Note that digital data is transferred as 16-bit data, with the least-significant bit representing D0, and the most-significant bit representing D15.

<wfm> can consist of the following:

CH<x> selects the specified analog channel as the source.

MATH<x> selects the specified reference waveform as the source. The reference number is specified by x, which ranges from 1 through 4.

REF<x> selects the specified reference waveform as the source. The reference number is specified by x, which ranges from 1 through 8.

CH<x>_D<x> selects the specified digital channel.

CH<x>_DA11 selects the specified supper channel group of digital channels.

DIGITALALL selects digital waveforms as the source. The Digital data is transferred as 16-bit data, with the least-significant bit representing D0, and the most-significant bit representing D15. The LSB always contains D0-D7 and MSB always contains D8-D15 data.

Examples `DATA:SOURCE CH1` specifies that the CH1 waveforms will be transferred in the next `CURVe?` query.

`DATA:SOURCE?` might return `:DATA:SOURCE REF3`, indicating that the source for the waveform data which is transferred using a `CURVe?` query is reference 3.

DATA:SOURCE:AVAILable? (Query Only)

This query returns a list of enumerations representing the source waveforms that are currently available for `:CURVe?` queries. This means that the waveforms have been acquired. If there are none, NONE is returned.

Group Waveform Transfer

Syntax `DATA:SOURCE:AVAILable?`

Related Commands [CURVe](#)

Returns Returns a list of source waveforms that are currently available for `:CURVe?` queries.

Examples `DATA:SOURCE:AVAILABLE?` might return `:DATA:SOURCE:AVAILABLE CH2,CH3,CH6,CH7,MATH1,REF1` indicating that CH2, CH3, CH6, CH7, MATH1, and REF1 are available.

DATA:START

This command sets or queries the starting data point for waveform transfer. This command allows for the transfer of partial waveforms to and from the instrument.

Group Waveform Transfer

Syntax `DATA:START <NR1>`
`DATA:START?`

Related Commands [CURVe](#)
[DATAa](#)
[DATAa:STOP](#)
[WFMOutpre:NR_Pt?](#)

Arguments <NR1> is the first data point that will be transferred, which ranges from 1 to the record length. Data will be transferred from <NR1> to DATAa:STOP or the record length, whichever is less. If <NR1> is greater than the record length, the last data point in the record is transferred.

DATAa:STARTt and DATAa:STOP are order independent. When DATAa:STOP is greater than DATAa:STARTt, the values will be swapped internally for the CURVe? query.

Examples DATA:START 10 specifies that the waveform transfer will begin with data point 10.

DATA:START? might return :DATA:START 214, indicating that data point 214 is the first waveform data point that will be transferred.

DATAa:STOP

This command sets or queries the last data point that will be transferred when using the [CURVe?](#) query. This command allows for the transfer of partial waveforms to the controller.

Changes to the record length value are not automatically reflected in the data:stop value. As record length is varied, the DATAa:STOP value must be explicitly changed to ensure the entire record is transmitted. In other words, curve results will not automatically and correctly reflect increases in record length if the distance from DATAa:STARTt to DATAa:STOP stays smaller than the increased record length.

NOTE. When using the [CURVe](#) command, DATAa:STOP is ignored.

Group Waveform Transfer

Syntax DATAa:STOP <NR1>
DATAa:STOP?

Related Commands [CURVe](#)
[DATAa](#)
[DATAa:STARTt](#)

[WFMOutpre:NR_Pt?](#)

Arguments	<NR1> is the last data point that will be transferred, which ranges from 1 to the record length. If <NR1> is greater than the record length, then data will be transferred up to the record length. If both DATA:START and DATA:STOP are greater than the record length, the last data point in the record is returned. DATA:START and DATA:STOP are order independent. When DATA:STOP is less than DATA:START, the values will be swapped internally for the CURVE? query. If you always want to transfer complete waveforms, set DATA:START to 1 and DATA:STOP to the maximum record length, or larger.
Examples	DATA:STOP 15000 specifies that the waveform transfer will stop at data point 15000. DATA:STOP? might return :DATA:STOP 14900, indicating that 14900 is the last waveform data point that will be transferred.

DATA:WIDTH

This command specifies the width, in bytes per point, for waveform data transferred from the oscilloscope via the CURVE? query. (This command is synonymous with WFMOutpre:BYT_Nr.)

Group	Waveform Transfer
Syntax	DATA:WIDTH <NR1> DATA:WIDTH?
Related Commands	WFMOutpre:BYT_Nr
Arguments	<NR1> is an integer that indicates the number of bytes per point for the outgoing waveform data when queried using the CURVE? command. For analog channels the values can be 1 or 2. For digital channels, the values can be 1 or 2. For the digital collection, the values can be 4 or 8.
Examples	DATA:WIDTH 1 sets the data width to 1 byte. DATA:WIDTH? might return :DATA:WIDTH 1 indicating the width, in bytes per point, for waveform data transferred by the CURVE? query is 1 byte.

DATE? (Query Only)

This command queries the date that the instrument displays.

Group Miscellaneous

Syntax DATE?

Related Commands [TIMe?](#)

Returns <QString> is a date in the form “yyyy-mm-dd” where yyyy refers to a four-digit year number, mm refers to a two-digit month number from 01 to 12, and dd refers to a two-digit day number in the month.

Examples DATE? might return :DATE 2000-01-24, indicating the current date is set to January 24, 2000.

*DDT

This command allows you to specify a command or a list of commands that are executed when the instrument receives a [*TRG](#) command. Define Device Trigger (*DDT) is a special alias that the *TRG command uses.

Group Miscellaneous

Syntax *DDT {<Block>|<QString>}
*DDT?

Related Commands [ALias](#)
[*TRG](#)

Arguments <Block> is a complete sequence of program messages. The messages can contain only valid commands that must be separated by semicolons and must follow all rules for concatenating commands. The sequence must be less than or equal to 80 characters. The format of this argument is always returned as a query.

<QString> is a complete sequence of program messages. The messages can contain only valid commands that must be separated by semicolons and must follow all rules for concatenating commands. The sequence must be less than or equal to 80 characters.

Examples *DDT #217ACQUIRE:STATE RUN specifies that the acquisition system will be started each time a *TRG command is sent.

DESE

This command sets and queries the bits in the Device Event Status Enable Register (DESER). The DESER is the mask that determines whether events are reported to the Standard Event Status Register (SESR), and entered into the Event Queue. For a more detailed discussion of the use of these registers, see Registers.

Group Status and Error

Syntax DESE <NR1>
DESE?

Related Commands [*CLS](#)
[*ESE](#)
[*ESR?](#)
[EVENT?](#)
[EVMsg?](#)
[*SRE](#)
[*STB?](#)

Arguments <NR1> The binary bits of the DESER are set according to this value, which ranges from 1 through 255. For example, DESE 209 sets the DESER to the binary value 11010001 (that is, the most significant bit in the register is set to 1, the next most significant bit to 1, the next bit to 0, etc.).

The power-on default for DESER is all bits set if *PSC is 1. If *PSC is 0, the DESER maintains the previous power cycle value through the current power cycle.

NOTE. *Setting the DESER and ESER to the same value allows only those codes to be entered into the Event Queue and summarized on the ESB bit (bit 5) of the Status Byte Register. Use the [*ESE](#) command to set the ESER.*

Examples DESE 209 sets the DESER to binary 11010001, which enables the PON, URQ, EXE and OPC bits.

DESE? might return :DESE 186, showing that the DESER contains the binary value 10111010.

DIAG:LOOP:OPTion

This command sets or queries the type of looping desired.

Group	Self Test
Syntax	DIAG:LOOP:OPTion {FAIL ONCE ALWAYS ONFAIL NTIMES} DIAG:LOOP:OPTion?
Arguments	Fail - run until a failure is found, then halt. Once - run through one loop. Always - run forever. Onfail - run until a failure is found, then loop on it. Ntimes - run n number of loops.
Examples	DIAG:LOOP:OPTION ALWAYS sets loop option to ALWAYS. DIAG:LOOP:OPTION? returns current looping option value.

DIAG:LOOP:OPTion:NTIMes

This command sets or queries how many loops to run, if N-times is being used.

Group	Self Test
Syntax	DIAG:LOOP:OPTion:NTIMes <NR1> DIAG:LOOP:OPTion:NTIMes?
Arguments	<NR1> is how many loops to run.
Examples	DIAG:LOOP:OPTION:NTIMES 2 sets diagnostics to loop 2 times. DIAG:LOOP:OPTION:NTIMES? might return :DIAG:LOOP:OPTION:NTIMES 1 indicating diagnostics will run once.

DIAG:LOOP:STOP (No Query Form)

Request that diagnostics stop looping.

Group	Self Test
Syntax	DIAG:LOOP:STOP
Examples	DIAG:LOOP:STOP stops diagnostics looping.

DIAG:MODE

This command sets or queries the diagnostics mode.

Group	Self Test
Syntax	DIAG:MODE {POST EXTENDED SERVICE} DIAG:MODE?
Arguments	POST specifies the power on self test diagnostics. EXTENDED specifies the extended diagnostics. SERVICE specifies the service diagnostics.
Examples	DIAG:MODE POST specifies the power on self test diagnostics.

DIAG:RESULT? (Query Only)

This query returns both the overall diagnostics test results and the results of each individual test area.

Group	Self Test
Syntax	DIAG:RESULT?
Returns	The diagnostics results.
Examples	DIAG:RESULT? might return :DIAG:RESULT:FLAG "NOT RUN";LOG "PASS--IO,PASS--ANALOG,PASS--SYSTEM,PASS--ASIC,PASS--ACQ, PASS--SIGNAL,PASS--MEMORY" indicating the diagnostics results.

DIAG:RESUlt:FLAg? (Query Only)

This query returns the status of the diagnostic test area that has been selected.

Group Self Test

Syntax DIAG:RESUlt:FLAg?

Related Commands [DIAG:RESUlt:FLAg?](#)

Returns The status of the diagnostics (single area).

Examples DIAG:RESULT:FLAG? might return :DIAG:RESULT:FLAG "NOT RUN" indicating the diagnostics have not been run.

DIAG:RESUlt:LOG? (Query Only)

This query returns the test Pass/Fail status of each diagnostic area. It does not return the overall status.

Group Self Test

Syntax DIAG:RESUlt:LOG?

Returns The status of the diagnostic area.

Examples DIAG:RESULT:LOG? might return :DIAG:RESULT:LOG "PASS--IO, PASS--ANALOG, PASS--SYSTEM, PASS--ASIC, PASS--ACQ, PASS--SIGNAL, PASS--MEMORY" indicating the diagnostics result stored in the log.

DIAG:SELEct (No Query Form)

This command selects or queries an available diagnostic area.

Group Self Test

Syntax	<code>DIAG:SELEct {ALL IO ANALOG SYSTEM ASIC ACQ SIGNAL MEMORY}</code> <code>DIAG:SELEct?</code>
Arguments	The argument is the desired diagnostic area.
Examples	<code>DIAG:SELECT IO</code> will select the IO group.

DIAG:STATE (No Query Form)

This command starts or aborts Self Test. Abort happens after group under test completes.

Group	Self Test
Syntax	<code>DIAG:STATE {EXECute ABOrt}</code>
Arguments	<code>EXECUTE</code> starts execution of the diagnostics. <code>ABOrt</code> disables diagnostics capabilities and returns the instrument to a normal operating state.
Examples	<code>DIAG:STATE ABORT</code> turns off diagnostics capabilities and returns the instrument to a normal operating state. <code>DIAG:STATE?</code> might return <code>:DIAG:STATE ABORT</code> , indicating that diagnostics are disabled.

DIGGRP<x>:D<x>:THReshold

Sets or queries the threshold level in volts for the specified digital channel. If the source channel doesn't exist, a hardware missing error event is set.

Group	Digital
Syntax	<code>DIGGRP<x>:D<x>:THReshold <NR3></code> <code>DIGGRP<x>:D<x>:THReshold?</code>
Arguments	<code>DIGGRP<x></code> is the channel number. <code>D<x></code> is the digital channel bit number (0-7).

<NR3> is the threshold level in volts for the specified digital channel.

Examples DIGGRP1:D1:THReshoId 30.0E-3 sets the threshold to 30 mV.
DIGGRP1:D1:THReshoId? might return :DIGGRP1:D1:THRESHOLD 0.0E+0 indicating the threshold is 0.0 V.

DISplay? (Query Only)

This query-only command returns the current Display settings.

Group Display Control

Syntax DISplay?

Examples DISPLAY? might return :DISPLAY:CLOCK 1;COLOR:PALETTE:IMAGEVIEW
MONOGREEN; RECORDVIEW TEMPERATURE;USER:GRATICULE
165,50,15;CH1 180,50,100; CH2 300,50,100;CH3
60,50,100;CH4 240,50,100;REF1 0,90,0; REF2
0,90,100;REF3 60,90,100;REF4 240,90,100;MATH1
160,50,100; MATH2 40,60,100;MATH3 120,60,100;MATH4
195,50,100; HISTOGRAM 320,50,100;CARET 150,50,100;MASK
0,25,75;MASKHIGHLIGHT 140,50,100;:DISPLAY:COLOR:MATHCOLOR
DEFAULT;REFCOLOR DEFAULT; :DISPLAY:FILTER SINX;FORMAT
YT;GRATICULE IRE;INTENSITY :WAVEFORM:IMAGEVIEW
81.0000;RECORDVIEW 81.0000;:DISPLAY :INTENSITY:AUTOBRIGHT
0; :DISPLAY:PERSISTENCE OFF;STYLE DOTS;TRIGBAR OFF;TRIGT
1; CURSORTICK LONG;VARPERSIST 2.6000;SCREENTEXT:STATE
1; LABEL1:NAME "";XPOS 100;YPOS 5;:DISPLAY:SCREENTEXT
:LABEL2:NAME "THIS IS SCREEN TEXT";XPOS 100;YPOS 20;
:DISPLAY:SCREENTEXT:LABEL3:NAME "";XPOS 100;YPOS 35;
:DISPLAY:SCREENTEXT:LABEL4:NAME "";XPOS 100;YPOS 50;
:DISPLAY:SCREENTEXT:LABEL5:NAME "";XPOS 100;YPOS 343;
:DISPLAY:SCREENTEXT:LABEL6:NAME "";XPOS 100;YPOS 80;
:DISPLAY:SCREENTEXT:LABEL7:NAME "";XPOS 100;YPOS 95;
:DISPLAY:SCREENTEXT:LABEL8:NAME "";XPOS 100;YPOS 110;
:DISPLAY:WAVEFORM 1.

DISplay:COLors

Sets or queries the color mode for the graticule and waveform display.

Group	Display Control
Syntax	<code>DISPlay:COLors {NORMal INVERTed}</code> <code>DISPlay:COLors?</code>
Arguments	<code>NORMal</code> specifies normal color mode. <code>INVERTed</code> specifies inverted color mode.
Examples	<code>DISPLAY:COLORS NORMAL</code> sets the display colors to normal. <code>DISPLAY:COLORS?</code> might return <code>:DISPLAY:COLORS INVERTED</code> indicating the display colors are inverted.

DISPlay:GLObal:B<x>:STATE

This command sets or queries the global state (display mode On or Off) of the specified bus. Setting this value true (On or `NR1 ≠ 0`) turns on the source in the waveform view. Setting this value false (Off or `NR1 = 0`) turns off the source in the waveform view. This command only works if the specified bus is added already.

Group	Display Control
Syntax	<code>DISPlay:GLObal:B<x>:STATE {<NR1> OFF ON}</code>
Arguments	<code><NR1> = 0</code> disables the display of the specified bus; any other value enables display of the bus. <code>ON</code> enables display of the specified bus. <code>OFF</code> disables display of the specified bus.
Examples	<code>DISPlay:GLObal:B1:STATE OFF</code> turns off the display of bus 1. <code>DISPlay:GLObal:B1:STATE?</code> might return <code>:DISPLAY:GLOBAL:B1:STATE 1</code> indicating the bus is displayed.

DISPlay:GLObal:CH<x>:STATE

This command sets or queries the global state (display mode On or Off) of the specified channel (both analog and digital). Setting this value true (On or `NR1`

$\neq 0$) turns on the source in the waveform view. Setting this value false (Off or $\text{NR1} = 0$) turns off the source in the waveform view. This command only works if the specified channel is added already.

Group Display Control

Syntax `DISPlay:GLObal:CH<x>:STATE {<NR1>|OFF|ON}`

Arguments $\text{<NR1>} = 0$ disables the display of the specified channel; any other value enables display of the channel.

ON enables display of the specified channel.

OFF disables display of the specified channel.

Examples `DISPlay:GLObal:CH1:STATE OFF` turns off the display of channel 1.

`DISPlay:GLObal:CH1:STATE?` might return `:DISPLAY:GLOBAL:CH1:STATE 0` indicating that channel 1 is not displayed.

DISPlay:GLObal:MATH<x>:STATE

This command sets or queries the global state (display mode On or Off) of the specified math. Setting this value true (On or $\text{NR1} \neq 0$) turns on the source in the waveform view. Setting this value false (Off or $\text{NR1} = 0$) turns off the source in the waveform view. This command only works if the specified math waveform is added already.

Group Display Control

Syntax `DISPlay:GLObal:MATH<x>:STATE {<NR1>|OFF|ON}`

Arguments $\text{<NR1>} = 0$ disables the display of the specified math; any other value enables display of the math.

ON enables display of the specified math.

OFF disables display of the specified math.

Examples `DISPlay:GLObal:MATH1:STATE 1` displays math 1.

`DISPlay:GLObal:MATH1:STATE?` might return `:DISPLAY:GLOBAL:MATH1:STATE 1` indicating that math1 is displayed.

DISplay:GLObal:PLOT<x>:STATE

This command sets or queries the global state (display mode On or Off) of the specified time trend plot. Setting this value true (On or $\text{NR1} \neq 0$) turns on the source in the waveform view. Setting this value false (Off or $\text{NR1} = 0$) turns off the source in the waveform view. This command only works if the specified plot is added already.

Group Display Control

Syntax DISplay:GLObal:PLOT<x>:STATE {<NR1>|OFF|ON}

Arguments PLOT<x> is the plot number.

<NR1> = 0 disables the display of the specified plot; any other value enables display of the plot.

ON enables display of the specified plot.

OFF disables display of the specified plot.

Examples DISplay:GLObal:PLOT1:STATE ON displays plot 1.

DISplay:GLObal:PLOT1:STATE? might return
:DISPLAY:GLOBAL:PLOT1:STATE 1 indicating plot 1 is displayed.

DISplay:GLObal:REF<x>:STATE

this command sets or queries the global state (display mode On or Off) of the specified reference waveform. Setting this value true (On or $\text{NR1} \neq 0$) turns on the source in the waveform view. Setting this value false (Off or $\text{NR1} = 0$) turns off the source in the waveform view. This command only works if the specified reference waveform is added already.

Group Display Control

Syntax DISplay:GLObal:REF<x>:STATE {<NR1>|OFF|ON}

Arguments REF<x> is the Reference waveform number.

<NR1> = 0 disables the display of the specified reference; any other value enables display of the reference.

ON enables display of the specified reference.

OFF disables display of the specified reference.

Examples `DISPlay:GLObal:REF1:STATE 1` displays reference 1.
`DISPlay:GLObal:REF1:STATE?` might return
`:DISPLAY:GLOBAL:REF1:STATE 1` indicating that reference 1 is displayed.

DISPlay:INTENSITY? (Query Only)

This query-only command returns the waveform saturation level and screen saver settings.

Group Display Control

Syntax `DISPlay:INTENSITY?`

Related Commands

Arguments None

Examples `DISPLAY:INTENSITY?` might return `:DISPLAY:INTENSITY:BACKLIGHT 1`, indicating the intensity is set to 1.

DISPlay:INTENSITY:BACKLight

This command sets or queries the display backlight intensity setting.

Group Display Control

Syntax `DISPlay:INTENSITY:BACKLight {LOW|MEDium|HIGH}`
`DISPlay:INTENSITY:BACKLight?`

Arguments LOW selects a low brightness level.
 MEDium selects a moderate brightness level.
 HIGH selects a full brightness level.

Examples `DISPLAY:INTENSITY:BACKLIGHT LOW` sets the display backlight to low brightness level.

DISPLAY:INTENSITY:BACKLIGHT? might return
DISPLAY:INTENSITY:BACKLIGHT HIGH, indicating that the display backlight
is set to full brightness level.

DISplay:INTENSITy:BACKLight:AUTODim:ENABle

Sets or queries the state of the display auto-dim feature. The default is enabled.
Once the backlight has dimmed, any button push, knob turn or mouse movement
returns the backlight value to the value set by :DISplay:INTENSITy:BACKLight.

Group Display Control

Syntax DISplay:INTENSITY:BACKLight:AUTODim:ENABle {ON|OFF}
DISplay:INTENSITY:BACKLight:AUTODim:ENABle?

Related Commands [DISplay:INTENSITy:BACKLight](#)

Arguments ON enables the display auto-dim feature.
OFF disables the display auto-dim feature.

Examples DISPLAY:INTENSITY:BACKLIGHT:AUTODIM:ENABLE ON enables
auto-dimming of the backlight.

DISPLAY:INTENSITY:BACKLIGHT:AUTODIM:ENABLE? might return
DISPLAY:INTENSITY:BACKLIGHT:AUTODIM:ENABLE ON, indicating that
auto-dimming of the backlight is enabled.

DISplay:INTENSITy:BACKLight:AUTODim:TIME

Sets or queries the amount of time, in minutes, to wait for no user interface activity
before automatically dimming the display. The time can range from a minimum of
10 minutes to a maximum of 1440 minutes (24 hours). The default is 10 minutes.

Group Display Control

Syntax DISplay:INTENSITY:BACKLight:AUTODim:TIME <NR1>
DISplay:INTENSITY:BACKLight:AUTODim:TIME?

Arguments <NR1> is the amount of time, in minutes, to wait for no user interface activity
before automatically dimming the display.

- Examples** `DISPLAY:INTENSITY:BACKLIGHT:AUTODIM:TIME 30` sets the backlight to autodim in 30 minutes.
- `DISPLAY:INTENSITY:BACKLIGHT:AUTODIM:TIME?` might return `DISPLAY:INTENSITY:BACKLIGHT:AUTODIM:TIME 60`, indicating the backlight will auto-dim in 60 minutes.

DISplay:MATHFFTView<x>:AUTOScale

This command sets or returns the enabled state of autoscale for Math/FFT waveforms.

- Group** Display Control
- Syntax** `DISplay:MATHFFTView<x>:AUTOScale {OFF|ON|<NR1>}`
 `DISplay:MATHFFTView<x>:AUTOScale?`
- Arguments** `MATHFFTView<x>` is the Math-FFT waveform number.
- `OFF` disables the autoscale feature.
- `ON` enables the autoscale feature.
- `<NR1> = 0` disables the autoscale feature; any other value enables the autoscale feature.
- Examples** `DISplay:MATHFFTView1:AUTOScale OFF` disables the autoscale feature.
- `DISplay:MATHFFTView3:AUTOScale?` might return `DISPLAY:MATHFFTVIEW3:AUTOSCALE 1`, indicating that autoscale is on for the Math3 FFT waveform.

DISplay:MATHFFTView<x>:CURSor:ASOUrce? (Query Only)

This command queries the Math-FFT waveform view source for cursor A.

- Group** Cursor
- Syntax** `DISplay:MATHFFTView<x>:CURSor:ASOUrce?`
- Arguments** `MATHFFTView<x>` is the Math-FFT waveform number.

Examples `DISPlay:MATHFFTView1:CUSor:ASource?` might return
`DISPLAY:MATHFFTVIEW1:CUSOR:ASOURCE MATH1`, indicating the source
is MATH 1.

DISPlay:MATHFFTView<x>:CURSor:BSOURce? (Query Only)

This command queries the Math-FFT waveform view source for cursor B.

Group Cursor

Syntax `DISPlay:MATHFFTView<x>:CURSor:BSOURce?`

Arguments `MATHFFTView<x>` is the Math-FFT waveform number.

Examples `DISPlay:MATHFFTView5:CUSor:BSOURce?` might return
`DISPLAY:MATHFFTVIEW5:CUSOR:BSOURCE MATH5`, indicating the source
is Math waveform MATH 5.

DISPlay:MATHFFTView<x>:CURSor:DDT? (Query Only)

This command queries the delta Y over delta X ($\Delta Y/\Delta X$) cursor readout value of the specified cursor in the specified Math-FFT view.

NOTE. *If the current cursor is set to horizontal mode, this command will time out.*

Group Cursor

Syntax `DISPlay:MATHFFTView<x>:CURSor:DDT?`

Arguments `MATHFFTView<x>` is the Math-FFT waveform number.

Examples `DISPlay:MATHFFTView<x>:CURSor:DDT?` might return
`DISPLAY:MATHFFTVIEW1:CUSOR:DDT 14.0056E-9`, indicating
the $\Delta Y/\Delta X$ cursor readout value is 14.0056E-9.

DISplay:MATHFFTView<x>:CURSor:FUNction

This command sets or queries the cursor type for the specified Math-FFT view.

Group	Cursor
Syntax	DISplay:MATHFFTView<x>:CURSor:FUNction {WAVEform VBArS HBArS SCREEN} DISplay:MATHFFTView<x>:CURSor:FUNction?
Arguments	MATHFFTView<x> is the Math-FFT waveform number. WAVEFORM specifies to display the paired vertical cursors in YT display format for measuring waveform amplitude and time. Measurements are taken at where the cursor intersects the waveform, and tracks waveform changes. VBArS specifies vertical bar cursors, which measure in horizontal units. HBArS specifies horizontal bar cursors, which measure in vertical units. SCREEN specifies to display both horizontal and vertical bar cursors, which display the horizontal and vertical positions of the cursors, not waveform levels. Use these cursors to measure anywhere in the waveform display area.
Examples	DISplay:MATHFFTView5:CURSor:FUNction HBARS sets Math-FFT waveform 5 to use horizontal bar cursors. DISplay:MATHFFTView1:CURSor:FUNction? might return DISplay:MATHFFTView1:CURSor:FUNction? SCREEN, indicating that Math-FFT waveform 2 is set to use both horizontal and vertical cursors.

DISplay:MATHFFTView<x>:CURSor:HBArS:APOsition

This command sets or returns the position of horizontal cursor A for the specified Math-FFT view.

Group	Cursor
Syntax	DISplay:MATHFFTView<x>:CURSor:HBArS:APOsition <NR3> DISplay:MATHFFTView<x>:CURSor:HBArS:APOsition?
Arguments	MATHFFTView<x> is the Math-FFT waveform number. <NR3> is the cursor position of the specified cursor in the specified view.

Examples `DISPlay:MATHFFTView1:CUSor:HBArS:APOSition 50.0e-3` sets the position to 50 mV.

`DISPlay:MATHFFTView4:CUSor:HBArS:APOSition?` might return `DISPLAY:MATHFFTVIEW4:CUSOR:HBARS:APOSITION -274.2247E-3`, indicating the A cursor position is $-274.2247\text{E-}3$.

DISPlay:MATHFFTView<x>:CURSor:HBArS:AUNIts? (Query Only)

This command queries the vertical units of horizontal cursor A for the specified Math-FFT view.

Group Cursor

Syntax `DISPlay:MATHFFTView<x>:CURSor:HBArS:AUNIts?`

Arguments `MATHFFTView<x>` is the Math-FFT waveform number.

Examples `DISPlay:MATHFFTView1:CUSor:HBArS:AUNIts?` might return `DISPLAY:MATHFFTVIEW1:CUSOR:HBARS:AUNITS "dBm"`, indicating the vertical units are dBm.

DISPlay:MATHFFTView<x>:CURSor:HBArS:BPOSition

This command sets or returns the position of horizontal cursor B for the specified Math-FFT view.

Group Cursor

Syntax `DISPlay:MATHFFTView<x>:CURSor:HBArS:BPOSition <NR3>`
`DISPlay:MATHFFTView<x>:CURSor:HBArS:BPOSition?`

Arguments `MATHFFTView<x>` is the Math-FFT waveform number.

`<NR3>` is the vertical cursor B position for the specified Math-FFT view.

Examples `DISPlay:MATHFFTView1:CUSor:HBArS:BPOSition 50e-3` sets the position to 0.005 vertical units.

DISplay:MATHFFTView1:CURSor:HBArS:BPOStion? might return
DISPLAY:MATHFFTVIEW1:CURSOR:HBARS:BPOSITION -40.4061, indicating
the position is -40.4061 vertical units.

DISplay:MATHFFTView<x>:CURSor:HBArS:BUNIts? (Query Only)

This command queries the vertical units of horizontal cursor B for the specified Math-FFT view.

Group	Cursor
Syntax	DISplay:MATHFFTView<x>:CURSor:HBArS:BUNIts?
Arguments	MATHFFTView<x> is the Math-FFT waveform number.
Examples	DISplay:MATHFFTView3:CURSor:HBArS:BUNIts? might return DISPLAY:MATHFFTVIEW3:CURSOR:HBARS:BUNITS "dBm", indicating the cursor units are dBm.

DISplay:MATHFFTView<x>:CURSor:HBArS:DELTA? (Query Only)

This command queries the horizontal cursor's delta value of the specified Math-FFT view.

Group	Cursor
Syntax	DISplay:MATHFFTView<x>:CURSor:HBArS:DELTA?
Arguments	MATHFFTView<x> is the Math-FFT waveform number.
Examples	DISplay:MATHFFTView1:CURSor:HBArS:DELTA? might return DISPLAY:MATHFFTVIEW1:CURSOR:HBARS:DELTA 30.61448, indicating the cursor readout is 30.61488 units.

DISplay:MATHFFTView<x>:CURSor:MODE

This command sets or queries the cursor tracking mode of the specified Math-FFT view.

Group	Cursor
Syntax	<code>DISPlay:MATHFFTView<x>:CURSor:MODE {INDEPENDENT TRACK}</code>
Arguments	<code>MATHFFTView<x></code> is the Math-FFT waveform number. <code>INDEPENDENT</code> allows independent adjustment of the two cursors. <code>TRACK</code> ties the navigational functionality of the two cursors together. For cursor 1 adjustments, this ties the movement of the two cursors together; however, cursor 2 continues to move independently of cursor 1.
Examples	<code>DISPlay:MATHFFTView1:CURSor:MODE INDEPENDENT</code> allows independent adjustment of the two cursors. <code>DISPlay:MATHFFTView1:CURSor:MODE?</code> might return <code>:DISPLAY:MATHFFTVIEW1:CURSOR:MODE INDEPENDENT</code>, indicating the mode is set to independent.

`DISPlay:MATHFFTView<x>:CURSor:ONEOVERDELTATVALUE? (Query Only)`

This command queries the one over delta cursor readout value of the specified Math-FFT view.

Group	Cursor
Syntax	<code>DISPlay:MATHFFTView<x>:CURSor:ONEOVERDELTATVALUE?</code>
Arguments	<code>MATHFFTView<x></code> is the Math-FFT waveform number.
Examples	<code>DISPlay:MATHFFTView1:CURSor:ONEOVERDELTATVALUE?</code> might return <code>DISPLAY:MATHFFTVIEW1:CURSOR:ONEOVERDELTATVALUE 2.6005E-9</code>, indicating the one over delta time value is 2.6005 nanoseconds.

`DISPlay:MATHFFTView<x>:CURSor:SCREEN:AXPOSiTion`

This command sets or returns the vertical cursor A x-axis waveform measurement position of the specified Math-FFT view.

Group	Cursor
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Syntax	<code>DISPlay:MATHFFTView<x>:CURSor:SCREEN:AXPOsition <NR3></code> <code>DISPlay:MATHFFTView<x>:CURSor:SCREEN:AXPOsition?</code>
Arguments	<code>MATHFFTView<x></code> is the Math-FFT waveform number. <code><NR3></code> is the cursor position in MHz.
Examples	<code>DISPLAY:MATHFFTVIEW1:CURSOR:SCREEN:AXPOSITION 700.000E+6</code> sets the cursor position to 700 MHz. <code>DISPLAY:MATHFFTVIEW3:CURSOR:SCREEN:AXPOSITION?</code> might return <code>DISPLAY:MATHFFTVIEW3:CURSOR:SCREEN:AXPOSITION 125.000E+6</code> , indicating the x-axis value of cursor A is 125 MHz.

DISPlay:MATHFFTView<x>:CURSor:SCREEN:AYPOsition

This command sets or returns the vertical cursor A y-axis amplitude measurement value of the specified Math-FFT view.

Group	Cursor
Syntax	<code>DISPlay:MATHFFTView<x>:CURSor:SCREEN:AYPOsition <NR3></code> <code>DISPlay:MATHFFTView<x>:CURSor:SCREEN:AYPOsition?</code>
Arguments	<code>MATHFFTView<x></code> is the Math-FFT waveform number. <code><NR3></code> is the cursor A position of the specified cursor in the specified view.
Examples	<code>DISPlay:MATHFFTView2:CURSor:SCREEN:AYPOsition 1.0e0</code> sets the cursor position to 1.0 dBm. <code>DISPlay:MATHFFTView5:CURSor:SCREEN:AYPOsition?</code> might return <code>DISPLAY:MATHFFTVIEW5:CURSOR:SCREEN:AYPOSITION 1.1741714106633</code> , indicating the cursor position is 1.174 vertical units (such as dBm).

DISPlay:MATHFFTView<x>:CURSor:SCREEN:BXPOsition

This command sets or returns the vertical cursor Bx-axis waveform time measurement position of the specified Math-FFT view.

Group	Cursor
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Syntax	<code>DISPlay:MATHFFTView<x>:CURSor:SCREEN:BXPOsition <NR3></code> <code>DISPlay:MATHFFTView<x>:CURSor:SCREEN:BXPOsition?</code>
Arguments	<code>MATHFFTView<x></code> is the Math-FFT waveform number. <code><NR3></code> is the horizontal cursor B position of the specified cursor in the specified view.
Examples	<code>DISPlay:MATHFFTView1:CURSor:SCREEN:BXPOsition 3.5e9</code> sets the cursor position to 3.5 GHz. <code>DISPlay:MATHFFTView1:CURSor:SCREEN:BXPOsition?</code> might return <code>DISPLAY:MATHFFTVIEW1:CURSOR:SCREEN:BXPOSITION 2.50E+9</code> , indicating the cursor position is 2.5 GHz.

DISPlay:MATHFFTView<x>:CURSor:SCREEN:BYPOsition

This command sets or returns the vertical cursor B y-axis amplitude measurement value of the specified Math-FFT view.

Group	Cursor
Syntax	<code>DISPlay:MATHFFTView<x>:CURSor:SCREEN:BYPOsition <NR3></code> <code>DISPlay:MATHFFTView<x>:CURSor:SCREEN:BYPOsition?</code>
Arguments	<code>MATHFFTView<x></code> is the Math-FFT waveform number. <code><NR3></code> is the vertical cursor B position of the specified cursor in the specified view.
Examples	<code>DISPlay:MATHFFTView1:CURSor:SCREEN:BYPOsition -50.0e0</code> sets the cursor position to -50.0. <code>DISPlay:MATHFFTView4:CURSor:SCREEN:BYPOsition?</code> might return <code>DISPLAY:MATHFFTVIEW4:CURSOR:SCREEN:BYPOSITION -59.0460929065173</code> , indicating the cursor position is at -59.0 dBm.

DISPlay:MATHFFTView<x>:CURSor:STATE

This command sets or queries the visible state of cursors for the specified Math-FFT view.

Group	Cursor
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Syntax	<code>DISPlay:MATHFFTView<x>:CURSor:STATE {ON OFF <NR1>}</code> <code>DISPlay:MATHFFTView<x>:CURSor:STATE?</code>
Arguments	<code>MATHFFTView<x></code> is the Math-FFT waveform number. <code>1</code> or <code>ON</code> enables the cursors. <code>0</code> or <code>OFF</code> disables the cursors. <code><NR1> = 0</code> turns off cursors; any other value displays cursors.
Examples	<code>DISPlay:MATHFFTView2:CURSor:STATE OFF</code> turns off the cursors for Math-FFT waveform 2. <code>DISPlay:MATHFFTView1:CURSor:STATE?</code> might return <code>DISPLAY:MATHFFTVIEW1:CURSOR:STATE 1</code> indicating the cursors are enabled.

DISPlay:MATHFFTView<x>:CURSor:VBArS:APOsition

This command sets or queries the horizontal cursor A position for the specified Math-FFT view.

Group	Cursor
Syntax	<code>DISPlay:MATHFFTView<x>:CURSor:VBArS:APOsition <NR3></code> <code>DISPlay:MATHFFTView<x>:CURSor:VBArS:APOsition?</code>
Arguments	<code>MATHFFTView<x></code> is the Math-FFT waveform number. <code><NR3></code> sets the vertical cursor A position in the specified view.
Examples	<code>DISPlay:MATHFFTView1:CURSor:VBArS:APOsition 500.0e6</code> sets the cursor A position to 500 MHz. <code>DISPlay:MATHFFTView1:CURSor:VBArS:APOsition?</code> might return <code>DISPLAY:MATHFFTVIEW1:CURSOR:VBARS:APOSITION 585.533869115958E+6</code>, indicating cursor position is 585.5 MHz.

DISPlay:MATHFFTView<x>:CURSor:VBArS:BPOsition

This command sets or queries the vertical cursor B position for the specified Math-FFT view.

Group	Cursor
Syntax	<code>DISPlay:MATHFFTView<x>:CURSor:VBArS:BPOStion <NR3></code> <code>DISPlay:MATHFFTView<x>:CURSor:VBArS:BPOStion?</code>
Arguments	MATHFFTView<x> is the Math-FFT waveform number. <NR3> sets the vertical cursor B position in the specified view.
Examples	<code>DISPlay:MATHFFTView1:CURSor:VBArS:BPOStion 2.0e9</code> sets the cursor B position to 2.0 GHz. <code>DISPlay:MATHFFTView1:CURSor:VBArS:BPOStion?</code> might return <code>DISPLAY:MATHFFTVIEW1:CURSOR:VBARS:BPOSITION 2.50E+9</code>, indicating the cursor position is at 2.5 GHz.

DISPlay:MATHFFTView<x>:CURSor:VBArS:DELTA? (Query Only)

This command queries the vertical cursor's delta T readout value for the specified Math-FFT view.

Group	Cursor
Syntax	<code>DISPlay:MATHFFTView<x>:CURSor:VBArS:DELTA?</code>
Arguments	MATHFFTView<x> is the Math-FFT waveform number.
Examples	<code>DISPlay:MATHFFTView1:CURSor:VBArS:DELTA?</code> might return <code>DISPLAY:MATHFFTVIEW1:CURSOR:VBARS:DELTA 1.9144661308840E+9</code>, indicating the delta T cursor value is 1.9 GHz.

DISPlay:MATHFFTView<x>:CURSor:VBArS:AUNIts? (Query Only)

This command queries the vertical cursor A measurement units for the specified Math-FFT view.

Group	Cursor
Syntax	<code>DISPlay:MATHFFTView<x>:CURSor:VBArS:AUNIts?</code>

Arguments	MATHFFTView<x> is the Math-FFT waveform number.
Examples	DISplay:MATHFFTView1:CURSor:VBArS:AUNIts? might return :DISPLAY:MATHFFTVIEW1:CURSOR:VBARS:UNITS "dBm", indicating the cursor units are dBm.

DISplay:MATHFFTView<x>:CURSor:VBArS:BUNIts? (Query Only)

This command queries the vertical cursor B measurement units for the specified Math-FFT view.

Group	Cursor
Syntax	DISplay:MATHFFTView<x>:CURSor:VBArS:BUNIts?
Arguments	MATHFFTView<x> is the Math-FFT waveform number.
Examples	DISplay:MATHFFTView1:CURSor:VBArS:BUNIts? might return :DISPLAY:MATHFFTVIEW1:CURSOR:VBARS:BUNITS "Hz" indicating the cursor units are Hz.

DISplay:MATHFFTView<x>:CURSor:WAVEform:APOSition

This command sets or queries the waveform cursor A position for the specified Math-FFT view.

Group	Cursor
Syntax	DISplay:MATHFFTView<x>:CURSor:WAVEform:APOSition <NR3> DISplay:MATHFFTView<x>:CURSor:WAVEform:APOSition?
Arguments	MATHFFTView<x> is the Math-FFT waveform number. <NR3> is the waveform cursor A position in the specified plot view.
Examples	DISplay:MATHFFTView1:CURSor:WAVEform:APOSition 600.0e6 sets the position to 600 MHz.

DISPlay:MATHFFTView1:CURSor:WAVEform:APOsition? might return
 DISPLAY:MATHFFTVIEW1:CURSOR:WAVEFORM:APOSITION 25.000E+6,
 indicating the cursor position is 25 MHz.

DISPlay:MATHFFTView<x>:CURSor:WAVEform:BPOsition

This command sets or queries the waveform cursor B position for the specified Math-FFT view.

Group	Cursor
Syntax	DISPlay:MATHFFTView<x>:CURSor:WAVEform:BPOsition <NR3> DISPlay:MATHFFTView<x>:CURSor:WAVEform:BPOsition?
Arguments	MATHFFTView<x> is the Math-FFT waveform number. <NR3> is the waveform cursor B position in the specified plot view.
Examples	DISPlay:MATHFFTView1:CURSor:WAVEform:BPOsition 2.0E+9 sets the cursor position to 2.0 GHz. DISPlay:MATHFFTView1:CURSor:WAVEform:BPOsition? might return DISPLAY:MATHFFTVIEW1:CURSOR:WAVEFORM:BPOSITION 250.0E+6, indicating the waveform cursor B position is 250 MHz.

DISPlay:MATHFFTView<x>:GRIDlines

This command sets or queries the grid lines setting for the specified Math-FFT view.

Group	Display Control
Syntax	DISPlay:MATHFFTView<x>:GRIDlines {HORIZontal VERTical BOTH} DISPlay:MATHFFTView<x>:GRIDlines?
Arguments	MATHFFTView<x> is the Math-FFT waveform number. HORIZontal specifies horizontal grid lines. VERTical specifies vertical grid lines. BOTH specifies both vertical and horizontal grid lines.

Examples `DISPlay:MATHFFTView1:GRIDlines VERTICAL` specifies vertical grid lines.
`DISPlay:MATHFFTView1:GRIDlines?` might return
`:DISPLAY:MATHFFTVIEW1:GRIDLINES BOTH` indicating both
horizontal and vertical grid lines are displayed.

DISPlay:MATHFFTView<x>:MATH:MATH<x>:STATE

This command sets or queries the display state of the specified math waveform for the specified Math-FFT view.

Group Display Control

Syntax `DISPlay:MATHFFTView<x>:MATH:MATH<x>:STATE {OFF|ON|<NR1>}`

Arguments `MATHFFTView<x>` is the Math-FFT waveform number.
`OFF` disables displaying the specified Math-FFT view.
`ON` enables displaying the specified Math-FFT view.
`<NR1> = 0` disables the specified Math-FFT view; any other value enables the specified Math-FFT view.

Examples `DISPlay:MATHFFTView1:MATH:MATH1:STATE ON` enables the specified Math-FFT view.
`DISPlay:MATHFFTView4:MATH:MATH4:STATE?` might return
`:DISPLAY:MATHFFTVIEW1:MATH:MATH1:STATE 1`, indicating the waveform Math-FFT 1 is displayed.

DISPlay:MATHFFTView<x>:XAXIS:SCALE

This command sets or queries the x-axis scale (Linear or Log) for the specified Math-FFT view.

Group Display Control

Syntax `DISPlay:MATHFFTView<x>:XAXIS:SCALE {LINEAR|LOG}`
`DISPlay:MATHFFTView<x>:XAXIS:SCALE?`

Arguments	MATHFFTView<x> is the Math-FFT waveform number. LINEAr specifies a linear scale. LOG specifies a logarithmic scale.
Examples	DISPlay:MATHFFTView1:XAXIS:SCALE LOG specifies a logarithmic scale. DISPlay:MATHFFTView1:XAXIS:SCALE? might return DISPLAY:MATHFFTVIEW1:XAXIS:SCALE LINEAR, indicating that the scale setting is linear.

DISPlay:MATHFFTView<x>:YAXIS:SCALE

This command sets or queries the vertical scale setting (Linear or dBm) for the specified Math-FFT view.

Group	Display Control
Syntax	DISPlay:MATHFFTView<x>:YAXIS:SCALE {LINEAr DBM} DISPlay:MATHFFTView<x>:YAXIS:SCALE?
Arguments	MATHFFTView<x> is the Math-FFT waveform number. LINEAr specifies a linear scale. DBM specifies a dBm scale.
Examples	DISPlay:MATHFFTView1:YAXIS:SCALE LINEAR specifies a linear scale. DISPlay:MATHFFTView1:YAXIS:SCALE? might return DISPLAY:MATHFFTVIEW1:YAXIS:SCALE DBM, indicating that the scale is dBm.

DISPlay:MATHFFTView<x>:ZOOM:XAXIS:FROM

This command sets or queries the value of the left edge of the zoom area for the specified Math-FFT view.

Group	Zoom
Syntax	DISPlay:MATHFFTView<x>:ZOOM:XAXIS:FROM <NR3> DISPlay:MATHFFTView<x>:ZOOM:XAXIS:FROM?

Arguments	MATHFFTView<x> is the Math-FFT waveform number. <NR3> is the value of the left edge of the zoom x axis in the specified plot view.
Examples	DISplay:MATHFFTView1:ZOOM:XAXIS:FROM 800.e6 sets the x axis value of the left edge of the plot view to 800 MHz. DISplay:MATHFFTView1:ZOOM:XAXIS:FROM? might return DISPLAY:MATHFFTVIEW1:ZOOM:XAXIS:FROM 781.0E+6, indicating the x axis value of the left edge of the plot view is at 781 MHz.

DISplay:MATHFFTView<x>:ZOOM:XAXIS:TO

This command sets or queries the value of the right edge value of the zoom area for the specified Math-FFT view.

Group	Zoom
Syntax	DISplay:MATHFFTView<x>:ZOOM:XAXIS:TO <NR3>
Arguments	MATHFFTView<x> is the Math-FFT waveform number. <NR3> is the value of the right edge of the zoom x axis in the specified plot view.
Examples	DISplay:MATHFFTView1:ZOOM:XAXIS:TO 2.3E+9 sets the value of the right edge of the x axis to 2.3 GHz. DISplay:MATHFFTView1:ZOOM:XAXIS:TO? might return :DISPLAY:MATHFFTVIEW1:ZOOM:XAXIS:TO 2.343750E+9, indicating the value of the right edge of the zoom x axis is 2.34 GHz.

DISplay:MATHFFTView<x>:ZOOM:YAXIS:FROM

This command sets or queries the bottom edge value of the zoom y-axis area for the specified Math-FFT view.

Group	Zoom
Syntax	DISplay:MATHFFTView<x>:ZOOM:YAXIS:FROM <NR3> DISplay:MATHFFTView<x>:ZOOM:YAXIS:FROM?

Arguments	MATHFFTView<x> is the Math-FFT waveform number. <NR3> is the bottom value of the zoom y axis in the specified plot view.
Examples	DISPlay:MATHFFTView1:ZOOM:YAXIS:FROM -75.0e0 sets the bottom value of the y axis to -75 dBm. DISPlay:MATHFFTView1:ZOOM:YAXIS:FROM? might return DISPLAY:MATHFFTVIEW1:ZOOM:YAXIS:FROM -78.8258285893367 indicating the bottom value of the y axis is -78.8 dBm.

DISPlay:MATHFFTView<x>:ZOOM:YAXIS:TO

This command sets or queries the top edge value of the zoom y-axis area for the specified Math-FFT view.

Group	Zoom
Syntax	DISPlay:MATHFFTView<x>:ZOOM:YAXIS:TO <NR3> DISPlay:MATHFFTView<x>:ZOOM:YAXIS:TO?
Arguments	MATHFFTView<x> is the Math-FFT waveform number. <NR3> is the top value of the zoom y axis in the specified plot view.
Examples	DISPlay:MATHFFTView1:ZOOM:YAXIS:TO 21.0E0 sets the top value of the zoom y axis in the specified plot view to 21 dBm. DISPlay:MATHFFTView1:ZOOM:YAXIS:TO? might return DISPLAY:MATHFFTVIEW1:ZOOM:YAXIS:TO 21.1741714106633, indicating the top value of the zoom y axis in the specified plot view is 21.17 dBm.

DISPlay:PERStence

This command sets or queries the display persistence for analog waveforms. Persistence is valid for wave views only.

Group	Display Control
Syntax	DISPlay:PERStence {OFF AUTO INFPersist INFinite VARpersist CLEAR} DISPlay:PERStence?

Related Commands [DISplay:VARpersist](#)

Arguments OFF disables the persistence aspect of the display.
 AUTO automatically set the persistence.
 INFPersist sets a display mode where any pixels, once touched by samples, remain set until cleared by a mode change.
 INFINite sets a display mode where any pixels, once touched by samples, remain set until cleared by a mode change.
 VARPersist sets a display mode where set pixels are gradually dimmed.
 CLEAR resets the persist time count down and clears the display of acquired points.

Examples DISPLAY:PERSISTENCE VARPERSIST sets the persistence aspect of the display to fade set pixels according to the time set in the [DISplay:VARpersist](#) command.
 DISPLAY:PERSISTENCE? might return :DISPLAY:PERSISTENCE OFF, indicating that the persistence aspect of the display is disabled.

DISplay:PERStence:RESET (No Query Form)

This command controls the clearing of persistence data that has been built up over time. Persistence is valid for wave views only.

Group Display Control

Syntax DISplay:PERStence:RESET

Related Commands [DISplay:PERStence](#)
 [DISplay:VARpersist](#)

Examples DISPLAY:PERSISTENCE:RESET clears the display of persistence data.

DISplay:PLOTView<x>:AUTOScale

This command sets or queries the enabled state of autoscale for the specified plot.

Group Display Control

Syntax	<code>DISplay:PLOTView<x>:AUTOScale {OFF ON <NR1>}</code> <code>DISplay:PLOTView<x>:AUTOScale?</code>
Arguments	<code>PLOTView<x></code> is the Plot waveform number. <code>OFF</code> disables the autoscale feature. <code>ON</code> enables the autoscale feature. <code><NR1> = 0</code> disables the autoscale feature; any other value enables the autoscale feature.
Examples	<code>DISplay:PLOTView1:AUTOScale OFF</code> turns off plot autoscale. <code>DISplay:PLOTView3:AUTOScale?</code> might return <code>:DISPLAY:PLOTVIEW3:AUTOSCALE 0</code>, indicating the plot autoscale is off for plot 3.

DISplay:PLOTView<x>:CURSor:ASOURce? (Query Only)

This command queries the cursor source for plot cursor A.

Group	Cursor
Syntax	<code>DISplay:PLOTView<x>:CURSor:ASOURce?</code>
Arguments	<code>PLOTView<x></code> is the Plot waveform number.
Examples	<code>DISplay:PLOTView1:CURSor:ASOURce?</code> might return <code>DISPLAY:PLOTVIEW1:CURSOR:ASOURCE PLOT1</code> indicating the source of the cursor is plot 1.

DISplay:PLOTView<x>:CURSor:BSOURce? (Query Only)

This command queries the cursor source for plot cursor B.

Group	Cursor
Syntax	<code>DISplay:PLOTView<x>:CURSor:BSOURce?</code>

Arguments	PLOTView<x> is the Plot waveform number.
Examples	DISplay:PLOTView1:CURSor:BSOURce? might return :DISPLAY:PLOTVIEW1:CURSOR:BSOURCE PLOT1 indicating the B cursor source is plot 1.

DISplay:PLOTView<x>:CURSor:DDT? (Query Only)

This command returns the delta V over delta T cursor readout value for the specified Plot view.

Group	Cursor
Syntax	DISplay:PLOTView<x>:CURSor:DDT?
Arguments	PlotView<x> is the Plot waveform number.
Examples	DISplay:PLOTView1:CURSor:DDT? might return :DISPLAY:PLOTVIEW1:CURSOR:DDT 131.0470E+6 indicating the cursor readout value is 131.0470E+6.

DISplay:PLOTView<x>:CURSor:FUNCTION

This command sets or queries the cursor mode for the specified Plot view.

Group	Cursor
Syntax	DISplay:PLOTView<x>:CURSor:FUNCTION {WAVEFORM VBArS HBArS SCREEN} DISplay:PLOTView<x>:CURSor:FUNCTION?
Arguments	PLOTView<x> is the Plot waveform number. WAVEFORM specifies to display the paired cursors in YT display format for measuring waveform amplitude and time. VBArS specifies vertical bar cursors, which measure in horizontal units. HBArS specifies horizontal bar cursors, which measure in vertical units.

SCREEN specifies to display both horizontal and vertical bar cursors, which measure the selected waveform in horizontal and vertical units. Use these cursors to measure anywhere in the waveform display area.

Examples `DISplay:PLOTview1:CUSor:FUNctioN VBARS` displays vertical bar cursors.
`DISplay:PLOTview1:CUSor:FUNctioN?` might return
`:DISplay:PLOTVIEW1:CUSOR:FUNCTION SCREEN` indicating that both horizontal and vertical bar cursors are displayed.

DISplay:PLOTView<x>:CURSor:HBArs:APOSition

This command sets or queries the horizontal cursor A position for the specified Plot view.

Group Cursor

Syntax `DISplay:PLOTview<x>:CURSor:HBArs:APOSition <NR3>`
`DISplay:PLOTview<x>:CURSor:HBArs:APOSition?`

Arguments `PLOTview<x>` is the Plot waveform number.
`<NR3>` is the cursor position.

Examples `DISplay:PLOTview<x>:CURSor:HBArs:APOSition 2.0` sets the cursor to 2 Volts.
`DISplay:PLOTview<x>:CURSor:HBArs:APOSition?` might return
`:DISplay:PLOTVIEW1:CUSOR:HBArs:APOSITION 2.2186` indicating the HBArs cursors are at 2.2186 Volts.

DISplay:PLOTView<x>:CURSor:HBArs:AUNIts? (Query Only)

This command queries the horizontal cursor A vertical units for the specified Plot view.

Group Cursor

Syntax `DISplay:PLOTview<x>:CURSor:HBArs:AUNIts?`

Returns `PLOTview<x>` is the Plot waveform number.

Examples `DISPlay:PLOTView1:CURSor:HBArS:AUNIts?` might return
 `:DISPlay:PLOTVIEW1:CURSOR:HBARS:AUNITS "V"`, indicating the A units
 are volts.

DISPlay:PLOTView<x>:CURSor:HBArS:BPOStion

This command sets or queries the horizontal cursor B position for the specified Plot view.

Group Cursor

Syntax `DISPlay:PLOTView<x>:CURSor:HBArS:BPOStion <NR3>`
 `DISPlay:PLOTView<x>:CURSor:HBArS:BPOStion?`

Arguments `PLOTView<x>` is the Plot waveform number.
 `<NR3>` is the HBArS vertical position.

Examples `DISPlay:PLOTView<x>:CURSor:HBArS:BPOStion 2.0` sets the cursor to
 2 Volts.
 `DISPlay:PLOTView<x>:CURSor:HBArS:BPOStion?` might return
 `:DISPlay:PLOTVIEW1:CURSOR:HBARS:BPOSITION 2.2186` indicating the
 HBArS cursors are at 2.2186 Volts.

DISPlay:PLOTView<x>:CURSor:HBArS:BUNIts? (Query Only)

This command queries the cursor B vertical units for the specified Plot view.

Group Cursor

Syntax `DISPlay:PLOTView<x>:CURSor:HBArS:BUNIts??`

Arguments `PLOTView<x>` is the Plot waveform number.

Examples `DISPlay:PLOTView1:CURSor:HBArS:BUNIts?` might return
 `:DISPlay:PLOTVIEW1:CURSOR:HBARS:BUNITS "V"` indicating the units are
 Volts.

DISplay:PLOTView<x>:CURSor:HBArS:DELTA? (Query Only)

This command queries the delta V cursor readout value for the specified Plot view.

Group	Cursor
Syntax	DISplay:PLOTView<x>:CURSor:HBArS:DELTA??
Arguments	PLOTView<x> is the Plot waveform number.
Returns	Returns an <NR3> that is the delta V cursor value.
Examples	DISplay:PLOTView1:CURSor:HBArS:DELTA? might return :DISPLAY:PLOTVIEW1:CURSOR:HBARS:DELTA 1.2543, indicating the delta value is 1.2543 V.

DISplay:PLOTView<x>:CURSor:MODE

This command sets or queries the cursor tracking mode for the specified Plot view.

Group	Cursor
Syntax	DISplay:PLOTView<x>:CURSor:MODE {INDEPENDENT TRACK} DISplay:PLOTView<x>:CURSor:MODE?
Arguments	PLOTView<x> is the Plot waveform number. INDEPENDENT allows independent adjustment of the two cursors. TRACK ties the navigational functionality of the two cursors together. For cursor A adjustments, this ties the movement of the two cursors together; however, cursor B continues to move independently of cursor A.
Examples	DISplay:PLOTView1:CURSor:MODE TRACK sets the cursor to track together. DISplay:PLOTView1:CURSor:MODE? might return :DISPLAY:PLOTVIEW1:CURSOR:MODE INDEPENDENT, indicating independent adjustment of the cursors is allowed.

DISplay:PLOTView<x>:CURSor:ONEOVERDELTATVALUE? (Query Only)

This command sets or queries the one over delta T cursor readout value for the specified Plot view.

Group	Cursor
Syntax	DISplay:PLOTView<x>:CURSor:ONEOVERDELTATVALUE?
Arguments	PLOTView<x> is the Plot waveform number.
Returns	Returns an <NR3> that is the one over delta T cursor readout value (it may not be delta T, depending on the plot units).
Examples	DISplay:PLOTView<x>:CURSor:ONEOVERDELTATVALUE? might return :DISPLAY:PLOTVIEW1:CURSOR:ONEOVERDELTATVALUE 88.2295E+6 indicating the 1 over delta time value is 88.2295 MHz.

DISplay:PLOTView<x>:CURSor:SCREEN:AXPOSITION

This command sets or queries the horizontal cursor A position of the specified cursor in the specified view.

Group	Cursor
Syntax	DISplay:PLOTView<x>:CURSor:SCREEN:AXPOSITION <NR3> DISplay:PLOTView<x>:CURSor:SCREEN:AXPOSITION?
Arguments	PLOTView<x> is the Plot waveform number. <NR3> is the horizontal cursor A position.
Examples	DISplay:PLOTView1:CURSor:SCREEN:AXPOSITION -5.0 sets the A X cursor position to —5.0 ns. DISplay:PLOTView1:CURSor:SCREEN:AXPOSITION? might return :DISPLAY:PLOTVIEW1:CURSOR:SCREEN:AXPOSITION -5.5460E-9 indication the cursor A X position is -5.546 ns.

DISplay:PLOTView<x>:CURSor:SCREEN:AYPOSition

This command sets or queries the vertical cursor A position of the specified cursor in the specified view.

Group	Cursor
Syntax	DISplay:PLOTView<x>:CURSor:SCREEN:AYPOSition <NR3> DISplay:PLOTView<x>:CURSor:SCREEN:AYPOSition?
Arguments	PLOTView<x> is the Plot waveform number. <NR3> is the vertical cursor A position.
Examples	DISplay:PLOTView1:CURSor:SCREEN:AYPOSition 2.0 set the A cursor Y position to 2.0 Volts. DISplay:PLOTView1:CURSor:SCREEN:AYPOSition? might return :DISPLAY:PLOTVIEW1:CURSOR:SCREEN:AYPOSITION 1.9035 indicating the A cursor Y position is 1.9035 V.

DISplay:PLOTView<x>:CURSor:SCREEN:BXPOSition

This command sets or queries the horizontal cursor B position of the specified cursor in the specified view.

Group	Cursor
Syntax	DISplay:PLOTView<x>:CURSor:SCREEN:BXPOSition <NR3> DISplay:PLOTView<x>:CURSor:SCREEN:BXPOSition?
Arguments	PLOTView<x> is the Plot waveform number. <NR3> is the horizontal cursor B position.
Examples	DISplay:PLOTView1:CURSor:SCREEN:BXPOSition 59.0E-9 sets the B cursor X position to 59.0 ns. DISplay:PLOTView1:CURSor:SCREEN:BXPOSition? might return :DISPLAY:PLOTVIEW1:CURSOR:SCREEN:BXPOSITION 59.6802E-9 indicating the B cursor X position is 59.6802 ns.

DISplay:PLOTView<x>:CURSor:SCREEN:BYPOSition

This command sets or queries the vertical cursor B position of the specified cursor in the specified view.

Group	Cursor
Syntax	DISplay:PLOTView<x>:CURSor:SCREEN:BYPOSition <NR3> DISplay:PLOTView<x>:CURSor:SCREEN:BYPOSition?
Arguments	PLOTView<x> is the Plot waveform number. <NR3> is the vertical cursor B position.
Examples	DISplay:PLOTView1:CURSor:SCREEN:BYPOSition 0.589 sets the B cursor Y position to 589 mV. DISplay:PLOTView1:CURSor:SCREEN:BYPOSition? might return :DISPLAY:PLOTVIEW1:CURSOR:SCREEN:BYPOSITION 589.0696E-3 indicating the B cursor Y position is 589.0696 mV.

DISplay:PLOTView<x>:CURSor:SPLITMODE

This command sets or queries the cursor source mode in the specified view.

Group	Cursor
Syntax	DISplay:PLOTView<x>:CURSor:SPLITMODE {SAME SPLIT} DISplay:PLOTView<x>:CURSor:SPLITMODE?
Arguments	PLOTView<x> is the Plot waveform number. SAME specifies that both cursors are on the same waveform. SPLIT specifies that the cursors can be on different waveforms.
Examples	DISplay:PLOTView1:CURSor:SPLITMODE SAME sets the cursors to be on the same waveform. DISplay:PLOTView1:CURSor:SPLITMODE? might return :DISPLAY:PLOTVIEW1:CURSOR:SPLITMODE SAME indicating both cursors are on the same waveform.

DISplay:PLOTView<x>:CURSor:STATE

This command sets or queries the visible state of the cursor of the specified cursor in the specified view.

Group	Cursor
Syntax	DISplay:PLOTView<x>:CURSor:STATE {OFF ON <NR1>} DISplay:PLOTView<x>:CURSor:STATE?
Arguments	PLOTView<x> is the Plot waveform number. OFF disables the specified cursor. ON enables the specified cursor. <NR1> = 0 disables the specified cursor; any other value enables the specified cursor.
Examples	DISplay:PLOTView1:CURSor:STATE OFF disables the specified cursor. DISplay:PLOTView5:CURSor:STATE? might return :DISPLAY:PLOTVIEW5:CURSOR:STATE 1 indicating cursors are on.

DISplay:PLOTView<x>:CURSor:VBArS:APOsition

This command sets or queries the vertical cursor A position of the specified cursor in the specified view.

Group	Cursor
Syntax	DISplay:PLOTView<x>:CURSor:VBArS:APOsition <NR3> DISplay:PLOTView<x>:CURSor:VBArS:APOsition?
Arguments	PLOTView<x> is the Plot waveform number. <NR3> is the vertical cursor A position.
Examples	DISplay:PLOTView1:CURSor:VBArS:APOsition 50.0E-9 sets the VBARS APOSITION to 50 ns.

DISplay:PLOTView1:CURSor:VBArS:APOSition? might return
:DISPLAY:PLOTVIEW1:CURSOR:VBARS:APOSITION -60.0796E-9
indicating the VBARS APOSITION is set to -60.08 ns.

DISplay:PLOTView<x>:CURSor:VBArS:BPOSition

This command sets or queries the vertical cursor B position of the specified cursor in the specified view.

Group Cursor

Syntax DISplay:PLOTView<x>:CURSor:VBArS:BPOSition <NR3>
DISplay:PLOTView<x>:CURSor:VBArS:BPOSition?

Arguments PLOTView<x> is the Plot waveform number.
<NR3> is the vertical cursor B position.

Examples :DISPLAY:PLOTVIEW1:CURSOR:VBARS:BPOSITION 50.0E-9 sets the VBARS
BPOSITION to 50 ns.
:DISPLAY:PLOTVIEW1:CURSOR:VBARS:BPOSITION? might return
:DISPLAY:PLOTVIEW1:CURSOR:VBARS:BPOSITION -60.0796E-9
indicating the VBARS BPOSITION is set to 60.08 ns.

DISplay:PLOTView<x>:CURSor:VBArS:DELTA? (Query Only)

This command queries the delta T cursor readout value of the specified cursor in the specified view.

Group Cursor

Syntax DISplay:PLOTView<x>:CURSor:VBArS:DELTA?

Arguments PLOTView<x> is the Plot waveform number.

Returns The delta T cursor readout value is returned.

Examples `DISPlay:PLOTView1:CURSOR:VBARS:DELTA?` might return
 `:DISPLAY:PLOTVIEW1:CURSOR:VBARS:DELTA 119.7593E-9` indicating
 the VBARS DELTA is set to 119.76 ns.

DISPlay:PLOTView<x>:CURSOR:VBARS:UNIts? (Query Only)

This command queries the VBARS cursor readout units of the specified cursor in the specified view.

Group Cursor

Syntax `DISPlay:PLOTView<x>:CURSOR:VBARS:UNIts?`

Arguments `PLOTView<x>` is the Plot waveform number.

Returns The VBARS cursor readout units are returned.

Examples `DISPlay:PLOTView<x>:CURSOR:VBARS:UNIts?` might return
 `:DISPLAY:PLOTVIEW5:CURSOR:VBARS:UNITS "V"` indicating the units are
 Volts.

DISPlay:PLOTView<x>:CURSOR:WAVEform:APOSition

This command sets or queries the waveform cursor A horizontal position of the specified cursor in the specified view.

NOTE. *In case of XY plot, this command has no effect when used to set the value. In case of bathtub plot, this command sets or returns the cursor A vertical position. For all other plots, this command sets or returns the cursor A horizontal position.*

Group Cursor

Syntax `DISPlay:PLOTView<x>:CURSOR:WAVEform:APOSition <NR3>`
 `DISPlay:PLOTView<x>:CURSOR:WAVEform:APOSition?`

Arguments `PLOTView<x>` is the Plot waveform number.
 `<NR3>` is the horizontal cursor A position.

Examples `DISPlay:PLOTView1:CUSor:WAVEform:APOSition -50.0E-9` sets the waveform APOSITION to -50.0 ns.

`DISPlay:PLOTView1:CUSor:WAVEform:APOSition?` might return `:DISPLAY:PLOTVIEW1:CUSOR:WAVEFORM:APOSITION -60.0796E-9` indicating the waveform APOSITION is set to -60.08 ns.

DISplay:PLOTView<x>:CURSor:WAVEform:BPOSition

This command sets or queries the waveform cursor B horizontal position of the specified cursor in the specified view.

NOTE. *In case of XY plot, this command has no effect when used to set the value. In case of bathtub plot, this command has no effect. Query returns invalid values. For all other plots, this command sets or returns the cursor B horizontal position.*

Group Cursor

Syntax `DISPlay:PLOTView<x>:CURSor:WAVEform:BPOSition <NR3>`
`DISPlay:PLOTView<x>:CURSor:WAVEform:BPOSition?`

Arguments `PLOTView<x>` is the Plot waveform number.
`<NR3>` is the horizontal cursor B position.

Examples `DISPlay:PLOTView1:CUSor:WAVEform:BPOSition 3.0` sets the position to 3.0 V.

`DISPlay:PLOTView5:CUSor:WAVEform:BPOSition?` might return `:DISPLAY:PLOTVIEW5:CUSOR:WAVEFORM:BPOSITION 2.600517840` indicating the position is 2.5 V.

DISplay:PLOTView<x>:GRIDlines

This command sets or queries the Grid (graticule) lines setting of the specified plot. This command works for plots that have vertical and horizontal units associated with the graticule. For example, this command does not work for XY or XYZ plots.

Group Display Control

Syntax	<code>DISPlay:PLOTView<x>:GRIDlines {HORIZONTAL VERTICAL BOTH}</code> <code>DISPlay:PLOTView<x>:GRIDlines?</code>
Arguments	<code>PLOTView<x></code> is the Plot waveform number. <code>HORIZONTAL</code> specifies horizontal grid lines. <code>VERTICAL</code> specifies vertical grid lines. <code>BOTH</code> specifies both vertical and horizontal grid lines.
Examples	<code>DISPlay:PLOTView1:GRIDlines vertical</code> sets the vertical grid lines to display. <code>DISPlay:PLOTView1:GRIDlines?</code> might return <code>:DISPLAY:PLOTVIEW1:GRIDLINES BOTH</code> indicating that both vertical and horizontal grid lines are displayed.

DISPLAY:PLOTVIEW<x>:XAXIS:SCALE

This command sets or queries the horizontal scale setting for applicable plots (Linear or Log) for the specified plot view.

Group	Display
Syntax	<code>DISPLAY:PLOTVIEW<x>:XAXIS:SCALE {LINEAR LOG}</code>
Arguments	<code>PLOTView<x></code> is the Plot waveform number. <code>LINEAR</code> creates a plot with linear scales. <code>LOG</code> creates a plot with logarithmic scales.
Examples	<code>DISPLAY:PLOTVIEW1:XAXIS:SCALE LOG</code> creates a plot with a logarithmic horizontal scale. <code>DISPLAY:PLOTVIEW1:XAXIS:SCALE?</code> might return <code>:DISPLAY:PLOTVIEW1:XAXIS:SCALE LINEAR</code> indicating plots will have linear scales.

DISPLAY:PLOTVIEW<x>:YAXIS:SCALE

This command sets or queries the vertical scale setting for applicable plots (Linear or Log) in the specified plot view.

Group	Display
Syntax	<code>DISPLAY:PLOTVIEW<x>:YAXIS:SCALE {LINEAR LOG}</code>
Arguments	<code>PLOTView<x></code> is the Plot waveform number. <code>LINEAR</code> specifies a linear vertical scale. <code>LOG</code> specifies a logarithmic vertical scale.
Examples	<code>DISPLAY:PLOTVIEW1:YAXIS:SCALE LINEAR</code> sets the plot to use a linear vertical scale. <code>DISPLAY:PLOTVIEW1:YAXIS:SCALE?</code> might return <code>:DISPLAY:PLOTVIEW1:YAXIS:SCALE LOG</code> indicating a logarithmic vertical scale is used.

DISplay:PLOTView<x>:ZOOM:XAXIS:FROM

This command sets or queries the value of the left edge of the specified plot.

Group	Zoom
Syntax	<code>DISplay:PLOTView<x>:ZOOM:XAXIS:FROM <NR3></code> <code>DISplay:PLOTView<x>:ZOOM:XAXIS:FROM?</code>
Arguments	<code>PLOTView<x></code> is the Plot waveform number. <code><NR3></code> is start of the zoom x-axis.
Examples	<code>DISplay:PLOTView1:ZOOM:XAXIS:FROM 2.0</code> sets the start of the zoom x-axis to 2.0 V. <code>DISplay:PLOTView3:ZOOM:XAXIS:FROM?</code> might return <code>DISPLAY:PLOTVIEW3:ZOOM:XAXIS:FROM 1.9844803459459</code> indicating the start of the zoom x-axis is 1.9845 V.

DISplay:PLOTView<x>:ZOOM:XAXIS:TO

This command sets or queries the value of the right edge of the specified plot.

Group	Zoom
Syntax	DISPlay:PLOTView<x>:ZOOM:XAXIS:TO <NR3> DISPlay:PLOTView<x>:ZOOM:XAXIS:TO?
Arguments	PLOTView<x> is the Plot waveform number. <NR3> is the end of the zoom x-axis.
Examples	DISPlay:PLOTView1:ZOOM:XAXIS:TO 2.5 sets the end of the zoom x-axis to 2.5 V. DISPlay:PLOTView3:ZOOM:XAXIS:TO? might return DISPLAY:PLOTVIEW3:ZOOM:XAXIS:TO 2.2144883507508 indicating the end of the zoom x-axis is 2.21 V.

DISPlay:PLOTView<x>:ZOOM:YAXIS:FROM

This command sets or queries the bottom value of the zoom y-axis in the specified plot view.

Group	Zoom
Syntax	DISPlay:PLOTView<x>:ZOOM:YAXIS:FROM <NR3> DISPlay:PLOTView<x>:ZOOM:YAXIS:FROM?
Arguments	PLOTView<x> is the Plot waveform number. <NR3> is the bottom value of the zoom y-axis.
Examples	DISPlay:PLOTView1:ZOOM:YAXIS:FROM -1.0E+20 sets the bottom value of the zoom y-axis to -1E+20. DISPlay:PLOTView1:ZOOM:YAXIS:FROM? might return :DISPLAY:PLOTVIEW1:ZOOM:YAXIS:FROM -1.0E+21 indicating the bottom value of the zoom y-axis is -1E+21.

DISPlay:PLOTView<x>:ZOOM:YAXIS:TO

This command sets or queries the top value of the zoom y-axis in the specified plot view.

Group	Zoom
Syntax	DISPlay:PLOTView<x>:ZOOM:YAXIS:TO <NR3> DISPlay:PLOTView<x>:ZOOM:YAXIS:TO?
Arguments	PLOTView<x> is the Plot waveform number. <NR3> is the top value of the zoom y-axis.
Examples	DISPlay:PLOTView1:ZOOM:YAXIS:TO 100 sets the top value of the zoom y-axis to 100. DISPlay:PLOTView1:ZOOM:YAXIS:TO? might return :DISPLAY:PLOTVIEW3:ZOOM:YAXIS:FROM 0.0E+0 indicating the top value of the zoom y-axis is 0.0 hits (for a histogram plot, or it could be Volts for an XY plot).

DISPlay:REFFFTView<x>:AUTOScale

This command sets or queries the enabled state of auto-scale for plots.

Group	Display Control
Syntax	DISPlay:REFFFTView<x>:AUTOScale {OFF ON 0 1} DISPlay:REFFFTView<x>:AUTOScale?
Arguments	REFFFTView<x> is the plot number. <NR1> = 0 disables auto-scale on the specified reffftview; any other value turns this feature on. OFF disables auto-scale on the specified reffftview. ON enables the specified channel on the specified waveview.
Examples	DISPlay:REFFFTView1:AUTOScale 1 enables auto-scale on the specified view. DISPlay:REFFFTView5:AUTOScale? might return :DISPLAY:REFFFTVIEW5:AUTOSCALE 1 indicating auto-scale is on.

DISplay:REFFFTView<x>:CURSor:ASOURce? (Query Only)

This command returns the cursor source for plot cursor A

Group Cursor

Syntax DISplay:REFFFTView<x>:CURSor:ASOURce?

Arguments PLOTview<x> is the Plot waveform number.

Returns Returns the cursor source for plot cursor A.

Examples DISplay:REFFFTview5:CURSor:ASOURce? might return
:DISPLAY:REFFFTVIEW5:CURSOR:ASOURCE REF5 indicating the A cursor
source is reference 5.

DISplay:REFFFTView<x>:CURSor:BSOURce? (Query Only)

This command returns the cursor source for plot cursor B.

Group Cursor

Syntax DISplay:REFFFTView<x>:CURSor:BSOURce?

Returns Returns the cursor source for plot cursor B.

Examples DISplay:REFFFTview5:CURSor:BSOURce? might return
:DISPLAY:REFFFTVIEW5:CURSOR:BSOURCE REF5 indicating the source of the
B cursor is reference 5.

DISplay:REFFFTView<x>:CURSor:DDT? (Query Only)

This command returns the delta V over delta T cursor readout value of the
specified cursor in the specified view.

Group Cursor

Syntax	<code>DISPlay:REFFFTView<x>:CURSor:DDT?</code>
Returns	Returns the delta V over delta T cursor readout value
Examples	<code>DISPlay:REFFFTView5:CURSor:DDT?</code> might return : <code>DISPLAY:REFFFTVIEW5:CURSOR:DDT 9.91E+37</code> indicating the delta V over delta T cursor readout value is 9.91E+37.

DISplay:REFFFTView<x>:CURSor:FUNCTION

This command sets or queries the cursor type of the specified cursor in the specified view.

Group	Cursor
Syntax	<code>DISPlay:REFFFTView<x>:CURSor:FUNCTION</code> { <code>WAVEform</code> <code>VBArs</code> <code>HBArs</code> <code>SCREEN</code> } <code>DISPlay:REFFFTView<x>:CURSor:FUNCTION?</code>
Arguments	<code>HBArs</code> specifies horizontal bar cursors, which measure in vertical units. <code>VBArs</code> specifies vertical bar cursors, which measure in horizontal units. <code>SCREEN</code> specifies both horizontal and vertical bar cursors, which measure in horizontal and vertical units specified by the cursor sources. Use these cursors to measure anywhere in the waveform display area. <code>WAVEform</code> specifies paired or split cursors in YT display format for measuring waveform amplitude and time. In XY and XYZ format, these cursors indicate the amplitude positions of an XY pair (Ch1 vs Ch2 voltage, where Ch1 is the X axis and Ch2 is the Y axis) relative to the trigger.
Examples	<code>DISPlay:REFFFTView5:CURSor:FUNCTION screen</code> sets the cursor function to screen. <code>DISPlay:REFFFTView5:CURSor:FUNCTION?</code> might return :<code>DISPLAY:REFFFTVIEW5:CURSOR:FUNCTION WAVEFORM</code> indicating the cursor function is set to waveform.

DISplay:REFFFTView<x>:CURSor:HBArs:APOsition

This command sets or queries the vertical cursor A position of the specified cursor in the specified view.

Group	Cursor
Syntax	<pre>DISPlay:REFFFTView<x>:CURSor:HBArS:APOSition <NR3> DISPlay:REFFFTView<x>:CURSor:HBArS:APOSition?</pre>
Arguments	<NR3> is the vertical cursor A position of the specified cursor in the specified view.
Examples	DISPlay:REFFFTView5:CURSor:HBArS:APOSition -20 sets the A cursor position to -20. DISPlay:REFFFTView5:CURSor:HBArS:APOSition? might return :DISPLAY:REFFFTVIEW5:CURSOR:HBARS:APOSITION -14.4762489421361 indicating the A cursor position is -14.48 dBm.

DISPlay:REFFFTView<x>:CURSor:HBArS:AUNIts? (Query Only)

This command returns cursor A vertical units of the specified cursor in the specified view.

Group	Cursor
Syntax	DISPlay:REFFFTView<x>:CURSor:HBArS:AUNIts?
Returns	Returns the cursor A vertical units of the specified cursor in the specified view.
Examples	DISPlay:REFFFTView5:CURSor:HBArS:AUNIts? might return :DISPLAY:REFFFTVIEW5:CURSOR:HBARS:AUNITS "dBm" indicating the A cursor units are dBm.

DISPlay:REFFFTView<x>:CURSor:HBArS:BPOSition

This command sets or queries the vertical cursor B position of the specified cursor in the specified view.

Group	Cursor
Syntax	<pre>DISPlay:REFFFTView<x>:CURSor:HBArS:BPOSition <NR3> DISPlay:REFFFTView<x>:CURSor:HBArS:BPOSition?</pre>

Arguments <NR3> is the vertical cursor B position of the specified cursor in the specified view.

Examples `DISPlay:REFFFTView5:CUSor:HBArS:BPOsition -90` sets the B cursor position to -90.

`DISPlay:REFFFTView5:CUSor:HBArS:BPOsition?` might return `:DISPlay:REFFFTVIEW5:CUSOR:HBARS:BPOSITION -100.6262359333038` indicating the B cursor position is -100.6.

DISPlay:REFFFTView<x>:CURSor:HBArS:BUNIts? (Query Only)

This command returns the cursor B vertical units of the specified cursor in the specified view.

Group Cursor

Syntax `DISPlay:REFFFTView<x>:CURSor:HBArS:BUNIts?`

Returns Returns the cursor B vertical units of the specified cursor in the specified view.

Examples `DISPlay:REFFFTView5:CUSor:HBArS:BUNIts?` might return `:DISPlay:REFFFTVIEW5:CUSOR:HBARS:BUNITS "dBm"` indicating the units are dBm.

DISPlay:REFFFTView<x>:CURSor:HBArS:DELTA? (Query Only)

This command returns the delta V cursor readout value of the specified cursor in the specified view.

Group Cursor

Syntax `DISPlay:REFFFTView<x>:CURSor:HBArS:DELTA?`

Returns Returns the delta V cursor readout value of the specified cursor in the specified view.

Examples `DISPlay:REFFFTView5:CUSor:HBArS:DELTA?` might return `:DISPlay:REFFFTVIEW5:CUSOR:HBARS:DELTA 59.9867396737569` indicating the delta V cursor readout value is 59.99.

DISPlay:REFFFTView<x>:CURSor:MODE

This command sets or queries the cursor tracking mode of the specified cursor in the specified view.

Group	Cursor
Syntax	DISPlay:REFFFTView<x>:CURSor:MODE {INDEPENDENT TRACK} DISPlay:REFFFTView<x>:CURSor:MODE?
Arguments	TRACK ties the navigational functionality of the two cursors together. For cursor A adjustments, this ties the movement of the two cursors together; however, cursor B continues to move independently of cursor A. INDEPENDENT allows independent adjustment of the two cursors.
Examples	DISPlay:REFFFTView5:CURSor:MODE TRACK sets the cursors to track together. DISPlay:REFFFTView5:CURSor:MODE? might return :DISPLAY:REFFFTVIEW5:CURSOR:MODE INDEPENDENT indicating the cursors move independently.

DISPlay:REFFFTView<x>:CURSor:ONEOVERDELTATVALUE? (Query Only)

This command returns the one over delta T cursor readout value of the specified cursor in the specified view.

Group	Cursor
Syntax	DISPlay:REFFFTView<x>:CURSor:ONEOVERDELTATVALUE?
Examples	DISPlay:REFFFTView5:CURSor:ONEOVERDELTATVALUE? might return :DISPLAY:REFFFTVIEW5:CURSOR:ONEOVERDELTATVALUE 2.540935140340E-9 indicating the one over delta T cursor readout value is 2.54 ns.

DISPlay:REFFFTView<x>:CURSor:SCREEN:AXPOsition

This command sets or queries the horizontal cursor A position of the specified cursor in the specified view.

Group	Cursor
Syntax	DISplay:REFFFTView<x>:CURSor:SCREEN:AXPOsition <NR3> DISplay:REFFFTView<x>:CURSor:SCREEN:AXPOsition?
Arguments	<NR3> is the horizontal cursor A position of the specified cursor in the specified view.
Examples	DISplay:REFFFTView5:CURSor:SCREEN:AXPOsition 1.0E+6 sets the cursor position to 1.0 MHz. DISplay:REFFFTView5:CURSor:SCREEN:AXPOsition? might return :DISPLAY:REFFFTVIEW5:CURSOR:SCREEN:AXPOSITION 792.4465962305570E+3 indicating the A cursor horizontal position is 792.447 kHz.

DISplay:REFFFTView<x>:CURSor:SCREEN:AYPOsition

This command sets or queries the vertical cursor A position of the specified cursor in the specified view.

Group	Cursor
Syntax	DISplay:REFFFTView<x>:CURSor:SCREEN:AYPOsition <NR3> DISplay:REFFFTView<x>:CURSor:SCREEN:AYPOsition?
Arguments	<NR3> is the vertical cursor A position of the specified cursor in the specified view.
Examples	DISplay:REFFFTView5:CURSor:SCREEN:AYPOsition -20.9 sets the position to -20 dBm. DISplay:REFFFTView5:CURSor:SCREEN:AYPOsition? might return :DISPLAY:REFFFTVIEW5:CURSOR:SCREEN:AYPOSITION -30.0132603262431 indicating the A cursor vertical position is -30.01 dBm.

DISplay:REFFFTView<x>:CURSor:SCREEN:BXPOsition

This command sets or queries the horizontal cursor B position of the specified cursor in the specified view.

Group	Cursor
Syntax	<pre>DISplay:REFFFTView<x>:CURSor:SCREEN:BXPOsition <NR3> DISplay:REFFFTView<x>:CURSor:SCREEN:BXPOsition?</pre>
Arguments	<NR3> is the horizontal cursor B position of the specified cursor in the specified view.
Examples	DISplay:REFFFTView5:CURSor:SCREEN:BXPOsition 300 sets the position to 300 MHz. DISplay:REFFFTView5:CURSor:SCREEN:BXPOsition? might return :DISPLAY:REFFFTVIEW5:CURSOR:SCREEN:BXPOSITION 394.3483403001212E+6 indicating the B cursor horizontal position is 394.348 MHz.

DISplay:REFFFTView<x>:CURSor:SCREEN:BYPOsition

This command sets or queries the vertical cursor B position of the specified cursor in the specified view.

Group	Cursor
Syntax	<pre>DISplay:REFFFTView<x>:CURSor:SCREEN:BYPOsition <NR3> DISplay:REFFFTView<x>:CURSor:SCREEN:BYPOsition?</pre>
Arguments	<NR3> is the vertical cursor B position of the specified cursor in the specified view.
Examples	DISplay:REFFFTView5:CURSor:SCREEN:BYPOsition -80 sets the cursor position to -90 dBm. DISplay:REFFFTView5:CURSor:SCREEN:BYPOsition? might return :DISPLAY:REFFFTVIEW5:CURSOR:SCREEN:BYPOSITION -90.0 indicating the B cursor vertical position is -90 dBm.

DISplay:REFFFTView<x>:CURSor:SPLITMODE

This command sets or queries whether both cursors have same or different source.

Group	Cursor
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Syntax	<code>DISplay:REFFFTView<x>:CURSor:SPLITMODE {SAME SPLIT}</code> <code>DISplay:REFFFTView<x>:CURSor:SPLITMODE?</code>
Arguments	<code>SAME</code> specifies both cursors have the same sources. <code>SPLIT</code> specifies both cursors have different sources.
Examples	<code>DISplay:REFFFTView5:CURSor:SPLITMODE SPLIT</code> specified that the cursors have different sources. <code>DISplay:REFFFTView5:CURSor:SPLITMODE?</code> might return <code>:DISPLAY:REFFFTVIEW5:CURSOR:SPLITMODE SAME</code> indicating both cursors have the same source.

DISplay:REFFFTView<x>:CURSor:STATE

This command sets or queries the visible state of the cursor of the specified cursor n the specified view.

Group	Cursor
Syntax	<code>DISplay:REFFFTView<x>:CURSor:STATE {OFF ON 0 1}</code> <code>DISplay:REFFFTView<x>:CURSor:STATE?</code>
Arguments	<code><NR1> = 0</code> specifies the cursor is not visible; any other value displays the cursor. <code>OFF</code> specifies the cursor is not visible. <code>ON</code> displays the cursor.
Examples	<code>DISplay:REFFFTView5:CURSor:STATE ON</code> specifies the cursor is visible. <code>DISplay:REFFFTView5:CURSor:STATE?</code> might return <code>:DISPLAY:REFFFTVIEW5:CURSOR:STATE 1</code> indicating the cursor is visible.

DISplay:REFFFTView<x>:CURSor:VBArS:APOsition

This command sets or queries the horizontal cursor A position of the specified cursor in the specified view.

Group	Cursor
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Syntax	<code>DISPlay:REFFFTView<x>:CURSor:VBArS:APOSition <NR3></code> <code>DISPlay:REFFFTView<x>:CURSor:VBArS:APOSition?</code>
Arguments	<NR3> is the horizontal cursor A position of the specified cursor in the specified view.
Examples	<code>DISPlay:REFFFTView5:CURSor:VBArS:APOSition 1.0E+6</code> sets the cursor to 1 MHz. <code>DISPlay:REFFFTView5:CURSor:VBArS:APOSition?</code> might return <code>:DISPlay:REFFFTVIEW5:CURSOR:VBARS:APOSITION 792.4465962305570E+3</code> indicating the cursor position is 792.447 kHz.

DISPlay:REFFFTView<x>:CURSor:VBArS:BPOSition

This command sets or queries the horizontal cursor B position of the specified cursor in the specified view.

Group	Cursor
Syntax	<code>DISPlay:REFFFTView<x>:CURSor:VBArS:BPOSition <NR3></code> <code>DISPlay:REFFFTView<x>:CURSor:VBArS:BPOSition?</code>
Arguments	<NR3> is the horizontal cursor B position of the specified cursor in the specified view.
Examples	<code>DISPlay:REFFFTView5:CURSor:VBArS:BPOSition 300.0E+6</code> sets the cursor position to 300 MHz. <code>DISPlay:REFFFTView5:CURSor:VBArS:BPOSition?</code> might return <code>:DISPlay:REFFFTVIEW5:CURSOR:VBARS:BPOSITION 394.3483403001212E+6</code> indicating the B cursor horizontal position is 394.348 MHz.

DISPlay:REFFFTView<x>:CURSor:VBArS:DELTA? (Query Only)

This command returns the delta T cursor readout value of the specified cursor in the specified view.

Group	Cursor
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Syntax	<code>DISPlay:REFFFTView<x>:CURSor:VBArS:DELTA?</code>
Returns	Returns the delta T cursor readout value of the specified cursor in the specified view.
Examples	<code>DISPlay:REFFFTView5:CURSor:VBArS:DELTA?</code> might return <code>:DISPlay:REFFFTVIEW5:CURSOR:VBARS:DELTA 393.5558937038906E+6</code> indicating the delta T cursor readout value is 393.556 MHz.

DISPlay:REFFFTView<x>:CURSor:VBArS:UNIts? (Query Only)

This command returns cursor A vertical units of the specified cursor in the specified view.

Group	Cursor
Syntax	<code>DISPlay:REFFFTView<x>:CURSor:VBArS:UNIts?</code>
Returns	Returns cursor A vertical units of the specified cursor in the specified view.
Examples	<code>DISPlay:REFFFTView5:CURSor:VBArS:UNIts?</code> might return <code>:DISPlay:REFFFTVIEW5:CURSOR:VBARS:UNITS "Hz"</code> indicating the A cursor vertical units are Hz.

DISPlay:REFFFTView<x>:CURSor:WAVEform:AHPOSition? (Query Only)

This query-only command returns the value of the cursor A horizontal position.

Group	Cursor
Syntax	<code>DISPlay:REFFFTView<x>:CURSor:WAVEform:AHPOSition?</code>
Examples	<code>DISPlay:REFFFTView1:CURSor:WAVEform:AHPOSition?</code> might return <code>:DISPlay:REFFFTVIEW1:CURSOR:WAVEFORM:AHPOSITION 9.91E+37.</code>

DISplay:REFFFTView<x>:CURSor:WAVEform:APOSition

Sets or returns the waveform cursor A position in the specified plot view.

NOTE. *In case of an XY plot, this command has no effect when used to set the value. In the case of a bathtub plot, this command sets or queries the cursor A vertical position. For all other plots, this command sets or queries the cursor A horizontal position.*

Group	Cursor
Syntax	DISplay:REFFFTView<x>:CURSor:WAVEform:APOSition <NR3> DISplay:REFFFTView<x>:CURSor:WAVEform:APOSition?
Arguments	<NR3> is the waveform cursor A position in the specified plot view.
Examples	DISplay:REFFFTView5:CURSor:WAVEform:APOSition 7E+5 sets the position to 700 kHz. DISplay:REFFFTView5:CURSor:WAVEform:APOSition? might return :DISplay:REFFFTView5:CURSor:WAVEform:APOSITION 792.4465962305570E+3 indicating the A cursor waveform position is 792.4 kHz.

DISplay:REFFFTView<x>:CURSor:WAVEform:AVPOSition? (Query Only)

This query-only command returns the value of the cursor A vertical position.

Group	Cursor
Syntax	DISplay:REFFFTView<x>:CURSor:WAVEform:AVPOSition?
Examples	DISplay:REFFFTView<x>:CURSor:WAVEform:AVPOSition? might return :DISplay:REFFFTView1:CURSor:WAVEform:AVPOSITION 1.605E+0 indicating the cursor vertical position is 1.6 dBm.

DISplay:REFFFTView<x>:CURSor:WAVEform:BHPOSition? (Query Only)

This query-only command returns the value of the cursor B horizontal position.

Group	Cursor
Syntax	DISplay:REFFFTView<x>:CURSor:WAVEform:BHPOSition?
Examples	DISplay:REFFFTView<x>:CURSor:WAVEform:BHPOSition? might return :DISPLAY:REFFFTVIEW1:CURSOR:WAVEFORM:BHPOSITION 9.91E+37.

DISplay:REFFFTView<x>:CURSor:WAVEform:BPOSition

Sets or returns the waveform cursor B position in the specified plot view.

NOTE. In case of an XY plot, this command has no effect when used to set the value. In the case of a bathtub plot, this command has no effect. Queries return invalid values. For all other plots, this command sets or queries the cursor B horizontal position.

Group	Cursor
Syntax	DISplay:REFFFTView<x>:CURSor:WAVEform:BPOSition <NR3> DISplay:REFFFTView<x>:CURSor:WAVEform:BPOSition?
Arguments	<NR3> is the waveform cursor B position in the specified plot view.
Examples	DISplay:REFFFTView5:CURSor:WAVEform:BPOSition 4E+8 sets the cursor position to 400 MHz. DISplay:REFFFTView5:CURSor:WAVEform:BPOSition? might return :DISPLAY:REFFFTVIEW5:CURSOR:WAVEFORM:BPOSITION 394.3483403001212E+6 indicating the B waveform curosr position is 394.35 MHz.

DISplay:REFFFTView<x>:CURSor:WAVEform:BVPOSition? (Query Only)

This query-only command returns the value of the cursor B vertical position.

Group	Cursor
Syntax	DISplay:REFFFTView<x>:CURSor:WAVEform:BVPOSition?

Examples `DISPlay:REFFFTView1:CUSor:WAVEform:BVPOSition?` might return `:DISPlay:REFFFTView<x>:CUSor:WAVEform:BVPOSition -119.866E+0` indicating the cursor vertical position is -119.9 dBm.

DISPlay:REFFFTView<x>:GRIDlines

This command sets or returns the grid lines setting of the plot.

Group Display Control

Syntax `DISPlay:REFFFTView<x>:GRIDlines {HORizontal|VERTical|BOTH}`
`DISPlay:REFFFTView<x>:GRIDlines?`

Arguments HORizontal specifies horizontal grid lines.
 VERTical specifies vertical grid lines.
 BOTH specifies both horizontal and vertical grid lines.

Examples `DISPlay:REFFFTView5:GRIDlines hhorizontal` specifies horizontal grid lines.
`DISPlay:REFFFTView5:GRIDlines?` might return `:DISPLAY:REFFFTVIEW5:GRIDLINES BOTH` indicating that both vertical and horizontal grid lines are displayed

DISPlay:REFFFTView<x>:REF:REF<x>:STATE

This command sets or queries the state of the specified reference waveform in the specified waveview.

Group Display Control

Syntax `DISPlay:REFFFTView<x>:REF:REF<x>:STATE boolean`
`DISPlay:REFFFTView<x>:REF:REF<x>:STATE?`

Arguments <NR1> = 0 disables the specified reference; any other value turns this feature on.
 OFF disables the display the specified reference.
 ON enables the specified reference.

Examples `DISPlay:REFFFTView5:REF:REF5:STATE 0` turns display of the reference off.
`DISPlay:REFFFTView5:REF:REF5:STATE?` might return
`:DISPLAY:REFFFTVIEW5:REF:REF5:STATE 1` indicating the
reference is displayed.

DISPlay:REFFFTView<x>:XAXIS:SCALE

This command sets or queries the x-axis scale setting for Ref FFT.

Group Display Control

Syntax `DISPlay:REFFFTView<x>:XAXIS:SCALE {LINEAR|LOG}`
`DISPlay:REFFFTView<x>:XAXIS:SCALE?`

Arguments `LINEAR` specifies a linear scale.
`LOG` specifies a logarithmic scale.

Examples `DISPlay:REFFFTView5:XAXIS:SCALE 1` linear sets the x-axis scale to linear.
`DISPlay:REFFFTView5:XAXIS:SCALE?` might return
`:DISPLAY:REFFFTVIEW5:XAXIS:SCALE LOG` indicating x-axis
scale for the specified reference view is logarithmic.

DISPlay:REFFFTView<x>:ZOOM:XAXIS:FROM

This command sets or returns the left edge of the zoom x-axis in the specified plot view.

Group Zoom

Syntax `DISPlay:REFFFTView<x>:ZOOM:XAXIS:FROM <NR3>`
`DISPlay:REFFFTView<x>:ZOOM:XAXIS:FROM?`

Arguments `<NR3>` is the left edge of the zoom x-axis in the specified plot view.

Examples `DISPlay:REFFFTView5:ZOOM:XAXIS:FROM 2.0E+6` sets the left edge of
the specified view to 2.0 MHz.

DISplay:REFFFTView5:ZOOM:XAXIS:FROM? might return
:DISPLAY:REFFFTVIEW5:ZOOM:XAXIS:FROM 1.3295739742362E+6
indicating the left edge of the specified view is 1.33 MHz.

DISplay:REFFFTView<x>:ZOOM:XAXIS:TO

This command sets or queries the right edge of the zoom x-axis in the specified plot view.

Group	Zoom
Syntax	DISplay:REFFFTView<x>:ZOOM:XAXIS:TO <NR3> DISplay:REFFFTView<x>:ZOOM:XAXIS:TO?
Arguments	<NR3> is the right edge of the zoom x-axis in the specified plot view.
Examples	DISplay:REFFFTView5:ZOOM:XAXIS:TO 2.0E+8 sets the right edge of the zoom x-axis to 200 MHz. DISplay:REFFFTView5:ZOOM:XAXIS:TO? might return :DISPLAY:REFFFTVIEW5:ZOOM:XAXIS:TO 235.0376933178995E+6 indicating the right edge of the zoom x-axis is 235.0 MHz.

DISplay:REFFFTView<x>:ZOOM:YAXIS:FROM

This command sets or queries the bottom value of the zoom y-axis in the specified plot view.

Group	Zoom
Syntax	DISplay:REFFFTView<x>:ZOOM:YAXIS:FROM <NR3> DISplay:REFFFTView<x>:ZOOM:YAXIS:FROM?
Arguments	<NR3> is the bottom value of the zoom y-axis in the specified plot view.
Examples	DISplay:REFFFTView5:ZOOM:YAXIS:FROM -120.0 sets the bottom value of the zoom y-axis to -120.0 dBm.

DISplay:REFFFTView5:ZOOM:YAXIS:FROM? might return
:DISPLAY:REFFFTVIEW5:ZOOM:YAXIS:FROM -129.7533120632172
indicating the bottom value of the zoom y-axis is -129.8 dBm.

DISplay:REFFFTView<x>:ZOOM:YAXIS:TO

This command sets or queries the top value of the zoom y-axis in the specified plot view.

Group	Zoom
Syntax	DISplay:REFFFTView<x>:ZOOM:YAXIS:TO <NR3> DISplay:REFFFTView<x>:ZOOM:YAXIS:TO?
Arguments	<NR3> is the top value of the zoom y-axis in the specified plot view.
Examples	DISplay:REFFFTView5:ZOOM:YAXIS:TO 0.0 sets the top value of the zoom y-axis to 0.0 dBm. DISplay:REFFFTView5:ZOOM:YAXIS:TO? might return :DISPLAY:REFFFTVIEW5:ZOOM:YAXIS:TO 15.8820685863495 indicating the top value of the zoom y-axis is 15.9 dBm.

DISplay:SElect:BUS

This command sets or queries the overall selected bus. Sets are applied to all views that contain the source and the selected view is changed. When multiple buses are open, querying the command gives the correct result, but the bus cannot set.

Group	Display Control
Syntax	DISplay:SElect:BUS BUS<x> DISplay:SElect:BUS?
Arguments	BUS<x> is the selected bus.
Examples	DISplay:SElect:BUS Bus2 sets Bus 2 as the selected bus. DISplay:SElect:BUS? might return :DISPLAY:SELECT:BUS BUS1 indicating Bus 1 is the selected bus.

DISplay:SElect:MATH

This command sets or queries the overall selected math. Sets are applied to all views that contain the source and the selected view is changed. When multiple Math are open, querying the command gives the correct result, but the required Math cannot be set.

Group Display Control

Syntax DISplay:SElect:MATH MATH<x>
DISplay:SElect:MATH?

Arguments MATH<x> is the selected math.

Examples DISplay:SElect:MATH MATH1 selects Math 1.
DISplay:SElect:MATH? might display :DISPLAY:SELECT:MATH MATH2 indicating that Math 2 is selected.

DISplay:SElect:REference

This command sets or queries the overall selected reference waveform. Sets are applied to all views that contain the source and the selected view is changed.

Group Display Control

Syntax DISplay:SElect:REference {NONE|REF<x>}

Arguments Arguments are the selected reference.

Examples DISplay:SElect:REference REF2 selects reference 2.
DISplay:SElect:REFERENCE? might return DISPLAY:SELECT:REFERENCE NONE indicating no reference is selected.

DISplay:SElect:SOUrce

This command sets or queries the overall selected source. Sets are applied to all views that contain the source and the selected view is changed.

Group	Display Control
Syntax	<code>DISplay:SElect:SOURce</code> {NONE CH<x> BUS<x> MATH<x> PLOT<x> REF<x>}
Arguments	Arguments are the selected source.
Examples	<code>DISplay:SElect:SOURce Bus1</code> selects Bus1 as the selected source. <code>DISplay:SElect:SOURce?</code> might return <code>:DISPLAY:SELECT:SOURCE MATH1</code> indicating Math 1 is the selected source.

DISplay:SElect:VIEW

This command sets or queries the selected view.

Group	Display Control
Syntax	<code>DISplay:SElect:VIEW</code> {WAVEVIEW1 MATHFFT<x> PLOTVIEW<x> REFFFT<x>}
Arguments	Arguments are the selected view.
Examples	<code>DISplay:SElect:VIEW PLOTVIEW1</code> selects PLOTVIEW1 as the selected view. <code>DISplay:SElect:VIEW?</code> might return <code>:DISPLAY:SELECT:VIEW MATHFFT1</code> indicating MATHFFT1 is the selected view.

DISplay:SElect:WAVEView<x>:SOURce

This command sets or queries the selected source in the given waveview.

Group	Display Control
Syntax	<code>DISplay:SElect:WAVEView<x>:SOURce</code> {CH<x> MATH<x> BUS<x> REF<x> PLOT<x>}
Arguments	Arguments are the selected source.

Examples `DISPlay:SElect:WAVEView1:SOURce MATH1` sets MATH1 as the selected source.

`DISPlay:SElect:WAVEView1:SOURce?` might return
`:DISPLAY:SELECT:WAVEVIEW1:SOURCE CH2` indicating CH2 is the selected source.

DISplay:VARpersist

This command sets or queries display persistence decay time, which is the approximate decay time for a freshly struck persistence sample.

Group Display Control

Syntax `DISplay:VARpersist <NR3>`
`DISplay:VARpersist?`

Related Commands [DISplay:PERStistence](#)

Arguments <NR3> indicates the persistence decay time and ranges from 0.5 to 100.

Examples `DISPLAY:VARPERSIST 5` sets the persistence decay time to 5.

`DISPLAY:VARPERSIST?` might return `:DISPLAY:VARPERSIST 3.0000E-01`, indicating that persistence decay time is currently set to 0.300.

DISplay:WAVEView<x>:BUS:B<x>:STATE

Sets or queries the state of the specified bus in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.*

Group Display Control

Syntax `DISplay:WAVEView<x>:BUS:B<x>:STATE {OFF|ON|0|1}`
`DISplay:WAVEView<x>:BUS:B<x>:STATE?`

Arguments 0 turns specified bus off.
 1 turns the specified bus on.

ON turns the specified bus on.

OFF turns specified bus off.

Examples `DISPlay:WAVEView1:BUS:B1:STATE OFF` turns specified bus off.
 `DISPlay:WAVEView1:BUS:B1:STATE?` might return
 `:DISPlay:WAVEVIEW1:BUS:B1:STATE 1` indicating the specified bus is on.

DISPlay:WAVEView<x>:BUS:B<x>:VERTical:POSition

Sets or queries the vertical position of the specified bus in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.*

Group Display Control

Syntax `DISPlay:WAVEView<x>:BUS:B<x>:VERTical:POSition <NR3>`
 `DISPlay:WAVEView<x>:BUS:B<x>:VERTical:POSition?`

Arguments <NR3> is the vertical position of the specified bus.

Examples `DISPlay:WAVEView1:BUS:B1:VERTical:POSition 1.0e0` sets the position to 1 division.
 `DISPlay:WAVEView1:BUS:B1:VERTical:POSition?` might return
 `:DISPlay:WAVEVIEW1:BUS:B1:VERTICAL:POSITION 0.0E+0` indicating the position is 0.0 divisions.

DISPlay:WAVEView<x>:CH<x>:STATE

Sets or queries the state of the specified channel in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.*

Group Display Control

Syntax `DISPlay:WAVEView<x>:CH<x>:STATE {<NR1>|OFF|ON}`
 `DISPlay:WAVEView<x>:CH<x>:STATE?`

Arguments	<NR1> = 0 disables the specified channel on the specified waveview; any other value turns this feature on. OFF disables the display the specified channel on the specified waveview. ON enables the specified channel on the specified waveview.
Examples	DISplay:WAVEview1:CH1:STATE ON turns on channel 1. DISplay:WAVEview1:CH1:STATE? might return :DISPLAY:WAVEVIEW1:CH1:STATE 0 indicating channel 1 is off.

DISplay:WAVEView<x>:CH<x>:VERTical:POSition

Sets or queries the vertical position of the specified channel in the specified waveview in divisions. 0.0 divisions is center, 5.0 top of the window, and -5.0 the bottom of the window.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.*

Group	Display Control
Syntax	DISplay:WAVEview<x>:CH<x>:VERTical:POSition <NR3> DISplay:WAVEview<x>:CH<x>:VERTical:POSition?
Arguments	<NR3> is the vertical position in divisions. 0.0 divisions is center, 5.0 top of the window, and -5.0 the bottom of the window.
Examples	DISplay:WAVEview1:CH1:VERTical:POSition 1.0e0 sets the position to 1 division. DISplay:WAVEview1:CH1:VERTical:POSition? might return :DISPLAY:WAVEVIEW1:CH1:VERTICAL:POSITION -3.4400 indicating the channel 1 vertical position is -3.44 divisions.

DISplay:WAVEView<x>:CH<x>:VERTical:SCALE

Sets or queries the vertical scale of the specified channel in volts per division within the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.*

Group	Display Control
Syntax	<pre>DISplay:WAVEview<x>:CH<x>:VERTical:SCALE <NR3> DISplay:WAVEview<x>:CH<x>:VERTical:SCALE?</pre>
Arguments	<NR3> is the vertical scale of the specified channel.
Examples	DISplay:WAVEview1:CH1:VERTical:SCALE 2.0e2 sets the vertical scale to 200 mV per division. DISplay:WAVEview1:CH1:VERTical:SCALE? might return :DISPLAY:WAVEVIEW1:CH1:VERTICAL:SCALE 200.0000E-3 indicating the scale is 200 mV per division.

DISplay:WAVEView<x>:CH<x>_DALL:STATE

This command sets or queries the display state of the specified digital channel in the specified waveview.

Group	Display Control
Syntax	<pre>DISplay:WAVEview<x>:CH<x>_DALL:STATE {<NR1> OFF ON} DISplay:WAVEview<x>:CH<x>_DALL:STATE?</pre>
Arguments	<NR1> = 0 disables the display of the specified channels on the specified waveview; any other value turns this feature on. OFF disables the display of the specified channels on the specified waveview. ON enables the display of the specified channels on the specified waveview.
Examples	DISplay:WAVEview1:CH1_DALL:STATE ON enables the display of the specified channels on the specified waveview. DISplay:WAVEview1:CH1_DALL:STATE? might return :DISPLAY:WAVEVIEW1:CH1_DALL:STATE 0 indicating specified digital channels in the specified waveview are off.

DISplay:WAVEView<x>:CH<x>_DALL:VERTical:POSition

This command sets or queries the vertical position of the specified digital channel in the specified waveview in divisions. The position ranges from 5.0 to -5.0 divisions.

Group	Display Control
Syntax	DISplay:WAVEView<x>:CH<x>_DALL:VERTical:POSition <NR3> DISplay:WAVEView<x>:CH<x>_DALL:VERTical:POSition?
Arguments	<NR3> is the vertical position of the specified digital channel in the specified waveview in divisions.
Examples	DISplay:WAVEView1:CH1_DALL:VERTical:POSition 1.0e0 sets the vertical position to 1.0 divisions. DISplay:WAVEView1:CH1_DALL:VERTical:POSition? might return :DISPLAY:WAVEVIEW1:CH1_DALL:VERTICAL:POSITION 0.0E+0 indicating the vertical position is 0.0 divisions.

DISplay:WAVEView<x>:CH<x>_D<x>:STATE

This command sets or queries the display state of the specified digital channel in the specified waveview.

Group	Display Control
Syntax	DISplay:WAVEView<x>:CH<x>_D<x>:STATE {<NR1> OFF ON} DISplay:WAVEView<x>:CH<x>_D<x>:STATE?
Arguments	<NR1> = 0 disables the display of the specified channel on the specified waveview; any other value turns this feature on. OFF disables the display of the specified channel on the specified waveview. ON enables the display of the specified channel on the specified waveview.
Examples	DISplay:WAVEView1:CH1_D1:STATE OFF turns off the specified digital channel.

DISPlay:WAVEView1:CH1_D1:STATE? might return
:DISPLAY:WAVEVIEW1:CH1_D1:STATE 1 indicating the specified digital
channel is on.

DISPlay:WAVEView<x>:CURSor? (Query Only)

This query returns the cursor parameters for the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.*

Group	Cursor
Syntax	DISPlay:WAVEView<x>:CURSor?
Returns	Returns the cursor parameters for the specified waveview.
Examples	DISPlay:WAVEView1:CURSor? might return WAVEFORM;AUTO;2.4993299784074;2.4958885115371; -3.4414668702805E-3;"A";"A";300.0E-6;2.23696699999970; 256.967000E-3;-300.0E-6;-300.0E-6;300.0E-6; "???? ????";"s";"???? ????";600.0E-6;1.66666666666667E+3; 1;300.0E-6;-300.0E-6;SAME;4.1655499640123E+3;INDEPENDENT;AUTO

DISPlay:WAVEView<x>:CURSor:CURSOR<x>? (Query Only)

This query returns the cursor parameters for the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.
Cursor<x> is the specified cursor and must be CURSOR1.*

Group	Cursor
Syntax	DISPlay:WAVEView<x>:CURSor:CURSOR<x>?
Returns	Returns the cursor parameters for the specified cursor in the specified waveview.

Examples DISplay:WAVEView1:CURSor:CURSOR1? might return
:DISPLAY:WAVEVIEW1:CURSOR:CURSOR:WAVEFORM:BPOSITION
5.0E-6;APOSITION
-5.0E-6;:DISPLAY:WAVEVIEW1:CURSOR:CURSOR:BSOURCE
AUTO;HBARS:BPOSITION 9.91E+37;BUNITS
"V";AUNITS "V";APOSITION 9.91E+37;DELTA
9.91E+37;:DISPLAY:WAVEVIEW1:CURSOR:CURSOR:VBA
RS:BPOSITION 5.0E-6;APOSITION -5.0E-6;UNITS "s";DELTA
10.0E-6;ALTERNATEB "???? ????";ALTERNATEA
"???? ????";:DISPLAY:WAVEVIEW1:CURSOR:CURSOR:STATE
1;FUNCTION WAVEFORM;SCREEN:AYPOSITION
-2.0E-3;BXPOSITION -5.0E-6;BYPOSITION -2.0E-3;AXPOSITIO N
-5.0E-6;:DISPLAY:WAVEVIEW1:CURSOR:CURSOR:MODE
INDEPENDENT;ASOURCE AUTO;DDT
9.91E+37;ONEOVERDELTATVALUE 100.0E+3;SPLITM ODE
SAME;LINESTYLE SOLID.

DISplay:WAVEView<x>:CURSor:CURSOR<x>:ASOURce

This command sets or queries the cursor A source of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.
Cursor<x> is the specified cursor and must be CURSOR1.*

Group	Cursor
Syntax	DISplay:WAVEView<x>:CURSor:CURSOR<x>:ASOURce {AUTO CH<x> BUS<x> MATH<x> REF<x> PLOT<x>} DISplay:WAVEView<x>:CURSor:CURSOR<x>:ASOURce?
Arguments	Arguments are the specified source waveform.
Examples	DISplay:WAVEView1:CURSor:CURSOR1:ASOURce CH1 sets the cursor 1 source to channel 1. DISplay:WAVEView1:CURSor:CURSOR1:ASOURce? might return :DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:ASOURCE REF1 indicating the cursor A source is reference 1.

DISplay:WAVEView<x>:CURSor:CURSOR<x>:BSOURce

This command sets or queries the cursor B source of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.
Cursor<x> is the specified cursor and must be CURSOR1.*

Group	Cursor
Syntax	DISplay:WAVEView<x>:CURSor:CURSOR<x>:BSOURce {CH<x> BUS<x> MATH<x> REF<x> PLOT<x>} DISplay:WAVEView<x>:CURSor:CURSOR<x>:BSOURce?
Arguments	Arguments are the specified source waveform.
Examples	DISplay:WAVEView1:CURSor:CURSOR1:BSOURce CH1 sets the cursor B source to channel 1. DISplay:WAVEView1:CURSor:CURSOR1:BSOURce? might return :DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:BSOURCE REF1 indicating the cursor b source is reference 1.

DISplay:WAVEView<x>:CURSor:CURSOR<x>:DDT? (Query Only)

This query returns the delta V over delta T cursor readout value of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.
Cursor<x> is the specified cursor and must be CURSOR1.*

Group	Cursor
Syntax	DISplay:WAVEView<x>:CURSor:CURSOR<x>:DDT?
Returns	The delta V over delta T cursor readout value of the specified cursor in the specified waveview.

Examples `DISPlay:WAVEView1:CURSor:CURSOR1:DDT?` might return
`:DISPlay:WAVEVIEW1:CURSOR:CURSOR1:DDT 9.91E+37` indicating the delta
V over delta T cursor readout value of the specified cursor is 9.91E+37.

DISPlay:WAVEView<x>:CURSor:CURSOR<x>:FUNctIon

This command sets or queries the cursor type of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.
CURSor<x> is the specified cursor and must be CURSOR1.*

Group Cursor

Syntax `DISPlay:WAVEView<x>:CURSor:CURSOR<x>:FUNctIon`
`{SCREEN|WAVEFORM|VBArS|HBArS}`
`DISPlay:WAVEView<x>:CURSor:CURSOR<x>:FUNctIon?`

Arguments HBArS specifies horizontal bar cursors, which measure in vertical units.
VBArS specifies vertical bar cursors, which measure in horizontal units.
SCREEN specifies both horizontal and vertical bar cursors, which measure in horizontal and vertical units specified by the Cursor 1 and Cursor 2 Sources. Use these cursors to measure anywhere in the waveform display area.
WAVEform specifies paired or split cursors in YT display format for measuring waveform amplitude and time. In XY and XYZ format, these cursors indicate the amplitude positions of an XY pair (Ch1 vs Ch2 voltage, where Ch1 is the X axis and Ch2 is the Y axis) relative to the trigger.

Examples `DISPlay:WAVEVIEW1:CURSOR:CURSOR1:FUNCTION VBArS` selects the vertical bar cursor type for the specified waveview and cursor.
`DISPlay:WAVEVIEW1:CURSOR:CURSOR1:FUNCTION?` might return
`:DISPlay:WAVEVIEW1:CURSOR:CURSOR1:FUNCTION HBArS` indicating that the specified cursor is set to HBArS.

DISPlay:WAVEView<x>:CURSor:CURSOR<x>:HBArS:APOSition

Sets or queries the HBArS vertical A position of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.
Cursor<x> is the specified cursor and must be CURSOR1.*

Group	Cursor
Syntax	DISplay:WAVEView<x>:CURSor:CURSOR<x>:HBArS:APOSition <NR3> DISplay:WAVEView<x>:CURSor:CURSOR<x>:HBArS:APOSition?
Arguments	<NR3> is the vertical cursor A position of the specified cursor in the specified waveview. 0.0 divisions is center, 5.0 top of the waveview, and -5.0 the bottom of the waveview.
Examples	DISplay:WAVEView1:CURSor:CURSOR1:HBArS:APOSition 3.0e0 sets the cursor position to 3 V. DISplay:WAVEView1:CURSor:CURSOR1:HBArS:APOSition? might return :DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:HBARS:APOSITION 2.9303448275862 indicating the cursor position is 2.93 V.

DISplay:WAVEView<x>:CURSor:CURSOR<x>:HBArS:AUNIts? (Query Only)

This command queries the cursor A vertical units of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.
Cursor<x> is the specified cursor and must be CURSOR1.*

Group	Cursor
Syntax	DISplay:WAVEView<x>:CURSor:CURSOR<x>:HBArS:AUNIts?
Arguments	<QString> is the cursor A vertical units of the specified cursor in the specified waveview.
Examples	DISplay:WAVEView1:CURSor:CURSOR1:HBArS:AUNIts? might return :DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:HBARS:AUNITS "V" indicating the AUNITS are set to Volts.

DISplay:WAVEView<x>:CURSor:CURSOR<x>:HBArS:BPOStion

Sets or queries the HBArS vertical B position of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1. CURSor<x> is the specified cursor and must be CURSOR1.*

Group	Cursor
Syntax	DISplay:WAVEView<x>:CURSor:CURSOR<x>:HBArS:BPOStion <NR3> DISplay:WAVEView<x>:CURSor:CURSOR<x>:HBArS:BPOStion?
Arguments	<NR3> is the vertical cursor B position of the specified cursor in the specified waveview.
Examples	DISplay:WAVEView1:CURSor:CURSOR1:HBArS:BPOStion 4.0e-1 sets the position to 400 mV. DISplay:WAVEView1:CURSor:CURSOR1:HBArS:BPOStion? might return :DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:HBARS:BPOSITION 387.9148706896567E-3 indicating the B cursor position is 387.9 mV.

DISplay:WAVEView<x>:CURSor:CURSOR<x>:HBArS:BUNIts? (Query Only)

This command queries the cursor B vertical units of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1. CURSor<x> is the specified cursor and must be CURSOR1.*

Group	Cursor
Syntax	DISplay:WAVEView<x>:CURSor:CURSOR<x>:HBArS:BUNIts?
Returns	<QString> is the cursor B vertical units of the specified cursor in the specified waveview.

Examples `DISPlay:WAVEView1:CURSor:CURSOR1:HBArS:BUNItS?` might return
`:DISPlay:WAVEVIEW1:CURSOR:CURSOR1:HBARS:BUNITS` "V" indicating
the BUNITS are set to Volts.

DISPlay:WAVEView<x>:CURSor:CURSOR<x>:HBArS:DELTA? (Query Only)

This command queries the delta V cursor readout value of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.
CURSor<x> is the specified cursor and must be CURSOR1.*

Group Cursor

Syntax `DISPlay:WAVEView<x>:CURSor:CURSOR<x>:HBArS:DELTA?`

Returns The delta V cursor readout value of the specified cursor in the specified waveview.

Examples `DISPlay:WAVEView1:CURSor:CURSOR1:HBArS:DELTA?` might return
`:DISPlay:WAVEVIEW1:CURSOR:CURSOR1:HBARS:DELTA` 2.612085129310
indicating the delta between the cursors is 2.61 V.

DISPlay:WAVEView<x>:CURSor:CURSOR<x>:MODE

Sets or queries the cursor tracking mode of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.
CURSor<x> is the specified cursor and must be CURSOR1.*

Group Cursor

Syntax `DISPlay:WAVEView<x>:CURSor:CURSOR<x>:MODE`
`{INDEPENDENT|TRACK}`

Arguments TRACK ties the navigational functionality of the two cursors together. For cursor 1 adjustments, this ties the movement of the two cursors together; however, cursor 2 continues to move independently of cursor 1.

INDEPENDENT allows independent adjustment of the two cursors.

Examples `DISPLAY:WAVEVIEW1:CUSOR:CURSOR1:MODE TRACK` specifies that the cursor positions move in unison.

`DISPLAY:WAVEVIEW1:CUSOR:CURSOR1:MODE?` might return
`:DISPLAY:WAVEVIEW1:CUSOR:CURSOR1:MODE TRACK`, indicating that the two cursors move in unison.

DISplay:WAVEView<x>:CURSor:CURSOR<x>:ONEOVERDELTATVALUE? (Query Only)

This query returns the one over delta T cursor readout value of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1. Cursor<x> is the specified cursor and must be CURSOR1.*

Group Cursor

Syntax `DISplay:WAVEView<x>:CURSor:CURSOR<x>:ONEOVERDELTATVALUE?`

Returns The one over delta T cursor readout value of the specified cursor in the specified waveview.

Examples `DISplay:WAVEView1:CUSor:CURSOR1:ONEOVERDELTATVALUE?` might return
`:DISPLAY:WAVEVIEW1:CUSOR:CURSOR1:ONEOVERDELTATVALUE 179.3188E+3` indicating the one over delta time value is 179.3 kHz.

DISplay:WAVEView<x>:CURSor:CURSOR<x>:SCREEN:AXPOSition

Sets or queries the horizontal cursor A position of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1. Cursor<x> is the specified cursor and must be CURSOR1.*

Group Cursor

Syntax	<code>DISPlay:WAVEView<x>:CURSOR:CURSOR<x>:SCREEN:AXPOSITION <NR3></code> <code>DISPlay:WAVEView<x>:CURSOR:CURSOR<x>:SCREEN:AXPOSITION?</code>
Arguments	<NR3> is the horizontal cursor A position of the specified cursor in the specified waveview.
Examples	<code>DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:SCREEN:AXPOSITION 15.0E-3</code> sets the x position of screen cursor1 in waveview1 to 15 mV. <code>DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:SCREEN:AXPOSITION?</code> might return <code>:DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:SCREEN:AXPOSITION -64.0000E-03</code> indicating that x position of the specified screen cursor in the specified waveview is set to -64 mV.

DISPlay:WAVEView<x>:CURSOR:CURSOR<x>:SCREEN:AYPOSITION

This command sets or queries the vertical cursor A position of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.
CURSOR<x> is the specified cursor and must be CURSOR1.*

Group	Cursor
Syntax	<code>DISPlay:WAVEView<x>:CURSOR:CURSOR<x>:SCREEN:AYPOSITION <NR3></code> <code>DISPlay:WAVEView<x>:CURSOR:CURSOR<x>:SCREEN:AYPOSITION?</code>
Arguments	<NR3> the vertical cursor A position of the specified cursor in the specified waveview.
Examples	<code>DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:SCREEN:AYPOSITION 25.0E-3</code> sets the y position of the specified screen cursor of the specified waveview to 25 mV. <code>DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:SCREEN:AYPOSITION?</code> might return <code>:DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:SCREEN:AYPOSITION -53.0000E-03</code> indicating that y position of the specified screen cursor of the specified waveview is set to -53 mV.

DISplay:WAVEView<x>:CURSor:CURSOR<x>:SCREEN:BXPOsition

Sets or queries the horizontal cursor B position of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1. CURSor<x> is the specified cursor and must be CURSOR1.*

Group	Cursor
Syntax	DISplay:WAVEView<x>:CURSor:CURSOR<x>:SCREEN:BXPOsition <NR3> DISplay:WAVEView<x>:CURSor:CURSOR<x>:SCREEN:BXPOsition?
Arguments	<NR3> is the horizontal cursor B position of the specified cursor in the specified waveview.
Examples	DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:SCREEN:AXPOSITION 15.0E-3 sets the x position of the specified screen cursor in the specified waveview to 15 mV. DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:SCREEN:AXPOSITION? might return :DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:SCREEN:AXPOSITION -64.0000E-03 indicating that x position of the specified screen cursor in the specified waveview is set to -64 mV.

DISplay:WAVEView<x>:CURSor:CURSOR<x>:SCREEN:BYPOsition

This command sets or queries the vertical cursor B position of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1. CURSor<x> is the specified cursor and must be CURSOR1.*

Group	Cursor
Syntax	DISplay:WAVEView<x>:CURSor:CURSOR<x>:SCREEN:BYPOsition <NR3> DISplay:WAVEView<x>:CURSor:CURSOR<x>:SCREEN:BYPOsition?
Arguments	<NR3> the vertical cursor B position of the specified cursor in the specified waveview.

Examples `DISPLAY:WAVEVIEW1:Cursors:CURSOR1:SCREEN:BYPOSITION 25.0E-3`
 sets the y position of the screen cursor of waveview1 to 25 mV.

`DISPLAY:WAVEVIEW1:Cursors:CURSOR1:SCREEN:BYPOSITION?` might
 return `:DISPLAY:WAVEVIEW1:Cursors:CURSOR1:SCREEN:BYPOSITION`
`-53.0000E-03` indicating that y position of the specified screen cursor of the
 specified waveview is set to -53 mV.

DISplay:WAVEView<x>:CURSor:CURSOR<x>:SPLITMODE

This command sets or queries whether both cursors have the same or different sources.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.
 CURsor<x> is the specified cursor and must be CURSOR1.*

Group Cursor

Syntax `DISplay:WAVEview<x>:CURSor:CURSOR<x>:SPLITMODE {SAME|SPLIT}`
`DISplay:WAVEview<x>:CURSor:CURSOR<x>:SPLITMODE?`

Arguments SAME specifies both cursors have the same source.
 SPLIT specifies the cursors have different sources.

Examples `DISplay:WAVEview1:CURSor:CURSOR1:SPLITMODE SPLIT` specifies the
 cursors have different sources.

`DISplay:WAVEview1:CURSor:CURSOR1:SPLITMODE?` might return
`:DISPLAY:WAVEVIEW1:Cursors:CURSOR1:SPLITMODE SAME` indicating the
 cursors have the same source.

DISplay:WAVEView<x>:CURSor:CURSOR<x>:STATE

This command sets or queries the visible state of the specified cursor in the
 specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.
 CURsor<x> is the specified cursor and must be CURSOR1.*

Group Cursor

Syntax	DISPlay:WAVEView<x>:CURSor:CURSOR<x>:STATE {<NR1> OFF ON} DISPlay:WAVEView<x>:CURSor:CURSOR<x>:STATE?
Arguments	<NR1> = 0 disables the specified cursor in the specified waveview; any other value turns this feature on. OFF disables the specified cursor in the specified waveview. ON enables the specified cursor in the specified waveview.
Examples	DISPlay:WAVEView1:CURSor:CURSOR1:STATE OFF turns the specified cursor off. DISPlay:WAVEView1:CURSor:CURSOR1:STATE? might return :DISPlay:WAVEVIEW1:CURSOR:CURSOR1:STATE 1 indicating the specified cursor in the specified waveview is on.

DISPlay:WAVEView<x>:CURSor:CURSOR<x>:VBArS:APOsition

This command sets or queries the cursor A horizontal position of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.
CURSor<x> is the specified cursor and must be CURSOR1.*

Group	Cursor
Syntax	DISPlay:WAVEView<x>:CURSor:CURSOR<x>:VBArS:APOsition <NR3> DISPlay:WAVEView<x>:CURSor:CURSOR<x>:VBArS:APOsition?
Arguments	<NR3> is the horizontal cursor A position of the specified cursor in the specified waveview.
Examples	DISPlay:WAVEView1:CURSor:CURSOR1:VBArS:APOsition -4.0e-6 set the cursor position to 4.0 μs. DISPlay:WAVEView1:CURSor:CURSOR1:VBArS:APOsition? might return :DISPlay:WAVEVIEW1:CURSOR:CURSOR1:VBARS:APOSITION -5.0E-6 indicating the cursor A position is -5.0 μs.

DISplay:WAVEView<x>:CURSor:CURSOR<x>:VBArS:BPOStion

This command sets or queries the cursor B horizontal position of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1. CURSor<x> is the specified cursor and must be CURSOR1.*

Group	Cursor
Syntax	DISplay:WAVEView<x>:CURSor:CURSOR<x>:VBArS:BPOStion <NR3> DISplay:WAVEView<x>:CURSor:CURSOR<x>:VBArS:BPOStion?
Arguments	<NR3> is the horizontal cursor B position of the specified cursor in the specified waveview.
Examples	DISplay:WAVEView1:CURSor:CURSOR1:VBArS:BPOStion 4.0e-6 sets the cursor position to 4.0 μ s. DISplay:WAVEView1:CURSor:CURSOR1:VBArS:BPOStion? might return :DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:VBARS:BPOSITION 3.2736951319481E-6 indicating the B cursor position is 3.27 μ s.

DISplay:WAVEView<x>:CURSor:CURSOR<x>:VBArS:DELTA? (Query Only)

This query sets or returns the delta T cursor readout value of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1. CURSor<x> is the specified cursor and must be CURSOR1.*

Group	Cursor
Syntax	DISplay:WAVEView<x>:CURSor:CURSOR<x>:VBArS:DELTA?
Returns	The delta T cursor readout value of the specified cursor in the specified waveview.

Examples `DISPlay:WAVEview1:CURSOR:CURSOR1:VBARS:DELTA?` might return `:DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:VBARS:DELTA 6.7926388747725E-6` indicating the delta T cursor readout value is 6.79 μ s.

`DISPlay:WAVEView<x>:CURSOR:CURSOR<x>:VBARS:UNIts? (Query Only)`

This query returns cursor A vertical units of the specified cursor in the specified waveview.

NOTE. *WAVEview<x> is the specified waveview and must be WAVEview1. Cursor<x> is the specified cursor and must be CURSOR1.*

Group Cursor

Syntax `DISPlay:WAVEview<x>:CURSOR:CURSOR<x>:VBARS:UNIts?`

Returns The cursor A vertical units of the specified cursor in the specified waveview.

Examples `DISPlay:WAVEview1:CURSOR:CURSOR1:VBARS:UNIts?` might return `:DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:VBARS:UNITS "s"` indicating the cursor units are seconds.

`DISPlay:WAVEView<x>:CURSOR:CURSOR<x>:WAVEform:APOsition`

This command sets or queries the horizontal cursor A position of the specified cursor in the specified waveview.

NOTE. *WAVEview<x> is the specified waveview and must be WAVEview1. Cursor<x> is the specified cursor and must be CURSOR1.*

Group Cursor

Syntax `DISPlay:WAVEview<x>:CURSOR:CURSOR<x>:WAVEform:APOsition <NR3>`
`DISPlay:WAVEview<x>:CURSOR:CURSOR<x>:WAVEform:APOsition?`

Arguments `<NR3>` is the horizontal cursor A position of the specified cursor in the specified waveview.

Examples `DISPlay:WAVEView1:CURSor:CURSOR1:WAVEform:APOSition 1.5e-6`
 sets the position to 1.5 μ s.

`DISPlay:WAVEView1:CURSor:CURSOR1:WAVEform:APOSition?` might
 return `:DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:WAVEFORM:APOSITION`
`-1.2667480236557E-6` indicating the position is 1.27 μ s.

DISplay:WAVEView<x>:CURSor:CURSOR<x>:WAVEform:BPOSition

This command sets or queries the horizontal cursor B position of the specified cursor in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.
 CURsor<x> is the specified cursor and must be CURSOR1.*

Group Cursor

Syntax `DISPlay:WAVEView<x>:CURSor:CURSOR<x>:WAVEform:BPOSition`
`<NR3>`
`DISPlay:WAVEView<x>:CURSor:CURSOR<x>:WAVEform:BPOSition?`

Arguments `<NR3>` is the horizontal cursor B position of the specified cursor in the specified waveview.

Examples `DISPlay:WAVEView1:CURSor:CURSOR1:WAVEform:BPOSition 8.0e-9`
 sets the position to 8.0 ns.

`DISPlay:WAVEView1:CURSor:CURSOR1:WAVEform:BPOSition?` might
 return `:DISPLAY:WAVEVIEW1:CURSOR:CURSOR1:WAVEFORM:BPOSITION`
`711.6310718892624E-9` indicating the cursor position is 711.6 ns.

DISplay:WAVEView<x>:FILTer

This command sets or queries the type of interpolation filter for the display.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.*

Group Display Control

Syntax `DISPlay:WAVEView<x>:FILTer {SINX|LINEar}`

Arguments	LINEAR specifies linear interpolation, where acquired points are connected with straight lines. SINX specifies sin(x)/x interpolation, where acquired points are fit to a curve.
Examples	DISPlay:WAVEView1:FILTer SINX specifies sine-curve interpolation, when magnifying waveforms. DISPlay:WAVEView1:FILTer? might return :DISPlay:WAVEView1:FILTer LINEAR, indicating that straight-line interpolation is specified for magnifying waveforms.

DISPlay:WAVEView<x>:GRAticule

This command selects or queries the type of graticule that is displayed.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.*

Group	Display Control
Syntax	DISPlay:WAVEView<x>:GRAticule {GRID TIME FULl NONE} DISPlay:WAVEView<x>:GRAticule?
Arguments	GRID specifies a frame and grid only. TIME specifies a time graticule only. FULl specifies a frame, a grid and cross hairs. NONE specified no graticule.
Examples	DISPlay:WAVEView1:GRAticule TIME specifies a time graticule. DISPlay:WAVEView1:GRAticule? might return :DISPLAY:WAVEVIEW1:GRATICULE GRID indicating the graticule is a grid.

DISPlay:WAVEView<x>:INTENSITY:GRATICule

This command sets or queries the graticule saturation level.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.*

Group	Display Control
Syntax	DISplay:WAVEview<x>:INTENSITY:GRATicule <NR2> DISplay:WAVEview<x>:INTENSITY:GRATicule?
Arguments	<NR2> is the graticule saturation level.
Examples	DISplay:WAVEview1:INTENSITY:GRATicule 75 sets the saturation level to 75%. DISplay:WAVEview1:INTENSITY:GRATicule? might return :DISPLAY:WAVEVIEW1:INTENSITY:GRATICULE 66.0000 indicating the saturation level is at 66%.

DISplay:WAVEView<x>:INTENSITY:WAVEform

This command sets or queries the waveform saturation level.

NOTE. *WAVEview<x> is the specified waveview and must be WAVEview1.*

Group	Display Control
Syntax	DISplay:WAVEview<x>:INTENSITY:WAVEform <NR2> DISplay:WAVEview<x>:INTENSITY:WAVEform?
Arguments	<NR2> is the waveform saturation level.
Examples	DISplay:WAVEview1:INTENSITY:WAVEform 75 sets the saturation level to 75%. DISplay:WAVEview1:INTENSITY:WAVEform? might return :DISPLAY:WAVEVIEW1:INTENSITY:WAVEFORM 62.0000 indicating the saturation level is 62%.

DISplay:WAVEView<x>:MATH:MATH<x>:AUTOScale

This command sets or queries whether the specified math gets auto-scaled when the math equation changes within the specified waveview.

Group	Display Control
Syntax	<code>DISPlay:WAVEView<x>:MATH:MATH<x>:AUTOScale {<NR1> OFF ON}</code> <code>DISPlay:WAVEView<x>:MATH:MATH<x>:AUTOScale?</code>
Arguments	<NR1> = 0 disables the autoscaling the math in the specified waveview; any other value turns this feature on. OFF disables the autoscaling the math in the specified waveview. ON enables the autoscaling the math in the specified waveview.
Examples	<code>DISPlay:WAVEView1:MATH:MATH3:AUTOScale 1</code> enables the autoscaling the math in the specified waveview. <code>DISPlay:WAVEView1:MATH:MATH3:AUTOScale?</code> might return <code>:DISPlay:WAVEVIEW1:MATH:MATH3:AUTOSCALE 0</code> indicating that Math 3 will not auto-scale.

DISPlay:WAVEView<x>:MATH:MATH<x>:STATE

This command sets or queries the state of the specified math waveform in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.*

Group	Display Control
Syntax	<code>DISPlay:WAVEView<x>:MATH:MATH<x>:STATE {<NR1> OFF ON}</code> <code>DISPlay:WAVEView<x>:MATH:MATH<x>:STATE?</code>
Arguments	<NR1> = 0 disables the specified math in the specified waveview; any other value turns this feature on. OFF disables the specified math in the specified waveview. ON enables the specified math in the specified waveview.
Examples	<code>DISPlay:WAVEView1:MATH:MATH1:STATE OFF</code> disables the specified math in the specified waveview.

DISPlay:WAVEView1:MATH:MATH2:STATE? might return
:DISPLAY:WAVEVIEW1:MATH:MATH2:STATE 1 indicating the math
is displayed.

DISPlay:WAVEView<x>:MATH:MATH<x>:VERTical:POStion

This command sets or queries the vertical position in divisions of the specified math waveform in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.*

Group	Display Control
Syntax	DISPlay:WAVEView<x>:MATH:MATH<x>:VERTical:POStion <NR3> DISPlay:WAVEView<x>:MATH:MATH<x>:VERTical:POStion?
Arguments	<NR3> is the vertical position in divisions of the specified math waveform.
Examples	DISPlay:WAVEView1:MATH:MATH1:VERTical:POStion 0 sets the position to 0 divisions. DISPlay:WAVEView1:MATH:MATH1:VERTical:POStion? might return :DISPLAY:WAVEVIEW1:MATH:MATH1:VERTICAL:POSITION -2.2800 indicating the position is at -2.28 divisions.

DISPlay:WAVEView<x>:MATH:MATH<x>:VERTical:SCAle

Sets or queries the vertical scale of the specified math in volts per division within the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.*

Group	Display Control
Syntax	DISPlay:WAVEView<x>:MATH:MATH<x>:VERTical:SCAle <NR3> DISPlay:WAVEView<x>:MATH:MATH<x>:VERTical:SCAle?
Arguments	<NR3> is the vertical scale of the specified math waveform.

Examples `DISPlay:WAVEView1:MATH:MATH1:VERTical:SCAle 400e-3` sets the scale to 400 mV per division.

`DISPlay:WAVEView1:MATH:MATH1:VERTical:SCAle?` might return
`:DISPlay:WAVEVIEW1:MATH:MATH1:VERTICAL:SCALE 395.0000E-3`
 indicating the scale is 395 mV.

DISPlay:WAVEView<x>:PLOT:PLOT<x>:AUTOScale

This command sets or queries whether the specified trend gets auto-scaled when the new data is available within the specified waveview.

Group Display Control

Syntax `DISPlay:WAVEView<x>:PLOT:PLOT<x>:AUTOScale {<NR1>|OFF|ON}`
`DISPlay:WAVEView<x>:PLOT:PLOT<x>:AUTOScale?`

Arguments `<NR1> = 0` disables auto-scaling the specified plot in the specified waveview; any other value turns this feature on.

`OFF` disables auto-scaling the specified plot in the specified waveview.

`ON` enables auto-scaling the specified plot in the specified waveview.

Examples `DISPlay:WAVEView1:PLOT:PLOT1:AUTOScale 1` enables auto-scaling the specified plot in the specified waveview.

`DISPlay:WAVEView1:PLOT:PLOT1:AUTOScale?` might return
`:DISPlay:WAVEVIEW1:PLOT:PLOT1:AUTOSCALE 0` indicating that auto-scaling for plot 1 is off.

DISPlay:WAVEView<x>:PLOT:PLOT<x>:STATE

This command sets or queries the state of the specified time trend plot waveform in the specified waveview.

Group Display Control

Syntax `DISPlay:WAVEView<x>:PLOT:PLOT<x>:STATE {<NR1>|OFF|ON}`
`DISPlay:WAVEView<x>:PLOT:PLOT<x>:STATE?`

Arguments	<NR1> = 0 disables the specified plot in the specified waveview; any other value turns this feature on. OFF disables the specified plot in the specified waveview. ON enables the specified plot in the specified waveview.
Examples	DISplay:WAVEview1:PLOT:PLOT1:STATE OFF disables the specified plot in the specified waveview. DISplay:WAVEview1:PLOT:PLOT1:STATE? might return :DISPLAY:WAVEVIEW1:PLOT:PLOT1:STATE 1 indicating the specified plot in the specified waveview is on.

DISplay:WAVEView<x>:PLOT:PLOT<x>:VERTical:POSition

This command sets or queries the vertical position of the specified time trend in the specified waveview in absolute units.

Group	Display Control
Syntax	DISplay:WAVEview<x>:PLOT:PLOT<x>:VERTical:POSition <NR3> DISplay:WAVEview<x>:PLOT:PLOT<x>:VERTical:POSition?
Arguments	<NR3> is the vertical position.
Examples	DISplay:WAVEview1:PLOT:PLOT1:VERTical:POSition 2.0e0 sets the position to 2.0 units. DISplay:WAVEview1:PLOT:PLOT1:VERTical:POSition? might return :DISPLAY:WAVEVIEW1:PLOT:PLOT1:VERTICAL:POSITION 1.9918 indicating the position is 1.99 units.

DISplay:WAVEView<x>:PLOT:PLOT<x>:VERTical:SCAle

This command sets or queries the vertical scale of the specified time trend in units per division in the specified waveview.

Group	Display Control
Syntax	DISplay:WAVEview<x>:PLOT:PLOT<x>:VERTical:SCAle <NR3> DISplay:WAVEview<x>:PLOT:PLOT<x>:VERTical:SCAle?

Arguments	<NR3> is the vertical scale.
Examples	DISplay:WAVEview1:PLOT:PLOT1:VERTical:SCAle 725.0e-3 sets the scale to 725 mUnits. DISplay:WAVEview1:PLOT:PLOT1:VERTical:SCAle? might return :DISPLAY:WAVEVIEW1:PLOT:PLOT1:VERTICAL:SCALE 730.0000E-3 indicating the scale is 730.0 mUnits.

DISplay:WAVEView<x>:REF:REF<x>:STATE

This command sets or queries the state of the specified reference waveform in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.*

Group	Display Control
Syntax	DISplay:WAVEview<x>:REF:REF<x>:STATE {<NR1> OFF ON} DISplay:WAVEview<x>:REF:REF<x>:STATE?
Arguments	<NR1> = 0 disables the specified reference in the specified waveview; any other value turns this feature on. OFF disables the specified reference in the specified waveview. ON enables the specified reference in the specified waveview.
Examples	DISplay:WAVEview1:REF:REF1:STATE OFF disables the specified reference in the specified waveview. DISplay:WAVEview1:REF:REF1:STATE? might return :DISPLAY:WAVEVIEW1:REF:REF1:STATE 1 indicating the specified reference in the specified waveview is on.

DISplay:WAVEView<x>:REF:REF<x>:VERTical:POSition

This command sets or queries the vertical position in divisions of the specified reference in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.*

Group	Display Control
Syntax	DISplay:WAVEview<x>:REF:REF<x>:VERTical:POSition <NR3> DISplay:WAVEview<x>:REF:REF<x>:VERTical:POSition?
Arguments	<NR3> is the vertical position in divisions.
Examples	DISplay:WAVEview1:REF:REF1:VERTical:POSition -2.5e0 sets the position to -2.5 divisions. DISplay:WAVEview1:REF:REF1:VERTical:POSition? might return :DISPLAY:WAVEVIEW1:REF:REF1:VERTICAL:POSITION -2.6400 indicating the position is set to -2.64 divisions.

DISplay:WAVEView<x>:REF:REF<x>:VERTical:SCALE

This command sets or queries the vertical scale of the specified reference in volts per div within the specified waveview.

NOTE. *WAVEview<x> is the specified waveview and must be WAVEview1.*

Group	Display Control
Syntax	DISplay:WAVEview<x>:REF:REF<x>:VERTical:SCALE <NR3> DISplay:WAVEview<x>:REF:REF<x>:VERTical:SCALE?
Arguments	<NR3> is the vertical scale of the specified reference waveform.
Examples	DISplay:WAVEview1:REF:REF1:VERTical:SCALE 400e-3 sets the scale to 400 mV per division. DISplay:WAVEview1:REF:REF1:VERTical:SCALE? might return :DISPLAY:WAVEVIEW1:REF:REF1:VERTICAL:SCALE 378.0000E-3 indicating the scale is 378 mV per division.

DISplay:WAVEView<x>:STYLE

This command sets or queries how the waveforms are displayed for analysis mode.

NOTE. *WAVEview<x> is the specified waveview and must be WAVEview1.*

Group	Display Control
Syntax	<code>DISplay:WAVEview<x>:STYLE {VECTors DOTsonly}</code> <code>DISplay:WAVEview<x>:STYLE?</code>
Arguments	<code>DOTs</code> displays individual data points. New points immediately replace old ones. <code>VECTors</code> connects adjacent data points. New points immediately replace old ones.
Examples	<code>DISplay:WAVEview1:STYLE VECTORS</code> sets the display to connect adjacent data points. <code>DISplay:WAVEview1:STYLE?</code> might return <code>:DISplay:WAVEview1:STYLE DOTs</code>, indicating that data points are not connected.

DISplay:WAVEView<x>:VIEWStyle

The command sets or queries the waveform layout style used by the display.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEview1.*

Group	Display Control
Syntax	<code>DISplay:WAVEview<x>:VIEWStyle {OVERlay STACKed}</code> <code>DISplay:WAVEview<x>:VIEWStyle?</code>
Arguments	<code>OVERlay</code> specifies that the display view style used by the specified waveview is overlay. <code>STACKed</code> specifies that the display view style used by the specified waveview is stacked.
Examples	<code>DISplay:WAVEview1:VIEWStyle OVERLAY</code> sets the view style to overlay. <code>DISplay:WAVEview1:VIEWStyle?</code> might return <code>:DISplay:WAVEview1:VIEWStyle STACKed</code> indicating the view style is stacked.

DISplay:WAVEView<x>:Zoom? (Query Only)

This query returns the zoom parameters of the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.*

Group	Zoom
Syntax	DISplay:WAVEView<x>:Zoom?
Returns	Returns the zoom parameters of the specified waveview.
Examples	DISplay:WAVEView1:Zoom? might return :DISPLAY:WAVEVIEW1:ZOOM:ZOOM1:HORIZONTAL:SCALE 2.5000;POSITION 70.0000;WINSCALE 400.0000E-9;:DISPLAY:WAVEVIEW1:ZOOM:ZOOM 1:VERTICAL:SCALE 3.8000;POSITION 2.5951;:DISPLAY:WAVEVIEW1:ZOOM:ZOOM1:STATE 0.

DISplay:WAVEView<x>:ZOOM:ZOOM<x>? (Query Only)

This query returns the zoom parameters of the specified zoom in the specified waveview. <x> must be 1.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.
ZOOM<x> is the specified zoom and must be ZOOM1.*

Group	Zoom
Syntax	DISplay:WAVEView<x>:ZOOM:ZOOM<x>?
Returns	Returns the zoom parameters of the specified zoom in the specified waveview.
Examples	DISplay:WAVEView1:ZOOM:ZOOM1? might return :DISPLAY:WAVEVIEW1:ZOOM:ZOOM1:HORIZONTAL:SCALE 2.5000;POSITION 70.0000;WINSCALE 400.0000E-9;:DISPLAY:WAVEVIEW1:ZOOM:ZOOM 1:VERTICAL:SCALE 3.8000;POSITION 2.5951;:DISPLAY:WAVEVIEW1:ZOOM:ZOOM1:STATE 0.

DISplay:WAVEView<x>:ZOOM:ZOOM<x>:HORizontal:POSition

Sets or queries the horizontal zoom position (of the specified zoom in the specified waveview) of the zoomed waveform or zoom waveform in the display, around

which the zoom waveform displays. It is freely movable around the acquisition settings (horizontal span). An acquired waveform or reference could extend off screen. The valid zoom area does not care about the waveform itself, only the user setting for acquisition.

For example, if horizontal scale is set to 1 second, position to 50, then the acquisition area will go from -5 s to +5 s. Zoom window 0 will focus on -5 s and zoom area 100 will focus on +5 s. If the instrument is stopped and the scale changed to 0.5 s, there will be data off the ends of the display. However, 0% zoom will put the user focus on -2.5 s, the lower bound of the acquisition span.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1. ZOOM<x> is the specified zoom and must be ZOOM1.*

Group	Zoom
Syntax	DISplay:WAVEView<x>:ZOOM:ZOOM<x>:HORizontal:POSition <NR3> DISplay:WAVEView<x>:ZOOM:ZOOM<x>:HORizontal:POSition?
Arguments	<NR3> is a value from 0 to 100.00 and is the percent of the waveform that is to the left of screen center, when the zoom factor is 2× or greater.
Examples	DISPLAY:WAVEVIEW1:ZOOM:ZOOM1:HORIZONTAL:POSITION 50 sets the horizontal position of the zoom box of waveview1 to 50 so that it is centered horizontally on the screen. DISPLAY:WAVEVIEW1:ZOOM:ZOOM1:HORIZONTAL:POSITION? might return :DISPLAY:WAVEVIEW1:ZOOM:ZOOM1:HORIZONTAL:POSITION 10.0000, indicating that the horizontal position of the zoom box of waveview 1 is centered over the first major graticule division.

DISplay:WAVEView<x>:ZOOM:ZOOM<x>:HORizontal:SCALE

This command sets or queries the horizontal zoom factor of the specified zoom in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1. ZOOM<x> is the specified zoom and must be ZOOM1.*

Group Zoom

Syntax	<code>DISPlay:WAVEView<x>:ZOOM:ZOOM<x>:HORIZontal:SCALE <NR3></code> <code>DISPlay:WAVEView<x>:ZOOM:ZOOM<x>:HORIZontal:SCALE?</code>
Arguments	<NR3> is the amount of expansion in the horizontal direction in 1-2-4 increments of the specified zoom in the specified waveview.
Examples	<code>DISPlay:WAVEView1:ZOOM:ZOOM1:HORIZontal:SCALE 5</code> sets the horizontal zoom factor of zoom1 in waveview1 to 5x. <code>DISPlay:WAVEView1:ZOOM:ZOOM1:HORIZontal:SCALE?</code> might return <code>:DISPlay:WAVEView1:ZOOM:ZOOM1:HORIZontal:SCALE 10</code> , indicating that the zoom factor of waveview1 is set to 10x.

DISPlay:WAVEView<x>:ZOOM:ZOOM<x>:HORIZontal:WINSCALE

This command sets or queries the overview window horizontal scale in the specified waveview.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.
ZOOM<x> is the specified zoom and must be ZOOM1.*

Group	Zoom
Syntax	<code>DISPlay:WAVEView<x>:ZOOM:ZOOM<x>:HORIZontal:WINSCALE <NR3></code> <code>DISPlay:WAVEView<x>:ZOOM:ZOOM<x>:HORIZontal:WINSCALE?</code>
Arguments	<NR3> is the horizontal scale of the zoom window.
Examples	<code>DISPlay:WAVEView1:ZOOM:ZOOM1:HORIZontal:WINSCALE?</code> might return <code>:DISPlay:WAVEView1:ZOOM:ZOOM1:HORIZontal:WINSCALE 80.0000E-9</code> indicating that the horizontal scale of the overview window is 80 ns per division.

DISPlay:WAVEView<x>:ZOOM:ZOOM<x>:STATe

This command sets or queries the zoom display state of the specified zoom in the specified waveview. This command is equivalent to pushing the zoom button on the front panel.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1.
ZOOM<x> is the specified zoom and must be ZOOM1.*

Group	Zoom
Syntax	DISPlay:WAVEView<x>:ZOOM:ZOOM<x>:STATE {ON OFF <NR1>} DISPlay:WAVEView<x>:ZOOM:ZOOM<x>:STATE?
Arguments	ON turns the specified zoom on. OFF turns specified zoom off. <NR1> = 0 disables the specified zoom; any other value enables the specified zoom.
Examples	DISPlay:WAVEView1:ZOOM:ZOOM1:STATE ON turns the specified zoom on. DISPlay:WAVEView1:ZOOM:ZOOM1:STATE? might return :DISPlay:WAVEView1:ZOOM:ZOOM1:STATE 1, indicating that specified zoom is on.

DISPlay:WAVEView<x>:ZOOM:ZOOM<x>:VERTical:POSition

This command sets or queries the vertical position of the specified zoom in the specified waveview. It is freely movable within the confines of the acquired waveform. It is measured from the top to bottom of the acquisition window. The top of the zoom window is $-5 * \text{vertical zoom factor}$. The bottom of the zoom window is $+5 * \text{the vertical zoom factor}$. For a zoom of 5x, the position ranges from -25 to 25.

NOTE. *WAVEView<x> is the specified waveview and must be WAVEView1. ZOOM<x> is the specified zoom and must be ZOOM1.*

Group	Zoom
Syntax	DISPlay:WAVEView<x>:ZOOM:ZOOM<x>:VERTical:POSition <NR3> DISPlay:WAVEView<x>:ZOOM:ZOOM<x>:VERTical:POSition?
Arguments	NR3 is the vertical position of the specified zoom in the specified waveview. It is freely movable within the confines of the acquired waveform. The top of the zoom window is $-5 * \text{vertical zoom factor}$. The bottom of the zoom window is $+5 * \text{the vertical zoom factor}$. For a vertical zoom of 5x, the position ranges from -25 to 25.
Examples	DISPlay:WAVEView1:ZOOM:ZOOM1:VERTical:POSition 50 sets the vertical position of the specified zoom in the specified waveview to 50, where the vertical

zoom factor is 10x, top of screen is -50 and bottom if +50, and 0 is the vertical center.

DISplay:WAVEview1:ZOOM:ZOOM1:VERTical:POSition? might return :DISplay:WAVEview1:ZOOM:ZOOM1:VERTical:POSition 23.90000, indicating that the vertical position of the specified zoom in the specified waveview is set to 23.9, where the vertical zoom factor is 10x, top of screen is -50 and bottom if +50, and 0 is the vertical center.

DISplay:WAVEView<x>:ZOOM:ZOOM<x>:VERTical:SCALE

This command sets or queries the vertical zoom factor of the specified zoom in the specified waveview.

NOTE. *WAVEview<x> is the specified waveview and must be WAVEview1. ZOOM<x> is the specified zoom and must be ZOOM1.*

Group Zoom

Syntax DISplay:WAVEview<x>:ZOOM:ZOOM<x>:VERTical:SCALE <NR3>
DISplay:WAVEview<x>:ZOOM:ZOOM<x>:VERTical:SCALE?

Arguments <NR3> is the amount of vertical expansion or compression. Based on the value that you entered, this command uses the nearest scale factor. Setting the vertical scale to 1 indicates unity (no zoom).

Examples DISplay:WAVEview1:ZOOM:ZOOM2:VERTical:SCALE 5 sets the vertical scale of zoom1 of waveview1 to 5x.

DISplay:WAVEview1:ZOOM:ZOOM2:VERTical:SCALE? might return :DISplay:WAVEview1:ZOOM:ZOOM2:VERTical:SCALE 2.0000, indicating that the vertical scale of zoom1 of waveview1 is 2x.

DISplay:WAVEView<y>:REF:REF<x>:FRAME

This command sets or returns the selected frame of the specified analog ref. Each ref has a unique selected frame.

Group Display Control

Syntax	<code>DISplay:WAVEview<y>:REF:REF<x>:FRAME <NR1></code> <code>DISplay:WAVEview<y>:REF:REF<x>:FRAME?</code>
Arguments	<NR1> is the selected frame of the specified analog ref.
Examples	<code>DISplay:WAVEview1:REF:REF2:FRAME?</code> might return <code>:DISPLAY:WAVEVIEW1:REF:REF2:FRAME 1</code> , indicating the selected frame is 1.

DISplay:WAVEView<y>:REF<x>_DALL:FRAMe

This command sets or returns the selected frame of the specified digital ref. Each ref has a unique selected frame.

Group	Display Control
Syntax	<code>DISplay:WAVEview<y>:REF<x>_DALL:FRAME <NR1></code> <code>DISplay:WAVEview<y>:REF<x>_DALL:FRAME?</code>
Arguments	<NR1> is the selected frame of the specified digital ref.
Examples	<code>DISplay:WAVEview<y>:REF<x>_DALL:FRAME?</code> might return <code>:DISPLAY:WAVEVIEW1:REF1_DALL:FRAME 1</code> , indicating the selected frame is 1.

DVM (No Query Form)

Resets the Digital Voltmeter measurements and history.

Conditions	Requires DVM option (free with product registration).
Group	DVM
Syntax	<code>DVM RESET</code>
Arguments	RESET specifies resetting DVM measurements and history.

Examples DVM RESET resets the DVM measurement and history.

DVM:AUTORange

Sets (or queries) the autorange state for the Digital Voltmeter.

NOTE. *the DVM will not autorange as long as the DVM source is the same channel as the trigger source.*

Conditions Requires DVM option (free with product registration).

Group DVM

Syntax DVM:AUTORange {0|1|OFF|ON}
DVM:AUTORange?

Arguments 1 or ON turns on autorange for the Digital Voltmeter.
0 or OFF turns autorange off.

Examples DVM:AUTO ON turns on autorange for the Digital Voltmeter.

DVM:MEASUrement:FREQuency? (Query Only)

This command returns the current frequency value for the DVM.

Conditions Requires DVM option (free with product registration).

Group DVM

Syntax DVM:MEASUrement:FREQuency?

Related Commands [DVM:TRIGger:FREQuency:COUNTer](#)

Examples DVM:MEASU:FREQ? might return 100.0000E+3, which represents 100 kHz as the current frequency value for the DVM.

DVM:MEASUrement:HIStory:AVERage? (Query Only)

Returns the average DVM readout value over the history period. The history period is a constant period of 5 seconds.

Conditions Requires DVM option (free with product registration).

Group DVM

Syntax DVM:MEASUrement:HIStory:AVERage?

Examples DVM:MEASU:HIStory:AVE? might return :DVM:MEASUREMENT:HISTORY:AVERAGE 429.3000E-3 which represents the average DVM readout value over the history period in volts.

DVM:MEASUrement:HIStory:MAXimum? (Query Only)

Returns the maximum readout value for the DVM function over the history period. The history period is a constant period of 5 seconds.

Conditions Requires DVM option (free with product registration).

Group DVM

Syntax DVM:MEASUrement:HIStory:MAXimum?

Examples DVM:MEASU:HIStory:MAX? might return :DVM:MEASUREMENT:HISTORY:MAXIMUM 431.9000E-3, which represents the maximum readout value for the DVM function over the history period in volts.

DVM:MEASUrement:HIStory:MINimum? (Query Only)

Returns the minimum readout value for the DVM over the history period. The history period is a constant period of 5 seconds.

Conditions Requires DVM option (free with product registration).

Group DVM

Syntax DVM:MEASUREMENT:HISTORY:MINIMUM?

Examples DVM:MEASU:HIS:MINI? might return
:DVM:MEASUREMENT:HISTORY:MINIMUM 430.9000E-3, which represents the minimum readout value for the DVM function over the history period in volts.

DVM:MEASUREMENT:INFMAXimum? (Query Only)

Returns the maximum DVM readout value over the entire time that the DVM has been on since the last change using the [DVM:MODE](#) or [DVM:SOURCE](#) commands or [DVM RESET](#).

Conditions Requires DVM option (free with product registration).

Group DVM

Syntax DVM:MEASUREMENT:INFMAXimum?

Examples DVM:MEASU:INFMAX? might return :DVM:MEASUREMENT:INFMAXIMUM
432.9000E-3, which represents the maximum readout value (in volts) of the DVM function over the entire time that the DVM has been on since the last change using the [DVM:MODE](#) or [DVM:SOURCE](#) commands or DVM RESET.

DVM:MEASUREMENT:INFMINimum? (Query Only)

Returns the minimum readout value of the DVM over the entire time that the DVM has been on since the last change using the [DVM:MODE](#) or [DVM:SOURCE](#) commands or DVM RESET.

Conditions Requires DVM option (free with product registration).

Group DVM

Syntax DVM:MEASUREMENT:INFMINimum?

Examples DVM:MEASU:INMIN? might return :DVM:MEASUREMENT:INFMINIMUM
427.3000E-3, which represents the minimum readout value of the DVM

function (in volts) over the entire time that the DVM has been on since the last change using the **DVM:MODE** or **DVM:SOURce** commands or **DVM RESET**.

DVM:MEASUrement:VALue? (Query Only)

Returns the DVM readout value (the largest displayed value at the top of the DVM screen).

Conditions Requires DVM option (free with product registration).

Group DVM

Syntax DVM:MEASUrement:VALue?

Examples DVM:MEASU:VAL? might return :DVM:MEASUREMENT:VALUE 430.7000E-3, which represents the DVM value.

DVM:MODE

This command specifies (or queries) the mode to use for the Digital Voltmeter.

Conditions Requires DVM option (free with product registration).

Group DVM

Syntax DVM:MODE {ACRMS | ACDCRMS | DC | OFF}
DVM:MODE?

Arguments ACRMS – displays the root-mean-square value of the acquired data, with the DC component removed.

ACDCRMS – displays the RMS value of the acquired data.

DC – displays the DC value of the acquired data.

OFF

Examples DVM:MODE DC sets the mode for the DVM to DC, which displays the DC value of the acquired data.

DVM:MOD? might return ACRMS, which indicates the mode is currently set to ACRMS.

DVM:SOUrce

This command sets (or queries) the source for the DVM.

Conditions	Requires DVM option (free with product registration).
Group	DVM
Syntax	DVM:SOUrce {CH<x>} DVM:SOUrce?
Arguments	CH<x> specify which channel to use as the source for the DVM.
Examples	DVM:SOURCE CH4 sets the DVM source to Channel 4.

DVM:TRIGger:FREQuency:COUNTer

This command sets or queries the state of the trigger frequency counter readout in the trigger badge.

Conditions	Requires DVM option (free with product registration).
Group	DVM
Syntax	DVM:TRIGger:FREQuency:COUNTer {0 1 OFF ON} DVM:TRIGger:FREQuency:COUNTer?
Arguments	1 or ON turns on the trigger frequency counter for the Digital Voltmeter. 0 or OFF turns it off.
Examples	DVM:TRIGGER:FREQUENCY:COUNTER 0 turns off the counter. DVM:TRIGGER:FREQUENCY:COUNTER? might return :DVM:TRIGGER:FREQUENCY:COUNTER 1 indicating the counter is on.

***ESE**

This command sets and queries the bits in the Event Status Enable Register (ESER). The ESER prevents events from being reported to the Status Byte Register (STB). For a more detailed discussion of the use of these registers, see Registers.

Group Status and Error

Syntax *ESE <NR1>
 *ESE?

Related Commands [*CLS](#)
 [DESE](#)
 [*ESR?](#)
 [EVENT?](#)
 [EVMsg?](#)
 [*SRE](#)
 [*STB?](#)

Arguments <NR1> specifies the binary bits of the ESER according to this value, which ranges from 0 through 255.

The power-on default for the ESER is 0 if *PSC is 1. If *PSC is 0, the ESER maintains the previous power cycle value through the current power cycle.

NOTE. *Setting the DESER and the ESER to the same values allows only those codes to be entered into the Event Queue and summarized on the ESB bit (bit 5) of the Status Byte Register. Use the [DESE](#) command to set the DESER.*

Examples *ESE 209 sets the ESER to binary 11010001, which enables the PON, URQ, EXE, and OPC bits.

*ESE? might return 186, showing that the ESER contains [DESE](#) the binary value 10111010.

*ESR? (Query Only)

This query-only command returns the contents of the Standard Event Status Register (SESR). *ESR? also clears the SESR (since reading the SESR clears it). For a more detailed discussion of the use of these registers, see Registers.

Group	Status and Error
Syntax	*ESR?
Related Commands	ALLEv? *CLS DESE *ESE EVENT? EVMsg? *SRE *STB?
Examples	*ESR? might return *ESR 213, showing that the SESR contains the binary value 11010101.

ETHERnet:DHCPbootp

This command sets the network configuration method to DHCP (that is ON) or static IP address (that is OFF).

Group	Ethernet
Syntax	ETHERnet:DHCPbootp {ON OFF} ETHERnet:DHCPbootp?
Arguments	ON enables the oscilloscope to search the network for a DHCP server in order to automatically assign a dynamic IP address to the oscilloscope.

NOTE. Do not use DHCP searching if your oscilloscope has been assigned a static address on a network. If you set this command to ON, the DHCP search will delete or change your static IP address information.

OFF disables the oscilloscope to search the network for a DHCP server.

Examples `ETHERNET:DHCPBOOTP ON` sets the oscilloscope to search for a DHCP server and assign a dynamic IP address to the oscilloscope.

ETHERnet:DNS:IPADdress

This command specifies the network Domain Name Server (DNS) IP address.

Group Ethernet

Syntax `ETHERnet:DNS:IPADdress <QString>`
`ETHERnet:DNS:IPADdress?`

Arguments `<QString>` is a standard IP address value, enclosed in quotes.

Examples `ETHERnet:DNS:IPADdress "128.196.13.252"` sets the DNS IP address that the oscilloscope uses to communicate with the network.

ETHERnet:DOMAINname

This command specifies the network domain name.

Group Ethernet

Syntax `ETHERnet:DOMAINname <QString>`
`ETHERnet:DOMAINname?`

Arguments `<QString>` is the network domain name, enclosed in quotes.

Examples `ETHERnet:DOMAINNAME "Alpha1.Mycorp.com"` sets the domain name that the oscilloscope uses to communicate with the network.

ETHERnet:ENET:ADdResS? (Query Only)

Returns the Ethernet address (MAC address) value assigned to the oscilloscope. This is assigned at the factory and can not be changed.

Group Ethernet

Syntax ETHERnet:ENET:ADdResS?

Examples ETHERNET:ENET:ADDRESS? returns an Ethernet address such as 08:00:11:01:02:03

ETHERnet:GATEWay:IPADdResS

This command specifies the network gateway IP address.

Group Ethernet

Syntax ETHERnet:GATEWay:IPADdResS <QString>
ETHERnet:GATEWay:IPADdResS?

Related Commands [ETHERnet:NETWORKCONFig](#)
[ETHERnet:IPADdResS](#)
[ETHERnet:SUBNETMask](#)

Arguments <QString> is a standard IP address value, enclosed in quotes.

Examples ETHERNET:GATEWAY:IPADDRESS "128.143.16.1" sets the gateway IP address.

ETHERnet:IPADdResS

This command sets the IP address assigned to the oscilloscope.

Group Ethernet

Syntax `ETHERnet:IPADDRESS <QString>`
`ETHERnet:IPADDRESS?`

Related Commands [ETHERnet:NETWORKCONFig](#)
[ETHERnet:SUBNETMask](#)
[ETHERnet:GATEWay:IPADDRESS](#)

Arguments `<QString>` is a standard IP address value, enclosed in quotes.

Examples `ETHERNET:IPADDRESS "123.121.13.214"` sets the oscilloscope's IP address.

ETHERnet:LXI:LAN:RESET (No Query Form)

This command resets the LXI local area network.

Group Ethernet

Syntax `ETHERnet:LXI:LAN:RESET`

Examples `ETHERnet:LXI:LAN:RESET` resets the LXI local area network.

ETHERnet:LXI:LAN:SERVICENAME

This command sets or queries the service name used for the LXI interface.

Group Ethernet

Syntax `ETHERnet:LXI:LAN:SERVICENAME <QString>`
`ETHERnet:LXI:LAN:SERVICENAME?`

Arguments `<QString>` is a quoted string of up to 64 characters that specifies the mDNS service name used for the LXI interface.

Examples `ETHERnet:LXI:LAN:SERVICENAME?` might return "Tektronix Oscilloscope MSO5 053CVV"

ETHERnet:LXI:LAN:STATus? (Query Only)

This query returns the LXI network status: one of OK, FAULT, or IDENTIFY. IDENTIFY indicates that the device identify mode is enabled.

Group	Ethernet
Syntax	ETHERnet:LXI:LAN:STATus?
Related Commands	ETHERnet:PING ETHERnet:PING:STATus?
Returns	OK — indicates the network is running and the oscilloscope can “see” the network. FAULT — indicates the network is not visible, or the network settings are incorrect. IDENTIFY indicates that a message is being displayed on the oscilloscope’s front panel indicating which scope the user is currently accessing.
Examples	ETHER:LXI:LAN:STAT? might return FAULT, indicating the network is not visible, or the network settings are incorrect.

ETHERnet:NAME

This command sets or queries the instrument Ethernet hostname assigned to the oscilloscope.

Group	Ethernet
Syntax	ETHERnet:NAME <QString> ETHERnet:NAME?
Arguments	<QString> is the network name assigned to the oscilloscope, enclosed in quotes.
Examples	ETHERnet:NAME "labcscope1" sets the oscilloscope's network name.

ETHERnet:NETWORKCONFig

This command specifies the Ethernet network configuration setting.

Group	Ethernet
Syntax	<code>ETHERnet:NETWORKCONFig {AUTOMATIC MANual}</code> <code>ETHERnet:NETWORKCONFig?</code>
Related Commands	ETHERnet:NAME ETHERnet:IPADdress ETHERnet:SUBNETMask ETHERnet:GATEWay:IPADdress ETHERnet:DHCPbootp
Arguments	<code>AUTOMATIC</code> specifies that the oscilloscope's IP address, subnet mask and gateway settings will be received from a DHCP server on the local network. <code>MANual</code> specifies that the Ethernet settings will be configured manually, using ETHERnet:IPADdress, ETHERnet:SUBNETMask, and ETHERnet:GATEWay:IPADdress.
Examples	<code>ETHERnet:NETWORKCONFig MANual</code> specifies to configure the Ethernet settings manually. <code>ETHERnet:NETWORKCONFig?</code> might return <code>AUTOMATIC</code>, indicating the settings are being configured automatically.

ETHERnet:PING (No Query Form)

Sends a ping packet to the instrument gateway and sets the status accordingly.

Group	Ethernet
Syntax	<code>ETHERnet:PING EXECute</code>
Examples	<code>ETHERnet:PING EXECute</code> causes the oscilloscope to ping the gateway IP address.

ETHERnet:PING:STATus? (Query Only)

Returns the results of sending the [ETHERnet:PING](#) command to ping the gateway IP address.

Group	Ethernet
Syntax	ETHERnet:PING:STATUS?
Returns	OK is returned if the computer at the gateway IP address answers. NORESPONSE is returned if the computer at the gateway IP address does not answer. TRYING is returned if the ping operation is still executing. NEVER is returned if ETHERnet:PING EXECute has not been previously sent, indicating that no ping response has ever been received in response to an ETHERnet:PING EXECute command since the instrument was powered on.

ETHERnet:SUBNETMask

This command sets or queries the instrument subnet mask value.

Group	Ethernet
Syntax	ETHERnet:SUBNETMask <QString> ETHERnet:SUBNETMask?
Related Commands	ETHERnet:NETWORKCONFig ETHERnet:IPADdress ETHERnet:GATEWay:IPADdress
Arguments	<QString> is the subnet mask value, enclosed in quotes.
Examples	ETHERNET:SUBNETMASK "255.255.255.0" sets the subnet mask value using standard IP address notation format.

EVENT? (Query Only)

This query-only command returns an event code from the Event Queue that provides information about the results of the last [*ESR?](#) read. EVENT? also removes the returned value from the Event Queue.

Group	Status and Error
--------------	------------------

Syntax	EVENT?
Related Commands	ALLEv? *CLS DESE *ESE *ESR? EVMsg? *SRE *STB?
Examples	EVENT? might return :EVENT 110, showing that there was an error in a command header.

EVMsg? (Query Only)

This query-only command removes a single event code from the Event Queue that is associated with the results of the last [*ESR?](#) read and returns the event code with an explanatory message. For more information, see Event Handling.

Group	Status and Error
Syntax	EVMsg?
Related Commands	ALLEv? *CLS DESE *ESE *ESR? EVENT? *SRE *STB?
Returns	The event code and message in the following format:

```
<Event Code><Comma><QString>[<Event  
Code><Comma><QString>...]<QString>::= <Message>;[<Command>]
```

where <Command> is the command that caused the error and may be returned when a command error is detected by the instrument. As much of the command will be returned as possible without exceeding the 60 character limit of the <Message> and <Command> string combined. The command string is right-justified.

Examples EVMSG? might return :EVMSG 110,"Command header error".

EVQty? (Query Only)

This query-only command returns the number of events that are enabled in the queue. This is useful when using the [ALLEv?](#) query, since it lets you know exactly how many events will be returned.

Group Status and Error

Syntax EVQty?

Related Commands [ALLEv?](#)
[EVENT?](#)
[EVMsg?](#)

Examples EVQTY? might return :EVQTY 3, indicating the number of event codes in the Event Queue.

EYEMASK:MASK<x>:COUNT:HITS? (Query Only)

This command returns the total number of hit violations for all segments in the specified mask test.

Conditions Requires option 5-DJA or 6-DJA (Advanced Jitter Analysis).

Group Mask

Syntax EYEMASK:MASK<x>:COUNT:HITS?

Related Commands [PLOT:PLOT<x>:MASK?](#)

Arguments MASK<x> is the number of the specified mask test in an eye diagram plot.

Examples EYEMASK:MASK2:COUNT:HITS? might return 211, indicating that mask 2 has a total of 211 violations for all segments in the mask.

EYEMASK:MASK<x>:COUNT:SEG<y>:HITS? (Query Only)

This command returns the number of hit violations for the specified segment (area).

Conditions Requires option 5-DJA or 6-DJA (Advanced Jitter Analysis).

Group Mask

Syntax EYEMASK:MASK<x>:COUNT:SEG<y>:HITS?

Related Commands [PLOT:PLOT<x>:MASK?](#)

Arguments MASK<x> is the number of the specified mask test (or mask test plot?).
SEG<y> is the number of the mask segment for which to return hit violations data.

Returns Returns an integer number representing the number of mask test violations for the specified mask test and mask segment.

Examples EYEMASK:MASK1:COUNT:SEG3:HITS? might return 435, indicating that segment 3 of mask test 1 has 435 hit violations.

EYEMASK:MASK<x>:CREATOR? (Query Only)

This query-only command returns the name of the eye diagram plot that created the mask.

Conditions Requires option 5-DJA or 6-DJA (Advanced Jitter Analysis).

Group	Mask
Syntax	EYEMASK:MASK<x>:CREATOR?
Related Commands	PLOT:PLOT<x>:MASK?
Arguments	MASK<x> is the number of the specified mask test (or mask test plot?).
Returns	<QString> is a quoted string that contains the name of the eye diagram plot associated with the specified mask.
Examples	EYEMASK:MASK4:CREATOR? might return "plot2".

EYEMASK:MASK<x>:ENAbled

This command enables or disables eye mask testing in the specified plot.

Conditions	Requires option 5-DJA or 6-DJA (Advanced Jitter Analysis).
Group	Mask?
Syntax	EYEMASK:MASK<x>:ENAbled {ON OFF} EYEMASK:MASK<x>:ENAbled?
Related Commands	PLOT:PLOT<x>:MASK?
Arguments	MASK<x> is the number of the specified mask test (or mask test plot?).
Returns	Returns an integer number representing enabled (1) or disabled (0).
Examples	EYEMASK:MASK4:ENAbled ON enables the eye mask test in the specified mask plot. EYEMASK:MASK2:ENAbled? might return 0, indicating that mask testing is disabled in the specified mask plot.

EYEMASK:MASK<x>:MASKfile

This command sets or queries the current mask definition file name for the specified mask test.

Conditions Requires option 5-DJA or 6-DJA (Advanced Jitter Analysis).

Group Mask

Syntax EYEMASK:MASK<x>:MASKfile <Qstring>
EYEMASK:MASK<x>:MASKfile?

Related Commands [PLOT:PLOT<x>:MASK?](#)

Arguments <x> is the number of the specified mask test (or mask test plot?).

<Qstring> is a quoted string that defines the file path that specifies the location of the mask file to use, in the format '[<path>]<filename.ext>'. Specifying a path is optional. If no path is entered, the instrument will search in the current working directory as set in [FILESystem:CWD](#).

- If the file argument begins with a file path separator (forward slash character) or a drive designator such as C:, then the file name is interpreted as a full path.
- If the file argument begins with "." or ".." or has a file path separator appearing anywhere other than the first character position, then the file name is treated as a path that is relative to the current working directory.

Examples EYEMASK:MASK1:MASKfile "/C:/Applications/Masks/Serial
RapidIO/RIO Serial 2_5G.msk" sets mask1 to use the RIO serial 2.5G mask file at the specified location

EYEMASK:MASK4:MASKfile? might return
"/media/C:/Applications/Masks/1394b/s400b T1.msk"

EYEMASK:MASK<x>:TEST:SAMple:THReshold

This command sets or queries the total number of hit violations that will cause a mask test failure.

Conditions Requires option 5-DJA or 6-DJA (Advanced Jitter Analysis).

Group	Mask
Syntax	EYEMASK:MASK<x>:TEST:SAMPle:THReshoId <NR1> EYEMASK:MASK<x>:TEST:SAMPle:THReshoId?
Related Commands	PLOT:PLOT<x>:MASK?
Arguments	MASK<x> is the number of the specified mask test (or mask test plot?). <NR1> is a positive integer indicating the number of mask hits required to cause a fail condition for that mask test.
Examples	EYEMASK:MASK1:TEST:SAMPle:THReshoId 15 sets the specified mask test to fail if 15 or more mask violations occur. EYEMASK:MASK3:TEST:SAMPle:THReshoId? might return 20, indicating that the specified mask test is set to fail if 20 or more mask violations occur.

EYEMASK:MASK<x>:TEST:STATUS? (Query Only)

This query-only command returns the mask hit test status.

Conditions	Requires option 5-DJA or 6-DJA (Advanced Jitter Analysis).
Group	Mask
Syntax	EYEMASK:MASK<x>:TEST:STATUS?
Related Commands	PLOT:PLOT<x>:MASK?
Arguments	MASK<x> is the number of the specified mask test (or mask test plot?).
Returns	This command returns a string with the mask test status. Valid status strings are: ■ PASS. The number of mask hits is less than the target fail threshold. ■ FAIL. The number of mask hits is greater than or equal to the target fail threshold. ■ OFF. Mask testing disabled on the specified mask plot.

Examples EYEMASK:MASK2:TEST:STATUS? might return Fail, indicating that mask testing has failed in the specified mask plot.

FACTory (No Query Form)

This command (no query form) resets the instrument to its factory default settings. This command is equivalent to pressing the DEFAULT SETUP button located on the instrument front panel or selecting Default Setup from the File menu.

This command Performs the following in addition to what is done for the *RST command:

- Clears any pending OPC operations.
- Resets the following IEEE488.2 registers:
 - *ESE 0 (Event Status Enable Register)
 - *SRE 0 (Service Request Enable Register)
 - DESE 255 (Device Event Status Enable Register)
 - *PSC 1 (Power-on Status Clear Flag)
- Deletes all defined aliases.
- Enables command headers (:HEADer 1).

Group Save and Recall

Syntax FACTory

Related Commands [*PSC](#)
[RECALL:SETUp](#)
[*RST](#)

Arguments None

Examples FACTORY resets the instrument to its factory default settings.

FILESystem? (Query Only)

This query-only command returns the directory listing of the current working directory. This query is the same as the FILESystem:DIR? query.

Group	File System
Syntax	FILESystem?
Related Commands	FILESystem:COPy FILESystem:CWD FILESystem:DELEte FILESystem:DIR? FILESystem:READFile FILESystem:REName FILESystem:WRITEFile
Arguments	None.
Examples	FILESYSTEM? might return :FILESYSTEM:DIR "myFile.txt", "mywaveform.wfm".

FILESystem:COPy (No Query Form)

This command (no query form) copies a named file to a new file. The new file might be in a totally separate directory than the old file. You can only copy one file at a time using this command. Wild card characters are not allowed.

Group	File System
Syntax	FILESystem:COPy {<source_file_path>,<destination_file_path>}
Related Commands	FILESystem:CWD FILESystem:DELEte
Arguments	<source_file_path> is a quoted string that defines the file name and path or directory. If the file path is within the current working directory, you need only specify the file name. <destination_file_path> is a quoted string that defines the file name and path. If the path is within the current working directory, you need only specify the file name.

Examples `FILESYSTEM:COPY "E:/setup1.set", "E:/SETUPS/setup1.set"` copies the file named setup1.set, on the E drive to a file named setup1.set in the SETUPS directory on the E drive.

`:FILESystem:CWD "E:/SETUPS"`

`:FILESystem:COPy ".", "I:/Archive/SETUPS"`

`:FILESystem:COPy "../ch1.isf", "I:/SavedWfms/ch1_new.isf"`

FILESystem:CWD

This command sets or queries the current working directory. CWD is short for Current Working Directory. It changes the directory (folder) that the other FILESystem commands operate on.

NOTE. *There are three host ports on the front panel (E:, F:, and G:), and two on the back panel (H: and I:).*

Group	File System
Syntax	<code>FILESystem:CWD {<new_working_directory_path>}</code> <code>FILESystem:CWD?</code>
Arguments	<code><new_working_directory_path></code> is a quoted string that defines the current working; a directory name can be up to 128 characters.
Examples	<code>FILESYSTEM:CWD "E:/waveforms/"</code> changes the current working directory to a folder named "waveforms" on the USB flash drive installed in the "E:" USB Host port. <code>FILESYSTEM:CWD?</code> might return <code>:FILESYSTEM:CWD "E:/"</code> Indicates the current working directory is the root folder of the E: USB flash drive port.

FILESystem:DELEte (No Query Form)

This command (no query form) deletes a named file or directory from a mass storage device. Once removed, the data in that file or directory can no longer be accessed. If the specified file is a directory, it must be empty before it can be deleted.

Group	File System
--------------	-------------

Syntax	<code>FILESystem:DELEte <file_path></code>
Related Commands	FILESystem:COPy FILESystem:CWD FILESystem:RMDir
Arguments	<code><file_path></code> is a quoted string that defines the file name and path. If the file path is within the current working directory, you need only specify the file name.
Examples	<code>FILESYSTEM:DELETE "NOT_MINE.SET"</code> deletes the file named NOT_MINE.SET from the folder referred to by the <code>FILESYSTEM:CWD</code> FILESystem:CWD command.

FILESystem:DIR? (Query Only)

This query-only command returns a comma separated list of quoted strings. Each string contains the name of a file or directory in the folder referred to by the `FILESYSTEM:CWD`[FILESystem:CWD](#) command.

Group	File System
Syntax	<code>FILESystem:DIR?</code>
Related Commands	FILESystem:CWD FILESystem:MKDir
Arguments	None
Examples	<code>FILESYSTEM:DIR?</code> might return <code>:FILESYSTEM:DIR</code> <code>"161012_132039_000.wfm","161012_132039_001.wfm","161220_191452.png",</code> <code>"161220_191554.csv","170320_132925_000.wfm","170320_132929_000.set",</code> <code>"Ch2-Ch3_000.set","E:","F:","G:","H:","I:","J:","Rgjtest_000.set",</code> <code>"RgjSetupRuntTrig4chnl</code> <code>s_000.set","RgjSetup_000.set","ScrnShot1jeh2_000.wfm","ScrnShot1jeh_000.wfm",</code> <code>"ScrnShot_000.png","ScrnShot_000.wfm","Scrn</code> <code>_000.set","Scrn_001.set","Scrn_002.set","Scrn_003.set","Scrn_004.set",</code> <code>"Wfm_000.wfm","Wfm_001.wfm","Wfm_002.wfm","Wfm_003</code> <code>.wfm","Wfm_004.wfm","Wfm_005.wfm","Wfm_006.wfm","digChans_000.wfm".</code>

FILESystem:HOMEDir? (Query Only)

This query returns the current user's home directory.

Group	File System
Syntax	FILESystem:HOMEDir?
Returns	The current user's home directory as a quoted string.
Examples	FILESYSTEM:HOMEDIR? might return: "C:".

FILESystem:LDIR? (Query Only)

Returns a comma separated list of every file, file size, type, modification date and time, and directory in the folder referred to by the FILESYSTEM:CWD[FILESystem:CWD](#) command. This is different than the :DIR query in that it provides a long output format with the file size, type, and modification date/time. Each entry is a semicolon separated list: < file name>;<type>;<size in bytes>;<date>;<time>

Group	File System
Syntax	FILESystem:LDIR?
Returns	A comma separated list of every file, file size, type, modification date and time, and directory in the folder referred to by the FILESystem:CWD command.
Examples	FILESYSTEM:LDIR? might return "tek0000CH1.isf;FILE;20342;2009-05-21;13:58:24", "TEMP;DIR;4096;2009-09-15;06:20:44".

FILESystem:MKDir (No Query Form)

This command (no query form) creates a new directory.

Group	File System
Syntax	FILESystem:MKDir <directory_path>

Related Commands [FILESystem:CWD](#)
[FILESystem:DIR?](#)

Arguments <directory_path> is a quoted string that specifies the directory to create.

Examples `FILESYSTEM:MKDIR "E:\NewDirectory"` creates the directory named *NewDirectory* at the root of the E drive.

These two commands create the directory MyNewSubDirectory within the existing directory MyDirectory at the root of the E drive:

```
FILESYSTEM:CWD "E:/MyDirectory"; FILESYSTEM:MKDIR  
"MyNewSubDirectory"
```

FILESystem:READFile (No Query Form)

This command writes the contents of the specified file to the current interface. If the specified file does not exist or is not readable, an appropriate error event is posted.

Group File System

Syntax `FILESystem:READFile <QString>`

Related Commands [FILESystem:CWD](#)

Arguments <QString> is a quoted string that defines the file name and path. If the file path is within the current working directory, you need only specify the file name.

Examples `FILESYSTEM:READFILE "E:/test_data/tek00016CH1.csv"` reads the content of the specified file, if the file exists and is readable, and sends the content of the file to the current interface.

FILESystem:REName (No Query Form)

This command (no query form) assigns a new name to an existing file or folder.

Group File System

Syntax `FILESystem:REName <old_file_path>,<new_file_path>`

Related Commands [FILESystem:CWD](#)

Arguments `<old_file_path>` is a quoted string that defines the file or folder name and path. If the path is within the current working directory, you need only specify the file or folder name.

`<new_file_path>` is a quoted string that defines the file or folder name and path. If the path is within the current working directory, you need only specify the file or folder name.

Examples `FILESYSTEM:RENAME "E:/TEK00000.SET", "E:/MYSETTING.SET"` gives the file named TEK00000.SET the new name of MYSETTING.SET. The file remains in the root directory on the E drive.

`FILESYSTEM:RENAME
"e:/mySettings/tek00000.set", "e:/setup1.set"`. This example illustrates how to move a file from one folder to another.

FILESystem:RMDir (No Query Form)

This command (no query form) deletes a named directory. The directory must be empty.

Group File System

Syntax `FILESystem:RMDir <directory_path>`

Related Commands [FILESystem:CWD](#)

Arguments `<directory_path>` is a quoted string that defines the folder name and path. If the folder path is within the current working directory, you need only specify the folder name.

Examples `FILESYSTEM:RMDIR "E:/OldDirectory"` removes the directory named OldDirectory from the root of the E drive.

FILESystem:UNMOUNT:DRive (No Query Form)

This command unmounts the USB drive specified by the quoted string argument.

Group File System

Syntax FILESystem:UNMOUNT:DRive <QString>

Related Commands

Arguments <QString> is a quoted string that specifies which USB drive to unmount. String is a case insensitive single letter followed by a colon.

Examples FILESYSTEM:UNMOUNT:DRIVE "G:" specifies to unmount the flash drive installed in the right-most USB slot on the front of the instrument.

FILESystem:WRITEFile (No Query Form)

This command (no query form) writes the specified block data to the specified file on the instruments file system. If the destination file cannot be written, an error event is posted.

Group File System

Syntax FILESystem:WRITEFile <file_path>,<data>

Related Commands [FILESystem:CWD](#)

Arguments <file path> is a quoted string that defines the file name and path. If the file path is within the current working directory, you need only specify the file name.
<data> is the specified block data to be written.

FPAnel:PRESS (No Query Form)

This command is used to emulate a button press. When used with knob enumerations, this command pushes the knob. Use the [FPAnel:TURN](#) command to emulate knob turns.

Group	Miscellaneous
Syntax	<code>FPANEL:PRESS {AUTOset BUS CH1<x> CLEAR CURSor DEFaultsetup FASTAcq FORCetrig GPKNOB1 GPKNOB2 HIGHRES HORZPOS HORZScale MATH NEXT PREV REF RUNSTop SETTO50 SINGl eseq TOUCHSCreen TRIGMode TRIGSl ope USER VERTPOS VERTSCALE ZOOM}</code>
Arguments	Arguments are instrument buttons.
Examples	<code>FPANEL:PRESS FORCETRIG</code> emulates pressing the Force trigger button.

FPANEL:TURN

This command is used to emulate a knob turn. The optional NR1 specifies the number of clicks where negative values indicate counter clockwise. If not specified, the default of 1 click is used indicating the knob is turned clockwise 1 click.

Group	Miscellaneous
Syntax	<code>FPANEL:TURN {GPKNOB1 GPKNOB2 HORZPOS HORZScale PANKNOB TRIGLevel VERTPOS VERTSCALE ZOOM} [, <NR1>]</code>
Arguments	Arguments are knobs that turn obtained during startup. <NR1> is the number of clicks to turn the knob.
Examples	<code>FPANEL:TURN TRIGLEVEL , 3</code> emulates turning the trigger Level knob 3 clicks in the clockwise direction.

HEADer

This command sets or queries the Response Header Enable State that causes the instrument to either include or omit headers on query responses.

NOTE. *This command does not affect IEEE Std 488.2-1987 Common Commands (those starting with an asterisk); these commands never return headers.*

Whether the long or short form of header keywords and enumerations are returned is dependent upon the state of :VERBoSe.

Group	Miscellaneous
Syntax	<pre>HEADer {<NR1> OFF ON} HEADer?</pre>
Related Commands	VERBose
Arguments	<NR1> = 0 sets the Response Header Enable State to false; any other value sets this state to true. OFF sets the Response Header Enable State to false. This causes the instrument to omit headers on query responses, so that only the argument is returned. ON sets the Response Header Enable State to true. This causes the instrument to include headers on applicable query responses. You can then use the query response as a command.
Examples	HEADER OFF specifies that the instrument omits headers on query responses, so that only the argument is returned. HEADER? might return :HEADER 1, indicating that the instrument is including headers on applicable query responses. With :VERBose ON and :HEADer ON, the :ACQuire:MODE? query might return: :ACQUIRE:MODE SAMPLE. With :VERBose OFF and :HEADer ON, the :ACQuire:MODE? query might return: :ACQ:MOD SAM With :VERBose ON and :HEADer OFF, the :ACQuire:MODE? query might return: SAMPLE With :VERBose OFF and :HEADer OFF, the :ACQuire:MODE? query might return: SAM

HORizontal? (Query Only)

Queries the current horizontal settings.

Group	Horizontal
Syntax	HORizontal?

Examples HORIZONTAL? might return :HORIZONTAL:SAMPLERATE
6.2500E+9;SAMPLERATE:ANALYZEMODE:MINIMUM:VALUE
AUTOMATIC;OVERRIDE 1;:HORIZONTAL:MODE
AUTO;MODE:AUTOMATIC:FASTACQ:RECORDLENGTH:MAXIMUM:ZOOMOVERRIDE
1;VALUE 1250;:HORIZONTAL:MODE:MANUAL:CONFIGURE
HORIZONTALSCALE;:HORIZONTAL:MAIN:UNITS
S;:HORIZONTAL:DIVISIONS 10;RECORDLENGTH 62500;DELAY:TIME
0.0E+0;MODE 1;:HORIZONTAL:POSITION 50.0000;SCALE
1.0000E-6;ACQDURATION 10.0000E-6.

HORizontal:ACQDURATION? (Query Only)

This query returns the timebase duration.

Group Horizontal

Syntax HOrizontal:ACQDURATION?

Returns <NR3> returns the duration of the acquisition.

Examples HORIZONTAL:ACQDURATION? might return :HORIZONTAL:ACQDURATION
5.0E-9, indicating the acquisition duration is 5.0 us.

HORizontal:DELay:MODE

This command sets or queries the horizontal delay mode.

Group Horizontal

Syntax HOrizontal:DELay:MODE {OFF|ON|<NR1>}
HOrizontal:DELay:MODE?

Related Commands [HORizontal:POSition](#)

Arguments OFF sets the Horizontal Delay Mode to off. This causes the HOrizontal:POSition command to operate like the HORIZONTAL POSITION knob on the front panel.

ON sets the Horizontal Delay Mode to on. This causes the HOrizontal:DELay:TIME command to operate like the HORIZONTAL POSITION knob on the front panel.

<NR1> = 0 sets the Horizontal Delay Mode to off; any other value sets this mode to on.

- Examples** `HORIZONTAL:DELAY:MODE OFF` sets the Horizontal Delay Mode to off, allowing the horizontal position command to operate like the HORIZONTAL POSITION knob on the front panel.
- `HORIZONTAL:DELAY:MODE?` might return `HORIZONTAL:DELAY:MODE OFF` indicating that the Horizontal Delay Mode is off and that the horizontal position command operates like the HORIZONTAL POSITION knob on the front panel.

HORizontal:DElay:TIME

This command sets or queries the horizontal delay time that is used when delay mode is on.

- Group** Horizontal
- Syntax** `HORizontal:DElay:TIME <NR3>`
`HORizontal:DElay:TIME?`
- Related Commands** `HORizontal:POSition`
- Arguments** NR3 is the delay in seconds.
- Examples** `HORizontal:DElay:TIME 0.3` sets the delay of acquisition data so that the resulting waveform is centered 300 ms after the trigger occurs.

HORizontal:DIVisions? (Query Only)

This query-only command returns the number of graticule divisions.

- Group** Horizontal
- Syntax** `HORizontal:DIVisions?`
- Examples** `HORIZONTAL:DIVISIONS?` might return `:HORIZONTAL:DIVISIONS 10.0000`, indicating that the waveform is displayed across ten divisions.

HORizontal:FASTframe? (Query Only)

This query returns all information under horizontal:fastframe.

Group Horizontal

Syntax HORizontal:FASTframe?

Examples HORIZONTAL:FASTFRAME? might return
:HORIZONTAL:FASTFRAME:SELECTED 1;SUMFRAME:STATE
0;:HORIZONTAL:FASTFRAME:COUNT 2;MULTIPLEFRAMES:MODE
OFF;:HORIZONTAL:FASTFRAME:STATE 0;REF:INCLUDE 0;FRAME 1, the
current FastFrame settings.

HORizontal:FASTframe:COUNT

This command sets or returns the number of frames.

Group Horizontal

Syntax HORizontal:FASTframe:COUNT <NR1>
HORizontal:FASTframe:COUNT?

Arguments <NR1> is the number of frames.

Examples :HORizontal:FASTframe:COUNT 10 sets the number of frames to be acquired to 10. If this is beyond the maximum number of frames, the value will be constrained. For example, if 8 is the current maximum (based on record length) 8 will be set instead of 10.

HORizontal:FASTframe:MAXFRames? (Query Only)

This query returns the maximum number of frames.

Group Horizontal

Syntax HORizontal:FASTframe:MAXFRames?

Examples `HORIZONTAL:FASTFRAME:MAXFRAMES?` might return
 `:HORIZONTAL:FASTFRAME:MAXFRAMES 2`, indicating the maximum number
 of frames is 2.

HORizontal:FASTframe:MULTipleframes:MODE

This command sets or returns the overlay display type.

Group Horizontal

Syntax `HORizontal:FASTframe:MULTipleframes:MODE {OFF|OVERlay}`
 `HORizontal:FASTframe:MULTipleframes:MODE?`

Arguments `OFF` specifies only displaying the selected frame.

 `OVERlay` specifies overlaying all frames with the temperature palette. The
 summary frame is not included in the overlay. The selected frame is drawn in
 blue on top of all other frames.

Examples `:HORizontal:FASTframe:MULTipleframes:MODE OVERlay` will cause the
 entire acquired set of frames to be rendered in a single image, similar to how
 FastAcq appears. The currently selected frame is drawn on top in blue.

HORizontal:FASTframe:REF:FRame

This command sets or returns the reference frame number.

Group Horizontal

Syntax `HORizontal:FASTframe:REF:FRame <NR1>`
 `HORizontal:FASTframe:REF:FRame?`

Arguments `<NR1>` is the reference frame number.

Examples `HORIZONTAL:FASTFRAME:REF:FRAME?` might return
 `:HORIZONTAL:FASTFRAME:REF:FRAME 1`, indicating the reference frame
 number is 1.

HORizontal:FASTframe:REF:INCLUde

This command sets or returns whether the reference frame delta information is shown in the display.

Group Horizontal

Syntax HORizontal:FASTframe:REF:INCLUde {<NR1>|OFF|ON}
HORizontal:FASTframe:REF:INCLUde?

Arguments ON displays the delta information.
OFF does not display the delta information.
<NR1> a 0 indicates the delta information is off; any other value displays the delta information.

HORizontal:FASTframe:SELECTED

This command sets or returns the selected frame number for acquired frames. Refs have their own selected frames.

Group Horizontal

Syntax HORizontal:FASTframe:SELECTED <NR1>
HORizontal:FASTframe:SELECTED?

Arguments <NR1> is the selected frame number for acquired frames.

Examples HORIZONTAL:FASTFRAME:SELECTED? might return
:HORIZONTAL:FASTFRAME:SELECTED 1, indicating the selected frame number is 1.

HORizontal:FASTframe:STATE

This command sets or returns the state of FastFrame. Acquisition modes Envelope and Average are not compatible with FastFrame. If FastFrame is on, an attempted set to those acquisition modes will fail and revert to Sample mode. If FastFrame is turned on while in one of those acquisition modes, the acquisition mode is changed to Sample.

Group	Horizontal
Syntax	<code>HORizontal:FASTframe:STATE {<NR1> OFF ON}</code> <code>HORizontal:FASTframe:STATE?</code>
Arguments	ON indicates FastFrame is active. OFF indicates that FastFrame is off. <NR1> A 0 turns off FastFrame; any other value activates FastFrame.
Examples	<code>HORIZONTAL:FASTFRAME:STATE?</code> might return <code>:HORIZONTAL:FASTFRAME:STATE 0</code>, indicating FastFrame is off.

HORizontal:FASTframe:SUMFrame? (Query Only)

This command sets or returns the summary frame type. Turning on Summary Frame does not adjust the numberFrames value as long as there is room for an additional frame. If there is not enough room then numberFrames will be reduced by 1. The numberFrames value is always the number of frames to acquire.

Group	Horizontal
Syntax	<code>HORizontal:FASTframe:SUMFrame? {NONE AVERage ENVElope}</code> <code>HORizontal:FASTframe:SUMFrame??</code>
Arguments	NONE sets the Summary frame to off. AVERage sets the Summary frame to average of all acquired frames. ENVElope sets the Summary frame to envelope of all acquired frames.
Examples	<code>HORIZONTAL:FASTFRAME:SUMFRAME?</code> might return <code>:HORIZONTAL:FASTFRAME:SUMFRAME NONE</code>, indicating the summary frame is off.

HORizontal:FASTframe:SUMFrame:STATE

This command sets or returns the state of FastFrame summary frame. Summary frame mode is set automatically based on the acquisition mode. When in Sample mode, the summary frame type is set to Average. When in Peak Detect mode, the

summary frame type is set to Envelope. When in High Res mode, the summary frame type is set to Average.

Group	Horizontal
Syntax	<code>HORizontal:FASTframe:SUMFrame:STATE {<NR1> OFF ON}</code> <code>HORizontal:FASTframe:SUMFrame:STATE?</code>
Arguments	ON indicates summary frame is active. OFF indicates that summary frame is off. <NR1> a 0 turns off summary frame; any other value activates the summary frame.
Examples	<code>HORIZONTAL:FASTFRAME:SUMFRAME:STATE?</code> might return <code>:HORIZONTAL:FASTFRAME:SUMFRAME:STATE 0</code> , indicating the summary frame is off.

HORizontal:FASTframe:TIMESTAMP:ALL? (Query Only)

This query-only command returns the time stamp of all frames. The format is (Frame #: TimeStamp, Frame #: TimeStamp, and so on). Each time-stamp string is of the form `DD.MM.YYYY.HH:MM:SS.aaaaaaaaaaaa`.

Group	Horizontal
Syntax	<code>HORizontal:FASTframe:TIMESTAMP:ALL?</code>
Returns	Returns all of the timestamps.
Examples	<code>HORIZONTAL:FASTFRAME:TIMESTAMP:ALL?</code> might return <code>:HORIZONTAL:FASTFRAME:TIMESTAMP:ALL "1: 31.12.1969.16:00:00.000000000000"</code> , indicating the only time

HORizontal:FASTframe:TIMESTAMP:DELTA? (Query Only)

This query returns the time difference between the Selected and Reference time-stamps.

Group	Horizontal
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Syntax	<code>HORizontal:FASTframe:TIMEStamp:DELTA?</code>
Returns	Return value is in seconds and fraction of a second.
Examples	<code>HORIZONTAL:FASTFRAME:TIMESTAMP:DELTA?</code> might return <code>:HORIZONTAL:FASTFRAME:TIMESTAMP:DELTA "0.000000000000"</code> indicating the time difference is 0.0 seconds.

HORizontal:FASTframe:TIMEStamp:REFerence? (Query Only)

This query returns the time-stamp of the FastFrame Reference frame.

Group	Horizontal
Syntax	<code>HORizontal:FASTframe:TIMEStamp:REFerence?</code>
Examples	<code>HORIZONTAL:FASTFRAME:TIMESTAMP:REFERENCE?</code> might return <code>:HORIZONTAL:FASTFRAME:TIMESTAMP:REFERENCE</code> <code>"31.12.1969.17:00:00.000000000000"</code> indicating the reference timestamp is 31.12.1969.17:00:00.000000000000.

HORizontal:FASTframe:TIMEStamp:SELECTED? (Query Only)

This query returns the time-stamp of the FastFrame Selected acquired frame.

Group	Horizontal
Syntax	<code>HORizontal:FASTframe:TIMEStamp:SELECTED?</code>
Examples	<code>HORIZONTAL:FASTFRAME:TIMESTAMP:SELECTED?</code> might return <code>:HORIZONTAL:FASTFRAME:TIMESTAMP:SELECTED</code> <code>"31.12.1969.17:00:00.000000000000"</code> indicating the timestamp of the selected frame is 31.12.1969.17:00:00.000000000000.

HORizontal:FASTframe:XZEro:ALL? (Query Only)

This query-only command returns the sub-sample time between the trigger sample (designated by PT_OFF) and the occurrence of the actual trigger for the

waveform specified by the DATA:SOURCE command for all frames. This value is in units of WFMOutpre:XUNit. The format is a string of the form (frame #:xzero, frame #:xzero, and so on).

Group	Horizontal
Syntax	HORizontal:FASTframe:XZERO:ALL?
Returns	This returns the XZERO values for all of the frames.
Examples	HORIZONTAL:FASTFRAME:XZERO:ALL? might return :HORIZONTAL:FASTFRAME:XZERO:ALL "1: 1.6000E-10", indicating the time between the trigger sample and the actual trigger.

HORizontal:FASTframe:XZERO:REF? (Query Only)

This query-only command returns the sub-sample time between the trigger sample (designated by PT_OFF) and the occurrence of the actual trigger for the waveform specified by the DATA:SOURCE command for the reference frame. This value is in units of WFMOutpre:XUNit.

Group	Horizontal
Syntax	HORizontal:FASTframe:XZERO:REF?
Examples	HORIZONTAL:FASTFRAME:XZERO:REF? might return :HORIZONTAL:FASTFRAME:XZERO:REF 1.0 indicating the sub-sample time is 1.0.

HORizontal:FASTframe:XZERO:SELECTED? (Query Only)

This query-only command returns the sub-sample time between the trigger sample (designated by PT_OFF) and the occurrence of the actual trigger for the waveform specified by the DATA:SOURCE command for the selected frame. This value is in units of WFMOutpre:XUNit.

Group	Horizontal
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Syntax `HORizontal:FASTframe:XZERO:SELECTED?`

Examples `HORIZONTAL:FASTFRAME:XZERO:SELECTED?` might return
 `:HORIZONTAL:FASTFRAME:XZERO:SELECTED 1.0` indicating the sub-sample
 time is 1.0.

HORizontal:MAIn:INTERPRatio? (Query Only)

This query-only command returns the Horizontal interpolation ratio.

Group Horizontal

Syntax `HORizontal:MAIn:INTERPRatio?`

Examples `HORIZONTAL:MAIN:INTERPRATIO?` might return
 `:HORIZONTAL:MAIN:INTERPRATIO 10.0`.

HORizontal:MODE

This command set or queries the horizontal operating mode.

Group Horizontal

Syntax `HORizontal:MODE {AUTO|MANual}`
 `HORizontal:MODE?`

Related Commands [HORizontal:MODE:RECOrdlength](#)
 [HORizontal:MODE:SAMPLERate](#)
 [HORizontal:MODE:SCALE](#)

Arguments `AUTO` selects the automatic horizontal model. Auto mode automatically adjusts the sample rate and record length to provide a high acquisition rate in Fast Acq or signal fidelity in analysis. Record length is read only.

`MANUAL` selects the manual horizontal model. Manual mode lets you change the sample rate, horizontal scale, and record length. These values interact. For example, when you change record length then the horizontal scale also changes.

Examples `HORIZONTAL:MODE AUTO` sets the horizontal mode to auto.

`HORIZONTAL:MODE?` might return `:HORIZONTAL:MODE MANUAL`, indicating that the horizontal mode is manual.

HORizontal:MODE:AUToMatic:FASTAcq:RECOrdlength:MAXimum:VALue

Sets or queries the horizontal FastAcq maximum record length.

Group Horizontal

Syntax `HORizontal:MODE:AUToMatic:FASTAcq:RECOrdlength:MAXimum:VALue`
 `<NR1>`
 `HORizontal:MODE:AUToMatic:FASTAcq:RECOrdlength:MAXimum:VALue?`

Arguments `<NR1>` is the horizontal FastAcq maximum record length.

Examples `HORIZONTAL:MODE:AUTOMATIC:FASTACQ:RECORDLENGTH:MAXIMUM:VALUE`
 `6250` sets the maximum value to 6.25 k.

HORizontal:MODE:AUToMatic:FASTAcq:RECOrdlength:MAXimum:ZOOMOVERride

Sets or queries the flag which allows override of the horizontal FastAcq maximum record length.

Group Horizontal

Syntax `HORizontal:MODE:AUToMatic:FASTAcq:RECOrdlength:MAXimum:ZOOMOVERride {OFF|ON|0|1}`
 `HORizontal:MODE:AUToMatic:FASTAcq:RECOrdlength:MAXimum:ZOOMOVERride?`

Arguments `OFF` does not allow override of the horizontal FastAcq maximum record length.

`ON` allows override of the horizontal FastAcq maximum record length.

`0` does not allow override of the horizontal FastAcq maximum record length.

`1` allows override of the horizontal FastAcq maximum record length.

Examples `HORIZONTAL:MODE:AUTOMATIC:FASTACQ:RECORDLENGTH:MAXIMUM:ZOOMOVERRIDE 0` does not allow override of the horizontal FastAcq maximum record length.

`HORIZONTAL:MODE:AUTOMATIC:FASTACQ:RECORDLENGTH:MAXIMUM:ZOOMOVERRIDE?` might return
`:HORIZONTAL:MODE:AUTOMATIC:FASTACQ:RECORDLENGTH:MAXIMUM:ZOOMOVERRIDE 1` indicating that override of the horizontal FastAcq maximum record length is allowed.

HORizontal:MODE:MANual:CONFIgure

Sets or queries which horizontal control (scale or record length) will primarily change when the sample rate is changed in Manual mode. If the selected control (scale or record length) reaches a limit then the unselected control (record length or scale) may also change.

Group Horizontal

Syntax `HORizontal:MODE:MANual:CONFIgure`
`{HORizontal:scale|RECORDLength}`
`HORizontal:MODE:MANual:CONFIgure?`

Arguments `HORizontal:scale` will change when sample rate is adjusted.
`RECORDLength` will change when sample rate is adjusted.

Examples `HORIZONTAL:MODE:MANUAL:CONFIGURE RECORDLength` allows the record length be adjusted when sample rate is changed in Manual mode.

HORizontal:MODE:RECOrdlength

This command sets or queries the record length.

Group Horizontal

Syntax `HORizontal:MODE:RECOrdlength <NR1>`
`HORizontal:MODE:RECOrdlength?`

Arguments `<NR1>` is the record length in samples. Manual mode lets you change the record length, while the record length is read only for Automatic mode.

Examples `HORIZONTAL:MODE:RECORDLENGTH 1000` sets the record length to 1000 samples.

`HORIZONTAL:MODE:RECORDLENGTH?` might return
`:HORIZONTAL:MODE:RECORDLENGTH 1000000`, indicating that the record length is set to 1,000,000 samples

HORizontal:MODE:SAMPLERate

This command sets or queries the sample rate.

Group Horizontal

Syntax `HORizontal:MODE:SAMPLERate <NR1>`
`HORizontal:MODE:SAMPLERate?`

Arguments `<NR1>` is the sample rate in samples per second.

Examples `HORIZONTAL:MODE:SAMPLERATE 1e6` sets the sample rate to 1 million samples per second.

`HORIZONTAL:MODE:SAMPLERATE?` might return
`:HORIZONTAL:MODE:SAMPLERATE 5.0000E+6`, indicating that the sample rate is set to 5 million samples per second.

HORizontal:MODE:SCALE

This command sets or queries the horizontal scale.

Group Horizontal

Syntax `HORizontal:MODE:SCALE <NR1>`
`HORizontal:MODE:SCALE?`

Arguments `<NR1>` is the horizontal scale in seconds per division.

Examples `HORIZONTAL:MODE:SCALE 2e-9` sets the horizontal scale to 2 ns per division.

`HORIZONTAL:MODE:SCALE?` might return `:HORIZONTAL:MODE:SCALE 20.0000E-6`, indicating that the horizontal scale is set to 10 μ s per division.

HORizontal:POSition

This command sets or queries the horizontal position as a percent of screen width. When Horizontal Delay Mode is turned off, this command is equivalent to adjusting the HORIZONTAL POSITION knob on the front panel. When Horizontal Delay Mode is turned on, the horizontal position is forced to 50%.

Group	Horizontal
Syntax	<code>HORizontal:POSition <NR3></code> <code>HORizontal:POSition?</code>
Arguments	<NR3> is from 0 to ≈ 100 and is the position of the trigger point on the screen (0 = left edge, 100 = right edge).
Examples	<code>HORIZONTAL:POSITION 10</code> sets the trigger position of the waveform such that 10% of the display is to the left of the trigger position. <code>HORIZONTAL:POSITION?</code> might return <code>:HORIZONTAL:POSITION 5.0000E+01</code> indicates the trigger point displayed on the screen.

HORizontal:PREViewstate? (Query Only)

This query returns the display system preview state.

Group	Horizontal
Syntax	<code>HORizontal:PREViewstate?</code>
Returns	<NR1> = 1 if the system is in the preview state. <NR1> = 0 if the system is not in the preview state.
Examples	<code>HORIZONTAL:PREVIEWSTATE?</code> might return <code>:HORIZONTAL:PREVIEWSTATE 0</code> indicating the system is not in the preview state.

HORizontal:RECOrdlength

This command sets or queries the horizontal record length. To change the record length the Horizontal Mode must be set to Manual.

Group	Horizontal
Syntax	<code>HORizontal:RECOrdlength <NR1></code> <code>HORizontal:RECOrdlength?</code>
Arguments	<NR1> is the horizontal record length.
Examples	<code>HORIZONTAL:RECORDLENGTH 1000</code> sets the record length to 1000 samples. <code>HORIZONTAL:RECORDLENGTH?</code> might return <code>:HORIZONTAL:RECORDLENGTH 1000000</code> , indicating that the record length is set to 1,000,000 samples

HORizontal:ROLL? (Query Only)

Queries the horizontal roll mode status.

Group	Horizontal
Syntax	<code>HORizontal:ROLL?</code>
Returns	<code>ON</code> indicates roll mode is active. <code>OFF</code> indicates that summary frame is off. <NR1> a 0 indicates roll mode is off; any other value activates roll mode.
Examples	<code>HORIZONTAL:ROLL?</code> might return <code>:HORIZONTAL:ROLL 0</code> indicating roll mode is off.

HORizontal:SAMPLERate

This command sets or queries the horizontal sample rate.

Group	Horizontal
Syntax	<code>HORizontal:SAMPLERate <NR3></code> <code>HORizontal:SAMPLERate?</code>
Arguments	<NR3> is the horizontal sample rate in samples per second.

Examples `HORIZONTAL:SAMPLERATE 1e6` sets the sample rate to 1 million samples per second.

`HORIZONTAL:SAMPLERATE?` might return `:HORIZONTAL:SAMPLERATE 5.0000E+6`, indicating that the sample rate is set to 5 million samples per second.

HORizontal:SAMPLERate:ANALYZemode:MINimum:OVERRide

Sets or queries the flag which allows override of the horizontal analyze minimum sample rate.

Group Horizontal

Syntax `HORizontal:SAMPLERate:ANALYZemode:MINimum:OVERRide`
 `{OFF|ON|0|1}`
 `HORizontal:SAMPLERate:ANALYZemode:MINimum:OVERRide?`

Arguments 0 does not allow override of the horizontal analyze minimum sample rate.

 1 allows override of the horizontal analyze minimum sample rate.

 OFF does not allow override of the horizontal analyze minimum sample rate.

 ON allows override of the horizontal analyze minimum sample rate.

Examples `HORIZONTAL:SAMPLERATE:ANALYZEMODE:MINIMUM:OVERRIDE OFF` does not allow override of the horizontal analyze minimum sample rate.

`HORIZONTAL:SAMPLERATE:ANALYZEMODE:MINIMUM:OVERRIDE?` might return `:HORIZONTAL:SAMPLERATE:ANALYZEMODE:MINIMUM:OVERRIDE 1` indicating that override of the horizontal analyze minimum sample rate is allowed.

HORizontal:SAMPLERate:ANALYZemode:MINimum:VALue

Sets or queries the minimum sample rate used by Analysis Automatic horizontal mode.

Group Horizontal

Syntax `HORizontal:SAMPLERate:ANALYZemode:MINimum:VALue`
 `{AUTOMATIC|<NR3>}`
 `HORizontal:SAMPLERate:ANALYZemode:MINimum:VALue?`

Related Commands

Arguments	AUTOMATIC allows the oscilloscope to set the minimum value. <NR3> is the minimum sample rate.
Examples	HORIZONTAL:SAMPLERate:ANALYZemode:MINimum:VALue AUTOMATIC allows the oscilloscope to set the minimum value. HORIZONTAL:SAMPLERate:ANALYZemode:MINimum:VALue? might return :HORIZONTAL:SAMPLERATE:ANALYZEMODE:MINIMUM:VALUE AUTOMATIC indicating the oscilloscope automatically sets the minimum value.

HORIZONTAL:SCALE

This command sets or queries the horizontal scale.

Group	Horizontal
Syntax	HORIZONTAL:SCALE <NR3> HORIZONTAL:SCALE?
Arguments	<NR3> is the horizontal scale in time per division.
Returns	The current horizontal scale is returned.
Examples	HORIZONTAL:SCALE 20e-9 sets the horizontal scale to 20 ns/division. HORIZONTAL:SCALE? might return :HORIZONTAL:SCALE 20e-9 indicating the horizontal scale is set to 20 ns/division.

ID? (Query Only)

This query-only command returns identifying information about the instrument and related firmware similar to that returned by the *IDN? IEEE488.2 common query but does not include the instrument serial number.

Group	Miscellaneous
Syntax	ID?

Related Commands [*IDN?](#)

Examples ID? might return ID TEK/MSO54,CF:91.1CT,FV:1.2.0.2886, indicating that the instrument model number is set to MSO54, codes and formats is CF:91.1CT, and firmware version is FV:1.2.0.

***IDN? (Query Only)**

This query-only command returns the instrument identification code.

Group Miscellaneous

Syntax *IDN?

Related Commands [ID?](#)

Examples *IDN? might return TEKTRONIX,MSO54,C100123,CF:91.1CT
FV:1.2.0.2886, indicating the instrument model number, serial number, codes and formats number, and firmware version number.

LIC:UNINSTALL? (Query Only)

Returns the exit license indicated for the user to return to their TekAMS account. Active licenses can be specified by their nomenclature. TransactionIDs can be used to specify an active license or a previously uninstalled license. In either case, the exit-license is returned as block-data.

Group Miscellaneous

Syntax LIC:UNINSTALL? <QString>

Arguments <QString> is the nomenclature of an active license or a TransactionIDs to specify an active license or a previously uninstalled license.

Returns The exit-license is returned as block-data.

Examples LIC:UNINSTALL? "LIC5-SRAERO" uninstalls the given license and returns the license block data.

LIC:UNINSTALL? “569765772” uninstalls the license with the given transaction ID and returns the license block data.

LICense? (Query Only)

This query-only command returns all license parameters.

Group Miscellaneous

Syntax LICense?

Examples **LICENSE?** might return **:LICENSE:GMT**
"2016-05-23T17:05:10-07:00";**HID** "TMS-AAA9CS4US5SGJN6X";**LIST**
"5-BW-1000";**COUNT** 1.

LICENSE:APPID? (Query Only)

This query returns a comma-separated list of the active application IDs. If a string argument is provided, a “0” or “1” is returned, according to whether the string matches an active application ID.

Group Miscellaneous

Syntax **LICENSE:APPID?** {<QString>}

Returns This query returns a comma-separated list of the active application IDs. If a string argument is provided, a “0” or “1” is returned, according to whether the string matches an active application ID.

Examples **LIC:APPID?** might return “BW5-2000, AFG, DVM, DJA”, which is a complete list of the active applications.
LIC:APPID? “AFG” would return “1” because the app is active.

LICense:COUNT? (Query Only)

This query returns a count of the number of active licenses installed.

Group Miscellaneous

Syntax `LICense:COUNT?`

Returns A count of the number of active licenses installed.

Examples `LICENSE:COUNT?` might return `:LICENSE:COUNT 2` indicating that 2 active licenses are installed

LICense:ERRor? (Query Only)

This query-only command prompts the instrument to return all events and their messages (delimited by commas), and removes the returned events from the Event Queue. This command is an alias for `ALLEV?`.

Group Miscellaneous

Syntax `LICense:ERRor?`

Related Commands [ALLEv?](#)
 [*ESR?](#)
 [EVMsg?](#)

Examples `LICense:ERRor?` might return 0, "No events to report- queue empty".

LICense:GMT? (Query Only)

This query returns the GMT time in ISO 8601 format, the local date, 24 hour time and time-zone offset.

Group Miscellaneous

Syntax `LICense:GMT?`

Returns The GMT time in ISO 8601 format, the local date, 24 hour time and time-zone offset.

Examples `LICENSE:GMT?` might return `:LICENSE:GMT`
 `"2016-05-23T17:05:10-07:00"` indicating the local GMT time.

LICense:HID? (Query Only)

This query returns the instrument HostID unique identifier.

Group Miscellaneous

Syntax `LICense:HID?`

Returns The instrument HostID unique identifier.

Examples `LICENSE:HID?` might return `:LICENSE:HID "TMS-9CS4US5SGJN6X"`

LICense:INSTall (No Query Form)

This command accepts a `<block_data>` license and installs it on the instrument. Restarting the instrument may be necessary to fully activate the additional capabilities.

Group Miscellaneous

Syntax `LICense:INSTall <block_data>`

Arguments `<block_data>` is the license in block data format.

Examples `LICENSE:INSTALL <block_data>`

LICense:ITEM? (Query Only)

This query returns the details pertaining to a specific license. The NR1 argument is zero-indexed. If no argument is provided, zero is assumed.

Group Miscellaneous

Syntax `LICense:ITEM? <NR1>`

Arguments `<NR1>` is the zero-indexed argument specifying a specific license.

Returns The details pertaining to a specific license.

Examples `LICENSE:ITEM? 0` might return
"5-BW-1000,Fixed,2116-06-15T14:55:54-07:00,11870047,BW5-1000,1 GHz
bandwidth on 5 series oscilloscopes"

LICense:LIST? (Query Only)

This query returns the active license nomenclatures as a comma-separated list of strings. Duplicate nomenclatures, that is, the same license but with different expiration dates, are included.

Group Miscellaneous

Syntax `LICense:LIST?`

Returns The active license nomenclatures as a comma-separated list of strings.

Examples `LICENSE:LIST?` might return `:LICENSE:LIST`
"5-BW-1000","SUP5-RL125M".

LICense:VALidate? (Query Only)

This query accepts a license nomenclature as an argument and returns True (1) if that nomenclature is active and any required hardware is installed, or False (0) if either the nomenclature is not active or required hardware is not installed.

Group Miscellaneous

Syntax `LICense:VALidate? <QString>`

Arguments `<QString>` is the license nomenclature.

Returns	True (1) if that nomenclature is active and any required hardware is installed. False (0) if either the nomenclature is not active or required hardware is not installed.
Examples	<code>LICENSE:VALIDATE? "AFG"</code> might return <code>:LICENSE:VALIDATE "AFG",0</code> indicating the license is not active.

LOCK

This command enables or disables all front panel buttons and knobs. There is no front panel equivalent.

To completely disable front panel operation, combine two commands as follows: `LOCK ALL; :TOUCHSCREEN:STATE OFF`. To re-enable the front panel, send these two commands: `LOCK NONE; :TOUCHSCREEN:STATE ON`. The commands must be sent in that order.

When the front panel is locked, the front panel commands will not work and will not generate error events. You can work around a locked front panel, by using the appropriate programmatic interface commands, instead of the front-panel commands. For example, to set the trigger level to 50%, you could use `TRIGGER:A SETLevel`. To force a trigger, you could use `TRIGGER FORCE`.

Group	Miscellaneous
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Syntax	<code>LOCK {ALL NONE}</code> <code>LOCK?</code>
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Related Commands	UNLock TOUCHSCREEN:STATe
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Arguments	<code>ALL</code> disables all front panel controls. <code>NONE</code> enables all front panel controls. This is equivalent to the <code>UNLock ALL</code> command. If the instrument is in the Remote With Lockout State (RWLS), the <code>LOCK NONE</code> command has no effect. For more information, see the ANSI/IEEE Std 488.1-1987 Standard Digital Interface for Programmable Instrumentation, section 2.8.3 on RL State Descriptions.
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Examples LOCK ALL locks the front panel controls.

LOCK? might return :LOCK NONE, indicating that the front panel controls are enabled.

*LRN? (Query Only)

This query-only command returns the commands that list the instrument settings, allowing you to record or “learn” the current instrument settings. You can use these commands to return the instrument to the state it was in when you made the *LRN? query. This command is identical to the [SET?](#) command.

Group Miscellaneous

Syntax *LRN?

Related Commands [SET?](#)

Examples *LRN? might return the following response: *RST;;PARAMBATCHING 0;;HEADER 1;;VERBOSE 1;;ALIAS:STATE 0;;ACQUIRE:STATE 1;;ACQUIRE:NUMENV INFINITE;;FASTACQ:STATE 0;;FASTACQ:PALETTE TEMPERATURE;;HORIZONTAL:SAMPLERATE:ANALYZEMODE:MINIMUM:VALUE AUTOMATIC;;SELECT:CH1 1;;SELECT:CH2 0;;SELECT:CH3 0;;SELECT:CH4 0;;SELECT:CH5 0;;SELECT:CH6 0;;SELECT:CH7 0;;SELECT:CH8 0;;CH1:BANDWIDTH 250.0000E+6;;CH2:BANDWIDTH 500.0000E+6;;CH3:BANDWIDTH 500.0000E+6;;CH4:BANDWIDTH 500.0000E+6;;CH5:BANDWIDTH 500.0000E+6;;CH6:BANDWIDTH 500.0000E+6;;CH7:BANDWIDTH 120.0000E+6;;CH8:BANDWIDTH 500.0000E+6;;CH1:BANDWIDTH:ENHANCED 0;;CH2:BANDWIDTH:ENHANCED 0;;CH3:BANDWIDTH:ENHANCED 0;;CH4:BANDWIDTH:ENHANCED 0;;CH5:BANDWIDTH:ENHANCED 0;;CH6:BANDWIDTH:ENHANCED 0;;CH7:BANDWIDTH:ENHANCED 0;;CH8:BANDWIDTH:ENHANCED 0;;DATA:DESTINATION REF1;;DATA:ENCDG ASCII;;DATA:SOURCE CH1;;DATA:START 1;;DATA:STOP 1000000;;CH1:PROBETYPE ANALOG;;CH2:PROBETYPE ANALOG;;CH3:PROBETYPE ANALOG;;CH4:PROBETYPE ANALOG;;CH5:PROBETYPE DIGITAL;;CH6:PROBETYPE ANALOG;;CH7:PROBETYPE ANALOG;;CH8:PROBETYPE ANALOG;;HORIZONTAL:MODE AUTO;;HORIZONTAL:SAMPLERATE:ANALYZEMODE:MINIMUM:OVERRIDE 1;;HORIZONTAL:MODE:AUTOMATIC:FASTACQ:RECORDLENGTH:MAXIMUM: ZOOMOVERRIDE 1;;HORIZONTAL:MODE:AUTOMATIC:FASTACQ:RECORDLENGTH: MAXIMUM:VALUE 1250;;HORIZONTAL:MODE:MANUAL:CONFIGURE

```
HORIZONTALSCALE;:HORIZONTAL:SAMPLERATE
6.2500E+9;:DISPLAY:WAVEVIEW1:ZOOM:ZOOM1:VERTICAL:SCALE
1.0000;:TRIGGER:B:TYPE EDGE;:TRIGGER:A:TYPE
EDGE;:ACQUIRE:SEQUENCE:NUMSEQUENCE
1;:ACQUIRE:SEQUENCE:MODE NUMACQS;:ACQUIRE:MODE
SAMPLE;:ACQUIRE:STOPAFTER RUNSTOP;:ACQUIRE:FASTACQ:PALETTE
TEMPERATURE;:ACQUIRE:FASTACQ:STATE 0;:ACQUIRE:NUMAVG 16T.
```

MAINWindow:RRBDisplaystate (No Query Form)

Sets the display state of the Results readout bar to ON (displayed) or OFF (not displayed).

Group Miscellaneous

Syntax MAINWindow:RRBDisplaystate {1|ON|0|OFF}

Arguments 1 or ON enables Autoset to change vertical settings.
0 or OFF disables Autoset from changing vertical settings.

Examples MAINWindow:RRBDisplaystate 1 expands (displays) the Results readout bar.

MATH:ADDNew (No Query Form)

This command adds the specified math.

Group Math

Syntax MATH:ADDNew <QString>

Related Commands [MATH:LIST?](#)
[MATH:DELeTe](#)

Arguments <QString> is the quoted string specifying the math waveform to add. The argument is of the form "MATH<NR1>", where <NR1> is ≥ 1 .

Examples MATH:ADDNEW "MATH2" adds MATH2.

MATH:DELeTe (No Query Form)

This command deletes the specified math.

Group Math

Syntax MATH:DELeTe <QString>

Related Commands [MATH:ADDNew](#)
[MATH:LIST?](#)

Arguments <QString> is a quoted string specifying the math waveform to delete. The quoted string is of the form "MATH<NR1>", where <NR1> is ≥ 1 .

Examples MATH:DELETE "MATH1", which deletes the MATH1 waveform.

MATH:LIST? (Query Only)

This query returns a comma separated list of all currently defined math waveforms.

Group Math

Syntax MATH:LIST?

Related Commands [MATH:ADDNew](#)
[MATH:DELeTe](#)

Returns All currently defined math waveforms.

Examples MATH:LIST? Might return :MATH:LIST MATH1,MATH4, indicating MATH1 and MATH4 are the currently defined math waveforms.

MATH:MATH<x>:AVG:MODE

This command sets or queries the math average mode flag. If the flag is set to 1, math averaging is turned on. The math waveform is specified by x.

Group	Math
Syntax	MATH:MATH<x>:AVG:MODE {<NR1> OFF ON}
Related Commands	MATH:MATH<x>:AVG:WEIGHT
Arguments	<NR1> = 0 turns off average mode, and any other integer turns on average mode. OFF turns off average mode. ON turns on average mode.
Examples	MATH:MATH2:AVG:MODE ON sets the average mode on. MATH:MATH1:AVG:MODE? might return :MATH:MATH1:AVG:MODE 0, indicating average mode is off.

MATH:MATH<x>:AVG:WEIGHT

This command sets or queries the number of acquisitions at which the averaging algorithm will begin exponential averaging. The math waveform is specified by x.

Group	Math
Syntax	MATH:MATH<x>:AVG:WEIGHT <NR1>
Related Commands	MATH:MATH<x>:AVG:MODE
Arguments	<NR1> is the number of acquisitions at which the averaging algorithm will begin exponential averaging.
Examples	MATH:MATH2:AVG:WEIGHT 143 sets the number of acquisitions needed to begin averaging to 20 MATH:MATH3:AVG:WEIGHT? might return :MATH:MATH3:AVG:WEIGHT 20 indicating the number of acquisitions needed to begin averaging is 8.

MATH:MATH<x>:DEFine

This command allows you to define new waveforms using mathematical expressions. The query form of this command returns the math definition for the specified math waveform. The math waveform is specified by x.

You can specify a math expression from waveforms, measurements and scalar sources, functions, operands, and numerical constants.

Math expressions can be simple, such as Ch1, which specifies that a waveform should show the signal source of Channel 1 with no mathematical computation. Math expressions can also be complex, consisting of 100 plus characters and comprising many sources (including other math waveforms), functions, and operands. As an example, you can enter the expression Log(Ch1+Ch2), which specifies that the signals from channels 1 and 2 are to be algebraically added, and the base 10 log of the sum is to be shown as the final math waveform.

Group	Math
Syntax	MATH:MATH<x>:DEFine <QString> MATH:MATH<x>:DEFine?
Arguments	<QString> quoted string argument is the mathematical expression that defines the waveform. MATH:MATH<x>:DEFINE? is for use when the MATH:MATH<x>:TYPE is ADVANCED.
Examples	MATH:MATH2:DEFine "CH1+CH2" adds the Channel 1 and Channel 2, defines the Math2 waveform to be Channel1 and Channel2 added together. MATH:MATH1:DEFine? might return :MATH:MATH1:DEFine "CH2*REF2" as the expression that defines Math 1 waveform.

MATH:MATH<x>:FUNCTION

This command sets or queries the basic math arithmetic function. The math waveform is specified by x.

NOTE. This command does not affect the same Math equation in Advanced math (also accessed via the command MATH:MATH<x>:DEFINE).

Group	Math
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Syntax	<code>MATH:MATH<x>:FUNCTION {ADD SUBtract MULTiply DIVide}</code>
Arguments	<code>ADD</code> sets the basic math function to add. <code>SUBtract</code> sets the basic math function to subtract. <code>MULTiply</code> sets the basic math function to multiply. <code>DIVide</code> sets the basic math function to divide.
Examples	<code>MATH:MATH2:FUNCTION MULTIPLY</code> sets the basic math function to multiply. <code>MATH:MATH1:FUNCTION?</code> might return <code>:MATH:MATH1:FUNCTION ADD</code> indicating the current basic math function is addition.

MATH:MATH<x>:GATing

This command specifies or returns the gating setting. It only applies to Math FFT plots. The math waveform is specified by x.

Group	Math
Syntax	<code>MATH:MATH<x>:GATing {NONE SCREEN CURSor}</code> <code>MATH:MATH<x>:GATing?</code>
Related Commands	MATH:MATH<x>:TYPE
Arguments	<code>NONE</code> turns off math gating. <code>SCREEN</code> turns on gating, using the left and right edges of the screen. <code>CURSor</code> limits math to the portion of the waveform between the vertical bar cursors, even if they are off screen.
Examples	<code>MATH:MATH3:GATING CURSOR</code> sets the spectral math plot to be gated by the cursors. <code>MATH:MATH2:GATING?</code> might return <code>:MATH:MATH:2:GATING SCREEN</code> which indicates the spectral math plot is gated by the screen.

MATH:MATH<x>:LABel:COLor

This command sets or queries color of the specified math's label. The math waveform is specified by x.

Group	Math
Syntax	<code>MATH:MATH<x>:LABel:COLor <QString></code>
Arguments	<code><QString></code> is the color of the label. To return the color to the default color, send an empty string as in this example: <code>:MATH:MATH1:LABEL:COLOR ""</code> .
Examples	<code>MATH:MATH3:LABEL:COLOR "GREEN"</code> sets the Math 3 label color to green. <code>MATH:MATH1:LABEL:COLOR?</code> might return <code>:MATH:MATH1:LABEL:COLOR "BLUE"</code> indicating the color of the Math1 label is blue.

MATH:MATH<x>:LABel:FONT:BOLD

This command sets or queries the bold state of the specified math label. The math waveform is specified by x.

Group	Math
Syntax	<code>MATH:MATH<x>:LABel:FONT:BOLD {<NR1> OFF ON}</code>
Arguments	<code><NR1> = 0</code> turns off bold, and any other integer turns on bold. <code>OFF</code> turns off bold. <code>ON</code> turns on bold.
Examples	<code>MATH:MATH3:FONT:BOLD ON</code> set the math 3 label to bold. <code>MATH:MATH2:FONT:BOLD?</code> might return <code>:MATH:MATH2:FONT:BOLD 0</code> indicating the math 2 label is not currently bold.

MATH:MATH<x>:LABel:FONT:ITALic

This command sets or queries italic state of the specified math label. The math waveform is specified by x.

Group	Math
Syntax	<code>MATH:MATH<x>:LABel:FONT:ITALic {<NR1> OFF ON}</code>

Arguments	<code><NR1> = 0</code> turns off italic, and any other integer turns on italic. <code>OFF</code> turns off italic. <code>ON</code> turns on italic.
Examples	<code>MATH:MATH4:FONT:ITALIC ON</code> set the math 4 label to be italic. <code>MATH:MATH1:FONT:ITALIC?</code> might return <code>:MATH:MATH1:FONT:ITALIC 0</code> indicating the math 1 label is not currently italic. <code><NR1></code> is the font size of the label.

MATH:MATH<x>:LABel:FONT:SIZE

This command sets or queries font size of the specified math label. The math waveform is specified by x.

Group	Math
Syntax	<code>MATH:MATH<x>:LABel:FONT:SIZE <NR1></code>
Arguments	
Examples	<code>MATH:MATH4:LABEL:FONT:SIZE 32</code> sets the math 4 label size to 32. <code>MATH:MATH2:LABEL:FONT:SIZE?</code> might return <code>:MATH:MATH2:LABEL:FONT:SIZE 14</code> indicating the math 2 label size is currently 14.

MATH:MATH<x>:LABel:FONT:TYPE

This command sets or queries font type of the specified math label, such as Arial or Times New Roman. The math waveform is specified by x.

Group	Math
Syntax	<code>MATH:MATH<x>:LABel:FONT:TYPE <QString></code>
Arguments	<code><QString></code> is the name of the font type.

Examples `MATH:MATH2:LABEL:FONT:TYPE "Serif"` sets the math 2 label font type to Serif.
`MATH:MATH3:LABEL:FONT:TYPE?` might return
`:MATH:MATH3:LABEL:FONT:TYPE "Monospace"` indicating the math 3 font type is currently Monospace.

MATH:MATH<x>:LABel:FONT:UNDERline

This command sets or queries the underline state of the specified math label. The math waveform is specified by x.

Group Math

Syntax `MATH:MATH<x>:LABel:FONT:UNDERline {<NR1>|OFF|ON}`

Arguments `<NR1> = 0` turns off underline, and any other integer turns on underline.
`OFF` turns off underline.
`ON` turns on underline.

Examples `MATH:MATH3:FONT:UNDERLINE ON` sets the math 3 label to be underlined.
`MATH:MATH2:FONT:UNDERLINE?` might return
`:MATH:MATH2:FONT:UNDERLINE 0` indicating the math 2 label is currently not underlined.

MATH:MATH<x>:LABel:NAME

This command sets or queries the label string, which is used for annotating the math waveform on the screen. The math waveform to which the label is attached is specified by x.

Group Math

Syntax `MATH:MATH<x>:LABel:NAME <QString>`
`MATH:MATH<x>:LABel:NAME?`

Arguments `<QString>` specifies the label to annotate the math waveform.

Examples `MATH:MATH2:LABel:NAME "PROBE POINT7"` assigns "Probe point7" Math 2 waveform.

`MATH:MATH2:LABel:NAME?` might return `:MATH:MATH2:LABel:NAME "Probe point7"`, indicating that Probe point 7 is the label for the Math 2 waveform.

MATH:MATH<x>:LABel:XPOS

This command sets or queries the X position of the specified math label. Maths are specified by x.

Group Math

Syntax `MATH:MATH<x>:LABel:XPOS <NR1>`
`MATH:MATH<x>:LABel:XPOS?`

Related Commands [MATH:MATH<x>:LABel:YPOS](#)

Arguments <NR1> is the location (in pixels) where the label for the selected math waveform is displayed, relative to the left edge of the display.

Examples `MATH:MATH2:LABel:XPOS 5` moves the waveform label for the Math 2 waveform so that it begins 5 pixels to the right of the left edge of the screen.

`MATH:MATH2:LABel:XPOS?` might return `:MATH:MATH2:LABel:XPOS 2.5`, indicating that the waveform label for the Math 2 waveform is currently 2.5 pixels to the right of the left edge of the display.

MATH:MATH<x>:LABel:YPOS

This command sets or queries the y-position of the specified math label. The Math waveform is specified by x.

Group Math

Syntax `MATH:MATH<x>:LABel:YPOS <NR1>`
`MATH:MATH<x>:LABel:YPOS?`

Related Commands [MATH:MATH<x>:LABel:XPOS](#)

Arguments	<NR1> is the location (in pixels) where the label for the selected math waveform is displayed, relative to the baseline of the waveform.
Examples	MATH:MATH2:LABEL:YPOS -2.5 moves the waveform label for the Math 2 waveform to 2.5 pixels below the baseline of the waveform. MATH:MATH2:LABEL:YPOS? might return :MATH:MATH2:LABEL:YPOS 0, indicating that the waveform label for the Math 2 waveform is currently located at the baseline of the waveform.

MATH:MATH<x>:SOURCE<x>

This command sets or queries the specified math source. The source in the command can be either 1 or 2. This command sets the Basic Math components in the user interface, with two sources and a function. You would also need to set the math type to Basic to see the change in the user interface but this will not effect the programmable interface. The math waveform and source are specified by x.

Group	Math
Syntax	MATH:MATH<x>:SOURCE<x> {CH<x> MATH<x> REF<x>}
Related Commands	MATH:MATH<x>:TYPE MATH:MATH<x>:FUNCTION
Arguments	Arguments are possible math sources. SOURCE1 and SOURCE2 are for use when the MATH:MATH<x>:TYPE is BASIC.
Examples	MATH:MATH2:SOURCE1 CH1 sets the first source of math 2 to Channel 1. MATH:MATH3:SOURCE2? might return :MATH:MATH3:SOURCE2 REF3 indicating the second source of Math 3 is Ref 3.

MATH:MATH<x>:SPECTral:HORZ

This command sets or queries the horizontal display scale of the spectral math waveform. The math waveform is specified by x.

Group	Math
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Syntax	<code>MATH:MATH<x>:SPECTral:HORZ {LOG LINEAr}</code>
Arguments	<code>LINEAr</code> sets the SpectralMag units to linear. <code>LOG</code> sets the SpectralMag units to log.
Examples	<code>MATH:MATH2:SPECTRAL:HORZ LOG</code> sets the horizontal display scale of the spectral math waveform to log. <code>MATH:MATH3:SPECTRAL:HORZ?</code> might return <code>:MATH:MATH3:SPECTRAL:HORZ LINEAr</code> indicating the horizontal display scale of the Math 3 spectral math waveform is currently set to linear.

MATH:MATH<x>:SPECTral:MAG

This command sets or queries the units of the SpectralMag function in the specified math definition string. The Math waveform is specified by x.

Group	Math
Syntax	<code>MATH:MATH<x>:SPECTral:MAG {LINEAr DBM}</code> <code>MATH:MATH<x>:SPECTral:MAG?</code>
Arguments	<code>LINEAr</code> sets the SpectralMag units to linear. <code>DBM</code> sets the SpectralMag units to decibels. It also sets the Ref Level Offset to a value that is the equivalent of 1 mW into 50 Ω.
Examples	<code>MATH:MATH2:SPECTral:MAG DBM</code> sets the SpectralMag units for Math 2 waveform to decibels. <code>MATH:MATH2:SPECTral:MAG?</code> might return <code>:MATH:MATH2:SPECTral:MAG DBM</code>, indicating that the SpectralMag units for Math 2 waveform are set to decibels.

MATH:MATH<x>:SPECTral:PHASE

This command sets or queries the units of a SpectralPhase function in the specified math definition string. The Math waveform is specified by x.

Group	Math
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Syntax `MATH:MATH<x>:SPECTral:PHASE {DEGrees|RADians|GROUPDeLay}`
`MATH:MATH<x>:SPECTral:PHASE?`

Arguments DEGREES sets the SpectralPhase units to degrees.
 RADIANS sets the SpectralPhase units to radians.
 GROUPDELAY sets the SpectralPhase units to groupdelay, which computes the derivative of unwrapped phase spectrum. Units are expressed in seconds.

Examples `MATH:MATH2:SPECTral:PHASE DEGREES` sets the SpectralPhase units for Math 2 waveform to degrees.
 `MATH:MATH2:SPECTral:PHASE?` might return
 `:MATH:MATH2:SPECTral:PHASE RADIANS`, indicating that the SpectralPhase units for Math 2 waveform are set to radians.

MATH:MATH<x>:SPECTral:SOURce

This command sets or queries the specified spectral math source. This only works with a math of type FFT. The math waveform is specified by x.

Group Math

Syntax `MATH:MATH<x>:SPECTral:SOURce {CH<x>|MATH<x>|REF<x>}`
`MATH:MATH<x>:SPECTral:SOURce?`

Related Commands [MATH:MATH<x>:TYPE](#)

Arguments Arguments are math sources. `MATH:MATH<x>:SPECTRAL:SOURCE` is for use when the `MATH:MATH<x>:TYPE` is FFT.

Examples `MATH:MATH3:SPECTRAL:SOURCE REF3` sets the source of the Spectral Math waveform to Ref 3.
 `MATH:MATH2:SPECTRAL:SOURCE?` might return
 `:MATH:MATH2:SPECTRAL:SOURCE CH4` indicating the Math 2 spectral source is Channel 4.

MATH:MATH<x>:SPECTral:SUPPress

This command sets or queries whether suppression threshold for the specified math waveform is enabled. This is only applied when Spectral Plot type is Phase. The math waveform is specified by x.

Group Math

Syntax MATH:MATH<x>:SPECTral:SUPPress {OFF|ON|0|1}
MATH:MATH<x>:SPECTral:SUPPress?

Related Commands [MATH:MATH<x>:SPECTral:TYPE](#)

Arguments 0 disables suppression threshold for the specified math waveform.
1 enables suppression threshold for the specified math waveform.
ON enables suppression threshold for the specified math waveform.
OFF disables suppression threshold for the specified math waveform.

Examples MATH:MATH3:SPECTRAL:SUPPRESS ON enables the suppression threshold on the Math 3 spectral waveform.
MATH:MATH2:SPECTRAL:SUPPRESS? might return
:MATH:MATH2:SPECTRAL:SUPPRESS 0 indicating the suppression threshold is disabled on the Math 2 spectral waveform.

MATH:MATH<x>:SPECTral:SUPPress:VALue

This command sets or queries in volts the value of suppression threshold of the specified math waveform. This requires the Spectral type to be Phase and the Suppression to be enabled for this PI command to have any affect. The math waveform is specified by x.

Group Math

Syntax MATH:MATH<x>:SPECTral:SUPPress:VALue <NR3>
MATH:MATH<x>:SPECTral:SUPPress:VALue?

Related Commands [MATH:MATH<x>:SPECTral:TYPE](#)
[MATH:MATH<x>:SPECTral:SUPPress](#)

Arguments <NR3> is the value of suppression threshold of the specified math waveform in volts.

Examples MATH:MATH3:SPECTRAL:SUPPRESS:VALUE 100.0E-3 sets the suppression threshold of Math 3 to 100 mV.

MATH:MATH1:SPECTRAL:SUPPRESS:VALUE? might return
MATH:MATH1:SPECTRAL:SUPPRESS:VALUE 10.0000E+3 indicating
the suppression threshold of Math 1 is currently set to 10 kV.

MATH:MATH<x>:SPECTral:TYPE

This command sets or queries the FFT type selected for spectral analysis. The math waveform is specified by x.

Group Math

Syntax MATH:MATH<x>:SPECTral:TYPE {MAGNitude|PHASe|REAL|IMAGinary}
MATH:MATH<x>:SPECTral:TYPE?

Arguments MAGNitude specifies the magnitude spectral function.

PHASe specifies the phase spectral function.

REAL specifies the real spectral function.

IMAGinary specifies the imaginary spectral function.

Examples MATH:MATH1:SPECTral:TYPE REAL specifies the real spectral function.

MATH:MATH1:SPECTral:TYPE? might return :MATH:MATH1:SPECTRAL:TYPE
MAGNITUDE indicating the math is the magnitude spectral function.

MATH:MATH<x>:SPECTral:UNWRap

This command sets or queries whether phase unwrap of the spectral analyzer output data is enabled. The Math waveform is specified by x.

Group Math

Syntax MATH:MATH<x>:SPECTral:UNWRap {OFF|ON|0|1}
MATH:MATH<x>:SPECTral:UNWRap?

Arguments	0 disables phase unwrap for the specified math waveform.
	1 enables phase unwrap for the specified math waveform.
	ON enables phase unwrap for the specified math waveform.
	OFF disables phase unwrap for the specified math waveform.
Examples	MATH1:SPECTRAL:UNWRAP ON enables phase unwrap of the spectral analyzer output data.
	MATH1:SPECTRAL:UNWRAP? might return :MATH1:SPECTRAL:UNWRAP 0, indicating that the phase unwrap of the spectral analyzer output data is disabled.

MATH:MATH<x>:SPECTral:UNWRap:DEGrees

This command sets or queries how many degrees adjacent phase values can jump before being unwrapped. This requires the Spectral type to be Phase and the UNWRAP to be enabled for this PI command to have any affect. The math waveform is specified by x.

Group	Math
Syntax	MATH:MATH<x>:SPECTral:UNWRap:DEGrees <NR3> MATH:MATH<x>:SPECTral:UNWRap:DEGrees?
Related Commands	MATH:MATH<x>:SPECTral:TYPE MATH:MATH<x>:SPECTral:UNWRap
Arguments	<NR3> is the value of unwrap phase in degrees.
Examples	MATH:MATH2:SPECTRAL:UNWRAP:DEGREES 90 sets the unwrap phase of the spectral Math 2 to 90 degrees. MATH:MATH2:SPECTRAL:UNWRAP:DEGREES? might return :MATH:MATH2:SPECTRAL:UNWRAP:DEGREES 180 indicating unwrap phase of the spectral math 3 waveform is 180 degrees.

MATH:MATH<x>:SPECTral:WINDow

This command sets or queries the window function used to apply the specified FFT window to the input data for the specified math waveform. The Math waveform is specified by x. A spectral window determines what the filter shape

of the spectral analyzer will be in the frequency domain. It can be described by a mathematical function that is multiplied point-by-point times the input data to the spectral analyzer.

Following is a list of arguments that specify the window function used to multiply the input data. The windows are listed in the order of their ability to resolve frequencies (resolution bandwidth).

Group	Math
Syntax	MATH:MATH<x>:SPECTRa1:WINDow {RECTANGUlar HAMMing HANNing BLACKMANHarris KAISERBesse1 GAUSSian FLATTOP2 TEKEXPonential} MATH:MATH<x>:SPECTRa1:WINDow?
Related Commands	MATH:MATH<x>:TYPE
Arguments	RECTANGUlar window function is equivalent to multiplying all gate data by one. HAMMing window function is based on a cosine series. HANNing window function is based on a cosine series. BLACKMANHarris window function is based on a cosine series. KAISERBesse1 window function is based on a cosine series. GAUSSian window function has the best localization characteristics in the joint time/frequency plane FLATTOP2 window function is a cosine series window with a flattened frequency response lobe. TEKEXPonential window has an exponential nonsymmetrical shape in the time domain and a triangular shape in the frequency domain.
Examples	MATH2:SPECTRAL:WINDOW GAUSSIAN applies a Gaussian window to the spectral analyzer input data. MATH2:SPECTRAL:WINDOW? might return :MATH2:SPECTRAL:WINDOW BLACKMANHarris, indicating that the window function used to multiply the spectral analyzer input data is the BLACKMANHarris function.

MATH:MATH<x>:TYPE

This command sets or queries the math type. The math waveform is specified by x.

Group	Math
Syntax	MATH:MATH<x>:TYPE {BASiC FFT ADVanced}
Arguments	BASiC set the type to basic math. FFT sets the type to FFT math, which can use any live analog or reference waveform in the time domain. NOTE. You can also use FFT as part of a math expression by declaring the type ADVanced. See examples for the command MATH:MATH<x>:DEFine. ADVanced sets the type to advanced math.
Examples	MATH:MATH2:TYPE BASIC sets the type of Math 2 to basic. MATH:MATH2:TYPE? might return :MATH:MATH2:TYPE FFT indicating the type of Math 4 is currently FFT.

MATH:MATH<x>:VUNIT

This command specifies or returns the math custom vertical units. The math waveform is specified by x.

Group	Math
Syntax	MATH:MATH<x>:VUNIT <QString>
Arguments	<QString> is the custom vertical units.
Examples	MATH:MATH2:VUNIT "Small" MATH:MATH2:VUNIT? might return :MATH:MATH2:VUNIT "Large" indicating the Math 2 vertical unit is set to "Large".

MATHArbflt<x>:FILEpath

This command or query sets the file path for a file of filter coefficients for the specified arbitrary filter. Setting a path will read that file and load the filter for ARBFLT<x>. Access these filters using a Math with an expression of the form "ARBFLT<x>()".

Group	Math
Syntax	MATHArbflt<x>:FILEpath <QString> MATHArbflt<x>:FILEpath?
Arguments	<QString> specifies the path to the file of filter coefficients.
Examples	MATHARBFLT8:FILEPATH "c:/myfilters/20mhz.flr" reads filter coefficients from the file and loads the filter for ARBFLT8. MATHARBFLT1:FILEPATH? might return :MATHARBFLT1:FILEPATH "C:\Users\Public\Tektronix\TekScope\Math Arbitrary Filters\LowPass-Norm\lowpass_0.bw.flr", indicating that the path to the file of filter coefficients is set to "C:\Users\Public\Tektronix\TekScope\Math Arbitrary Filters\LowPass-Norm\lowpass_0.bw.flr".

MEASUrement? (Query Only)

This query-only command returns all measurement parameters.

Group	Measurement
Syntax	MEASUrement?
Examples	MEASUREMENT? might return :MEASUREMENT:GATING OFF;IMMED:TYPE UNDEFINED;UNITS "V";SOURCE1 CH1;SOURCE2 CH1;SOURCE1:SIGTYPE PULSE;;MEASUREMENT:IMMED:SOURCE2:SIGTYPE PULSE;;MEASUREMENT:IMMED:DELAY:EDGE1 RISE;EDGE2 RISE;DIRECTION FORWARDS;;MEASUREMENT:IMMED:REFLEVEL:METHO D PERCENT;ABSOLUTE:HIGH 0.0000;LOW 0.0000;MID1 0.0000;MID2 0.0000;;MEASUREMENT:IMMED:REFLEVEL:PERCENT:HIGH 90.0000;LOW 10.0000;MID1 50.0000;MID2 50.0000;;MEASUREMENT:IMMED:METHO D HISTOGRAM;NOISE HIGH;;MEASUREMENT:MEAS1:STATE 0;TYPE UNDEFINED;UNITS "V";SOURCE1 CH1;SOURCE2 CH1;SOURCE1:SIGTYPE PULSE;;MEASUREMENT:MEAS1:SOURCE2:SIGTYPE PULSE;;MEASUREMENT:MEAS1:DELAY:EDGE1 RISE;EDGE2 RISE;DIRECTION FORWARDS;;MEASUREMENT:MEAS1:REFLEVEL:METHO D PERCENT;ABSOLUTE:HIGH 0.0000;LOW 0.0000;MID1 0.0000;MID2 0.0000;;MEASUREMENT:MEAS1:REFLEVEL:PERCENT:HIGH 90.0000;LOW 10.0000;MID1 50.0000;MID2 50.0000;;MEASUREMENT:MEAS1:METHO D HISTOGRAM;NOISE HIGH;;MEASUREMENT:MEAS2:STATE 0;TYPE UNDEFINED;UNITS "V";SOURCE1 CH1;SOURCE2

```
CH1;SOURCE1:SIGTYPE PULSE;;MEASUREMENT:MEAS2:SOURCE2:SIGTYPE
PULSE;;MEASUREMENT:MEAS2:DELAY:EDGE1 RISE;EDGE2
RISE;DIRECTION.
```

MEASUrement:ADDMEAS (No Query Form)

This command adds a measurement.

Group Measurement

Syntax MEASUrement:ADDMEAS {ACCOMMONMODE|ACRMS|AMP|ITUDE|AREA|
BASE|BITAMPLITUDE|BITHIGH|BITLOW|BURSTWIDTH|COMMONMODE|
DATARATE|DCD|DDJ|DDRAOS|DDRAOSPRTCK|DDRAOSPERUI|DDRAUS|
DDRAUSPERTCK|DDRAUSPERUI|DDRHOLDDIFF|DDRSETUPDIFF|
DDRTCHABS|DDRTCHAVERAGE|DDRTCKAVERAGE|DDRTCLABS|
DDRTCLAVERAGE|DDRTERRMN|DDRTERRN|DDRTJITCC|DDRTJITDUTY|
DDRTJITPER|DDRTPST|DDRTRPRE|DDRTWPRE|DDRVIAX|DDRTDQSK|DELAY|DJ|DJJ|
FALLSLEWRATE|FALLTIME|FREQUENCY|F2|F4|F8|HEIGHT|HEIGHTBER|
HIGH|HIGHTIME|HOLD|JITTERSUMMARY|J2|J9|LOW|LOWTIME|
MAXIMUM|MEAN|MINIMUM|NDuty|NOVERSHOOT|NPERIOD|NPJ|
NWIDTTH|PDUTY|PERIOD|PHASE|PHASENOISE|PJ|PK2Pk|
POVERSHOOT|PWIDTH|QFACTOR|RISESLEWRATE|RISETIME|RJ|
RJDIRAC|RMS|SETUP|SKEW|SRJ|SSCFREQDEV|SSCMODRATE|TIE|
TIMEOUTSIDELEVEL|TJBER|TNTRATIO|TOP|UNITINTERVAL|
VDIFFXOVR|WIDTH|WIDTHBER}

Arguments ACCOMMONMODE AC Common Mode (Pk-Pk) is the peak-to-peak of the common mode voltage of two sources. This measurement is made across the entire record.

ACRMS (AC RMS) is the true Root Mean Square of the data points, about the Mean. This measurement can be made across the entire record, or on each cycle in the record.

AMPLITUDE is the difference between the Top value and the Base value. This measurement can be made across the entire record, or on each cycle in the record.

$$\text{Amplitude} = \text{High} - \text{Low}$$

AREA is the area under the curve, calculated by integrating the data points. The area measured above ground is positive. The area measured below ground is negative. This measurement can be made across the entire record, or on each cycle in the record.

BASE is the most common data value below the midpoint of the waveform. This measurement can be made across the entire record, or on each cycle in the record.

BITAMPLITUDE (Bit Amplitude) is the difference between the amplitudes of the 1 bit and the 0 bit surrounding a transition. The amplitude is measured over a user

specified portion at the center of the recovered unit interval. This measurement is made on each transition bit in the record (Mean) or across the entire record (Mode).

BITHIGH (Bit High) is the amplitude of a 1 bit. The amplitude is measured over a user specified portion at the center of the recovered unit interval. This measurement is made on each high bit in the record (Mean) or across the entire record (Mode).

BITLOW (Bit Low) is the amplitude of a 0 bit. The amplitude is measured over a user specified portion at the center of the recovered unit interval. This measurement is made on each high bit in the record (Mean) or across the entire record (Mode).

BURSTWIDTH (Burst Width) is the duration of a series of adjacent crossings of the Mid reference level (RM). Bursts are separated by a user-defined idle time (tI). This measurement is made on each burst in the record.

COMMONMODE (DC Common Mode) is the arithmetic mean of the common mode voltage of two sources. This measurement is made across the entire record.

DATARATE (Data Rate) is the reciprocal of Unit Interval. This measurement is made on each bit in the record.

DCD (duty cycle distortion) is the peak-to-peak amplitude of the component of the deterministic jitter correlated with the signal polarity. This measurement is made across the entire record.

DDJ (data dependent jitter) is the peak-to-peak amplitude of the component of the deterministic jitter correlated with the data pattern in the waveform. This measurement is made across the entire record.

DDRAOS (area above signal) is the total area of the signal above a specified reference level. This measurement is made across the entire record.

DDRAOSPERTCK (area over signal for tCK events) is the total area of the signal above a specified reference level calculated over consecutive tCK intervals. It is applicable to clock and address/command waveforms.

DDRAOSPERUI (area over signal for UI events) is the total area of the signal above a specified reference level calculated over consecutive unit intervals. It is applicable to data and data strobe waveforms.

DDRAUS (area under signal) is the total area of the signal below a specified reference level. This measurement is made across the entire record.

DDRAUSPERTCK (area under signal for tCK events) is the total area of the signal below a specified reference level calculated over consecutive tCK intervals. It is applicable to clock and address/command waveforms.

DDRAUSPERUI (area under signal for UI events) is the total area of the signal below a specified reference level calculated over consecutive unit intervals. It is applicable to data and data strobe waveforms.

DDRHOLDDIFF (hold difference) is the elapsed time between the specified edge of a single-ended clock waveform and the specified edge of a differential data waveform. The measurement uses the closest respective waveform edges that fall within the range limits. This measurement is made across the entire record.

DDRSETUPDIFF (setup difference) is the elapsed time between the specified edge of a single-ended clock waveform and when the specified edge of a differential data waveform crosses a specified level. The measurement uses the closest respective waveform edges that fall within the range limits. This measurement is made across the entire record.

DDRTCHABS (absolute high pulse width) is the absolute value of the high pulse width as measured from one rising edge to the next falling edge.

DDRTCHAVERAGE (average high pulse width) is the average value of the high pulse width as measured from one rising edge to the next falling edge, across 200 consecutive cycles. This measurement is made across the entire record.

DDRTCKAVERAGE (average clock period) is the average clock period calculated from rising edge to rising edge, across 200 consecutive cycles. This measurement is made across the entire record.

DDRTCLABS (absolute low pulse width) is the absolute value of the low pulse width as measured from a falling edge to the next rising edge. This measurement is made across the entire record.

DDRTCLAVERAGE (average low pulse width) is the average value of the low pulse width as measured from one falling edge to the next rising edge, across 200 consecutive cycles.

DDRTERRMN (cumulative error) is the cumulative error across multiple consecutive defined cycles from $t_{CK}(avg)$.

DDRTERRN (cumulative error) is the cumulative error across specified consecutive cycles from $t_{CK}(avg)$. In other words, this measures the time difference between the sum of the clock period from a 200 cycle window and n times $t_{CK}(avg)$.

DDRTJITCC (cycle to cycle jitter period) is the absolute difference in clock period between two consecutive clock cycles. This measurement is made across the entire record.

DDRTJITDUTY (half period jitter) is the largest elapsed time between t_{CH} and $t_{CH}(avg)$, and t_{CL} and $t_{CL}(avg)$, over 200 consecutive cycles.

DDRTJITPER (clock period jitter) is the largest deviation of any t_{CK} signal from $t_{CK}(avg)$. This measurement is made across the entire record.

DDRTPST (read/write burst postamble) is the width of the Read or Write burst postamble, measured from the last falling edge of the mid reference level to the start of an undriven state. This measurement is made across the entire record.

DDRTRPRE (read burst preamble) is the width of the Read burst preamble, measured from exiting tristate levels to the first driving edge of the differential strobe. This measurement is made across the entire record.

DDRTWPRE is the width of the Write burst preamble, measured from exiting tristate levels to the first driving edge of the differential strobe. This measurement is made across the entire record.

DDRVIXAC is the differential input cross-point voltage measured from the true state transition (and it's compliment) to a specified reference level, measured on a single-ended signal.

DDRTDQSCK is the strobe output access time, measured between the rising edge of the clock and before or after the differential strobe Read preamble time. Signal edges are determined by the mid-ref threshold level settings.

DElay is the time between the specified Mid reference level (RM) crossing on one source to a specified Mid reference level (RM) crossing on a second source. This measurement is made on the first occurrence in the record.

DJ (deterministic jitter) is the peak-to-peak amplitude of all timing errors that exhibit deterministic behavior. This measurement is made across the entire record.

DJDIRAC (dual-dirac deterministic jitter) is deterministic jitter based on a simplifying assumption that the histogram of all deterministic jitter can be modeled as a pair of equal-magnitude Dirac functions. This measurement is made across the entire record.

EYEHIGH (Eye High) is the amplitude of a high (1) bit measured at a user specified location within the recovered unit interval. This measurement is made on each high bit in the record.

EYELow (Eye Low) is the amplitude of a low (0) bit measured at a user specified location within the recovered unit interval. This measurement is made on each low bit in the record.

FALLSLEWRATE (Falling Slew Rate) is the rate of change in voltage as an edge transitions from the Top reference level (RT) to the Bottom reference level (RB). This measurement is made on each cycle in the record.

FALLTIME (Fall Time) is the time required for an edge to fall from the Top reference level (RT) to the Base reference level (RB). This measurement is made on each cycle in the record.

FREQUENCY is the reciprocal of Period. This measurement is made on each cycle in the record.

F2 is the peak-to-peak amplitude of the periodic jitter occurring at a rate of Fb (data rate) divided by 2. This measurement is made across the entire record.

F4 is the peak-to-peak amplitude of the periodic jitter occurring at a rate of Fb (data rate) divided by 4. This measurement is made across the entire record.

F8 is the peak-to-peak amplitude of the periodic jitter occurring at a rate of Fb (data rate) divided by 8. This measurement is made across the entire record.

HIGH (Eye High) is the amplitude of a high (1) bit measured at a user specified location within the recovered unit interval. This measurement is made on each high bit in the record.

HEIGHT (Eye Height) is the minimum vertical eye opening at the center of the recovered unit interval. This measurement is made across the entire record.

HEIGHTBER (Eye Height@BER) is the predicted vertical eye opening that will be violated with a probability equal to the bit error rate. This measurement is made across the entire record.

HIGH

HIGHTIME (High Time) is the time the signal remains above the Top reference level (RT). This measurement is made on each cycle in the record.

HOLD (Hold Time) is the time between the specified Mid reference level crossing (RM) on the Clock source to the closest specified Mid reference level (RM) crossing on the Data source. This measurement is made on each specified Clock edge in the record.

JITTERSUMMARY (Jitter Summary) is a group consisting of the following measurements: TIE, TJ@BER, Eye Width@BER, Eye Height@BER, RJ- $\delta\delta$, DJ- $\delta\delta$, PJ, DDJ, DcD, F/2, F/4, F/8.

J2 is the total jitter at a bit error rate of $2.5e-3$ (TJ@ $2.5e-3$). This measurement is made across the entire record.

J9 is the total jitter at a bit error rate of $2.5e-10$ (TJ@ $2.5e-10$). This measurement is made across the entire record.

LOW (Eye Low) is the amplitude of a low (0) bit measured at a user specified location within the recovered unit interval. This measurement is made on each low bit in the record.

LOWTIME (Low Time) is the time the signal remains below the Base reference level (RB). This measurement is made on each cycle in the record.

MAXimum is the maximum data point. This measurement can be made across the entire record, or on each cycle in the record.

MEAN is the arithmetic mean of the data points. This measurement can be made across the entire record, or on each cycle in the record.

MINimum is the minimum data point. This measurement can be made across the entire record, or on each cycle in the record.

NDuty (Negative Duty Cycle) is the ratio of the Negative Pulse Width to the Period. This measurement is made on each cycle in the record.

$$\text{Negative Duty Cycle} = (\text{Negative Width}) / \text{Period} \times 100\%$$

NPERIOD (Duration N-Periods) is the time required to complete N cycles. A cycle is the time between two adjacent (same direction) crossings of the Mid reference level (RM). This measurement is made on each cycle in the record.

NPJ (non-periodic jitter) is the portion of the BUJ (bounded uncorrelated jitter) that is random. BUJ excludes DDJ, DCD and RJ. This measurement is made across the entire record.

NOvershoot (Negative Overshoot) is the difference between Minimum and Base, divided by the Amplitude. This measurement can be made across the entire record, or on each cycle in the record.

$$\text{Negative Overshoot} = (\text{Base} - \text{Minimum}) / \text{Amplitude} \times 100\%$$

NWidth (Negative Pulse Width) is the time the signal remains below the Mid reference level (RM). This measurement is made on each cycle in the record.

PDUTY (Positive Duty Cycle) is the ratio of the Positive Pulse Width to the Period. This measurement is made on each cycle in the record.

$$\text{Positive Duty Cycle} = (\text{Positive Width}) / \text{Period} \times 100\%$$

PERIOD is the time required to complete a cycle. A cycle is the time between two adjacent (same direction) crossings of the Mid reference level (RM). This measurement is made on each cycle in the record.

PHASE is the ratio of the Skew between two sources to the Period of the first source. This measurement is made on each cycle in the record.

PHASENOISE (Phase Noise) is the RMS magnitude of all integrated jitter falling within a user specified offset range of the fundamental clock frequency. This measurement is made across the entire record.

PJ (periodic jitter) is the peak-to-peak amplitude of the uncorrelated sinusoidal components of the deterministic jitter. This measurement is made across the entire record.

PK2Pk (Peak-to-peak) is the difference between Maximum and Minimum. This measurement can be made across the entire record, or on each cycle in the record.

POVERSHOOT (Positive Overshoot) is the difference between Maximum and Top, divided by the Amplitude. This measurement can be made across the entire record, or on each cycle in the record.

$$\text{Positive Overshoot} = (\text{Maximum} - \text{Top}) / \text{Amplitude} \times 100\%$$

PWIDTH (Positive Pulse Width) is the time the signal remains above the Mid reference level (RM). This measurement is made on each cycle in the record.

QFACTOR (Q-Factor) is the ratio of the vertical eye opening to RMS vertical noise measured at a user specified location within the recovered unit interval. This measurement is made across the entire record.

RISESLEWRATE (Rising Slew Rate) is the rate of change in voltage as an edge transitions from the Base reference level (RB) to the Top reference level (RT). This measurement is made on each cycle in the record.

RISETIME Rise Time is the time required for an edge to rise from the Base reference level (RB) to the Top reference level (RT). This measurement is made on each cycle in the record.

RJ (random jitter) is the RMS magnitude of all random timing errors following a Gaussian distribution. This measurement is made across the entire record.

RJDIRAC (dual-dirac random jitter) is random jitter based on a simplifying assumption that the histogram of all deterministic jitter can be modeled as a pair of equal-magnitude Dirac functions. This measurement is made across the entire record.

RMS is the true Root Mean Square of the data points. This measurement can be made across the entire record, or on each cycle in the record.

SRJ (sub-rate jitter) is the composite jitter due to periodic components at 1/2, 1/4 and 1/8 of the data rate. This measurement is made across the entire record.

SSCFREQDEV (SSC Frequency Deviation) is the spread spectrum clock frequency deviation. This measurement enables a time trend plot of the spread spectrum clock modulation profile. This measurement is made on each cycle in the record.

SSCMODRATE (SSC Modulation Rate) is the modulating frequency of a spread spectrum clock. This measurement is made on each cycle in the record.

SETUP (Setup Time) is the time between the specified Mid reference level (RM) crossing on the Data source to the closest specified Mid reference level (RM) crossing on the Clock source. This measurement is made on each specified Clock edge in the record.

SKEW Skew is the time between the specified Mid reference level (RM) crossing on one source to the following specified Mid reference level (RM) crossing on a second source. This measurement is made on each cycle in the record.

TIE (time interval error) is the difference, in time, between an edge in the source waveform and the corresponding edge in a recovered reference clock. This measurement is made on each edge in the waveform.

TIMEOUTSIDELEVEL Time Outside Level is the time the signal remains above the Top reference level (RT) and/or below the Base reference level (RB). This measurement is made on each occurrence in the record.

TJBER (total jitter at a specified bit error rate) is the predicted peak-to-peak amplitude of jitter that will only be exceeded with a probability equal to the bit error rate. This measurement is made across the entire record.

TNTRATIO T/nT Ratio is the ratio of a non-transition bit voltage (2nd and subsequent bit voltage after a transition) to its nearest preceding transition bit voltage (1st bit voltage after the transition). Bit voltages are measured at the

interpolated midpoint of the recovered unit interval. This measurement is made on each non-transition bit in the record.

TOP is the most common data value above the midpoint of the waveform. This measurement can be made across the entire record, or on each cycle in the record.

UNITINTERVAL (Unit Interval) is the time difference between two successive bits. This measurement is made on each bit in the record.

VDIFFXOVR (Differential Crossover) is the voltage level of a differential signal pair at the crossover points. This measurement is made at each crossover point in the record.

WIDTH (Eye Width) is the minimum horizontal eye opening at the user specified reference level. This measurement is made across the entire record.

WIDTHBER (Eye Width@BER) is the predicted horizontal eye opening that will be violated with a probability equal to the bit error rate. This measurement is made across the entire record.

Examples `MEASUREMENT:ADDMEAS FREQUENCY` adds a frequency measurement.

MEASUrement:ADDNew (No Query Form)

This command adds the specified measurement.

Group Measurement

Syntax `MEASUrement:ADDNew "QString"`

Arguments "QString" is the measurement to add. The argument is of the form "MEAS<NR1>" where $NR1 \geq 1$.

Examples `MEASUrement:ADDNew "MEAS11"` adds measurement 11.

MEASUrement:ANNOtate

This command sets or queries the annotation state for measurements.

Group Measurement

Syntax	<code>MEASUREMENT:ANNOtate {OFF AUTO}</code> <code>MEASUREMENT:ANNOtate?</code>
Arguments	OFF turns off measurement annotations. AUTO turns on visible measurement annotations.
Examples	<code>MEASUREMENT:ANNOtATE OFF</code> turns off measurement annotations. <code>MEASUREMENT:ANNOtATE?</code> might return <code>:MEASUREMENT:ANNOtATION:STATE OFF</code> , indicating that no measurement annotations are active.

MEASUREMENT:AUTOset (No Query Form)

This command performs an analysis jitter autoset.

Group	Measurement
Syntax	<code>MEASUREMENT:AUTOset EXECute</code>
Examples	<code>MEASUREMENT:AUTOSET EXECute</code> performs an analysis jitter autoset.

MEASUREMENT:CH<x>:REFLevels:ABSolute:FALLHigh

This command sets or queries the value used as the high reference level of the falling edge when the source ref level method is set to absolute. The channel number is specified by x.

Group	Measurement
Syntax	<code>MEASUREMENT:CH<x>:REFLevels:ABSolute:FALLHigh <NR3></code> <code>MEASUREMENT:CH<x>:REFLevels:ABSolute:FALLHigh?</code>
Arguments	<NR3> is the high reference level, and is the zero percent level when <code>MEASUREMENT:IMMed:REFLevel:METHod</code> is set to Absolute.
Examples	<code>MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:FALLHIGH 1.5</code> sets the high reference level of the falling edge to 1.5 V.

MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:FALLHIGH? might return
:MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:FALLHIGH 1.0000 indicating
that the high reference level of the falling edge is 1.0 V.

MEASUREMENT:CH<x>:REFLEVELS:ABSOLUTE:FALLLOW

This command sets or queries the value used as the low reference level of the falling edge when the source ref level method is set to absolute. The channel number is specified by x.

Group	Measurement
Syntax	MEASUREMENT:CH<x>:REFLEVELS:ABSOLUTE:FALLLOW <NR3> MEASUREMENT:CH<x>:REFLEVELS:ABSOLUTE:FALLLOW?
Arguments	<NR3> is the high reference level, and is the zero percent level when MEASUREMENT:IMMED:REFLEVEL:METHOD is set to Absolute.
Examples	MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:FALLLOW 1.5 sets the low reference level of the falling edge to 1.5 V. MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:FALLLOW? might return :MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:FALLLOW 1.5000 indicating that the low reference level of the falling edge is 1.5 V.

MEASUREMENT:CH<x>:REFLEVELS:ABSOLUTE:FALLMID

This command sets or queries the value used as the mid reference level of the falling edge when the source ref level method is set to absolute. The channel number is specified by x.

Group	Measurement
Syntax	MEASUREMENT:CH<x>:REFLEVELS:ABSOLUTE:FALLMID <NR3> MEASUREMENT:CH<x>:REFLEVELS:ABSOLUTE:FALLMID?
Arguments	<NR3> is the mid reference level used to calculate the mid reference level when the measurement's Ref level method is set to Absolute.

Examples `MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:FALLMID 0` sets the mid reference level of the falling edge to 0.0 V.

`MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:FALLMID?` might return
`:MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:FALLMID 0.0E+0` indicating that the mid reference level of the falling edge is 0.0 V.

MEASUREMENT:CH<x>:REFLEVELS:ABSOLUTE:HYSTERESIS

This command sets or queries the value of the hysteresis of the reference level when the source ref level method is set to absolute. The channel number is specified by x.

Group Measurement

Syntax `MEASUREMENT:CH<x>:REFLEVELS:ABSOLUTE:HYSTERESIS <NR3>`
`MEASUREMENT:CH<x>:REFLEVELS:ABSOLUTE:HYSTERESIS?`

Arguments <NR3> is the hysteresis value used for autoset.

Examples `MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:HYSTERESIS 30.0000E-3` sets the reference hysteresis level to 30 mV.

`:MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:HYSTERESIS?` might return
`:MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:HYSTERESIS 30.0000E-3` indicating that reference hysteresis level is set to 30 mV.

MEASUREMENT:CH<x>:REFLEVELS:ABSOLUTE:RISEHIGH

This command sets or queries the value used as the high reference level of the rising edge when the source ref level method is set to absolute. The channel number is specified by x.

Group Measurement

Syntax `MEASUREMENT:CH<x>:REFLEVELS:ABSOLUTE:RISEHIGH <NR3>`
`MEASUREMENT:CH<x>:REFLEVELS:ABSOLUTE:RISEHIGH?`

Arguments <NR3> is the high reference level of the rising edge when the source ref level method is set to absolute.

Examples `MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:RISEHIGH 1.5` sets the high reference level of the rising edge to 1.5 V.

`MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:RISEHIGH?` might return
`:MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:RISEHIGH 1.0000` indicating
that the high reference level of the rising edge is 1.0 V.

MEASUrement:CH<x>:REFLeveLs:ABSolute:RISELow

This command sets or queries the value used as the low reference level of the rising edge when the source ref level method is set to absolute. The channel number is specified by x.

Group Measurement

Syntax `MEASUrement:CH<x>:REFLeveLs:ABSolute:RISELow <NR3>`
`MEASUrement:CH<x>:REFLeveLs:ABSolute:RISELow?`

Arguments <NR3> is the low reference level of the rising edge when the source ref level method is set to absolute.

Examples `MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:RISELOW 1.5` sets the low reference level of the rising edge to 1.5 V.

`MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:RISELOW?` might return
`:MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:RISELOW 1.0000` indicating
that the low reference level of the rising edge is 1.0 V.

MEASUrement:CH<x>:REFLeveLs:ABSolute:RISEMid

This command sets or queries the value used as the mid reference level of the rising edge when the source ref level method is set to absolute. The channel number is specified by x.

Group Measurement

Syntax `MEASUrement:CH<x>:REFLeveLs:ABSolute:RISEMid <NR3>`
`MEASUrement:CH<x>:REFLeveLs:ABSolute:RISEMid?`

Arguments <NR3> is the mid reference level of the rising edge when the source ref level method is set to absolute.

Examples `MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:RISEMID 0` sets the mid reference level of the rising edge to 0.0 V.

`MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:RISEMID?` might return
`:MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:RISEMID 0.0E+0` indicating that the mid reference level of the rising edge is 0.0 V.

MEASUREMENT:CH<x>:REFLEVELS:ABSOLUTE:TYPE

This command sets or queries the reference level type for the source. The channel number is specified by x.

Group Measurement

Syntax `MEASUREMENT:CH<x>:REFLEVELS:ABSOLUTE:TYPE {SAME|UNIQUE}`
`MEASUREMENT:CH<x>:REFLEVELS:ABSOLUTE:TYPE?`

Arguments `SAME` specifies that the absolute reference levels for the specified measurement channel are the same.

`UNIQUE` specifies that the absolute reference levels for the specified measurement channel are not the same.

Examples `MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:TYPE SAME` set the reference levels to be the same.

`MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:TYPE?` might return
`:MEASUREMENT:CH2:REFLEVELS:ABSOLUTE:TYPE SAME` indicating that the reference levels for absolute measurements are the same.

MEASUREMENT:CH<x>:REFLEVELS:BASETOP

This command sets or queries the method used to calculate the TOP and BASE, used to calculate reference levels for the measurement. The channel number is specified by x.

Group Measurement

Syntax `MEASUREMENT:CH<x>:REFLEVELS:BASETOP`
`{AUTO|MINMAX|MEANHISTOGRAM|MODEHISTOGRAM|EYEHISTOGRAM}`
`MEASUREMENT:CH<x>:REFLEVELS:BASETOP?`

Arguments	AUTO automatically chooses a reference level method.
	MINMax specifies that reference levels are relative to the measurement MIN and MAX.
	MEANHistogram specifies that reference levels are relative to the histogram mean BASE and TOP.
	MODEHistogram specifies that reference levels are relative to the histogram mode BASE and TOP.
	EYEHistogram specifies that reference levels are relative to the eye histogram BASE and TOP.
Examples	MEASUREMENT:CH2:REFLEVELS:BASETOP MINMAX specifies that reference levels are relative to the measurement MIN and MAX.
	MEASUREMENT:CH2:REFLEVELS:BASETOP? might return :MEASUREMENT:CH2:REFLEVELS:BASETOP AUTO indicating the reference levels are chosen automatically.

MEASUREMENT:CH<x>:REFLEVELS:METHOd

This command sets or queries the method used to calculate reference levels for the measurement. The channel number is specified by x.

Group	Measurement
Syntax	MEASUREMENT:CH<x>:REFLEVELS:METHOd {PERCENT ABSOLUTE} MEASUREMENT:CH<x>:REFLEVELS:METHOd?
Arguments	PERCENT specifies percent reference level units.
	ABSOLUTE specifies absolute reference level units.
Examples	MEASUREMENT:CH2:REFLEVELS:METHOd PERCENT sets reference levels to be calculated in percent.
	MEASUREMENT:CH2:REFLEVELS:METHOd? might return :MEASUREMENT:CH2:REFLEVELS:METHOd PERCENT indicating the reference levels are calculated in percent.

MEASUREMENT:CH<x>:REFLEVELS:PERCENT:FALLHIGH

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the high reference level of the falling edge when the source ref level method is set to percent. The channel number is specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:CH<x>:REFLEVELS:PERCENT:FALLHIGH <NR3> MEASUREMENT:CH<x>:REFLEVELS:PERCENT:FALLHIGH?</pre>
Arguments	<NR3> is the percentage (where 100% is equal to TOP) used to calculate the high reference level when the measurement's Ref level method is set to Percent.
Examples	MEASUREMENT:CH2:REFLEVELS:PERCENT:FALLHIGH 90.0000 sets the high reference level for the falling edge to 90%. MEASUREMENT:CH2:REFLEVELS:PERCENT:FALLHIGH? might return :MEASUREMENT:CH2:REFLEVELS:PERCENT:FALLHIGH 90.0000 indicating the high reference level for the falling edge is set to 90%.

MEASUREMENT:CH<x>:REFLEVELS:PERCENT:FALLLOW

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the low reference level of the falling edge when the source ref level method is set to percent. The channel number is specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:CH<x>:REFLEVELS:PERCENT:FALLLOW <NR3> MEASUREMENT:CH<x>:REFLEVELS:PERCENT:FALLLOW?</pre>
Arguments	<NR3> is the percentage (where 100% is equal to TOP) used to calculate the low reference level when the measurement Ref level method is set to Percent.
Examples	MEASUREMENT:CH2:REFLEVELS:PERCENT:FALLLOW 10.0000 sets the reference levels for the falling edge to 10%.

MEASUREMENT:CH2:REFLEVELS:PERCENT:FALLLOW? might return
:MEASUREMENT:CH2:REFLEVELS:PERCENT:FALLLOW 10.0000 indicating the
reference levels for the falling edge is set to 10%.

MEASUREMENT:CH<x>:REFLEVELS:PERCENT:FALLMID

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the falling edge when the source ref level method is set to percent. The channel number is specified by x.

Group	Measurement
Syntax	MEASUREMENT:CH<x>:REFLEVELS:PERCENT:FALLMID <NR3> MEASUREMENT:CH<x>:REFLEVELS:PERCENT:FALLMID?
Arguments	<NR3> is the percentage (where 50% is equal to MID) used to calculate the mid reference level when the measurement's Ref level method is set to Percent.
Examples	MEASUREMENT:CH2:REFLEVELS:PERCENT:FALLMID 50.0000 sets the MID reference level for the falling edge to 50%. MEASUREMENT:CH2:REFLEVELS:PERCENT:FALLMID? might return :MEASUREMENT:CH2:REFLEVELS:PERCENT:FALLMID 50.0000 indicating the MID reference level for the falling edge is set to 50%.

MEASUREMENT:CH<x>:REFLEVELS:PERCENT:HYSTERESIS

This command sets or queries the percentage (where 100% is equal to MAX and 0% is equal to MIN) used to calculate the hysteresis of the reference level when the measurement ref level method is set to percent. The channel number is specified by x.

Group	Measurement
Syntax	MEASUREMENT:CH<x>:REFLEVELS:PERCENT:HYSTERESIS <NR3> MEASUREMENT:CH<x>:REFLEVELS:PERCENT:HYSTERESIS?
Arguments	<NR3> is the hysteresis value used for the autoset.

Examples `MEASUREMENT:CH2:REFLEVELS:PERCENT:HYSTERESIS 5.0000` sets the reference level hysteresis to 5.0 mV.

`MEASUREMENT:CH2:REFLEVELS:PERCENT:HYSTERESIS?` might return
`:MEASUREMENT:CH2:REFLEVELS:PERCENT:HYSTERESIS 5.0000` indicating the reference level hysteresis is set to 5.0 mV.

MEASUREMENT:CH<x>:REFLEVELS:PERCENT:RISEHigh

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the high reference level of the rising edge when the measurement ref level method is set to percent. The channel number is specified by x.

Group Measurement

Syntax `MEASUREMENT:CH<x>:REFLEVELS:PERCENT:RISEHigh <NR3>`
`MEASUREMENT:CH<x>:REFLEVELS:PERCENT:RISEHigh?`

Arguments <NR3> is the percentage (where 100% is equal to TOP) used to calculate the high reference level when the measurement's Ref level method is set to Percent.

Examples `MEASUREMENT:CH2:REFLEVELS:PERCENT:RISEHIGH 90.0000` sets the high reference level for the rising edge to 90%.

`MEASUREMENT:CH2:REFLEVELS:PERCENT:RISEHIGH?` might return
`:MEASUREMENT:CH2:REFLEVELS:PERCENT:RISEHIGH 90.0000` indicating the high reference level for the rising edge is set to 90%.

MEASUREMENT:CH<x>:REFLEVELS:PERCENT:RISELow

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the low reference level of the rising edge when the measurement ref level method is set to percent. The channel number is specified by x.

Group Measurement

Syntax `MEASUREMENT:CH<x>:REFLEVELS:PERCENT:RISELow <NR3>`
`MEASUREMENT:CH<x>:REFLEVELS:PERCENT:RISELow?`

Arguments	<NR3> is the percentage (where 100% is equal to TOP) used to calculate the mid reference level when the measurement's Ref level method is set to Percent.
Examples	MEASUREMENT:CH2:REFLEVELS:PERCENT:RISELOW 10.0000 sets the reference levels for the rising edge to 10%. MEASUREMENT:CH2:REFLEVELS:PERCENT:RISELOW? might return :MEASUREMENT:CH2:REFLEVELS:PERCENT:RISELOW 10.0000 indicating the reference levels for the rising edge is set to 10%.

MEASUREMENT:CH<x>:REFLEVELS:PERCENT:RISEMid

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the rising edge when the measurement ref level method is set to percent. The channel number is specified by x.

Group	Measurement
Syntax	MEASUREMENT:CH<x>:REFLEVELS:PERCENT:RISEMid <NR3> MEASUREMENT:CH<x>:REFLEVELS:PERCENT:RISEMid?
Arguments	<NR3> is the percentage (where 50% is equal to MID) used to calculate the mid reference level when the measurement's Ref level method is set to Percent.
Examples	MEASUREMENT:CH2:REFLEVELS:PERCENT:RISEMid 50.0000 sets the MID reference level for the rising edge to 50%. MEASUREMENT:CH2:REFLEVELS:PERCENT:RISEMid? might return :MEASUREMENT:CH2:REFLEVELS:PERCENT:RISEMid 50.0000 indicating the MID reference level for the rising edge is set to 50%.

MEASUREMENT:CH<x>:REFLEVELS:PERCENT:TYPE

This command specifies or queries the reference level percent type for the measurement. The channel number is specified by x.

Group	Measurement
Syntax	MEASUREMENT:CH<x>:REFLEVELS:PERCENT:TYPE {TENNINety TWENTyeighty CUSTOM}

MEASUREMENT:CH<x>:REFLEVELS:PERCENT:TYPE?

Arguments	TENNINETY specifies reference levels at the 10 and 90% levels.
	TWENTYEIGHTY specifies reference levels at the 20 and 80% levels.
	CUSTOM specifies custom reference levels.
Examples	MEASUREMENT:CH2:REFLEVELS:PERCENT:TYPE TENNINETY sets the reference levels to the 10 and 90% levels.
	MEASUREMENT:CH2:REFLEVELS:PERCENT:TYPE? might return :MEASUREMENT:CH2:REFLEVELS:PERCENT:TYPE TWENTYEIGHTY indicating the reference levels are set to the 20 and 80% levels.

MEASUREMENT:CLOCKRECOVERY:ADVANCED:METHOd

This command sets or queries the global advanced clock recovery method. This will affect measurements whose :MEASUREMENT:MEAS<x>:CLOCKRECOVERY:GLOBAl flag is set to 1.

Group	Measurement
Syntax	MEASUREMENT:CLOCKRECOVERY:ADVANCED:METHOd {NONE NOMInAl PATTern} MEASUREMENT:CLOCKRECOVERY:ADVANCED:METHOd?
Arguments	NONE sets to use no advanced CRD method.
	NOMInAl sets the advanced CRD method to Nominal Data Rate.
	PATTern sets the advanced CDR method to use a Known Data Pattern.
Examples	MEASUREMENT:CLOCKRECOVERY:ADVANCED:METHOd NOMINAL sets the CDR method to use a Known Data Pattern.
	MEASUREMENT:CLOCKRECOVERY:ADVANCED:METHOd? might return :MEASUREMENT:CLOCKRECOVERY:ADVANCED:METHOd NONE indicating that no advanced CRD method will be used.

MEASUrement:CLOCKRecovery:CLOCKFrequency

This command sets or queries the global clock frequency used when fixed constant clock recovery is used for the measurement. This will affect measurements whose :MEASUrement:MEAS<x>:CLOCKRecovery:GLOBal flag is set to 1.

Group	Measurement
Syntax	MEASUrement:CLOCKRecovery:CLOCKFrequency <NR3> MEASUrement:CLOCKRecovery:CLOCKFrequency?
Arguments	<NR3> is the global clock frequency used with Constant Clock - Fixed clock recovery method.
Examples	MEASUREMENT:CLOCKRECOVERY:CLOCKFREQUENCY 2.0E+9 sets the clock frequency to 2.0 GHz. :MEASUREMENT:CLOCKRECOVERY:CLOCKFREQUENCY? might return :MEASUREMENT:CLOCKRECOVERY:CLOCKFREQUENCY 2.5000E+9 indicating the clock frequency is set to 2.5 GHz.

MEASUrement:CLOCKRecovery:CLOCKMultiplier

This command sets or queries the global clock multiplier used when explicit clock recovery is used for the measurement. This will affect measurements whose :MEASUrement:MEAS<x>:CLOCKRecovery:GLOBal flag is set to 1.

Group	Measurement
Syntax	MEASUrement:CLOCKRecovery:CLOCKMultiplier <NR3> MEASUrement:CLOCKRecovery:CLOCKMultiplier?
Arguments	<NR3> is the global clock multiplier.
Examples	MEASUREMENT:CLOCKRECOVERY:CLOCKMULTIPLIER 1.000 sets the clock multiplier to 1.000. MEASUREMENT:CLOCKRECOVERY:CLOCKMULTIPLIER? might return :MEASUREMENT:CLOCKRECOVERY:CLOCKMULTIPLIER 1.0000 indicating the clock multiplier is set to 1.0000.

MEASUrement:CLOCKRecovery:CONSTCLOCKMODE

This command sets or queries the global constant clock mode used when constant clock recovery is used for the measurement. This will affect measurements whose :MEASUrement:MEAS<x>:CLOCKRecovery:GLOBal flag is set to 1.

Group	Measurement
Syntax	MEASUrement:CLOCKRecovery:CONSTCLOCKMODE {MEAN MEDIan FIXed} MEASUrement:CLOCKRecovery:CONSTCLOCKMODE?
Arguments	MEAN sets the constant clock mode to MEAN. MEDIan sets the constant clock mode to MEDIan. FIXed sets the constant clock mode to FIXed.
Examples	MEASUREMENT:CLOCKRECOVERY:CONSTCLOCKMODE MEAN sets the constant clock mode to MEAN. MEASUREMENT:CLOCKRECOVERY:CONSTCLOCKMODE? might return :MEASUREMENT:CLOCKRECOVERY:CONSTCLOCKMODE MEAN indicating the constant clock mode is set to MEAN.

MEASUrement:CLOCKRecovery:DAMPing

This command sets or queries the global damping value used when PLL clock recovery is used for the measurement. This will affect measurements whose :MEASUrement:MEAS<x>:CLOCKRecovery:GLOBal flag is set to 1.

Group	Measurement
Syntax	MEASUrement:CLOCKRecovery:DAMPing <NR3> MEASUrement:CLOCKRecovery:DAMPing?
Arguments	<NR3> is the global clock recovery damping value.
Examples	MEASUREMENT:CLOCKRECOVERY:DAMPING :MEASUREMENT:CLOCKRECOVERY:DAMPING? might return :MEASUREMENT:CLOCKRECOVERY:DAMPING 700.0000E-3 indicating the damping value is set to

MEASUrement:CLOCKRecovery:DATAPath

This command sets or queries the global file containing the data pattern used when known data pattern clock recovery is used for the measurement. This will affect measurements whose :MEASUrement:MEAS<x>:CLOCKRecovery:GLOBal flag is set to 1.

Group	Measurement
Syntax	MEASUrement:CLOCKRecovery:DATAPath <QString> MEASUrement:CLOCKRecovery:DATAPath?
Arguments	<QString> is the path and filename, in quotes, of the file containing the data pattern.
Examples	MEASUREMENT:CLOCKRECOVERY:DATAPATH "C:/E:" sets the data path to C:/E:. MEASUREMENT:CLOCKRECOVERY:DATAPATH? might return :MEASUREMENT:CLOCKRECOVERY:DATAPATH "C:/E".

MEASUrement:CLOCKRecovery:DATARate

This command sets or queries the global nominal data bit rate used when nominal data rate clock recovery is used for the measurement. This will affect measurements whose :MEASUrement:MEAS<x>:CLOCKRecovery:GLOBal flag is set to 1.

Group	Measurement
Syntax	MEASUrement:CLOCKRecovery:DATARate <NR3> MEASUrement:CLOCKRecovery:DATARate?
Arguments	<NR3> is the global value for the Nominal data rate.
Examples	MEASUREMENT:CLOCKRECOVERY:DATARATE 2.0e+9 sets the data rate for clock recovery to 2.0 GHz. MEASUREMENT:CLOCKRECOVERY:DATARATE? might return :MEASUREMENT:CLOCKRECOVERY:DATARATE 2.5000E+9 indicating the data rate is 2.5 GHz.

MEASUrement:CLOCKRecovery:EXPLICITCLOCKMODE

This command sets or queries the global explicit clock mode used when explicit clock recovery is used for the measurement. This will affect measurements whose :MEASUrement:MEAS<x>:CLOCKRecovery:GLOBal flag is set to 1.

Group	Measurement
Syntax	MEASUrement:CLOCKRecovery:EXPLICITCLOCKMODE {EDGE PLL} MEASUrement:CLOCKRecovery:EXPLICITCLOCKMODE?
Arguments	EDGE sets the clock mode to clock edge. PLL sets the clock mode to phase locked loop.
Examples	MEASUREMENT:CLOCKRECOVERY:EXPLICITCLOCKMODE EDGE sets the clock mode to edge. MEASUREMENT:CLOCKRECOVERY:EXPLICITCLOCKMODE? might return :MEASUREMENT:CLOCKRECOVERY:EXPLICITCLOCKMODE EDGE indicating the clock mode is edge.

MEASUrement:CLOCKRecovery:JTFBandwidth

This command sets or queries the global JTF bandwidth used when PLL clock recovery is used for the measurement. This will affect measurements whose :MEASUrement:MEAS<x>:CLOCKRecovery:GLOBal flag is set to 1.

Group	Measurement
Syntax	MEASUrement:CLOCKRecovery:JTFBandwidth <NR3> MEASUrement:CLOCKRecovery:JTFBandwidth?
Arguments	<NR3> is the global clock recovery JTF bandwidth.
Examples	MEASUREMENT:CLOCKRECOVERY:JTFBANDWIDTH 2.0e6 sets the bandwidth to 2.0 MHz. MEASUREMENT:CLOCKRECOVERY:JTFBANDWIDTH? might return :MEASUREMENT:CLOCKRECOVERY:JTFBANDWIDTH 1.0000E+6 indicating the bandwidth is 1.0 MHz.

MEASUrement:CLOCKRecovery:LOOPBandwidth

This command sets or queries the global loop bandwidth used when PLL clock recovery is used for the measurement. This will affect measurements whose :MEASUrement:MEAS<x>:CLOCKRecovery:GLOBal flag is set to 1.

Group	Measurement
Syntax	MEASUrement:CLOCKRecovery:LOOPBandwidth <NR3> MEASUrement:CLOCKRecovery:LOOPBandwidth?
Arguments	<NR3> is the global loop bandwidth.
Examples	MEASUREMENT:CLOCKRECOVERY:LOOPBANDWIDTH 2.0e6 sets the loop bandwidth to 2.0 MHz. MEASUREMENT:CLOCKRECOVERY:LOOPBANDWIDTH? might return :MEASUREMENT:CLOCKRECOVERY:LOOPBANDWIDTH 1.0000E+6 indicating the loop bandwidth is 1.0 MHz.

MEASUrement:CLOCKRecovery:MEANAUTOCalculate

This command sets or queries how often the clock is calculated when constant clock recovery is used for the measurement. This will affect measurements whose :MEASUrement:MEAS<x>:CLOCKRecovery:GLOBal flag is set to 1.

Group	Measurement
Syntax	MEASUrement:CLOCKRecovery:MEANAUTOCalculate {FIRST EVERY} MEASUrement:CLOCKRecovery:MEANAUTOCalculate?
Arguments	FIRST calculates the clock on the first acquisition. EVERY calculates the clock on every acquisition.
Examples	MEASUREMENT:CLOCKRECOVERY:MEANAUTOCALCULATE EVERY calculates the clock on every acquisition. MEASUREMENT:CLOCKRECOVERY:MEANAUTOCALCULATE? might return :MEASUREMENT:CLOCKRECOVERY:MEANAUTOCALCULATE FIRST indicating the clock is calculated on the first acquisition.

MEASUrement:CLOCKRecovery:METHod

This command sets or queries the global clock recovery method for the measurement. This will affect measurements whose :MEASUrement:MEAS<x>:CLOCKRecovery:GLOBal flag is set to 1.

Group	Measurement
Syntax	<pre>MEASUrement:CLOCKRecovery:METHod {PLL CONSTANTCLOCK EXPLICITCLOCK} MEASUrement:CLOCKRecovery:METHod?</pre>
Arguments	PLL specifies using the phase locked loop. CONSTANTCLOCK specifies using a constant clock. EXPLICITCLOCK specifies using an explicit clock.
Examples	MEASUREMENT:CLOCKRECOVERY:METHOD EXPLICITCLOCK specifies using an explicit clock. MEASUREMENT:CLOCKRECOVERY:METHOD? might return :MEASUREMENT:CLOCKRECOVERY:METHOD PLL indicating the method is set to PLL.

MEASUrement:CLOCKRecovery:MODEl

This command sets or queries the global phase locked loop (PLL) clock recovery model used when PLL clock recovery is used for the measurement. This will affect measurements whose :MEASUrement:MEAS<x>:CLOCKRecovery:GLOBal flag is set to 1.

Group	Measurement
Syntax	<pre>MEASUrement:CLOCKRecovery:MODEl {TYPE1 TYPE2} MEASUrement:CLOCKRecovery:MODEl?</pre>
Arguments	TYPE1 PLL clock recovery uses PLL model type I. TYPE2 PLL clock recovery uses PLL model type II.

Examples `MEASUREMENT:CLOCKRECOVERY:MODEL TYPE2` sets the model to type 2.

`MEASUREMENT:CLOCKRECOVERY:MODEL?` might return
`:MEASUREMENT:CLOCKRECOVERY:MODEL TYPE1` indicating the
recovery model is type 1.

MEASUrement:CLOCKRecovery:NOMINALOFFset

This command sets or queries the global offset value used when explicit clock recovery is used for the measurement. This will affect measurements whose
:MEASUrement:MEAS<x>:CLOCKRecovery:GLOBal flag is set to 1.

Group Measurement

Syntax `MEASUrement:CLOCKRecovery:NOMINALOFFset <NR3>`
`MEASUrement:CLOCKRecovery:NOMINALOFFset?`

Arguments <NR3> is the global clock offset.

Examples `MEASUREMENT:CLOCKRECOVERY:NOMINALOFFSET 1.0` sets the offset to 1.0.

`MEASUREMENT:CLOCKRECOVERY:NOMINALOFFSET?` might return
`:MEASUREMENT:CLOCKRECOVERY:NOMINALOFFSET 0.0E+0` indicating the
offset is set to 0.0.

MEASUrement:CLOCKRecovery:NOMINALOFFset:SELECTIONtype

This command sets or queries the global offset type used when explicit clock recovery is used for the measurement. This will affect measurements whose
:MEASUrement:MEAS<x>:CLOCKRecovery:GLOBal flag is set to 1.

Group Measurement

Syntax `MEASUrement:CLOCKRecovery:NOMINALOFFset:SELECTIONtype`
`{AUTO|MANUAL}`
`MEASUrement:CLOCKRecovery:NOMINALOFFset:SELECTIONtype?`

Arguments `AUTO` sets the selection type to `AUTO`.

`MANUAL` sets the selection type to `MANUAL`.

- Examples** `MEASUREMENT:CLOCKRECOVERY:NOMINALOFFSET:SELECTIONTYPE AUTO`
sets the selection type to auto.
- `MEASUREMENT:CLOCKRECOVERY:NOMINALOFFSET:SELECTIONTYPE?` might
return `:MEASUREMENT:CLOCKRECOVERY:NOMINALOFFSET:SELECTIONTYPE`
`MANUAL` indicating the selection type is manual.

MEASUrement:CLOCKRecovery:STANDARD

This command sets or queries the global communications standard used when PLL clock recovery is used for the measurement. This will affect measurements whose `:MEASUrement:MEAS<x>:CLOCKRecovery:GLOBal` flag is set to 1.

Group Measurement

Syntax `MEASUrement:CLOCKRecovery:STANDARD`
`{CUSTOm|ENET100|FW1394BS400B|FW1394BS800B|`
`FW1394BS1600B|FBD1|FBD2|FBD3|FC133|FC266|`
`FC531|FC1063|FC2125|FC4250|FC8500|ENET1000|`
`IBA2500|IBA_GEN2|OC1|OC3|OC12|OC48|PCIE_GEN1|`
`PCIE_GEN2|PCIE_GEN3|RIO125|RIO250|RIO3125|`
`SAS15_NOSSC|SAS3_NOSSC|SAS6_NOSSC|SAS12_NOSSC|`
`SAS15_SSC|SAS3_SSC|SAS6_SSC|SAS12_SSC|SATA_GEN1|`
`SATA_GEN2|SATA_GEN3|USB3|XAUI|XAUI_GEN2}`
`MEASUrement:CLOCKRecovery:STANDARD?`

Arguments Arguments are the global clock recovery standards.

- Examples** `MEASUREMENT:CLOCKRECOVERY:STANDARD PCIE_GEN3` sets the standard to
`PCIE_GEN3`.
- `MEASUREMENT:CLOCKRECOVERY:STANDARD?` might return
`:MEASUREMENT:CLOCKRECOVERY:STANDARD PCIE_GEN1` indicating
the standard is set to `PCIE_GEN1`.

MEASUrement:DELeTe (No Query Form)

The command deletes the specified measurement.

Group Measurement

Syntax `MEASUrement:DELeTe <QString>`

Arguments <QString> is the measurement to delete. Argument is of the form "MEAS<NR1>" where <NR1> is ≥ 1 .

Examples MEASUREMENT:DELETE "MEAS4" deletes measurement 4.

MEASUrement:DIRacmodel

This command sets or queries the dirac model used to separate random from deterministic jitter for jitter measurements.

Group Measurement

Syntax MEASUrement:DIRacmodel {PCIExpress|FIBREchannel}
MEASUrement:DIRacmodel?

Arguments PCIExpress specifies the PCIExpress dirac model.
FIBREchannel specifies the FIBREchannel dirac model.

Examples MEASUREMENT:DIRACMODEL FIBRECHANNEL sets the dirac model to FIBRECHANNEL.
MEASUREMENT:DIRACMODEL? might return :MEASUREMENT:DIRACMODEL PCIEXPRESS indicating the dirac model is set to PCIECPRESS.

MEASUrement:DISPLAYUnits

This command sets or queries the display units used for jitter summary measurements.

Group Measurement

Syntax MEASUrement:DISPLAYUnits {SECONDS|UNITINTERVALS}

Arguments SECONDS set the display units to seconds.
UNITINTERVALS set the display units to unit intervals.

Examples MEASUREMENT:DISPLAYUNITS UNITINTERVALS sets the display units to unit intervals.

MEASUREMENT:DISPLAYUNITS? might return :MEASUREMENT:DISPLAYUNITS SECONDS indicating the display units are seconds.

MEASUrement:EDGE<x>

This command sets or queries the type of the edge for the measurement.

Group	Measurement
Syntax	MEASUrement:EDGE<x> {RISE FALL BOTH}
Arguments	FALL specifies the falling edge of the waveform. RISE specifies the rising edge of the waveform. BOTH specifies both a rising and falling edge of the waveform.
Examples	MEASUrement:EDGE2 MEASUrement:EDGE2? might return :MEASUREMENT:EDGE2 BOTH indicating

MEASUrement:EYERENDER

This command sets or queries the state of high-performance eye rendering for an eye diagram.

Group	Measurement
Syntax	MEASUrement:EYERENDER {<NR1> OFF ON} MEASUrement:EYERENDER?
Arguments	<NR1> = 0 disables high-performance eye rendering for an eye diagram; any other value turns this feature on. OFF disables high-performance eye rendering for an eye diagram. ON enables high-performance eye rendering for an eye diagram.

Examples `MEASUREMENT:EYERENDER`

`MEASUREMENT:EYERENDER?` might return `:MEASUREMENT:EYERENDER 1` indicating high-performance eye rendering is on.

MEASUrement:FILTERs:BLANKingtime

This command sets or queries the global filter blanking time.

Group Measurement

Syntax `MEASUrement:FILTERs:BLANKingtime <NR3>`
`MEASUrement:FILTERs:BLANKingtime?`

Arguments `<NR3>` is the current filter blanking time.

Examples `MEASUREMENT:FILTERS:BLANKINGTIME 3.0` sets the blanking time to 3.0.

`MEASUREMENT:FILTERS:BLANKINGTIME?` might return `:MEASUREMENT:FILTERS:BLANKINGTIME 4.0000` indicating the blanking time is 4.0.

MEASUrement:FILTERs:HIGHPass:FREQ

This command sets or queries the global high pass filter frequency for the measurement.

Group Measurement

Syntax `MEASUrement:FILTERs:HIGHPass:FREQ <NR3>`
`MEASUrement:FILTERs:HIGHPass:FREQ?`

Arguments `<NR3>` is the current high pass filter frequency.

Examples `MEASUREMENT:FILTERS:HIGHPASS:FREQ 20.0e6` sets the frequency to 20 MHz.

`MEASUREMENT:FILTERS:HIGHPASS:FREQ?` might return `:MEASUREMENT:FILTERS:HIGHPASS:FREQ 10.0000E+6` indicating the filter frequency is set to 10.0 MHz.

MEASUREMENT:FILTERS:HIGHPASS:SPEC

This command sets or queries the global high pass filter order for the measurement.

Group	Measurement
Syntax	MEASUREMENT:FILTERS:HIGHPASS:SPEC {NONE FIRST SECOND THIRD} MEASUREMENT:FILTERS:HIGHPASS:SPEC?
Arguments	NONE specifies no high pass filter. FIRST specifies a first order high pass filter. SECOND specifies a second order high pass filter. THIRD specifies a third order high pass filter.
Examples	MEASUREMENT:FILTERS:HIGHPASS:SPEC NONE specifies no high pass filter. MEASUREMENT:FILTERS:HIGHPASS:SPEC? might return :MEASUREMENT:FILTERS:HIGHPASS:SPEC SECOND indicating a second order high pass filter.

MEASUREMENT:FILTERS:LOWPASS:FREQ

This command sets or queries the global low pass filter cutoff frequency for the measurement.

Group	Measurement
Syntax	MEASUREMENT:FILTERS:LOWPASS:FREQ <NR3> MEASUREMENT:FILTERS:LOWPASS:FREQ?
Arguments	<NR3> is the current low pass filter frequency.
Examples	MEASUREMENT:FILTERS:LOWPASS:FREQ 20.0e6 sets the low pass frequency to 20 MHz. MEASUREMENT:FILTERS:LOWPASS:FREQ? might return :MEASUREMENT:FILTERS:LOWPASS:FREQ 10.0000E+6 indicating the low pass frequency is 10.0 MHz.

MEASUrement:FILTers:LOWPass:SPEC

This command sets or queries the global low pass filter order for the measurement.

Group	Measurement
Syntax	MEASUrement:FILTers:LOWPass:SPEC {NONE FIRST SECOND THIRD} MEASUrement:FILTers:LOWPass:SPEC?
Arguments	NONE specifies no low pass filter. FIRST specifies a first order low pass filter. SECOND specifies a second order low pass filter. THIRD specifies a third order low pass filter.
Examples	MEASUREMENT:FILTERS:LOWPASS:SPEC SECOND specifies a second order low pass filter. MEASUREMENT:FILTERS:LOWPASS:SPEC? might return :MEASUREMENT:FILTERS:LOWPASS:SPEC NONE indicating no low pass filter.

MEASUrement:FILTers:RAMPTime

This command sets or queries the global filter ramp time for the measurement.

Group	Measurement
Syntax	MEASUrement:FILTers:RAMPTime <NR3> MEASUrement:FILTers:RAMPTime?
Arguments	<NR3> is the current filter ramp time.
Examples	MEASUREMENT:FILTERS:RAMPTIME 3.0 sets the ramp time to 3.0. MEASUREMENT:FILTERS:RAMPTIME? might return :MEASUREMENT:FILTERS:RAMPTIME 2.0000 indicating the ramp time is 2.0.

MEASUrement:GATing

This command sets or queries the global gating type for the measurement.

Group	Measurement
Syntax	<code>MEASUrement:GATing {NONE SCREEN CURSor LOGic SEARCh}</code> <code>MEASUrement:GATing?</code>
Arguments	NONE turns off measurement gating. SCREEn turns on gating, using the left and right edges of the screen. CURSor limits measurements to the portion of the waveform between the vertical bar cursors, even if they are off screen. LOGic specifies that measurements are taken only on the portion of the waveform where the logic source is in the active state. SEARCH specifies that measurements are taken based on search criteria.
Examples	<code>MEASUREMENT:GATING SCREEN</code> turns on measurement gating, using the left and right edges of the screen. <code>MEASUREMENT:GATING?</code> might return <code>:MEASUREMENT:GATING CURSOR</code>, indicating that measurements are limited to the portion of the waveform between the vertical bar cursors.

MEASUrement:GATing:ACTive

This command sets or queries the global gating active level used for logic gating.

Group	Measurement
Syntax	<code>MEASUrement:GATing:ACTive {HIGH LOW}</code> <code>MEASUrement:GATing:ACTive?</code>
Arguments	HIGH specifies the gate is HIGH. LOW specifies the gate is LOW.

Examples `MEASUREMENT:GATING:ACTIVE LOW` specifies a low gate.

`MEASUREMENT:GATING:ACTIVE?` might return
 `:MEASUREMENT:GATING:ACTIVE HIGH` indicating the gate is high.

MEASUREMENT:GATING:HYSTERESIS

This command sets or queries the global gating hysteresis value used for logic gating.

Group Measurement

Syntax `MEASUREMENT:GATING:HYSTERESIS <NR3>`
 `MEASUREMENT:GATING:HYSTERESIS?`

Arguments `<NR3>` is the gating hysteresis.

Examples `MEASUREMENT:GATING:HYSTERESIS 40.0e-3` sets the hysteresis to 40 mV.

`MEASUREMENT:GATING:HYSTERESIS?` might return
 `:MEASUREMENT:GATING:HYSTERESIS 30.0000E-3` indicating the hysteresis is 30 mV.

MEASUREMENT:GATING:LOGICSOURCE

This command sets or queries the gating data source used for logic gating.

Group Measurement

Syntax `MEASUREMENT:GATING:LOGICSOURCE {CH<x>|MATH<x>|REF<x>}`
 `MEASUREMENT:GATING:LOGICSOURCE?`

Arguments Arguments are the logic gating source.

Examples `MEASUREMENT:GATING:LOGICSOURCE MATH1` sets the logic source to MATH 1.

`MEASUREMENT:GATING:LOGICSOURCE?` might return
 `:MEASUREMENT:GATING:LOGICSOURCE CH6` indicating the logic source is channel 6.

MEASUrement:GATing:MIDRef

This command sets or queries the global gating mid ref value used for logic gating.

Group	Measurement
Syntax	MEASUrement:GATing:MIDRef <NR3> MEASUrement:GATing:MIDRef?
Arguments	<NR3> is the mid ref value for gating.
Examples	MEASUREMENT:GATING:MIDREF 2.0 set the gating mid ref to 2.0 V. MEASUREMENT:GATING:MIDREF? might return :MEASUREMENT:GATING:MIDREF 1.5000 indicating the gating mid ref is 1.5 V.

MEASUrement:GATing:SEARCHSource

This command sets or queries the global gating search source when the gating type is search.

Group	Measurement
Syntax	MEASUrement:GATing:SEARCHSource SEARCH1 MEASUrement:GATing:SEARCHSource?
Arguments	Argument is the search source.
Examples	MEASUREMENT:GATING:SEARCHSOURCE SEARCH1 set the search source to search 1. MEASUREMENT:GATING:SEARCHSOURCE? might return :MEASUREMENT:GATING:SEARCHSOURCE UNDEFINED indicating the search source is not defined.

MEASUrement:INTERp

This command sets or queries the interpolation mode used to locate edge crossings.

Group	Measurement
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Syntax `MEASUREMENT:INTERP {AUTO|SINX|LINEAR}`
`MEASUREMENT:INTERP?`

Arguments `AUTO` automatically selects the interpolation mode.

`SINX` specifies $\sin(x)/x$ interpolation, where acquired points are fit to a curve.

`LINEAR` specifies linear interpolation, where acquired points are connected with straight lines.

Examples `MEASUREMENT:INTERP SINX` set the interpolation mode to $\sin(x)/x$.

`MEASUREMENT:INTERP?` might return `:MEASUREMENT:INTERP AUTO` indicating the interpolation mode is auto.

MEASUREMENT:JITTERmodel

This command sets or queries the model used to separate random from deterministic jitter for jitter measurements.

Group Measurement

Syntax `MEASUREMENT:JITTERmodel {SPECTRAL|SPECTRALBUJ}`
`MEASUREMENT:JITTERmodel?`

Arguments `SPECTRAL` sets the jitter separation model to spectral only.

`SPECTRALBUJ` sets the jitter separation model to spectral plus BUJ.

Examples `MEASUREMENT:JITTERMODEL SPECTRAL` set the jitter model to spectral.

`MEASUREMENT:JITTERMODEL?` might return `:MEASUREMENT:JITTERMODEL SPECTRAL` indicating the jitter model is set to spectral.

MEASUREMENT:LIST? (Query Only)

This query returns a comma separated list of all currently defined measurements.

Group Measurement

Syntax `MEASUREMENT:LIST?`

Returns	Returns a list of all currently defined measurements.
Examples	MEASUREMENT:LIST? might return :MEASUREMENT:LIST MEAS1,MEAS2,MEAS3,MEAS4,MEAS5,MEAS6 indicating 6 measurements are defined.

MEASUrement:LOCKRJ

This command sets or queries the state of RJ locking.

Group	Measurement
Syntax	MEASUrement:LOCKRJ {OFF ON <NR1>} MEASUrement:LOCKRJ?
Arguments	ON indicates that RJ locking is active. OFF indicates that RJ locking is off. <NR1> a 0 turns off RJ locking; any other value activates RJ locking.
Examples	MEASUREMENT:LOCKRJ ON activates RJ locking

MEASUrement:LOCKRJValue

This command sets or queries the RJ lock value.

Group	Measurement
Syntax	MEASUrement:LOCKRJValue <NR3> MEASUrement:LOCKRJValue?
Arguments	<NR3> default value = 1e-12; minimum value = 1e-15; maximum value = 1.
Examples	MEASUREMENT:LOCKRJVALUE 1 sets the lock value to the maximum value of 1. MEASUREMENT:LOCKRJVALUE? might return :MEASUREMENT:LOCKRJVALUE 1.0000E-12, indicating the lock value is set to the default value of 1e-12.

MEASUREMENT:MATH<x>:REFLEVELS:ABSOLUTE:FALLHIGH

This command sets or queries the value used as the high reference level of the falling edge when the measurement ref level method is set to absolute. The math number is specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MATH<x>:REFLEVELS:ABSOLUTE:FALLHIGH <NR3> MEASUREMENT:MATH<x>:REFLEVELS:ABSOLUTE:FALLHIGH?</pre>
Arguments	<NR3> is the high reference level, and is the zero percent level when MEASUREMENT:IMMED:REFLEVEL:METHOD is set to Absolute.
Examples	MEASUREMENT:MATH1:REFLEVELS:ABSOLUTE:FALLHIGH 1.5 sets the reference level to 1.5 V. MEASUREMENT:MATH2:REFLEVELS:ABSOLUTE:FALLHIGH? might return :MEASUREMENT:MATH2:REFLEVELS:ABSOLUTE:FALLHIGH 1.0000 indicating the reference level is 1.0 V.

MEASUREMENT:MATH<x>:REFLEVELS:ABSOLUTE:FALLLOW

This command sets or queries the value used as the low reference level of the falling edge when the measurement ref level method is set to absolute. The math number is specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MATH<x>:REFLEVELS:ABSOLUTE:FALLLOW <NR3> MEASUREMENT:MATH<x>:REFLEVELS:ABSOLUTE:FALLLOW?</pre>
Arguments	<NR3> is the high reference level, and is the zero percent level when MEASUREMENT:IMMED:REFLEVEL:METHOD is set to Absolute.
Examples	MEASUREMENT:MATH2:REFLEVELS:ABSOLUTE:FALLLOW -1.5 sets the reference level to -1.5 V. MEASUREMENT:MATH2:REFLEVELS:ABSOLUTE:FALLLOW? might return :MEASUREMENT:MATH2:REFLEVELS:ABSOLUTE:FALLLOW -1.0000 indicating the reference level is -1.0 V.

MEASUREMENT:MATH<x>:REFLEVELS:ABSOLUTE:FALLMID

This command sets or queries the value used as the mid reference level of the falling edge when the measurement ref level method is set to absolute. The math number is specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MATH<x>:REFLEVELS:ABSOLUTE:FALLMID <NR3> MEASUREMENT:MATH<x>:REFLEVELS:ABSOLUTE:FALLMID?</pre>
Arguments	<NR3> is the mid reference level (where 50% is equal to MID) used to calculate the mid reference level when the measurement's Ref level method is set to Absolute.
Examples	MEASUREMENT:MATH2:REFLEVELS:ABSOLUTE:FALLMID 0.0 sets the reference level to 0.0 V. MEASUREMENT:MATH2:REFLEVELS:ABSOLUTE:FALLMID? might return :MEASUREMENT:MATH2:REFLEVELS:ABSOLUTE:FALLMID 0.0E+0 indicating the reference level is 0.0 V.

MEASUREMENT:MATH<x>:REFLEVELS:ABSOLUTE:HYSTERESIS

This command sets or queries the value of the hysteresis of the reference level when the measurement ref level method is set to absolute. The math number is specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MATH<x>:REFLEVELS:ABSOLUTE:HYSTERESIS <NR3> MEASUREMENT:MATH<x>:REFLEVELS:ABSOLUTE:HYSTERESIS?</pre>
Arguments	<NR3> is the hysteresis value used for the autoset.
Examples	MEASUREMENT:MATH2:REFLEVELS:ABSOLUTE:HYSTERESIS 20.0E-3 sets the hysteresis to 20.0 mV. MEASUREMENT:MATH2:REFLEVELS:ABSOLUTE:HYSTERESIS? might return :MEASUREMENT:MATH2:REFLEVELS:ABSOLUTE:HYSTERESIS 30.0000E-3 indicating the hysteresis value is 30.0 mV.

MEASUrement:MATH<x>:REFLevels:ABSolute:RISEHigh

This command sets or queries the value used as the high reference level of the rising edge when the measurement ref level method is set to absolute. The math number is specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MATH<x>:REFLevels:ABSolute:RISEHigh <NR3> MEASUrement:MATH<x>:REFLevels:ABSolute:RISEHigh?</pre>
Arguments	<NR3> is the high reference level, and is the zero percent level when MEASUrement:IMMed:REFLevel:METHOD is set to Absolute.
Examples	MEASUrement:MATH2:REFLevels:ABSolute:RISEHigh 1.5 sets the reference level to 1.5 V. MEASUrement:MATH2:REFLevels:ABSolute:RISEHigh? might return :MEASUREMENT:MATH2:REFLEVELS:ABSOLUTE:RISEHIGH 1.0000 indicating the reference level is 1.0 V.

MEASUrement:MATH<x>:REFLevels:ABSolute:RISELow

This command sets or queries the value used as the low reference level of the rising edge when the measurement ref level method is set to absolute. The math number is specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MATH<x>:REFLevels:ABSolute:RISELow <NR3> MEASUrement:MATH<x>:REFLevels:ABSolute:RISELow?</pre>
Arguments	<NR3> is the high reference level, and is the zero percent level when MEASUrement:IMMed:REFLevel:METHOD is set to Absolute.
Examples	MEASUrement:MATH2:REFLevels:ABSolute:RISELow -1.5 sets the reference level to -1.5 V. MEASUrement:MATH2:REFLevels:ABSolute:RISELow? might return :MEASUREMENT:MATH2:REFLEVELS:ABSOLUTE:RISELOW -1.0000 indicating the reference level is -1.0 V.

MEASUrement:MATH<x>:REFLevels:ABSolute:RISEMid

This command sets or queries the value used as the mid reference level of the rising edge when the measurement ref level method is set to absolute. The math number is specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MATH<x>:REFLevels:ABSolute:RISEMid <NR3> MEASUrement:MATH<x>:REFLevels:ABSolute:RISEMid?</pre>
Arguments	<NR3> is the mid reference level (where 50% is equal to MID) used to calculate the mid reference level when the measurement Ref level method is set to Absolute.
Examples	<pre>MEASUrement:MATH2:REFLevels:ABSolute:RISEMid 30.0E-3</pre> sets the reference level to 30 mV. <pre>MEASUrement:MATH2:REFLevels:ABSolute:RISEMid?</pre> might return <pre>:MEASUREMENT:MATH2:REFLEVELS:ABSOLUTE:RISEMID 0.0E+0</pre> indicating the reference level is 0.0 V.

MEASUrement:MATH<x>:REFLevels:ABSolute:TYPE

This command sets or queries the reference level type for the measurement. The math number is specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MATH<x>:REFLevels:ABSolute:TYPE {SAME UNIQUE} MEASUrement:MATH<x>:REFLevels:ABSolute:TYPE?</pre>
Arguments	SAME specifies that the absolute levels are set the same. UNIQUE specifies that the absolute levels can be set independently.
Examples	<pre>MEASUrement:MATH2:REFLevels:ABSolute:TYPE UNIQUE</pre> specifies that the absolute levels can be set independently. <pre>MEASUrement:MATH2:REFLevels:ABSolute:TYPE?</pre> might return <pre>:MEASUREMENT:MATH2:REFLEVELS:ABSOLUTE:TYPE SAME</pre> indicating that the absolute levels are set the same.

MEASUrement:MATH<x>:REFLevels:BASETop

This command sets or queries the method used to calculate the TOP and BASE used to calculate reference levels for the measurement. The math number is specified by x.

Group Measurement

Syntax MEASUrement:MATH<x>:REFLevels:BASETop
{AUTO|MINMax|MEANhistogram| MODEhistogram|EYEHistogram}
MEASUrement:MATH<x>:REFLevels:BASETop?

Arguments AUTO automatically chooses a reference level method.

MINMax specifies that reference levels are relative to the measurement MIN and MAX.

MEANhistogram specifies that reference levels are relative to the histogram mean BASE and TOP.

MODEhistogram specifies that reference levels are relative to the histogram mode BASE and TOP.

EYEHistogram specifies that reverence levels are relative to the eye histogram BASE and TOP.

Examples MEASUrement:MATH1:REFLevels:BASETop MINMax specifies that reference levels are relative to the measurement MIN and MAX.

MEASUrement:MATH1:REFLevels:BASETop? might return
:MEASUREMENT:MATH2:REFLEVELS:BASETOP AUTO indicating the instrument automatically chooses a reference level method.

MEASUrement:MATH<x>:REFLevels:METHod

This command sets or queries the method used to calculate reference levels for the measurement. The math number is specified by x.

Group Measurement

Syntax MEASUrement:MATH<x>:REFLevels:METHod {PERCent|ABSolute}
MEASUrement:MATH<x>:REFLevels:METHod?

Arguments	PERCent specifies that the reference levels are calculated as a percent relative to HIGH and LOW. The percentages are defined using the MEASUREMENT:MATH<x>:REFLevel:PERCent commands. ABSolute specifies that the reference levels are set explicitly using the MEASUREMENT:MATH<x>:REFLevel:ABSolute commands. This method is useful when precise values are required.
Examples	MEASUREMENT:MATH2:REFLevels:METHOD ABSOLUTE specifies that the reference levels are set explicitly MEASUREMENT:MATH2:REFLevels:METHOD? might return :MEASUREMENT:MATH2:REFLEVELS:METHOd PERCENT indicating reference levels are in percent relative to HIGH and LOW.

MEASUREMENT:MATH<x>:REFLevels:PERCent:FALLHigh

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the high reference level of the falling edge when the measurement ref level method is set to percent. The math number is specified by x.

Group	Measurement
Syntax	MEASUREMENT:MATH<x>:REFLevels:PERCent:FALLHigh <NR3> MEASUREMENT:MATH<x>:REFLevels:PERCent:FALLHigh?
Arguments	<NR3> is the percentage (where 100% is equal to HIGH) used to calculate the high reference level when the measurement Ref level method is set to Percent.
Examples	MEASUREMENT:MATH2:REFLevels:PERCent:FALLHigh 95 sets the reference level to 95% of TOP. MEASUREMENT:MATH2:REFLevels:PERCent:FALLHigh? might return :MEASUREMENT:MATH2:REFLEVELS:PERCENT:FALLHIGH 90.0000 indicating the reference level is set to 90% of TOP.

MEASUREMENT:MATH<x>:REFLevels:PERCent:FALLLow

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the low reference level of the falling edge when the measurement ref level method is set to percent. The math number is specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MATH<x>:REFLeve1s:PERCent:FALLLow <NR3> MEASUrement:MATH<x>:REFLeve1s:PERCent:FALLLow?</pre>
Arguments	<NR3> is the percentage (where 100% is equal to HIGH) used to calculate the mid reference level when the measurement's Ref level method is set to Percent.
Examples	MEASUrement:MATH2:REFLeve1s:PERCent:FALLLow 5 sets the reference level to 5% of TOP. MEASUrement:MATH2:REFLeve1s:PERCent:FALLLow? might return :MEASUREMENT:MATH2:REFLEVELS:PERCENT:FALLLOW 10.0000 indicating the reference level is set to 10% of TOP.

MEASUrement:MATH<x>:REFLevels:PERCent:FALLMid

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the falling edge when the measurement ref level method is set to percent. The math number is specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MATH<x>:REFLeve1s:PERCent:FALLMid <NR3> MEASUrement:MATH<x>:REFLeve1s:PERCent:FALLMid?</pre>
Arguments	<NR3> is the percentage (where 50% is equal to MID) used to calculate the mid reference level when the measurement Ref level method is set to Percent.
Examples	MEASUrement:MATH2:REFLeve1s:PERCent:FALLMid 50 sets the reference level to 50% of TOP. MEASUrement:MATH2:REFLeve1s:PERCent:FALLMid? might return :MEASUREMENT:MATH2:REFLEVELS:PERCENT:FALLMID 50.0000 indicating the reference level is set to 50% of TOP.

MEASUrement:MATH<x>:REFLevels:PERCent:HYSteresis

This command sets or queries the percentage (where 100% is equal to MAX and 0% is equal to MIN) used to calculate the hysteresis of the reference level

when the measurement ref level method is set to percent. The math number is specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MATH<x>:REFLEVELS:PERCENT:HYSTERESIS <NR3> MEASUREMENT:MATH<x>:REFLEVELS:PERCENT:HYSTERESIS?</pre>
Arguments	<NR3> is the hysteresis value used for the autoset.
Examples	MEASUREMENT:MATH2:REFLEVELS:PERCENT:HYSTERESIS 2.5 sets the hysteresis to 2.5% of MAX MEASUREMENT:MATH2:REFLEVELS:PERCENT:HYSTERESIS? might return :MEASUREMENT:MATH2:REFLEVELS:PERCENT:HYSTERESIS 5.0000 indicating the hysteresis is set to 5% of MAX.

MEASUREMENT:MATH<x>:REFLEVELS:PERCENT:RISEHIGH

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the high reference level of the rising edge when the measurement ref level method is set to percent. The math number is specified by x. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MATH<x>:REFLEVELS:PERCENT:RISEHIGH <NR3> MEASUREMENT:MATH<x>:REFLEVELS:PERCENT:RISEHIGH?</pre>
Arguments	<NR3> is the percentage (where 100% is equal to TOP) used to calculate the high reference level when the measurement's Ref level method is set to Percent.
Examples	MEASUREMENT:MATH2:REFLEVELS:PERCENT:RISEHIGH 95 sets the reference level to 95% of TOP. MEASUREMENT:MATH2:REFLEVELS:PERCENT:RISEHIGH? might return :MEASUREMENT:MATH2:REFLEVELS:PERCENT:RISEHIGH 90.0000 indicating the reference level is set to 90% of TOP.

MEASUrement:MATH<x>:REFLeveLs:PERCent:RISELow

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the low reference level of the rising edge when the measurement ref level method is set to percent. The math number is specified by x.

Group Measurement

Syntax MEASUrement:MATH<x>:REFLeveLs:PERCent:RISELow <NR3>
MEASUrement:MATH<x>:REFLeveLs:PERCent:RISELow?

Arguments <NR3> is the percentage (where 100% is equal to TOP) used to calculate the mid reference level when the measurement Ref level method is set to Percent.

Examples MEASUrement:MATH2REFLeveLs:PERCent:RISELow 5 sets the reference level to 5% of TOP.

MEASUrement:MATH2REFLeveLs:PERCent:RISELow? might return
:MEASUREMENT:MATH2:REFLEVELS:PERCENT:RISELOW 10.0000 indicating the reference level is set to 10% of TOP.

MEASUrement:MATH<x>:REFLeveLs:PERCent:RISEMid

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the rising edge when the measurement ref level method is set to percent. The math number is specified by x.

Group Measurement

Syntax MEASUrement:MATH<x>:REFLeveLs:PERCent:RISEMid <NR3>
MEASUrement:MATH<x>:REFLeveLs:PERCent:RISEMid?

Arguments <NR3> is the percentage (where 50% is equal to MID) used to calculate the mid reference level when the measurement Ref level method is set to Percent.

Examples MEASUrement:MATH2:REFLeveLs:PERCent:RISEMid 50 sets the reference level to 50% of TOP.

MEASUREMENT:MATH2:REFLEVELS:PERCENT:RISEMID? might return
:MEASUREMENT:MATH2:REFLEVELS:PERCENT:RISEMID 50.0000 indicating
the reference level is set to 50% of TOP.

MEASUREMENT:MATH<x>:REFLEVELS:PERCENT:TYPE

This command specifies or queries the reference level percent type for the measurement. The math number is specified by x.

Group	Measurement
Syntax	MEASUREMENT:MATH<x>:REFLEVELS:PERCENT:TYPE {TENNINETY TWENTYEIGHTY CUSTOM} MEASUREMENT:MATH<x>:REFLEVELS:PERCENT:TYPE?
Arguments	TENNINETY sets the values for Low, Mid and High Ref to 10%, 50% and 90% respectively. TWENTYEIGHTY sets the values for Low, Mid and High Ref are set to 20%, 50% and 80% respectively. CUSTOM allows setting other reference level percents.
Examples	MEASUREMENT:MATH2:REFLEVELS:PERCENT:TYPE TWENTYEIGHTY sets the values for Low, Mid and High Ref are set to 20%, 50% and 80% respectively. MEASUREMENT:MATH2:REFLEVELS:PERCENT:TYPE? might return :MEASUREMENT:MATH2:REFLEVELS:PERCENT:TYPE TENNINETY indicating the reference levels are set to 10%, 50% and 90%.

MEASUREMENT:MEASRange:MAX

This command sets or queries the global range maximum value.

Group	Measurement
Syntax	MEASUREMENT:MEASRange:MAX <NR3> MEASUREMENT:MEASRange:MAX?
Arguments	<NR3> is the maximum measurement range limit value.

Examples `MEASUREMENT:MEASRANGE:MAX 1.5` sets the maximum range limit to 1.5 V.
`MEASUREMENT:MEASRANGE:MAX?` might return
`:MEASUREMENT:MEASRANGE:MAX 1.0000` indicating the maximum range limit is 1.0 V.

MEASUrement:MEASRange:MIN

This command sets or queries the global range minimum value.

Group Measurement

Syntax `MEASUrement:MEASRange:MIN <NR3>`
`MEASUrement:MEASRange:MIN?`

Arguments `<NR3>` is the minimum measurement range limit value.

Examples `MEASUREMENT:MEASRANGE:MIN 30.0E-3` sets the minimum measurement range limit to 30.0 mV.
`MEASUREMENT:MEASRANGE:MIN?` might return
`:MEASUREMENT:MEASRANGE:MIN 0.0E+0` indicating minimum range limit is 0.0 V.

MEASUrement:MEASRange:STATE

This command sets or queries the global range state.

Group Measurement

Syntax `MEASUrement:MEASRange:STATE {OFF|ON|0|1}`
`MEASUrement:MEASRange:STATE?`

Arguments OFF specifies that the measurement results are not limited.

ON specifies that the measurement results are limited to results with values between the range minimum and maximum.

0 specifies that the measurement results are not limited.

1 specifies that the measurement results are limited to results with values between the range minimum and maximum.

Examples MEASUREMENT:MEASRANGE:STATE 1 specifies that the measurement results are limited to results with values between the range minimum and maximum.

MEASUREMENT:MEASRANGE:STATE? might return
:MEASUREMENT:MEASRANGE:STATE 0 indicating that the measurement results are not limited.

MEASUREMENT:MEAS<x>:BER

This command sets or queries BER value for the measurement. Measurements are specified by x.

Group Measurement

Syntax MEASUREMENT:MEAS<x>:BER <NR3>
MEASUREMENT:MEAS<x>:BER?

Arguments <NR3> is BER value for the measurement.

Examples MEASUREMENT:MEAS2:BER 10.0 sets the BER for the measurement to 10.0.

MEASUREMENT:MEAS2:BER? might return :MEASUREMENT:MEAS2:BER
12.0000 indicating the measurement BER is 12.0.

MEASUREMENT:MEAS<x>:BER:TARGETBER

This command sets or queries the target BER value for the measurement. Measurements are specified by x.

Group Measurement

Syntax MEASUREMENT:MEAS<x>:BER:TARGETBER <NR3>
MEASUREMENT:MEAS<x>:BER:TARGETBER?

Arguments <NR3> is the target BER value.

Examples MEASUREMENT:MEAS1:BER:TARGETBER 14.0 sets the target BER to 14.0.

MEASUREMENT:MEAS1:BER:TARGETBER? might return
:MEASUREMENT:MEAS1:BER:TARGETBER 12.0000 indicating the
target BER is 12.0.

MEASUrement:MEAS<x>:BIN

This command sets or queries the bin count for the measurement. Measurements are specified by x.

Group Measurement

Syntax MEASUrement:MEAS<x>:BIN <NR3>
MEASUrement:MEAS<x>:BIN?

Arguments <NR3> is the bin count.

Examples MEASUrement:MEAS1:BIN 2 sets the bin count to 2.
MEASUrement:MEAS1:BIN? might return :MEASUREMENT:MEAS1:BIN 1 indicating the bin count is 1.

MEASUrement:MEAS<x>:BITCfgmode

This command sets or queries whether the measurement returns the mean or mode statistic result when the measurement type is bit amplitude/high/low. Measurements are specified by x.

Group Measurement

Syntax MEASUrement:MEAS<x>:BITCfgmode {MEAN|MODE}
MEASUrement:MEAS<x>:BITCfgmode?

Arguments MEAN specifies that the measurement returns results for each bit.
MODE specifies that the measurement returns the mode of its results.

Examples MEASUrement:MEAS1:BITCfgmode MODE specifies that the measurement returns the mode of its results.
MEASUrement:MEAS1:BITCfgmode? might return :MEASUREMENT:MEAS1:BITCFGMODE MEAN indicating the configuration mode is set to MEAN.

MEASUrement:MEAS<x>:BITEnd

This command sets or queries the bit end as a percentage of the unit interval. Measurements are specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:BITEnd <NR3> MEASUrement:MEAS<x>:BITEnd?
Arguments	<NR3> is the bit end.
Examples	MEASUrement:MEAS2:BITEnd 60 sets the bit end to 60% of the unit interval. MEASUrement:MEAS2:BITEnd? might return :MEASUREMENT:MEAS2:BITEND 50.0000 indicating the bit end is 50%.

MEASUrement:MEAS<x>:BITPcnt

This command sets or queries the bit center as a percentage of the unit interval. Measurements are specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:BITPcnt <NR3> MEASUrement:MEAS<x>:BITPcnt?
Arguments	<NR3> is the bit center percentage value to be measured for the bit type selected.
Examples	MEASUrement:MEAS2:BITPcnt 55 sets the bit center to 55% of the unit interval. MEASUrement:MEAS2:BITPcnt? might return :MEASUREMENT:MEAS2:BITPCNT 50.0000 indicating bit center is 50% of the unit interval.

MEASUrement:MEAS<x>:BITStart

This command sets or queries the bit start as a percentage of the unit interval. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:BITStart <NR3> MEASUREMENT:MEAS<x>:BITStart?
Arguments	<NR3> is the bit start.
Examples	MEASUREMENT:MEAS2:BITStart 45 sets the bit start to 45% of the unit interval. MEASUREMENT:MEAS2:BITStart? might return :MEASUREMENT:MEAS2:BITSTART 50.0000 indicating the bit start is 50% of the unit interval.

MEASUREMENT:MEAS<x>:BITType

This command sets or queries the bit type for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:BITType {ALLBits TRANSITION NONTRANSITION} MEASUREMENT:MEAS<x>:BITType?
Arguments	ALLBits specifies that the measurement returns results for all bits. TRANSITION specifies that the measurement returns results for transitions bit only. NONTRANSITION specifies that the measurement returns results for non-transition bits only.
Examples	MEASUREMENT:MEAS1:BITType TRANSITION specifies that the measurement returns results for transition bits only. MEASUREMENT:MEAS1:BITType? might return :MEASUREMENT:MEAS1:BITTYPE ALLBITS indicating that measurements return results for all bits.

MEASUREMENT:MEAS<x>:BURSTEDGTYPe

This command sets or queries the burst edge type for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	<pre>MEASurement:MEAS<x>:BURSTEDGTYPE {RISE FALL} MEASurement:MEAS<x>:BURSTEDGTYPE?</pre>
Arguments	RISE specifies a burst with a rising edge. FALL specifies a burst with a falling edge.
Examples	MEASurement:MEAS2:BURSTEDGTYPE FALL specifies a burst with a falling edge. MEASurement:MEAS2:BURSTEDGTYPE? might return :MEASUREMENT:MEAS2:BURSTEDGTYPE RISE indicating a burst with a rising edge.

MEASurement:MEAS<x>:CCRESUltS:ALLAcqs:MAXimum? (Query Only)

This query-only command returns the maximum cycle-cycle value for the specified measurement for all acquisitions. Measurements are specified by x.

Group	Measurement
Syntax	<pre>MEASurement:MEAS<x>:CCRESUltS:ALLAcqs:MAXimum?</pre>
Returns	The maximum cycle-cycle statistic value for the specified measurement for all acquisitions.
Examples	MEASurement:MEAS2:CCRESUltS:ALLAcqs:MAXimum? might return :MEASUREMENT:MEAS2:CCRESULTS:ALLACQS:MAXIMUM 9.91E+37.

MEASurement:MEAS<x>:CCRESUltS:ALLAcqs:MEAN? (Query Only)

This query-only command returns the mean cycle-cycle value for the specified measurement for all acquisitions. Measurements are specified by x.

Group	Measurement
Syntax	<pre>MEASurement:MEAS<x>:CCRESUltS:ALLAcqs:MEAN?</pre>

Returns The mean cycle-cycle statistic value for the specified measurement for all acquisitions.

Examples MEASUREMENT:MEAS2:CCRESULTS:ALLACQS:MEAN? might return
:MEASUREMENT:MEAS2:CCRESULTS:ALLACQS:MEAN 9.91E+37.

MEASUREMENT:MEAS<x>:CCRESULTS:ALLACQS:MINIMUM? (Query Only)

This query-only command returns the minimum cycle-cycle value for the specified measurement for all acquisitions. Measurements are specified by x.

Group Measurement

Syntax MEASUREMENT:MEAS<x>:CCRESULTS:ALLACQS:MINIMUM?

Returns The minimum cycle-cycle statistic value for the specified measurement for all acquisitions.

Examples MEASUREMENT:MEAS2:CCRESULTS:ALLACQS:MINIMUM? might return
:MEASUREMENT:MEAS2:CCRESULTS:ALLACQS:MINIMUM 9.91E+37.

MEASUREMENT:MEAS<x>:CCRESULTS:ALLACQS:PK2PK? (Query Only)

This query-only command returns the peak to peak cycle-cycle statistic for the specified measurement for all acquisitions. Measurements are specified by x.

Group Measurement

Syntax MEASUREMENT:MEAS<x>:CCRESULTS:ALLACQS:PK2PK?

Returns The peak to peak cycle-cycle statistic value for the specified measurement all acquisitions.

Examples MEASUREMENT:MEAS2:CCRESULTS:ALLACQS:PK2PK? might return
:MEASUREMENT:MEAS2:CCRESULTS:ALLACQS:PK2PK 9.91E+37.

MEASUrement:MEAS<x>:CCRESUlts:ALLAcqs:POPULation? (Query Only)

This query-only command returns the population of all cycle-cycle statistics for the specified measurement for all acquisitions accumulated since statistics were last reset. Measurements are specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:CCRESUlts:ALLAcqs:POPULation?
Returns	The population of all cycle-cycle statistics for the specified measurement accumulated over all acquisitions since statistics were last reset.
Examples	MEASUrement:MEAS2:CCRESUlts:ALLAcqs:POPULation? might return :MEASUREMENT:MEAS2:CCRESULTS:ALLACQS:POPULATION 0 indicating no statistics for the measurement.

MEASUrement:MEAS<x>:CCRESUlts:ALLAcqs:STDDev? (Query Only)

This query-only command returns the standard deviation cycle-cycle for the specified measurement for all acquisitions. Measurements are specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:CCRESUlts:ALLAcqs:STDDev?
Returns	The standard deviation cycle-cycle statistic value for the specified measurement all acquisitions.
Examples	MEASUrement:MEAS2:CCRESUlts:ALLAcqs:STDDev? might return :MEASUREMENT:MEAS2:CCRESULTS:ALLACQS:STDDEV 9.91E+37.

MEASUrement:MEAS<x>:CCRESUlts:CURRentacq:MAXimum? (Query Only)

This query-only command returns the maximum cycle-cycle value for the specified measurement for the current acquisition. Measurements are specified by x.

Group	Measurement
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Syntax	<code>MEASUREMENT:MEAS<x>:CCRESULTS:CURRENTACQ:MAXIMUM?</code>
Returns	The maximum cycle-cycle statistic value for the specified measurement for the current acquisition.
Examples	<code>MEASUREMENT:MEAS2:CCRESULTS:CURRENTACQ:MAXIMUM?</code> might return <code>:MEASUREMENT:MEAS1:CCRESULTS:CURRENTACQ:MAXIMUM 9.91E+37.</code>

MEASUREMENT:MEAS<x>:CCRESULTS:CURRENTACQ:MEAN? (Query Only)

This query-only command returns the mean cycle-cycle value for the specified measurement for the current acquisition. Measurements are specified by x.

Group	Measurement
Syntax	<code>MEASUREMENT:MEAS<x>:CCRESULTS:CURRENTACQ:MEAN?</code>
Returns	The mean cycle-cycle statistic value for the specified measurement for the current acquisition.
Examples	<code>MEASUREMENT:MEAS1:CCRESULTS:CURRENTACQ:MEAN?</code> might return <code>:MEASUREMENT:MEAS1:CCRESULTS:CURRENTACQ:MEAN 9.91E+37.</code>

MEASUREMENT:MEAS<x>:CCRESULTS:CURRENTACQ:MINIMUM? (Query Only)

This query-only command returns the minimum cycle-cycle value for the specified measurement for the current acquisition. Measurements are specified by x.

Group	Measurement
Syntax	<code>MEASUREMENT:MEAS<x>:CCRESULTS:CURRENTACQ:MINIMUM?</code>
Returns	The minimum cycle-cycle statistic value for the specified measurement for the current acquisition.
Examples	<code>MEASUREMENT:MEAS2:CCRESULTS:CURRENTACQ:MINIMUM?</code> might return <code>:MEASUREMENT:MEAS1:CCRESULTS:CURRENTACQ:MINIMUM 9.91E+37.</code>

MEASUrement:MEAS<x>:CCRESUIts:CURRentacq:PK2PK? (Query Only)

This query-only command returns the peak to peak cycle-cycle statistic for the specified measurement for the current acquisition. Measurements are specified by x.

Group Measurement

Syntax MEASUrement:MEAS<x>:CCRESUIts:CURRentacq:PK2PK?

Returns The peak to peak cycle-cycle statistic value for the specified measurement for the current acquisition.

Examples MEASUrement:MEAS1:CCRESUIts:CURRentacq:PK2PK? might return
:MEASUREMENT:MEAS1:CCRESULTS:CURRENTACQ:PK2PK 9.91E+37.

MEASUrement:MEAS<x>:CCRESUIts:CURRentacq:POPulation? (Query Only)

This query-only command returns the population of the cycle-cycle statistics for the specified measurement for the current acquisition. Measurements are specified by x.

Group Measurement

Syntax MEASUrement:MEAS<x>:CCRESUIts:CURRentacq:POPulation?

Returns The population of the cycle-cycle statistics for the specified measurement for the current acquisition.

Examples MEASUrement:MEAS1:CCRESUIts:CURRentacq:POPulation? might return
:MEASUREMENT:MEAS1:CCRESULTS:CURRENTACQ:POPULATION 0 indicating the population is 0.

MEASUrement:MEAS<x>:CCRESUIts:CURRentacq:STDDev? (Query Only)

This query-only command returns the standard deviation cycle-cycle for the specified measurement for the current acquisition. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:CCRESULTS:CURRENTACQ:STDDEV?
Returns	The standard deviation cycle-cycle statistic value for the specified measurement for the current acquisition.
Examples	MEASUREMENT:MEAS1:CCRESULTS:CURRENTACQ:STDDEV? might return :MEASUREMENT:MEAS1:CCRESULTS:CURRENTACQ:STDDEV 9.91E+37.

MEASUREMENT:MEAS<x>:CLOCKRECOVERY:ADVANCED:METHOd

This command sets or queries the advanced clock recovery method when advanced clock recovery is used for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:CLOCKRECOVERY:ADVANCED:METHOd {NONE NOMiNaL PATTERN} MEASUREMENT:MEAS<x>:CLOCKRECOVERY:ADVANCED:METHOd?
Arguments	NOMiNaL sets the advanced CRD method to Nominal Data Rate. PATTERN sets the advanced CDR method to use a Known Data Pattern. NONE sets no advanced CRD method.
Examples	MEASUREMENT:MEAS1:CLOCKRECOVERY:ADVANCED:METHOd PATTERN sets the advanced CDR method to use a Known Data Pattern. MEASUREMENT:MEAS1:CLOCKRECOVERY:ADVANCED:METHOd? might return :MEASUREMENT:MEAS1:CLOCKRECOVERY:ADVANCED:METHOd NONE indicating no advanced CRD method.

MEASUREMENT:MEAS<x>:CLOCKRECOVERY:CLOCKFrequency

This command sets or queries the clock frequency used when fixed constant clock recovery is used for the measurement. Measurements are specified by x.

Group	Measurement
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Syntax	<pre>MEASUREMENT:MEAS<x>:CLOCKRECOVERY:CLOCKFREQUENCY <NR3> MEASUREMENT:MEAS<x>:CLOCKRECOVERY:CLOCKFREQUENCY?</pre>
Arguments	<NR3> is the clock frequency.
Examples	MEASUREMENT:MEAS1:CLOCKRECOVERY:CLOCKFREQUENCY 2.0E+9 sets the frequency to 2.0 GHz. MEASUREMENT:MEAS1:CLOCKRECOVERY:CLOCKFREQUENCY? might return :MEASUREMENT:MEAS1:CLOCKRECOVERY:CLOCKFREQUENCY 2.5000E+9 indicating the frequency is 2.5 GHz.

MEASUREMENT:MEAS<x>:CLOCKRECOVERY:CLOCKMULTIPLIER

This command sets or queries the clock multiplier used when explicit clock recovery is used for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:CLOCKRECOVERY:CLOCKMULTIPLIER <NR3> MEASUREMENT:MEAS<x>:CLOCKRECOVERY:CLOCKMULTIPLIER?</pre>
Arguments	<NR3> is the clock multiplier.
Examples	MEASUREMENT:MEAS1:CLOCKRECOVERY:CLOCKMULTIPLIER 1.5 sets the clock multiplier to 1.5. MEASUREMENT:MEAS1:CLOCKRECOVERY:CLOCKMULTIPLIER? might return :MEASUREMENT:MEAS1:CLOCKRECOVERY:CLOCKMULTIPLIER 1.0000 indicating the clock multiplier is 1.0.

MEASUREMENT:MEAS<x>:CLOCKRECOVERY:CONSTCLOCKMODE

This command sets or queries the constant clock mode used when constant clock recovery is used for the measurement. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:CLOCKRECOVERY:CONSTCLOCKMODE {MEAN MEDIan FIXed}</pre>

Arguments	MEAN specifies that clock recovery uses the mean of the clock signal as the clock frequency.
	MEDian specifies that clock recovery uses the mode of the clock signal as the clock frequency.
	FIXed specifies that clock recovery uses the value set by the user as the clock frequency.
Examples	MEASUREMENT:MEAS1:CLOCKRECOVERY:CONSTCLOCKMODE FIXED specifies that clock recovery uses the value set by the user as the clock frequency.
	MEASUREMENT:MEAS1:CLOCKRECOVERY:CONSTCLOCKMODE? might return :MEASUREMENT:MEAS1:CLOCKRECOVERY:CONSTCLOCKMODE MEAN indicating that clock recovery uses the mean of the clock signal as the clock frequency.

MEASUREMENT:MEAS<x>:CLOCKRECOVERY:DAMPing

This command sets or queries the damping value used when PLL clock recovery is used for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:CLOCKRECOVERY:DAMPing <NR3> MEASUREMENT:MEAS<x>:CLOCKRECOVERY:DAMPing?
Arguments	<NR3> is the clock recovery damping value.
Examples	MEASUREMENT:MEAS1:CLOCKRECOVERY:DAMPing 700.0E-3 sets the damping value to 0.70.
	MEASUREMENT:MEAS1:CLOCKRECOVERY:DAMPing? might return :MEASUREMENT:MEAS1:CLOCKRECOVERY:DAMPING 700.0000E-3 indicating the damping value is 0.70.

MEASUREMENT:MEAS<x>:CLOCKRECOVERY:DATAPath

This command sets or queries the file containing the data pattern used when known data pattern clock recovery is used for the measurement. Measurements are specified by x.

Group	Measurement
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Syntax	<code>MEASUREMENT:MEAS<x>:CLOCKRECOVERY:DATAPATH <QString></code> <code>MEASUREMENT:MEAS<x>:CLOCKRECOVERY:DATAPATH?</code>
Arguments	<code><QString></code> is the file containing the data pattern to be used for known data pattern clock recovery.
Examples	<code>MEASUREMENT:MEAS1:CLOCKRECOVERY:DATAPATH "TEST125.txt"</code> specifies the file containing the data pattern is TEST125.txt. <code>MEASUREMENT:MEAS1:CLOCKRECOVERY:DATAPATH?</code> might return <code>:MEASUREMENT:MEAS1:CLOCKRECOVERY:DATAPATH "PRBS127.txt"</code> indicating the file containing the data pattern is PRBS127.txt.

MEASUREMENT:MEAS<x>:CLOCKRECOVERY:DATARATE

This command sets or queries the nominal data bit rate when nominal data rate clock recovery is used for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	<code>MEASUREMENT:MEAS<x>:CLOCKRECOVERY:DATARATE <NR3></code> <code>MEASUREMENT:MEAS<x>:CLOCKRECOVERY:DATARATE?</code>
Arguments	<code><NR3></code> is the value for the Nominal data rate.
Examples	<code>MEASUREMENT:MEAS1:CLOCKRECOVERY:DATARATE 2.0000E+9</code> sets the data rate to 2.0 Gb/s. <code>MEASUREMENT:MEAS1:CLOCKRECOVERY:DATARATE?</code> might return <code>:MEASUREMENT:MEAS1:CLOCKRECOVERY:DATARATE 2.5000E+9</code> indicating the data rate is 2.5 Gb/s.

MEASUREMENT:MEAS<x>:CLOCKRECOVERY:EXPLICITCLOCKMODE

This command sets or queries the explicit clock mode used when explicit clock recovery is used for the measurement. The measurement number is specified by x.

Group	Measurement
Syntax	<code>MEASUREMENT:MEAS<x>:CLOCKRECOVERY:EXPLICITCLOCKMODE {EDGE PLL}</code>

MEASUREMENT:MEAS<x>:CLOCKRECOVERY:EXPLICITCLOCKMODE?

Arguments EDGE specifies the clock edge.
 PLL specifies the phase locked loop.

Examples **MEASUREMENT:MEAS1:CLOCKRECOVERY:EXPLICITCLOCKMODE** EDGE sets the clock mode to edge.
 MEASUREMENT:MEAS1:CLOCKRECOVERY:EXPLICITCLOCKMODE? might return **:MEASUREMENT:MEAS1:CLOCKRECOVERY:EXPLICITCLOCKMODE** EDGE indicating the explicit clock mode is edge.

MEASUREMENT:MEAS<x>:CLOCKRECOVERY:GLOBAL

This command sets or queries the clock recovery settings global flag for the measurement. Measurements are specified by x.

Group Measurement

Syntax **MEASUREMENT:MEAS<x>:CLOCKRECOVERY:GLOBAL** {OFF|ON|0|1}
 MEASUREMENT:MEAS<x>:CLOCKRECOVERY:GLOBAL?

Arguments OFF clock recovery settings are changed independently for each individual measurement.
 ON applies global clock recovery settings to all the measurements' clock recovery settings.
 0 clock recovery settings are changed independently for each individual measurement.
 1 applies global clock recovery settings to all the measurements' clock recovery settings.

Examples **MEASUREMENT:MEAS1:CLOCKRECOVERY:GLOBAL** 1 applies global clock recovery settings to all the measurements' clock recovery settings.
 MEASUREMENT:MEAS1:CLOCKRECOVERY:GLOBAL? might return **:MEASUREMENT:MEAS1:CLOCKRECOVERY:GLOBAL** 0 indicating clock recovery settings are changed independently for each individual measurement.

MEASUREMENT:MEAS<x>:CLOCKRECOVERY:JTFBANDWIDTH

This command sets or queries the JTF bandwidth used when PLL clock recovery is used for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:CLOCKRECOVERY:JTFBANDWIDTH <NR3> MEASUREMENT:MEAS<x>:CLOCKRECOVERY:JTFBANDWIDTH?</pre>
Arguments	<NR3> is the clock recovery JTF bandwidth.
Examples	MEASUREMENT:MEAS1:CLOCKRECOVERY:JTFBANDWIDTH 1.10E+6 sets the bandwidth to 1.1 MHz. MEASUREMENT:MEAS1:CLOCKRECOVERY:JTFBANDWIDTH? might return :MEASUREMENT:MEAS1:CLOCKRECOVERY:JTFBANDWIDTH 1.0000E+6 indicating the bandwidth is 1.0 MHz.

MEASUREMENT:MEAS<x>:CLOCKRECOVERY:LOOPBANDWIDTH

This command sets or queries the loop bandwidth used when PLL clock recovery is used for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:CLOCKRECOVERY:LOOPBANDWIDTH <NR3> MEASUREMENT:MEAS<x>:CLOCKRECOVERY:LOOPBANDWIDTH?</pre>
Arguments	<NR3> is the clock recovery loop bandwidth.
Examples	MEASUREMENT:MEAS1:CLOCKRECOVERY:LOOPBANDWIDTH 1.10E+6 sets the bandwidth to 1.1 Mhz. MEASUREMENT:MEAS1:CLOCKRECOVERY:LOOPBANDWIDTH? might return :MEASUREMENT:MEAS1:CLOCKRECOVERY:LOOPBANDWIDTH 1.0000E+6 indicating the bandwidth is 1.0 MHz.

MEASUrement:MEAS<x>:CLOCKRecovery:MEANAUTOCalculate

This command sets or queries how often the clock is calculated when constant clock recovery is used for the measurement. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MEAS<x>:CLOCKRecovery:MEANAUTOCalculate {FIRST EVERY} MEASUrement:MEAS<x>:CLOCKRecovery:MEANAUTOCalculate?</pre>
Arguments	FIRST calculates the clock on the first acquisition. EVERY calculates the clock on every acquisition.
Examples	MEASUrement:MEAS1:CLOCKRecovery:MEANAUTOCalculate EVERY causes the clock to be calculated on every acquisition. MEASUrement:MEAS1:CLOCKRecovery:MEANAUTOCalculate? might return :MEASUREMENT:MEAS1:CLOCKRECOVERY:MEANAUTOCALCULATE FIRST indicating the clock is calculated on the first acquisition.

MEASUrement:MEAS<x>:CLOCKRecovery:METHod

This command sets or queries the clock recovery method for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MEAS<x>:CLOCKRecovery:METHod {PLL CONSTANTCLOCK EXPLICITCLOCK} MEASUrement:MEAS<x>:CLOCKRecovery:METHod?</pre>
Arguments	PLL specifies a phase locked loop clock recovery method. CONSTANTCLOCK specifies using a constant clock. EXPLICITCLOCK specifies using an explicit clock.
Examples	MEASUrement:MEAS1:CLOCKRecovery:METHod PLL specifies a phase locked loop clock recovery method.

MEASUREMENT:MEAS1:CLOCKRECOVERY:METHOd? might return
:MEASUREMENT:MEAS1:CLOCKRECOVERY:METHOd CONSTANTCLOCK
indicating a constant clock is in use.

MEASUREMENT:MEAS<x>:CLOCKRECOVERY:MODEl

This command sets or queries the PLL clock recovery model used when PLL clock recovery is used for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:CLOCKRECOVERY:MODEl {TYPE1 TYPE2} MEASUREMENT:MEAS<x>:CLOCKRECOVERY:MODEl?
Arguments	Argument is the clock recovery model type.
Examples	MEASUREMENT:MEAS1:CLOCKRECOVERY:MODEl TYPE1 set the clock recovery model to type 1. MEASUREMENT:MEAS1:CLOCKRECOVERY:MODEl? might return :MEASUREMENT:MEAS1:CLOCKRECOVERY:MODEl TYPE1 indicating the clock recovery model is type 1.

MEASUREMENT:MEAS<x>:CLOCKRECOVERY:NOMINALOFFset

This command sets or queries the offset value used when explicit clock recovery is used for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:CLOCKRECOVERY:NOMINALOFFset <NR3> MEASUREMENT:MEAS<x>:CLOCKRECOVERY:NOMINALOFFset?
Arguments	<NR3> is the clock offset.
Examples	MEASUREMENT:MEAS1:CLOCKRECOVERY:NOMINALOFFset 1.0e-9 sets the clock offset to 1 ns.

MEASUrement:MEAS1:CLOCKRecovery:NOMINALOFFset? might return
:MEASUREMENT:MEAS1:CLOCKRECOVERY:NOMINALOFFSET
0.0E+0 indicating the offset is 0.0 ns.

MEASUrement:MEAS<x>:CLOCKRecovery:NOMINALOFFset:SELECTIONtype

This command sets or queries the offset type used when explicit clock recovery is used for the measurement. The measurement number is specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:CLOCKRecovery:NOMINALOFFset: SELECTIONtype {AUTO MANUAL} MEASUrement:MEAS<x>:CLOCKRecovery:NOMINALOFFset: SELECTIONtype?
Arguments	AUTO automatically calculates the offset. MANUAL allows the user to set the offset.
Examples	MEASUrement:MEAS1:CLOCKRecovery:NOMINALOFFset:SELECTIONtype AUTO sets the offset selection type to auto. MEASUrement:MEAS1:CLOCKRecovery:NOMINALOFFset:SELECTIONtype? might return :MEASUREMENT:MEAS1:CLOCKRECOVERY:NOMINALOFFSET:SELECTIONTYPE MANUAL indicating the selection type is manual.

MEASUrement:MEAS<x>:CLOCKRecovery:STANDARD

This command sets or queries the communications standard when PLL clock recovery is used for the measurement. The measurement number is specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:CLOCKRecovery:STANDARD {CUSTOM ENET100 FW1394BS400B FW1394BS800B FW1394BS1600B FBD1 FBD2 FBD3 FC133 FC266 FC531 FC1063 FC2125 FC4250 FC8500 ENET1000 IBA2500 IBA_GEN2 OC1 OC3 OC12 OC48 PCIE_GEN1 PCIE_GEN2 PCIE_GEN3 RIO125 RIO250 RIO3125 SAS15_NOSSC SAS3_NOSSC SAS6_NOSSC SAS12_NOSSC SAS15_SSC SAS3_SSC SAS6_SSC SAS12_SSC SATA_GEN1 SATA_GEN2 SATA_GEN3 USB3 XAUI XAUI_GEN2}

Arguments	Arguments are the clock recovery standards.
Examples	MEASUREMENT:MEAS1:CLOCKRECOVERY:STANDARD PCIE_GEN2 sets the standard to PCIE_GEN2. MEASUREMENT:MEAS1:CLOCKRECOVERY:STANDARD? might return :MEASUREMENT:MEAS1:CLOCKRECOVERY:STANDARD PCIE_GEN1 indicating the standard is PCIE_GEN1.

MEASUREMENT:MEAS<x>:COMMONMODE:FILTERs:STATE

This command sets or queries whether a filter is used for the measurement when the measurement type is AC common mode. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:COMMONMODE:FILTERs:STATE {OFF ON 0 1} MEASUREMENT:MEAS<x>:COMMONMODE:FILTERs:STATE?
Arguments	OFF turns the filter off. ON turns the filter on. 0 turns the filter off. 1 turns the filter on.
Examples	MEASUREMENT:MEAS1:COMMONMODE:FILTERs:STATE ON sets the filter to on. MEASUREMENT:MEAS1:COMMONMODE:FILTERs:STATE? might return :MEASUREMENT:MEAS1:COMMONMODE:FILTERs:STATE 0 indicating the filter is off.

MEASUREMENT:MEAS<x>:COMMONMODE:SOURCEs

This command sets or queries the number of sources for the measurement when the measurement type is AC common mode. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:COMMONMODE:SOURCEs {SINGLE DOUBLE} MEASUREMENT:MEAS<x>:COMMONMODE:SOURCEs?

Arguments	SINGLE specifies a single source. DOUBLE specifies double sources.
Examples	MEASUREMENT:MEAS1:COMMONMODE:SOURCES SINGLE specifies a single source. MEASUREMENT:MEAS1:COMMONMODE:SOURCES? might return :MEASUREMENT:MEAS1:COMMONMODE:SOURCES DOUBLE indicating sources are set to double.

MEASUREMENT:MEAS<x>:CYCLEmode

This command sets or queries the cycle mode for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:CYCLEmode {RECORD CYCLE} MEASUREMENT:MEAS<x>:CYCLEmode?
Arguments	RECORD specifies that the measurement is taken over the whole record. CYCLE specifies that measurements are taken on each cycle of the source.
Examples	MEASUREMENT:MEAS1:CYCLEmode CYCLE sets the measurement to be taken over each cycle of the source. MEASUREMENT:MEAS1:CYCLEmode? might return :MEASUREMENT:MEAS1:CYCLEMODE RECORD indicating the measurement is taken over the whole record.

MEASUREMENT:MEAS<x>:DELAY:EDGE<x>

This command sets or queries the 'to edge' type when EDGE<x> is EDGE1 and the 'from edge' type when EDGE<x> is EDG2, for the measurement when the measurement type is DELAY. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:DELAY:EDGE<x> {FALL RISe BOTH SAMEas OPPositeas}

MEASUREMENT:MEAS<x>:DELAY:EDGE<x>?

Related Commands [MEASUREMENT:MEAS<x>:TOEDGE](#)

Arguments FALL specifies the falling edge of the waveform.
 RISE specifies the rising edge of the waveform.
 BOTH specifies both a rising and falling edge of the waveform.
 SAMEas specifies that both edges of the waveform are the same.
 OPPOSITEas specifies that the edges of the waveform are not the same.

Examples MEASUREMENT:MEAS3:DELAY:EDGE2 RISE specifies that the rising edge of the waveform be used for measurement 3.
 MEASUREMENT:MEAS1:DELAY:EDGE2? might return
 :MEASUREMENT:MEAS1:DELAY:EDGE2 FALL, indicating that the falling edge of the waveform is being used for measurement 1.

MEASUREMENT:MEAS<x>:DISPLAYSTAT:ENABLE

This command turns on and off display of statistics in measurement badges in the user interface. This command affects only the display of statistics, basic-statistics are computed regardless of the state of this command. Measurements are specified by x.

Group Measurement

Syntax MEASUREMENT:MEAS<x>:DISPLAYSTAT:ENABLE {OFF|ON|<NR1>}
 MEASUREMENT:MEAS<x>:DISPLAYSTAT:ENABLE?

Arguments OFF turns off the display of statistics in measurement badges.
 ON turns on the display of statistics in measurement badges.
 <NR1> = 0 turns off the display of statistics in the measurement badge, any other value turns on the display of statistics.

Examples MEASUREMENT:MEAS<x>:DISPLAYSTAT:ENABLE 0 turns off the display of statistics in the measurement badge.

MEASUREMENT:MEAS<x>:DISPLAYSTAT:ENABLE? might return
:MEASUREMENT:MEAS1:DISPLAYSTAT:ENABLE 1 indicating that the display of statistics in the measurement badge is on.

MEASUREMENT:MEAS<x>:EDGEINCRE

This command sets or queries the edge increment value for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:EDGEINCRE <NR3> MEASUREMENT:MEAS<x>:EDGEINCRE?
Arguments	<NR3> is the measurements edge increment value.
Examples	:MEASUREMENT:MEAS2:EDGEINCRE? might return :MEASUREMENT:MEAS2:EDGEINCRE 1.0000 indicating the edge increment value is set to 1.0000.

MEASUREMENT:MEAS<x>:EDGES:FROMLEVEL

This command sets or queries the 'from level' edge for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:EDGES:FROMLEVEL {MID LOW HIGH} MEASUREMENT:MEAS<x>:EDGES:FROMLEVEL?
Arguments	MID specifies the MID level. HIGH specifies the HIGH level. LOW specifies the LOW level.
Examples	MEASUREMENT:MEAS1:EDGES:FROMLEVEL HIGH set the from level to the high level

MEASUREMENT:MEAS1:EDGES:FROMLEVEL? might return
:MEASUREMENT:MEAS1:EDGES:FROMLEVEL LOW indicating the from
level is the low level.

MEASUREMENT:MEAS<x>:EDGES:LEVEL

This sets or queries the level type for the 'time outside level' measurement.
Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:EDGES:LEVEL {HIGH LOW BOTH} MEASUREMENT:MEAS<x>:EDGES:LEVEL?
Arguments	HIGH specifies the HIGH level. LOW specifies the LOW level. BOTH specifies both the HIGH and LOW level.
Examples	MEASUREMENT:MEAS1:EDGES:LEVEL LOW sets the level to the low level. MEASUREMENT:MEAS1:EDGES:LEVEL? might return :MEASUREMENT:MEAS1:EDGES:LEVEL HIGH indicating the level is set to the high level.

MEASUREMENT:MEAS<x>:EDGES:LOWERFREQUENCY

This command sets or queries the lower frequency for the measurement when the
measurement type is phase noise. Lower frequencies are ignored. Measurements
are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:EDGES:LOWERFREQUENCY <NR3> MEASUREMENT:MEAS<x>:EDGES:LOWERFREQUENCY?
Arguments	<NR3> is the lower frequency of the edge.
Examples	MEASUREMENT:MEAS1:EDGES:LOWERFREQUENCY 1.0e3 sets the lower frequency to 1 kHz.

MEASUREMENT:MEAS1:EDGES:LOWERFREQUENCY? might return
:MEASUREMENT:MEAS1:EDGES:LOWERFREQUENCY 0.0E+0 indicating the
lower frequency is set to 0.0 Hertz.

MEASUREMENT:MEAS<x>:EDGES:N

The command sets or queries the number of accumulation cycles for the measurement when the measurement type is nperiod. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:EDGES:N <NR3> MEASUREMENT:MEAS<x>:EDGES:N?
Arguments	<NR3> is the maximum number of edges used by the measurement.
Examples	measurement:meas1:edges:n 2 sets the number of edges to 2. measurement:meas1:edges:n? might return :MEASUREMENT:MEAS1:EDGES:N 6.0000.

MEASUREMENT:MEAS<x>:EDGES:SLEWRATEMethod

This command sets or queries the slew rate method for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:EDGES:SLEWRATEMethod {NOMINAL DDR} MEASUREMENT:MEAS<x>:EDGES:SLEWRATEMethod?
Arguments	NOMINAL specifies the nominal slew rate method. DDR specifies the DDR slew rate method.
Examples	measurement:meas1:edges:slewratemethod NOMINAL specifies the nominal slew rate method.

`measurement:meas1:edges:slewratemethod?` might return
`:MEASUREMENT:MEAS1:EDGES:SLEWRATEMETHOD NOMINAL` indicating the
slew rate method is set to NOMINAL.

MEASUrement:MEAS<x>:EDGES:TOLevel

This command sets or queries the 'to level' edge for the measurement.
Measurements are specified by x.

Group	Measurement
Syntax	<code>MEASUrement:MEAS<x>:EDGES:TOLevel {HIGH MID LOW}</code> <code>MEASUrement:MEAS<x>:EDGES:TOLevel?</code>
Arguments	HIGH specifies the HIGH level. MID specifies the MID level. LOW specifies the LOW level.
Examples	<code>MEASUrement:MEAS1:EDGES:TOLevel HIGH</code> specifies the High to level. <code>MEASUrement:MEAS1:EDGES:TOLevel?</code> might return <code>:MEASUREMENT:MEAS1:EDGES:TOLEVEL LOW</code> indicating the to level edge is set to the Low level.

MEASUrement:MEAS<x>:EDGES:UPPERFREQUENCY

This command sets or queries the upper frequency for the measurement when the
measurement type is phase noise. Higher frequencies are ignored. Measurements
are specified by x.

Group	Measurement
Syntax	<code>MEASUrement:MEAS<x>:EDGES:UPPERFREQUENCY <NR3></code> <code>MEASUrement:MEAS<x>:EDGES:UPPERFREQUENCY?</code>
Arguments	<NR3> is the upper frequency of the edge.
Examples	<code>MEASUrement:MEAS1:EDGES:UPPERFREQUENCY 1.5</code> sets the upper frequency to 1.5 MHz.

MEASUREMENT:MEAS1:EDGES:UPPERFREQUENCY? might return
:MEASUREMENT:MEAS1:EDGES:UPPERFREQUENCY 1.0000E+6 indicating the
upper frequency is 1.0 MHz.

MEASUREMENT:MEAS<x>:EDGE<x>

This command sets or queries the type of the specified edge, rise or fall, for the measurement. The measurement number is specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:EDGE<x> {RISE FALL BOTH}
Arguments	RISE specifies the rising edge. FALL specifies the falling edge. BOTH specifies either the rising or falling edge.
Examples	MEASUREMENT:MEAS2:EDGE1? might return :MEASUREMENT:MEAS2:EDGE1 RISE indicating that edge 1 of measurement 2 is the rising edge.

MEASUREMENT:MEAS<x>:FILTERS:BLANKINGtime

This command sets or queries the filter blanking time for the measurement.
Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:FILTERS:BLANKINGtime <NR3> MEASUREMENT:MEAS<x>:FILTERS:BLANKINGtime?
Arguments	<NR3> is the current filter blanking time.
Examples	MEASUREMENT:MEAS1:FILTERS:BLANKINGtime 3.5 sets the blanking time to 3.5. MEASUREMENT:MEAS1:FILTERS:BLANKINGtime? might return :MEASUREMENT:MEAS1:FILTERS:BLANKINGTIME 4.0000 indicating the filter blanking time is 4.0.

MEASUrement:MEAS<x>:FILTerS:GLOBal

This command sets or queries the global flag for filter settings for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MEAS<x>:FILTerS:GLOBal {OFF ON 0 1} MEASUrement:MEAS<x>:FILTerS:GLOBal?</pre>
Arguments	OFF causes filter settings to be changed independently for each individual measurement. ON applies global filter settings to all the measurements' filter settings. 0 causes filter settings to be changed independently for each individual measurement. 1 applies global filter settings to all the measurements' filter settings.
Examples	MEASUrement:MEAS<x>:FILTerS:GLOBal OFF causes filter settings to be changed independently for each individual measurement. MEASUrement:MEAS<x>:FILTerS:GLOBal? might return :MEASUREMENT:MEAS1:FILTERS:GLOBAL 1 indicating filter settings are for all measurements.

MEASUrement:MEAS<x>:FILTerS:HIGHPass:FREQ

This command sets or queries the high pass filter frequency for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MEAS<x>:FILTerS:HIGHPass:FREQ <NR3> MEASUrement:MEAS<x>:FILTerS:HIGHPass:FREQ?</pre>
Arguments	<NR3> is the current high pass filter frequency.
Examples	MEASUrement:MEAS1:FILTerS:HIGHPass:FREQ 15.0E+6 sets the high pass frequency to 15.0 MHz.

MEASUREMENT:MEAS1:FILTERS:HIGHPASS:FREQ? might return
:MEASUREMENT:MEAS1:FILTERS:HIGHPASS:FREQ 10.0000E+6 indicating
the high pass frequency is set to 10.0 MHz.

MEASUREMENT:MEAS<x>:FILTERS:HIGHPASS:SPEC

This command sets or queries the high pass filter order for the measurement.
Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:FILTERS:HIGHPASS:SPEC {NONE FIRST SECOND THIRD} MEASUREMENT:MEAS<x>:FILTERS:HIGHPASS:SPEC?
Arguments	NONE specifies no filter. FIRST specifies a first-order filter. SECOND specifies a second-order filter. THIRD specifies a third-order filter.
Examples	MEASUREMENT:MEAS1:FILTERS:HIGHPASS:SPEC FIRST specifies a first-order filter. MEASUREMENT:MEAS1:FILTERS:HIGHPASS:SPEC? might return :MEASUREMENT:MEAS1:FILTERS:HIGHPASS:SPEC NONE indicating no high pass filter is used.

MEASUREMENT:MEAS<x>:FILTERS:LOWPASS:FREQ

This command sets or queries the low pass filter cutoff frequency for the
measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:FILTERS:LOWPASS:FREQ <NR3> MEASUREMENT:MEAS<x>:FILTERS:LOWPASS:FREQ?
Arguments	<NR3> is the current low pass filter frequency.

Examples `MEASUREMENT:MEAS1:FILTERs:LOWPass:FREQ 5.0E+6` sets the low pass frequency to 5.0 MHz.

`MEASUREMENT:MEAS1:FILTERs:LOWPass:FREQ?` might return
`:MEASUREMENT:MEAS1:FILTERs:LOWPASS:FREQ 10.0000E+6` indicating
the low pass frequency is 10.0 MHz.

MEASUREMENT:MEAS<x>:FILTERs:LOWPass:SPEC

This command sets or queries the low pass filter order for the measurement.
Measurements are specified by x.

Group Measurement

Syntax `MEASUREMENT:MEAS<x>:FILTERs:LOWPass:SPEC`
`{NONE|FIRST|SECOND|THIRD}`
`MEASUREMENT:MEAS<x>:FILTERs:LOWPass:SPEC?`

Arguments `NONE` specifies no filter.
`FIRST` specifies a first-order filter.
`SECOND` specifies a second-order filter.
`THIRD` specifies a third-order filter.

Examples `MEASUREMENT:MEAS1:FILTERs:LOWPass:SPEC FIRST` specifies a first-order filter.

`MEASUREMENT:MEAS1:FILTERs:LOWPass:SPEC?` might return
`:MEASUREMENT:MEAS1:FILTERs:LOWPASS:SPEC NONE` indicating no low
pass filter is used.

MEASUREMENT:MEAS<x>:FILTERs:RAMptime

This command sets or queries the filter ramp time for the measurement.
Measurements are specified by x.

Group Measurement

Syntax `MEASUREMENT:MEAS<x>:FILTERs:RAMptime <NR3>`
`MEASUREMENT:MEAS<x>:FILTERs:RAMptime?`

Arguments	<NR3> is the current filter ramp time.
Examples	MEASUREMENT:MEAS1:FILTERS:RAMPtime 1.5 sets the ramp time to 1.5. MEASUREMENT:MEAS1:FILTERS:RAMPtime? might return :MEASUREMENT:MEAS1:FILTERS:RAMPtime 2.0000 indicating the ramp time is 2.0.

MEASUREMENT:MEAS<x>:FROMedge

This command sets or queries the from edge type for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:FROMedge {RISe FALL BOTH} MEASUREMENT:MEAS<x>:FROMedge?
Arguments	FALL specifies the falling edge of the waveform. RISe specifies the rising edge of the waveform. BOTH specifies both the rising and falling edges of the waveform.
Examples	MEASUREMENT:MEAS1:FROMedge RISe specifies the rising edge of the waveform. MEASUREMENT:MEAS1:FROMedge? might return :MEASUREMENT:MEAS1:FROMedge BOTH indicating both the rising and falling edges of the waveform.

MEASUREMENT:MEAS<x>:FROMEDGESEARCHDIRect

This command sets or queries the from edge search direction for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:FROMEDGESEARCHDIRect {FORward BACKward} MEASUREMENT:MEAS<x>:FROMEDGESEARCHDIRect?

Arguments	FORWARD specifies a forward search from the edge. BACKWARD specifies a backward search from the edge.
Examples	MEASUREMENT:MEAS1:FROMEDGESEARCHDIRECT BACKWARD specifies a backward search from the edge. MEASUREMENT:MEAS1:FROMEDGESEARCHDIRECT? might return :MEASUREMENT:MEAS1:FROMEDGESEARCHDIRECT FORWARD indicating a forward search from the edge.

MEASUREMENT:MEAS<x>:GATING

This command sets or queries the gating type for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:GATING {NONE SCREEN CURSOR LOGIC SEARCH} MEASUREMENT:MEAS<x>:GATING?
Arguments	NONE specifies measurements are taken across the entire record. SCREEN turns on gating, using the left and right edges of the screen. CURSOR limits measurements to the portion of the waveform between the vertical bar cursors, even if they are off screen. LOGIC specifies that measurements are taken only when the logical state of other waveforms is true. SEARCH specifies that measurements are taken only where the results of a user specified search are found.
Examples	MEASUREMENT:MEAS1:GATING CURSOR limits measurements to the portion of the waveform between the vertical bar cursor. MEASUREMENT:MEAS1:GATING? might return :MEASUREMENT:MEAS1:GATING NONE indicating measurements are taken across the entire record.

MEASUREMENT:MEAS<x>:GATING:ACTIVE

This command sets or queries the gating active level when the gating type is logic. Measurements are specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:GATING:ACTIVE {HIGH LOW} MEASUREMENT:MEAS<x>:GATING:ACTIVE?</pre>
Arguments	HIGH takes a measurement when logic gating is High. LOW takes a measurement when logic gating Low.
Examples	MEASUREMENT:MEAS1:GATING:ACTIVE LOW specifies taking measurements when logic gating is Low. MEASUREMENT:MEAS1:GATING:ACTIVE? might return :MEASUREMENT:MEAS1:GATING:ACTIVE HIGH indicating measurements are taken when logic gating is High.

MEASUREMENT:MEAS<x>:GATING:GLOBAL

This command sets or queries the gating settings global flag. Measurements are specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:GATING:GLOBAL {OFF ON 0 1} MEASUREMENT:MEAS<x>:GATING:GLOBAL?</pre>
Arguments	OFF specifies gate settings can be changed independently for each individual measurement. ON applies global gate settings to all the measurements' gate settings. 0 specifies gate settings can be changed independently for each individual measurement. 1 applies global gate settings to all the measurements' gate settings.
Examples	MEASUREMENT:MEAS1:GATING:GLOBAL OFF specifies gate settings can be changed independently for each individual measurement. MEASUREMENT:MEAS1:GATING:GLOBAL? might return :MEASUREMENT:MEAS1:GATING:GLOBAL 1 indicating that gating settings apply to all measurements.

MEASUREMENT:MEAS<x>:GATING:HYSTERESIS

This command sets or queries the gating hysteresis value when the gating type is logic. Measurements are specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:GATING:HYSTERESIS <NR3> MEASUREMENT:MEAS<x>:GATING:HYSTERESIS?</pre>
Arguments	<NR3> is the gating hysteresis.
Examples	MEASUREMENT:MEAS1:GATING:HYSTERESIS 25.0E-3 sets the hysteresis to 25 mV. MEASUREMENT:MEAS1:GATING:HYSTERESIS? might return :MEASUREMENT:MEAS1:GATING:HYSTERESIS 30.0000E-3 indicating the hysteresis is set to 30.0 mV.

MEASUREMENT:MEAS<x>:GATING:LOGICSOURCE

This command sets or queries the gating data source when the gating type is logic. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:GATING:LOGICSOURCE {CH<x> MATH<x> REF<x>}</pre>
Arguments	Arguments are the sources for logic gating.
Examples	MEASUREMENT:MEAS1:GATING:LOGICSOURCE CH3 sets the gating logic source to channel 3. MEASUREMENT:MEAS1:GATING:LOGICSOURCE? might return :MEASUREMENT:MEAS1:GATING:LOGICSOURCE CH2 indicating the gating logic source is channel 2.

MEASUrement:MEAS<x>:GATing:MIDRef

This command sets or queries the gating mid ref value when the gating type is logic. Measurements are specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:GATing:MIDRef <NR3> MEASUrement:MEAS<x>:GATing:MIDRef?
Arguments	<NR3> is the mid ref value for gating.
Examples	MEASUrement:MEAS1:GATing:MIDRef 1.0E+0 sets the gating midref to 1.0. MEASUrement:MEAS1:GATing:MIDRef? might return :MEASUREMENT:MEAS1:GATING:MIDREF 0.0E+0 indicating the midref value is set to 0.0.

MEASUrement:MEAS<x>:GATing:SEARCHSource

This command sets or queries the gating search source when the gating type is search. The measurement number is specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:GATing:SEARCHSource SEARCH1
Arguments	SEARCH1 is the gating source for search gating.
Examples	MEASUrement:MEAS1:GATing:SEARCHSource SEARCH1 sets the gating search source to SEARCH1. MEASUrement:MEAS1:GATing:SEARCHSource? might return :MEASUREMENT:MEAS1:GATING:SEARCHSOURCE SEARCH1 indicating the gating search source is SEARCH1.

MEASUrement:MEAS<x>:GLOBalref

This command sets or queries the reference levels global flag for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:GLOBALref {OFF ON 0 1} MEASUREMENT:MEAS<x>:GLOBALref?</pre>
Arguments	OFF allows ref levels to be set separately for each measurement. ON applies the same ref levels to all measurements. 0 allows ref levels to be set separately for each measurement. 1 applies the same ref levels to all measurements.
Examples	MEASUREMENT:MEAS1:GLOBALref 0 allows ref levels to be set separately for each measurement. MEASUREMENT:MEAS1:GLOBALref? might return :MEASUREMENT:MEAS1:GLOBALREF 1 indicating the same ref levels apply to all measurements.

MEASUREMENT:MEAS<x>:HIGHREFVOLTAGE

This command sets or queries the high reference voltage value for the 'time outside level' measurement. Measurements are specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:HIGHREFVOLTAGE <NR3> MEASUREMENT:MEAS<x>:HIGHREFVOLTAGE?</pre>
Arguments	<NR3> is the high reference voltage value for the selected configuration.
Examples	MEASUREMENT:MEAS1:HIGHREFVOLTAGE 1.5 sets the high reference voltage to 1.5 V. MEASUREMENT:MEAS1:HIGHREFVOLTAGE? might return :MEASUREMENT:MEAS1:HIGHREFVOLTAGE 1.0000 indicating the high reference voltage is set to 1.0 V.

MEASUrement:MEAS<x>:IDLETime

This command sets or queries the idle time for the measurement when the measurement type is burst width. Measurements are specified by x.

Group Measurement

Syntax MEASUrement:MEAS<x>:IDLETime <NR3>
MEASUrement:MEAS<x>:IDLETime?

Arguments <NR3> is the idle time.

Examples MEASUrement:MEAS1:IDLETime 40.0E-6 sets the idle time to 40.0 μ s.
MEASUrement:MEAS1:IDLETime? might return
:MEASUREMENT:MEAS1:IDLETIME 50.0000E-6 indicating the idle time
is 50.0 μ s.

MEASUrement:MEAS<x>:JITTERSummary:DCD

This command sets or queries whether DCD is included in the jitter summary for the measurement. Measurements are specified by x.

Group Measurement

Syntax MEASUrement:MEAS<x>:JITTERSummary:DCD {0|1}
MEASUrement:MEAS<x>:JITTERSummary:DCD?

Arguments 1 add the DCD measurement as part of jitter summary.
0 do not add the DCD measurement as part of jitter summary.

Examples MEASUrement:MEAS1:JITTERSummary:DCD 0 specifies that the DCD
measurement is not part of the jitter summary.
MEASUrement:MEAS1:JITTERSummary:DCD? might return
:MEASUREMENT:MEAS4:JITTERSUMMARY:DCD 1 indicating the DCD
measurement is part of the jitter summary.

MEASUrement:MEAS<x>:JITTERSummary:DDJ

This command sets or queries whether DDJ is included in the jitter summary for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:JITTERSummary:DDJ {0 1} MEASUrement:MEAS<x>:JITTERSummary:DDJ?
Arguments	1 add the DDJ measurement as part of jitter summary. 0 do not add the DDJ measurement as part of jitter summary.
Examples	MEASUrement:MEAS4:JITTERSummary:DDJ 0 specifies that the DDJ measurement is not part of the jitter summary. MEASUrement:MEAS4:JITTERSummary:DDJ? might return :MEASUREMENT:MEAS4:JITTERSUMMARY:DDJ 1 indicating the DDJ measurement is part of the jitter summary.

MEASUrement:MEAS<x>:JITTERSummary:DJDD

This command sets or queries whether DJ-dd is included in the jitter summary for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:JITTERSummary:DJDD {0 1} MEASUrement:MEAS<x>:JITTERSummary:DJDD?
Arguments	1 add the DJDD measurement as part of jitter summary. 0 do not add the DJDD measurement as part of jitter summary.
Examples	MEASUrement:MEAS4:JITTERSummary:DJDD 0 specifies the DJDD measurement is not part of the jitter summary. MEASUrement:MEAS4:JITTERSummary:DJDD? might return :MEASUREMENT:MEAS4:JITTERSUMMARY:DJDD 1 indicating the DJDD measurement is part of the jitter summary.

MEASUrement:MEAS<x>:JITTERSummary:EYEWIDTHBER

This command sets or queries whether EyeWidth@BER is included in the jitter summary for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:JITTERSummary:EYEWIDTHBER {0 1} MEASUrement:MEAS<x>:JITTERSummary:EYEWIDTHBER?
Arguments	1 add the EyeWidth@BER measurement as part of jitter summary. 0 do not add the EyeWidth@BER measurement as part of jitter summary.
Examples	MEASUrement:MEAS4:JITTERSummary:EYEWIDTHBER 0 specifies that the EyeWidth@BER measurement is not part of the jitter summary. MEASUrement:MEAS4:JITTERSummary:EYEWIDTHBER? might return :MEASUREMENT:MEAS4:JITTERSUMMARY:EYEWIDTHBER 1 indicating EyeWidth@BER is part of the jitter summary

MEASUrement:MEAS<x>:JITTERSummary:NPJ

This command sets or queries whether NPJ is included in the jitter summary for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:JITTERSummary:NPJ {0 1} MEASUrement:MEAS<x>:JITTERSummary:NPJ?
Arguments	1 add the NPJ measurement as part of jitter summary. 0 do not add the NPJ measurement as part of jitter summary.
Examples	MEASUrement:MEAS4:JITTERSummary:NPJ 0 specifies that the NPJ measurement is not part of jitter summary. MEASUrement:MEAS4:JITTERSummary:NPJ? might return :MEASUREMENT:MEAS4:JITTERSUMMARY:NPJ 1 indicating the NPJ measurement is part of the jitter summary.

MEASUrement:MEAS<x>:JITTERSummary:PJ

This command sets or queries whether PJ is included in the jitter summary for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	<code>MEASUrement:MEAS<x>:JITTERSummary:PJ {0 1}</code> <code>MEASUrement:MEAS<x>:JITTERSummary:PJ?</code>
Arguments	1 add the PJ measurement as part of jitter summary. 0 do not add the PJ measurement as part of jitter summary.
Examples	<code>MEASUrement:MEAS4:JITTERSummary:PJ 0</code> specifies the PJ measurement is not part of jitter summary. <code>MEASUrement:MEAS4:JITTERSummary:PJ?</code> might return <code>:MEASUREMENT:MEAS4:JITTERSUMMARY:PJ 1</code> indicating the PJ measurement is part of jitter summary.

MEASUrement:MEAS<x>:JITTERSummary:RJDD

This command sets or queries whether RJ-dd is included in the jitter summary for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	<code>MEASUrement:MEAS<x>:JITTERSummary:RJDD {0 1}</code> <code>MEASUrement:MEAS<x>:JITTERSummary:RJDD?</code>
Arguments	1 add the RJ-dd measurement as part of jitter summary. 0 do not add the RJ-dd measurement as part of jitter summary.
Examples	<code>MEASUrement:MEAS4:JITTERSummary:RJDD 0</code> specifies the RJ-dd measurement is not part of the jitter summary. <code>MEASUrement:MEAS4:JITTERSummary:RJDD?</code> might return <code>:MEASUREMENT:MEAS4:JITTERSUMMARY:RJDD 1</code> indicating the RJ-dd measurement is part of the jitter summary.

MEASUrement:MEAS<x>:JITTERSummary:TIE

This command sets or queries whether TIE is included in the jitter summary for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:JITTERSummary:TIE {0 1} MEASUrement:MEAS<x>:JITTERSummary:TIE?
Arguments	1 add the TIE measurement as part of jitter summary. 0 do not add the TIE measurement as part of jitter summary.
Examples	MEASUrement:MEAS4:JITTERSummary:TIE 0 specifies the TIE measurement is not part of jitter summary. MEASUrement:MEAS4:JITTERSummary:TIE? might return :MEASUREMENT:MEAS4:JITTERSUMMARY:TIE 1 indicating the TIE measurement is part of jitter summary.

MEASUrement:MEAS<x>:JITTERSummary:TJBER

This command sets or queries whether TJ@BER is included in the jitter summary for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:JITTERSummary:TJBER {0 1} MEASUrement:MEAS<x>:JITTERSummary:TJBER?
Arguments	1 add the TJ@BER measurement as part of jitter summary. 0 do not add the TJ@BER measurement as part of jitter summary.
Examples	MEASUrement:MEAS4:JITTERSummary:TJBER 0 specifies that the TJ@BER measurement is not part of jitter summary. MEASUrement:MEAS4:JITTERSummary:TJBER? might return :MEASUREMENT:MEAS4:JITTERSUMMARY:TJBER 1 indicating the TJ@BER measurement is part of jitter summary.

MEASUrement:MEAS<x>:LABel

This command sets or queries the label for the measurement. As the label can contain non 7-bit ASCII text, it is stored in Percent Encoding format. The measurement number is specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:LABel <QString>
Arguments	<QString> is the measurement label.
Examples	MEASUrement:MEAS1:LABel "Delay" sets the label to Delay. MEASUrement:MEAS1:LABel? might return :MEASUREMENT:MEAS1:LABEL "Peak-to-Peak" indicating that the measurement 1 label is Peak-to-peak.

MEASUrement:MEAS<x>:LOWREFVoltage

This command sets or queries the low reference voltage value for the 'time outside level' measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:LOWREFVoltage <NR3> MEASUrement:MEAS<x>:LOWREFVoltage?
Arguments	<NR3> is the low reference voltage value for the selected configuration.
Examples	MEASUrement:MEAS1:LOWREFVoltage -1.30 sets the low reference voltage to -1.3 V. MEASUrement:MEAS1:LOWREFVoltage? might return :MEASUREMENT:MEAS1:LOWREFVOLTAGE -1.0000 indicating the low ref voltage is -1.0 V.

MEASUrement:MEAS<x>:MAXCycle

This command sets or queries the maximum cycle value for the DDRTERRN and DDRTERRMN measurements.

Conditions	Requires option 6-DBDDR3 Requires 6 Series MSO oscilloscope
Group	Measurements
Syntax	MEASurement:MEAS<x>:MAXCyc1e <NR1> MEASurement:MEAS<x>:MAXCyc1e?
Arguments	<NR1> is the maximum cycle range limit value in the range or 2 to 50.
Examples	MEASurement:MEAS2:MAXCyc1e 45 sets the maximum cycle value for measurement 2 to 45. MEASurement:MEAS1:MAXCyc1e? might return MEASurement:MEAS1:MAXCyc1e 17, indicating the maximum cycle value is set to 17 for measurement 1.

MEASurement:MEAS<x>:MEASRange:GLOBal

This command sets or queries the range settings global flag for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASurement:MEAS<x>:MEASRange:GLOBal {OFF ON 0 1} MEASurement:MEAS<x>:MEASRange:GLOBal?
Arguments	OFF specifies that range settings can be set independently for each individual measurement. ON applies global measurement range settings to all the measurements' range settings. 0 specifies that range settings can be set independently for each individual measurement. 1 applies global measurement range settings to all the measurements' range settings.
Examples	MEASurement:MEAS1:MEASRange:GLOBal 1 applies global measurement range settings to all the measurements' range settings.

MEASUREMENT:MEAS1:MEASRange:GLOBal? might return
:MEASUREMENT:MEAS1:MEASRange:GLOBAL 0 indicating that range settings
can be set independently for each individual measurement.

MEASUREMENT:MEAS<x>:MEASRange:MAX

This command sets or queries the range maximum value for the measurement.
Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:MEASRange:MAX <NR3> MEASUREMENT:MEAS<x>:MEASRange:MAX?
Arguments	<NR3> is the maximum measurement range limit value.
Examples	MEASUREMENT:MEAS1:MEASRange:MAX 2.50 sets the maximum range to 2.5 V. MEASUREMENT:MEAS1:MEASRange:MAX? might return :MEASUREMENT:MEAS1:MEASRange:MAX 1.0000 indicating the maximum range is 1.0 V.

MEASUREMENT:MEAS<x>:MEASRange:MIN

This command sets or queries the range minimum value for the measurement.
Measurements are specified by x.

Group	Measurement
Syntax	MEASUREMENT:MEAS<x>:MEASRange:MIN <NR3> MEASUREMENT:MEAS<x>:MEASRange:MIN?
Arguments	<NR3> is the minimum measurement range limit value.
Examples	MEASUREMENT:MEAS1:MEASRange:MIN -1.0 sets the minimum measurement range to -1.0 V. MEASUREMENT:MEAS1:MEASRange:MIN? might return :MEASUREMENT:MEAS1:MEASRange:MIN 0.0E+0 indicating the minimum range is 0.0 V.

MEASUrement:MEAS<x>:MEASRange:STATE

This command sets or queries the range state for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:MEASRange:STATE {OFF ON 0 1} MEASUrement:MEAS<x>:MEASRange:STATE?
Arguments	OFF turns off the measurement range limits. ON turns on the measurement range limits. 0 turns off the measurement range limits. 1 turns on the measurement range limits.
Examples	MEASUrement:MEAS1:MEASRange:STATE 0 turns off the measurement range limits. MEASUrement:MEAS1:MEASRange:STATE? might return :MEASUREMENT:MEAS1:MEASRANGE:STATE 1 indicating the measurement range limits are on.

MEASUrement:MEAS<x>:MINCycle

This command sets or queries the minimum cycle value for the DDRTERRN and DDRTERRMN measurements.

Conditions	Requires option 6-DBDDR3 Requires 6 Series MSO oscilloscope
Group	Measurements
Syntax	MEASUrement:MEAS<x>:MINCycle <NR1> MEASUrement:MEAS<x>:MINCycle?
Arguments	<NR1> is the minimum cycle range limit value in the range or 2 to 50.

Examples `MEASurement:MEAS2:MINCycle 20` sets the minimum cycle value for measurement 2 to 20.

`MEASurement:MEAS1:MINCycle?` might return
 `MEASurement:MEAS1:MINCycle 17`, indicating the minimum cycle value is set to 17 for measurement 1.

MEASUrement:MEAS<x>:PATTERNDETECTION

This command sets or queries the pattern detection type for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	<code>MEASurement:MEAS<x>:PATTERNDETECTION {AUTO MANUAL}</code> <code>MEASurement:MEAS<x>:PATTERNDETECTION?</code>
Arguments	AUTO automatically detects the pattern. MANUAL requires manually detecting the pattern.
Examples	<code>MEASurement:MEAS1:PATTERNDETECTION MANUAL</code> specifies manually detecting the pattern. <code>MEASurement:MEAS1:PATTERNDETECTION?</code> might return <code>:MEASUREMENT:MEAS1:PATTERNDETECTION AUTO</code> indicating patterns are automatically detected.

MEASUrement:MEAS<x>:PATTERNLENgth

This command sets or queries the pattern length for the measurement. Measurements are specified by x.

Group	Measurement
Syntax	<code>MEASurement:MEAS<x>:PATTERNLENgth <NR3></code> <code>MEASurement:MEAS<x>:PATTERNLENgth?</code>
Arguments	<NR3> is the pattern length.

Examples `MEASUREMENT:MEAS1:PATTERNLENGTH 3.0` sets the pattern length to 3.0.
 `MEASUREMENT:MEAS1:PATTERNLENGTH?` might return
 `:MEASUREMENT:MEAS1:PATTERNLENGTH 2.0000` indicating the
 patternlength is 2.0.

MEASUREMENT:MEAS<x>:PATTERNType

This command sets or queries the pattern type for the measurement. Measurements are specified by x.

Group Measurement

Syntax `MEASUREMENT:MEAS<x>:PATTERNType {REPeating|ARBiTrary}`
 `MEASUREMENT:MEAS<x>:PATTERNType?`

Arguments `REPeating` specifies a repeating pattern.
 `ARBiTrary` specifies an arbitrary pattern.

Examples `MEASUREMENT:MEAS1:PATTERNType ARBITRARY` specifies an arbitrary pattern.
 `MEASUREMENT:MEAS1:PATTERNType?` might return
 `:MEASUREMENT:MEAS1:PATTERNType REPEATING` indicating a repeating
 pattern.

MEASUREMENT:MEAS<x>:PERFREQ:EDGE

This command sets or queries the edge type of a Period/Frequency measurement. The measurement number is specified by x.

Group Measurement

Syntax `MEASUREMENT:MEAS<x>:PERFREQ:EDGE {FIRST|RISE|FALL}`

Arguments `FIRST` computes the measurement between Rising edges if the first edge is Rising.
 Computes the measurement between Falling edges if the first edge is Falling.
 `RISE` computes the measurement between Rising edges.
 `FALL` computes the measurement between Falling edges.

Examples `MEASUREMENT:MEAS1:PERFREQ:EDGE RISE` specifies computing the measurement between Rising edges.

`measurement:meas1:perfreq:edge?` might return
`:MEASUREMENT:MEAS1:PERFREQ:EDGE FIRST` if the measurement is computed between edges of the first type edge found.

MEASUREMENT:MEAS<x>:POLARITY

This command sets or queries the polarity for the measurement when the measurement type is burst width. Measurements are specified by x.

Group Measurement

Syntax `MEASUREMENT:MEAS<x>:POLARITY {NORMAL|INVERTED}`
`MEASUREMENT:MEAS<x>:POLARITY?`

Arguments `NORMAL` specifies normal polarity.
`INVERTED` specifies inverted polarity.

Examples `MEASUREMENT:MEAS1:POLARITY INVERTED` specifies inverted polarity.
`MEASUREMENT:MEAS1:POLARITY?` might return
`:MEASUREMENT:MEAS1:POLARITY NORMAL` indicating normal polarity.

MEASUREMENT:MEAS<x>:POPULATION:GLOBAL

This command sets or queries the population settings global flag. The measurement number is specified by x.

Group Measurement

Syntax `MEASUREMENT:MEAS<x>:POPULATION:GLOBAL {OFF|ON|0|1}`
`MEASUREMENT:MEAS<x>:POPULATION:GLOBAL?`

Arguments `OFF` specifies that population settings can be changed independently for each individual measurement.

`ON` applies the global population settings to all the measurements' population settings.

0 specifies that population settings can be changed independently for each individual measurement.

1 applies the global population settings to all the measurements' population settings.

Examples `MEASUREMENT:MEAS1:POPULATION:GLOBAL 1` applies the global population settings to all the measurements' population settings.

`MEASUREMENT:MEAS1:POPULATION:GLOBAL?` might return
`:MEASUREMENT:MEAS1:POPULATION:GLOBAL 0` indicating that population settings can be changed independently for each individual measurement.

MEASUREMENT:MEAS<x>:POPULATION:LIMIT:STATE

This command sets or queries the population limit state for the measurement. The measurement number is specified by x.

Group Measurement

Syntax `MEASUREMENT:MEAS<x>:POPULATION:LIMIT:STATE {OFF|ON|0|1}`
`MEASUREMENT:MEAS<x>:POPULATION:LIMIT:STATE?`

Arguments OFF turns off the population limit.
 ON turns on the population limit.
 0 turns off the population limit.
 1 turns on the population limit.

Examples `MEASUREMENT:MEAS1:POPULATION:LIMIT:STATE 0` turns off the population limit.

`MEASUREMENT:MEAS1:POPULATION:LIMIT:STATE?` might return
`:MEASUREMENT:MEAS1:POPULATION:LIMIT:STATE 1` indicating the population limit is on.

MEASUREMENT:MEAS<x>:POPULATION:LIMIT:VALUE

This command sets or queries the population limit value for the measurement. The measurement number is specified by x.

Group Measurement

Syntax	<pre>MEASUREMENT:MEAS<x>:POPULATION:LIMIT:VALUE <NR3> MEASUREMENT:MEAS<x>:POPULATION:LIMIT:VALUE?</pre>
Arguments	<NR3> the current limit value.
Examples	MEASUREMENT:MEAS1:POPULATION:LIMIT:VALUE 2000 sets the population limit to 2000. MEASUREMENT:MEAS1:POPULATION:LIMIT:VALUE? might return :MEASUREMENT:MEAS1:POPULATION:LIMIT:VALUE 1000 indicating the limit is set to 1000.

MEASUREMENT:MEAS<x>:REFLEVELS:ABSOLUTE:FALLHIGH

This command sets or queries the value used as the high reference level of the falling edge when the measurement's ref level method is set to absolute. Measurements are specified by x.

NOTE. *This command affects the results of rise and fall measurements*

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:REFLEVELS:ABSOLUTE:FALLHIGH <NR3> MEASUREMENT:MEAS<x>:REFLEVELS:ABSOLUTE:FALLHIGH?</pre>
Arguments	<NR3> is the high reference level in volts. The default is 0.0 V.
Examples	MEASUREMENT:MEAS1:REFLEVELS:ABSOLUTE:FALLHIGH 1.71 sets the high reference level to 1.71 V. MEASUREMENT:MEAS1:REFLEVELS:ABSOLUTE:FALLHIGH? might return :MEASUREMENT:MEAS1:REFLEVELS:ABSOLUTE:FALLHIGH 1.7100E+00, indicating that the absolute high reference level is set to 1.71 V.

MEASUREMENT:MEAS<x>:REFLEVELS<x>:ABSOLUTE:FALLLOW

This command sets or queries the value used as the low reference level of the falling edge when the measurement's ref level method is set to absolute. Measurements are specified by x.

NOTE. *This command affects the results of rise and fall measurements.*

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:REFLEVELS<x>:ABSOLUTE:FALLLOW <NR3> MEASUREMENT:MEAS<x>:REFLEVELS<x>:ABSOLUTE:FALLLOW?</pre>
Arguments	<NR3> is the low reference level in volts. The default is 0.0 V.
Examples	MEASUREMENT:MEAS1:REFLEVELS1:ABSOLUTE:FALLLOW 0.0 V sets the low reference level to 0.0 V. MEASUREMENT:MEAS1:REFLEVELS1:ABSOLUTE:FALLLOW? might return :MEASUREMENT:MEAS1:REFLEVELS1:ABSOLUTE:FALLLOW 0.0000E+00, indicating that the absolute low reference level is set to 0.0 V.

MEASUREMENT:MEAS<x>:REFLEVELS<x>:ABSOLUTE:FALLMID

This command sets or queries the value used as the mid reference level of the falling edge when the measurement's ref level method is set to absolute. Measurements are specified by x.

NOTE. *This command affects the results of period, frequency, delay, and all cyclic measurements.*

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:REFLEVELS<x>:ABSOLUTE:FALLMID <NR3> MEASUREMENT:MEAS<x>:REFLEVELS<x>:ABSOLUTE:FALLMID?</pre>
Arguments	<NR3> is the mid reference level in volts. The default is 0.0 V.
Examples	MEASUREMENT:MEAS1:REFLEVELS1:ABSOLUTE:FALLMID 0.5 sets the mid reference level for the delay waveform to 0.5 V. MEASUREMENT:MEAS1:REFLEVELS1:ABSOLUTE:FALLMID? might return :MEASUREMENT:MEAS1:REFLEVELS1:ABSOLUTE:FALLMID 0.5000E+00, indicating that the absolute mid reference level is set to 0.5 V.

MEASUREMENT:MEAS<x>:REFLEVELS<x>:ABSOLUTE:HYSTERESIS

This command sets or queries the value of the hysteresis of the reference level when the measurement's ref level method is set to absolute. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:REFLEVELS<x>:ABSOLUTE:HYSTERESIS <NR3> MEASUREMENT:MEAS<x>:REFLEVELS<x>:ABSOLUTE:HYSTERESIS?</pre>
Arguments	<NR3> is the hysteresis value used for autoset.
Examples	MEASUREMENT:MEAS1:REFLEVELS1:ABSOLUTE:HYSTERESIS 25.0E-3 sets the hysteresis to 25 mV. MEASUREMENT:MEAS1:REFLEVELS1:ABSOLUTE:HYSTERESIS? might return :MEASUREMENT:MEAS1:REFLEVELS1:ABSOLUTE:HYSTERESIS 30.0000E-3 indicating the hysteresis is set to 30 mV.

MEASUREMENT:MEAS<x>:REFLEVELS<x>:ABSOLUTE:RISEHIGH

This command sets or queries the value used as the high reference level of the rising edge when the measurement's ref level method is set to absolute. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:REFLEVELS<x>:ABSOLUTE:RISEHIGH <NR3> MEASUREMENT:MEAS<x>:REFLEVELS<x>:ABSOLUTE:RISEHIGH?</pre>
Arguments	<NR3> is the high reference level, and is the zero percent level when the measurement's Ref level method is set to Absolute.
Examples	MEASUREMENT:MEAS1:REFLEVELS1:ABSOLUTE:RISEHIGH 1.50 sets the high reference level to 1.5 V. MEASUREMENT:MEAS1:REFLEVELS1:ABSOLUTE:RISEHIGH? might return :MEASUREMENT:MEAS1:REFLEVELS1:ABSOLUTE:RISEHIGH 1.0000 indicating the high reference level is set to 1.0 V.

MEASUrement:MEAS<x>:REFLevels<x>:ABSolute:RISELow

This command sets or queries the value used as the low reference level of the rising edge when the measurement's ref level method is set to absolute. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MEAS<x>:REFLeveLs<x>:ABSolute:RISELow <NR3> MEASUrement:MEAS<x>:REFLeveLs<x>:ABSolute:RISELow?</pre>
Arguments	<NR3> is the low reference level, and is the zero percent level when the measurement's Ref level method is set to Absolute.
Examples	MEASUrement:MEAS1:REFLeveLs1:ABSolute:RISELow -1.50 sets the low reference level to -1.5 V. MEASUrement:MEAS1:REFLeveLs1:ABSolute:RISELow? might return :MEASUREMENT:MEAS1:REFLEVELS1:ABSOLUTE:RISELOW -1.0000 indicating the low reference level is -1.0 V.

MEASUrement:MEAS<x>:REFLevels<x>:ABSolute:RISEMid

This command sets or queries the value used as the mid reference level of the rising edge when the measurement's ref level method is set to absolute. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MEAS<x>:REFLeveLs<x>:ABSolute:RISEMid <NR3> MEASUrement:MEAS<x>:REFLeveLs<x>:ABSolute:RISEMid?</pre>
Arguments	<NR3> is the mid reference level (where 50% is equal to MID) used to calculate the mid reference level when the measurement's Ref level method is set to Absolute.
Examples	MEASUrement:MEAS1:REFLeveLs1:ABSolute:RISEMid 30.0E-3 sets the mid reference level to 30 mV. MEASUrement:MEAS1:REFLeveLs1:ABSolute:RISEMid? might return :MEASUREMENT:MEAS1:REFLEVELS1:ABSOLUTE:RISEMID 0.0E+0 indicating the mid reference level is 0.0 V.

MEASUREMENT:MEAS<x>:REFLEVELS<x>:ABSOLUTE:TYPE

This command sets or queries the reference level type for the measurement. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:REFLEVELS<x>:ABSOLUTE:TYPE {SAME UNIQUE} MEASUREMENT:MEAS<x>:REFLEVELS<x>:ABSOLUTE:TYPE?</pre>
Arguments	SAME specifies that the absolute levels are set the same. UNIQUE specifies that the absolute levels can be set independently.
Examples	MEASUREMENT:MEAS1:REFLEVELS1:ABSOLUTE:TYPE UNIQUE specifies that the absolute levels can be set independently. MEASUREMENT:MEAS1:REFLEVELS1:ABSOLUTE:TYPE? might return :MEASUREMENT:MEAS1:REFLEVELS1:ABSOLUTE:TYPE SAME indicating the absolute levels are set the same.

MEASUREMENT:MEAS<x>:REFLEVELS<x>:BASETOP

This command sets or queries the method used to calculate the TOP and BASE used to calculate reference levels for the measurement. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:REFLEVELS<x>:BASETOP {AUTO MINMAX MEANHISTOGRAM MODEHISTOGRAM EYEHISTOGRAM} MEASUREMENT:MEAS<x>:REFLEVELS<x>:BASETOP?</pre>
Arguments	AUTO automatically chooses a reference level method. MINMAX specifies that reference levels are relative to the measurement MIN and MAX. MEANHISTOGRAM specifies that reference levels are relative to the histogram mean BASE and TOP. MODEHISTOGRAM specifies that reference levels are relative to the histogram mode BASE and TOP.

EYEHistogram specifies that reference levels are relative to the eye histogram BASE and TOP.

Examples `MEASUREMENT:MEAS1:REFLEVELS1:BASETOP MINMAX` specifies that reference levels are relative to the measurement MIN and MAX.

`MEASUREMENT:MEAS1:REFLEVELS1:BASETOP?` might return
`:MEASUREMENT:MEAS1:REFLEVELS1:BASETOP AUTO` indicating the instrument automatically chooses a reference level method.

MEASUREMENT:MEAS<x>:REFLEVELS<x>:METHOD

This command sets or queries the method used to calculate reference levels for the measurement. The measurement number is specified by x.

Group Measurement

Syntax `MEASUREMENT:MEAS<x>:REFLEVELS<x>:METHOD {PERCENT|ABSOLUTE}`
`MEASUREMENT:MEAS<x>:REFLEVELS<x>:METHOD?`

Arguments **PERCENT** specifies that the reference levels are calculated as a percent relative to HIGH and LOW. The percentages are defined using the `MEASUREMENT:MEAS<x>:REFLEVEL:PERCENT` commands.

ABSOLUTE specifies that the reference levels are set explicitly using the `MEASUREMENT:MEAS<x>:REFLEVEL:ABSOLUTE` commands. This method is useful when precise values are required.

Examples `MEASUREMENT:MEAS1:REFLEVELS1:METHOd ABSOLUTE` specifies that explicit user-defined values are used for the reference levels.

`MEASUREMENT:MEAS1:REFLEVELS1:METHOd?` might return
`:MEASUREMENT:MEAS1:REFLEVELS1:METHOd PERCENT`, indicating that the reference level units used are calculated as a percent relative to HIGH and LOW.

MEASUREMENT:MEAS<x>:REFLEVELS<x>:PERCENT:FALLHigh

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the high reference level of the falling edge when the measurement's ref level method is set to percent. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:REFLEVELS<x>:PERCENT:FALLHIGH <NR3> MEASUREMENT:MEAS<x>:REFLEVELS<x>:PERCENT:FALLHIGH?</pre>
Arguments	<NR3> is the percentage (where 100% is equal to HIGH) used to calculate the high reference level.
Examples	MEASUREMENT:MEAS1:REFLEVELS1:PERCENT:FALLHIGH 95.0 sets the high reference level of the falling edge to 95%. MEASUREMENT:MEAS1:REFLEVELS1:PERCENT:FALLHIGH? might return :MEASUREMENT:MEAS1:REFLEVELS1:PERCENT:FALLHIGH 90.0000 indicating the high reference level is set to 90%.

MEASUREMENT:MEAS<x>:REFLEVELS<x>:PERCENT:FALLLOW

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the low reference level of the falling edge when the measurement's ref level method is set to percent. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:REFLEVELS<x>:PERCENT:FALLLOW <NR3> MEASUREMENT:MEAS<x>:REFLEVELS<x>:PERCENT:FALLLOW?</pre>
Arguments	<NR3> is the percentage (where 100% is equal to HIGH) used to calculate the mid reference level.
Examples	MEASUREMENT:MEAS1:REFLEVELS1:PERCENT:FALLLOW 5.0 sets the low reference level of the falling edge is 5%. MEASUREMENT:MEAS1:REFLEVELS1:PERCENT:FALLLOW? might return :MEASUREMENT:MEAS1:REFLEVELS1:PERCENT:FALLLOW 10.0000 indicating the low reference level of the falling edge is 10%.

MEASUREMENT:MEAS<x>:REFLEVELS<x>:PERCENT:FALLMID

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the falling edge

when the measurement's ref level method is set to percent. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MEAS<x>:REFLeveLs<x>:PERCent:FALLMid <NR3> MEASUrement:MEAS<x>:REFLeveLs<x>:PERCent:FALLMid?</pre>
Arguments	<NR3> is the percentage (where 50% is equal to MID) used to calculate the mid reference level.
Examples	MEASUrement:MEAS1:REFLeveLs1:PERCent:FALLMid 50.0 sets the mid reference level of the falling edge to 50%. MEASUrement:MEAS1:REFLeveLs1:PERCent:FALLMid? might return :MEASUREMENT:MEAS1:REFLEVELS1:PERCENT:FALLMID 50.0000 indicating the mid reference level of the falling edge is set to 50%.

MEASUrement:MEAS<x>:REFLeveLs<x>:PERCent:HYSTeresis

This command sets or queries the percentage (where 100% is equal to MAX and 0% is equal to MIN) used to calculate the hysteresis of the reference level when the measurement's ref level method is set to percent. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MEAS<x>:REFLeveLs<x>:PERCent:HYSTeresis <NR3> MEASUrement:MEAS<x>:REFLeveLs<x>:PERCent:HYSTeresis?</pre>
Arguments	<NR3> is the hysteresis value used for the autoset.
Examples	MEASUrement:MEAS1:REFLeveLs1:PERCent:HYSTeresis 5.0 sets the hysteresis level to 5%. MEASUrement:MEAS1:REFLeveLs1:PERCent:HYSTeresis? might return :MEASUREMENT:MEAS1:REFLEVELS1:PERCENT:HYSTERESIS 5.0000 indicating the hysteresis is 5.0%.

MEASUrement:MEAS<x>:REFLevels<x>:PERCent:RISEHigh

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the high reference level of the rising edge when the measurement's ref level method is set to percent. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MEAS<x>:REFLevels<x>:PERCent:RISEHigh <NR3> MEASUrement:MEAS<x>:REFLevels<x>:PERCent:RISEHigh?</pre>
Arguments	<NR3> is the percentage (where 100% is equal to TOP) used to calculate the high reference level when the measurement's Ref level method is set to Percent.
Examples	MEASUrement:MEAS1:REFLevels1:PERCent:RISEHigh 95.0 sets the high reference level of the rising edge is set to 95%. MEASUrement:MEAS1:REFLevels1:PERCent:RISEHigh? might return :MEASUREMENT:MEAS1:REFLEVELS1:PERCENT:RISEHIGH 90.0000 indicating the high reference level of the rising edge is set to 90%.

MEASUrement:MEAS<x>:REFLevels<x>:PERCent:RISELow

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the low reference level of the rising edge when the measurement's ref level method is set to percent. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MEAS<x>:REFLevels<x>:PERCent:RISELow <NR3> MEASUrement:MEAS<x>:REFLevels<x>:PERCent:RISELow?</pre>
Arguments	<NR3> is the percentage (where 100% is equal to TOP) used to calculate the mid reference level when the measurement's Ref level method is set to Percent.
Examples	MEASUrement:MEAS1:REFLevels1:PERCent:RISELow 5.0 sets the low reference level is set to 5.0%.

MEASUrement:MEAS1:REFLeve1s1:PERCent:RISELow? might return
:MEASUREMENT:MEAS1:REFLEVELS1:PERCENT:RISELOW 10.0000
indicating the low reference level is set to 10.0%.

MEASUrement:MEAS<x>:REFLevels<x>:PERCent:RISEMid

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the rising edge when the measurement's ref level method is set to percent. The measurement number is specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:REFLeve1s<x>:PERCent:RISEMid <NR3> MEASUrement:MEAS<x>:REFLeve1s<x>:PERCent:RISEMid?
Arguments	<NR3> the percentage (where 50% is equal to MID) used to calculate the mid reference level when the measurement Ref level method is set to Percent.
Examples	MEASUrement:MEAS1:REFLeve1s1:PERCent:RISEMid 50.0000 sets the mid reference level of the rising edge is set to 50.0%. MEASUrement:MEAS1:REFLeve1s1:PERCent:RISEMid? might return :MEASUREMENT:MEAS1:REFLEVELS1:PERCENT:RISEMID 50.0000 indicating the mid reference level of the rising edge is set to 50.0%.

MEASUrement:MEAS<x>:REFLevels<x>:PERCent:TYPE

This command specifies or queries the reference level percent type for the measurement. The measurement number is specified by x.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:REFLeve1s<x>:PERCent:TYPE {TENNinety TWENTyeighty CUSTOM} MEASUrement:MEAS<x>:REFLeve1s<x>:PERCent:TYPE?
Arguments	TENNinety sets the values for Low, Mid and High Ref to 10%, 50% and 90% respectively.

TWENTyeighty sets the values for Low, Mid and High Ref are set to 20%, 50% and 80% respectively.

CUSTom allows setting other reference level percents.

Examples MEASUREMENT:MEAS1:REFLEVELS1:PERCENT:TYPE TWENTyeighty sets the reference levels percent to 20%, 50% and 80%.

MEASUREMENT:MEAS1:REFLEVELS1:PERCENT:TYPE? might return
:MEASUREMENT:MEAS1:REFLEVELS1:PERCENT:TYPE TENNINETY indicating the reference levels percent type is 10%, 50% and 90% respectively

MEASUREMENT:MEAS<x>:REFMode

This command sets or queries the reference level mode for the measurement. The measurement number is specified by x.

Group Measurement

Syntax MEASUREMENT:MEAS<x>:REFMode {AUTO|MANual}
MEASUREMENT:MEAS<x>:REFMode?

Arguments AUTO sets the reference level for the measurement automatically.
MANual allows the user to set the reference level for the measurement.

Examples MEASUREMENT:MEAS1:REFMode Manua1 allows the user to set the reference level for the measurement.
MEASUREMENT:MEAS1:REFMode? might return
:MEASUREMENT:MEAS1:REFMODE AUTO indicating the reference levels for the measurement are set automatically.

MEASUREMENT:MEAS<x>:REFVoltage

This command sets or queries the reference voltage value for the measurement. The measurement number is specified by x.

Group Measurement

Syntax MEASUREMENT:MEAS<x>:REFVoltage <NR3>
MEASUREMENT:MEAS<x>:REFVoltage?

Arguments	<NR3> is the reference voltage value for the selected configuration.
Examples	<pre>MEASUREMENT:MEAS1:REFVOLTAGE 30.E-3</pre> sets the reference voltage value for the measurement to 30 mV. <pre>MEASUREMENT:MEAS1:REFVOLTAGE?</pre> might return <pre>:MEASUREMENT:MEAS1:REFVOLTAGE 0.0E+0</pre> indicating the reference voltage value for the measurement is 0.0 V.

MEASUREMENT:MEAS<x>:RESULTS:ALLACQS:MAXIMUM? (Query Only)

This query-only command returns the maximum value for all accumulated measurement acquisitions of the specified measurement. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:RESULTS:ALLACQS:MAXIMUM?</pre>
Returns	The maximum value for all accumulated measurement acquisitions of the specified measurement.
Examples	<pre>MEASUREMENT:MEAS1:RESULTS:ALLACQS:MAXIMUM?</pre> might return <pre>:MEASUREMENT:MEAS1:RESULTS:ALLACQS:MAXIMUM 2.420</pre> indicating the maximum measurement value is 2.420 V.

MEASUREMENT:MEAS<x>:RESULTS:ALLACQS:MEAN? (Query Only)

This query-only command returns the mean value for all accumulated measurement acquisitions for measurement <x>.

Group	Measurement
Syntax	<pre>MEASUREMENT:MEAS<x>:RESULTS:ALLACQS:MEAN?</pre>
Returns	The mean value for all accumulated measurement acquisitions for measurement <x>.

Examples `MEASUREMENT:MEAS1:RESULTS:ALLACQS:MEAN?` might return
 `:MEASUREMENT:MEAS1:RESULTS:ALLACQS:MEAN 2.2807617754647`
 indicating the measurement mean is 2.28 V.

MEASUREMENT:MEAS<x>:RESULTS:ALLACQS:MINIMUM? (Query Only)

This query-only command returns the minimum value for all accumulated measurement acquisitions for measurement <x>.

Group Measurement

Syntax `MEASUREMENT:MEAS<x>:RESULTS:ALLACQS:MINIMUM?`

Returns The minimum value for all accumulated measurement acquisitions for measurement <x>.

Examples `MEASUREMENT:MEAS1:RESULTS:ALLACQS:MINIMUM?` might return
 `:MEASUREMENT:MEAS1:RESULTS:ALLACQS:MINIMUM 2.220` indicating the
 measurement minimum value is 2.220 V.

MEASUREMENT:MEAS<x>:RESULTS:ALLACQS:PK2PK? (Query Only)

This query-only command returns the peak-to-peak value for all accumulated measurement acquisitions for measurement <x>.

Group Measurement

Syntax `MEASUREMENT:MEAS<x>:RESULTS:ALLACQS:PK2PK?`

Returns The peak-to-peak value for all accumulated measurement acquisitions for measurement <x>.

Examples `MEASUREMENT:MEAS1:RESULTS:ALLACQS:PK2PK?` might return
 `:MEASUREMENT:MEAS1:RESULTS:ALLACQS:PK2PK 200.0E-3` indicating the
 measurement peak-to-peak value is 200 mV.

MEASUrement:MEAS<x>:RESUlts:ALLAcqs:POPULation? (Query Only)

This query-only command returns the population measurement value for measurement <x>.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:RESUlts:ALLAcqs:POPULation?
Returns	The population measurement value for measurement <x>.
Examples	MEASUrement:MEAS1:RESUlts:ALLAcqs:POPULation? might return :MEASUREMENT:MEAS1:RESULTS:ALLACQS:POPULATION 29825 indicating the measurement population is 29825 measurements.

MEASUrement:MEAS<x>:RESUlts:ALLAcqs:STDDev? (Query Only)

This query-only command returns the standard deviation for all accumulated measurement acquisitions for measurement <x>.

Group	Measurement
Syntax	MEASUrement:MEAS<x>:RESUlts:ALLAcqs:STDDev?
Returns	The standard deviation for all accumulated measurement acquisitions for measurement <x>.
Examples	MEASUrement:MEAS1:RESUlts:ALLAcqs:STDDev? might return :MEASUREMENT:MEAS1:RESULTS:ALLACQS:STDDEV 23.5741246494459E-3 indicating the measurement standard deviation is 23.57 mV.

MEASUrement:MEAS<x>:RESUlts:CURRentacq:MAXimum? (Query Only)

This query-only command returns the maximum value found for the specified measurement since the last statistical reset. The measurement number is specified by x.

Group	Measurement
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Syntax	<code>MEASurement:MEAS<x>:RESUlts:CURRentacq:MAXimum?</code>
Returns	The maximum value found for the specified measurement since the last statistical reset.
Examples	<code>MEASurement:MEAS1:RESUlts:CURRentacq:MAXimum?</code> might return <code>:MEASUREMENT:MEAS1:RESULTS:CURRENTACQ:MAXIMUM 2.280</code> indicating the current measurement maximum value is 2.28 V.

MEASurement:MEAS<x>:RESUlts:CURRentacq:MEAN? (Query Only)

This query-only command returns the mean value for the measurement for the current acquisition.

Group	Measurement
Syntax	<code>MEASurement:MEAS<x>:RESUlts:CURRentacq:MEAN?</code>
Arguments	<code>MEAS<x></code> is the measurement number for which to return a value. This is the equivalent of the number shown in the measurement badge on the UI.
Returns	The mean value accumulated for the specified measurement since the last statistical reset.
Examples	<code>MEASurement:MEAS3:RESUlts:CURRentacq:MEAN?</code> might return <code>:MEASUREMENT:MEAS3:RESULTS:CURRENTACQ:MEAN 2.30</code> indicating the mean of measurement 3 in the current acquisition is 2.3 V.

MEASurement:MEAS<x>:RESUlts:CURRentacq:MINimum? (Query Only)

This query-only command returns the minimum value found for the specified measurement since the last statistical reset. The measurement number is specified by `x`.

Group	Measurement
Syntax	<code>MEASurement:MEAS<x>:RESUlts:CURRentacq:MINimum?</code>

Returns The minimum value found for the specified measurement since the last statistical reset.

Examples `MEASUREMENT:MEAS1:RESULTS:CURRENTACQ:MINIMUM?` might return `:MEASUREMENT:MEAS1:RESULTS:CURRENTACQ:MINIMUM 2.260` indicating the minimum value of the measurement in the current acquisition is 2.26 V.

MEASUREMENT:MEAS<x>:RESULTS:CURRENTACQ:PK2PK? (Query Only)

This query-only command returns the peak-to-peak value for the specified measurement for the current acquisition. The measurement number is specified by x.

Group Measurement

Syntax `MEASUREMENT:MEAS<x>:RESULTS:CURRENTACQ:PK2PK?`

Returns The peak-to-peak value for the specified measurement.

Examples `MEASUREMENT:MEAS1:RESULTS:CURRENTACQ:PK2PK?` might return `:MEASUREMENT:MEAS1:RESULTS:CURRENTACQ:PK2PK 0.0E+0` indicating the peak-to-peak value for the specified measurement for the current acquisition is 0.0 V.

MEASUREMENT:MEAS<x>:RESULTS:CURRENTACQ:POPULATION? (Query Only)

This query-only command returns the population for the specified measurement for the current acquisition. The measurement number is specified by x.

Group Measurement

Syntax `MEASUREMENT:MEAS<x>:RESULTS:CURRENTACQ:POPULATION?`

Returns The population measurement value for the specified measurement.

Examples `MEASUREMENT:MEAS<x>:RESULTS:CURRENTACQ:POPULATION?` might return `:MEASUREMENT:MEAS1:RESULTS:CURRENTACQ:POPULATION 1` indicating the population for the current measurement is 1 measurement.

MEASUrement:MEAS<x>:RESUlts:CURRentacq:STDDev? (Query Only)

This query-only command returns the standard deviation for the specified measurement for all acquisitions accumulated since statistics were last reset. The measurement number is specified by x.

Group Measurement

Syntax MEASUrement:MEAS<x>:RESUlts:CURRentacq:STDDev?

Returns The standard deviation of values accumulated for the specified measurement since the last statistical reset.

Examples MEASUrement:MEAS<x>:RESUlts:CURRentacq:STDDev? might return
:MEASUREMENT:MEAS1:RESULTS:CURRENTACQ:STDDEV 0.0E+0 indicating
the standard deviation for the measurement is 0.0 V.

MEASUrement:MEAS<x>:SIGNALType

This command sets or queries the signal type of source 1 for the measurement. The measurement number is specified by x.

Group Measurement

Syntax MEASUrement:MEAS<x>:SIGNALType {CLOCK|DATA|AUTO}
MEASUrement:MEAS<x>:SIGNALType?

Arguments CLOCK specifies a clock signal type.
DATA specifies a data signal type.
AUTO automatically selects the signal type.

Examples MEASUrement:MEAS1:SIGNALType CLOCK specifies a clock signal type.
MEASUrement:MEAS1:SIGNALType? might return
:MEASUREMENT:MEAS1:SIGNALTYPE AUTO indicating the instrument
automatically selects the signal type.

MEASUrement:MEAS<x>:SOUrce<x>

This command sets or queries the measurement source. The measurement number and source are specified by x.

Group Measurement

Syntax MEASUrement:MEAS<x>:SOUrce<x>

Arguments

Examples

MEASUrement:MEAS<x>:SSC:NOMinalfreq

This command sets or queries the user-defined frequency for the measurement when the measurement type is SSC. The measurement number is specified by x.

Group Measurement

Syntax MEASUrement:MEAS<x>:SSC:NOMinalfreq <NR3>
MEASUrement:MEAS<x>:SSC:NOMinalfreq?

Arguments <NR3> is the user-defined nominal frequency type for SSC configurations.

Examples MEASUrement:MEAS1:SSC:NOMinalfreq 2.0E+9 sets the frequency to 2.0 GHz.

MEASUrement:MEAS1:SSC:NOMinalfreq? might return
:MEASUREMENT:MEAS1:SSC:NOMINALFREQ 2.5000E+9 indicating
the frequency is 2.5 GH.

MEASUrement:MEAS<x>:SSC:NOMinalfreq:SELECTIONtype

This command sets or queries the frequency detection type for the measurement when the measurement type is SSC. The measurement number is specified by x.

Group Measurement

Syntax	<pre>MEASurement:MEAS<x>:SSC:NOMinalfreq:SELECTIONtype {AUTO MANual} MEASurement:MEAS<x>:SSC:NOMinalfreq:SELECTIONtype?</pre>
Arguments	AUTO automatically sets the detection type. MANual specifies the manual detection type.
Examples	MEASurement:MEAS1:SSC:NOMinalfreq:SELECTIONtype AUTO specifies the auto-detection type. MEASurement:MEAS1:SSC:NOMinalfreq:SELECTIONtype? might return :MEASUREMENT:MEAS1:SSC:NOMINALFREQ:SELECTIONTYPE AUTO indicating the auto-detection type is selected.

MEASurement:MEAS<x>:TCKAVG

This command sets or queries the average clock period value used in DDR measurements.

Group	Measurement
Syntax	<pre>MEASurement:MEAS<x>:TCKAVG {EACHCLOCKCYCLE TWOLOCKCYCLES} MEASurement:MEAS<x>:TCKAVG?</pre>
Related Commands	MEASurement:MEAS<x>:TIMINGMode
Arguments	EACHCLOCKCYCLE sets the DDR Timing Mode to use each clock cycle at a time. TWOLOCKCYCLES sets the DDR Timing Mode to use two cycles at a time.
Examples	MEASurement:MEAS1:TIMINGMode EACHCLOCKCYCLE sets the Timing Mode to consider each clock cycle. MEASurement:MEAS1:TIMINGMode? might return MEASurement:MEAS1:TIMINGMode EACHCLOCKCYCLE, indicating that Timing Mode is set to EACHCLOCKCYCLE.

MEASurement:MEAS<x>:TIMINGMode

This command sets or queries the Timing mode for the specified DDR measurement.

Group	Measurement
Syntax	<pre>MEASurement:MEAS<x>:TIMINGMode <NR3> MEASurement:MEAS<x>:TIMINGMode?</pre>
Related Commands	MEASurement:MEAS<x>:TCKAVG
Arguments	NR3 is a floating point number that represents the DDR average clock period in seconds.
Examples	MEASurement:MEAS7:TCKAVG 2.5E-9 sets the average clock period value as 3.2 ns for measurement 7. MEASurement:MEAS2:TCKAVG? might return MEASurement:MEAS2:TCKAVG 2.0E-9 indicating that average clock period is set to 2.0 ns for measurement 2.

MEASurement:MEAS<x>:TOEdge

This command sets or queries the 'to edge' type for the measurement. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASurement:MEAS<x>:TOEdge {SAMEas OPPositeas RISe FALL BOTH} MEASurement:MEAS<x>:TOEdge?</pre>
Arguments	FALL specifies the falling edge of the waveform. RISe specifies the rising edge of the waveform. BOTH specifies both a rising and falling edge of the waveform. SAMEas specifies that both edges of the waveform are the same. OPPositeas specifies that the edges of the waveform are not the same.
Examples	MEASurement:MEAS1:TOEdge FALL specifies the to edge is the falling edge of the waveform. MEASurement:MEAS1:TOEdge? might return :MEASUREMENT:MEAS1:TOEDGE SAMEAS indicating that both edges of the waveform are the same.

MEASUrement:MEAS<x>:TOEDGESEARCHDIRect

This command sets or queries the to edge search direction for the measurement. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MEAS<x>:TOEDGESEARCHDIRect {FORWARD BACKWARD} MEASUrement:MEAS<x>:TOEDGESEARCHDIRect?</pre>
Arguments	FORWARD specifies a forward search to the edge. BACKWARD specifies a backward search to the edge.
Examples	MEASUrement:MEAS1:TOEDGESEARCHDIRect FORWARD specifies a forward search to the edge. MEASUrement:MEAS1:TOEDGESEARCHDIRect? might return :MEASUREMENT:MEAS1:TOEDGESEARCHDIRECT FORWARD indicating the instrument will search in the forward direction for the to edge.

MEASUrement:MEAS<x>:TRANSition

This command sets or queries the transition edges flag for the measurement. The measurement number is specified by x.

Group	Measurement
Syntax	<pre>MEASUrement:MEAS<x>:TRANSition {<NR1> OFF ON}</pre>
Arguments	<NR1> = 1, the measurement is computed on rising (if measurement type is rise time) or falling edges (if measurement type is fall time) following a double transition only. If it is set to 0, the measurement is computed on all rising (if measurement type is rise time) or falling (if measurement type is fall time) edges. OFF computes the measurement on all rising (if measurement type is rise time) or falling (if measurement type is fall time) edges. ON computes the measurement on rising (if measurement type is rise time) or falling edges (if measurement type is fall time) following a double transition only.

Examples `MEASUREMENT:MEAS1:TRANSITION 1` specifying the measurement is computed on rising (if measurement type is rise time) or falling edges (if measurement type is fall time) following a double transition only.

`MEASUREMENT:MEAS1:TRANSITION?` might return
`:MEASUREMENT:MEAS1:TRANSITION 0` indicating the measurement is computed on all rising (if measurement type is rise time) or falling (if measurement type is fall time) edges.

MEASUREMENT:MEAS<x>:TYPE

This command sets or queries the measurement type for the measurement specified by x.

Group Measurement

Syntax `MEASUREMENT:MEAS<x>:TYPE`
`{ACCOMMONMODE|ACRMS|AMPLITUDE|AREA|BASE|BITAMPLITUDE|BITHIGH|`
`BITLOW|BURSTWIDTH|COMMONMODE|DATARATE|DCD|DDJ|DDRAOS|`
`DDRAOSPERTCK|DDRAOSPERUI|DDRAUS|DDRAUSPERTCK|DDRAUSPERUI|`
`DDRHOLDDIFF|DDRSETUPDIFF|DDRTCHABS|DDRTCHAVERAGE|`
`DDRTCKAVERAGE|DDRTCLABS|DDRTCLAVERAGE|DDRTERRMN|DDRTERRN|`
`DDRTJITCC|DDRTJITDUTY|DDRTJITPER|DDRTPST|DDRTRPRE|`
`DDRTWPRE|DDRVIXAC|DDRTDQSCK|DELAY|DJ|DJDIRAC|`
`EYEHIGH|EYELOW|FALLSLEWRATE|FALLTIME|FREQUENCY|F2|F4|F8|`
`HIGH|HEIGHT|HEIGHTBER|HIGHTIME|HOLD|JITTERSUMMARY|`
`J2|J9|LOW|LOWTIME|MAXIMUM|MEAN|MINIMUM|NDUTY|NPERIOD|`
`NPJ|NOVERSHOOT|NWIDTH|PDUTTY|PERIOD|PHASE|PHASENOISE|PJ|`
`PK2PK|POVERSHOOT|PWIDTH|QFACTOR|RISESLEWRATE|RISETIME|`
`RJ|RJDIRAC|RMS|SRJ|SSCFREQDEV|SSCMODRATE|SETUP|SKEW|TIE|`
`TIMEOUTSIDELEVEL|TJBER|TNTRATIO|TOP|UNITINTERVAL|`
`VDIFFXOVR|WIDTH|WIDTHBER}`
`MEASUREMENT:MEAS<x>:TYPE?`

Arguments `ACCOMMONMODE` AC Common Mode (Pk-Pk) is the peak-to-peak of the common mode voltage of two sources. This measurement is made across the entire record.

`ACRMS` (AC RMS) is the true Root Mean Square of the data points, about the Mean. This measurement can be made across the entire record, or on each cycle in the record.

`AMPLITUDE` is the difference between the Top value and the Base value. This measurement can be made across the entire record, or on each cycle in the record.

$$\textit{Amplitude} = \textit{High} - \textit{Low}$$

`AREA` is the area under the curve, calculated by integrating the data points. The area measured above ground is positive. The area measured below ground is

negative. This measurement can be made across the entire record, or on each cycle in the record.

BASE is the most common data value below the midpoint of the waveform. This measurement can be made across the entire record, or on each cycle in the record.

BITAMPLITUDE (Bit Amplitude) is the difference between the amplitudes of the 1 bit and the 0 bit surrounding a transition. The amplitude is measured over a user specified portion at the center of the recovered unit interval. This measurement is made on each transition bit in the record (Mean) or across the entire record (Mode).

BITHIGH (Bit High) is the amplitude of a 1 bit. The amplitude is measured over a user specified portion at the center of the recovered unit interval. This measurement is made on each high bit in the record (Mean) or across the entire record (Mode).

BITLOW (Bit Low) is the amplitude of a 0 bit. The amplitude is measured over a user specified portion at the center of the recovered unit interval. This measurement is made on each high bit in the record (Mean) or across the entire record (Mode).

BURSTWIDTH (Burst Width) is the duration of a series of adjacent crossings of the Mid reference level (RM). Bursts are separated by a user-defined idle time (tI). This measurement is made on each burst in the record.

COMMONMODE (DC Common Mode) is the arithmetic mean of the common mode voltage of two sources. This measurement is made across the entire record.

DATARATE (Data Rate) is the reciprocal of Unit Interval. This measurement is made on each bit in the record.

DCD (duty cycle distortion) is the peak-to-peak amplitude of the component of the deterministic jitter correlated with the signal polarity. This measurement is made across the entire record.

DDJ (data dependent jitter) is the peak-to-peak amplitude of the component of the deterministic jitter correlated with the data pattern in the waveform. This measurement is made across the entire record.

DDRAOS (area above signal) is the total area of the signal above a specified reference level. This measurement is made across the entire record.

DDRAOSPERTCK (area over signal for tCK events) is the total area of the signal above a specified reference level calculated over consecutive tCK intervals. It is applicable to clock and address/command waveforms.

DDRAOSPERUI (area over signal for UI events) is the total area of the signal above a specified reference level calculated over consecutive unit intervals. It is applicable to data and data strobe waveforms.

DDRAUS (area under signal) is the total area of the signal below a specified reference level. This measurement is made across the entire record.

DDRAUSPERTCK (area under signal for tCK events) is the total area of the signal below a specified reference level calculated over consecutive tCK intervals. It is applicable to clock and address/command waveforms.

DDRAUSPERUI (area under signal for UI events) is the total area of the signal below a specified reference level calculated over consecutive unit intervals. It is applicable to data and data strobe waveforms.

DDRHOLDDIFF (hold difference) is the elapsed time between the specified edge of a single-ended clock waveform and the specified edge of a differential data waveform. The measurement uses the closest respective waveform edges that fall within the range limits. This measurement is made across the entire record.

DDRSETUPDIFF (setup difference) is the elapsed time between the specified edge of a single-ended clock waveform and when the specified edge of a differential data waveform crosses a specified level. The measurement uses the closest respective waveform edges that fall within the range limits. This measurement is made across the entire record.

DDRTCHABS (absolute high pulse width) is the absolute value of the high pulse width as measured from one rising edge to the next falling edge.

DDRTCHAVERAGE (average high pulse width) is the average value of the high pulse width as measured from one rising edge to the next falling edge, across 200 consecutive cycles. This measurement is made across the entire record.

DDRTCKAVERAGE (average clock period) is the average clock period calculated from rising edge to rising edge, across 200 consecutive cycles. This measurement is made across the entire record.

DDRTCLABS (absolute low pulse width) is the absolute value of the low pulse width as measured from a falling edge to the next rising edge. This measurement is made across the entire record.

DDRTCLAVERAGE (average low pulse width) is the average value of the low pulse width as measured from one falling edge to the next rising edge, across 200 consecutive cycles.

DDRTERRMN (cumulative error) is the cumulative error across multiple consecutive defined cycles from tCK(avg).

DDRTERRN (cumulative error) is the cumulative error across specified consecutive cycles from tCK(avg). In other words, this measures the time difference between the sum of the clock period from a 200 cycle window and n times tCK(avg).

DDRTJITCC (cycle to cycle jitter period) is the absolute difference in clock period between two consecutive clock cycles. This measurement is made across the entire record.

DDRTJITDUTY (half period jitter) is the largest elapsed time between tCH and tCH(avg), and tCL and tCL(avg), over 200 consecutive cycles.

DDRTJITPER (clock period jitter) is the largest deviation of any tCK signal from tCK(avg). This measurement is made across the entire record.

DDRTPST (read/write burst postamble) is the width of the Read or Write burst postamble, measured from the last falling edge of the mid reference level to the start of an undriven state. This measurement is made across the entire record.

DDRTRPRE (read burst preamble) is the width of the Read burst preamble, measured from exiting tristate levels to the first driving edge of the differential strobe. This measurement is made across the entire record.

DDRTWPRE is the width of the Write burst preamble, measured from exiting tristate levels to the first driving edge of the differential strobe. This measurement is made across the entire record.

DDRVIXAC is the differential input cross-point voltage measured from the true state transition (and it's compliment) to a specified reference level, measured on a single-ended signal.

DDRTDQSCK is the strobe output access time, measured between the rising edge of the clock and before or after the differential strobe Read preamble time. Signal edges are determined by the mid-ref threshold level settings.

DElAy is the time between the specified Mid reference level (RM) crossing on one source to a specified Mid reference level (RM) crossing on a second source. This measurement is made on the first occurrence in the record.

DJ (deterministic jitter) is the peak-to-peak amplitude of all timing errors that exhibit deterministic behavior. This measurement is made across the entire record.

DJDIRAC (dual-dirac deterministic jitter) is deterministic jitter based on a simplifying assumption that the histogram of all deterministic jitter can be modeled as a pair of equal-magnitude Dirac functions. This measurement is made across the entire record.

EYEHIGH (Eye High) is the amplitude of a high (1) bit measured at a user specified location within the recovered unit interval. This measurement is made on each high bit in the record.

EYELow (Eye Low) is the amplitude of a low (0) bit measured at a user specified location within the recovered unit interval. This measurement is made on each low bit in the record.

FALLSLEWRATE (Falling Slew Rate) is the rate of change in voltage as an edge transitions from the Top reference level (RT) to the Bottom reference level (RB). This measurement is made on each cycle in the record.

FALLTIME (Fall Time) is the time required for an edge to fall from the Top reference level (RT) to the Base reference level (RB). This measurement is made on each cycle in the record.

FREQuency is the reciprocal of Period. This measurement is made on each cycle in the record.

F2 is the peak-to-peak amplitude of the periodic jitter occurring at a rate of F_b (data rate) divided by 2. This measurement is made across the entire record.

F4 is the peak-to-peak amplitude of the periodic jitter occurring at a rate of F_b (data rate) divided by 4. This measurement is made across the entire record.

F8 is the peak-to-peak amplitude of the periodic jitter occurring at a rate of F_b (data rate) divided by 8. This measurement is made across the entire record.

HIGH (Eye High) is the amplitude of a high (1) bit measured at a user specified location within the recovered unit interval. This measurement is made on each high bit in the record.

HEIGHT (Eye Height) is the minimum vertical eye opening at the center of the recovered unit interval. This measurement is made across the entire record.

HEIGHTBER (Eye Height@BER) is the predicted vertical eye opening that will be violated with a probability equal to the bit error rate. This measurement is made across the entire record.

HIGH

HIGHTIME (High Time) is the time the signal remains above the Top reference level (RT). This measurement is made on each cycle in the record.

HOLD (Hold Time) is the time between the specified Mid reference level crossing (RM) on the Clock source to the closest specified Mid reference level (RM) crossing on the Data source. This measurement is made on each specified Clock edge in the record.

JITTERSUMMARY (Jitter Summary) is a group consisting of the following measurements: TIE, TJ@BER, Eye Width@BER, Eye Height@BER, RJ- $\delta\delta$, DJ- $\delta\delta$, PJ, DDJ, DcD, F/2, F/4, F/8.

J2 is the total jitter at a bit error rate of $2.5e-3$ (TJ@ $2.5e-3$). This measurement is made across the entire record.

J9 is the total jitter at a bit error rate of $2.5e-10$ (TJ@ $2.5e-10$). This measurement is made across the entire record.

LOW (Eye Low) is the amplitude of a low (0) bit measured at a user specified location within the recovered unit interval. This measurement is made on each low bit in the record.

LOWTIME (Low Time) is the time the signal remains below the Base reference level (RB). This measurement is made on each cycle in the record.

MAXimum is the maximum data point. This measurement can be made across the entire record, or on each cycle in the record.

MEAN is the arithmetic mean of the data points. This measurement can be made across the entire record, or on each cycle in the record.

MINimum is the minimum data point. This measurement can be made across the entire record, or on each cycle in the record.

NDuty (Negative Duty Cycle) is the ratio of the Negative Pulse Width to the Period. This measurement is made on each cycle in the record.

$$\text{Negative Duty Cycle} = (\text{Negative Width}) / \text{Period} \times 100\%$$

NPERIOD (Duration N-Periods) is the time required to complete N cycles. A cycle is the time between two adjacent (same direction) crossings of the Mid reference level (RM). This measurement is made on each cycle in the record.

NPJ (non-periodic jitter) is the portion of the BUJ (bounded uncorrelated jitter) that is random. BUJ excludes DDJ, DCD and RJ. This measurement is made across the entire record.

NOvershoot (Negative Overshoot) is the difference between Minimum and Base, divided by the Amplitude. This measurement can be made across the entire record, or on each cycle in the record.

$$\text{Negative Overshoot} = (\text{Base} - \text{Minimum}) / \text{Amplitude} \times 100\%$$

NWidth (Negative Pulse Width) is the time the signal remains below the Mid reference level (RM). This measurement is made on each cycle in the record.

PDUTY (Positive Duty Cycle) is the ratio of the Positive Pulse Width to the Period. This measurement is made on each cycle in the record.

$$\text{Positive Duty Cycle} = (\text{Positive Width}) / \text{Period} \times 100\%$$

PERIOD is the time required to complete a cycle. A cycle is the time between two adjacent (same direction) crossings of the Mid reference level (RM). This measurement is made on each cycle in the record.

PHASE is the ratio of the Skew between two sources to the Period of the first source. This measurement is made on each cycle in the record.

PHASENOISE (Phase Noise) is the RMS magnitude of all integrated jitter falling within a user specified offset range of the fundamental clock frequency. This measurement is made across the entire record.

PJ (periodic jitter) is the peak-to-peak amplitude of the uncorrelated sinusoidal components of the deterministic jitter. This measurement is made across the entire record.

PK2Pk (Peak-to-peak) is the difference between Maximum and Minimum. This measurement can be made across the entire record, or on each cycle in the record.

POVERSHOOT (Positive Overshoot) is the difference between Maximum and Top, divided by the Amplitude. This measurement can be made across the entire record, or on each cycle in the record.

$$\text{Positive Overshoot} = (\text{Maximum} - \text{Top}) / \text{Amplitude} \times 100\%$$

PWIDTH (Positive Pulse Width) is the time the signal remains above the Mid reference level (RM). This measurement is made on each cycle in the record.

QFACTOR (Q-Factor) is the ratio of the vertical eye opening to RMS vertical noise measured at a user specified location within the recovered unit interval. This measurement is made across the entire record.

RISESLEWRATE (Rising Slew Rate) is the rate of change in voltage as an edge transitions from the Base reference level (RB) to the Top reference level (RT). This measurement is made on each cycle in the record.

RISETIME Rise Time is the time required for an edge to rise from the Base reference level (RB) to the Top reference level (RT). This measurement is made on each cycle in the record.

RJ (random jitter) is the RMS magnitude of all random timing errors following a Gaussian distribution. This measurement is made across the entire record.

RJDIRAC (dual-dirac random jitter) is random jitter based on a simplifying assumption that the histogram of all deterministic jitter can be modeled as a pair of equal-magnitude Dirac functions. This measurement is made across the entire record.

RMS is the true Root Mean Square of the data points. This measurement can be made across the entire record, or on each cycle in the record.

SRJ (sub-rate jitter) is the composite jitter due to periodic components at 1/2, 1/4 and 1/8 of the data rate. This measurement is made across the entire record.

SSCFREQDEV (SSC Frequency Deviation) is the spread spectrum clock frequency deviation. This measurement enables a time trend plot of the spread spectrum clock modulation profile. This measurement is made on each cycle in the record.

SSCMODRATE (SSC Modulation Rate) is the modulating frequency of a spread spectrum clock. This measurement is made on each cycle in the record.

SETUP (Setup Time) is the time between the specified Mid reference level (RM) crossing on the Data source to the closest specified Mid reference level (RM) crossing on the Clock source. This measurement is made on each specified Clock edge in the record.

SKEW Skew is the time between the specified Mid reference level (RM) crossing on one source to the following specified Mid reference level (RM) crossing on a second source. This measurement is made on each cycle in the record.

TIE (time interval error) is the difference, in time, between an edge in the source waveform and the corresponding edge in a recovered reference clock. This measurement is made on each edge in the waveform.

TIMEOUTSIDELEVEL Time Outside Level is the time the signal remains above the Top reference level (RT) and/or below the Base reference level (RB). This measurement is made on each occurrence in the record.

TJBER (total jitter at a specified bit error rate) is the predicted peak-to-peak amplitude of jitter that will only be exceeded with a probability equal to the bit error rate. This measurement is made across the entire record.

TNTRATIO T/nT Ratio is the ratio of a non-transition bit voltage (2nd and subsequent bit voltage after a transition) to its nearest preceding transition bit voltage (1st bit voltage after the transition). Bit voltages are measured at the interpolated midpoint of the recovered unit interval. This measurement is made on each non-transition bit in the record.

TOP is the most common data value above the midpoint of the waveform. This measurement can be made across the entire record, or on each cycle in the record.

UNITINTERVAL (Unit Interval) is the time difference between two successive bits. This measurement is made on each bit in the record.

VDIFFXOVR (Differential Crossover) is the voltage level of a differential signal pair at the crossover points. This measurement is made at each crossover point in the record.

WIDTH (Eye Width) is the minimum horizontal eye opening at the user specified reference level. This measurement is made across the entire record.

WIDTHBER (Eye Width@BER) is the predicted horizontal eye opening that will be violated with a probability equal to the bit error rate. This measurement is made across the entire record.

Examples `MEASUREMENT:MEAS2:TYPE FREQUENCY` defines measurement 2 as a measurement of the frequency of a waveform.

`MEASUREMENT:MEAS1:TYPE?` might return `:MEASUREMENT:MEAS1:TYPE RMS`, indicating that measurement 1 is defined to measure the RMS value of a waveform.

MEASUrement:MEAS<x>:WINDOWLENgth

This command sets or queries the window length for the measurement. The measurement number is specified by x.

Group Measurement

Syntax `MEASUrement:MEAS<x>:WINDOWLENgth <NR3>`
`MEASUrement:MEAS<x>:WINDOWLENgth?`

Arguments <NR3> is the value for the window length.

Examples `MEASUrement:MEAS1:WINDOWLENgth 10` sets the window length to 10.

`MEASUrement:MEAS1:WINDOWLENgth?` might return `:MEASUREMENT:MEAS1:WINDOWLENGTH 10.0000` indicating the window length is 10.

MEASUrement:MINUI

This command sets or queries the minimum number of unit intervals required for BUJ analysis.

Group	Measurement
Syntax	MEASUrement:MINUI <NR3> MEASUrement:MINUI?
Arguments	<NR3> is the minimum number of unit intervals required for BUJ analysis.
Examples	MEASUREMENT:MINUI 1.0000E+6 sets the minimum number of unit intervals required for BUJ analysis to 1,000,000. MEASUREMENT:MINUI? might return :MEASUREMENT:MINUI 1.0000E+6 indicating the minimum number of unit intervals required for BUJ analysis is 1,000,000.

MEASUrement:POPULATION:LIMIT:STATE

This command sets or queries the global population limit state for the measurement.

Group	Measurement
Syntax	MEASUrement:POPULATION:LIMIT:STATE {OFF ON 0 1 } MEASUrement:POPULATION:LIMIT:STATE?
Arguments	OFF turns off the population limit. ON turns on the population limit. 0 turns off the population limit. 1 turns on the population limit.
Examples	MEASUREMENT:POPULATION:LIMIT:STATE 1 turns on the population limit. MEASUREMENT:POPULATION:LIMIT:STATE? might return :MEASUREMENT:POPULATION:LIMIT:STATE 0 indicating the population limit is off.

MEASUrement:POPULation:LIMIT:VALue

This command sets or queries the global population limit value for the measurement.

Group	Measurement
Syntax	<code>MEASUrement:POPULation:LIMIT:VALue <NR1></code> <code>MEASUrement:POPULation:LIMIT:VALue?</code>
Arguments	<NR1> is the current limit value.
Examples	<code>MEASUREMENT:POPULATION:LIMIT:VALUE 2000</code> sets the population limit to 2000. <code>MEASUREMENT:POPULATION:LIMIT:VALUE?</code> might return <code>:MEASUREMENT:POPULATION:LIMIT:VALUE 1000</code> indicating the population limit value is 1000.

MEASUrement:REFLevels:ABSolute:FALLHigh

This command sets or queries the value used as the high reference level of the falling edge when the measurement's ref level method is set to absolute.

Group	Measurement
Syntax	<code>MEASUrement:REFLevels:ABSolute:FALLHigh <NR3></code> <code>MEASUrement:REFLevels:ABSolute:FALLHigh?</code>
Arguments	<NR3> is the value used as the high reference level of the falling edge when the measurement's ref level method is set to absolute.
Examples	<code>MEASUREMENT:REFLEVELS:ABSOLUTE:FALLHIGH 1.5</code> sets the high reference level to 1.5 V. <code>MEASUREMENT:REFLEVELS:ABSOLUTE:FALLHIGH?</code> might return <code>:MEASUREMENT:REFLEVELS:ABSOLUTE:FALLHIGH 1.0000</code> indicating the high reference level is 1.0 V.

MEASUrement:REFLevels:ABSolute:FALLLow

This command sets or queries the value used as the low reference level of the falling edge when the measurement's ref level method is set to absolute.

Group	Measurement
Syntax	<pre>MEASUrement:REFLevels:ABSolute:FALLLow <NR3> MEASUrement:REFLevels:ABSolute:FALLLow?</pre>
Arguments	<NR3> is the value used as the low reference level of the falling edge.
Examples	MEASUrement:REFLevels:ABSolute:FALLLow -1.5 sets the low reference level to -1.5 V. MEASUrement:REFLevels:ABSolute:FALLLow? might return :MEASUREMENT:REFLEVELS:ABSOLUTE:FALLLOW -1.0000 indicating the low reference level is -1.0 V.

MEASUrement:REFLevels:ABSolute:FALLMid

This command sets or queries the value used as the mid reference level of the falling edge when the measurement's ref level method is set to absolute.

Group	Measurement
Syntax	<pre>MEASUrement:REFLevels:ABSolute:FALLMid <NR3> MEASUrement:REFLevels:ABSolute:FALLMid?</pre>
Arguments	<NR3> is the value used as the mid reference level of the falling edge.
Examples	MEASUrement:REFLevels:ABSolute:FALLMid 10.0E-3 sets the mid reference level to 10.0 mV. MEASUrement:REFLevels:ABSolute:FALLMid? might return :MEASUREMENT:REFLEVELS:ABSOLUTE:FALLMID 0.0E+0 indicating the mid reference level is 0.0 V.

MEASUrement:REFLevels:ABSolute:HYSTeresis

This command sets or queries the value of the hysteresis of the reference level when the measurement's ref level method is set to absolute.

Group	Measurement
Syntax	<code>MEASUrement:REFLevels:ABSolute:HYSTeresis <NR3></code> <code>MEASUrement:REFLevels:ABSolute:HYSTeresis?</code>
Arguments	<NR3> is the value of the hysteresis of the reference level.
Examples	<code>MEASUrement:REFLevels:ABSolute:HYSTeresis 20.0E-3</code> sets the hysteresis to 20.0 mV. <code>MEASUrement:REFLevels:ABSolute:HYSTeresis?</code> might return <code>:MEASUREMENT:REFLEVELS:ABSOLUTE:HYSTERESIS 30.0000E-3</code> indicating the hysteresis is set to 30.0 mV.

MEASUrement:REFLevels:ABSolute:RISEHigh

This command sets or queries the value used as the high reference level of the rising edge when the measurement's ref level method is set to absolute.

Group	Measurement
Syntax	<code>MEASUrement:REFLevels:ABSolute:RISEHigh <NR3></code> <code>MEASUrement:REFLevels:ABSolute:RISEHigh?</code>
Arguments	<NR3> is the value used as the high reference level of the rising edge.
Examples	<code>MEASUrement:REFLevels:ABSolute:RISEHigh 1.5</code> sets the high reference to 1.5 V. <code>MEASUrement:REFLevels:ABSolute:RISEHigh?</code> might return <code>:MEASUREMENT:REFLEVELS:ABSOLUTE:RISEHIGH 1.0000</code> indicating the high reference is 1.0 V.

MEASUrement:REFLevels:ABSolute:RISELow

This command sets or queries the value used as the low reference level of the rising edge when the measurement's ref level method is set to absolute.

Group	Measurement
Syntax	MEASUrement:REFLeveLs:ABSolute:RISELow <NR3> MEASUrement:REFLeveLs:ABSolute:RISELow?
Arguments	<NR3> is the value used as the the low reference level of the rising edge
Examples	MEASUrement:REFLeveLs:ABSolute:RISELow -1.5 sets the low reference level to -1.5 V. MEASUrement:REFLeveLs:ABSolute:RISELow? might return :MEASUREMENT:REFLEVELS:ABSOLUTE:RISELOW -1.0000 indicating the low reference level is -1.0 V.

MEASUrement:REFLevels:ABSolute:RISEMid

This command sets or queries the value used as the mid reference level of the rising edge when the measurement's ref level method is set to absolute.

Group	Measurement
Syntax	MEASUrement:REFLeveLs:ABSolute:RISEMid <NR3> MEASUrement:REFLeveLs:ABSolute:RISEMid?
Arguments	<NR3> is the mid reference level of the rising edge.
Examples	MEASUrement:REFLeveLs:ABSolute:RISEMid 10.0E-3 sets the mid reference to 10.0 mV. MEASUrement:REFLeveLs:ABSolute:RISEMid? might return :MEASUREMENT:REFLEVELS:ABSOLUTE:RISEMID 0.0E+0 indicating the mid reference is 0.0 V.

MEASUrement:REFLevels:ABSolute:TYPE

This command sets or queries the reference level type for the measurement.

Group	Measurement
Syntax	<pre>MEASUREMENT:REFLEVELS:ABSOLUTE:TYPE {SAME UNIQUE} MEASUREMENT:REFLEVELS:ABSOLUTE:TYPE?</pre>
Arguments	SAME specifies that the absolute levels are set the same. UNIQUE specifies that the absolute levels can be set independently.
Examples	MEASUREMENT:REFLEVELS:ABSOLUTE:TYPE UNIQUE specifies that the absolute levels can be set independently. MEASUREMENT:REFLEVELS:ABSOLUTE:TYPE? might return :MEASUREMENT:REFLEVELS:ABSOLUTE:TYPE SAME indicating the absolute levels are set the same.

MEASUREMENT:REFLEVELS:BASETOP

This command sets or queries the method used to calculate the TOP and BASE, used to calculate reference levels for the measurement.

Group	Measurement
Syntax	<pre>MEASUREMENT:REFLEVELS:BASETOP {AUTO MINMAX MEANHISTOGRAM MODEHISTOGRAM EYEHISTOGRAM} MEASUREMENT:REFLEVELS:BASETOP?</pre>
Arguments	Arguments are the base top methods.
Examples	MEASUREMENT:REFLEVELS:BASETOP MINMAX selects the MINMAX base top method. MEASUREMENT:REFLEVELS:BASETOP? might return :MEASUREMENT:REFLEVELS:BASETOP MINMAX indicating the base top method is MINMAX.

MEASUREMENT:REFLEVELS:JITTERMODE

This command sets or queries how often reference levels are calculated on Jitter measurements. If the mode is set to Latch, ref levels are calculated only on the first acquisition after a statistics reset. If it is set to Continuous, reference levels are calculated on every acquisition.

Group	Measurement
Syntax	<pre>MEASUREMENT:REFLEVELS:JITTERMODE {CONTINUOUS LATCH} MEASUREMENT:REFLEVELS:JITTERMODE?</pre>
Arguments	CONTINUOUS specifies that reference levels are calculated on every acquisition. LATCH specifies that reference levels are calculated only on the first acquisition after a statistics reset.
Examples	MEASUREMENT:REFLEVELS:JITTERMODE CONTINUOUS specifies that reference levels are calculated on every acquisition. MEASUREMENT:REFLEVELS:JITTERMODE? might return LATCH indicating that reference levels are calculated only on the first acquisition after a statistics reset.

MEASUREMENT:REFLEVELS:METHOd

This command sets or queries the method used to calculate reference levels for the measurement.

Group	Measurement
Syntax	<pre>MEASUREMENT:REFLEVELS:METHOd {PERCENT ABSOLUTE} MEASUREMENT:REFLEVELS:METHOd?</pre>
Arguments	PERCENT specifies that the reference levels are calculated as a percent relative to HIGH and LOW. The percentages are defined using the MEASUREMENT:REFLEVELS:REFLEVEL:PERCENT commands. ABSOLUTE specifies that the reference levels are set explicitly using the MEASUREMENT:REFLEVELS:REFLEVEL:ABSOLUTE commands. This method is useful when precise values are required.
Examples	MEASUREMENT:REFLEVELS:METHOd ABSOLUTE specifies that the reference levels are set explicitly. MEASUREMENT:REFLEVELS:METHOd? might return :MEASUREMENT:REFLEVELS:METHOd PERCENT indicating the reference levels are calculated as a percent relative to HIGH and LOW.

MEASUREMENT:REFLEVELS:MODE

This command sets or queries how often reference levels are calculated.

Group	Measurement
Syntax	<code>MEASUREMENT:REFLEVELS:MODE {LATCH CONTINUOUS}</code> <code>MEASUREMENT:REFLEVELS:MODE?</code>
Arguments	LATCH calculates reference levels only on the first acquisition after a statistics reset. CONTINUOUS calculates reference levels on every acquisition.
Examples	<code>MEASUREMENT:REFLEVELS:MODE CONTINUOUS</code> calculates reference levels on every acquisition. <code>MEASUREMENT:REFLEVELS:MODE?</code> might return <code>:MEASUREMENT:REFLEVELS:MODE LATCH</code> indicating reference levels are calculated only on the first acquisition after a statistics reset.

MEASUREMENT:REFLEVELS:PERCENT:FALLHIGH

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the high reference level of the falling edge when the measurement's ref level method is set to percent.

Group	Measurement
Syntax	<code>MEASUREMENT:REFLEVELS:PERCENT:FALLHIGH <NR3></code> <code>MEASUREMENT:REFLEVELS:PERCENT:FALLHIGH?</code>
Arguments	<NR3> is the percentage used to calculate the high reference level of the falling edge
Examples	<code>MEASUREMENT:REFLEVELS:PERCENT:FALLHIGH 90</code> sets the high reference level to 90%. <code>MEASUREMENT:REFLEVELS:PERCENT:FALLHIGH?</code> might return <code>:MEASUREMENT:REFLEVELS:PERCENT:FALLHIGH 80.0000</code> indicating the high reference level is 80%.

MEASUrement:REFLeveLs:PERCent:FALLLow

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the falling edge when the measurement's ref level method is set to percent.

Group	Measurement
Syntax	<pre>MEASUrement:REFLeveLs:PERCent:FALLLow <NR3> MEASUrement:REFLeveLs:PERCent:FALLLow?</pre>
Arguments	<NR3> is the percentage used to calculate the mid reference level of the falling edge.
Examples	MEASUrement:REFLeveLs:PERCent:FALLLow 10 sets the low reference level to 10%. MEASUrement:REFLeveLs:PERCent:FALLLow? might return :MEASUREMENT:REFLEVELS:PERCENT:FALLLOW 20.0000 indicating the low reference level is 20%.

MEASUrement:REFLeveLs:PERCent:FALLMid

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the falling edge when the measurement's ref level method is set to percent.

Group	Measurement
Syntax	<pre>MEASUrement:REFLeveLs:PERCent:FALLMid <NR3> MEASUrement:REFLeveLs:PERCent:FALLMid?</pre>
Arguments	<NR3> is the percentage used to calculate the mid reference level of the falling edge
Examples	MEASUrement:REFLeveLs:PERCent:FALLMid 55 sets the mid reference level to 55%. MEASUrement:REFLeveLs:PERCent:FALLMid? might return :MEASUREMENT:REFLEVELS:PERCENT:FALLMID 50.0000 indicating the mid reference level is 50%.

MEASUREMENT:REFLEVELS:PERCENT:HYSTeresis

This command sets or queries the percentage (where 100% is equal to MAX and 0% is equal to MIN) used to calculate the hysteresis of the reference level when the measurement's ref level method is set to percent.

Group	Measurement
Syntax	<pre>MEASUREMENT:REFLEVELS:PERCENT:HYSTeresis <NR3> MEASUREMENT:REFLEVELS:PERCENT:HYSTeresis?</pre>
Arguments	<NR3> is the percentage used to calculate the hysteresis of the reference level.
Examples	MEASUREMENT:REFLEVELS:PERCENT:HYSTeresis 3 sets the hysteresis to 3%. MEASUREMENT:REFLEVELS:PERCENT:HYSTeresis? might return :MEASUREMENT:REFLEVELS:PERCENT:HYSTeresis 5.0000 indicating the hysteresis is set to 5%.

MEASUREMENT:REFLEVELS:PERCENT:RISEHigh

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the high reference level of the rising edge when the measurement's ref level method is set to percent.

Group	Measurement
Syntax	<pre>MEASUREMENT:REFLEVELS:PERCENT:RISEHigh <NR3> MEASUREMENT:REFLEVELS:PERCENT:RISEHigh?</pre>
Arguments	<NR3> is the percentage used to calculate the high reference level of the rising edge.
Examples	MEASUREMENT:REFLEVELS:PERCENT:RISEHigh 90 sets the high reference level to 90%. MEASUREMENT:REFLEVELS:PERCENT:RISEHigh? might return :MEASUREMENT:REFLEVELS:PERCENT:RISEHIGH 80.0000 indicating the high ref level is 80%.

MEASUrement:REFLeveLs:PERCent:RISELow

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the low reference level of the rising edge when the measurement's ref level method is set to percent.

Group	Measurement
Syntax	<pre>MEASUrement:REFLeveLs:PERCent:RISELow <NR3> MEASUrement:REFLeveLs:PERCent:RISELow?</pre>
Arguments	<NR3> is the percentage used to calculate the low reference level of the rising edge.
Examples	MEASUrement:REFLeveLs:PERCent:RISELow 10 sets the low reference level to 10%. MEASUrement:REFLeveLs:PERCent:RISELow? might return :MEASUREMENT:REFLEVELS:PERCENT:RISELOW 20.0000 indicating the low reference level is 20%.

MEASUrement:REFLeveLs:PERCent:RISEMid

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the rising edge when the measurement's ref level method is set to percent.

Group	Measurement
Syntax	<pre>MEASUrement:REFLeveLs:PERCent:RISEMid <NR3> MEASUrement:REFLeveLs:PERCent:RISEMid?</pre>
Arguments	<NR3> is the percentage used to calculate the mid reference level of the rising edge.
Examples	MEASUrement:REFLeveLs:PERCent:RISEMid 55 sets the mid reference level to 55%. MEASUrement:REFLeveLs:PERCent:RISEMid? might return :MEASUREMENT:REFLEVELS:PERCENT:RISEMID 50.0000 indicating the mid reference level is 50%.

MEASUrement:REFLevels:PERCent:TYPE

This command sets or queries the reference level percent type for the measurement.

Group	Measurement
Syntax	<pre>MEASUrement:REFLevels:PERCent:TYPE {TENNinety TWENTyeighty CUSTom} MEASUrement:REFLevels:PERCent:TYPE?</pre>
Arguments	TENNinety sets the values for Low, Mid and High Ref to 10%, 50% and 90% respectively. TWENTyeighty sets the values for Low, Mid and High Ref are set to 20%, 50% and 80% respectively. CUSTom allows setting other reference level percents.
Examples	MEASUrement:REFLevels:PERCent:TYPE TENNINETY sets the values for Low, Mid and High Ref to 10%, 50% and 90% respectively. MEASUrement:REFLevels:PERCent:TYPE? might return :MEASUREMENT:REFLEVELS:PERCENT:TYPE CUSTOM indicating that custom reference levels can be set.

MEASUrement:REFLevels:TYPE

This command sets or queries the shared reference level method used for sources of measurement calculations.

Group	Measurement
Syntax	<pre>MEASUrement:REFLevels:TYPE {GLOBa1 PERSource} MEASUrement:REFLevels:TYPE?</pre>
Arguments	GLOBa1 shares reference levels across measurements. PERSource causes reference levels to be used on individual measurements.
Examples	MEASUREMENT:REFLEVELS:TYPE PERSource causes reference levels to be used on individual measurements.

MEASUREMENT:REFLEVELS:TYPE? might return
:MEASUREMENT:REFLEVELS:TYPE GLOBAL indicating reference levels are
shared across measurements.

MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:FALLHIGH

This command sets or queries the value used as the high reference level of the falling edge when the measurement's ref level method is set to absolute.

Group	Measurement
Syntax	MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:FALLHIGH <NR3> MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:FALLHIGH?
Arguments	<NR3> is the value used as the high reference level of the falling edge when the measurement's ref level method is set to absolute.
Examples	MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:FALLHIGH 1.5 sets the reference level to 1.5 V. MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:FALLHIGH? might return :MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:FALLHIGH 1.0000 indicating the reference level is 1.0 V.

MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:FALLLOW

This command sets or queries the value used as the low reference level of the falling edge when the measurement's ref level method is set to absolute.

Group	Measurement
Syntax	MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:FALLLOW <NR3> MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:FALLLOW?
Arguments	<NR3> is the value used as the low reference level of the falling edge when the measurement's ref level method is set to absolute.
Examples	MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:FALLLOW -1.5 sets the low reference level to -1.5 V.

MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:FALLLOW? might return
:MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:FALLLOW -1.0000 indicating
the low reference level is -1.0 V.

MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:FALLMID

This command sets or queries the value used as the mid reference level of the falling edge when the measurement's ref level method is set to absolute.

Group	Measurement
Syntax	MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:FALLMID <NR3> MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:FALLMID?
Arguments	<NR3> is the value used as the mid reference level of the falling edge when the measurement's ref level method is set to absolute.
Examples	MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:FALLMID 50.0E-3 sets the mid reference level to 50.0 mV. MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:FALLMID? might return :MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:FALLMID 0.0E+0 indicating the reference level is 0.0 V.

MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:HYSTERESIS

This command sets or queries the value of the hysteresis of the reference level when the measurement's ref level method is set to absolute.

Group	Measurement
Syntax	MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:HYSTERESIS <NR3> MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:HYSTERESIS?
Arguments	<NR3> is the value of the hysteresis of the reference level when the measurement's ref level method is set to absolute.
Examples	MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:HYSTERESIS 20.0E-3 sets the hysteresis to 20.0 mV.

MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:HYSTERESIS? might return
:MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:HYSTERESIS 30.0000E-3
indicating the hysteresis is 30.0 mV.

MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:RISEHigh

This command sets or queries the value used as the high reference level of the rising edge when the measurement's ref level method is set to absolute.

Group	Measurement
Syntax	MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:RISEHigh <NR3> MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:RISEHigh?
Arguments	<NR3> is the value used as the high reference level of the rising edge when the measurement's ref level method is set to absolute.
Examples	MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:RISEHigh 1.5 sets the reference level to 1.5 V. MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:RISEHigh? might return :MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:RISEHIGH 1.0000 indicating the reference level is 1.0 V.

MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:RISELow

This command sets or queries the value used as the low reference level of the rising edge when the measurement's ref level method is set to absolute.

Group	Measurement
Syntax	MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:RISELow <NR3> MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:RISELow?
Arguments	<NR3> is the value used as the low reference level of the rising edge when the measurement's ref level method is set to absolute.
Examples	MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:RISELow -1.5 sets the reference level to -1.5 V.

MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:RISELOW? might return
:MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:RISELOW -1.0000 indicating
the reference level is -1.0 V.

MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:RISEMid

This command sets or queries the value used as the mid reference level of the rising edge when the measurement's ref level method is set to absolute.

Group	Measurement
Syntax	MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:RISEMid <NR3> MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:RISEMid?
Arguments	<NR3> is the value used as the mid reference level of the rising edge when the measurement's ref level method is set to absolute.
Examples	MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:RISEMid 10.0E-3 sets the reference level to 10.0 mV. MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:RISEMid? might return :MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:RISEMid 0.0E+0 indicating the mid reference level is 0.0 V.

MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:TYPE

This command sets or queries the reference level type for the measurement.

Group	Measurement
Syntax	MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:TYPE {SAME UNIQUE} MEASUREMENT:REF<x>:REFLEVELS:ABSOLUTE:TYPE?
Arguments	SAME specifies that the absolute levels are set the same. UNIQUE specifies that the absolute levels can be set independently.
Examples	MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:TYPE UNIQUE specifies that the absolute levels can be set independently.

MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:TYPE? might return
:MEASUREMENT:REF1:REFLEVELS:ABSOLUTE:TYPE SAME indicating that the
absolute levels are set the same.

MEASUREMENT:REF<x>:REFLEVELS:BASETOP

This command sets or queries the method used to calculate the TOP and BASE,
used to calculate reference levels for the measurement.

Group	Measurement
Syntax	MEASUREMENT:REF<x>:REFLEVELS:BASETOP
Arguments	AUTO automatically chooses a reference level method. MINMAX specifies that reference levels are relative to the measurement MIN and MAX. MEANHISTOGRAM specifies that reference levels are relative to the histogram mean BASE and TOP. MODEHISTOGRAM specifies that reference levels are relative to the histogram mode BASE and TOP. EYEHISTOGRAM specifies that reference levels are relative to the eye histogram BASE and TOP.
Examples	MEASUREMENT:REF1:REFLEVELS:BASETOP MINMAX specifies that reference levels are relative to the measurement MIN and MAX. MEASUREMENT:REF1:REFLEVELS:BASETOP? might return :MEASUREMENT:REF1:REFLEVELS:BASETOP AUTO indicating the instrument automatically chooses a reference level method.

MEASUREMENT:REF<x>:REFLEVELS:METHOD

This command sets or queries the method used to calculate reference levels for
the measurement.

Group	Measurement
Syntax	MEASUREMENT:REF<x>:REFLEVELS:METHOD {PERCENT ABSOLUTE} MEASUREMENT:REF<x>:REFLEVELS:METHOD?

Arguments **PERCent** specifies that the reference levels are calculated as a percent relative to HIGH and LOW. The percentages are defined using the **MEASUREMENT:REF<x>:REFLevel:PERCent** commands.

ABSolute specifies that the reference levels are set explicitly using the **MEASUREMENT:REF<x>:REFLevel:ABSolute** commands. This method is useful when precise values are required.

Examples **MEASUREMENT:REF1:REFLevels:METHOD ABSOLUTE** specifies that the reference levels are set explicitly.

MEASUREMENT:REF1:REFLevels:METHOD? might return
:MEASUREMENT:REF1:REFLEVELS:METHOd PERCENT indicating the reference levels are calculated as a percent relative to HIGH and LOW.

MEASUREMENT:REF<x>:REFLevels:PERCent:FALLHigh

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the high reference level of the falling edge when the measurement's ref level method is set to percent.

Group Measurement

Syntax **MEASUREMENT:REF<x>:REFLevels:PERCent:FALLHigh <NR3>**
MEASUREMENT:REF<x>:REFLevels:PERCent:FALLHigh?

Arguments **<NR3>** is the percentage used to calculate the high reference level of the falling edge

Examples **MEASUREMENT:REF1:REFLevels:PERCent:FALLHigh 95** sets the reference level to 95%.

MEASUREMENT:REF1:REFLevels:PERCent:FALLHigh? might return
:MEASUREMENT:REF1:REFLEVELS:PERCENT:FALLHIGH 90.0000 indicating the reference level is 90%.

MEASUREMENT:REF<x>:REFLevels:PERCent:FALLLow

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the low reference level of the falling edge when the measurement's ref level method is set to percent.

Group	Measurement
Syntax	<pre>MEASUREMENT:REF<x>:REFLEVELS:PERCENT:FALLLOW <NR3> MEASUREMENT:REF<x>:REFLEVELS:PERCENT:FALLLOW?</pre>
Arguments	<NR3> is the percentage used to calculate the low reference level
Examples	MEASUREMENT:REF1:REFLEVELS:PERCENT:FALLLOW 5 sets the low reference level to 5%. MEASUREMENT:REF1:REFLEVELS:PERCENT:FALLLOW? might return :MEASUREMENT:REF1:REFLEVELS:PERCENT:FALLLOW 10.0000 indicating the low reference level is 10%.

MEASUREMENT:REF<x>:REFLEVELS:PERCENT:FALLMID

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the falling edge when the measurement's ref level method is set to percent.

Group	Measurement
Syntax	<pre>MEASUREMENT:REF<x>:REFLEVELS:PERCENT:FALLMID <NR3> MEASUREMENT:REF<x>:REFLEVELS:PERCENT:FALLMID?</pre>
Arguments	<NR3> is the percentage used to calculate the mid reference level of the falling edge.
Examples	MEASUREMENT:REF1:REFLEVELS:PERCENT:FALLMID 55 sets the reference level to 50%. MEASUREMENT:REF1:REFLEVELS:PERCENT:FALLMID? might return :MEASUREMENT:REF1:REFLEVELS:PERCENT:FALLMID 50.0000 indicating the mid reference level is 50%.

MEASUREMENT:REF<x>:REFLEVELS:PERCENT:HYSTeresis

This command sets or queries the percentage (where 100% is equal to MAX and 0% is equal to MIN) used to calculate the hysteresis of the reference level when the measurement's ref level method is set to percent.

Group	Measurement
Syntax	<pre>MEASUREMENT:REF<x>:REFLEVELS:PERCENT:HYSTERESIS <NR3> MEASUREMENT:REF<x>:REFLEVELS:PERCENT:HYSTERESIS?</pre>
Arguments	<NR3> is the percentage used to calculate the hysteresis of the reference level.
Examples	MEASUREMENT:REF1:REFLEVELS:PERCENT:HYSTERESIS 2 sets the hysteresis to 2%. MEASUREMENT:REF1:REFLEVELS:PERCENT:HYSTERESIS? might return :MEASUREMENT:REF1:REFLEVELS:PERCENT:HYSTERESIS 5.0000 indicating the hysteresis is 5%.

MEASUREMENT:REF<x>:REFLEVELS:PERCENT:RISEHigh

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the high reference level of the rising edge when the measurement's ref level method is set to percent.

Group	Measurement
Syntax	<pre>MEASUREMENT:REF<x>:REFLEVELS:PERCENT:RISEHigh <NR3> MEASUREMENT:REF<x>:REFLEVELS:PERCENT:RISEHigh?</pre>
Arguments	<NR3> is the percentage used to calculate the high reference level of the rising edge
Examples	MEASUREMENT:REF1:REFLEVELS:PERCENT:RISEHigh 95 sets the reference level to 95%. MEASUREMENT:REF1:REFLEVELS:PERCENT:RISEHigh? might return :MEASUREMENT:REF1:REFLEVELS:PERCENT:RISEHIGH 90.0000 indicating the reference level is 90%.

MEASUREMENT:REF<x>:REFLEVELS:PERCENT:RISELow

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the low reference level of the rising edge when the measurement's ref level method is set to percent.

Group	Measurement
Syntax	<pre>MEASUREMENT:REF<x>:REFLEVELS:PERCENT:RISELOW <NR3> MEASUREMENT:REF<x>:REFLEVELS:PERCENT:RISELOW?</pre>
Arguments	<NR3> is the percentage used to calculate the low reference level of the rising edge.
Examples	MEASUREMENT:REF1:REFLEVELS:PERCENT:RISELOW 5 sets the low reference level to 5%. MEASUREMENT:REF1:REFLEVELS:PERCENT:RISELOW? might return :MEASUREMENT:REF1:REFLEVELS:PERCENT:RISELOW 10.0000 indicating the reference level is 10%.

MEASUREMENT:REF<x>:REFLEVELS:PERCENT:RISEMID

This command sets or queries the percentage (where 100% is equal to TOP and 0% is equal to BASE) used to calculate the mid reference level of the rising edge when the measurement's ref level method is set to percent.

Group	Measurement
Syntax	<pre>MEASUREMENT:REF<x>:REFLEVELS:PERCENT:RISEMID <NR3> MEASUREMENT:REF<x>:REFLEVELS:PERCENT:RISEMID?</pre>
Arguments	<NR3> is the percentage used to calculate the mid reference level of the rising edge.
Examples	MEASUREMENT:REF1:REFLEVELS:PERCENT:RISEMID 55 sets the mid reference level to 55%. MEASUREMENT:REF1:REFLEVELS:PERCENT:RISEMID? might return :MEASUREMENT:REF1:REFLEVELS:PERCENT:RISEMID 50.0000 indicating the mid reference level is 50%.

MEASUREMENT:REF<x>:REFLEVELS:PERCENT:TYPE

This command sets or queries the reference level percent type for the measurement.

Group	Measurement
--------------	-------------

Syntax	<pre>MEASUREMENT:REF<x>:REFLEVELS:PERCENT:TYPE {TENNINETY TWENTYEIGHTY CUSTOM} MEASUREMENT:REF<x>:REFLEVELS:PERCENT:TYPE?</pre>
Arguments	TENNINETY sets the values for Low, Mid and High Ref to 10%, 50% and 90% respectively. TWENTYEIGHTY sets the values for Low, Mid and High Ref are set to 20%, 50% and 80% respectively. CUSTOM allows setting other reference level percents.
Examples	MEASUREMENT:REF1:REFLEVELS:PERCENT:TYPE TWENTYEIGHTY sets the values for Low, Mid and High Ref are set to 20%, 50% and 80% respectively. MEASUREMENT:REF1:REFLEVELS:PERCENT:TYPE? might return :MEASUREMENT:REF1:REFLEVELS:PERCENT:TYPE TENNINETY indicating the values for Low, Mid and High Ref to 10%, 50% and 90% respectively.

MEASUREMENT:STATISTICS:CYCLEMODE

This command turns on and off cycle to cycle measurement statistics tracking and affects computation and display of cycle-cycle statistics in the Measurement Result table. It affects measurement statistics after being enabled and after new data is acquired and measured.

Group	Measurement
Syntax	<pre>MEASUREMENT:STATISTICS:CYCLEMODE {OFF ON 0 1} MEASUREMENT:STATISTICS:CYCLEMODE?</pre>
Arguments	OFF turns off statistics for all measurements. This is the default value. ON turns on statistics and displays all statistics for each measurement. 0 turns off statistics for all measurements. 1 turns on statistics and displays all statistics for each measurement.
Examples	MEASUREMENT:STATISTICS:CYCLEMODE OFF turns off statistics for all measurements. MEASUREMENT:STATISTICS:CYCLEMODE? might return :MEASUREMENT:STATISTICS:CYCLEMODE 1 indicating that statistics are displayed for each measurement.

NEWpass (No Query Form)

This command (no query form) changes the password that enables access to password protected data. The PASSWord command must be successfully executed before using this command or an execution error will be generated.

Group	Miscellaneous
Syntax	NEWpass <QString>
Related Commands	PASSWord *PUD
Arguments	<QString> is the new password, which can contain up to 10 characters.
Examples	NEWPASS"mypassword" creates a new password (mypassword) for accessing your protected data.

*OPC

This command generates the operation complete message in the Standard Event Status Register (SESR) when all pending commands that generate an OPC message are complete. The *OPC? query places the ASCII character “1” into the output queue when all such OPC commands are complete. The *OPC? response is not available to read until all pending operations finish. For a complete discussion of the use of these registers and the output queue, see Registers and Queues.

The *OPC command allows you to synchronize the operation of the instrument with your application program. For more information, see Synchronization Methods.

Table 2-46: Commands that Generate an OPC Message

Command
ACQuire:STATE ON or ACQuire:STATE RUN or ACQuire:STATE 1 (when ACQuire:STOPAfter is set to SEquence)
AUTOSet , CALibrate:INTERNaL , CALibrate:INTERNaL:START Refer to the Service Manual.
DIAg:STATE , SAVe:IMAGe , SAVe:WAVEform , CH<x>:PRObe:AUTOZero , CH<x>:PRObe:DEGAUSS
FACtory , RECAll:SETUp , *RST , TEKSecure

Group	Status and Error
Syntax	*OPC *OPC?
Related Commands	BUSY? *WAI
Examples	*OPC generates the operation complete message in the SESR at the completion of all pending OPC operations. *OPC? might return 1 to indicate that all pending OPC operations are finished.

*OPT? (Query Only)

This query-only command returns a comma separated list of installed options as an arbitrary ASCII string (no quotes) of the form:

<optionCode>:<optionDescription>,<optionCode>:<optionDescription>...

The last section of each entry (the text following the last hyphen) indicates the license type.

If no options are found, NONE is returned.

Group	Status and Error
Syntax	*OPT?
Examples	*OPT? response (with each option listed on a separate line for clarity): 5-BW-1000 – 1 GHz , 5-DJA – Advanced Jitter and Eye Analysis– NONE.

PASSWord (No Query Form)

This command (no query form) enables the *PUD and NEWpass set commands. Sending PASSWord without any arguments disables these same commands. Once the password is successfully entered, the *PUD and NEWpass commands are enabled until the instrument is powered off, or until the FACTory command, the PASSWord command with no arguments, or the *RST command is issued.

To change the password, you must first enter the valid password with the PASSWord command and then change to your new password with the NEWpass command. Remember that the password is case sensitive.

Group	Miscellaneous
Syntax	PASSWord <QString>
Related Commands	NEWpass *PUD
Arguments	<QString> is the password, which can contain up to 10 characters. The factory default password is "XYZZY" and is always valid.
Examples	PASSWORD "XYZZY" enables the *PUD and NEWPass set commands. PASSWORD disables the *PUD and NEWPass set commands. You can still use the query version of *PUD.

PAUSE (No Query Form)

This command causes the interface to pause the specified number of seconds before processing any other commands.

Group	Miscellaneous
Syntax	PAUSE <NR3>
Arguments	<NR3> is the specified number of seconds the interface is to pause before processing any other commands. The pause time is specified as a floating point value in seconds and must be > 0.0 and ≥ 1800.0 .
Examples	PAUSE 10.0e0; :ACQUIRE:NUMACQ causes the interface to pause 10 seconds before returning the number of acquisitions.

PLOT:ADDNew (No Query Form)

This command adds the specified plot.

Group	Plot
Syntax	PLOT:ADDNew <QString>
Arguments	<QString> is the specified plot. The argument is of the form "PLOT<NR1>", where $\text{NR1} \geq 1$.
Examples	PLOT:ADDNEW "PLOT1" adds PLOT1.

PLOT:DELeTe (No Query Form)

This command deletes the specified plot.

Group	Plot
Syntax	PLOT:DELeTe <QString>
Arguments	<QString> is the specified plot. Argument is of the form "PLOT<NR1>", where $\text{NR1} \geq 1$).
Examples	PLOT:DELETE "PLOT1" deletes PLOT1.

PLOT:LIST? (Query Only)

This command lists all currently defined plots.

Group	Plot
Syntax	PLOT:LIST?
Returns	A list of all currently defined plots is returned.
Examples	PLOT:LIST? might return :PLOT:LIST PLOT1,PLOT3,PLOT4,PLOT5,PLOT6,PLOT7 listing all currently defined plots.

PLOT:PLOT<x>:BAThtub:BER

This command sets or queries the bathtub BER value.

Group	Plot
Syntax	PLOT:PLOT<x>:BAThtub:BER <NR1> PLOT:PLOT<x>:BAThtub:BER?
Arguments	<NR1> is the bathtub BER value.
Examples	PLOT:PLOT1:BAThtub:BER 16 sets the BER value to 16. PLOT:PLOT1:BAThtub:BER? might return :PLOT:PLOT1:BATHTUB:BER 14 indicating the BER value is 14.

PLOT:PLOT<x>:BAThtub:XAXISUnits

This command sets or queries the X-Axis unit, either unit intervals or seconds.

Group	Plot
Syntax	PLOT:PLOT<x>:BAThtub:XAXISUnits {UNITIntervals SECONDS} PLOT:PLOT<x>:BAThtub:XAXISUnits?
Arguments	UNITIntervals specifies units as unit intervals. SECONDS specifies units as seconds.
Examples	PLOT:PLOT1:BAThtub:XAXISUnits SECONDS sets the units to seconds. PLOT:PLOT1:BAThtub:XAXISUnits? might return SECONDS.

PLOT:PLOT<x>:BITType

This command sets or queries the bit type to display for the specified eye diagram plot.

Group	Plot
--------------	------

Syntax	<code>PLOT:PLOT<x>:BITType {ALLBits TRANSition NONTRANSition}</code> <code>PLOT:PLOT<x>:BITType?</code>
Arguments	<code>PLOT<x></code> is the plot number. <code>ALLBits</code> sets the eye diagram plot to show both transition and nontransition bits. <code>TRANSition</code> sets the eye diagram plot to show only bits where a logic level transition occurs. <code>NONTRANSition</code> sets the eye diagram plot to show only bits where no logic level transition occurs.
Examples	<code>PLOT:PLOT1:BITType TRANSition</code> sets the eye diagram in Plot 1 to only show transition bits. <code>PLOT:PLOT3:BITType?</code> might return <code>ALLBITS</code>, indicating that the eye diagram in Plot 3 is set to show both transition and nontransition bits in the plot.

PLOT:PLOT<x>:LABel:COLor

This command sets or queries the color of the specified trend label. This command/query only applies to Time Trend plots.

Group	Plot
Syntax	<code>PLOT:PLOT<x>:LABel:COLor <QString></code>
Arguments	<code><QString></code> is the label color. The default color is specified by a quoted empty string, and is the only available color.
Examples	<code>PLOT:PLOT2:LABel:COLor " "</code> sets the plot color to the default color. <code>PLOT:PLOT2:LABel:COLor?</code> might return <code>:PLOT:PLOT2:LABel:COLor ""</code> indicating the plot color is the default color.

PLOT:PLOT<x>:LABel:FONT:BOLD

This command sets or queries the bold state of the specified trend label. This command/query only applies to Time Trend plots.

Group	Plot
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Syntax `PLOT:PLOT<x>:LABel:FONT:BOLD {<NR1>|OFF|ON}`

Arguments `<NR1> = 0` disables bold font; any other value turns this feature on.
 `OFF` disables bold font.
 `ON` enables bold font.

Examples `PLOT:PLOT1:LABEL:FONT:BOLD 1` sets the label to a bold font.
 `PLOT:PLOT1:LABEL:FONT:BOLD?` might return
 `:PLOT:PLOT1:LABEL:FONT:BOLD 1` indicating the label is bold.

PLOT:PLOT<x>:LABel:FONT:ITALic

This command sets or queries the italic state of the specified trend label. This command/query only applies to Time Trend plots.

Group Plot

Syntax `PLOT:PLOT<x>:LABel:FONT:ITALic {<NR1>|OFF|ON}`

Arguments `<NR1> = 0` disables italic font; any other value turns this feature on.
 `OFF` disables italic font.
 `ON` enables italic font.

Examples `PLOT:PLOT1:LABel:FONT:ITALic 1` sets the font to italic.
 `PLOT:PLOT1:LABel:FONT:ITALic?` might return
 `:PLOT:PLOT1:LABEL:FONT:ITALIC 0` indicating the font in not italic.

PLOT:PLOT<x>:LABel:FONT:SIZE

This command sets or queries the font size of the specified trend label. This command/query only applies to Time Trend plots.

Group Plot

Syntax `PLOT:PLOT<x>:LABel:FONT:SIZE <NR1>`

Arguments	<NR1> is the font size.
Examples	<pre>PLOT:PLOT1:LABel:FONT:SIZE 12</pre> sets the font size to 12 points. <pre>PLOT:PLOT1:LABel:FONT:SIZE?</pre> might return <pre>:PLOT:PLOT1:LABEL:FONT:SIZE 72</pre> indicating the font size is 72 points.

PLOT:PLOT<x>:LABel:FONT:TYPE

This command sets or queries the font type of the specified trend label, such as Arial or Times New Roman. This command/query only applies to Time Trend plots.

Group	Plot
Syntax	<pre>PLOT:PLOT<x>:LABel:FONT:TYPE <QString></pre>
Arguments	<QString> is the font type: Times New Roman, Arial, Frutiger LT Std 55 Roman, DejaVu Sans, DejaVu Sans Mono, Frutiger LT Std, Monospace, Sans Serif, Serif, Ubuntu, Ubuntu Condensed, and Ubuntu Mono.
Examples	<pre>PLOT:PLOT1:LABel:FONT:TYPE Arial</pre> sets the font type to Arial. <pre>PLOT:PLOT1:LABel:FONT:TYPE?</pre> might return <pre>:PLOT:PLOT1:LABEL:FONT:TYPE "Frutiger LT Std 55 Roman"</pre> indicating the font type is Frutiger LT Std 55 Roman.

PLOT:PLOT<x>:LABel:FONT:UNDERline

This command sets or queries the underline state of the specified trend label. This command/query only applies to Time Trend plots.

Group	Plot
Syntax	<pre>PLOT:PLOT<x>:LABel:FONT:UNDERline {<NR1> OFF ON}</pre>
Arguments	<NR1> = 0 disables underline font; any other value turns this feature on. OFF disables underline font. ON enables underline font.

Examples `PLOT:PLOT1:LABel:FONT:UNDERline 1` set the font to underlined.
`PLOT:PLOT1:LABel:FONT:UNDERline?`
`:PLOT:PLOT1:LABel:FONT:UNDERLINE 0` indicating the font is underlined.

PLOT:PLOT<x>:LABel:NAME

This command sets or queries the specified trend's label. This command/query only applies to Time Trend plots.

Group Plot

Syntax `PLOT:PLOT<x>:LABel:NAME <QString>`

Arguments `<QString>` is the label.

Examples `PLOT:PLOT1:LABel:NAME "Time Trend number 1"` sets the plot name to Time Trend number 1.
`PLOT:PLOT1:LABel:NAME?` might return `:PLOT:PLOT1:LABel:NAME "This is a label test plot"` indicating the label name is This is a label test plot.

PLOT:PLOT<x>:LABel:XPOS

This command sets or queries the x-position of the specified trend label. This command/query only applies to Time Trend plots.

Group Plot

Syntax `PLOT:PLOT<x>:LABel:XPOS <NR3>`

Arguments `<NR3>` is the y-position, in pixels relative to the left edge of the display, of the label.

Examples `PLOT:PLOT1:LABel:XPOS 200` sets the X position to 200.
`PLOT:PLOT1:LABel:XPOS?` might return `:PLOT:PLOT1:LABel:XPOS 45.0000` indicating the X position is 45.0 pixels to the right of the left edge of the display.

PLOT:PLOT<x>:LABEL:YPOS

This command sets or queries the y-position of the specified trend label. This command/query only applies to Time Trend plots.

Group	Plot
Syntax	PLOT:PLOT<x>:LABEL:YPOS <NR3>
Arguments	<NR3> is the x-position, in pixels relative to the baseline of the waveform, of the label.
Examples	PLOT:PLOT1:LABEL:YPOS 100 sets the Y position to 100. PLOT:PLOT1:LABEL:YPOS? might return :PLOT:PLOT1:LABEL:YPOS 0.0E+0 indicating the Y position of the label is at the baseline of the waveform.

PLOT:PLOT<x>:MASK? (Query Only)

This command returns the name of the mask test associated with the specified eye diagram plot.

Group	Plot
Syntax	PLOT:PLOT<x>:MASK?
Arguments	PLOT<x> is the plot number.
Returns	<String> is a quoted string that contains the name of the eye diagram mask test associated with the specified plot. If a plot does not have an eye diagram mask test, the command returns an empty string "".
Examples	PLOT:PLOT1:MASK? might return "mask2", indicating that the eye diagram mask test mask2 is associated with plot 1.

PLOT:PLOT<x>:NUMBins

This command sets or queries the current histogram resolution.

Group	Plot
Syntax	PLOT:PLOT<x>:NUMBins {TWENTYfive FIFTY HUNDred TWOOfifty FIVEHundred TWOThousand MAXimum}
Arguments	Arguments are the number of bins.
Examples	PLOT:PLOT4:NUMBins TWENTYFIVE sets the number of bins to 25. PLOT:PLOT4:NUMBins? might return :PLOT:PLOT4:NUMBINS TWOFIFTY indicating the histogram resolution is 250 bins.

PLOT:PLOT<x>:SOURce<x>

This command sets or queries the plot source.

Group	Plot
Syntax	PLOT:PLOT<x>:SOURce<x> MEAS<x> PLOT:PLOT<x>:SOURce<x>?
Arguments	MEAS<x> is the specified measurement source for the specified plot.
Examples	PLOT:PLOT1:SOURce1 MEAS2 sets source 1 of plot 1 to measurement 2. PLOT:PLOT1:SOURce1? might return :PLOT:PLOT1:SOURCE1 MEAS1 indicating the specified source of the specified plot is measurement 1.

PLOT:PLOT<x>:SPECTrum:BASE

This command sets or queries the spectrum base. Undefined for non-spectrum plots.

Group	Plot
Syntax	PLOT:PLOT<x>:SPECTrum:BASE <NR1> PLOT:PLOT<x>:SPECTrum:BASE?
Arguments	<NR1> is the spectrum base.

Examples `PLOT:PLOT3:SPECTrum:BASE -10` sets the base to -10.

`PLOT:PLOT3:SPECTrum:BASE?` might return `:PLOT:PLOT3:SPECTRUM:BASE -15` indicating the base is -15.

PLOT:PLOT<x>:SPECTrum:DYNRange

This command sets or queries the dynamic range value.

Group Plot

Syntax `PLOT:PLOT<x>:SPECTrum:DYNRange <NR3>`
`PLOT:PLOT<x>:SPECTrum:DYNRange?`

Arguments <NR3> is the dynamic range value.

Examples `PLOT:PLOT3:SPECTrum:DYNRange 150` sets the dynamic range to 150 dB.

`PLOT:PLOT3:SPECTrum:DYNRange?` might return
`:PLOT:PLOT3:SPECTRUM:DYNRange 100` indicating the dynamic range
is 100 dB.

PLOT:PLOT<x>:TYPE

This command sets or returns the current plot type of the specified plot.

Group Plot

Syntax `PLOT:PLOT<x>:TYPE`
`{NONE | BATHTUB | EYEDIAGRAM | HARMONICS | HISTOGRAM |`
`INDUCTANCE | IVSINTEGRALV | MAGPROPERTY | PHASENOISE | SOA |`
`SPECTRUM | SSCPROFILE | SWL | TIEHISTOGRAM | TIETIMETREND |`
`TIESPECTRUM | TIMETREND | XY | XYZ}`

Arguments <x> is the plot number. This is the equivalent of the number shown on a plot heading in the UI.

`NONE` does not create a plot.

`BATHTUB` creates a bathtub plot.

`EYEDIAGRAM` creates an eye diagram.

HARMONICS creates a harmonics bar graph
HISTOGRAM creates a histogram plot.
INDUCTANCE creates a inductance plot
IVSINTEGRALV creates a I vs. $\int V$ plot
PHASENOISE creates a phase noise plot.
MAGPROPERTY creates a BH curve
SOA creates an SOA plot
SPECTRUM creates a spectrum plot.
SSCPROFILE creates a SSC profile plot.
SWL creates a Switching Loss plot
TIEHISTOGRAM creates a TIE histogram plot.
TIESPECTRUM creates a TIE spectrum plot.
TIETIMETREND creates a TIE time trend plot.
TIMETREND creates a time trend plot.
XY creates a XY plot.
XYZ creates a XYZ plot.

Examples PLOT:PLOT2:TYPE BATHTUB creates a bathtub plot.
PLOT:PLOT2:TYPE? might return :PLOT:PLOT2:TYPE EYEDIAGRAM
indicating the plot is an eye diagram.

POWER:ADDNew (No Query Form)

This command adds the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWER:ADDNew "POWER<x>"

Examples POWER:ADDNEW "POWER1" adds POWER1 measurement badge and selects
Switching Loss measurement by default.

POWer:DELeTe (No Query Form)

This command deletes the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWer:DELeTe "POWER<x>"`

Examples `POWer:DELETE "POWER1"` deletes the POWER1 measurement badge.

POWer:POWer<x>:AUTOSet (No Query Form)

This command executes power autoset for the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWer:POWer<x>:AUTOSet EXECute`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

Examples `POWer:Power1:AUTOSet EXECute` executes the power autoset for the power measurement Power 1.

POWer:POWer<x>:CLRESPONSE:AMP[1-10]Val

This command sets or queries the generator amplitude value of the specified configuration step for the Control Loop Response power measurement.

Conditions Requires option 5-PWR or 6-PWR

Group	Power
Syntax	Power:Power<x>:CLRESPONSE:AMP[1-10]Va1 <NR3> Power:Power<x>:CLRESPONSE:AMP[1-10]Va1?
Arguments	Power<x> sets the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. AMP[1-10] sets the configuration step number, in the range of 1 to 10. Values outside this range will report an error. <NR3> sets the generator amplitude for the specified configuration step, in the range of -100 V to 100 V.
Examples	Power:Power1:CLRESPONSE:AMP3Va1 20 sets the generator output amplitude for configuration step 3 to 20 volts, for power measurement 1. Power:Power2:CLRESPONSE:AMP8Va1? might return 60, indicating that the generator output amplitude setting of configuration step 8 is 60 volts, for power measurement 2.

POWER:POWER<x>:CLRESPONSE:AMPMode

This command sets or queries the amplitude mode for the Control Loop Response power measurement.

Conditions	Requires option 5-PWR or 6-PWR
Group	Power
Syntax	Power:Power<x>:CLRESPONSE:AMPMode {CONSTant PROFile} Power:Power<x>:CLRESPONSE:AMPMode?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. CONSTant sets the amplitude mode to output a constant amplitude signal from the DUT stimulus generator for all frequency bands. PROFile enables configuring the generator to set amplitude values for each frequency band.

Examples `Power:Power1:CLRESPONSE:AMPMode Constant` sets the amplitude mode to Constant for power measurement 1.

`Power:Power2:CLRESPONSE:AMPMode?` might return `Profile`, indicating that the amplitude mode power measurement 2 is set to Profile.

POWer:POWer<x>:CLRESPONSE:CONSTAMPlitude

This command sets or queries the constant amplitude voltage for the Control Loop Response power measurement.

Conditions Requires option 5-PWR or 6-PWR

Group Power

Syntax `Power:Power<x>:CLRESPONSE:CONSTAMPlitude <NR3>`
`Power:Power<x>:CLRESPONSE:CONSTAMPlitude?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

`<NR3>` is the constant amplitude voltage value for the measurement, in the range of -100 V to 100 V.

Examples `Power:Power3:CLRESPONSE:CONSTAMPlitude 120` sets the constant amplitude voltage for Power measurement 3 to be 120 volts.

`Power:Power5:CLRESPONSE:CONSTAMPlitude?` might return `-15`, indicating that the constant amplitude voltage for power measurement 5 is -15 volts.

POWer:POWer<x>:CLRESPONSE:FREQ[1-11]Val

This command sets or queries the generator frequency value of the specified configuration step for the Control Loop Response power measurement.

Conditions Requires option 5-PWR or 6-PWR

Group Power

Syntax `Power:Power<x>:CLRESPONSE:FREQ[1-11]Val <NR3>`
`Power:Power<x>:CLRESPONSE:FREQ[1-11]Val?`

Arguments **Power**<x> sets the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

FREQ[1–11] sets the configuration step number, in the range of 1 to 11. Values outside this range will report an error.

<NR3> sets the frequency of the specified configuration step number, in the range of 10 Hz to 50 MHz.

Examples **Power:Power5:CLRESPONSE:FREQ1Val 200** sets the generator frequency value for frequency band 1 to 200 Hz, for power measurement 5.

Power:Power2:CLRESPONSE:FREQ3Val? might return **2.000E+6**, indicating that the generator frequency output for frequency band 3 is 2.0 MHz, for power measurement 2.

POWER:POWER<x>:CLRESPONSE:GENERator

This command sets or queries the generator source used to send stimulus signals to the DUT, for the Control Loop Response power measurement.

Conditions Requires option 5-PWR or 6-PWR

Group Power

Syntax **Power:Power<x>:CLRESPONSE:GENERator {INTERNAL}**
Power:Power<x>:CLRESPONSE:GENERator?

Arguments **Power**<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

INTERNAL sets the generator to the oscilloscope AFG. This is the only valid argument.

Examples **Power:Power3:CLRESPONSE:GENERator INTERNAL** sets the generator to the oscilloscope AFG for power measurement 3.

Power:Power2:CLRESPONSE:GENERator? might return **INTERNAL**, indicating that the generator source for power measurement 2 is the oscilloscope AFG.

POWer:POWer<x>:CLRESPONSE:IMPEDance

This command sets or queries the vertical termination impedance for the Control Loop Response power measurement.

Conditions Requires option 5-PWR or 6-PWR

Group Power

Syntax `POWer:POWer<x>:CLRESPONSE:IMPEDance {FIFTy|ONEMEGa}`
`POWer:POWer<x>:CLRESPONSE:IMPEDance?`

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

FIFTy sets the impedance to be 50 Ω .

ONEMEGa sets the impedance to be 1 M Ω .

Examples `POWer:POWer3:CLRESPONSE:IMPEDance FIFTy` sets the vertical termination impedance for power measurement 3 to be 50 Ω .

`POWer:POWer2:CLRESPONSE:IMPEDance?` might return CH6, indicating that the vertical termination impedance for power measurement 2 is 1 M Ω .

POWer:POWer<x>:CLRESPONSE:INPUTSource

This command sets or queries the input source for the Control Loop Response power measurement.

Conditions Requires option 5-PWR or 6-PWR

Group Power

Syntax `POWer:POWer<x>:CLRESPONSE:INPUTSource CH<x>`
`POWer:POWer<x>:CLRESPONSE:INPUTSource?`

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

CH<x> sets the channel to use for the output signal source.

- Examples** `Power:Power2:CLRESPONSE:INPUTSource CH8` sets the input for power measurement 2 to be the channel 8 waveform.
- `Power:Power2:CLRESPONSE:INPUTSource?` might return CH1, indicating that the input source for power measurement 1 is channel 1.

POWer:POWer<x>:CLRESPONSE:OUTPUTSource

This command sets or queries the output source for the Control Loop Response power measurement.

- Conditions** Requires option 5-PWR or 6-PWR
- Group** Power
- Syntax** `Power:Power<x>:CLRESPONSE:OUTPUTSource CH<x>`
`Power:Power<x>:CLRESPONSE:OUTPUTSource?`
- Arguments** `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

`CH<x>` sets the channel to use for the output signal source.
- Examples** `Power:Power1:CLRESPONSE:OUTPUTSource CH5` sets the output source for power measurement 1 to be the channel 5 waveform.
- `Power:Power3:CLRESPONSE:OUTPUTSource?` might return CH6, indicating that the output source for power measurement 3 is channel 6.

POWer:POWer<x>:CLRESPONSE:PPD

This command sets or queries the points per decade (PPD) value for the Control Loop Response power measurement.

- Conditions** Requires option 5-PWR or 6-PWR
- Group** Power
- Syntax** `Power:Power<x>:CLRESPONSE:PPD <NR3>`
`Power:Power<x>:CLRESPONSE:PPD?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

`<NR3>` is the PPD value for the measurement, in the range of 10 to 100 points.

Examples `Power:Power3:CLRESPONSE:PPD 30` sets the PPD for Power measurement 3 to be 30 points.

`Power:Power5:CLRESPONSE:PPD?` might return 20, indicating that the PPD value for power measurement 5 is 20 points.

POWER:POWER<x>:CLRESPONSE:STARTFREQUENCY

This command sets or queries the start frequency value for the Control Loop Response power measurement.

Conditions Requires option 5-PWR or 6-PWR

Group Power

Syntax `Power:Power<x>:CLRESPONSE:STARTFREQUENCY <NR3>`
`Power:Power<x>:CLRESPONSE:STARTFREQUENCY?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

`<NR3>` is the starting frequency for the measurement, in the range of 10 Hz to 50 MHz.

Examples `Power:Power8:CLRESPONSE:STARTFREQUENCY 12` sets the starting frequency for power measurement 8 to be 12 Hz.

`Power:Power3:CLRESPONSE:STARTFREQUENCY?` might return 2400, indicating that the starting frequency for power measurement 3 is 2400 Hz.

POWER:POWER<x>:CLRESPONSE:STOPFREQUENCY

This command sets or queries the stop frequency value for the Control Loop Response power measurement.

Conditions Requires option 5-PWR or 6-PWR

Group	Power
Syntax	Power:Power<x>:CLRESPONSE:STOPFREQUENCY <NR3> Power:Power<x>:CLRESPONSE:STOPFREQUENCY?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <NR3> is the stop frequency for the measurement, in the range of 10 Hz to 50 MHz.
Examples	Power:Power3:CLRESPONSE:STOPFREQUENCY 120 sets the stop frequency for Power measurement 8 to be 120 Hz. Power:Power5:CLRESPONSE:STOPFREQUENCY? might return 2000, indicating that the stop frequency for power measurement 5 is 2000 Hz.

POWER:POWER<x>:CYCLEAMP:INPUTSOURCE

This command sets or queries the input source for cycle amplitude measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:CYCLEAMP:INPUTSOURCE {CH<x> MATH<x> REF<x>} Power:Power<x>:CYCLEAMP:INPUTSOURCE?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument. MATH<x> = A math waveform specifier; <x> is ≥ 1. REF<x> = A reference waveform specifier; <x> is ≥ 1.
Examples	Power:Power1:CYCLEAMP:INPUTSOURCE CH1 sets the input source for cycle amplitude measurement as CH1 for the power measurement badge Power 1.

POWer:POWer<x>:CYCLEBase:INPUTSource

This command sets or queries the input source for cycle base measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	<code>POWer:POWer<x>:CYCLEBase:INPUTSource {CH<x> MATH<x> REF<x>}</code> <code>POWer:POWer<x>:CYCLEBase:INPUTSource?</code>
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <code>CH<x></code> = A channel specifier; <code><x></code> is 1 through 8 and is limited by the number of FlexChannels in your instrument. <code>MATH<x></code> = A math waveform specifier; <code><x></code> is ≥ 1. <code>REF<x></code> = A reference waveform specifier; <code><x></code> is ≥ 1.
Examples	<code>POWer:POWer1:CYCLEBase:INPUTSource CH2</code> sets the input source for cycle base measurement as CH2 for the power measurement badge Power 1.

POWer:POWer<x>:CYCLEMAX:INPUTSource

This command sets or queries the input source for cycle maximum measurement in the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	<code>POWer:POWer<x>:CYCLEMAX:INPUTSource {CH<x> MATH<x> REF<x>}</code> <code>POWer:POWer<x>:CYCLEMAX:INPUTSource?</code>
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <code>CH<x></code> = A channel specifier; <code><x></code> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

MATH<x> = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples `Power:Power1:CYCMAx:INPUTSource CH2` sets the input source for cycle maximum measurement as CH2 for the power measurement badge Power 1.

POWER:POWER<x>:CYCLEMin:INPUTSource

This command sets or queries the input source for cycle minimum measurement in the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:CYCLEMin:INPUTSource {CH<x>|MATH<x>|REF<x>}`
`Power:Power<x>:CYCLEMin:INPUTSource?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

`CH<x>` = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

`MATH<x>` = A math waveform specifier; <x> is ≥ 1 .

`REF<x>` = A reference waveform specifier; <x> is ≥ 1 .

Examples `Power:Power1:CYCMin:INPUTSource CH2` sets the input source for cycle minimum measurement as CH2 for the power measurement badge Power 1.

POWER:POWER<x>:CYCLEPKPK:INPUTSource

This command sets or queries the input source for cycle peak-to-peak measurement in the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax	<code>Power:Power<x>:CYCLEPKPK:INPUTSource {CH<x> MATH<x> REF<x>}</code> <code>Power:Power<x>:CYCLEPKPK:INPUTSource?</code>
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <code>CH<x></code> = A channel specifier; <code><x></code> is 1 through 8 and is limited by the number of FlexChannels in your instrument. <code>MATH<x></code> = A math waveform specifier; <code><x></code> is ≥ 1. <code>REF<x></code> = A reference waveform specifier; <code><x></code> is ≥ 1.
Examples	<code>Power:Power1:CYCPKPK:INPUTSource REF1</code> sets the input source for cycle peak-to-peak measurement as REF1 for the power measurement badge Power 1.

POWER:POWER<x>:CYCLETop:INPUTSource

This command sets or queries the input source for cycle top measurement in the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	<code>Power:Power<x>:CYCLETop:INPUTSource {CH<x> MATH<x> REF<x>}</code> <code>Power:Power<x>:CYCLETop:INPUTSource?</code>
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <code>CH<x></code> = A channel specifier; <code><x></code> is 1 through 8 and is limited by the number of FlexChannels in your instrument. <code>MATH<x></code> = A math waveform specifier; <code><x></code> is ≥ 1. <code>REF<x></code> = A reference waveform specifier; <code><x></code> is ≥ 1.
Examples	<code>Power:Power1:CYCTop:INPUTSource REF3</code> sets the input source for cycle top measurement as REF3 for the power measurement badge Power 1.

POWER:POWER<x>:DIDT:INPUTSource

This command sets or queries the input source for di/dt measurement in the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWER:POWER<x>:DIDT:INPUTSource {CH<x>|MATH<x>|REF<x>}
POWER:POWER<x>:DIDT:INPUTSource?

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

MATH<x> = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples POWER:POWER1:DIDT:INPUTSource CH6 sets the input source for di/dt measurement as channel 2.

POWER:POWER<x>:DIDT:SOURCEEDGEType

This command sets or queries the edge type for di/dt measurement of the specified power measurement number. <x> specifies the number of the power measurement badge.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWER:POWER<x>:DIDT:SOURCEEDGEType {RISE|FALL}
POWER:POWER<x>:DIDT:SOURCEEDGEType?

Examples POWER:POWER1:DIDT:SOURCEEDGEType RISE sets the edge type as rise for the di/dt measurement.

POWER:POWER<x>:DVDT:INPUTSource

This command sets or queries the input source for dv/dt measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:DVDT:INPUTSource {CH<x>|MATH<x>|REF<x>}`
`Power:Power<x>:DVDT:INPUTSource?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

`CH<x>` = A channel specifier; `<x>` is 1 through 8 and is limited by the number of FlexChannels in your instrument.

`MATH<x>` = A math waveform specifier; `<x>` is ≥ 1 .

`REF<x>` = A reference waveform specifier; `<x>` is ≥ 1 .

Examples `Power:Power1:DVDT:INPUTSource CH1` sets the input source for dv/dt measurement as channel 1.

POWER:POWER<x>:DVDT:SOURCEEDGEType

This command sets or queries the edge type for dv/dt measurement in the specified power measurement number. The power measurement number is specified by `<x>`.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:DVDT:SOURCEEDGEType {RISE|FALL}`
`Power:Power<x>:DVDT:SOURCEEDGEType?`

Examples `Power:Power1:DVDT:SOURCEEDGEType RISE` sets the edge type as rise for the dv/dt measurement.

POWer:POWer<x>:EFFICIENCY:INPUTType

This command sets or queries the input type (AC or DC) for power Efficiency measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:EFFICIENCY:INPUTType {AC|DC}`
`Power:Power<x>:EFFICIENCY:INPUTType?`

Related Commands [POWer:POWer<x>:EFFICIENCY:OUTPUTType](#)

Arguments Power<x> is the number of a power efficiency measurement. This is the equivalent of the number shown on the power measurement badge of the UI.

AC sets the input voltage type to AC.

DC sets the input voltage type to DC.

Examples `Power:Power1:EFFICIENCY:INPUTType DC` sets the input voltage type to DC for power Efficiency measurement 1

`Power:Power2:EFFICIENCY:INPUTType?` might return AC, indicating that the input voltage type for power Efficiency measurement 2 is AC.

POWer:POWer<x>:EFFICIENCY:IOUT1Source

This command sets or queries the output 1 current source for the power Efficiency measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:EFFICIENCY:IOUT1Source {CH<x>|MATH<x>|REF<x>}`
`Power:Power<x>:EFFICIENCY:IOUT1Source?`

Related Commands [POWER:POWER<x>:EFFICIENCY:IOUT2SOURCE](#)
[POWER:POWER<x>:EFFICIENCY:IOUT3SOURCE](#)

Arguments Power<x> is the number of a power efficiency measurement. This is the equivalent of the number shown in the power measurement badge on the UI.

CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

MATH<x> = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples Power:Power1:EFFICIENCY:IOUT1SOURCE CH4 sets the source for the output 1 current efficiency measurement of power measurement 1 to Channel 4.

Power:Power1:EFFICIENCY:IOUT1SOURCE? might return CH6, indicating that the source for the output 1 current efficiency measurement of power measurement 1 is Channel 6.

POWER:POWER<x>:EFFICIENCY:IOUT2SOURCE

This command sets or queries the output 2 current source for the power Efficiency measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax Power:Power<x>:EFFICIENCY:IOUT2SOURCE {CH<x>|MATH<x>|REF<x>}
 Power:Power<x>:EFFICIENCY:IOUT2SOURCE?

Related Commands [POWER:POWER<x>:EFFICIENCY:IOUT1SOURCE](#)
[POWER:POWER<x>:EFFICIENCY:IOUT3SOURCE](#)

Arguments Power<x> is the number of a power efficiency measurement. This is the equivalent of the number shown in the power measurement badge on the UI.

CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

MATH<x> = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples `Power:Power3:EFFICIENCY:IOUT2SOURCE REF2` sets the source for the output 2 current efficiency measurement of power measurement 3 to Reference waveform 2.

`Power:Power1:EFFICIENCY:IOUT2SOURCE?` might return `MATH1`, indicating that the source for the output 2 current efficiency measurement of power measurement 1 is Math waveform 1.

POWER:POWER<x>:EFFICIENCY:IOUT3SOURCE

This command sets or queries the output 3 current source for the power Efficiency measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:EFFICIENCY:IOUT3SOURCE {CH<x>|MATH<x>|REF<x>}`
`Power:Power<x>:EFFICIENCY:IOUT3SOURCE?`

Related Commands [POWER:POWER<x>:EFFICIENCY:IOUT1SOURCE](#)
[POWER:POWER<x>:EFFICIENCY:IOUT2SOURCE](#)

Arguments `Power<x>` is the number of a power efficiency measurement. This is the equivalent of the number shown on the power measurement badge in the UI.

`CH<x>` = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

`MATH<x>` = A math waveform specifier; <x> is ≥ 1 .

`REF<x>` = A reference waveform specifier; <x> is ≥ 1 .

Examples `Power:Power1:EFFICIENCY:IOUT3SOURCE REF2` sets the source for the output 3 current efficiency measurement of power measurement 1 to Reference waveform 2.

`Power:Power4:EFFICIENCY:IOUT3SOURCE?` might return `CH8`, indicating that the source for the output 3 current efficiency measurement of power measurement 4 is Channel 8.

POWer:POWer<x>:EFFICIENCY:ISOUrce

This command sets or queries the current source for the power Efficiency measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWer:POWer<x>:EFFICIENCY:ISOUrce`
`POWer:POWer<x>:EFFICIENCY:ISOUrce?`

Related Commands [POWer:POWer<x>:EFFICIENCY:VSOUrce](#)

Arguments Power<x> is the number of a power efficiency measurement. This is the equivalent of the number shown in the power measurement badge on the UI.

CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

MATH<x> = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples `POWer:POWer1:EFFICIENCY:ISOUrce CH2` sets the input current source for power Efficiency measurement 1 to Channel 2.

`POWer:POWer6:EFFICIENCY:ISOUrce?` might return CH2, indicating that the input current source for power Efficiency measurement 6 is Channel 2.

POWer:POWer<x>:EFFICIENCY:NUMOFOutputs

This command sets or queries the number of outputs for the power Efficiency measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWer:POWer<x>:EFFICIENCY:NUMOFOutputs {ONE|TWO|THREE}`
`POWer:POWer<x>:EFFICIENCY:NUMOFOutputs?`

Related Commands	POWer:POWer<x>:EFFICIENCY:IOUT1SOUrce POWer:POWer<x>:EFFICIENCY:IOUT2SOUrce POWer:POWer<x>:EFFICIENCY:IOUT3SOUrce POWer:POWer<x>:EFFICIENCY:VOUT1SOUrce POWer:POWer<x>:EFFICIENCY:VOUT2SOUrce POWer:POWer<x>:EFFICIENCY:VOUT3SOUrce
Arguments	Power<x> is the number of a power efficiency measurement. This is the equivalent of the number shown in the power measurement badge on the UI. ONE, TWO, THREE sets the number of outputs to test in the power efficiency measurement.
Examples	POWer:POWer4:EFFICIENCY:NUMOFOutputs TWO sets the number of outputs to measure in power measurement 4 to two. POWer:POWer3:EFFICIENCY:NUMOFOutputs? might return ONE, indicating that the number of outputs being measured in power measurement 3 is one.

POWer:POWer<x>:EFFICIENCY:OUTPUT1Type

This command sets or queries the Output1 type (AC or DC) for the power Efficiency measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	POWer:POWer<x>:EFFICIENCY:OUTPUT1Type {AC DC} POWer:POWer<x>:EFFICIENCY:OUTPUT1Type?
Related Commands	POWer:POWer<x>:EFFICIENCY:INPUTType
Arguments	Power<x> is the number of a power efficiency measurement. This is the equivalent of the number shown on the power measurement badge of the UI. AC sets the output1 voltage type to AC. DC sets the output1 voltage type to DC.

- Examples** `Power:Power1:EFFICIENCY:OUTPUT1Type DC` sets the output type to DC for power Efficiency measurement 1.
- `Power:Power3:EFFICIENCY:OUTPUT1Type?` might return AC, indicating that the power output type for power Efficiency measurement 3 is AC.

POWer:POWer<x>:EFFICIENCY:OUTPUT2Type

This command sets or queries the Output2 type (AC or DC) for the power Efficiency measurement of the specified power measurement number.

- Conditions** Requires option PWR or PS2.

- Group** Power

- Syntax** `Power:Power<x>:EFFICIENCY:OUTPUT2Type {AC|DC}`
`Power:Power<x>:EFFICIENCY:OUTPUT2Type?`

- Related Commands** [POWer:POWer<x>:EFFICIENCY:INPUTType](#)

- Arguments** `Power<x>` is the number of a power efficiency measurement. This is the equivalent of the number shown on the power measurement badge of the UI.
- AC sets the Output2 voltage type to AC.
- DC sets the Output2 voltage type to DC.

- Examples** `Power:Power1:EFFICIENCY:OUTPUT2Type DC` sets the Output2 type to DC for power Efficiency measurement 1.
- `Power:Power3:EFFICIENCY:OUTPUT2Type?` might return AC, indicating that the power Output2 type for power Efficiency measurement 3 is AC.

POWer:POWer<x>:EFFICIENCY:OUTPUT3Type

This command sets or queries the Output3 type (AC or DC) for the power Efficiency measurement of the specified power measurement number.

- Conditions** Requires option PWR or PS2.

- Group** Power

Syntax	<code>Power:Power<x>:EFFICIENCY:OUTPUT3Type {AC DC}</code> <code>Power:Power<x>:EFFICIENCY:OUTPUT3Type?</code>
Related Commands	<code>POWer:POWer<x>:EFFICIENCY:INPUTType</code>
Arguments	<code>Power<x></code> is the number of a power efficiency measurement. This is the equivalent of the number shown on the power measurement badge of the UI. AC sets the Output3 voltage type to AC. DC sets the Output3 voltage type to DC.
Examples	<code>Power:Power1:EFFICIENCY:OUTPUT3Type DC</code> sets the Output3 type to DC for power Efficiency measurement 1. <code>Power:Power3:EFFICIENCY:OUTPUT3Type?</code> might return AC, indicating that the power Output3 type for power Efficiency measurement 3 is AC.

POWer:POWer<x>:EFFICIENCY:OUTPUTType

This command sets or queries the output type (AC or DC) for power Efficiency measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	<code>Power:Power<x>:EFFICIENCY:OUTPUTType {AC DC}</code> <code>Power:Power<x>:EFFICIENCY:OUTPUTType?</code>
Related Commands	<code>POWer:POWer<x>:EFFICIENCY:INPUTType</code>
Arguments	<code>Power<x></code> is the number of a power efficiency measurement. This is the equivalent of the number shown on the power measurement badge of the UI. AC sets the output voltage type to AC. DC sets the output voltage type to DC.
Examples	<code>Power:Power1:EFFICIENCY:OUTPUTType DC</code> sets the output type to DC for power Efficiency measurement 1

`Power:Power3:EFFICIENCY:OUTPUTType?` might return `AC`, indicating that the power output type for power Efficiency measurement 3 is `AC`.

POWer:POWer<x>:EFFICIENCY:VOUT1SOUrce

This command sets or queries the output 1 voltage source for the power Efficiency measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:EFFICIENCY:VOUT1SOUrce {CH<x>|MATH<x>|REF<x>}`
`Power:Power<x>:EFFICIENCY:VOUT1SOUrce?`

Related Commands [POWer:POWer<x>:EFFICIENCY:VOUT2SOUrce](#)
[POWer:POWer<x>:EFFICIENCY:VOUT3SOUrce](#)

Arguments `Power<x>` is the number of a power efficiency measurement. This is the equivalent of the number shown in the power measurement badge on the UI.

`CH<x>` = A channel specifier; `<x>` is 1 through 8 and is limited by the number of FlexChannels in your instrument.

`MATH<x>` = A math waveform specifier; `<x>` is ≥ 1 .

`REF<x>` = A reference waveform specifier; `<x>` is ≥ 1 .

Examples `Power:Power4:EFFICIENCY:VOUT1SOUrce CH4` sets the source for the output 1 voltage efficiency measurement of power measurement 4 to Channel 4.

`Power:Power1:EFFICIENCY:VOUT1SOUrce?` might return `CH3`, indicating that the source for the output 1 voltage efficiency measurement of power measurement 1 is Channel 3.

POWer:POWer<x>:EFFICIENCY:VOUT2SOUrce

This command sets or queries the output 2 voltage source for the power Efficiency measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group	Power
Syntax	Power:Power<x>:EFFICIENCY:VOUT2Source {CH<x> MATH<x> REF<x>} Power:Power<x>:EFFICIENCY:VOUT2Source?
Related Commands	POWer:POWer<x>:EFFICIENCY:VOUT1Source POWer:POWer<x>:EFFICIENCY:VOUT3Source
Arguments	Power<x> is the number of a power efficiency measurement. This is the equivalent of the number shown in the power measurement badge on the UI. CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument. MATH<x> = A math waveform specifier; <x> is ≥ 1. REF<x> = A reference waveform specifier; <x> is ≥ 1.
Examples	Power:Power4:EFFICIENCY:VOUT2Source CH5 sets the source for the output 2 voltage efficiency measurement of power measurement 4 to Channel 5. Power:Power1:EFFICIENCY:VOUT2Source? might return CH3, indicating that the source for the output 2 voltage efficiency measurement of power measurement 1 is Channel 3.

POWer:POWer<x>:EFFICIENCY:VOUT3Source

This command sets or queries the output 3 voltage source for the power Efficiency measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:EFFICIENCY:VOUT3Source {CH<x> MATH<x> REF<x>} Power:Power<x>:EFFICIENCY:VOUT3Source?
Related Commands	POWer:POWer<x>:EFFICIENCY:VOUT1Source POWer:POWer<x>:EFFICIENCY:VOUT2Source

Arguments	Power<x> is the number of a power efficiency measurement. This is the equivalent of the number shown on the power measurement badge in the UI. CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument. MATH<x> = A math waveform specifier; <x> is ≥ 1. REF<x> = A reference waveform specifier; <x> is ≥ 1.
Examples	Power:Power4:EFFICIENCY:VOUT3SOURCE REF7 sets the source for the output 3 voltage efficiency measurement of power measurement 4 to Reference waveform 7. Power:Power1:EFFICIENCY:VOUT3SOURCE? might return CH5, indicating that the source for the output 3 voltage efficiency measurement of power measurement 1 is Channel 5.

POWER:POWER<x>:EFFICIENCY:VSOURCE

This command sets or queries the voltage source for the power Efficiency measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:EFFICIENCY:VSOURCE Power:Power<x>:EFFICIENCY:VSOURCE?

Related Commands [POWER:POWER<x>:EFFICIENCY:ISOURCE](#)

Arguments	Power<x> is the number of a power efficiency measurement. This is the equivalent of the number shown in the power measurement badge on the UI. CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument. MATH<x> = A math waveform specifier; <x> is ≥ 1. REF<x> = A reference waveform specifier; <x> is ≥ 1.
Examples	Power:Power2:EFFICIENCY:VSOURCE CH4 sets the input voltage source for power Efficiency measurement 2 to Channel 4.

`Power:Power1:EFFICIENCY:VSOURCE?` might return CH3, indicating that the input voltage source for power Efficiency measurement 1 is Channel 3.

POWER:POWER<x>:FREQUENCY:EDGE

This command sets or queries the edge type for frequency measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:FREQUENCY:EDGE {RISE|FALL}`
`Power:Power<x>:FREQUENCY:EDGE?`

Related Commands [`POWER:POWER<x>:FREQUENCY:INPUTSOURCE`](#)

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI.

Examples `Power:Power1:FREQUENCY:EDGE FALL` sets the edge type as fall for the frequency measurement.

POWER:POWER<x>:FREQUENCY:INPUTSOURCE

This command sets or queries the input source for frequency measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:FREQUENCY:INPUTSOURCE {CH<x>|MATH<x>|REF<x>}`
`Power:Power<x>:FREQUENCY:INPUTSOURCE?`

Related Commands [`POWER:POWER<x>:FREQUENCY:EDGE`](#)

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument. MATH<x> = A math waveform specifier; <x> is ≥ 1. REF<x> = A reference waveform specifier; <x> is ≥ 1.
Examples	Power:Power1:FREQUENCY:INPUTSource REF1 sets the input source for frequency measurement as REF1.

POWer:POWer<x>:GATing

This command sets or queries the gating type for the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:POWer<x>:GATing {NONE CURSOR SCREEN LOGIC} Power:POWer<x>:GATing?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. NONE makes measurement across the entire waveform record. CURSOR makes measurements on that portion of the waveform between the cursors. Selecting Cursors opens cursors on the measurement source. Set the cursors so that the waveform area of interest is in between the cursors. SCREEN takes measurements on that portion of the waveform shown in the display. When Zoom is on, the display is the zoom window. LOGIC takes measurements only when the logical state of a specified waveform is true.
Examples	Power:Power1:GATing CURSOR sets the gating type for the power measurement badge 1 as Cursor. Power:Power2:GATing? might return SCREEN indicating the configured gating type for the power measurement badge 2.

POWer:POWer<x>:GATing:GLOBal

This command sets or queries the gating settings for the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:POWer<x>:GATing:GLOBal {ON|OFF|1|0}`
`Power:POWer<x>:GATing:GLOBal?`

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

1 | ON selects the gating settings as Global.

0 | OFF selects the gating settings as Local.

Examples `Power:POWer1:GATing:GLOBal 1` selects the gating settings as Global for the power measurement badge 1.

POWer:POWer<x>:HARMONICS:CLASSs

This command sets or queries the class type for the harmonics measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:POWer<x>:HARMONICS:CLASS {CLASSA | CLASSB | CLASSC | CLASSD}`
`Power:POWer<x>:HARMONICS:CLASS?`

Related Commands [POWer:POWer<x>:HARMONICS:STANDard](#)

Examples `Power:POWer1:HARMONICS:CLASS CLASSB` sets the class type for harmonics measurement as CLASSB.

POWer:POWer<x>:HARMONICS:CMETHod

This command sets or queries the fundamental current method for the harmonics measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWer:POWer<x>:HARMONICS:CMETHod {RATed | MEASured}`
`POWer:POWer<x>:HARMONICS:CMETHod?`

Related Commands [POWer:POWer<x>:HARMONICS:STANDard](#)

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

RATed: select to use the standard input current values in the measurement.

MEASured: select to use the measured the input current values in the measurement.

Examples `POWer:POWer1:HARMONICS:CMETHod RATed` sets the current method for harmonics measurement as rated.

POWer:POWer<x>:HARMONICS:FUNDCURRent

This command sets or queries the fundamental current value for the harmonics measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWer:POWer<x>:HARMONICS:FUNDCURRent <NR1>`
`POWer:POWer<x>:HARMONICS:FUNDCURRent?`

Related Commands [POWer:POWer<x>:HARMONICS:CLASs](#)

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <NR1> ranges from 0 to 16.
Examples	Power:Power<x>:HARMONICS:FUNDCURRENT 1.5 sets the fundamental current for the harmonics measurement as 1.5.

POWER:POWER<x>:HARMONICS:HORDER

This command sets or queries the order value for the harmonics measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:HARMONICS:HORDER <NR1> Power:Power<x>:HARMONICS:HORDER?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <NR1> ranges from 40 to 100.
Examples	Power:Power1:HARMONICS:HORDER 70 sets the order value for the harmonics measurement as 70.

POWER:POWER<x>:HARMONICS:HSOURCE

This command sets or queries the source type for the harmonics measurement of the specified power measurement number. The power measurement number is specified by <x>.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:HARMONICS:HSOURCE {CURRENT VOLTAGE} Power:Power<x>:HARMONICS:HSOURCE?

Related Commands [POWer:POWer<x>:HARMONICS:STANDard](#)

Examples `POWer:POWer1:HARMONICS:HSOURCE CURRENT` sets the source as current for the harmonics measurement.

POWer:POWer<x>:HARMONICS:IPOWer

This command sets or queries the input power value for the harmonics measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWer:POWer<x>:HARMONICS:IPOWer <NR1>`
`POWer:POWer<x>:HARMONICS:IPOWer?`

Related Commands [POWer:POWer<x>:HARMONICS:CLASSs](#)

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

`<NR1>` ranges from 0 to 600.

Examples `POWer:POWer<x>:HARMONICS:IPOWer 150` sets the input power for the harmonics measurement as 150.

POWer:POWer<x>:HARMONICS:ISOURce

This command sets or queries the current source for SOA measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWer:POWer<x>:HARMONICS:ISOURce {CH<x>|MATH<x>|REF<x>}`
`POWer:POWer<x>:HARMONICS:ISOURce?`

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument. MATH<x> = A math waveform specifier; <x> is ≥ 1. REF<x> = A reference waveform specifier; <x> is ≥ 1.
Examples	Power:Power1:HARMONICS:ISOURCE CH2 sets the current source for harmonics measurement as channel 2.

POWER:POWER<x>:HARMONICS:LINEFREQUEncy

This command sets or queries the value for the line frequency for the Harmonics measurement.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:HARMONICS:LINEFREQUEncy {Auto FIFTyhz SIXTyhz THREESIXTyhz FOURHUNDREdHz SIXFIFTyhz EIGHTHUNDREdHz CUSTOM Power:Power<x>:HARMONICS:LINEFREQUEncy?
Related Commands	POWER:POWER<x>:HARMONICS:LINEFREQUEncy
Arguments	Power<x> is the Power measurement identifier number. Auto automatically detects and sets the line frequency value. FIFTyhz sets the line frequency value to 50 Hz. SIXTyhz sets the line frequency value to 60 Hz. THREESIXTyhz sets the line frequency value to 360 Hz. FOURHUNDREdHz sets the line frequency value to 400 Hz. SIXFIFTyhz sets the line frequency value to 650 Hz. EIGHTHUNDREdHz sets the line frequency value to 800 Hz. CUSTOM sets the line frequency value to Custom. The default value for custom is 100 Hz. Use the POWER:POWER<x>:HARMONICS:LINEFREQUEncy command to set a custom line frequency value.

Examples `Power:Power3:HARMONICS:LINEFREQUENCY FIFTYHz` sets the line frequency value for the Harmonics power measurement 3 to 50 Hz.

`Power:Power1:HARMONICS:LINEFREQUENCY?` might return
 `Power:Power1:HARMONICS:LINEFREQUENCY CUSTOM`, indicating that the line frequency setting for Harmonics power measurement 1 is a custom value.

Power:Power<x>:HARMONICS:ODDEven

This command sets or queries the harmonics value analysis format of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:HARMONICS:ODDEven {ALL | ODD | EVEN}`
 `Power:Power<x>:HARMONICS:ODDEven?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

 ALL to display all harmonics values

 ODD to display only the odd values of harmonics

 EVEN to display only the even values of harmonics

Examples `Power:Power3:HARMONICS:ODDEven ALL` sets the harmonics value analysis format to display all result values for harmonics measurement 3.

Power:Power<x>:HARMONICS:PFACTOR

This command sets or queries the value of power factor for the harmonics measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:HARMONICS:PFACTOR <NR1>`
`Power:Power<x>:HARMONICS:PFACTOR?`

Related Commands [POWER:Power<x>:HARMONICS:CLASs](#)

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.
`<NR1>` ranges from 0 to 1.

Examples `Power:Power2:HARMONICS:PFACTOR 0.7` sets the power factor as 0.7 for harmonics measurement 2.

POWER:Power<x>:HARMONICS:POWERRating

This command sets or queries the power level for the harmonics measurement of the specified power measurement number. The power measurement number is specified by `<x>`.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:HARMONICS:POWERRating {HIGH | LOW}`
`Power:Power<x>:HARMONICS:POWERRating?`

Related Commands [POWER:Power<x>:HARMONICS:STANDARD](#)

Examples `Power:Power1:HARMONICS:POWERRating HIGH` sets the power level as high for harmonics measurement 1.

POWER:Power<x>:HARMONICS:RCURRENT

This command sets or queries the rated current for the harmonics measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group	Power
Syntax	Power:Power<x>:HARMONICS:RCURRENT <NR1> Power:Power<x>:HARMONICS:RCURRENT?
Related Commands	Power:Power<x>:HARMONICS:CMETHOD
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <NR1> ranges from 0 to 100
Examples	Power:Power2:HARMONICS:RCURRENT 1.5 sets the rated current as 1.5 for harmonics measurement 2.

Power:Power<x>:HARMONICS:STANDARD

This command sets or queries the test mode for harmonics measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:HARMONICS:STANDARD {NONE IEC MIL AM14} Power:Power<x>:HARMONICS:STANDARD?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. NONE = No standard IEC = IEC 61000-3-2 standard MIL = MIL-STD-1399 standard AM14 = AM14 standard
Examples	Power:Power3:HARMONICS:STANDARD AM14 sets the test mode for harmonics measurement as AM14 for harmonics measurement 3.

POWer:POWer<x>:HARMONICS:STARTFREQUEncy

This command sets or queries the value for the start frequency for the Harmonics measurement. in the range of 1 Hz to 1 GHz.

Conditions Requires option PWR or PS2.

Group Power

Syntax Power:POWer<x>:HARMONICS:STARTFREQUEncy <NR2>
Power:POWer<x>:HARMONICS:STARTFREQUEncy?

Arguments Power<x> is the Power measurement.
<NR2> sets the starting frequency, in hertz.

Examples Power:POWer1:HARMONICS:STARTFREQUEncy 1000 sets the Harmonics measurement start frequency to 1 kHz for power measurement 1.
Power:POWer4:HARMONICS:STARTFREQUEncy? might return
Power:POWer4:HARMONICS:STARTFREQUEncy 60. indicating that the start frequency for Harmonics power measurement 4 is 60 Hz.

POWer:POWer<x>:HARMONICS:UNITs

This command sets or queries the harmonics results units of the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax Power:POWer<x>:HARMONICS:UNITs {LOG | LINEar}
Power:POWer<x>:HARMONICS:UNITs?

Examples Power:POWer1:HARMONICS:UNITs LOG sets the harmonics results units as logarithmic for harmonics measurement 1.

POWer:POWer<x>:HARMONICS:VSOURce

This command sets or queries the voltage source for SOA measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWer:POWer<x>:HARMONICS:VSOURce {CH<x>|MATH<x>|REF<x>}`
`POWer:POWer<x>:HARMONICS:VSOURce?`

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

MATH<x> = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples `POWer:POWer4:HARMONICS:VSOURce CH1` sets the voltage source for harmonics measurement 4 as channel 1.

POWer:POWer<x>:INDUCTANCE:EDGESource

This command sets or queries the edge source for the power inductance measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWer:POWer<x>:INDUCTANCE:EDGESource {CH<x>|MATH<x>|REF<x>}`
`POWer:POWer<x>:INDUCTANCE:EDGESource?`

Related Commands [POWer:POWer<x>:INDUCTANCE:ISOURce](#)
[POWer:POWer<x>:INDUCTANCE:VSOURce](#)

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument. MATH<x> = A math waveform specifier; <x> is ≥ 1. REF<x> = A reference waveform specifier; <x> is ≥ 1.
Examples	Power:Power1:HARMONICS:INDUCTANCE:EDGESource CH2 sets the signal edge source for inductance measurement 1 as Channel 2. Power:Power3:HARMONICS:INDUCTANCE:EDGESource? might return CH4, indicating that the signal edge source for inductance measurement 3 is Channel 4.

POWER:POWER<x>:INDUCTANCE:ISource

This command sets or queries the current signal source for the inductance measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:INDUCTANCE:ISource Power:Power<x>:INDUCTANCE:ISource?
Related Commands	POWER:POWER<x>:INDUCTANCE:EDGESource POWER:POWER<x>:INDUCTANCE:VSOURCE
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument. MATH<x> = A math waveform specifier; <x> is ≥ 1. REF<x> = A reference waveform specifier; <x> is ≥ 1.
Examples	Power:Power7:HARMONICS:INDUCTANCE:ISource CH3 sets the current signal source for inductance measurement 7 as Channel 3.

`Power:Power3:HARMONICS:INDUCTANCE:ISource?` might return CH4, indicating that the current signal source for inductance measurement 3 is Channel 4.

POWer:POWer<x>:INDUCTANCE:VSOURce

This command sets or queries the voltage source for inductance measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:INDUCTANCE:VSOURce`
`Power:Power<x>:INDUCTANCE:VSOURce?`

Related Commands [`POWer:POWer<x>:INDUCTANCE:EDGESource`](#)
[`POWer:POWer<x>:INDUCTANCE:ISOURce`](#)

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

`CH<x>` = A channel specifier; `<x>` is 1 through 8 and is limited by the number of FlexChannels in your instrument.

`MATH<x>` = A math waveform specifier; `<x>` is ≥ 1 .

`REF<x>` = A reference waveform specifier; `<x>` is ≥ 1 .

Examples `Power:Power7:HARMONICS:INDUCTANCE:VSource CH3` sets the voltage signal source for inductance measurement 7 as Channel 3.

`Power:Power1:HARMONICS:INDUCTANCE:VSource?` might return MATH2, indicating that the voltage signal source for inductance measurement 1 is Math waveform 2.

POWer:POWer<x>:INPUTCAP:ISOURce

This command sets or queries the inrush current input source of the specified Input Capacitance measurement.

Conditions Requires option PWR or PS2.

Group	Power
Syntax	Power:Power<x>:INPUTCAP:ISOURce {CH<x> REF<x> MATH<x>} Power:Power<x>:INPUTCAP:ISOURce?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. CH<x> is the channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels. REF<x> is the Reference waveform specifier ≥ 1. This is the equivalent of the number shown on a Reference waveform badge in the UI. MATH<x> is the Math waveform specifier ≥ 1. This is the equivalent of the number shown on a Math waveform badge in the UI.
Examples	Power:Power1:INRUSHcurrent:ISOURce REF3 sets the current source of Input Capacitance 1 to Reference waveform 3. Power:Power2:INRUSHcurrent:ISOURce? might return Power:Power2:INRUSHcurrent:ISOURce CH6, indicating that the current source of the Input Capacitance measurement 2 is Channel 6.

POWER:POWER<x>:INPUTCAP:PEAKCURREnt

This command sets or queries the peak current value of the specified Input Capacitance measurement.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:INPUTCAP:PEAKCURREnt <NR3> Power:Power<x>:INPUTCAP:PEAKCURREnt?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <NR3> is a floating point number that represents the peak current value, in amps, in the range -100 A to 100 A.

- Examples** `Power:Power4:INPUTCAP:PEAKCURRENT 12.5` sets the peak current value of Input Capacitance measurement 4 to 12.5 amps.
- `Power:Power2:INPUTCAP:PEAKCURRENT?` might return
`Power:Power2:INPUTCAP:PEAKCURRENT -85`, indicating that the peak current setting of Input Capacitance measurement 2 is -85 amps.

POWER:POWER<x>:INPUTCAP:PEAKVOLTage

This command sets or queries the peak voltage value of the specified Input Capacitance measurement.

- Conditions** Requires option PWR or PS2.

- Group** Power

- Syntax** `Power:Power<x>:INPUTCAP:PEAKVOLTage <NR3>`
`Power:Power<x>:INPUTCAP:PEAKVOLTage?`

- Arguments** `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.
- `<NR3>` is a floating point number that represents the peak voltage value in the range -100 V to 100 V.

- Examples** `Power:Power4:INPUTCAP:PEAKVOLTage 122.5` sets the peak voltage value of Input Capacitance measurement 4 to 122.5 volts.
- `Power:Power2:INPUTCAP:PEAKVOLTage?` might return
`Power:Power2:INPUTCAP:PEAKVOLTage 955`, indicating that the peak voltage setting of Input Capacitance measurement 2 to 955 volts.

POWER:POWER<x>:INPUTCAP:VSOURCE

This command sets or queries the input voltage source of the specified Input Capacitance measurement.

- Conditions** Requires option PWR or PS2.

- Group** Power

Syntax `Power:Power<x>:INPUTCAP:VSOURCE {CH<x>|REF<x>|MATH<x>}`
`Power:Power<x>:INPUTCAP:VSOURCE?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

`CH<x>` is the channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels.

`REF<x>` is the Reference waveform specifier ≥ 1 . This is the equivalent of the number shown on a Reference waveform badge in the UI.

`MATH<x>` is the Math waveform specifier ≥ 1 . This is the equivalent of the number shown on a Math waveform badge in the UI.

Examples `Power:Power1:INPUTCAP:VSOURCE CH2` sets the voltage source of the input capacitance measurement 1 to Reference Channel 2.

`Power:Power1:INPUTCAP:VSOURCE?` might return
`Power:Power2:INPUTCAP:VSOURCE MATH6`, indicating that the voltage source of Input Capacitance measurement 2 is Math waveform 6.

POWER:POWER<x>:INRUSHcurrent:INPUTSource

This command sets or returns the input source of the specified Inrush Current measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:INRUSHcurrent:INPUTSource`
`Power:Power<x>:INRUSHcurrent:INPUTSource?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

`CH<x>` is the channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels.

`REF<x>` is the Reference waveform specifier ≥ 1 . This is the equivalent of the number shown on a Reference waveform badge in the UI.

`MATH<x>` is the Math waveform specifier ≥ 1 . This is the equivalent of the number shown on a Math waveform badge in the UI.

Examples `Power:Power3:INRUSHcurrent:INPUTSource MATH1` sets the input source of Inrush Current measurement 3 to Math waveform 1.

`Power:Power2:INRUSHcurrent:INPUTSource?` might return
`Power:Power2:INRUSHcurrent:INPUTSource CH7`, indicating that the input source of Inrush Current measurement 2 is Channel 7.

POWer:POWer<x>:INRUSHcurrent:PEAKCURREnt

This command sets or returns the peak current value of the specified Inrush Current measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:INRUSHcurrent:PEAKCURREnt <NR3>`
`Power:Power<x>:INRUSHcurrent:PEAKCURREnt?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

`<NR3>` is a floating point number that represents the peak current value, in amps, in the range -100 A to 100 A.

Examples `Power:Power4:INRUSHcurrent:PEAKCURREnt 12.5` sets the peak current value of Inrush Current measurement 4 to 12.5 amps.

`Power:Power2:INRUSHcurrent:PEAKCURREnt?` might return
`Power:Power2:INRUSHcurrent:PEAKCURREnt 75`, indicating that the peak current setting of Inrush Current measurement 2 is 75 amps.

POWer:POWer<x>:IVSINTEGRALV:ISOURce

This command sets or queries the current source for I vs Integral V measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax	<code>Power:Power<x>:IVSINTEGRALV:ISOURce {CH<x> MATH<x> REF<x>}</code> <code>Power:Power<x>:IVSINTEGRALV:ISOURce?</code>
Related Commands	POWer:POWer<x>:IVSINTEGRALV:VSOURce
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI. <code>CH<x></code> sets the channel specifier; <code><x></code> is 1 through 8 and is limited by the number of FlexChannels in your instrument. <code>MATH<x></code> = A math waveform specifier; <code><x></code> is ≥ 1. <code>REF<x></code> = A reference waveform specifier; <code><x></code> is ≥ 1.
Examples	<code>Power:Power1:IVSINTEGRALV:ISOURce CH7</code> sets power measurement 1 to use Channel 7 as the current signal source for the measurement. <code>Power:Power2:IVSINTEGRALV:ISOURce?</code> might return CH1, indicating that channel 1 is the current signal source for power measurement number 2.

POWer:POWer<x>:IVSINTEGRALV:VSOURce

This command sets or queries the voltage source for I vs Integral V measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	<code>Power:Power<x>:IVSINTEGRALV:VSOURce {CH<x> MATH<x> REF<x>}</code> <code>Power:Power<x>:IVSINTEGRALV:VSOURce?</code>
Related Commands	POWer:POWer<x>:IVSINTEGRALV:ISOURce
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI. <code>CH<x></code> sets the channel specifier; <code><x></code> is 1 through 8 and is limited by the number of FlexChannels in your instrument. <code>MATH<x></code> = A math waveform specifier; <code><x></code> is ≥ 1. <code>REF<x></code> = A reference waveform specifier; <code><x></code> is ≥ 1.

- Examples** `Power:Power2:IVSINTEGRALV:VSOURCE CH1` sets I vs Integral V power measurement 2 to use Channel 1 as the voltage signal source for the measurement.
- `Power:Power4:IVSINTEGRALV:VSOURCE?` might return CH1, indicating that channel 1 is the voltage source for I vs Integral V power measurement number 4.

POWER:POWER<x>:LABEL

This command sets or queries the label for the specified power measurement. As the label can contain non 7-bit ASCII text, it is stored in Percent Encoding format. The power measurement badge is specified by x.

- Conditions** Requires option PWR or PS2.

- Group** Power

- Syntax** `Power:Power<x>:LABEL <QString>`
`Power:Power<x>:LABEL?`

- Examples** `Power:Power1:LABEL "NewMeasurement"` sets the custom measurement name for the measurement in Power 1 badge as New Measurement.

POWER:POWER<x>:LINERIPPLE:INPUTSource

This command sets or queries the input source for line ripple measurement of the specified power measurement number.

- Conditions** Requires option PWR or PS2.

- Group** Power

- Syntax** `Power:Power<x>:LINERIPPLE:INPUTSource {CH<x>|MATH<x>|REF<x>}`
`Power:Power<x>:LINERIPPLE:INPUTSource?`

- Arguments** `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.
- `CH<x>` = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.
- `MATH<x>` = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples `Power:Power1:LINERIPPLE:INPUTSource CH1` sets channel 1 as the input source for line ripple measurement of power measurement badge Power 1.

POWER:POWER<x>:LINERIPPLE:LFREQUENCY

This command sets or queries the frequency present for line ripple measurement of the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:LINERIPPLE:LFREQUENCY {FIFTY | SIXTY | FOURHUNDRED}`
`Power:Power<x>:LINERIPPLE:LFREQUENCY?`

Examples `Power:Power2:LINERIPPLE:LFREQUENCY FIFTY` sets the frequency present for line ripple measurement as 50 Hz for power measurement 2.

POWER:POWER<x>:MAGNETICLOSS:ISOURCE

This command sets or queries the current source for the magnetic loss measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:MAGNETICLOSS:ISOURCE {CH<x> | MATH<x> | REF<x>}`
`Power:Power<x>:MAGNETICLOSS:ISOURCE?`

Related Commands [POWER:Power<x>:MAGNETICLOSS:VSOURCE](#)

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI.

CH<x> sets the channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

MATH<x> = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples `Power:Power1:MAGNETICLOSS:ISOURce CH7` sets power measurement 1 to use Channel 7 as the current signal source for the magnetic loss measurement.

`Power:Power2:MAGNETICLOSS:ISOURce?` might return CH1, indicating that channel 1 is the current signal source for the magnetic loss measurement number 2.

POWER:POWER<x>:MAGNETICLOSS:VSOURce

This command sets or queries the voltage source for magnetic measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:MAGNETICLOSS:VSOURce {CH<x>|MATH<x>|REF<x>}`
`Power:Power<x>:MAGNETICLOSS:VSOURce?`

Related Commands [POWER:POWER<x>:MAGNETICLOSS:ISOURce](#)

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI.

CH<x> sets the channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

MATH<x> = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples `Power:Power2:MAGNETICLOSS:VSOURce CH4` sets power measurement 2 to use Channel 4 as the voltage signal source for the magnetic loss measurement.

`Power:Power1:MAGNETICLOSS:VSOURce?` might return CH1, indicating that channel 1 is the voltage source for the magnetic loss measurement number 1.

POWer:POWer<x>:MAGPROPERTY:AREAofcrosssection

This command sets or queries the coil cross section area for magnetic measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWer:POWer<x>:MAGPROPERTY:AREAofcrosssection <NR2>`
`POWer:POWer<x>:MAGPROPERTY:AREAofcrosssection?`

Related Commands [POWer:POWer<x>:MAGPROPERTY:UNITs](#)

Arguments `Power<x>` is the magnetic property power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI.
`<NR2>` is the cross section area in square meters, in the range of 1 nanometer² to 1 M².

Examples `POWer:POWer2:MAGPROPERTY:AREAofcrosssection .005` sets the primary winding cross section area to 5 for power measurement 2.

`POWer:POWer4:MAGPROPERTY:AREAofcrosssection?` might return 10.54E-3, indicating that the coil cross section area for power measurement 4 is 0.01054.

NOTE. The coil Cross Section Area unit depends on the setting of the Units value (see [POWer:POWer<x>:MAGPROPERTY:UNITs](#)). The command returns the same numeric value regardless of the units setting.

POWer:POWer<x>:MAGPROPERTY:EDGESOURCe

This command sets or queries the edge source type for the magnetic property measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax	Power:Power<x>:MAGPROPERTY:EDGESOURCE {Current VOLTAGE} Power:Power<x>:MAGPROPERTY:EDGESOURCE?
Arguments	Power<x> is the magnetic property power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. Current sets the measurement to use the primary voltage source as the signal edge for the magnetic property measurement. VOLTAGE sets the measurement to use the primary current source as the signal edge for the magnetic property measurement.
Examples	Power:Power1:MAGPROPERTY:EDGESource VOLTAGE sets the signal edge source for power measurement 1 to use the primary voltage source. Power:Power3:MAGPROPERTY:EDGESource? might return CURRENT, indicating that the signal edge source for power measurement 3 is the primary current source.

POWer:POWer<x>:MAGPROPERTY:ISOURce

This command sets or queries the current source for the magnetic property measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:MAGPROPERTY:ISOURce {CH<x> MATH<x> REF<x>} Power:Power<x>:MAGPROPERTY:ISOURce?
Related Commands	POWer:POWer<x>:MAGPROPERTY:VSOURce
Arguments	Power<x> is the magnetic property power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI. CH<x> sets the channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument. MATH<x> = A math waveform specifier; <x> is ≥ 1. REF<x> = A reference waveform specifier; <x> is ≥ 1.

- Examples** `Power:Power1:MAGPROPERTY:ISOURce CH4` sets power measurement 1 to use channel 4 as the primary winding current source for the magnetic power measurement.
- `Power:Power2:MAGPROPERTY:ISOURce?` might return CH1, indicating that channel 1 is the primary winding current source for the magnetic power measurement number 2.

POWER:POWER<x>:MAGPROPERTY:LENGth

This command sets or queries the conductor length of the primary winding for magnetic measurement of the specified power measurement number.

- Conditions** Requires option PWR or PS2.
- Group** Power
- Syntax** `Power:Power<x>:MAGPROPERTY:LENGth <NR2>`
`Power:Power<x>:MAGPROPERTY:LENGth?`
- Related Commands** [POWER:POWER<x>:MAGPROPERTY:UNITs](#)
- Arguments** `Power<x>` is the magnetic property power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI.
- `<NR2>` is the magnetic length, in the range of 1.00E-09 through 1,000,000.
- Examples** `Power:Power2:MAGPROPERTY:LENGth 2.5` sets the primary winding length to 2.5 for power measurement 2.
- `Power:Power5:MAGPROPERTY:LENGth?` might return `10.54E-3`, indicating that the primary winding length for power measurement 5 is 0.01054.

NOTE. The Magnetic Length unit depends on the setting of the Units value (see [POWER:POWER<x>:MAGPROPERTY:UNITs](#)). The command returns the same numeric value regardless of the units setting.

POWER:POWER<x>:MAGPROPERTY:PRIMARYTURNs

This command sets or queries the number of primary turns for magnetic measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	<pre>Power:Power<x>:MAGPROPERTY:PRIMARYTURNS <NR1> Power:Power<x>:MAGPROPERTY:PRIMARYTURNS?</pre>
Related Commands	Power:Power<x>:MAGPROPERTY:LENgth Power:Power<x>:MAGPROPERTY:AREAOfcrosssection
Arguments	Power<x> is the magnetic property power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI. <NR1> is the integer number of turns in the primary winding, in the range of 1 to 1 M.
Examples	Power:Power2:MAGPROPERTY:PRIMARYTURNS 150 sets the primary winding turn count to 150 for power measurement 2. Power:Power5:MAGPROPERTY:PRIMARYTURNS? might return 50, indicating that the primary winding has 50 turns for power measurement 5.

Power:Power<x>:MAGPROPERTY:SEC1SOURCE

This command sets or queries the current source channel for secondary winding 1 for magnetic measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	<pre>Power:Power<x>:MAGPROPERTY:SEC1SOURCE {CH<x> MATH<x> REF<x>} Power:Power<x>:MAGPROPERTY:SEC1SOURCE?</pre>
Related Commands	Power:Power<x>:MAGPROPERTY:SEC2SOURCE Power:Power<x>:MAGPROPERTY:SEC3SOURCE Power:Power<x>:MAGPROPERTY:SEC4SOURCE Power:Power<x>:MAGPROPERTY:SEC5SOURCE

[POWer:POWer<x>:MAGPROPERTY:SEC6SOURce](#)

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI. CH<x> sets the channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument. MATH<x> = A math waveform specifier; <x> is ≥ 1. REF<x> = A reference waveform specifier; <x> is ≥ 1.
Examples	Power:Power3:MAGPROPERTY:SEC1SOURce CH7 sets power measurement number 3 to use channel 7 as the current source for measuring secondary winding 1. Power:Power1:MAGPROPERTY:SEC1SOURce? might return CH3, indicating that channel 3 is the source for measuring secondary winding 1 of magnetic power measurement number 1.

POWer:POWer<x>:MAGPROPERTY:SEC1TURNs

This command sets or queries the number of turns of secondary winding 1 for magnetic measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:MAGPROPERTY:SEC1TURNs <NR1> Power:Power<x>:MAGPROPERTY:SEC1TURNs?
Related Commands	POWer:POWer<x>:MAGPROPERTY:SEC2TURNs POWer:POWer<x>:MAGPROPERTY:SEC3TURNs POWer:POWer<x>:MAGPROPERTY:SEC4TURNs POWer:POWer<x>:MAGPROPERTY:SEC5TURNs POWer:POWer<x>:MAGPROPERTY:SEC6TURNs
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI.

<NR1> is the number of turns on the secondary winding, and ranges from 0 to 1,000,000.

- Examples** `Power:Power3:MAGPROPERTY:SEC1TURNS 500` sets the number of turns on secondary winding 1 to 500 for magnetic power measurement number 3.
- `Power:Power3:MAGPROPERTY:SEC1TURNS?` might return 2500, indicating that their are 2,500 turns on secondary winding 4 for magnetic power measurement number 3.

POWer:POWer<x>:MAGPROPERTY:SEC2SOURce

This command sets or queries the current source for secondary winding2 for magnetic measurement of the specified power measurement number.

- Conditions** Requires option PWR or PS2.
- Group** Power
- Syntax** `Power:Power<x>:MAGPROPERTY:SEC2SOURce {CH<x>|MATH<x>|REF<x>}`
`Power:Power<x>:MAGPROPERTY:SEC2SOURce?`
- Related Commands** [POWer:POWer<x>:MAGPROPERTY:SEC1SOURce](#)
[POWer:POWer<x>:MAGPROPERTY:SEC3SOURce](#)
[POWer:POWer<x>:MAGPROPERTY:SEC4SOURce](#)
[POWer:POWer<x>:MAGPROPERTY:SEC5SOURce](#)
[POWer:POWer<x>:MAGPROPERTY:SEC6SOURce](#)
- Arguments** `Power<x>` is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI.
- `CH<x>` sets the channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.
- `MATH<x>` = A math waveform specifier; <x> is ≥ 1 .
- `REF<x>` = A reference waveform specifier; <x> is ≥ 1 .
- Examples** `Power:Power1:MAGPROPERTY:SEC2SOURce CH2` sets power measurement number 1 to use channel 2 as the current source for measuring secondary winding 2.

Power:Power3:MAGPROPERTY:SEC2SOURCE? might return MATH2, indicating that math waveform 2 is the source for measuring secondary winding 2 of magnetic power measurement number 3.

POWER:POWER<x>:MAGPROPERTY:SEC2TURNS

This command sets or queries the number of turns of secondary winding 2 for magnetic measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax **Power:Power<x>:MAGPROPERTY:SEC2TURNS <NR1>**
Power:Power<x>:MAGPROPERTY:SEC2TURNS?

Related Commands [POWER:POWER<x>:MAGPROPERTY:SEC1TURNS](#)
[POWER:POWER<x>:MAGPROPERTY:SEC3TURNS](#)
[POWER:POWER<x>:MAGPROPERTY:SEC4TURNS](#)
[POWER:POWER<x>:MAGPROPERTY:SEC5TURNS](#)
[POWER:POWER<x>:MAGPROPERTY:SEC6TURNS](#)

Arguments **Power<x>** is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI.
<NR1> is the number of turns on the secondary winding, and ranges from 0 to 1,000,000.

Examples **Power:Power1:MAGPROPERTY:SEC2TURNS 50** sets the number of turns on secondary winding 2 to 50 for magnetic power measurement number 3.
Power:Power7:MAGPROPERTY:SEC2TURNS? might return 250, indicating that there are 250 turns on secondary winding 2 for magnetic power measurement number 7.

POWER:POWER<x>:MAGPROPERTY:SEC3SOURCE

This command sets or queries the current source channel for secondary winding 3 for magnetic measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:MAGPROPERTY:SEC3SOURCE {CH<x> MATH<x> REF<x>} Power:Power<x>:MAGPROPERTY:SEC3SOURCE?
Related Commands	Power:Power<x>:MAGPROPERTY:SEC1SOURCE Power:Power<x>:MAGPROPERTY:SEC2SOURCE Power:Power<x>:MAGPROPERTY:SEC4SOURCE Power:Power<x>:MAGPROPERTY:SEC5SOURCE Power:Power<x>:MAGPROPERTY:SEC6SOURCE
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI. CH<x> sets the channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument. MATH<x> = A math waveform specifier; <x> is ≥ 1. REF<x> = A reference waveform specifier; <x> is ≥ 1.
Examples	Power:Power1:MAGPROPERTY:SEC3SOURCE CH1 sets power measurement number 1 to use channel 1 as the current source for measuring secondary winding 3. Power:Power1:MAGPROPERTY:SEC3SOURCE? might return CH1, indicating that channel 1 is the source for measuring secondary winding 3 of magnetic power measurement number 1.

Power:Power<x>:MAGPROPERTY:SEC3TURNS

This command sets or queries the number of turns of secondary winding 3 for magnetic measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax	Power:POWer<x>:MAGPROPERTY:SEC3TURNS <NR1> Power:POWer<x>:MAGPROPERTY:SEC3TURNS?
Related Commands	POWer:POWer<x>:MAGPROPERTY:SEC1TURNS POWer:POWer<x>:MAGPROPERTY:SEC2TURNS POWer:POWer<x>:MAGPROPERTY:SEC4TURNS POWer:POWer<x>:MAGPROPERTY:SEC5TURNS POWer:POWer<x>:MAGPROPERTY:SEC6TURNS
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI. <NR1> is the number of turns on the secondary winding, and ranges from 0 to 1,000,000.
Examples	Power:POWer1:MAGPROPERTY:SEC3TURNS 15 sets the number of turns on secondary winding 3 to 15 for magnetic power measurement number 1. Power:POWer3:MAGPROPERTY:SEC3TURNS? might return 2500, indicating that there are 2,500 turns on secondary winding 3 for magnetic power measurement number 3.

POWer:POWer<x>:MAGPROPERTY:SEC4SOURce

This command sets or queries the current source for secondary winding 4 for magnetic measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:POWer<x>:MAGPROPERTY:SEC4SOURce {CH<x> MATH<x> REF<x>} Power:POWer<x>:MAGPROPERTY:SEC4SOURce?
Related Commands	POWer:POWer<x>:MAGPROPERTY:SEC1SOURce POWer:POWer<x>:MAGPROPERTY:SEC2SOURce POWer:POWer<x>:MAGPROPERTY:SEC3SOURce POWer:POWer<x>:MAGPROPERTY:SEC5SOURce

[POWER:POWER<x>:MAGPROPERTY:SEC6SOURCE](#)

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI. CH<x> sets the channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument. MATH<x> = A math waveform specifier; <x> is ≥ 1. REF<x> = A reference waveform specifier; <x> is ≥ 1.
Examples	Power:Power3:MAGPROPERTY:SEC4SOURCE CH7 sets power measurement number 3 to use channel 7 as the current source for measuring secondary winding 4. Power:Power12:MAGPROPERTY:SEC4SOURCE? might return CH1, indicating that Channel 1 is the source for measuring secondary winding 4 of magnetic power measurement number 12.

POWER:POWER<x>:MAGPROPERTY:SEC4TURNS

This command sets or queries the number of turns of secondary winding 4 for magnetic measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	<pre>Power:Power<x>:MAGPROPERTY:SEC4TURNS <NR1> Power:Power<x>:MAGPROPERTY:SEC4TURNS?</pre>

Related Commands

[POWER:POWER<x>:MAGPROPERTY:SEC1TURNS](#)

[POWER:POWER<x>:MAGPROPERTY:SEC2TURNS](#)

[POWER:POWER<x>:MAGPROPERTY:SEC3TURNS](#)

[POWER:POWER<x>:MAGPROPERTY:SEC5TURNS](#)

[POWER:POWER<x>:MAGPROPERTY:SEC6TURNS](#)

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI.
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<NR1> is the number of turns on the secondary winding, and ranges from 0 to 1,000,000.

Examples `Power:Power4:MAGPROPERTY:SEC4TURNS 500` sets the number of turns on secondary winding 4 to 500 for magnetic power measurement number 4.

`Power:Power1:MAGPROPERTY:SEC4TURNS?` might return 90, indicating that there are 90 turns on secondary winding 4 for magnetic power measurement number 3.

POWER:POWER<x>:MAGPROPERTY:SEC5SOURCE

This command sets or queries the current source for secondary winding 5 for magnetic measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:MAGPROPERTY:SEC5SOURCE {CH<x>|MATH<x>|REF<x>}`
`Power:Power<x>:MAGPROPERTY:SEC5SOURCE?`

Related Commands [POWER:POWER<x>:MAGPROPERTY:SEC1SOURCE](#)
[POWER:POWER<x>:MAGPROPERTY:SEC2SOURCE](#)
[POWER:POWER<x>:MAGPROPERTY:SEC3SOURCE](#)
[POWER:POWER<x>:MAGPROPERTY:SEC4SOURCE](#)
[POWER:POWER<x>:MAGPROPERTY:SEC6SOURCE](#)

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI.

`CH<x>` sets the channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

`MATH<x>` = A math waveform specifier; <x> is ≥ 1 .

`REF<x>` = A reference waveform specifier; <x> is ≥ 1 .

Examples `Power:Power3:MAGPROPERTY:SEC5SOURCE MATH1` sets power measurement number 3 to use math waveform 1 as the current source for measuring secondary winding 5.

`Power:Power4:MAGPROPERTY:SEC5SOURCE?` might return CH2, indicating that channel 2 is the source for measuring secondary winding 5 of magnetic power measurement number 4.

Power:Power<x>:MAGPROPERTY:SEC5TURNS

This command sets or queries the number of turns of secondary winding 5 for magnetic measurement of the specified power measurement badge.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:MAGPROPERTY:SEC5TURNS <NR1>`
`Power:Power<x>:MAGPROPERTY:SEC5TURNS?`

Related Commands [Power:Power<x>:MAGPROPERTY:SEC1TURNS](#)
[Power:Power<x>:MAGPROPERTY:SEC2TURNS](#)
[Power:Power<x>:MAGPROPERTY:SEC3TURNS](#)
[Power:Power<x>:MAGPROPERTY:SEC4TURNS](#)
[Power:Power<x>:MAGPROPERTY:SEC6TURNS](#)

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI.
`<NR1>` is the number of turns on the secondary winding, and ranges from 0 to 1,000,000.

Examples `Power:Power3:MAGPROPERTY:SEC5TURNS 250` sets the number of turns on secondary winding 5 to 250 for magnetic power measurement number 3.
`Power:Power1:MAGPROPERTY:SEC5TURNS?` might return 150, indicating that there are 150 turns on secondary winding 5 for magnetic power measurement number 1.

Power:Power<x>:MAGPROPERTY:SEC6SOURCE

This command sets or queries the current source for secondary winding 6 for magnetic measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:MAGPROPERTY:SEC6SOURCE {CH<x> MATH<x> REF<x>} Power:Power<x>:MAGPROPERTY:SEC6SOURCE?
Related Commands	Power:Power<x>:MAGPROPERTY:SEC1SOURCE Power:Power<x>:MAGPROPERTY:SEC2SOURCE Power:Power<x>:MAGPROPERTY:SEC3SOURCE Power:Power<x>:MAGPROPERTY:SEC4SOURCE Power:Power<x>:MAGPROPERTY:SEC5SOURCE
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI. CH<x> sets the channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument. MATH<x> = A math waveform specifier; <x> is ≥ 1. REF<x> = A reference waveform specifier; <x> is ≥ 1.
Examples	Power:Power3:MAGPROPERTY:SEC6SOURCE CH4 sets power measurement number 3 to use channel 4 as the current source for measuring secondary winding 6. Power:Power2:MAGPROPERTY:SEC6SOURCE? might return CH4, indicating that channel 4 is the source for measuring secondary winding 3 of magnetic power measurement number 2.

Power:Power<x>:MAGPROPERTY:SEC6TURNS

This command sets or queries the number of turns of secondary winding 6 for magnetic measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:MAGPROPERTY:SEC6TURNS <NR1>`
`Power:Power<x>:MAGPROPERTY:SEC6TURNS?`

Related Commands [Power:Power<x>:MAGPROPERTY:SEC1TURNS](#)
[Power:Power<x>:MAGPROPERTY:SEC2TURNS](#)
[Power:Power<x>:MAGPROPERTY:SEC3TURNS](#)
[Power:Power<x>:MAGPROPERTY:SEC4TURNS](#)
[Power:Power<x>:MAGPROPERTY:SEC5TURNS](#)

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI.
`<NR1>` is the number of turns on the secondary winding, and ranges from 0 to 1,000,000.

Examples `Power:Power8:MAGPROPERTY:SEC6TURNS 22` sets the number of turns on secondary winding 6 to 22 for magnetic power measurement number 8.
`Power:Power9:MAGPROPERTY:SEC6TURNS?` might return 7000, indicating that their are 7,000 turns on secondary winding 6 for magnetic power measurement number 9.

Power:Power<x>:MAGPROPERTY:SECWINDings

This command sets or queries the number of secondary windings for the magnetic property measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:MAGPROPERTY:SECWINDings`
`{None|ONE|TWO|THREE|FOUR|FIVE|SIX}`
`Power:Power<x>:MAGPROPERTY:SECWINDings?`

Related Commands [Power:Power<x>:MAGPROPERTY:SEC1SOURCE](#)
[Power:Power<x>:MAGPROPERTY:SEC1TURNS](#)
[Power:Power<x>:MAGPROPERTY:SEC2SOURCE](#)
[Power:Power<x>:MAGPROPERTY:SEC2TURNS](#)

`POWer:POWer<x>:MAGPROPERTY:SEC3SOURce`

`POWer:POWer<x>:MAGPROPERTY:SEC3TURNs`

`POWer:POWer<x>:MAGPROPERTY:SEC4SOURce`

`POWer:POWer<x>:MAGPROPERTY:SEC4TURNs`

`POWer:POWer<x>:MAGPROPERTY:SEC5SOURce`

`POWer:POWer<x>:MAGPROPERTY:SEC5TURNs`

`POWer:POWer<x>:MAGPROPERTY:SEC6SOURce`

`POWer:POWer<x>:MAGPROPERTY:SEC6TURNs`

Arguments	<code>Power<x></code> is the magnetic property power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI. None, ONE, TWO, THREE, FOUR, FIVE, SIX sets the number of secondary windings to the specified value.
Examples	<code>Power:Power2:MAGPROPERTY:SECWINDings 4</code> sets power measurement 2 to use 4 secondary transformer windings. <code>Power:Power11:MAGPROPERTY:SECWINDings?</code> might return FOUR, indicating that there are four secondary transformer windings for power measurement 11.

POWer:POWer<x>:MAGPROPERTY:UNITs

This command sets or queries the units for magnetic measurements of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	<code>Power:Power<x>:MAGPROPERTY:UNITs {SI CGS}</code> <code>Power:Power<x>:MAGPROPERTY:UNITs?</code>
Related Commands	<code>POWer:POWer<x>:MAGPROPERTY:AREAOfcrosssection</code> <code>POWer:POWer<x>:MAGPROPERTY:LENgth</code>
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI.

SI sets the measurement to International System of Units.

CGS sets the measurement to Gaussian units.

Examples `Power:Power5:MAGPROPERTY:UNITs` SI sets magnetic property measurement 5 to use SI units.

`Power:Power1:MAGPROPERTY:UNITs?` might return CGS, indicating that the measurement unit for power measurement 1 is set to CGS.

POWer:POWer<x>:MAGPROPERTY:VSOURce

This command sets or queries the primary winding voltage source for the magnetic measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:MAGPROPERTY:VSOURce {CH<x>|MATH<x>|REF<x>}`
`Power:Power<x>:MAGPROPERTY:VSOURce?`

Related Commands [POWer:POWer<x>:MAGPROPERTY:ISOURce](#)

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI.

`CH<x>` sets the channel specifier; `<x>` is 1 through 8 and is limited by the number of FlexChannels in your instrument.

`MATH<x>` = A math waveform specifier; `<x>` is ≥ 1 .

`REF<x>` = A reference waveform specifier; `<x>` is ≥ 1 .

Examples `Power:Power1:MAGPROPERTY:VSOURce CH4` sets power measurement 1 to use channel 4 as the primary winding voltage source for the magnetic power measurement.

`Power:Power2:MAGPROPERTY:VSOURce?` might return CH1, indicating that channel 1 is the primary winding voltage source for the magnetic power measurement number 2.

POWER:POWER<x>:NDUTYCYCLE:EDGEType

This command sets or queries the clock edge type for negative duty cycle measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWER:POWER<x>:NDUTYCYCLE:EDGEType {RISE | FALL | BOTH}
POWER:POWER<x>:NDUTYCYCLE:EDGEType?

Examples POWER:POWER1:NDUTYCYCLE:EDGEType RISE sets the clock edge type as rise for the negative duty cycle measurement.

POWER:POWER<x>:NDUTYCYCLE:INPUTSource

This command sets or queries the input source for negative duty cycle measurement in the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWER:POWER<x>:NDUTYCYCLE:INPUTSource {CH<x> | MATH<x> | REF<x>}
POWER:POWER<x>:NDUTYCYCLE:INPUTSource?

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

MATH<x> = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples POWER:POWER1:NDUTYCYCLE:INPUTSource CH3 sets the input source for negative duty cycle measurement as CH3 for the power measurement badge Power 1.

POWer:POWer<x>:NPULSEWIDTH:INPUTSource

This command sets or queries the input source for negative pulse width measurement in the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax Power:POWer<x>:NPULSEWIDTH:INPUTSource
{CH<x>|MATH<x>|REF<x>}
Power:POWer<x>:NPULSEWIDTH:INPUTSource?

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

MATH<x> = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples Power:POWer1:NPULSEWIDTH:INPUTSource CH1 sets the input source for negative pulse width measurement as channel 1.

POWer:POWer<x>:PDUTYCYCLE:EDGEType

This command sets or queries the clock edge type for positive duty cycle measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax Power:POWer<x>:PDUTYCYCLE:EDGEType {CH<x>|MATH<x>|REF<x>}
Power:POWer<x>:PDUTYCYCLE:EDGEType?

Examples Power:POWer1:PDUTYCYCLE:EDGEType BOTH sets the clock edge type as both (rise and fall) for the positive duty cycle measurement.

POWer:POWer<x>:PDUTYCYCLE:INPUTSource

This command sets or queries the input source for positive duty cycle measurement in the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	POWer:POWer<x>:PDUTYCYCLE:INPUTSource {CH<x> MATH<x> REF<x>} POWer:POWer<x>:PDUTYCYCLE:INPUTSource?
Arguments	POWer<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument. MATH<x> = A math waveform specifier; <x> is ≥ 1. REF<x> = A reference waveform specifier; <x> is ≥ 1.
Examples	POWer:POWer1:PDUTYCYCLE:INPUTSource CH4 sets the input source for positive duty cycle measurement as channel 4.

POWer:POWer<x>:PERIOD:EDGE

This command sets or queries the edge type for period measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	POWer:POWer<x>:PERIOD:EDGE {RISE FALL} POWer:POWer<x>:PERIOD:EDGE?
Examples	POWer:POWer1:PERIOD:EDGE RISE sets the edge type as rise for the period measurement.

POWer:POWer<x>:PERIOD:INPUTSource

This command sets or queries the input source for period measurement in the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWer:POWer<x>:PERIOD:INPUTSource {CH<x>|MATH<x>|REF<x>}`
`POWer:POWer<x>:PERIOD:INPUTSource?`

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

MATH<x> = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples `POWer:POWer1:PERIOD:INPUTSource CH6` sets the input source for period measurement as channel 6.

POWer:POWer<x>:POWERQUALITY:CCYCLes

This command sets or queries the calculate cycles over full cycles settings for the specified power quality measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWer:POWer<x>:POWERQUALITY:CCYCLes {ON | OFF 1 | 0}`
`POWer:POWer<x>:POWERQUALITY:CCYCLes?`

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

1 | ON selects the calculate cycles over full cycles.

0 | OFF unselects the calculate cycles over full cycles.

Examples `Power:Power1:POWERQUALITY:CCYCles 1` selects the calculate cycles over full cycles for the power measurement badge 1.

POWER:POWER<x>:POWERQUALITY:FREFerence

This command sets or queries the frequency reference type for power quality measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:POWERQUALITY:FREFerence {VOLTage | CURRENT}`
`Power:Power<x>:POWERQUALITY:FREFerence?`

Examples `Power:Power1:POWERQUALITY:FREFerence CURRENT` sets the frequency reference type for power quality measurement as current.

POWER:POWER<x>:POWERQUALITY:ISOURce

This command sets or queries the current source for power quality measurement in the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:POWERQUALITY:ISOURce {CH<x> | MATH<x> | REF<x>}`
`Power:Power<x>:POWERQUALITY:ISOURce?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

`CH<x>` = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

MATH<x> = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples `Power:Power1:POWERQUALITY:VSOURCE CH2` sets the current source for power quality measurement as channel 2.

POWer:POWer<x>:POWERQUALITY:VSOURce

This command sets or queries the voltage source for power quality measurement in the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:POWERQUALITY:VSOURCE {CH<x>|MATH<x>|REF<x>}`
`Power:Power<x>:POWERQUALITY:VSOURCE?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

`CH<x>` = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

`MATH<x>` = A math waveform specifier; <x> is ≥ 1 .

`REF<x>` = A reference waveform specifier; <x> is ≥ 1 .

Examples `Power:Power1:POWERQUALITY:VSOURCE CH1` sets the voltage source for power quality measurement as channel 1.

POWer:POWer<x>:PPULSEWIDTH:INPUTSOurce

This command sets or queries the input source for positive pulse width measurement in the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:PPULSEWIDTH:INPUTSource`
 `{CH<x>|MATH<x>|REF<x>}`
 `Power:Power<x>:PPULSEWIDTH:INPUTSource?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

`CH<x>` = A channel specifier; `<x>` is 1 through 8 and is limited by the number of FlexChannels in your instrument.

`MATH<x>` = A math waveform specifier; `<x>` is ≥ 1 .

`REF<x>` = A reference waveform specifier; `<x>` is ≥ 1 .

Examples `Power:Power1:PPULSEWIDTH:INPUTSource CH5` sets channel 5 as the input source for the positive pulse width measurement 1.

POWER:POWER<x>:PRESET (No Query Form)

This command runs a power preset action for the specified power measurement number.

Conditions Requires option 5-PWR, 6-PWR, or PS2.

Group Power

Syntax `Power:Power<x>:PRESET {EXECute}`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

`EXECute` runs the power preset action.

Examples `Power:Power5:PRESET EXECute` runs the power preset action for power measurement 5.

POWER:POWER<x>:PSRR:AMP[1-10]Val

This command sets or queries the generator amplitude value of the specified configuration step for the Power Supply Rejection Ratio (PSRR) power measurement.

Conditions	Requires option 5-PWR or 6-PWR
Group	Power
Syntax	<pre>Power:Power<x>:PSRR:AMP[1-10]VaN1 <NR3> Power:Power<x>:PSRR:AMP[1-10]VaN1?</pre>
Arguments	Power<x> sets the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. AMP[1-10] sets the configuration step number, in the range of 1 to 10. Values outside this range will report an error. <NR3> sets the generator amplitude for the specified configuration step, in the range of -100 V to 100 V.
Examples	Power:Power1:PSRR:AMP3VaN1 20 sets the generator output amplitude for configuration step 3 to 20 volts, for power measurement 1. Power:Power2:PSRR:AMP8VaN1? might return 60, indicating that the generator output amplitude setting of configuration step 8 is 60 volts, for power measurement 2.

POWer:POWer<x>:PSRR:AMPMode

This command sets or queries the amplitude mode for the Power Supply Rejection Ratio (PSRR) power measurement.

Conditions	Requires option 5-PWR or 6-PWR
Group	Power
Syntax	<pre>Power:Power<x>:PSRR:AMPMode {CONSTant PROFile} Power:Power<x>:PSRR:AMPMode?</pre>
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. CONSTant sets the amplitude mode to output a constant amplitude signal from the DUT stimulus generator for all frequency bands. PROFile enables configuring the generator to set amplitude values for each frequency band.

- Examples** `Power:Power1:PSRR:AMPMode Constant` sets the amplitude mode to Constant for power measurement 1.
- `Power:Power2:PSRR:AMPMode?` might return `Profile`, indicating that the amplitude mode power measurement 2 is set to Profile.

POWer:POWer<x>:PSRR:CONSTAMPlitude

This command sets or queries the constant amplitude voltage for the Power Supply Rejection Ratio (PSRR) power measurement.

- Conditions** Requires option 5-PWR or 6-PWR
- Group** Power
- Syntax** `Power:Power<x>:PSRR:CONSTAMPlitude <NR3>`
`Power:Power<x>:PSRR:CONSTAMPlitude?`
- Arguments** `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.
- `<NR3>` is the constant amplitude voltage value for the measurement, in the range of -100 V to 100 V.
- Examples** `Power:Power3:PSRR:CONSTAMPlitude 120` sets the constant amplitude voltage for Power measurement 3 to be 120 volts.
- `Power:Power5:PSRR:CONSTAMPlitude?` might return `-15`, indicating that the constant amplitude voltage for power measurement 5 is -15 volts.

POWer:POWer<x>:PSRR:FREQ[1-11]Val

This command sets or queries the generator frequency value of the specified configuration step for the Power Supply Rejection Ratio (PSRR) power measurement.

- Conditions** Requires option 5-PWR or 6-PWR
- Group** Power

Syntax	Power:Power<x>:PSRR:FREQ[1-11]Val <NR3> Power:Power<x>:PSRR:FREQ[1-11]Val?
Arguments	Power<x> sets the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. FREQ[1-11] sets the configuration step number, in the range of 1 to 11. Values outside this range will report an error. <NR3> sets the frequency of the specified configuration step number, in the range of 10 Hz to 50 MHz.
Examples	Power:Power5:PSRR:FREQ1Val 200 sets the generator frequency value for frequency band 1 to 200 Hz, for power measurement 5. Power:Power2:PSRR:FREQ3Val? might return 2.000E+6, indicating that the generator frequency output for frequency band 3 is 2.0 MHz, for power measurement 2.

POWer:POWer<x>:PSRR:GENerator

This command sets or queries the generator source used to send stimulus signals to the DUT, for the Power Supply Rejection Ratio (PSRR) power measurement.

Conditions	Requires option 5-PWR or 6-PWR
Group	Power
Syntax	Power:Power<x>:PSRR:GENerator {INTERNAL} Power:Power<x>:PSRR:GENerator?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. INTERNAL sets the generator to the oscilloscope AFG signal source. This is the only valid argument.
Examples	Power:Power3:PSRR:GENerator INTERNAL sets the generator to the oscilloscope AFG for power measurement 3. Power:Power2:PSRR:GENerator? might return INTERNAL, indicating that the generator source for power measurement 2 is the oscilloscope AFG.

POWer:POWer<x>:PSRR:IMPEDance

This command sets or queries the vertical termination impedance for the Power Supply Rejection Ratio (PSRR) power measurement.

Conditions	Requires option 5-PWR or 6-PWR
Group	Power
Syntax	<code>POWer:POWer<x>:PSRR:IMPEDance {FIFTy ONEMEGa}</code> <code>POWer:POWer<x>:PSRR:IMPEDance?</code>
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <code>FIFTy</code> sets the impedance to be 50 Ω. <code>ONEMEGa</code> sets the impedance to be 1 MΩ.
Examples	<code>POWer:POWer3:PSRR:IMPEDance FIFTy</code> sets the vertical termination impedance for power measurement 3 to be 50 Ω. <code>POWer:POWer2:PSRR:IMPEDance?</code> might return CH6, indicating that the vertical termination impedance for power measurement 2 is 1 MΩ.

POWer:POWer<x>:PSRR:INPUTSource

This command sets or queries the input source for the Power Supply Rejection Ratio (PSRR) power measurement.

Conditions	Requires option 5-PWR or 6-PWR
Group	Power
Syntax	<code>POWer:POWer<x>:PSRR:INPUTSource CH<x></code> <code>POWer:POWer<x>:PSRR:INPUTSource?</code>
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <code>CH<x></code> sets the channel to use for the input source.

- Examples** `Power:Power2:PSRR:INPUTSource CH3` sets the input for power measurement 2 to be the channel 3 waveform.
- `Power:Power2:PSRR:INPUTSource?` might return CH1, indicating that the input source for power measurement 1 is channel 1.

POWer:POWer<x>:PSRR:OUTPUTSource

This command sets or queries the output source for the Power Supply Rejection Ratio (PSRR) power measurement.

- Conditions** Requires option 5-PWR or 6-PWR
- Group** Power
- Syntax** `Power:Power<x>:PSRR:OUTPUTSource CH<x>`
`Power:Power<x>:PSRR:OUTPUTSource?`
- Arguments** `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.
- `CH<x>` sets the channel to use for the output source.
- Examples** `Power:Power1:PSRR:OUTPUTSource CH3` sets the output source for power measurement 1 to be the channel 3 waveform.
- `Power:Power3:PSRR:OUTPUTSource?` might return CH6, indicating that the output source for power measurement 3 is channel 6.

POWer:POWer<x>:PSRR:PPD

This command sets or queries the points per decade (PPD) value for the Power Supply Rejection Ratio (PSRR) power measurement.

- Conditions** Requires option 5-PWR or 6-PWR
- Group** Power
- Syntax** `Power:Power<x>:PSRR:PPD <NR3>`
`Power:Power<x>:PSRR:PPD?`

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <NR3> is the PPD value for the measurement, in the range of 10 to 100 points.
Examples	Power:Power3:PSRR:PPD 30 sets the PPD for Power measurement 3 to be 30 points. Power:Power5:PSRR:PPD? might return 20, indicating that the PPD value for power measurement 5 is 20 points.

POWER:POWER<x>:PSRR:STARTFREQuency

This command sets or queries the start frequency value for the Power Supply Rejection Ratio (PSRR) power measurement.

Conditions	Requires option 5-PWR or 6-PWR
Group	Power
Syntax	Power:Power<x>:PSRR:STARTFREQuency <NR3> Power:Power<x>:PSRR:STARTFREQuency?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <NR3> is the starting frequency for the measurement, in the range of 10 Hz to 50 MHz.
Examples	Power:Power8:PSRR:STARTFREQuency 12 sets the starting frequency for power measurement 8 to be 12 Hz. Power:Power3:PSRR:STARTFREQuency? might return 2400, indicating that the starting frequency for power measurement 3 is 2400 Hz.

POWER:POWER<x>:PSRR:STOPFREQuency

This command sets or queries the stop frequency value for the Power Supply Rejection Ratio (PSRR) power measurement.

Conditions	Requires option 5-PWR or 6-PWR
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Group	Power
Syntax	<code>POWER:POWER<x>:PSRR:STOPFREQUENCY <NR3></code> <code>POWER:POWER<x>:PSRR:STOPFREQUENCY?</code>
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <code><NR3></code> is the stop frequency for the measurement, in the range of 10 Hz to 50 MHz.
Examples	<code>POWER:POWER3:PSRR:STOPFREQUENCY 120</code> sets the stop frequency for Power measurement 8 to be 120 Hz. <code>POWER:POWER5:PSRR:STOPFREQUENCY?</code> might return 2000, indicating that the stop frequency for power measurement 5 is 2000 Hz.

POWER:POWER<x>:RDSOn:DEVICEType

This command sets or queries the device type for the power drain source on resistance measurement for RDSon measurement of the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	<code>POWER:POWER<x>:RDSOn:DEVICEType {SWITCHING PNJUNCTION}</code> <code>POWER:POWER<x>:RDSOn:DEVICEType?</code>
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI. <code>SWITCHING</code> sets the Device Type to a switching device (v/i). <code>PNJUNCTION</code> sets the Device Type to a PN Junction device (dv/di).
Examples	<code>POWER:POWER3:RDSOn:DEVICEType PNJUNCTION</code> sets the RDSon power measurement number 3 to measure a PN junction device. <code>POWER:POWER1:RDSOn:DEVICEType?</code> might return <code>SWITCHING</code>, indicating that RDSon power measurement number 1 is set to measure a switching device.

POWer:POWer<x>:RDSOn:ISOURce

This command sets or queries the current source for RDSon measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax Power:POWer<x>:RDSOn:ISOURce {CH<x>|MATH<x>|REF<x>}
Power:POWer<x>:RDSOn:ISOURce?

Related Commands [POWer:POWer<x>:RDSOn:VSOURce](#)

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI.

CH<x> sets the channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

MATH<x> = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples Power:POWer1:RDSOn:ISOURce REF2 sets RDSon power measurement number 1 to use Reference waveform 2 as the current signal source.

Power:POWer1:RDSOn:ISOURce? might return CH1, indicating that channel 1 is the current signal source for the RDSon power measurement number 1.

POWer:POWer<x>:RDSOn:VSOURce

This command sets or queries the voltage source for RDSon measurement of the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax Power:POWer<x>:RDSOn:VSOURce {CH<x>|MATH<x>|REF<x>}
Power:POWer<x>:RDSOn:VSOURce?

Related Commands [POWer:POWer<x>:RDSON:ISOURce](#)

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the power measurement badge on the UI.

CH<x> sets the channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

MATH<x> = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples Power:POWer2:RDSON:VSOURce CH7 sets RDSon power measurement number 2 to use channel 7 as the voltage source.

Power:POWer1:RDSON:VSOURce? might return CH7, indicating that channel 1 is the voltage signal source for RDSon power measurement number 1.

POWer:POWer<x>:REFLevels:ABSolute:FALLHigh

This command sets or queries the falling edge for high reference level in absolute units for the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax Power:POWer<x>:REFLevels:ABSolute:FALLHigh <NR1>
Power:POWer<x>:REFLevels:ABSolute:FALLHigh?

Related Commands [POWer:POWer<x>:REFLevels:METHod](#)
[POWer:POWer<x>:REFLevels:ABSolute:TYPE](#)

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

<NR1> ranges from -40000 to 40000

Examples Power:POWer1:REFLevels:ABSolute:FALLHigh 1 sets the high reference level for falling edge as 1 volt.

POWer:POWer<x>:REFLevels:ABSolute:FALLLow

This command sets or queries the falling edge for low reference level in absolute units for the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWer:POWer<x>:REFLevels:ABSolute:FALLLow <NR1>`
`POWer:POWer<x>:REFLevels:ABSolute:FALLLow?`

Related Commands [POWer:POWer<x>:REFLevels:METHod](#)
[POWer:POWer<x>:REFLevels:ABSolute:TYPE](#)

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.
`<NR1>` ranges from -40000 to 40000

Examples `POWer:POWer1:REFLevels:ABSolute:FALLLow -1` sets the low reference level for falling edge as -1 volt.

POWer:POWer<x>:REFLevels:ABSolute:FALLMid

This command sets or queries the falling edge for mid reference level in absolute units for the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWer:POWer<x>:REFLevels:ABSolute:FALLMid <NR1>`
`POWer:POWer<x>:REFLevels:ABSolute:FALLMid?`

Related Commands [POWer:POWer<x>:REFLevels:METHod](#)
[POWer:POWer<x>:REFLevels:ABSolute:TYPE](#)

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <NR1> ranges from -40000 to 40000
Examples	Power:Power1:REFLevels:ABSolute:FALLMid 0 sets the mid reference level for falling edge as 0 volt.

Power:Power<x>:REFLevels:ABSolute:HYSTeresis

This command sets or queries the absolute hysteresis value for the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:REFLevels:ABSolute:HYSTeresis <NR1> Power:Power<x>:REFLevels:ABSolute:HYSTeresis?

Related Commands [Power:Power<x>:REFLevels:METHod](#)

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <NR1> ranges from 0.0000005 to 10
Examples	Power:Power1:REFLevels:ABSolute:HYSTeresis 10 sets the reference level hysteresis value in absolute to 10.

Power:Power<x>:REFLevels:ABSolute:RISEHigh

This command sets or queries the rising edge for high reference level in absolute units for the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power

Syntax	<code>Power:Power<x>:REFLevels:ABSolute:RISEHigh <NR1></code> <code>Power:Power<x>:REFLevels:ABSolute:RISEHigh?</code>
Related Commands	POWer:POWer<x>:REFLevels:METHod POWer:POWer<x>:REFLevels:ABSolute:TYPE
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <code><NR1></code> ranges from –40000 to 40000
Examples	<code>Power:Power1:REFLevels:ABSolute:RISEHigh 1</code> sets the high reference level for rising edge as 1 volt.

POWer:POWer<x>:REFLevels:ABSolute:RISELow

This command sets or queries the rising edge for low reference level in absolute units for the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	<code>Power:Power<x>:REFLevels:ABSolute:RISELow <NR1></code> <code>Power:Power<x>:REFLevels:ABSolute:RISELow?</code>
Related Commands	POWer:POWer<x>:REFLevels:METHod POWer:POWer<x>:REFLevels:ABSolute:TYPE
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <code><NR1></code> ranges from –40000 to 40000.
Examples	<code>Power:Power1:REFLevels:ABSolute:RISELow -1</code> sets the low reference level for rising edge as –1 volt.

POWER:POWER<x>:REFLevels:ABSolute:RISEMid

This command sets or queries the rising edge for mid reference level in absolute units for the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWER:POWER<x>:REFLevels:ABSolute:RISEMid <NR1>`
`POWER:POWER<x>:REFLevels:ABSolute:RISEMid?`

Related Commands [POWER:POWER<x>:REFLevels:METHod](#)
[POWER:POWER<x>:REFLevels:ABSolute:TYPE](#)

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.
`<NR1>` ranges from -40000 to 40000.

Examples `POWER:POWER1:REFLevels:ABSolute:RISEMid 0` sets the mid reference level for rising edge as 0 volt.

POWER:POWER<x>:REFLevels:ABSolute:TYPE

This command sets or queries the type of measurement levels when reference level is set to absolute for the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWER:POWER<x>:REFLevels:ABSolute:TYPE {SAME | UNIQUE}`
`POWER:POWER<x>:REFLevels:ABSolute:TYPE?`

Related Commands [POWER:POWER<x>:REFLevels:METHod](#)

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. SAME: select when the rising edge and falling edge reference levels are same. UNIQUE: select when the rising edge and falling edge reference levels are different.
Examples	Power:Power1:REFLevels:ABSolute:TYPE UNIQUE sets the type of measurement levels as unique for the specified power measurement badge.

POWer:POWer<x>:REFLevels:BASETop

This command sets or queries the reference level base top method for the specified power measurement number. The power measurement number is specified by <x>.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:REFLevels:BASETop {AUTO MINMax MEANhistogram MODEhistogram EYEhistogram} Power:Power<x>:REFLevels:BASETop?
Related Commands	POWer:POWer<x>:REFLevels:METHod
Examples	Power:Power1:REFLevels:BASETop AUTO sets the reference level base top method as auto for the power measurement badge 1.

POWer:POWer<x>:REFLevels:METHod

This command sets or queries the method to configure reference level values for the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:REFLevels:METHod {PERCent ABSolute} Power:Power<x>:REFLevels:METHod?

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. PERCent sets the power measurement to use absolute values to configure reference level values. ABSolute sets the power measurement to use percentage to configure reference level values.
Examples	Power:Power3:REFLevels:METHOD PERCent sets power measurement 3 to use percentage to configure reference level values.

POWer:POWer<x>:REFLevels:PERCent:FALLHigh

This command sets or queries the falling edge for high reference level in percentage for the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:REFLevels:PERCent:FALLHigh <NR1> Power:Power<x>:REFLevels:PERCent:FALLHigh?

Related Commands

[POWer:POWer<x>:REFLevels:METHod](#)

[POWer:POWer<x>:REFLevels:PERCent:TYPE](#)

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <NR1> ranges from 1 to 99.
Examples	Power:Power1:REFLevels:PERCent:FALLHigh 1 sets the high reference level for falling edge as 1 percentage.

POWer:POWer<x>:REFLevels:PERCent:FALLLow

This command sets or queries the falling edge for low reference level in percentage for the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	<pre>Power:Power<x>:REFLevels:PERCent:FALLLow <NR1> Power:Power<x>:REFLevels:PERCent:FALLLow?</pre>
Related Commands	POWer:POWer<x>:REFLevels:METHod POWer:POWer<x>:REFLevels:PERCent:TYPE
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <NR1> ranges from 1 to 99.
Examples	<pre>Power:Power1:REFLevels:PERCent:FALLLow 0</pre> sets the low reference level for falling edge as 0 percentage.

POWer:POWer<x>:REFLevels:PERCent:FALLMid

This command sets or queries the falling edge for mid reference level in percentage for the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	<pre>Power:Power<x>:REFLevels:PERCent:FALLMid <NR1> Power:Power<x>:REFLevels:PERCent:FALLMid?</pre>
Related Commands	POWer:POWer<x>:REFLevels:METHod POWer:POWer<x>:REFLevels:PERCent:TYPE
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <NR1> ranges from 1 to 99.

Examples `Power:Power1:REFLevels:PERCent:FALLMid 50` sets the mid reference level for falling edge as 50 percentage.

POWer:POWer<x>:REFLevels:PERCent:HYSTeresis

This command sets or queries the hysteresis in percentage for the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:REFLevels:PERCent:HYSTeresis <NR1>`
`Power:Power<x>:REFLevels:PERCent:HYSTeresis?`

Related Commands [POWer:POWer<x>:REFLevels:METHod](#)

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.
`<NR1>` ranges from 1 to 99.

Examples `Power:Power1:REFLevels:PERCent:HYSTeresis 25` sets the hysteresis as 25 percentage.

POWer:POWer<x>:REFLevels:PERCent:RISEHigh

This command sets or queries the rising edge for high reference level in percentage for the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:REFLevels:PERCent:RISEHigh <NR1>`
`Power:Power<x>:REFLevels:PERCent:RISEHigh?`

Related Commands [POWer:POWer<x>:REFLevels:METHod](#)

[POWer:POWer<x>:REFLevels:PERCent:TYPE](#)

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

<NR1> ranges from 1 to 99.

Examples Power:POWer1:REFLevels:PERCent:RISEHigh 85 sets the high reference level for rising edge as 85 percentage.

POWer:POWer<x>:REFLevels:PERCent:RISELow

This command sets or queries the rising edge for low reference level in percentage for the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax Power:POWer<x>:REFLevels:PERCent:RISELow <NR1>
Power:POWer<x>:REFLevels:PERCent:RISELow?

Related Commands [POWer:POWer<x>:REFLevels:METHod](#)

[POWer:POWer<x>:REFLevels:PERCent:TYPE](#)

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

<NR1> ranges from 1 to 99.

Examples Power:POWer1:REFLevels:PERCent:RISELow 10 sets the low reference level for rising edge as 10 percentage.

POWer:POWer<x>:REFLevels:PERCent:RISEMid

This command sets or queries the rising edge for mid reference level in percentage for the specified power measurement number. The power measurement number is specified by <x>.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:REFLevels:PERCent:RISEMid <NR1> Power:Power<x>:REFLevels:PERCent:RISEMid?
Related Commands	POWER:Power<x>:REFLevels:METHod POWER:Power<x>:REFLevels:PERCent:TYPE
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <NR1> ranges from 1 to 99.
Examples	Power:Power1:REFLevels:PERCent:RISEMid 55 sets the mid reference level for rising edge as 55 percentage.

POWER:Power<x>:REFLevels:PERCent:TYPE

This command sets or queries the reference levels for the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:REFLevels:PERCent:TYPE {TENNinety TWENTyeighty CUSTOM} Power:Power<x>:REFLevels:PERCent:TYPE?
Related Commands	POWER:Power<x>:REFLevels:METHod POWER:Power<x>:REFLevels:PERCent:FALLHigh POWER:Power<x>:REFLevels:PERCent:FALLLow POWER:Power<x>:REFLevels:PERCent:FALLMid POWER:Power<x>:REFLevels:PERCent:RISEHigh POWER:Power<x>:REFLevels:PERCent:RISELow

POWER:POWER<x>:REFLevels:PERCent:RISEMid

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. TENNinety to set the low reference levels as 10% and high reference levels as 90% TWENTyeighty to set the low reference levels as 20% and high reference levels as 80% CUSTom to set the custom low, high, and mid reference levels for rising and falling edges
Examples	Power:Power1:REFLevels:PERCent:TYPE TENNinety sets the low reference levels as 10% and high reference levels as 90% for the power measurement badge 1.

POWER:POWER<x>:RESUltS:ALLAcqs:MAXimum? (Query Only)

This command queries the maximum value of all acquisitions for the measurement parameter in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax **Power:Power<x>:RESUltS:ALLAcqs:MAXimum?** {InputPwr|Output1Pwr|Output2Pwr|Output3Pwr|Efficiency1|Efficiency2|Efficiency3|TotalEfficiency|INDUCT|IVSINTV|MAGLOSS|Bpeak|Br|Hc|Hmax|IRipple|DeltaB|DeltaH|Permeability|RDS|TRUEPWR|APPPWR|REPWR|PWRFACTOR|PHASE|PWRFREQ|ICFACTOR|VCFACTOR|IRMS|VRMS|TONENRG|TONLOSS|TOFFENRG|TOFFLOSS|CONDENRG|CONDLOSS|TTLLOSS|TTLENRG|DVBYDT|DIBYDT|SOAHITSCNT|LRIPRMS|LRIPPKPK|SWRIPRMS|SWRIPPKPK|PRIOD|FREQ|PDUTY|NDUTY|PPULSE|NPULSE|AMPL|PKPK|HIGH|LOW|MAX|MIN|INRUSH|CAPACITANCE|OUTPUT1|OUTPUT2|OUTPUT3|OUTPUT4|OUTPUT5|OUTPUT6|OUTPUT7|GAINCROSSOVERFREQ|PHASECROSSOVERFREQ|GM|PM|MAXPSRR|MAXPSRRFREQ|MINPSRR|MINPSRRFREQ}

NOTE. Above entries are <QString> entries, and must be entered in enclosing quotes.

Arguments

Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

<QString> = the measurement result that you want to return from the specified power measurement number. Available results depend on the power measurement being taken in the specified measurement number. The valid <Qstring> arguments are:

For the Efficiency measurement, the parameters are "InputPwr", "Output1Pwr", "Output2Pwr", "Output3Pwr", "Efficiency1", "Efficiency2", "Efficiency3", "TotalEfficiency".

For Inductance measurement, the parameter is "INDUCT".

For IVSIntegralV measurement, the parameter is "IVSINTV".

For Magnetic Loss measurement, the parameter is "MAGLOSS".

For Magnetic Property measurement, the parameters are "Bpeak", "Br", "Hc", "Hmax", "IRipple", "DeltaB", "DeltaH", "Permeability".

For RDSon measurement, the parameter is "RDS".

For Power Quality measurement, the parameters are "TRUEPWR", "APPPWR", "REPWR", "PWRFACTOR", "PHASE", "PWRFREQ", "ICFACTOR", "VCFACTOR", "IRMS", "VRMS".

For Switching Loss measurement, the parameters are "TONENRG", "TONLOSS", "TOFFENRG", "TOFFLOSS", "CONDENRG", "CONDLOSS", "TTLLOSS", "TTLENRG".

"DVBYDT" is the parameter for the dV by dt measurement.

"DIBYDT" is the parameter for the dI by dt measurement.

"SOAHITSCNT" is the parameter for the SOA measurement.

"LRIPRMS" and "LRIPPKPK" are the parameters for the Line Ripple measurement.

"SWRIPRMS" and "SWRIPPKPK" are the parameters for the Switching Ripple measurement.

"PRIOD" is the parameter for the Cycle Period measurement.

"FREQ" is the parameter for the Cycle Frequency measurement.

"PDUTY" is the parameter for the Positive Duty Cycle measurement.

"NDUTY" is the parameter for the Negative Duty Cycle measurement.

"PPULSE" is the parameter for the Positive Pulse Width measurement.

"NPULSE" is the parameter for the Negative Pulse Width measurement.

"AMPL" is the parameter for the Cycle Amplitude measurement.

"PKPK" is the parameter for the Cycle Peak–Peak measurement.

"HIGH" is the parameter for the Cycle Top measurement.

"LOW" is the parameter for the Cycle Base measurement.

"Max" is the parameter for the Cycle Max measurement.

"MIN" is the parameter for the Cycle Min measurement.

"INRUSH" is the parameter for the Inrush Current measurement.

"CAPACITANCE" is the parameter for the Input Capacitance measurement.

"OUTPUT1" - "OUTPUT7" are the parameters for the Turn On Time and Turn Off Time measurements.

"GAINCROSSOVERFREQ", "PHASECROSSOVERFREQ", "GM", "PM" are the parameters for the Control Loop Response measurement.

"MAXPSRR", "MAXPSRRFREQ", "MINPSRR", "MINPSRRFREQ" are the parameters for the PSRR measurement

Examples `Power:Power4:RESULTS:ALLAcqs:MAXimum?` "PKPK" might return 28.56, indicating the maximum value of the Peak-to-Peak measurement for power measurement 4.

POWER:POWER<x>:RESULTS:ALLAcqs:MEAN? (Query Only)

This command queries the mean value of all acquisitions for the measurement parameter in the specified power measurement number <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:RESULTS:ALLAcqs:MEAN?`
`{InputPwr|Output1Pwr|Output2Pwr|`
`Output3Pwr|Efficiency1|Efficiency2|`
`Efficiency3|TotalEfficiency|`
`INDUCT|IVSINTV|MAGLOSS|Bpeak|Br|Hc|Hmax|`
`IRipple|DeltaB|DeltaH|Permeability|RDS|TRUEPWR|`
`APPPWR|REPWR|PWRFACTOR|PHASE|PWRFREQ|`
`ICFACTOR|VCFACOR|IRMS|VRMS|TONENRG|`
`TONLOSS|TOFFENRG|TOFFLOSS|CONDENRG|CONDLOSS|`

TTLLOSS | TTLENRG | DVBYDT | DIBYDT | SOAHITSCNT |
 LRIPRMS | LRIPPKPK | SWRIPRMS | SWRIPPKPK | PRIOD |
 FREQ | PDUTY | NDUTY | PPULSE | NPULSE | AMPL |
 PKPK | HIGH | LOW | MAX | MIN | INRUSH | CAPACITANCE | OUTPUT1 |
 OUTPUT2 | OUTPUT3 | OUTPUT4 | OUTPUT5 | OUTPUT6 | OUTPUT7 |
 GAINCROSSOVERFREQ | PHASECROSSOVERFREQ | GM | PM |
 MAXPSRR | MAXPSRRFREQ | MINPSRR | MINPSRRFREQ }

NOTE. Above entries are <QString> entries, and must be entered in enclosing quotes.

Arguments

Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

<QString> = the measurement result that you want to return from the specified power measurement number. Available results depend on the power measurement being taken in the specified measurement number. The valid <Qstring> arguments are:

For Efficiency measurement, the parameters are "InputPwr", "Output1Pwr", "Output2Pwr", "Output3Pwr", "Efficiency1", "Efficiency2", "Efficiency3", "TotalEfficiency".

For Inductance measurement, the parameter is "INDUCT".

For IVSIntegralV measurement, the parameter is "IVSINTV".

For Magnetic Loss measurement, the parameter is "MAGLOSS".

For Magnetic Property measurement, the parameters are "Bpeak", "Br", "Hc", "Hmax", "IRipple", "DeltaB", "DeltaH", "Permeability".

For RDSon measurement, the parameter is "RDS".

For Power Quality measurement, the parameters are "TRUEPWR", "APPPWR", "REPWR", "PWRFACTOR", "PHASE", "PWRFREQ", "ICFACTOR", "VCFACTOR", "IRMS", "VRMS".

For Switching Loss measurement, the parameters are "TONENRG", "TONLOSS", "TOFFENRG", "TOFFLOSS", "CONDENRG", "CONDLOSS", "TTLLOSS", "TTLENRG".

"DVBYDT" is the parameter for the dV by dt measurement.

"DIBYDT" is the parameter for the dI by dt measurement.

"SOAHITSCNT" is the parameter for the SOA measurement.

"LRIPRMS" and "LRIPPKPK" are the parameters for the Line Ripple measurement.

"SWRIPRMS" and "SWRIPPKPK" are the parameters for the Switching Ripple measurement.

"PRIOD" is the parameter for the Cycle Period measurement.

"FREQ" is the parameter for the Cycle Frequency measurement.

"PDUTY" is the parameter for the Positive Duty Cycle measurement.

"NDUTY" is the parameter for the Negative Duty Cycle measurement.

"PPULSE" is the parameter for the Positive Pulse Width measurement.

"NPULSE" is the parameter for the Negative Pulse Width measurement.

"AMPL" is the parameter for the Cycle Amplitude measurement.

"PKPK" is the parameter for the Cycle Peak–Peak measurement.

"HIGH" is the parameter for the Cycle Top measurement.

"LOW" is the parameter for the Cycle Base measurement.

"Max" is the parameter for the Cycle Max measurement.

"MIN" is the parameter for the Cycle Min measurement.

"INRUSH" is the parameter for the Inrush Current measurement.

"CAPACITANCE" is the parameter for the Input Capacitance measurement.

"OUTPUT1" - "OUTPUT7" are the parameters for the Turn On Time and Turn Off Time measurements.

"GAINCROSSOVERFREQ", "PHASECROSSOVERFREQ", "GM", "PM" are the parameters for the Control Loop Response measurement.

"MAXPSRR", "MAXPSRRFREQ", "MINPSRR", "MINPSRRFREQ" are the parameters for the PSRR measurement

Examples `Power:Power2:RESULTS:ALLAcqs:MEAN?` "AMPL" might return 57.45, indicating the mean value of the Amplitude measurement for all acquisitions of power measurement 2.

POWER:POWER<x>:RESULTS:ALLAcqs:MINimum? (Query Only)

This command queries the minimum value of all acquisitions for the measurement parameter of the specified power measurement <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:RESULTS:ALLAcqs:
 MINimum? {InputPwr|Output1Pwr|
 Output2Pwr|Output3Pwr|Efficiency1|Efficiency2|Efficiency3|
 TotalEfficiency|INDUCT|IVSINTV|MAGLOSS|
 Bpeak|Br|Hc|Hmax|IRipple|DeltaB|DeltaH|
 Permeability|RDS|TRUEPWR|APPPWR|REPWR|
 PWRFACTOR|PHASE|PWRFREQ|ICFACTOR|VCFACTOR|
 IRMS|VRMS|TONENRG|TONLOSS|TOFFENRG|
 TOFFLOSS|CONDENRG|CONDLOSS|TTLLOSS|TTLENRG|
 DVBYDT|DIBYDT|SOAHITSCNT|LRIPRMS|LRIPPKPK|
 SWRIPRMS|SWRIPPKPK|PRIOD|FREQ|PDUTY|
 NDUTY|PPULSE|NPULSE|AMPL|PKPK|
 HIGH|LOW|MAX|MIN|INRUSH|CAPACITANCE|OUTPUT1|
 OUTPUT2|OUTPUT3|OUTPUT4|OUTPUT5|OUTPUT6|OUTPUT7|
 GAINCROSSOVERFREQ|PHASECROSSOVERFREQ|GM|PM|
 MAXPSRR|MAXPSRRFREQ|MINPSRR|MINPSRRFREQ}`

NOTE. Above entries are <QString> entries, and must be entered in enclosing quotes.

Arguments

`Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

<Qstring> = the measurement result that you want to return from the specified power measurement number. Available results depend on the power measurement being taken in the specified measurement number. The valid <Qstring> arguments are:

For Efficiency measurement, the parameters are "InputPwr", "Output1Pwr", "Output2Pwr", "Output3Pwr", "Efficiency1", "Efficiency2", "Efficiency3", "TotalEfficiency".

For Inductance measurement, the parameter is "INDUCT".

For IVSIntegralV measurement, the parameter is "IVSINTV".

For Magnetic Loss measurement, the parameter is "MAGLOSS".

For Magnetic Property measurement, the parameters are "Bpeak", "Br", "Hc", "Hmax", "IRipple", "DeltaB", "DeltaH", "Permeability".

For RDSon measurement, the parameter is "RDS".

For Power Quality measurement, the parameters are "TRUEPWR", "APPPWR", "REPWR", "PWRFACTOR", "PHASE", "PWRFREQ", "ICFACTOR", "VCFACTOR", "IRMS", "VRMS".

For Switching Loss measurement, the parameters are "TONENRG", "TONLOSS", "TOFFENRG", "TOFFLOSS", "CONDENRG", "CONDLOSS", "TTLLOSS", "TTLENRG".

"DVBYDT" is the parameter for the dV by dt measurement.

"DIBYDT" is the parameter for the dI by dt measurement.

"SOAHITSCNT" is the parameter for the SOA measurement.

"LRIPRMS" and "LRIPPKPK" are the parameters for the Line Ripple measurement.

"SWRIPRMS" and "SWRIPPKPK" are the parameters for the Switching Ripple measurement.

"PRIOD" is the parameter for the Cycle Period measurement.

"FREQ" is the parameter for the Cycle Frequency measurement.

"PDUTY" is the parameter for the Positive Duty Cycle measurement.

"NDUTY" is the parameter for the Negative Duty Cycle measurement.

"PPULSE" is the parameter for the Positive Pulse Width measurement.

"NPULSE" is the parameter for the Negative Pulse Width measurement.

"AMPL" is the parameter for the Cycle Amplitude measurement.

"PKPK" is the parameter for the Cycle Peak–Peak measurement.

"HIGH" is the parameter for the Cycle Top measurement.

"LOW" is the parameter for the Cycle Base measurement.

"Max" is the parameter for the Cycle Max measurement.

"MIN" is the parameter for the Cycle Min measurement.

"INRUSH" is the parameter for the Inrush Current measurement.

"CAPACITANCE" is the parameter for the Input Capacitance measurement.

"OUTPUT1" - "OUTPUT7" are the parameters for the Turn On Time and Turn Off Time measurements.

"GAINCROSSOVERFREQ", "PHASECROSSOVERFREQ", "GM", "PM" are the parameters for the Control Loop Response measurement.

"MAXPSRR", "MAXPSRRFREQ", "MINPSRR", "MINPSRRFREQ" are the parameters for the PSRR measurement

Examples `Power:Power5:RESULTS:ALLAcqs:MINimum?` "FREQ" might return 5.1307829019093E6, indicating the minimum frequency measurement for all acquisitions of power measurement 5.

POWER:POWER<x>:RESULTS:ALLAcqs:PK2PK? (Query Only)

This command queries the peak-to-peak value of all acquisitions for the measurement parameter in the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:RESULTS:ALLAcqs:PK2PK? {InputPwr Output1Pwr Output2Pwr Output3Pwr Efficiency1 Efficiency2 Efficiency3 TotalEfficiency INDUCT IVSINTV MAGLOSS Bpeak Br Hc Hmax IRipple DeltaB DeltaH Permeability RDS TRUEPWR APPPWR REPWR PWRFACTOR PHASE PWRFREQ ICFACTOR VCFACtor IRMS VRMS TONENRG TONLOSS TOFFENRG TOFFLOSS CONDENRG CONDLOSS TTLLOSS TTLENRG DVBYDT DIBYDT SOAHITSCNT LRIPRMS LRIPPKPK SWRIPRMS SWRIPPKPK PRIOD FREQ PDUTY NDUTY PPULSE NPULSE AMPL PKPK HIGH LOW MAX MIN INRUSH CAPACITANCE OUTPUT1 OUTPUT2 OUTPUT3 OUTPUT4 OUTPUT5 OUTPUT6 OUTPUT7 GAINCROSSOVERFREQ PHASECROSSOVERFREQ GM PM MAXPSRR MAXPSRRFREQ MINPSRR MINPSRRFREQ}

NOTE. Above entries are <QString> entries, and must be entered in enclosing quotes.

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <QString> = the measurement result that you want to return from the specified power measurement number. Available results depend on the power measurement being taken in the specified measurement number. The valid <Qstring> arguments are: For Efficiency measurement, the parameters are "InputPwr", "Output1Pwr", "Output2Pwr", "Output3Pwr", "Efficiency1", "Efficiency2", "Efficiency3", "TotalEfficiency". For Inductance measurement, the parameter is "INDUCT". For IVSIntegralV measurement, the parameter is "IVSINTV". For Magnetic Loss measurement, the parameter is "MAGLOSS". For Magnetic Property measurement, the parameters are "Bpeak", "Br", "Hc", "Hmax", "IRipple", "DeltaB", "DeltaH", "Permeability". For RDSon measurement, the parameter is "RDS". For Power Quality measurement, the parameters are "TRUEPWR", "APPPWR", "REPWR", "PWRFACTOR", "PHASE", "PWRFREQ", "ICFACTOR", "VCFACtor", "IRMS", "VRMS".
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For Switching Loss measurement, the parameters are "TONENRG", "TONLOSS", "TOFFENRG", "TOFFLOSS", "CONDENRG", "CONDLOSS", "TTLLOSS", "TTLENRG".

"DVBYDT" is the parameter for the dV by dt measurement.

"DIBYDT" is the parameter for the dI by dt measurement.

"SOAHITSCNT" is the parameter for the SOA measurement.

"LRIPRMS" and "LRIPPKPK" are the parameters for the Line Ripple measurement.

"SWRIPRMS" and "SWRIPPKPK" are the parameters for the Switching Ripple measurement.

"PRIOD" is the parameter for the Cycle Period measurement.

"FREQ" is the parameter for the Cycle Frequency measurement.

"PDUTY" is the parameter for the Positive Duty Cycle measurement.

"NDUTY" is the parameter for the Negative Duty Cycle measurement.

"PPULSE" is the parameter for the Positive Pulse Width measurement.

"NPULSE" is the parameter for the Negative Pulse Width measurement.

"AMPL" is the parameter for the Cycle Amplitude measurement.

"PKPK" is the parameter for the Cycle Peak–Peak measurement.

"HIGH" is the parameter for the Cycle Top measurement.

"LOW" is the parameter for the Cycle Base measurement.

"Max" is the parameter for the Cycle Max measurement.

"MIN" is the parameter for the Cycle Min measurement.

"INRUSH" is the parameter for the Inrush Current measurement.

"CAPACITANCE" is the parameter for the Input Capacitance measurement.

"OUTPUT1" - "OUTPUT7" are the parameters for the Turn On Time and Turn Off Time measurements.

"GAINCROSSOVERFREQ", "PHASECROSSOVERFREQ", "GM", "PM" are the parameters for the Control Loop Response measurement.

"MAXPSRR", "MAXPSRRFREQ", "MINPSRR", "MINPSRRFREQ" are the parameters for the PSRR measurement

Examples

POWER:POWER1:RESULTS:ALLAcqs:PK2PK? "TONLoss" might return 9.91, indicating the peak-to-peak value of T-On Energy for all acquisitions.

POWER:POWER<x>:RESULTS:ALLAcqs:POPULATION? (Query Only)

This command queries the population (number of complete cycles) of all acquisitions for the measurement parameter in the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWER:Power<x>:RESULTS:ALLAcqs:
POPULATION? {InputPwr|Output1Pwr|
Output2Pwr|Output3Pwr|Efficiency1|Efficiency2|Efficiency3|
TotalEfficiency|INDUCT|IVSINTV|MAGLOSS|Bpeak|
Br|Hc|Hmax|IRipple|DeltaB|DeltaH|
Permeability|RDS|TRUEPWR|APPPWR|REPWR|PWRFACTOR|
PHASE|PWRFREQ|ICFACTOR|VCFACTOR|IRMS|VRMS|
TONENRG|TONLOSS|TOFFENRG|TOFFLOSS|CONDENRG|CONDLOSS|
TTLLOSS|TTLENRG|DVBYDT|DIBYDT|SOAHITSCNT|
LRIPRMS|LRIPPKPK|SWRIPRMS|SWRIPPKPK|PRIOD|
FREQ|PDUTY|NDUTY|PPULSE|NPULSE|
AMPL|PKPK|HIGH|LOW|MAX|MIN|INRUSH|CAPACITANCE|OUTPUT1|
OUTPUT2|OUTPUT3|OUTPUT4|OUTPUT5|OUTPUT6|OUTPUT7|
GAINCROSSOVERFREQ|PHASECROSSOVERFREQ|GM|PM|
MAXPSRR|MAXPSRRFREQ|MINPSRR|MINPSRRFREQ}

NOTE. Above entries are <QString> entries, and must be entered in enclosing quotes.

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

<QString> = the measurement result that you want to return from the specified power measurement number. Available results depend on the power measurement being taken in the specified measurement number. The valid <Qstring> arguments are:

For Efficiency measurement, the parameters are "InputPwr", "Output1Pwr", "Output2Pwr", "Output3Pwr", "Efficiency1", "Efficiency2", "Efficiency3", "TotalEfficiency".

For Inductance measurement, the parameter is "INDUCT".

For IVSIntegralV measurement, the parameter is "IVSINTV".

For Magnetic Loss measurement, the parameter is "MAGLOSS".

For Magnetic Property measurement, the parameters are "Bpeak", "Br", "Hc", "Hmax", "IRipple", "DeltaB", "DeltaH", "Permeability".

For RDSon measurement, the parameter is "RDS".

For Power Quality measurement, the parameters are "TRUEPWR", "APPPWR", "REPWR", "PWRFACTOR", "PHASE", "PWRFREQ", "ICFACTOR", "VCFACTOR", "IRMS", "VRMS".

For Switching Loss measurement, the parameters are "TONENRG", "TONLOSS", "TOFFENRG", "TOFFLOSS", "CONDENRG", "CONDLOSS", "TTLLOSS", "TTLENRG".

"DVBYDT" is the parameter for the dV by dt measurement.

"DIBYDT" is the parameter for the dI by dt measurement.

"SOAHITSCNT" is the parameter for the SOA measurement.

"LRIPRMS" and "LRIPPKPK" are the parameters for the Line Ripple measurement.

"SWRIPRMS" and "SWRIPPKPK" are the parameters for the Switching Ripple measurement.

"PRIOD" is the parameter for the Cycle Period measurement.

"FREQ" is the parameter for the Cycle Frequency measurement.

"PDUTY" is the parameter for the Positive Duty Cycle measurement.

"NDUTY" is the parameter for the Negative Duty Cycle measurement.

"PPULSE" is the parameter for the Positive Pulse Width measurement.

"NPULSE" is the parameter for the Negative Pulse Width measurement.

"AMPL" is the parameter for the Cycle Amplitude measurement.

"PKPK" is the parameter for the Cycle Peak–Peak measurement.

"HIGH" is the parameter for the Cycle Top measurement.

"LOW" is the parameter for the Cycle Base measurement.

"Max" is the parameter for the Cycle Max measurement.

"MIN" is the parameter for the Cycle Min measurement.

"INRUSH" is the parameter for the Inrush Current measurement.

"CAPACITANCE" is the parameter for the Input Capacitance measurement.

"OUTPUT1" - "OUTPUT7" are the parameters for the Turn On Time and Turn Off Time measurements.

"GAINCROSSOVERFREQ", "PHASECROSSOVERFREQ", "GM", "PM" are the parameters for the Control Loop Response measurement.

"MAXPSRR", "MAXPSRRFREQ", "MINPSRR", "MINPSRRFREQ" are the parameters for the PSRR measurement

Examples `Power:Power1:RESULTS:ALLAcqs:POPulation?` "CondEnrg" might return 4.91E-3, indicating the population (number of complete cycles) of conduction energy for all acquisitions.

POWER:POWER<x>:RESULTS:ALLAcqs:STDDDev? (Query Only)

This command queries the standard deviation value of all acquisitions for the measurement parameter in the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:RESULTS:ALLAcqs:STDDDev? {InputPwr|Output1Pwr|Output2Pwr|Output3Pwr|Efficiency1|Efficiency2|Efficiency3|TotalEfficiency|INDUCT|IVSINTV|MAGLOSS|Bpeak|Br|Hc|Hmax|IRipple|DeltaB|DeltaH|Permeability|RDS|TRUEPWR|APPPWR|REPWR|PWRFACTOR|PHASE|PWRFREQ|ICFACTOR|VCFACTOR|IRMS|VRMS|TONENRG|TONLOSS|TOFFENRG|TOFFLOSS|CONDENRG|CONDLOSS|TTLLOSS|TTLENRG|DVBYDT|DIBYDT|SOAHITSCNT|LRIPRMS|LRIPPKPK|SWRIPRMS|SWRIPPKPK|PRIOD|FREQ|PDUTY|NDUTY|PPULSE|NPULSE|AMPL|PKPK|HIGH|LOW|MAX|MIN|INRUSH|CAPACITANCE|OUTPUT1|OUTPUT2|OUTPUT3|OUTPUT4|OUTPUT5|OUTPUT6|OUTPUT7|GAINCROSSOVERFREQ|PHASECROSSOVERFREQ|GM|PM|MAXPSRR|MAXPSRRFREQ|MINPSRR|MINPSRRFREQ}`

NOTE. Above entries are <QString> entries, and must be entered in enclosing quotes.

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

<QString> = the measurement result that you want to return from the specified power measurement number. Available results depend on the power measurement being taken in the specified measurement number. The valid <Qstring> arguments are:

For Efficiency measurement, the parameters are "InputPwr", "Output1Pwr", "Output2Pwr", "Output3Pwr", "Efficiency1", "Efficiency2", "Efficiency3", "TotalEfficiency".

For Inductance measurement, the parameter is "INDUCT".

For IVSIntegralV measurement, the parameter is "IVSINTV".

For Magnetic Loss measurement, the parameter is "MAGLOSS".

For Magnetic Property measurement, the parameters are "Bpeak", "Br", "Hc", "Hmax", "IRipple", "DeltaB", "DeltaH", "Permeability".

For RDSon measurement, the parameter is "RDS".

For Power Quality measurement, the parameters are "TRUEPWR", "APPPWR", "REPWR", "PWRFACTOR", "PHASE", "PWRFREQ", "ICFACTOR", "VCFACTOR", "IRMS", "VRMS".

For Switching Loss measurement, the parameters are "TONENRG", "TONLOSS", "TOFFENRG", "TOFFLOSS", "CONDENRG", "CONDLOSS", "TTLLOSS", "TTLENRG".

"DVBYDT" is the parameter for the dV by dt measurement.

"DIBYDT" is the parameter for the dI by dt measurement.

"SOAHITSCNT" is the parameter for the SOA measurement.

"LRIPRMS" and "LRIPPKPK" are the parameters for the Line Ripple measurement.

"SWRIPRMS" and "SWRIPPKPK" are the parameters for the Switching Ripple measurement.

"PRIOD" is the parameter for the Cycle Period measurement.

"FREQ" is the parameter for the Cycle Frequency measurement.

"PDUTY" is the parameter for the Positive Duty Cycle measurement.

"NDUTY" is the parameter for the Negative Duty Cycle measurement.

"PPULSE" is the parameter for the Positive Pulse Width measurement.

"NPULSE" is the parameter for the Negative Pulse Width measurement.

"AMPL" is the parameter for the Cycle Amplitude measurement.

"PKPK" is the parameter for the Cycle Peak–Peak measurement.

"HIGH" is the parameter for the Cycle Top measurement.

"LOW" is the parameter for the Cycle Base measurement.

"Max" is the parameter for the Cycle Max measurement.

"MIN" is the parameter for the Cycle Min measurement.

"INRUSH" is the parameter for the Inrush Current measurement.

"CAPACITANCE" is the parameter for the Input Capacitance measurement.

"OUTPUT1" - "OUTPUT7" are the parameters for the Turn On Time and Turn Off Time measurements.

"GAINCROSSOVERFREQ", "PHASECROSSOVERFREQ", "GM", "PM" are the parameters for the Control Loop Response measurement.

"MAXPSRR", "MAXPSRRFREQ", "MINPSRR", "MINPSRRFREQ" are the parameters for the PSRR measurement

Examples `Power:Power1:RESULTS:ALLAcqs:STDDev?` "TONLoss" might return 4.25, indicating the standard deviation value of T-On Energy loss for all acquisitions.

POWER:POWER<x>:RESULTS:CURREntacq:F1MAG? (Query Only)

This command queries the first harmonics magnitude value for the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:RESULTS:CURREntacq:F1MAG?` "harmonics"

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

Examples `Power:Power1:RESULTS:CURREntacq:F1MAG?` "harmonics" might return 1.4151834770090, indicating the value of the first harmonics magnitude for the power measurement badge 1.

POWER:POWER<x>:RESULTS:CURREntacq:F3MAG? (Query Only)

This command queries the third harmonics magnitude value for the specified power measurement number.

Conditions Requires option PWR or PS2.

Group	Power
Syntax	<code>Power:Power<x>:RESULTS:CURRENTacq:F3MAG? "harmonics"</code>
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.
Examples	<code>Power:Power1:RESULTS:CURRENTacq:f3MAG? "harmonics"</code> might return 234.0187140104806E-6, indicating the value of third harmonics magnitude for the power measurement badge 1.

POWER:Power<x>:RESULTS:CURRENTacq:FREQUENCY? (Query Only)

This command queries the fundamental frequency for the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	<code>Power:Power<x>:RESULTS:CURRENTacq:FREQUENCY? "harmonics"</code>
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.
Examples	<code>Power:Power1:RESULTS:CURRENTacq:FREQUENCY? "harmonics"</code> might return 100.0067656931537E+3, indicating the fundamental frequency for the power measurement badge 1.

POWER:Power<x>:RESULTS:CURRENTacq:IRMS? (Query Only)

This command queries the RMS current value for the specified power measurement number.

Conditions	Requires option PWR or PS2.
Group	Power

Syntax	<code>Power:Power<x>:RESULTS:CURRENTacq:IRMS? "harmonics"</code>
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.
Examples	<code>Power:Power1:RESULTS:CURRENTacq:IRMS? "harmonics"</code> might return 1.4149980733491, indicating the RMS current value for the power measurement badge 1.

POWER:POWER<x>:RESULTS:CURRENTacq:MAXimum? (Query Only)

This command queries the maximum value of the current acquisition for the measurement parameter in the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:RESULTS:CURRENTacq:MAXimum? {InputPwr|Output1Pwr|Output2Pwr|Output3Pwr|Efficiency1|Efficiency2|Efficiency3|TotalEfficiency|INDUCT|IVSINTV|MAGLOSS|Bpeak|Br|Hc|Hmax|IRipple|DeltaB|DeltaH|Permeability|RDS|TRUEPWR|APPPWR|REPWR|PWRFACTOR|PHASE|PWRFREQ|ICFACTOR|VCFACTOR|IRMS|VRMS|TONENRG|TONLOSS|TOFFENRG|TOFFLOSS|CONDENRG|CONDLOSS|TTLLLOSS|TTLENRG|DVBYDT|DIBYDT|SOAHITSCNT|LRIPRMS|LRIPPKPK|SWRIPRMS|SWRIPPKPK|PRIOD|FREQ|PDUTY|NDUTY|PPULSE|NPULSE|AMPL|PKPK|HIGH|LOW|MAX|MIN|INRUSH|CAPACITANCE|OUTPUT1|OUTPUT2|OUTPUT3|OUTPUT4|OUTPUT5|OUTPUT6|OUTPUT7|GAINCROSSOVERFREQ|PHASECROSSOVERFREQ|GM|PM|MAXPSRR|MAXPSRRFREQ|MINPSRR|MINPSRRFREQ}`

NOTE. Above entries are <QString> entries, and must be entered in enclosing quotes.

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

<QString> = the measurement result that you want to return from the specified power measurement number. Available results depend on the power measurement being taken in the specified measurement number. The valid <Qstring> arguments are:

For Efficiency measurement, the parameters are "InputPwr", "Output1Pwr", "Output2Pwr", "Output3Pwr", "Efficiency1", "Efficiency2", "Efficiency3", "TotalEfficiency".

For Inductance measurement, the parameter is "INDUCT".

For IVSIntegralV measurement, the parameter is "IVSINTV".

For Magnetic Loss measurement, the parameter is "MAGLOSS".

For Magnetic Property measurement, the parameters are "Bpeak", "Br", "Hc", "Hmax", "IRipple", "DeltaB", "DeltaH", "Permeability".

For RDSon measurement, the parameter is "RDS".

For Power Quality measurement, the parameters are "TRUEPWR", "APPPWR", "REPWR", "PWRFACTOR", "PHASE", "PWRFREQ", "ICFACTOR", "VCFACTOR", "IRMS", "VRMS".

For Switching Loss measurement, the parameters are "TONENRG", "TONLOSS", "TOFFENRG", "TOFFLOSS", "CONDENRG", "CONDLOSS", "TTLLOSS", "TTLENRG".

"DVBYDT" is the parameter for the dV by dt measurement.

"DIBYDT" is the parameter for the dI by dt measurement.

"SOAHITSCNT" is the parameter for the SOA measurement.

"LRIPRMS" and "LRIPPKPK" are the parameters for the Line Ripple measurement.

"SWRIPRMS" and "SWRIPPKPK" are the parameters for the Switching Ripple measurement.

"PRIOD" is the parameter for the Cycle Period measurement.

"FREQ" is the parameter for the Cycle Frequency measurement.

"PDUTY" is the parameter for the Positive Duty Cycle measurement.

"NDUTY" is the parameter for the Negative Duty Cycle measurement.

"PPULSE" is the parameter for the Positive Pulse Width measurement.

"NPULSE" is the parameter for the Negative Pulse Width measurement.

"AMPL" is the parameter for the Cycle Amplitude measurement.

"PKPK" is the parameter for the Cycle Peak–Peak measurement.

"HIGH" is the parameter for the Cycle Top measurement.

"LOW" is the parameter for the Cycle Base measurement.

"Max" is the parameter for the Cycle Max measurement.

"MIN" is the parameter for the Cycle Min measurement.

"INRUSH" is the parameter for the Inrush Current measurement.

"CAPACITANCE" is the parameter for the Input Capacitance measurement.

"OUTPUT1" - "OUTPUT7" are the parameters for the Turn On Time and Turn Off Time measurements.

"GAINCROSSOVERFREQ", "PHASECROSSOVERFREQ", "GM", "PM" are the parameters for the Control Loop Response measurement.

"MAXPSRR", "MAXPSRRFREQ", "MINPSRR", "MINPSRRFREQ" are the parameters for the PSRR measurement

Examples `Power:Power1:RESULTS:CURRENTacq:MAXimum?` "TONEnrg" might return 32.8, indicating the maximum value of Ton Energy for the current acquisition.

POWER:POWER<x>:RESULTS:CURRENTacq:MEAN? (Query Only)

This command queries the mean value of the current acquisition for the measurement parameter of the specified power measurement <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:RESULTS:CURRENTacq:MEAN? {InputPwr|Output1Pwr|Output2Pwr|Output3Pwr|Efficiency1|Efficiency2|Efficiency3|TotalEfficiency|INDUCT|IVSINTV|MAGLOSS|Bpeak|Br|Hc|Hmax|IRipple|DeltaB|DeltaH|Permeability|RDS|TRUEPWR|APPPWR|REPWR|PWRFACTOR|PHASE|PWRFREQ|ICFACTOR|VCFACOR|IRMS|VRMS|TONENRG|TONLOSS|TOFFENRG|TOFFLOSS|CONDENRG|CONDLOSS|TTLLLOSS|TTLENRG|DVBYDT|DIBYDT|SOAHITSCNT|LRIPRMS|LRIPPKPK|SWRIPRMS|SWRIPPKPK|PRIOD|FREQ|PDUTY|NDUTY|PPULSE|NPULSE|AMPL|PKPK|HIGH|LOW|MAX|MIN|INRUSH|CAPACITANCE|OUTPUT1|OUTPUT2|OUTPUT3|OUTPUT4|OUTPUT5|OUTPUT6|OUTPUT7|GAINCROSSOVERFREQ|PHASECROSSOVERFREQ|GM|PM|MAXPSRR|MAXPSRRFREQ|MINPSRR|MINPSRRFREQ}`

NOTE. Above entries are <QString> entries, and must be entered in enclosing quotes.

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

<Qstring> = the measurement result that you want to return from the specified power measurement number. Available results depend on the power measurement being taken in the specified measurement number. The valid <Qstring> arguments are:

For Efficiency measurement, the parameters are "InputPwr", "Output1Pwr", "Output2Pwr", "Output3Pwr", "Efficiency1", "Efficiency2", "Efficiency3", "TotalEfficiency".

For Inductance measurement, the parameter is "INDUCT".

For IVSIntegralV measurement, the parameter is "IVSINTV".

For Magnetic Loss measurement, the parameter is "MAGLOSS".

For Magnetic Property measurement, the parameters are "Bpeak", "Br", "Hc", "Hmax", "IRipple", "DeltaB", "DeltaH", "Permeability".

For RDSon measurement, the parameter is "RDS".

For Power Quality measurement, the parameters are "TRUEPWR", "APPPWR", "REPWR", "PWRFACTOR", "PHASE", "PWRFREQ", "ICFACTOR", "VCFACTOR", "IRMS", "VRMS".

For Switching Loss measurement, the parameters are "TONENRG", "TONLOSS", "TOFFENRG", "TOFFLOSS", "CONDENRG", "CONDLOSS", "TTLLOSS", "TTLENRG".

"DVBYDT" is the parameter for the dV by dt measurement.

"DIBYDT" is the parameter for the dI by dt measurement.

"SOAHITSCNT" is the parameter for the SOA measurement.

"LRIPRMS" and "LRIPPKPK" are the parameters for the Line Ripple measurement.

"SWRIPRMS" and "SWRIPPKPK" are the parameters for the Switching Ripple measurement.

"PRIOD" is the parameter for the Cycle Period measurement.

"FREQ" is the parameter for the Cycle Frequency measurement.

"PDUTY" is the parameter for the Positive Duty Cycle measurement.

"NDUTY" is the parameter for the Negative Duty Cycle measurement.

"PPULSE" is the parameter for the Positive Pulse Width measurement.

"NPULSE" is the parameter for the Negative Pulse Width measurement.

"AMPL" is the parameter for the Cycle Amplitude measurement.

"PKPK" is the parameter for the Cycle Peak–Peak measurement.

"HIGH" is the parameter for the Cycle Top measurement.

"LOW" is the parameter for the Cycle Base measurement.

"Max" is the parameter for the Cycle Max measurement.

"MIN" is the parameter for the Cycle Min measurement.

"INRUSH" is the parameter for the Inrush Current measurement.

"CAPACITANCE" is the parameter for the Input Capacitance measurement.

"OUTPUT1" - "OUTPUT7" are the parameters for the Turn On Time and Turn Off Time measurements.

"GAINCROSSOVERFREQ", "PHASECROSSOVERFREQ", "GM", "PM" are the parameters for the Control Loop Response measurement.

"MAXPSRR", "MAXPSRRFREQ", "MINPSRR", "MINPSRRFREQ" are the parameters for the PSRR measurement

Examples `Power:Power2:RESULTS:CURRENTacq:MEAN?` "TruePwr" might return `42.6097255943E-2`, indicating the mean value of true power for the current acquisition of power measurement 2.

POWER:POWER<x>:RESULTS:CURRENTacq:MINimum? (Query Only)

This command queries the minimum value of the current acquisition for the measurement parameter in the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:RESULTS:CURRENTacq:MINimum? {InputPwr|Output1Pwr|Output2Pwr|Output3Pwr|Efficiency1|Efficiency2|Efficiency3|TotalEfficiency|INDUCT|IVSINTV|MAGLOSS|Bpeak|Br|Hc|Hmax|IRipple|DeltaB|DeltaH|Permeability|RDS|TRUEPWR|APPPWR|REPWR|PWRFACTOR|PHASE|PWRFREQ|ICFACTOR|VCFACTOR|IRMS|VRMS|TONENRG|TONLOSS|TOFFENRG|TOFFLOSS|CONDENRG|CONDLOSS|TTLLOSS|TTLENRG|DVBYDT|DIBYDT|SOAHITSCNT|LRIPRMS|LRIPPKPK|SWRIPRMS|SWRIPPKPK|PRIOD|FREQ|PDUTY|NDUTY|PPULSE|NPULSE|AMPL|PKPK|HIGH|LOW|MAX|MIN|INRUSH|CAPACITANCE|OUTPUT1|OUTPUT2|OUTPUT3|OUTPUT4|OUTPUT5|OUTPUT6|OUTPUT7|GAINCROSSOVERFREQ|PHASECROSSOVERFREQ|GM|PM|MAXPSRR|MAXPSRRFREQ|MINPSRR|MINPSRRFREQ}`

NOTE. Above entries are <QSTring> entries, and must be entered in enclosing quotes.

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <QString> = the measurement result that you want to return from the specified power measurement number. Available results depend on the power measurement being taken in the specified measurement number. The valid <Qstring> arguments are: For Efficiency measurement, the parameters are "InputPwr", "Output1Pwr", "Output2Pwr", "Output3Pwr", "Efficiency1", "Efficiency2", "Efficiency3", "TotalEfficiency". For Inductance measurement, the parameter is "INDUCT". For IVSIntegralV measurement, the parameter is "IVSINTV". For Magnetic Loss measurement, the parameter is "MAGLOSS". For Magnetic Property measurement, the parameters are "Bpeak", "Br", "Hc", "Hmax", "IRipple", "DeltaB", "DeltaH", "Permeability". For RDSon measurement, the parameter is "RDS". For Power Quality measurement, the parameters are "TRUEPWR", "APPPWR", "REPWR", "PWRFACTOR", "PHASE", "PWRFREQ", "ICFACTOR", "VCFACTOR", "IRMS", "VRMS". For Switching Loss measurement, the parameters are "TONENRG", "TONLOSS", "TOFFENRG", "TOFFLOSS", "CONDENRG", "CONDLOSS", "TTLLOSS", "TTLENRG". "DVBYDT" is the parameter for the dV by dt measurement. "DIBYDT" is the parameter for the dI by dt measurement. "SOAHITSCNT" is the parameter for the SOA measurement. "LRIPRMS" and "LRIPPKPK" are the parameters for the Line Ripple measurement. "SWRIPRMS" and "SWRIPPKPK" are the parameters for the Switching Ripple measurement. "PRIOD" is the parameter for the Cycle Period measurement. "FREQ" is the parameter for the Cycle Frequency measurement. "PDUTY" is the parameter for the Positive Duty Cycle measurement. "NDUTY" is the parameter for the Negative Duty Cycle measurement. "PPULSE" is the parameter for the Positive Pulse Width measurement. "NPULSE" is the parameter for the Negative Pulse Width measurement. "AMPL" is the parameter for the Cycle Amplitude measurement.
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"PKPK" is the parameter for the Cycle Peak–Peak measurement.

"HIGH" is the parameter for the Cycle Top measurement.

"LOW" is the parameter for the Cycle Base measurement.

"Max" is the parameter for the Cycle Max measurement.

"MIN" is the parameter for the Cycle Min measurement.

"INRUSH" is the parameter for the Inrush Current measurement.

"CAPACITANCE" is the parameter for the Input Capacitance measurement.

"OUTPUT1" - "OUTPUT7" are the parameters for the Turn On Time and Turn Off Time measurements.

"GAINCROSSOVERFREQ", "PHASECROSSOVERFREQ", "GM", "PM" are the parameters for the Control Loop Response measurement.

"MAXPSRR", "MAXPSRRFREQ", "MINPSRR", "MINPSRRFREQ" are the parameters for the PSRR measurement

Examples `Power:Power1:RESULTS:CURRENTacq:MINimum?` "TruePwr" might return 4.17829019093E–9, indicating the minimum value of true power for the current acquisition.

POWER:POWER<x>:RESULTS:CURRENTacq:PK2PK? (Query Only)

This command queries the peak-to-peak value of the current acquisition for the measurement parameter in the specified power measurement number.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:RESULTS:CURRENTacq:PK2PK? {InputPwr|Output1Pwr|Output2Pwr|Output3Pwr|Efficiency1|Efficiency2|Efficiency3|TotalEfficiency|INDUCT|IVSINTV|MAGLOSS|Bpeak|Br|Hc|Hmax|IRipple|DeltaB|DeltaH|Permeability|RDS|TRUEPWR|APPPWR|REPWR|PWRFACTOR|PHASE|PWRFREQ|ICFACTOR|VCFACTOR|IRMS|VRMS|TONENRG|TONLOSS|TOFFENRG|TOFFLOSS|CONDENRG|CONDLOSS|TTLLOSS|TTLENRG|DVBYDT|DIBYDT|SOAHITSCNT|LRIPRMS|LRIPKPK|SWRIPRMS|SWRIPKPK|PRIOD|FREQ|PDUTY|NDUTY|PPULSE|NPULSE|AMPL|PKPK|HIGH|LOW|MAX|MIN|INRUSH|CAPACITANCE|OUTPUT1|OUTPUT2|OUTPUT3|OUTPUT4|OUTPUT5|OUTPUT6|OUTPUT7|GAINCROSSOVERFREQ|PHASECROSSOVERFREQ|GM|PM|MAXPSRR|MAXPSRRFREQ|MINPSRR|MINPSRRFREQ}`

NOTE. Above entries are <QString> entries, and must be entered in enclosing quotes.

Arguments

Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

<Qstring> = the measurement result that you want to return from the specified power measurement number. Available results depend on the power measurement being taken in the specified measurement number. The valid <Qstring> arguments are:

For Efficiency measurement, the parameters are "InputPwr", "Output1Pwr", "Output2Pwr", "Output3Pwr", "Efficiency1", "Efficiency2", "Efficiency3", "TotalEfficiency".

For Inductance measurement, the parameter is "INDUCT".

For IVSIntegralV measurement, the parameter is "IVSINTV".

For Magnetic Loss measurement, the parameter is "MAGLOSS".

For Magnetic Property measurement, the parameters are "Bpeak", "Br", "Hc", "Hmax", "IRipple", "DeltaB", "DeltaH", "Permeability".

For RDSon measurement, the parameter is "RDS".

For Power Quality measurement, the parameters are "TRUEPWR", "APPPWR", "REPWR", "PWRFACTOR", "PHASE", "PWRFREQ", "ICFACTOR", "VCFACOR", "IRMS", "VRMS".

For Switching Loss measurement, the parameters are "TONENRG", "TONLOSS", "TOFFENRG", "TOFFLOSS", "CONDENRG", "CONDLOSS", "TTLLOSS", "TTLENRG".

"DVBYDT" is the parameter for the dV by dt measurement.

"DIBYDT" is the parameter for the dI by dt measurement.

"SOAHITSCNT" is the parameter for the SOA measurement.

"LRIPRMS" and "LRIPPKPK" are the parameters for the Line Ripple measurement.

"SWRIPRMS" and "SWRIPPKPK" are the parameters for the Switching Ripple measurement.

"PRIOD" is the parameter for the Cycle Period measurement.

"FREQ" is the parameter for the Cycle Frequency measurement.

"PDUTY" is the parameter for the Positive Duty Cycle measurement.

"NDUTY" is the parameter for the Negative Duty Cycle measurement.

"PPULSE" is the parameter for the Positive Pulse Width measurement.

"NPULSE" is the parameter for the Negative Pulse Width measurement.

"AMPL" is the parameter for the Cycle Amplitude measurement.

"PKPK" is the parameter for the Cycle Peak–Peak measurement.

"HIGH" is the parameter for the Cycle Top measurement.

"LOW" is the parameter for the Cycle Base measurement.

"Max" is the parameter for the Cycle Max measurement.

"MIN" is the parameter for the Cycle Min measurement.

"INRUSH" is the parameter for the Inrush Current measurement.

"CAPACITANCE" is the parameter for the Input Capacitance measurement.

"OUTPUT1" - "OUTPUT7" are the parameters for the Turn On Time and Turn Off Time measurements.

"GAINCROSSOVERFREQ", "PHASECROSSOVERFREQ", "GM", "PM" are the parameters for the Control Loop Response measurement.

"MAXPSRR", "MAXPSRRFREQ", "MINPSRR", "MINPSRRFREQ" are the parameters for the PSRR measurement

Examples `Power:Power1:RESULTS:CURRENTacq:PK2PK?` "TONLoss" might return 9.91, indicating the peak-to-peak value of Ton Energy for the current acquisition.

Power:Power<x>:RESULTS:CURRENTacq:POHCL? (Query Only)

This command queries the limit of partial odd harmonic current for the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:RESULTS:CURRENTacq:POHCL?` "harmonics"

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

Examples `Power:Power1:RESULTS:CURRENTacq:POHCL?` "harmonics" might return 251.3529788962128E-3, indicating the limit of partial odd harmonic current for the power measurement badge 1.

POWER:POWER<x>:RESULTS:CURRENtAcq:POHCM? (Query Only)

This command queries the measured value of partial odd harmonic current for the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWER:POWER<x>:RESULTS:CURRENtAcq:POHCM? "harmonics"

Examples POWER:POWER1:RESULTS:CURRENtAcq:POHCM? "harmonics" might return 515.422617782020E-6, indicating the measured value of partial odd harmonic current for the power measurement badge 1.

POWER:POWER<x>:RESULTS:CURRENtAcq:POHCS? (Query Only)

This command queries the status of partial odd harmonic current for the specified power measurement number. The power measurement number is specified by x.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWER:POWER<x>:RESULTS:CURRENtAcq:POHCS? "harmonics"

Arguments POWER<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

Examples POWER:POWER1:RESULTS:CURRENtAcq:POHCS? "harmonics" might return Pass, indicating the status of partial odd harmonic current for power measurement 1.

POWER:POWER<x>:RESULTS:CURRENtAcq:POPULATION? (Query Only)

This command queries the population (number of complete cycles) of the current acquisition for the measurement parameter in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:RESuLts:CURRentacq:POPuLation? {InputPwr|Output1Pwr|Output2Pwr|Output3Pwr|Efficiency1|Efficiency2|Efficiency3|TotalEfficiency|INDUCT|IVSINTV|MAGLOSS|Bpeak|Br|Hc|Hmax|IRipple|DeltaB|DeltaH|Permeability|RDS|TRUEPWR|APPPWR|REPWR|PWRFACTOR|PHASE|PWRFREQ|ICFACTOR|VCFACTOR|IRMS|VRMS|TONENRG|TONLOSS|TOFFENRG|TOFFLOSS|CONDENRG|CONDLOSS|TTLLLOSS|TTLENRG|DVBYDT|DIBYDT|SOAHITSCNT|LRIPRMS|LRIPPKPK|SWRIPRMS|SWRIPPKPK|PRIOD|FREQ|PDUTY|NDUTY|PPULSE|NPULSE|AMPL|PKPK|HIGH|LOW|MAX|MIN|INRUSH|CAPACITANCE|OUTPUT1|OUTPUT2|OUTPUT3|OUTPUT4|OUTPUT5|OUTPUT6|OUTPUT7|GAINCROSSOVERFREQ|PHASECROSSOVERFREQ|GM|PM|MAXPSRR|MAXPSRRFREQ|MINPSRR|MINPSRRFREQ}`

NOTE. Above entries are <QString> entries, and must be entered in enclosing quotes.

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

<Qstring> = the measurement result that you want to return from the specified power measurement number. Available results depend on the power measurement being taken in the specified measurement number. The valid <Qstring> arguments are:

For Efficiency measurement, the parameters are "InputPwr", "Output1Pwr", "Output2Pwr", "Output3Pwr", "Efficiency1", "Efficiency2", "Efficiency3", "TotalEfficiency".

For Inductance measurement, the parameter is "INDUCT".

For IVSIntegralV measurement, the parameter is "IVSINTV".

For Magnetic Loss measurement, the parameter is "MAGLOSS".

For Magnetic Property measurement, the parameters are "Bpeak", "Br", "Hc", "Hmax", "IRipple", "DeltaB", "DeltaH", "Permeability".

For RDSon measurement, the parameter is "RDS".

For Power Quality measurement, the parameters are "TRUEPWR", "APPPWR", "REPWR", "PWRFACTOR", "PHASE", "PWRFREQ", "ICFACTOR", "VCFACTOR", "IRMS", "VRMS".

For Switching Loss measurement, the parameters are "TONENRG", "TONLOSS", "TOFFENRG", "TOFFLOSS", "CONDENRG", "CONDLOSS", "TTLLOSS", "TTLENRG".

"DVBYDT" is the parameter for the dV by dt measurement.

"DIBYDT" is the parameter for the dI by dt measurement.

"SOAHITSCNT" is the parameter for the SOA measurement.

"LRIPRMS" and "LRIPPKPK" are the parameters for the Line Ripple measurement.

"SWRIPRMS" and "SWRIPPKPK" are the parameters for the Switching Ripple measurement.

"PRIOD" is the parameter for the Cycle Period measurement.

"FREQ" is the parameter for the Cycle Frequency measurement.

"PDUTY" is the parameter for the Positive Duty Cycle measurement.

"NDUTY" is the parameter for the Negative Duty Cycle measurement.

"PPULSE" is the parameter for the Positive Pulse Width measurement.

"NPULSE" is the parameter for the Negative Pulse Width measurement.

"AMPL" is the parameter for the Cycle Amplitude measurement.

"PKPK" is the parameter for the Cycle Peak–Peak measurement.

"HIGH" is the parameter for the Cycle Top measurement.

"LOW" is the parameter for the Cycle Base measurement.

"Max" is the parameter for the Cycle Max measurement.

"MIN" is the parameter for the Cycle Min measurement.

"INRUSH" is the parameter for the Inrush Current measurement.

"CAPACITANCE" is the parameter for the Input Capacitance measurement.

"OUTPUT1" - "OUTPUT7" are the parameters for the Turn On Time and Turn Off Time measurements.

"GAINCROSSOVERFREQ", "PHASECROSSOVERFREQ", "GM", "PM" are the parameters for the Control Loop Response measurement.

"MAXPSRR", "MAXPSRRFREQ", "MINPSRR", "MINPSRRFREQ" are the parameters for the PSRR measurement

Examples

Power:Power1:RESULTS:CURRENTacq:POPULATION? "CondEnrg" might return 9.24E+4, indicating the population (number of complete cycles) of conduction energy for the current acquisition.

POWER:POWER<x>:RESULTS:CURRENtAcq:RMS? (Query Only)

This command queries the RMS value of the source selected for the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWER:POWER<x>:RESULTS:CURRENtAcq:RMS? "harmonics"

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

Examples POWER:POWER1:RESULTS:CURRENtAcq:RMS? "harmonics" might return 1.4143420437461, indicating the RMS value of the source selected for power measurement 1.

POWER:POWER<x>:RESULTS:CURRENtAcq:STATUS? (Query Only)

This command queries the status of the measurement for the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWER:POWER<x>:RESULTS:CURRENtAcq:STATUS? "harmonics"

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

Examples POWER:POWER1:RESULTS:CURRENtAcq:STATUS? "harmonics" might return Pass, indicating that the measurement has passed according to the standard for power measurement 1.

POWER:POWER<x>:RESULTS:CURRENTacq:STDDDev? (Query Only)

This command queries the standard deviation value of the current acquisition for the measurement parameter in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWER:POWER<x>:RESULTS:CURRENTacq:STDDDev?
 {InputPwr|Output1Pwr|
 Output2Pwr|Output3Pwr|Efficiency1|Efficiency2|Efficiency3|
 TotalEfficiency|INDUCT|IVSINTV|MAGLOSS|Bpeak|Br|Hc|Hmax|IRipple|
 DeltaB|DeltaH|Permeability|RDS|TRUEPWR|APPPWR|REPWR|PWRFACTOR|
 PHASE|PWRFREQ|ICFACTOR|VCFACOR|IRMS|VRMS|TONENRG|TONLOSS|
 TOFFENRG|TOFFLOSS|CONDENRG|CONDLOSS|TTLLOSS|TTLENRG|DVBYDT|
 DIBYDT|SOAHITSNT|LRIPRMS|LRIPPKPK|SWRIPRMS|SWRIPPKPK|PRIOD|
 FREQ|PDUTY|NDUTY|PPULSE|NPULSE|AMPL|PKPK|HIGH|LOW|MAX|MIN
 |INRUSH|CAPACITANCE|OUTPUT1|
 OUTPUT2|OUTPUT3|OUTPUT4|OUTPUT5|OUTPUT6|OUTPUT7|
 GAINCROSSOVERFREQ|PHASECROSSOVERFREQ|GM|PM|
 MAXPSRR|MAXPSRRFREQ|MINPSRR|MINPSRRFREQ}

NOTE. Above entries are <QString> entries, and must be entered in enclosing quotes.

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

<QString> = the measurement result that you want to return from the specified power measurement number. Available results depend on the power measurement being taken in the specified measurement number. The valid <Qstring> arguments are:

For Efficiency measurement, the parameters are "InputPwr", "Output1Pwr", "Output2Pwr", "Output3Pwr", "Efficiency1", "Efficiency2", "Efficiency3", "TotalEfficiency".

For Inductance measurement, the parameter is "INDUCT".

For IVSIntegralV measurement, the parameter is "IVSINTV".

For Magnetic Loss measurement, the parameter is "MAGLOSS".

For Magnetic Property measurement, the parameters are "Bpeak", "Br", "Hc", "Hmax", "IRipple", "DeltaB", "DeltaH", "Permeability".

For RDSon measurement, the parameter is "RDS".

For Power Quality measurement, the parameters are "TRUEPWR", "APPPWR", "REPWR", "PWRFACTOR", "PHASE", "PWRFREQ", "ICFACTOR", "VCFACTOR", "IRMS", "VRMS".

For Switching Loss measurement, the parameters are "TONENRG", "TONLOSS", "TOFFENRG", "TOFFLOSS", "CONDENRG", "CONDLOSS", "TTLLOSS", "TTLENRG".

"DVBYDT" is the parameter for the dV by dt measurement.

"DIBYDT" is the parameter for the dI by dt measurement.

"SOAHITSCNT" is the parameter for the SOA measurement.

"LRIPRMS" and "LRIPPKPK" are the parameters for the Line Ripple measurement.

"SWRIPRMS" and "SWRIPPKPK" are the parameters for the Switching Ripple measurement.

"PRIOD" is the parameter for the Cycle Period measurement.

"FREQ" is the parameter for the Cycle Frequency measurement.

"PDUTY" is the parameter for the Positive Duty Cycle measurement.

"NDUTY" is the parameter for the Negative Duty Cycle measurement.

"PPULSE" is the parameter for the Positive Pulse Width measurement.

"NPULSE" is the parameter for the Negative Pulse Width measurement.

"AMPL" is the parameter for the Cycle Amplitude measurement.

"PKPK" is the parameter for the Cycle Peak–Peak measurement.

"HIGH" is the parameter for the Cycle Top measurement.

"LOW" is the parameter for the Cycle Base measurement.

"Max" is the parameter for the Cycle Max measurement.

"MIN" is the parameter for the Cycle Min measurement.

"INRUSH" is the parameter for the Inrush Current measurement.

"CAPACITANCE" is the parameter for the Input Capacitance measurement.

"OUTPUT1" - "OUTPUT7" are the parameters for the Turn On Time and Turn Off Time measurements.

"GAINCROSSOVERFREQ", "PHASECROSSOVERFREQ", "GM", "PM" are the parameters for the Control Loop Response measurement.

"MAXPSRR", "MAXPSRRFREQ", "MINPSRR", "MINPSRRFREQ" are the parameters for the PSRR measurement

Examples `Power:Power2:RESULTS:CURRENTacq:STDDev?` "TONLoss" might return 1.31E+2, indicating the standard deviation value of Ton Energy for current acquisition of power measurement 2.

POWER:POWER<x>:RESULTS:CURRENTacq:THDF? (Query Only)

This command queries the total harmonic distortion (fundamental) value for the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:RESULTS:CURRENTacq:THDF?` "harmonics"

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

Examples `Power:Power1:RESULTS:CURRENTacq:THDF?` "harmonics" might return 96.9846996670887E-3, indicating the value of total harmonic distortion (fundamental) for power measurement 1.

POWER:POWER<x>:RESULTS:CURRENTacq:THDR? (Query Only)

This command queries the total harmonic distortion (RMS) value for the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:RESULTS:CURRENTacq:THDR?` "harmonics"

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

Examples `Power:Power1:RESULTS:CURRENTacq:THDR?` "harmonics" might return 107.1384597967292E-3, indicating the value of total harmonic distortion (fundamental) for power measurement 1.

POWER:POWER<x>:RESULTS:CURRENTacq:TRPWR? (Query Only)

This command queries the true power value for the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:RESULTS:CURRENTacq:TRPWR?` "harmonics"

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

Examples `Power:Power1:RESULTS:CURRENTacq:TRPWR?` "harmonics" might return 2.0002612633993, indicating the true power value for power measurement 1.

POWER:POWER<x>:RESULTS:CURRENTacq:VRMS? (Query Only)

This command queries the RMS voltage value for the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:RESULTS:CURRENTacq:VRMS?` "harmonics"

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

Examples `Power:Power3:RESULTS:CURRENTacq:VRMS?` "harmonics" might return 1.4117680233354, indicating the RMS voltage value for the power measurement 3.

POWer:POWer<x>:SEQSETup (No Query Form)

This command sets up the oscilloscope's horizontal, vertical, and trigger parameters to optimize for taking the specified power measurement.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	POWer:POWer<x>:SEQSETup Execute
Arguments	POWer<x> is the Power measurement identifier number. The number must be for a power measurement that requires a single sequence acquisition. RUN sets the measurement to run an acquisition and acquire data for the specified single sequence power measurement.
Examples	POWer:POWer3:SEquence Execute runs the power measurement 3 sequence setup.

POWer:POWer<x>:SEQuence

This command sets or queries the run state of a single sequence power measurement.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	POWer:POWer<x>:SEQuence {RUN RERUN} POWer:POWer<x>:SEQuence?
Arguments	POWer<x> is the Power measurement identifier number. The number must be for a power measurement that requires a single sequence acquisition. RUN sets the measurement to run an acquisition and acquire data for the specified single sequence power measurement. RERUN sets the measurement to rerun an acquisition and acquire data for the specified single sequence power measurement.

- Examples** `Power:Power3:Sequence RUN` sets power measurement 3 to run the measurement.
- `Power:Power1:Sequence?` might return `Power:Power1:Sequence RERUN`, indicating that the current state of the power measurement is rerun.

POWer:POWer<x>:SOA:ISOURce

This command sets or queries the current source for SOA measurement in the specified power measurement number. The power measurement number is specified by <x>.

- Conditions** Requires option PWR or PS2.

- Group** Power

- Syntax** `Power:Power<x>:SOA:ISOURce {CH<x>|MATH<x>|REF<x>}`
`Power:Power<x>:SOA:ISOURce?`

- Arguments** `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.
- `CH<x>` = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels on your instrument.
- `MATH<x>` = A math waveform specifier; <x> is ≥ 1 .
- `REF<x>` = A reference waveform specifier; <x> is ≥ 1 .

- Examples** `Power:Power1:SOA:ISOURce CH2` sets the current source for the SOA measurement 1 as channel 2.

POWer:POWer<x>:SOA:POINT

This command sets or queries the X or Y coordinate value for an SOA mask of a specified power measurement.

- Conditions** Requires option PWR or PS2.

- Group** Power

Syntax `Power:Power<x>:SOA:POINT <QString> <NR1>`
 `Power:Power<x>:SOA:POINT <QString>?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

`<QString>` = 1X | 2X | 3X | 4X | 5X | 6X | 7X | 8X | 9X | 10X | 11X | 12X | 13X | 14X | 15X | 16X | 17X | 18X | 19X | 20X | 21X | 22X | 23X | 24X | 25X | 26X | 27X | 28X | 29X | 30X | 31X | 32X | 1Y | 2Y | 3Y | 4Y | 5Y | 6Y | 7Y | 8Y | 9Y | 10Y | 11Y | 12Y | 13Y | 14Y | 15Y | 16Y | 17Y | 18Y | 19Y | 20Y | 21Y | 22Y | 23Y | 24Y | 25Y | 26Y | 27Y | 28Y | 29Y | 30Y | 31Y | 32Y

`<NR2>` sets the specified SOA mask X or Y point value, as a floating number, in the range from -5000 to 5000.

Examples `Power:Power1:SOA:POINT6Y 0.125` sets the #6Y coordinate value of the SOA mask of power measurement 1 to 0.125.

`Power:Power3:SOA:POINT12X?` might return
`Power:Power3:SOA:POINT12X 1.435`, indicating that the #12X coordinate value of the SOA mask of power measurement 3 is 1.435.

POWER:POWER<x>:SOA:RECALLmask

This command recalls or queries the recall mask file name in the specified power measurement number. The power measurement number is specified by `<x>`.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:SOA:RECALLmask`
 `Power:Power<x>:SOA:RECALLmask?`

Examples `Power:Power1:SOA:RECALLmask?` might return `Tek000.msk`, indicating the file name of the mask that will be recalled.

POWER:POWER<x>:SOA:RECALLmask:FILENAME

This command sets or queries the file name for saving SOA mask file name in the specified power measurement number. The power measurement number is specified by `<x>`.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:SOA:RECallmask:FILENAME Power:Power<x>:SOA:RECallmask:FILENAME?
Examples	Power:Power1:SOA:RECallmask:FILENAME

Power:Power<x>:SOA:SAVemask

This command saves the mask file as per the name configured and at the configured path or queries the mask file name, path, and file type for the SOA measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:SOA:SAVemask Power:Power<x>:SOA:SAVemask?
Related Commands	Power:Power<x>:SOA:SAVemask:FILENAME
Examples	Power:Power1:SOA:SAVemask saves the mask file of SOA measurement as the configured file name at the configured path.

Power:Power<x>:SOA:SAVemask:AUTOINCrement

This command sets or queries the state of auto-increment for saved SOA mask file names in the specified power measurement number. The power measurement number is specified by <x>.

Conditions	Requires option PWR or PS2.
Group	Power

Syntax `Power:Power<x>:SOA:SAVemask:AUTOINCrement`
 `Power:Power<x>:SOA:SAVemask:AUTOINCrement?`

POWER:POWER<x>:SOA:SAVemask:FILENAME

This command sets or queries the mask file name for SOA measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:SOA:SAVemask:FILENAME`
 `Power:Power<x>:SOA:SAVemask:FILENAME?`

Examples `Power:Power1:SOA:SAVemask:FILENAME "Tek001.msk"` sets the mask file name for SOA measurement as Tek001.msk.

POWER:POWER<x>:SOA:SAVemask:FOLDER

This command sets or queries the mask file folder path for SOA measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:SOA:SAVemask:FOLDER`
 `Power:Power<x>:SOA:SAVemask:FOLDER?`

Examples `Power:Power1:SOA:SAVemask:FOLDER "C:"` sets the mask file folder path for SOA measurement as C drive.

POWER:POWER<x>:SOA:VSOURCE

This command sets or queries the voltage source for SOA measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWER:POWER<x>:SOA:VSOURCE {CH<x> | MATH<x> | REF<x>}
POWER:POWER<x>:SOA:VSOURCE?

Arguments POWER<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

MATH<x> = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples POWER:POWER1:SOA:VSOURCE CH1 sets the voltage source for SOA measurement as channel 1.

POWER:POWER<x>:SWITCHINGLOSS:DEVICETYPE

This command sets or queries the conduction calculation method for switching loss measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWER:POWER<x>:SWITCHINGLOSS:DEVICETYPE {MOSFET | BJT}
POWER:POWER<x>:SWITCHINGLOSS:DEVICETYPE?

Examples `Power:Power1:SWITCHINGLOSS:DEVICETYPE MOSFET` sets the conduction calculation method as mosfet for switching loss measurement of the power measurement badge Power 1.

POWER:POWER<x>:SWITCHINGLOSS:GATESource

This command sets or queries the gate voltage (V_g) for the switching loss measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:SWITCHINGLOSS:GATESource {CH<x> | MATH<x> | REF<x>}`
`Power:Power<x>:SWITCHINGLOSS:GATESource?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

`CH<x>` = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

`MATH<x>` = A math waveform specifier; <x> is ≥ 1 .

`REF<x>` = A reference waveform specifier; <x> is ≥ 1 .

Examples `Power:Power1:SWITCHINGLOSS:GATESource MATH1` sets the gate voltage (V_g) source for switching loss measurement of power measurement 1 as MATH1.

`Power:Power2:SWITCHINGLOSS:VSOURCE?` might return MATH3 indicating the gate voltage (V_g) for switching loss measurement of power measurement 2.

POWER:POWER<x>:SWITCHINGLOSS:ILEVELAbs

This command sets or queries the current level (Ton-Start & Stop) in absolute units for switching loss measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group	Power
Syntax	Power:Power<x>:SWITCHINGLOSS:ILEVELAbs <NR1> Power:Power<x>:SWITCHINGLOSS:ILEVELAbs?
Related Commands	Power:Power<x>:SWITCHINGLOSS:LEVELUNIts
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <NR1> ranges from –100 to 100
Examples	Power:Power1:SWITCHINGLOSS:ILEVELAbs 1.2 sets the current level (Ton-Start & Stop) value as 1.2 for the switching loss measurement in the power measurement badge 1.

Power:Power<x>:SWITCHINGLOSS:ILEVELPct

This command sets or queries the current level (Ton-Start & Stop) in percentage for switching loss measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:SWITCHINGLOSS:ILEVELPct <NR1> Power:Power<x>:SWITCHINGLOSS:ILEVELPct?
Related Commands	Power:Power<x>:SWITCHINGLOSS:LEVELUNIts
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <NR1> ranges from 0.0001 to 90
Examples	Power:Power1:SWITCHINGLOSS:ILEVELPct 6 sets the current level (Ton-Start & Stop) value as 6 percentage for the switching loss measurement in the power measurement badge 1.

POWer:POWer<x>:SWITCHINGLOSS:ISOURce

This command sets or queries the current source for the switching loss measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax Power:POWer<x>:SWITCHINGLOSS:ISOURce {CH<x> | MATH<x> | REF<x>}
Power:POWer<x>:SWITCHINGLOSS:ISOURce?

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

MATH<x> = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples Power:POWer1:SWITCHINGLOSS:ISOURce CH2 sets the current source for the switching loss measurement in the specified power measurement badge as CH2.

Power:POWer2:SWITCHINGLOSS:ISOURce? might return CH1 indicating the current source for switching loss measurement of Power 2 power measurement badge.

POWer:POWer<x>:SWITCHINGLOSS:LEVELUNIts

This command sets or queries the level units for switching loss measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax Power:POWer<x>:SWITCHINGLOSS:LEVELUNIts {PERCent | ABSolute}
Power:POWer<x>:SWITCHINGLOSS:LEVELUNIts?

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. PERCent to set the High, Mid, and Low reference levels in percentage. ABSolute to set the High, Mid, and Low reference levels to specific signal levels.
Examples	Power:Power1:SWITCHINGLOSS:LEVELUNITS ABSolute sets the level units as Absolute for switching loss measurement of the power measurement badge Power 1.

POWer:POWer<x>:SWITCHINGLOSS:RDSOn

This command sets or queries the RDS(on) value for switching loss measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions	Requires option PWR or PS2.
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Group	Power
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Syntax	Power:Power<x>:SWITCHINGLOSS:RDSOn <NR1> Power:Power<x>:SWITCHINGLOSS:RDSOn?
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Related Commands	POWer:POWer<x>:SWITCHINGLOSS:DEVICeType
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Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <NR1> ranges from 0 to 100
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Examples	Power:Power1:SWITCHINGLOSS:RDSOn 2 sets the switching loss RDSOn value as 2 for switching loss measurement of the power measurement badge Power 1.
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POWer:POWer<x>:SWITCHINGLOSS:SWLCONFIGType

This command sets or queries the configuration type for the switching loss measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:SWITCHINGLOSS:SWLCONFIGType {SMPS PFC FLYBACK} Power:Power<x>:SWITCHINGLOSS:SWLCONFIGType?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. SMPS: Select this option in case of signals without noise and ringing. The V_g source is not required. Select V_g source (Source 3), in case of noisy signal. PFC: Select this option when input DUT signals are from Power Factor Correction Circuit. For this case, V_g source is mandatory. FLYBACK: Select this option when input signals are ringing. This option does not require a V_g source.
Examples	Power:Power1:SWITCHINGLOSS:SWLCONFIGType PFC sets the configuration type as PFC for the switching loss measurement of the power measurement badge Power 1.

POWER:POWER<x>:SWITCHINGLOSS:VCESat

This command sets or queries the value for the VCE(sat) value for switching loss measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:SWITCHINGLOSS:VCESat <NR1> Power:Power<x>:SWITCHINGLOSS:VCESat?
Related Commands	POWER:POWER<x>:SWITCHINGLOSS:DEVICETYPE
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

<NR1> ranges from 0.001 to 100

Examples `Power:Power1:SWITCHINGLOSS:VCESat 6` sets VCE(sat) value as 6 for switching loss measurement of the power measurement badge Power 1.

POWer:POWer<x>:SWITCHINGLOSS:VGLeve1

This command sets or queries the gate voltage value (V_g Level Ton-Start) for the switching loss measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:SWITCHINGLOSS:VGLeve1 <NR1>`
`Power:Power<x>:SWITCHINGLOSS:VGLeve1?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

<NR1> ranges from -100 to 100

Examples `Power:Power1:SWITCHINGLOSS:VGLeve1 1.2` sets the gate voltage value (V_g Level Ton-Start) for the switching loss measurement of the power measurement badge Power 1.

POWer:POWer<x>:SWITCHINGLOSS:VLEVELAbs

This command sets or queries the voltage level (Ton-Start & Stop) in absolute units for switching loss measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:SWITCHINGLOSS:VLEVELAbs <NR1>`
`Power:Power<x>:SWITCHINGLOSS:VLEVELAbs?`

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <NR1> ranges from –100 to 100
Examples	Power:Power1:SWITCHINGLOSS:VLEVELAbs 2 sets the voltage level (Ton-Start & Stop) value as 2 for the switching loss measurement in the power measurement badge 1.

POWer:POWer<x>:SWITCHINGLOSS:VLEVELPct

This command sets or queries the voltage level (Ton-Start & Stop) in percentage for switching loss measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:SWITCHINGLOSS:VLEVELPct <NR1> Power:Power<x>:SWITCHINGLOSS:VLEVELPct?
Related Commands	POWer:POWer<x>:SWITCHINGLOSS:LEVELUNIts
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. <NR1> ranges from 0.0001 to 90
Examples	Power:Power1:SWITCHINGLOSS:VLEVELPct 7 sets the voltage level (Ton-Start & Stop) value as 7 percentage for the switching loss measurement in the power measurement badge 1.

POWer:POWer<x>:SWITCHINGLOSS:VSOURCE

This command sets or queries the voltage source for the switching loss measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions	Requires option PWR or PS2.
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Group	Power
Syntax	Power:Power<x>:SWITCHINGLOSS:VSOURCE {CH<x> MATH<x> REF<x>} Power:Power<x>:SWITCHINGLOSS:VSOURCE?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument. MATH<x> = A math waveform specifier; <x> is ≥ 1. REF<x> = A reference waveform specifier; <x> is ≥ 1.
Examples	Power:Power1:SWITCHINGLOSS:VSOURCE REF1 sets the voltage source for the switching loss measurement in the specified power measurement badge as REF1. Power:Power2:SWITCHINGLOSS:VSOURCE? might return REF3 indicating the voltage source for switching loss measurement of Power 2 power measurement badge.

POWER:POWER<x>:SWITCHINGRIPPLE:INPUTSource

This command sets or queries the input source for switching ripple measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:SWITCHINGRIPPLE:INPUTSource {CH<x> MATH<x> REF<x>} Power:Power<x>:SWITCHINGRIPPLE:INPUTSource?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge. CH<x> = A channel specifier; <x> is 1 through 8 and is limited by the number of FlexChannels in your instrument.

MATH<x> = A math waveform specifier; <x> is ≥ 1 .

REF<x> = A reference waveform specifier; <x> is ≥ 1 .

Examples `Power:Power1:SWITCHINGRIPPLE:INPUTSource CH5` sets the input source for switching ripple measurement as CH5 for the power measurement badge Power 1.

POWER:POWER<x>:SWITCHINGRIPPLE:LFREQuency

This command sets or queries the switching frequency for switching ripple measurement in the specified power measurement number. The power measurement number is specified by <x>.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:SWITCHINGRIPPLE:LFREQuency <NR1>`
`Power:Power<x>:SWITCHINGRIPPLE:LFREQuency?`

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.
<NR1> ranges from 50 to 1000000

Examples `Power:Power1:SWITCHINGRIPPLE:SFREQuency 60000` sets the frequency present for line ripple measurement as 60000 Hz.

POWER:POWER<x>:TURNOFFtime:FREQuency

This command sets or queries the input frequency used by the AC or DC converter of the specified Turn Off Time measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:TURNOFFtime:FREQuency <NR3>`
`Power:Power<x>:TURNOFFtime:FREQuency?`

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <NR3> is a floating point number that represents the frequency, in Hertz, from 1 Hz to 500 Hz.
Examples	Power:Power1:TURNOfftime:FREQUENCY 50 sets the frequency value of Turn Off Time measurement 1 to 50 Hz. Power:Power4:TURNOfftime:FREQUENCY? might return Power:Power4:TURNOfftime:FREQUENCY 350, indicating that the frequency value of Turn Off Time measurement 4 is set to 350 Hz.

POWer:POWer<x>:TURNOfftime:INPUTLeVel

This command sets or queries the input voltage level of the specified Turn Off Time measurement.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:TURNOfftime:INPUTLeVel <NR3> Power:Power<x>:TURNOfftime:INPUTLeVel?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <NR3> is a floating point number that represents the voltage level, in volts, from -500 V to 500 V.
Examples	Power:Power1:TURNOfftime:INPUTLeVel -350 sets the input voltage level of Turn Off Time measurement 1 to -350 V. Power:Power4:TURNOfftime:INPUTLeVel? might return Power:Power4:TURNOfftime:INPUTLeVel 200, indicating that the input voltage level value of Turn Off Time measurement 4 is set to 200 V.

POWer:POWer<x>:TURNOfftime:INPUTSource

This command sets or queries the input source of the specified Turn Off Time measurement.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	<code>Power:Power<x>:TURNOfftime:INPUTSource</code> <code>{CH<x> REF<x> MATH<x>}</code> <code>Power:Power<x>:TURNOfftime:INPUTSource?</code>
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <code>CH<x></code> is the channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels. <code>REF<x></code> is the Reference waveform specifier ≥ 1. This is the equivalent of the number shown on a Reference waveform badge in the UI. <code>MATH<x></code> is the Math waveform specifier ≥ 1. This is the equivalent of the number shown on a Math waveform badge in the UI.
Examples	<code>Power:Power1:TURNOfftime:INPUTSource CH4</code> sets the input source of Turn Off Time measurement 1 to Channel 4. <code>Power:Power6:TURNOfftime:INPUTSource?</code> might return <code>Power:Power6:TURNOfftime:INPUTSource MATH2</code>, indicating that the input source of Turn Off Time measurement 6 is set to Math waveform 2.

POWER:POWER<x>:TURNOfftime:MAXTIME

This command sets or queries the maximum turn off time of the specified Turn Off Time measurement.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	<code>Power:Power<x>:TURNOfftime:MAXTIME <NR3></code> <code>Power:Power<x>:TURNOfftime:MAXTIME?</code>
Arguments	<code>Power<x></code> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

<NR3> is a floating point number that represents the maximum time value, in seconds, in the range 1 second to 500 seconds.

Examples `Power:Power1:TURNOfftime:MAXTime 70` sets the maximum time value of Turn Off Time measurement 1 to 70 seconds.

`Power:Power5:TURNOfftime:MAXTime?` might return
`Power:Power5:TURNOfftime:MAXTime 120`, indicating that the maximum time value of Turn Off Time measurement 5 is 120 seconds.

POWer:POWer<x>:TURNOfftime:MAXVoltage

This command sets or queries the maximum voltage of the specified Turn OffTime measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:TURNOfftime:MAXVoltage <NR3>`
`Power:Power<x>:TURNOfftime:MAXVoltage?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

`<NR3>` is a floating point number that represents the maximum voltage in the range 1 V to 500 V.

Examples `Power:Power1:TURNOfftime:MAXVoltage 50` sets the maximum voltage value of Turn Off Time measurement 1 to 50 volts.

`Power:Power4:TURNOfftime:MAXVoltage?` might return
`Power:Power4:TURNOfftime:MAXVoltage 12`, indicating that the maximum voltage value of Turn Off Time measurement 4 is 12 volts.

POWer:POWer<x>:TURNOfftime:NUMOUTputs

This command sets or queries the number of outputs of the specified Turn Off Time power measurement.

Conditions Requires option PWR or PS2.

Group	Power
Syntax	Power:POWER<x>:TURNOFFtime:NUMOUTputs {ONE TWO THREE FOUR FIVE SIX SEVEN} Power:POWER<x>:TURNOFFtime:NUMOUTputs?
Related Commands	POWER:POWER<x>:TURNOFFtime:OUTPUT1SOURce POWER:POWER<x>:TURNOFFtime:OUTPUT1VOLTage POWER:POWER<x>:TURNOFFtime:OUTPUT2SOURce POWER:POWER<x>:TURNOFFtime:OUTPUT2VOLTage POWER:POWER<x>:TURNOFFtime:OUTPUT3SOURce POWER:POWER<x>:TURNOFFtime:OUTPUT3VOLTage POWER:POWER<x>:TURNOFFtime:OUTPUT4SOURce POWER:POWER<x>:TURNOFFtime:OUTPUT4VOLTage POWER:POWER<x>:TURNOFFtime:OUTPUT5SOURce POWER:POWER<x>:TURNOFFtime:OUTPUT5VOLTage POWER:POWER<x>:TURNOFFtime:OUTPUT6SOURce POWER:POWER<x>:TURNOFFtime:OUTPUT6VOLTage POWER:POWER<x>:TURNOFFtime:OUTPUT7SOURce POWER:POWER<x>:TURNOFFtime:OUTPUT7VOLTage
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. ONE through SEVEN sets the number of outputs for the Turn Off Time measurement.
Examples	Power:POWER1:TURNOFFtime:NUMOUTputs TWO sets the number of outputs of Turn Off Time measurement 1 to two. Power:POWER3:TURNOFFtime:NUMOUTputs? might return Power:POWER3:TURNOFFtime:NUMOUTputs FOUR, indicating that the number of outputs of Turn Off Time measurement 3 is set to four inputs.

POWER:POWER<x>:TURNOFFtime:OUTPUT1SOURce

This command sets or queries the output 1 source of the specified Turn Off Time measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWER:POWER<x>:TURNOFFtime:OUTPUT1SOURce
{CH<x>|REF<x>|MATH<x>}
POWER:POWER<x>:TURNOFFtime:OUTPUT1SOURce?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

`CH<x>` A channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels.

`REF<x>` A Reference waveform specifier ≥ 1 . This is the equivalent of the number shown on a Reference waveform badge in the UI.

`MATH<x>` A Math waveform specifier ≥ 1 . This is the equivalent of the number shown on a Math waveform badge in the UI.

Examples `Power:Power2:TURNOFFtime:OUTPUT1SOURce CH2` sets the output 1 source of Turn Off Time measurement 2 to Channel 2.

`Power:Power1:TURNOFFtime:OUTPUT1SOURce?` might return `Power:Power1:TURNOFFtime:OUTPUT1SOURce CH4`, indicating that the output 1 source of Turn Off Time measurement 1 is Channel 4.

POWER:POWER<x>:TURNOFFtime:OUTPUT1VOLTage

This command sets or queries the output 1 voltage level of the of the specified Turn Off Time power measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:POWER<x>:TURNOFFtime:OUTPUT1VOLTage <NR2>
Power:POWER<x>:TURNOFFtime:OUTPUT1VOLTage?`

Related Commands	POWer:POWer<x>:TURNOFFtime:NUMOUTputs POWer:POWer<x>:TURNOFFtime:OUTPUT1SOURce
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <NR2> sets the output voltage value, in the range of –6,000 volts to +6,000 volts.
Examples	POWer:POWer7:TURNOFFtime:OUTPUT1VOLTage -2550 sets the output 1 voltage value of Turn Off Time power measurement 7 to –2,550 volts. POWer:POWer3:TURNOFFtime:OUTPUT1VOLTage? might return POWer:POWer3:TURNOFFtime:OUTPUT1VOLTage 1000, indicating that the output 1 voltage value of Turn Off Time power measurement 3 is set to 1,000 volts.

POWer:POWer<x>:TURNOFFtime:OUTPUT2SOURce

This command sets or queries the output 2 source of the specified Turn Off Time measurement.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	POWer:POWer<x>:TURNOFFtime:OUTPUT2SOURce {CH<x> REF<x> MATH<x>} POWer:POWer<x>:TURNOFFtime:OUTPUT2SOURce?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. CH<x> A channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels. REF<x> A Reference waveform specifier ≥ 1. This is the equivalent of the number shown on a Reference waveform badge in the UI. MATH<x> A Math waveform specifier ≥ 1. This is the equivalent of the number shown on a Math waveform badge in the UI.
Examples	POWer:POWer3:TURNOFFtime:OUTPUT2SOURce CH2 sets the output 2 source of Turn Off Time measurement 3 to Channel 2.

Power:Power5:TURNOfftime:OUTPUT2SOURCE? might return
 Power:Power5:TURNOfftime:OUTPUT2SOURCE CH8, indicating that the
 output 2 source of Turn Off Time measurement 5 is Channel 8.

POWer:POWer<x>:TURNOfftime:OUTPUT2VOLTage

This command sets or queries the output 2 voltage level of the specified Turn Off Time power measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax Power:Power<x>:TURNOfftime:OUTPUT2VOLTage <NR2>
 Power:Power<x>:TURNOfftime:OUTPUT2VOLTage?

Related Commands [Power:Power<x>:TURNOfftime:NUMOUTputs](#)
[Power:Power<x>:TURNOfftime:OUTPUT2SOURCE](#)

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.
 <NR2> sets the output voltage value, in the range of –6,000 volts to +6,000 volts.

Examples Power:Power3:TURNOfftime:OUTPUT2VOLTage -550 sets the output 2 voltage value of Turn Off Time power measurement 3 to –550 volts.
 Power:Power1:TURNOfftime:OUTPUT2VOLTage? might return
 Power:Power1:TURNOfftime:OUTPUT2VOLTage 100, indicating that the output 2 voltage value of Turn Off Time power measurement 1 is set to 100 volts.

POWer:POWer<x>:TURNOfftime:OUTPUT3SOURCE

This command sets or queries the output 3 source of the specified Turn Off Time measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax	Power:Power<x>:TURNOfftime:OUTPUT3SOURCE {CH<x> REF<x> MATH<x>} Power:Power<x>:TURNOfftime:OUTPUT3SOURCE?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. CH<x> A channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels. REF<x> A Reference waveform specifier ≥ 1. This is the equivalent of the number shown on a Reference waveform badge in the UI. MATH<x> A Math waveform specifier ≥ 1. This is the equivalent of the number shown on a Math waveform badge in the UI.
Examples	Power:Power3:TURNOfftime:OUTPUT3SOURCE CH8 sets the output 3 source of Turn Off Time measurement 3 to Channel 8. Power:Power1:TURNOfftime:OUTPUT3SOURCE? might return Power:Power1:TURNOfftime:OUTPUT3SOURCE CH4, indicating that the output 3 source of Turn Off Time measurement 1 is Channel 4.

POWer:POWer<x>:TURNOfftime:OUTPUT3VOLTage

This command sets or queries the output 3 voltage level of the specified Turn Off Time power measurement.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:TURNOfftime:OUTPUT3VOLTage <NR2> Power:Power<x>:TURNOfftime:OUTPUT3VOLTage?
Related Commands	POWer:POWer<x>:TURNOfftime:NUMOUTputs POWer:POWer<x>:TURNOfftime:OUTPUT3SOURCE
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <NR2> sets the output voltage value, in the range of –6,000 volts to +6,000 volts.

Examples `Power:Power4:TURNOfftime:OUTPUT3VOLTage -50` sets the output 3 voltage value of Turn Off Time power measurement 4 to -50 volts.

`Power:Power8:TURNOfftime:OUTPUT3VOLTage?` might return `Power:Power8:TURNOfftime:OUTPUT3VOLTage -200`, indicating that the output 3 voltage value of Turn Off Time power measurement 8 is set to -200 volts.

POWer:POWer<x>:TURNOfftime:OUTPUT4SOURce

This command sets or queries the output 4 source of the specified Turn Off Time measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:TURNOfftime:OUTPUT4SOURce`
`{CH<x>|REF<x>|MATH<x>}`
`Power:Power<x>:TURNOfftime:OUTPUT4SOURce?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

`CH<x>` A channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels.

`REF<x>` A Reference waveform specifier ≥ 1 . This is the equivalent of the number shown on a Reference waveform badge in the UI.

`MATH<x>` A Math waveform specifier ≥ 1 . This is the equivalent of the number shown on a Math waveform badge in the UI.

Examples `Power:Power3:TURNOfftime:OUTPUT4SOURce CH2` sets the output 4 source of Turn Off Time measurement 3 is Channel 2.

`Power:Power10:TURNOfftime:OUTPUT4SOURce?` might return `Power:Power10:TURNOfftime:OUTPUT4SOURce REF4`, indicating that the output 4 source of Turn Off Time measurement 10 is Reference waveform 4.

POWer:POWer<x>:TURNOfftime:OUTPUT4VOLTage

This command sets or queries the output 4 voltage level of the specified Turn Off Time power measurement.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:TURNOfftime:OUTPUT4VOLTage <NR2> Power:Power<x>:TURNOfftime:OUTPUT4VOLTage?
Related Commands	POWer:POWer<x>:TURNOfftime:NUMOUTputs POWer:POWer<x>:TURNOfftime:OUTPUT4SOURce
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <NR2> sets the output voltage value, in the range of –6,000 volts to +6,000 volts.
Examples	Power:Power8:TURNOfftime:OUTPUT4VOLTage 50 sets the output 4 voltage value of Turn Off Time power measurement 8 to 50 volts. Power:Power4:TURNOfftime:OUTPUT4VOLTage? might return Power:Power4:TURNOfftime:OUTPUT4VOLTage 1000, indicating that the output 4 voltage value of Turn Off Time power measurement 4 is set to 1,000 volts.

POWer:POWer<x>:TURNOfftime:OUTPUT5SOURce

This command sets or queries the output 5 source of the specified Turn Off Time measurement.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:TURNOfftime:OUTPUT5SOURce {CH<x> REF<x> MATH<x>} Power:Power<x>:TURNOfftime:OUTPUT5SOURce?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. CH<x> A channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels.

REF<x> A Reference waveform specifier ≥ 1 . This is the equivalent of the number shown on a Reference waveform badge in the UI.

MATH<x> A Math waveform specifier ≥ 1 . This is the equivalent of the number shown on a Math waveform badge in the UI.

Examples `Power:Power3:TURNOfftime:OUTPUT5SOURCE CH2` sets the output 5 source of Turn Off Time measurement 3 to Channel 2.

`Power:Power11:TURNOfftime:OUTPUT5SOURCE?` might return `Power:Power11:TURNOfftime:OUTPUT5SOURCE CH4`, indicating that the output 5 source of Turn Off Time measurement 11 is Channel 4.

POWer:POWer<x>:TURNOfftime:OUTPUT5VOLTage

This command sets or queries the output 5 voltage level of the specified Turn Off Time power measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:TURNOfftime:OUTPUT5VOLTage <NR2>`
`Power:Power<x>:TURNOfftime:OUTPUT5VOLTage?`

Related Commands [POWer:POWer<x>:TURNOfftime:NUMOUTputs](#)
[POWer:POWer<x>:TURNOfftime:OUTPUT5SOURCE](#)

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.
`<NR2>` sets the output voltage value, in the range of –6,000 volts to +6,000 volts.

Examples `Power:Power2:TURNOfftime:OUTPUT5VOLTage -255` sets the output 5 voltage value of Turn Off Time power measurement 2 to –250 volts.
`Power:Power3:TURNOfftime:OUTPUT5VOLTage?` might return `Power:Power3:TURNOfftime:OUTPUT5VOLTage 100`, indicating that the output 5 voltage value of Turn Off Time power measurement 3 is set to 100 volts.

POWer:POWer<x>:TURNOfftime:OUTPUT6SOURce

This command sets or queries the output 6 source of the specified Turn Off Time measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax Power:POWer<x>:TURNOfftime:OUTPUT6SOURce
{CH<x>|REF<x>|MATH<x>}
Power:POWer<x>:TURNOfftime:OUTPUT6SOURce?

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

CH<x> A channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels.

REF<x> A Reference waveform specifier ≥ 1 . This is the equivalent of the number shown on a Reference waveform badge in the UI.

MATH<x> A Math waveform specifier ≥ 1 . This is the equivalent of the number shown on a Math waveform badge in the UI.

Examples Power:POWer3:TURNOfftime:OUTPUT6SOURce CH2 sets the output 6 source of Turn Off Time measurement 3 to Channel 2.

Power:POWer1:TURNOfftime:OUTPUT6SOURce? might return
Power:POWer1:TURNOfftime:OUTPUT6SOURce CH4, indicating that the output 6 source of Turn Off Time measurement 1 is Channel 4.

POWer:POWer<x>:TURNOfftime:OUTPUT6VOLTage

This command sets or queries the output 6 voltage level of the specified Turn Off Time power measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax Power:POWer<x>:TURNOfftime:OUTPUT6VOLTage <NR2>
Power:POWer<x>:TURNOfftime:OUTPUT6VOLTage?

Related Commands [POWER:Power<x>:TURNOFFtime:NUMOUTputs](#)
[POWER:Power<x>:TURNOFFtime:OUTPUT6SOURce](#)

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.
 <NR2> sets the output voltage value, in the range of –6,000 volts to +6,000 volts.

Examples `Power:Power5:TURNOFFtime:OUTPUT6VOLTage 1550` sets the output 6 voltage value of Turn Off Time power measurement 5 to 1,550 volts.
`Power:Power2:TURNOFFtime:OUTPUT6VOLTage?` might return
`Power:Power2:TURNOFFtime:OUTPUT6VOLTage -100`, indicating that the output 6 voltage value of Turn Off Time power measurement 2 is set to –100 volts.

POWER:Power<x>:TURNOFFtime:OUTPUT7SOURce

This command sets or queries the output 7 source of the specified Turn Off Time measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:TURNOFFtime:OUTPUT7SOURce`
`{CH<x>|REF<x>|MATH<x>}`
`Power:Power<x>:TURNOFFtime:OUTPUT7SOURce?`

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.
 CH<x> A channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels.
 REF<x> A Reference waveform specifier ≥ 1 . This is the equivalent of the number shown on a Reference waveform badge in the UI.
 MATH<x> A Math waveform specifier ≥ 1 . This is the equivalent of the number shown on a Math waveform badge in the UI.

Examples `Power:Power3:TURNOFFtime:OUTPUT7SOURce CH2` sets the output 7 source of Turn Off Time measurement 3 is Channel 2.

`Power:Power1:TURNOfftime:OUTPUT7SOURCE?` might return
`Power:Power1:TURNOfftime:OUTPUT7SOURCE CH4`, indicating that the
output 7 source of Turn Off Time measurement 1 is channel 4.

Power:Power<x>:TURNOfftime:OUTPUT7VOLTage

This command sets or queries the output 7 voltage level of the specified Turn Off Time power measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:TURNOfftime:OUTPUT7VOLTage <NR2>`
`Power:Power<x>:TURNOfftime:OUTPUT7VOLTage?`

Related Commands [`Power:Power<x>:TURNOfftime:NUMOUTputs`](#)
[`Power:Power<x>:TURNOfftime:OUTPUT7SOURCE`](#)

Arguments `Power<x>` is the power measurement number. This is the equivalent of the
number shown on a power measurement badge in the UI.

`<NR2>` sets the output voltage value, in the range of –6,000 volts to +6,000 volts.

Examples `Power:Power2:TURNOfftime:OUTPUT7VOLTage -2100` sets the output 7
voltage value of Turn Off Time power measurement 2 to –2,100 volts.

`Power:Power3:TURNOfftime:OUTPUT7VOLTage?` might return
`Power:Power3:TURNOfftime:OUTPUT7VOLTage 450`, indicating that the
output 7 voltage value of Turn Off Time power measurement 3 is set to 450 volts.

Power:Power<x>:TURNOfftime:TYPE

This command sets or queries the type of AC/DC converter used in the specified
Turn Off Time power measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax	Power:Power<x>:TURNOFFtime:TYPE {DCDC ACDC} Power:Power<x>:TURNOFFtime:TYPE?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. DCDC sets the measurement to use a DC to DC converter. ACDC sets the measurement to use an AC to DC converter.
Examples	Power:Power1:TURNOFFtime:TYPE ACDC sets Turn Off time measurement 1 to use an AC to DC converter. Power:Power5:TURNOFFtime:TYPE? might return Power:Power5:TURNOFFtime:TYPE DCDC, indicating that the converter type for Turn Off Time measurement 5 is set to use a DC to DC converter.

POWER:POWER<x>:TURNONtime:FREQUENCY

This command sets or queries the input frequency used by the AC or DC converter of the specified Turn On Time measurement.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:Power<x>:TURNONtime:FREQUENCY <NR3> Power:Power<x>:TURNONtime:FREQUENCY?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <NR3> is a floating point number that represents the frequency, in Hertz, from 1 Hz to 500 Hz.
Examples	Power:Power1:TURNONtime:FREQUENCY 50 sets the frequency value of the AC/DC converter in Turn On Time measurement 1 to 50 Hz. Power:Power4:TURNONtime:FREQUENCY? might return Power:Power4:TURNONtime:FREQUENCY 200, indicating that the frequency value of the AC/DC converter in Turn On Time measurement 4 is set to 200 Hz.

POWER:POWER<x>:TURNONtime:INPUTLeve1

This command sets or returns the input voltage level of the specified Turn On Time measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWER:POWER<x>:TURNONtime:INPUTLeve1 <NR3>
POWER:POWER<x>:TURNONtime:INPUTLeve1?

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

<NR3> is a floating point number that represents the voltage level, in volts, from –500 V to 500 V.

Examples POWER:POWER12:TURNONtime:INPUTLeve1 –350 sets the input voltage level of Turn On Time measurement 12 to –350 V.

POWER:POWER4:TURNONtime:INPUTLeve1? might return
POWER:POWER4:TURNONtime:INPUTLeve1 230, indicating that the input voltage level value of Turn ON Time measurement 4 is set to 230 V.

POWER:POWER<x>:TURNONtime:INPUTSource

This command sets or queries the input source of the specified Turn On Time measurement.

Group Power

Syntax POWER:POWER<x>:TURNONtime:INPUTSource
POWER:POWER<x>:TURNONtime:INPUTSource?

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

CH<x> is the channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels.

REF<x> is the Reference waveform specifier ≥ 1 . This is the equivalent of the number shown on a Reference waveform badge in the UI.

MATH<x> is the Math waveform specifier ≥ 1 . This is the equivalent of the number shown on a Math waveform badge in the UI.

Examples `Power:Power1:TURNONtime:INPUTSource CH2` sets the input source of Turn On Time measurement 1 to Channel 2.

`Power:Power2:TURNONtime:INPUTSource?` might return `Power:Power2:TURNONtime:INPUTSource MATH6`, indicating that the input source of Turn On Time measurement 2 is Math waveform 6.

POWer:POWer<x>:TURNONtime:MAXTime

This command sets or returns the maximum turn on time of the specified Turn On Time measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:TURNONtime:MAXTime <NR3>`
`Power:Power<x>:TURNONtime:MAXTime?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

`<NR3>` is a floating point number that represents the maximum time value, in seconds, in the range 1 second to 500 seconds.

Examples `Power:Power3:TURNONtime:MAXTime 30` sets the maximum time value of Turn On Time measurement 3 to 30 seconds.

`Power:Power5:TURNONtime:MAXTime?` might return `Power:Power5:TURNONtime:MAXTime 120`, indicating that the maximum time value of Turn On Time measurement 5 is set to 120 seconds.

POWer:POWer<x>:TURNONTime:MAXVoltage

This command sets or returns the maximum voltage setting of the specified Turn On Time measurement.

Conditions Requires option PWR or PS2.

Group	Power
Syntax	<pre>Power:Power<x>:TURNONtime:MAXVoltage <NR3> Power:Power<x>:TURNONtime:MAXVoltage?</pre>
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <NR3> is a floating point number that represents the maximum voltage in the range 1 V to 500 V.
Examples	Power:Power1:TURNONtime:MAXVoltage 50 sets the maximum voltage value of Turn On Time measurement 1 to 50 volts. Power:Power4:TURNONtime:MAXVoltage? might return Power:Power4:TURNONtime:MAXVoltage 12, indicating that the maximum voltage value of Turn On Time measurement 4 is set to 12 volts.

POWER:Power<x>:TURNONtime:NUMOUTputs

This command sets or queries the number of outputs for the specified Turn On Time power measurement.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	<pre>Power:Power<x>:TURNONtime:NUMOUTputs {ONE TWO THREE FOUR FIVE SIX SEVEN} Power:Power<x>:TURNONtime:NUMOUTputs?</pre>
Related Commands	Power:Power<x>:TURNONtime:INPUTSource Power:Power<x>:TURNONtime:OUTPUT1Source Power:Power<x>:TURNONtime:OUTPUT1VOLTage Power:Power<x>:TURNONtime:OUTPUT2Source Power:Power<x>:TURNONtime:OUTPUT2VOLTage Power:Power<x>:TURNONtime:OUTPUT3Source Power:Power<x>:TURNONtime:OUTPUT3VOLTage

POWer:POWer<x>:TURNONtime:OUTPUT4SOURce
 POWer:POWer<x>:TURNONtime:OUTPUT4VOLTage
 POWer:POWer<x>:TURNONtime:OUTPUT5SOURce
 POWer:POWer<x>:TURNONtime:OUTPUT5VOLTage
 POWer:POWer<x>:TURNONtime:OUTPUT6SOURce
 POWer:POWer<x>:TURNONtime:OUTPUT6VOLTage
 POWer:POWer<x>:TURNONtime:OUTPUT7SOURce
 POWer:POWer<x>:TURNONtime:OUTPUT7VOLTage

Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. ONE through SEVEN sets the number of outputs for the specified Turn On Time power measurement.
Examples	Power:Power1:TURNONtime:NUMOUTputs TWO sets the number of outputs of Turn On Time measurement 1 to two. Power:Power3:TURNONtime:NUMOUTputs? might return Power:Power3:TURNONtime:NUMOUTputs FOUR, indicating that the number of outputs of Turn On Time measurement 3 is set to four inputs.

POWer:POWer<x>:TURNONtime:OUTPUT1SOURce

This command sets or queries the output 1 source of the specified Turn On Time measurement.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:POWer<x>:TURNONtime:OUTPUT1SOURce {CH<x> REF<x> MATH<x>} Power:POWer<x>:TURNONtime:OUTPUT1SOURce?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. CH<x> A channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels.

REF<x> A Reference waveform specifier ≥ 1 . This is the equivalent of the number shown on a Reference waveform badge in the UI.

MATH<x> A Math waveform specifier ≥ 1 . This is the equivalent of the number shown on a Math waveform badge in the UI.

Examples `Power:Power5:TURNONtime:OUTPUT1SOURCE CH2` sets the output 1 source of Turn On Time measurement 5 to Channel 2.

`Power:Power1:TURNONtime:OUTPUT1SOURCE?` might return
`Power:Power1:TURNONtime:OUTPUT1SOURCE CH4`, indicating that the output 1 source of Turn On Time measurement 1 is Channel 4.

POWER:POWER<x>:TURNONtime:OUTPUT1VOLTage

This command sets or queries the output 1 voltage level of the specified Turn On Time power measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:TURNONtime:OUTPUT1VOLTage <NR2>`
`Power:Power<x>:TURNONtime:OUTPUT1VOLTage?`

Related Commands [POWER:POWER<x>:TURNONtime:NUMOUTputs](#)
[POWER:POWER<x>:TURNONtime:OUTPUT1SOURCE](#)

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

`<NR2>` sets the output voltage value, in the range of –6,000 volts to +6,000 volts.

Examples `Power:Power1:TURNONtime:OUTPUT1VOLTage -2550` sets the output 1 voltage value of Turn On Time power measurement 1 to –2,550 volts.

`Power:Power3:TURNONtime:OUTPUT1VOLTage?` might return
`Power:Power3:TURNONtime:OUTPUT1VOLTage 1000`, indicating that the output 1 voltage value of Turn On Time power measurement 3 is set to 1,000 volts.

POWER:POWER<x>:TURNONtime:OUTPUT2SOURce

This command sets or queries the output 2 source of the specified Turn On Time measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWER:POWER<x>:TURNONtime:OUTPUT2SOURce
{CH<x>|REF<x>|MATH<x>}
POWER:POWER<x>:TURNONtime:OUTPUT2SOURce?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

`CH<x>` A channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels.

`REF<x>` A Reference waveform specifier ≥ 1 . This is the equivalent of the number shown on a Reference waveform badge in the UI.

`MATH<x>` A Math waveform specifier ≥ 1 . This is the equivalent of the number shown on a Math waveform badge in the UI.

Examples `Power:Power2:TURNONtime:OUTPUT2SOURce CH2` sets the output 2 source of Turn On Time measurement 2 to Channel 2.

`Power:Power1:TURNONtime:OUTPUT2SOURce?` might return
`Power:Power1:TURNONtime:OUTPUT2SOURce CH4`, indicating that the output 2 source of Turn On Time measurement 1 is Channel 4.

POWER:POWER<x>:TURNONtime:OUTPUT2VOLTage

This command sets or queries the output 2 voltage level of the specified Turn On Time power measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:POWER<x>:TURNONtime:OUTPUT2VOLTage <NR2>
Power:POWER<x>:TURNONtime:OUTPUT2VOLTage?`

Related Commands [POWer:POWer<x>:TURNONtime:NUMOUTputs](#)
[POWer:POWer<x>:TURNONtime:OUTPUT2SOURce](#)

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.
 <NR2> sets the output voltage value, in the range of –6,000 volts to +6,000 volts.

Examples `POWer:POWer7:TURNONtime:OUTPUT2VOLTage -2550` sets the output 2 voltage value of Turn On Time power measurement 7 to –2,550 volts.
`POWer:POWer1:TURNONtime:OUTPUT2VOLTage?` might return
`POWer:POWer1:TURNONtime:OUTPUT2VOLTage 120`, indicating that the output 2 voltage value of Turn On Time power measurement 1 is set to 120 volts.

POWer:POWer<x>:TURNONtime:OUTPUT3SOURce

This command sets or queries the output 3 source of the specified Turn On Time measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWer:POWer<x>:TURNONtime:OUTPUT3SOURce`
`{CH<x>|REF<x>|MATH<x>}`
`POWer:POWer<x>:TURNONtime:OUTPUT3SOURce?`

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.
 CH<x> A channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels.
 REF<x> A Reference waveform specifier ≥ 1 . This is the equivalent of the number shown on a Reference waveform badge in the UI.
 MATH<x> A Math waveform specifier ≥ 1 . This is the equivalent of the number shown on a Math waveform badge in the UI.

Examples `POWer:POWer3:TURNONtime:OUTPUT3SOURce CH2` sets the output 3 source of Turn On Time measurement 3 to Channel 2.

Power:Power13:TURNONtime:OUTPUT3SOURCE? might return
 Power:Power13:TURNONtime:OUTPUT3SOURCE REF3, indicating that the
 output 3 source of Turn On Time measurement 13 is Reference waveform 3.

POWer:POWer<x>:TURNONtime:OUTPUT3VOLTage

This command sets or queries the output 3 voltage level of the specified Turn On Time power measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax Power:Power<x>:TURNONtime:OUTPUT3VOLTage <NR2>
 Power:Power<x>:TURNONtime:OUTPUT3VOLTage?

Related Commands [Power:Power<x>:TURNONtime:NUMOUTputs](#)
[Power:Power<x>:TURNONtime:OUTPUT3SOURCE](#)

Arguments Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.
 <NR2> sets the output voltage value, in the range of –6,000 volts to +6,000 volts.

Examples Power:Power7:TURNONtime:OUTPUT3VOLTage –550 sets the output 3 voltage value of Turn On Time power measurement 7 to –550 volts.
 Power:Power3:TURNONtime:OUTPUT3VOLTage? might return
 Power:Power3:TURNONtime:OUTPUT3VOLTage –200, indicating that the output 3 voltage value of Turn On Time power measurement 3 is set to –200 volts.

POWer:POWer<x>:TURNONtime:OUTPUT4SOURCE

This command sets or queries the output 4 source of the specified Turn On Time measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:TURNONtime:OUTPUT4SOURCE`
 `{CH<x>|REF<x>|MATH<x>}`
 `Power:Power<x>:TURNONtime:OUTPUT4SOURCE?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

`CH<x>` A channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels.

`REF<x>` A Reference waveform specifier ≥ 1 . This is the equivalent of the number shown on a Reference waveform badge in the UI.

`MATH<x>` A Math waveform specifier ≥ 1 . This is the equivalent of the number shown on a Math waveform badge in the UI.

Examples `Power:Power3:TURNONtime:OUTPUT4SOURCE CH2` sets the output 4 source of Turn On Time measurement 3 to Channel 2.

`Power:Power1:TURNONtime:OUTPUT4SOURCE?` might return
`Power:Power1:TURNONtime:OUTPUT4SOURCE MATH3`, indicating that the output 4 source of Turn On Time measurement 1 is Math waveform 3.

POWER:POWER<x>:TURNONtime:OUTPUT4VOLTage

This command sets or queries the output 4 voltage level of the specified Turn On Time power measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:TURNONtime:OUTPUT4VOLTage <NR2>`
 `Power:Power<x>:TURNONtime:OUTPUT4VOLTage?`

Related Commands [POWER:POWER<x>:TURNONtime:NUMOUTputs](#)
 [POWER:POWER<x>:TURNONtime:OUTPUT4SOURCE](#)

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

`<NR2>` sets the output voltage value, in the range of –6,000 volts to +6,000 volts.

- Examples** `Power:Power8:TURNONtime:OUTPUT4VOLTage 50` sets the output 4 voltage value of Turn On Time power measurement 8 to 50 volts.
- `Power:Power2:TURNONtime:OUTPUT4VOLTage?` might return
`Power:Power2:TURNONtime:OUTPUT4VOLTage 1000`, indicating that the output 4 voltage value of Turn On Time power measurement 2 is set to 1,000 volts.

POWER:POWER<x>:TURNONtime:OUTPUT5SOURCE

This command sets or queries the output 5 source of the specified Turn On Time measurement.

- Conditions** Requires option PWR or PS2.

- Group** Power

- Syntax** `Power:Power<x>:TURNONtime:OUTPUT5SOURCE`
`{CH<x>|REF<x>|MATH<x>}`
`Power:Power<x>:TURNONtime:OUTPUT5SOURCE?`

- Arguments** `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.
- `CH<x>` A channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels.
- `REF<x>` A Reference waveform specifier ≥ 1 . This is the equivalent of the number shown on a Reference waveform badge in the UI.
- `MATH<x>` A Math waveform specifier ≥ 1 . This is the equivalent of the number shown on a Math waveform badge in the UI.

- Examples** `Power:Power3:TURNONtime:OUTPUT5SOURCE CH2` sets the output 5 source of Turn On Time measurement 3 to Channel 2.
- `Power:Power1:TURNONtime:OUTPUT5SOURCE?` might return
`Power:Power1:TURNONtime:OUTPUT5SOURCE CH4`, indicating that the output 5 source of Turn On Time measurement 1 is Channel 4.

POWER:POWER<x>:TURNONtime:OUTPUT5VOLTage

This command sets or queries the output 5 voltage level of the specified Turn On Time power measurement.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:POWER<x>:TURNONtime:OUTPUT5VOLTage <NR2> Power:POWER<x>:TURNONtime:OUTPUT5VOLTage?
Related Commands	POWer:POWer<x>:TURNONtime:NUMOUTputs POWer:POWer<x>:TURNONtime:OUTPUT5SOURce
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <NR2> sets the output voltage value, in the range of –6,000 volts to +6,000 volts.
Examples	Power:POWER2:TURNONtime:OUTPUT5VOLTage -255 sets the output 5 voltage value of Turn On Time power measurement 2 to –250 volts. Power:POWER3:TURNONtime:OUTPUT5VOLTage? might return Power:POWER3:TURNONtime:OUTPUT5VOLTage 100, indicating that the output 5 voltage value of Turn On Time power measurement 3 is set to 100 volts.

POWer:POWer<x>:TURNONtime:OUTPUT6SOURce

This command sets or queries the output 6 source of the specified Turn On Time measurement.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	Power:POWER<x>:TURNONtime:OUTPUT6SOURce {CH<x> REF<x> MATH<x>} Power:POWER<x>:TURNONtime:OUTPUT6SOURce?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. CH<x> A channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels.

REF<x> A Reference waveform specifier ≥ 1 . This is the equivalent of the number shown on a Reference waveform badge in the UI.

MATH<x> A Math waveform specifier ≥ 1 . This is the equivalent of the number shown on a Math waveform badge in the UI.

Examples `Power:Power3:TURNONtime:OUTPUT6SOURCE CH2` sets the output 6 source of Turn On Time measurement 3 to Channel 2.

`Power:Power8:TURNONtime:OUTPUT6SOURCE?` might return
`Power:Power8:TURNONtime:OUTPUT6SOURCE CH4`, indicating that the output 6 source of Turn On Time measurement 8 is set to Channel 4.

POWER:POWER<x>:TURNONtime:OUTPUT6VOLTage

This command sets or queries the output 6 voltage level of the specified Turn On Time power measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:TURNONtime:OUTPUT6VOLTage <NR2>`
`Power:Power<x>:TURNONtime:OUTPUT6VOLTage?`

Related Commands [POWER:POWER<x>:TURNONtime:NUMOUTputs](#)
[POWER:POWER<x>:TURNONtime:OUTPUT6SOURCE](#)

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

`<NR2>` sets the output voltage value, in the range of –6,000 volts to +6,000 volts.

Examples `Power:Power5:TURNONtime:OUTPUT6VOLTage 1550` sets the output 6 voltage value of Turn On Time power measurement 5 to 1,550 volts.

`Power:Power2:TURNONtime:OUTPUT6VOLTage?` might return
`Power:Power2:TURNONtime:OUTPUT6VOLTage -100`, indicating that the output 6 voltage value of Turn On Time power measurement 2 is set to –100 volts.

POWER:POWER<x>:TURNONtime:OUTPUT7SOURCE

This command sets or queries the output 7 source of the specified Turn On Time measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWER:POWER<x>:TURNONtime:OUTPUT7SOURCE
{CH<x>|REF<x>|MATH<x>}
POWER:POWER<x>:TURNONtime:OUTPUT7SOURCE?

Arguments POWER<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI.

CH<x> A channel specifier in the range of 1 through 8 and is limited by the number of instrument input channels.

REF<x> A Reference waveform specifier ≥ 1 . This is the equivalent of the number shown on a Reference waveform badge in the UI.

MATH<x> A Math waveform specifier ≥ 1 . This is the equivalent of the number shown on a Math waveform badge in the UI.

Examples POWER:POWER3:TURNONtime:OUTPUT6SOURCE CH1 sets the output 7 source of Turn On Time measurement 3 to Channel 1.

POWER:POWER7:TURNONtime:OUTPUT6SOURCE? might return
POWER:POWER7:TURNONtime:OUTPUT6SOURCE REF4, indicating that the output 7 source of Turn On Time measurement 7 is set to Reference waveform 4.

POWER:POWER<x>:TURNONtime:OUTPUT7VOLTage

This command sets or queries the output 7 voltage level of the specified Turn On Time power measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWER:POWER<x>:TURNONtime:OUTPUT7VOLTage <NR2>
POWER:POWER<x>:TURNONtime:OUTPUT7VOLTage?

Related Commands	POWER:Power<x>:TURNONtime:NUMOUTputs POWER:Power<x>:TURNONtime:OUTPUT7SOURCE
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. <NR2> sets the output voltage value, in the range of –6,000 volts to +6,000 volts.
Examples	POWER:Power12:TURNONtime:OUTPUT7VOLTage -2100 sets the output 7 voltage value of Turn On Time power measurement 12 to –2,100 volts. POWER:Power3:TURNONtime:OUTPUT7VOLTage? might return POWER:Power3:TURNONtime:OUTPUT7VOLTage 450, indicating that the output 7 voltage value of Turn On Time power measurement 3 is set to 450 volts.

POWER:Power<x>:TURNONtime:TYPE

This command sets or queries the type of AC/DC converter used in the specified Turn On Time power measurement.

Conditions	Requires option PWR or PS2.
Group	Power
Syntax	POWER:Power<x>:TURNONtime:TYPE {DCDC ACDC} POWER:Power<x>:TURNONtime:TYPE?
Arguments	Power<x> is the power measurement number. This is the equivalent of the number shown on a power measurement badge in the UI. DCDC sets the measurement to use a DC to DC converter. ACDC sets the measurement to use an AC to DC converter.
Examples	POWER:Power3:TURNONtime:TYPE ACDC sets Turn On time measurement 3 to use an AC to DC converter. POWER:Power12:TURNONtime:TYPE? might return POWER:Power12:TURNONtime:TYPE DCDC, indicating that the converter type for Turn On Time measurement 12 is set to use a DC to DC converter.

POWer:POWer<x>:TYPe

This command sets or queries the measurement type of the specified power measurement number. If the measurement number does not exist, this command creates a new power measurement, assigns the specified measurement number to the new measurement, and then assigns the measurement type to the new measurement.

Conditions Requires option PWR or PS2.

Group Power

Syntax `Power:Power<x>:TYPe <Measurement Type>`
`Power:Power<x>:TYPe?`

Arguments `Power<x>` is the power measurement number. This is the equivalent of the number shown in the UI for a power measurement badge.

`<Measurement Type>` = CYCLEAmp | CYCLEBase | CYCLEMAx | CYCLEMIn | CYCLEPKpk | CYCLETop | DIDT | DVDT | EFFICIENCY | FREQuency | HARMonics | INDUCTANCE | INPUTCAP | INRUSHcurrent | IVSINTEGRALV | LINERipple | MAGNETICLOSS | MAGPROPERTY | NDUTYCycle | NPULSEWidth | PDUTYCycle | PERIOD | POWERQUALity | PPULSEWidth | RDSON | SOA | SWITCHINGLOSS | SWITCHINGRIpple | TURNOFFtime | TURNONtime | CLRESPONSE | PSRR

Examples `Power:Power6:TYPe INDUCTANCE` sets power measurement 6 to be an Inductance measurement.

`Power:Power1:TYPe?` might return `INPUTCAP`, indicating that power measurement 1 is an Input Capacitance measurement.

POWERTABle:ADDNew (No Query Form)

This command adds the power harmonics table. The power measurement badge is specified by x.

Conditions Requires option PWR or PS2.

Group Power

Syntax `POWERTABle:ADDNew TABLE<x>`

Arguments <x> must be greater than or equal to one.

Examples POWTABLE:ADDNew "TABLE1" adds power harmonics table.

POWERTABLE:DELeTe (No Query Form)

This command deletes the power harmonics table.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWERTABLE:DELeTe TABLE<x>

Arguments <x> must be greater than or equal to one.

Examples POWTABLE:DELeTe "TABLE1" deletes TABLE1 power harmonics table.

POWERTABLE:LIST? (Query Only)

This command lists all defined power harmonics table.

Conditions Requires option PWR or PS2.

Group Power

Syntax POWERTABLE:LIST?

Examples POWERTABLE:LIST? might return TABLE1 indicating the defined power harmonics table.

*PSC

This command sets and queries the power-on status flag that controls the automatic power-on handling of the DESER, SRER, and ESER registers. When *PSC is true, the DESER register is set to 255 and the SRER and ESER registers

are set to 0 at power-on. When *PSC is false, the current values in the DESER, SRER, and ESER registers are preserved in nonvolatile memory when power is shut off and are restored at power-on.

Group	Status and Error
Syntax	*PSC {<NR1> OFF ON} *PSC?
Related Commands	DESE *ESE FACtory *RST *SRE
Arguments	<NR1> = 0 sets the power-on status clear flag to false, disables the power-on clear and allows the instrument to possibly assert SRQ after power-on; any other value sets the power-on status clear flag to true, enabling the power-on status clear and prevents any SRQ assertion after power on. OFF sets the power-on status clear flag to false, disables the power-on clear and allows the instrument to possibly assert SRQ after power-on. ON sets the power-on status clear flag to true, enabling the power-on status clear and prevents any SRQ assertion after power on.
Examples	*PSC 0 sets the power-on status clear flag to false. *PSC? might return 1 to indicate that the power-on status clear flag is set to true.

*PUD

This command sets or queries a string of Protected User Data. This data is protected by the PASSWord command. You can modify it only by first entering the correct password. This password is not necessary to query the data.

Group	Status and Error
Syntax	*PUD {<Block> <QString>} *PUD?

Related Commands [PASSWord](#)

Arguments <Block> is a block containing up to 100 characters.
 <QString> is a string containing up to 100 characters.

Examples *PUD #229This instrument belongs to me stores the string "This instrument belongs to me" in the user protected data area.
 *PUD? might return #221PROPERTY OF COMPANY X.

RECALL:SESSion (No Query Form)

Restores the state of the instrument, including reference waveforms, from a saved session file.

Group Save and Recall

Syntax RECALL:SESSION <file_path>

Arguments <file_path> is the file path that specifies the location of the specified instrument session file.

If a file name or path is specified, the file is expected to be located in a directory relative to the current working directory (specified by [FILESystem:CWD](#)) unless a complete path is specified:

- Complete path specification. If the file argument begins with a drive designator (such as C), then the file name is interpreted as a full path.
- Relative path specification. If the file argument begins with "." or ".." or has a file path separator appearing anywhere other than the first character position, then the file name is treated as a path that is relative to the current working directory.

Returns Recalls the instrument session from the specified session file.

Examples RECALL:SESSION "TEK00000.TSS" recalls the setup from the file TEK00000.TSS in the current working directory.

RECALL:SETUp (No Query Form)

This command (no query form) returns stored or factory settings to the instrument from a copy of the settings stored in memory. This command performs the same function as selecting Recall from the File menu, and then choosing the Setup button.

Group Save and Recall

Syntax RECALL:SETUp {FACTory|<file_path>}

Related Commands [FACTory](#)
[*RST](#)

Arguments FACTory restores the factory setup. Performs the same operation as the :FACTory command.

<file_path> specifies a location for an instrument setup file. <file path> is a quoted string that defines the file name and path. If a file name or path is specified, the file is expected to be located in a directory relative to the current working directory (specified by [FILESystem:CWD](#)) unless a complete path is specified:

- Complete path specification. If the file argument begins with drive designator (such as C:), then the file name is interpreted as a full path.
- Relative path specification. If the file argument begins with "." or ".." or has a file path separator appearing anywhere other than the first character position, then the file name is treated as a path that is relative to the current working directory.

Examples RECALL:SETUP FACTORY recalls (and makes current) the instrument setup to its factory defaults.

RECALL:SETUP "TEK00000.SET" recalls the setup from the file TEK00000.SET in the default directory for setups.

RECALL:WAVEform (No Query Form)

This command recalls a stored waveform to a reference memory location.

Group Save and Recall

Syntax	<code>RECALL:WAVEform <source file>,<destination></code>
Arguments	<code><source file></code> is the source file. The file is expected to be located in a directory relative to the current working directory (specified by FILESystem:CWD) unless a complete path is specified: ■ Complete path specification. If the file argument begins with drive designator (such as C:), then the file name is interpreted as a full path. ■ Relative path specification. If the file argument begins with "." or ".." or has a file path separator appearing anywhere other than the first character position, then the file name is treated as a path that is relative to the current working directory. <code><destination></code> is REF<x> which specifies a reference to create from the recalled waveform data file.
Examples	<code>RECALL:WAVEFORM "TEK00000.ISF"</code> , REF1 recalls the waveform stored in the file named TEK00000.ISF from the current directory to reference 1.

REF:ADDNew (No Query Form)

This command adds the specified reference. Argument is of the form "REF<NR1>", where $NR1 \geq 1$.

Group	Vertical
Syntax	<code>REF:ADDNew <QString></code>
Arguments	<code><QString></code> is the specified reference. Argument is of the form "REF<NR1>", where $NR1 \geq 1$.
Examples	<code>REF:ADDNEW "REF2"</code> adds reference 2 to the display.

REF:DELeTe (No Query Form)

Deletes the specified reference. Argument is of the form "REF<NR1>", where $NR1 \geq 1$.

Conditions	Vertical
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Group	Vertical
Syntax	REF:DELEte <QString>
Arguments	<QString> is the specified reference. Argument is of the form "REF<NR1>", where $NR1 \geq 1$.
Examples	REF:DELETE "REF2" deletes reference 2 from the display.

REF:LIST? (Query Only)

This command returns a comma separated list of all currently defined references.

Group	Vertical
Syntax	REF:LIST?
Returns	All currently defined references.
Examples	REF:LIST? might return :REF:LIST REF1,REF2 indicating references 1 and 2 are defined.

REF:REF<x>:DESKew

This command sets or queries the deskew value used for the specified reference.

Group	Vertical
Syntax	REF:REF<x>:DESKew <NR3>
Arguments	<NR3> is the deskew value used for the specified reference.
Examples	REF:REF1:DESKew -1.5e-9 sets the deskew value to -1.5 ns. REF:REF1:DESKew? might return :REF:REF1:DESKEW 1.5200E-9 indicating the deskew value is 1.52 ns.

REF:REF<x>:LABel:COLor

This command sets or queries the color of the specified ref label.

Group Vertical

Syntax REF:REF<x>:LABel:COLor <QString>

Arguments <QString> is the label. To return the color to the default color, send an empty string as in this example: :REF:REF1:LABel:COLor "".

Examples REF:REF1:LABel:COLor "#FFFF00" sets the font color to yellow.
REF:REF1:LABel:COLor? might return :REF:REF1:LABel:COLor "#FF0000" indicating the font color is red.

REF:REF<x>:LABel:FONT:BOLD

This command sets or queries the bold state of the specified reference label.

Group Vertical

Syntax REF:REF<x>:LABel:FONT:BOLD {<NR1>|OFF|ON}

Arguments <NR1> = 0 disables bold font; any other value turns this feature on.
OFF disables bold font.
ON enables bold font.

Examples REF:REF1:LABel:FONT:BOLD ON turns on the bold font.
REF:REF1:LABel:FONT:BOLD? might return :REF:REF1:LABel:FONT:BOLD 0 indicating the bold font is off.

REF:REF<x>:LABel:FONT:ITALic

This command sets or queries the italic state of the specified reference label.

Group Vertical

Syntax `REF:REF<x>:LABeI:FONT:ITAlIc {<NR1>|OFF|ON}`

Arguments `<NR1> = 0` disables italic font; any other value turns this feature on.
 `OFF` disables italic font.
 `ON` enables italic font.

Examples `REF:REF1:LABeI:FONT:ITAlIc ON` turns on the italic font.
 `REF:REF1:LABeI:FONT:ITAlIc?` might return
 `:REF:REF1:LABeI:FONT:ITAlIc 0` indicating the italic font is off.

REF:REF<x>:LABeI:FONT:SIZE

This command sets or queries the font size of the specified reference label.

Group Vertical

Syntax `REF:REF<x>:LABeI:FONT:SIZE <NR1>`

Arguments `<NR1>` is the font size of the label.

Examples `REF:REF1:LABeI:FONT:SIZE 20` sets the font size to 20 points.
 `REF:REF1:LABeI:FONT:SIZE?` might return `:REF:REF1:LABeI:FONT:SIZE 14` indicating that the font size is 14 points.

REF:REF<x>:LABeI:FONT:TYPE

This command sets or queries the font type of the specified reference label, such as Arial or Times New Roman.

Group Vertical

Syntax `REF:REF<x>:LABeI:FONT:TYPE <QString>`

Arguments `<QString>` is the font type.

Examples `REF:REF1:LABel:FONT:TYPE "Monospace"` specifies a mono spaced font.
`REF:REF1:LABel:FONT:TYPE?` might return `:REF:REF1:LABEL:FONT:TYPE "Frutiger LT Std 55 Roman"`.

REF:REF<x>:LABel:FONT:UNDERline

This command sets or queries the underline state of the specified reference label.

Group Vertical

Syntax `REF:REF<x>:LABel:FONT:UNDERline {<NR1>|OFF|ON}`

Arguments `<NR1> = 0` disables underline font; any other value turns this feature on.
`OFF` disables underline font.
`ON` enables underline font.

Examples `REF:REF1:LABel:FONT:UNDERline ON` turns on the underline font.
`REF:REF1:LABel:FONT:UNDERline?` might return
`:REF:REF1:LABEL:FONT:UNDERLINE 0` indicating that underline is off.

REF:REF<x>:LABel:NAMe

This command sets or queries the label of the specified reference. The reference waveform is specified by x.

Group Vertical

Syntax `REF:REF<x>:LABel:NAMe <QString>`
`REF:REF<x>:LABel:NAMe?`

Arguments `<QString>` is the character string that will be used for the reference waveform label name.

Examples `REF:REF4:LABEL:NAMe "My Reference"` sets the label name of Reference 4 waveform to "My Reference".

REF:REF3:LABEL:NAME? might return :REF:REF3:LABEL:NAME "Signal2", indicating that the label name for Reference 3 waveform is currently set to "Signal2".

REF:REF<x>:LABel:XPOS

This command sets or queries the X-position at which the label (attached to the displayed waveform of the specified reference) is displayed, relative to the left edge of the waveview. The reference waveform is specified by x.

Group	Vertical
Syntax	REF:REF<x>:LABel:XPOS <NR1> REF:REF<x>:LABel:XPOS?
Arguments	<NR1> is the location (control in divisions) where the waveform label for the selected reference is displayed, relative to the left edge of the screen.
Examples	REF:REF4:LABEL:XPOS 10 moves the waveform label for the Reference 3 waveform, so that it begins 10 divisions to the right of the left edge of the screen. REF:REF2:LABEL:XPOS? might return :REF:REF2:LABEL:XPOS 1.5, indicating that the x-axis for the Reference 2 waveform is currently 1.5 divisions to the right of the left edge of the screen.

REF:REF<x>:LABel:YPOS

This command sets or queries the Y-position of the label (attached to the displayed waveform of the specified reference), relative to the baseline of the waveform. The reference waveform is specified by x.

Group	Vertical
Syntax	REF:REF<x>:LABel:YPOS <NR1> REF:REF<x>:LABel:YPOS?
Arguments	<NR1> is the location where the waveform label for the selected reference is displayed, relative to the baseline of the waveform.

Examples `REF:REF3:LABEL:YPOS -10` moves the waveform label for the Reference 3 waveform 10 vertical units below the baseline of the waveform.

`REF:REF2:LABEL:YPOS?` might return `:REF:REF2:LABEL:YPOS 0`, indicating that the waveform label for the Reference 2 waveform is currently located at the baseline of the waveform.

REF:REF<x>:SOUrce

This command sets or queries the filename used by the given reference.

Group Vertical

Syntax `REF:REF<x>:SOUrce <QString>`

Arguments `<QString>` is the reference file name.

Examples `REF:REF1:SOUrce`
 `"/home/guest/.local/share/Tektronix/TekScope/`
 `FirstRecalledSession/161012_132000_000.wfm"` sets the source of
 the reference.

`REF:REF1:SOUrce?` might return `:REF:REF1:SOUrce`
 `"/home/guest/.local/share/Tektronix/TekScope/`
 `LastRecalledSession/161012_132039_000.wfm"`.

REF<x>_DALL:LABel:COLor

This command sets or queries the color of the specified digital group. The reference is specified by x.

Group Digital

Syntax `REF<x>_DALL:LABel:COLor <QString>`

Arguments `<QString>` is the color of the digital group label. To return the color to the default color, send an empty string as in this example: `:REF5_DALL:LABel:COLOR ""`.

Examples `REF1_DALL:LABe1:COLor "#FF0000"` sets the font color to red.
`REF1_DALL:LABe1:COLor?` might return `:REF1_DALL:LABEL:COLOR "#FFFF00"` indicating the font color is yellow.

REF<x>_DALL:LABe1:FONT:BOLD

This command sets or queries the bold state of the specified digital group. The reference is specified by x.

Group Digital

Syntax `REF<x>_DALL:LABe1:FONT:BOLD {ON|OFF|<NR1>}`

Arguments OFF argument turns off bold font.
ON argument turns on bold font.
<NR1> = 0 turns off bold font; any other value turns on bold font.

Examples `REF1_DALL:LABe1:FONT:BOLD ON` sets the font to bold.
`REF1_DALL:LABe1:FONT:BOLD?` might return
`:REF1_DALL:LABEL:FONT:BOLD 0` indicating the font is not bold.

REF<x>_DALL:LABe1:FONT:ITALic

This command sets or queries the italic state of the specified digital group. The reference is specified by x.

Group Digital

Syntax `REF<x>_DALL:LABe1:FONT:ITALic {ON|OFF|<NR1>}`

Arguments OFF argument turns off italic font.
ON argument turns on italic font.
<NR1> = 0 turns off italic font; any other value turns on italic font.

Examples `REF1_DALL:LABel:FONT:ITALic 1` turns on italic font.
`REF1_DALL:LABel:FONT:ITALic?` might return
`:REF1_DALL:LABEL:FONT:ITALIC 0` indicating the font is not italic.

REF<x>_DALL:LABel:FONT:SIZE

This command sets or queries the font size of the specified digital group. The reference is specified by x.

Group Digital

Syntax `REF<x>_DALL:LABel:FONT:SIZE <NR1>`

Arguments <NR1> is the font size.

Examples `REF1_DALL:LABel:FONT:SIZE 16` sets the font size to 16 points.
`REF1_DALL:LABel:FONT:SIZE?` might return
`:REF1_DALL:LABEL:FONT:SIZE 20` indicating the font size is 20 points.

REF<x>_DALL:LABel:FONT:TYPE

This command sets or queries the font type of the specified digital group, such as Arial or Times New Roman. The reference is specified by x.

Group Digital

Syntax `REF<x>_DALL:LABel:FONT:TYPE <QString>`

Arguments <QString> is the font type.

Examples `REF1_DALL:LABel:FONT:TYPE "Monospace"` sets the font to a monospace font.
`REF1_DALL:LABel:FONT:TYPE?` might return
`:REF1_DALL:LABEL:FONT:TYPE "Frutiger LT Std 55 Roman".`

REF<x>_DALL:LABel:FONT:UNDERline

This command sets or queries the underline state of the specified digital group. The reference is specified by x.

Group Digital

Syntax REF<x>_DALL:LABel:FONT:UNDERline {ON|OFF|<NR1>}

Arguments OFF argument turns off underline font.
ON argument turns on underline font.
<NR1> = 0 turns off underline font; any other value turns on underline font.

Examples REF1_DALL:LABel:FONT:UNDERline ON specifies an underlined font.
REF1_DALL:LABel:FONT:UNDERline? might return
:REF1_DALL:LABEL:FONT:UNDERLINE 0 indicating underline is off.

REF<x>_DALL:LABel:NAME

This command sets or queries the label of the specified digital group. The reference is specified by x.

Group Digital

Syntax REF<x>_DALL:LABel:NAME <QString>

Arguments <QString> is the name of the group.

Examples REF1_DALL:LABel:NAME "Clock Out" sets the label name to Clock Out.
REF1_DALL:LABel:NAME? might return :REF1_DALL:LABEL:NAME "This is the digital name".

REF<x>_DALL:LABel:XPOS

This command sets or queries the x-position of the label of the specified digital group. The reference is specified by x.

Group	Digital
Syntax	REF<x>_DALL:LABel:XPOS <NR3>
Arguments	<NR3> is the x-position, in pixels relative to the left edge of the display, of the group.
Examples	REF1_DALL:LABel:XPOS 90.0e0 sets the x position of the label to 90. REF1_DALL:LABel:XPOS? might return :REF1_DALL:LABEL:XPOS 45.0000 indicating the x position of the label is at 45 pixels to the right of the left edge of the display.

REF<x>_DALL:LABel:YPOS

This command sets or queries the y-position of the label of the specified digital group. The reference is specified by x.

Group	Digital
Syntax	REF<x>_DALL:LABel:YPOS <NR3>
Arguments	<NR3> is the y-position, in pixels relative to the baseline of the waveform, of the group.
Examples	REF1_DALL:LABel:YPOS 50 sets the y position to 50. REF1_DALL:LABel:YPOS? might return :REF1_DALL:LABEL:YPOS 0.0E+0 indicating the y position of the label is at the baseline of the waveform.

REF<x>_D<x>:LABel:COLor

This command sets or queries the color of the label of the specified digital bit. The reference is specified by x.

Group	Digital
Syntax	REF<x>_D<x>:LABel:COLor <QString>

Arguments	<QString> is the label color. To return the color to the default color, send an empty string as in this example: <code>:REF5_D1:LABEL:COLOR ""</code> .
Examples	<code>REF1_D1:LABe1:COLor "#FF0000"</code> sets the color to red. <code>REF1_D1:LABe1:COLor?</code> might return <code>:REF1_D1:LABEL:COLOR "#FFFF00"</code> indicating the color is yellow.

REF<x>_D<x>:LABe1:FONT:BOLD

This command sets or queries the bold state of the label of the specified digital bit. The reference is specified by x.

Group	Digital
Syntax	<code>REF<x>_D<x>:LABe1:FONT:BOLD {ON OFF <NR1>}</code>
Arguments	OFF argument turns off bold font. ON argument turns on bold font. <NR1> = 0 turns off bold font; any other value turns on bold font.
Examples	<code>REF1_D1:LABe1:FONT:BOLD ON</code> sets the font to bold. <code>REF1_D1:LABe1:FONT:BOLD?</code> might return <code>:REF1_D1:LABEL:FONT:BOLD 0</code> indicating the font is not bold.

REF<x>_D<x>:LABe1:FONT:ITALic

This command sets or queries the italic state of the label of the specified digital bit. The reference is specified by x.

Group	Digital
Syntax	<code>REF<x>_D<x>:LABe1:FONT:ITALic {ON OFF <NR1>}</code>
Arguments	OFF argument turns off italic font. ON argument turns on italic font. <NR1> = 0 turns off italic font; any other value turns on italic font.

Examples `REF1_D1:LABel:FONT:ITALic OFF` turns off italic font.
`REF1_D1:LABel:FONT:ITALic?` might return
`:REF1_D1:LABEL:FONT:ITALIC 1` indicating the font is italic.

REF<x>_D<x>:LABel:FONT:SIZE

This command sets or queries the font size of the label of the specified digital bit. The reference is specified by x.

Group Digital

Syntax `REF<x>_D<x>:LABel:FONT:SIZE <NR1>`

Arguments <NR1> is the font size.

Examples `REF1_D1:LABel:FONT:SIZE 16` sets the font size to 16 points.
`REF1_D1:LABel:FONT:SIZE?` might return `:REF1_D1:LABEL:FONT:SIZE 20` indicating the font size is 20 points.

REF<x>_D<x>:LABel:FONT:TYPE

This command sets or queries the font type of the label of the specified digital bit, such as Arial or Times New Roman. The reference is specified by x.

Group Digital

Syntax `REF<x>_D<x>:LABel:FONT:TYPE <QString>`

Arguments <QString> is the font type of the label.

Examples `REF1_D1:LABel:FONT:TYPE "Monospace"` sets the font to Monospace.
`CHREF1_D1:LABel:FONT:TYPE?` might return `:REF1_D1:LABEL:FONT:TYPE "Frutiger LT Std 55 Roman"`.

REF<x>_D<x>:LABel:FONT:UNDERline

This command sets or queries the underline state of the label of the specified digital bit. The reference is specified by x.

Group Digital

Syntax REF<x>_D<x>:LABel:FONT:UNDERline {ON|OFF|<NR1>}

Arguments OFF argument turns off underline font.
ON argument turns on underline font.
<NR1> = 0 turns off underline font; any other value turns on underline font.

Examples REF1_D1:LABel:FONT:UNDERline ON turns on underline font.
REF1_D1:LABel:FONT:UNDERline? might return
:REF1_D1:LABel:FONT:UNDERLINE 0 indicating the underline font is off.

REF<x>_D<x>:LABel:NAME

Sets or queries the label of the specified digital bit. The channel is specified by x.

Group Digital

Syntax REF<x>_D<x>:LABel:NAME <QString>

Arguments <QString> is the label.

Examples REF1_D1:LABel:NAME "Clock in" sets the name to Clock in.
REF1_D1:LABel:NAME? might return :REF1_D1:LABel:NAME "Digital 1".

REF<x>_D<x>:LABel:XPOS

This command sets or queries the x-position of the label of the specified digital bit. The reference is specified by x.

Group Digital

Syntax	REF<x>_D<x>:LABe1:XPOS <NR3>
Arguments	<NR3> is the x-position, in pixels relative to the left edge of the display, of the label.
Examples	REF1_D1:LABe1:XPOS 90 sets the x position to 90. REF1_D1:LABe1:XPOS? might return :REF1_D1:LABEL:XPOS 45.0000 indicating the position is 45 pixels to the right of the left edge of the waveform.

REF<x>_D<x>:LABe1:YPOS

This command sets or queries the y-position of the label of the specified digital bit. The channel is specified by x.

Group	Digital
Syntax	REF<x>_D<x>:LABe1:YPOS <NR3>
Arguments	<NR3> is the y-position, in pixels relative to the baseline of the waveform, of the label.
Examples	REF1_D1:LABe1:YPOS 10 sets the y position to 10. REF1_D1:LABe1:YPOS? might return :REF1_D1:LABEL:YPOS 0.0E+0 indicating the y position of the label is at the baseline of the waveform.

REM (No Query Form)

This command (no query form) embeds a comment within programs as a means of internally documenting the programs. This is how to embed comments in a .set file. The instrument ignores these embedded comment lines.

Group	Miscellaneous
Syntax	REM <QString>
Arguments	<QString> is a string that can contain a maximum of 80 characters.

Examples REM "This is a comment" is a comment string that the instrument will ignore.

ROSc:SOURce

This command sets or queries the selected source for the time base reference oscillator. The reference oscillator locks to this source. Depending on the command argument that you specify, you can use an external reference or use the internal crystal oscillator as the time base reference.

Group Miscellaneous

Syntax ROSC:SOURce {INTERNAL|EXTERNAL}
ROSc:SOURce?

Related Commands [ROSc:STATE?](#)

Arguments INTERNAL specifies the internal 10 MHz crystal oscillator as the time base reference.
EXTERNAL specifies the user-supplied external signal as the time base reference.

Examples ROSC:SOURCE INTERNAL specifies the internal 10 MHz crystal oscillator as the time base reference.
ROSc:SOURCE? might return :ROSc:SOURCE INTERNAL, indicating that the 10 MHz crystal oscillator is being used as the time base reference.

ROSc:STATE? (Query Only)

This query-only command returns whether the time base reference oscillator is locked. This command will return either LOCKED or UNLOCKED.

Group Miscellaneous

Syntax ROSC:STATE?

Returns LOCKED indicates the reference oscillator is locked.
UNLOCKED indicates the reference oscillator is not locked.

Examples `ROSC:STATE?` might return `:ROSC:STATE LOCKED`, indicating that the time base reference is locked.

***RST (No Query Form)**

This command (no query form) resets the instrument to the factory default settings.

This command does the following:

- Recalls the default instrument setup.
- Clears the current *DDT command.
- Disables aliases (:ALIAS:STATE 0).
- Disables the user password (for the *PUD command).

The *RST command does not change the following:

- The current working directory (:FILESystem:CWD command).
- The state of command headers (:HEADer command).
- The state of keyword and enumeration verbosity (:VERBoSe command).
- The Power-on Status Clear Flag (*PSC command).
- The Event Status Enable Register (*ESE command).
- The Service Request Enable Register (*SRE command).
- The Device Event Status Enable Register (DESE command).
- The user password for protected user data (:PASSword command).
- The content of protected user data (*PUD command).
- The enabled state of the socket server (:SOCKETServer:ENABle command).
- The socket server port number (:SOCKETServer:PORT command).
- The socket server protocol (:SOCKETServer:PROTOCOL command).
- The USBTMC port configuration (:USBDevice:CONFIgure command).
- The destination reference waveform or file path for the :CURVe command (:DATA:DESTInation command).
- The source waveform for the :CURVe? or :WAVFrm? queries (:DATA:SOURce command).
- The waveform data encoding for the :CURVe command or query or the :WAVFrm? query (:DATA:ENCdg command).
- The starting point for :CURVe? queries (:DATA:STARt command).
- The ending point for :CURVe? queries (:DATA:STOP command).
- All settings associated the :WFMInpre commands.
- All user settable settings associated with the WFMOutpre commands.

*RST only resets the programmable interface settings, it does not change the user interface settings.

Group Status and Error

Syntax *RST

Related Commands [FACtory](#)
[RECAIl:SETUp](#)

Arguments None

Examples *RST resets the instrument settings to factory defaults.

SAVe:EVENTtable:BUS (No Query Form)

This command saves bus results table to the specified file.

Group Save and Recall

Syntax SAvE:EVENTtable:BUS <QString>

Arguments <QString> is the specified file. If a file name or path is specified, the file is expected to be located in a directory relative to the current working directory (specified by [FILESystem:CWD](#)) unless a complete path is specified.

- Complete path specification. If the file argument begins with drive designator (such as C:), then the file name is interpreted as a full path.
- Relative path specification. If the file argument begins with "." or ".." or has a file path separator appearing anywhere other than the first character position, then the file name is treated as a path that is relative to the current working directory.

Examples SAvE:EVENTtable:BUS "TEK000.CSV" saves the bus decode event table in the file named TEK000.CSV.

SAVe:EVENTtable:CUSTom (No Query Form)

This command saves the results table to the specified file path and name.

Group Save and Recall

Syntax SAvE:EVENTtable:CUSTom <QString>

Arguments <QString> is the specified file. If a file name or path is specified, the file is expected to be located in a directory relative to the current working directory (specified by [FILESystem:CWD](#)) unless a complete path is specified.

- Complete path specification. If the file argument begins with drive designator (such as C:), then the file name is interpreted as a full path.
- Relative path specification. If the file argument begins with "." or ".." or has a file path separator appearing anywhere other than the first character position, then the file name is treated as a path that is relative to the current working directory.

Examples `SAVE:EVENTtable:CUSTom "C:/TEK000.CSV"` sets the oscilloscope to save the results table in the top level C: directory in a file named TEK000.CSV.

SAVE:EVENTtable:CUSTom:COMMeNts

This command sets or queries comments to be included in saved results table files.

Group Save and Recall

Syntax `SAVE:EVENTtable:CUSTom:COMMeNts <Qstring>`
`SAVE:EVENTtable:CUSTom:COMMeNts?`

Arguments `<Qstring>` sets the oscilloscope to save the quoted string as a comment in the saved results table file.

Examples `SAVE:EVENTtable:CUSTom:COMMeNts "PSRR batch 2 testing."` sets the oscilloscope to save the quoted string in all following saved results table files.

`SAVE:EVENTtable:CUSTom:COMMeNts?` might return "", indicating that the there is no comment set to be saved in results table files.

SAVE:EVENTtable:CUSTom:DATAFormat

This command sets or queries the data format to use for saving results table data.

Group Save and Recall

Syntax `SAVE:EVENTtable:CUSTom:DATAFormat [SCIENTific|ENGINEERING]`

Arguments `SCIENTific` sets the oscilloscope to save results tables data in scientific notation (for example, 5.0100E-12).

ENGINEERING sets the oscilloscope to save results tables data in engineering notation (for example, 5.0100ps).

- Examples** `SAVE:EVENTtable:CUSTom:DATAFormat SCIENTific` sets the oscilloscope to save results tables data in scientific notation.
- `SAVE:EVENTtable:CUSTom:DATAFormat SCIENTific` might return `ENGINEERING`, indicating that the oscilloscope is set to save results tables data in engineering notation.

SAVe:EVENTtable:CUSTom:INCLUDEREFS

This command sets or queries whether to include displayed reference waveforms with saved results table files.

- Group** Save and Recall
- Syntax** `SAVe:EVENTtable:CUSTom:INCLUDEREFS {1|0}`
`SAVe:EVENTtable:CUSTom:INCLUDEREFS?`
- Arguments** 1 sets the oscilloscope to save all displayed reference waveforms as part of a saved results table file.
- 0 sets the oscilloscope to not save all displayed reference waveforms as part of a saved results table file.
- Examples** `SAVe:EVENTtable:CUSTom:INCLUDEREFS 1` sets the oscilloscope to save all displayed reference waveforms as part of a saved results table file.
- `SAVe:EVENTtable:CUSTom:INCLUDEREFS?` might return 0, indicating that the oscilloscope will not save all displayed reference waveforms as part of a saved results table file.

SAVe:EVENTtable:MEASUrement (No Query Form)

This command saves data (measurement) results to the specified file.

- Group** Save and Recall
- Syntax** `SAVe:EVENTtable:MEASUrement <QString>`

Arguments <QString> is the specified file. If a file name or path is specified, the file is expected to be located in a directory relative to the current working directory (specified by [FILESystem:CWD](#)) unless a complete path is specified:

- Complete path specification. If the file argument begins with drive designator (such as C:), then the file name is interpreted as a full path.
- Relative path specification. If the file argument begins with "." or ".." or has a file path separator appearing anywhere other than the first character position, then the file name is treated as a path that is relative to the current working directory.

Examples `SAVE:EVENTtable:MEASurement "TEK000.CSV"` saves the measurement in the file named TEK000.CSV.

SAVE:IMAGe (No Query Form)

Saves a capture of the screen contents to the specified image file. Supported image formats are PNG, Windows Bitmap, and JPEG.

Group Save and Recall

Syntax `SAVE:IMAGe <QString>`

Arguments <QString> is the file name and location used to store the image file.

When specifying the file name with this command, use the correct file extension (".png" for PNG format, ".bmp" for BMP format, or ".jpg" for JPEG format). If a file name or path is specified, the file is expected to be located in a directory relative to the current working directory (specified by [:FILESystem:CWD](#)[FILESystem:CWD](#)) unless a complete path is specified:

- Complete path specification. If the file argument begins with drive designator (such as C:), then the file name is interpreted as a full path.
- Relative path specification. If the file argument begins with "." or ".." or has a file path separator appearing anywhere other than the first character position, then the file name is treated as a path that is relative to the current working directory.

Examples `SAVE:IMAGe "c:/rose_was_here.png"` saves the image at the location specified.

SAVe:REPOrt (No Query Form)

This command saves a report to the specified file. Supported report formats are PDF and MHT (web page archive file).

Group Save and Recall

Syntax SAvE:REPOrt <QString>

Related Commands [SAVe:REPOrt:COMMeNts](#)

Arguments <QString> is the complete path specification. When specifying the file name with this command, use the correct file extension (.pdf for PDF format, or .mht for MHT format).

If a file name or path is specified, the file is expected to be located in a directory relative to the current working directory (specified by [FILESystem:CWD](#)) unless a complete path is specified:

- Complete path specification. If the file argument begins with a file path separator (forward slash character) or a Windows drive designator such as C:), then the file name is interpreted as a full path.
- Relative path specification. If the file argument begins with "." or ".." or has a file path separator appearing anywhere other than the first character position, then the file name is treated as a path that is relative to the current working directory.

Examples SAvE:REPORT "report.pdf" creates a report in PDF format, in the location specified.

SAVe:REPOrt:COMMeNts

This command sets or queries the comments to be included in saved report files.

Group Save and Recall

Syntax SAvE:REPOrt:COMMeNts <QString>
SAVe:REPOrt:COMMeNts?

Arguments <QString> is the comments to be included in saved report files.

Examples `SAVE:REPORT:COMMENTS "Test 3"` adds comments to the report.

`SAVE:REPORT:COMMENTS?` might return `:SAVE:REPORT:COMMENTS "High Temp Test 1"`.

SAVe:SESSion (No Query Form)

Saves the state of the instrument, including reference waveforms, to a saved session file.

Group Save and Recall

Syntax `SAVe:SESSion <QString>`

Arguments `<QString>` is the file path that specifies the location to save the specified instrument session file. If a file name or path is specified, the file is expected to be located in a directory relative to the current working directory (specified by `FILESystem:CWD`[FILESystem:CWD](#)) unless a complete path is specified:

- Complete path specification. If the file argument begins with a file path separator (forward slash character) or a Windows drive designator such as `C:`, then the file name is interpreted as a full path.
- Relative path specification. If the file argument begins with `."` or `.."` or has a file path separator appearing anywhere other than the first character position, then the file name is treated as a path that is relative to the current working directory.

Examples `SAVE:SESSION "c:/rose_was_here.tss"` saves the instrument state in the specified file.

SAVe:SETUp (No Query Form)

Saves the current instrument state to the specified file.

Group Save and Recall

Syntax `SAVe:SETUp <QString>`

Related Commands [SAVe:SETUp:INCLUDEREFs](#)

Arguments	<QString> is a quoted string that is the complete path specification. If a file name or path is specified, the file is expected to be located in a directory relative to the current working directory (specified by FILESystem:CWD) unless a complete path is specified: ■ Complete path specification. If the file argument begins with a file path separator (forward slash character) or a Windows drive designator such as C:), then the file name is interpreted as a full path. ■ Relative path specification. If the file argument begins with "." or ".." or has a file path separator appearing anywhere other than the first character position, then the file name is treated as a path that is relative to the current working directory.
Examples	SAVE:SETUP "c:/rose_was_here.set" saves the instrument setup in the specified file.

SAVe:SETUp:INCLUDEREFS

This command sets or queries whether displayed reference waveforms are to be included in saved setups.

Group	Save and Recall
Syntax	SAVe:SETUp:INCLUDEREFS {OFF ON 0 1} SAVe:SETUp:INCLUDEREFS?
Arguments	OFF specifies not including displayed reference waveforms in saved setups. ON specifies including displayed reference waveforms in saved setups. 0 specifies not including displayed reference waveforms in saved setups. 1 specifies including displayed reference waveforms in saved setups.
Examples	SAVE:SETUP:INCLUDEREFS 0 sets reference waveforms not to be included in saved setups. SAVE:SETUP:INCLUDEREFS? might return :SAVE:SETUP:INCLUDEREFS 1 indicating that reference waveforms are to be included in saved setups.

SAVe:WAVEform (No Query Form)

This command saves the specified waveform(s) to the specified destination file. The waveform source or sources must be active (turned on) to save data to a file.

Group	Save and Recall
Syntax	SAVe:WAVEform {CH<x>[_DALL] MATH<x> REF<x> ALL}, <Qstring>
Related Commands	FILESystem:CWD SAVe:WAVEform:SOURCEList?
Arguments	<x> is the number of the analog channel, math, or reference waveform source used to save the waveform data. _DALL saves the digital channel waveform data of the specified channel. This argument is required if the channel specified is a digital channel. ALL saves all displayed analog, math, and reference waveforms to individual files. Each file name created includes the name of the source (ch1, math3, and so on) used to create that file. <Qstring> is a quoted string that defines the path and file name to use to save the specified file, in the format '[<path>]<filename.ext>'. Specifying a path is optional. If no path is entered, the file is saved to the current working directory as set in FILESystem:CWD. <path> uses the form '<drive>/<dir>.../'. You can specify a relative path or a complete path: ■ Relative path specification. If the file argument begins with "." or ".." or has a file path separator appearing anywhere other than the first character position, then the file name is treated as a path that is relative to the current working directory. ■ Complete path specification. If the file argument begins with a file path separator (forward slash character) or a drive designator (such as C:), then the file name is interpreted as a full path from the specified drive. <filename> sets the file name to use to create the file. A file can have up to 125 characters. When using the ALL argument to save multiple files, each filename has the filename appended with the source used to create that file. For example, a filename of QualTest can create QualTest_ch1.xxx, QualTest_ref1.xxx, and so on. <.ext> sets the file format to which to save the data: ■ Use the .wfm extension to save waveform data to a Tektronix Internal format. ■ Use the .csv extension to save waveform data to a comma separated values spreadsheet format.

Examples `SAVE:WAVEFORM MATH1, "TEK0000.WFM"` saves the Math1 waveform to the file TEK00000.WFM in the current working directory.

SAVe:WAVEform:SOURCEList? (Query Only)

This query returns a list of the available waveforms that can be specified as the source for the SAvE:WAVEform command. Source waveforms must have their display mode set to On to appear in this list and to be saved.

Group Save and Recall

Syntax `SAVe:WAVEform:SOURCEList?`

Examples `SAVE:WAVEFORM:SOURCELIST?` might return
`ALL, CH1_DALL, CH2, CH3, CH8, MATH1, MATH2, REF2, REF4.`

SAVEON:FILE:DEST

This command sets or queries the location where files are saved when SAVEON:TRIGGER is ON and SAVEON:WAVEFORM is ON. You can save the files to a local drive or network path by entering the desired location in <QString>. You can also select to save the files to a USB drive.

Group Save On

Syntax `SAVEON:FILE:DEST <QString>`
`SAVEON:FILE:DEST?`

Related Commands [SAVEON:FILE:NAME](#)

Arguments <QString> specifies the location to store files.

Examples `SAVEON:FILE:DEST`
 `"C:\users\username\Tektronix\TekScope\SaveOnTrigger"` sets this as the location to save files (named by the `SAVEON:FILE:NAME` command), when there is a trigger.

`SAVEON:FILE:DEST?` might return `:SAVEON:FILE:DEST`
 `"C:\users\username\Tektronix\TekScope\SaveOnEvent"`,
 indicating the drive location where files will be saved when there is a trigger.

SAVEON:FILE:NAME

Sets or queries the file name to use when SAVEON:TRIGger is ON.

Group Save On

Syntax SAVEON:FILE:NAME <QString>
SAVEON:FILE:NAME?

Related Commands [SAVEON:FILE:DEST](#)

Arguments <QString> is the file name you want to use.

Examples SAVEON:FILE:NAME "MaskFailure" sets the name of the file to MaskFailure.
SAVEON:FILE:NAME? might return ":SAVEON:FILE:NAME MaskFailure5", indicating the name you set for the oscilloscope to use, with the autoincrement number (5) appended.

SAVEON:IMAGe

This command sets or queries whether to save a screen capture when a trigger occurs and SAVEON:TRIGger is ON and SAVEON:IMAGe is ON.

Group Save On

Syntax SAVEON:IMAGe {<NR1>|OFF|ON}
SAVEON:IMAGe?

Related Commands [SAVEON:FILE:DEST](#)
[SAVEON:FILE:NAME](#)
[SAVEON:TRIGger](#)
[SAVEON:IMAGe](#)

Arguments <NR1> = 0 disables Save On Image; any other value turns this feature on.
OFF disables Save On Image.
ON enables Save On Image.

- Examples** `SAVEON:IMAGE ON` sets the oscilloscope to save a screen capture on a specified trigger.
- `SAVEON:IMAGE?` might return `:SAVEON:IMAGE 1`, indicating that the oscilloscope will save a screen capture when the specified trigger occurs.

SAVEON:IMAGe:FILEFormat

This command sets or queries the file format to be used for saved image files when `:SAVEON:IMAGe` is set to 1.

Group Save On

Syntax `SAVEON:IMAGe:FILEFormat {PNG|BMP|JPG}`
`SAVEON:IMAGe:FILEFormat?`

- Arguments** PNG specifies using PNG format for saved image files.
- BMP specifies using BMP format for saved image files.
- JPG specifies using JPEG format for saved image files.
- When specifying the file name with this command, use the correct file extension (".png" for PNG format, ".bmp" for BMP format, or ".jpg" for JPEG format). If a file name or path is specified, the file is expected to be located in a directory relative to the current working directory (specified by [FILESystem:CWD](#)) unless a complete path is specified:
- Complete path specification. If the file argument begins with drive designator (such as C:), then the file name is interpreted as a full path.
 - Relative path specification. If the file argument begins with "." or ".." or has a file path separator appearing anywhere other than the first character position, then the file name is treated as a path that is relative to the current working directory.

- Examples** `SAVEON:IMAGe:FILEFORMAT JPG` sets the image file format to JPEG.
- `SAVEON:IMAGe:FILEFORMAT?` might return `:SAVEON:IMAGe:FILEFORMAT PNG` indicating that the file format is set to PNG.

SAVEON:TRIGger

Sets or queries whether to save a file when a trigger occurs. You can define the trigger using Trigger commands or the oscilloscope user interface.

The trigger will cause the instrument to save an image or a waveform to a file, depending on what you specified. For example, if you have set [SAVEON:IMAGe](#) to On, and a trigger event occurs, the instrument will save a screen capture. You can set options for file storage (such as file name, file destination, and auto increment), using the SAVEON:FILE commands.

Use the oscilloscope interface to select whether to save one or more analog channels, digital channels, or math waveforms

Analog and math waveforms are saved using one file per waveform. Digital waveforms are all saved to a single file.

Group	Save On
Syntax	SAVEON:TRIGger {<NR1> ON OFF} SAVEON:TRIGger?
Related Commands	SAVEON:IMAGe SAVEON:WAVEform SAVEON:FILE:DEST SAVEON:FILE:NAME
Arguments	<NR1> = 0 disables Save On Trigger; any other value turns this feature on. OFF disables Save On Trigger. ON enables Save On Trigger.
Examples	SAVEON:TRIGGER ON sets the oscilloscope to save an image, measurement, and/or waveform when a trigger occurs. SAVEON:TRIGGER? might return :SAVEON:TRIGGER ON, indicating that a file will be saved on triggering.

SAVEON:WAVEform

Sets or queries whether to save a waveform when a trigger occurs when [SAVEON:TRIGger](#) is ON.

The waveform will be saved to the file you selected with [SAVEON:FILE:NAME](#), in the location that you selected using [SAVEON:FILE:DEST](#). You can set options for file storage (such as file name, file destination, and autoincrement), using the SAVEON:FILE commands.

Group	Save On
Syntax	<code>SAVEON:WAVEform {<NR1> ON OFF}</code> <code>SAVEON:WAVEform?</code>
Related Commands	SAVEON:FILE:DEST SAVEON:FILE:NAME SAVEON:TRIGger
Arguments	<code><NR1> = 0</code> disables Save On Waveform; any other value turns this feature on. <code>OFF</code> disables Save On Waveform. <code>ON</code> enables Save On Waveform.
Examples	<code>SAVEON:WAVEFORM ON</code> turns on the Save On Waveform feature, so that a waveform will be saved when a selected trigger occurs. <code>SAVEON:WAVEFORM?</code> might return <code>:SAVEON:WAVEFORM ON</code> , indicating that a waveform will be saved when a selected trigger occurs.

SAVEON:WAVEform:FILEFormat

This command sets or queries the file format for saving waveforms when `:SAVEON:WAVEform` is set to 1.

Group	Save On
Syntax	<code>SAVEON:WAVEform:FILEFormat {INTERNAL SPREADSheet}</code> <code>SAVEON:WAVEform:FILEFormat?</code>
Arguments	<code>INTERNAL</code> specifies saving the waveform in the oscilloscope internal format. <code>SPREADSheet</code> specifies saving the waveform in comma separated values format.
Examples	<code>SAVEON:WAVEFORM:FILEFORMAT SPREADSheet</code> sets the file format to spreadsheet. <code>SAVEON:WAVEFORM:FILEFORMAT?</code> might return <code>:SAVEON:WAVEFORM:FILEFORMAT INTERNAL</code> indicating the file format is set to <code>INTERNAL</code> .

SAVEON:WAVEform:SOURce

This command sets or queries the sources for saving waveforms when [SAVEON:TRIGger](#) is ON.

Group	Save On
Syntax	SAVEON:WAVEform:SOURce {CH<x> MATH<x> REF<x> ALL} SAVEON:WAVEform:SOURce?
Arguments	Arguments are the available sources.
Examples	SAVEON:WAVEform:SOURce MATH1 specifies MATH 1 as the save on source. SAVEON:WAVEform:SOURce? might return :SAVEON:WAVEFORM:SOURCE REF1 indicating the save on source is REF1.

SEARCH:ADDNew (No Query Form)

This command adds the specified search.

Group	Search and Mark
Syntax	SEARCH:ADDNew <QString>
Arguments	<QString> is the specified search. The argument is of the form "SEARCH<NR1>", where <NR1> is ≥ 1 .
Examples	SEARCH:ADDNEW "SEARCH2" adds a new search named SEARCH 2.

SEARCH:DELeTe (No Query Form)

This command deletes the specified search.

Group	Search and Mark
Syntax	SEARCH:DELeTe <QString>

Arguments <QString> is the specified search. The argument is of the form "SEARCH<NR1>", where <NR1> is ≥ 1).

Examples SEARCH:DELETE "SEARCH3" deletes SEARCH 3.

SEARCH:LIST? (Query Only)

This command returns a comma separated list of all currently defined searches.

Group Search and Mark

Syntax SEARCH:LIST?

Returns All currently defined searches.

Examples SEARCH:LIST? might return :SEARCH:LIST SEARCH1, SEARCH2 indicating that Search 1 and Search 2 are defined.

SEARCH:SEARCH<x>:COPy (No Query Form)

This command (no query form) copies the search criteria to or from the trigger. The search number is specified by <x>.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:COPy {SEARCHtotrigger|TRIGgertosearch}

Arguments SEARCHtotrigger copies the search criteria to the trigger.
TRIGgertosearch copies the trigger criteria to the search.

Examples SEARCH:SEARCH1:COPy TRIGGERTOSEARCH copies the trigger criteria to the search 1 criteria.

SEARCH:SEARCH1:COPy SEARCHTOTRIGGER copies the search criteria to the trigger.

SEARCH:SEARCH<x>:NAVigate (No Query Form)

This command sets the navigation action for search marks. The **NONE** action is the default setting when no action is being taken. The search number is specified by <x>.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:NAVigate {NEXT|PREVIOUS|MIN|NONE|MAX}

Arguments NEXT goes to the next search mark.
PREVIOUS goes to the previous search mark.
MIN goes to the first search mark.
NONE is the default setting when no action is being taken.
MAX goes to the last search mark.

Examples SEARCH:SEARCH1:NAVigate NEXT goes to the next search mark.

SEARCH:SEARCH<x>:TOTAL? (Query Only)

This query-only command returns the total number of found search marks for this search. The search number is specified by <x>.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TOTAL?

Examples SEARCH:SEARCH1:TOTAL? might return SEARCH:SEARCH1:TOTAL 7, indicating that there are 7 matches for search 1.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:CONDition

This command specifies a field or condition for an ARINC429 bus to search on. The search number is specified by x.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:CONDition {SOW LABe1 DATa LABELANDDATA EOW ERROR} SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:CONDition?</pre>
Arguments	SOW specifies a search for the first bit of a word. LABe1 specifies a search for a matching label. DATa specifies a search for matching packet data fields. LABELANDDATA specifies a search for a matching label and matching packet data field(s). EOW specifies a search for the 32nd bit of a word. ERROR specifies a search for a specified error condition. NOTE. The type of error searched for is specified by <code>SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:ERRType</code>.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:CONDITION DATA specifies finding packets that contain matching data field(s). SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:CONDITION? might return SOW, indicating that the bus is being searched for the first bit of each word in the packet.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:DATa:HIVALue

This command sets or queries the high value when searching on an ARINC429 data field. The search number is specified by x. The search condition must be set to DATa or LABELANDDATA, and the data qualifier must be INrange or OUTrange.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:DATa:HIVALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:DATa:HIVALue?</pre>

Arguments	<QString> is the label value.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:DATA:HIVALUE "XXXXXXXXXXXXXXXX1000" sets the value to XXXXXXXXXXXXXXXXXXX1000. SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:DATA:HIVALUE? might return "XXXXXXXXXXXXXXXXXXXX", indicating that the value is XXXXXXXXXXXXXXXXXXXX.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:DATA:QUALifier

This command sets or queries the qualifier to be used when searching on data in the DATA field for an ARINC429 bus signal. The search number is specified by x. The search condition must be set to DATA or LABELANDDATA.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:DATA: QUALifier {EQUAL UNEQUAL LESSThan MOREthan LESSEQUAL MOREEQUAL INrange OUTrange} SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:DATA:QUALifier?
Arguments	Arguments are the available data qualifiers.

NOTE. The search qualifier only applies to the bits defined as the data field via the bus data field format specifier (using *BUS:B<x>:ARINC429A:DATAFormat*).

Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:DATA:QUALIFIER LESSTHAN sets the data qualifier to less than. SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:DATA:QUALIFIER? might return EQUAL, indicating that the data qualifier is set to equal.
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SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:DATA:VALue

This command sets or queries the low value when searching on an ARINC429 data field. The search number is specified by x. The search condition must be set to DATA or LABELANDDATA.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:DATA:VALUE <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:DATA:VALUE?
Arguments	<QString> is the label value.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:DATA:VALUE "XXXXXXXXXXXXXXXXX1000" sets the value to XXXXXXXXXXXXXXXXXXXX1000. SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:DATA:VALUE? might return "XXXXXXXXXXXXXXXXXXXX", indicating that the value is XXXXXXXXXXXXXXXXXXXX.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:ERRType

This command sets or queries the error type when searching on an ARINC429 bus signal. The search number is specified by x. The search condition must be set to ERRor.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:ERRType {ANY PARity WORD GAP} SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:ERRType?
Arguments	ANY sets the error type to match any of the other available error types. PARity sets the error type to match on parity errors (parity value results in even parity count for a word). WORD sets the error type to match on word errors (any unframed or unknown decode data). GAP sets the error type to match on gap violations (less than 4 bits idle time between two packets on the bus).

Examples `SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:ERRTYPE` `PARITY` sets the error type to match on parity errors.

`SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:ERRTYPE?` might return `ANY`, indicating that any error condition will produce a match.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:LABel:HIVALue

This command sets or queries the high value when searching on an ARINC429 label field. The search number is specified by `x`. The search condition must be set to `LABel`, and the label qualifier must be `INrange` or `OUTrange`.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:LABel:HIVALue`
 `<QString>`
 `SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:LABel:HIVALue?`

Arguments `<QString>` is the label value.

Examples `SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:LABEL:HIVALUE`
 `"XXXX1010"` sets the value to `XXXX1010`.

`SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:LABEL:HIVALUE?` might return `"XXXXXXXX"`, indicating that the value is `XXXXXXXX`.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:LABel:QUALifier

This command sets or queries the qualifier to be used when searching on label data for an ARINC429 bus signal. The search number is specified by `x`. The search condition must be set to `LABel` or `LABELANDDATA`.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:LABel:
QUALifier {Equal|UNEQual|LESSthan|MOREthan
|LESSEQual|MOREEQual|INrange|OUTrange}
SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:LABel:QUALifier?

Arguments Arguments are the available data qualifiers.

NOTE. If the search condition is set to LABELANDDATA, the label qualifier will be locked to Equal until the search condition is changed again.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:LABEL:QUALIFIER
LESSTHAN sets the label qualifier to less than.

SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:LABEL:QUALIFIER? might
return EQUAL, indicating that the label qualifier is set to equal.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:LABel:VALue

This command sets or queries the low value when searching on an ARINC429 label field. The search number is specified by x. The search condition must be set to LABEL or LABELANDDATA.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:LABel:VALue
<QString>

Arguments <QString> is the label value.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:LABEL:VALUE
"XXXX1010" sets the value to XXXX1010.

SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:LABEL:VALUE? might return
"XXXXXXXX", indicating that the value is XXXXXXXX.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:SDI:VALue

This command sets or queries the label when searching on an ARINC429 SDI field. The search number is specified by x. The search condition must be set to DATA or LABELANDDATA, and the data format must be set to DATA.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:SDI:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:SDI:VALue?
Arguments	<QString> is the label value.

NOTE. The SDI field is only present when the selected data field format is DATA (using *BUS:B<x>:ARINC429A:DATAFormat*). Also, the stored QString is reset to its default value whenever the data field format is changed.

Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:SDI:VALUE "X0" sets the value to X0. SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:SDI:VALUE? might return "XX", indicating that the value is XX.
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SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:SSM:VALue

This command sets or queries the label when searching on an ARINC429 SSM field. The search number is specified by x. The search condition must be set to DATA or LABELANDDATA, and the data format must be set to DATA or SDIDATA.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:SSM:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:ARINC429A:SSM:VALue?

Arguments	<QString> is the label value.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:SSM:VALUE "X0" sets the value to X0. SEARCH:SEARCH1:TRIGGER:A:BUS:ARINC429A:SSM:VALUE? might return "XX", indicating that the value is XX.

SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:CONDition

This command sets or queries the condition (word select / frame sync, or matching data) to be used when searching on an audio bus signal. The search number is specified by <x>.

Conditions	Requires option 5-SRAUDIO or SUP5-SRAUDIO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:CONDition {SOF DATA} SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:CONDition?
Arguments	SOF specifies to search on a word select or start of frame (depending on Audio Type). DATA specifies to search on matching data.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:AUDio:CONDition SOF sets conditions to search on start of frame. SEARCH:SEARCH1:TRIGger:A:BUS:AUDio:CONDition? might return SEARCH:SEARCH1:TRIGGER:A:BUS:AUDIO:CONDITION DATA indicating settings for a search on DATA.

SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATa:HITDMVALue

This command sets or queries the binary data string for the high data word to be used when searching on an TDM audio bus signal. The search condition must be set to DATa using SEARCH:SEARCH{x}:TRIGger:A:BUS:AUDio:CONDition.

Group	Search and Mark
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Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATa:HITDMVALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATa:HITDMVALue?
Arguments	<QString> is the binary data string for the high data word to be used when searching on an TDM audio bus signal.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:AUDio:DATa:HITDMVALue "XXXXXXXXXXXXXXXXXXXX1100" sets the high value to 1100. SEARCH:SEARCH1:TRIGger:A:BUS:AUDio:DATa:HITDMVALue? might return SEARCH:SEARCH1:TRIGGER:A:BUS:AUDIO:DATA:HITDMVALUE "XXXXXXXXXXXXXXXXXXXX1010" indicating the high value is set to 1010.

SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATa:HIVALue

This command sets or queries the binary data string for the high data word to be used when searching on an audio bus signal. The search condition must be set to DATa using :SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:CONDition. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATa:HIVALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATa:HIVALue?
Arguments	<QString> specifies the upper word value.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:AUDio:DATa:HIVALue "XXXX" sets the HIVALUE to XXXX. SEARCH:SEARCH1:TRIGger:A:BUS:AUDio:DATa:HIVALue? might return SEARCH:SEARCH1:TRIGGER:A:BUS:AUDIO:DATA:HIVALUE "TEST_001101", indicating the HIVALUE is set to TEST_001101.

SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATa:OFFSet

This command sets or queries the data offset value (TDM channel) to be used when searching on a TDM type audio bus signal. The search condition must be set to DATa using SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:TDM:CONDition. The search number is specified by <x>.

Conditions	Requires option 5-SRAUDIO or SUP5-SRAUDIO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATA:OFFSet <NR1> SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATA:OFFSet?
Arguments	<NR1> is the data offset value.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:AUDio:DATA:OFFSet 2 sets the data offset value to 2. SEARCH:SEARCH1:TRIGger:A:BUS:AUDio:DATA:OFFSet? might return SEARCH:SEARCH1:TRIGGER:A:BUS:AUDIO:DATA:OFFSET 1 indicating data offset value is set to TDM channel 1.

SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATA:QUALifier

This command sets or queries the qualifier to be used when searching on an audio bus signal. The search condition must be set to DATA using SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:{NONTdm|TDM}:CONDition. The search number is specified by <x>.

Conditions	Requires option 5-SRAUDIO or SUP5-SRAUDIO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS: AUDio:DATA:QUALifier {EQual UNEQual LESSthan MOREthan LESSEQual MOREEQual INrange OUTrange} SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATA:QUALifier?
Arguments	LESSthan sets the qualifier to less than. MOREthan sets the qualifier to greater than. EQual sets the qualifier to equal. UNEQual sets the qualifier to not equal. LESSEQual sets the qualifier to less than or equal.

MOREEqual sets the qualifier to greater than or equal.

INrange sets the qualifier to in range.

OUTrange sets the qualifier to out of range.

Examples **SEARCH:SEARCH1:TRIGger:A:BUS:AUDio:DATa:QUALifier LESSthan**
 sets the qualifier to LESSthan.

SEARCH:SEARCH1:TRIGger:A:BUS:AUDio:DATa:QUALifier? might return
SEARCH:SEARCH1:TRIGGER:A:BUS:AUDIO:DATA:QUALIFIER EQUAL
 indicating the qualifier is set to EQUAL.

SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATa:TDMVALue

This command sets or queries the binary data string for the single or low data word to be used when searching on an TDM audio bus signal. The search condition must be set to DATA using **SEARCH:SEARCH{x}:TRIGger:A:BUS:AUDio:CONDition**.

Group Search and Mark

Syntax **SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATa:TDMVALue <QString>**
SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATa:TDMVALue?

Arguments <QString> is the binary data string for the single or low data word to be used when searching on an TDM audio bus signal.

Examples **SEARCH:SEARCH1:TRIGger:A:BUS:AUDio:DATa:TDMVALue "1100"** sets
 the TDMVALUE to 1100.

SEARCH:SEARCH1:TRIGger:A:BUS:AUDio:DATa:TDMVALue? might
 return **SEARCH:SEARCH1:TRIGGER:A:BUS:AUDIO:DATA:TDMVALUE**
 "XXXXXXXXXXXXXXXXXXXX1010", indicating the TDM value is 1010.

SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATa:VALue

This command sets or queries the binary data string for the single or low data word to be used when searching on an audio bus signal. The search condition must be set to DATA using **SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:{NONTdm|TDM}:CONDition**. The search number is specified by <x>.

Conditions	Requires option 5-SRAUDIO or SUP5-SRAUDIO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATA:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATA:VALue?
Arguments	<QString> is the lower word value.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:AUDio:DATA:VALue "X0X011" sets the data value to X0X011. SEARCH:SEARCH1:TRIGger:A:BUS:AUDio:DATA:VALue? might return SEARCH:SEARCH1:TRIGGER:A:BUS:AUDIO:DATA:VALUE "XXXXXXXXXXXXXXXXXXXXXXXXXXXX", indicating the data value is set to XXXXXXXXXXXXXXXXXXXXXXXXXXXX.

SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATA:WORD

This command sets or queries the alignment of the data (left, right or either) to be used when searching on a non-TDM type audio bus signal. The search condition must be set to DATA using SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:NONTdm:CONDition. The search number is specified by <x>.

Conditions	Requires option 5-SRAUDIO or SUP5-SRAUDIO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATA:WORD {EITHer LEFt RIGht} SEARCH:SEARCH<x>:TRIGger:A:BUS:AUDio:DATA:WORD?
Arguments	EITHer aligns the data to either left or right. LEFt aligns the data to the left. RIGht aligns the data to the right.

- Examples** `SEARCH:SEARCH1:TRIGGER:A:BUS:AUDIO:DATA:WORD LEFT` aligns the data to the left.
- `SEARCH:SEARCH1:TRIGGER:A:BUS:AUDIO:DATA:WORD?` might return `SEARCH:SEARCH1:TRIGGER:A:BUS:AUDIO:DATA:WORD EITHER` indicating the data is aligned either to the left or right.

SEARCH:SEARCH<x>:TRIGGER:A:BUS:CAN:CONDITION

This command sets or queries the search condition for a CAN bus. The search number is specified by <x>.

- Conditions** Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
- Group** Search and Mark
- Syntax** `SEARCH:SEARCH<x>:TRIGGER:A:BUS:CAN:CONDITION`
 `{SOF|FRAMETYPE|IDENTIFIER|DATA|IDANDDATA|EOF|ERROR|FDBITS}`
 `SEARCH:SEARCH<x>:TRIGGER:A:BUS:CAN:CONDITION?`
- Arguments** Arguments specify the CAN bus trigger condition.
- Examples** `SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:CONDITION EOF` sets the CAN bus trigger condition to end of frame.
- `SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:CONDITION?` might return `SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:CONDITION SOF`, indicating the CAN bus trigger condition is set to start of frame.

SEARCH:SEARCH<x>:TRIGGER:A:BUS:CAN:DATA:DIRECTION

This command specifies the CAN search type to be valid on a Read, Write, or Either condition. The search number is specified by <x>.

- Group** Search and Mark
- Syntax** `SEARCH:SEARCH<x>:TRIGGER:A:BUS:CAN:DATA:DIRECTION`
 `{READ|WRITE|NOCARE}`
 `SEARCH:SEARCH<x>:TRIGGER:A:BUS:CAN:DATA:DIRECTION?`

Arguments	READ specifies the read direction. WRITE specifies the write direction. NOCARE specifies either data direction.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:CAN:DATA:DIRection READ sets the data direction to READ. SEARCH:SEARCH1:TRIGger:A:BUS:CAN:DATA:DIRection? might return SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:DATA:DIRECTION NOCARE, indicating the data direction is set to either data direction.

SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:OFFSet

This command sets or queries the data offset value, in bytes, to use when searching on the CAN data field. The search number is specified by x. The search condition must be set to DATA or IDANDDATA.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:OFFSet <NR1> SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:OFFSet?
Related Commands	BUS:B<x>:CAN:STANDard SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:SIZE SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:VALue

Arguments	<NR1> is an integer whose minimum and default values are -1 (don't care) and maximum is up to 7 (for CAN 2.0) or up to 63 (for ISO CAN FD and Non-ISO CAN FD). The maximum is dependent on the number of bytes being matched and the CAN standard selected. Its value is calculated as [Absolute Maximum] - [Data Match Size]. For CAN 2.0, the absolute maximum is 8 bytes. For ISO CAN FD and Non-ISO CAN FD, the absolute maximum is 64 bytes. The minimum data match size is 1 byte, which produces the ranges listed above. Increasing the data match size above 1 byte will adjust the range of valid data offset values accordingly.
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- Examples** `SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:DATA:OFFSET 5` sets the CAN data offset to 5 bytes.
- `SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:DATA:OFFSET?` might return 7, indicating the CAN data offset is 7 bytes. If the CAN standard is set for CAN 2.0 and the search data size is set to 3, the maximum value for the data offset will be 5 ($8 - 3 = 5$). If the CAN standard is set for ISO CAN FD or Non-ISO CAN FD and the search data size is set to 8, the maximum value for the data offset will be 56 ($64 - 8 = 56$).

SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:QUALifier

This command sets or queries the CAN bus trigger data qualifier to be used when searching on a CAN bus signal. The search number is specified by <x>.

- Conditions** Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
- Group** Search and Mark
- Syntax** `SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:QUALifier {EQUa|LESSEQua|MOREEQua|UNEQua|LESSthan|MOREthan}`
`SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:QUALifier?`
- Arguments** Arguments are the data qualifier types.
- Examples** `SEARCH:SEARCH1:TRIGger:A:BUS:CAN:DATA:QUALifier UNEQua` sets the data qualifier to unequal.
- `SEARCH:SEARCH1:TRIGger:A:BUS:CAN:DATA:QUALifier?` might return `SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:DATA:QUALIFIER EQUAL`, indicating that the data qualifier is set to equal.

SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:SIZE

This command sets or queries the length of the data string, in bytes, to be used when searching on a CAN bus signal. The search condition must be set to IDANDDATA or DATA. The search number is specified by <x>.

- Conditions** Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:SIZE <NR1> SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:SIZE?
Arguments	<NR1> specifies the data size.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:CAN:DATA:SIZE 1 sets the data size to 1. SEARCH:SEARCH1:TRIGger:A:BUS:CAN:DATA:SIZE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:DATA:SIZE 1, indicating the data size is set to 1.

SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:VALue

This command sets or queries the binary data value to be used when searching on a CAN bus signal. The search condition must be set to IDANDDATA OR DATA.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:DATA:VALue <QString>
Arguments	<QString>
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:CAN:DATA:VALue "1111" sets the data value to 1111. SEARCH:SEARCH1:TRIGger:A:BUS:CAN:DATA:VALue? might return SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:DATA:VALUE "1010" indicating the data value is 1010.

SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:ERRType

This command sets or queries the type of error condition for a CAN bus to search on. The search number is specified by x. The search condition must be set to ERRor.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
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Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:ERRType {ACKMISS BITSTUFFing FORMERror ANYERror} SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:ERRType?</pre>
Arguments	ACKMISS specifies a search based on a missing ACK field. BITSTUFFing specifies a search based on a bit stuffing error. FORMERror specifies a search based on a CAN FD form error. To use this option, the CAN standard must be set to FDISO or FDNONISO. ANYERror specifies a search based on any error type.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:ERRTYPE ACKMISS specifies searching for any missing ACK fields. SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:ERRTYPE? might return ANYERROR, indicating that the bus is being searched for all error types.

SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:FD:BRSBit

This command sets or queries the value of the bit rate switch bit (BRS bit) for a CAN bus to search on. The search number is specified by x. The search condition must be set to FDBITS, and the CAN standard must be FDISO or FDNONISO.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:FD:BRSBit {ONE ZERO NOCARE} SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:FD:BRSBit?</pre>
Arguments	ONE filters CAN FD packets to only match those where the BRS bit has a value of 1 (fast data enabled). ZERO filters CAN FD packets to only match those where the BRS bit has a value of 0 (fast data disabled). NOCARE disables filtering of CAN FD packets on the BRS bit.

- Examples** `SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:FD:BRSBit` ONE specifies filtering CAN FD packets for those where the BRS bit has a value of 1.
- `SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:FD:BRSBIT?` might return NOCARE, indicating that CAN FD packets are not being filtered based on the BRS bit value.

SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:FD:ESIBit

This command sets or queries the value of the error state indicator bit (ESI bit) for a CAN bus to search on. The search number is specified by x. The search condition must be set to FDBITS, and the CAN standard must be FDISO or FDNONISO.

- Conditions** Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

- Group** Search and Mark

- Syntax** `SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:FD:ESIBit`
 `{ONE|ZERO|NOCARE}`
 `SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:FD:ESIBit?`

- Arguments** ONE filters CAN FD packets to only match those where the ESI bit has a value of 1 (recessive).
- ZERO filters CAN FD packets to only match those where the ESI bit has a value of 0 (dominant).
- NOCARE disables filtering of CAN FD packets on the ESI bit.

- Examples** `SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:FD:ESIBit` ONE specifies filtering CAN FD packets for those where the ESI bit has a value of 1.
- `SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:FD:ESIBIT?` might return NOCARE, indicating that CAN FD packets are not being filtered based on the ESI bit value.

SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:FRAMEtype

This command sets or queries CAN bus trigger frame type to be used when searching on a CAN bus signal. The search condition must be set to FRAMEtype. The search number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:FRAMetype {DATA ERRor OVERLoad REMote} SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:FRAMetype?
Arguments	Arguments are the available frame types.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:CAN:FRAMetype ERROR sets the frame type to error. SEARCH:SEARCH1:TRIGger:A:BUS:CAN:FRAMetype? might return SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:FRAMETYPE DATA, indicates the frame type is set to data.

SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:IDentifier:MODE

This command sets or queries the CAN bus trigger identifier (address) mode to be used when searching on a CAN bus signal. The search number is specified by <x>. The search condition must be set to IDANDDATA or DATA.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:IDentifier:MODE {EXTENDED STANDARD} SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:IDentifier:MODE?
Arguments	EXTENDED specifies the extended identifier mode. STANDARD specifies the standard identifier mode.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:CAN:IDentifier:MODE EXTEND sets the identifier mode to extended.

SEARCH:SEARCH1:TRIGger:A:BUS:CAN:IDentifier:MODE? might return
 SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:IDENTIFIER:MODE STANDARD,
 indicating the identifier mode is set to standard.

SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:IDentifier:VALue

This command sets or queries CAN bus trigger identifier (address) value to be used when searching on a CAN bus signal. The search number is specified by <x>. The search condition must be set to IDANDDATA or DATA.

Conditions Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:IDentifier:VALue
 <QString>
 SEARCH:SEARCH<x>:TRIGger:A:BUS:CAN:IDentifier:VALue?

Arguments <QString> is the identifier value.

Examples SEARCH:SEARCH1:TRIGger:A:BUS:CAN:IDentifier:VALue "1010" sets the identifier value to 1010.
 SEARCH:SEARCH1:TRIGger:A:BUS:CAN:IDentifier:VALue? might return
 SEARCH:SEARCH1:TRIGGER:A:BUS:CAN:IDENTIFIER:VALUE "101011",
 indicating the identifier value is 101011.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition

This command specifies a field or condition for an Ethernet bus to search on. The search number is specified by <x>.

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition
 {SFD|MACADDRESS|MACLENGTH|IPHeader|TCPHeader|DATA|EOP|

IDLe|FCSError|QTAG}

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition?

Related Commands Most of the other TRIGger:A:BUS:B<x>:ETHERnet commands are impacted by the setting of this command.

Arguments

- SFD – Start of frame delimiter.
- MACADDRESS – MAC addresses field.
- MACLEngth – MAC length/type field.
- IPHeader – IP header field.
- TCPHeader – TCP header field.
- DATA – TCP/IPv4 or MAC protocol client data field.
- EOP – End of Packet field.
- IDLe – Idle field.
- FCSError – Frame Check Sequence Error (CRC) field.
- QTAG – IEEE 802.1Q (VLAN) control information field.

Examples

SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:CONDITION MACADDRESS specifies MACADDRESS as the field within an Ethernet frame to search on.

SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:CONDITION? might return DATA, indicating that DATA is the currently specified field within an Ethernet frame to search on.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:HIVALue

This command sets or queries the binary data value to be used when searching on an Ethernet bus signal. The search condition must be set to DATA and the data qualifier to inside or outside range. The search number is specified by <x>.

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:HIVALue <QString>

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:HIVALue?

Arguments <QString> is the binary data value to be used when searching on an Ethernet bus signal.

Examples SEARCH:SEARCH1:TRIGger:A:BUS:ETHERnet:DATA:HIVALue "101011" sets the data HIVALue to 101011.

SEARCH:SEARCH1:TRIGger:A:BUS:ETHERnet:DATA:HIVALue? might return SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:DATA:HIVALUE "10101".

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:OFFSet

This command specifies the data offset value, in bytes, to use when searching on the Ethernet data field. The search condition needs to be set to DATA. The search number is specified by <x>.

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:OFFSet <NR1>
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:OFFSet?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition](#)
[SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:VALue](#)
[SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:SIZE](#)

Arguments <NR1> is an integer whose minimum and default values are -1 (don't care) and maximum is 1,499.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:DATA:OFFSet 36 sets the data offset to 36 bytes.

SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:DATA:OFFSet? might return -1, indicating that the data offset value is the default value, -1, meaning "don't care".

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:QUALifier

This command specifies the qualifier to be used when searching on an Ethernet bus signal. The search condition must be set to DATA. The search number is specified by <x>.

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:QUALifier
{LESSthan|MOREthan|EQua1|UNEQua1|LESSEQua1|MOREEQua1|INrange|OUTrange}

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:QUALifier?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition](#)

Arguments LESSthan sets the qualifier to less than.
MOREthan sets the qualifier to greater than.
EQua1 sets the qualifier to equal.
UNEQua1 sets the qualifier to not equal.
LESSEQua1 sets the qualifier to less than or equal.
MOREEQua1 sets the qualifier to greater than or equal.
INrange sets the qualifier to in range.
OUTrange sets the qualifier to out of range.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:DATA:QUALIFIER
LESSTHAN sets the qualifier to "less than".

SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:DATA:QUALIFIER? might
return SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:DATA:QUALIFIER
EQUAL indicating that the qualifier is set to EQUAL.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:SIZE

This command specifies the length of the data string, in bytes, to use when searching on the Ethernet bus signal. The search condition needs to be set to DATA. The search number is specified by <x>.

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:SIZE <NR1>
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:SIZE?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition](#)

Arguments The minimum and default values are 1, and the maximum value is 16.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:DATA:SIZE 4 specifies 4 as the number of contiguous TCP/IPv4/MAC client bytes to use when searching on the Ethernet data field.

SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:DATA:SIZE? might return 6, indicating that 6 is the currently specified number of contiguous TCP/IPv4/MAC client bytes to use when searching on the Ethernet data field.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:VALue

This command specifies the binary value to use when searching on the Ethernet bus signal. The search condition needs to be set to DATA. The search number is specified by <x>.

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:VALue <QString>
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:VALue?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition](#)
[SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:OFFSet](#)
[SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:SIZE](#)

Arguments <QString> is a quoted string where the allowable characters are 0, 1, and X. The allowable number of characters depends on the setting for size (using [SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:DATA:SIZE](#)). The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:DATA:VALUE "00001000" specifies 00001000 as the value to use when triggering on the Ethernet binary data field, assuming that TRIGGER:A:BUS:ETHERNET:DATA:SIZE is set to 1 byte.

SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:DATA:VALUE? might return "00001000", indicating 00001000 is the currently specified value used when searching on the Ethernet binary data field.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:IPHeader:DESTinationaddr:VALue

This command specifies the binary destination address value to use when searching on an Ethernet bus signal. The search condition needs to be set to IPHeader. The search number is specified by <x>.

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:IPHeader:DESTinationaddr:VALue <QString>
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:IPHeader:DESTinationaddr:VALue?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition](#)

Arguments <QString> is a quoted string of up to 32 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.

Examples

```
SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:IPHEADER:
DESTINATIONADDR:VALUE "11000000101010000000000000000001"
specifies 192.168.0.1 as the value to use when searching on the Ethernet IPv4
header address destination field.
```

```
SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:IPHEADER:
DESTINATIONADDR:VALUE? might return
"11000000101010000000000000000001", indicating that 192.168.0.1 is
the currently specified value used when searching on the Ethernet IPv4 header
address destination field.
```

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:IPHeader:PROTOcol:VALue

This command specifies the binary protocol value to use when searching on the Ethernet bus signal. The search condition needs to be set to IPHeader. The search number is specified by <x>.

NOTE. Commonly used protocol values are 1 (ICMP), 2 (IGMP), 6 (TCP) and 17 (UDP).

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Search and Mark

Syntax

```
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:IPHeader:PROTOcol:
VALue <QString>
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:IPHeader:PROTOcol:
VALue?
```

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition](#)

Arguments <QString> is a quoted string of up to 8 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.

Examples

```
SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:IPHEADER:PROTOCOL:VALUE
"01010010" specifies 01010010 as the value to use when searching on
the Ethernet IP header protocol field.
```

SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:IPHEADER:PROTOCOL:VALUE?
might return "XXXXXXX", indicating that the trigger value has been set to
"don't care".

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:IPHeader:SOURceaddr:VALue

This command specifies the binary source address value to use when searching on an Ethernet bus signal. The search condition needs to be set to **IPHeader**. The search number is specified by <x>.

Conditions	Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:IPHeader:SOURceaddr:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:IPHeader:SOURceaddr:VALue?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition](#)

Arguments QString is a quoted string of up to 32 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:IPHEADER:SOURCEADDR:VALUE
"11000000101010000000000000000001" specifies to use the value of
192.168.0.1 when searching on the Ethernet IP header address source field.

SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:IPHEADER:SOURCEADDR:VALUE
might return "11000000101010000000000000000001", indicating that the
search value has been set to 192.168.0.1.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:MAC:ADDRes:DESTination:VALue

This command specifies the binary MAC address destination value to use when searching on an Ethernet bus signal. The search condition needs to be set to **MACADDRESS**. The search number is specified by <x>.

NOTE. MAC Addresses are 48-bit values such as 08:00:11:1E:C9:AE hex.

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:MAC:ADDRESS:DESTination:VALue <QString>
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:MAC:ADDRESS:DESTination:VALue?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition](#)

Arguments <QString> is a quoted string of up to 48 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:MAC:ADDRESS:DESTINATION:VALUE
"XXXXXXXX0011010111111100000001111010101011001000" specifies to use the value of XX:35:FC:07:AA:C8 hex when searching on the Ethernet MAC address destination field.

SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:MAC:ADDRESS:DESTINATION:VALUE? might return
"XXXXXXXX0011010111111100000001111010101011001000", indicating a MAC address destination field value of XX:35:FC:07:AA:C8 hex.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:MAC:ADDReSS:SOUrce:VALue

This command specifies the binary MAC address source value to use when searching on an Ethernet bus signal. The search condition needs to be set to MACADDRESS. The search number is specified by <x>.

NOTE. MAC Addresses are 48-bit values such as 08:00:11:1E:C9:AE hex.

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Search and Mark

Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:MAC:ADDRESS:SOURce: VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:MAC:ADDRESS:SOURce: VALue?</pre>
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition
Arguments	<QString> is a quoted string of up to 48 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:MAC:ADDRESS:SOURCE:VALUE "XXXXXXXX001101011111100000001111010101011001000" specifies to use the value of XX:35:FC:07:AA:C8 hex when searching on the Ethernet MAC address source field. SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:MAC:ADDRESS:SOURCE:VALUE? might return "XXXXXXXX001101011111100000001111010101011001000", indicating a MAC address source field value of XX:35:FC:07:AA:C8 hex.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:MAC:LENGth:HIVALue

This command specifies the binary MAC length high value to use when searching on an Ethernet bus signal. The search condition needs to be set to **MACADDRESS**. The search number is specified by <x>.

Conditions	Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:MAC:LENGth:HIVALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:MAC:LENGth:HIVALue?</pre>
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition
Arguments	<QString> is a quoted string of up to 16 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.

- Examples** `SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:MAC:LENGTH:HIVALUE "XXXXXXXX00001000"` specifies to use the hexadecimal value XX08 when searching on the Ethernet MAC length.
- `SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:MAC:LENGTH:HIVALUE?` might return `"XXXXXXXX00001000"`, indicating an Ethernet MAC length value of XX08 hex.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:MAC:LENgth:VALue

This command specifies the MAC length value to use when searching on an Ethernet bus signal. The search condition needs to be set to `MACADDRESS`. The search number is specified by `<x>`.

- Conditions** Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

- Group** Search and Mark

- Syntax** `SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:MAC:LENgth:VALue <QString>`
`SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:MAC:LENgth:VALue?`

- Related Commands** [SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition](#)

- Arguments** QString is a quoted string of up to 16 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.

- Examples** `SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:MAC:LENGTH:HIVALUE "XXXXXXXX00001000"` specifies to use the hexadecimal value XX08 when searching on the Ethernet MAC length.
- `SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:MAC:LENGTH:HIVALUE?` might return `"XXXXXXXX00001000"`, indicating an Ethernet MAC length value of XX08 hex.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:QTAG:VALue

This command specifies the binary Q-tag information to use when searching on an Ethernet bus signal. The search condition needs to be set to `QTAG`. The search number is specified by `<x>`.

Conditions	Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:QTAG:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:QTAG:VALue?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition
Arguments	<QString> is a quoted string of up to 32 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:QTAG:VALUE "XXXXXXXXXXXXXXXXXXXX010010001010" specifies to use the value of hexadecimal XXXXX48A when searching on the Ethernet Q-Tag field. SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:QTAG:VALUE? might return "XXXXXXXXXXXXXXXXXXXX010010001010", indicating that hexadecimal XXXXX48A has been set as the Ethernet Q-Tag field search value.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:TCPHeader:ACKnum:VALue

This command specifies the binary ack number value to use when searching on an Ethernet bus signal. The default is all X's (don't care). The search condition needs to be set to TCPHeader. The search number is specified by <x>.

Conditions	Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:TCPHeader:ACKnum: VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:TCPHeader:ACKnum: VALue?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition

Arguments QString is a quoted string of up to 32 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:TCPHEADER:ACKNUM:VALUE "XXXXXXXXXXXXXXXXXXXX00001000" specifies hexadecimal XXXXXX08 as the value to use when searching on the Ethernet TCP header acknowledgement number.

TRIGGER:A:BUS:ETHERNET:TCPHEADER:ACKNUM:VALUE? might return "XXXXXXXXXXXXXXXXXXXX00001000", indicating that hexadecimal XXXXXX08 has been specified as the value to use when searching on the Ethernet TCP header acknowledgement number.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:TCPHeader:DESTinationport:VALue

This command specifies the binary destination port value to use when searching on the Ethernet TCP header destination port number. The search condition needs to be set to TCPHeader. The search number is specified by <x>.

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:TCPHeader:DESTinationport:VALue <QString>
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:TCPHeader:DESTinationport:VALue?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition](#)

Arguments <QString> is a quoted string of up to 16 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:TCPHEADER:DESTINATIONPORT:VALUE "XXXXXXXX00100010" specifies to use the value of hexadecimal XX22 when searching on the Ethernet TCP header destination port number.

SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:TCPHEADER:DESTINATIONPORT:VALUE? might return "XXXXXXXXXXXXXXXXXXXX", indicating that hexadecimal XX22 has been set as the value to use when searching on the Ethernet TCP header destination port number.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:TCPHeader:SEQnum:VALue

This command specifies the binary sequence number value to use when searching on an Ethernet bus signal. The default is all X's (don't care). The search condition needs to be set to TCPHeader. The search number is specified by <x>.

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:TCPHeader:SEQnum:VALue <QString>
SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:TCPHeader:SEQnum:VALue?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition](#)

Arguments QString is a quoted string of up to 32 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:TCPHEADER:SEQNUM:VALUE "XXXXXXXXXXXXXXXXXXXX000100010001" specifies to use the value of hexadecimal XXXXX111 when searching on the Ethernet TCP header sequence number.

SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:TCPHEADER:SEQNUM:VALUE? might return "XXXXXXXXXXXXXXXXXXXX0010010100", indicating that hexadecimal XXXXX111 has been specified as the value to use when searching on the Ethernet TCP header sequence number.

SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:TCPHeader:SOUrceport:VALue

This command specifies the binary source port value to use when searching on an Ethernet bus signal. The search condition needs to be set to TCPHeader. The search number is specified by <x>.

Conditions	Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:TCPHeader: Sourceport:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:TCPHeader: Sourceport:VALue?</pre>
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:BUS:ETHERnet:CONDition
Arguments	<QString> is a quoted string of up to 16 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.
Examples	<pre>SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:TCPHEADER: SOURCEPORT:VALUE "XXXX000010100110" specifies to use the value of hexadecimal X0A6 when searching on the Ethernet TCP header source port number.</pre> <pre>SEARCH:SEARCH1:TRIGGER:A:BUS:ETHERNET:TCPHEADER: SOURCEPORT:VALUE? might return "XXXXX01001010110", indicating that hexadecimal X0A6 has been specified as the value to use when searching on the Ethernet TCP header source port number.</pre>

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:CONDition

This command sets or queries the FlexRay bus search condition. The search number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS: FLEXRAY:CONDition {SOF FRAMEType IDentifier CYCLEcount HEADER DATA IDANDDATA EOF ERROR} SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:CONDition?</pre>

Arguments Arguments are the available trigger conditions.

Examples SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:CONDition SOF sets the trigger condition to start of frame.

SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:CONDition? might return SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:CONDITION SOF, indicating the FlexRay trigger condition is start of frame.

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:CYCLEcount:HIVALue

This command sets or queries the heigh value when searching on a FlexRay bus cycle count field. The search number is specified by <x>. The search condition must be set to CYCLEcount.

Conditions Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:CYCLEcount:HIVALue <QString>
SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:CYCLEcount:HIVALue?

Arguments <QString> specifies the cycle count value.

Examples SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:CYCLEcount:HIVALue "1010" sets the cycle count value to XX1010.

SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:CYCLEcount:HIVALue? might return SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:CYCLECOUNT:HIVALUE "XXXXXX", indicating the cycle count value is don't care, and it will trigger on any cycle count.

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:CYCLEcount:QUALifier

This command sets or queries the qualifier to be used when searching on a FlexRay bus search cycle count field. The search number is specified by <x>. The search condition must be set to CYCLEcount.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:CYCLEcount:QUALifier {EQUAL LESSEqual MOREEqual UNEQUAL LESSthan MOREthan INrange OUTrange} SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:CYCLEcount:QUALifier?</pre>
Arguments	Arguments are the available cycle count qualifiers.
Examples	<pre>SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:CYCLEcount:QUALifier LESSthan sets the cycle count qualifier to less than. SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:CYCLEcount:QUALifier? might return SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:CYCLECOUNT:QUALIFIER EQUAL, indicating the cycle count qualifier is set to equal.</pre>

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:CYCLEcount:VALue

This command sets or queries the low value when searching on a FlexRay bus cycle count field. The search number is specified by <x>. The search condition must be set to CYCLEcount.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:CYCLEcount:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:CYCLEcount:VALue?</pre>
Arguments	<QString> specifies the cycle count value.
Examples	<pre>SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:CYCLEcount:VALue "1010" sets the cycle count value to XX1010.</pre>

SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:CYCLEcount:VALue? might return SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:CYCLECOUNT:VALUE "XXXXXX", indicating the cycle count value is don't care, and it will trigger on any cycle count.

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATA:HIVALue

This command sets or queries the high value when searching on a FlexRay bus data field. The search number is specified by <x>. The search condition must be set to IDANDDATA OR DATA.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATA:HIVALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATA:HIVALue?
Arguments	<QString> specifies the data value.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:DATA:HIVALue "1010" sets the data value to XXXXXX1010. SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:DATA:HIVALue? might return SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:DATA:HIVALUE "XXXXXXXXXX", indicating the data value is a don't care.

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATA:OFFSet

This command sets or queries the offset of the data string, in bytes, when searching on a FlexRay bus data field. The search number is specified by <x>. The search condition must be set to IDANDDATA OR DATA.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark

Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATA:OFFSet <NR3> SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATA:OFFSet?
Arguments	<NR3> specifies the data offset in bytes. A setting of X/Don't care is accomplished by setting the offset to -1.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:DATA:OFFSet 2 sets the data offset to 2 bytes. SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:DATA:OFFSet? might return SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:DATA:OFFSET -1, indicating the data offset is don't care.

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATA:QUALifier

This command sets or queries the qualifier to be used when searching on a FlexRay bus signal. The search number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATA:QUALifier {EQUAL LESSEQUAL MOREEQUAL UNEQUAL LESSthan MOREthan INrange OUTrange} SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATA:QUALifier?
Arguments	Arguments are the available data qualifiers.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:DATA:QUALifier LESSTHAN sets the data qualifier to less than. SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:DATA:QUALifier? might return SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:DATA:QUALIFIER EQUAL, indicating the data qualifier is equal.

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATA:SIZE

This command sets or queries the length of the data string, in bytes, to be used when searching on a FlexRay bus data field. The search number is specified by <x>. The search condition must be set to IDANDDATA OR DATA.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATA:SIZE <NR1> SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATA:SIZE?
Arguments	<NR1> specifies the data size in bytes. A data size of -1 is don't care.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:DATA:SIZE 2 sets the data size to 2. SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:DATA:SIZE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:DATA:SIZE 1, indicating the data size is 1 byte.

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATA:VALue

This command sets or queries the low value to be used when searching on a FlexRay bus data field. The search number is specified by <x>. The search condition must be set to IDANDDATA or DATA.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATA:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:DATA:VALue?
Arguments	<QString> specifies the data value.

- Examples** `SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:DATA:VALUE "1010"` sets the data value to `XXXXXX1010`.
- `SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:DATA:VALUE?` might return `SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:DATA:VALUE "XXXXXXXXXX"`, indicating the data value is a don't care.

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:EOFTYPE

This command sets or queries the end of frame type when searching on a FlexRay bus signal. The search number is specified by <x>. The search condition must be set to EOF.

- Conditions** Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

- Group** Search and Mark

- Syntax** `SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:EOFTYPE`
 `{ANY|STATIC|DYNAMIC}`
 `SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:EOFTYPE?`

- Arguments** ALL specifies either end of file type.
- STATIC specifies the static end of file type.
- DYNAMIC specifies the static end of file type.

- Examples** `SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:EOFTYPE STATIC` sets the end of file type to static.
- `SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:EOFTYPE?` might return `SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:EOFTYPE ANY`, indicating either end of file type will cause a trigger.

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:ERRTYPE

This command sets or queries the error type when searching on a FlexRay bus signal. The search number is specified by <x>. The search condition must be set to ERROR.

- Conditions** Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:ERRTYPE {CRCHeader CRCTrailer NULLFRStatic NULLFRDynamic SYNCFrame STARTupnosync} SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:ERRTYPE?</pre>
Arguments	Arguments are the available error types.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:ERRTYPE SYNCFRAME sets the error type to SYNCFRAME. SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:ERRTYPE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:ERRTYPE CRCHEADER, indicating the error type is CRCHEADER.

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXray:FRAMEID:HIVALue

This command sets or queries the high value when searching on a FlexRay bus frame id field. The search condition must be set to IDentifier. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXray:FRAMEID:HIVALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXray:FRAMEID:HIVALue?</pre>
Arguments	<QString> is the frame id high value.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:FLEXray:FRAMEID:HIVALue "XXXXXXXX1010" sets the HIVALUE to XXXXXXXX1010. SEARCH:SEARCH1:TRIGger:A:BUS:FLEXray:FRAMEID:HIVALue? might return SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:FRAMEID:HIVALUE "XXXXXXXXXXXX" indicating the HIVALUE is XXXXXXXXXXXX.

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXray:FRAMEID:QUALifier

This command sets the qualifier to be used when searching on a FlexRay bus signal. The search condition must be set to Identifier. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXray:FRAMEID:QUALifier {EQUAL UNEQUAL LESSthan MOREthan LESSEQUAL MOREEQUAL INrange OUTrange} SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXray:FRAMEID:QUALifier?</pre>
Arguments	Arguments are the available data qualifiers.
Examples	<pre>SEARCH:SEARCH1:TRIGger:A:BUS:FLEXray:FRAMEID:QUALifier UNEQUAL sets the qualifier to unequal. SEARCH:SEARCH1:TRIGger:A:BUS:FLEXray:FRAMEID:QUALifier? might return SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:FRAMEID:QUALIFIER EQUAL indicating the qualifier is set to equal.</pre>

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXray:FRAMEID:VALue

This command sets the low value when searching on a FlexRay bus id field. The search condition must be set to IDentifier. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXray:FRAMEID:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXray:FRAMEID:VALue?</pre>
Arguments	<QString> is the frame id value.
Examples	<pre>SEARCH:SEARCH1:TRIGger:A:BUS:FLEXray:FRAMEID:VALue "XXXXXXXX1010" sets the value to XXXXXXXX1010. SEARCH:SEARCH1:TRIGger:A:BUS:FLEXray:FRAMEID:VALue? might return SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:FRAMEID:VALUE "XXXXXXXXXXXX" indicating the value is XXXXXXXXXXXX.</pre>

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXray:FRAMEType

This command sets or queries the FlexRay bus search frame type. The search number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXray:FRAMEType {NORMa1 PAYload NULL SYNC STARTUp} SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXray:FRAMEType?
Arguments	Arguments are the available frame types.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:FRAMEType startup sets the frame type to startup. SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:FRAMEType? might return SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:FRAMETYPE NORMAL, indicating the frame type is normal.

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:CRC

This command sets or queries the CRC portion of the binary header string when searching on a FlexRay bus header. The search number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:CRC <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:CRC?
Arguments	<QString> specifies the CRC.
Examples	SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:CRC "1010" sets the header CRC to XXXXXX101. SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:CRC? might return SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:HEADER:CRC "XXXXXXXXXXXX", indicating the header CRC is a don't care.

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:CYCLECount

This command sets or queries the cycle count portion of the binary header string when searching on a FlexRay bus header. The search number is specified by <x>. The search condition must be set to HEADer.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:CYCLECount <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:CYCLECount?
Arguments	<QString> specifies the header cycle count.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:HEADER:CYCLECount "1010" sets the header cycle count to XX1010. SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:HEADER:CYCLECount? might return SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:HEADER:CYCLECOUNT "XXXXXX" indicating the cycle count has not been set.

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:FRAMEID

This command sets or queries the frame id portion of the binary header string when searching on a FlexRay bus header. The search number is specified by <x>. The search condition must be set to HEADer or IDANDDATA.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:FRAMEID <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:FRAMEID?
Arguments	<QString> specifies the frame ID.

Examples `SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:HEADER:FRAMEID "1010"`
 sets the header frame ID to XXXXXXXX1010.

`SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:HEADER:FRAMEID?` might
 return `SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:HEADER:FRAMEID`
 "XXXXXXXXXX", indicating the frame ID is a don't care.

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:INDBits

This command sets or queries the Indicator bits portion of the binary header string when searching on a FlexRay bus header. The search number is specified by <x>. The search condition must be set to HEADer.

Conditions Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:INDBits`
 <QString>
 `SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:INDBits?`

Arguments <QString> specifies the header Indicator Bits.

Examples `SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:INDBits "1010"`
 sets the header Indicator Bits to X1010.

`SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:INDBits?` might
 return `SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:HEADER:INDBITS`
 "XXXXX", indicating the Indicator Bits have not been set.

SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:PAYLENgth

This command sets or queries the payload length portion of the binary header string when searching on a FlexRay bus search trigger header. The search number is specified by <x>. The search condition must be set to HEADer.

Conditions Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group Search and Mark

Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:PAYLEnGth <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:FLEXRAY:HEADER:PAYLEnGth?</pre>
Arguments	<QString> specifies the header Payload Length.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:HEADER:PAYLEnGth "1010" sets the Payload Length to 1010. SEARCH:SEARCH1:TRIGger:A:BUS:FLEXRAY:HEADER:PAYLEnGth? might return SEARCH:SEARCH1:TRIGGER:A:BUS:FLEXRAY:HEADER:PAYLENGTH "XXXXXXX", indicating the Payload Length has not been set.

SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:ADDReSS:MODe

This command sets or queries the I2C address mode for the specified bus search to determine where to place a mark. The search number is specified by <x>.

Conditions	Requires option 5-SREMBD or SUP5-SREMBD Triggering and Analysis application.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:ADDReSS:MODe {ADDR10 ADDR7} SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:ADDReSS:MODe?</pre>
Arguments	ADDR10 specifies the address mode as ADDR10. ADDR7 specifies the address mode as ADDR7.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:I2C:ADDRESS:MODE ADDR10 sets the address mode for I2C bus trigger search 1 to ADDR10. SEARCH:SEARCH2:TRIGGER:A:BUS:I2C:ADDRESS:MODE? might return :SEARCH:SEARCH2:TRIGGER:A:BUS:I2C:ADDRESS:MODE ADDR7, indicating that the address mode for I2C bus trigger search 2 is set to ADDR7.

SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:ADDReSS:VALue

This command sets or queries the binary address string used for the I2C search the specified search condition is Address or AddressData. The search number is specified by <x>.

Conditions Requires option 5-SREMBD or SUP5-SREMBD Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:ADDReSS:VALue <QString>
SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:ADDReSS:VALue?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:ADDReSS:MODE](#)

Arguments <QString> specifies the address value. This is either a 7-bit or 10-bit value depending on the address mode. The valid characters are 0-9, A-F, and X for addresses in hexadecimal format; and 0, 1, and X otherwise.

Examples SEARCH:SEARCH2:TRIGGER:A:BUS:I2C:ADDRESS:VALUE "01XXXXX" sets the address value to "01XXXXX" when the mode is ADDR7 and the format is binary.

SEARCH:SEARCH1:TRIGGER:A:BUS:I2C:ADDRESS:VALUE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:I2C:ADDRESS:VALUE "XX", indicating that the address value is "XX" when the address mode is set to ADDR7 and the address format is hexadecimal.

SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:CONDition

This command sets or queries the search condition for an I2C bus. The search number is specified by <x>.

Conditions Requires option 5-SREMBD or SUP5-SREMBD Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:CONDition
{ADDRESS|ADDRANDDATA|DATA|ACKMISS|REPEATstart|START|STOP}

SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:CONDition?

- Arguments**
- ADDRESS specifies the trigger condition as Address.
 - ADDRANDDATA specifies the trigger condition as Address and Data.
 - DATA specifies the trigger condition as Data.
 - ACKMISS specifies the trigger condition as Missing of Acknowledgement.
 - REPEATstart specifies the trigger condition as Repeat of Start.
 - START specifies the trigger condition as Start.
 - STOP specifies the trigger condition as Stop.

- Examples**
- SEARCH:SEARCH1:TRIGGER:A:BUS:I2C:CONDITION ADDRESS sets the trigger condition for I2C bus trigger search 1 to Address.
- SEARCH:SEARCH1:TRIGGER:A:BUS:I2C:CONDITION? might return
SEARCH:SEARCH1:TRIGGER:A:BUS:I2C:CONDITION ADDRANDDATA,
indicating that the trigger condition for I2C bus trigger search 1 is set to Address and Data.

SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:DATA:DIRection

This command sets or queries the direction of the data for the I2C bus search to determine where to place a mark. The search number is specified by <x>. Read or write is indicated by the R/W bit in the I2C protocol.

- Conditions** Requires option 5-SREMBD or SUP5-SREMBD Triggering and Analysis application.

- Group** Search and Mark

- Syntax**
- SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:DATA:DIRection
{NOCARE|READ|WRITE}
SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:DATA:DIRection?

- Arguments**
- NOCARE specifies the direction of data as Don't Care.
 - READ specifies the direction of data as Read.
 - WRITE specifies the direction of data as Write.

Examples `SEARCH:SEARCH1:TRIGGER:A:BUS:I2C:DATA:DIRECTION READ` sets the data direction for I2C bus trigger search 1 to READ.

`SEARCH:SEARCH2:TRIGGER:A:BUS:I2C:DATA:DIRECTION?` might return `:SEARCH:SEARCH2:TRIGGER:A:BUS:I2C:DATA:DIRECTION DONTCARE`, indicating that the data direction of the I2C bus trigger search 2 is DONTCare.

SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:DATA:SIZE

This command sets or queries the length of the data string in bytes used for an I2C bus search to determine where to place a mark. The search number is specified by <x>. The search condition must be DATA or ADDRANDDATA.

Conditions Requires option 5-SREMBD or SUP5-SREMBD Triggering and Analysis application.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:DATA:SIZE <NR1>`
`SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:DATA:SIZE?`

Arguments <NR1> specifies the data size in bytes.

Examples `SEARCH:SEARCH1:TRIGGER:A:BUS:I2C:DATA:SIZE 1` sets the length of the data string for I2C bus trigger search 1 to 1 byte.

`SEARCH:SEARCH1:TRIGGER:A:BUS:I2C:DATA:SIZE?` might return `SEARCH:SEARCH1:TRIGGER:A:BUS:I2C:DATA:SIZE 3`, indicating that the length of the data string for I2C bus trigger search 1 is 3 bytes.

SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:DATA:VALue

This command sets or queries the binary data string used for I2C bus search to determine where to place a mark. The search number is specified by <x>. The search condition must be DATA or ADDRANDDATA.

Conditions Requires option 5-SREMBD or SUP5-SREMBD Triggering and Analysis application.

Group Search and Mark

Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:DATA:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:I2C:DATA:VALue?
Arguments	<QString> specifies the data value. The valid characters are 0, 1, or X for binary format; and A-F, 0-9, and X for hexadecimal format.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:I2C:DATA:VALUE "1001" sets the data value for I2C bus trigger search 1 to "1001". SEARCH:SEARCH2:TRIGGER:A:BUS:I2C:DATA:VALUE? might return :SEARCH:SEARCH2:TRIGGER:A:BUS:I2C:DATA:VALUE "XX", indicating that the data value for I2C bus trigger search is "XX" in hexadecimal format.

SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:CONDition

This command sets or queries the condition for a LIN bus search. The search number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:CONDition {DATA IDANDDATA ERRor IDentifier SLEEP SYNCfield WAKEup} SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:CONDition?
Arguments	Arguments are the available trigger conditions.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:LIN:CONDition DATA sets the trigger condition to data. SEARCH:SEARCH1:TRIGger:A:BUS:LIN:CONDition? might return SEARCH:SEARCH1:TRIGGER:A:BUS:LIN:CONDITION SYNC, indicating the trigger condition is sync.

SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:DATA:HIVALue

This command sets or queries the high data value string used in a LIN bus search. The search number is specified by <x>. The search condition must be DATA or IDANDDATA and the data qualifier must be INRANGE or OUTRANGE.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:DATA:HIVALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:DATA:HIVALue?
Arguments	<QString> is a quoted string of 1s, 0s, or Xs representing the binary data string to be used in a LIN search if the search condition is IDentifier or IDANDDATA (identifier and data).
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:LIN:DATA:HIVALUE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:LIN:DATA:HIVALUE "XXX XXXXXXXXXXXX" indicating the high value is "don't care".

SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:DATA:QUALifier

This command sets or queries the data qualifier used in a LIN bus search. The search number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:DATA:QUALifier {Equal LESSEqual MOREEqual UNEQual LESSthan MOREthan INrange OUTrange} SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:DATA:QUALifier?
Arguments	Arguments are the available data qualifiers.
Examples	SEARCH:SEARCH2:TRIGger:A:BUS:LIN:DATA:QUALifier LESS sets the data qualifier to less than. SEARCH:SEARCH1:TRIGger:A:BUS:LIN:DATA:QUALifier? might return SEARCH:SEARCH1:TRIGGER:A:BUS:LIN:DATA:QUALIFIER EQUAL, indicating the data qualifier is equal.

SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:DATA:SIZE

This command sets or queries the length of the stat string in bytes used for a LIN bus search. The search number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:DATA:SIZE <NR1> SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:DATA:SIZE?
Arguments	<NR1> specifies the data size.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:LIN:DATA:SIZE 1.0 sets the data size to 1. SEARCH:SEARCH1:TRIGger:A:BUS:LIN:DATA:SIZE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:LIN:DATA:SIZE 1, indicating the data size is 1.

SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:DATA:VALue

This command sets or queries the data string used for a LIN bus search. The search number is specified by <x>. The search condition must be DATA or IDANDDATA.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:DATA:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:DATA:VALue?
Arguments	<QString> specifies the data value.
Examples	SEARCH:SEARCH2:TRIGger:A:BUS:LIN:DATA:VALue 1010 sets the data value to XXXXXX1010.

SEARCH:SEARCH1:TRIGger:A:BUS:LIN:DATA:VALue? might return
 SEARCH:SEARCH1:TRIGGER:A:BUS:LIN:DATA:VALUE "XXXXXXXXXX",
 indicating that the data value is a don't care.

SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:ERRTYPE

This command sets or queries the error type for a LIN bus search. The search number is specified by <x>. The search condition must be set to ERROR.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:ERRTYPE {Checksum PARity SYNC} SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:ERRTYPE?
Arguments	Checksum specifies the error type is checksum. PARity specifies the error type is parity. SYNC specifies the error type is sync.
Examples	SEARCH:SEARCH2:TRIGger:A:BUS:LIN:ERRTYPE Parity sets the error type to parity. SEARCH:SEARCH2:TRIGger:A:BUS:LIN:ERRTYPE? might return :SEARCH:SEARCH2:TRIGGER:A:BUS:LIN:ERRTYPE SYNC, indicating that the error type is sync.

SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:IDentifier:VALue

This command sets or queries the string used for a LIN bus identifier value. The search number is specified by <x>. The search condition must be IDENTIFIER or IDANDDATA.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Search and Mark

Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:IDentifier:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:LIN:IDentifier:VALue?
Arguments	<QString> specifies the identifier value.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:LIN:IDentifier:VALue 1010 sets the identifier value to XX1010. SEARCH:SEARCH1:TRIGger:A:BUS:LIN:IDentifier:VALue? might return SEARCH:SEARCH1:TRIGGER:A:BUS:LIN:IDENTIFIER:VALUE "XXXXXX", indicates the identifier value is don't care.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:ADDRess:HIVALue

This command sets or queries the high value when searching on command word addresses for a MIL-STD-1553 bus. The search number is specified by x. The search condition must be set to COMMAND, and the address qualifier must be INrange or OUTrange.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:ADDRess:HIVALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:ADDRess:HIVALue?
Arguments	<QString> is the address value.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:COMMAND:ADDRESS:HIVALUE "X1000" sets the value to X1000. SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:COMMAND:ADDRESS:HIVALUE? might return "XXXXXX", indicating that the value is XXXXX.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:ADDRess:QUALifier

This command sets or queries the qualifier to be used when searching on command word addresses for a MIL-STD-1553 bus. The search number is specified by x. The search condition must be set to COMMAND.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:ADDRess:QUALifier {EQUAL UNEQUAL LESSThan MOREthan LESSEQUAL MOREEQUAL INrange OUTrange} SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:ADDRess:QUALifier?
Arguments	Arguments are the available address qualifiers.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:COMMAND:ADDRESS:QUALIFIER LESSTHAN sets the address qualifier to less than. SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:COMMAND:ADDRESS:QUALIFIER? might return EQUAL, indicating that the address qualifier is set to equal.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:ADDRess:VALue

This command sets or queries the low value when searching on command word addresses for a MIL-STD-1553 bus. The search number is specified by x. The search condition must be set to COMMAND.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:ADDRess:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:ADDRess:VALue?

Arguments <QString> is the address value.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:COMMAND:ADDRESS:VALUE "X1000" sets the value to X1000.

SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:COMMAND:ADDRESS:VALUE? might return "XXXXX", indicating that the value is XXXXX.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:COUNT

This command sets or queries the value of the command word "word count" field for a MIL-STD-1553 bus to search on. The search number is specified by x. The search condition must be set to COMMAND.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:COUNT <QString>
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:COUNT?

Arguments <QString> is the word count value.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:COMMAND:COUNT "X1000" sets the value to X1000.

SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:COMMAND:COUNT? might return "XXXXX", indicating that the value is XXXXX.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:PARity

This command sets or queries the value of the command word parity bit for a MIL-STD-1553 bus to search on. The search number is specified by x. The search condition must be set to COMMAND.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:PARity {ONE ZERO NOCARE} SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:PARity?</pre>
Arguments	ONE filters command words to only match those where the parity bit has a value of 1. ZERO filters command words to only match those where the parity bit has a value of 0. NOCARE disables filtering of command words on the parity bit.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:COMMAND:PARITY ONE specifies filtering command words for those where the parity bit has a value of 1. SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:COMMAND:PARITY? might return NOCARE, indicating that command words are not being filtered based on the parity bit value.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:SUBADdress

This command sets or queries the value of the command word subaddress field for a MIL-STD-1553 bus to search on. The search number is specified by x. The search condition must be set to COMMAND.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:SUBADdress <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:SUBADdress?</pre>
Arguments	<QString> is the word count value.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:COMMAND:SUBADDRESS "X1000" sets the value to X1000. SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:COMMAND:SUBADDRESS? might return "XXXXX", indicating that the value is XXXXX.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:TRBit

This command sets or queries the value of the command word Transmit / Receive bit for a MIL-STD-1553 bus to search on. The search number is specified by x. The search condition must be set to COMMAND.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:TRBit
{RX|TX|X}
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:COMMAND:TRBit?

Arguments RX filters command words to only match those that are receive packets.
TX filters command words to only match those that are transmit packets.
X disables filtering of command words on the R/T bit.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:COMMAND:TRBIT TX
specifies filtering command words for only transmit messages.

SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:COMMAND:TRBIT? might
return X, indicating that command words are not being filtered based on the R/T
bit value.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:CONDition

This command sets or queries the field or condition for a MIL-STD-1553 bus to search on. The search number is specified by x.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:CONDition
{SYNC|COMMAND|STATUS|DATA|ERROR}
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:CONDition?

Arguments	SYNC specifies a search for the sync pulse of any word. COMMAND specifies a search for a matching command word. STATUS specifies a search for a matching status word. DATA specifies a search for a matching data word. ERROR specifies a search for a specified error condition.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:CONDITION DATA specifies finding matching data word(s). SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:CONDITION? might return SYNC, indicating that the bus is being searched for sync pulses found in any word.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:DATA:PARity

This command sets or queries the value of the command word parity bit for a MIL-STD-1553 bus to search on. The search number is specified by x. The search condition must be set to DATA.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:DATA:PARity {ONE ZERO NOCARE} SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:DATA:PARity?
Arguments	ONE filters data words to only match those where the parity bit has a value of 1. ZERO filters data words to only match those where the parity bit has a value of 0. NOCARE disables filtering of data words on the parity bit.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:DATA:PARITY ONE specifies filtering data words for those where the parity bit has a value of 1. SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:DATA:PARITY? might return NOCARE, indicating that data words are not being filtered based on the parity bit value.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:DATa:VALue

This command sets or queries the value when searching on data words for a MIL-STD-1553 bus. The search number is specified by x. The search condition must be set to DATA.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:DATa:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:DATa:VALue?
Arguments	<QString> is the data value.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:DATA:VALUE "XXXXXXXXXXXX1000" sets the value to XXXXXXXXXXXXXXX1000. SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:DATA:VALUE? might return "XXXXXXXXXXXXXXXXXXXX", indicating that the value is XXXXXXXXXXXXXXXXXXXX.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:ERRTYPe

This command sets or queries the type of error condition for a MIL-STD-1553 bus to search on. The search number is specified by x. The search condition must be set to ERRor.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:ERRTYPe {PARity SYNC DATA} SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:ERRTYPe?

Arguments **PARity** specifies searching for an incorrectly calculated parity bit in any word.

SYNC specifies searching for any sync pulse that does not transition in the middle of the pulse as required.

DATA specifies searching for any non-contiguous data words.

Examples **SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:ERRTYPE DATA** specifies searching for non-contiguous data words.

SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:ERRTYPE? might return **PARITY**, indicating that the bus is being searched for parity errors in any word.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:ADDRes:HIVALue

This command sets or queries the high value when searching on status word addresses for a MIL-STD-1553 bus. The search number is specified by x. The search condition must be set to **STATus** and the address qualifier must be **INrange** or **OUTrange**.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Search and Mark

Syntax **SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:ADDRes:HIVALue <QString>**
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:ADDRes:HIVALue?

Arguments <QString> is the address value.

Examples **SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:ADDRESS:HIVALUE "X1000"** sets the value to X1000.

SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:ADDRESS:HIVALUE? might return "XXXXX", indicating that the value is XXXXX.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:ADDRes:QUALifier

This command sets or queries the qualifier used when searching on status word addresses for a MIL-STD-1553 bus. The search number is specified by x. The search condition must be set to **STATus**.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS: MIL1553B:STATus:ADdResS:QUALifier {EQUAL UNEQUAL LESSthan MOPethan LESSEQUAL MOREEQUAL INrange OUTrange} SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:ADdResS: QUALifier?</pre>
Arguments	Arguments are the available address qualifiers.
Examples	<pre>SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:ADDRESS:QUALIFIER LESSTHAN</pre> sets the address qualifier to less than. <pre>SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:ADDRESS:QUALIFIER?</pre> might return EQUAL, indicating that the address qualifier is set to equal.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:ADdResS:VALue

This command sets or queries the low value when searching on status word addresses for a MIL-STD-1553 bus. The search number is specified by x. The search condition must be set to STATUS.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:ADdResS:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:ADdResS: VALue?</pre>
Arguments	<QString> is the address value.
Examples	<pre>SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:ADDRESS:VALUE "X1000"</pre> sets the value to X1000.

SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:ADDRESS:VALUE?
might return "XXXXX", indicating that the value is XXXXX.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:BCR

This command sets or queries the value of the broadcast command received bit (BCR bit, bit 15) in a status word for a MIL-STD-1553 bus to search on. The search number is specified by x. The search condition must be set to STATus.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:BCR {ONE ZERo NOCARE} SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:BCR?
Arguments	ONE filters status words to only match those where the BCR bit has a value of 1. ZERo filters status words to only match those where the BCR bit has a value of 0. NOCARE disables filtering of status words on the BCR bit.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:BIT:BCR ONE specifies filtering status words for those where the BCR bit has a value of 1. SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:BIT:BCR? might return NOCARE, indicating that status words are not being filtered based on the BCR bit value.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:BUSY

This command sets or queries the value of the busy bit (BUSY bit, bit 16) in a status word for a MIL-STD-1553 bus to search on. The search number is specified by x. The search condition must be set to STATus.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark

Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:BUSY {ONE ZERO NOCARE} SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:BUSY?</pre>
Arguments	ONE filters status words to only match those where the BUSY bit has a value of 1. ZERO filters status words to only match those where the BUSY bit has a value of 0. NOCARE disables filtering of status words on the BUSY bit.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:BIT:BUSY ONE specifies filtering status words for those where the BUSY bit has a value of 1. SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:BIT:BUSY? might return NOCARE, indicating that status words are not being filtered based on the BUSY bit value.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:DBCA

This command sets or queries the value of the dynamic bus control acceptance bit (DBCA bit, bit 18) in a status word for a MIL-STD-1553 bus to search on. The search number is specified by x. The search condition must be set to STATUS.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:DBCA {ONE ZERO NOCARE} SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:DBCA?</pre>
Arguments	ONE filters status words to only match those where the DBCA bit has a value of 1. ZERO filters status words to only match those where the DBCA bit has a value of 0. NOCARE disables filtering of status words on the DBCA bit.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:BIT:DBCA ONE specifies filtering status words for those where the DBCA bit has a value of 1. SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:BIT:DBCA? might return NOCARE, indicating that status words are not being filtered based on the DBCA bit value.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:INSTR

This command sets or queries the value of the instrumentation bit (INSTR bit, bit 10) in a status word for a MIL-STD-1553 bus to search on. The search number is specified by x. The search condition must be set to STATUS.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:INSTR {ONE ZERO NOCARE} SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:INSTR?
Arguments	ONE filters status words to only match those where the INSTR bit has a value of 1. ZERO filters status words to only match those where the INSTR bit has a value of 0. NOCARE disables filtering of status words on the INSTR bit.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:BIT:INSTR ONE specifies filtering status words for those where the INSTR bit has a value of 1. SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:BIT:INSTR? might return NOCARE, indicating that status words are not being filtered based on the INSTR bit value.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:ME

This command sets or queries the value of the message error bit (ME bit, bit 9) in a status word for a MIL-STD-1553 bus to search on. The search number is specified by x. The search condition must be set to STATUS.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:ME {ONE ZERO NOCARE} SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:ME?

Arguments ONE filters status words to only match those where the ME bit has a value of 1.
ZERO filters status words to only match those where the ME bit has a value of 0.
NOCARE disables filtering of status words on the ME bit.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:BIT:ME ONE
specifies filtering status words for those where the ME bit has a value of 1.
SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:BIT:ME? might
return NOCARE, indicating that status words are not being filtered based on
the ME bit value.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:SRQ

This command sets or queries the value of the status word service request bit (SRQ bit, bit 11) in a status word for a MIL-STD-1553 bus to search on. The search number is specified by x. The search condition must be set to STATUS.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:SRQ
{ONE|ZERO|NOCARE}
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:SRQ?

Arguments ONE filters status words to only match those where the SRQ bit has a value of 1.
ZERO filters status words to only match those where the SRQ bit has a value of 0.
NOCARE disables filtering of status words on the SRQ bit.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:BIT:SRQ ONE
specifies filtering status words to those where the SRQ bit has a value of 1.
SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:BIT:SRQ? might
return NOCARE, indicating that status words are not being filtered based on
the SRQ bit value.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:SUBSF

This command sets or queries the value of the subsystem flag bit (SUBSF bit, bit 17) in a status word for a MIL-STD-1553 bus to search on. The search number is specified by x>. The search condition must be set to STATus.

Conditions	Requires SR-AERO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:SUBSF {ONE ZERO NOCARE} SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:SUBSF?
Arguments	ONE filters status words to only match those where the SUBSF bit has a value of 1. ZERO filters status words to only match those where the SUBSF bit has a value of 0. NOCARE disables filtering of status words on the SUBSF bit.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATus:BIT:SUBSF ONE specifies filtering status words for those where the SUBSF bit has a value of 1. SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATus:BIT:SUBSF? might return NOCARE, indicating that status words are not being filtered based on the SUBSF bit value.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:TF

This command sets or queries the value of the terminal flag bit (TF bit, bit 19) in a status word for a MIL-STD-1553 bus to search on. The search number is specified by x. The search condition must be set to STATus.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:TF {ONE ZERO NOCARE} SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:BIT:TF?

Arguments ONE filters status words to only match those where the TF bit has a value of 1.
ZERO filters status words to only match those where the TF bit has a value of 0.
NOCARE disables filtering of status words on the TF bit.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:BIT:TF ONE specifies filtering status words for those where the TF bit has a value of 1.
SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:BIT:TF? might return NOCARE, indicating that status words are not being filtered based on the TF bit value.

SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATus:PARity

This command sets or queries the value of the status word parity bit for a MIL-STD-1553 bus to search on. The search number is specified by x. The search condition must be set to STATus.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATUS:PARity
{ONE | ZERO | NOCARE}
SEARCH:SEARCH<x>:TRIGger:A:BUS:MIL1553B:STATUS:PARity?

Arguments ONE filters status words to only match those where the parity bit has a value of 1.
ZERO filters status words to only match those where the parity bit has a value of 0.
NOCARE disables filtering of status words on the parity bit.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:PARITY ONE specifies filtering status words for those where the parity bit has a value of 1.
SEARCH:SEARCH1:TRIGGER:A:BUS:MIL1553B:STATUS:PARITY? might return NOCARE, indicating that status words are not being filtered based on the parity bit value.

SEARCH:SEARCH<x>:TRIGger:A:BUS:PARAllel:DATa:VALue

This command sets or queries the binary data string used for a parallel bus search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:PARAllel:DATa:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:PARAllel:DATa:VALue?
Arguments	<QString> specifies the data value in a valid format. Valid characters are 0-9,
Examples	SEARCH:SEARCH2:TRIGGER:A:BUS:PARALLEL:DATA:VALUE "XXXXXXXX" sets the string data value to "XXXXXXXX" in binary format. SEARCH:SEARCH1:TRIGGER:A:BUS:PARALLEL:DATA:VALUE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:PARALLEL:DATA:VALUE "XXXXXXXXXXXXXXXX10010", indicating that the data value string is set to "XXXXXXXXXXXXXXXX10010" in binary format.

SEARCH:SEARCH<x>:TRIGger:A:BUS:RS232C:CONDition

This command sets or queries the condition for an RS232C bus search to determine where to place a mark. The search number is specified by <x>.

Conditions	Requires option 5-SRCOMP or SUP5-SRCOMP Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:RS232C:CONDition {DATa EOp PARITyerror START} SEARCH:SEARCH<x>:TRIGger:A:BUS:RS232C:CONDition?
Arguments	DATa specifies the search condition as Data. EOp specifies the search condition as End of Packet. PARITyerror specifies the search condition as Parity Error. START specifies the search condition as Start.

Examples `SEARCH:SEARCH1:TRIGGER:A:BUS:RS232C:CONDITION DATA` sets the search condition for RS232C bus trigger search 1 to Data.

`SEARCH:SEARCH2:TRIGGER:A:BUS:RS232C:CONDITION?` might return
`:SEARCH:SEARCH2:TRIGGER:A:BUS:RS232C:CONDITION EOP`, indicating that the search condition for RS232C bus trigger search 2 is set to End of Packet.

SEARCH:SEARCH<x>:TRIGger:A:BUS:RS232C:DATA:SIZE

This command sets or queries the length of the data string in bytes to be used for an RS232 bus search to determine where to place a mark when the search condition is Data. The search number is specified by <x>.

Conditions Requires option 5-SRCOMP or SUP5-SRCOMP Triggering and Analysis application.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:BUS:RS232C:DATA:SIZE <NR3>`
`SEARCH:SEARCH<x>:TRIGger:A:BUS:RS232C:DATA:SIZE?`

Arguments <NR3> is the number of bits per word in the data string, from 1 to 8.

Examples `SEARCH:SEARCH1:TRIGGER:A:BUS:RS232C:DATA:SIZE 1` sets the number of bits per word in RS232C bus trigger search 1 data string to 1.

`SEARCH:SEARCH2:TRIGGER:A:BUS:RS232C:DATA:SIZE?` might return
`:SEARCH:SEARCH2:TRIGGER:A:BUS:RS232C:DATA:SIZE 8`, indicating that the bits per word for RS232C bus trigger search 2 is set to 8.

SEARCH:SEARCH<x>:TRIGger:A:BUS:RS232C:DATA:VALue

This command sets or queries the data string used for the specified RS232C bus trigger search to determine where to place a mark. The search condition must be Data. The search number is specified by <x>.

Conditions Requires option 5-SRCOMP or SUP5-SRCOMP Triggering and Analysis application.

Group Search and Mark

Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:RS232C:DATA:VALUE <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:RS232C:DATA:VALUE?
Arguments	<QString> specifies the value of the data string. The valid characters are 0, 1, and X for values in binary format; and A-F, 0-9, and X for values in hexadecimal format.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:RS232C:DATA:VALUE "01" sets the value of the data string for RS232C bus trigger search 1 to "01" when the format is hexadecimal. SEARCH:SEARCH1:TRIGGER:A:BUS:RS232C:DATA:VALUE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:RS232C:DATA:VALUE "XXXXXXX1", indicating that the data string value for RS232C bus trigger search 1 is set to "XXXXXXX1" when the format is binary.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition

This command sets or queries the search condition for a SENT bus.

Conditions	Requires option SRAUTOSEN.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition {START FAST SLOW PAUSE ERROR} SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition?
Arguments	Search<x> is the number of the search. START specifies searching for start of packet. FAST specifies searching for fast channel data. SLOW specifies searching for slow channel data. PAUSE specifies searching for pause pulses. ERROR specifies searching on errors.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:CONDITION SLOW specifies searching for slow channel packet.

SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:CONDITION? might return
SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:CONDITION PAUSE, indicating
the search is set to find pause pulses.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:ERRType

This command sets or queries the error type to be used when searching on SENT data.

Conditions Requires option SRAUTOSEN.
SENT bus trigger condition is set to ERRor.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:ERRType
{FRAMELEngth|CRC}
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:ERRType?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition](#)

Arguments Search<x> is the number of the search.
FRAMELEngth specifies searching for SENT frame length errors.
CRC specifies searching for CRC errors.

Examples SEARCH:SEARCH1:TRIGger:A:BUS:SENT:ERRType FRAMELENGTH sets the
search to find SENT frame length errors.

SEARCH:SEARCH1:TRIGger:A:BUS:SENT:ERRType? might return
SEARCH:SEARCH1:TRIGger:A:BUS:SENT:ERRType CRC to indicate the
search is set to find CRC errors.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:ERRType:CRC

This command sets or queries the CRC error type to be used when searching on SENT data.

Conditions Requires option SRAUTOSEN.
SENT bus trigger condition is set to ERRor and ERRType is set to CRC.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:ERRType:CRC {FAST SLOW} SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:ERRType:CRC?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:ERRType
Arguments	Search<x> is the number of the search. FAST specifies searching for CRC errors only in the fast channel. SLOW specifies searching for CRC errors only in the slow channel.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:ERRTYPE:CRC SLOW sets the search to find SENT slow channel CRC errors. SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:ERRTYPE:CRC? might return SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:ERRTYPE:CRC FAST, indicating that the SENT fast channel is being searched for CRC errors.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN1A:HIVALue

This command sets or queries the high binary fast channel 1 value to use when searching on a SENT bus signal.

Conditions	Requires option SRAUTOSEN. The SENT bus search condition must be set to FAST.
Group	Trigger
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN1A:HIVALue <Qstring> SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN1A:HIVALue?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition
Arguments	Search<x> is the Search identifier number. <Qstring> sets the Fast Channel 1 binary data high value.

Examples `SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:CHAN1A:HIVALue`
`"XXXXXXXXXXXX"` sets the Fast Channel 1 high value on which to search to
`XXXXXXXXXXXX`, or "don't care."

`SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:CHAN1A:HIVALue?` might
return `SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:CHAN1A:HIVALUE`
`"0101XXXX1111"`, indicating the Fast Channel 1 high value on which to search is
set to the binary value 0101XXXX1111.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN1A:QUALifier

This command sets or queries the qualifier to be used when searching on SENT
fast packet bus data for device channel 1.

Conditions Requires option SRAUTOSEN
The SENT bus trigger condition must be set to FAST.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN1A:`
`QUALifier {EQual|UNEQual|LESSthan|MOREthan|`
`LESSEQual|MOREEQual|INrange|OUTrange}`
`SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN1A:QUALifier?`

Related Commands [`SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition`](#)
[`SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN1A:VALue`](#)

Arguments Search<x> is the number of the search.

`EQual` specifies the qualifier as Equal.

`LESSEQual` specifies the qualifier as Less Than or Equal to.

`LESSThan` specifies the qualifier as Less Than.

`MOREEQual` specifies the qualifier as More Than or Equal to.

`MOREThan` specifies the qualifier as More Than.

`UNEQual` specifies the qualifier as Unequal.

`INrange` sets the qualifier to inside a range.

`OUTrange` sets the qualifier to outside a range.

Examples `SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:CHAN1A:QUALIFIER`
`UNEQUAL` sets the fast channel 1 data qualifier to not equal for search 1.

`SEARCH:SEARCH3:TRIGGER:A:BUS:SENT:FAST:CHAN1A:QUALIFIER?`
might return
`SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:CHAN1A:QUALIFIER`
`MOREEQUAL` to indicate that the fast channel 1 data qualifier is set
to greater than or equal for search 3.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN1A:VALue

This command sets or queries the binary fast channel 1 value to be used when searching on a SENT bus signal.

Conditions Requires option SRAUTOSEN.
The trigger condition must be set to FAST.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN1A:VALue`
`<Qstring>`
`SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN1A:VALue?`

Related Commands [`SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition`](#)

Arguments `Search<x>` is the number of the search.
`<Qstring>` is the Fast Channel 1 binary value.

Examples `SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:CHAN1A:VALUE`
`"XXXXXXXXXXXX"` sets the Fast Channel 1 value to `XXXXXXXXXXXX`, or
`"don't care."`

`SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:CHAN1A:VALUE?` might
return `SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:CHAN1A:VALUE`
`"0000XXXX1111"` to indicate the binary value `0000XXXX1111`.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN2B:HIVALue

This command sets or queries the high binary fast channel 2 value to use when searching on a SENT bus signal.

Conditions	Requires option SRAUTOSEN. The SENT bus search condition must be set to FAST.
Group	Trigger
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN2B:HIVALue <Qstring> SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN2B:HIVALue?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition
Arguments	Search<x> is the Search identifier number. <Qstring> sets the Fast Channel 2 high binary data value.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:CHAN2B:HIVALue "100000000000" sets the Fast Channel 2 high value on which to search to 100000000000. SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:CHAN2B:HIVALue? might return SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:CHAN2B:HIVALUE "0101XXXX1111", indicating the Fast Channel 2 high value on which to search is set to the binary value 0101XXXX1111.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN2B:QUALifier

This command sets or queries the qualifier to be used when searching on SENT fast packet bus data for device channel 2.

Conditions	Requires option SRAUTOSEN The SENT bus trigger condition must be set to FAST.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN2B: QUALifier {EQUAL UNEQUAL LESSTHAN MORETHAN LESSEQUAL MOREEQUAL INrange OUTrange} SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN2B:QUALifier?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition

[SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN2B:VALue](#)

Arguments	Search<x> is the number of the search. EQUa1 specifies the qualifier as Equal. LESSEQUa1 specifies the qualifier as Less Than or Equal to. LESSThan specifies the qualifier as Less Than. MOREEQUa1 specifies the qualifier as More Than or Equal to. MOREThan specifies the qualifier as More Than. UNEQUa1 specifies the qualifier as Unequal. INrange sets the qualifier to inside a range. OUTrange sets the qualifier to outside a range.
Examples	SEARCH:SEARCH5:TRIGGER:A:BUS:SENT:FAST:CHAN2B:QUALIFIER UNEQUAL sets the fast channel 2 data qualifier to not equal for search 5. SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:CHAN2B:QUALIFIER? might return SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:CHAN2B:QUALIFIER MOREEQUAL to indicate that the fast channel 2 data qualifier is set to greater than or equal for search 1.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN2B:VALue

This command sets or queries the binary fast channel 2 value to be used when searching on a SENT bus signal.

Conditions	Requires option SRAUTOSEN. The trigger condition must be set to FAST.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN2B:VALue <Qstring> SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:CHAN2B:VALue?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition](#)

Arguments Search<x> is the number of the search.
 <Qstring> is the Fast Channel 2 binary value.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:CHAN2B:VALUE
 "111111111111" sets the Fast Channel 2 value to 111111111111.
 SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:CHAN2B:VALUE? might
 return SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:CHAN2B:VALUE
 "000000000000" to indicate the binary value 000000000000.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:COUNTER:HIVALue

This command sets or queries the high binary fast message counter value to use when searching on a SENT bus signal.

Conditions Requires option SRAUTOSEN.
 The SENT bus search condition must be set to FAST.
 The number of channels must be set to 1.
 The nibble count must be set to 6.

Group Trigger

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:COUNTER:HIVALue
 <QString>
 SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:COUNTER:HIVALue?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition](#)
 [BUS:B<x>:SENT:NUMCHANnel](#)
 [BUS:B<x>:SENT:NIBBLECount](#)

Arguments Search<x> is the Search identifier number.
 <Qstring> sets the Fast Channel 1 counter binary value.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:COUNTER:HIVALUE
 "XXXXXXXX" sets the Fast Channel secure counter high value on which to search
 to "don't care."

SEARCH:SEARCH4:TRIGGER:A:BUS:SENT:FAST:COUNTER:HIVALUE? might return SEARCH:SEARCH4:TRIGGER:A:BUS:SENT:FAST:COUNTER:HIVALUE "11110000" to indicate a binary value of 11110000 on which to search.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:COUNTER:QUALifier

This command sets or queries the qualifier to be used when searching on SENT fast packet bus data for the secure format counter.

Conditions Requires option SRAUTOSEN

The SENT bus trigger condition must be set to FAST.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:COUNTER:QUALifier {Equal|UNEQual|LESSthan|MOREthan|LESSEQual|MOREEQual|INrange|OUTrange}
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:COUNTER:QUALifier?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition](#)
[SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:COUNTER:VALue](#)

Arguments Search<x> is the number of the search.
Equal specifies the qualifier as Equal.
LESSEQual specifies the qualifier as Less Than or Equal to.
LESSThan specifies the qualifier as Less Than.
MOREEQual specifies the qualifier as More Than or Equal to.
MOREThan specifies the qualifier as More Than.
UNEQual specifies the qualifier as Unequal.
INrange sets the qualifier to inside a range.
OUTrange sets the qualifier to outside a range.

Examples SEARCH:SEARCH5:TRIGGER:A:BUS:SENT:FAST:COUNTER:QUALifier
LESSThan sets the fast channel 2 data qualifier on search 5 to less than.
SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:COUNTER:QUALifier?
might return

SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:COUNTER:QUALIFIER
EQUAL to indicate that the fast channel 2 data qualifier on search 1 is set to equal.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:COUNTER:VALue

This command sets or queries the binary fast message counter value to be used when searching on a SENT bus signal.

Conditions	Requires option SRAUTOSEN. The search condition must be set to FAST. The number of channels must be set to 1. The nibble count must be set to 6.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:COUNTER:VALue <Qstring> SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:COUNTER:VALue?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition BUS:B<x>:SENT:NIBBLECount BUS:B<x>:SENT:NUMCHANnel
Arguments	Search<x> is the number of the search. <Qstring> is the Fast Channel 1 counter value.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:COUNTER:VALUE "XXXXXXXX" sets the Fast Channel 1 secure counter value to "don't care." SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:COUNTER:VALUE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:COUNTER:VALUE "00001111" to search for a binary counter value of 00001111.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:INVERTNIBble:VALue

This command sets or queries the binary fast message inverted nibble value to be used when searching on a SENT bus signal.

Conditions	Requires option SRAUTOSEN. The search condition must be set to FAST. The number of channels must be set to 1. The nibble count must be set to 6.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:INVERTNIBble:VALue <Qstring> SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:INVERTNIBble:VALue?</pre>
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition BUS:B<x>:SENT:NIBBLECount BUS:B<x>:SENT:NUMCHANnel
Arguments	Search<x> is the number of the search. <Qstring> is the fast message inverted nibble binary value.
Examples	TRIGGER:A:BUS:B1:SENT:FAST:INVERTNIBBLE:VALUE "XXXX" sets the Fast Channel 1 secure inverted nibble value to "don't care." TRIGGER:A:BUS:B1:SENT:FAST:INVERTNIBBLE:VALUE? might return :TRIGGER:A:BUS:B1:SENT:FAST:INVERTNIBBLE:VALUE "1111" to search for a binary inverted nibble value of 1111.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:STATus:VALue

This command sets or queries the binary status value to be used when searching on a SENT bus signal.

Conditions	Requires option SRAUTOSEN. The trigger condition must be set to FAST.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:STATus:VALue <Qstring></pre>

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:FAST:STATUS:VALUE?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition](#)

Arguments Search<x> is the number of the search.
<Qstring> is the binary status binary value.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:STATUS:VALUE "XXXX" sets the binary status value to XXXX.

SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:STATUS:VALUE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:FAST:STATUS:VALUE "1010" to indicate searching on a binary status value of 1010.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:PAUSE:QUALifier

This command sets or queries the qualifier to be used when searching on SENT pause pulses.

Conditions Requires option SRAUTOSEN
The SENT bus trigger condition must be set to PAUSE.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:PAUSE:
QUALifier {Equal|UNEQual|LESSthan|MOREthan|
LESSEQual|MOREEQual|INrange|OUTrange}
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:PAUSE:QUALifier?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition](#)
[SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:PAUSE:TICKs:VALue](#)
[SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:PAUSE:TICKs:HIVALue](#)

Arguments Search<x> is the number of the search.

Equal specifies the qualifier as Equal.

INrange sets the qualifier to be within a range.

LESSEQual sets the qualifier as Less Than or Equal to.

LESSThan sets the qualifier as Less Than.

MOREEqual sets the qualifier as More Than or Equal to.

MOREThan sets the qualifier as More Than.

OUTrange sets the qualifier to be outside a range.

UNEQUAL specifies the qualifier as Unequal.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:PAUSE:QUALIFIER LESSTHAN
sets the pause pulse qualifier to less than.

SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:PAUSE:QUALIFIER? might return
SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:PAUSE:QUALIFIER INRANGE to
indicate that the pause pulse qualifier is set to search inside the range of ticks
low and ticks high.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:PAUSE:TICKs:HIValue

This command sets or queries the maximum number of pause clock ticks to be used when searching on a SENT bus signal.

Conditions Requires option SRAUTOSEN.
The trigger condition must be set to PAUSE.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:PAUSE:TICKs:HIValue
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:PAUSE:TICKs:HIValue ?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition](#)

Arguments Search<x> is the number of the search.

<NR1> is the maximum number of pause clock ticks to be used when searching.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:PAUSE:TICKs:HIValue 15 sets
the maximum pause tick count for which to search to 15.

SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:PAUSE:TICKs:HIValue? might
return SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:PAUSE:TICKs:HIValue
20 to indicate the maximum number of pause ticks to search for is 20.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:PAUSE:TICKs:VALue

This command sets or queries the minimum number of pause clock ticks to be used when searching on a SENT bus signal.

Conditions Requires option SRAUTOSEN.

The trigger condition must be set to PAUSE.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:PAUSE:TICKs:VALue <NR1>
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:PAUSE:TICKs:VALue?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition](#)

Arguments Search<x> is the number of the search.

<NR1> is the minimum number of pause clock ticks to be used when searching.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:PAUSE:TICKS:VALUE 12 sets the minimum pause tick count on which to search to 12.

SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:PAUSE:TICKS:VALUE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:PAUSE:TICKS:VALUE 15 to indicate the minimum number of pause ticks to search for is 15.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:SLOW:DATA:HIVALue

This command sets or queries the high binary Slow channel data value to use when searching on a SENT bus signal.

Conditions Requires option SRAUTOSEN.

The SENT bus search condition must be set to SLOW.

Group Trigger

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:SLOW:DATA:HIVALue
<QString>
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:SLOW:DATA:HIVALue?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition](#)

Arguments Search<x> is the Search identifier number.
 <Qstring> sets the binary Slow channel data value.

Examples SEARCH:SEARCH2:TRIGGER:A:BUS:SENT:SLOW:DATA:HIVALUE
 "XXXXXXXX" sets the Slow data high value on which to search to "don't care"
 for Search 2.
 SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:SLOW:DATA:HIVALUE? might
 return SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:SLOW:DATA:HIVALUE
 "11110001", to indicate searching on slow data values that match 11110001.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:SLOW:DATA:QUALifier

This command sets or queries the qualifier to be used when searching on SENT slow packet bus data.

Conditions Requires option SRAUTOSEN.
 The SENT bus trigger condition must be set to SLOW.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:SLOW:DATA:
 QUALifier {EQUAL|UNEQUAL|LESSThan|MOREthan|
 LESSEQUAL|MOREEQUAL|INrange|OUTrange}
 SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:SLOW:DATA:QUALifier?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition](#)
 [SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:SLOW:DATA:VALue](#)

Arguments Search<x> is the number of the search.
 EQUAL specifies the qualifier as Equal.
 LESSEQUAL sets the qualifier as Less Than or Equal to.
 LESSThan sets the qualifier as Less Than.
 MOREEQUAL sets the qualifier as More Than or Equal to.
 MOREThan sets the qualifier as More Than.

UNEQUAL specifies the qualifier as Unequal.

INrange sets the search qualifier to inside a range.

OUTrange sets the search qualifier to outside a range.

Examples

SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:SLOW:DATA:QUALIFIER
LESSEQUAL sets the slow channel data qualifier to less than or equal.

SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:SLOW:DATA:QUALIFIER? might
return SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:SLOW:DATA:QUALIFIER
EQUAL to indicate that the slow channel data qualifier is set to equal.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:SLOW:DATA:VALue

This command sets or queries the binary slow channel data value to be used when searching on a SENT bus signal.

Conditions

Requires option SRAUTOSEN.

The trigger condition must be set to SLOW.

Group

Search and Mark

Syntax

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:SLOW:DATA:VALue
<Qstring>
SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:SLOW:DATA:VALue?

Related Commands

[SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition](#)

Arguments

Search<x> is the number of the search.

<Qstring> is the slow channel data binary value.

Examples

SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:SLOW:DATA:VALUE "XXXXXXXX"
sets the slow data search value to "don't care."

SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:SLOW:DATA:VALUE? might return
"XXXX1111", indicating the search is marking slow data values that match
XXXX1111.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:SLOW:IDentifier:VALue

This command sets or queries the binary slow identifier value to be used when searching on a SENT bus signal.

Conditions	Requires option SRAUTOSEN. The trigger condition must be set to SLOW.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:SLOW:IDentifier:VALue <Qstring> SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:SLOW:IDentifier:VALue?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:CONDition
Arguments	Search<x> is the number of the search. <Qstring> is the slow channel identifier binary value.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:SLOW:IDENTIFIER:VALUE "XXXX" sets the search identifier value to "don't care." SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:SLOW:IDENTIFIER:VALUE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:SENT:SLOW:IDENTIFIER:VALUE "0001" indicating to search for the slow channel identifier value of 0001.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SOURce

This command sets or queries the bus source for the bus search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:SOURce {B0 B1 B2 B3 B4 B5 B6 B7 B8 B9 B10 B11 B12 B13 B14 B15 B16} SEARCH:SEARCH<x>:TRIGger:A:BUS:SOURce?
Arguments	B0–B16 specifies the bus source as a bus number from B01 to B16.

Examples `SEARCH:SEARCH1:TRIGGER:A:BUS:SOURCE B1` sets the bus source for bus trigger search 1 to B1.

`SEARCH:SEARCH1:TRIGGER:A:BUS:SOURCE?` might return
`SEARCH:SEARCH1:TRIGGER:A:BUS:SOURCE B15`, indicating that the bus source for bus trigger search 1 is B15.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SPI:CONDition

This command sets or queries the search condition for an SPI bus search to determine where to place a mark. The search number is specified by <x>.

Conditions Requires option 5-SRCOMP or SUP5-SRCOMP Triggering and Analysis application.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:BUS:SPI:CONDition`
 `{DATA|SS|STARTOfframe}`
 `SEARCH:SEARCH<x>:TRIGger:A:BUS:SPI:CONDition?`

Arguments DATA specifies the trigger condition as Data.

 SS specifies the trigger condition as Slave Selection.

 STARTOfframe specifies the trigger condition as start of frame.

Examples `SEARCH:SEARCH1:TRIGGER:A:BUS:SPI:CONDITION SS` sets the trigger condition for SPI bus trigger search 1 to SS.

`SEARCH:SEARCH1:TRIGGER:A:BUS:SPI:CONDITION?` might return
`SEARCH:SEARCH1:TRIGGER:A:BUS:SPI:CONDITION DATA`, indicating that the trigger condition for SPI bus trigger search 1 is set to Data.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SPI:DATA:SIZE

This command sets or queries the length of the data string in bytes used for the specified SPI bus trigger search to determine where to place a mark. The search condition must be DATA. The search number is specified by <x>.

Conditions Requires option 5-SRCOMP or SUP5-SRCOMP Triggering and Analysis application.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:SPI:DATA:SIZE <NR1> SEARCH:SEARCH<x>:TRIGger:A:BUS:SPI:DATA:SIZE?
Arguments	<NR1> specifies the number of contiguous data bytes.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:SPI:DATA:SIZE 1 sets the length of the data string for SPI bus trigger search 1 to 1 byte. SEARCH:SEARCH2:TRIGGER:A:BUS:SPI:DATA:SIZE? might return :SEARCH:SEARCH2:TRIGGER:A:BUS:SPI:DATA:SIZE 4, indicating that the length of the data string for SPI bus trigger search 2 is 4 bytes.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SPI:DATA:VALue

The command sets or queries the binary data string used for an SPI bus search to determine where to place a mark. The search number is specified by <x>. The search condition must be DATA.

Conditions	Requires option 5-SRCOMP or SUP5-SRCOMP Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:SPI:DATA:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:SPI:DATA:VALue?
Arguments	<QString> specifies the data value in the specified valid format. The valid characters are 0, 1, and X for binary format; and A-F, 0-9, and X for hexadecimal format.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:SPI:DATA:VALue "00001111" sets the data value for SPI bus trigger search 1 to "00001111" in binary format. SEARCH:SEARCH1:TRIGGER:A:BUS:SPI:DATA:VALue? might return SEARCH:SEARCH1:TRIGGER:A:BUS:SPI:DATA:VALue "11", indicating that the data value for SPI bus trigger search 1 is set to "11" in hexadecimal format.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:CONDition

This command sets or queries the search condition for an SPMI bus.

Conditions Requires option SRPM

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:
SPMI:CONDition {SSC|RESet|SLEEp|
SHUTdown|WAKEup|MASTERREAd|MASTERWRite|REGREAd|REGWRite|
DEVIDESCMASrERREAd|DEVIDESCSLAVErEAd|EXTREGREAd|
EXTREGWRite|LONGEXTREGREAd|LONGEXTREGWRite|REG0WRite|
AUTHenticate|TRANSferbusownership|PARityerror}
SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:CONDition?

Arguments Search<x> is the search identifier number.

SSC specifies searching for the Sequence Start Condition.

RESet specifies searching for the Reset command sequence.

SLEEp specifies searching for the Sleep command sequence.

SHUTdown specifies searching for the Shutdown command sequence.

WAKEup specifies searching for the Wakeup command sequence.

MASTERREAd specifies searching for the Master Read command sequence.

MASTERWRite specifies searching for the Master Write command sequence.

REGREAd specifies searching for the Register Read command sequence.

REGWRite specifies searching for the Register Write command sequence.

DEVIDESCMASrERREAd specifies searching for the Device Descriptor Block Master Read command sequence

DEVIDESCSLAVErEAd specifies searching for the Device Descriptor Block Slave Read command sequence.

EXTREGREAd specifies triggering on the Extended Register Read command sequence.

EXTREGWRite specifies searching for the Extended Register Write command sequence.

LONGEXTREGREAd specifies searching for the Extended Register Read Long command sequence.

LONGEXTREGWRItE specifies searching for the Extended Register Write Long command sequence.

REG0WRItE specifies searching for the Register 0 Write command sequence.

AUTHenticate specifies searching for the Authentication command sequence.

TRANSferbusownership specifies searching for the Transfer Bus Ownership (TBO) command sequence.

PARItYerror specifies searching for the parity errors.

Examples SEARCH:SEARCH4:TRIGger:A:BUS:SPMI:CONDItION SLEEP specifies searching for a Sleep command sequence in search Search4.

SEARCH:SEARCH2:TRIGger:A:BUS:SPMI:CONDItION? might return
:SEARCH:SEARCH2:TRIGger:A:BUS:SPMI:CONDItION MASTERREAD,
indicating that search Search2 is searching for Master Read command sequences.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:DATa:SIZE

This command sets or queries the length of the data string, in bytes, to be used when searching on an SPMI bus signal.

Conditions Requires option SRPM.

The Search condition must be set to EXTREGREAd, EXTREGWRItE, LONGEXTREGREAd, or LONGEXTREGWRItE.

Note that this a floating point value. If the condition is set to REG0WRItE, the query return value will be a fractional value (0.875) because the data size is less than a byte.

Group Trigger

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:DATa:SIZE <NR2>
SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:DATa:SIZE?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:CONDItion](#)
[SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:DATa:VALue](#)

Arguments Search<x> is the Search identifier number.
<NR2> is the size of the data string in bytes.

Examples `SEARCH:SEARCH2:TRIGger:A:BUS:SPMI:DATA:SIZE 1` sets the data size to 1 byte.

`SEARCH:SEARCH2:TRIGger:A:BUS:SPMI:DATA:SIZE?` might return `SEARCH:SEARCH2:TRIGger:A:BUS:SPMI:DATA:SIZE 4` to indicate the search data pattern has four bytes of data.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:DATA:VALue

This command specifies the binary data string used for SPMI when the search condition is MASTERREAd, MASTERWRItE, REGREAd, REGWRItE, EXTREGREAd, EXTREGWRItE, LONGEXTREGREAd, LONGEXTREGWRItE, or REG0WRItE.

Conditions Requires option SRPM

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:DATA:VALue <Qstring>`
`SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:DATA:VALue?`

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:CONDition](#)

Arguments Search<x> is the search identifier number.

`<Qstring>` is the binary data string that identifies the data value for which to search.

Examples `SEARCH:SEARCH2:TRIGger:A:BUS:SPMI:DATA:VALUE "XXXXXXXX"` sets the binary data value to "don't care" for Search 2.

`SEARCH:SEARCH2:TRIGger:A:BUS:SPMI:DATA:VALUE?` might return `SEARCH:SEARCH2:TRIGger:A:BUS:SPMI:DATA:VALUE "XXXX1111"` to indicate Search 2 is searching for binary data value XXXX1111.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:MASTERADDReSS:VALue

This command sets or queries the binary data string that identifies the master address used in SPMI when the search condition is MASTERREAd, MASTERWRItE, or DEVIDESCMASREAd.

Conditions Requires option SRPM

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:MASTERADDRESS:VALUE <Qstring> SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:MASTERADDRESS:VALUE?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:CONDition
Arguments	Search<x> is the search identifier number. <Qstring> is the binary data string that identifies the master address for which to search.
Examples	SEARCH:SEARCH2:TRIGger:A:BUS:SPMI:MASTERADDRESS:VALUE "XX" sets the binary master address on which to search to "don't care." SEARCH:SEARCH3:TRIGger:A:BUS:SPMI:MASTERADDRESS:VALUE? might return SEARCH:SEARCH3:TRIGger:A:BUS:SPMI:MASTERADDRESS:VALUE "11" to indicate that Search 3 is searching for binary master address 11.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:NORESPonse

This command sets or queries whether or not to search for No Response frames.

Conditions	Requires option SRPM
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:NORESPonse <Qstring> SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:NORESPonse?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:CONDition
Arguments	Search<x> is the search identifier number. <Qstring> is either "TRUE" or "FALSE" boolean value for searching on SPMI No Response frames.
Examples	SEARCH:SEARCH6:TRIGger:A:BUS:SPMI:NORESPONSE TRUE sets Search 6 to look for No Response frames.

SEARCH:SEARCH4:TRIGger:A:BUS:SPMI:NORESPONSE? might return
 SEARCH:SEARCH4:TRIGger:A:BUS:SPMI:NORESPONSE FALSE to indicate
 that the Search 4 is not looking for No Response frames.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:REGISTERADDRESS:VALue

This command sets or queries the binary data string that identifies the register address used in SPMI triggering if the trigger condition is MASTERREAd, MASTERWRItE, REGREAd, REGWRItE, EXTREGREAd, EXTREGWRItE, LONGEXTREGREAd, or LONGEXTREGWRItE.

Conditions Requires option SRPM

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:REGISTERADDRESS:VALue
 <Qstring>
 SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:REGISTERADDRESS:VALue?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:CONDition](#)

Arguments Search<x> is the search identifier number.
 <Qstring> is the binary data string that identifies the register address for which to search.

Examples SEARCH:SEARCH2:TRIGger:A:BUS:SPMI:REGISTERADDRESS:VALUE
 "XXXXXXXX" sets the binary register address to "don't care."
 SEARCH:SEARCH5:TRIGger:A:BUS:SPMI:REGISTERADDRESS:VALUE?
 might return
 SEARCH:SEARCH5:TRIGger:A:BUS:SPMI:REGISTERADDRESS:VALUE
 "XXXX1111" to indicate Search 5 is searching for binary register address is
 XXXX1111.

SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:SLAVEADDRESS:VALue

This command sets or queries the binary data string that identifies the slave address used in SPMI when the search condition is RESet, SLEEp, SHUTdown, WAKEup, AUTHenticate, REGREAd, REGWRItE, EXTREGREAd, EXTREGWRItE, LONGEXTREGREAd, LONGEXTREGWRItE, DEVICEDESCSLAVEREAd, or REG0WRItE.

Conditions	Requires option SRPM
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:SLAVEADDRESS:VALue <Qstring> SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:SLAVEADDRESS:VALue?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:BUS:SPMI:CONDition
Arguments	Search<x> is the search identifier number. <Qstring> is the binary data string that identifies the slave address for which to search.
Examples	SEARCH:SEARCH3:TRIGger:A:BUS:SPMI:SLAVEADDRESS:VALUE "0011" sets the binary slave address to 0011 on Search 3. SEARCH:SEARCH1:TRIGger:A:BUS:SPMI:SLAVEADDRESS:VALUE? might return SEARCH:SEARCH1:TRIGger:A:BUS:SPMI:SLAVEADDRESS:VALUE "1010" to indicate the binary slave address is 1010 for Search 1.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ADDRESS:HIVALue

This command sets or queries the high binary address value used when searching on a USB bus signal to determine where to place a mark. The search number is specified by <x>. The search condition must be set to TOKEN.

The VALue and HIVALue set a range that the INrange and OUTrange qualifiers use to decide when to trigger. For example, if the QUALIFER is set to INrange, and the address is within the range set by VALue and HIVALue, then a trigger can be generated.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ADDRESS:HIVALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ADDRESS:HIVALue?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ADDress:VALue](#)

Arguments <QString> specifies the data value in the specified valid format. The valid characters are 0, 1, and X for binary; for hexadecimal; and symbolic).

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:USB:ADDRESS:HIVALUE

SEARCH:SEARCH1:TRIGGER:A:BUS:USB:ADDRESS:HIVALUE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:USB:ADDRESS:HIVALUE "XX", indicating that the address value for normal token for USB bus trigger search 1 is set to "XX," in hexadecimal format.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ADDress:VALue

This command sets or queries the binary address value used for a USB bus search to determine where to place a mark. The search number is specified by <x>. The search condition must be set to TOKEN.

Conditions Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ADDRESS:VALUE <QString>
SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ADDRESS:VALUE?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ADDress:HIVALue](#)

Arguments <QString> specifies the data value in the specified valid format. Valid characters are 0, 1, and X for binary; A-F, 0-9 and X for hexadecimal; and for symbolic.

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:USB:ADDRESS:VALUE "1110000" sets the address value for the normal token for USB bus trigger search 1 to "1110000" in binary format.

SEARCH:SEARCH1:TRIGGER:A:BUS:USB:ADDRESS:VALUE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:USB:ADDRESS:VALUE "11", indicating that the address value for normal token for USB bus trigger search 1 is set to "11" hexadecimal format.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:CONDition

This command sets or queries the search condition for a USB bus search to determine where to place a mark. The search number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:CONDition {DATAPacket EOP ERROR HANDSHAKEPacket RESET RESUME SPECIALPacket SUSPEND SYNC TOKENPacket} SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:CONDition?</pre>
Arguments	DATAPacket specifies the search condition as Data Packet. EOP specifies the search condition as End of Packet. ERROR specifies the search condition as Error. HANDSHAKEPacket specifies the search condition as Handshake Packet. RESET specifies the search condition as Reset. RESUME specifies the search condition as Resume. SPECIALPacket specifies the search condition as Special Packet. SUSPEND specifies the search condition as Suspend. SYNC specifies the search condition as Sync. TOKENPacket specifies the search condition as Token (Address) Packet.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:USB:CONDITION EOP sets the search condition for USB bus trigger search 1 to End of Packet. SEARCH:SEARCH1:TRIGGER:A:BUS:USB:CONDITION? might return SEARCH:SEARCH1:TRIGGER:A:BUS:USB:CONDITION SYNC, indicating that the search condition for USB bus trigger search 1 is set to Sync.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:HIVALue

This command sets or queries the high binary data value used with In Range and Out of Range qualifiers for a USB bus search to determine where to place a mark. The search number is specified by <x>. The search condition must be set to DATA.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:HIVALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:HIVALue?
Arguments	<QString> specifies the data value in the specified valid format (binary, hexadecimal, or symbolic). The valid characters for binary are 0, 1, and X;
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:USB:DATA:HIVALUE "XXXXX111" sets the data value for USB bus trigger search 1 to "XXXXX111" in binary format. SEARCH:SEARCH1:TRIGGER:A:BUS:USB:DATA:HIVALUE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:USB:DATA:HIVALUE "11", indicating that the data value for USB bus trigger search 1 is set to "11" in hexadecimal format.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:OFFSet

This command sets the byte offset to look for a data pattern at, in bytes, to be used when searching on a USB bus signal. The search number is specified by <x>. The search condition must be set to DATA.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:OFFSet <NR1> SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:OFFSet?
Arguments	<NR1> specifies the data offset in bytes.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:USB:DATA:OFFSET 5 sets the data offset for USB bus trigger search 1 to 5 bytes. SEARCH:SEARCH1:TRIGGER:A:BUS:USB:DATA:OFFSET? might return SEARCH:SEARCH1:TRIGGER:A:BUS:USB:DATA:OFFSET DONTcare, indicating that the data offset for USB bus trigger search 1 is set to DONTcare.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:QUALifier

This command sets or queries the qualifier to be used when searching on a USB bus signal. The search condition must be set to IDANDDATA OR DATA. The search number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:QUALifier {EQUAL UNEQUAL LESSThan MOREthan LESSEQUAL MOREEQUAL INrange OUTrange}
Arguments	EQUAL specifies the qualifier as Equal. INrange specifies the qualifier as Inside Range. LESSEQUAL specifies the qualifier as Less Than or Equal to. MOREEQUAL specifies the qualifier as More Than or Equal to. OUTrange specifies the qualifier as Out of Range. UNEQUAL specifies the qualifier as Unequal. LESSThan specifies the qualifier as Less Than. MOREThan specifies the qualifier as More Than.
Examples	SEARCH:SEARCH1:TRIGger:A:BUS:USB:DATA:QUALifier INRANGE sets the qualifier to INRANGE. SEARCH:SEARCH1:TRIGger:A:BUS:USB:DATA:QUALifier? might return SEARCH:SEARCH1:TRIGGER:A:BUS:USB:DATA:QUALIFIER EQUAL indicating the qualifier is set to equal.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:SIZE

This command sets or queries the length of the data string, in bytes, used for a USB bus search to determine where to place a mark. The search number is specified by <x>. The search condition must be set to DATA.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
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Group	Search and Mark
Syntax	<code>SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:SIZE <NR1></code> <code>SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:SIZE?</code>
Arguments	<NR1> specifies the data size in bytes.
Examples	<code>SEARCH:SEARCH1:TRIGGER:A:BUS:USB:DATA:SIZE 1</code> sets the size of the data string for USB bus trigger search 1 to 1 byte. <code>SEARCH:SEARCH1:TRIGGER:A:BUS:USB:DATA:SIZE?</code> might return <code>SEARCH:SEARCH1:TRIGGER:A:BUS:USB:DATA:SIZE 3</code> , indicating that the size of the data string for USB bus trigger search 1 is set to 3 bytes.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:TYPE

This command sets or queries the USB bus search type. The search number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark
Syntax	<code>SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:TYPE</code> <code>{ANY DATA0 DATA1 DATA2 MDATA}</code> <code>SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:TYPE?</code>
Arguments	ANY specifies the data packet type as Any. DATA0 specifies the data packet type as DATA0. DATA1 specifies the data packet type as DATA1. DATA2 specifies the data packet type as DATA2. MDATA specifies the data packet type as MDATA.
Examples	<code>SEARCH:SEARCH1:TRIGGER:A:BUS:USB:DATA:TYPE ANY</code> sets the data packet type for USB bus trigger search 1 to ANY.

SEARCH:SEARCH2:TRIGGER:A:BUS:USB:DATA:TYPE? might return
:SEARCH:SEARCH2:TRIGGER:A:BUS:USB:DATA:TYPE MDADA, indicating that
the packet type for USB bus trigger search 2 is set to MDATA.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:VALue

This command sets or queries the binary data value used for a USB bus search to determine where to place a mark. The search number is specified by <x>. The search condition must be set to DATA.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:DATA:VALue?
Arguments	<QString> specifies the data value. The valid characters are 0, 1, and X for binary format; and A-F, 0-9, and X for hexadecimal format.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:USB:DATA:VALUE "AB" sets the data value for data token for USB bus trigger search 1 to "AB" in hexadecimal format. SEARCH:SEARCH1:TRIGGER:A:BUS:USB:DATA:VALUE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:USB:DATA:VALUE "XXXXXXXX", indicating that the data value for data token for USB bus trigger search 1 is set to "XXXXXXXX" in binary format.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ENDPoint:VALue

This command sets or queries the endpoint binary value used for a USB bus search to determine where to place a mark. The search number is specified by <x>. The search condition must be set to TOKEN.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark

Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ENDPoint:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ENDPoint:VALue?
Arguments	<QString> specifies the data value in the specified valid format. The valid characters are 0, 1, and X for binary format; A-F, 0-9 and X for hexadecimal format.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:USB:ENDPOINT:VALUE "1111" sets the endpoint value for normal token for USB bus trigger search 1 to "1111" in binary format. SEARCH:SEARCH1:TRIGGER:A:BUS:USB:ENDPOINT:VALUE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:USB:ENDPOINT:VALUE "1", indicating that the endpoint value for normal token for USB bus trigger search 1 is set to "1" in hexadecimal format.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ERRTYPE

This command sets or queries the error type for a USB bus search to determine where to place a mark. The search number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ERRTYPE {BITSTUFFing CRC5 CRC16 PID} SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:ERRTYPE?
Arguments	BITSTUFFing specifies the error type as Bit Stuffing. CRC5 specifies the error type as Token CRC5 (Cyclic Redundancy Check 5). CRC16 specifies the error type as Data CRC16 (Cyclic Redundancy Check 16). PID specifies the error type as PID Check Bits.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:USB:ERRTYPE CRC5 sets the error type for USB bus trigger search 1 to Token CRC5. SEARCH:SEARCH1:TRIGGER:A:BUS:USB:ERRTYPE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:USB:ERRTYPE PID, indicating that the error type for USB bus trigger search 1 is set to PID Check Bits.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:HANDSHAKEType

This command sets or queries the handshake type for the specified USB bus trigger search to determine where to place a mark. The search number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:HANDSHAKEType {ACK ANY NAK NYET STALL} SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:HANDSHAKEType?
Arguments	ACK specifies the handshake type as Acknowledgement (positive) (XX10). ANY specifies the handshake type as Any (0010). NAK specifies the handshake type as Negative Acknowledgment (1010). NYET specifies the handshake type as No response Yet (0110). STALL specifies the handshake type as Stall (endpoint is halted or control pipe request not supported) (1110).
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:USB:HANDSHAKETYPE NAK sets the handshake type for USB bus trigger search 1 to NAK. SEARCH:SEARCH2:TRIGGER:A:BUS:USB:HANDSHAKETYPE? might return :SEARCH:SEARCH2:TRIGGER:A:BUS:USB:HANDSHAKETYPE NYET, indicating that the handshake type for USB bus trigger search 2 is set to NYET.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SOFFRAMENUMber

This command sets or queries the frame number string to use for the Start of Frame for the specified USB bus trigger search to determine where to place a mark. The search number is specified by <x>. The search condition must be set to TOKEN.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark

Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SOFFRAMENUMber <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SOFFRAMENUMber?
Arguments	<QString> specifies the frame number string for the Start of Frame in a valid format (binary, hexadecimal, or symbolic).
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SOFFRAMENUMBER "1001" sets the frame number string for the SOF for USB bus trigger search 1 to "1001" in symbolic format. SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SOFFRAMENUMBER? might return SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SOFFRAMENUMBER "xxxxxxx1001", indicating that the frame number string for the SOF for USB bus trigger search 1 is set to "XXXXXXXX1001" in binary format.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPECIALType

This command sets or queries the USB search type for the specified USB bus trigger search to determine where to place a mark. The search number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPECIALType {ANY ERROR PING PREAmble RESERVED SPLit} SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPECIALType?
Arguments	ANY specifies the PID value as Any (XX00). ERROR specifies the PID value as ERR (1100). PING specifies the PID value as PING (0100). PREAmble specifies the PID value as PRE (1100). RESERVED specifies the PID value as Reserved (0000). SPLit specifies the PID value as Split (1000).
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SPECIALTYPE ERROR sets the PID value for USB bus trigger search 1 to ERROR.

SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SPECIALTYPE? might return
SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SPECIALTYPE SPLIT, indicating
that the PID value for USB bus trigger search 1 is set to SPLIT.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:ET:VALue

This command sets or queries the Endpoint Type value for the specified USB bus trigger search on split token field to determine where to place a mark. The search number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:ET:VALue {BULK CONTROL NOCARE INTERRUPT ISOchronous} SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:ET:VALue?
Arguments	BULK specifies the Endpoint Type value as Bulk (10). CONTROL specifies the Endpoint Type value as Control (00). NOCARE specifies the Endpoint Type value as Don't Care. INTERRUPT specifies the Endpoint Type value as Interrupt (11). ISOchronous specifies the Endpoint Type value as Isochronous (01).
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SPLIT:ET:VALUE BULK sets the Endpoint Type value for USB bus trigger search 1 to Bulk. SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SPLIT:ET:VALUE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SPLIT:ET:VALUE DONTCARE, indicating that the Endpoint Type value for USB bus trigger search 1 is set to DONTCare.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:HUB:VALue

This command sets or queries the binary hub address value to be used when searching on a USB bus signal. The search number is specified by <x>. The search condition must be set to Special with packet type SPLIT.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:HUB:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:HUB:VALue?
Arguments	<QString> specifies the hub address. The valid characters are .
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SPLIT:HUB:VALUE "1001" sets the hub address to "XXX1001". SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SPLIT:HUB:VALUE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SPLIT:HUB:VALUE "XX" (for binary format), indicating that the hub address for the USB bus trigger search on split token field is "XX".

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:PORT:VALue

This command sets or queries the binary port address used when searching on a USB bus signal. The search number is specified by <x>. The search condition must be set to Special with a packet type SPLIT.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:PORT:VALue <QString> SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:PORT:VALue?
Arguments	<QString> specifies the port address in the valid format. The valid characters are
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SPLIT:PORT:VALUE "1111" sets the port address for USB bus trigger search 1 to "XXX1111" for binary format. SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SPLIT:PORT:VALUE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SPLIT:PORT:VALUE "XX",

indicating that the port address for USB hub trigger search 1 is set to "XX" in hexadecimal format.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:SC:VALue

This command sets or queries the Start/Complete value for the specified USB bus trigger on split token field search to determine where to place a mark. The search number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:SC:VALue {CSPLIT NOCARE SSPLIT} SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:SC:VALue?
Arguments	CSPLIT specifies Complete (CSPLIT)(1) Start/Complete value. NOCARE specifies the Start/Complete value as Don't Care (X). SSPLIT specifies Start (SSPLIT) (0) Start/Complete value.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SPLIT:SC:VALUE DONTCARE sets the Start/Complete value for USB bus trigger search 1 to DONTCare. SEARCH:SEARCH2:TRIGGER:A:BUS:USB:SPLIT:SC:VALUE? might return :SEARCH:SEARCH2:TRIGGER:A:BUS:USB:SPLIT:SC:VALUE SSPLIT, indicating that the Start/Complete value for USB bus trigger search 2 is set to SSPLIT.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:SE:VALue

This command sets or queries the Start/End value for the specified USB bus trigger on split token field search to determine where to place a mark. The search number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark

Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:SE:VALue {NOCARE FULLSPEED ISOALL ISOEND ISOMID ISOSTART LOWSPEED} SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:SPLIT:SE:VALue?</pre>
Arguments	NOCARE specifies Don't Care (X) Start/End bit value. FULLSPEED specifies Control/Bulk/Interrupt Full Speed device (0X) Start/End bit value. ISOALL specifies Isochronous Data is All (11) Start/End bit value. ISOEND specifies Isochronous Data is End (01) Start/End bit value. ISOMID specifies Isochronous Data is Middle (00) Start/End bit value. ISOSTART specifies Isochronous Data is Start (10) Start/End bit value. LOWSPEED specifies Control/Bulk/Interrupt Low Speed device (1X) Start/End bit value.
Examples	SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SPLIT:SE:VALUE ISOSTART sets the Start/End value for USB bus trigger on split token field search 1 to ISOSTART. SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SPLIT:SE:VALUE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:USB:SPLIT:SE:VALUE DONTCARE, indicating that the Start/End value for USB bus trigger on split token field search 1 is set to DONTCare.

SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:TOKENType

This command sets or queries the token type used to search a USB bus signal. The search number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:TOKENType {ANY IN OUT SETUP SOF} SEARCH:SEARCH<x>:TRIGger:A:BUS:USB:TOKENType?</pre>
Arguments	ANY specifies the token type as Any (XX01). IN specifies the token type as IN (1001). OUT specifies the token type as OUT (0001).

SETUP specifies the token type as SETUP (1101).

SOF specifies the token type as Start Of Frame (0101).

Examples SEARCH:SEARCH1:TRIGGER:A:BUS:USB:TOKENTYPE IN sets the token type for USB bus trigger search 1 to IN.

SEARCH:SEARCH1:TRIGGER:A:BUS:USB:TOKENTYPE? might return SEARCH:SEARCH1:TRIGGER:A:BUS:USB:TOKENTYPE SOF, indicating that the token type for USB bus trigger search 1 is set to Start Of Frame.

SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:DATARate

This command sets or queries the DDR read search data rate for DDR3 and LPDDR3 standards of the specified search.

Conditions Requires option 6-DBDDR3.

Requires a 6 Series MSO oscilloscope.

Set the DDR Read standard (DDR3 or LPDDR3) before using this command. See Related Commands.

Group Search

Syntax SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:DATARate {333|800|1066|1200|1333|1466|1600|1866|2133}

SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:DATARate?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:STANdard](#)

Arguments 333–2133 sets the data rate to the specified value.

The DDR3 standard supports the following data rates:
800|1066|1333|1600|1866|2133.

The LPDDR3 standard supports the following data rates:
333|800|1066|1200|1333|1466|1600|1866|2133.

Examples SEARCH:SEARCH2:TRIGger:A:DDRREAD:DATARate 1333 sets the data rate to 1333 for DDR read Search 2.

SEARCH:SEARCH1:TRIGger:A:DDRREAD:DATARate? might return SEARCH:SEARCH1:TRIGger:A:DDRREAD:DATARate 800 indicating that the data rate for DDR read Search 1 is 800.

SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:DATASource

This command sets or queries the DDR read data source when the search type is DDR READ.

Conditions	Requires option 6-DBDDR3 Requires 6 Series MSO oscilloscope Load a reference waveform on the instrument before using this command to set the search data source to a reference waveform.
Group	Search
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:DATASource {CH<x> CH<x>_D<x> Math<x> REF<x> REF<x>_D<x>} SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:DATASource?</pre>
Arguments	CH<x> specifies channel <x> as the DDR read data source for the specified search <x>. CH<x>_D<x> specifies digital waveform <x> of channel <x> as the DDR read data source for the specified search <x>. Math<x> specifies math waveform <x> as the DDR read data source for the specified search <x>. REF<x> specifies reference waveform <x> as the DDR read data source for the specified search <x>. REF<x>_D<x> specifies digital waveform <x> of reference waveform <x> as the DDR read data source for the specified search <x>.
Examples	SEARCH:SEARCH1:TRIGGER:A:DDRREAD:DATASource CH2 sets the source waveform for the A trigger to Channel 2. SEARCH:SEARCH1:TRIGGER:A:DDRREAD:DATASource? might return SEARCH:SEARCH1:TRIGGER:A:DDRREAD:DATASource REF1, indicating that the Reference waveform 1 DDR read trigger data is the source for search 1.

SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:HYSteresis

This command sets or queries the DDR read hysteresis reference level value, when the search type is DDR READ.

Conditions	Requires option 6-DBDDR3.
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Requires a 6 Series MSO oscilloscope.

Set the DDR read reference level mode to Manual before using this command.
See Related Commands.

Group Search

Syntax SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:HYSteresis <NR3>
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:HYSteresis?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLEVELMode](#)

Arguments NR3 sets the DDR read search hysteresis percent value in the range of 0% to 50%.

Examples SEARCH:SEARCH2:TRIGger:A:DDRREAD:HYSteresis 15 sets the DDR read search hysteresis value to 15%.

SEARCH:SEARCH1:TRIGger:A:DDRREAD:HYSteresis? might return
SEARCH:SEARCH1:TRIGger:A:DDRREAD:HYSteresis 10, indicating that the
DDR read search hysteresis value is set to 10%.

SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:MARgin

This command sets or queries the DDR read margin reference level value, when the search type is DDR READ.

Conditions Requires option 6-DBDDR3.

Requires a 6 Series MSO oscilloscope.

Set the DDR read reference level mode to Manual before using this command.
See Related Commands.

Group Search

Syntax SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:MARgin <NR3>
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:MARgin?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLEVELMode](#)

Arguments NR3 sets the DDR read search margin percent value in the range of 0% to 100%.

- Examples** `SEARCH:SEARCH2:TRIGger:A:DDRREAD:MARgin 12` sets the DDR read search margin value to 12%.
- `SEARCH:SEARCH1:TRIGger:A:DDRREAD:MARgin?` might return `SEARCH:SEARCH1:TRIGger:A:DDRREAD:MARgin 10`, indicating that the DDR read search margin value is set to 10%.

SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:POSTAMBLE:LENGTH

This command sets or queries the DDR read postamble length when the search type is DDR READ.

- Conditions** Requires option 6-DBDDR3.
Requires a 6 Series MSO oscilloscope.
- Group** Search
- Syntax** `SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:POSTAMBLE:LENGTH {500E-3|500E-1.5|500E-1}`
`SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:POSTAMBLE:LENGTH?`
- Arguments** `500E-1.5` sets the postamble length to 1.5 tCK (number of clock cycles).
`500E-1` sets the postamble length to 1 tCK (number of clock cycles).
`500E-3` sets the postamble length to 0.5 tCK (number of clock cycles).
- Examples** `SEARCH:SEARCH2:TRIGger:A:DDRREAD:POSTAMBLE:LENGTH 500E-3` sets the DDR read postamble length for Search 2 to 0.5 tCK.
- `SEARCH:SEARCH1:TRIGger:A:DDRREAD:POSTAMBLE:LENGTH?` might return `SEARCH:SEARCH1:TRIGger:A:DDRREAD:POSTAMBLE:LENGTH 500E-1`, indicating that the read postamble length for Search 1 is set to 1 tCK.

SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:PREAMBLE:TYPE

This command sets or queries the DDR read preamble type when the search type is DDR READ.

- Conditions** Requires option 6-DBDDR3.
Requires a 6 Series MSO oscilloscope.

Group	Search
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:PREAMBLE:TYPE {STATIC DYNAMIC} SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:PREAMBLE:TYPE?</pre>
Arguments	STATIC sets the DDR Read preamble type to Static. DYNAMIC sets the DDR Read preamble type to Dynamic.
Examples	SEARCH:SEARCH2:TRIGger:A:DDRREAD:PREAMBLE:TYPE STATIC sets the DDR read preamble type to Static for Search 2. SEARCH:SEARCH4:TRIGger:A:DDRREAD:PREAMBLE:TYPE? might return SEARCH:SEARCH4:TRIGger:A:DDRREAD:PREAMBLE:TYPE DYNAMIC, indicating that the DDR read preamble type for Search 4 is set to dynamic mode.

SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:HIGH

This command sets or queries the DDR read data high reference level value, when the search type is DDR READ.

Conditions	Requires option 6-DBDDR3. Requires a 6 Series MSO oscilloscope. Set the DDR read reference level mode to Manual before using this command. See Related Commands.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:HIGH <NR3> SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:HIGH?</pre>
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLEVELMode
Arguments	NR3 sets the DDR read search data high reference value.
Examples	SEARCH:SEARCH2:TRIGger:A:DDRREAD:REFLevel:DATA:HIGH 3.8E-3 sets the DDR read search data high reference value.

SEARCH:SEARCH1:TRIGger:A:DDRREAD:REFLevel:DATA:HIGh? might return SEARCH:SEARCH1:TRIGger:A:DDRREAD:REFLevel:DATA:HIGh 50E-3, indicating that the DDR read search data high reference value is set to 50E-3.

SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:LOW

This command sets or queries the DDR read data low reference level value, when the search type is DDR READ.

Conditions	Requires option 6-DBDDR3.
	Requires a 6 Series MSO oscilloscope.
	Set the DDR read reference level mode to Manual before using this command. See Related Commands.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:LOW <NR3>
	SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:LOW?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLEVELMode
Arguments	NR3 sets the DDR read search data low reference value.
Examples	SEARCH:SEARCH2:TRIGger:A:DDRREAD:REFLevel:DATA:LOW 3.8E-3 sets the DDR read search data low reference value.
	SEARCH:SEARCH1:TRIGger:A:DDRREAD:REFLevel:DATA:LOW? might return SEARCH:SEARCH1:TRIGger:A:DDRREAD:REFLevel:DATA:LOW 50E-3, indicating that the DDR read search data low reference value is set to 50E-3.

SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:MID

This command sets or queries the DDR read data mid reference level value, when the search type is DDR READ.

Conditions	Requires option 6-DBDDR3.
	Requires a 6 Series MSO oscilloscope.

Set the DDR read reference level mode to Manual before using this command.
See Related Commands.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLeve1:DATA:MID <NR3>
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLeve1:DATA:MID?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLEVELMode](#)

Arguments NR3 sets the DDR read search data mid reference value.

Examples SEARCH:SEARCH2:TRIGger:A:DDRREAD:REFLeve1:DATA:MID 3.8E-3 sets the DDR read search data mid reference value.

SEARCH:SEARCH1:TRIGger:A:DDRREAD:REFLeve1:DATA:MID? might return SEARCH:SEARCH1:TRIGger:A:DDRREAD:REFLeve1:DATA:MID 50E-3, indicating that the DDR read search data mid reference value is set to 50E-3.

SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLeve1:STROBE:HIGH

This command sets or queries the DDR read reference level strobe high value, when the search type is DDR READ.

Conditions Requires option 6-DBDDR3.

Requires a 6 Series MSO oscilloscope.

Set the DDR read reference level mode to Manual before using this command.
See Related Commands.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLeve1:STROBE:HIGH <NR3>
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLeve1:STROBE:HIGH?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLEVELMode](#)

Arguments NR3 sets the DDR read search strobe high value.

Examples `SEARCH:SEARCH2:TRIGger:A:DDRREAD:REFLevel:STROBE:HIGH 3.8E-3`
sets the DDR read search strobe high value.

`SEARCH:SEARCH1:TRIGger:A:DDRREAD:REFLevel:STROBE:HIGH?` might
return `SEARCH:SEARCH1:TRIGger:A:DDRREAD:REFLevel:STROBE:HIGH`
`50E-3`, indicating that the DDR read search strobe high value is set to 50E-3.

SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:STROBE:LOW

This command sets or queries the DDR read reference level strobe low value, when the search type is DDR READ.

Conditions Requires option 6-DBDDR3.
Requires a 6 Series MSO oscilloscope.
Set the DDR read reference level mode to Manual before using this command.
See Related Commands.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:STROBE:LOW <NR3>`
`SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:STROBE:LOW?`

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLEVELMode](#)

Arguments NR3 sets the DDR read search strobe low reference value.

Examples `SEARCH:SEARCH2:TRIGger:A:DDRREAD:REFLevel:STROBE:LOW 3.8E-3`
sets the DDR read search strobe low reference value.

`SEARCH:SEARCH1:TRIGger:A:DDRREAD:REFLevel:STROBE:LOW?` might
return `SEARCH:SEARCH1:TRIGger:A:DDRREAD:REFLevel:STROBE:MID`
`50E-3`, indicating that the DDR read search strobe low reference value is set
to 50E-3.

SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:STROBE:MID

This command sets or queries the DDR read reference level strobe mid value, when the search type is DDR READ.

Conditions Requires option 6-DBDDR3.

Requires a 6 Series MSO oscilloscope.

Set the DDR read reference level mode to Manual before using this command.
See Related Commands.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLeve1:STROBE:MID <NR3>
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLeve1:STROBE:MID?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLEVELMode](#)

Arguments NR3 sets the DDR read search strobe mid reference value.

Examples SEARCH:SEARCH2:TRIGger:A:DDRREAD:REFLeve1:STROBE:MID 3.8E-3
sets the DDR read search strobe mid reference value.

SEARCH:SEARCH1:TRIGger:A:DDRREAD:REFLeve1:STROBE:MID? might
return SEARCH:SEARCH1:TRIGger:A:DDRREAD:REFLeve1:STROBE:MID
50E-3, indicating that the DDR read search strobe mid reference value is set
to 50E-3.

SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLEVELMode

This command sets or queries the DDR read reference level mode to auto or manual, when the search type is DDR READ.

Conditions Requires option 6-DBDDR3.
Requires a 6 Series MSO oscilloscope.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLEVELMode
{AUTO|MANUAL}
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLEVELMode?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:HIGH](#)
[SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:LOW](#)
[SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:MID](#)

Arguments	AUTO sets the DDR read reference level mode to auto.
	MANUAL sets the DDR read reference level mode to manual. Use the Related Commands to set the Reference levels when in Manual mode.
Examples	SEARCH:SEARCH3:TRIGger:A:DDRREAD:REFLEVELMode MANUAL sets the DDR read search reference level mode to Manual for Search 3.
	SEARCH:SEARCH1:TRIGger:A:DDRREAD:REFLEVELMode? might return SEARCH:SEARCH1:TRIGger:A:DDRREAD:REFLEVELMode MANUAL, indicating that the DDR read search reference level mode set to automatic for Search 1.

SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:STANDard

This command sets or queries the DDR read search standard as DDR3 or LPDDR3.

Conditions	Requires option 6-DBDDR3
	Requires 6 Series MSO oscilloscope
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:STANDard {DDR3 LPDDR3} SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:STANDard?
Arguments	DDR3 sets the DDR read search standard as DDR3.
	LPDDR3 sets the DDR read search standard as LPDDR3.
Examples	SEARCH:SEARCH2:TRIGger:A:DDRREAD:STANDard DDR3 sets the DDR standard to DDR3 for DDR read Search 2.
	SEARCH:SEARCH5:TRIGger:A:DDRREAD:STANDard? might return SEARCH:SEARCH5:TRIGger:A:DDRREAD:STANDard LPDDR3, indicating that the DDR standard is set to LPDDR3 for DDR read Search 5.

SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:STROBESource

This command sets or queries the DDR read strobe source when the search type is DDR READ.

Conditions	Requires option 6-DBDDR3
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Requires 6 Series MSO oscilloscope

Load a reference waveform on the instrument before using this command to set the search data source to a reference waveform.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:STROBESource
{CH<x>|CH<x>_D<x>|Math<x>|REF<x>|REF<x>_D<x>}
SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:STROBESource?

Arguments CH<x> specifies channel <x> as the DDR read strobe trigger source for the specified search <x>.

CH<x>_D<x> specifies digital waveform <x> of channel <x> as the DDR read strobe trigger source for the specified search <x>.

Math<x> specifies math waveform <x> as the DDR read strobe trigger source for the specified search <x>.

REF<x> specifies reference waveform <x> as the DDR read strobe trigger source for the specified search <x>.

REF<x>_D<x> specifies digital waveform <x> of reference waveform <x> as the DDR read strobe trigger source for the specified search <x>.

Examples SEARCH:SEARCH3:TRIGGER:A:DDRREAD:STROBESource MAT25 sets the Search 3 source waveform to Math 2 for the DDR read strobe search.

SEARCH:SEARCH1:TRIGGER:A:DDRREAD:STROBESource? might return SEARCH:SEARCH1:TRIGGER:A:DDRREAD:STROBESource CH3, indicating that the Channel 3 waveform is the source for the DDR read strobe of Search 1.

SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:DATARate

This command sets or queries the DDR read/write data rate for DDR3 and LPDDR3 standards of the specified search.

Conditions Requires option 6-DBDDR3.

Requires a 6 Series MSO oscilloscope.

Set the DDR Read/Write standard (DDR3 or LPDDR3) before using this command. See Related Commands.

Group Search and Mark

Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:DAtARate {333 800 1066 1200 1333 1466 1600 1866 2133} SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:DAtARate?</pre>
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:STANdard
Arguments	333–2133 sets the data rate to the specified value. The DDR3 standard supports the following data rates: 800 1066 1333 1600 1866 2133. The LPDDR3 standard supports the following data rates: 333 800 1066 1200 1333 1466 1600 1866 2133.
Examples	SEARCH:SEARCH2:TRIGger:A:DDRREADWRITE:DAtARate 1333 sets the data rate to 1333 for DDR read Search 2. SEARCH:SEARCH5:TRIGger:A:DDRREADWRITE:DAtARate? might return SEARCH:SEARCH5:TRIGger:A:DDRREADWRITE:DAtARate 1466 indicating that the data rate for DDR read Search 5 is 1466.

SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:DAtASource

This command sets or queries the DDR read/write data source when the search type is DDR READWRITE.

Conditions	Requires option 6-DBDDR3 Requires 6 Series MSO oscilloscope Load a reference waveform on the instrument before using this command to set the search data source to a reference waveform.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:DAtASource {CH<x> CH<x>_D<x> Math<x> REF<x> REF<x>_D<x>} SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:DAtASource?</pre>
Arguments	CH<x> specifies channel <x> as the DDR read/write data source for the specified search <x>. CH<x>_D<x> specifies digital waveform <x> of channel <x> as the DDR read/write data source for the specified search <x>.

Math<x> specifies math waveform <x> as the DDR read/write data source for the specified search <x>.

REF<x> specifies reference waveform <x> as the DDR read/write data source for the specified search <x>.

REF<x>_D<x> specifies digital waveform <x> of reference waveform <x> as the DDR read/write data source for the specified search <x>.

Examples **SEARCH:SEARCH3:TRIGGER:A:DDRREADWRITE:DATASource CH2** sets the source waveform of Search 3 to Channel 2 for the DDR read/write search.

SEARCH:SEARCH1:TRIGGER:A:DDRREADWRITE:DATASource? might return **SEARCH:SEARCH1:TRIGGER:A:DDRREADWRITE:DATASource CH2**, indicating that the Channel 2 waveform is the source for the DDR read/write data of Search 1.

SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:HYSteresis

This command sets or queries the DDR read/write hysteresis reference level value, when the search type is DDR READWRITE.

Conditions	Requires option 6-DBDDR3. Requires a 6 Series MSO oscilloscope. Set the DDR read/write reference level mode to Manual before using this command. See Related Commands.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:HYSteresis <NR3> SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:HYSteresis?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLEVELMode
Arguments	NR3 sets the DDR read/write search hysteresis percent value in the range of 0% to 50%.
Examples	SEARCH:SEARCH2:TRIGger:A:DDRREADWRITE:HYSteresis 10 sets the DDR read/write search hysteresis value to 10%. SEARCH:SEARCH1:TRIGger:A:DDRREADWRITE:HYSteresis? might return SEARCH:SEARCH1:TRIGger:A:DDRREADWRITE:HYSteresis 5, indicating that the DDR read/write search hysteresis value is set to 5%.

SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:MARgin

This command sets or queries the DDR read/write margin reference level value, when the search type is DDR READWRITE.

Conditions Requires option 6-DBDDR3.

Requires a 6 Series MSO oscilloscope.

Set the DDR read/write reference level mode to Manual before using this command. See Related Commands.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:MARgin <NR3>
SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:MARgin?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLEVELMode](#)

Arguments NR3 sets the DDR read/write search margin percent value in the range of 0% to 100%.

Examples SEARCH:SEARCH2:TRIGger:A:DDRREADWRITE:MARgin 15 sets the DDR read/write search margin value to 15%.

SEARCH:SEARCH1:TRIGger:A:DDRREADWRITE:MARgin? might return
SEARCH:SEARCH1:TRIGger:A:DDRREADWRITE:MARgin 10, indicating that the DDR read/write search margin value is set to 10%.

SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:POSTAMBLE:LENGth

This command sets or queries the DDR read/write postamble length when the search type is DDR READWRITE.

Conditions Requires option 6-DBDDR3.

Requires a 6 Series MSO oscilloscope.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:POSTAMBLE:LENGth
{500E-3|1.5|1}

SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:POSTAMBLE:LENGTH?

Arguments 500E-1.5 sets the postamble length to 1.5 tCK (number of clock cycles).
500E-1 sets the postamble length to 1 tCK (number of clock cycles).
500E-3 sets the postamble length to 0.5 tCK (number of clock cycles).

Examples SEARCH:SEARCH3:TRIGger:A:DDRREADWRITE:POSTAMBLE:LENGTH
500E-1.5 sets the DDR read/write postamble length for Search 3 to 1.5 tCK.

SEARCH:SEARCH2:TRIGger:A:DDRREADWRITE:POSTAMBLE:LENGTH? might
return SEARCH:SEARCH2:TRIGger:A:DDRREADWRITE:POSTAMBLE:LENGTH
500E-1, indicating that the read/write postamble length for Search 2 is set to 1
tCK.

SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:PREAMBLE:TYPE

This command sets or queries the DDR read/write preamble type when the search
type is DDR READWRITE.

Conditions Requires option 6-DBDDR3.
Requires a 6 Series MSO oscilloscope.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:PREAMBLE:TYPE
{STATIC|DYNAMIC}
SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:PREAMBLE:TYPE?

Arguments STATIC sets the DDR Read/Write preamble type to Static.
DYNAMIC sets the DDR Read/Write preamble type to Dynamic.

Examples SEARCH:SEARCH2:TRIGger:A:DDRREADWRITE:PREAMBLE:TYPE STATIC
sets the search 2 DDR read preamble type to Static.

SEARCH:SEARCH5:TRIGger:A:DDRREADWRITE:PREAMBLE:TYPE? might
return SEARCH:SEARCH5:TRIGger:A:DDRREADWRITE:PREAMBLE:TYPE
DYNAMIC, indicating that the DDR read/write preamble type for Search 5 is set
to dynamic mode.

SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLevel:DATA:HIGh

This command sets or queries the DDR read/write data high reference level value, when the search type is DDR READWRITE.

Conditions	Requires option 6-DBDDR3. Requires a 6 Series MSO oscilloscope. Set the DDR read/write reference level mode to Manual before using this command. See Related Commands.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLevel:DATA:HIGh <NR3> SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLevel:DATA:HIGh?</pre>
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLEVELMode
Arguments	NR3 sets the DDR read/write search data high reference value.
Examples	<pre>SEARCH:SEARCH2:TRIGger:A:DDRREADWRITE:REFLevel:DATA:HIGh 3.8E-3</pre> sets the DDR read/write search data high reference value. <pre>SEARCH:SEARCH1:TRIGger:A:DDRREADWRITE:REFLevel:DATA:HIGh? might return SEARCH:SEARCH1:TRIGger:A:DDRREADWRITE:REFLevel:DATA:HIGh 50E-3,</pre> indicating that the DDR read/write search data high reference value is set to 50E-3.

SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLevel:DATA:LOW

This command sets or queries the DDR read/write data low reference level value, when the search type is DDR READWRITE.

Conditions	Requires option 6-DBDDR3. Requires a 6 Series MSO oscilloscope. Set the DDR read/write reference level mode to Manual before using this command. See Related Commands.
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Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLeve1:DATA:LOW <NR3> SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLeve1:DATA:LOW?</pre>
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLEVELMode
Arguments	NR3 sets the DDR read/write search data low reference value.
Examples	<pre>SEARCH:SEARCH2:TRIGger:A:DDRREADWRITE:REFLeve1:DATA:LOW 3.8E-3 sets the DDR read/write search data low reference value. SEARCH:SEARCH1:TRIGger:A:DDRREADWRITE:REFLeve1:DATA:LOW? might return SEARCH:SEARCH1:TRIGger:A:DDRREADWRITE:REFLeve1:DATA:LOW 50E-3, indicating that the DDR read/write search data low reference value is set to 50E-3.</pre>

SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLevel:DATA:MID

This command sets or queries the DDR read/write data mid reference level value, when the search type is DDR READWRITE.

Conditions	Requires option 6-DBDDR3. Requires a 6 Series MSO oscilloscope. Set the DDR read/write reference level mode to Manual before using this command. See Related Commands.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLeve1:DATA:MID <NR3> SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLeve1:DATA:MID?</pre>
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLEVELMode
Arguments	NR3 sets the DDR read/write search data mid reference value.

Examples `SEARCH:SEARCH2:TRIGger:A:DDRREADWRITE:REFLevel:DATA:MID 3.8E-3` sets the DDR read/write search data mid reference value.

`SEARCH:SEARCH1:TRIGger:A:DDRREAWRITE:REFLevel:DATA:MID?` might return `SEARCH:SEARCH1:TRIGger:A:DDRREADWRITE:REFLevel:DATA:MID 50E-3`, indicating that the DDR read/write search data mid reference value is set to 50E-3.

SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLevel:STROBE:HIGH

This command sets or queries the DDR read/write reference level strobe high value, when the search type is DDR READWRITE.

Conditions Requires option 6-DBDDR3.

Requires a 6 Series MSO oscilloscope.

Set the DDR read/write reference level mode to Manual before using this command. See Related Commands.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLevel:STROBE:HIGH <NR3>`
`SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLevel:STROBE:HIGH?`

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLEVELMode](#)

Arguments NR3 sets the DDR read search strobe high value.

Examples `SEARCH:SEARCH2:TRIGger:A:DDRREADWRITE:REFLevel:STROBE:HIGH 3.8E-3` sets the DDR read/write search strobe high value.

`SEARCH:SEARCH1:TRIGger:A:DDRREADWRITE:REFLevel:STROBE:HIGH?` might return `SEARCH:SEARCH1:TRIGger:A:DDRREADWRITE:REFLevel:STROBE:HIGH 50E-3`, indicating that the DDR read/write search strobe high value is set to 50E-3.

SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLevel:STROBE:LOW

This command sets or queries the DDR read/write reference level strobe low value, when the search type is DDR READWRITE.

Conditions	Requires option 6-DBDDR3. Requires a 6 Series MSO oscilloscope. Set the DDR read/write reference level mode to Manual before using this command. See Related Commands.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLeve1:STROBE:LOW <NR3> SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLeve1:STROBE:LOW?</pre>
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLEVELMode
Arguments	NR3 sets the DDR read/write search strobe low reference value.
Examples	<pre>SEARCH:SEARCH2:TRIGger:A:DDRREADWRITE:REFLeve1:STROBE:LOW 3.8E-3 sets the DDR read/write search strobe low reference value. SEARCH:SEARCH1:TRIGger:A:DDRREAWRITE:REFLeve1:STROBE:LOW? might return SEARCH:SEARCH1:TRIGger:A:DDRREADWRITE:REFLeve1:STROBE:LOW 50E-3, indicating that the DDR read/write search strobe low reference value is set to 50E-3.</pre>

SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLevel:STROBE:MID

This command sets or queries the DDR read/write reference level strobe mid value, when the search type is DDR READWRITE.

Conditions	Requires option 6-DBDDR3. Requires a 6 Series MSO oscilloscope. Set the DDR read/write reference level mode to Manual before using this command. See Related Commands.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLeve1:STROBE:MID <NR3> SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLeve1:STROBE:MID?</pre>

Related Commands	SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLEVELMode
Arguments	NR3 sets the DDR read/write search strobe mid reference value.
Examples	SEARCH:SEARCH2:TRIGger:A:DDRREADWRITE:REFLevel:STROBE:MID 3.8E-3 sets the DDR read/write search strobe mid reference value. SEARCH:SEARCH1:TRIGger:A:DDRREAWRITE:REFLevel:STROBE:MID? might return SEARCH:SEARCH1:TRIGger:A:DDRREADWRITE:REFLevel:STROBE:MID 50E-3, indicating that the DDR read/write search strobe mid reference value is set to 50E-3.

SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLEVELMode

This command sets or queries the DDR read/write reference level mode to auto or manual, when the search type is DDR READWRITE.

Conditions	Requires option 6-DBDDR3. Requires a 6 Series MSO oscilloscope.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLEVELMode {AUTO MANUAL} SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:REFLEVELMode?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:HIGH SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:LOW SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:MID
Arguments	AUTO sets the DDR read/write reference level mode to auto. MANUAL sets the DDR read/write reference level mode to manual. Use the Related Commands to set the Reference levels when in Manual mode.
Examples	SEARCH:SEARCH3:TRIGger:A:DDRREADWRITE:REFLEVELMode MANUAL sets the DDR read/write search reference level mode to Manual for Search 3.

SEARCH:SEARCH1:TRIGger:A:DDRREADWRITE:REFLEVELMode? might return SEARCH:SEARCH1:TRIGger:A:DDRREADWRITE:REFLEVELMode MANUAL, indicating that the DDR read/write search reference level mode set to automatic for Search 1.

SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:STANdard

This command sets or queries the DDR read/write search standard as DDR3 or LPDDR3.

Conditions	Requires option 6-DBDDR3. Requires a 6 Series MSO oscilloscope.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:STANdard {DDR3 LPDDR3} SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:STANdard?
Arguments	DDR3 sets the DDR read/write search standard as DDR3. LPDDR3 sets the DDR read/write search standard as LPDDR3.
Examples	SEARCH:SEARCH2:TRIGger:A:DDRREADWRITE:STANdard DDR3 sets the DDR standard to DDR3 for DDR read/write Search 2. SEARCH:SEARCH2:TRIGger:A:DDRREADWRITE:STANdard? might return SEARCH:SEARCH2:TRIGger:A:DDRREADWRITE:STANdard LPDDR3 indicating that the DDR standard is set to LPDDR3 for DDR read/write Search 2.

SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:STROBESource

This command sets or queries the DDR readwrite strobe source when the search type is DDR READWRITE.

Conditions	Requires option 6-DBDDR3. Requires a 6 Series MSO oscilloscope. Load a reference waveform on the instrument before using this command to set the search data source to a reference waveform.
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Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:STROBESource {CH<x> CH<x>_D<x> Math<x> REF<x> REF<x>_D<x>} SEARCH:SEARCH<x>:TRIGger:A:DDRREADWRITE:STROBESource?</pre>
Arguments	CH<x> specifies channel <x> as the DDR read/write strobe source for the specified search <x>. CH<x>_D<x> specifies digital waveform <x> of channel <x> as the DDR read/write strobe source for the specified search <x>. Math<x> specifies math waveform <x> as the DDR read/write strobe source for the specified search <x>. REF<x> specifies reference waveform <x> as the DDR read/write strobe source for the specified search <x>. REF<x>_D<x> specifies digital waveform <x> of reference waveform <x> as the DDR read/write strobe source for the specified search <x>.
Examples	SEARCH:SEARCH5:TRIGGER:A:DDRREADWRITE:STROBESource CH8 sets the Search 5 source waveform to Channel 8 for the DDR read/write strobe search. SEARCH:SEARCH1:TRIGGER:A:DDRREADWRITE:STROBESource? might return SEARCH:SEARCH1:TRIGGER:A:DDRREADWRITE:STROBESource CH3, indicating that the Channel 3 waveform is the source for the DDR read/write strobe of Search 1.

SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:DATARate

This command sets or queries the DDR write search data rate for DDR3 and LPDDR3 standards of the specified search.

Conditions	Requires option 6-DBDDR3. Requires a 6 Series MSO oscilloscope. Set the DDR Write standard (DDR3 or LPDDR3) before using this command. See Related Commands.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:DATARate {333 800 1066 1200 1333 1466 1600 1866 2133} SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:DATARate?</pre>

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:STANDARD](#)

Arguments 333–2133 sets the data rate to the specified value.

The DDR3 standard supports the following data rates:
800|1066|1333|1600|1866|2133.

The LPDDR3 standard supports the following data rates:
333|800|1066|1200|1333|1466|1600|1866|2133.

Examples `SEARCH:SEARCH2:TRIGger:A:DDRWRITE:DATARate 1333` sets the data rate to 1333 for DDR read Search 2.

`SEARCH:SEARCH7:TRIGger:A:DDRWRITE:DATARate?` might return
`SEARCH:SEARCH7:TRIGger:A:DDRWRITE:DATARate 1466` indicating that the data rate for DDR read Search 7 is 1466.

SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:DATASource

This command sets or queries the DDR write data source when the search type is DDR Write.

Conditions Requires option 6-DBDDR3

Requires 6 Series MSO oscilloscope

Load a reference waveform on the instrument before using this command to set the search data source to a reference waveform.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:DATASource`
 `{CH<x>|CH<x>_D<x>|Math<x>|REF<x>|REF<x>_D<x>}`
 `SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:DATASource?`

Arguments `CH<x>` specifies channel <x> as the DDR write data source for the specified search <x>.

`CH<x>_D<x>` specifies digital waveform <x> of channel <x> as the DDR write data source for the specified search <x>.

`Math<x>` specifies math waveform <x> as the DDR write data source for the specified search <x>.

`REF<x>` specifies reference waveform <x> as the DDR write data source for the specified search <x>.

REF<x>_D<x> specifies digital waveform <x> of reference waveform <x> as the DDR write data source for the specified search <x>.

Examples SEARCH:SEARCH1:TRIGGER:A:WRITE:DATASource CH2 sets the search 1 source waveform to Channel 2 for the DDR write search.

SEARCH:SEARCH1:TRIGGER:A:DDRWRITE:DATASource? might return SEARCH:SEARCH1:TRIGGER:A:DDRWRITE:DATASource RMATH3, indicating that the Math 1 waveform DDR write trigger data is the source for search 1.

SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:HYSteresis

This command sets or queries the DDR write hysteresis reference level value, when the search type is DDR WRITE.

Conditions Requires option 6-DBDDR3.

Requires a 6 Series MSO oscilloscope.

Set the DDR write reference level mode to Manual before using this command. See Related Commands.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:HYSteresis <NR3>
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:HYSteresis?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLEVELMode](#)

Arguments NR3 sets the DDR write search hysteresis percent value in the range of 0% to 50%.

Examples SEARCH:SEARCH2:TRIGger:A:DDRWRITE:HYSteresis 7 sets the DDR write search hysteresis value to 7%.

SEARCH:SEARCH1:TRIGger:A:DDRWRITE:HYSteresis? might return SEARCH:SEARCH1:TRIGger:A:DDRWRITE:HYSteresis 10, indicating that the DDR write search hysteresis value is set to 10%.

SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:MARgin

This command sets or queries the DDR write margin reference level value, when the search type is DDR WRITE.

Conditions	Requires option 6-DBDDR3. Requires a 6 Series MSO oscilloscope. Set the DDR write reference level mode to Manual before using this command. See Related Commands.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:MARgin <NR3> SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:MARgin?</pre>
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLEVELMode
Arguments	NR3 sets the DDR write search margin percent value in the range of 0% to 100%.
Examples	SEARCH:SEARCH2:TRIGger:A:DDRWRITE:MARgin 11 sets the DDR write search margin value to 11%. SEARCH:SEARCH1:TRIGger:A:DDRWRITE:MARgin? might return SEARCH:SEARCH1:TRIGger:A:DDRWRITE:MARgin 5, indicating that the DDR write search margin value is set to 5%.

SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:POSTAMBLE:LENGth

This command sets or queries the DDR write postamble length when the search type is DDR WRITE.

Conditions	Requires option 6-DBDDR3. Requires a 6 Series MSO oscilloscope.
Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:POSTAMBLE:LENGth {500E-3 1.5 1} SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:POSTAMBLE:LENGth?</pre>
Arguments	500E-1.5 sets the postamble length to 1.5 tCK (number of clock cycles). 500E-1 sets the postamble length to 1 tCK (number of clock cycles). 500E-3 sets the postamble length to 0.5 tCK (number of clock cycles).

Examples `SEARCH:SEARCH6:TRIGger:A:DDRWRITE:POSTAMBLE:LENGTH 500E-3` sets the DDR write postamble length for Search 6 to 0.5 tCK.

`SEARCH:SEARCH2:TRIGger:A:DDRWRITE:POSTAMBLE:LENGTH?` might return `SEARCH:SEARCH2:TRIGger:A:DDRWRITE:POSTAMBLE:LENGTH 500E-1`, indicating that the write postamble length for Search 2 is set to 1 tCK.

SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:PREAMBLE:TYPE

This command sets or queries the DDR write preamble type when the search type is DDR WRITE.

Conditions Requires option 6-DBDDR3.
Requires a 6 Series MSO oscilloscope.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:PREAMBLE:TYPE {STATIC|DYNAMIC}`
`SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:PREAMBLE:TYPE?`

Arguments `STATIC` sets the DDR Write preamble type to Static.
`DYNAMIC` sets the DDR Write preamble type to Dynamic.

Examples `SEARCH:SEARCH1:TRIGger:A:DDRWRITE:PREAMBLE:TYPE STATIC` sets the DDR write preamble type to Static for Search 1.

`SEARCH:SEARCH4:TRIGger:A:DDRWRITE:PREAMBLE:TYPE?` might return `SEARCH:SEARCH4:TRIGger:A:DDRWRITE:PREAMBLE:TYPE DYNAMIC`, indicating that the DDR write preamble type for Search 4 is set to dynamic mode.

SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLevel:DATA:HIGH

This command sets or queries the DDR write data high reference level value, when the search type is DDR WRITE.

Conditions Requires option 6-DBDDR3.
Requires a 6 Series MSO oscilloscope.
Set the DDR write reference level mode to Manual before using this command.
See Related Commands.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLeve1:DATA:HIGH <NR3> SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLeve1:DATA:HIGH?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLEVELMode
Arguments	NR3 sets the DDR read search data high reference value.
Examples	SEARCH:SEARCH2:TRIGger:A:DDRWRITE:REFLeve1:DATA:HIGH 3.8E-3 sets the DDR write search data high reference value. SEARCH:SEARCH1:TRIGger:A:DDRWRITE:REFLeve1:DATA:HIGH? might return SEARCH:SEARCH1:TRIGger:A:DDRWRITE:REFLeve1:DATA:HIGH 50E-3, indicating that the DDR write search data high reference value is set to 50E-3.

SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLevel:DATA:LOW

This command sets or queries the DDR write data low reference level value, when the search type is DDR WRITE.

Conditions	Requires option 6-DBDDR3. Requires a 6 Series MSO oscilloscope. Set the DDR write reference level mode to Manual before using this command. See Related Commands.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLeve1:DATA:LOW <NR3> SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLeve1:DATA:LOW?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLEVELMode
Arguments	NR3 sets the DDR read search data low reference value.
Examples	SEARCH:SEARCH2:TRIGger:A:DDRWRITE:REFLeve1:DATA:LOW 3.8E-3 sets the DDR write search data low reference value.

SEARCH:SEARCH1:TRIGger:A:DDRWRITE:REFLevel:DATA:LOW? might return SEARCH:SEARCH1:TRIGger:A:DDRWRITE:REFLevel:DATA:LOW 50E-3, indicating that the DDR write search data low reference value is set to 50E-3.

SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLevel:DATA:MID

This command sets or queries the DDR write data mid reference level value, when the search type is DDR WRITE.

Conditions Requires option 6-DBDDR3.
Requires a 6 Series MSO oscilloscope.
Set the DDR write reference level mode to Manual before using this command. See Related Commands.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLevel:DATA:MID <NR3>
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLevel:DATA:MID?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLEVELMode](#)

Arguments NR3 sets the DDR read search data mid reference value.

Examples SEARCH:SEARCH2:TRIGger:A:DDRWRITE:REFLevel:DATA:MID 3.8E-3 sets the DDR write search data mid reference value.

SEARCH:SEARCH1:TRIGger:A:DDRWRITE:REFLevel:DATA:MID? might return SEARCH:SEARCH1:TRIGger:A:DDRWRITE:REFLevel:DATA:MID 50E-3, indicating that the DDR write search data mid reference value is set to 50E-3.

SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLevel:STROBE:HIG

This command sets or queries the DDR write reference level strobe high value, when the search type is DDR WRITE.

Conditions Requires option 6-DBDDR3.
Requires a 6 Series MSO oscilloscope.

Set the DDR write reference level mode to Manual before using this command.
See Related Commands.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLeve1:STROBE:HIGH
<NR3>
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLeve1:STROBE:HIGH?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLEVELMode](#)

Arguments NR3 sets the DDR write search strobe high value.

Examples SEARCH:SEARCH2:TRIGger:A:DDRWRITE:REFLeve1:STROBE:HIGH
3.8E-3 sets the DDR write search strobe high value.

SEARCH:SEARCH1:TRIGger:A:DDRWRITE:REFLeve1:STROBE:HIGH? might
return SEARCH:SEARCH1:TRIGger:A:DDRWRITE:REFLeve1:STROBE:HIGH
50E-3, indicating that the DDR write search strobe high value is set to 50E-3.

SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLevel:STROBE:LOW

This command sets or queries the DDR write reference level strobe low value,
when the search type is DDR WRITE.

Conditions Requires option 6-DBDDR3.

Requires a 6 Series MSO oscilloscope.

Set the DDR write reference level mode to Manual before using this command.
See Related Commands.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLeve1:STROBE:LOW
<NR3>
SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLeve1:STROBE:LOW?

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLEVELMode](#)

Arguments NR3 sets the DDR read search strobe low reference value.

Examples `SEARCH:SEARCH2:TRIGger:A:DDRWRITE:REFLevel:STROBE:LOW 3.8E-3`
sets the DDR write search strobe low reference value.

`SEARCH:SEARCH1:TRIGger:A:DDRWRITE:REFLevel:STROBE:LOW?` might
return `SEARCH:SEARCH1:TRIGger:A:DDRWRITE:REFLevel:STROBE:LOW`
`50E-3`, indicating that the DDR write search strobe low reference value is set
to `50E-3`.

SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLevel:STROBE:MID

This command sets or queries the DDR write reference level strobe mid value,
when the search type is DDR WRITE.

Conditions Requires option 6-DBDDR3.
Requires a 6 Series MSO oscilloscope.
Set the DDR write reference level mode to Manual before using this command.
See Related Commands.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLevel:STROBE:MID`
`<NR3>`
`SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLevel:STROBE:MID?`

Related Commands [SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLEVELMode](#)

Arguments NR3 sets the DDR read search strobe mid reference value.

Examples `SEARCH:SEARCH2:TRIGger:A:DDRWRITE:REFLevel:STROBE:MID 3.8E-3`
sets the DDR write search strobe mid reference value.

`SEARCH:SEARCH1:TRIGger:A:DDRWRITE:REFLevel:STROBE:MID?` might
return `SEARCH:SEARCH1:TRIGger:A:DDRWRITE:REFLevel:STROBE:MID`
`50E-3`, indicating that the DDR write search strobe mid reference value is set
to `50E-3`.

SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLEVELMode

This command sets or queries the DDR write reference level mode to auto or
manual, when the search type is DDR WRITE.

Conditions	Requires option 6-DBDDR3. Requires a 6 Series MSO oscilloscope.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLEVELMode {AUTO MANUAL} SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:REFLEVELMode?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:HIGH SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:LOW SEARCH:SEARCH<x>:TRIGger:A:DDRREAD:REFLevel:DATA:MID
Arguments	AUTO sets the DDR write reference level mode to auto. MANUAL sets the DDR write reference level mode to manual. Use the Related Commands to set the Reference levels when in Manual mode.
Examples	SEARCH:SEARCH3:TRIGger:A:DDRWRITE:REFLEVELMode MANUAL sets the DDR write search reference level mode to Manual for Search 3. SEARCH:SEARCH1:TRIGger:A:DDRWRITE:REFLEVELMode? might return SEARCH:SEARCH1:TRIGger:A:DDRWRITE:REFLEVELMode MANUAL, indicating that the DDR write search reference level mode set to automatic for Search 1.

SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:STANdard

This command sets or queries the DDR write search standard as DDR3 or LPDDR3.

Conditions	Requires option 6-DBDDR3 Requires 6 Series MSO oscilloscope
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:STANdard {DDR3 LPDDR3} SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:STANdard?

Arguments	DDR3 sets the DDR write search standard as DDR3. LPDDR3 sets the DDR write search standard as LPDDR3.
Examples	SEARCH:SEARCH2:TRIGger:A:DDRWRITE:STANDARD DDR3 sets the DDR standard to DDR3 for DDR write Search 2. SEARCH:SEARCH5:TRIGger:A:DDRWRITE:STANDARD? might return SEARCH:SEARCH5:TRIGger:A:DDRWRITE:STANDARD LPDDR3 indicating that the DDR standard is set to LPDDR3 for DDR write Search 5.

SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:STROBESource

This command sets or queries the DDR write strobe source when the search type is DDR Write.

Conditions	Requires option 6-DBDDR3 Requires 6 Series MSO oscilloscope Load a reference waveform on the instrument before using this command to set the search data source to a reference waveform.
Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:STROBESource {CH<x> CH<x>_D<x> Math<x> REF<x> REF<x>_D<x>} SEARCH:SEARCH<x>:TRIGger:A:DDRWRITE:STROBESource?
Arguments	CH<x> specifies channel <x> as the DDR write strobe source for the specified search <x>. CH<x>_D<x> specifies digital waveform <x> of channel <x> as the DDR write strobe source for the specified search <x>. Math<x> specifies math waveform <x> as the DDR write strobe source for the specified search <x>. REF<x> specifies reference waveform <x> as the DDR write strobe source for the specified search <x>. REF<x>_D<x> specifies digital waveform <x> of reference waveform <x> as the DDR write strobe source for the specified search <x>.
Examples	SEARCH:SEARCH2:TRIGGER:A:DDRWRITE:STROBESource MATH4 sets the Search 2 source waveform to Math 4 for the DDR write strobe search.

SEARCH:SEARCH1:TRIGGER:A:DDRWRITE:STROBESource? might return
SEARCH:SEARCH1:TRIGGER:A:DDRWRITE:STROBESource CH3, indicating
that the Channel 3 waveform is the source for the DDR write strobe of Search 1.

SEARCH:SEARCH<x>:TRIGger:A:EDGE:SLOpe

This command sets or queries the slope for an edge trigger search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:EDGE:SLOpe {RISe FALL EITHer} SEARCH:SEARCH<x>:TRIGger:A:EDGE:SLOpe?
Arguments	RISe specifies a rising edge. FALL specifies a falling edge. EITHer specifies either rising or falling edge.
Examples	SEARCH:SEARCH1:TRIGGER:A:EDGE:SLOPE RISE sets the slope for search 1 to rise. SEARCH:SEARCH1:TRIGGER:A:EDGE:SLOPE? might return SEARCH:SEARCH1:TRIGGER:A:EDGE:SLOPE RISE, indicating that the slope for the trigger for search 1 is rise.

SEARCH:SEARCH<x>:TRIGger:A:EDGE:SOUrce

This command sets or queries the source waveform for an edge trigger search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:EDGE:SOUrce {CH<x> CH<x>_D<x> MATH<x> REF<x> REF<x>_D<x>} SEARCH:SEARCH<x>:TRIGger:A:EDGE:SOUrce?
Arguments	CH<x> specifies one input channel as the edge source, where the channel number is specified by <x>.

CH<x>_D<x> specifies a digital reference waveform as the source waveform for the specified search.

MATH<x> specifies the math waveform as the search source, where the math number is specified by <x>.

REF<x> specifies the reference waveform as the search source, where the reference number is specified by <x>.

REF<x>_D<x> specifies a digital reference waveform as the source waveform for the specified search.

Examples SEARCH:SEARCH1:TRIGGER:A:EDGE:SOURCE CH2 sets the source waveform for the A trigger to Channel 2.

SEARCH:SEARCH1:TRIGGER:A:EDGE:SOURCE? might return
SEARCH:SEARCH1:TRIGGER:A:EDGE:SOURCE CH2, indicating that the Channel 2 edge trigger is the source for search 1.

SEARCH:SEARCH<x>:TRIGger:A:EDGE:THReshold

This command sets or queries the source threshold level for an edge trigger search to determine where to place a mark. The search number is specified by <x>.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:EDGE:THReshold <NR3>
SEARCH:SEARCH<x>:TRIGger:A:EDGE:THReshold?

Arguments <NR3> is the source threshold level for an edge trigger search.

Examples SEARCH:SEARCH1:TRIGger:A:EDGE:THReshold 50.0e-3 sets the threshold to 50 mV.

SEARCH:SEARCH1:TRIGger:A:EDGE:THReshold? might return
SEARCH:SEARCH1:TRIGGER:A:EDGE:THRESHOLD 0.0E+0 indicating the threshold is 0.0 V.

SEARCH:SEARCH<x>:TRIGger:A:LOGIc:CLOCK:THReshold

This command sets or queries the logic clock threshold for a logic trigger search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:LOGic:CLOCK:THResho1d <NR3> SEARCH:SEARCH<x>:TRIGger:A:LOGic:CLOCK:THResho1d?</pre>
Arguments	<NR3> is the logic clock threshold.
Examples	SEARCH:SEARCH1:TRIGger:A:LOGic:CLOCK:THResho1d 50.0e-3 sets the threshold to 50 mV. SEARCH:SEARCH1:TRIGger:A:LOGic:CLOCK:THResho1d? might return SEARCH:SEARCH1:TRIGGER:A:LOGIC:CLOCK:THRESHOLD 0.0E+0 indicating the threshold is set to 0.0 V.

SEARCH:SEARCH<x>:TRIGger:A:LOGic:DELTatime

This command specifies the Logic search delta time value. The time value is used as part of the Logic search condition to determine if the duration of a logic pattern meets the specified time constraints. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:LOGic:DELTatime <NR3> SEARCH:SEARCH<x>:TRIGger:A:LOGic:DELTatime?</pre>
Arguments	<NR3> is delta time value.
Examples	SEARCH:SEARCH1:TRIGger:A:LOGic:DELTatime 1.5e-9 sets the delta time to 1.5 ns. SEARCH:SEARCH1:TRIGger:A:LOGic:DELTatime? might return SEARCH:SEARCH1:TRIGGER:A:LOGIC:DELTATIME 1.0000E-9 indicating the delta time is 1.0 ns.

SEARCH:SEARCH<x>:TRIGger:A:LOGic:FUNCTion

This command sets or queries the logic operator for a pattern or state trigger search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
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Syntax	SEARCH:SEARCH<x>:TRIGger:A:LOGic:FUNctIon {AND NANd NOR OR} SEARCH:SEARCH<x>:TRIGger:A:LOGic:FUNctIon?
Arguments	AND places a mark if all conditions are true. NAND places a mark if any of the conditions are false. NOR places a mark if all conditions are false. OR places a mark if any of the conditions are true.
Examples	SEARCH:SEARCH1:TRIGGER:A:LOGIC:FUNCTION AND sets the trigger a logic function for search 1 to AND. SEARCH:SEARCH1:TRIGGER:A:LOGIC:FUNCTION? might return SEARCH:SEARCH1:TRIGGER:A:LOGIC:FUNCTION NOR, indicating that the logic function for search 1 is set to NOR.

SEARCH:SEARCH<x>:TRIGger:A:LOGic:INPUT:CLOCK:SOURce

This command specifies or queries the channel to use as the clock source for logic trigger. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:LOGic:INPUT:CLOCK:SOURce {CH<x> Ch<x>_D<x> REF<x>_D<x>} SEARCH:SEARCH<x>:TRIGger:A:LOGic:INPUT:CLOCK:SOURce?
Arguments	Arguments are the possible input channels.
Examples	SEARCH:SEARCH1:TRIGger:A:LOGic:INPUT:CLOCK:SOURce CH2 sets the clock source to channel 2. SEARCH:SEARCH1:TRIGger:A:LOGic:INPUT:CLOCK:SOURce? might return SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:CLOCK:SOURCE UNDEFINED indicating the clock source is not defined.

SEARCH:SEARCH<x>:TRIGger:A:LOGic:LEVel:CH<x>

This command sets or queries the voltage level to use for logic trigger search. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:LOGic:LEVe1:CH<x> <NR3> SEARCH:SEARCH<x>:TRIGger:A:LOGic:LEVe1:CH<x>?</pre>
Arguments	<NR3> is the voltage level to use for logic trigger search.
Examples	SEARCH:SEARCH1:TRIGger:A:LOGic:LEVe1:CH2 50.0e-3 sets the level to 50 mV. SEARCH:SEARCH1:TRIGger:A:LOGic:LEVe1:CH2? might return SEARCH:SEARCH1:TRIGGER:A:LOGIC:LEVEL:CH2 0.0E+0 indicating the level is set to 0.0 V.

SEARCH:SEARCH<x>:TRIGger:A:LOGic:LEVe1:MATH<x>

This command sets the voltage level to use for logic trigger search. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:LOGic:LEVe1:MATH<x> <NR3> SEARCH:SEARCH<x>:TRIGger:A:LOGic:LEVe1:MATH<x>?</pre>
Arguments	<NR3> is the voltage level to use for logic trigger search.
Examples	SEARCH:SEARCH1:TRIGger:A:LOGic:LEVe1:MATH1 50.0e-3 sets the level to 50.0 mV. SEARCH:SEARCH1:TRIGger:A:LOGic:LEVe1:MATH1? might return SEARCH:SEARCH1:TRIGGER:A:LOGIC:LEVEL:MATH1 0.0E+0 indicating the level is set to 0.0 V.

SEARCH:SEARCH<x>:TRIGger:A:LOGic:LEVe1:REF<x>

This command sets the voltage level to use for logic trigger search. The search number is specified by <x>.

Group	Search and Mark
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Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:LOGic:LEVel:REF<x> <NR3> SEARCH:SEARCH<x>:TRIGger:A:LOGic:LEVel:REF<x>?</pre>
Arguments	<NR3> is the voltage level to use for logic trigger search.
Examples	<pre>SEARCH:SEARCH1:TRIGger:A:LOGic:LEVel:REF1 50.0e-3 sets the level to 50.0 mV. SEARCH:SEARCH1:TRIGger:A:LOGic:LEVel:REF1? might return SEARCH:SEARCH1:TRIGGER:A:LOGIC:LEVEL:REF1 0.0E+0 indicating the level is set to 0.0 V.</pre>

SEARCH:SEARCH<x>:TRIGger:A:LOGic:LOGICPattern:CH<x>

This command sets or queries the conditions used for generating an A logic pattern, with respect to the defined input pattern, and identifies the time that the selected pattern may be true and still generate the trigger. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:LOGic:LOGICPattern:CH<x> {H L X} SEARCH:SEARCH<x>:TRIGger:A:LOGic:LOGICPattern:CH<x>?</pre>
Arguments	H specifies triggering when the pattern is high. L specifies triggering when the pattern is low. X specifies triggering when the pattern is high or low.
Examples	<pre>SEARCH:SEARCH1:TRIGger:A:LOGic:LOGICPattern:CH2 H sets the channel 2 pattern to a high. SEARCH:SEARCH1:TRIGger:A:LOGic:LOGICPattern:CH2? might return SEARCH:SEARCH1:TRIGGER:A:LOGIC:LOGICPATTERN:CH2 X indicating channel 2 is a don't care.</pre>

SEARCH:SEARCH<x>:TRIGger:A:LOGic:LOGICPattern:CH<x>_D<x>

This command sets or queries the conditions used for generating an A logic pattern, with respect to the defined input pattern, and identifies the time that the selected pattern may be true and still generate the trigger. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:LOGic:LOGICPattern:CH<x>_D<x> {H L X} SEARCH:SEARCH<x>:TRIGger:A:LOGic:LOGICPattern:CH<x>_D<x>?</pre>
Arguments	H specifies triggering when the pattern is high. L specifies triggering when the pattern is low. X specifies triggering when the pattern is high or low.
Examples	SEARCH:SEARCH1:TRIGger:A:LOGic:LOGICPattern:CH1_D1 H sets the channel patter to a high. SEARCH:SEARCH1:TRIGger:A:LOGic:LOGICPattern:CH1_D1? might return SEARCH:SEARCH1:TRIGGER:A:LOGIC:LOGICPATTERN:CH1_D1 X indicating the channel patter is a don't care.

SEARCH:SEARCH<x>:TRIGger:A:LOGic:LOGICPattern:MATH<x>

This command sets or queries the conditions used for generating an A logic pattern, with respect to the defined input pattern, and identifies the time that the selected pattern may be true and still generate the trigger. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:LOGic:LOGICPattern:MATH<x> {H L X} SEARCH:SEARCH<x>:TRIGger:A:LOGic:LOGICPattern:MATH<x>?</pre>
Arguments	H specifies triggering when the pattern is high. L specifies triggering when the pattern is low. X specifies triggering when the pattern is high or low.
Examples	SEARCH:SEARCH1:TRIGger:A:LOGic:LOGICPattern:MATH1 H sets the pattern to a high. SEARCH:SEARCH1:TRIGger:A:LOGic:LOGICPattern:MATH1? might return SEARCH:SEARCH1:TRIGGER:A:LOGIC:LOGICPATTERN:MATH1 X indicating the pattern is a don't care.

SEARCH:SEARCH<x>:TRIGger:A:LOGic:LOGICPattern:REF<x>

This command sets or queries the conditions used for generating an A logic pattern, with respect to the defined input pattern, and identifies the time that the selected pattern may be true and still generate the trigger. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:LOGic:LOGICPattern:REF<x> {H L X} SEARCH:SEARCH<x>:TRIGger:A:LOGic:LOGICPattern:REF<x>?
Arguments	H specifies triggering when the pattern is high. L specifies triggering when the pattern is low. X specifies triggering when the pattern is high or low.
Examples	SEARCH:SEARCH1:TRIGger:A:LOGic:LOGICPattern:REF1 H sets the pattern to a high. SEARCH:SEARCH1:TRIGger:A:LOGic:LOGICPattern:REF1? might return SEARCH:SEARCH1:TRIGGER:A:LOGIC:LOGICPATTERN:REF1 X indicating the pattern is a don't care.

SEARCH:SEARCH<x>:TRIGger:A:LOGic:POLarity

This command sets or queries the polarity for the clock channel when Use Clock Edge is set to Yes for Logic search type. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:LOGic:POLarity {POSitive NEGative EITher} SEARCH:SEARCH<x>:TRIGger:A:LOGic:POLarity?
Arguments	POSitive specifies using the positive clock edge. NEGative specifies using negative clock edge. EITher specifies using either the positive or negative clock edge.

Examples `SEARCH:SEARCH1:TRIGger:A:LOGic:POLarity` `NEGATIVE` sets the polarity to negative.

`SEARCH:SEARCH1:TRIGger:A:LOGic:POLarity?` might return `SEARCH:SEARCH1:TRIGGER:A:LOGIC:POLARITY POSITIVE` indicating the polarity is set to positive.

SEARCH:SEARCH<x>:TRIGger:A:LOGic:USEClOCKedge

This command specifies whether or not Logic search uses a clock source. The search number is specified by <x>.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:LOGic:USEClOCKedge {OFF|ON|0|1}`
`SEARCH:SEARCH<x>:TRIGger:A:LOGic:USEClOCKedge?`

Arguments `OFF` specifies not to use the clock source.

`ON` specifies to use the clock source.

`0` specifies not to use the clock source.

`1` specifies to use the clock source.

Examples `SEARCH:SEARCH1:TRIGger:A:LOGic:USEClOCKedge` `ON` specifies to use the clock source.

`SEARCH:SEARCH1:TRIGger:A:LOGic:USEClOCKedge?` might return `SEARCH:SEARCH1:TRIGGER:A:LOGIC:USECLOCKEDGE 0` indicating not to use the clock source.

SEARCH:SEARCH<x>:TRIGger:A:LOGic:WHEn

This command sets or queries the condition for generating an A or B logic search with respect to the defined input pattern.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:LOGic:WHEn`
`{TRUE|FALSE|MOREThan|LESSThan|EQua1|UNEQua1}`
`SEARCH:SEARCH<x>:TRIGger:A:LOGic:WHEn?`

Arguments	TRUE searches on an input value that is true.
	FALSE searches on an input value that is false.
	MOREthan searches on an input value that is greater than a set value.
	LESSthan searches on an input value that is less than a set value.
	Equal searches on an input value that is equal to a set value.
	UNEQUAL searches on an input value that is not equal to a set value.
Examples	SEARCH:SEARCH1:TRIGGER:A:LOGIC:WHEN FALSE specifies a search on an input value that is false.
	SEARCH:SEARCH1:TRIGGER:A:LOGIC:WHEN? might return SEARCH:SEARCH1:TRIGGER:A:LOGIC:WHEN TRUE indicating a search on an input value that is true.

SEARCH:SEARCH<x>:TRIGGER:A:PULSEWidth:HIGHLimit

This command specifies the upper limit to use, in seconds, when searching for a pulse whose duration is inside or outside a range of two values. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGGER:A:PULSEwidth:HIGHLimit <NR3> SEARCH:SEARCH<x>:TRIGGER:A:PULSEwidth:HIGHLimit?
Arguments	<NR3> is the upper limit to use, in seconds, when searching for a pulse.
Examples	SEARCH:SEARCH1:TRIGGER:A:PULSEwidth:HIGHLimit 2.5e-9 sets the high limit to 2.5 ns.
	SEARCH:SEARCH1:TRIGGER:A:PULSEwidth:HIGHLimit? might return SEARCH:SEARCH1:TRIGGER:A:PULSEWIDTH:HIGHLIMIT 2.0000E-9 indicating the pulse width high limit is 2.0 ns.

SEARCH:SEARCH<x>:TRIGGER:A:PULSEWidth:LOGICQUALification

This command specifies whether or not to use logic qualification for a pulse width search. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:PULSEwidth:LOGICQUALification {ON OFF} SEARCH:SEARCH<x>:TRIGger:A:PULSEwidth:LOGICQUALification?</pre>
Arguments	ON specifies to use logic qualification. OFF specifies not to use logic qualification.
Examples	SEARCH:SEARCH1:TRIGger:A:PULSEwidth:LOGICQUALification ON turns on logic qualification. SEARCH:SEARCH1:TRIGger:A:PULSEwidth:LOGICQUALification? might return SEARCH:SEARCH1:TRIGGER:A:PULSEWIDTH:LOGICQUALIFICATION OFF indicating logic qualification is off.

SEARCH:SEARCH<x>:TRIGger:A:PULSEwidth:LOWLimit

This command specifies the lower limit to use, in seconds, when searching for a pulse whose duration is inside or outside a range of two values. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:PULSEwidth:LOWLimit <NR3> SEARCH:SEARCH<x>:TRIGger:A:PULSEwidth:LOWLimit?</pre>
Arguments	<NR3> is the lower limit to use, in seconds, when searching for a pulse.
Examples	SEARCH:SEARCH1:TRIGger:A:PULSEwidth:LOWLimit 0.5e-9 sets the low limit to 0.5 ns. SEARCH:SEARCH1:TRIGger:A:PULSEwidth:LOWLimit? might return SEARCH:SEARCH1:TRIGGER:A:PULSEWIDTH:LOWLIMIT 1.0000E-9 indicating the low limit is 2.0 ns.

SEARCH:SEARCH<x>:TRIGger:A:PULSEwidth:POLarity

This command specifies the polarity for a pulse width search. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:PULSEwidth:POLarity {POSitive NEGative} SEARCH:SEARCH<x>:TRIGger:A:PULSEwidth:POLarity?</pre>
Arguments	POSitive specifies positive polarity for a pulse width search. NEGative specifies negative polarity for a pulse width search.
Examples	SEARCH:SEARCH1:TRIGger:A:PULSEwidth:POLarity NEGATIVE sets the polarity to negative. SEARCH:SEARCH1:TRIGger:A:PULSEwidth:POLarity? might return SEARCH:SEARCH1:TRIGGER:A:PULSEWIDTH:POLARITY POSITIVE indicating the polarity is positive.

SEARCH:SEARCH<x>:TRIGger:A:PULSEwidth:SOURce

This command sets and queries the source for the pulse width search input. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:PULSEwidth:SOURce {CH<x> CH<x>_D<x> REF<x> REF<x>_D<x>} SEARCH:SEARCH<x>:TRIGger:A:PULSEwidth:SOURce?</pre>
Arguments	Arguments are possible sources.
Examples	SEARCH:SEARCH1:TRIGger:A:PULSEwidth:SOURce CH1 sets the source to channel 1. SEARCH:SEARCH1:TRIGger:A:PULSEwidth:SOURce? might return SEARCH:SEARCH1:TRIGGER:A:PULSEWIDTH:SOURCE CH2 indicating channel 2 is the source.

SEARCH:SEARCH<x>:TRIGger:A:PULSEwidth:THReshold

Sets or queries the source threshold level for a pulse width trigger search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:PULSEwidth:THResho1d <NR3> SEARCH:SEARCH<x>:TRIGger:A:PULSEwidth:THResho1d?</pre>
Arguments	<NR3> is the source threshold level for a pulse width trigger search.
Examples	SEARCH:SEARCH1:TRIGger:A:PULSEwidth:THResho1d 1.0e-9 sets to 1.0 ns. SEARCH:SEARCH1:TRIGger:A:PULSEwidth:THResho1d? might return SEARCH:SEARCH1:TRIGGER:A:PULSEWIDTH:THRESHOLD 0.0E+0 indicating the threshold is 0.0 s.

SEARCH:SEARCH<x>:TRIGger:A:PULSEWidth:WHEn

This command specifies to search for a pulse with a width (duration) that is less than, greater than, equal to, or unequal to a specified value (set using SEARCH:A:PULSEwidth:WIDTH), OR whose SEARCH:A:PULSEwidth:LOWLimit and SEARCH:A:PULSEwidth:HIGHLimit). The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:PULSEwidth:WHEn {LESSthan MORethan EQua1 UNEQua1 WITHin OUTside} SEARCH:SEARCH<x>:TRIGger:A:PULSEwidth:WHEn?</pre>
Arguments	LESSthan causes a search when a pulse is detected with a width less than the time set by the SEARCh:A:PULSEWidth:WIDTh command. MORethan causes a search when a pulse is detected with a width greater than the time set by the SEARCh:A:PULSEWidth:WIDTh command. EQua1 causes a search when a pulse is detected with a width equal to the time period specified in SEARCh:A:PULSEWidth:WIDTh within a $\pm 5\%$ tolerance. UNEQua1 causes a search when a pulse is detected with a width greater than or less than (but not equal) the time period specified in SEARCh:A:PULSEWidth:WIDTh within a $\pm 5\%$ tolerance. WITHin causes a search when a pulse is detected that is within a range set by two values.

OUTside causes a search when a pulse is detected that is outside of a range set by two values.

Examples SEARCH:SEARCH1:TRIGger:A:PULSEwidth:WHEN OUTside causes a search when a pulse is detected that is outside the set range.

SEARCH:SEARCH1:TRIGger:A:PULSEwidth:WHEN? might return
SEARCH:SEARCH1:TRIGGER:A:PULSEWIDTH:WHEN WITHIN indicating that a search will occur when a pulse is detected that is within a set range.

SEARCH:SEARCH<x>:TRIGger:A:RUNT:LOGICQUALification

This command specifies whether or not to use logic qualification for a runt search. The search number is specified by <x>.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:RUNT:LOGICQUALification {ON|OFF}
SEARCH:SEARCH<x>:TRIGger:A:RUNT:LOGICQUALification?

Arguments ON specifies to use logic qualification for a runt search.
OFF specifies not to use logic qualification for a runt search.

Examples SEARCH:SEARCH1:TRIGger:A:RUNT:LOGICQUALification ON turns on logic qualification.

SEARCH:SEARCH1:TRIGger:A:RUNT:LOGICQUALification? might return
SEARCH:SEARCH1:TRIGGER:A:RUNT:LOGICQUALIFICATION OFF
indicating logic qualification is off.

SEARCH:SEARCH<x>:TRIGger:A:RUNT:POLarity

This command specifies the polarity for the runt search. The search number is specified by <x>.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:RUNT:POLarity
{POSitive|NEGative|EITHer}
SEARCH:SEARCH<x>:TRIGger:A:RUNT:POLarity?

Arguments	POSitive specifies using positive polarity for the runt search. NEGative specifies using negative polarity for the runt search. EITher specifies using either positive or negative polarity for the runt search.
Examples	SEARCH:SEARCH1:TRIGger:A:RUNT:POLArity NEGATIVE specifies using negative polarity for the runt search. SEARCH:SEARCH1:TRIGger:A:RUNT:POLArity? might return SEARCH:SEARCH1:TRIGGER:A:RUNT:POLARITY POSITIVE indicating the polarity is set to positive.

SEARCH:SEARCH<x>:TRIGger:A:RUNT:SOURce

This command sets and queries the source for the Runt search input. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:RUNT:SOURce {CH<x> REF<x>}
Arguments	Arguments are the available sources.
Examples	SEARCH:SEARCH1:TRIGger:A:RUNT:SOURce CH1 sets the source to channel 1. SEARCH:SEARCH1:TRIGger:A:RUNT:SOURce? might return SEARCH:SEARCH1:TRIGGER:A:RUNT:SOURCE CH2 indicating the source is set to channel 2.

SEARCH:SEARCH<x>:TRIGger:A:RUNT:THReshold:HIGH

This command sets or queries the source threshold HIGH level for a runt trigger search to determine where to place a mark.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:RUNT:THReshoId:HIGH <NR3> SEARCH:SEARCH<x>:TRIGger:A:RUNT:THReshoId:HIGH?

Arguments	<NR3> is the source threshold HIGH level for a runt trigger search.
Examples	<pre>SEARCH:SEARCH1:TRIGger:A:RUNT:THReshold:HIGH 50.0E-3</pre> sets the high threshold to 50 mV. <pre>SEARCH:SEARCH1:TRIGger:A:RUNT:THReshold:HIGH?</pre> might return <pre>SEARCH:SEARCH1:TRIGGER:A:RUNT:THRESHOLD:HIGH 0.0E+0</pre> indicating the high threshold is 0.0 V.

SEARCH:SEARCH<x>:TRIGger:A:RUNT:THReshold:LOW

Sets or queries the source threshold LOW level for a runt trigger search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:RUNT:THReshold:LOW <NR3></pre> <pre>SEARCH:SEARCH<x>:TRIGger:A:RUNT:THReshold:LOW?</pre>
Arguments	<NR3> is the source threshold LOW level for a runt trigger search.
Examples	<pre>SEARCH:SEARCH1:TRIGger:A:RUNT:THReshold:LOW 50.0e-3</pre> sets the threshold to 50 mV. <pre>SEARCH:SEARCH1:TRIGger:A:RUNT:THReshold:LOW?</pre> might return <pre>SEARCH:SEARCH1:TRIGGER:A:RUNT:THRESHOLD:LOW 0.0E+0</pre> indicating the threshold is 0.0 V.

SEARCH:SEARCH<x>:TRIGger:A:RUNT:WHEn

This command sets or queries the condition setting for a runt trigger search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:RUNT:WHEn</pre> <pre>{OCCURS LESSthan MOREthan EQual}NOTEQual}</pre> <pre>SEARCH:SEARCH<x>:TRIGger:A:RUNT:WHEn?</pre>

Arguments	LESSthan argument sets the oscilloscope to search if the a runt pulse is detected with width less than the time set by the <code>SEARCH:SEARCH<x>:TRIGger:A:RUNT:WIDTH</code> command. MOREthan argument sets the oscilloscope to search if the a runt pulse is detected with width more than the time set by the <code>SEARCH:SEARCH<x>:TRIGger:A:RUNT:WIDTH</code> command. EQual argument sets the oscilloscope to search when the pattern is true for a time period equal to the time period specified in <code>SEARCH:SEARCH<x>:TRIGger:A:RUNT:WIDTH</code> within a $\pm 5\%$ tolerance. NOTEQua1 argument sets the oscilloscope to search when the pattern is true for a time period greater than or less than (but not equal) the time period specified in <code>SEARCH:SEARCH<x>:TRIGger:A:RUNT:WIDTH</code> within a $\pm 5\%$ tolerance. OCCURS argument specifies a search event if a runt of any detectable width occurs.
Examples	<code>SEARCH:SEARCH1:TRIGGER:A:RUNT:WHEN MORETHAN</code> sets the oscilloscope to trigger when a runt pulse is detected with width wider than the time set by the <code>SEARCH:SEARCH<x>:TRIGger:A:RUNT:WIDTH</code> command. <code>SEARCH:SEARCH1:TRIGGER:A:RUNT:WHEN?</code> might return <code>SEARCH:SEARCH1:TRIGGER:A:RUNT:WHEN OCCURS</code>, indicating that a mark is placed if a runt trigger event occurs.

SEARCH:SEARCH<x>:TRIGger:A:RUNT:WIDTH

This command sets or queries the width setting for a runt trigger search to determine where to place a mark. The search number is specified by `<x>`.

Group	Search and Mark
Syntax	<code>SEARCH:SEARCH<x>:TRIGger:A:RUNT:WIDTH <NR3></code> <code>SEARCH:SEARCH<x>:TRIGger:A:RUNT:WIDTH?</code>
Arguments	<code><NR3></code> specifies the minimum width in seconds.
Examples	<code>SEARCH:SEARCH1:TRIGGER:A:RUNT:WIDTH 400E-12</code> sets the runt trigger minimum width for search 1 to 0.4 nanoseconds. <code>SEARCH:SEARCH1:TRIGGER:A:RUNT:WIDTH?</code> might return <code>SEARCH:SEARCH1:TRIGGER:A:RUNT:WIDTH 500.0000E-12</code>, indicating that the runt trigger minimum width for search 1 is set to 0.5 nanoseconds.

SEARCH:SEARCH<x>:TRIGger:A:SETHold:CLOCK:EDGE

This command sets or queries the clock slope setting for a setup/hold trigger search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:SETHold:CLOCK:EDGE {FALL RISe} SEARCH:SEARCH<x>:TRIGger:A:SETHold:CLOCK:EDGE?
Arguments	FALL specifies the polarity as the clock falling edge. RISe specifies the polarity as the clock rising edge.
Examples	SEARCH:SEARCH1:TRIGGER:A:SETHOLD:CLOCK:EDGE FALL sets the setup/hold trigger clock slope polarity for search 1 to falling edge. SEARCH:SEARCH1:TRIGGER:A:SETHOLD:CLOCK:EDGE? might return SEARCH:SEARCH1:TRIGGER:A:SETHOLD:CLOCK:EDGE RISE, indicating that the setup/hold trigger clock slope polarity for search 1 is set to rising edge.

SEARCH:SEARCH<x>:TRIGger:A:SETHold:CLOCK:SOURce

This command sets or queries the clock source setting for a setup/hold trigger search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:SETHold:CLOCK:SOURce {CH<x> CH<x>_D<x> MATH<x> REF<x> REF<x>_D<x>} SEARCH:SEARCH<x>:TRIGger:A:SETHold:CLOCK:SOURce SEARCH:SEARCH<x>:TRIGger:A:SETHold:CLOCK:SOURce?
Arguments	CH<x> specifies an input channel as the edge source, where <x> = 1, 2, 3, 4, 5, 6, 7, or 8, depending on the number of channels in your instrument. CH<x>_D<x> specifies a digital waveform as the setup and hold clock source waveform for the specified search. MATH<x> specifies the math waveform as the search source, where <x> = ≥ 1 . REF<x> specifies the reference waveform as the search source, where <x> = ≥ 1 .

REF<x>_D<x> specifies a digital reference waveform as the setup and hold clock source waveform for the specified search.

Examples SEARCH:SEARCH1:TRIGGER:A:SETHOLD:CLOCK:SOURCE MATH1 sets the setup/hold trigger clock source setting for search 1 to MATH1.

SEARCH:SEARCH1:TRIGGER:A:SETHOLD:CLOCK:SOURCE? might return
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:CLOCK:SOURCE CH1, indicating that the setup/hold trigger clock source setting for search 1 is set to CH1.

SEARCH:SEARCH<x>:TRIGger:A:SETHold:CLOCK:THReshold

This command sets or queries the clock threshold setting for a setup/hold trigger search to determine where to place a mark. The search number is specified by <x>.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:SETHold:CLOCK:THReshold <NR3>
SEARCH:SEARCH<x>:TRIGger:A:SETHold:CLOCK:THReshold?

Arguments <NR3> the clock threshold setting for a setup/hold trigger search.

Examples SEARCH:SEARCH1:TRIGGER:A:SETHOLD:CLOCK:THRESHOLD -1.3 sets the setup/hold trigger clock threshold setting for search 1 to -1.3 volts.

SEARCH:SEARCH1:TRIGGER:A:SETHOLD:CLOCK:THRESHOLD? might return
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:CLOCK:THRESHOLD -1.3000, indicating that the setup/hold trigger clock threshold setting for search 1 is set to -1.3 volts.

** CANNOT USE LocalCmd HERE BECAUSE OF VARIABLES **

SEARCH:SEARCH<x>:TRIGger:A:SETHold:HOLDTime

This command sets or queries the hold time setting for a setup/hold trigger search to determine where to place a mark. The search number is specified by <x>.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:SETHold:HOLDTime <NR3>
SEARCH:SEARCH<x>:TRIGger:A:SETHold:HOLDTime?

Arguments	<NR3> specifies the hold time setting in seconds. Positive values for hold time occur after the clock edge. Negative values occur before the clock edge.
Examples	SEARCH:SEARCH1:TRIGGER:A:SETHOLD:HOLDTIME 400E-12 sets the setup/hold trigger hold time setting for search 1 to 400 ps. SEARCH:SEARCH1:TRIGGER:A:SETHOLD:HOLDTIME? might return SEARCH:SEARCH1:TRIGGER:A:SETHOLD:HOLDTIME 500.0000E-12, indicating that the setup/hold trigger hold time setting for search 1 is set to 0.5 ns.

SEARCH:SEARCH<x>:TRIGger:A:SETHold:LEVel:CH<x>

This command sets or queries the voltage level to use for setup & hold trigger search. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:SETHold:LEVel:CH<x> <NR3> SEARCH:SEARCH<x>:TRIGger:A:SETHold:LEVel:CH<x>?
Arguments	<NR3> the voltage level to use for setup & hold trigger search.
Examples	SEARCH:SEARCH1:TRIGger:A:SETHold:LEVel:CH1 50.0e-3 sets the level to 50.0 mV. SEARCH:SEARCH1:TRIGger:A:SETHold:LEVel:CH1? might return SEARCH:SEARCH1:TRIGGER:A:SETHOLD:LEVEL:CH1 0.0E+0 indicating the level is 0.0 V.

SEARCH:SEARCH<x>:TRIGger:A:SETHold:LEVel:MATH<x>

This command sets or queries the voltage level to use for setup & hold trigger search. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:SETHold:LEVel:MATH<x> <NR3> SEARCH:SEARCH<x>:TRIGger:A:SETHold:LEVel:MATH<x>?
Arguments	<NR3> isi the voltage level to use for setup & hold trigger search.

Examples `SEARCH:SEARCH1:TRIGger:A:SETHold:LEVel:MATH1 50.0e-3` sets the level to 50.0 mV.

`SEARCH:SEARCH1:TRIGger:A:SETHold:LEVel:MATH1?` might return `SEARCH:SEARCH1:TRIGGER:A:SETHOLD:LEVEL:MATH1 0.0E+0` indicating the level is 0.0 V.

SEARCH:SEARCH<x>:TRIGger:A:SETHold:LEVel:REF<x>

This command sets or queries the voltage level to use for setup & hold trigger search. The search number is specified by <x>.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:SETHold:LEVel:REF<x> <NR3>`
`SEARCH:SEARCH<x>:TRIGger:A:SETHold:LEVel:REF<x>?`

Arguments <NR3> is the voltage level to use for setup & hold trigger search.

Examples `SEARCH:SEARCH1:TRIGger:A:SETHold:LEVel:REF1 50.0e-3` sets the level to 50.0 mV.

`SEARCH:SEARCH1:TRIGger:A:SETHold:LEVel:REF1?` might return `SEARCH:SEARCH1:TRIGGER:A:SETHOLD:LEVEL:REF1 0.0E+0` indicating the level is 0.0 V.

SEARCH:SEARCH<x>:TRIGger:A:SETHold:LOGICPattern:CH<x>

This command sets or queries the conditions used for generating an A logic pattern, with respect to the defined input pattern, and identifies the time that the selected pattern may be true and still generate the trigger. The search number is specified by <x>.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:SETHold:LOGICPattern:CH<x>`
`{INCLUDE|DONTInclude}`
`SEARCH:SEARCH<x>:TRIGger:A:SETHold:LOGICPattern:CH<x>?`

Arguments INCLUDE specifies including the specified channel SETHOLD inputs in the specified search.

DONTINclude specifies not including the specified channel SETHOLD inputs in the specified search.

Examples SEARCH:SEARCH1:TRIGger:A:SETHold:LOGICPattern:CH1 INCLUDE specifies including the specified channel SETHOLD inputs in the specified search.

SEARCH:SEARCH1:TRIGger:A:SETHold:LOGICPattern:CH1? might return SEARCH:SEARCH1:TRIGGER:A:SETHOLD:LOGICPATTERN:CH1 DONTINCLUDE indicating the specified channel SETHOLD inputs will not be included in the specified search.

SEARCH:SEARCH<x>:TRIGger:A:SETHold:LOGICPattern:CH<x>_D<x>

This command sets or queries the conditions used for generating an A logic pattern, with respect to the defined input pattern, and identifies the time that the selected pattern may be true and still generate the trigger. The search number is specified by <x>.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:SETHold:LOGICPattern:CH<x>_D<x>
{INCLUDE|DONTINclude}
SEARCH:SEARCH<x>:TRIGger:A:SETHold:LOGICPattern:CH<x>_D<x>?

Arguments INCLUDE specifies including the specified digital channel SETHOLD inputs in the specified search.

DONTINclude specifies not including the specified digital channel SETHOLD inputs in the specified search.

Examples SEARCH:SEARCH1:TRIGger:A:SETHold:LOGICPattern:CH1_D0 INCLUDE specifies including the specified digital channel SETHOLD inputs in the specified search.

SEARCH:SEARCH1:TRIGger:A:SETHold:LOGICPattern:CH1_D0? might return SEARCH:SEARCH1:TRIGGER:A:SETHOLD:LOGICPATTERN:CH1_D0 DONTINCLUDE indicating the specified digital channel SETHOLD inputs will not be included in the specified search.

SEARCH:SEARCH<x>:TRIGger:A:SETHold:LOGICPattern:MATH<x>

This command sets or queries the conditions used for generating an A logic pattern, with respect to the defined input pattern, and identifies the time that the

selected pattern may be true and still generate the trigger. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:SETHold:LOGICPattern:MATH<x> {INCLude DONTIncLude} SEARCH:SEARCH<x>:TRIGger:A:SETHold:LOGICPattern:MATH<x>?</pre>
Arguments	INCLude specifies including the specified math SETHOLD inputs in the specified search. DONTIncLude specifies not including the specified math SETHOLD inputs in the specified search.
Examples	SEARCH:SEARCH1:TRIGger:A:SETHold:LOGICPattern:MATH1 INCLUDE specifies including the specified math SETHOLD inputs in the specified search. SEARCH:SEARCH1:TRIGger:A:SETHold:LOGICPattern:MATH1? might return SEARCH:SEARCH1:TRIGGER:A:SETHOLD:LOGICPATTERN:MATH1 DONTINCLUDE indicating the specified math SETHOLD inputs will not be included in the specified search.

SEARCH:SEARCH<x>:TRIGger:A:SETHold:LOGICPattern:REF<x>

This command sets and returns the conditions used for generating an A logic pattern, with respect to the defined input pattern, and identifies the time that the selected pattern may be true and still generate the trigger. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:SETHold:LOGICPattern:REF<x> {INCLude DONTIncLude} SEARCH:SEARCH<x>:TRIGger:A:SETHold:LOGICPattern:REF<x>?</pre>
Arguments	INCLude specifies including the specified reference SETHOLD inputs in the specified search. DONTIncLude specifies not including the specified reference SETHOLD inputs in the specified search.

Examples `SEARCH:SEARCH1:TRIGger:A:SETHold:LOGICPattern:REF1 INCLUDE` specifies including the specified reference SETHOLD inputs in the specified search.

`SEARCH:SEARCH1:TRIGger:A:SETHold:LOGICPattern:REF1?` might return `SEARCH:SEARCH1:TRIGGER:A:SETHOLD:LOGICPATTERN:REF1 DONTINCLUDE` indicating the specified reference SETHOLD inputs will not be included in the specified search.

SEARCH:SEARCH<x>:TRIGger:A:SETHold:SETTime

This command sets or queries the setup time setting for a setup/hold trigger search to determine where to place a mark. The search number is specified by <x>.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:SETHold:SETTime <NR3>`
`SEARCH:SEARCH<x>:TRIGger:A:SETHold:SETTime?`

Arguments <NR3> specifies the setup time for setup and hold violation triggering.

Examples `SEARCH:SEARCH1:TRIGGER:A:SETHOLD:SETTIME 2E-9` sets the setup/hold trigger setup time setting for search 1 to 2 ns.

`SEARCH:SEARCH1:TRIGGER:A:SETHOLD:SETTIME?` might return `SEARCH:SEARCH1:TRIGGER:A:SETHOLD:SETTIME 1.0000E-9`, indicating that the setup/hold trigger setup time setting for search 1 is set to 1.0 ns.

SEARCH:SEARCH<x>:TRIGger:A:STATE

This command sets or queries the enabled state of the search. The search number is specified by <x>.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:STATE {<NR1>|OFF|ON}`

Arguments <NR1> = 1 enables the search. Any other character disables the search.
 ON enables the search.
 OFF disables the search.

Examples `SEARCH:SEARCH1:TRIGger:A:STATE OFF` disables the search.
 `SEARCH:SEARCH1:TRIGger:A:STATE?` might return
 `SEARCH:SEARCH1:TRIGGER:A:STATE 1` indicating search 1 is enabled.

SEARCH:SEARCH<x>:TRIGger:A:STOPAcq

This command sets or queries whether acquisitions are stopped when a search hit is found. The search number is specified by <x>.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:STOPAcq {<NR1>|OFF|ON}`
 `SEARCH:SEARCH<x>:TRIGger:A:STOPAcq?`

Arguments <x> is the number of the search on which to enable or disable the stop acquisition function.

 <NR1> = 1 enables stopping when a search hit is found. Any other character disables the feature.

 ON enables stopping when a search hit is found.

 OFF disables stopping on a search hit.

Examples `SEARCH:SEARCH1:TRIGger:A:STOPAcq ON` enables stopping when a search hit is found.

`SEARCH:SEARCH3:TRIGger:A:STOPAcq?` might return
 `:SEARCH:SEARCH3:TRIGGER:A:STOPACQ 1` indicating acquisitions are stopped when a search hit is found on search number 3.

SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:LOGICQUALification

This command specifies whether or not to use logic qualification for a timeout search. The search number is specified by <x>.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:LOGICQUALification {ON|OFF}`
 `SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:LOGICQUALification?`

Arguments	ON specifies to use logic qualification. OFF specifies not to use logic qualification.
Examples	SEARCH:SEARCH1:TRIGger:A:TIMEOut:LOGICQUALification ON specifies to use logic qualification. SEARCH:SEARCH1:TRIGger:A:TIMEOut:LOGICQUALification? might return SEARCH:SEARCH1:TRIGGER:A:TIMEOUT:LOGICQUALIFICATION OFF indicating logic qualification is off.

SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:POLarity

The polarity to be used for a Timeout search. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:POLarity {STAYSHigh STAYSLow EITHer} SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:POLarity?
Arguments	STAYSHigh specifies the polarity stays HIGH. STAYSLow specifies the polarity stays LOW. EITHer specifies the polarity stays HIGH or stays LOW.
Examples	SEARCH:SEARCH1:TRIGger:A:TIMEOut:POLarity STAYSLow specifies the polarity stays LOW. SEARCH:SEARCH1:TRIGger:A:TIMEOut:POLarity? might return SEARCH:SEARCH1:TRIGGER:A:TIMEOUT:POLARITY STAYSHIGH indicating the polarity stays HIGH.

SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:SOUrce

This command sets and queries the source for timeout search input. The search number is specified by <x>.

Group	Search and Mark
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Syntax `SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:SOURce
{CH<x>|CH<x>_D<x>|MATH<x>REF<x>|REF<x>_D<x>}]
SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:SOURce?`

Arguments Arguments are the available sources.

Examples `SEARCH:SEARCH1:TRIGger:A:TIMEOut:SOURce`
`SEARCH:SEARCH1:TRIGger:A:TIMEOut:SOURce?` might return
`SEARCH:SEARCH1:TRIGGER:A:TIMEOUT:SOURCE CH1_D0` indicating
CH1_D0 is the source.

SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:THReshold

Sets or queries the source threshold level for a timeout trigger search to determine where to place a mark. The search number is specified by <x>.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:THReshold <NR3>
SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:THReshold?`

Arguments <NR3> is the source threshold level for a timeout trigger search.

Examples `SEARCH:SEARCH1:TRIGger:A:TIMEOut:THReshold 50.0e-3` sets the
threshold to 50.0 mV.
`SEARCH:SEARCH1:TRIGger:A:TIMEOut:THReshold?` might return
`SEARCH:SEARCH1:TRIGGER:A:TIMEOUT:THRESHOLD 0.0E+0` indicating
the threshold is 0.0 V.

SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:TIME

This command sets or queries the time setting for a timeout trigger search to determine where to place a mark. The search number is specified by <x>.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:TIME <NR3>
SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:TIME?`

Arguments <NR3> is the time in seconds.

Examples SEARCH:SEARCH1:TRIGGER:A:TIMEOUT:TIME 400E-9 sets the timeout trigger time setting for search 1 to 400 ns.

SEARCH:SEARCH1:TRIGGER:A:TIMEOUT:TIME? might return
SEARCH:SEARCH1:TRIGGER:A:TIMEOUT:TIME 500.0000E-12, indicating that the timeout trigger time setting for search 1 is set to 500 ns.

SEARCH:SEARCH<x>:TRIGger:A:TRANSition:DELTATime

This command sets or queries the transition time setting for a transition trigger search to determine where to place a mark. The search number is specified by <x>.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:TRANSition:DELTATime <NR3>
SEARCH:SEARCH<x>:TRIGger:A:TRANSition:DELTATime?

Arguments <NR3> specifies the transition time in seconds.

Examples SEARCH:SEARCH1:TRIGGER:A:TIMEOUT:TIME 400E-9 sets the transition trigger time setting for search 1 to 400 ns.

SEARCH:SEARCH1:TRIGGER:A:TRANSITION:DELTATIME? might return
SEARCH:SEARCH1:TRIGGER:A:TRANSITION:DELTATIME 500.0000E-12, indicating that the transition trigger time setting for search 1 is set to 500 ps.

SEARCH:SEARCH<x>:TRIGger:A:TRANSition:LOGICQUALification

This command specifies whether or not to use logic qualification for a transition search. The search number is specified by <x>.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:TRANSition:LOGICQUALification
{ON|OFF}
SEARCH:SEARCH<x>:TRIGger:A:TRANSition:LOGICQUALification?

Arguments	ON specifies to use logic qualification for a transition search. OFF specifies not to use logic qualification for a transition search.
Examples	SEARCH:SEARCH1:TRIGger:A:TRANSition:LOGICQUALification ON specifies to use logic qualification for a transition search. SEARCH:SEARCH1:TRIGger:A:TRANSition:LOGICQUALification? might return SEARCH:SEARCH1:TRIGGER:A:TRANSITION:LOGICQUALIFICATION OFF indicating logic qualification is off.

SEARCH:SEARCH<x>:TRIGger:A:TRANSition:POLarity

This command specifies the polarity for the transition search. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:TRANSition:POLarity {POSitive NEGative EITher} SEARCH:SEARCH<x>:TRIGger:A:TRANSition:POLarity?
Arguments	EITHer places a mark on a transition of either polarity. NEGative places a mark on a transition of negative polarity. POSitive places a mark on a transition of positive polarity.
Examples	SEARCH:SEARCH1:TRIGger:A:TRANSition:POLarity NEGATIVE places a mark on a transition of negative polarity. SEARCH:SEARCH1:TRIGger:A:TRANSition:POLarity? might return SEARCH:SEARCH1:TRIGGER:A:TRANSITION:POLARITY POSITIVE indicating the polaity is positive.

SEARCH:SEARCH<x>:TRIGger:A:TRANSition:SOUrce

This command sets and queries the source for the transition search input. The search number is specified by <x>.

Group	Search and Mark
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Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:TRANSition:SOUrce {CH<x> MATH<x> REF<x>} SEARCH:SEARCH<x>:TRIGger:A:TRANSition:SOUrce?</pre>
Arguments	Arguments are the available sources.
Examples	SEARCH:SEARCH1:TRIGger:A:TRANSition:SOUrce CH1 sets the source to channel 1. SEARCH:SEARCH1:TRIGger:A:TRANSition:SOUrce? might return SEARCH:SEARCH1:TRIGGER:A:TRANSITION:SOURCE CH2 indicating channel 2 is the source.

SEARCH:SEARCH<x>:TRIGger:A:TRANSition:THReshold:HIGH

Sets or queries the source threshold HIGH level for a transition trigger search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:TRANSition:THReshold:HIGH <NR3> SEARCH:SEARCH<x>:TRIGger:A:TRANSition:THReshold:HIGH?</pre>
Arguments	<NR3> the source threshold HIGH level for a transition trigger search.
Examples	SEARCH:SEARCH1:TRIGger:A:TRANSition:THReshold:HIGH 50.0e-3 sets the high threshold to 50.0 mV. SEARCH:SEARCH1:TRIGger:A:TRANSition:THReshold:HIGH? might return SEARCH:SEARCH1:TRIGGER:A:TRANSITION:THRESHOLD:HIGH 0.0E+0 indicating the high threshold is 0.0 V.

SEARCH:SEARCH<x>:TRIGger:A:TRANSition:THReshold:LOW

Sets or queries the source threshold LOW level for a transition trigger search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:TRANSition:THReshold:LOW <NR3> SEARCH:SEARCH<x>:TRIGger:A:TRANSition:THReshold:LOW?</pre>

Arguments	<NR3> is the source threshold LOW level for a transition trigger searchk.
Examples	SEARCH:SEARCH1:TRIGger:A:TRANSition:THReshoId:LOW -50.0e-3 sets the LOW threshold to -50.0 mV. SEARCH:SEARCH1:TRIGger:A:TRANSition:THReshoId:LOW? might return SEARCH:SEARCH1:TRIGGER:A:TRANSITION:THRESHOLD:LOW 0.0E+0 indicating the LOW threshold is 0.0 V.

SEARCH:SEARCH<x>:TRIGger:A:TRANSition:WHEn

This command sets or queries the condition setting for a transition trigger search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:TRANSition:WHEn {FASTERthan SLOWERthan EQua1 UNEQua1} SEARCH:SEARCH<x>:TRIGger:A:TRANSition:WHEn?
Arguments	FASTERthan sets the trigger to occur when the transitioning signal is faster than the set volts/second rate. SLOWERthan sets the trigger to occur when the transitioning signal is slower than the set volts/second rate. EQua1 sets the trigger to occur when the transitioning signal is equal to the set volts/second rate. UNEQua1 sets the trigger to occur when the transitioning signal is not equal to the set volts/second rate.
Examples	SEARCH:SEARCH1:TRIGGER:A:TRANSITION:WHEn SLOWERTHAN sets the transition trigger condition setting for search 1 to SLOWERTHAN. SEARCH:SEARCH1:TRIGGER:A:TRANSITION:WHEn? might return SEARCH:SEARCH1:TRIGGER:A:TRANSITION:WHEn FASTERTHAN, indicating that the transition trigger condition setting for search 1 is set to FASTERTHAN.

SEARCH:SEARCH<x>:TRIGger:A:TYPE

This command sets or queries the trigger type setting for a search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
Syntax	<pre>SEARCH:SEARCH<x>:TRIGger:A:TYPe {EDGE RUNT TRANSition PULSEwidth TIMEOut LOGIc SETHold WINDow Bus DDRRead DDRREADWrite DDRWrite} SEARCH:SEARCH<x>:TRIGger:A:TYPe?</pre>
Arguments	EDGE triggers when the source input signal amplitude crosses the specified level in the direction given by the slope.

NOTE. *Some trigger types are optional.*

RUNT triggers when a pulse crosses the first preset voltage threshold but does not cross the second preset threshold before recrossing the first. The thresholds are set with the SEARCH:SEARCH<x>:TRIGger:A:RUNt:HIGH and SEARCH:SEARCH<x>:TRIGger:A:RUNt:LOW THRESHOLD commands.

TRANSition triggers when a pulse crosses both thresholds in the same direction as the specified polarity and the transition time between the two threshold crossings is greater or less than the specified time delta.

PULSEwidth triggers on input signal source pulses that are inside or outside of the given time range specified by SEARCH:SEARCH<x>:TRIGger:A:LOGIc:PATtern:WHEn:LESSLimit and SEARCH:SEARCH<x>:TRIGger:A:LOGIc:PATtern:WHEn:MORELimit. The polarity is selected using the SEARCH:SEARCH<x>:TRIGger:A:RUNt:POLarity command.

TIMEOut triggers on an input signal source that stays above, stays below, or stays either above or below the trigger level for a given time.

LOGIc specifies that a search occurs when specified conditions are met, and is controlled by the SEARCH:A:LOGIc commands.

SETHold triggers on a functional pattern combination of one to three data sources at the time of the clock transition.

WINDow triggers on an input signal source that enters or exits the horizontal band defined by the two trigger levels.

Bus specifies that a search occurs when a communications signal is found.

DDRRead triggers on a DDR Read event.

DDRREADWrite triggers on a DDR Read or Write event.

DDRWrite triggers on a DDR Write event.

Examples `SEARCH:SEARCH1:TRIGGER:A:TYPE RUNT` sets the trigger type setting for search 1 to RUNT.

`SEARCH:SEARCH1:TRIGGER:A:TYPE?` might return
`SEARCH:SEARCH1:TRIGGER:A:TYPE EDGE`, indicating that the trigger type setting for search 1 is set to EDGE.

SEARCH:SEARCH<x>:TRIGger:A:WINDow:CROSSIng

This command sets or queries the window trigger threshold crossing of the selected trigger Source. The threshold crossing selection is only effective when :TRIGger:A:WINDow:WHEN is INSIDEGreater or OUTSIDEGreater. The search number is specified by <x>.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:WINDow:CROSSIng`
`{UPPer|LOWer|EITHer|NONE}`

Arguments **UPPer** if :TRIGger:A:WINDow:WHEN is INSIDEGreater, the instrument triggers when the signal remains between the upper and lower thresholds for longer than the time limit (:TRIGger:A:WINDow:WIDTH) and then exits through the upper threshold. If :TRIGger:A:WINDow:WHEN is OUTSIDEGreater, the instrument triggers when the signal remains above the upper threshold for longer than the time limit (:TRIGger:A:WINDow:WIDTH) and then crosses downward through the upper threshold.

LOWer if :TRIGger:A:WINDow:WHEN is INSIDEGreater, the instrument triggers when the signal remains between the upper and lower thresholds for longer than the time limit (:TRIGger:A:WINDow:WIDTH) and then exits through the lower threshold. If :TRIGger:A:WINDow:WHEN is OUTSIDEGreater, the instrument triggers when the signal remains below the lower threshold for longer than the time limit (:TRIGger:A:WINDow:WIDTH) and then crosses upwards through the lower threshold.

EITHer if :TRIGger:A:WINDow:WHEN is INSIDEGreater, the instrument triggers when the signal remains between the upper and lower thresholds for longer than the time limit (:TRIGger:A:WINDow:WIDTH) and then exits through either the upper or lower threshold. If :TRIGger:A:WINDow:WHEN is OUTSIDEGreater, the instrument triggers when the signal remains either above the upper threshold or below the lower threshold for longer than the time limit (:TRIGger:A:WINDow:WIDTH) and then crosses a threshold.

NONE if :TRIGger:A:WINDow:WHEN is INSIDEGreater, the instrument triggers when the signal remains between the upper and lower thresholds for longer than the time limit (:TRIGger:A:WINDow:WIDTH) without crossing through either the

upper or lower threshold. If `:TRIGger:A:WINDOW:WHEN` is `OUTSIDEGreater`, the instrument triggers when the signal remains outside the upper and lower thresholds for longer than the time limit (`:TRIGger:A:WINDOW:WIDTH`) without crossing through either the upper or lower threshold.

Examples `SEARCH:SEARCH1:TRIGger:A:WINDOW:CROSSing LOWER` sets the `CROSSING` to `LOWER`.

`SEARCH:SEARCH1:TRIGger:A:WINDOW:CROSSing?` might return `SEARCH:SEARCH1:TRIGGER:A:WINDOW:CROSSING UPPER` indicating that `CROSSING` is set to `UPPER`.

SEARCH:SEARCH<x>:TRIGger:A:WINDow:LOGICQUALification

This command specifies or queries whether or not to use logic qualification for a window search. The search number is specified by `<x>`.

Group Search and Mark

Syntax `SEARCH:SEARCH<x>:TRIGger:A:WINDow:LOGICQUALification {ON|OFF}`
`SEARCH:SEARCH<x>:TRIGger:A:WINDow:LOGICQUALification?`

Arguments `ON` specifies to use logic qualification for a window search.
`OFF` specifies not to use logic qualification for a window search.

Examples `SEARCH:SEARCH1:TRIGger:A:WINDow:LOGICQUALification ON` turns `ON` logic qualification for a window search.

`SEARCH:SEARCH1:TRIGger:A:WINDow:LOGICQUALification?` might return `SEARCH:SEARCH1:TRIGGER:A:WINDOW:LOGICQUALIFICATION OFF` indicating logic qualification is off.

SEARCH:SEARCH<x>:TRIGger:A:WINDow:POLarity

This command sets or queries the window trigger threshold crossing of the selected trigger Source. The search number is specified by `<x>`.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:WINDow:POLarity
 {UPPer|LOWer|EITHer|NONE}
 SEARCH:SEARCH<x>:TRIGger:A:WINDow:POLarity?

Arguments UPPer specifies that the instrument triggers when the signal remains above the upper threshold for longer than the time limit and then crosses downward through the upper threshold.

 LOWer specifies that the instrument triggers when the signal remains below the lower threshold for longer than the time limit and then crosses upwards through the lower threshold.

 EITHer specifies that the instrument triggers when the signal remains either above the upper threshold or below the lower threshold for longer than the time limit and then crosses a threshold.

 NONE specifies that the instrument triggers when the signal remains outside the upper and lower thresholds for longer than the time limit without crossing through either the upper or lower threshold.

Examples SEARCH:SEARCH1:TRIGger:A:WINDow:POLarity LOWER sets the polarity to lower.

 SEARCH:SEARCH1:TRIGger:A:WINDow:POLarity? might return
 SEARCH:SEARCH1:TRIGGER:A:WINDOW:POLARITY UPPER indicating the polarity is set to upper.

SEARCH:SEARCH<x>:TRIGger:A:WINDow:SOURce

This command sets and queries the source for the window search input. The search number is specified by <x>.

Group Search and Mark

Syntax SEARCH:SEARCH<x>:TRIGger:A:WINDow:SOURce
 {CH<x>|MATH<x>|REF<x>}
 SEARCH:SEARCH<x>:TRIGger:A:WINDow:SOURce?

Arguments Arguments are the available sources.

Examples SEARCH:SEARCH1:TRIGger:A:WINDow:SOURce CH2 sets the source to channel 2.

SEARCH:SEARCH1:TRIGger:A:WINDow:Source? might return
SEARCH:SEARCH1:TRIGGER:A:WINDOW:SOURCE CH3 indicating the source
is channel 3.

SEARCH:SEARCH<x>:TRIGger:A:WINDow:THReshold:HIGH

This command sets or queries the source threshold HIGH level for a transition trigger search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:WINDow:THReshold:HIGH <NR3> SEARCH:SEARCH<x>:TRIGger:A:WINDow:THReshold:HIGH?
Arguments	<NR3> is the source threshold HIGH level for a transition trigger search.
Examples	SEARCH:SEARCH1:TRIGger:A:WINDow:THReshold:HIGH 50.0e-3 sets the high threshold to 50.0 mV. SEARCH:SEARCH1:TRIGger:A:WINDow:THReshold:HIGH? might return SEARCH:SEARCH1:TRIGGER:A:WINDOW:THRESHOLD:HIGH 0.0E+0 indicating the high threshold is 0.0 V.

SEARCH:SEARCH<x>:TRIGger:A:WINDow:THReshold:LOW

This command sets or queries the source threshold LOW level for a transition trigger search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:WINDow:THReshold:LOW <NR3> SEARCH:SEARCH<x>:TRIGger:A:WINDow:THReshold:LOW?
Arguments	<NR3> the source threshold LOW level for a transition trigger search
Examples	SEARCH:SEARCH1:TRIGger:A:WINDow:THReshold:LOW -50.0e-3 sets the low threshold to -50.0 mV.

SEARCH:SEARCH1:TRIGger:A:WINDow:THResho1d:LOW? might return
 SEARCH:SEARCH1:TRIGGER:A:WINDOW:THRESHOLD:LOW 0.0E+0 indicating
 the low threshold is 0.0 V.

SEARCH:SEARCH<x>:TRIGger:A:WINDow:WHEn

This command sets or queries the window search event. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:WINDow:WHEn {ENTERSWindow EXITSWindow INSIDEGreater OUTSIDEGreater} SEARCH:SEARCH<x>:TRIGger:A:WINDow:WHEn?
Arguments	ENTERSWindow specifies a window search when the signal enters the window. EXITSWindow specifies a window search when the signal exits the window. OUTSIDEGreater specifies a search when the signal leaves the window defined by the threshold levels for the time specified by Width. INSIDEGreater specifies a search when the signal enters the window defined by the threshold levels for the time specified by Width.
Examples	SEARCH:SEARCH1:TRIGger:A:WINDow:WHEn EXITSWINDOW specifies a window search when the signal exits the window. SEARCH:SEARCH1:TRIGger:A:WINDow:WHEn? might return SEARCH:SEARCH1:TRIGGER:A:WINDOW:WHEn ENTERSWINDOW indicating a window search when the signal enters the window.

SEARCH:SEARCH<x>:TRIGger:A:WINDow:WIDTh

This command sets or queries the width setting for a window trigger search to determine where to place a mark. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SEARCH<x>:TRIGger:A:WINDow:WIDTh <NR3> SEARCH:SEARCH<x>:TRIGger:A:WINDow:WIDTh?

Arguments	<NR3> specifies the minimum width in seconds. ECL specifies a preset high level of -1.3 V and TTL specifies a preset high level of +1.4V.
Examples	SEARCH:SEARCH1:TRIGGER:A:WINDOW:WIDTH 400E-12 sets the window trigger width setting for search 1 to 0.4 ns. SEARCH:SEARCH1:TRIGGER:A:WINDOW:WIDTH? might return SEARCH:SEARCH1:TRIGGER:A:WINDOW:WIDTH 500.0000E-12, indicating that the window trigger width setting for search 1 is set to 0.5 ns.

SEARCH:SELEcted

This command sets or queries the selected search, for example SEARCH1. The search number is specified by <x>.

Group	Search and Mark
Syntax	SEARCH:SELEcted SEARCH1
Arguments	SEARCH1 is the specified search.
Examples	SEARCH:SELEcted SEARCH1 specifies a search 1 search. SEARCH:SELEcted? might return :SEARCH:SELECTED SEARCH1 indicating search 1 is selected.

SEARCHTABLE (No Query Form)

This command adds or deletes a new search event table in an Option 5-WIN (Microsoft Windows 10 OS) TekExpress compliance testing application.

Group	Search and Mark
Syntax	SEARCHTABLE {ADDNew DELEte} <qstring>
Arguments	ADDNew adds a new search events table in the display area. DELEte removes a displayed search events table from the display area. <qstring> contains the name of the search table.

- Examples** SEARCHTABLE:ADDNew "SEARCHTABLE1" adds the SEARCHTABLE1 in the display area.
- SEARCHTABLE:DELeTe "SEARCHTABLE1" removes the SEARCHTABLE1 from the display area.

SET? (Query Only)

This query-only command returns the commands that list the instrument settings, except for configuration information for the calibration values. You can use these commands to return the instrument to the state it was in when you made the SET? query. The SET? query always returns command headers, regardless of the setting of the [HEADer](#) command. This is because the returned commands are intended to be sent back to the instrument as a command string. The [VERBose](#) command can still be used to specify whether the returned headers should be abbreviated or full-length.

This command is identical to the [*LRN?](#) command.

Group Miscellaneous

Syntax SET?

Related Commands [HEADer](#)
 [*LRN?](#)
 [VERBose](#)

- Examples** SET? might return the following response: :ACQUIRE:STOPIAFTER
 RUNSTOP;STATE 1;MODE SAMPLE;NUMENV 10;NUMAVG 16;REPET
 1;:FASTACQ:STATE 0;:APPLICATION:GPKNOB1:ACTIVE
 0;:APPLICATION:GPKNOB2:ACTIVE 0;:APPLICATION:WINDOW:HEIGHT
 236;WIDTH 640;:APPLICATION:SCOPEAPP:STATE
 RUNNING;WINDOW FULLSCREEN;:APPLICATION:EXTAPP:STATE
 NOTRUNNING;:AUXOUT:SOURCE ATRIGGER;EDGE FALLING;:CMBATCH
 1;:HEADER 1;:LOCK NONE;:ROSC:SOURCE INTERNAL;:VERBOSE
 1;:ALIAS:STATE 0;:DISPLAY:CLOCK 1;COLOR:PALETTE
 NORMAL;MATHCOLOR DEFAULT;REFCOLOR DEFAULT;:DISPLAY:FILTER
 SINX;FORMAT YT;GRATICULE FULL;INTENSITY:WAVEFORM
 75.0000;AUTOBRIGHT 1;:DISPLAY:PERSISTENCE
 OFF;STYLE VECTORS;TRIGBAR SHORT;TRIGT 1;VARPERSIST
 500.0000E-3;PORT FILE;:DIAG:LEVEL SUBSYS;:TRIGGER:A:MODE
 AUTO;TYPE EDGE;LEVEL 0.0000;HOLDOFF:BY DEFAULT;TIME
 1.5000E-6;:TRIGGER:A:EDGE:SOURCE CH1;COUPLING DC;SLOPE RISE;.

SOCKETServer:ENAbLe

This command enables or disables the socket server which supports a telnet or other TCPIP socket connection to send commands and queries to the instrument. The default state is enabled.

Group	Miscellaneous
Syntax	<code>SOCKETServer:ENAbLe {0 1 OFF ON}</code> <code>SOCKETServer:ENAbLe?</code>
Arguments	1 enables the socket server. If the state is 0 (disabled) and this command is sent to enable the socket server when the port is in use by another service, then the error event code 221 (Settings conflict) is posted to the event queue and the socket server remains disabled. In this case, select a different port number and attempt to enable the socket server again. 0 disables the socket server. ON enables the socket server. OFF disables the socket server.
Examples	<code>SOCKETSERVER:ENABLE On</code> enables the socket server. <code>SOCKETSERVER:ENABLE?</code> might return <code>:SOCKETSERVER:ENABLE 1</code> indicating the socket server is enabled.

SOCKETServer:PORT

This command sets the TCPIP port for the socket server connection.

Group	Miscellaneous
Syntax	<code>SOCKETServer:PORT <NR1></code> <code>SOCKETServer:PORT?</code>
Arguments	<code><NR1></code> is the TCPIP port for the socket server connection. If the socket server is enabled and the port specified is in use by another service, then the error event code 221 (Settings conflict) is posted to the event queue and the socket server remains in its current state (i.e. enabled/disabled and port address remain unchanged).

Similarly, if any sessions are active when the port is changed, this same error event may be posted to the event queue and the port will remain unchanged. In this case, exit all current sessions and send the :SOCKETServer:PORT command again.

Examples SOCKETSERVER:PORT 4000 sets the socket server port number to 4000.

SOCKETSERVER:PORT? might return :SOCKETSERVER:PORT 4000 indicating that the port number is 4000.

SOCKETServer:PROTOCOL

This command sets or queries the protocol for the socket server.

Group Miscellaneous

Syntax SOCKETServer:PROTOCOL {TERMiNaL|NONE}
SOCKETServer:PROTOCOL?

Arguments TERMiNaL specifies terminal protocol for the socket server. When set to TERMiNaL, a session startup message is sent to the socket and a command prompt is provided.

NONE disables the terminal features, allowing the server to be used for raw socket transactions, such as with a VISA socket server. The default setting is NONE.

Examples SOCKETSERVER:PROTOCOL NONE sets the protocol to none.

SOCKETSERVER:PROTOCOL? might return :SOCKETSERVER:PROTOCOL TERMINAL indicating the protocol is set to terminal.

*SRE

The *SRE (Service Request Enable) command sets and queries the bits in the Service Request Enable Register. For more information, refer to Registers.

Group Status and Error

Syntax *SRE <NR1>
*SRE?

Related Commands

[*CLS](#)
[DESE](#)
[*ESE](#)
[*ESR?](#)
[EVENT?](#)
[EVMsg?](#)
[FACtory](#)
[*STB?](#)

Arguments

<NR1> is a value in the range from 0 through 255. The binary bits of the SRER are set according to this value. Using an out-of-range value causes an execution error. The power-on default for SRER is 0 if *PSC is 1. If *PSC is 0, the SRER maintains the previous power cycle value through the current power cycle.

Examples

*SRE 48 sets the bits in the SRER to binary 00110000.

*SRE? might return 32, showing that the bits in the SRER have the binary value of 00100000.

***STB? (Query Only)**

The *STB? (Read Status Byte) query returns the contents of the Status Byte Register (SBR) using the Master Summary Status (MSS) bit. For more information, refer to Registers.

Group

Status and Error

Syntax

*STB?

Related Commands

[*CLS](#)
[DESE](#)
[*ESE](#)
[*ESR?](#)
[EVENT?](#)
[EVMsg?](#)
[FACtory](#)

[*SRE](#)

Returns <NR1>

Examples *STB? might return 96, showing that the SBR contains the binary value 01100000.

TEKSecure (No Query Form)

This command initializes, for the current user, both waveform and setup memories, overwriting any previously stored data.

Equivalent to invoking Teksecure from the Utility menu. This is a time-consuming operation (3 to 5 minutes) and the instrument is inoperable until the TekSecure operation is complete.

Group Miscellaneous

Syntax TEKSecure

Examples TEKSECURE initializes both waveform and setup memories.

TIME? (Query Only)

This command queries the time that the instrument displays.

Group Miscellaneous

Syntax TIME?

Related Commands [DATE?](#)

Returns <QString> is a time in the form “hh:mm:ss” where hh refers to a two-digit hour number, mm refers to a two-digit minute number from 01 to 60, and ss refers to a two-digit second number from 01 to 60.

Examples TIME?? might return :TIME "14:05:17", indicating the current time is set to 2:05 p.m. and 17 seconds.

TIME:ZONE

This command sets the time zone to the one specified.

Group	Miscellaneous
Syntax	TIME:ZONE <QString> TIME:ZONE?
Arguments	<QString> is a quoted string representing the desired time zone.
Examples	TIME:ZONE "America/Yellowknife" sets the time zone to UTC delta -7. TIME:ZONE? might return :TIME:ZONE "America/Los_Angeles".

TIME:ZONE:UTCDELTA

This command sets or queries the time zone using the difference between the desired time zone and UTC.

Group	Miscellaneous
Syntax	TIME:ZONE:UTCDELTA <NR3> TIME:ZONE:UTCDELTA?
Arguments	<NR3> is the specified number of hours difference between the desired time zone and UTC which is equivalent to GMT. The deltas supported are: -12.00, -11.00, -10.00, -9.30, -9.00, -8.30, -8.00, -7.00, -6.00, -5.00, -4.00, -3.30, -3.00, -2.00, -1.00, 0.0, 1.00, 2.00, 3.00, 3.30, 4.00, 4.30, 5.00, 5.30, 6.00, 6.30, 7.00, 8.00, 9.00, 9.30, 10.00, 10.30, 11.00, 11.30, 12.00
Examples	TIME:ZONE:UTCDELTA -7.0e0 sets the time zone to America/Yellowknife. TIME:ZONE:UTCDELTA? might return :TIME:ZONE:UTCDELTA -8.0000.

TOTaluptime? (Query Only)

Total number of hours the oscilloscope has been turned on since the NV memory was last programmed, usually during the initial manufacturing process.

Group	Miscellaneous
Syntax	TOTALuptime?
Returns	The total number of hours the instrument has been turned on since the NV memory was last programmed.
Examples	TOTALUPTIME? might return :TOTALUPTIME 756 indicating the up time is 756 minutes.

TOUCHSCReen:CALibrate (No Query Form)

This command launches the touchscreen calibration procedure. This command is equivalent to tapping the Calibrate Touchscreen control in the Utility->Self Test menu.

Group	Self Test
Syntax	TOUCHSCReen:CALibrate START
Arguments	START launches the touchscreen calibration procedure.
Examples	TOUCHSCREEN:CALIBRATE START launches the touchscreen calibration procedure.

TOUCHSCReen:STATe

This sets or queries the enabled state of the touch screen. This command is equivalent to pushing the Touch Off button on the front panel.

To completely disable front panel operation, use the following two commands: LOCK ALL; :TOUCHSCREEN:STATE OFF. To re-enable the front panel, send these two commands: LOCK NONE; :TOUCHSCREEN:STATE ON. The commands must be sent in that order.

Group	Miscellaneous
Syntax	TOUCHSCReen:STATe {0 1 OFF ON} TOUCHSCReen:STATe?

Related Commands [LOCK](#)

Arguments 0 disables the touch screen.
ON enables the touch screen.
OFF disables the touch screen.

Examples TOUCHSCREEN:STATE OFF disables the touch screen.
TOUCHSCREEN:STATE? might return :TOUCHSCREEN:STATE 1 indicating the touch screen is enabled.

*TRG (No Query Form)

Performs a group execute trigger on commands defined by *DDT.

Group Miscellaneous

Syntax *TRG

Related Commands [*DDT](#)

Examples *TRG immediately executes all commands that have been defined by *DDT.

TRIGger

This command forces a trigger event to occur. The query returns the current trigger parameters for the instrument.

Group Trigger

Syntax TRIGger FORCE
TRIGger?

Arguments FORCE creates a trigger event. If TRIGger:STATE is set to READy, the acquisition will complete. Otherwise, this command will be ignored. This is equivalent to pressing the Force button on the front panel.

- Examples** **TRIGGER FORCE** forces a trigger event to occur.
- TRIGGER?** returns the current trigger parameters for the instrument.

TRIGger:{A|B|B:RESET}

This command sets the A, B, or B Reset trigger level automatically to 50% of the range of the minimum and maximum values of the trigger input signal. The query returns current trigger parameters. The trigger level is the voltage threshold through which the trigger source signal must pass to generate a trigger event. This command is equivalent to pushing the LEVEL knob on the front panel.

- Group** Trigger
- Syntax** **TRIGger:{A|B|B:RESET} SETLeve1**
TRIGger:{A|B|B:RESET}?
- Arguments** **SETLeve1** sets the trigger level to 50% of the range of the minimum and maximum values of the trigger input signal.
- Examples** **TRIGger:A SETLEVEL** sets the trigger level to 50% of the range of the minimum and maximum values of the trigger input signal.
- TRIGger:A?** returns current trigger parameters.

TRIGger:{A|B}:BUS:B<x>:ARINC429A:CONDition

This command specifies a field or condition for an ARINC429 bus to trigger on. The bus number is specified by x.

- Conditions** Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
- Group** Trigger
- Syntax** **TRIGger:{A|B}:BUS:B<x>:ARINC429A:CONDition**
{SOW|LABe1|DATA|LABELANDDATA |EOW|ERROR}
TRIGger:{A|B}:BUS:B<x>:ARINC429A:CONDition?

Arguments	SOW specifies triggering on the first bit of a word.
	LABe1 specifies triggering on a matching label.
	DATA specifies triggering on matching packet data field(s).
	LABELANDDATA specifies triggering on a matching label and matching packet data field(s).
	EOW specifies triggering on the 32nd bit of a word.
	ERROR specifies triggering on a specified error condition.
NOTE. <i>The type of error triggered on is specified by TRIGger:{A B}:BUS:B<x>:ARINC429A:ERRType.</i>	
Examples	TRIGGER:A:BUS:B1:ARINC429A:CONDITION DATA specifies triggering on packets that contain matching data field(s).
	TRIGGER:A:BUS:B1:ARINC429A:CONDITION? might return SOW, indicating that the bus is triggering on the first bit of each word in the packet.

TRIGger:{A|B}:BUS:B<x>:ARINC429A:DATA:HIVALue

This command sets or queries the high value when trigger on an ARINC429 data field. The bus number is specified by x. The trigger condition must be set to DATA or LABELANDDATA, and the data qualifier must be INrange or OUTrange.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:ARINC429A:DATA:HIVALue <QString> TRIGger:{A B}:BUS:B<x>:ARINC429A:DATA:HIVALue?
Arguments	<QString> is the label value.
NOTE. <i>The size of the QString is dependent on the data field format selected using BUS:Bx:ARINC429A:DATAFormat. Also, the stored QString is reset to its default value whenever the data field format is changed.</i>	

Examples `TRIGGER:A:BUS:B1:ARINC429A:DATA:HIVALUE`
 `"XXXXXXXXXXXXXXXXX1000"` sets the value to `XXXXXXXXXXXXXXXXX1000`.
 `TRIGGER:A:BUS:B1:ARINC429A:DATA:HIVALUE?` might return
 `"XXXXXXXXXXXXXXXXXXXXX"`, indicating that the value is
 `XXXXXXXXXXXXXXXXXXXXX`.

TRIGger:{A|B}:BUS:B<x>:ARINC429A:DATA:QUALifier

This command sets or queries the qualifier to be used when triggering on data in the DATA field for an ARINC429 bus signal. The bus number is specified by x. The trigger condition must be set to DATA or LABELANDDATA.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Trigger

Syntax `TRIGger:{A|B}:BUS:B<x>:ARINC429A:DATA:`
 `QUALifier {Equal|UNEQual|LESSthan|MOREthan`
 `|LESSEqual|MOREEqual|INrange|OUTrange}`
 `TRIGger:{A|B}:BUS:B<x>:ARINC429A:DATA:QUALifier?`

Arguments Arguments are the available data qualifiers.

NOTE. *The trigger qualifier only applies to the bits defined as the data field via the bus data field format specifier (using `BUS:B<x>:ARINC429A:DATAFormat`)*

Examples `TRIGGER:A:BUS:B1:ARINC429A:DATA:QUALIFIER LESSTHAN` sets the data
 qualifier to less than.
 `TRIGGER:A:BUS:B1:ARINC429A:DATA:QUALIFIER?` might return `EQUAL`,
 indicating that the data qualifier is set to equal.

TRIGger:{A|B}:BUS:B<x>:ARINC429A:DATA:VALue

This command sets or queries the low value when triggering on an ARINC429 data field. The bus number is specified by x. The trigger condition must be set to DATA or LABELANDDATA.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:ARINC429A:DATA:VALUE <QString> TRIGger:{A B}:BUS:B<x>:ARINC429A:DATA:VALUE?</pre>
Arguments	<QString> is the label value.
NOTE. The size of the QString is dependent on the data field format selected using <i>BUS:B<x>:ARINC429A:DATAFormat</i>. Also, the stored QString is reset to its default value whenever the data field format is changed.	
Examples	<pre>TRIGGER:A:BUS:B1:ARINC429A:DATA:VALUE "XXXXXXXXXXXXXXXX1000" sets the value to XXXXXXXXXXXXXXXXXXX1000.</pre> <pre>TRIGGER:A:BUS:B1:ARINC429A:DATA:VALUE? might return "XXXXXXXXXXXXXXXXXXXX", indicating that the value is XXXXXXXXXXXXXXXXXXXX.</pre>

TRIGger:{A|B}:BUS:B<x>:ARINC429A:ERRType

This command sets or queries the error type when triggering on an ARINC429 bus signal. The bus number is specified by x. The trigger condition must be set to ERRor.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:ARINC429A:ERRType {ANY PARity WORD GAP} TRIGger:{A B}:BUS:B<x>:ARINC429A:ERRType?</pre>
Arguments	ANY sets the error type to match any of the other available error types. PARity sets the error type to match on parity errors (parity value results in even parity count for a word). WORD sets the error type to match on word errors (any unframed or unknown decode data). GAP sets the error type to match on gap violations (less than 4 bits idle time between two packets on the bus).

- Examples** `TRIGGER:A:BUS:B1:ARINC429A:ERRTYPE PARITY` sets the error type to match on parity errors.
- `TRIGGER:A:BUS:B1:ARINC429A:ERRTYPE?` might return `ANY`, indicating that any error condition will produce a match.

TRIGger:{A|B}:BUS:B<x>:ARINC429A:LABel:HIVALue

This command sets or queries the high value when triggering on an ARINC429 label field. The bus number is specified by x. The trigger condition must be set to LABel, and the label qualifier must be INrange or OUTrange.

- Conditions** Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
- Group** Trigger
- Syntax** `TRIGger:{A|B}:BUS:B<x>:ARINC429A:LABel:HIVALue <QString>`
`TRIGger:{A|B}:BUS:B<x>:ARINC429A:LABel:HIVALue?`
- Arguments** <QString> is the label value.
- Examples** `TRIGGER:A:BUS:B1:ARINC429A:LABEL:HIVALUE "XXXX1010"` sets the value to XXXX1010.
- `TRIGGER:A:BUS:B1:ARINC429A:LABEL:HIVALUE?` might return "XXXXXXXX", indicating that the value is XXXXXXXX.

TRIGger:{A|B}:BUS:B<x>:ARINC429A:LABel:QUALifier

This command sets or queries the qualifier to be used when triggering on label data for an ARINC429 bus signal. The bus number is specified by x. The trigger condition must be set to LABel or LABELANDDATA.

- Conditions** Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
- Group** Trigger

Syntax	TRIGger:{A B}:BUS:B<x>:ARINC429A:LABel: QUALifier {Equal UNEQual LESSthan MOREthan LESSEQual MOREEQual INrange OUTrange} TRIGger:{A B}:BUS:B<x>:ARINC429A:LABel:QUALifier?
Arguments	Arguments are the available data qualifiers. NOTE. <i>If the trigger condition is set to LABELANDDATA, the label qualifier will be locked to Equal until the trigger condition is changed again.</i>
Examples	TRIGGER:A:BUS:B1:ARINC429A:LABEL:QUALIFIER LESSTHAN sets the label qualifier to less than. TRIGGER:A:BUS:B1:ARINC429A:LABEL:QUALIFIER? might return EQUAL, indicating that the label qualifier is set to equal.

TRIGger:{A|B}:BUS:B<x>:ARINC429A:LABel:VALue

This command sets or queries the low value when triggering on an ARINC429 label field. The bus number is specified by x. The trigger condition must be set to LABel or LABELANDDATA.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:ARINC429A:LABel:VALue <QString> TRIGger:{A B}:BUS:B<x>:ARINC429A:LABel:VALue?
Arguments	<QString> is the label value.
Examples	TRIGGER:A:BUS:B1:ARINC429A:LABEL:VALUE "XXXX1010" sets the value to XXXX1010. TRIGGER:A:BUS:B1:ARINC429A:LABEL:VALUE? might return "XXXXXXXX", indicating that the value is XXXXXXXX.

TRIGger:{A|B}:BUS:B<x>:ARINC429A:SDI:VALue

This command sets or queries the label when triggering on an ARINC429 SDI field. The bus number is specified by x. The trigger condition must be set to DATA or LABELANDDATA, and the data format must be set to DATA.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:ARINC429A:SDI:VALue <QString> TRIGger:{A B}:BUS:B<x>:ARINC429A:SDI:VALue?
Arguments	<QString> is the label value.
Examples	TRIGGER:A:BUS:B1:ARINC429A:SDI:VALUE "X0" sets the value to X0. TRIGGER:A:BUS:B1:ARINC429A:SDI:VALUE? might return "XX", indicating that the value is XX.

TRIGger:{A|B}:BUS:B<x>:ARINC429A:SSM:VALue

This command sets or queries the label value when triggering on an ARINC429 SSM field. The bus number is specified by x. The trigger condition must be set to DATA or LABELANDDATA, and the data format must be set to DATA or SDIDATA.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:ARINC429A:SSM:VALue <QString> TRIGger:{A B}:BUS:B<x>:ARINC429A:SSM:VALue?

Arguments <QString> is the label value.

NOTE. The SSM field is only present when the selected data field format is DATA or SDIDATA (using *BUS:B<x>:ARINC429A:DATAFormat*). Also, the stored QString is reset to its default value whenever the data field format is changed.

Examples TRIGGER:A:BUS:B1:ARINC429A:SSM:VALUE "X0" sets the value to X0.
TRIGGER:A:BUS:B1:ARINC429A:SSM:VALUE? might return "XX", indicating that the value is XX.

TRIGger:{A|B}:BUS:B<x>:AUDio:CONDition

This command sets the condition (word select, start of frame, or matching data) to be used when triggering on an audio bus signal. The bus number is specified by <x>.

Conditions Requires option 5-SRAUDIO or SUP5-SRAUDIO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:AUDio:CONDition {SOF|DATA}
TRIGger:{A|B}:BUS:B<x>:AUDio:CONDition?

Arguments SOF enables triggering on a word select or start of frame (depending on Audio Type).
DATA enables triggering on matching data.

Examples TRIGger:A:BUS:B1:AUDio:CONDition SOF sets the condition to start of frame.
TRIGger:A:BUS:B1:AUDio:CONDition? might return
:TRIGGER:A:BUS:B1:AUDIO:CONDITION DATA indicating the condition is data.

TRIGger:{A|B}:BUS:B<x>:AUDio:DATa:HITDMVALue

This command sets or queries the binary data string for the high data word to be used when triggering on an TDM audio bus signal. The trigger condition must be set to DATA using TRIGger:{A|B}:BUS:B<x>:AUDio:CONDition.

Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:AUDio:DATa:HITDMVALue <QString> TRIGger:{A B}:BUS:B<x>:AUDio:DATa:HITDMVALue?</pre>
Arguments	<QString> is the binary data string for the high data word to be used when triggering on an TDM audio bus signal.
Examples	TRIGger:A:BUS:B1:AUDio:DATa:HITDMVALue "1100" sets the high value to 1100. TRIGger:A:BUS:B1:AUDio:DATa:HITDMVALue? might return :TRIGGER:A:BUS:B1:AUDIO:DATA:HITDMVALUE "XXXXXXXXXXXXXXXXXXXX1010" indicating the high value is 1010.

TRIGger:{A|B}:BUS:B<x>:AUDio:DATa:HIVALue

This command sets the upper word value to be used when triggering on an audio bus signal. The trigger condition must be set to DATA using [TRIGger:{A|B}:BUS:B<x>:AUDio:CONDition](#).

The bus number is specified by <x>.

Conditions	Requires option 5-SRAUDIO or SUP5-SRAUDIO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:AUDio:DATa:HIVALue <QString> TRIGger:{A B}:BUS:B<x>:AUDio:DATa:HIVALue?</pre>
Arguments	<QString> is the upper word value to be used when triggering on an audio bus signal.
Examples	TRIGger:A:BUS:B1:AUDio:DATa:HIVALue "11001101" sets the hi value to XXXXXXXXXXXXXXXXXXXX11001101. TRIGger:A:BUS:B1:AUDio:DATa:HIVALue? might return :TRIGGER:A:BUS:B1:AUDIO:DATA:HIVALUE "XXXXXXXXXXXXXXXXXXXXXXX" indicating the hi value is set to XXXXXXXXXXXXXXXXXXXXXXXX.

TRIGger:{A|B}:BUS:B<x>:AUDio:DATa:OFFSet

This command sets the data offset value to be used when triggering on an audio bus signal. The trigger condition must be set to DATA using [TRIGger:{A|B}:BUS:B<x>:AUDio:CONDition](#).

The bus number is specified by <x>.

Conditions	Requires option 5-SRAUDIO or SUP5-SRAUDIO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:AUDio:DATa:OFFSet <NR1> TRIGger:{A B}:BUS:B<x>:AUDio:DATa:OFFSet?
Arguments	<NR1> is the data offset value.
Examples	TRIGger:A:BUS:B1:AUDio:DATa:OFFSet 2 sets the data offset to 2. TRIGger:A:BUS:B1:AUDio:DATa:OFFSet? might return :TRIGGER:A:BUS:B1:AUDIO:DATA:OFFSET 1 indicating the data offset value is 1.

TRIGger:{A|B}:BUS:B<x>:AUDio:DATa:QUALifier

This command sets the qualifier to be used when triggering on an audio bus signal. The trigger condition must be set to DATA using [TRIGger:{A|B}:BUS:B<x>:AUDio:CONDition](#).

The bus number is specified by <x>.

Conditions	Requires option 5-SRAUDIO or SUP5-SRAUDIO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:AUDio:DATa:QUALifier {LESSthan MOREthan Equal UNEQual LESSEQual MOREEQual INrange OUTrange} TRIGger:{A B}:BUS:B<x>:AUDio:DATa:QUALifier?

Arguments	LESSthan sets the qualifier to less than. MOREthan sets the qualifier to greater than. EQua1 sets the qualifier to equal. UNEQua1 sets the qualifier to not equal. LESSEQua1 sets the qualifier to less than or equal. MOREEQua1 sets the qualifier to greater than or equal. INrange sets the qualifier to in range. OUTrange sets the qualifier to out of range.
Examples	TRIGger:A:BUS:B1:AUDio:DATa:QUALifier LESSTHAN sets the qualifier to less than. TRIGger:A:BUS:B1:AUDio:DATa:QUALifier? might return :TRIGGER:A:BUS:B1:AUDIO:DATA:QUALIFIER EQUAL indicating the qualifies is set to equal.

TRIGger:{A|B}:BUS:B<x>:AUDio:DATa:TDMVALue

This command sets or queries the binary data string for the single or low data word to be used when triggering on an TDM audio bus signal. The trigger condition must be set to DATa using TRIGger:{A|B}:BUS:B{x}:AUDio:CONDition.

Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:AUDio:DATa:TDMVALue <QString> TRIGger:{A B}:BUS:B<x>:AUDio:DATa:TDMVALue?
Arguments	<QString> is the binary data string for the single or low data word to be used when triggering on an TDM audio bus signal.
Examples	TRIGger:A:BUS:B1:AUDio:DATa:TDMVALue "1100" sets the TDMVALUE to 1100. TRIGger:A:BUS:B1:AUDio:DATa:TDMVALue? might return :TRIGGER:A:BUS:B1:AUDIO:DATA:TDMVALUE "XXXXXXXXXXXXXXXXXXXX1010" indicating the TDM value is 1010.

TRIGger:{A|B}:BUS:B<x>:AUDio:DATa:VALue

This command sets the lower word value to be used when triggering on an audio bus signal. The trigger condition must be set to DATA using [TRIGger:{A|B}:BUS:B<x>:AUDio:CONDition](#).

The bus number is specified by <x>.

Conditions Requires option 5-SRAUDIO or SUP5-SRAUDIO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:AUDio:DATa:VALue <QString>
TRIGger:{A|B}:BUS:B<x>:AUDio:DATa:VALue?

Arguments <QString> specifies the trigger data lower word.

Examples TRIGger:A:BUS:B1:AUDio:DATa:VALue "11001100101" sets the data value to XXXXXXXXXXXXXXXX11001100101.

TRIGger:A:BUS:B1:AUDio:DATa:VALue? might
return :TRIGGER:A:BUS:B1:AUDIO:DATA:VALUE
"XXXXXXXXXXXXXXXXXXXXXXXXXXXX" indicating the data value is
XXXXXXXXXXXXXXXXXXXXXXXXXXXX.

TRIGger:{A|B}:BUS:B<x>:AUDio:DATa:WORD

This command sets the alignment of the data (left, right or either) to be used to trigger on an audio bus signal. The trigger condition must be set to DATA using [TRIGger:{A|B}:BUS:B<x>:AUDio:CONDition](#).

The bus number is specified by <x>.

Conditions Requires option 5-SRAUDIO or SUP5-SRAUDIO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:AUDio:DATa:WORD {EITHer|LEFt|RIGHt}
TRIGger:{A|B}:BUS:B<x>:AUDio:DATa:WORD?

Arguments	Either aligns the trigger data to either left or right. Left aligns the trigger data to the left. Right aligns the trigger data to the right.
Examples	TRIGGER:A:BUS:B1:AUDIO:DATA:WORD Left sets the word alignment to the left. TRIGGER:A:BUS:B1:AUDIO:DATA:WORD? might return :TRIGGER:A:BUS:B1:AUDIO:DATA:WORD Either indicating the trigger data is aligned to either left or right.

TRIGGER:{A|B}:BUS:B<x>:CAN:CONDITION

This command sets the condition (start of frame, frame type, identifier, matching data, EOF, missing ACK field, bit-stuffing error) to be used when triggering on a CAN bus signal. The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGGER:{A B}:BUS:B<x>:CAN:CONDITION {SOF FRAMETYPE IDENTIFIER DATA IDANDDATA EOF ERROR FDBITS} TRIGGER:{A B}:BUS:B<x>:CAN:CONDITION?
Arguments	SOF enables triggering on the start of frame. FDBITS enables triggering on the values of the BRS and ESI bits in an FD packet. FRAMETYPE enables triggering on the type of frame. IDENTIFIER enables triggering on a matching identifier. DATA enables triggering on matching data. IDANDDATA enables triggering on a matching identifier and matching data. EOF enables triggering on the end of frame. ERROR enables triggering on a specified error condition.
Examples	TRIGGER:A:BUS:B1:CAN:CONDITION? might return :TRIGGER:A:BUS:B1:CAN:CONDITION EOF indicating an end of file condition.

TRIGGER:A:BUS:B1:CAN:CONDITION DATA enables triggering on matching CAN data.

TRIGger:{A|B}:BUS:B<x>:CAN:DATA:DIRection

This command sets the data direction (read, write or “nocare”) to be used to search on a CAN bus signal. The trigger condition must be set to IDentifier (using [TRIGger:{A|B}:BUS:B<x>:CAN:CONDition](#)). The bus number is specified by <x>.

Conditions Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:CAN:DATA:DIRection
{READ|WRITE|NOCARE}
TRIGger:{A|B}:BUS:B<x>:CAN:DATA:DIRection?

Arguments READ sets the CAN data direction to READ.
WRITE sets the CAN data direction to WRITE.
NOCARE sets the CAN data direction to either.

Examples TRIGGER:A:BUS:B1:CAN:DATA:DIRECTION WRITE sets the CAN data direction to Write.
TRIGGER:A:BUS:B1:CAN:DATA:DIRECTION? might return
:TRIGGER:A:BUS:B1:CAN:DATA:DIRECTION NOCARE indicating the data direction can be either read or write.

TRIGger:{A|B}:BUS:B<x>:CAN:DATA:OFFSet

This command sets or queries the data offset value, in bytes, to use when triggering on the CAN data field. The bus number is specified by x. The trigger condition must be set to DATA or IDANDDATA.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Trigger

Syntax	<pre>TRIGger:{A B}:BUS:B<x>:CAN:DATA:OFFSet <NR1> TRIGger:{A B}:BUS:B<x>:CAN:DATA:OFFSet?</pre>
Related Commands	<pre>BUS:B<x>:CAN:STANDARD TRIGger:{A B}:BUS:B<x>:CAN:DATA:SIZE TRIGger:{A B}:BUS:B<x>:CAN:DATA:VALUE</pre>
Arguments	<NR1> is an integer whose minimum and default values are -1 (don't care), and the maximum is up to 7 (for CAN 2.0) or up to 63 (for ISO CAN FD and Non-ISO CAN FD). The maximum is dependent on the number of bytes being matched and the CAN standard selected. Its value is calculated as [Absolute Maximum] - [Data Match Size]. For CAN 2.0, the absolute maximum is 8 bytes. For ISO CAN FD and Non-ISO CAN FD, the absolute maximum is 64 bytes. The minimum data match size is 1 byte, which produces the ranges listed above. Increasing the data match size above 1 byte will adjust the range of valid data offset values accordingly.
Examples	TRIGGER:A:BUS:B1:CAN:DATA:OFFSET 5 sets the CAN data offset to 5 bytes. TRIGGER:A:BUS:B1:CAN:DATA:OFFSET? might return 7, indicating the CAN data offset is 7 bytes. If the CAN standard is set for CAN 2.0, and the trigger data size is set to 3, the maximum value for the data offset will be 5 ($8 - 3 = 5$). If the CAN standard is set for ISO CAN FD or Non-ISO CAN FD, and the trigger data size is set to 8, the maximum value for the data offset will be 56 ($64 - 8 = 56$).

TRIGger:{A|B}:BUS:B<x>:CAN:DATA:QUALifier

This command sets the qualifier (<, >, =, ≠, ≤, ≥) to be used when triggering on a CAN bus signal. The trigger condition must be set to IDANDDATA or DATA (using [TRIGger:{A|B}:BUS:B<x>:CAN:CONDition](#)). The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:CAN:DATA:QUALifier {LESSthan MOREthan Equal UNEQual LESSEQual MOREEqual}</pre>

TRIGger:{A|B}:BUS:B<x>:CAN:DATA:QUALifier?

Arguments	LESSthan sets the oscilloscope to trigger when the data is less than the qualifier value. MOREthan sets the oscilloscope to trigger when the data is greater than the qualifier value. EQua1 sets the oscilloscope to trigger when the data is equal to the qualifier value. UNEQua1 sets the oscilloscope to trigger when the data is not equal to the qualifier value. LESSEQua1 sets the oscilloscope to trigger when the data is less than or equal to the qualifier value. MOREEQua1 sets the oscilloscope to trigger when the data is greater than or equal to the qualifier value.
Examples	TRIGGER:A:BUS:B1:CAN:DATA:QUALIFIER LESSTHAN sets the oscilloscope to trigger when the data is less than the qualifier value. TRIGGER:A:BUS:B1:CAN:DATA:QUALIFIER? might return :TRIGGER:A:BUS:B1:CAN:DATA:QUALIFIER LESSTHAN, indicating that the oscilloscope is set to trigger when the data is less than the qualifier value.

TRIGger:{A|B}:BUS:B<x>:CAN:DATA:SIZE

This command sets the length of the data string, in bytes, to be used when triggering on a CAN bus signal. The trigger condition must be set to IDANDDATA or DATA (using [TRIGger:{A|B}:BUS:B<x>:CAN:CONDition](#)). The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:CAN:DATA:SIZE <NR1> TRIGger:{A B}:BUS:B<x>:CAN:DATA:SIZE?
Arguments	<NR1> is the length of the data string in bytes.

Examples `TRIGger:A:BUS:B1:CAN:DATA:SIZE 2` sets the data size to 2 bytes.
 `TRIGger:A:BUS:B1:CAN:DATA:SIZE?` might return
 `:TRIGGER:A:BUS:B1:CAN:DATA:SIZE 1` indicating the data size is
 set to 1 byte.

TRIGger:{A|B}:BUS:B<x>:CAN:DATA:VALue

This command sets the binary data value to be used when triggering on a CAN bus signal. The trigger condition must be set to IDANDDATA or DATA (using [TRIGger:{A|B}:BUS:B<x>:CAN:CONDition](#)). The bus number is specified by <x>.

Conditions Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group Trigger

Syntax `TRIGger:{A|B}:BUS:B<x>:CAN:DATA:VALue <QString>`
 `TRIGger:{A|B}:BUS:B<x>:CAN:DATA:VALue?`

Arguments <QString> is the data value in binary format. The only allowed characters in the QString are 0, 1, and X.

Examples `TRIGGER:A:BUS:B1:CAN:DATA:VALUE "1011"` sets the CAN data value to 1011.
 `TRIGGER:A:BUS:B1:CAN:DATA:VALUE?` might return
 `:TRIGGER:A:BUS:B1:CAN:DATA:VALUE "XXXXXXXX"` indicating
 the data valule is set to XXXXXXXX.

TRIGger:{A|B}:BUS:B<x>:CAN:ERRType

This command sets or queries the type of error condition for a CAN bus to triggering on. The bus number is specified by x. The trigger condition must be set to ERRor.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Trigger

Syntax	TRIGger: {A B}:BUS:B<x>:CAN:ERRTYPE {ACKMISS BITSTUFFing FORMERError ANYERError} TRIGger: {A B}:BUS:B<x>:CAN:ERRTYPE?
Arguments	ACKMISS specifies triggering on a missing ACK field. BITSTUFFing specifies triggering on a bit stuffing error. FORMERError specifies triggering on a CAN FD form error. To use this option, the CAN standard must be set to FDISO or FDNONISO. ANYERError specifies triggering on any error type.
Examples	TRIGGER:A:BUS:B1:CAN:ERRTYPE ACKMISS specifies triggering on any missing ACK fields. TRIGGER:A:BUS:B1:CAN:ERRTYPE? might return ANYERROR, indicating that the bus is triggering on all error types.

TRIGger:{A|B}:BUS:B<x>:CAN:FD:BRSBit

This command sets or queries the value of the bit rate switch bit (BRS bit) for a CAN bus to triggering on. The bus number is specified by x. The trigger condition must be set to FDBITS, and the CAN standard must be FDISO or FDNONISO.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger: {A B}:BUS:B<x>:CAN:FD:BRSBit {ONE ZERO NOCARE} TRIGger: {A B}:BUS:B<x>:CAN:FD:BRSBit?
Arguments	ONE filters CAN FD packets to only match those where the BRS bit has a value of 1 (fast data enabled). ZERO filters CAN FD packets to only match those where the BRS bit has a value of 0 (fast data disabled). NOCARE disables filtering of CAN FD packets on the BRS bit.
Examples	TRIGGER:A:BUS:B1:CAN:FD:BRSBit ONE specifies filtering CAN FD packets for those where the BRS bit has a value of 1.

`TRIGGER:A:BUS:B1:CAN:FD:BRSBIT?` might return `NOCARE`, indicating that CAN FD packets are not being filtered based on the BRS bit value.

TRIGger:{A|B}:BUS:B<x>:CAN:FD:ESIBit

This command sets or queries the value of the error state indicator bit (ESI bit) for a CAN bus to triggering on. The bus number is specified by x. The trigger condition must be set to `FDBITS`, and the CAN standard must be `FDISO` or `FDNONISO`.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger
Syntax	<code>TRIGger:{A B}:BUS:B<x>:CAN:FD:ESIBit {ONE ZERO NOCARE}</code> <code>TRIGger:{A B}:BUS:B<x>:CAN:FD:ESIBit?</code>
Arguments	<code>ONE</code> filters CAN FD packets to only match those where the ESI bit has a value of 1 (recessive). <code>ZERO</code> filters CAN FD packets to only match those where the ESI bit has a value of 0 (dominant). <code>NOCARE</code> disables filtering of CAN FD packets on the ESI bit.
Examples	<code>TRIGGER:A:BUS:B1:CAN:FD:ESIBit ONE</code> specifies filtering CAN FD packets for those where the ESI bit has a value of 1. <code>TRIGGER:A:BUS:B1:CAN:FD:ESIBIT?</code> might return <code>NOCARE</code>, indicating that CAN FD packets are not being filtered based on the ESI bit value.

TRIGger:{A|B}:BUS:B<x>:CAN:FRAMetype

This command sets the frame type (data, remote, error or overload) to be used when triggering on a CAN bus signal. The trigger condition must be set to `FRAMetype` (using [TRIGger:{A|B}:BUS:B<x>:CAN:CONDition](#)). B<x> is the bus number. The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
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Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:CAN:FRAMetype {DATa REMOte ERRor OVERLoad} TRIGger:{A B}:BUS:B<x>:CAN:FRAMetype?</pre>
Arguments	DATa specifies a data frame type. REMOte specifies a remote frame type. ERRor specifies an error frame type. OVERLoad specifies an overload frame type.
Examples	<pre>TRIGGER:A:BUS:B1:CAN:FRAMETYPE DATA</pre> sets the CAN trigger frame type to DATA. <pre>TRIGGER:A:BUS:B1:CAN:FRAMETYPE?</pre> might return <pre>:TRIGGER:A:BUS:B1:CAN:FRAMETYPE DATA</pre> indicating the frame type is data.

TRIGger:{A|B}:BUS:B<x>:CAN:IDentifier:MODE

This command sets the addressing mode (standard or extended format) to be used when triggering on a CAN bus signal. The trigger condition must be set to IDANDDATA or DATa (using [TRIGger:{A|B}:BUS:B<x>:CAN:CONDition](#)). The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:CAN:IDentifier:MODE {STandard EXTended} TRIGger:{A B}:BUS:B<x>:CAN:IDentifier:MODE?</pre>
Arguments	STandard specifies the standard addressing mode. EXTended specifies the extended addressing mode.
Examples	<pre>TRIGger:A:BUS:B1:CAN:IDentifier:MODE EXTENDED</pre> sets the addressing mode to extended.

TRIGger:A:BUS:B1:CAN:IDentifier:MODE? might return
:TRIGGER:A:BUS:B1:CAN:IDENTIFIER:MODE STANDARD indicating the
address mode is standard.

TRIGger:{A|B}:BUS:B<x>:CAN:IDentifier:VALue

This command sets the binary address value to be used when triggering on a CAN bus signal. The trigger condition must be set to IDANDDATA or DATA (using [TRIGger:{A|B}:BUS:B<x>:CAN:CONDition](#)). The bus number is specified by <x>.

Conditions Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:CAN:IDentifier:VALue <QString>
TRIGger:{A|B}:BUS:B<x>:CAN:IDentifier:VALue?

Arguments <QString> is up to 29 bits specifying the binary identifier value. The only allowed characters in the QString are 0, 1, and X.

Examples TRIGGER:A:BUS:B1:CAN:IDENTIFIER:VALUE "1011" sets the identifier value to 1011.

TRIGGER:A:BUS:B1:CAN:IDENTIFIER:VALUE? might return
:TRIGGER:A:BUS:B1:CAN:IDENTIFIER:VALUE "XXXXXXXXXXXX" indicating the identifier values is XXXXXXXXXXXX.

TRIGger:{A|B}:BUS:B<x>:ETHERnet:CONDition

This command specifies a field or condition within an Ethernet frame to trigger on. The bus number is specified by <x>.

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Trigger

Syntax	<pre>TRIGger:{A B}:BUS:B<x>:ETHERnet:CONDition {SFD MACADDRESS MACLENGTH IPHeader TCPHeader DATA EOP IDLe FCSError QTAG} TRIGger:{A B}:BUS:B<x>:ETHERnet:CONDition?</pre>
Related Commands	Most of the other <code>TRIGger:A:BUS:B<x>:ETHERnet</code> commands are impacted by the setting of this command.
Arguments	<code>SFD</code> — Start of frame delimiter. <code>MACADDRESS</code> — MAC addresses field. <code>MACLENGTH</code> — MAC length/type field. <code>IPHeader</code> — IP header field. This argument is only available when <code>PROTOCOL</code> is set to IPv4. <code>TCPHeader</code> — TCP header field. This argument is only available when <code>PROTOCOL</code> is set to IPv4. <code>DATA</code> — TCP/IPv4 or MAC protocol client data field. If the protocol is set to <code>OTHER</code>, then <code>DATA</code> refers to the MAC client data. <code>EOP</code> — End of Packet field. <code>IDLe</code> — Idle field. <code>FCSError</code> — Frame Check Sequence Error (CRC) field. <code>QTAG</code> — IEEE 802.1Q (VLAN) control information field. In order to use <code>QTAG</code> as a trigger condition, the frame type must be set to <code>QTAG</code>).
Examples	<pre>TRIGger:A:BUS:B1:ETHERnet:CONDition MACADDRESS sets the trigger field to MACADDRESS. TRIGger:A:BUS:B1:ETHERnet:CONDition? might return DATA.</pre>

TRIGger:{A|B}:BUS:B<x>:ETHERnet:DATA:HIValue

When the Ethernet trigger condition is set to `DATA`, and the qualifier is set to either `INrange` or `OUTrange`, this command specifies the upper data value of the range. (Use the command `TRIGger:{A|B}:BUS:B<x>:ETHERnet:DATA:VALue` to specify the lower limit of the range.) The default is all X's (don't care). The bus number is specified by `<x>`.

Conditions	Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.
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Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:ETHERnet:DATA:HIVALue <QString> TRIGger:{A B}:BUS:B<x>:ETHERnet:DATA:HIVALue?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:ETHERnet:DATA:VALue TRIGger:{A B}:BUS:B<x>:ETHERnet:DATA:SIZE TRIGger:{A B}:BUS:B<x>:ETHERnet:CONDition
Arguments	<QString> is a quoted string whose length varies depending on the size setting, up to 32 bits. (Use the command TRIGger:{A B}:BUS:B<x>:ETHERnet:DATA:SIZE to specify the size.) The allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.
Examples	<pre>TRIGger:A:BUS:B1:ETHERnet:DATA:HIVALue "XXXXXXXXXXXXXXXXXXXXXXXXXXXX00001000" sets the upper limit of the range to XXXXXXXXXXXXXXXXXXXXXXXXXXXX00001000(when the trigger condition is set to DATA, and the qualifier is set to INrange or OUTrange) . TRIGger:A:BUS:B1:ETHERnet:DATA:HIVALue? might return "XXXXXXXXXXXXXXXXXXXXXXXXXXXX"</pre>

TRIGger:{A|B}:BUS:B<x>:ETHERnet:DATA:OFFSet

When the Ethernet trigger condition is set to DATA, this command specifies where in the data field to look for the data trigger value. It specifies the offset into the data field, in bytes, where the value will be matched. The default is -1 (don't care). The bus number is specified by <x>.

Conditions	Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:ETHERnet:DATA:OFFSet <NR1> TRIGger:{A B}:BUS:B<x>:ETHERnet:DATA:OFFSet?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:ETHERnet:CONDition

TRIGger:{A|B}:BUS:B<x>:ETHERnet:DATA:VALue

Arguments <NR1> is an integer whose minimum and default values are -1 (don't care) and maximum is 1,499.

Examples TRIGger:A:BUS:B1:ETHERnet:DATA:OFFSet 36 sets the data offset to 36 bytes.

TRIGger:A:BUS:B1:ETHERnet:DATA:OFFSet? might return -1, indicating that the data offset value is the default value, -1, meaning "don't care".

TRIGger:{A|B}:BUS:B<x>:ETHERnet:DATA:QUALifier

This command sets or queries the qualifier to be used when triggering on an Ethernet bus signal. The trigger condition must be set to DATA. The bus number is specified by <x>.

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:ETHERnet:DATA:QUALifier {EQUAL|UNEQUAL|LESSThan|MOREthan|LESSEQUAL|MOREEQUAL|INrange|OUTrange}
TRIGger:{A|B}:BUS:B<x>:ETHERnet:DATA:QUALifier?

Arguments LESSThan sets the qualifier to less than.
MOREthan sets the qualifier to greater than.
EQUAL sets the qualifier to equal.
UNEQUAL sets the qualifier to not equal.
LESSEQUAL sets the qualifier to less than or equal.
MOREEQUAL sets the qualifier to greater than or equal.
INrange sets the qualifier to in range.
OUTrange sets the qualifier to out of range.

Examples TRIGger:A:BUS:B1:ETHERnet:DATA:QUALifier LESSTHAN sets the qualifier to less than.

TRIGger:A:BUS:B1:ETHERnet:DATA:QUALifier? might return
:TRIGGER:A:BUS:B1:ETHERNET:DATA:QUALIFIER EQUAL indicating the
qualifier is set to equal.

TRIGger:{A|B}:BUS:B<x>:ETHERnet:DATA:SIZE

When the Ethernet trigger condition is set to DATA, this command specifies the number of contiguous TCP/IPv4/MAC client data bytes to trigger on. The bus number is specified by <x>.

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:ETHERnet:DATA:SIZE <NR1>
TRIGger:{A|B}:BUS:B<x>:ETHERnet:DATA:SIZE?

Related Commands [TRIGger:{A|B}:BUS:B<x>:ETHERnet:CONDition](#)

Arguments The minimum and default values are 1 and maximum is 16, except when the qualifier is set to Inside Range or Outside Range. In these cases, the maximum size is 4.

Examples TRIGger:A:BUS:B1:ETHERnet:DATA:SIZE 4 sets the oscilloscope to trigger on 4 contiguous data bytes.

TRIGger:A:BUS:B1:ETHERnet:DATA:SIZE? might return 6, indicating that the oscilloscope is set to trigger on 6 contiguous data bytes.

TRIGger:{A|B}:BUS:B<x>:ETHERnet:DATA:VALue

When the Ethernet trigger condition is set to DATA, and the qualifier is set to LESSthan, MOREthan, EQUAL, UNEQUAL, LESSEQUAL or MOREEQUAL, this command specifies the value to trigger on. When the Ethernet trigger condition is set to DATA, and the qualifier is set to INrange or OUTrange, this command specifies the lower limit of the range. (Use the command [TRIGger:{A|B}:BUS:B<x>:ETHERnet:DATA:HIVALue](#) to set the upper limit of the range.) The default is all X's (don't care). The bus number is specified by <x>.

Conditions	Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:ETHERnet:DATA:VALue <QString> TRIGger:{A B}:BUS:B<x>:ETHERnet:DATA:VALue?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:ETHERnet:CONDition TRIGger:{A B}:BUS:B<x>:ETHERnet:DATA:OFFSet TRIGger:{A B}:BUS:B<x>:ETHERnet:DATA:SIZE TRIGger:{A B}:BUS:B<x>:ETHERnet:DATA:HIVALue
Arguments	QString is a quoted string where the allowable characters are 0, 1, and X. The allowable number of characters depends on the setting for size (using TRIGger:A:BUS:B<x>:ETHERnet:DATA:SIZE). The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.
Examples	TRIGger:A:BUS:B1:ETHERnet:DATA:VALue "00001000" sets the binary data to trigger on to 00001000, assuming the qualifier is set to LESSthan, MOREthan, EQUAL, UNEQUAL, LESSEQUAL or MOREEQUAL, and DATA:SIZE is set to 1 byte. TRIGger:A:BUS:B1:ETHERnet:DATA:VALue "00001000" sets the lower limit of the range to 00001000, assuming the qualifier is set to INrange or OUTrange, and DATA:SIZE is set to 1 byte.

TRIGger:{A|B}:BUS:B<x>:ETHERnet:IPHeader:DESTinationaddr:VALue

When the Ethernet trigger condition is set to IPHeader, this command specifies the value of the 32-bit destination address that is to be used in the trigger (along with the source address and protocol value). The IP destination address is a standard IP address such as 192.168.0.1. The default is all X's (don't care). The bus number is specified by <x>.

Conditions	Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.
Group	Trigger

Syntax	<pre>TRIGger:{A B}:BUS:B<x>:ETHERnet:IPHeader:DESTinationaddr: VALue <QString> TRIGger:{A B}:BUS:B<x>:ETHERnet:IPHeader:DESTinationaddr: VALue?</pre>
Related Commands	<pre>TRIGger:{A B}:BUS:B<x>:ETHERnet:CONDition TRIGger:{A B}:BUS:B<x>:ETHERnet:IPHeader:SOUrceaddr:VALue TRIGger:{A B}:BUS:B<x>:ETHERnet:IPHeader:PROTOcol:VALue</pre>
Arguments	<QString> is a quoted string of up to 32 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.
Examples	<pre>TRIGger:A:BUS:B1:ETHERnet:IPHeader:DESTinationaddr:VALue "00011001001000010110100000000001" sets the IP destination address to trigger on to 192.168.0.1. TRIGger:A:BUS:B1:ETHERnet:IPHeader:DESTinationaddr:VALue might return "XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX01".</pre>

TRIGger:{A|B}:BUS:B<x>:ETHERnet:IPHeader:PROTOcol:VALue

When the Ethernet trigger condition is set to **IPHeader**, this command specifies the value of the 8-bit protocol field that is to be used in the trigger (along with the source and destination addresses). The default is all X's (don't care). The bus number is specified by <x>.

NOTE. Commonly used protocol values are 1 (ICMP), 2 (IGMP), 6 (TCP) and 17 (UDP).

Conditions	Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:ETHERnet:IPHeader:PROTOcol:VALue <QString> TRIGger:{A B}:BUS:B<x>:ETHERnet:IPHeader:PROTOcol:VALue?</pre>
Related Commands	<pre>TRIGger:{A B}:BUS:B<x>:ETHERnet:CONDition</pre>

TRIGger:{A|B}:BUS:B<x>:ETHERnet:IPHeader:SOURceaddr:VALue

TRIGger:{A|B}:BUS:B<x>:ETHERnet:IPHeader:DESTinationaddr:VALue

Arguments <QString> is a quoted string of up to 8 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.

Examples TRIGGER:A:BUS:B1:ETHERNET:IPHEADER:PROTOCOL:VALUE "01010010" would set the value to be used in the trigger to 01010010.

TRIGger:A:BUS:B1:ETHERnet:IPHeader:PROTOCOL:VALue? might return "XXXXXXXX".

TRIGger:{A|B}:BUS:B<x>:ETHERnet:IPHeader:SOURceaddr:VALue

When the Ethernet trigger condition is set to **IPHeader**, this command specifies the value of the 32-bit source address that is to be used in the trigger (along with the destination address and protocol value). The IP source address is a standard IP address such as 192.168.0.1. The default is all X's (don't care). The bus number is specified by <x>.

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:ETHERnet:IPHeader:SOURceaddr:VALue
<QString>
TRIGger:{A|B}:BUS:B<x>:ETHERnet:IPHeader:SOURceaddr:VALue?

Related Commands TRIGger:{A|B}:BUS:B<x>:ETHERnet:CONDition
TRIGger:{A|B}:BUS:B<x>:ETHERnet:IPHeader:DESTinationaddr:VALue
TRIGger:{A|B}:BUS:B<x>:ETHERnet:IPHeader:PROTOCOL:VALue

Arguments <QString> is a quoted string of up to 32 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.

Examples `TRIGger:A:BUS:B1:ETHERnet:IPHeader:SOURceaddr:VALue`
`"00011001001000010110100000000001"` sets the IP source address to trigger on to 192.168.0.1.

`TRIGger:A:BUS:B1:ETHERnet:IPHeader:SOURceaddr:VALue` might return `"XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX01"`.

TRIGger:{A|B}:BUS:B<x>:ETHERnet:MAC:ADDReSS:DESTination:VALue

When the Ethernet trigger condition is set to `MACADDRESS`, this command specifies the 48-bit MAC destination address that is to be used in the trigger (along with the source address value). The default is all X's (don't care). The bus number is specified by `<x>`.

NOTE. *MAC Addresses are 48-bit values such as 08:00:11:1E:C9:AE hex.*

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Trigger

Syntax `TRIGger:{A|B}:BUS:B<x>:ETHERnet:MAC:ADDReSS:DESTination:VALue <QString>`
`TRIGger:{A|B}:BUS:B<x>:ETHERnet:MAC:ADDReSS:DESTination:VALue?`

Related Commands [TRIGger:{A|B}:BUS:B<x>:ETHERnet:CONDition](#)
[TRIGger:{A|B}:BUS:B<x>:ETHERnet:MAC:ADDReSS:SOURce:VALue](#)

Arguments `<QString>` is a quoted string of up to 48 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.

Examples `TRIGger:A:BUS:B1:ETHERnet:MAC:ADDReSS:DESTination:VALue`
`"XXXXXXXXXX010101111111000000011110101010101000"` would set the MAC destination address to trigger on to XX:35:FC:07:AA:C8 hex.

`TRIGger:A:BUS:B1:ETHERnet:MAC:ADDReSS:DESTination:VALue?`
might return `"XXXXXXXXXX01010111111100000001111010101010101000"`.

TRIGger:{A|B}:BUS:B<x>:ETHERnet:MAC:ADDReSS:SOUrce:VALue

When the Ethernet trigger condition is set to **MACADDRESS**, this command specifies the 48-bit MAC source address value that is to be used in the trigger (along with the destination address value). The default is all X's (don't care). The bus number is specified by <x>.

NOTE. MAC Addresses are 48-bit values such as 08:00:11:1E:C9:AE hex.

Conditions	Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:ETHERnet:MAC:ADDReSS:SOUrce:VALue <QString> TRIGger:{A B}:BUS:B<x>:ETHERnet:MAC:ADDReSS:SOUrce:VALue?
Related Commands	TRIGger:{A B}:BUS:B<x>:ETHERnet:CONDition TRIGger:{A B}:BUS:B<x>:ETHERnet:MAC:ADDReSS:DESTination:VALue
Arguments	<QString> is a quoted string of up to 48 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.
Examples	TRIGger:A:BUS:B1:ETHERnet:MAC:ADDReSS:SOUrce:VALue "XXXXXXXXXX010101111111000000011110101010101000" would set the MAC destination address to trigger on to XX:35:FC:07:AA:C8 hex. TRIGger:A:BUS:B1:ETHERnet:MAC:ADDReSS:SOUrce:VALue? might return "XXXXXXXXXX01010111111100000001111010101010101000".

TRIGger:{A|B}:BUS:B<x>:ETHERnet:MAC:LENGth:HIVALue

When the Ethernet trigger condition is set to **MACLENGTH**, and the qualifier is set to **INrange** or **OUTrange**, this command specifies the upper data value of the range. (Use the command [TRIGger:{A|B}:BUS:B<x>:ETHERnet:MAC:LENGth:VALue](#) to specify the lower limit of the range.) The default is all X's (don't care). The bus number is specified by <x>.

Conditions	Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:ETHERnet:MAC:LENGth:HIVALue <QString> TRIGger:{A B}:BUS:B<x>:ETHERnet:MAC:LENGth:HIVALue?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:ETHERnet:CONDition TRIGger:{A B}:BUS:B<x>:ETHERnet:MAC:LENGth:VALue
Arguments	<QString> is a quoted string of up to 16 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.
Examples	<pre>TRIGger:A:BUS:B1:ETHERnet:MAC:LENGth:HIVALue "XXXXXXXX00001000" sets the upper limit of the range to the hexadecimal value XX08 (when the trigger condition is set to MACLENGth, and the qualifier is set to INrange or OUTrange) . TRIGGER:A:BUS:B1:ETHERNET:MAC:LENGTH:HIVALUE? might return "XXXXXXXX00001000".</pre>

TRIGger:{A|B}:BUS:B<x>:ETHERnet:MAC:LENGth:VALue

When the Ethernet trigger condition is set to MACLENGth, and the qualifier is set to LESSthan, MOREthan, EQUAL, UNEQUAL, LESSEQUAL or MOREEQUAL, this command specifies the 16-bit value to trigger on. When the qualifier is set to INrange or OUTrange, this command specifies the lower limit of the range. (Use the command [TRIGger:{A|B}:BUS:B<x>:ETHERnet:MAC:LENGth:HIVALue](#) to set the upper limit of the range.) The default is all X's (don't care). The bus number is specified by <x>.

Conditions	Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:ETHERnet:MAC:LENGth:VALue <QString> TRIGger:{A B}:BUS:B<x>:ETHERnet:MAC:LENGth:VALue?</pre>

Related Commands	TRIGger:{A B}:BUS:B<x>:ETHERnet:CONDition TRIGger:{A B}:BUS:B<x>:ETHERnet:MAC:LENGth:HIVALue
Arguments	<QString> is a quoted string of up to 16 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.
Examples	TRIGger:A:BUS:B1:ETHERnet:MAC:LENGth:VALue "xxxxxxxx00001000" sets the MAC length/type value to trigger on the hexadecimal value XX08, assuming the qualifier is set to LESSthan, MOREthan, EQUAL, UNEQUAL, LESSEQUAL or MOREEQUAL. TRIGger:A:BUS:B1:ETHERnet:MAC:LENGth:VALue "xxxxxxxx00001000" sets the lower limit of the range to the hexadecimal value XX08, assuming the qualifier is set to INrange or OUTrange.

TRIGger:{A|B}:BUS:B<x>:ETHERnet:QTAG:VALue

When the Ethernet trigger condition is set to QTAG, this command specifies the 32-bit Q-Tag value to trigger on. The default is all X's (don't care). The bus number is specified by <x>.

Conditions	Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:ETHERnet:QTAG:VALue <QString> TRIGger:{A B}:BUS:B<x>:ETHERnet:QTAG:VALue?
Related Commands	TRIGger:{A B}:BUS:B<x>:ETHERnet:CONDition
Arguments	<QString> is a quoted string of up to 32 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.
Examples	TRIGGER:A:BUS:B1:ETHERNET:QTAG:VALUE "xxxxxxxxxxxxxxxxxxxx010010001010" would specify to trigger on the Q-Tag value of hexadecimal XXXXX48A.

TRIGGER:A:BUS:B1:ETHERNET:QTAG:VALUE? might return
"XXXXXXXXXXXXXXXXXXXX010010001010".

TRIGger:{A|B}:BUS:B<x>:ETHERnet:TCPHeader:ACKnum:VALue

When the Ethernet trigger condition is set to TCPHeader, this command specifies the 32-bit acknowledgement number that is to be used in the trigger (along with the destination and source port addresses and the sequence number). The default is all X's (don't care). The bus number is specified by <x>.

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:ETHERnet:TCPHeader:ACKnum:VALue
<QString>
TRIGger:{A|B}:BUS:B<x>:ETHERnet:TCPHeader:ACKnum:VALue?

Related Commands [TRIGger:{A|B}:BUS:B<x>:ETHERnet:CONDition](#)
[TRIGger:{A|B}:BUS:B<x>:ETHERnet:TCPHeader:DESTinationport:VALue](#)
[TRIGger:{A|B}:BUS:B<x>:ETHERnet:TCPHeader:SOURceport:VALue](#)
[TRIGger:{A|B}:BUS:B<x>:ETHERnet:TCPHeader:SEQnum:VALue](#)

Arguments <QString> is a quoted string of up to 32 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.

Examples TRIGger:A:BUS:B1:ETHERnet:TCPHeader:ACKnum:VALue
"XXXXXXXXXXXXXXXXXXXX00001000" sets the acknowledgement number to be used in the trigger to hexadecimal XXXXXX08.

TRIGger:A:BUS:B1:ETHERnet:TCPHeader:ACKnum:VALue? might
return :TRIGGER:A:BUS:B1:ETHERNET:TCPHEADER:ACKNUM:VALUE
"XXXXXXXXXXXXXXXXXXXX00001000" indicating the value is set to
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX.

TRIGger:{A|B}:BUS:B<x>:ETHERnet:TCPHeader:DESTinationport:VALue

When the Ethernet trigger condition is set **TCPHeader**, this command specifies the 16-bit destination port address value that is to be used in the trigger (along with the acknowledgement value, source port address and the sequence number). The default is all X's (don't care). The bus number is specified by <x>.

Conditions Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:ETHERnet:TCPHeader:DESTinationport:VALue <QString>
TRIGger:{A|B}:BUS:B<x>:ETHERnet:TCPHeader:DESTinationport:VALue?

Related Commands [TRIGger:{A|B}:BUS:B<x>:ETHERnet:CONDition](#)
[TRIGger:{A|B}:BUS:B<x>:ETHERnet:TCPHeader:ACKnum:VALue](#)
[TRIGger:{A|B}:BUS:B<x>:ETHERnet:TCPHeader:SOURceport:VALue](#)
[TRIGger:{A|B}:BUS:B<x>:ETHERnet:TCPHeader:SEQnum:VALue](#)

Arguments <QString> is a quoted string of up to 16 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.

Examples TRIGger:A:BUS:B1:ETHERnet:TCPHeader:DESTinationport:VALue "XXXXXXXX00100010" would set the destination port address value that is to be used in the trigger to hexadecimal XX22.

TRIGger:A:BUS:B1:ETHERnet:TCPHeader:DESTinationport:VALue? might return "XXXXXXXXXXXXXXXXXX".

TRIGger:{A|B}:BUS:B<x>:ETHERnet:TCPHeader:SEQnum:VALue

When the Ethernet trigger condition is set to **TCPHeader**, this command specifies the 32-bit sequence number that is to be used in the trigger (along with the destination and source port addresses and the acknowledgement value). The default is all X's (don't care). The bus number is specified by <x>.

Conditions	Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger: {A B}:BUS:B<x>:ETHERnet:TCPHeader:SEQnum:VALue <QString> TRIGger: {A B}:BUS:B<x>:ETHERnet:TCPHeader:SEQnum:VALue?</pre>
Related Commands	<pre>TRIGger: {A B}:BUS:B<x>:ETHERnet:CONDition TRIGger: {A B}:BUS:B<x>:ETHERnet:TCPHeader:ACKnum:VALue TRIGger: {A B}:BUS:B<x>:ETHERnet:TCPHeader:SOURceport:VALue TRIGger: {A B}:BUS:B<x>:ETHERnet:TCPHeader:DESTinationport:VALue</pre>
Arguments	<QString> is a quoted string of up to 32 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.
Examples	<pre>TRIGger:A:BUS:B1:ETHERnet:TCPHeader:SEQnum:VALue "XXXXXXXXXXXXXXXXXXXX000100010001" would set the sequence number that is to be used in the trigger to hexadecimal XXXXX111. TRIGger:A:BUS:B1:ETHERnet:TCPHeader:SEQnum:VALue? might return "XXXXXXXXXXXXXXXXXXXX0010010100".</pre>

TRIGger:{A|B}:BUS:B<x>:ETHERnet:TCPHeader:SOURceport:VALue

When the Ethernet trigger condition is set to TCPHeader, this command specifies the 16-bit source port address that is to be used in the trigger (along with the destination port address, the sequence number and the acknowledgement number). The default is all X's (don't care). The bus number is specified by <x>.

Conditions	Requires option 5-SRENET or SUP5-SRENET Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger: {A B}:BUS:B<x>:ETHERnet:TCPHeader:SOURceport:VALue <QString> TRIGger: {A B}:BUS:B<x>:ETHERnet:TCPHeader:SOURceport:VALue?</pre>

Related Commands	TRIGger:{A B}:BUS:B<x>:ETHERnet:CONDition TRIGger:{A B}:BUS:B<x>:ETHERnet:TCPHeader:ACKnum:VALue TRIGger:{A B}:BUS:B<x>:ETHERnet:TCPHeader:SEQnum:VALue TRIGger:{A B}:BUS:B<x>:ETHERnet:TCPHeader:DESTinationport:VALue
Arguments	<QString> is a quoted string of up to 16 characters where the allowable characters are 0, 1, and X. The bits specified in the quoted string replace the least significant bits, leaving any unspecified upper bits unchanged.
Examples	TRIGger:A:BUS:B1:ETHERnet:TCPHeader:Sourceport:VALue "XXXX000010100110" would set the source port address that is to be used in the trigger to hexadecimal X0A6. TRIGger:A:BUS:B1:ETHERnet:TCPHeader:Sourceport:VALue? might return "XXXXX01001010110".

TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition

This command specifies the condition to use when triggering on a FlexRay bus signal (start of frame, frame type, ID, cycle count, header, data, ID and data, EOF, error). The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:FLEXray:CONDition {SOF FRAMEType IDentifier CYCLEcount HEADer DATA IDANDDATA EOF ERROR} TRIGger:{A B}:BUS:B<x>:FLEXray:CONDition?
Arguments	SOF sets the trigger condition to start of frame. FRAMEType sets the trigger condition to frame type. IDentifier sets the trigger condition to identifier. CYCLEcount sets the trigger condition to cycle count. HEADer sets the trigger condition to header.

DATA sets the trigger condition to data.

IDANDDATA sets the trigger condition to id and data.

EOF sets the trigger condition to end of frame.

ERROR sets the trigger condition to error.

Examples TRIGGER:A:BUS:B1:FLEXRAY:CONDITION FRAMETYPE sets the FlexRay condition to frame type.

TRIGGER:A:BUS:B1:FLEXRAY:CONDITION? might return
TRIGGER:A:BUS:B1:FLEXRAY:CONDITION SOF indicating the FlexRay condition is start of frame.

TRIGger:{A|B}:BUS:B<x>:FLEXray:CYCLEcount:HIVALue

This command specifies the high value when triggering on a FlexRay bus cycle count field. (Use TRIGger:{A|B}:BUS:B<x>:FLEXray:CYCLEcount:VALue to set the low value.) The trigger condition must be set to CYCLEcount (using TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition). The bus number is specified by <x>.

Conditions Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:FLEXray:CYCLEcount:HIVALue <QString>
TRIGger:{A|B}:BUS:B<x>:FLEXray:CYCLEcount:HIVALue?

Arguments <QString> is a quoted string that is the cycle count high value.

Examples TRIGGER:A:BUS:B1:FLEXRAY:CYCLECOUNT:HIVALUE "110010" sets the cycle count high value to 110010.

TRIGGER:A:BUS:B1:FLEXRAY:CYCLECOUNT:HIVALUE? might return
TRIGGER:A:BUS:B1:FLEXRAY:CYCLECOUNT:HIVALUE "XXXXXX" indicating the cycle count high value is don't care.

TRIGger:{A|B}:BUS:B<x>:FLEXray:CYCLEcount:QUALifier

This command specifies the qualifier (<, >, =, <=, >=, not =, in range, out of range) to use when triggering on the FlexRay bus cycle count field. The trigger condition must be set to CYCLEcount (using [TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by <x>.

Conditions Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:FLEXray:CYCLEcount:QUALifier
{LESSthan|MOREthan|Equal|UNEQual|LESSEQual|MOREEQual|INrange|OUTrange}

TRIGger:{A|B}:BUS:B<x>:FLEXray:CYCLEcount:QUALifier?

Arguments LESSthan sets the cycle count qualifier to less than.

MOREthan sets the cycle count qualifier to more than.

Equal sets the cycle count qualifier to equal.

UNEQual sets the cycle count qualifier to not equal.

LESSEQual sets the cycle count qualifier to less than or equal.

MOREEQual sets the cycle count qualifier to greater than or equal.

INrange sets the cycle count qualifier to in range.

OUTrange sets the cycle count qualifier to out of range.

Examples TRIGGER:A:BUS:B1:FLEXRAY:CYCLECOUNT:QUALIFIER LESSTHAN sets the cycle count qualifier to LESSTHAN.

TRIGGER:A:BUS:B1:FLEXRAY:CYCLECOUNT:QUALIFIER? might return
TRIGGER:A:BUS:B1:FLEXRAY:CYCLECOUNT:QUALIFIER EQUAL indicating that the cycle count qualifier is set to EQUAL.

TRIGger:{A|B}:BUS:B<x>:FLEXray:CYCLEcount:VALue

This command specifies the low value when triggering on the FlexRay bus cycle count field. (Use [TRIGger:{A|B}:BUS:B<x>:FLEXray:CYCLEcount:HIVALue](#) to set the upper value.) The trigger condition must be set to CYCLEcount (using

[TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:FLEXray:CYCLEcount:VALue <QString> TRIGger:{A B}:BUS:B<x>:FLEXray:CYCLEcount:VALue?</pre>
Arguments	<QString> is a quoted binary data string that represents the cycle count low value.
Examples	TRIGGER:A:BUS:B1:FLEXRAY:CYCLECOUNT:VALUE "11001101" sets the cycle count value to 11001101. TRIGGER:A:BUS:B1:FLEXRAY:CYCLECOUNT:VALUE? might return TRIGGER:A:BUS:B1:FLEXRAY:CYCLECOUNT:VALUE "XXXXXX" indicating the cycle count value is don't care.

TRIGger:{A|B}:BUS:B<x>:FLEXray:DATA:HIVALue

This command specifies the high value when triggering on the FlexRay bus data field. (Use [TRIGger:{A|B}:BUS:B<x>:FLEXray:DATA:VALue](#) to set the lower value.) The trigger condition needs to be set to ID or IDANDDATA (using [TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:FLEXray:DATA:HIVALue <QString> TRIGger:{A B}:BUS:B<x>:FLEXray:DATA:HIVALue?</pre>
Arguments	<QString> is a quoted string that is the binary data high value.

Examples

```

TRIGGER:A:BUS:B1:FLEXRAY:DATA:HIVALUE
"11001101XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXXXX1" sets the binary data string high value to
"11001101XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXXXX1".

TRIGGER:A:BUS:B1:FLEXRAY:DATA:HIVALUE? might
return TRIGGER:A:BUS:B1:FLEXRAY:DATA:HIVALUE
"XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXXXX" indicating the binary data string high value is don't care.

```

TRIGger:{A|B}:BUS:B<x>:FLEXray:DATA:OFFSet

This command specifies the offset of the data string, in bytes, when triggering on the FlexRay bus data field. The trigger condition needs to be set to ID or IDANDDATA (using [TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by <x>.

Conditions Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group Trigger

Syntax

```

TRIGger:{A|B}:BUS:B<x>:FLEXray:DATA:OFFSet <NR1>
TRIGger:{A|B}:BUS:B<x>:FLEXray:DATA:OFFSet?

```

Arguments <NR1> is the offset of the data string in bytes. A byte offset of -1 signifies “don't care”, and no byte offset is used. The instrument will trigger on or match any byte value that fits.

Examples

```

TRIGGER:A:BUS:B1:FLEXRAY:DATA:OFFSET 1 sets the offset to 1.

TRIGGER:A:BUS:B1:FLEXRAY:DATA:OFFSET? might return
TRIGGER:A:BUS:B1:FLEXRAY:DATA:OFFSET 0 indicating that a data offset
of 0.

```

TRIGger:{A|B}:BUS:B<x>:FLEXray:DATA:QUALifier

This command specifies the qualifier (<, >, =, <=, >=, not =, in range, out of range) to use when triggering on the FlexRay bus data field. The trigger condition needs to be set to ID or IDANDDATA (using [TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:FLEXray:DATA:QUALifier {LESSthan MOREthan EQua1 UNEQua1 LESSEQua1 MOREEQua1 INrange OUTrange} TRIGger:{A B}:BUS:B<x>:FLEXray:DATA:QUALifier?
Arguments	LESSthan sets the data qualifier to less than. MOREthan sets the data qualifier to greater than. EQua1 sets the data qualifier to equal. UNEQua1 sets the data qualifier to not equal. LESSEQua1 sets the data qualifier to less than or equal. MOREEQua1 sets the data qualifier to greater than or equal. INrange sets the data qualifier to in range. OUTrange sets the data qualifier to out of range.
Examples	TRIGGER:A:BUS:B1:FLEXRAY:DATA:QUALIFIER LESSTHAN sets the data qualifier to LESSTHAN. TRIGGER:A:BUS:B1:FLEXRAY:DATA:QUALIFIER? might return TRIGGER:A:BUS:B1:FLEXRAY:DATA:QUALIFIER EQUAL indicating the data qualifier is EQUAL.

TRIGger:{A|B}:BUS:B<x>:FLEXray:DATA:SIZE

This command specifies the length of the data string, in bytes, when triggering on the FlexRay bus data field. The trigger condition needs to be set to ID or IDANDDATA (using [TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger

Syntax	<pre>TRIGger:{A B}:BUS:B<x>:FLEXray:DATA:SIZE <NR1> TRIGger:{A B}:BUS:B<x>:FLEXray:DATA:SIZE?</pre>
Arguments	<NR1> is the FlexRay data string length, in bytes.
Examples	TRIGGER:A:BUS:B1:FLEXRAY:DATA:SIZE 8 sets the data string size to 8 bytes. TRIGGER:A:BUS:B1:FLEXRAY:DATA:SIZE? might return TRIGGER:A:BUS:B1:FLEXRAY:DATA:SIZE 1 indicating the data size is 1 byte.

TRIGger:{A|B}:BUS:B<x>:FLEXray:DATA:VALue

This command specifies the low value when triggering on the FlexRay bus data field. (Use [TRIGger:{A|B}:BUS:B<x>:FLEXray:DATA:HIVALue](#) to set the upper value.) The trigger condition needs to be set to ID or IDANDDATA (using [TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:FLEXray:DATA:VALue <QString> TRIGger:{A B}:BUS:B<x>:FLEXray:DATA:VALue?</pre>
Arguments	<QString> is a quoted string.
Examples	TRIGGER:A:BUS:B1:FLEXRAY:DATA:VALUE "11001101" sets the FlexRay data value for triggering to 11001101. TRIGGER:A:BUS:B1:FLEXRAY:DATA:VALUE? might return TRIGGER:A:BUS:B1:FLEXRAY:DATA:VALUE "XXXXXXXX" indicating the FlexRay data value is don't care.

TRIGger:{A|B}:BUS:B<x>:FLEXray:EOFTYPE

This command specifies the end of file type (static, dynamic or any) when triggering on the FlexRay bus EOF field. The trigger condition needs to be set

to EOF (using [TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:FLEXray:EOFTYPE {STATic DYNAMiC ANY} TRIGger:{A B}:BUS:B<x>:FLEXray:EOFTYPE?</pre>
Arguments	STATic specifies triggering on the STATIC end of file type. DYNAMiC specifies triggering on the DYNAMIC end of file type. ANY specifies triggering on a STATIC or DYNAMIC end of file type.
Examples	<pre>TRIGGER:A:BUS:B1:FLEXRAY:EOFTYPE ANY sets the FlexRay end of file type to ANY. TRIGGER:A:BUS:B1:FLEXRAY:EOFTYPE? might return TRIGGER:A:BUS:B1:FLEXRAY:EOFTYPE STATIC indicating the FlexRay end of file type is STATIC</pre>

TRIGger:{A|B}:BUS:B<x>:FLEXray:ERRTYPE

This command specifies the error type when triggering on the FlexRay bus signal. The trigger condition needs to be set to ERROR (using [TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:FLEXray:ERRTYPE {CRCHeader CRCTrailer SYNCFRAME STARTUPnosync NULLFRStatic NULLFRDynamic} TRIGger:{A B}:BUS:B<x>:FLEXray:ERRTYPE?</pre>

Arguments	<code>CRCHeader</code> sets the error type to <code>CRCHeader</code>. <code>CRCTrailer</code> sets the error type to <code>CRCTrailer</code>. <code>SYNCFrame</code> sets the error type to <code>SYNCFrame</code>. <code>STARTupnosync</code> sets the error type to <code>STARTupnosync</code>. <code>NULLFRStatic</code> sets the error type to <code>NULLFRStatic</code>. <code>NULLFRDynamic</code> sets the error type to <code>NULLFRDynamic</code>.
Examples	<code>TRIGGER:A:BUS:B1:FLEXRAY:ERRTYPE SYNCFRAME</code> sets the trigger type is <code>SYNCFRAME</code>. <code>TRIGGER:A:BUS:B1:FLEXRAY:ERRTYPE?</code> might return <code>TRIGGER:A:BUS:B1:FLEXRAY:ERRTYPE CRCHEADER</code> indicating the FlexRay trigger type is <code>CRCHeader</code>.

TRIGger:{A|B}:BUS:B<x>:FLEXray:FRAMEID:HIVALue

This command specifies the high value when triggering on the FlexRay bus frame ID field. (Use [TRIGger:{A|B}:BUS:B<x>:FLEXray:FRAMEID:VALue](#) to set the low value.) The trigger condition needs to be set to `Identifier` (using [TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by `<x>`.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	<code>TRIGger:{A B}:BUS:B<x>:FLEXray:FRAMEID:HIVALue <QString></code> <code>TRIGger:{A B}:BUS:B<x>:FLEXray:FRAMEID:HIVALue?</code>
Arguments	<code><QString></code> is a quoted string that is the binary frame ID high value.
Examples	<code>TRIGGER:A:BUS:B1:FLEXRAY:FRAMEID:HIVALUE "11001100101"</code> sets the frame ID high value to <code>11001100101</code>. <code>TRIGGER:A:BUS:B1:FLEXRAY:FRAMEID:HIVALUE?</code> might return <code>TRIGGER:A:BUS:B1:FLEXRAY:FRAMEID:HIVALUE "XXXXXXXXXXXX"</code> indicating the frame ID high value is "don't care".

TRIGger:{A|B}:BUS:B<x>:FLEXray:FRAMEID:QUALifier

This command specifies the qualifier to use when triggering on the FlexRay bus frame ID field. The trigger condition needs to be set to Identifier (using [TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:FLEXray:FRAMEID:QUALifier {LESSthan MOREthan EQUAL UNEQUAL LESSEQUAL MOREEQUAL INrange OUTrange} TRIGger:{A B}:BUS:B<x>:FLEXray:FRAMEID:QUALifier?
Arguments	LESSthan sets the frame ID qualifier to less than. MOREthan sets the frame ID qualifier to greater than. EQUAL sets the frame ID qualifier to equal. UNEQUAL sets the frame ID qualifier to not equal. LESSEQUAL sets the frame ID qualifier to less than or equal. MOREEQUAL sets the frame ID qualifier to greater than or equal. INrange sets the frame ID qualifier to in range. OUTrange sets the frame ID qualifier to out of range.
Examples	TRIGGER:A:BUS:B1:FLEXRAY:FRAMEID:QUALIFIER LESSTHAN sets the frame ID qualifier to less than. TRIGGER:A:BUS:B1:FLEXRAY:FRAMEID:QUALIFIER? might return TRIGGER:A:BUS:B1:FLEXRAY:FRAMEID:QUALIFIER EQUAL indicating the frame ID qualifier is set to equal.

TRIGger:{A|B}:BUS:B<x>:FLEXray:FRAMEID:VALue

This command specifies the low value when triggering on the FlexRay bus frame ID field. (Use [TRIGger:{A|B}:BUS:B<x>:FLEXray:FRAMEID:HIVALue](#) to set the high value.) The trigger condition needs to be set to Identifier (using [TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:FLEXray:FRAMEID:VALUE <QString> TRIGger:{A B}:BUS:B<x>:FLEXray:FRAMEID:VALUE?</pre>
Arguments	<QString> is a quoted string that is the FlexRay frame ID low value.
Examples	TRIGGER:A:BUS:B1:FLEXRAY:FRAMEID:VALUE "11001100101" sets the frame ID value to 11001100101. TRIGGER:A:BUS:B1:FLEXRAY:FRAMEID:VALUE? might return TRIGGER:A:BUS:B1:FLEXRAY:FRAMEID:VALUE "XXXXXXXXXXXX" indicating the frame ID value is don't care.

TRIGger:{A|B}:BUS:B<x>:FLEXray:FRAMEType

This command specifies the frame type (normal, payload, null, sync or startup) when triggering on the FlexRay bus signal. The trigger condition needs to be set to FRAMEType (using [TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:FLEXray:FRAMEType {NORMa1 PAYLoad NULL SYNC STARTUp} TRIGger:{A B}:BUS:B<x>:FLEXray:FRAMEType?</pre>
Arguments	NORMa1 specifies the normal frame type. PAYLoad specifies the payload frame type. NULL specifies the null frame type. SYNC specifies the sync frame type. STARTUp specifies the startup frame type.

Examples `TRIGGER:A:BUS:B1:FLEXRAY:FRAMETYPE PAYLOAD` sets the frame type to payload.

`TRIGGER:A:BUS:B1:FLEXRAY:FRAMETYPE?` might return
`TRIGGER:A:BUS:B1:FLEXRAY:FRAMETYPE NORMAL` indicating the frame type is set to normal.

TRIGger:{A|B}:BUS:B<x>:FLEXray:HEADER:CRC

This command specifies the CRC portion of the binary header string when triggering on the FlexRay bus signal. The trigger condition needs to be set to HEADER (using [TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by <x>.

Conditions Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group Trigger

Syntax `TRIGger:{A|B}:BUS:B<x>:FLEXray:HEADER:CRC <QString>`
`TRIGger:{A|B}:BUS:B<x>:FLEXray:HEADER:CRC?`

Arguments <QString> is a quoted string that is the CRC portion of the binary header string.

Examples `TRIGGER:A:BUS:B1:FLEXRAY:HEADER:CRC "11001100101"` sets the CRC portion of the binary header string to 11001100101.

`TRIGGER:A:BUS:B1:FLEXRAY:HEADER:CRC?` might return
`TRIGGER:A:BUS:B1:FLEXRAY:HEADER:CRC "XXXXXXXXXXXX"` indicating the CRC portion of the binary header string is don't care.

TRIGger:{A|B}:BUS:B<x>:FLEXray:HEADER:CYCLEcount

This command specifies the cycle count portion of the binary header string when triggering on the FlexRay bus header. The trigger condition needs to be set to HEADER (using [TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by <x>.

Conditions Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:FLEXray:HEADER:CYCLEcount <QString> TRIGger:{A B}:BUS:B<x>:FLEXray:HEADER:CYCLEcount?</pre>
Arguments	<QString> is a quoted string that is the cycle count portion of the binary header string.
Examples	TRIGGER:A:BUS:B1:FLEXRAY:HEADER:CYCLECOUNT "110010" sets the cycle count to 110010. TRIGGER:A:BUS:B1:FLEXRAY:HEADER:CYCLECOUNT? might return TRIGGER:A:BUS:B1:FLEXRAY:HEADER:CYCLECOUNT "XXXXXX" indicating the cycle count is don't care.

TRIGger:{A|B}:BUS:B<x>:FLEXray:HEADER:FRAMEID

This command specifies the frame ID portion of the binary header string when triggering on the FlexRay bus header. The trigger condition needs to be set to HEADER (using [TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:FLEXray:HEADER:FRAMEID <QString> TRIGger:{A B}:BUS:B<x>:FLEXray:HEADER:FRAMEID?</pre>
Arguments	<QString> is a quoted string that represents the frame ID portion of the binary header string.
Examples	TRIGGER:A:BUS:B1:FLEXRAY:HEADER:FRAMEID "11001100101" sets the frame ID portion of the binary header string to 11001100101. TRIGGER:A:BUS:B1:FLEXRAY:HEADER:FRAMEID? might return TRIGGER:A:BUS:B1:FLEXRAY:HEADER:FRAMEID "XXXXXXXXXXXX" indicating the frame ID portion of the binary header string is "don't care".

TRIGger:{A|B}:BUS:B<x>:FLEXray:HEADER:INDBits

This command specifies the indicator bits portion of the binary header string when triggering on the FlexRay bus header. The trigger condition needs to be set to HEADER (using [TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by <x>.

Conditions Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:FLEXray:HEADER:INDBits <QString>
TRIGger:{A|B}:BUS:B<x>:FLEXray:HEADER:INDBits?

Arguments <QString> is a quoted string that is the indicator bits portion of the binary header string.

Examples TRIGGER:A:BUS:B1:FLEXRAY:HEADER:INDBITS "11001" sets the indicator bits portion of the header string to 11001.

TRIGGER:A:BUS:B1:FLEXRAY:HEADER:INDBITS? might return
TRIGGER:A:BUS:B1:FLEXRAY:HEADER:INDBITS "XXXXX" indicating that the indicator bits portion of the header string are "don't cares".

TRIGger:{A|B}:BUS:B<x>:FLEXray:HEADER:PAYLength

This command specifies the payload length portion of the binary header string when triggering on the FlexRay bus header. The trigger condition needs to be set to HEADER (using [TRIGger:{A|B}:BUS:B<x>:FLEXray:CONDition](#)). The bus number is specified by <x>.

Conditions Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:FLEXray:HEADER:PAYLength <QString>
TRIGger:{A|B}:BUS:B<x>:FLEXray:HEADER:PAYLength?

Arguments	<QString> is the length of the payload portion of the Binary header string.
Examples	TRIGGER:A:BUS:B1:FLEXRAY:HEADER:PAYLENGTH "1100101" sets the FlexRay header paylength to 1100101. TRIGGER:A:BUS:B1:FLEXRAY:HEADER:PAYLENGTH? might return TRIGGER:A:BUS:B1:FLEXRAY:HEADER:PAYLENGTH "XXXXXXX" indicating the FlexRay header paylength is don't care.

TRIGger:{A|B}:BUS:B<x>:I2C:ADDRess:MODE

This command specifies the I²C address mode to 7 or 10-bit. The bus number is specified by <x>.

Conditions	Requires option 5-SREMBD or SUP5-SREMBD Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:I2C:ADDRess:MODE {ADDR7 ADDR10} TRIGger:{A B}:BUS:B<x>:I2C:ADDRess:MODE?
Arguments	ADDR7 specifies the 7-bit I²C address mode. ADDR10 specifies the 10-bit I²C address mode.
Examples	TRIGGER:A:BUS:B1:I2C:ADDRESS:MODE ADDR10 sets the I2C address mode to 10-bit. TRIGGER:A:BUS:B1:I2C:ADDRESS:MODE? might return :TRIGGER:A:BUS:B1:I2C:ADDRESS:MODE ADDR7 indicating the address mode is set to the 7-bit mode.

TRIGger:{A|B}:BUS:B<x>:I2C:ADDRess:VALue

This command specifies the binary address string used for the I²C trigger if the trigger condition is ADDRESS or ADDRANDDATA. The bus number is specified by <x>.

Conditions	Requires option 5-SREMBD or SUP5-SREMBD Triggering and Analysis application.
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Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:I2C:ADDReSS:VALue <QString> TRIGger:{A B}:BUS:B<x>:I2C:ADDReSS:VALue?</pre>
Arguments	<QString> is up to 7 or 10-bits depending on the address mode that specifies the address. The only allowed characters in the QString are 0, 1, and X.
Examples	TRIGGER:A:BUS:B1:I2C:ADDRESS:VALUE "1011" sets the I²C address value to XXX1011. TRIGGER:A:BUS:B1:I2C:ADDRESS:VALUE? might return :TRIGGER:A:BUS:B1:I2C:ADDRESS:VALUE "XXXXXXX" indicating the address value is set to XXXXXXXX.

TRIGger:{A|B}:BUS:B<x>:I2C:CONDition

This command specifies the trigger condition for an I²C trigger. The bus number is specified by <x>.

Conditions	Requires option 5-SREMBD or SUP5-SREMBD Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:I2C:CONDition {START STOP REPEATstart ACKMISS ADDRESS DATA ADDRANDDATA} TRIGger:{A B}:BUS:B<x>:I2C:CONDition?</pre>
Arguments	START specifies a search based on start condition. STOP specifies a search based on stop condition. REPEATstart specifies a search based on repeat of start condition. ACKMISS specifies a search based on missing acknowledgement condition. ADDRESS specifies a search based on address. DATA specifies a search based on data. ADDRANDDATA specifies a search based on address and data.

Examples `TRIGGER:A:BUS:B1:I2C:CONDITION START` specifies start as the I²C trigger condition.

`TRIGGER:A:BUS:B1:I2C:CONDITION?` might return
`:TRIGGER:A:BUS:B1:I2C:CONDITION START` indicating the condition is set to the start condition.

TRIGger:{A|B}:BUS:B<x>:I2C:DATA:DIRection

This command specifies the I²C trigger type to be valid on a Read, Write, or Either condition. Read or write is indicated by the R/W bit in the I²C protocol. The bus number is specified by <x>.

Conditions Requires option 5-SREMBD or SUP5-SREMBD Triggering and Analysis application.

Group Trigger

Syntax `TRIGger:{A|B}:BUS:B<x>:I2C:DATA:DIRection`
`{READ|WRITE|NOCARE}`
`TRIGger:{A|B}:BUS:B<x>:I2C:DATA:DIRection?`

Arguments `READ` specifies read as the data direction.
`WRITE` specifies write as the data direction.
`NOCARE` specifies either as the data direction.

Examples `TRIGGER:A:BUS:B1:I2C:DATA:DIRECTION WRITE` specifies write as the I²C data direction.

`TRIGGER:A:BUS:B1:I2C:DATA:DIRECTION?` might return
`:TRIGGER:A:BUS:B1:I2C:DATA:DIRECTION NOCARE` indicating the data direction is either read or write.

TRIGger:{A|B}:BUS:B<x>:I2C:DATA:SIZE

This command specifies the length of the data string in bytes to be used for an I²C trigger if the trigger condition is DATA or ADDRANDDATA. Applies to bus <x>, where the bus number is specified by <x>.

Conditions Requires option 5-SREMBD or SUP5-SREMBD Triggering and Analysis application.

Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:I2C:DATa:SIZE <NR1> TRIGger:{A B}:BUS:B<x>:I2C:DATa:SIZE?</pre>
Arguments	<NR1> is the length of the data string in bytes.
Examples	TRIGger:A:BUS:B1:I2C:DATa:SIZE 1 sets the data size to 1 byte. TRIGger:A:BUS:B1:I2C:DATa:SIZE? might return :TRIGGER:A:BUS:B1:I2C:DATa:SIZE 1 indicating the size is set to 1 byte.

TRIGger:{A|B}:BUS:B<x>:I2C:DATa:VALue

This command specifies the binary data string used for I2C triggering if the trigger condition is DATA or ADDRANDDATA. The bus number is specified by <x>.

Conditions	Requires option 5-SREMBD or SUP5-SREMBD Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:I2C:DATa:VALue <QString> TRIGger:{A B}:BUS:B<x>:I2C:DATa:VALue?</pre>
Arguments	<QString> is the binary data string, where the number of bits is 8 times the number of bytes specified. The only allowed characters in the string are 0, 1, and X.
Examples	TRIGger:A:BUS:B1:I2C:DATa:VALue "11001101" sets the data value to 1100101. TRIGger:A:BUS:B1:I2C:DATa:VALue? might return :TRIGGER:A:BUS:B1:I2C:DATa:VALue "XXXXXXXX" indicating the data value is XXXXXXXX.

TRIGger:{A|B}:BUS:B<x>:LIN:CONDition

This command specifies the trigger condition for LIN. The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:LIN:CONDition {SYNCfield Identifier DATA IDANDDATA WAKEup SLEEP ERROR} TRIGger:{A B}:BUS:B<x>:LIN:CONDition?
Arguments	SYNCfield sets the LIN trigger condition to sync field. Identifier sets the LIN trigger condition to identifier. DATA sets the LIN trigger condition to data. IDANDDATA sets the LIN trigger condition to id and data. WAKEup sets the LIN trigger condition to wake up. SLEEP sets the LIN trigger condition to sleep. ERROR sets the LIN trigger condition to error.
Examples	TRIGGER:A:BUS:B1:LIN:CONDITION ERROR sets the LIN trigger condition to error. TRIGGER:A:BUS:B1:LIN:CONDITION? might return TRIGGER:A:BUS:B1:LIN:CONDITION SYNCFIELD indicating the LIN trigger condition is sync field.

TRIGger:{A|B}:BUS:B<x>:LIN:DATA:HIVALue

This command specifies the high data value string used for a LIN bus trigger when the trigger condition is DATA or IDANDDATA and the data qualifier is INRANGE or OUTRANGE. The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
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Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:LIN:DATA:HIVALue <QString> TRIGger:{A B}:BUS:B<x>:LIN:DATA:HIVALue?</pre>
Arguments	<QString> is a quoted string that is the binary data string used for LIN trigger if the trigger condition is ID or IDANDDATA.
Examples	TRIGGER:A:BUS:B1:LIN:DATA:HIVALUE "11001010" sets the high value to 11001010. TRIGGER:A:BUS:B1:LIN:DATA:HIVALUE? might return TRIGGER:A:BUS:B1:LIN:DATA:HIVALUE "XXXXXXXX" indicating the high value is don't care.

TRIGger:{A|B}:BUS:B<x>:LIN:DATA:QUALifier

This command specifies the LIN data qualifier. This only applies if the trigger condition is IDANDDATA or DATA. The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:LIN:DATA:QUALifier {LESSthan MOREthan Equal UNEQual LESSEQual MOREEQual INrange OUTrange} TRIGger:{A B}:BUS:B<x>:LIN:DATA:QUALifier?</pre>
Arguments	LESSthan sets the LIN data qualifier to less than. MOREthan sets the LIN data qualifier to greater than. Equal sets the LIN data qualifier to equal. UNEQual sets the LIN data qualifier to not equal. LESSEQual sets the LIN data qualifier to less than or equal. MOREEQual sets the LIN data qualifier to greater than or equal. INrange sets the LIN data qualifier to in range.

OUTRange sets the LIN data qualifier to out of range.

Examples TRIGGER:A:BUS:B1:LIN:DATA:QUALIFIER OUTRANGE sets the data qualifier to out of range.

TRIGGER:A:BUS:B1:LIN:DATA:QUALIFIER? might return
 TRIGGER:A:BUS:B1:LIN:DATA:QUALIFIER EQUAL indicating the data qualifier is set to equal.

TRIGger:{A|B}:BUS:B<x>:LIN:DATA:SIZE

This command specifies the length of the data string in bytes to be used for LIN trigger. The bus number is specified by <x>.

Conditions Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:LIN:DATA:SIZE <NR1>
 TRIGger:{A|B}:BUS:B<x>:LIN:DATA:SIZE?

Arguments <NR1> is the size of the data string in bytes.

Examples TRIGGER:A:BUS:B1:LIN:DATA:SIZE 8 sets the data size to 8 bytes.

TRIGGER:A:BUS:B1:LIN:DATA:SIZE? might return
 TRIGGER:A:BUS:B1:LIN:DATA:SIZE 1 indicating the data size is 1 byte.

TRIGger:{A|B}:BUS:B<x>:LIN:DATA:VALue

This command specifies the binary data string to be used for LIN trigger condition if trigger condition is ID or IDANDDATA. The bus number is specified by <x>.

Conditions Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.

Group Trigger

Syntax	<pre>TRIGger:{A B}:BUS:B<x>:LIN:DATA:VALue <QString> TRIGger:{A B}:BUS:B<x>:LIN:DATA:VALue?</pre>
Arguments	<QString> is a quoted string that is the LIN trigger data value.
Examples	TRIGGER:A:BUS:B1:LIN:DATA:VALUE "11001101" sets the data value to 11001101. TRIGGER:A:BUS:B1:LIN:DATA:VALUE? might return TRIGGER:A:BUS:B1:LIN:DATA:VALUE "XXXXXXXX" indicating the data value is don't care.

TRIGger:{A|B}:BUS:B<x>:LIN:ERRTYPE

This command specifies the error type be used for LIN trigger. The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:LIN:ERRTYPE {SYNC PARity Checksum} TRIGger:{A B}:BUS:B<x>:LIN:ERRTYPE?</pre>
Arguments	SYNC sets the LIN error type to SYNC. PARity sets the LIN error type to parity. Checksum sets the LIN error type to checksum.
Examples	TRIGGER:A:BUS:B1:LIN:ERRTYPE CHECKSUM sets the LIN error type to checksum. TRIGGER:A:BUS:B1:LIN:ERRTYPE? might return TRIGGER:A:BUS:B1:LIN:ERRTYPE SYNC indicating the LIN error type is SYNC.

TRIGger:{A|B}:BUS:B<x>:LIN:IDentifier:VALue

This command specifies the binary address string used for LIN bus trigger if the trigger condition is ID or IDANDDATA. The bus number is specified by <x>.

Conditions	Requires option 5-SRAUTO or SUP5-SRAUTO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:LIN:IDentifier:VALue <QString> TRIGger:{A B}:BUS:B<x>:LIN:IDentifier:VALue?
Arguments	<QString> is the binary address string used for LIN trigger if the trigger condition is ID or IDANDDATA.
Examples	TRIGGER:A:BUS:B1:LIN:IDENTIFIER:VALUE "110010" sets the identifier value to 110010. TRIGGER:A:BUS:B1:LIN:IDENTIFIER:VALUE? might return TRIGGER:A:BUS:B1:LIN:IDENTIFIER:VALUE "XXXXXX" indicating the identifier value is XXXXXX.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:COMMAND:ADDRess:HIVALue

This command sets or queries the high value when triggering on command word addresses for a MIL-STD-1553 bus. The bus number is specified by x. The trigger condition must be set to COMMAND, and the address qualifier must be INrange or OUTrange.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:MIL1553B:COMMAND:ADDRess:HIVALue <QString> TRIGger:{A B}:BUS:B<x>:MIL1553B:COMMAND:ADDRess:HIVALue?
Arguments	<QString> is the address value.
Examples	TRIGGER:A:BUS:B1:MIL1553B:COMMAND:ADDRESS:HIVALUE "X1000" sets the value to X1000. TRIGGER:A:BUS:B1:MIL1553B:COMMAND:ADDRESS:HIVALUE? might return "XXXXX", indicating that the value is XXXXX.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:COMMAND:ADDRess:QUALifier

This command sets or queries the qualifier to be used when triggering on command word addresses for a MIL-STD-1553 bus. The bus number is specified by x. The trigger condition must be set to COMMAND.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:MIL1553B:COMMAND:ADDRess:QUALifier {Equal|UNEQual|LESSthan|MOREthan|LESSEQual|MOREEQual|INrange|OUTrange}
TRIGger:{A|B}:BUS:B<x>:MIL1553B:COMMAND:ADDRess:QUALifier?

Arguments Arguments are the available address qualifiers.

Examples TRIGGER:A:BUS:B1:MIL1553B:COMMAND:ADDRESS:QUALIFIER LESSTHAN sets the address qualifier to less than.

TRIGGER:A:BUS:B1:MIL1553B:COMMAND:ADDRESS:QUALIFIER? might return EQUAL, indicating that the address qualifier is set to equal.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:COMMAND:ADDRess:VALue

This command sets or queries the low value when triggering on command word addresses for a MIL-STD-1553 bus. The bus number is specified by x. The trigger condition must be set to COMMAND

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:MIL1553B:COMMAND:ADDRess:VALue <QString>
TRIGger:{A|B}:BUS:B<x>:MIL1553B:COMMAND:ADDRess:VALue?

Arguments <QString> is the address value.

- Examples** `TRIGGER:A:BUS:B1:MIL1553B:COMMAND:ADDRESS:VALUE "X1000"` sets the value to X1000.
- `TRIGGER:A:BUS:B1:MIL1553B:COMMAND:ADDRESS:VALUE?` might return "XXXXX", indicating that the value is XXXXX.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:COMMAND:COUNT

This command sets or queries the value of the command word "word count" field for a MIL-STD-1553 bus to triggering on. The bus number is specified by x. The trigger condition must be set to COMMAND.

- Conditions** Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

- Group** Trigger

- Syntax** `TRIGger:{A|B}:BUS:B<x>:MIL1553B:COMMAND:COUNT <QString>`
`TRIGger:{A|B}:BUS:B<x>:MIL1553B:COMMAND:COUNT?`

- Arguments** <QString> is the word count value.

- Examples** `TRIGGER:A:BUS:B1:MIL1553B:COMMAND:COUNT "X1000"` sets the value to X1000.
- `TRIGGER:A:BUS:B1:MIL1553B:COMMAND:COUNT?` might return "XXXXX", indicating that the value is XXXXX.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:COMMAND:PARity

This command sets or queries the value of the command word parity bit for a MIL-STD-1553 bus to triggering on. The bus number is specified by x. The trigger condition must be set to COMMAND.

- Conditions** Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

- Group** Trigger

Syntax	TRIGger:{A B}:BUS:B<x>:MIL1553B:COMMAND:PARity {ONE ZERo NOCARE} TRIGger:{A B}:BUS:B<x>:MIL1553B:COMMAND:PARity?
Arguments	ONE filters command words to only match those where the parity bit has a value of 1. ZERo filters command words to only match those where the parity bit has a value of 0. NOCARE disables filtering of command words on the parity bit.
Examples	TRIGGER:A:BUS:B1:MIL1553B:COMMAND:PARITY ONE specifies filtering command words for those where the parity bit has a value of 1. TRIGGER:A:BUS:B1:MIL1553B:COMMAND:PARITY? might return NOCARE, indicating that command words are not being filtered based on the parity bit value.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:COMMAND:SUBADdress

This command sets or queries the value of the command word subaddress field for a MIL-STD-1553 bus to triggering on. The bus number is specified by x. The trigger condition must be set to COMMAND.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:MIL1553B:COMMAND:SUBAddress <QString> TRIGger:{A B}:BUS:B<x>:MIL1553B:COMMAND:SUBAddress?
Arguments	<QString> is the subaddress value.
Examples	TRIGGER:A:BUS:B1:MIL1553B:COMMAND:SUBADDRESS "X1000" sets the value to X1000. TRIGGER:A:BUS:B1:MIL1553B:COMMAND:SUBADDRESS? might return "XXXXX", indicating that the value is XXXXX.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:COMMAND:TRBit

This command sets or queries the value of the command word Transmit / Receive bit for a MIL-STD-1553 bus to trigger on. The bus number is specified by x. The trigger condition must be set to COMMAND.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:MIL1553B:COMMAND:TRBit {RX|TX|X}
TRIGger:{A|B}:BUS:B<x>:MIL1553B:COMMAND:TRBit?

Arguments RX filters command words to only match those that are receive packets.
TX filters command words to only match those that are transmit packets.
X disables filtering of command words on the R/T bit.

Examples TRIGGER:A:BUS:B1:MIL1553B:COMMAND:TRBIT TX specifies filtering command words for only transmit messages.
TRIGGER:A:BUS:B1:MIL1553B:COMMAND:TRBIT? might return X, indicating that command words are not being filtered based on the R/T bit value.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:CONDition

This command sets or queries the field or condition for a MIL-STD-1553 bus to trigger on. The bus number is specified by x.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:MIL1553B:CONDition
{SYNC|COMMAND|STATUS|DATA |TIME|ERROR}
TRIGger:{A|B}:BUS:B<x>:MIL1553B:CONDition?

Arguments SYNC specifies triggering on the sync pulse of any word.
 COMMAND specifies triggering on a matching command word.
 STATUS specifies triggering on a matching status word.
 DATA specifies triggering on a matching data word.
 TIME specifies triggering on the response time or intermessage gap between words.
 ERROR specifies triggering on a specified error condition.

Examples TRIGGER:A:BUS:B1:MIL1553B:CONDITION DATA specifies finding matching data word(s).
 TRIGGER:A:BUS:B1:MIL1553B:CONDITION? might return SYNC, indicating that the bus is triggering on sync pulses found in any word.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:DATA:PARity

This command sets or queries the value of the command word parity bit for a MIL-STD-1553 bus to triggering on. The bus number is specified by x. The trigger condition must be set to DATA.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:MIL1553B:DATA:PARity
 {ONE|ZERO|NOCARE}
 TRIGger:{A|B}:BUS:B<x>:MIL1553B:DATA:PARity?

Arguments ONE filters data words to only match those where the parity bit has a value of 1.
 ZERO filters data words to only match those where the parity bit has a value of 0.
 NOCARE disables filtering of data words on the parity bit.

Examples TRIGGER:A:BUS:B1:MIL1553B:DATA:PARITY ONE specifies filtering data words for those where the parity bit has the value 1.
 TRIGGER:A:BUS:B1:MIL1553B:DATA:PARITY? might return NOCARE, indicating that data words are not being filtered based on the parity bit value.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:DATa:VALue

This command sets or queries the value when triggering on data words for a MIL-STD-1553 bus. The bus number is specified by x. The trigger condition must be set to DATA.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:MIL1553B:DATa:VALue <QString>
TRIGger:{A|B}:BUS:B<x>:MIL1553B:DATa:VALue?

Arguments <QString> is the data value.

Examples TRIGGER:A:BUS:B1:MIL1553B:DATA:VALUE "XXXXXXXXXXXX1000" sets the value to XXXXXXXXXXXXXXX1000.

TRIGGER:A:BUS:B1:MIL1553B:DATA:VALUE? might return "XXXXXXXXXXXXXXXXXXXX", indicating that the value is XXXXXXXXXXXXXXXXXXXX.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:ERRTYPe

This command sets or queries the type of error condition for a MIL-STD-1553 bus to trigger on. The bus number is specified by x. The trigger condition must be set to ERRor.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:MIL1553B:ERRTYPe {PARity|SYNC|DATA}
TRIGger:{A|B}:BUS:B<x>:MIL1553B:ERRTYPe?

Arguments PARity specifies triggering on an incorrectly calculated parity bit in any word.
SYNC specifies triggering on any sync pulse that does not transition in the middle of the pulse as required.

DATA specifies triggering on any non-contiguous data words.

- Examples** `TRIGGER:A:BUS:B1:MIL1553B:ERRTYPE` DATA specifies triggering on non-contiguous data words.
- `TRIGGER:A:BUS:B1:MIL1553B:ERRTYPE?` might return `PARITY`, indicating that the bus is triggering on parity errors in any word.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:STATus:ADDRess:HIVALue

This command sets or queries the high value when triggering on status word addresses for a MIL-STD-1553 bus. The bus number is specified by x. The trigger condition must be set to `STATUS` and the address qualifier must be `INRange` or `OUTrange`.

- Conditions** Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
- Group** Trigger
- Syntax** `TRIGger:{A|B}:BUS:B<x>:MIL1553B:STATus:ADDRess:HIVALue <QString>`
`TRIGger:{A|B}:BUS:B<x>:MIL1553B:STATus:ADDRess:HIVALue?`
- Arguments** `<QString>` is the address value.
- Examples** `TRIGGER:A:BUS:B1:MIL1553B:STATUS:ADDRESS:HIVALUE "X1000"` sets the value to X1000.
- `TRIGGER:A:BUS:B1:MIL1553B:STATUS:ADDRESS:HIVALUE?` might return `"XXXXXX"`, indicating that the value is XXXXXX.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:STATus:ADDRess:QUALifier

This command sets or queries the qualifier to be used when triggering on status word addresses for a MIL-STD-1553 bus. The bus number is specified by x. The trigger condition must be set to `STATUS`.

- Conditions** Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:MIL1553B:STATUS: ADDRESS:QUALifier {EQUAL UNEQUAL LESSThan MOREthan LESSEQUAL MOREEQUAL INrange OUTrange} TRIGger:{A B}:BUS:B<x>:MIL1553B:STATUS:ADDRESS:QUALifier?</pre>
Arguments	Arguments are the available address qualifiers.
Examples	TRIGGER:A:BUS:B1:MIL1553B:STATUS:ADDRESS:QUALIFIER LESSTHAN sets the address qualifier to less than. TRIGGER:A:BUS:B1:MIL1553B:STATUS:ADDRESS:QUALIFIER? might return EQUAL, indicating that the address qualifier is set to equal.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:STATUS:ADDRESS:VALUE

This command sets or queries the low value when triggering on status word addresses for a MIL-STD-1553 bus. The bus number is specified by x. The trigger condition must be set to STATUS.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:MIL1553B:STATUS:ADDRESS:VALUE <QString> TRIGger:{A B}:BUS:B<x>:MIL1553B:STATUS:ADDRESS:VALUE?</pre>
Arguments	<QString> is the address value.
Examples	TRIGGER:A:BUS:B1:MIL1553B:STATUS:ADDRESS:VALUE "X1000" sets the value to X1000. TRIGGER:A:BUS:B1:MIL1553B:STATUS:ADDRESS:VALUE? might return "XXXXX", indicating that the value is XXXXX.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:STATus:BIT:BCR

This command sets or queries the value of the broadcast command received bit (BCR bit, bit 15) in a status word for a MIL-STD-1553 bus to triggering on. The bus number is specified by x. The trigger condition must be set to STATus.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:BIT:BCR {ONE ZERO NOCARE} TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:BIT:BCR?
Arguments	ONE filters status words to only match those where the BCR bit has a value of 1. ZERO filters status words to only match those where the BCR bit has a value of 0. NOCARE disables filtering of status words on the BCR bit.
Examples	TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:BCR ONE specifies filtering status words for those where the BCR bit has a value of 1. TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:BCR? might return NOCARE, indicating that status words are not being filtered based on the BCR bit value.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:STATus:BIT:BUSY

This command sets or queries the value of the busy bit (BUSY bit, bit 16) in a status word for a MIL-STD-1553 bus to triggering on. The bus number is specified by x. The trigger condition must be set to STATus.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:BIT:BUSY {ONE ZERO NOCARE} TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:BIT:BUSY?

Arguments	ONE filters status words to only match those where the BUSY bit has a value of 1. ZERO filters status words to only match those where the BUSY bit has a value of 0. NOCARE disables filtering of status words on the BUSY bit.
Examples	TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:BUSY ONE specifies filtering status words for those where the BUSY bit has a value of 1. TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:BUSY? might return NOCARE, indicating that status words are not being filtered based on the BUSY bit value.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:STATus:BIT:DBCA

This command sets or queries the value of the dynamic bus control acceptance bit (DBCA bit, bit 18) in a status word for a MIL-STD-1553 bus to triggering on. The bus number is specified by x. The trigger condition must be set to STATUS.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:BIT:DBCA {ONE ZERO NOCARE} TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:BIT:DBCA?
Arguments	ONE filters status words to only match those where the DBCA bit has a value of 1. ZERO filters status words to only match those where the DBCA bit has a value of 0. NOCARE disables filtering of status words on the DBCA bit.
Examples	TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:DBCA ONE specifies filtering status words for those where the DBCA bit has a value of 1. TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:DBCA? might return NOCARE, indicating that status words are not being filtered based on the DBCA bit value.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:STATus:BIT:INSTR

This command sets or queries the value of the instrumentation bit (INSTR bit, bit 10) in a status word for a MIL-STD-1553 bus to triggering on. The bus number is specified by x. The trigger condition must be set to STATUS.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:MIL1553B:STATUS:BIT:INSTR {ONE ZERO NOCARE} TRIGger:{A B}:BUS:B<x>:MIL1553B:STATUS:BIT:INSTR?</pre>
Arguments	ONE filters status words to only match those where the INSTR bit has a value of 1. ZERO filters status words to only match those where the INSTR bit has a value of 0. NOCARE disables filtering of status words on the INSTR bit.
Examples	TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:INSTR ONE specifies filtering status words for those where the INSTR bit has a value of 1. TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:INSTR? might return NOCARE, indicating that status words are not being filtered based on the INSTR bit value.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:STATUS:BIT:ME

This command sets or queries the value of the message error bit (ME bit, bit 9) in a status word for a MIL-STD-1553 bus to triggering on. The bus number is specified by x. The trigger condition must be set to STATUS.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:MIL1553B:STATUS:BIT:ME {ONE ZERO NOCARE} TRIGger:{A B}:BUS:B<x>:MIL1553B:STATUS:BIT:ME?</pre>
Arguments	ONE filters status words to only match those where the ME bit has a value of 1. ZERO filters status words to only match those where the ME bit has a value of 0. NOCARE disables filtering of status words on the ME bit.

- Examples** `TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:ME` ONE specifies filtering status words for those where the ME bit has a value of 1.
- `TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:ME?` might return NOCARE, indicating that status words are not being filtered based on the ME bit value.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:STATus:BIT:SRQ

This command sets or queries the value of the status word service request bit (SRQ bit, bit 11) in a status word for a MIL-STD-1553 bus to triggering on. The bus number is specified by x. The trigger condition must be set to STATUS.

- Conditions** Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
- Group** Trigger
- Syntax** `TRIGger:{A|B}:BUS:B<x>:MIL1553B:STATus:BIT:SRQ`
`{ONE|ZERO|NOCARE}`
`TRIGger:{A|B}:BUS:B<x>:MIL1553B:STATus:BIT:SRQ?`
- Arguments** ONE filters status words to only match those where the SRQ bit has a value of 1.
 ZERO filters status words to only match those where the SRQ bit has a value of 0.
 NOCARE disables filtering of status words on the SRQ bit.
- Examples** `TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:SRQ` ONE specifies filtering status words for those where the SRQ bit has a value of 1.
- `TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:SRQ?` might return NOCARE, indicating that status words are not being filtered based on the SRQ bit value.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:STATus:BIT:SBSF

This command sets or queries the value of the subsystem flag bit (SBSF bit, bit 17) in a status word for a MIL-STD-1553 bus to triggering on. The bus number is specified by x. The trigger condition must be set to STATUS.

- Conditions** Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:MIL1553B:STATUS:BIT:SBSF {ONE ZERO NOCARE} TRIGger:{A B}:BUS:B<x>:MIL1553B:STATUS:BIT:SBSF?
Arguments	ONE filters status words to only match those where the SBSF bit has a value of 1. ZERO filters status words to only match those where the SBSF bit has a value of 0. NOCARE disables filtering of status words on the SBSF bit.
Examples	TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:SBSF ONE specifies filtering status words for those where the SBSF bit has a value of 1. TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:SBSF? might return NOCARE, indicating that status words are not being filtered based on the SBSF bit value.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:STATUS:BIT:TF

This command sets or queries the value of the terminal flag bit (TF bit, bit 19) in a status word for a MIL-STD-1553 bus to triggering on. The bus number is specified by x. The trigger condition must be set to STATUS.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:MIL1553B:STATUS:BIT:TF {ONE ZERO NOCARE} TRIGger:{A B}:BUS:B<x>:MIL1553B:STATUS:BIT:TF?
Arguments	ONE filters status words to only match those where the TF bit has a value of 1. ZERO filters status words to only match those where the TF bit has a value of 0. NOCARE disables filtering of status words on the TF bit.
Examples	TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:TF ONE specifies filtering status words for those where the TF bit has a value of 1.

TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:TF? might return NOCARE, indicating that status words are not being filtered based on the TF bit value.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:STATus:PARity

This command sets or queries the value of the status word parity bit for a MIL-STD-1553 bus to triggering on. The bus number is specified by x. The trigger condition must be set to STATUS.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:PARity {ONE ZERO NOCARE} TRIGger:{A B}:BUS:B<x>:MIL1553B:STATus:PARity?
Arguments	ONE filters status words to only match those where the parity bit has a value of 1. ZERO filters status words to only match those where the parity bit has a value of 0. NOCARE disables filtering of status words on the parity bit.
Examples	TRIGGER:A:BUS:B1:MIL1553B:STATUS:PARITY ONE specifies filtering status words for those where the parity bit has a value of 1. TRIGGER:A:BUS:B1:MIL1553B:STATUS:PARITY? might return NOCARE, indicating that status words are not being filtered based on the parity bit value.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:TIME:LESSLimit

This command sets or queries the lower limit to be used when triggering on response time / inter message gap time for a MIL-STD-1553 bus. The bus number is specified by x. The trigger condition must be set to TIME.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger

Syntax	<pre>TRIGger:{A B}:BUS:B<x>:MIL1553B:TIME:LESSLimit <NR3> TRIGger:{A B}:BUS:B<x>:MIL1553B:TIME:LESSLimit?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:MIL1553B:TIME:QUALifier
Arguments	<NR3> is a floating point number that specifies the lower bound for measuring remote terminal response time (RT) or the inter-message gap (IMG) in seconds.
Examples	TRIGGER:A:BUS:B1:MIL1553B:TIME:LESSLIMIT 2.0000e-6 sets the lower bound for comparison to 2 microseconds. TRIGGER:A:BUS:B1:MIL1553B:TIME:LESSLIMIT? might return 4.0000E-6, indicating that the lower bound for comparison is set to 4 microseconds.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:TIME:MORELimit

This command sets or queries the upper limit to be used when triggering on response time / inter message gap time for a MIL-STD-1553 bus. The bus number is specified by x. The trigger condition must be set to TIME.

Conditions	Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:MIL1553B:TIME:MORELimit <NR3> TRIGger:{A B}:BUS:B<x>:MIL1553B:TIME:MORELimit?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:MIL1553B:TIME:QUALifier
Arguments	<NR3> is a floating point number that specifies the upper bound for measuring remote terminal response time (RT) or the inter-message gap (IMG) in seconds.
Examples	TRIGGER:A:BUS:B1:MIL1553B:TIME:MORELIMIT 15.0000e-6 sets the upper bound for comparison to 15 microseconds. TRIGGER:A:BUS:B1:MIL1553B:TIME:MORELIMIT? might return 12.0000E-6, indicating that the upper bound for comparison is set to 12 microseconds.

TRIGger:{A|B}:BUS:B<x>:MIL1553B:TIME:QUALifier

This command sets or queries the qualifier to be used when triggering on response time / inter message gap time for a MIL-STD-1553 bus. The bus number is specified by x. The trigger condition must be set to TIME.

Conditions Requires option 5-SRAERO or SUP5-SRAERO Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:MIL1553B:TIME:QUALifier
{LESSthan|MOREthan|INrange|OUTrange}
TRIGger:{A|B}:BUS:B<x>:MIL1553B:TIME:QUALifier?

Related Commands [TRIGger:{A|B}:BUS:B<x>:MIL1553B:TIME:LESSLimit](#)
[TRIGger:{A|B}:BUS:B<x>:MIL1553B:TIME:MORELimit](#)

Arguments Arguments are the available comparison qualifiers.

Examples TRIGGER:A:BUS:B1:MIL1553B:TIME:QUALIFIER LESSTHAN sets the time comparison qualifier to less than.

TRIGGER:A:BUS:B1:MIL1553B:TIME:QUALIFIER? might return OUTRANGE, indicating that the qualifier is set to out-of-range comparison.

TRIGger:{A|B}:BUS:B<x>:PARallel:DATA:VALue

This command specifies the binary data string used for a Parallel Bus trigger. The bus number is specified by <x>.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:PARallel:DATA:VALue <QString>
TRIGger:{A|B}:BUS:B<x>:PARallel:DATA:VALue?

Arguments <QString> is the binary data string used for a Parallel Bus trigger.

Examples `TRIGger:A:BUS:B1:PARAllel:DATA:VALue "11001101"` sets the data value to 11001101.

`TRIGger:A:BUS:B1:PARAllel:DATA:VALue?` might return
`:TRIGGER:A:BUS:B1:PARALLEL:DATA:VALUE "X"` indicating the data value is X.

TRIGger:{A|B}:BUS:B<x>:RS232C:CONDition

This command specifies the condition for an RS-232C trigger, where the bus number is specified by <x>.

Conditions Requires option 5-SRCOMP or SUP5-SRCOMP Triggering and Analysis application.

Group Trigger

Syntax `TRIGger:{A|B}:BUS:B<x>:RS232C:CONDition`
`{START|EOp|DATA|PARityerror}`
`TRIGger:{A|B}:BUS:B<x>:RS232C:CONDition?`

Arguments `START` sets the Trigger on condition to Start.

`EOp` sets the Trigger on condition to End of Packet.

`DATA` sets the Trigger on condition to Data.

`PARityerror` sets the Trigger on condition to Parity Error.

Examples `TRIGger:A:BUS:B1:RS232C:CONDition DATA` sets the trigger on condition to data.

`TRIGger:A:BUS:B1:RS232C:CONDition?` might return
`:TRIGGER:A:BUS:B1:RS232C:CONDITION START` indicating start is the trigger condition.

TRIGger:{A|B}:BUS:B<x>:RS232C:DATA:SIZE

This command sets or queries the length of the data string in bytes to be used for an RS-232C trigger when the trigger condition is Data. The bus number is specified by <x>.

Conditions	Requires option 5-SRCOMP or SUP5-SRCOMP Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:RS232C:DATA:SIZE <NR3> TRIGger:{A B}:BUS:B<x>:RS232C:DATA:SIZE?</pre>
Arguments	<NR3> specifies the data size in bytes.
Examples	TRIGGER:A:BUS:B1:RS232C:DATA:SIZE 3 sets three bytes data size for the RS-232C bus B1 trigger. TRIGGER:A:BUS:B1:RS232C:DATA:SIZE? might return :TRIGGER:A:BUS:B1:RS232C:DATA:SIZE 2, indicating that the data size for the RS-232C bus B1 trigger is set to two bytes.

TRIGger:{A|B}:BUS:B<x>:RS232C:DATA:VALue

This command sets or queries the data address string used for the RS-232 bus trigger when the trigger condition is set to Data. The bus number is specified by <x>.

Conditions	Requires option 5-SRCOMP or SUP5-SRCOMP Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:RS232C:DATA:VALue <QString> TRIGger:{A B}:BUS:B<x>:RS232C:DATA:VALue?</pre>
Arguments	<QString> specifies the address value. The argument is a string of 0, 1, or X representing a binary number.
Examples	TRIGGER:A:BUS:B1:RS232C:DATA:VALue "011XX11" sets the data address string used for the RS-232 bus trigger to 011XX11. TRIGGER:A:BUS:B1:RS232C:DATA:VALue? might return :TRIGGER:A:BUS:B1:RS232C:DATA:VALue "XXXXXX01", indicating that the data address string used for the RS-232 bus trigger is set to "XXXXXX01"

TRIGger:{A|B}:BUS:B<x>:SENT:CONDition

This command sets or queries the trigger condition for a SENT bus.

Conditions	Requires option SRAUTOSEN.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:SENT:CONDition {START FAST SLOW ERRor} TRIGger:{A B}:BUS:B<x>:SENT:CONDition?</pre>
Arguments	B<x> is the number of the bus waveform. START sets triggering on start of packet. FAST sets triggering on fast channel packets. SLOW sets triggering on slow channel packets. ERRor sets triggering on errors.
Examples	TRIGGER:A:BUS:B1:SENT:CONDITION START specifies triggering on start of packet. TRIGGER:A:BUS:B1:SENT:CONDITION? might return TRIGGER:A:BUS:B1:SENT:CONDITION ERROR, indicating the bus is triggering on errors.

TRIGger:{A|B}:BUS:B<x>:SENT:ERRType

This command sets or queries the error type to be used when triggering on SENT data.

Conditions	Requires option 5SRAUTOSEN. The SENT bus trigger condition is set to ERRor.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:SENT:ERRType CRC TRIGger:{A B}:BUS:B<x>:SENT:ERRType?</pre>

Related Commands	TRIGger:{A B}:BUS:B<x>:SENT:CONDition
Arguments	B<x> is the number of the bus waveform. CRC specifies triggering on CRC errors.
Examples	TRIGGER:A:BUS:B1:SENT:ERRTYPE CRC sets the bus to trigger on CRC errors. TRIGGER:A:BUS:B1:SENT:ERRTYPE? might return :TRIGGER:A:BUS:B1:SENT:ERRTYPE CRC to indicate the bus is triggering on CRC errors.

TRIGger:{A|B}:BUS:B<x>:SENT:ERRType:CRC

This command sets or queries the CRC error type to be used when triggering on SENT data.

Conditions	Requires option SRAUTOSEN. The SENT bus trigger condition is set to ERRor. The ERRType is set to CRC.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:SENT:ERRType:CRC {FAST SLOW} TRIGger:{A B}:BUS:B<x>:SENT:ERRType:CRC?
Related Commands	TRIGger:{A B}:BUS:B<x>:SENT:CONDition TRIGger:{A B}:BUS:B<x>:SENT:ERRType
Arguments	B<x> is the number of the bus waveform. FAST specifies triggering on CRC errors in only the Fast Channel. SLOW specifies triggering on CRC errors in only the slow channel.
Examples	TRIGGER:A:BUS:B1:SENT:ERRTYPE:CRC FAST sets the bus to trigger on Fast Channel CRC errors. TRIGGER:A:BUS:B1:SENT:ERRTYPE:CRC? might return TRIGGER:A:BUS:B1:SENT:ERRTYPE:CRC FAST to indicate that the bus is triggering on Fast Channel CRC errors.

TRIGger:{A|B}:BUS:B<x>:SENT:FAST:CHAN1A:HIVALue

This command sets or queries the high binary fast channel 1 value to use when triggering on a SENT bus signal.

Conditions Requires option SRAUTOSEN.
The SENT bus trigger condition must be set to FAST.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:SENT:FAST:CHAN1A:HIVALue <Qstring>
TRIGger:{A|B}:BUS:B<x>:SENT:FAST:CHAN1A:HIVALue?

Related Commands [TRIGger:{A|B}:BUS:B<x>:SENT:CONDition](#)

Arguments Bus<x> is the bus identifier number.
<Qstring> sets the Fast Channel 1 binary data high value.

Examples TRIGger:A:BUS:B12:SENT:FAST:CHAN1A:HIVALue "XXXXXXXXXXXXX"
sets the SENT bus B12 Fast Channel 1 high value to XXXXXXXXXXXXXXXX, or
"don't care."

TRIGger:A:BUS:B3:SENT:FAST:CHAN1A:HIVALue? might return
TRIGGER:A:BUS:B3:SENT:FAST:CHAN1A:HIVALUE "0101XXXX1111",
indicating the SENT bus B3 Fast Channel 1 high is set to the binary value
0101XXXX1111.

TRIGger:{A|B}:BUS:B<x>:SENT:FAST:CHAN1A:QUALifier

This command sets or queries the qualifier to be used when triggering on SENT fast packet bus data for device channel 1.

Conditions Requires option SRAUTOSEN
The SENT bus trigger condition must be set to FAST.

Group Trigger

Syntax `TRIGger:{A|B}:BUS:B<x>:SENT:FAST:CHAN1A:QUALifier {Equal|UNEQual|LESSthan|MOREthan|LESSEQual|MOREEQual|INrange|OUTrange}`
`TRIGger:{A|B}:BUS:B<x>:SENT:FAST:CHAN1A:QUALifier?`

Related Commands [TRIGger:{A|B}:BUS:B<x>:SENT:CONDition](#)
[TRIGger:{A|B}:BUS:B<x>:SENT:FAST:CHAN1A:VALue](#)

Arguments B<x> is the number of the bus.
Equal sets the qualifier as Equal.
LESSEQual sets the qualifier as Less Than or Equal to.
LESSThan sets the qualifier as Less Than.
MOREEQual sets the qualifier as More Than or Equal to.
MOREThan sets the qualifier as More Than.
UNEQual sets the qualifier as Unequal.
INrange sets the qualifier to inside a range.
OUTrange sets the qualifier to outside a range.

Examples `TRIGGER:A:BUS:B2:SENT:FAST:CHAN1A:QUALIFIER UNEQUAL` sets the fast channel 1 data qualifier trigger to not equal for bus 2.
`TRIGGER:A:BUS:B5:SENT:FAST:CHAN1A:QUALIFIER?` might return
`TRIGGER:A:BUS:B1:SENT:FAST:CHAN1A:QUALIFIER MOREEQUAL` to indicate that the fast channel 1 data qualifier trigger is set to greater than or equal for bus 5.

TRIGger:{A|B}:BUS:B<x>:SENT:FAST:CHAN1A:VALue

This command sets or queries the binary fast channel 1 value to be used when triggering on a SENT bus signal.

Conditions Requires option SRAUTOSEN.
The search trigger condition must be set to FAST.

Group Trigger

Syntax	<pre>TRIGger:{A B}:BUS:B<x>:SENT:FAST:CHAN1A:VALue <Qstring> TRIGger:{A B}:BUS:B<x>:SENT:FAST:CHAN1A:VALue?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:SENT:CONDition TRIGger:{A B}:BUS:B<x>:SENT:FAST:CHAN1A:QUALifier
Arguments	B<x> is the number of the bus waveform. <Qstring> is the Fast Channel 1 value on which to trigger.
Examples	TRIGGER:A:BUS:B1:SENT:FAST:CHAN1A:VALUE "XXXXXXXXXXXX" sets the Fast Channel 1 value to trigger on XXXXXXXXXXXXX, or "don't care." TRIGGER:A:BUS:B1:SENT:FAST:CHAN1A:VALUE? might return :TRIGGER:A:BUS:B1:SENT:FAST:CHAN1A:VALUE "0000XXXX1111" to indicate the Fast Channel 1 binary trigger value is 0000XXXX1111.

TRIGger:{A|B}:BUS:B<x>:SENT:FAST:CHAN2B:HIVALue

This command sets or queries the high binary fast channel 2 value to use when triggering on a SENT bus signal.

Conditions	Requires option SRAUTOSEN. The SENT bus trigger condition must be set to FAST.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:SENT:FAST:CHAN2B:HIVALue <QString> TRIGger:{A B}:BUS:B<x>:SENT:FAST:CHAN2B:HIVALue?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:SENT:CONDition
Arguments	BUS<x> is the bus identifier number. <Qstring> sets the Fast Channel 2 high binary data value.
Examples	TRIGger:A:BUS:B2:SENT:FAST:CHAN2B:HIVALue "100000000000" sets the SENT bus B2 Fast Channel 2 high value to 100000000000.

TRIGger:A:BUS:B4:SENT:FAST:CHAN2B:HIVALue? might return
 TRIGGER:A:BUS:B4:SENT:FAST:CHAN2B:HIVALUE "0101XXXX1111",
 indicating the SENT bus B4 Fast Channel 2 high is set to the binary value
 0101XXXX1111.

TRIGger:{A|B}:BUS:B<x>:SENT:FAST:CHAN2B:QUALifier

This command sets or queries the qualifier to be used when triggering on SENT fast packet bus data for device channel 2.

Conditions Requires option SRAUTOSEN.

The SENT bus trigger condition must be set to FAST.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:SENT:FAST:CHAN2B:
 QUALifier {EQual|UNEQual|LESSthan|MOREthan|
 LESSEQual|MOREEQual|INrange|OUTrange}
 TRIGger:{A|B}:BUS:B<x>:SENT:FAST:CHAN2B:QUALifier?

Related Commands [TRIGger:{A|B}:BUS:B<x>:SENT:CONDition](#)
[TRIGger:{A|B}:BUS:B<x>:SENT:FAST:CHAN2B:VALue](#)

Arguments B<x> is the number of the bus.
 EQual sets the qualifier as Equal.
 LESSEQual sets the qualifier as Less Than or Equal to.
 LESSThan sets the qualifier as Less Than.
 MOREEQual sets the qualifier as More Than or Equal to.
 MOREThan sets the qualifier as More Than.
 UNEQual sets the qualifier as Unequal.
 INrange sets the qualifier to inside a range.
 OUTrange sets the qualifier to outside a range.

Examples TRIGGER:A:BUS:B2:SENT:FAST:CHAN2B:QUALIFIER UNEQUAL sets the fast
 channel 2 data qualifier trigger to not equal for bus 2.

TRIGGER:A:BUS:B5:SENT:FAST:CHAN2B:QUALIFIER? might return
 TRIGGER:A:BUS:B1:SENT:FAST:CHAN2B:QUALIFIER MOREEQUAL to
 indicate that the fast channel 2 data qualifier trigger is set to greater than or equal
 for bus 5.

TRIGger:{A|B}:BUS:B<x>:SENT:FAST:CHAN2B:VALue

This command sets or queries the binary fast channel 2 value to be used when
 triggering on a SENT bus signal. The trigger condition must be set to FAST.

Conditions Requires option SRAUTOSEN.

The search condition must be set to FAST.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:SENT:FAST:CHAN2B:VALue <Qstring>
 TRIGger:{A|B}:BUS:B<x>:SENT:FAST:CHAN2B:VALue?

Related Commands [TRIGger:{A|B}:BUS:B<x>:SENT:CONDition](#)
[TRIGger:{A|B}:BUS:B<x>:SENT:PAUSE:QUALifier](#)

Arguments B<x> is the number of the bus waveform.
 <Qstring> is the Fast Channel 2 binary value on which to trigger.

Examples TRIGGER:A:BUS:B1:SENT:FAST:CHAN2B:VALUE "1111111111" sets the
 Fast Channel 2 value on which to trigger to 1111111111.
 TRIGGER:A:BUS:B1:SENT:FAST:CHAN2B:VALUE? might return
 :TRIGGER:A:BUS:B1:SENT:FAST:CHAN2B:VALUE "000000000000" to
 indicate the Fast Channel 2 binary value on which to trigger is 000000000000.

TRIGger:{A|B}:BUS:B<x>:SENT:FAST:COUNTER:HIVALue

This command sets or queries the high binary fast message counter value to be
 used when triggering on a SENT bus signal.

Conditions Requires option SRAUTOSEN.

The SENT bus trigger condition must be set to FAST.

The number of channels must be set to 1.

The nibble count must be set to 6.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:SENT:FAST:COUNTER:HIVALue <QString>
TRIGger:{A|B}:BUS:B<x>:SENT:FAST:COUNTER:HIVALue?

Related Commands [TRIGger:{A|B}:BUS:B<x>:SENT:CONDition](#)
[BUS:B<x>:SENT:NUMCHANnel](#)
[BUS:B<x>:SENT:NIBBLECount](#)

Arguments Bus<x> is the bus identifier number.
<Qstring> sets the Fast Channel 1 counter binary value.

Examples TRIGGER:A:BUS:B1:SENT:FAST:COUNTER:HIVALUE "XXXXXXXX" sets the Fast Channel 1 secure counter high value to "don't care."

TRIGGER:A:BUS:B3:SENT:FAST:COUNTER:HIVALUE? might return
TRIGGER:A:BUS:B3:SENT:FAST:COUNTER:HIVALUE "11110000" to indicate a binary value of 11110000.

TRIGger:{A|B}:BUS:B<x>:SENT:FAST:COUNTER:QUALifier

This command sets or queries the qualifier to be used when triggering on SENT fast packet bus data for the secure format counter.

Conditions Requires option SRAUTOSEN.
The SENT bus trigger condition must be set to FAST.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:SENT:FAST:COUNTER:
QUALifier {EQual|UNEQual|LESSthan|MOREthan|
LESSEQual|MOREEQual|INrange|OUTrange}
TRIGger:{A|B}:BUS:B<x>:SENT:FAST:COUNTER:QUALifier?

Related Commands [TRIGger:{A|B}:BUS:B<x>:SENT:CONDition](#)

[TRIGger:{A|B}:BUS:B<x>:SENT:FAST:COUNTER:VALue](#)

Arguments	B<x> is the number of the bus. EQUa1 sets the qualifier as Equal. LESSEQUa1 sets the qualifier as Less Than or Equal to. LESSThan sets the qualifier as Less Than. MOREEQUa1 sets the qualifier as More Than or Equal to. MOREThan sets the qualifier as More Than. UNEQUa1 sets the qualifier as Unequal. INrange sets the qualifier to inside a range. OUTrange sets the qualifier to outside a range.
Examples	TRIGGER:A:BUS:B2:SENT:FAST:COUNTER:QUALIFIER UNEQUAL sets the fast channel 2 counter qualifier trigger to not equal for bus 2. TRIGGER:A:BUS:B5:SENT:FAST:COUNTER:QUALIFIER? might return TRIGGER:A:BUS:B1:SENT:FAST:COUNTER:QUALIFIER MOREEQUAL to indicate that the fast channel 2 counter qualifier trigger is set to greater than or equal for bus 5.

TRIGger:{A|B}:BUS:B<x>:SENT:FAST:COUNTER:VALue

This command sets or queries the binary fast message counter value to be used when triggering on a SENT bus signal.

Conditions	Requires option SRAUTOSEN. The search condition must be set to FAST. The number of channels must be set to 1. The nibble count must be set to 6.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:SENT:FAST:COUNTER:VALue <Qstring> TRIGger:{A B}:BUS:B<x>:SENT:FAST:COUNTER:VALue?

Related Commands [TRIGger:{A|B}:BUS:B<x>:SENT:CONDition](#)

[BUS:B<x>:SENT:NIBBLECount](#)

[BUS:B<x>:SENT:NUMCHANnel](#)

Arguments	B<x> is the number of the bus waveform. <Qstring> is the Fast Channel 1 fast message counter binary value on which to trigger.
Examples	TRIGGER:A:BUS:B1:SENT:FAST:COUNTER:VALUE "XXXXXXX" sets the Fast Channel 1 fast message counter value to "don't care." TRIGGER:A:BUS:B1:SENT:FAST:COUNTER:VALUE? might return :TRIGGER:A:BUS:B1:SENT:FAST:COUNTER:VALUE "00001111" to indicate the Fast Channel 1 fast message counter binary value on which to trigger is 00001111.

TRIGger:{A|B}:BUS:B<x>:SENT:FAST:INVERTNIBble:VALue

This command sets or queries the binary fast message inverted nibble value to be used when triggering on a SENT bus signal.

Conditions	Requires option SRAUTOSEN. The search condition must be set to FAST. The number of channels must be set to 1. The nibble count must be set to 6.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:SENT:FAST:INVERTNIBble:VALue <Qstring> TRIGger:{A B}:BUS:B<x>:SENT:FAST:INVERTNIBble:VALue?

Related Commands

[TRIGger:{A|B}:BUS:B<x>:SENT:CONDition](#)

[BUS:B<x>:SENT:NIBBLECount](#)

[BUS:B<x>:SENT:NUMCHANnel](#)

Arguments	B<x> is the number of the bus waveform. <Qstring> is the Fast Channel 1 inverted nibble binary value on which to trigger.
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Examples `TRIGGER:A:BUS:B1:SENT:FAST:INVERTNIBBLE:VALUE "XXXX"` sets the Fast Channel 1 secure inverted nibble value to "don't care."

`TRIGGER:A:BUS:B1:SENT:FAST:INVERTNIBBLE:VALUE?` might return `:TRIGGER:A:BUS:B1:SENT:FAST:INVERTNIBBLE:VALUE "1111"` to indicate a binary value of 1111.

TRIGger:{A|B}:BUS:B<x>:SENT:FAST:STATUS:VALue

This command sets or queries the binary status value to be used when triggering on a SENT bus signal.

Conditions Requires option SRAUTOSEN.
The search trigger condition must be set to FAST.

Group Trigger

Syntax `TRIGger:{A|B}:BUS:B<x>:SENT:FAST:STATUS:VALue <Qstring>`
`TRIGger:{A|B}:BUS:B<x>:SENT:FAST:STATUS:VALue?`

Related Commands [TRIGger:{A|B}:BUS:B<x>:SENT:CONDition](#)

Arguments `B<x>` is the number of the bus waveform.
`<Qstring>` is the binary status value on which to trigger.

Examples `TRIGger:A:BUS:B1:SENT:FAST:STATUS:VALue "XXXX"` sets the binary status trigger value to XXXX.

`TRIGGER:A:BUS:B1:SENT:FAST:STATUS:VALUE?` might return `:TRIGGER:A:BUS:B1:SENT:FAST:STATUS:VALUE "1010"` to indicate a binary trigger value is 1010.

TRIGger:{A|B}:BUS:B<x>:SENT:PAUSE:QUALifier

This command sets or queries the qualifier to be used when triggering on SENT pause pulses.

Conditions Requires option SRAUTOSEN.
The SENT bus trigger condition must be set to PAUSE.

Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:SENT:PAUSE:QUALifier {Equal UNEQual LESSThan MOREthan LESSEQual MOREEQual Inrange OUTrange} TRIGger:{A B}:BUS:B<x>:SENT:PAUSE:QUALifier?
Related Commands	TRIGger:{A B}:BUS:B<x>:SENT:CONDition SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:PAUSE:TICKs:HIVALue SEARCH:SEARCH<x>:TRIGger:A:BUS:SENT:PAUSE:TICKs:VALue
Arguments	B<x> is the bus identifier number. EQua1 sets the qualifier as Equal. INrange sets the qualifier to in range. LESSEQual sets the qualifier as Less Than or Equal to. LESSThan sets the qualifier as Less Than. MOREEQual sets the qualifier as More Than or Equal to. MOREThan sets the qualifier as More Than. OUTrange sets the qualifier to out of range. UNEQual sets the qualifier as Unequal.
Examples	TRIGGER:A:BUS:B2:SENT:PAUSE:QUALifier UNEQUAL sets the pause pulse qualifier trigger to not equal for bus 2. TRIGGER:A:BUS:B1:SENT:PAUSE:QUALIFIER? might return :TRIGGER:A:BUS:B1:SENT:PAUSE:QUALIFIER INRANGE to indicate that the pause pulse qualifier is set to trigger inside the range of ticks low and ticks high on bus 1.

TRIGger:{A|B}:BUS:B<x>:SENT:SLOW:DATA:HIVALue

This command sets or queries the high binary Slow channel data value to use when triggering on a SENT bus signal.

Conditions	Requires option SRAUTOSEN. The SENT bus trigger condition must be set to SLOW.
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Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:SENT:SLOW:DATA:HIVALue <QString> TRIGger:{A B}:BUS:B<x>:SENT:SLOW:DATA:HIVALue?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:SENT:CONDition
Arguments	Bus<x> is the bus identifier number. <Qstring> sets the binary Slow channel data value.
Examples	TRIGGER:A:BUS:B1:SENT:SLOW:DATA:HIVALUE "XXXXXXXX" sets the Slow data high value to "don't care." TRIGGER:A:BUS:B1:SENT:SLOW:DATA:HIVALUE? might return TRIGGER:A:BUS:B1:SENT:SLOW:DATA:HIVALUE "11110001", to indicate triggering on Slow data values that match 11110001.

TRIGger:{A|B}:BUS:B<x>:SENT:SLOW:DATA:QUALifier

This command sets or queries the binary identifier value to use when triggering on a SENT bus signal.

Conditions	Requires option SRAUTOSEN The SENT bus trigger condition must be set to SLOW.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:SENT:SLOW:DATA:QUALifier {Equal UNEQual LESSthan MOREthan LESSEQual MOREEqual} TRIGger:{A B}:BUS:B<x>:SENT:SLOW:DATA:QUALifier?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:SENT:CONDition
Arguments	B<x> is the bus identifier number. EQual specifies the qualifier as Equal. UNEQual specifies the qualifier as Not Equal to. LESSthan specifies the qualifier as Less Than.

MOREthan specifies the qualifier as More Than.

LESSEqual specifies the qualifier as Less Than or Equal to.

MOREEqual specifies the qualifier as More Than or Equal to.

INrange sets the qualifier to inside a range.

OUTrange sets the qualifier to outside a range.

Examples TRIGger:{A|B}:BUS:B2:SENT:SLOW:DATA:QUALIFIER LESSEQUAL sets the slow channel data qualifier to less than or equal for bus 2.

TRIGger:{A|B}:BUS:B5:SENT:SLOW:DATA:QUALIFIER? might return :TRIGger:{A|B}:BUS:B5:SENT:SLOW:DATA:QUALIFIER EQUAL to indicate that the slow channel data qualifier is set to equal on bus 5.

TRIGger:{A|B}:BUS:B<x>:SENT:SLOW:DATA:VALue

This command sets or queries the binary slow channel value to use when triggering on a SENT bus signal.

Conditions Requires option SRAUTOSEN

The SENT bus trigger condition must be set to SLOW.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:SENT:SLOW:DATA:VALue <Qstring>
TRIGger:{A|B}:BUS:B<x>:SENT:SLOW:DATA:VALue?

Related Commands [TRIGger:{A|B}:BUS:B<x>:SENT:CONDition](#)

Arguments B<x> is the bus identifier number.

<Qstring> is the binary slow channel data value.

Examples TRIGGER:A:BUS:B2:SENT:SLOW:DATA:VALUE "XXXXXXXX" sets the slow data value to "don't care" for bus 2.

TRIGGER:A:BUS:B4:SENT:SLOW:DATA:VALUE? might return :TRIGGER:A:BUS:B4:SENT:SLOW:DATA:VALUE "XXXX1111", to indicate triggering on slow data values that match XXXX1111 on bus 4.

TRIGger:{A|B}:BUS:B<x>:SENT:SLOW:IDentifier:VALue

This command sets or queries the qualifier to use when triggering on SENT slow packet bus data.

Conditions Requires option SRAUTOSEN

The SENT bus trigger condition must be set to SLOW.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:SENT:SLOW:IDentifier:VALue <Qstring>
TRIGger:{A|B}:BUS:B<x>:SENT:SLOW:IDentifier:VALue

Related Commands [TRIGger:{A|B}:BUS:B<x>:SENT:CONDition](#)
[TRIGger:{A|B}:BUS:B<x>:SENT:SLOW:DATA:VALue](#)

Arguments B<x> is the bus identifier number.
<Qstring> is the binary identifier value.

Examples TRIGGER:A:BUS:B3:SENT:SLOW:IDENTIFIER:VALUE "XXXX" sets the identifier value to "don't care" for bus 3.
TRIGGER:A:BUS:B1:SENT:SLOW:IDENTIFIER:VALUE? might return
:TRIGGER:A:BUS:B1:SENT:SLOW:IDENTIFIER:VALUE "0001" to indicate the slow channel identifier is set to 0001 on bus 1.

TRIGger:{A|B}:BUS:B<x>:SPI:CONDition

This command specifies the trigger condition for a SPI trigger. The bus number is specified by <x>.

Conditions Requires option 5-SRCOMP or SUP5-SRCOMP Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:SPI:CONDition {SS|STARTofframe|DATA}
TRIGger:{A|B}:BUS:B<x>:SPI:CONDition?

Related Commands [BUS:B<x>:SPI:IDLETime](#)
[BUS:B<x>:SPI:FRAMING](#)

Arguments B<x> is the bus identifier number.
SS specifies the Slave Selection condition.
STARTofframe is applicable when BUS:B<x>:SPI:FRAMING is set to IDLEtime. When the trigger condition is set to STARTofframe, the instrument triggers on the first SPI clock after an idle time when there are no clocks.
DATA sets the trigger condition to Master-In Slave-Out and Master-Out Slave-In.

TRIGger:{A|B}:BUS:B<x>:SPI:DATA:SIZE

This command specifies the length of the data string to be used for a SPI trigger if the trigger condition is DATA. The bus number is specified by <x>.

Conditions Requires option 5-SRCOMP or SUP5-SRCOMP Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:SPI:DATA:SIZE <NR1>
TRIGger:{A|B}:BUS:B<x>:SPI:DATA:SIZE?

Arguments B<x> is the bus identifier number.
<NR1> is the length of the data string in bytes.

Examples TRIGger:A:BUS:B1:SPI:DATA:SIZE 1 sets the data size to 1 byte.
TRIGger:A:BUS:B1:SPI:DATA:SIZE? might return
:TRIGGER:A:BUS:B1:SPI:DATA:SIZE 1 indicating the data size is 1 byte.

TRIGger:{A|B}:BUS:B<x>:SPI:DATA:VALue

This command specifies the binary data string used for SPI triggering if the trigger condition is DATA. The bus number is specified by <x>.

Conditions	Requires option 5-SRCOMP or SUP5-SRCOMP Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:SPI:DATA:VALue <QString> TRIGger:{A B}:BUS:B<x>:SPI:DATA:VALue?</pre>
Arguments	B<x> is the bus identifier number. <QString> specifies the data value in the specified valid format. The valid characters are 0, 1, and X for binary format.
Examples	TRIGger:A:BUS:B1:SPI:DATA:VALue "11011010" sets the data value to 11011010. TRIGger:A:BUS:B1:SPI:DATA:VALue? might return :TRIGGER:A:BUS:B1:SPI:DATA:VALUE "XXXXXXXX" indicating the data value is XXXXXXXX.

TRIGger:{A|B}:BUS:B<x>:SPMI:CONDition

This command sets or queries the trigger condition for the specified SPMI bus.

Conditions	Requires option SRPM
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:SPMI:CONDition {SSC RESet SLEEp SHUTdown WAKEup MASTERREAD MASTERWRITE REGREAD REGWRITE DEVIDESCMASERREAD DEVIDESCSLAVEREAD EXTREGREAD EXTREGWRITE LONGEXTREGREAD LONGEXTREGWRITE REGOWRITE AUTHenticate TRANSFERbusownership PARityerror} TRIGger:{A B}:BUS:B<x>:SPMI:CONDition?</pre>
Arguments	Bus<x> is the bus identifier number. SSC specifies triggering on the Sequence Start Condition. RESet specifies triggering on the Reset command sequence. SLEEp specifies triggering on the Sleep command sequence.

SHUTdown specifies triggering on the Shutdown command sequence.

WAKEup specifies triggering on the Wakeup command sequence.

MASTERREAd specifies triggering on the Master Read command sequence.

MASTERWRITe specifies triggering on the Master Write command sequence.

REGREAd specifies triggering on the Register Read command sequence.

REGWRITe specifies triggering on the Register Write command sequence.

DEVICEDESCMASTERREAd specifies triggering on the Device Descriptor Block Master Read command sequence

DEVICEDESCSLAVEREAd specifies triggering on the Device Descriptor Block Slave Read command sequence.

EXTREGREAd specifies triggering on the Extended Register Read command sequence.

EXTREGWRITe specifies triggering on the Extended Register Write command sequence.

LONGEXTREGREAd specifies triggering on the Extended Register Read Long command sequence.

LONGEXTREGWRITe specifies triggering on the Extended Register Write Long command sequence.

REG0WRITe specifies triggering on the Register 0 Write command sequence.

AUTHenticate specifies triggering on the Authentication command sequence.

TRANSferbusownership specifies triggering on the Transfer Bus Ownership (TBO) command sequence.

PARityerror specifies triggering on the parity errors.

Examples `TRIGGER:A:BUS:B5:SPMI:CONDITION SLEEP` specifies triggering on a Sleep command sequence for bus B5.

`TRIGGER:A:BUS:B2:SPMI:CONDITION?` might return
`:TRIGGER:A:BUS:B2:SPMI:CONDITION MASTERREAD`, indicating that bus B2 is triggering on Master Read command sequences.

TRIGger:{A|B}:BUS:B<x>:SPMI:DATa:SiZe

This command sets or queries the length of the data string, in bytes, to be used when triggering on an SPMI bus signal.

Conditions Requires option SRPM.

The trigger condition must be set to EXTREGREAd, EXTREGWRItE, LONGEXTREGREAd, or LONGEXTREGWRItE.

Note that this a floating point value. If the condition is set to REG0WRItE, the query return value will be a fractional value (0.875) because the data size is less than a byte.

Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:SPMI:DATa:SIZE <NR2> TRIGger:{A B}:BUS:B<x>:SPMI:DATa:SIZE?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:SPMI:CONDition TRIGger:{A B}:BUS:B<x>:SPMI:DATa:VALue
Arguments	Bus<x> is the bus identifier number. <NR2> is the size of the data string in bytes.
Examples	TRIGGER:A:BUS:B1:SPMI:DATA:SIZE 1 sets the data value to 1 byte. TRIGGER:A:BUS:B1:SPMI:DATA:SIZE? might return :TRIGGER:A:BUS:B1:SPMI:DATA:SIZE 4 to indicate the data pattern has four bytes of data.

TRIGger:{A|B}:BUS:B<x>:SPMI:DATa:VALue

This command sets or queries the binary data string used for SPMI triggering if the trigger condition is MASTERREAd, MASTERWRItE, REGREAd, REGWRItE, EXTREGREAd, EXTREGWRItE, LONGEXTREGREAd, LONGEXTREGWRItE, or REG0WRItE.

Conditions	Requires option SRPM
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:SPMI:DATa:VALue <Qstring> TRIGger:{A B}:BUS:B<x>:SPMI:DATa:VALue?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:SPMI:CONDition

Arguments	Bus<x> is the bus identifier number. <Qstring> is the binary data string that identifies the data value on which to trigger.
Examples	TRIGGER:A:BUS:B1:SPMI:DATA:VALUE "XXXXXXXX" sets the binary data value on which to trigger for bus B1 to "don't care." TRIGGER:A:BUS:B5:SPMI:DATA:VALUE? might return TRIGGER:A:BUS:B5:SPMI:DATA:VALUE "XXXX1111" to indicate the binary data value on which to trigger for bus B5 is set to XXXX1111.

TRIGger:{A|B}:BUS:B<x>:SPMI:MASTERADDRess:VALue

This command sets or queries the binary data string that identifies the master address used in SPMI triggering if the trigger condition is MASTERREAd, MASTERWRItE, or DEVICEDeSCMASTERREAd.

Conditions	Requires option SRPM
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:SPMI:MASTERADDRess:VALue <Qstring> TRIGger:{A B}:BUS:B<x>:SPMI:MASTERADDRess:VALue?
Related Commands	TRIGger:{A B}:BUS:B<x>:SPMI:CONDition
Arguments	Bus<x> is the bus identifier number. <Qstring> is the binary data string that identifies the master address on which to trigger.
Examples	TRIGGER:A:BUS:B3:SPMI:MASTERADDRESS:VALUE "XX" sets the binary master address trigger of bus B3 to "don't care." TRIGGER:A:BUS:B1:SPMI:MASTERADDRESS:VALUE? might return TRIGGER:A:BUS:B1:SPMI:MASTERADDRESS:VALUE "11" to indicate the binary master address on which to trigger for bus B1 is set to binary 11.

TRIGger:{A|B}:BUS:B<x>:SPMI:NOReSPonse

This command sets or queries whether or not to trigger on No Response frames.

Conditions	Requires option SRPM
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:SPMI:NORESPonse <Qstring> TRIGger:{A B}:BUS:B<x>:SPMI:NORESPonse?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:SPMI:CONDition
Arguments	Bus<x> is the bus identifier number. <Qstring> is either TRUE or FALSE boolean value.
Examples	TRIGGER:A:BUS:B4:SPMI:NORESPONSE TRUE set the bus B4 to look for No Response frames. TRIGGER:A:BUS:B1:SPMI:NORESPONSE? might return :TRIGGER:A:BUS:B1:SPMI:NORESPONSE FALSE to indicate that bus B4 is not looking for No Response frames.

TRIGger:{A|B}:BUS:B<x>:SPMI:REGISTERADDReSS:VALue

This command sets or queries the binary data string that identifies the register address used in SPMI triggering if the trigger condition is MASTERREAd, MASTERWRItE, REGREAd, REGWRItE, EXTREGREAd, EXTREGWRItE, LONGEXTREGREAd, or LONGEXTREGWRItE.

Conditions	Requires option SRPM
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:SPMI:REGISTERADDReSS:VALue <Qstring> TRIGger:{A B}:BUS:B<x>:SPMI:REGISTERADDReSS:VALue?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:SPMI:CONDition
Arguments	Bus<x> is the bus identifier number. <Qstring> is the binary data string that identifies the register address.

Examples `TRIGGER:A:BUS:B2:SPMI:REGISTERADDRESS:VALUE "XXXXXXX"` sets the binary register address of bus B2 to "don't care."

`TRIGGER:A:BUS:B1:SPMI:REGISTERADDRESS:VALUE?` might return `TRIGGER:A:BUS:B1:SPMI:REGISTERADDRESS:VALUE "XXXX1111"` to indicate the binary register address on which to trigger for bus B1 is XXXX1111.

TRIGger:{A|B}:BUS:B<x>:SPMI:SLAVEADDRESS:VALue

This command sets or queries the binary data string that identifies the slave address used in SPMI triggering if the trigger condition is RESet, SLEEp, SHUTdown, WAKeup, AUTHenticate, REGREAd, REGWRItE, EXTREGREAd, EXTREGWRItE, LONGEXTREGREAd, LONGEXTREGWRItE, DEVIDESCSLAVEREAd, or REG0WRItE.

Conditions Requires option SRPM

Group Trigger

Syntax `TRIGger:{A|B}:BUS:B<x>:SPMI:SLAVEADDRESS:VALue <Qstring>`
`TRIGger:{A|B}:BUS:B<x>:SPMI:SLAVEADDRESS:VALue?`

Related Commands [TRIGger:{A|B}:BUS:B<x>:SPMI:CONDition](#)

Arguments Bus<x> is the bus identifier number.

<Qstring> is the binary data string that identifies the slave address on which to trigger.

Examples `TRIGGER:A:BUS:B1:SPMI:SLAVEADDRESS:VALUE "0011"` sets the binary slave address of bus B1 to 0011.

`:TRIGGER:A:BUS:B4:SPMI:SLAVEADDRESS:VALUE?` might return `TRIGGER:A:BUS:B4:SPMI:SLAVEADDRESS:VALUE "1010"` to indicate that bus B4 is triggering on binary slave address1010.

TRIGger:{A|B}:BUS:B<x>:USB:ADDRESS:HIVALue

This command specifies the binary address string for the upper limit for inside-of-range and outside-of-range qualifiers for the USB trigger. Use the command [TRIGger:{A|B}:BUS:B<x>:USB:ADDRESS:VALue](#) to set the lower limit. The bus number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:USB:ADDRESS:HIVALue <QString> TRIGger:{A B}:BUS:B<x>:USB:ADDRESS:HIVALue?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:USB:CONDition TRIGger:{A B}:BUS:B<x>:USB:ADDRes:VALue
Arguments	B<x> is the bus identifier number. <QString> within the range 0000000 to 1111111 (00 hex to 7F hex).
Examples	TRIGGER:A:BUS:B1:USB:ADDRESS:HIVALUE "0001000" sets the upper limit to binary 0001000 (08 hex). TRIGGER:A:BUS:B1:USB:ADDRESS:HIVALUE? might return :TRIGGER:A:BUS:B1:USB:ADDRESS:HIVALUE "1111111", which indicates that the upper limit is 1111111 (7F hex).

TRIGger:{A|B}:BUS:B<x>:USB:ADDRes:VALue

This command specifies the binary address string to be used for USB trigger.

The trigger condition must be set to TOKEN. The bus number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:USB:ADDRESS:VALue <QString> TRIGger:{A B}:BUS:B<x>:USB:ADDRESS:VALue?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:USB:CONDition

Arguments	B<x> is the bus identifier number. <QString> within the range 0000000 to 1111111 (00 hex to 7F hex).
Examples	TRIGGER:A:BUS:B1:USB:ADDRESS:VALUE "0001000" sets the binary address to 0001000 (08 hex). TRIGGER:A:BUS:B1:USB:ADDRESS:VALUE? might return :TRIGGER:A:BUS:B1:USB:ADDRESS:VALUE "1000000", which indicates that the binary address is 100000 (40 hex).

TRIGger:{A|B}:BUS:B<x>:USB:CONDition

This command specifies the trigger condition for the USB trigger. The bus number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:USB:CONDition {SYNC RESET SUSPEND RESUME EOP TOKENPacket DATAPacket HANDSHAKEPacket SPECIALPacket ERROR} TRIGger:{A B}:BUS:B<x>:USB:CONDition?
Arguments	SYNC indicates triggering on a Sync field of a packet. RESET sets triggering on a reset condition. SUSPEND sets triggering on a suspend condition. RESUME sets triggering on a resume condition. EOP indicates triggering on an end-of-packet signal. TOKENPacket indicates triggering on a token packet. DATAPacket indicates triggering on a data packet HANDSHAKEPacket indicates triggering on a handshake packet. SPECIALPacket indicates triggering on a special status packet. ERROR indicates triggering on an error condition.

Examples `TRIGGER:A:BUS:B1:USB:CONDITION TOKENPACKET` sets the trigger condition to be a token packet.

`TRIGGER:A:BUS:B1:USB:CONDITION?` might return
`:TRIGGER:A:BUS:B1:USB:CONDITION SYNC`, which indicates that the trigger condition is a sync field.

TRIGger:{A|B}:BUS:B<x>:USB:DATA:HIVALue

This command specifies the binary data string for the upper limit for inside-of-range and outside-of-range qualifiers for the USB trigger when the trigger condition is DATA. Use the command [TRIGger:{A|B}:BUS:B<x>:USB:DATA:VALue](#) to set the lower limit. The bus number is specified by <x>.

Conditions Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.

Group Trigger

Syntax `TRIGger:{A|B}:BUS:B<x>:USB:DATA:HIVALue <QString>`
`TRIGger:{A|B}:BUS:B<x>:USB:DATA:HIVALue?`

Related Commands [TRIGger:{A|B}:BUS:B<x>:USB:CONDition](#)
[TRIGger:{A|B}:BUS:B<x>:USB:DATA:VALue](#)

Arguments <QString> within the range 00000000 to 11111111 (00 hex to FF hex).

Examples `TRIGGER:A:BUS:B1:USB:DATA:HIVALUE "00001000"` sets the upper limit to binary 00001000 (08 hex).

`TRIGGER:A:BUS:B1:USB:DATA:HIVALUE?` might return
`:TRIGGER:A:BUS:B1:USB:DATA:HIVALUE "01111111"`, which indicates that the upper limit is 01111111 (7F hex).

TRIGger:{A|B}:BUS:B<x>:USB:DATA:OFFSet

This command specifies the data offset in bytes to trigger on. The bus number is specified by <x>.

Conditions Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.

Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:USB:DATA:OFFSet <NR1> TRIGger:{A B}:BUS:B<x>:USB:DATA:OFFSet?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:USB:CONDition
Arguments	<NR1> is number in the range 0 to 1024.
Examples	TRIGGER:A:BUS:B1:USB:DATA:OFFSET 36 sets the data offset to 36. TRIGGER:A:BUS:B1:USB:DATA:OFFSET? might return :TRIGGER:A:BUS:B1:USB:DATA:OFFSET 0, indicating that the data offset is the default value, 0.

TRIGger:{A|B}:BUS:B<x>:USB:DATA:QUALifier

This command sets the qualifier to be used when triggering on a USB bus signal. The trigger condition must be set to DATAPACKET. The bus number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:USB:DATA:QUALifier {Equal UNEQual LESSthan MOREthan LESSEQual MOREEqual INrange OUTrange} TRIGger:{A B}:BUS:B<x>:USB:DATA:QUALifier?</pre>
Arguments	EQUa1 specifies the qualifier as Equal. INrange specifies the qualifier as Inside Range. LESSEQUa1 specifies the qualifier as Less Than or Equal to. MOREEQUa1 specifies the qualifier as More Than or Equal to. OUTrange specifies the qualifier as Out of Range. UNEQUa1 specifies the qualifier as Not Equal to. LESSthan specifies the qualifier as Less Than.

MORE specifies the qualifier as More Than.

Examples **TRIGger:A:BUS:B1:USB:DATA:QUALifier LESSTHAN** sets the qualifier to less than.

TRIGger:A:BUS:B1:USB:DATA:QUALifier? might return
:TRIGGER:A:BUS:B1:USB:DATA:QUALIFIER EQUAL indicating the qualifier is set to equal.

TRIGger:{A|B}:BUS:B<x>:USB:DATA:SIZE

This command specifies the number of contiguous data bytes to trigger on. The bus number is specified by <x>.

Conditions Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.

Group Trigger

Syntax **TRIGger:{A|B}:BUS:B<x>:USB:DATA:SIZE <NR1>**
TRIGger:{A|B}:BUS:B<x>:USB:DATA:SIZE?

Arguments <NR1> is number in the range 1 to 16.

Examples **TRIGGER:A:BUS:B1:USB:DATA:SIZE 4** sets the oscilloscope to trigger on four contiguous data bytes.

TRIGGER:A:BUS:B1:USB:DATA:SIZE? might return
:TRIGGER:A:BUS:B1:USB:DATA:SIZE 6 indicating that the oscilloscope will trigger on 6 contiguous data bytes.

TRIGger:{A|B}:BUS:B<x>:USB:DATA:TYPE

This command specifies the data type for a USB trigger. The bus number is specified by <x>.

Conditions Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.

Group Trigger

Syntax	<pre>TRIGger:{A B}:BUS:B<x>:USB:DATA:TYPE {ANY DATA0 DATA1 DATA2 MDATA} TRIGger:{A B}:BUS:B<x>:USB:DATA:TYPE?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:USB:CONDition
Arguments	ANY indicates either a DATA0 or DATA1 data packet type. DATA0 indicates a DATA0 data packet type. DATA1 indicates a DATA1 data packet type. DATA2 indicates a DATA2 data packet type when on HIGH speed. MDATA indicates a MDATA data packet type when on HIGH speed.
Examples	TRIGGER:A:BUS:B1:USB:DATA:TYPE DATA0 sets the oscilloscope to trigger on a DATA0 data packet type. TRIGGER:A:BUS:B1:USB:DATA:TYPE? might return :TRIGGER:A:BUS:B1:USB:DATA:TYPE DATA1, indicating that the data packet type is set to DATA1.

TRIGger:{A|B}:BUS:B<x>:USB:DATA:VALue

This command specifies the binary data string to be used when triggering on a USB trigger.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:USB:DATA:VALue <QString> TRIGger:{A B}:BUS:B<x>:USB:DATA:VALue?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:USB:CONDition
Arguments	<QString> within the range 00000000 to 11111111 (00 hex to FF hex).
Examples	TRIGGER:A:BUS:B1:USB:DATA:VALUE "00001000" sets the binary address to 00001000 (08 hex).

TRIGGER:A:BUS:B1:USB:DATA:VALUE? might return
:TRIGGER:A:BUS:B1:USB:DATA:VALUE "01000000", which
indicates that the binary address is 0100000 (40 hex).

TRIGger:{A|B}:BUS:B<x>:USB:ENDPoint:VALue

This command specifies the binary endpoint string to be used for the USB trigger.
The bus number is specified by <x>.

Conditions Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:USB:ENDPoint:VALue <QString>
TRIGger:{A|B}:BUS:B<x>:USB:ENDPoint:VALue?

Related Commands [TRIGger:{A|B}:BUS:B<x>:USB:CONDition](#)

Arguments <QString> within the range 0000 to 1111 (00 hex to 0F hex).

Examples TRIGGER:A:BUS:B1:USB:ENDPOINT:VALUE "1000" sets the binary address to 1000 (08 hex).

TRIGGER:A:BUS:B1:USB:ENDPOINT:VALUE? might return
:TRIGGER:A:BUS:B1:USB:ENDPOINT:VALUE "1001", which indicates that
the binary address is 1001 (09 hex).

TRIGger:{A|B}:BUS:B<x>:USB:ERRType

This command specifies the error type to be used when the trigger condition is set to ERRor. The bus number is specified by <x>.

Conditions Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.

Group Trigger

Syntax	TRIGger: {A B}:BUS:B<x>:USB:ERRType {PID CRC5 CRC16 BITSTUFFing} TRIGger: {A B}:BUS:B<x>:USB:ERRType?
Related Commands	TRIGger: {A B}:BUS:B<x>:USB:CONDition
Arguments	PID indicates the error type is set to packet ID. CRC5 indicates the error type is set to 5-bit CRC. CRC16 indicates the error type is set to 16-bit CRC. BITSTUFFing indicates the error type is set to bit stuffing.
Examples	TRIGGER:A:BUS:B1:USB:ERRTYPE PID sets the error trigger condition to packet ID. TRIGGER:A:BUS:B1:USB:ERRTYPE? might return :TRIGGER:A:BUS:B1:USB:TYPE PID, indicating the error trigger condition is packet ID.

TRIGger:{A|B}:BUS:B<x>:USB:HANDSHAKEType

This command specifies the handshake type for the USB trigger. The bus number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger: {A B}:BUS:B<x>:USB:HANDSHAKEType {ANY NAK ACK STALL NYET} TRIGger: {A B}:BUS:B<x>:USB:HANDSHAKEType?
Related Commands	TRIGger: {A B}:BUS:B<x>:USB:CONDition
Arguments	ANY indicates the oscilloscope will trigger on any handshake type. NAK indicates the oscilloscope will trigger when a device cannot send or receive data. ACK indicates the oscilloscope will trigger when a packet is successfully received.

STALL indicates the oscilloscope will trigger when a device requires intervention from the host.

NYET specifies the handshake type as No response Yet (0110).

Examples TRIGGER:A:BUS:B1:USB:HANDSHAKETYPE ACK sets the handshake type to acknowledgement.

TRIGGER:A:BUS:B1:USB:HANDSHAKETYPE? might return
:TRIGGER:A:BUS:B1:USB:HANDSHAKETYPE STALL, indicating that the handshake type is set to stall.

TRIGger:{A|B}:BUS:B<x>:USB:SOFFRAMENUMber

This command specifies the binary data string to be used for start of frame number, when the trigger condition is Token Packet and the token type is Start of Frame. The bus number is specified by <x>.

Conditions Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:USB:SOFFRAMENUMber <QString>
TRIGger:{A|B}:BUS:B<x>:USB:SOFFRAMENUMber?

Related Commands [TRIGger:{A|B}:BUS:B<x>:USB:CONDition](#)

Arguments <QString> within the range 000 0000 0000 to 111 1111 1111 (000 hex to 7FF hex).

Examples TRIGGER:A:BUS:B1:USB:SOFFRAMENUMBER "00000001000" sets the start of frame number to 00000001000 (008 hex).

TRIGGER:A:BUS:B1:USB:SOFFRAMENUMBER? might return
:TRIGGER:A:BUS:B1:USB:SOFFRAMENUMBER "00000001001", which indicates that the start of frame number is 00000001001 (009 hex).

TRIGger:{A|B}:BUS:B<x>:USB:SPECIALType

This command specifies the packet ID (PID) for the special packet. The bus number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:USB:SPECIALType {ANY ERROR PING PREAmble RESERVED SPLit} TRIGger:{A B}:BUS:B<x>:USB:SPECIALType?</pre>
Related Commands	TRIGger:{A B}:BUS:B<x>:USB:CONDition
Arguments	ANY specifies the PID value as Any (XX00). ERROR specifies the PID value as ERR (1100). PING specifies the PID value as PING (0100). PREAmble specifies the PID value as PRE (1100). RESERVED specifies the PID value as Reserved (0000). SPLit specifies the PID value as Split (1000).
Examples	TRIGGER:A:BUS1:USB:SPECIALTYPE PREAMBLE sets the special packet type to preamble. TRIGGER:A:BUS1:USB:SPECIALTYPE? might return :TRIGGER:A:BUS1:USB:SPECIALTYPE PREAMBLE, indicating that the special type is set to preamble.

TRIGger:{A|B}:BUS:B<x>:USB:SPLit:ET:VALue

When triggering on a high-speed USB split transaction, this command specifies the split transaction endpoint type value to trigger on. The bus number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Trigger
Syntax	<pre>TRIGger:{A B}:BUS:B<x>:USB:SPLit:ET:VALue {NOCARE CONTRol ISochronous BULK INTERRUPT} TRIGger:{A B}:BUS:B<x>:USB:SPLit:ET:VALue?</pre>

Related Commands [TRIGger:{A|B}:BUS:B<x>:USB:SPECIALType](#)

Arguments NOCARE — any endpoint type.
 CONTRo1 — control endpoint type.
 ISochronous — isochronous endpoint type.
 BULK — bulk endpoint type (BULK-IN or BULK-OUT).
 INTERRUPT — interrupt endpoint type (Interrupt-IN).

Examples TRIGger:A:BUS:B1:USB:SPLit:ET:VALue iso sets the endpoint type to trigger on to isochronous.
 TRIGger:A:BUS:B1:USB:SPLit:ET:VALue? might return BULK, indicating that the bulk endpoint type has been specified to trigger on.

TRIGger:{A|B}:BUS:B<x>:USB:SPLit:HUB:VALue

When triggering on a high-speed USB split transaction, this command specifies the split transaction hub address value to trigger on. The trigger condition must be set to Special with packet type SPLIT. The value can be up to 7 characters long. The default is all X's (don't care). The bus number is specified by <x>.

Conditions Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:USB:SPLit:HUB:VALue <QString>
 TRIGger:{A|B}:BUS:B<x>:USB:SPLit:HUB:VALue?

Related Commands [TRIGger:{A|B}:BUS:B<x>:USB:SPECIALType](#)

Arguments QString is a quoted string of up to 7 characters. The valid characters are 0 and 1.

Examples TRIGger:A:BUS:B1:USB:SPLit:HUB:VALue "001010" sets the split transaction hub address to trigger on to 001010.
 TRIGger:A:BUS:B1:USB:SPLit:HUB:VALue? might return XXXXXXXX, indicating that the hub address value to trigger on doesn't matter.

TRIGger:{A|B}:BUS:B<x>:USB:SPLit:PORT:VALue

When triggering on a high-speed USB split transaction, this command specifies the split transaction port address value to trigger on. The trigger condition must be set to Special with a packet type SPLIT. The value can be up to 7 characters long. The default is all X's (don't care). The bus number is specified by <x>.

Conditions Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:USB:SPLit:PORT:VALue <QString>
TRIGger:{A|B}:BUS:B<x>:USB:SPLit:PORT:VALue?

Related Commands [TRIGger:{A|B}:BUS:B<x>:USB:SPECIALType](#)

Arguments QString is a quoted string of up to 7 characters. The valid characters are 0 and 1.

Examples TRIGger:A:BUS:B1:USB:SPLit:PORT:VALue "001010" sets the split transaction hub address to trigger on to 001010.

TRIGger:A:BUS:B1:USB:SPLit:PORT:VALue? might return XXXXXXXX, indicating that the hub address value to trigger on doesn't matter.

TRIGger:{A|B}:BUS:B<x>:USB:SPLit:SC:VALue

When triggering on a high-speed USB split transaction, this command specifies whether to trigger on the start or complete phase of the split transaction, based on the Start/Complete bit field value. (0 = Start, 1 = Complete). The default is NOCARE. The bus number is specified by <x>.

Conditions Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.

Group Trigger

Syntax TRIGger:{A|B}:BUS:B<x>:USB:SPLit:SC:VALue
{NOCARE|SSPLIT|CSPLIT}
TRIGger:{A|B}:BUS:B<x>:USB:SPLit:SC:VALue?

Related Commands [TRIGger:{A|B}:BUS:B<x>:USB:SPECIALType](#)

Arguments NOCARE — trigger on either the start or complete phase of the split transaction.
 SSPLIT — trigger on the start phase of the split transaction.
 CSPLIT — trigger on the complete phase of the split transaction.

Examples TRIGger:A:BUS:B1:USB:SPLit:SC:VALue SSPLIT specifies to trigger on the start phase of the split transaction.
 TRIGger:A:BUS:B1:USB:SPLit:SC:VALue? might indicate NOCARE, specifying that it doesn't matter whether to trigger on the start or complete phase of the split transaction.

TRIGger:{A|B}:BUS:B<x>:USB:SPLit:SE:VALue

When triggering on a high-speed USB split transaction, this command specifies the split transaction start/end bit value to trigger on. The bus number is specified by <x>.

NOTE. *The start and end bits are interpreted based on the type of split transaction:*

For Interrupt and control transactions, the S bit means Speed: 0 = Full Speed, 1 = Low Speed.

For bulk IN/OUT and isochronous IN start-split transactions, the S field must be 0.

For bulk/control IN/OUT, interrupt IN/OUT, and isochronous IN start-split transactions, the E field must be 0.

For full-speed isochronous OUT start-split transactions, the S (Start) and E (End) fields specify how the high-speed data payload corresponds to data for a full-speed data packet as shown below:

S E High-speed to Full-speed Data Relation

0 0 High-speed data is the middle of the full-speed data payload.

0 1 High-speed data is the end of the full-speed data payload.

1 0 High-speed data is the beginning of the full-speed data payload.

1 1 High-speed data is all of the full speed data payload.

Conditions Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.

Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:USB:SPLit:SE:VALue {NOCARE FULLSPEED LOWSPEED ISOSTART ISOMID ISOEND ISOALL} TRIGger:{A B}:BUS:B<x>:USB:SPLit:SE:VALue?
Related Commands	TRIGger:{A B}:BUS:B<x>:USB:SPECIALType
Arguments	NOCARE — any combination of S and E bits. FULLSPEED — S bit = 0, E bit = 0. LOWSPEED — S bit = 1, E bit = 0. ISOSTART — S bit = 1, E bit = 0. ISOMID — see note above. ISOEND — see note above. ISOALL — see note above.
Examples	TRIGger:A:BUS:B1:USB:SPLit:SE:VALue ISOEND specifies to trigger on the ISOEND split transaction value. TRIGger:A:BUS:B1:USB:SPLit:SE:VALue? might return FULLSPEED.

TRIGger:{A|B}:BUS:B<x>:USB:TOKENType

This command specifies the token type for the USB trigger. The bus number is specified by <x>.

Conditions	Requires option 5-SRUSB2 or SUP5-SRUSB2 Triggering and Analysis application.
Group	Trigger
Syntax	TRIGger:{A B}:BUS:B<x>:USB:TOKENType {ANY SOF OUT IN SETUP} TRIGger:{A B}:BUS:B<x>:USB:TOKENType?
Related Commands	TRIGger:{A B}:BUS:B<x>:USB:CONDition

Arguments	ANY indicates any of the token types. SOF indicates a SOF (start-of-frame) token type OUT indicates an OUT token type. IN indicates an IN token type. SETUP indicates a SETUP token type.
Examples	TRIGGER:A:BUS:B1:USB:TOKENTYPE SETUP sets the token type to SETUP. TRIGGER:A:BUS:B1:USB:TOKENTYPE? might return :TRIGGER:A:BUS:B1:USB:TOKENTYPE SOF if the token type is SOF.

TRIGger:{A|B}:BUS:SOUrce

This command sets or queries the source bus for a bus trigger.

Group	Trigger
Syntax	TRIGger:{A B}:BUS:SOUrce B<x> TRIGger:{A B}:BUS:SOUrce?
Arguments	B<x> sets the selected source to the bus.
Examples	TRIGGER:A:BUS:SOURCE B2 sets the selected source for the bus trigger to Bus 2. TRIGGER:A:BUS:SOURCE? might return TRIGGER:A:BUS:SOURCE B1, indicating that the selected source for the bus trigger is set to Bus 1.

TRIGger:{A|B}:EDGE:COUPling

This command sets or queries the type of coupling for the edge trigger. This command is equivalent to selecting Edge from the Trigger Type drop-down in the Trigger setup context menu, and choosing from the Coupling drop-down list.

Group	Trigger
Syntax	TRIGger:{A B}:EDGE:COUPling {DC HFRej LFRej NOISerej} TRIGger:{A B}:EDGE:COUPling?

Related Commands	TRIGger:{A B}:EDGE:SOUrce TRIGger:{A B}:EDGE:SLOpe
Arguments	DC selects DC trigger coupling, which passes all input signals to the trigger circuitry. HFR ej coupling attenuates signals above 50 kHz before passing the signals to the trigger circuitry. LFR ej coupling attenuates signals below 80 kHz before passing the signals to the trigger circuitry. NOISER ej coupling provides stable triggering by increasing the trigger hysteresis. Increased hysteresis reduces the trigger sensitivity to noise but can require greater trigger signal amplitude.
Examples	TRIGGER:A:EDGE:COUPLING DC sets the A edge trigger coupling to DC. TRIGGER:A:EDGE:COUPLING? might return :TRIGGER:A:EDGE:COUPLING DC, indicating that the A edge trigger coupling is set to DC.

TRIGger:{A|B}:EDGE:SLOpe

This command sets or queries the slope for the edge trigger. This command is equivalent to selecting Edge from the Trigger Type drop-down in the Trigger setup context menu, and then choosing the desired Slope. This command is also equivalent to pressing the front-panel Slope button.

Group	Trigger
Syntax	TRIGger:{A B}:EDGE:SLOpe {RISe FALL EITHer} TRIGger:{A B}:EDGE:SLOpe?
Related Commands	TRIGger:{A B}:EDGE:SOUrce TRIGger:{A B}:EDGE:COUPling TRIGger:B:STATE
Arguments	RISe specifies to trigger on the rising or positive edge of a signal. FALL specifies to trigger on the falling or negative edge of a signal. EITHER specifies to trigger on either the rising or falling edge of a signal.

Examples `TRIGGER:A:EDGE:SLOPE RISE` sets the A edge trigger slope to positive, which triggers on the rising edge of the signal.

`TRIGGER:A:EDGE:SLOPE?` might return `:TRIGGER:A:EDGE:SLOPE FALL`, indicating that the A edge trigger slope is negative.

TRIGger:{A|B}:EDGE:SOURce

This command sets or queries the source for the edge trigger. For instruments that have an Auxiliary Input (such as the MSO58LP), AUXiliary can be selected as trigger source.

Group Trigger

Syntax `TRIGger:{A|B}:EDGE:SOURce {CH<x>|CH<x>_D<y>|LINE|AUXiliary}`
`TRIGger:{A|B}:EDGE:SOURce?`

Related Commands [TRIGger:{A|B}:EDGE:SLOpe](#)
[TRIGger:{A|B}:EDGE:COUPling](#)
[TRIGger:B:STATE](#)

Arguments `CH<x>` specifies an analog channel as the edge trigger source.
`CH<x>_D<y>` specifies a digital channel as the edge trigger source.
`LINE` specifies AC line voltage, and is a valid source when B trigger is inactive.
`AUXiliary` specifies the Auxiliary Input.

Examples `TRIGGER:A:EDGE:SOURCE CH2` sets the A edge trigger source to input channel 2.

`TRIGGER:A:EDGE:SOURCE?` might return `:TRIGGER:A:EDGE:SOURCE CH1`, indicating that the A edge trigger source is set to input channel 1.

TRIGger:{A|B}:LEVel:CH<x>

This command sets or queries the CH<x> trigger level for an Edge, Pulse Width, Runt or Rise/Fall (Transition and Slew Rate) trigger when triggering on an analog channel waveform. Each channel can have an independent trigger level. The <x> is the channel number.

Group	Trigger
Syntax	<pre>TRIGger:{A B}:LEVel:CH<x> <NR3> TRIGger:{A B}:LEVel:CH<x>?</pre>
Arguments	<NR3> specifies the trigger level in user units (usually volts).
Examples	<pre>TRIGGER:A:LEVEL:CH1 1.5</pre> sets the A trigger level for Channel 1 to 1.5 V. <pre>TRIGGER:A:LEVEL:CH2?</pre> might return <pre>:TRIGGER:A:LEVEL:CH2</pre> <pre>1.3000E+00</pre> , indicating that the A trigger level for Channel 2 is set to 1.3 V.

TRIGger:{A|B}:LOGIc:DELTatime

This command specifies or queries the Logic trigger delta time value. The time value is used as part of the Logic trigger condition to determine if the duration of a logic pattern meets the specified time constraints.

Group	Trigger
Syntax	<pre>TRIGger:{A B}:LOGIC:DELTatime <NR3> TRIGger:{A B}:LOGIC:DELTatime?</pre>
Arguments	<NR3> the Logic trigger delta time value.
Examples	<pre>TRIGger:A:LOGIC:DELTatime 4.5e-9</pre> sets the delta time to 4.5 ns. <pre>TRIGger:A:LOGIC:DELTatime?</pre> might return <pre>:TRIGGER:A:LOGIC:DELTATIME 4.0E-9</pre> indicating the delta time is set to 4 ns.

TRIGger:{A|B}:LOGIc:FUNCtion

This command sets or queries the logical combination of the input channels for logic triggers. This command is equivalent to selecting Logic for the Trigger Type, and setting or viewing the Define Logic.

Group	Trigger
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Syntax TRIGGER:{A|B}:LOGIC:FUNCTION {AND|NAND|NOR|OR}
TRIGGER:{A|B}:LOGIC:FUNCTION?

Arguments AND specifies to trigger if all conditions are true.
NAND specifies to trigger if any of the conditions are false.
NOR specifies to trigger if all conditions are false.
OR specifies to trigger if any of the conditions are true.

Examples TRIGGER:A:LOGIC:FUNCTION AND sets the logical combination of channels to be true when all conditions are true.
TRIGGER:A:LOGIC:FUNCTION? might return :TRIGGER:A:LOGIC:FUNCTION NAND, indicating that the instrument will trigger if the AND logic conditions are false.

TRIGGER:{A|B}:LOGIC:INPUT:CLOCK:SOURCE

This command specifies the channel to use as the clock source for logic trigger.

Group Trigger

Syntax TRIGGER:{A|B}:LOGIC:INPUT:CLOCK:SOURCE {CH<x>|CH<x>_D<y>}
TRIGGER:{A|B}:LOGIC:INPUT:CLOCK:SOURCE?

Arguments CH<x> specifies an analog channel as the clock source. Number of channels depends on instrument configuration.
CH<x>_D<y> specifies a digital channel as the clock source. Number of channels depends on instrument configuration.

Examples TRIGGER:A:LOGIC:INPUT:CLOCK:SOURCE CH3 sets the clock source to channel 3.
TRIGGER:A:LOGIC:INPUT:CLOCK:SOURCE? might return TRIGGER:A:LOGIC:INPUT:CLOCK:SOURCE CH3 indicating the clock source is set to channel 3.

TRIGger:{A|B}:LOGic:POLarity

This command sets or queries the polarity for the clock channel when Use Clock Edge is set to Yes for Logic trigger type.

Group	Trigger
Syntax	TRIGger:{A B}:LOGic:POLarity {POSitive NEGative EITher} TRIGger:{A B}:LOGic:POLarity?
Arguments	NEGative specifies negative polarity. POSITIVE specifies positive polarity. EITher specifies either polarity.
Examples	TRIGger:A:LOGic:POLarity EITHER sets the polarity to either positive or negative. TRIGger:A:LOGic:POLarity? might return :TRIGGER:A:LOGIC:POLARITY POSITIVE indicating the polarity is set to positive.

TRIGger:{A|B}:LOGic:USECLockededge

This command specifies whether or not Logic trigger type uses clock source.

Group	Trigger
Syntax	TRIGger:{A B}:LOGic:USECLockededge {OFF ON 0 1} TRIGger:{A B}:LOGic:USECLockededge?
Arguments	ON specifies that logic trigger type uses clock source. OFF specifies that logic trigger type does not use clock source. <NR1> = 0 specifies that logic trigger type does not use clock source; any other value uses clock source.
Examples	TRIGger:A:LOGic:USECLockededge OFF specifies that the clock edge will not be used.

TRIGger:A:LOGIC:USEClockedge? might return
:TRIGGER:A:LOGIC:USECLOCKEDGE 1 indicating that the clock edge will be used.

TRIGger:{A|B}:LOGIc:WHEn

This command sets or queries the condition for generating an A or B logic trigger with respect to the defined input pattern. This command is equivalent to selecting Logic for Trigger Type, Use Clock Edge to No, and choosing a trigger condition from the Logic Pattern drop-down list.

Group	Trigger
Syntax	TRIGger:{A B}:LOGIC:WHEn {TRUE FALSE MOREthan LESSThan EQua1 UNEQua1} TRIGger:{A B}:LOGIC:WHEn?
Arguments	TRUE triggers on an input pattern that is true. FALSE triggers on an input pattern that is false. MOREthan triggers on an input pattern that is true for a time period greater than a user defined Time Limit (DELTatime) value. LESSThan triggers on an input pattern that is true for a time period less than a user defined Time Limit (DELTatime) value. EQua1 triggers on an input pattern that is true for a time period equal to a user defined Time Limit (DELTatime) value. UNEQua1 triggers on an input pattern that is true for a time period not equal to a user defined Time Limit (DELTatime) value.
Examples	TRIGger:A:LOGIC:WHEn EQua1 specifies triggering when the input pattern is true for a time period equal to a user defined Time Limit value. TRIGger:A:LOGIC:WHEn? might return :TRIGGER:A:LOGIC:WHEn TRUE indicating a trigger when the logic is true.

TRIGger:{A|B}:LOGICPattern:{CH<x>|CH<x>_D<x>}

This command sets or queries the Logic Pattern that is used along with the Define Logic choice (LOGIc:FUNCTion) to determine when the logic trigger occurs.

Group	Trigger
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Syntax	<pre>TRIGger:{A B}:LOGICPattern:{CH<x> CH<x>_D<x>} {HIGH LOW X} TRIGger:{A B}:LOGICPattern:{CH<x> CH<x>_D<x>}?</pre>
Arguments	HIGH specifies the logic high. LOW specifies the logic low. X specifies a don't care state.
Examples	<pre>TRIGger:A:LOGICPattern:CH1 HIGH specifies triggering on a logic high. TRIGger:A:LOGICPattern:CH1? might return :TRIGGER:A:LOGICPATTERN:CH1 X indicating a don't care state for channel 1. TRIGger:A:LOGICPattern:CH1_D0 HIGH specifies a logic high. TRIGger:A:LOGICPattern:CH1_D0? might return :TRIGGER:A:LOGICPATTERN:CH1_D0 X indicating a don't care.</pre>

TRIGger:{A|B}:LOWerthreshold:CH<x>

This command sets or queries the A or B lower trigger level threshold for the channel, specified by x.

Group	Trigger
Syntax	<pre>TRIGger:{A B}:LOWerthreshold:CH<x> <NR3> TRIGger:{A B}:LOWerthreshold:CH<x>?</pre>
Related Commands	TRIGger:{A B}:UPPerthreshold:CH<x>
Arguments	<NR3> specifies the threshold voltage in user units.
Examples	<pre>TRIGGER:A:LOWERTHRESHOLD:CH2 1.3 sets the A trigger threshold voltage for Channel 2 to 1.3 V. TRIGGER:A:LOWERTHRESHOLD:CH3? might return TRIGGER:A:LOWERTHRESHOLD:CH3 1.2000E+00, indicating that the A trigger threshold voltage for Channel 3 is 1.2 V.</pre>

TRIGger:{A|B}:PULSEWidth:HIGHLimit

This command specifies the upper limit to use, in seconds, when triggering on detection of a pulse whose duration is inside or outside a range of two values. (Use [TRIGger:{A|B}:PULSEWidth:LOWLimit](#) to specify the lower value of the range.)

Group	Trigger
Syntax	<pre>TRIGger:{A B}:PULSEwidth:HIGHLimit <NR3> TRIGger:{A B}:PULSEwidth:HIGHLimit?</pre>
Related Commands	TRIGger:{A B}:PULSEWidth:WHEn TRIGger:{A B}:PULSEWidth:LOWLimit
Arguments	<NR3> is a floating point number that represents the higher value of the range.
Examples	<pre>TRIGger:A:PULSEwidth:HIGHLimit 110.0E-9</pre> sets the high limit to 110.0 ns. <pre>TRIGger:A:PULSEwidth:HIGHLimit?</pre> might return <pre>TRIGGER:A:PULSEWIDTH:HIGHLIMIT 178.88000E-9</pre> indicates the high limit is set to 178.88 ns.

TRIGger:{A|B}:PULSEWidth:LOGICQUALification

This command specifies whether or not to use logic qualification for a pulse width trigger.

Group	Trigger
Syntax	<pre>TRIGger:{A B}:PULSEwidth:LOGICQUALification {ON OFF} TRIGger:{A B}:PULSEwidth:LOGICQUALification?</pre>
Arguments	ON specifies that the pulse width trigger type uses logic qualification. OFF specifies that the pulse width trigger type does not use logic qualification.
Examples	<pre>TRIGger:A:PULSEwidth:LOGICQUALification ON</pre> turns on logic qualification.

TRIGger:A:PULSEwidth:LOGICQUALification? might return
TRIGGER:A:PULSEWIDTH:LOGICQUALIFICATION OFF indicating logic
qualification is off.

TRIGger:{A|B}:PULSEwidth:LOWLimit

This command specifies the lower limit to use, in seconds, when triggering on detection of a pulse whose duration is inside or outside a range of two values. (Use [TRIGger:{A|B}:PULSEwidth:HIGHLimit](#) to specify the upper limit of the range.)

This command also specifies the single limit to use, in seconds, when triggering on detection of a pulse whose duration is less than, greater than, equal to, or not equal to this time limit.

Group Trigger

Syntax TRIGger:{A|B}:PULSEwidth:LOWLimit <NR3>
TRIGger:{A|B}:PULSEwidth:LOWLimit?

Related Commands [TRIGger:{A|B}:PULSEwidth:WHEn](#)
[TRIGger:{A|B}:PULSEwidth:HIGHLimit](#)

Arguments <NR3> is a floating point number that represents the lower value of the range.

Examples TRIGger:A:PULSEwidth:LOWLimit 100.0E-9 sets the low limit to 100.0 ns.
TRIGger:A:PULSEwidth:HIGHLimit? might return
TRIGGER:A:PULSEWIDTH:LOWLIMIT 77.7600E-9 indicating the
low limit is set to 77.76 ns.

TRIGger:{A|B}:PULSEwidth:POLarity

This command specifies the polarity for a pulse width trigger.

Group Trigger

Syntax TRIGger:{A|B}:PULSEwidth:POLarity {NEGative|POSitive}
TRIGger:{A|B}:PULSEwidth:POLarity?

Arguments	NEGative specifies a negative pulse. POSitive specifies a positive pulse.
Examples	TRIGger:A:PULSEwidth:POLarity NEGATIVE sets the pulse polarity to negative. TRIGger:A:PULSEwidth:POLarity? might return TRIGGER:A:WIDTH:POLARITY POSITIVE indicating a positive pulse.

TRIGger:{A|B}:PULSEWidth:SOURce

This command specifies the source waveform for a pulse width trigger.

Group	Trigger
Syntax	TRIGger:{A B}:PULSEwidth:SOURce {CH<x> CH<x>_D<y>} TRIGger:{A B}:PULSEwidth:SOURce?
Arguments	CH<x> specifies an analog input channel as the pulse-width trigger source. CH<x>_D, y> specifies an digital input channel as the pulse-width trigger source.
Examples	TRIGGER:A:PULSEWIDTH:SOURCE CH1 sets channel 1 as the pulse width source. TRIGGER:A:PULSEWIDTH:SOURCE? might return :TRIGGER:A:PULSEWIDTH:SOURCE CH1 indicating that channel 1 is the pulse width trigger source.

TRIGger:{A|B}:PULSEWidth:WHEn

This command specifies to trigger when a pulse is detected with a width (duration) that is less than, greater than, equal to, or unequal to a specified value (set using [TRIGger:{A|B}:PULSEWidth:LOWLimit](#)), OR whose width falls outside of or within a specified range of two values (set using [TRIGger:{A|B}:PULSEWidth:LOWLimit](#) and [TRIGger:{A|B}:PULSEWidth:HIGHLimit](#)).

Group	Trigger
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Syntax	TRIGger:{A B}:PULSEwidth:WHEN {LESSthan MOREthan Equal UNEQual WITHin OUTside} TRIGger:{A B}:PULSEwidth:WHEN?
Related Commands	TRIGger:{A B}:PULSEwidth:LOWLimit TRIGger:{A B}:PULSEwidth:HIGHLimit TRIGger:{A B}:PULSEwidth:SOUrce
Arguments	LESSthan causes a trigger when a pulse is detected with a width less than the time set by the TRIGger:{A B}:PULSEwidth:LOWLimit command. MOREthan causes a trigger when a pulse is detected with a width greater than the time set by the TRIGger:{A B}:PULSEwidth:LOWLimit command. Equal causes a trigger when a pulse is detected with a width equal to the time period specified in TRIGger:{A B}:PULSEwidth:LOWLimit within a $\pm 5\%$ tolerance. UNEQual causes a trigger when a pulse is detected with a width greater than or less than (but not equal) the time period specified in TRIGger:{A B}:PULSEwidth:LOWLimit within a $\pm 5\%$ tolerance. WITHin causes a trigger when a pulse is detected that is within a range set by two values. OUTside causes a trigger when a pulse is detected that is outside of a range set by two values.
Examples	:TRIGger:B:PULSEwidth:WHEN LESSthan causes the oscilloscope to trigger when a pulse is detected that is shorter than the time specified by TRIGger:{A B}:PULSEwidth:LOWLimit. TRIGger:A:PULSEwidth:WHEN ? might return TRIGGER:A:PULSEWIDTH:WHEN GREATER THAN 2.0000E-9, indicating that a trigger is generated when a pulse is detected greater than 2 ns. TRIGger:B:PULSEwidth:WHEN? might return TRIGger:B:PULSEwidth:WHEN MOREthan indicating that a trigger is generated when a pulse is detected that is greater than the time specified by TRIGger:{A B}:PULSEwidth:LOWLimit.

TRIGger:{A|B}:RUNT:LOGICQUALification

This command specifies whether or not to use logic qualification for a runt trigger.

Group Trigger

Syntax	TRIGger:{A B}:RUNT:LOGICQUALification {ON OFF} TRIGger:{A B}:RUNT:LOGICQUALification?
Arguments	ON specifies that the runt trigger type uses logic qualification. OFF specifies that the runt trigger type does not use logic qualification.
Examples	TRIGger:A:RUNT:LOGICQUALification ON turns on logic qualification. TRIGger:A:RUNT:LOGICQUALification? might return :TRIGGER:A:RUNT:LOGICQUALIFICATION OFF indicating logic qualification is off.

TRIGger:{A|B}:RUNT:POLarity

This command specifies the polarity for the runt trigger.

Group	Trigger
Syntax	TRIGger:{A B}:RUNT:POLarity {EITHer NEGative POSitive} TRIGger:{A B}:RUNT:POLarity?
Arguments	POSitive indicates that the rising edge crosses the low threshold and the falling edge recrosses the low threshold without either edge ever crossing the high threshold. NEGative indicates that the falling edge crosses the high threshold and the rising edge recrosses the high threshold without either edge ever crossing the low threshold. EITHer triggers on a runt of either polarity.
Examples	TRIGger:A:RUNT:POLarity NEGATIVE specifies that the polarity of the A pulse runt trigger is negative. TRIGger:A:RUNT:POLarity? might return :TRIGGER:A:RUNT:POLARITY POSITIVE indicating that the polarity of the A pulse runt trigger is positive.

TRIGger:{A|B}:RUNT:SOURce

This command specifies the source waveform for the runt trigger.

NOTE. *Digital channels are not supported as runt trigger sources.*

Group	Trigger
Syntax	<pre>TRIGger:{A B}:RUNT:SOURce {CH<x>} TRIGger:{A B}:RUNT:SOURce?</pre>
Arguments	CH<x> specifies the analog channel number to use as the source waveform for the runt trigger. To specify the threshold levels when using CH<x> as the source, use TRIGger:{A B}:LOWerthreshold:CH<x> and TRIGger:{A B}:UPPerthreshold:CH<x>.
Examples	TRIGger:A:RUNT:SOURce CH4 sets channel 4 as the trigger source. TRIGger:A:RUNT:SOURce? might return :TRIGGER:A:RUNT:SOURCE CH2 indicating that channel 2 is the trigger source.

TRIGger:{A|B}:RUNT:WHEn

This command specifies the type of pulse width the trigger checks for when it detects a runt.

Group	Trigger
Syntax	<pre>TRIGger:{A B}:RUNT:WHEn {LESSthan MOREthan EQua1 UNEQua1 OCCURS} TRIGger:{A B}:RUNT:WHEn?</pre>
Related Commands	TRIGger:{A B}:RUNT:WIDth
Arguments	OCCURS argument specifies a trigger event if a runt of any detectable width occurs. LESSthan argument sets the oscilloscope to trigger if the a runt pulse is detected with width less than the time set by the TRIGger:{A B}:RUNT:WIDth command. MOREthan argument sets the oscilloscope to trigger if the a runt pulse is detected with width greater than the time set by the TRIGger:{A B}:RUNT:WIDth command. EQua1 argument sets the oscilloscope to trigger if a runt pulse is detected with width equal to the time period specified in TRIGger:{A B}:RUNT:WIDth within a $\pm 5\%$ tolerance. NOTEQua1 argument sets the oscilloscope to trigger if a runt pulse is detected with width greater than or less than (but not equal to) the time period specified in TRIGger:{A B}:RUNT:WIDth within a $\pm 5\%$ tolerance.

Examples `TRIGger:A:RUNT:WHEN MORETHAN` sets the runt trigger to occur when the oscilloscope detects a runt in a pulse wider than the specified width.

`TRIGger:A:RUNT:WHEN?` might return `:TRIGGER:A:PULSE:RUNT:WHEN OCCURS` indicating that a runt trigger will occur if the oscilloscope detects a runt of any detectable width.

TRIGger:{A|B}:RUNT:WIDTH

This command specifies the width, in seconds, for a runt trigger.

Group Trigger

Syntax `TRIGger:{A|B}:RUNT:WIDTH <NR3>`
`TRIGger:{A|B}:RUNT:WIDTH?`

Related Commands [TRIGger:{A|B}:RUNT:WHEn](#)

Arguments `<NR3>` is a floating point number that specifies the minimum width, in seconds.

Examples `TRIGger:A:RUNT:WIDTH 15E-6` sets the minimum width of the pulse runt trigger to 15 μ s.

`TRIGger:A:RUNT:WIDTH?` might return `:TRIGGER:A:PULSE:RUNT:WIDTH 2.0000E-09` indicating that the minimum width of a pulse runt trigger is 2 ns.

TRIGger:{A|B}:SETHold:CLOCK:EDGE

This command specifies the clock edge polarity for setup and hold triggering.

Group Trigger

Syntax `TRIGger:{A|B}:SETHold:CLOCK:EDGE {FALL|RISe}`
`TRIGger:{A|B}:SETHold:CLOCK:EDGE?`

Arguments `FALL` specifies polarity as the clock falling edge.
`RISe` specifies polarity as the clock rising edge.

Examples `TRIGger:A:SETHold:CLOCK:EDGE RISE` specifies the polarity as the clock rising edge.

`TRIGger:A:SETHold:CLOCK:EDGE?` might return
`:TRIGGER:A:SETHOLD:CLOCK:EDGE RISE` indicating that polarity is specified as the clock rising edge.

TRIGger:{A|B}:SETHold:CLOCK:SOURCE

This command specifies the clock source for the setup and hold triggering. You cannot specify the same source for both clock and data.

Group Trigger

Syntax `TRIGger:{A|B}:SETHold:CLOCK:SOURCE {CH<x>|CH<x>_D<y>}`
`TRIGger:{A|B}:SETHold:CLOCK:SOURCE?`

Arguments `CH<x>` specifies the analog channel to use as the clock source waveform.
`CH<x>_D<y>` specifies the digital channel to use as the clock source waveform.

Examples `TRIGger:A:SETHold:CLOCK:SOURCE CH1` specifies channel 1 as the clock source for a setup and hold trigger operation.

`TRIGger:A:SETHold:CLOCK:SOURCE?` might return
`:TRIGGER:A:SETHOLD:CLOCK:SOURCE CH4` indicating that channel 4 is the clock source for a setup and hold trigger operation.

TRIGger:{A|B}:SETHold:HOLDTime

This command specifies the hold time for setup and hold violation triggering. This command is equivalent to selecting Setup/Hold Setup from the Trig menu and then setting the desired Hold Time.

Group Trigger

Syntax `TRIGger:{A|B}:SETHold:HOLDTime <NR3>`
`TRIGger:{A|B}:SETHold:HOLDTime?`

Arguments	<NR3> is a floating point number that specifies the hold time setting, in seconds. Positive values for hold time occur after the clock edge. Negative values occur before the clock edge.
Examples	TRIGGER:A:SETHold:HOLDTime 3.0E-3 sets the hold time for the setup and hold trigger to 3 ms. TRIGGER:A:SETHold:HOLDTime? might return :TRIGGER:A:SETHOLD:HOLDTIME 2.0000E-09 indicating that the current hold time for the setup and hold trigger is 2 ns.

TRIGGER:{A|B}:SETHold:SETTime

This command specifies the setup time for setup and hold violation triggering. This command is equivalent to selecting Setup/Hold Setup from the Trig menu and then setting the desired Setup Time.

Group	Trigger
Syntax	TRIGGER:{A B}:SETHold:SETTime <NR3> TRIGGER:{A B}:SETHold:SETTime?
Arguments	<NR3> is a floating point number that specifies the setup time for setup and hold violation triggering.
Examples	TRIGGER:A:SETHold:SETTime 3.0E-6 specifies that the current setup time for setup and hold trigger is 3 μs. TRIGGER:A:SETHold:SETTime? might return :TRIGGER:A:LOGIC:SETHOLD:SETTIME 2.0000E-09 indicating that the current setup time for setup and hold trigger is 2 ns.

TRIGGER:{A|B}:SETHOLDLogicval:{CH<x>|CH<x>_D<x>}

This command sets or queries whether the specified channel is included (INCLUDE) or not included (DONTInclude) in the Setup & Hold trigger input configuration. The channel number is specified by <x>.

Group	Trigger
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Syntax	<pre>TRIGger:{A B}:SETHOLDLogicval:{CH<x> CH<x>_D<x>} {INCLude DONTInCLude} TRIGger:{A B}:SETHOLDLogicval:{CH<x> CH<x>_D<x>}?</pre>
Arguments	INCLude specifies including the specified channel in the Setup & Hold trigger input configuration. DONTInCLude specifies not including the specified channel in the Setup & Hold trigger input configuration.
Examples	TRIGger:A:SETHOLDLogicval:CH1 INCLUDE specifies including the specified channel in the Setup & Hold trigger input configuration. TRIGger:A:SETHOLDLogicval:CH1? might include :TRIGGER:A:SETHOLDLOGICVAL:CH1 DONTINCLUDE indicating not to include the channel in the configuration. TRIGger:A:SETHOLDLogicval:CH1_D0 include specifies including the specified channel in the setup and hold trigger input configuration. TRIGger:A:SETHOLDLogicval:CH1_D0? might return :TRIGGER:A:SETHOLDLOGICVAL:CH1_D0 DONTINCLUDE indicating the channel will not be included in the configuration.

TRIGger:{A|B}:TIMEOut:LOGICQUALification

This command specifies whether or not to use logic qualification for a timeout trigger.

Group	Trigger
Syntax	<pre>TRIGger:{A B}:TIMEOut:LOGICQUALification {ON OFF} TRIGger:{A B}:TIMEOut:LOGICQUALification?</pre>
Arguments	ON specifies that the timeout trigger type uses logic qualification. OFF specifies that the timeout trigger type does not use logic qualification.
Examples	TRIGger:A:TIMEOut:LOGICQUALification ON specifies using logic qualification. TRIGger:A:TIMEOut:LOGICQUALification? might include :TRIGGER:A:TIMEOUT:LOGICQUALIFICATION OFF indicating that logic qualification is off.

TRIGger:{A|B}:TIMEOut:POLarity

When triggering using the **TIMEOut** trigger type, this commands specifies the polarity to be used.

Group	Trigger
Syntax	TRIGger:{A B}:TIMEOut:POLarity {STAYSHigh STAYSLow EITHer} TRIGger:{A B}:TIMEOut:POLarity?
Related Commands	TRIGger:{A B}:TIMEOut:SOUrce TRIGger:{A B}:TIMEOut:TIME
Arguments	STAYSHigh — Trigger when the signal stays high during the timeout time specified by the command TRIGger:{A B}:TIMEOut:TIME. STAYSLow — Trigger when the signal stays low during the timeout time specified by the command TRIGger:{A B}:TIMEOut:TIME. EITHer — Trigger when the signal is either high or low during the timeout time specified by the command TRIGger:{A B}:TIMEOut:TIME.
Examples	TRIGger:A:TIMEOut:POLarity STAYSHigh specifies to trigger when the signal stays high during the timeout time when triggering using the TIMEOut trigger type. TRIGger:A:TIMEOut:POLarity? might return EITHER, indicating that the signal stays either high or low during the timeout time.

TRIGger:{A|B}:TIMEOut:SOUrce

When triggering using the **TIMEOut** trigger type, this command specifies the source. The available sources are live channels and digital channels. The default is channel 1. The timeout trigger type is selected using [TRIGger:{A|B}:TYPE TIMEOut](#).

Group	Trigger
Syntax	TRIGger:{A B}:TIMEOut:SOUrce {CH<x> CH<x>_D<y>} TRIGger:{A B}:TIMEOut:SOUrce?
Related Commands	SEARCH:SEARCH<x>:TRIGger:A:TIMEOut:SOUrce

[TRIGger:{A|B}:TIMEOut:TIME](#)

[TRIGger:{A|B}:TIMEOut:POLarity](#)

Arguments	CH<x> specifies an analog channel as the timeout trigger source. CH<x>_D<y> specifies a digital channel as the timeout trigger source.
Examples	TRIGger:A:TIMEOut:SOURce CH4 specifies to use channel 4 as the source for the timeout trigger. TRIGger:A:TIMEOut:SOURce? might return CH1, indicating that channel 1 has been set as the source for the timeout trigger.

TRIGger:{A|B}:TIMEOut:TIME

When triggering using the TIMEOut trigger type, this command specifies the timeout time, in seconds. This command is equivalent to selecting Timeout from the Trig menu and setting a value for Time Limit. The timeout trigger type is selected using [TRIGger:{A|B}:TYPe](#) TIMEOut

Group	Trigger
Syntax	TRIGger:{A B}:TIMEOut:TIME <NR3> TRIGger:{A B}:TIMEOut:TIME?
Related Commands	TRIGger:{A B}:TIMEOut:POLarity TRIGger:{A B}:TIMEOut:SOURce
Arguments	<NR3> is a floating point number that specifies the timeout time, in seconds.
Examples	:TRIGger:A:TIMEOut:TIME 4.0E-9 specifies the timeout time of 4.0 nsec. :TRIGger:A:TIMEOut:TIME? might return 8.0000E-9.

TRIGger:{A|B}:TRANSition:DELTatime

This command specifies the delta time (that is Time Limit) used in calculating the transition value for the transition (that is Rise or Fall Time) trigger.

Group	Trigger
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Syntax `TRIGger:{A|B}:TRANSition:DELTaTime <NR3>`
`TRIGger:{A|B}:TRANSition:DELTaTime?`

Arguments <NR3> is a floating point number that specifies the delta time, in seconds.

Examples `TRIGGER:A:TRANSITION:DELTATIME 15E-6` sets the delta time of the transition trigger to 15 μ s.
`TRIGGER:A:TRANSITION:DELTATIME?` might return
`:TRIGGER:A:TRANSITION:DELTATIME 2.0000E-09` indicating that the delta time of the transition trigger is set to 2 ns.

TRIGger:{A|B}:TRANSition:LOGICQUALification

This command specifies whether or not to use logic qualification for a transition trigger.

Group Trigger

Syntax `TRIGger:{A|B}:TRANSition:LOGICQUALification {ON|OFF}`
`TRIGger:{A|B}:TRANSition:LOGICQUALification?`

Arguments ON specifies that the transition trigger type uses logic qualification.
OFF specifies that the transition trigger type does not use logic qualification.

Examples `TRIGGER:A:TRANSITION:LOGICQUALIFICATION ON` specifies using logic qualification.
`TRIGGER:A:TRANSITION:LOGICQUALIFICATION?` might return
`:TRIGGER:A:TRANSITION:LOGICQUALIFICATION OFF` indicating logic qualification is off.

TRIGger:{A|B}:TRANSition:POLarity

This command specifies the polarity for the transition trigger.

Group Trigger

Syntax `TRIGger:{A|B}:TRANSition:POLarity {EITHER|NEGative|POSitive}`

TRIGger:{A|B}:TRANSition:POLarity?

Arguments	POSitive indicates that a pulse edge must traverse from the lower (most negative) to higher (most positive) level for transition triggering to occur. NEGative indicates that a pulse edge must traverse from the upper (most positive) to lower (most negative) level for transition triggering to occur. EITHer indicates either positive or negative polarity.
Examples	TRIGGER:A:TRANSITION:POLARITY NEGATIVE sets the transition polarity to negative. TRIGGER:A:TRANSITION:POLARITY? might return :TRIGGER:A:TRANSITION:POLARITY EITHER indicating that the polarity can be either positive or negative.

TRIGger:{A|B}:TRANSition:SOURce

This command specifies the source waveform for a transition trigger.

NOTE. *Digital channels are not supported as transition trigger sources.*

Group	Trigger
Syntax	TRIGger:{A B}:TRANSition:SOURce {CH<x>} TRIGger:{A B}:TRANSition:SOURce?
Arguments	CH<x> specifies one of the analog channels to be used as the source for a transition trigger.
Examples	TRIGGER:A:TRANSITION:SOURCE CH4 sets channel 4 as the source for the transition trigger. TRIGGER:A:TRANSITION:SOURCE? might return :TRIGGER:A:TRANSITION:SOURCE CH2 indicating that channel 2 is the source for the transition trigger.

TRIGger:{A|B}:TRANSition:WHEN

This command specifies whether to check for a transitioning signal that is faster or slower than the specified delta time.

Group	Trigger
Syntax	TRIGger:{A B}:TRANSition:WHEN {SLOWer FASTer EQua1 UNEQua1} TRIGger:{A B}:TRANSition:WHEN?
Arguments	FASTer sets the trigger to occur when the signal transition time is faster than the time set by TRIGger:A:TRANSition:DELTatime. SLOWer sets the trigger to occur when the signal transition time is slower than the time set by TRIGger:A:TRANSition:DELTatime. EQua1 sets the trigger to occur when the signal transition time is equal to the time set by TRIGger:A:TRANSition:DELTatime. UNEQua1 sets the trigger to occur when the signal transition time is not equal to the time set by TRIGger:A:TRANSition:DELTatime.
Examples	TRIGGER:A:TRANSITION:WHEN SLOWER sets the trigger to occur when the signal transition time is slower than the time set by the TRIGger:A:TRANSition:DELTatime command. TRIGGER:A:TRANSITION:WHEN? might return :TRIGGER:A:TRANSITION:WHEN FASTER

TRIGger:{A|B}:TYPE

This command sets or queries the type of A or B trigger.

Group	Trigger
Syntax	TRIGger:{A B}:TYPE {EDGE WIDTh TIMEOut RUNt WINDow LOGIC SETHo1d TRANSition BUS} TRIGger:{A B}:TYPE?
Arguments	EDGE is a normal trigger. A trigger event occurs when a signal passes through a specified voltage level in a specified direction and is controlled by the TRIGger:A:EDGE commands. WIDTh specifies that the trigger occurs when a pulse with a specified width is found. TIMEOut specifies that a trigger occurs when a pulse with the specified timeout is found.

RUNt specifies that a trigger occurs when a pulse with the specified parameters is found.

WINDow specifies that a trigger occurs when a signal with the specified window parameters is found.

LOGic specifies that a trigger occurs when specified conditions are met and is controlled by the **TRIGger:{A|B}:LOGic** commands.

SETHold specifies that a trigger occurs when a signal is found that meets the setup and hold parameters.

TransiTiOn specifies that a trigger occurs when a specified pulse is found that meets the transition trigger parameters.

BUS specifies that a trigger occurs when a signal is found that meets the specified bus setup parameters.

Examples **TRIGGER:A:TYPE** EDGE sets the A trigger type to EDGE.
TRIGGER:A:TYPE? might return **:TRIGGER:A:TYPE** RUNt indicating that the A trigger type is a runt trigger.

TRIGger:{A|B}:UPPerthreshold:CH<x>

This command sets or queries the specified channel upper trigger level. The CH<x> range is 1 to 8 and depends on the number of analog channels on your instrument.

Group Trigger

Syntax **TRIGger:{A|B}:UPPerthreshold:CH<x>** <NR3>
TRIGger:{A|B}:UPPerthreshold:CH<x>?

Related Commands [TRIGger:{A|B}:LOWerthreshold:CH<x>](#)

Arguments <NR3> specifies the trigger level in user units (usually volts).

Examples **TRIGGER:A:UPPERTHRESHOLD:CH1 1.3** This command sets the A trigger level for Channel 1 to 1.3 V level.
TRIGGER:A:UPPERTHRESHOLD:CH2? might return
TRIGGER:A:UPPERTHRESHOLD:CH2 1.3000E+00, indicating that the A trigger level for Channel 2 is set to 1.3 V.

TRIGger:{A|B}:WINDow:CROSSIng

This command sets or queries the window trigger threshold crossing of the selected trigger source. The threshold crossing selection is only effective when :TRIGger:{A|B}:WINDow:WHEN is INSIDEGreater or OUTSIDEGreater.

Group Trigger

Syntax TRIGger:{A|B}:WINDow:CROSSIng {UPPer|LOWer|EITHer|NONE}

Arguments UPPer if :TRIGger:{A|B}:WINDow:WHEN is INSIDEGreater, the instrument triggers when the signal remains between the upper and lower thresholds for longer than the time limit (:TRIGger:{A|B}:WINDow:WIDTH) and then exits through the upper threshold. If :TRIGger:{A|B}:WINDow:WHEN is OUTSIDEGreater, the instrument triggers when the signal remains above the upper threshold for longer than the time limit (:TRIGger:{A|B}:WINDow:WIDTH) and then crosses downward through the upper threshold.

LOWer if :TRIGger:{A|B}:WINDow:WHEN is INSIDEGreater, the instrument triggers when the signal remains between the upper and lower thresholds for longer than the time limit (:TRIGger:{A|B}:WINDow:WIDTH) and then exits through the lower threshold. If :TRIGger:{A|B}:WINDow:WHEN is OUTSIDEGreater, the instrument triggers when the signal remains below the lower threshold for longer than the time limit (:TRIGger:{A|B}:WINDow:WIDTH) and then crosses upwards through the lower threshold.

EITHer if :TRIGger:{A|B}:WINDow:WHEN is INSIDEGreater, the instrument triggers when the signal remains between the upper and lower thresholds for longer than the time limit (:TRIGger:{A|B}:WINDow:WIDTH) and then exits through either the upper or lower threshold. If :TRIGger:{A|B}:WINDow:WHEN is OUTSIDEGreater, the instrument triggers when the signal remains either above the upper threshold or below the lower threshold for longer than the time limit (:TRIGger:{A|B}:WINDow:WIDTH) and then crosses a threshold.

NONE if :TRIGger:{A|B}:WINDow:WHEN is INSIDEGreater, the instrument triggers when the signal remains between the upper and lower thresholds for longer than the time limit (:TRIGger:{A|B}:WINDow:WIDTH) without crossing through either the upper or lower threshold. If :TRIGger:{A|B}:WINDow:WHEN is OUTSIDEGreater, the instrument triggers when the signal remains outside the upper and lower thresholds for longer than the time limit (:TRIGger:{A|B}:WINDow:WIDTH) without crossing through either the upper or lower threshold.

Examples `TRIGger:A:WINDOW:CROSSing LOWER` sets crossing to lower.
`TRIGger:A:WINDOW:CROSSing?` might return
`TRIGGER:A:WINDOW:CROSSING UPPER` indicating the crossing is set to upper.

TRIGger:{A|B}:WINDOW:LOGICQUALification

This command specifies whether or not to use logic qualification for a window trigger.

Group Trigger

Syntax `TRIGger:{A|B}:WINDOW:LOGICQUALification {ON|OFF}`
`TRIGger:{A|B}:WINDOW:LOGICQUALification?`

Arguments `ON` specifies that the window trigger type uses logic qualification.
`OFF` specifies that the window trigger type does not use logic qualification.

Examples `TRIGger:A:WINDOW:LOGICQUALification ON` turns on logic qualification.
`TRIGger:A:WINDOW:LOGICQUALification?` might return
`TRIGGER:A:WINDOW:LOGICQUALIFICATION OFF` indicating logic qualification is off.

TRIGger:{A|B}:WINDOW:SOURce

This command sets or queries the source for a window trigger.

NOTE. *Digital channels are not supported as window trigger sources.*

Group Trigger

Syntax `TRIGger:{A|B}:WINDOW:SOURce {CH<x>}`
`TRIGger:{A|B}:WINDOW:SOURce?`

Arguments The window trigger source channel.

- Examples** `TRIGger:A:WINDOW:SOURce CH2` sets the source to channel 2.
- `TRIGger:A:WINDOW:SOURce?` might return `:TRIGGER:A:WINDOW:SOURCE CH3` indicating the source is channel 3.

TRIGger:{A|B}:WINDow:WHEn

This command sets or queries the window trigger event. This command is equivalent to selecting Window Setup from the Trig menu and selecting from the Window Trigger When box.

- Group** Trigger
- Syntax** `TRIGger:{A|B}:WINDow:WHEn {ENTERSWindow|EXITSWindow|INSIDEGreater|OUTSIDEGreater}`
`TRIGger:{A|B}:WINDow:WHEn?`
- Arguments** `OUTSIDEGreater` specifies a trigger event when the signal leaves the window defined by the threshold levels for the time specified by Width.
- `INSIDEGreater` specifies a trigger event when the signal enters the window defined by the threshold levels for the time specified by Width.
- `ENTERSWindow` specifies a trigger event when the signal enters the window defined by the threshold levels.
- `EXITSWindow` specifies a trigger event when the signal leaves the window defined by the threshold levels.
- Examples** `TRIGger:A:WINDOW:WHEn EXITSWINDOW` specifies a trigger event when the signal leaves the window defined by the threshold levels.
- `TRIGger:A:WINDOW:WHEn?` might return `:TRIGGER:A:WINDOW:WHEn ENTERSWINDOW` indicating a trigger when the signal enters the window.

TRIGger:{A|B}:WINDow:WIDTh

This command sets or queries the minimum width for a window violation. This command is equivalent to selecting Window Setup from the Trig menu, selecting Inside > Limit or Outside > Limit in the Trigger When box, and setting the Time Limit.

- Group** Trigger

Syntax	<code>TRIGger:{A B}:WINDOW:WIDTH <NR3></code>
Arguments	<NR3> is the minimum width for a window violation.
Examples	<code>TRIGger:A:WINDOW:WIDTH 4.5e-9</code> sets the minimum width for a window violation to 4.5 ns. <code>TRIGger:A:WINDOW:WIDTH?</code> might return <code>:TRIGGER:A:WINDOW:WIDTH 4.0E-9</code> indicating that 4.0 ns is the minimum width for a window violation.

TRIGger:A:HOLDoff:BY

This command sets or queries the type of holdoff for the A trigger. Holdoff types are expressed as either user-specified time (TIME) or by an internally calculated random time value (RANDOM). This command is equivalent to selecting Mode & Holdoff from the Trig menu and then setting the Holdoff type.

Group	Trigger
Syntax	<code>TRIGger:A:HOLDoff:BY {TIME RANDOM}</code> <code>TRIGger:A:HOLDoff:BY?</code>
Related Commands	TRIGger:A:HOLDoff:TIME
Arguments	TIME enables you to set the holdoff time via the TRIGger:A:HOLDoff:TIME command. RANDOM specifies a random time value.
Examples	<code>TRIGGER:A:HOLDOFF:BY TIME</code> sets the holdoff to the "by time" setting. This enables you to set the holdoff time. <code>TRIGGER:A:HOLDOFF:BY?</code> might return <code>:TRIGGER:A:HOLDOFF:BY TIME</code> , indicating that you will set the holdoff time.

TRIGger:A:HOLDoff:TIME

This command sets or queries the A trigger holdoff time. This command is equivalent to selecting Mode & Holdoff from the Trig menu, selecting Time, and then setting the desired Holdoff Time.

Group	Trigger
Syntax	TRIGger:A:HOLDoff:TIME <NR3> TRIGger:A:HOLDoff:TIME?
Related Commands	TRIGger:A:HOLDoff:BY
Arguments	<NR3> specifies the holdoff time in seconds. The range is from 0 seconds through 10 seconds.
Examples	TRIGGER:A:HOLDOFF:TIME 10 sets the A trigger holdoff time to 10 s. TRIGGER:A:HOLDOFF:TIME? might return :TRIGGER:A:HOLDOFFTIME 1.2000E-06, indicating that the A trigger holdoff time is set to 1.2 μ s.

TRIGger:A:LOGICQUALification

This command sets or queries the type of logic qualification to perform.

Group	Trigger
Syntax	TRIGger:A:LOGICQUALification {AND OR NAND NOR} TRIGger:A:LOGICQUALification?
Arguments	AND specifies to trigger if all conditions are true. NAND specifies to trigger if any of the conditions are false. NOR specifies to trigger if all conditions are false. OR specifies to trigger if any of the conditions are true.
Examples	TRIGGER:A:LOGICQUALIFICATION OR specifies to trigger if any of the conditions are true. TRIGGER:A:LOGICQUALIFICATION? might return TRIGGER:A:LOGICQUALIFICATION AND indicating the instrument will trigger if all conditions are true.

TRIGger:A:MODE

This command sets or queries the A trigger mode. This command is equivalent to pushing the Mode button on the front panel.

Group	Trigger
Syntax	TRIGger:A:MODE {AUTO NORMa1} TRIGger:A:MODE?
Arguments	AUTO generates a trigger if one is not detected within a specified time period. NORMa1 waits for a valid trigger event.
Examples	TRIGGER:A:MODE NORMAL sets the trigger mode to normal. TRIGGER:A:MODE? might return TRIGGER:A:MODE AUTO, indicating that the trigger mode is auto.

TRIGger:AUXLevel

For those instruments that have an Auxiliary Input (such as an MSO58LP), this command sets or queries the Auxiliary Input voltage level to use for an edge trigger.

Group	Trigger
Syntax	TRIGger:AUXLevel {<NR3> ECL TTL} TRIGger:AUXLevel?
Arguments	<NR3> is trigger level in Volts. ECL sets trigger level to -1.3 Volts. TTL sets trigger level to 1.4 Volts.
Returns	<NR3> is trigger level in Volts.
Examples	TRIGger:AUXLevel 1.2 sets trigger level for the Auxiliary Input to 1.2 Volts.

TRIGger:B:BY

This command selects or returns whether the B trigger occurs after a specified number of events or a specified period of time after the A trigger.

Group	Trigger
Syntax	TRIGger:B:BY {EVENTS TIME} TRIGger:B:BY?
Related Commands	TRIGger:B:EVENTS:COUNT TRIGger:B:TIME TRIGger:B:STATE
Arguments	EVENTS sets the B trigger to take place following a set number of trigger events after the A trigger occurs. The number of events is specified by TRIGger:B:EVENTS:COUNT. TIME sets the B trigger to occur a set time after the A trigger event. The time period is specified by TRIGger:B:TIME.
Examples	TRIGGER:B:BY TIME sets the B trigger to occur at a set time after the A trigger event. TRIGGER:B:BY? might return TRIGGER:B:BY EVENTS, indicating that the B trigger takes place following a set number of trigger events after the A trigger occurs.

TRIGger:B:EVENTS:COUNT

This command sets or queries the number of events that must occur before the B trigger. The B trigger event count applies only if TRIGger:B:BY is set to EVENTS.

Group	Trigger
Syntax	TRIGger:B:EVENTS:COUNT <NR1> TRIGger:B:EVENTS:COUNT?
Related Commands	TRIGger:B:STATE

Arguments <NR1> is the number of B trigger events, which can range from 1 to 65,471.

Examples TRIGGER:B:EVENTS:COUNT 4 sets the number of B trigger events to four.
TRIGGER:B:EVENTS:COUNT? might return TRIGGER:B:EVENTS:COUNT 2, indicating that two events must occur after the A trigger before the B trigger can occur.

TRIGger:B:RESET (No Query Form)

This command sets the B reset trigger level to 50%.

Group Trigger

Syntax TRIGger:B:RESET SETLeve1

Arguments SETLeve1 sets the B reset trigger level to 50%.

Examples TRIGger:B:RESET SETLeve1 sets the B reset trigger level to 50%.

TRIGger:B:RESET:EDGE:COUPling

Sets or queries the trigger coupling for a sequential edge trigger reset when the Source is set to an analog channel.

Group Trigger

Syntax TRIGger:B:RESET:EDGE:COUPling {DC|HFRej|LFRej|NOISerej}
TRIGger:B:RESET:EDGE:COUPling?

Related Commands

Arguments DC selects DC trigger coupling.
HFRej selects high frequency low sensitivity.
LFRej selects low frequency low sensitivity.
NOISerej selects DC low sensitivity.

Examples `TRIGGER:B:RESET:EDGE:COUPLING HFREJ` sets high frequency low sensitivity.

`TRIGGER:B:RESET:EDGE:COUPLING?` might return
`TRIGGER:B:RESET:EDGE:COUPLING DC` indicating DC trigger coupling is selected.

TRIGger:B:RESET:EDGE:LEVel

This command sets the voltage level to use for an Edge Reset trigger when triggering on an analog channel waveform.

Group Trigger

Syntax `TRIGger:B:RESET:EDGE:LEVel <NR3>`

Arguments `<NR3>` is the voltage level to use for an Edge Reset trigger when triggering on an analog channel waveform.

Examples `TRIGGER:B:RESET:EDGE:LEVEL 50.0e-3` sets the level to 50.0 mV.

`TRIGGER:B:RESET:EDGE:LEVEL?` might return
`TRIGGER:B:RESET:EDGE:LEVEL 0.0E+0` indicating the level is set to 0.0 V.

TRIGger:B:RESET:EDGE:SLOPe

This command sets or queries the trigger slope for a sequential edge trigger reset.

Group Trigger

Syntax `TRIGger:B:RESET:EDGE:SLOPe {RISe|FALL|EITHer}`
`TRIGger:B:RESET:EDGE:SLOPe?`

Arguments `RISe` specifies to reset the trigger on the rising or positive edge of a signal.

`FALL` specifies to reset the trigger on the falling or negative edge of a signal.

`EITHER` specified to reset the trigger on either the rising or falling edge of a signal.

Examples `TRIGGER:B:RESET:EDGE:SLOPE FALL` specifies to reset the trigger on the falling or negative edge of a signal.

TRIGGER:B:RESET:EDGE:SLOPE? might return
TRIGGER:B:RESET:EDGE:SLOPE RISE indicating the instrument is set to reset
the trigger on the rising or positive edge of a signal.

TRIGger:B:RESET:EDGE:SOURce

This command sets or queries the trigger source for the A→B sequential edge
trigger reset feature.

Group	Trigger
Syntax	TRIGger:B:RESET:EDGE:SOURce {CH<x> CH<x>_D<y>} TRIGger:B:RESET:EDGE:SOURce?
Arguments	The source channel for the trigger reset.
Examples	TRIGGER:B:RESET:EDGE:SOURCE CH4 sets Channel 4 as the input source for the trigger reset. TRIGGER:B:RESET:EDGE:SOURCE? might return TRIGGER:B:RESET:EDGE:SOURCE CH1, indicating that the current input source for the trigger reset is Channel 1.

TRIGger:B:RESET:TIMEOut:TIME

This command sets or queries the reset timer for a sequential timeout trigger reset.

Group	Trigger
Syntax	TRIGger:B:RESET:TIMEOut:TIME <NR3> TRIGger:B:RESET:TIMEOut:TIME?
Arguments	<NR3> is the reset timer for a sequential timeout trigger reset.
Examples	TRIGGER:B:RESET:TIMEOUT:TIME 100e-9 sets the time to 100 ns. TRIGGER:B:RESET:TIMEOUT:TIME? might return TRIGGER:B:RESET:TIMEOUT:TIME 20.0E-9 indicating the timeout time is set to 20 ns.

TRIGger:B:RESET:TYPe

This command sets or queries the type of A→B sequential trigger reset. If the B trigger reset is active, the reset criteria are part of the B triggering sequence. If the reset conditions defined by the reset type are met, the instrument must start over searching for a new occurrence of the A event.

You must identify a trigger Source and Threshold for each reset type, except for the Timeout trigger type.

NOTE. *If a reset condition occurs, the reset criteria itself is reset and must start over.*

Group	Trigger
Syntax	TRIGger:B:RESET:TYPe {NONE TIMEOut EDGE} TRIGger:B:RESET:TYPe?
Arguments	NONE defeats the trigger reset feature. TIMEOut initiates a reset if the timeout conditions specified by TRIGger:B:RESET:TIMEOut are met. EDGE initiates a reset if the edge trigger conditions are met.
Examples	TRIGGER:B:RESET:TYPE NONE deactivates the sequential trigger reset. TRIGGER:B:RESET:TYPE? This query might return TRIGGER:B:RESET:TYPE TIMEOUT, indicating that the sequential trigger reset is active following a timeout.

TRIGger:B:STATE

This command sets or queries the state of B trigger activity. If the B trigger state is on, the B trigger is part of the triggering sequence. If the B trigger state is off, then only the A trigger causes the trigger event.

Group	Trigger
Syntax	TRIGger:B:STATE {ON OFF <NR1>} TRIGger:B:STATE?
Related Commands	TRIGger:A:MODE

Arguments	ON indicates that the B trigger is active and causes trigger events with the A trigger. OFF indicates that only the A trigger causes trigger events. <NR1> is an integer number. 0 turns off the B trigger; any other value activates the B trigger.
Examples	TRIGGER:B:STATE ON sets the B trigger to active, making it capable of causing trigger events. TRIGGER:B:STATE? might return TRIGGER:B:STATE 0, indicating that the B trigger is inactive and that only the A trigger causes trigger events.

TRIGger:B:TIME

This command sets or queries B trigger delay time, in seconds. The B Trigger time applies only if TRIGger:B:BY is set to TIME.

Group	Trigger
Syntax	TRIGger:B:TIME <NR3> TRIGger:B:TIME?
Related Commands	TRIGger:B:BY TRIGger:B:EVENTS:COUNT TRIGger:B:STATE

Arguments	<NR3> is the B trigger delay time in seconds.
Examples	TRIGGER:B:TIME 4E-6 sets the B trigger delay time to 4 μs. TRIGGER:B:TIME? might return TRIGGER:B:TIME 16.0000E-9, indicating that the B trigger time is set to 16 ns.

TRIGger:STATE? (Query Only)

This query-only command returns the current state of the triggering system.

Group	Trigger
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Syntax `TRIGger:STATE?`

Related Commands [TRIGger:A:MODE](#)

Returns ARMED indicates that the instrument is acquiring pretrigger information.

AUTO indicates that the instrument is in the automatic mode and acquires data even in the absence of a trigger.

READY indicates that all pretrigger information is acquired and that the instrument is ready to accept a trigger.

SAVE indicates that the instrument is in save mode and is not acquiring data.

TRIGGER indicates that the instrument triggered and is acquiring the post trigger information.

Examples `TRIGGER:STATE?` might return `TRIGGER:STATE ARMED`, indicating that the pretrigger data is being acquired.

*TST? (Query Only)

Tests (self-test) the interface and returns a 0.

Group Miscellaneous

Syntax `*TST?`

Examples `*TST?` always returns 0.

UNDO (No Query Form)

Reverts the instrument settings to a state before the previous command or user interface action.

Group Miscellaneous

Syntax `UNDO`

Examples UNDO reverts the instrument settings to a state before the previous command or user interface action.

UNLock (No Query Form)

This command (no query form) unlocks the front panel. The command is equivalent to LOCK NONE.

NOTE. *If the instrument is in the Remote With Lockout State (RWLS), the UNLock command has no effect. For more information, see the ANSI-IEEE Std 488.1-1987 Standard Digital Interface for Programmable Instrumentation, section 2.8.3 on RL State Descriptions.*

Group Miscellaneous

Syntax UNLock ALL

Related Commands [LOCK](#)

Arguments ALL specifies that all front panel buttons and knobs are unlocked.

Examples UNLOCK ALL unlocks all front panel buttons and knobs.

USBDevice:CONFigure

This command may be used to configure the rear USB port to be off or enabled as a USBTMC device. Users should be cautious using this command via the USBTMC interface as a change to the configuration of this interface from a USBTMC device will cause USBTMC communication to cease. It is intended to be used via the Ethernet interface to control the USB device interface.

Group Miscellaneous

Syntax USBDevice:CONFigure {DISabled|USBTmc}
USBDevice:CONFigure?

Arguments DISabled will disable the rear USB port.
USBTmc enables the rear USB port.

Examples `USBDEVICE:CONFIGURE DISABLED` will disable the rear USB port.

`USBDEVICE:CONFIGURE?` might return `:USBDEVICE:CONFIGURE USBTMC` indicating the USB port is enabled.

VERBose

This command sets or queries the Verbose state that controls the length of keywords on query responses. Keywords can be both headers and arguments.

NOTE. *This command does not affect IEEE Std 488.2-1987 Common Commands (those starting with an asterisk).*

Group Miscellaneous

Syntax `VERBose {<NR1>|OFF|ON}`

Related Commands [HEADer](#)
 [*LRN?](#)
 [SET?](#)

Arguments `<NR1> = 0` disables Verbose, any other value enables Verbose.

`OFF` sets the Verbose state to false, which returns minimum-length keywords for applicable setting queries.

`ON` sets the Verbose state to true, which returns full-length keywords for applicable setting queries.

A 0 returns minimum-length keywords for applicable setting queries; any other value returns full-length keywords.

Examples `VERBOSE ON` sets the Verbose state to true and return the full length keyword for the applicable setting queries.

`VERBOSE?` might return `:VERBOSE OFF`, indicating that the Verbose state is set to false and return the minimum-length keywords for the applicable setting queries.

With `:HEADer ON` and `:VERBose ON`, the `:ACQuire:MODE?` query might return: `:ACQUIRE:MODE SAMPLE`

With `:HEADer ON` and `:VERBose OFF`, the `:ACQuire:MODE?` query might return: `:ACQ:MOD SAM`

With :HEADer OFF and :VERBoSe ON, the :ACQuire:MODE? query might return: SAMPLE

With :HEADer OFF and :VERBoSe OFF, the :ACQuire:MODE? query might return: SAM

VERTical:DESKew:FROM:CUSTOMPROPagation

This command sets or queries a target (FROM) delay that you can specify when the propagation delay of the target (FROM) probe used for deskew cannot be detected automatically.

Group	Vertical
Syntax	VERTical:DESKew:FROM:CUSTOMPROPagation <NR3> VERTical:DESKew:FROM:CUSTOMPROPagation?
Arguments	<NR3> is a target (FROM) delay that you can specify when the propagation delay of the target (FROM) probe used for deskew cannot be detected automatically.
Examples	VERTICAL:DESKew:FROM:CUSTOMPROPAGATION 0.25e-9 sets the custom delay to 250 ps. VERTICAL:DESKew:FROM:CUSTOMPROPAGATION? might return :VERTICAL:DESKew:FROM:CUSTOMPROPAGATION 0.0E+0 indicating the custom delay is 0.0 s.

VERTical:DESKew:FROM:SOUrce

This command sets or queries the source channel for performing channel-to-channel deskew adjustment. Sources can be any of the analog channels.

Group	Vertical
Syntax	VERTical:DESKew:FROM:SOUrce CH<x> VERTical:DESKew:FROM:SOUrce?
Arguments	The vertical deskew source.

Examples `VERTICAL:DESKEW:FROM:SOURCE CH1` sets channel 1 as the from source.
`VERTICAL:DESKEW:FROM:SOURCE?` might return
`:VERTICAL:DESKEW:FROM:SOURCE CH2` indicating channel 2 is the from source.

VERTical:DESKEW:STATIC (No Query Form)

This command executes static deskew using the deskew settings.

Group Vertical

Syntax `VERTical:DESKEW:STATIC EXECute`

Arguments `EXECute` will execute static deskew using the deskew settings.

Examples `VERTICAL:DESKEW:STATIC EXECUTE` will execute static deskew using the deskew settings.

VERTical:DESKew:TO:CUSTOMPROPagation

This command sets or queries a target (TO) delay that can be specified by the user when the propagation delay of the target (TO) probe used for deskew cannot be detected automatically.

Group Vertical

Syntax `VERTical:DESKew:TO:CUSTOMPROPagation <NR3>`

Arguments `<NR3>` is a target (TO) delay that can be specified by the user when the propagation delay of the target (TO) probe used for deskew cannot be detected automatically.

Examples `VERTICAL:DESKEW:TO:CUSTOMPROPAGATION 0.25e-9` sets the custom propagation to 250 ps.
`VERTICAL:DESKEW:TO:CUSTOMPROPAGATION?` might return
`:VERTICAL:DESKEW:TO:CUSTOMPROPAGATION 0.0E+0` indicating the custom propagation is set to 0.0 ns.

VERTical:DESKEW:TO:SOURce

This command sets or queries the target channel for performing channel-to-channel deskew adjustment. Target sources can be any of the live analog channels.

Group	Vertical
Syntax	VERTical:DESKEW:TO:SOURce CH<x> VERTical:DESKEW:TO:SOURce?
Arguments	Arguments are the live analog channels.
Examples	VERTICAL:DESKEW:TO:SOURCE CH4 sets the deskew to source is channel 4. VERTICAL:DESKEW:TO:SOURCE? might return :VERTICAL:DESKEW:TO:SOURCE CH2 indicating the deskew to source is channel 2.

VISual:AREA<x>:ASPEctratio

Sets or queries whether the aspect ratio of the specified Visual Trigger area is locked.

Group	Trigger
Syntax	VISual:AREA<x>:ASPEctratio {ON OFF <NR1>} VISual:AREA<x>:ASPEctratio?
Arguments	Area<x> specifies the integer number of a Visual Trigger area. ON locks the aspect ratio of the specified Visual Trigger area. The aspect ratio is kept constant when the height or width of the area changes. OFF unlocks the aspect ratio of the specified Visual Trigger area. <NR1> is an integer number. 1 locks the aspect ratio of the specified Visual Trigger area; any other value unlocks the aspect ratio of the specified Visual Trigger area
Examples	VISual:AREA6:ASPEctratio ON locks the aspect ratio of area 6. VISual:AREA2:ASPEctratio? might return VISual:AREA2:ASPEctratio OFF, indicating that the aspect ratio of area 2 is not locked.

VISual:AREA<x>:FLIP:HORizontal (No Query Form)

Flips the specified Visual Trigger area horizontally around its center point.

Group Trigger

Syntax VISual:AREA<x>:FLIP:HORizontal

Related Commands [VISual:AREA<x>:FLIP:VERTical](#)

Arguments Area<x> specifies the integer number of a Visual Trigger area.

Examples VISual:AREA1:FLIP:HORizontal horizontally flips area 1.

VISual:AREA<x>:FLIP:VERTical (No Query Form)

Flips the specified Visual Trigger area vertically around its center point.

Group Trigger

Syntax VISual:AREA<x>:FLIP:VERTical

Related Commands [VISual:AREA<x>:FLIP:HORizontal](#)

Arguments Area<x> specifies the integer number of a Visual Trigger area.

Examples VISual:AREA8:FLIP:VERTical vertically flips area 8.

VISual:AREA<x>:HEIGHT

Sets or queries the height of the specified Visual Trigger area.

Group Trigger

Syntax VISual:AREA<x>:HEIGHT <NR3>
VISual:AREA<x>:HEIGHT?

Arguments	<code>Area<x></code> specifies the integer number of a Visual Trigger area. <code><NR3></code> specifies the height of the Visual Trigger area in amplitude.
Examples	<code>VISua1:AREA51:HEIGHT 3.5</code> sets the height of area 51 to 3.5 (volts or Amps). <code>VISua1:AREA4:HEIGHT?</code> might return <code>VISua1:AREA4:HEIGHT 15</code> , indicating that the height of area 4 is 15.

VISual:AREA<x>:HITType

Sets or queries the area hit logic true condition for the specified Visual Trigger area.

Group	Trigger
Syntax	<code>VISua1:AREA<x>:HITType {IN OUT DONTcare}</code> <code>VISua1:AREA<x>:HITType?</code>
Arguments	<code>Area<x></code> specifies the integer number of a Visual Trigger area. <code>IN</code> specifies that the waveform must intrude anywhere into the specified area to be true. <code>OUT</code> specifies that the waveform must not intrude anywhere into the specified area to be true. <code>DONTcare</code> sets the area to a don't care state, causing the area to be ignored. This is useful when you are developing a Visual Trigger condition and need to change the hit logic type of an area while keeping the area.
Examples	<code>VISua1:AREA2:HITType OUT</code> sets the area 2 hit type to OUT. <code>VISua1:AREA5:HITType?</code> might return <code>VISua1:AREA5:HITType IN</code> , indicating that the waveform must intrude into area 5 to be true.

VISual:AREA<x>:RESET (No Query Form)

Sets the specified Visual Trigger area shape to a default-sized triangle.

Group	Trigger
Syntax	<code>VISua1:AREA<x>:RESET</code>

Arguments Area<x> specifies the integer number of a Visual Trigger area.

Examples VISual:AREA2:RESET changes area 2 to a default triangle shape.

VISual:AREA<x>:ROTation

Sets or queries the rotation angle of the specified Visual Trigger area.

Group Trigger

Syntax VISual:AREA<x>:ROTation <NR3>
VISual:AREA<x>:ROTation?

Arguments Area<x> specifies the integer number of a Visual Trigger area.

 <NR3> specifies the rotation angle of the Visual Trigger area, in positive degrees from 0 to 360. Zero degrees is referenced from when the area was created.

Examples VISual:AREA2:ROTation 45 rotates Visual Trigger area 2 by 45 degrees.

 VISual:AREA1:ROTation? might return VISual:AREA1:ROTation -60.
 indicating that area 4 rotation position is minus 60 degrees.

VISual:AREA<x>:SHAPE

Sets or queries the current shape of the area.

Group Trigger

Syntax VISual:AREA<x>:SHAPE {TRIANGLE|RECTANGLE|TRAPEZOID|HEXAGON}
VISual:AREA<x>:SHAPE?

Arguments Area<x> specifies the integer number of a Visual Trigger area.

 TRIANGLE sets the specified area to a triangular shape (three vertices). If the area does not exist, the instrument creates a new triangular area with the specified area number.

 RECTANGLE sets the specified area to a rectangular shape (four vertices, right angles at each corner). If the area does not exist, the instrument creates a new triangular area with the specified area number.

TRAPeZoid sets the specified area to a trapezoidal shape (four vertices, two parallel sides). If the area does not exist, the instrument creates a new triangular area with the specified area number.

HEXAgon sets the specified area to a hexagonal shape (six vertices). If the area does not exist, the instrument creates a new hexagonal area with the specified area number.

Returns CUSTOM indicates that the shape is a custom-created shape, or the positions of one or more vertices of a standard shape have been changed.

Examples VISua1:AREA1:SHAPE Hexagon sets area 1 to be a hexagonal shape, or creates a new area 1 hexagonal shape if the specified area does not exist.

VISua1:AREA2:SHAPE? might return VISua1:AREA2:SHAPE CUSTOM, indicating that area 2 is a custom shape.

VISua1:AREA<x>:SOUrce

Sets or queries the signal source for the specified Visual Trigger area. The source can only be an analog channel.

Group Trigger

Syntax VISua1:AREA<x>:SOUrce {CH1|CH2|CH3|CH4|CH5|CH6|CH7|CH8}
VISua1:AREA<x>:SOUrce?

Arguments Area<x> specifies the integer number of a Visual Trigger area.
CH1 through CH8 sets the source channel number for the specified area.

Examples VISua1:AREA1:SOUrce CH3 sets the source channel number for area 1 to Channel 3.

VISua1:AREA6:SOUrce? might return VISua1:AREA6:SOUrce CH8, indicating that the source for area 6 is Channel 8.

VISua1:AREA<x>:VERTICES

Sets or queries the X and Y vertex coordinate values for all vertices of the specified Visual Trigger area. Vertex values must be set in pairs.

Group	Trigger
Syntax	<code>VISual:AREA<x>:VERTICES "<NR3>, <NR3>, <NR3>, <NR3>, <NR3>, <NR3> [, <NR3>, <NR3>, ...]"</code> <code>VISual:AREA<x>:VERTICES?</code>
Arguments	Area<x> specifies the integer number of a Visual Trigger area. <NR3>, <NR3> specifies the X, Y coordinate pair of each vertex in an area. The first value is X (time) and the second value is Y (amplitude). There must be a minimum of three vertex pairs to create an area. If the specified area exists, the area is changed to the shape specified in the arguments. If the specified area does not exist, a new area is created and assigned the specified vertices.
Examples	<code>VISual:AREA3:VERTICES</code> <code>"400E-9,1,400E-9,0.5,600E-9,0.5,600E-9,1"</code> sets or creates area 3 as a rectangle that is 200 ns wide by 500 mV high. <code>VISual:AREA3:VERTICES?</code> might return <code>VISual:AREA3:VERTICES</code> <code>"171.1E-9,-141.7E-3,321.1E-9,-141.7E-3,283.6E-9,1.708,208.6E-9,1.708"</code>, indicating that area 3 is defined as a trapezoid that is 150 ns wide by 1.85 V high.

VISual:AREA<x>:WIDTH

Sets or queries the width of the specified Visual Trigger area.

Group	Trigger
Syntax	<code>VISual:AREA<x>:WIDTH <NR3></code> <code>VISual:AREA<x>:WIDTH?</code>
Arguments	Area<x> specifies the integer number of a Visual Trigger area. <NR3> specifies the width of the Visual Trigger area in seconds.
Examples	<code>VISual:AREA5:WIDTH 0.0000045</code> sets the width of area 5 to 4.5 μs. <code>VISual:AREA3:WIDTH?</code> might return <code>VISual:AREA3:WIDTH 0.000016</code>, indicating that the width of area 3 is 16 μs.

VISual:AREA<x>:XPOSition

Sets or queries the horizontal (X-axis) center position of the specified Visual Trigger area.

Group Trigger

Syntax VISual:AREA<x>:XPOSition <NR3>
VISual:AREA<x>:XPOSition?

Arguments Area<x> specifies the integer number of a Visual Trigger area.

<NR3> specifies the horizontal position of the center of the Visual Trigger area, in seconds.

Examples VISual:AREA1:XPOSition -6.9e-6 sets the horizontal center of area 1 to be -6.9 μ s from the trigger point.

VISual:AREA3:XPOSition? might return VISual:AREA3:XPOSition 8.0e-6, indicating that the center of area 3 is 8 μ s from the trigger point.

VISual:AREA<x>:YPOSition

Sets or queries the vertical (Y-axis) center position of the specified Visual Trigger area.

Group Trigger

Syntax VISual:AREA<x>:YPOSition <NR3>
VISual:AREA<x>:YPOSition?

Arguments Area<x> specifies the integer number of a Visual Trigger area.

<NR3> specifies the vertical position of the center of the Visual Trigger area, in amplitude (volts, amps).

Examples VISual:AREA1:YPOSition 0.5 sets the vertical center of area 1 to be 500 mV.

VISual:AREA3:YPOSition? might return VISual:AREA3:YPOSition 2, indicating that the vertical center of area 3 is 2 volts.

VISual:DELETEALL (No Query Form)

Deletes all Visual Trigger areas.

Group Trigger

Syntax VISual:DELETEALL

Examples VISUAL:DELETEALL deletes all Visual Trigger areas.

VISual:ENABle

Sets or queries the status (on or off) of the Visual Trigger mode.

Group Trigger

Syntax VISual:ENABle {ON|OFF|<NR1>}
VISual:ENABle?

Arguments ON enables the Visual Trigger mode.
OFF disables the Visual Trigger mode.
<NR1> is an integer number. 0 turns off the Visual Trigger mode; any other value enables Visual Trigger mode.

Examples VISual:ENABle ON enables the Visual Trigger function.
VISual:ENABle? might return VISual:ENABle OFF, indicating that Visual Trigger mode is not enabled.

VISual:EQUation

Sets or queries the Visual Trigger area combination logic equation.

Group Trigger

Syntax VISual:EQUation <Qstring>
VISual:EQUation?

Arguments	<Qstring> defines the Visual Trigger area combination logic equation. The equation is made up of area names (A<x>) combined with logic operators AND, OR, or XOR (&, , ^). It may also contain parentheses for grouping. The equation must be true to have a valid Visual Trigger event and display a waveform. Each area's true state depends on the area's condition setting (In, Out or Don't Care).
Examples	<code>VISual:EQUation "(A1 & A2) A3"</code> sets the combined area logic such that both areas 1 and 2 must be true, or area 3 must be true, to have a valid Visual Trigger event and display a waveform. <code>VISual:EQUation?</code> might return <code>VISual:EQUation "A1 & A2 & A3 & A4"</code>, indicating that the area combination logic equation requires that areas 1 through 4 must be true to have a valid Visual Trigger event.

VISual:SHOWAREas

Shows or hides all Visual Trigger areas.

Group	Trigger
Syntax	<code>VISual:SHOWAREas {ON OFF <NR1>}</code> <code>VISual:SHOWAREas?</code>
Arguments	ON shows all Visual Trigger areas. OFF hides all Visual Trigger areas. <NR1> is an integer number. 0 hides all Visual Trigger areas; any other value shows all Visual Trigger areas.
Examples	<code>VISual:SHOWAREas OFF</code> hides all Visual Trigger areas. <code>VISual:SHOWAREas?</code> might return <code>VISual:SHOWAREas ON</code>, indicating that Visual Trigger areas are displayed on the screen.

VISual:SHOWCRiteria

Sets or queries display of the area names and hit criteria for all visual trigger areas.

Group	Trigger
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Syntax	<code>VISual:SHOWCriteria {ON OFF <NR1>}</code> <code>VISual:SHOWCriteria?</code>
Arguments	<code>ON</code> enables display of the area name and hit criteria labels (In, Out, Don't care icons) of all Visual Trigger areas. <code>OFF</code> hides the area name and hit criteria labels (In, Out, Don't care icons) of all Visual Trigger areas. <code><NR1></code> is an integer number. 0 hides the area name and hit criteria of all Visual Trigger areas; any other value enables displaying the area name and hit criteria of all Visual Trigger areas.
Examples	<code>VISual:SHOWCriteria OFF</code> hides the name and hit criteria labels of all Visual Trigger areas. <code>VISual:SHOWCriteria?</code> might return <code>VISual:SHOWCriteria ON</code>, indicating that the name and hit criteria labels of all Visual Trigger areas are displayed.

VISual:SHOWEQuation

Shows or hides the Visual Trigger area combination logic equation.

Group	Trigger
Syntax	<code>VISual:SHOWEQuation {ON OFF <NR1>}</code> <code>VISual:SHOWEQuation?</code>
Arguments	<code>ON</code> shows the Visual Trigger area combination logic equation. <code>OFF</code> hides the Visual Trigger area combination logic equation. <code><NR1></code> is an integer number. 0 hides the area combination logic equation; any other value displays the area combination logic equation.
Examples	<code>VISual:SHOWEQuation ON</code> shows the Visual Trigger area combination logic equation. <code>VISual:SHOWEQuation?</code> might return <code>VISual:SHOWEQuation OFF</code>, indicating that the Visual Trigger area combination logic equation is not displayed on the screen.

*WAI (No Query Form)

The *WAI (Wait) command (no query form) prevents the instrument from executing further commands or queries until all pending commands that generate an OPC message are complete. This command allows you to synchronize the operation of the instrument with your application program. For more information, refer to Synchronization Methods.

Group Status and Error

Syntax *WAI

Related Commands [BUSY?](#)
[*OPC](#)

Examples *WAI prevents the instrument from executing any further commands or queries until all pending commands that generate an OPC message are complete.

WAVFrm? (Query Only)

This query-only command provides the Tektronix standard waveform query which returns the waveform preamble followed by the waveform data for the source specified by :DATA:SOURce using the :DATA settings for encoding, width, and so forth.

Group Waveform Transfer

Syntax WAVFrm?

Related Commands [CURVe](#)
[DATA:SOURce](#)
[WFMOutpre?](#)

Examples WAVFRM? might return the waveform data as: WFMOUTPRE:BIT_NR 8;BN_FMT RI;BYT_NR 1; BYT_OR MSB;ENCDG ASC;NR_PT 500;PT_FMT Y; PT_ORDER LINEAR;PT_OFF 0;XINCR 400.0000E-12; XZERO 0.0000;XUNIT "s";YMULT 4.0000E-3; YOFF 0.0000;YZERO 0.0000;YUNIT "V"; WFID "Ch1,DC coupling, 100.0mV/div, 200.0ns/div, 5000 points,sampl mode".

WFMOutpre? (Query Only)

This query-only command queries the waveform formatting data for the waveform specified by the [DATA:SOURce](#) command. The preamble components are considered to be of two types; formatting and interpretation. The formatting components are: ENCDg, BN_Fmt, BYT_Or, BYT_Nr, BIT_Nr. The interpretation components are derived from the DATA:SOURce specified waveform.

Group	Waveform Transfer
Syntax	WFMOutpre?
Examples	WFMOUTPRE? might return the waveform formatting data as: <pre>:WFMOUTPRE:BYT_NR 2;BIT_NR 16;ENCDG BINARY;BN_FMT RI;BYT_OR MSB;WFID "Ch1, DC coupling, 200.0mV/div, 10.00us/div, 1250 points, Sample mode";NR_PT 1000;PT_FMT Y;XUNIT "s";XINCR 80.0000E-9;XZERO 0.0000;PT_OFF 625;YUNIT "V";YMULT 31.2500E-6;YOFF 0.0000;YZERO 0.0000;NR_FR 3.</pre>

WFMOutpre:ASC_Fmt? (Query Only)

This query returns the format for ASCII data transferred from the instrument. No command form is provided as the format is determined by the data source type. Some waveforms are normalized vector data where the data points are 8-byte doubles in floating point format whereas other formats are 1-byte or 2-byte integers.

Group	Waveform Transfer
Syntax	WFMOutpre:ASC_Fmt?
Related Commands	DATA:SOURce WFMOutpre:BN_Fmt WFMOutpre:ENCDg
Returns	FP represents floating point ASCII data. The waveforms are normalized vector data where the data points are 8-byte doubles in floating point format.

INTEGER represents signed integer ASCII data. The waveform data are 1-byte or 2-byte integers.

Examples `WFMOutpre:ASC_Fmt?` might return `:WFMOUTPRE:ASC_FMT INTEGER` indicating the ASCII format is integer.

WFMOutpre:BIT_Nr

This command sets and queries the number of bits per waveform point that outgoing waveforms contain, as specified by the [DATA:SOURce](#) command. Note that values will be constrained according to the underlying waveform data. This specification is only meaningful when [WFMOutpre:ENCdg](#) is set to BIN and [WFMOutpre:BN_Fmt](#) is set to either RI or RP.

Group Waveform Transfer

Syntax `WFMOutpre:BIT_Nr <NR1>`
`WFMOutpre:BIT_Nr?`

Related Commands [DATA:SOURce](#)
[WFMOutpre:BN_Fmt](#)
[WFMOutpre:ENCdg](#)

Arguments `<NR1>` number of bits per data point can be 8 or 16.

Examples `WFMOUTPRE:BIT_NR 16` sets the number of bits per waveform point to 16 for incoming RI and RP binary format data.

`WFMOUTPRE:BIT_NR?` might return `:WFMOUTPRE:BIT_NR 8`, indicating that outgoing RI or RP binary format data uses 8 bits per waveform point.

WFMOutpre:BN_Fmt

This command sets or queries the format of binary data for outgoing waveforms specified by the [DATA:SOURce](#) command.

Group Waveform Transfer

Syntax	WFMOutpre:BN_Fmt {RI RP FP} WFMOutpre:BN_Fmt?
Related Commands	DATa:SOUrce
Arguments	RI specifies signed integer data point representation. RP specifies positive integer data point representation. FP specifies floating point representation.
Examples	WFMOUTPRE:BN_FMT FP specifies that outgoing waveform data will be in single-precision binary floating point format. WFMOUTPRE:BN_FMT? might return :WFMOUTPRE:BN_FMT RI, indicating that the outgoing waveform data is currently in signed integer format.

WFMOutpre:BYT_Nr

This command sets or queries the binary field data width (bytes per point) for the waveform specified by the [DATa:SOUrce](#) command. Note that values will be constrained according to the underlying waveform data. This specification is only meaningful when [WFMOutpre:ENCdg](#) is set to BIN, and [WFMOutpre:BN_Fmt](#) is set to either RI or RP.

Group	Waveform Transfer
Syntax	WFMOutpre:BYT_Nr <NR1> WFMOutpre:BYT_Nr?
Related Commands	DATa:SOUrce WFMOutpre:BN_Fmt WFMOutpre:ENCdg
Arguments	<NR1> is the number of bytes per data point and can be 1, 2 or 8. A value of 1 or 2 bytes per waveform point indicates channel data; 8 bytes per waveform point indicate pixel map (fast acquisition) data.
Examples	WFMOUTPRE:BYT_NR 1 sets the number of bytes per outgoing waveform data point to 1, which is the default setting.

WFMOUTPRE:BYT_NR? might return :WFMOUTPRE:BYT_NR 2, indicating that there are 2 bytes per outgoing waveform data point.

WFMOutpre:BYT_Or

This command sets or queries which byte of binary waveform data is transmitted first, during a waveform data transfer, when data points require more than one byte. This specification only has meaning when [WFMOutpre:ENCdg](#) is set to BIN.

Group Waveform Transfer

Syntax WFMOutpre:BYT_Or {LSB|MSB}
WFMOutpre:BYT_Or?

Related Commands [WFMOutpre:ENCdg](#)

Arguments LSB specifies that the least significant byte will be transmitted first.
MSB specifies that the most significant byte will be transmitted first.

Examples WFMOUTPRE:BYT_OR MSB sets the most significant outgoing byte of waveform data to be transmitted first.

WFMOUTPRE:BYT_OR? might return :WFMOUTPRE:BYT_OR LSB, indicating that the least significant data byte will be transmitted first.

WFMOutpre:DOMain? (Query Only)

This query returns the domain of the outgoing waveform.

Group Waveform Transfer

Syntax WFMOutpre:DOMain?

Related Commands [DATA:SOURce](#)
[WFMOutpre:BN_Fmt](#)
[WFMOutpre:ENCdg](#)

Returns Returns the domain of the outgoing waveform.

Examples `WFMOUTPRE:DOMAIN?` might return `:WFMOUTPRE:DOMAIN TIME`, indicating that the outgoing waveform is time domain trace.

WFMOutpre:ENCdg

This command sets or queries the type of encoding for outgoing waveforms.

Group Waveform Transfer

Syntax `WFMOutpre:ENCdg {ASCIi|BINary}`
`WFMOutpre:ENCdg?`

Related Commands [DATA:ENCdg](#)
[WFMOutpre:BYT_Nr](#)
[WFMOutpre:BYT_Or](#)
[WFMOutpre:BIT_Nr](#)
[WFMOutpre:BN_Fmt](#)

Arguments `ASCIi` specifies that the outgoing data is to be in ASCII format. Waveforms internally stored as integers will be sent as <NR1> numbers, while those stored as floating point will be sent as <NR3> numbers.

`BINary` specifies that outgoing data is to be in a binary format whose further specification is determined by [WFMOutpre:BYT_Nr](#), [WFMOutpre:BIT_Nr](#), [WFMOutpre:BN_Fmt](#) and [WFMOutpre:BYT_Or](#).

Examples `WFMOUTPRE:ENCDG ASCIi` specifies that the outgoing waveform data will be sent in ASCII format.

`WFMOUTPRE:ENCDG?` might return `:WFMOUTPRE:ENCDG BINary`, indicating that outgoing waveform data will be sent in binary format.

WFMOutpre:NR_Pt? (Query Only)

This query-only command returns the number of points for the [DATA:SOURce](#) waveform that will be transmitted in response to a [CURVe?](#) query.

Group	Waveform Transfer
Syntax	WFMOutpre:NR_Pt?
Related Commands	CURVe DATa DATa:STARt DATa:STOP SAVe:WAVEform SAVEON:WAVEform:FILEFormat
Examples	WFMOUTPRE:NR_PT? might return :WFMOUTPRE:NR_PT 5000, indicating that there are 5000 data points to be sent.

WFMOutpre:PT_Fmt? (Query Only)

This query-only command returns the point format for the waveform specified by the [DATa:SOUrce](#) command. The format specifies a set of equations describing how the scale factors in the preamble are used to give meaning to the [CURVe](#) data points.

An error is reported if the DATa:SOUrce waveform does not exist.

Group	Waveform Transfer
Syntax	WFMOutpre:PT_Fmt?
Related Commands	CURVe DATa:SOUrce
Examples	WFMOUTPRE:PT_FMT? might return :WFMOutpre:PT_Fmt ENV, indicating that the waveform data is a series of min-max pairs.

WFMOutpre:PT_Off? (Query Only)

This query-only command returns the trigger point relative to [DATa:STARt](#) for the waveform specified by the [DATa:SOUrce](#) command.

NOTE. This returned value is the point immediately following the actual trigger.

Group	Waveform Transfer
Syntax	WFMOutpre:PT_Off?
Related Commands	DATA:SOURce DATA:STARt WFMOutpre:XZEr0?
Examples	WFMOUTPRE:PT_OFF? might return :WFMOUTPRE:PT_OFF 251 specifying that the trigger actually occurred between points 250 and 251.

WFMOutpre:WFId? (Query Only)

This query-only command returns a string describing several aspects of the acquisition parameters for the waveform specified by the [DATA:SOURce](#) command.

An error is reported if the DATA:SOURce waveform does not exist.

Group	Waveform Transfer
Syntax	WFMOutpre:WFId?
Related Commands	DATA:SOURce
Returns	<QString> contains the following comma-separated fields documented in the following tables:

Table 2-47: Waveform Suffixes

Field	Description	Examples
Source	The source identification string as it appears in the front panel scale factor readouts.	"CH1-8" "Math<x>" "Ref<x>"
Coupling	A string describing the vertical coupling of the waveform (the Source1 waveform in the case of Dual Waveform Math).	"AC coupling" "DC coupling" "GND coupling"

Table 2-47: Waveform Suffixes (cont.)

Field	Description	Examples
Vert Scale	A string containing the vertical scale factor of the unzoomed waveform. The numeric portion will always be four digits. The examples cover all known internal units.	"100.0 mV/div" "20.00 dB/div" "45.00 deg/div" "785.4 mrad/div" "500.0 μ Vs/div" "10.00 kV/s/div" "200.0 mV/div" "50.00 unk/div"
Horiz Scale	A string containing the horizontal scale factor of the unzoomed waveform. The numeric portion will always be four digits. The examples cover all known internal units.	"100 ms/div" "10.00 kHz/div" "50.00 c/div"
Record Length	A string containing the number of waveform points available in the entire record. The numeric portion is given as an integer.	"500 points" "500000 points"
Acquisition Mode	A string describing the mode used to acquire the waveform.	"Sample mode" "Pk Detect mode" "Hi Res mode" "Envelope mode" "Average mode"

Examples WFMOUTPRE:WFID? might return :WFMOUTPRE:WFID "Ch1, DC coupling,100.0mVolts/div,500.0 μ s/div,500 points, Hi Res mode".

WFMOutpre:XINcr? (Query Only)

This query-only command returns the horizontal point spacing in units of WFMOutpre:XUNit for the waveform specified by the [DATA:SOURce](#) command. This value corresponds to the sampling interval.

An error is reported if the DATA:SOURce waveform does not exist.

Group Waveform Transfer

Syntax WFMOutpre:XINcr?

Related Commands [DATA:SOURce](#)
[WFMOutpre:XUNit?](#)

Examples `WFMOUTPRE:XINCR?` might return `WFMOUTPRE:XINCR 10.0000E-6`, indicating that the horizontal sampling interval is 10 μs /point (500 μs /div).

WFMOutpre:XUNit? (Query Only)

This query-only command returns the horizontal units for the waveform specified by the [DATA:SOURce](#) command.

An error is reported if the `DATA:SOURce` waveform does not exist.

Group Waveform Transfer

Syntax `WFMOutpre:XUNit?`

Related Commands [DATA:SOURce](#)

Examples `WFMOUTPRE:XUNIT?` might return `WFMOUTPRE:XUNIT "HZ"`, indicating that the horizontal units for the waveform are in Hertz.

WFMOutpre:XZZero? (Query Only)

This query-only command returns the sub-sample time between the trigger sample (designated by `PT_OFF`) and the occurrence of the actual trigger for the waveform specified by the [DATA:SOURce](#) command. This value is in units of `WFMOutpre:XUNit`.

An error is reported if the `DATA:SOURce` waveform does not exist.

NOTE. *During steady state operation, when all control changes have settled and triggers are arriving on a regular basis, this is the only part of the preamble that changes on each acquisition.*

Group Waveform Transfer

Syntax `WFMOutpre:XZZero?`

Related Commands [DATA:SOURce](#)
 [WFMOutpre:XUNit?](#)

Examples `WFMOUTPRE:XZERO?` might return `WFMOUTPRE:XZERO 5.6300E-9`, indicating that the trigger actually occurred 5.63 ns before the trigger sample.

WFMOutpre:YMUlt? (Query Only)

This query-only command returns the vertical scale factor per digitizing level in units specified by `WFMOutpre:YUNit` for the waveform specified by the [DATA:SOURce](#) command. For those formats in which [WFMOutpre:BYT_Nr](#) is important (all non-floating point formats), `WFMOutpre:YMUlt?` must take the location of the binary point implied by `BYT_NR` into consideration.

An error is reported if the `DATA:SOURce` waveform does not exist.

Group Waveform Transfer

Syntax `WFMOutpre:YMUlt?`

Related Commands [DATA:SOURce](#)

Examples `WFMOUTPRE:YMULT?` might return `WFMOUTPRE:YMULT 4.0000E-3`, indicating that the vertical scale for the corresponding waveform is 100 mV/div.

WFMOutpre:YOff? (Query Only)

This query-only command returns the vertical offset of the source specified by [DATA:SOURce](#). For this instrument family, the value returned is always 0.0 as the offset is combined with the `:YZero` value.

An error is reported if the [DATA:SOURce](#) waveform does not exist.

Group Waveform Transfer

Syntax `WFMOutpre:YOff?`

Related Commands [DATA:SOURce](#)
 [WFMOutpre:BYT_Nr](#)

Examples `WFMOUTPRE:YOFF?` might return `WFMOUTPRE:YOFF -50.0000E+0`, indicating that the position indicator for the waveform was 50 digitizing levels (2 divisions) below center screen.

WFMOutpre:YUNit? (Query Only)

This query-only command returns the vertical units for the waveform specified by the [DATA:SOURce](#) command.

An error is reported if the `DATA:SOURce` waveform does not exist.

Group Waveform Transfer

Syntax `WFMOutpre:YUNit?`

Related Commands [DATA:SOURce](#)

Examples `WFMOUTPRE:YUNIT?` might return `WFMOUTPRE:YUNIT "dB"`, indicating that the vertical units for the waveform are measured in decibels.

WFMOutpre:YZEro? (Query Only)

This query-only command returns the combined vertical position and offset for the source waveform specified by [DATA:SOURce](#). This represents a departure from previous instruments where the `:YZEro` value represented the vertical position in vertical units and the `:YOFF` value represented the vertical offset in digitizing levels. For this instrument family, the value of `:YOFF` is always 0.0.

An error is reported if the [DATA:SOURce](#) waveform does not exist.

Group Waveform Transfer

Syntax `WFMOutpre:YZEro?`

Related Commands [DATA:SOURce](#)
 [WFMOutpre:YUNit?](#)

Examples `WFMOUTPRE:YZERO?` might return `WFMOUTPRE:YZERO -100.0000E-3`, indicating that vertical offset is set to -100 mV.

Status and Events

The oscilloscope provides a status and event reporting system for the Ethernet and USB interfaces. This system informs you of certain significant events that occur within the oscilloscope.

The oscilloscope status handling system consists of five 8-bit registers and two queues for each interface. The remaining Status subtopics describe these registers and components. They also explain how the event handling system operates.

Registers

Overview

The registers in the event handling system fall into two functional groups:

- Status Registers contain information about the status of the oscilloscope. They include the Standard Event Status Register (SESR).
- Enable Registers determine whether selected types of events are reported to the Status Registers and the Event Queue. They include the Device Event Status Enable Register (DESER), the Event Status Enable Register (ESER), and the Service Request Enable Register (SRER).

Status Registers

The Standard Event Status Register (SESR) and the Status Byte Register (SBR) record certain types of events that may occur while the oscilloscope is in use. IEEE Std 488.2-1987 defines these registers.

Each bit in a Status Register records a particular type of event, such as an execution error or message available. When an event of a given type occurs, the oscilloscope sets the bit that represents that type of event to a value of one. (You can disable bits so that they ignore events and remain at zero. See Enable Registers). Reading the status registers tells you what types of events have occurred.

The Standard Event Status Register (SESR). The SESR records eight types of events that can occur within the oscilloscope. Use the *ESR? query to read the SESR register. Reading the register clears the bits of the register so that the register can accumulate information about new events.

NOTE. TekVISA applications use SESR bit 6 to respond to any of several events, including some front panel actions.

7	6	5	4	3	2	1	0
PON	URQ	CME	EXE	DDE	QYE	RQC	OPC

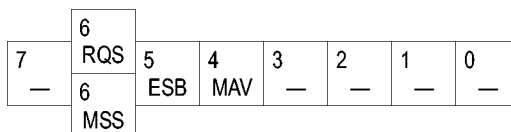
Figure 3-1: The Standard Event Status Register (SESR)

Table 3-1: SESR Bit Functions

Bit	Function	
7 (MSB)	PON	Power On. Shows that the oscilloscope was powered on. On completion, the diagnostic self tests also set this bit.
6	URQ	User Request. Indicates that an application event has occurred. *See note.
5	CME	Command Error. Shows that an error occurred while the oscilloscope was parsing a command or query.
4	EXE	Execution Error. Shows that an error executing a command or query.
3	DDE	Device Error. Shows that a device error occurred.
2	QYE	Query Error. Either an attempt was made to read the Output Queue when no data was present or pending, or that data in the Output Queue was lost.
1	RQC	Request Control. This is not used.
0 (LSB)	OPC	operation Complete. Shows that the operation is complete. This bit is set when all pending operations complete following an *OPC command.

The Status Byte Register (SBR). Records whether output is available in the Output Queue, whether the oscilloscope requests service, and whether the SESR has recorded any events.

Use a Serial Poll or the *STB? query to read the contents of the SBR. The bits in the SBR are set and cleared depending on the contents of the SESR, the Event Status Enable Register (ESER), and the Output Queue. When you use a Serial Poll to obtain the SBR, bit 6 is the RQS bit. When you use the *STB? query to obtain the SBR, bit 6 is the MSS bit. Reading the SBR does not clear the bits.

**Figure 3-2: The Status Byte Register (SBR)****Table 3-2: SBR Bit Functions**

Bit	Function	
7 (MSB)	—	Not used.
6	RQS	Request Service. Obtained from a serial poll. Shows that the oscilloscope requests service from the controller.
6	MSS	Master Status Summary. Obtained from *STB? query. Summarizes the ESB and MAV bits in the SBR.
5	ESB	Event Status Bit. Shows that status is enabled and present in the SESR.

Table 3-2: SBR Bit Functions (cont.)

Bit	Function
4	MAV Message Available. Shows that output is available in the Output Queue.
3	——— Not used.
2	——— Not used.
1–0	——— Not used.

Enable Registers

DESER, ESER, and SRER allow you to select which events are reported to the Status Registers and the Event Queue. Each Enable Register acts as a filter to a Status Register (the DESER also acts as a filter to the Event Queue) and can prevent information from being recorded in the register or queue.

Each bit in an Enable Register corresponds to a bit in the Status Register it controls. In order for an event to be reported to a bit in the Status Register, the corresponding bit in the Enable Register must be set to one. If the bit in the Enable Register is set to zero, the event is not recorded.

Various commands set the bits in the Enable Registers. The Enable Registers and the commands used to set them are described below.

The Device Event Status Enable Register (DESER). This register controls which types of events are reported to the SESR and the Event Queue. The bits in the DESER correspond to those in the SESR.

Use the DESE command to enable and disable the bits in the DESER. Use the DESE? query to read the DESER.

7	6	5	4	3	2	1	0
PON	URQ	CME	EXE	DDE	QYE	RQC	OPC

Figure 3-3: The Device Event Status Enable Register (DESER)

The Event Status Enable Register (ESER). This register controls which types of events are summarized by the Event Status Bit (ESB) in the SBR. Use the *ESE command to set the bits in the ESER. Use the *ESE? query to read it.

7	6	5	4	3	2	1	0
PON	URQ	CME	EXE	DDE	QYE	RQC	OPC

Figure 3-4: The Event Status Enable Register (ESER)

The Service Request Enable Register (SRER). This register controls which bits in the SBR generate a Service Request and are summarized by the Master Status Summary (MSS) bit.

Use the *SRE command to set the SRER. Use the *SRE? query to read the register. The RQS bit remains set to one until either the Status Byte Register is read with a Serial Poll or the MSS bit changes back to a zero.

7	6	5	4	3	2	1	0
—	—	ESB	MAV	—	—	—	—

Figure 3-5: The Service Request Enable Register (SRER)

***PSC Command** The *PSC command controls the Enable Registers contents at power-on. Sending *PSC 1 sets the Enable Registers at power on as follows:

- DESER 255 (equivalent to a DESe 255 command)
- ESER 0 (equivalent to an *ESE 0 command)
- SRER 0 (equivalent to an *SRE 0 command)

Sending *PSC 0 lets the Enable Registers maintain their values in nonvolatile memory through a power cycle.

NOTE. To enable the PON (Power On) event to generate a Service Request, send *PSC 0, use the DESe and *ESE commands to enable PON in the DESER and ESER, and use the *SRE command to enable bit 5 in the SRER. Subsequent power-on cycles will generate a Service Request.

Queues

The *PSC command controls the Enable Registers contents at power-on. Sending *PSC 1 sets the Enable Registers at power on as follows:

Output Queue The oscilloscope stores query responses in the Output Queue and empties this queue each time it receives a new command or query message after an <EOM>. The controller must read a query response before it sends the next command (or query) or it will lose responses to earlier queries.



CAUTION. When a controller sends a query, an <EOM>, and a second query, the oscilloscope normally clears the first response and outputs the second while reporting a Query Error (QYE bit in the ESER) to indicate the lost response. A fast controller, however, may receive a part or all of the first response as well. To avoid this situation, the controller should always read the response immediately after sending any terminated query message or send a DCL (Device Clear) before sending the second query.

Event Queue

The Event Queue stores detailed information on up to 33 events. If than 32 events stack up in the Event Queue, the 32nd event is replaced by event code 350, "Queue Overflow."

Read the Event Queue with the EVENT? query (which returns only the event number), with the EVMSG? query (which returns the event number and a text description of the event), or with the ALLEV? query (which returns all the event numbers along with a description of the event). Reading an event removes it from the queue.

Before reading an event from the Event Queue, you must use the *ESR? query to read the summary of the event from the SESR. This makes the events summarized by the *ESR? read available to the EVENT? and EVMSG? queries, and empties the SESR.

Reading the SESR erases any events that were summarized by previous *ESR? reads but not read from the Event Queue. Events that follow an *ESR? read are put in the Event Queue but are not available until *ESR? is used again.

Event Handling Sequence

The following figure shows how to use the status and event handling system. In the explanation that follows, numbers in parentheses refer to numbers in the figure.

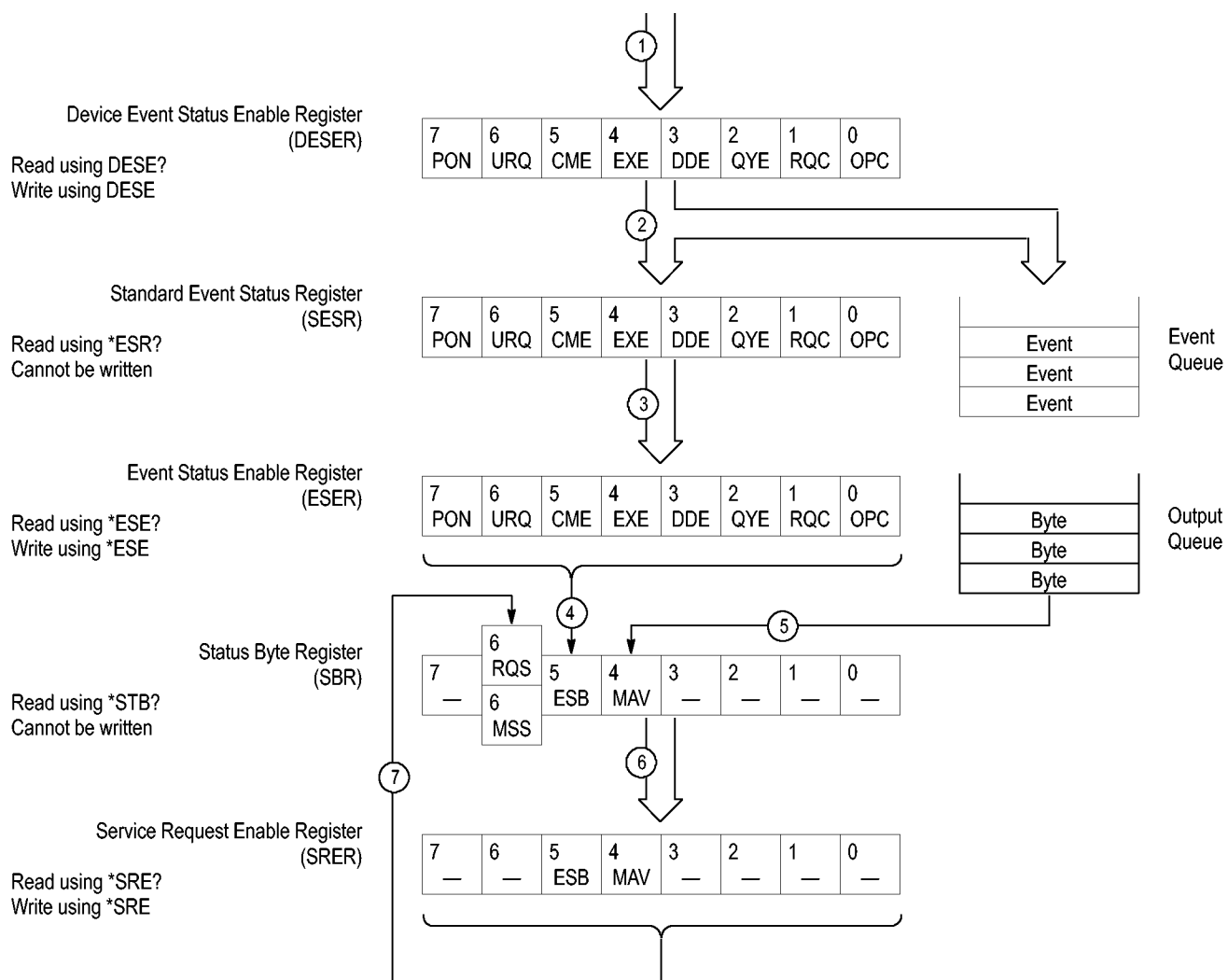


Figure 3-6: Status and Event Handling Process

When an event occurs, a signal is sent to the DESER (1). If that type of event is enabled in the DESER (that is, if the bit for that event type is set to 1), the appropriate bit in the SESR is set to one, and the event is recorded in the Event Queue (2). If the corresponding bit in the ESER is also enabled (3), then the ESB bit in the SBR is set to one (4).

When output is sent to the Output Queue, the MAV bit in the SBR is set to one (5).

When a bit in the SBR is set to one and the corresponding bit in the SRER is enabled (6), the MSS bit in the SBR is set to one and a service request is generated (7).

Synchronization Methods

Overview Although most commands are completed almost immediately after being received by the oscilloscope, some commands start a process that requires time. For example, once a single sequence acquisition command is executed, depending upon the applied signals and trigger settings, it may take an extended period of time before the acquisition is complete. Rather than remain idle while the operation is in process, the oscilloscope will continue processing other commands. This means that some operations will not be completed in the order that they were sent. Furthermore, sometimes the result of an operation depends upon the result of an earlier operation. A first operation must complete before the next one is processed.

In order to handle these situations, the oscilloscope status and event reporting system allows you to synchronize the operation of the oscilloscope with your application program, using the Operation Complete function. Note, however, that only some operations are able to take advantage of this function; a table is provided below of commands that support this.

The following commands are used to synchronize the oscilloscope functions using Operation Complete. See examples of how to use these commands later on in this section:

***OPC** — sending the *OPC command will set bit 0 of the SESR (Standard Events Status Register). The bit will only be set high when all pending operations that generate an OPC message have finished execution. (The SESR is queried using ***ESR?**) The *OPC? query form returns 1 only when all operations have completed, or a device clear is received.

***WAI** — prevents the oscilloscope from executing further commands or queries until all pending commands that generate an OPC message are complete.

BUSY? — returns the status of operations: 1 means there are pending operations, 0 means that all pending operations are complete.

NOTE. Some OPC operations may require an extended period of time to complete or may never complete. For example, a single sequence acquisition may never complete when no trigger event occurs. You should be aware of these conditions and tailor your program accordingly by:

— setting the timeout sufficiently for the anticipated maximum time for the operation and

— handle a timeout appropriately by querying the SESR (***ESR?**) and event queue (**EVMsg?** or **ALLEv?**).

NOTE. The *OPC command form can also be used to cause an SRQ to be generated upon completion of all pending operations. This requires that the ESB (Event Status Bit, bit 5) of the SRER (Service Request Enable Register) is set, and the OPC bit (bit 0) of the DESR (Device Event Status Enable Register) and the ESER (Event Status Enable Register) are set. (The SRER is set/queried using *SRE. The DESR is set/queried using DESE. The ESER is set/queried using *ESE.)

Only a subset of oscilloscope operations support the Operation Complete function (OPC):

Table 3-3: Oscilloscope operations that can generate OPC

Command	Conditions
ACQuire:STATE <non-zero NR1> ON RUN	Only when in single sequence acquisition mode
:AUTOset < EXECute >	
CALibrate:INTERNaI	
CALibrate:INTERNaI:STARt	
CALibrate:FACTory STARt	
CALibrate:FACTory CONTInue	
CALibrate:FACTory PREVious	
CH<x>:PRObe:AUTOZero EXECute	
CH<x>:PRObe:DEGAUss EXECute	
DIAG:STATE EXECute	
FACTory	
MEASUrement:MEAS<x> :RESUlts	When used in single sequence acquisition mode or during waveform recall.
RECAI:SETUp (<file as quoted string> FACTory)	
RECAI:WAVEform <.ISF or .CSV file >,< REF<x> >	
:RF:REFLevel AUTO	
*RST	
SAVe:IMAGe <file as quoted string>	
SAVe:SETUp <file as quoted string>	
SAVe:WAVEform < source wfm > , (< REF<x> > < file >)	
TEKSecure	
TRIGger:A SETLevel	

Example of Acquiring and Measuring a Single-Sequence Waveform

For example, a typical application might involve acquiring a single-sequence waveform and then taking a measurement on the acquired waveform. You could use the following command sequence to do this:

```
/** Set up conditional acquisition **/
ACQUIRE:STATE OFF
DISPLAY:WAVEVIEW1:CH1:STATE 1
HORIZONTAL:RECORDLENGTH 1000
ACQUIRE:MODE SAMPLE
ACQUIRE:STOPAFTER SEQUENCE
/** Acquire waveform data **/
ACQUIRE:STATE ON
/** Set up the measurement parameters **/
MEASUREMENT:MEAS1:TYPE AMPLITUDE
MEASUREMENT:MEAS1:SOURCE CH1
/** Take amplitude measurement **/
MEASUREMENT:MEAS1:RESULTS:CURRENTACQ:MEAN?
```

The acquisition of the waveform requires extended processing time. It may not finish before the oscilloscope takes an amplitude measurement (see the following figure). This can result in an incorrect amplitude value.

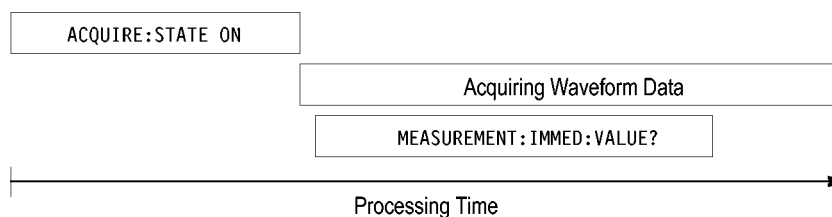


Figure 3-7: Command Processing Without Using Synchronization

To ensure the oscilloscope completes waveform acquisition before taking the measurement on the acquired data, you can synchronize the program using *WAI, BUSY, *OPC, and *OPC?.

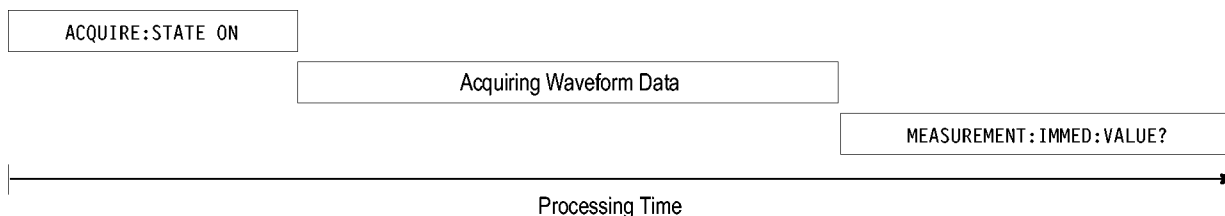


Figure 3-8: Processing Sequence With Synchronization

Example of Using the *OPC Command

If the corresponding status registers are enabled, the *OPC command sets the OPC bit in the Standard Event Status Register (SESR) when an operation is complete. You achieve synchronization by using this command with either a serial poll or service request handler.

Serial Poll Method: Enable the OPC bit in the Device Event Status Enable Register (DESER) and the Event Status Enable Register (ESER) using the DESE and *ESE commands.

When the operation is complete, the OPC bit in the Standard Event Status Register (SESR) will be enabled and the Event Status Bit (ESB) in the Status Byte Register will be enabled.

The same command sequence using the *OPC command for synchronization with serial polling looks like this:

```
/* Set up conditional acquisition */
ACQUIRE:STATE OFF
DISPLAY:WAVEVIEW1:CH1:STATE 1
HORIZONTAL:RECORDLENGTH 1000
ACQUIRE:MODE SAMPLE
ACQUIRE:STOPAFTER SEQUENCE
/* Enable the status registers */
DESE 1
*ESE 1
*SRE 0
/* Acquire waveform data */
ACQUIRE:STATE ON
/* Set up the measurement parameters on the channel we're
about to sequence */
MEASUREMENT:MEAS1:TYPE AMPLITUDE
MEASUREMENT:MEAS1:SOURCE CH1
/* wait until the acquisition is complete before taking the
measurement. */
*OPC
while serial poll = 0, keep looping
/* Take amplitude measurement */
MEASUREMENT:MEAS1:RESULTS:CURRENTacq:MEAN?
```

This technique requires less bus traffic than did looping on BUSY.

Service Request Method: Enable the OPC bit in the Device Event Status Enable Register (DESER) and the Event Status Enable Register (ESER) using the DESE and *ESE commands.

You can also enable service requests by setting the ESB bit in the Service Request Enable Register (SRER) using the *SRE command. When the operation is complete, the oscilloscope will generate a Service Request.

The same command sequence using the *OPC command for synchronization looks like this

```
/* Set up conditional acquisition */
ACQUIRE:STATE OFF
DISPLAY:WAVEVIEW1:CH1:STATE 1
HORIZONTAL:RECORDLENGTH 1000
```



```

ACQUIRE:MODE SAMPLE
ACQUIRE:STOPAFTER SEQUENCE
/* Enable the status registers */
DESE 1
*ESE 1
*SRE 32

/* Set up the measurement parameters on the channel we're
about to sequence */ :MEASUREMENT:MEAS1:TYPE AMPLITUDE
:MEASUREMENT:MEAS1:SOURCE CH1
/* Acquire waveform data */
ACQUIRE:STATE ON
/* wait until the acquisition is complete before taking the
measurement*/
*OPC

```

The program can now do different tasks such as talk to other devices. The SRQ, when it comes, interrupts those tasks and returns control to this task.

```

/* Take amplitude measurement */
MEASUREMENT:MEAS1:RESULTS:CURRENTACQ:MEAN?

```

Example of Using the *OPC? Query

The *OPC? query places a 1 in the Output Queue once an operation that generates an OPC message is complete. The *OPC? query does not return until all pending OPC operations have completed. Therefore, your time-out must be set to a time at least as long as the longest expected time for the operations to complete.

The same command sequence using the *OPC? query for synchronization looks like this:

```

/* Set up single sequence acquisition */
ACQUIRE:STATE OFF
DISPLAY:WAVEVIEW1:CH1:STATE 1
HORIZONTAL:RECORDLENGTH 1000
ACQUIRE:MODE SAMPLE
ACQUIRE:STOPAFTER SEQUENCE
/* Set up the measurement parameters on the channel we're
about to sequence */
MEASUREMENT:MEAS1:TYPE AMPLITUDE
MEASUREMENT:MEAS1:SOURCE CH1
/* Acquire waveform data */
ACQUIRE:STATE ON
/* wait until the acquisition is complete before taking the
measurement*/
*OPC?

```

Wait for read from Output Queue.

```

/* Take amplitude measurement */
MEASUREMENT:MEAS1:RESULTS:CURRENTACQ:MEAN?

```

This is the simplest approach. It requires no status handling or loops. However, you must set the controller time-out for longer than the acquisition operation.

Example of Using the *WAI Command

The *WAI command forces completion of previous commands that generate an OPC message. No commands after the *WAI are processed before the OPC message(s) are generated

The same command sequence using the *WAI command for synchronization looks like this:

```
/* Set up conditional acquisition */
ACQUIRE:STATE OFF
DISPLAY:WAVEVIEW1:CH1:STATE 1
HORIZONTAL:RECORDLENGTH 1000
ACQUIRE:MODE SAMPLE
ACQUIRE:STOPAFTER SEQUENCE
/* Set up the measurement parameters on the channel we're
about to sequence */
MEASUREMENT:MEAS1:TYPE AMPLITUDE
MEASUREMENT:MEAS1:SOURCE CH1
/* Acquire waveform data */
ACQUIRE:STATE ON
/* wait until the acquisition is complete before taking
the measurement*/
*/
*WAI
/* Take amplitude measurement */
MEASUREMENT:MEAS1:RESULTS:CURRENTACQ:MEAN?
```

The controller can continue to write commands to the input buffer of the oscilloscope, but the commands will not be processed by the oscilloscope until all in-process OPC operations are complete. If the input buffer becomes full, the controller will be unable to write commands to the buffer. This can cause a time-out.

Example of Using the BUSY? Query

The BUSY? query allows you to find out whether the oscilloscope is busy processing a command that has an extended processing time such as single-sequence acquisition.

The same command sequence, using the BUSY? query for synchronization, looks like this:

```
/* Set up conditional acquisition */
ACQUIRE:STATE OFF
DISPLAY:WAVEVIEW1:CH1:STATE 1
HORIZONTAL:RECORDLENGTH 1000
ACQUIRE:MODE SAMPLE
ACQUIRE:STOPAFTER SEQUENCE
```

```

/* Acquire waveform data */
ACQUIRE:STATE ON
/* Set up the measurement parameters */
MEASUREMENT:IMMED:TYPE AMPLITUDE
MEASUREMENT:IMMED:SOURCE CH1
/* wait until the acquisition is complete before taking
the measurement */
while BUSY? keep looping
/* Take amplitude measurement */
MEASUREMENT:IMMED:VALUE?

```

This sequence lets you create your own wait loop rather than using the *WAI command. The BUSY? query helps you avoid time-outs caused by writing too many commands to the input buffer. The controller is still tied up though, and the repeated BUSY? query will result in bus traffic.

Reference waveforms

Measurements on references also support OPC when used in conjunction with a :RECALL:WAVEFORM command.

```

/* Add a reference slot on which to measure */
:REF:ADDNEW "REF1"

/* Set up the measurement parameters on the reference */
:MEASUREMENT:MEAS1:TYPE AMPLITUDE
:MEASUREMENT:MEAS1:SOURCE REF1
/* Load the new waveform file */
:RECALL:WAVEFORM "E:\waveform.wfm",REF1
*OPC?
/* wait for read from Output Queue. */
/* Take amplitude measurement */
:MEASUREMENT:MEAS1:RESULTS:CURRENTACQ:MEAN?

```

Messages

The information contained in the topics above covers all the programming interface messages the oscilloscope generates in response to commands and queries.

For most messages, a secondary message from the oscilloscope gives detail about the cause of the error or the meaning of the message. This message is part of the message string and is separated from the main message by a semicolon.

Each message is the result of an event. Each type of event sets a specific bit in the SESR and is controlled by the equivalent bit in the DESER. Thus, each message

is associated with a specific SESR bit. In the message tables, the associated SESR bit is specified in the table title, with exceptions noted with the error message text.

No Event

The following table shows the messages when the system has no events or status to report. These have no associated SESR bit.

Table 3-4: No Event Messages

Code	Message
0	No events to report; queue empty
1	No events to report; new events pending *ESR?

Command Error

The following table shows the command error messages generated by improper syntax. Check that the command is properly formed and that it follows the rules in the section on command Syntax.

Table 3-5: Command Error Messages (CME Bit 5)

Code	Message
100	Command error
101	Invalid character
102	Syntax error
103	Invalid separator
104	Data type error
105	GET not allowed
108	Parameter not allowed
109	Missing parameter
110	Command header error
112	Program mnemonic too long
113	Undefined header
120	Numeric data error
121	Invalid character in numeric
123	Exponent too large
124	Too many digits
130	Suffix error
131	Invalid suffix
134	Suffix too long
140	Character data error
141	Invalid character data
144	Character data too long
150	String data error

Table 3-5: Command Error Messages (CME Bit 5) (cont.)

Code	Message
151	Invalid string data
152	String data too long
160	Block data error
161	Invalid block data
170	Command expression error
171	Invalid expression

Execution Error

The following table lists the execution errors that are detected during execution of a command.

Table 3-6: Execution Error Messages (EXE Bit 4)

Code	Message
200	Execution error
221	Settings conflict
222	Data out of range
224	Illegal parameter value
241	Hardware missing
250	Mass storage error
251	Missing mass storage
252	Missing media
253	Corrupt media
254	Media full
255	Directory full
256	File name not found
257	File name error
258	Media protected
259	File name too long
280	Program error
282	Insufficient network printer information
283	Network printer not responding
284	Network printer server not responding
286	Program runtime error
287	
2200	Measurement error, Measurement system error
2201	Measurement error, Zero period
2202	Measurement error, No period, second waveform

Table 3-6: Execution Error Messages (EXE Bit 4) (cont.)

Code	Message
2203	Measurement error, No period, second waveform
2204	Measurement error, Low amplitude, second waveform
2205	Measurement error, Low amplitude, second waveform
2206	Measurement error, Invalid gate
2207	Measurement error, Measurement overflow
2208	Measurement error, No backwards Mid Ref crossing
2209	Measurement error, No second Mid Ref crossing
2210	Measurement error, No Mid Ref crossing, second waveform
2211	Measurement error, No backwards Mid Ref crossing
2212	Measurement error, No negative crossing
2213	Measurement error, No positive crossing
2214	Measurement error, No crossing, target waveform
2215	Measurement error, No crossing, second waveform
2216	Measurement error, No crossing, target waveform
2217	Measurement error, Constant waveform
2219	Measurement error, No valid edge - No arm sample
2220	Measurement error, No valid edge - No arm cross
2221	Measurement error, No valid edge - No trigger cross
2222	Measurement error, No valid edge - No second cross
2223	Measurement error, Waveform mismatch
2224	Measurement error, WAIT calculating
2225	Measurement error, No waveform to measure
2226	Measurement error, Null Waveform
2227	Measurement error, Positive and Negative Clipping
2228	Measurement error, Positive Clipping
2229	Measurement error, Negative Clipping
2230	Measurement error, High Ref < Low Ref
2231	Measurement error, No statistics available
2233	Requested waveform is temporarily unavailable
2235	Math error, invalid math description
2240	Invalid password
2241	Waveform requested is invalid
2244	Source waveform is not active
2245	Saveref error, selected channel is turned off
2250	Reference error, the reference waveform file is invalid
2253	Reference error, too many points received
2254	Reference error, too few points received

Table 3-6: Execution Error Messages (EXE Bit 4) (cont.)

Code	Message
2259	File too big
2270	Alias error
2271	Alias syntax error
2273	Illegal alias label
2276	Alias expansion error
2277	Alias redefinition not allowed
2278	Alias header not found
2285	TekSecure(R) Pass
2286	TekSecure(R) Fail
2500	Setup error, file does not look like a setup file
2501	Setup warning, could not recall all values from external setup
2620	Mask error, too few points received
2760	Mark limit reached
2761	No mark present
2762	Search copy failed

Device Error

The following table lists the device errors that can occur during oscilloscope operation. These errors may indicate that the oscilloscope needs repair.

Table 3-7: Device Error Messages (DDE Bit 3)

Code	Message
310	System error
311	Memory error
312	PUD memory lost
314	Save/recall memory lost

System Event

The following table lists the system event messages. These messages are generated whenever certain system conditions occur.

Table 3-8: System Event Messages

Code	Message
400	Query event
401	Power on (PON bit 7 set)
402	Operation complete (OPC bit 0 set)
403	User request (URQ bit 6 set)
404	Power fail (DDE bit 3 set)

Table 3-8: System Event Messages (cont.)

Code	Message
405	Request control
410	Query INTERRUPTED (QYE bit 2 set)
420	Query UNTERMINATED (QYE bit 2 set)
430	Query DEADLOCKED (QYE bit 2 set)
440	Query UNTERMINATED after indefinite response (QYE bit 2 set)
468	Knob/Keypad value changed
472	Application variable changed

Execution Warning

The following table lists warning messages that do not interrupt the flow of command execution. They also notify you of possible unexpected results.

Table 3-9: Execution Warning Messages (EXE Bit 4)

Code	Message
528	Parameter out of range
532	Curve data too long, Curve truncated
533	Curve error, Preamble values are inconsistent
540	Measurement warning, Uncertain edge
541	Measurement warning, Low signal amplitude
542	Measurement warning, Unstable histogram
543	Measurement warning, Low resolution
544	Measurement warning, Uncertain edge
545	Measurement warning, Invalid in minmax
546	Measurement warning, Need 3 edges
547	Measurement warning, Clipping positive/negative
548	Measurement warning, Clipping positive
549	Measurement warning, Clipping negative

Table 3-10: Execution Warning Messages (EXE Bit 4)

Code	Message
540	Measurement warning
541	Measurement warning, Low signal amplitude
542	Measurement warning, Unstable histogram
543	Measurement warning, Low resolution
544	Measurement warning, Uncertain edge
545	Measurement warning, Invalid min max
546	Measurement warning, Need 3 edges

Table 3-10: Execution Warning Messages (EXE Bit 4) (cont.)

Code	Message
547	Measurement warning, Clipping positive/negative
548	Measurement warning, Clipping positive
549	Measurement warning, Clipping negative

Internal Warning

The following table shows internal errors that indicate an internal fault in the oscilloscope.

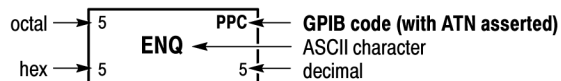
Table 3-11: Internal Warning Messages

Code	Message
630	Internal warning, 50Ω overload

Appendix A: Character Set

B7 B6 B5 BITS B4 B3 B2 B1	0 0 0		0 0 1		0 1 0		0 1 1		1 0 0		1 0 1		1 1 0		1 1 1	
	CONTROL				NUMBERS SYMBOLS				UPPER CASE				LOWER CASE			
0 0 0 0	0 NUL 0 0	20 DLE 10 16	40 SP 20 32	60 0 30 48	100 @ 40 64	120 P 50 80	140 , 60 96	160 p 70 112								
0 0 0 1	1 GTL 1 SOH 1 1	21 LL0 11 DC1 17	41 LA1 21 ! 33	61 LA17 31 1 49	101 TA1 41 A 65	121 TA17 51 Q 81	141 SA1 61 a 97	161 SA17 71 q 113								
0 0 1 0	2 STX 2 2	22 DC2 12 18	42 LA2 22 " 34	62 LA18 32 2 50	102 TA2 42 B 66	122 TA18 52 R 82	142 SA2 62 b 98	162 SA18 72 r 114								
0 0 1 1	3 ETX 3 3	23 DC3 13 19	43 LA3 23 # 35	63 LA19 33 3 51	103 TA3 43 C 67	123 TA19 53 S 83	143 SA3 63 c 99	163 SA19 73 s 115								
0 1 0 0	4 SDC 4 EOT 4 4	24 DCL 14 DC4 20	44 LA4 24 \$ 36	64 LA20 34 4 52	104 TA4 44 D 68	124 TA20 54 T 84	144 SA4 64 d 100	164 SA20 74 t 116								
0 1 0 1	5 PPC 5 ENQ 5 5	25 PPU 15 NAK 21	45 LA5 25 % 37	65 LA21 35 5 53	105 TA5 45 E 69	125 TA21 55 U 85	145 SA5 65 e 101	165 SA21 75 u 117								
0 1 1 0	6 ACK 6 6	26 SYN 16 22	46 LA6 26 & 38	66 LA22 36 6 54	106 TA6 46 F 70	126 TA22 56 V 86	146 SA6 66 f 102	166 SA22 76 v 118								
0 1 1 1	7 BEL 7 7	27 ETB 17 23	47 LA7 27 ' 39	67 LA23 37 7 55	107 TA7 47 G 71	127 TA23 57 W 87	147 SA7 67 g 103	167 SA23 77 w 119								
1 0 0 0	10 GET 8 BS 8 8	30 SPE 18 CAN 24	50 LA8 28 (40	70 LA24 38 8 56	110 TA8 48 H 72	130 TA24 58 X 88	150 SA8 68 h 104	170 SA24 78 x 120								
1 0 0 1	11 TCT 9 HT 9 9	31 SPD 19 EM 25	51 LA9 29) 41	71 LA25 39 9 57	111 TA9 49 I 73	131 TA25 59 Y 89	151 SA9 69 i 105	171 SA25 79 y 121								
1 0 1 0	12 LF A 10	32 SUB 1A 26	52 LA10 2A * 42	72 LA26 3A : 58	112 TA10 4A J 74	132 TA26 5A Z 90	152 SA10 6A j 106	172 SA26 7A z 122								
1 0 1 1	13 VT B 11	33 ESC 1B 27	53 LA11 2B + 43	73 LA27 3B ; 59	113 TA11 4B K 75	133 TA27 5B [91	153 SA11 6B k 107	173 SA27 7B { 123								
1 1 0 0	14 FF C 12	34 FS 1C 28	54 LA12 2C , 44	74 LA28 3C < 60	114 TA12 4C L 76	134 TA28 5C \ 92	154 SA12 6C l 108	174 SA28 7C ; 124								
1 1 0 1	15 CR D 13	35 GS 1D 29	55 LA13 2D - 45	75 LA29 3D = 61	115 TA13 4D M 77	135 TA29 5D] 93	155 SA13 6D m 109	175 SA29 7D } 125								
1 1 1 0	16 SO E 14	36 RS 1E 30	56 LA14 2E . 46	76 LA30 3E > 62	116 TA14 4E N 78	136 TA30 5E ^ 94	156 SA14 6E n 110	176 SA30 7E ~ 126								
1 1 1 1	17 SI F 15	37 US 1F 31	57 LA15 2F / 47	77 UNL 3F ? 63	117 TA15 4F O 79	137 UNT 5F - 95	157 SA15 6F o 111	177 RUBOUT (DEL) 7F 127								
	ADDRESSED COMMANDS		UNIVERSAL COMMANDS		LISTEN ADDRESSES		TALK ADDRESSES		SECONDARY ADDRESSES OR COMMANDS							

KEY



Tektronix

REF: ANSI STD X3.4-1977
IEEE STD 488.1-1987
ISO STD 646-2973

Appendix B: Reserved Words

This is a list of reserved words for your instrument. Capital letters identify the required minimum spelling.

*CAL	ADVanced	AUTOINCrement	BATHTUB
*CLS	AFG	AUTORange	BATHtub
*DDT	AHPOSition	AUTOSAVEPITIMEOUT	BCR
*ESE	ALiAs	AUTOSAVEUITIMEOUT	BDIFFBP
*ESR	ALL	AUTOScale	BEGIN
*IDN	ALLAcqs	AUTOSet	BELl
*LRN	ALLBits	AUTOZero	BER
*OPC	ALLEv	AUTOmatic	BHPOSition
*OPT	ALLQString	AUTOScale	BIAS
*PSC	ALLTHResholds	AUTOset	BIN
*PUD	ALTERNATEA	AUXLevel	BINARY
*RST	ALTERNATEB	AUXiliary	BiNary
*SRE	ALWAYS	AUXout	BIT
*STB	AM14	AVAILable	BITAMPLITUDE
*TRG	AMPLitude	AVERAge	BITCfmode
*TST	AMPlITUDE	AVERage	BITDelay
*WAI	ANALOG	AVG	BITEnd
0	ANALog	AVPOSITION	BITHIGH
1	ANALYZemode	AXPOSITION	BITLOW
7	AND	AYPOSITION	BITOrder
8	ANNOtate	B	BITPcnt
9	ANY	B0	BITRate
A	ANYERRor	B1	BITSTUFFing
ABOrt	AOFFSet	B10	BITStart
ABSolute	APOSition	B11	BITType
AC	APPID	B12	BIT_Nr
ACCOMMONMODE	APPPWR	B13	BJT
ACDCRMS	APPlY	B14	BLACKMANHarris
ACK	ARBitrary	B15	BLANKingtime
ACKMISS	AREA	B16	block_data
ACKnum	AREAofcrosssection	B2	BM
ACQ	ARINC429A	B3	BMP
ACQDURATION	ASCII	B4	BN_Fmt
ACQuire	ASCIi	B5	BOFFSet
ACQuisition	ASC_Fmt	B6	BOLD
ACRMS	ASCIi	B7	boolean
ACTive	asccurve	B8	BOTH
ACTivesourcecount	ASIC	B9	BPOSITION
ADD	ASIS	BACKLight	BRsBit
ADDMEAS	ASOURce	BACKward	BSOURce
ADDNew	ATRIGger	BANDwidth	BULK
ADDR10	AUDio	BANDwidth	BUNits
ADDR7	AUNIts	BASE	BURSTEDGTYPE
ADDRANDDATA	AUTHenticate	BASETop	BURSTWIDTH
ADDReSS	AUTO	BASEline	BUS
ADDReSS	AUTODim	BASic	BUSANDWAVEFORM

BUSTABle	CLOCKSOURCE	CYCLEMAX	DDRTERRN
BUSY	CLOCK	CYCLEMin	DDRTJITCC
BUSx	CLOCKSOURCE	CYCLEMode	DDRTJITDUTY
BVPOSITION	CLOCK	CYCLEPKPK	DDRTJITPER
BXPOSITION	CMETHOD	CYCLETOP	DDRTPST
BY	COFFSet	CYCLEcount	DDRTRPRE
BYPOSITION	COLor	CYCLEmode	DDRTWPRE
BYT_Nr	COLors	Chx_Dx	DDRVIxAC
BYT_Or	COMMAND	Current	DDRWRITE
Block	COMMONMODE	DAMPing	DDT
B	COMMONMode	DASHED	DECIMAL
Bpeak	COMMENTS	DASHSOLID	DEFAUltsetup
Br	COMPOSITION	DATA	DEFine
Bus	CONDENRG	DATA0	DEGAUSS
Bx	CONDition	DATA1	DEGrees
CALibrate	CONDLOSS	DATA2	DELAY
CAN	CONDition	DATABits	DELEte
CAN2X	CONFIGuration	DATAFormat	DELIMiter
CANH	CONFIGure	DATAFormat	DELTATime
CANL	CONFIGure	DATAMINUSTHRESHoId	DELTA
CAPACITANCE	CONSTANTCLOCK	DATAPLUSTHRESHoId	DELTAtime
CARDIac	CONSTCLOCKMODE	DATAPacket	DELAy
CATalog	CONTROL	DATAPath	DELeTe
CCRESUlts	CONTRol	DATARATE	DELeTeqstring
CCYClEs	CONTInuous	DATARESUlts	DESE
CENTERFREQUENCY	COPy	DATARate	DESKEW
CGS	COUNTer	DATE	DESKew
CH	COUNT	DATA	DEST
CH1	COUPLing	DAlI	DESTination
CHAN1A	CR	DBCA	DESTinationaddr
CHAN2B	CRC	DBM	DESTinationport
CHANwidth	CRC16	DC	DETECTPLEN
CHX	CRC5	DCD	DEVICEDESCMASTERREAD
Channel	CRCHeader	DCREJ	DEVICEDESCSLAVEREAD
CHECKsum	CRCTrailer	DDJ	DEVICEType
CH	CROSSIng	DDR	DHCPbootp
CH_DALL	CSPLIT	DDRAOS	DIAG
CH1_D	CURRent	DDRAOSPERTCK	DIDT
CHx	CURRentacq	DDRAOSPERUI	DIFF
CHx_DALL	CURSOR	DDRAUS	DIFFerential
CHx_Dx	CURSOR	DDRAUSPERTCK	DIGGRP
CHx_Dy	CURSor	DDRAUSPERUI	DIR
CLASSA	CURVe	DDRHOLDDIFF	DIRacmodel
CLASSB	CURrent	DDRREAD	DIREction
CLASSC	CURsor	DDRREADWRITE	DISPLAY
CLASSD	CUSTOM	DDRSETUPDIFF	DISPLAYUnits
CLASS	CUSTOMPROPagation	DDRTCHABS	DISPlaystat
CLEAR	CUSTOM	DDRTCHAVERAGE	DisAbled
CLOCK	CWD	DDRTCKAVERAGE	DISPlay
CLOCKBITSPERCHANNEL	CYCLE	DDRTCLABS	DISPlaymode
CLOCKFrequency	CYCLEAmp	DDRTCLAVERAGE	DIVide
CLOCKMultiplier	CYCLEBase	DDRTDQSCK	DIVisions
CLOCKRecovery	CYCLECount	DDRTERRMN	DJ

DJDD	EOW	FAIL	FOLDER
DJDIRAC	EOP	FALL	FOLDer
DMINus	Equal	FALLHigh	FONT
DNS	Equal	FALLING	FORCEDRange
DOFFSet	EqualNOTEQual	FALLLow	FORCe
DOMAINname	ERISe	FALLMid	FORCetrig
DOMain	ERROR	FALLSLEWRATE	FORMERror
DONTInclude	ERRTYPE	FALLTIME	FORMat
DOTsonly	ERRTYpe	FALSE	FORward
DOUBLE	ERRType	FALl	FOUR
DPLUS	ERRor	FALLing	FOURHundred
DRive	ESCOPEENABLe	FAST	FOURTEENten
DUTy	ESIBit	FASTAcq	FP
DVDT	ET	FASTERthan	FPAnel
DVM	ETHERNET	FASTer	FPBinary
DYNAMIC	ETHERnet	FASTframe	FRAME
DYNAMIC	ETHerNet	FBD1	FRAMEID
DYNRange	EVEN	FBD2	FRAMELEngth
DeltaB	EVENT	FBD3	FRAMEType
DeltaH	EVENTS	FC1063	FRAMEType
D	EVENTtable	FC133	FRAMING
EACHCLOCKCYCLE	EVERY	FC2125	FRame
ECL	EVMsg	FC266	FRame
EDECay	EVQty	FC4250	FREference
EDGE	EXECute	FC531	FREQ
EDGEIncre	EXITSwindow	FC8500	FREQUENCY
EDGES	EXPLICITCLOCK	FCSError	FREquency
EDGESOURce	EXPLICITCLOCKMODE	FD	FROM
EDGESource	EXTAtten	FDBITS	FROMEDGESEARCHDirect
EDGEType	EXTDBatten	FDISO	FROMLeve1
EDGE	EXTENDED	FDNONISO	FROMSearch
EDGE	EXTENDED	FFT	FROMedge
EFFICIENCY	EXTUnits	FIBREchannel	FRame
EITHER	EXTended	FIFTY	FULL
EITHer	EXTernal	FIFTy	FULLColor
ENABLe	EXTREGRead	FILE	FULLSPEED
ENABLe	EXTREGWrite	FILEFormat	FULLScreen
ENCdg	EYE	FILENAME	FULl
END	EYEDIAGRAM	FILESsystem	FUNCTion
ENDPoint	EYEHIGH	FILTER	FUNDcURRent
ENET	EYELOW	FILTERs	FW1394BS1600B
ENET100	EYERENDER	FILEpath	FW1394BS400B
ENET1000	EYEWIDTHBER	FIRST	FW1394BS800B
ENGineering	EYEhistogram	FIVE	GAIN
ENHANCED4	Efficiency1	FIVEHundred	GAP
ENHANCED8	Efficiency2	FIXed	GATESource
ENHanced	Efficiency3	FLATTOP2	GATEWay
ENTERSwindow	F1MAG	FLAg	GATING
ENV	F2	FLEXRAY	GAUSSian
ENVELOpe	F3MAG	FLEXray	GAUSSian
EOF	F4	FLYBACK	GLOBal
EOFTYPE	F8	FOCUS	GLOBalref
EOP	FActory	FOCus	GLOBal

GMT	HYSTERESIS	INVERTed	LFRej
GPKNOb1	HYSTeresis	INVerted	LIC
GPKNOb2	Hc	INVerted	LICENSE
GRATicule	Hmax	INrange	LICense
GRATicule	I2C	IO	LIMIT
GRIDlines	I2S	IOUT1SOurce	LIN
GRId	IBA2500	IOUT2SOurce	LINE
GROUPDelay	IBA_GEN2	IOUT3SOurce	LINEAR
H	ICFACTOR	IPAddress	LINEAR
HAMming	ID	IPHeader	LINERIPPLE
HANDSHAKEPacket	IDANDDATA	IPOwer	LINeStyle
HANDSHAKEType	IDentifier	IPVFOUR	LINEar
HANNing	IDFormat	IRMS	LIST
HARMONICS	IDLE	IRipple	LIVE
HAVERSINE	IDLETime	ISCLOCKED	LIVe
HBars	IDLe	ISOALL	LJ
HDR	IEC	ISOEND	LOAD
HEADER	ILEVELabs	ISOMID	LOCKRJ
HEADer	ILEVELPct	ISOSTART	LOCKRJValue
HEIGHT	IMAGe	ISOURce	LOCK
HEIGHTBER	IMAGinary	ISource	LOG
HEX	IMPEDance	ISOchronous	LOGIC
HFRej	IN	ITALic	LOGICPattern
HI	INCLUDEREFS	ITEM	LOGICQUALification
HID	INCLUde	IVSINTEGRALV	LOGICSource
HIGH	INCLUde	IVSINTV	LOGic
HIGHLevel	INDBits	InputPwr	LOGic
HIGHLimit	INDEPENDENT	J2	LONGEXTREGREAD
HIGHPass	INDUCT	J9	LONGEXTREGWRITE
HIGHREFVoltage	INDUCTANCE	JITTERMODE	LOOP
HIGHRES	INFinite	JITTERSUMMARY	LOOPBandwidth
HIGHTIME	INFMAximum	JITTERSummary	LOREntz
HIGHZ	INFMINimum	JITTermodel	LOW
HIRes	INFPersist	JPG	LOWERFREquency
HISTOGRAM	INIT	JTFBandwidth	LOWLevel
HIStory	INPUT	KAISERBessel	LOWLimit
HITDMVALue	INPUTCAP	L	LOWPass
HIVALue	INPUTMode	LABELANDDATA	LOWREFVoltage
HOLD	INPUTSource	LABel	LOWSPEED
HOLDTime	INPUTType	LAN	LOWTHRESHold
HOLDoff	INRUSH	LATCh	LOWTIME
HOMEDir	INRUSHcurrenr	LAYout	Lower
HORDER	INSIDEGreater	LDIR	Lowerthreshold
HORIZontalscale	INSTR	LEFt	LSB
HORZ	INSTAll	LENGth	LXI
HORZPOS	INTEGER	LESSEQual	MAC
HORZScale	INTENSITY	LESSLimit	MACADDRESS
HORizontAl	INTERNAL	LESSThan	MACLENGth
HSOURce	INTERPRatio	LESSThan	MAG
HTTPPort	INTERRUPT	LEVELUNits	MAGLOSS
HUB	INTERNAL	Leve1	MAGNETICLOSS
HUNDREDBASETX	INTERp	LF	MAGNitude
Hundred	INVERTNIBble	LFREquency	MAGPROPERTY

MAINwindow	MODe1	NR_Pt	OVERRide
MAIn	MOREEEqua	NTIMES	OVERlay
MANUAL	MOREEEQual	NTIMES	OVERlay
MANua1	MORELimit	NULL	Output1Pwr
MASTERADdress	MOREThan	NULLFRDynamic	Output2Pwr
MASTERREAd	MOREthan	NULLFRStatic	Output3Pwr
MASTERWRite	MOSFET	NUL1	PACKET
MATH	MSB	NUMACQs	PALEtte
MATHArbflt	MULTiply	NUMACq	PANKNOB
MATHFFTView	MULTipleframes	NUMAVg	PARItyleft
MATHFFTx	Mathx	NUMBins	PARalle1
MATH	N	NUMCHANne1	PARity
MATHx	NAK	NUMFRAMESACquired	PASSword
MATH	NAME	NUMOFOutputs	PATLen
MAX	NAME	NUMSEQUence	PATTERNDETECTION
MAXFRames	NAND	NUMBER	PATTERNLENGth
MAXIMUM	NANd	NWIDTH	PATTERNType
MAXSamplerate	NAVigate	NWIDTTH	PATtern
MAXimum	NDUTYCYCLE	NYET	PATtern
MDATA	NDuty	None	PAUSE
ME	NDuty	OC1	PAUSEPULSE
MEAN	NEGAtive	OC12	PAUSE
MEANAUTOCalculate	NEGAtive	OC3	PAYLength
MEANhistogram	NETWORKCONFig	OC48	PAYLength
MEASNAME	NEWpass	OCCURS	PAYLoad
MEASRange	NEXT	ODD	PAYload
MEASUREMENT	NEXt	ODDEven	PCIE_GEN1
MEAS	NIBBLECount	OFF	PCIE_GEN2
MEASured	NO	OFF1	PCIE_GEN3
MEASx	NOCARE	OFFSet	PCIExpress
MEDian	NOISEAdd	ON	PDF
MEDium	NOISErej	ONCE	PDUTTY
MEMORY	NOISe	ONE	PDUTY
METHod	NOMINALOFFset	ONEOVERDELTAVALUE	PDUTYCYCLE
MHT	NOMina1	ONFAIL	PEAKHo1d
MID	NOMina1freq	OPPOSITEas	PEAKdetect
MIDRef	NONE	OPTIMize	PERCent
MIL	NONTRANSition	OPTion	PERFREQ
MIL1553B	None	OR	PERIOD
MIN	NOPARity	OUT	PERIOD
MINIMUM	NOR	OUTPut	PERSistence
MINimum	NORESPonse	OUTPut1	PERSource
MINMax	NORMa1	OUTPut2	PFACTOR
MINUI	NORMa1	OUTPut3	PFC
MINimum	NOTE	OUTPut4	PHASE
MIXED	NOVERSHOOT	OUTPut5	PHASENOISE
MIXEDASCII	NOne	OUTPut6	PHASEnoise
MIXEDHEX	NPERIOD	OUTPut7	PHASe
MIXed	NPJ	OUTPUTType	PID
MKDir	NPULSEWIDTH	OUTSIDEGreater	PING
MODE	NR1	OUTrange	PJ
MODEhistogram	NR2	OUTside	PK2PK
Mode	NR3	OVERLoad	PK2Pk

PLL	QFACTOR	RATE9K	RI
PLOT	QString	RATE9M	RIBinary
PLOTData	QStringNR1	RATed	RIGHT
PLOTIMAGE	QStringQString	RCURRent	RIO125
PLOTVIEW	QTAG	RDS	RIO250
PLOTVIEW	QTAGGING	RDSON	RIO3125
PLOTview	QUALifier	RDson	RISE
PLOT	QUALifier	READ	RISEFall
PLOTx	Qstring	READFile	RISEHigh
PNG	RADians	REAL	RISELow
PNJUNCTION	RAMP	RECA11	RISEMid
POHCL	RAMPtime	RECA11mask	RISESLEWRATE
POHCM	RANdom	RECORD	RISETIME
POHCS	RATE100K	RECORDLength	RISING
POINT	RATE10K	RECOrdlength	RISe
POLARITY	RATE10M	RECTANGular	RISing
POLarity	RATE115K	REDO	RJ
POPulation	RATE11M	REF	RJDD
PORT	RATE125K	REFFFTView	RJDIRAC
POSITIVE	RATE12M	REFFFTx	RJDJ
POST	RATE13M	REFLEvel	RMDir
POStion	RATE14M	REFLevelS	RMS
POStive	RATE153K	REFLevelS	ROLL
POVERSHOOT	RATE15M	REFMode	ROSc
POWERQUALITY	RATE16M	REFOUT	RP
POWERRating	RATE19K	REFVo1tage	RPBinary
POWERTABLe	RATE1K	REFerence	RRBDisplaystate
Power	RATE1M	REF	RS232C
Power	RATE20K	REF_DALL	RUN
PPULSEWIDTH	RATE250K	REF_D	RUNSTop
PRESS	RATE25K	REFx	RUNT
PREViewstate	RATE2K	REFx_Dx	RUNT
PREvious	RATE2M	REGOWRite	RWINADDR
PREamble	RATE300	REGISTERADDDress	RX
PREv	RATE31K	REGREAd	SAME
PRIMARYTURNs	RATE33K	REGWRite	SAMEas
PROBEContro1	RATE38K	REM	SAMPLERate
PROBECa1	RATE3M	REMOte	SAMPLEpoint
PROBEFunc	RATE400K	REName	SAMp1e
PROTOCo1	RATE4K	REPEATstart	SAS12_NOSSC
PROTOco1	RATE4M	REPort	SAS12_SSC
Probe	RATE500K	REPWR	SAS15_NOSSC
PT_Fmt	RATE50K	REPeating	SAS15_SSC
PT_Order	RATE5M	RESERVed	SAS3_NOSSC
PT_off	RATE62K	RESET	SAS3_SSC
PULSEwidth	RATE68K	RESet	SAS6_NOSSC
PULSe	RATE6M	RESolution	SAS6_SSC
PULse	RATE7M	RESPonsetime	SATA_GEN1
PWIDTH	RATE800K	RESUME	SATA_GEN2
PWRFACTOR	RATE83K	RESUlt	SATA_GEN3
PWRFREQ	RATE8M	RESUltS	SAVEON
PWRUpstatus	RATE921K	RESistance	SAVe
Permeability	RATE92K	RESolution	SAVemask

SC	SETUP	SPAN	SUBAddress
SCALE	SFD	SPEC	SUBNETMask
SCALERATIo	SFPbinary	SPECIALPacket	SUBSF
SCALE	SHort	SPECIALType	SUBtract
SCAlē	SHUTdown	SPECTRAL	SUMFrame
SCientific	SI	SPECTRALBUJ	SUPPress
SCREEN	SIGNAL	SPECTRUM	SUSPEND
SDI	SIGNALType	SPECTraI	SWITCHING
SDIDATA	SIGNALType	SPECTraI	SWITCHINGLOSS
SDIDATASSM	SIGNaI	SPECTrum	SWITCHINGRIPPLE
SE	SIGNaI	SPI	SWLCONFIGType
SEARCH	SINC	SPLIT	SYMMetry
SEARCH1	SINE	SPLITMODE	SYNC
SEARCHSource	SINGLE	SPLit	SYNCFramē
SEARCHTABLe	SINGLEseq	SPREADSHEET	SYNCFiēd
SEARCH	SINX	SPREADSheet	SYSTEM
SEARCHtotrigger	SIX	SPace	TABLE
SEARCHX	SIXTEENeight	SQUare	TABLEX
SEARCh	SIXty	SRibinary	TARGETBER
SEC1SOURCe	SIZE	SRJ	TCPHeader
SEC1TURNS	SIZE	SRPbinary	TDM
SEC2SOURCe	SKEW	SRQ	TDMVAlue
SEC2TURNS	SLAVEADDRESS	SS	TEKEXponential
SEC3SOURCe	SCLK	SSC	TEKSecure
SEC3TURNS	SDATA	SSCFREQDEV	TEMPerature
SEC4SOURCe	SLEEP	SSCMODRATE	TENBASET
SEC4TURNS	SLEEp	SSCPROFILE	TENNinety
SEC5SOURCe	SLEWRATEMethod	SSM	TERMinal
SEC5TURNS	SLOW	SSPLIT	TERMination
SEC6SOURCe	SLOWERthan	STALL	TF
SEC6TURNS	SLOWer	STANDard	THDF
SECOND	SLOpe	START	THDR
SECOnds	SLOpe	STARTOfframe	THIRD
SECWINDings	SMPS	STARTup	THREE
SECOnds	SNAp	STARTupnosync	THRESHoId
SELECTED	SOA	START	THReshoId
SELECTIONtype	SOCKETServer	STATE	TICKTIME
SELEct	SOF	STATIC	TICKTOLerance
SELEcted	SOFFRAMENUMBER	STATISTICS	TICKS
SENT	SOLID	STATUS	TIE
SEQnum	SORTBY	STATE	TIEHISTOGRAM
SEQUence	SOURCE	STATic	TIEHISTOgram
SERVICE	SOURCEEDGEType	STATUS	TIESPECTRUM
SERVICENAME	SOURCEList	STAYSHigh	TIETIMETREND
SERnumber	SOURCEMODEA	STAYSLow	TIMEOUTSIDELEVEL
SESSion	SOURCEMODEB	STAcked	TIMEOut
SET	SOURCES	STAndard	TIMESamp
SETHOLDLogicval	SOURCe	STDDev	TIMETREND
SETHoId	SourCe	STOP	TIME
SETLevēl	Sourceaddr	STOPAcq	TJBER
SETTO50	Source	STOPAfter	TNTRATIO
SETTime	SOURCEport	STYle	TO
SETUP	SOW	STAndard	TOEDGESEARCHDIRECT

TOEdge	TX	VDIFFXOVR	WINDOWLENGTH
TOFFENRG	TXRX	VECTors	WINDOWlength
TOFFLOSS	TXRXTHRESHold	VERBoSe	WINSCALE
TOKENPacket	TYPE	VERTPOS	WINDow
TOKENType	TYPE1	VERTSCALE	WITHin
TOLevel	TYPE2	VERTical	WORD
TONENRG	TYPE	VGLevel	WORDSel
TONLOSS	TotalEfficiency	VIEW	WORDSize
TOP	UNDERline	VIEWStyle	WRITE
TOTAL	UNDO	VIEWType	WRITEFile
TOTALuptime	UNEQual	VISIBility	X
TOUCHSCREEN	UNINSTALL	VLEVELAbs	XAUI
TRACK	UNIQUE	VLEVELPct	XAUI_GEN2
TRANSFERbusownership	UNITINTERVAL	VOLTAGE	XAXIS
TRANSition	UNITIntervals	VOLTage	XAXISUnits
TRANSition	UNITIntervals	VOUT1Source	XFF
TRBit	UNITS	VOUT2Source	XINcr
TREND	UNIts	VOUT3Source	XPOS
TREND	UNLock	VRMS	XUNit
TRIGLevel	UNMOUNT	VSOURce	XY
TRIGMode	UNWrap	VSOURce	XYZ
TRIGSlope	UPPERFREquency	VUNIT	XZErO
TRIGger	UPPer	WAKEup	Y
TRIGgertosearch	UPPerthreshold	WAKEup	YAXIS
TRPWR	USB	WAVEFORM	YES
TRUEPWR	USB3	WAVEVIEW1	YMuLt
TRUe	USBDevice	WAVEView	YOff
TTL	USBTmc	WAVEviewlt;y>	YPOS
TTLENRG	USECLockedge	WAVEform	YUNit
TTLLOSS	USECLockedge	WAVFrm	YZErO
TURN	USER	WEIGHT	ZERo
TURNOfftime	UTCDELTA	WFId	ZONE
TURNONtime	V1X	WFMInpre	ZOOM
TWELVetwelve	V2X	WFMOutpre	ZOOMNR1
TWENTyeighty	VALidate	WFMPre	ZOOMOVERride
TWENTyfive	VALue	WFMTYPE	ZOOM
TWO	VARpersist	WHEn	Zoom
TWOCLOCKCYCLES	VBArs	WIDTH	
TWOFifty	VCESat	WIDTHBER	
TWOTHousand	VCFACTOR	WIDTh	

Appendix C: Factory Defaults

Default Setup

The following table lists the default values for each command.

NOTE. Find the most up-to-date default values for your instrument and software by performing a TekSecure command, saving the instrument setup and looking at the instrument or setup file.

Table C-1: Default Values

Item	Description
ACQUIRE:FASTACQ:PALETTE	TEMPERATURE
ACQUIRE:FASTACQ:STATE	0
ACQUIRE:MAGNIVU	0
ACQUIRE:MODE	SAMPLE
ACQUIRE:NUMAVG	16
ACQUIRE:NUMENV	INFINITE
ACQUIRE:STATE	1
ACQUIRE:STOPAFTER	RUNSTOP
ACTONEVENT:ACTION:AUXOUT:STATE	0
ACTONEVENT:ACTION:EMAIL:SETUP:TOADDRESS	"TestString"
ACTONEVENT:ACTION:EMAIL:STATE	0
ACTONEVENT:ACTION:PRINT:STATE	0
ACTONEVENT:ACTION:SAVEIMAGE:STATE	0
ACTONEVENT:ACTION:SAVEWFM:STATE	0
ACTONEVENT:ACTION:SRQ:STATE	0
ACTONEVENT:ACTION:STOPACQ:STATE	1
ACTONEVENT:ACTION:VISUAL:STATE	0
ACTONEVENT:EVENTTYPE	NONE
ACTONEVENT:NUMACQS	1
ACTONEVENT:REPEATCOUNT	1.0000
AFG:AMPLITUDE	500.0000E-3
AFG:ARBITRARY:EMEM:POINTS:ENCDG	ASCII
AFG:FREQUENCY	100.0E+3
AFG:FUNCTION	SINE
AFG:HIGLEVEL	250.0000E-3
AFG:LEVELPRESET	USER
AFG:LOWLEVEL	-250.0000E-3

Table C-1: Default Values (cont.)

Item	Description
AFG:NOISEADD:PERCENT	0.0E+0
AFG:NOISEADD:STATE	0
AFG:OFFSET	0.0E+0
AFG:OUTPUT:LOAD:IMPEDANCE	HIGHZ
AFG:PERIOD	9.99999999999916E-6
AFG:PHASE	0.0E+0
AFG:PULSE:WIDTH	1.0E-6
AFG:RAMP:SYMMETRY	50.0000
AFG:SQUARE:DUTY	50.0000
ALIAS:STATE	0
APPLICATION:TYPE	POWER
AUXOUT:EDGE	RISING
AUXOUT:SOURCE	ATRIGGER
BUS:B1:AUDIO:BITDELAY	1
BUS:B1:AUDIO:BITORDER	MSB
BUS:B1:AUDIO:CHANNEL:SIZE	24
BUS:B1:AUDIO:CLOCK:POLARITY	RISE
BUS:B1:AUDIO:CLOCK:SOURCE	CH1
BUS:B1:AUDIO:DATA:POLARITY	NORMAL
BUS:B1:AUDIO:DATA:SIZE	24
BUS:B1:AUDIO:DISPLAY:FORMAT	SIGNEDDECIMAL
BUS:B1:AUDIO:FRAME:SIZE	8
BUS:B1:AUDIO:FRAMESYNC:POLARITY	RISE
BUS:B1:AUDIO:FRAMESYNC:SOURCE	CH2
BUS:B1:AUDIO:TYPE	I2S
BUS:B1:AUDIO:WORDSEL:POLARITY	NORMAL
BUS:B1:AUDIO:WORDSEL:SOURCE	CH2
BUS:B1:CAN:BITRATE	500000
BUS:B1:CAN:FD:BITRATE	4000000
BUS:B1:CAN:FD:STANDARD	ISO
BUS:B1:CAN:PROBE	CANH
BUS:B1:CAN:SAMPLEPOINT	50
BUS:B1:CAN:SOURCE	CH1
BUS:B1:CAN:STANDARD	CAN2X
BUS:B1:DISPLAY:FORMAT	HEXADECIMAL
BUS:B1:DISPLAY:TYPE	BUS
BUS:B1:FLEXRAY:BITRATE	10000000

Table C-1: Default Values (cont.)

Item	Description
BUS:B1:FLEXRAY:CHANNEL	A
BUS:B1:FLEXRAY:SIGNAL	BDIFFBP
BUS:B1:FLEXRAY:SOURCE	CH1
BUS:B1:I2C:ADDRESS:RWINCLUDE	0
BUS:B1:I2C:CLOCK:SOURCE	CH1
BUS:B1:I2C:DATA:SOURCE	CH2
BUS:B1:LABEL	"Parallel"
BUS:B1:LIN:BITRATE	19200
BUS:B1:LIN:IDFORMAT	NOPARITY
BUS:B1:LIN:POLARITY	NORMAL
BUS:B1:LIN:SAMPLEPOINT	50
BUS:B1:LIN:SOURCE	CH1
BUS:B1:LIN:STANDARD	V2X
BUS:B1:MIL1553B:POLARITY	NORMAL
BUS:B1:MIL1553B:RESPONSETIME:MAXIMUM	12.0000E-6
BUS:B1:MIL1553B:RESPONSETIME:MINIMUM	4.0000E-6
BUS:B1:MIL1553B:SOURCE	CH1
BUS:B1:PARALLEL:BIT0:SOURCE	D0
BUS:B1:PARALLEL:BIT10:SOURCE	D10
BUS:B1:PARALLEL:BIT11:SOURCE	D11
BUS:B1:PARALLEL:BIT12:SOURCE	D12
BUS:B1:PARALLEL:BIT13:SOURCE	D13
BUS:B1:PARALLEL:BIT14:SOURCE	D14
BUS:B1:PARALLEL:BIT15:SOURCE	D15
BUS:B1:PARALLEL:BIT16:SOURCE	CH1
BUS:B1:PARALLEL:BIT17:SOURCE	CH2
BUS:B1:PARALLEL:BIT18:SOURCE	CH3
BUS:B1:PARALLEL:BIT19:SOURCE	CH4
BUS:B1:PARALLEL:BIT1:SOURCE	D1
BUS:B1:PARALLEL:BIT2:SOURCE	D2
BUS:B1:PARALLEL:BIT3:SOURCE	D3
BUS:B1:PARALLEL:BIT4:SOURCE	D4
BUS:B1:PARALLEL:BIT5:SOURCE	D5
BUS:B1:PARALLEL:BIT6:SOURCE	D6
BUS:B1:PARALLEL:BIT7:SOURCE	D7
BUS:B1:PARALLEL:BIT8:SOURCE	D8
BUS:B1:PARALLEL:BIT9:SOURCE	D9

Table C-1: Default Values (cont.)

Item	Description
BUS:B1:PARALLEL:CLOCK:EDGE	RISING
BUS:B1:PARALLEL:CLOCK:ISCLOCKED	NO
BUS:B1:PARALLEL:CLOCK:SOURCE	CH1
BUS:B1:PARALLEL:WIDTH	16
BUS:B1:POSITION	0.0E+0
BUS:B1:RS232C:BITRATE	9600
BUS:B1:RS232C:DATABITS	8
BUS:B1:RS232C:DELIMITER	LF
BUS:B1:RS232C:DISPLAYMODE	FRAME
BUS:B1:RS232C:PARITY	NONE
BUS:B1:RS232C:POLARITY	NORMAL
BUS:B1:RS232C:RX:SOURCE	OFF
BUS:B1:RS232C:TX:SOURCE	CH1
BUS:B1:SPI:BITORDER	MSB
BUS:B1:SPI:CLOCK:POLARITY	RISE
BUS:B1:SPI:CLOCK:SOURCE	CH1
BUS:B1:SPI:DATA:IN:POLARITY	HIGH
BUS:B1:SPI:DATA:IN:SOURCE	OFF
BUS:B1:SPI:DATA:OUT:POLARITY	HIGH
BUS:B1:SPI:DATA:OUT:SOURCE	CH3
BUS:B1:SPI:DATA:SIZE	8
BUS:B1:SPI:FRAMING	SS
BUS:B1:SPI:IDLETIME	5.0000E-6
BUS:B1:SPI:SELECT:POLARITY	LOW
BUS:B1:SPI:SELECT:SOURCE	CH2
BUS:B1:STATE	0
BUS:B1:TYPE	PARALLEL
BUS:B1:USB:BITRATE	FULL
BUS:B1:USB:PROBE	DIFFERENTIAL
BUS:B1:USB:SOURCE:DIFFERENTIAL	CH1
BUS:B1:USB:SOURCE:DMINUS	CH2
BUS:B1:USB:SOURCE:DPLUS	CH1
BUS:B2:AUDIO:BITDELAY	1
BUS:B2:AUDIO:BITORDER	MSB
BUS:B2:AUDIO:CHANNEL:SIZE	24
BUS:B2:AUDIO:CLOCK:POLARITY	RISE
BUS:B2:AUDIO:CLOCK:SOURCE	CH1

Table C-1: Default Values (cont.)

Item	Description
BUS:B2:AUDIO:DATA:POLARITY	NORMAL
BUS:B2:AUDIO:DATA:SIZE	24
BUS:B2:AUDIO:DISPLAY:FORMAT	SIGNEDDECIMAL
BUS:B2:AUDIO:FRAME:SIZE	8
BUS:B2:AUDIO:FRAMESYNC:POLARITY	RISE
BUS:B2:AUDIO:FRAMESYNC:SOURCE	CH2
BUS:B2:AUDIO:TYPE	I2S
BUS:B2:AUDIO:WORDSEL:POLARITY	NORMAL
BUS:B2:AUDIO:WORDSEL:SOURCE	CH2
BUS:B2:CAN:BITRATE	500000
BUS:B2:CAN:FD:BITRATE	4000000
BUS:B2:CAN:FD:STANDARD	ISO
BUS:B2:CAN:PROBE	CANH
BUS:B2:CAN:SAMPLEPOINT	50
BUS:B2:CAN:SOURCE	CH1
BUS:B2:CAN:STANDARD	CAN2X
BUS:B2:DISPLAY:FORMAT	HEXADECIMAL
BUS:B2:DISPLAY:TYPE	BUS
BUS:B2:FLEXRAY:BITRATE	10000000
BUS:B2:FLEXRAY:CHANNEL	A
BUS:B2:FLEXRAY:SIGNAL	BDIFFBP
BUS:B2:FLEXRAY:SOURCE	CH1
BUS:B2:I2C:ADDRESS:RWINCLUDE	0
BUS:B2:I2C:CLOCK:SOURCE	CH1
BUS:B2:I2C:DATA:SOURCE	CH2
BUS:B2:LABEL	"Parallel"
BUS:B2:LIN:BITRATE	19200
BUS:B2:LIN:IDFORMAT	NOPARITY
BUS:B2:LIN:POLARITY	NORMAL
BUS:B2:LIN:SAMPLEPOINT	50
BUS:B2:LIN:SOURCE	CH1
BUS:B2:LIN:STANDARD	V2X
BUS:B2:MIL1553B:POLARITY	NORMAL
BUS:B2:MIL1553B:RESPONSETIME:MAXIMUM	12.0000E-6
BUS:B2:MIL1553B:RESPONSETIME:MINIMUM	4.0000E-6
BUS:B2:MIL1553B:SOURCE	CH1
BUS:B2:PARALLEL:BIT0:SOURCE	D0

Table C-1: Default Values (cont.)

Item	Description
BUS:B2:PARALLEL:BIT10:SOURCE	D10
BUS:B2:PARALLEL:BIT11:SOURCE	D11
BUS:B2:PARALLEL:BIT12:SOURCE	D12
BUS:B2:PARALLEL:BIT13:SOURCE	D13
BUS:B2:PARALLEL:BIT14:SOURCE	D14
BUS:B2:PARALLEL:BIT15:SOURCE	D15
BUS:B2:PARALLEL:BIT16:SOURCE	CH1
BUS:B2:PARALLEL:BIT17:SOURCE	CH2
BUS:B2:PARALLEL:BIT18:SOURCE	CH3
BUS:B2:PARALLEL:BIT19:SOURCE	CH4
BUS:B2:PARALLEL:BIT1:SOURCE	D1
BUS:B2:PARALLEL:BIT2:SOURCE	D2
BUS:B2:PARALLEL:BIT3:SOURCE	D3
BUS:B2:PARALLEL:BIT4:SOURCE	D4
BUS:B2:PARALLEL:BIT5:SOURCE	D5
BUS:B2:PARALLEL:BIT6:SOURCE	D6
BUS:B2:PARALLEL:BIT7:SOURCE	D7
BUS:B2:PARALLEL:BIT8:SOURCE	D8
BUS:B2:PARALLEL:BIT9:SOURCE	D9
BUS:B2:PARALLEL:CLOCK:EDGE	RISING
BUS:B2:PARALLEL:CLOCK:ISCLOCKED	NO
BUS:B2:PARALLEL:CLOCK:SOURCE	CH1
BUS:B2:PARALLEL:WIDTH	16
BUS:B2:POSITION	0.0E+0
BUS:B2:RS232C:BITRATE	9600
BUS:B2:RS232C:DATABITS	8
BUS:B2:RS232C:DELIMITER	LF
BUS:B2:RS232C:DISPLAYMODE	FRAME
BUS:B2:RS232C:PARITY	NONE
BUS:B2:RS232C:POLARITY	NORMAL
BUS:B2:RS232C:RX:SOURCE	OFF
BUS:B2:RS232C:TX:SOURCE	CH1
BUS:B2:SPI:BITORDER	MSB
BUS:B2:SPI:CLOCK:POLARITY	RISE
BUS:B2:SPI:CLOCK:SOURCE	CH1
BUS:B2:SPI:DATA:IN:POLARITY	HIGH
BUS:B2:SPI:DATA:IN:SOURCE	OFF

Table C-1: Default Values (cont.)

Item	Description
BUS:B2:SPI:DATA:OUT:POLARITY	HIGH
BUS:B2:SPI:DATA:OUT:SOURCE	CH3
BUS:B2:SPI:DATA:SIZE	8
BUS:B2:SPI:FRAMING	SS
BUS:B2:SPI:IDLETIME	5.0000E-6
BUS:B2:SPI:SELECT:POLARITY	LOW
BUS:B2:SPI:SELECT:SOURCE	CH2
BUS:B2:STATE	0
BUS:B2:TYPE	PARALLEL
BUS:B2:USB:BITRATE	FULL
BUS:B2:USB:PROBE	DIFFERENTIAL
BUS:B2:USB:SOURCE:DIFFERENTIAL	CH1
BUS:B2:USB:SOURCE:DMINUS	CH2
BUS:B2:USB:SOURCE:DPLUS	CH1
BUS:LOWERTHRESHOLD:CH1	0.0E+0
BUS:LOWERTHRESHOLD:CH2	0.0E+0
BUS:LOWERTHRESHOLD:CH3	0.0E+0
BUS:LOWERTHRESHOLD:CH4	0.0E+0
BUS:LOWERTHRESHOLD:MATH	0.0E+0
BUS:LOWERTHRESHOLD:REF1	0.0E+0
BUS:LOWERTHRESHOLD:REF2	0.0E+0
BUS:LOWERTHRESHOLD:REF3	0.0E+0
BUS:LOWERTHRESHOLD:REF4	0.0E+0
BUS:UPPERTHRESHOLD:CH1	800.0000E-3
BUS:UPPERTHRESHOLD:CH2	800.0000E-3
BUS:UPPERTHRESHOLD:CH3	800.0000E-3
BUS:UPPERTHRESHOLD:CH4	800.0000E-3
BUS:UPPERTHRESHOLD:MATH	0.0E+0
BUS:UPPERTHRESHOLD:REF1	0.0E+0
BUS:UPPERTHRESHOLD:REF2	0.0E+0
BUS:UPPERTHRESHOLD:REF3	0.0E+0
BUS:UPPERTHRESHOLD:REF4	0.0E+0
CH1:AMPSVIAVOLTS:ENABLE	0
CH1:AMPSVIAVOLTS:FACTOR	10.0000
CH1:BANDWIDTH	1.0000E+9
CH1:COUPLING	DC
CH1:DESKEW	0.0E+0

Table C-1: Default Values (cont.)

Item	Description
CH1:INVERT	0
CH1:LABEL	""
CH1:OFFSET	0.0E+0
CH1:POSITION	0.0E+0
CH1:PROBE:FORCEDRANGE	0.0E+0
CH1:PROBE:GAIN	1.0000
CH1:PROBE:PROPDELAY	0.0E+0
CH1:SCALE	100.0000E-3
CH1:TERMINATION	1.0000E+6
CH1:YUNITS	"V"
CH2:AMPSVIAVOLTS:ENABLE	0
CH2:AMPSVIAVOLTS:FACTOR	10.0000
CH2:BANDWIDTH	1.0000E+9
CH2:COUPLING	DC
CH2:DESKEW	0.0E+0
CH2:INVERT	0
CH2:LABEL	""
CH2:OFFSET	0.0E+0
CH2:POSITION	0.0E+0
CH2:PROBE:FORCEDRANGE	0.0E+0
CH2:PROBE:GAIN	1.0000
CH2:PROBE:PROPDELAY	0.0E+0
CH2:SCALE	100.0000E-3
CH2:TERMINATION	1.0000E+6
CH2:YUNITS	"V"
CH3:AMPSVIAVOLTS:ENABLE	0
CH3:AMPSVIAVOLTS:FACTOR	10.0000
CH3:BANDWIDTH	1.0000E+9
CH3:COUPLING	DC
CH3:DESKEW	0.0E+0
CH3:INVERT	0
CH3:LABEL	""
CH3:OFFSET	0.0E+0
CH3:POSITION	0.0E+0
CH3:PROBE:FORCEDRANGE	0.0E+0
CH3:PROBE:GAIN	1.0000
CH3:PROBE:PROPDELAY	0.0E+0

Table C-1: Default Values (cont.)

Item	Description
CH3:SCALE	100.0000E-3
CH3:TERMINATION	1.0000E+6
CH3:YUNITS	"V"
CH4:AMPSVIAVOLTS:ENABLE	0
CH4:AMPSVIAVOLTS:FACTOR	10.0000
CH4:BANDWIDTH	1.0000E+9
CH4:COUPLING	DC
CH4:DESKEW	0.0E+0
CH4:INVERT	0
CH4:LABEL	""
CH4:OFFSET	0.0E+0
CH4:POSITION	0.0E+0
CH4:PROBE:FORCEDRANGE	0.0E+0
CH4:PROBE:GAIN	1.0000
CH4:PROBE:PROPDELAY	0.0E+0
CH4:SCALE	100.0000E-3
CH4:TERMINATION	1.0000E+6
CH4:YUNITS	"V"
CURSOR:FUNCTION	OFF
CURSOR:HBARS:POSITION1	0.0E+0
CURSOR:HBARS:POSITION2	0.0E+0
CURSOR:HBARS:UNITS	BASE
CURSOR:MODE	INDEPENDENT
CURSOR:SOURCE	AUTO
CURSOR:VBARS:POSITION1	-12.00E-6
CURSOR:VBARS:POSITION2	12.00E-6
CURSOR:VBARS:UNITS	SECONDS
CURSOR:XY:READOUT	RECTANGULAR
CURSOR:XY:RECTANGULAR:X:POSITION1	0.0E+0
CURSOR:XY:RECTANGULAR:X:POSITION2	0.0E+0
CURSOR:XY:RECTANGULAR:Y:POSITION1	0.0E+0
CURSOR:XY:RECTANGULAR:Y:POSITION2	0.0E+0
D0:LABEL	""
D0:POSITION	80.0000E-3
D0:THRESHOLD	1.4000
D10:LABEL	""
D10:POSITION	80.0000E-3

Table C-1: Default Values (cont.)

Item	Description
D10:THRESHOLD	1.4000
D11:LABEL	""
D11:POSITION	80.0000E-3
D11:THRESHOLD	1.4000
D12:LABEL	""
D12:POSITION	80.0000E-3
D12:THRESHOLD	1.4000
D13:LABEL	""
D13:POSITION	80.0000E-3
D13:THRESHOLD	1.4000
D14:LABEL	""
D14:POSITION	80.0000E-3
D14:THRESHOLD	1.4000
D15:LABEL	""
D15:POSITION	80.0000E-3
D15:THRESHOLD	1.4000
D1:LABEL	""
D1:POSITION	80.0000E-3
D1:THRESHOLD	1.4000
D2:LABEL	""
D2:POSITION	80.0000E-3
D2:THRESHOLD	1.4000
D3:LABEL	""
D3:POSITION	80.0000E-3
D3:THRESHOLD	1.4000
D4:LABEL	""
D4:POSITION	80.0000E-3
D4:THRESHOLD	1.4000
D5:LABEL	""
D5:POSITION	80.0000E-3
D5:THRESHOLD	1.4000
D6:LABEL	""
D6:POSITION	80.0000E-3
D6:THRESHOLD	1.4000
D7:LABEL	""
D7:POSITION	80.0000E-3
D7:THRESHOLD	1.4000

Table C-1: Default Values (cont.)

Item	Description
D8:LABEL	""
D8:POSITION	80.0000E-3
D8:THRESHOLD	1.4000
D9:LABEL	""
D9:POSITION	80.0000E-3
D9:THRESHOLD	1.4000
DATA:DESTINATION	REF1
DATA:ENCDG	RIBINARY
DATA:SOURCE	CH1
DATA:START	1
DATA:STOP	10000
DESKEW:DISPLAY	1
DISPLAY:CLOCK	1
DISPLAY:DIGITAL:ACTIVITY	0
DISPLAY:DIGITAL:HEIGHT	MEDIUM
DISPLAY:GRATICULE	FULL
DISPLAY:INTENSITY:BACKLIGHT	HIGH
DISPLAY:INTENSITY:BACKLIGHT:AUTODIM:ENABLE	1
DISPLAY:INTENSITY:BACKLIGHT:AUTODIM:TIME	60
DISPLAY:INTENSITY:GRATICULE	75
DISPLAY:INTENSITY:WAVEFORM	35
DISPLAY:PERSISTENCE	0.0E+0
DISPLAY:STYLE:DOTSONLY	0
DISPLAY:TRIGFREQUENCY	0
DISPLAY:XY	OFF
DISPLAY:XY:WITHYT	0
DVM:AUTORANGE	0
DVM:DISPLAYSTYLE	FULL
DVM:MODE	OFF
DVM:SOURCE	CH1
EMAIL:SETUP:FROMADDRESS	"TestString"
EMAIL:SETUP:HOSTALIASNAME	"TestString"
EMAIL:SETUP:SMTLOGIN	"TestString"
EMAIL:SETUP:SMTPPORT	25
EMAIL:SETUP:SMTPSERVER	"TestString"
FASTACQ:STATE	0
HEADER	0

Table C-1: Default Values (cont.)

Item	Description
HISTOGRAM:BOXPCNT	20.0000,20.0000,80.0000,80.0000
HISTOGRAM:DISPLAY	LINEAR
HISTOGRAM:MODE	OFF
HISTOGRAM:SOURCE	CH1
HORIZONTAL:DELAY:MODE	1
HORIZONTAL:DELAY:TIME	0.0E+0
HORIZONTAL:POSITION	50.0000
HORIZONTAL:RECORDLENGTH	10000
	4.0000E-6
LOCK	NONE
MARKER:M1:FREQUENCY:ABSOLUTE	0.0E+0
MARKER:M2:FREQUENCY:ABSOLUTE	0.0E+0
MARKER:MANUAL	0
MARKER:PEAK:EXCURSION	30.0000
MARKER:PEAK:MAXIMUM	5
MARKER:PEAK:STATE	1
MARKER:PEAK:THRESHOLD	-50.0000
MARKER:TYPE	ABSOLUTE
MASK:TEST:SAVEWFM	0
MASK:TEST:SRQ:COMPLETION	0
MASK:TEST:SRQ:FAILURE	0
MASK:TEST:STATE	0
MASK:TEST:STOP:FAILURE	0
MASK:TEST:THRESHOLD	1
MASK:TEST:TIME	INFINITE
MASK:TEST:WAVEFORM	INFINITE
MASK:USER:AMPLITUDE	1.0000
MASK:USER:HSCALE	400.0000E-6
MASK:USER:HTRIGPOS	500.0000E-3
MATH:AUTOSCALE	1
MATH:DEFINE	"CH1+CH2"
MATH:HORIZONTAL:POSITION	50.0000
MATH:	4.0000E-6
MATH:HORIZONTAL:UNITS	"s"
MATH:LABEL	""
MATH:SPECTRAL:MAG	DB
MATH:SPECTRAL:WINDOW	HANNING

Table C-1: Default Values (cont.)

Item	Description
MATH:TYPE	DUAL
MATH:VERTICAL:POSITION	0.0E+0
MATH:VERTICAL:SCALE	100.0000E-3
MATH:VERTICAL:UNITS	"V"
MATHVAR:VAR1	0.0E+0
MATHVAR:VAR2	0.0E+0
MEASUREMENT:GATING	SCREEN
MEASUREMENT:IMMED:DELAY:DIRECTION	FORWARDS
MEASUREMENT:IMMED:DELAY:EDGE1	RISE
MEASUREMENT:IMMED:DELAY:EDGE2	RISE
MEASUREMENT:IMMED:SOURCE1	CH1
MEASUREMENT:IMMED:SOURCE2	CH2
MEASUREMENT:IMMED:TYPE	PERIOD
MEASUREMENT:INDICATORS:STATE	OFF
MEASUREMENT:MEAS1:DELAY:DIRECTION	FORWARDS
MEASUREMENT:MEAS1:DELAY:EDGE1	RISE
MEASUREMENT:MEAS1:DELAY:EDGE2	RISE
MEASUREMENT:MEAS1:SOURCE1	CH1
MEASUREMENT:MEAS1:SOURCE2	CH2
MEASUREMENT:MEAS1:STATE	0
MEASUREMENT:MEAS1:TYPE	PERIOD
MEASUREMENT:MEAS2:DELAY:DIRECTION	FORWARDS
MEASUREMENT:MEAS2:DELAY:EDGE1	RISE
MEASUREMENT:MEAS2:DELAY:EDGE2	RISE
MEASUREMENT:MEAS2:SOURCE1	CH1
MEASUREMENT:MEAS2:SOURCE2	CH2
MEASUREMENT:MEAS2:STATE	0
MEASUREMENT:MEAS2:TYPE	PERIOD
MEASUREMENT:MEAS3:DELAY:DIRECTION	FORWARDS
MEASUREMENT:MEAS3:DELAY:EDGE1	RISE
MEASUREMENT:MEAS3:DELAY:EDGE2	RISE
MEASUREMENT:MEAS3:SOURCE1	CH1
MEASUREMENT:MEAS3:SOURCE2	CH2
MEASUREMENT:MEAS3:STATE	0
MEASUREMENT:MEAS3:TYPE	PERIOD
MEASUREMENT:MEAS4:DELAY:DIRECTION	FORWARDS
MEASUREMENT:MEAS4:DELAY:EDGE1	RISE

Table C-1: Default Values (cont.)

Item	Description
MEASUREMENT:MEAS4:DELAY:EDGE2	RISE
MEASUREMENT:MEAS4:SOURCE1	CH1
MEASUREMENT:MEAS4:SOURCE2	CH2
MEASUREMENT:MEAS4:STATE	0
MEASUREMENT:MEAS4:TYPE	PERIOD
MEASUREMENT:METHOD	AUTO
MEASUREMENT:REFLEVEL:ABSOLUTE:HIG	0.0E+0
MEASUREMENT:REFLEVEL:ABSOLUTE:LOW	0.0E+0
MEASUREMENT:REFLEVEL:ABSOLUTE:MID1	0.0E+0
MEASUREMENT:REFLEVEL:ABSOLUTE:MID2	0.0E+0
MEASUREMENT:REFLEVEL:METHOD	PERCENT
MEASUREMENT:REFLEVEL:PERCENT:HIG	90.0000
MEASUREMENT:REFLEVEL:PERCENT:LOW	10.0000
MEASUREMENT:REFLEVEL:PERCENT:MID1	50.0000
MEASUREMENT:REFLEVEL:PERCENT:MID2	50.0000
MEASUREMENT:STATISTICS:MODE	ALL
MEASUREMENT:STATISTICS:WEIGHTING	32
MESSAGE:BOX	167,67,167,83
MESSAGE:SHOW	""
MESSAGE:STATE	0
PICTBRIDGE:DATEPRINT	DEFLT
PICTBRIDGE:IDPRINT	OFF
PICTBRIDGE:IMAGESIZE	DEFLT
PICTBRIDGE:PAPERSIZE	DEFLT
PICTBRIDGE:PAPERTYPE	DEFLT
PICTBRIDGE:PRINTQUAL	DEFLT
REF1:HORIZONTAL:DELAY:TIME	16.0000E-6
REF1:	4.0000E-6
REF1:VERTICAL:POSITION	0.0E+0
REF1:VERTICAL:SCALE	100.0000E-3
REF2:HORIZONTAL:DELAY:TIME	16.0000E-6
REF2:	4.0000E-6
REF2:VERTICAL:POSITION	0.0E+0
REF2:VERTICAL:SCALE	100.0000E-3
REF3:HORIZONTAL:DELAY:TIME	16.0000E-6
REF3:	4.0000E-6
REF3:VERTICAL:POSITION	0.0E+0

Table C-1: Default Values (cont.)

Item	Description
REF3:VERTICAL:SCALE	100.0000E-3
REF4:HORIZONTAL:DELAY:TIME	16.0000E-6
REF4:	4.0000E-6
REF4:VERTICAL:POSITION	0.0E+0
REF4:VERTICAL:SCALE	100.0000E-3
ROSC:SOURCE	INTERNAL
SAVE:ASSIGN:TYPE	WAVEFORM
SAVE:IMAGE:FILEFORMAT	PNG
SAVE:IMAGE:INKSAVER	1
SAVE:IMAGE:LAYOUT	PORTRAIT
SAVE:WAVEFORM	WFM
SAVE:WAVEFORM:GATING	NONE
SEARCH:SEARCH1:STATE	0
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:AUDIO:CONDITION	SOF
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:AUDIO:DATA:HIVALUE	"XXXXXXXXXXXXXXXXXXXXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:AUDIO:DATA:OFFSET	0
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:AUDIO:DATA:QUALIFIER	EQUAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:AUDIO:DATA:VALUE	"XXXXXXXXXXXXXXXXXXXXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:AUDIO:DATA:WORD	EITHER
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:CAN:CONDITION	SOF
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:CAN:DATA:DIRECTION	NOCARE
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:CAN:DATA:OFFSET	0
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:CAN:DATA:QUALIFIER	EQUAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:CAN:DATA:SIZE	1
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:CAN:DATA:VALUE	"XXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:CAN:FD:BRSBIT	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:CAN:FD:ESIBIT	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:CAN:FRAMETYPE	DATA
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:CAN:IDENTIFIER:MODE	STANDARD
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:CAN:IDENTIFIER:VALUE	"XXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:CONDITION	SOF
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:CYCLECOUNT:HIVALUE	"XXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:CYCLECOUNT:QUALIFIER	EQUAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:CYCLECOUNT:VALUE	"XXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:DATA:HIVALUE	"XXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:DATA:OFFSET	-1
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:DATA:QUALIFIER	EQUAL

Table C-1: Default Values (cont.)

Item	Description
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:DATA:SIZE	1
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:DATA:VALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:EFTYPE	STATIC
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:ERRTYPE	CRCHEADER
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:FRAMEID:HIVALUE	"XXXXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:FRAMEID:QUALIFIER	EQUAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:FRAMEID:VALUE	"XXXXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:FRAMETYPE	NORMAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:HEADER:CRC	"XXXXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:HEADER:CYCLECOUNT	"XXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:HEADER:FRAMEID	"XXXXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:HEADER:INDBITS	"XXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:FLEXRAY:HEADER:PAYLENGTH	"XXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:I2C:ADDRESS:MODE	ADDR7
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:I2C:ADDRESS:TYPE	USER
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:I2C:ADDRESS:VALUE	"XXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:I2C:CONDITION	START
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:I2C:DATA:DIRECTION	NOCARE
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:I2C:DATA:SIZE	1
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:I2C:DATA:VALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:LIN:CONDITION	SYNCFIELD
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:LIN:DATA:HIVALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:LIN:DATA:QUALIFIER	EQUAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:LIN:DATA:SIZE	1
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:LIN:DATA:VALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:LIN:ERRTYPE	SYNC
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:LIN:IDENTIFIER:VALUE	"XXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:COMMAND:ADDRESS:HIVALUE	"XXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:COMMAND:ADDRESS:QUALIFIER	EQUAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:COMMAND:ADDRESS:VALUE	"XXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:COMMAND:COUNT	"XXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:COMMAND:PARITY	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:COMMAND:SUBADDRESS	"XXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:COMMAND:TRBIT	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:CONDITION	SYNC
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:DATA:PARITY	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:DATA:VALUE	"XXXXXXXXXXXXXXXXXX"

Table C-1: Default Values (cont.)

Item	Description
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:ERRTYPE	PARITY
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:STATUS:ADDRESS:HIVALUE	"XXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:STATUS:ADDRESS:QUALIFIER	EQUAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:STATUS:ADDRESS:VALUE	"XXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:BCR	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:BUSY	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:DBCA	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:INSTR	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:ME	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:SRQ	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:SBSF	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:STATUS:BIT:TF	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:STATUS:PARITY	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:TIME:LESSLIMIT	4.0000E-6
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:TIME:MORELIMIT	12.0000E-6
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:MIL1553B:TIME:QUALIFIER	MORETHAN
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:PARALLEL:VALUE	"XXXXXXXXXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:RS232C:CONDITION	TXSTART
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:RS232C:RX:DATA:SIZE	1
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:RS232C:RX:DATA:VALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:RS232C:TX:DATA:SIZE	1
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:RS232C:TX:DATA:VALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:SPI:CONDITION	SS
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:SPI:DATA:MISO:VALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:SPI:DATA:MOSI:VALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:SPI:DATA:SIZE	1
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:USB:ADDRESS:HIVALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:USB:ADDRESS:VALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:USB:CONDITION	SYNC
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:USB:DATA:HIVALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:USB:DATA:OFFSET	-1
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:USB:DATA:SIZE	1
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:USB:DATA:TYPE	ANY
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:USB:DATA:VALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:USB:ENDPOINT:VALUE	"XXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:USB:ERRTYPE	PID
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:USB:HANDSHAKETYPE	ANY

Table C-1: Default Values (cont.)

Item	Description
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:USB:QUALIFIER	EQUAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:USB:SOFFRAMENUMBER	"XXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:USB:SPECIALTYPE	ANY
SEARCH:SEARCH1:TRIGGER:A:BUS:B1:USB:TOKENTYPE	ANY
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:AUDIO:CONDITION	SOF
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:AUDIO:DATA:HIVALUE	"XXXXXXXXXXXXXXXXXXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:AUDIO:DATA:OFFSET	0
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:AUDIO:DATA:QUALIFIER	EQUAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:AUDIO:DATA:VALUE	"XXXXXXXXXXXXXXXXXXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:AUDIO:DATA:WORD	EITHER
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:CAN:CONDITION	SOF
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:CAN:DATA:DIRECTION	NOCARE
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:CAN:DATA:OFFSET	0
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:CAN:DATA:QUALIFIER	EQUAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:CAN:DATA:SIZE	1
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:CAN:DATA:VALUE	"XXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:CAN:FD:BRSBIT	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:CAN:FD:ESIBIT	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:CAN:FRAMETYPE	DATA
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:CAN:IDENTIFIER:MODE	STANDARD
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:CAN:IDENTIFIER:VALUE	"XXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:CONDITION	SOF
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:CYCLECOUNT:HIVALUE	"XXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:CYCLECOUNT:QUALIFIER	EQUAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:CYCLECOUNT:VALUE	"XXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:DATA:HIVALUE	"XXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:DATA:OFFSET	-1
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:DATA:QUALIFIER	EQUAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:DATA:SIZE	1
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:DATA:VALUE	"XXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:EOFTYPE	STATIC
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:ERRTYPE	CRCHEADER
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:FRAMEID:HIVALUE	"XXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:FRAMEID:QUALIFIER	EQUAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:FRAMEID:VALUE	"XXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:FRAMETYPE	NORMAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:HEADER:CRC	"XXXXXXXXXX"

Table C-1: Default Values (cont.)

Item	Description
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:HEADER:CYCLECOUNT	"XXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:HEADER:FRAMEID	"XXXXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:HEADER:INDBITS	"XXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:FLEXRAY:HEADER:PAYLENGTH	"XXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:I2C:ADDRESS:MODE	ADDR7
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:I2C:ADDRESS:TYPE	USER
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:I2C:ADDRESS:VALUE	"XXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:I2C:CONDITION	START
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:I2C:DATA:DIRECTION	NOCARE
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:I2C:DATA:SIZE	1
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:I2C:DATA:VALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:LIN:CONDITION	SYNCFIELD
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:LIN:DATA:HIVALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:LIN:DATA:QUALIFIER	EQUAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:LIN:DATA:SIZE	1
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:LIN:DATA:VALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:LIN:ERRTYPE	SYNC
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:LIN:IDENTIFIER:VALUE	"XXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:COMMAND:ADDRESS:HIVALUE	"XXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:COMMAND:ADDRESS:QUALIFIER	EQUAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:COMMAND:ADDRESS:VALUE	"XXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:COMMAND:COUNT	"XXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:COMMAND:PARITY	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:COMMAND:SUBADDRESS	"XXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:COMMAND:TRBIT	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:CONDITION	SYNC
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:DATA:PARITY	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:DATA:VALUE	"XXXXXXXXXXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:ERRTYPE	PARITY
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:STATUS:ADDRESS:HIVALUE	"XXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:STATUS:ADDRESS:QUALIFIER	EQUAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:STATUS:ADDRESS:VALUE	"XXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:STATUS:BIT:BCR	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:STATUS:BIT:BUSY	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:STATUS:BIT:DBCA	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:STATUS:BIT:INSTR	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:STATUS:BIT:ME	X

Table C-1: Default Values (cont.)

Item	Description
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:STATUS:BIT:SRQ	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:STATUS:BIT:SBSF	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:STATUS:BIT:TF	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:STATUS:PARITY	X
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:TIME:LESSLIMIT	4.0000E-6
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:TIME:MORELIMIT	12.0000E-6
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:MIL1553B:TIME:QUALIFIER	MORETHAN
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:PARALLEL:VALUE	"XXXXXXXXXXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:RS232C:CONDITION	TXSTART
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:RS232C:RX:DATA:SIZE	1
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:RS232C:RX:DATA:VALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:RS232C:TX:DATA:SIZE	1
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:RS232C:TX:DATA:VALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:SPI:CONDITION	SS
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:SPI:DATA:MISO:VALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:SPI:DATA:MOSI:VALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:SPI:DATA:SIZE	1
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:USB:ADDRESS:HIVALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:USB:ADDRESS:VALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:USB:CONDITION	SYNC
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:USB:DATA:HIVALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:USB:DATA:OFFSET	-1
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:USB:DATA:SIZE	1
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:USB:DATA:TYPE	ANY
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:USB:DATA:VALUE	"XXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:USB:ENDPOINT:VALUE	"XXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:USB:ERRTYPE	PID
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:USB:HANDSHAKETYPE	ANY
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:USB:QUALIFIER	EQUAL
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:USB:SOFFRAMENUMBER	"XXXXXXXXXXXX"
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:USB:SPECIALTYPE	ANY
SEARCH:SEARCH1:TRIGGER:A:BUS:B2:USB:TOKENTYPE	ANY
SEARCH:SEARCH1:TRIGGER:A:BUS:SOURCE	B1
SEARCH:SEARCH1:TRIGGER:A:EDGE:SLOPE	RISE
SEARCH:SEARCH1:TRIGGER:A:EDGE:SOURCE	CH1
SEARCH:SEARCH1:TRIGGER:A:LEVEL	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LEVEL:CH1	0.0E+0

Table C-1: Default Values (cont.)

Item	Description
SEARCH:SEARCH1:TRIGGER:A:LEVEL:CH2	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LEVEL:CH3	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LEVEL:CH4	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LEVEL:MATH	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LEVEL:REF1	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LEVEL:REF2	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LEVEL:REF3	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LEVEL:REF4	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOGIC:FUNCTION	AND
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:CH1	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:CH2	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:CH3	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:CH4	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:CLOCK:EDGE	RISE
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:CLOCK:SOURCE	NONE
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:D0	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:D1	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:D10	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:D11	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:D12	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:D13	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:D14	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:D15	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:D2	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:D3	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:D4	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:D5	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:D6	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:D7	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:D8	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:D9	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:MATH	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:REF1	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:REF2	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:REF3	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:INPUT:REF4	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:INPUT:D0	X

Table C-1: Default Values (cont.)

Item	Description
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:INPUT:D1	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:INPUT:D10	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:INPUT:D11	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:INPUT:D12	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:INPUT:D13	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:INPUT:D14	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:INPUT:D15	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:INPUT:D2	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:INPUT:D3	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:INPUT:D4	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:INPUT:D5	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:INPUT:D6	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:INPUT:D7	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:INPUT:D8	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:INPUT:D9	X
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:WHEN	TRUE
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:WHEN:LESSLIMIT	8.0000E-9
SEARCH:SEARCH1:TRIGGER:A:LOGIC:PATTERN:WHEN:MORELIMIT	8.0000E-9
SEARCH:SEARCH1:TRIGGER:A:LOGIC:THRESHOLD:CH1	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOGIC:THRESHOLD:CH2	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOGIC:THRESHOLD:CH3	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOGIC:THRESHOLD:CH4	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOGIC:THRESHOLD:MATH	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOGIC:THRESHOLD:REF1	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOGIC:THRESHOLD:REF2	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOGIC:THRESHOLD:REF3	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOGIC:THRESHOLD:REF4	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOWERTHRESHOLD:CH1	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOWERTHRESHOLD:CH2	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOWERTHRESHOLD:CH3	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOWERTHRESHOLD:CH4	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOWERTHRESHOLD:MATH	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOWERTHRESHOLD:REF1	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOWERTHRESHOLD:REF2	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOWERTHRESHOLD:REF3	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:LOWERTHRESHOLD:REF4	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:PULSEWIDTH:HIGHLIMIT	12.0000E-9

Table C-1: Default Values (cont.)

Item	Description
SEARCH:SEARCH1:TRIGGER:A:PULSEWIDTH:LOWLIMIT	8.0000E-9
SEARCH:SEARCH1:TRIGGER:A:PULSEWIDTH:POLARITY	POSITIVE
SEARCH:SEARCH1:TRIGGER:A:PULSEWIDTH:WHEN	LESSTHAN
SEARCH:SEARCH1:TRIGGER:A:PULSEWIDTH:WIDTH	8.0000E-9
SEARCH:SEARCH1:TRIGGER:A:RUNT:POLARITY	POSITIVE
SEARCH:SEARCH1:TRIGGER:A:RUNT:WHEN	OCCURS
SEARCH:SEARCH1:TRIGGER:A:RUNT:WIDTH	8.0000E-9
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:CLOCK:EDGE	RISE
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:CLOCK:SOURCE	CH1
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:CLOCK:THRESHOLD	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:DATA:SOURCE	NONE
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:DATA:THRESHOLD	9.91E+37
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:HOLDTIME	8.0000E-9
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:SETTIME	8.0000E-9
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:THRESHOLD:CH1	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:THRESHOLD:CH2	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:THRESHOLD:CH3	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:THRESHOLD:CH4	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:THRESHOLD:MATH	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:THRESHOLD:REF1	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:THRESHOLD:REF2	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:THRESHOLD:REF3	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:SETHOLD:THRESHOLD:REF4	0.0E+0
SEARCH:SEARCH1:TRIGGER:A:TIMEOUT:POLARITY	STAYSHIGH
SEARCH:SEARCH1:TRIGGER:A:TIMEOUT:TIME	8.0000E-9
SEARCH:SEARCH1:TRIGGER:A:TRANSITION:DELTATIME	8.0000E-9
SEARCH:SEARCH1:TRIGGER:A:TRANSITION:POLARITY	POSITIVE
SEARCH:SEARCH1:TRIGGER:A:TRANSITION:WHEN	SLOWER
SEARCH:SEARCH1:TRIGGER:A:TYPE	EDGE
SEARCH:SEARCH1:TRIGGER:A:UPPERTHRESHOLD:CH1	800.0000E-3
SEARCH:SEARCH1:TRIGGER:A:UPPERTHRESHOLD:CH2	800.0000E-3
SEARCH:SEARCH1:TRIGGER:A:UPPERTHRESHOLD:CH3	800.0000E-3
SEARCH:SEARCH1:TRIGGER:A:UPPERTHRESHOLD:CH4	800.0000E-3
SEARCH:SEARCH1:TRIGGER:A:UPPERTHRESHOLD:MATH	800.0000E-3
SEARCH:SEARCH1:TRIGGER:A:UPPERTHRESHOLD:REF1	800.0000E-3
SEARCH:SEARCH1:TRIGGER:A:UPPERTHRESHOLD:REF2	800.0000E-3
SEARCH:SEARCH1:TRIGGER:A:UPPERTHRESHOLD:REF3	800.0000E-3

Table C-1: Default Values (cont.)

Item	Description
SEARCH:SEARCH1:TRIGGER:A:UPPERTHRESHOLD:REF4	800.0000E-3
SEARCH:SEARCH<x>:TRIGGER:A:BUS:B<x>:CAN:DATA:OFFSet	0
SEARCH:SEARCH<x>:TRIGGER:A:BUS:B<x>:CAN:FD:BRSBIT	X
SEARCH:SEARCH<x>:TRIGGER:A:BUS:B<x>:CAN:FD:ESIBIT	X
TRIGGER:A:BUS:B1:AUDIO:CONDITION	SOF
TRIGGER:A:BUS:B1:AUDIO:DATA:HIVALUE	"XXXXXXXXXXXXXXXXXXXXXXX"
TRIGGER:A:BUS:B1:AUDIO:DATA:OFFSET	0
TRIGGER:A:BUS:B1:AUDIO:DATA:QUALIFIER	EQUAL
TRIGGER:A:BUS:B1:AUDIO:DATA:VALUE	"XXXXXXXXXXXXXXXXXXXXXXX"
TRIGGER:A:BUS:B1:AUDIO:DATA:WORD	EITHER
TRIGGER:A:BUS:B1:CAN:CONDITION	SOF
TRIGGER:A:BUS:B1:CAN:DATA:DIRECTION	NOCARE
TRIGGER:A:BUS:B1:CAN:DATA:OFFSET	0
TRIGGER:A:BUS:B1:CAN:DATA:QUALIFIER	EQUAL
TRIGGER:A:BUS:B1:CAN:DATA:SIZE	1
TRIGGER:A:BUS:B1:CAN:DATA:START	0.0E+0
TRIGGER:A:BUS:B1:CAN:FD:BRSBIT	X
TRIGGER:A:BUS:B1:CAN:FD:ESIBIT	X
TRIGGER:A:BUS:B1:CAN:DATA:VALUE	"XXXXXXX"
TRIGGER:A:BUS:B1:CAN:FRAMETYPE	DATA
TRIGGER:A:BUS:B1:CAN:IDENTIFIER:MODE	STANDARD
TRIGGER:A:BUS:B1:CAN:IDENTIFIER:VALUE	"XXXXXXXXXX"
TRIGGER:A:BUS:B1:FLEXRAY:CONDITION	SOF
TRIGGER:A:BUS:B1:FLEXRAY:CYCLECOUNT:HIVALUE	"XXXXXX"
TRIGGER:A:BUS:B1:FLEXRAY:CYCLECOUNT:QUALIFIER	EQUAL
TRIGGER:A:BUS:B1:FLEXRAY:CYCLECOUNT:VALUE	"XXXXXX"
TRIGGER:A:BUS:B1:FLEXRAY:DATA:HIVALUE	"XXXXXXX"
TRIGGER:A:BUS:B1:FLEXRAY:DATA:OFFSET	-1
TRIGGER:A:BUS:B1:FLEXRAY:DATA:QUALIFIER	EQUAL
TRIGGER:A:BUS:B1:FLEXRAY:DATA:SIZE	1
TRIGGER:A:BUS:B1:FLEXRAY:DATA:VALUE	"XXXXXXX"
TRIGGER:A:BUS:B1:FLEXRAY:EOfTYPE	STATIC
TRIGGER:A:BUS:B1:FLEXRAY:ERRTYPE	CRCHEADER
TRIGGER:A:BUS:B1:FLEXRAY:FRAMEID:HIVALUE	"XXXXXXXXXXXX"
TRIGGER:A:BUS:B1:FLEXRAY:FRAMEID:QUALIFIER	EQUAL
TRIGGER:A:BUS:B1:FLEXRAY:FRAMEID:VALUE	"XXXXXXXXXXXX"
TRIGGER:A:BUS:B1:FLEXRAY:FRAMETYPE	NORMAL

Table C-1: Default Values (cont.)

Item	Description
TRIGGER:A:BUS:B1:FLEXRAY:HEADER:CRC	"XXXXXXXXXXXX"
TRIGGER:A:BUS:B1:FLEXRAY:HEADER:CYCLECOUNT	"XXXXXX"
TRIGGER:A:BUS:B1:FLEXRAY:HEADER:FRAMEID	"XXXXXXXXXXXX"
TRIGGER:A:BUS:B1:FLEXRAY:HEADER:INDBITS	"XXXXXX"
TRIGGER:A:BUS:B1:FLEXRAY:HEADER:PAYLENGTH	"XXXXXXX"
TRIGGER:A:BUS:B1:I2C:ADDRESS:MODE	ADDR7
TRIGGER:A:BUS:B1:I2C:ADDRESS:TYPE	USER
TRIGGER:A:BUS:B1:I2C:ADDRESS:VALUE	"XXXXXXXX"
TRIGGER:A:BUS:B1:I2C:CONDITION	START
TRIGGER:A:BUS:B1:I2C:DATA:DIRECTION	NOCARE
TRIGGER:A:BUS:B1:I2C:DATA:SIZE	1
TRIGGER:A:BUS:B1:I2C:DATA:START	0.0E+0
TRIGGER:A:BUS:B1:I2C:DATA:VALUE	"XXXXXXXX"
TRIGGER:A:BUS:B1:LIN:CONDITION	SYNCFIELD
TRIGGER:A:BUS:B1:LIN:DATA:HIVALUE	"XXXXXXXX"
TRIGGER:A:BUS:B1:LIN:DATA:QUALIFIER	EQUAL
TRIGGER:A:BUS:B1:LIN:DATA:SIZE	1
TRIGGER:A:BUS:B1:LIN:DATA:VALUE	"XXXXXXXX"
TRIGGER:A:BUS:B1:LIN:ERRTYPE	SYNC
TRIGGER:A:BUS:B1:LIN:IDENTIFIER:VALUE	"XXXXXX"
TRIGGER:A:BUS:B1:PARALLEL:VALUE	"XXXXXXXXXXXXXXXXXX"
TRIGGER:A:BUS:B1:RS232C:CONDITION	TXSTART
TRIGGER:A:BUS:B1:RS232C:RX:DATA:SIZE	1
TRIGGER:A:BUS:B1:RS232C:RX:DATA:VALUE	"XXXXXXXX"
TRIGGER:A:BUS:B1:RS232C:TX:DATA:SIZE	1
TRIGGER:A:BUS:B1:RS232C:TX:DATA:VALUE	"XXXXXXXX"
TRIGGER:A:BUS:B1:SPI:CONDITION	SS
TRIGGER:A:BUS:B1:SPI:DATA:IN:VALUE	"XXXXXXXX"
TRIGGER:A:BUS:B1:SPI:DATA:OUT:VALUE	"XXXXXXXX"
TRIGGER:A:BUS:B1:SPI:DATA:SIZE	1
TRIGGER:A:BUS:B1:SPI:DATA:START	0.0E+0
TRIGGER:A:BUS:B1:USB:ADDRESS:HIVALUE	"XXXXXXX"
TRIGGER:A:BUS:B1:USB:ADDRESS:VALUE	"XXXXXXX"
TRIGGER:A:BUS:B1:USB:CONDITION	SYNC
TRIGGER:A:BUS:B1:USB:DATA:HIVALUE	"XXXXXXXX"
TRIGGER:A:BUS:B1:USB:DATA:OFFSET	-1
TRIGGER:A:BUS:B1:USB:DATA:SIZE	1

Table C-1: Default Values (cont.)

Item	Description
TRIGGER:A:BUS:B1:USB:DATA:TYPE	ANY
TRIGGER:A:BUS:B1:USB:DATA:VALUE	"XXXXXXXX"
TRIGGER:A:BUS:B1:USB:ENDPOINT:VALUE	"XXXX"
TRIGGER:A:BUS:B1:USB:ERRTYPE	PID
TRIGGER:A:BUS:B1:USB:HANDSHAKETYPE	ANY
TRIGGER:A:BUS:B1:USB:QUALIFIER	EQUAL
TRIGGER:A:BUS:B1:USB:SOFFRAMENUMBER	"XXXXXXXXXXXX"
TRIGGER:A:BUS:B1:USB:SPECIALTYPE	ANY
TRIGGER:A:BUS:B1:USB:TOKENTYPE	ANY
TRIGGER:A:BUS:B2:AUDIO:CONDITION	SOF
TRIGGER:A:BUS:B2:AUDIO:DATA:HIVALUE	"XXXXXXXXXXXXXXXXXXXXXXXXXXXX"
TRIGGER:A:BUS:B2:AUDIO:DATA:OFFSET	0
TRIGGER:A:BUS:B2:AUDIO:DATA:QUALIFIER	EQUAL
TRIGGER:A:BUS:B2:AUDIO:DATA:VALUE	"XXXXXXXXXXXXXXXXXXXXXXXXXXXX"
TRIGGER:A:BUS:B2:AUDIO:DATA:WORD	EITHER
TRIGGER:A:BUS:B2:CAN:CONDITION	SOF
TRIGGER:A:BUS:B2:CAN:DATA:DIRECTION	NOCARE
TRIGGER:A:BUS:B2:CAN:DATA:OFFSET	0
TRIGGER:A:BUS:B2:CAN:DATA:QUALIFIER	EQUAL
TRIGGER:A:BUS:B2:CAN:DATA:SIZE	1
TRIGGER:A:BUS:B2:CAN:DATA:START	0.0E+0
TRIGGER:A:BUS:B2:CAN:DATA:VALUE	"XXXXXXXX"
TRIGGER:A:BUS:B2:CAN:FD:BRBUT	X
TRIGGER:A:BUS:B2:CAN:FD:ESIBIT	X
TRIGGER:A:BUS:B2:CAN:FRAMETYPE	DATA
TRIGGER:A:BUS:B2:CAN:IDENTIFIER:MODE	STANDARD
TRIGGER:A:BUS:B2:CAN:IDENTIFIER:VALUE	"XXXXXXXXXXXX"
TRIGGER:A:BUS:B2:FLEXRAY:CONDITION	SOF
TRIGGER:A:BUS:B2:FLEXRAY:CYCLECOUNT:HIVALUE	"XXXXXX"
TRIGGER:A:BUS:B2:FLEXRAY:CYCLECOUNT:QUALIFIER	EQUAL
TRIGGER:A:BUS:B2:FLEXRAY:CYCLECOUNT:VALUE	"XXXXXX"
TRIGGER:A:BUS:B2:FLEXRAY:DATA:HIVALUE	"XXXXXXXX"
TRIGGER:A:BUS:B2:FLEXRAY:DATA:OFFSET	-1
TRIGGER:A:BUS:B2:FLEXRAY:DATA:QUALIFIER	EQUAL
TRIGGER:A:BUS:B2:FLEXRAY:DATA:SIZE	1
TRIGGER:A:BUS:B2:FLEXRAY:DATA:VALUE	"XXXXXXXX"
TRIGGER:A:BUS:B2:FLEXRAY:EOFTYPE	STATIC

Table C-1: Default Values (cont.)

Item	Description
TRIGGER:A:BUS:B2:FLEXRAY:ERRTYPE	CRCHEADER
TRIGGER:A:BUS:B2:FLEXRAY:FRAMEID:HIVALUE	"XXXXXXXXXX"
TRIGGER:A:BUS:B2:FLEXRAY:FRAMEID:QUALIFIER	EQUAL
TRIGGER:A:BUS:B2:FLEXRAY:FRAMEID:VALUE	"XXXXXXXXXX"
TRIGGER:A:BUS:B2:FLEXRAY:FRAMETYPE	NORMAL
TRIGGER:A:BUS:B2:FLEXRAY:HEADER:CRC	"XXXXXXXXXX"
TRIGGER:A:BUS:B2:FLEXRAY:HEADER:CYCLECOUNT	"XXXXXX"
TRIGGER:A:BUS:B2:FLEXRAY:HEADER:FRAMEID	"XXXXXXXXXX"
TRIGGER:A:BUS:B2:FLEXRAY:HEADER:INDBITS	"XXXXX"
TRIGGER:A:BUS:B2:FLEXRAY:HEADER:PAYLENGTH	"XXXXXXX"
TRIGGER:A:BUS:B2:I2C:ADDRESS:MODE	ADDR7
TRIGGER:A:BUS:B2:I2C:ADDRESS:TYPE	USER
TRIGGER:A:BUS:B2:I2C:ADDRESS:VALUE	"XXXXXXX"
TRIGGER:A:BUS:B2:I2C:CONDITION	START
TRIGGER:A:BUS:B2:I2C:DATA:DIRECTION	NOCARE
TRIGGER:A:BUS:B2:I2C:DATA:SIZE	1
TRIGGER:A:BUS:B2:I2C:DATA:START	0.0E+0
TRIGGER:A:BUS:B2:I2C:DATA:VALUE	"XXXXXXXX"
TRIGGER:A:BUS:B2:LIN:CONDITION	SYNCFIELD
TRIGGER:A:BUS:B2:LIN:DATA:HIVALUE	"XXXXXXXX"
TRIGGER:A:BUS:B2:LIN:DATA:QUALIFIER	EQUAL
TRIGGER:A:BUS:B2:LIN:DATA:SIZE	1
TRIGGER:A:BUS:B2:LIN:DATA:VALUE	"XXXXXXXX"
TRIGGER:A:BUS:B2:LIN:ERRTYPE	SYNC
TRIGGER:A:BUS:B2:LIN:IDENTIFIER:VALUE	"XXXXXX"
TRIGGER:A:BUS:B2:PARALLEL:VALUE	"XXXXXXXXXXXXXXXXXX"
TRIGGER:A:BUS:B2:RS232C:CONDITION	TXSTART
TRIGGER:A:BUS:B2:RS232C:RX:DATA:SIZE	1
TRIGGER:A:BUS:B2:RS232C:RX:DATA:VALUE	"XXXXXXXX"
TRIGGER:A:BUS:B2:RS232C:TX:DATA:SIZE	1
TRIGGER:A:BUS:B2:RS232C:TX:DATA:VALUE	"XXXXXXXX"
TRIGGER:A:BUS:B2:SPI:CONDITION	SS
TRIGGER:A:BUS:B2:SPI:DATA:IN:VALUE	"XXXXXXXX"
TRIGGER:A:BUS:B2:SPI:DATA:OUT:VALUE	"XXXXXXXX"
TRIGGER:A:BUS:B2:SPI:DATA:SIZE	1
TRIGGER:A:BUS:B2:SPI:DATA:START	0.0E+0
TRIGGER:A:BUS:B2:USB:ADDRESS:HIVALUE	"XXXXXXX"

Table C-1: Default Values (cont.)

Item	Description
TRIGGER:A:BUS:B2:USB:ADDRESS:VALUE	"XXXXXXX"
TRIGGER:A:BUS:B2:USB:CONDITION	SYNC
TRIGGER:A:BUS:B2:USB:DATA:HIVALUE	"XXXXXXXX"
TRIGGER:A:BUS:B2:USB:DATA:OFFSET	-1
TRIGGER:A:BUS:B2:USB:DATA:SIZE	1
TRIGGER:A:BUS:B2:USB:DATA:TYPE	ANY
TRIGGER:A:BUS:B2:USB:DATA:VALUE	"XXXXXXXX"
TRIGGER:A:BUS:B2:USB:ENDPOINT:VALUE	"XXXX"
TRIGGER:A:BUS:B2:USB:ERRTYPE	PID
TRIGGER:A:BUS:B2:USB:HANDSHAKETYPE	ANY
TRIGGER:A:BUS:B2:USB:QUALIFIER	EQUAL
TRIGGER:A:BUS:B2:USB:SOFFRAMENUMBER	"XXXXXXXXXXXX"
TRIGGER:A:BUS:B2:USB:SPECIALTYPE	ANY
TRIGGER:A:BUS:B2:USB:TOKENTYPE	ANY
TRIGGER:A:BUS:B<x>:CAN:DATA:OFFSet	0
TRIGGER:A:BUS:B<x>:CAN:FD:BRsBIT	X
TRIGGER:A:BUS:B<x>:CAN:FD:ESIBIT	X
TRIGGER:A:BUS:SOURCE	B1
TRIGGER:A:EDGE:COUPLING	DC
TRIGGER:A:EDGE:SLOPE	RISE
TRIGGER:A:EDGE:SOURCE	CH1
TRIGGER:A:HOLDOFF:TIME	20.0000E-9
TRIGGER:A:LEVEL	0.0E+0
TRIGGER:A:LEVEL:CH1	0.0E+0
TRIGGER:A:LEVEL:CH2	0.0E+0
TRIGGER:A:LEVEL:CH3	0.0E+0
TRIGGER:A:LEVEL:CH4	0.0E+0
TRIGGER:A:LEVEL:D0	1.4000
TRIGGER:A:LEVEL:D1	1.4000
TRIGGER:A:LEVEL:D10	1.4000
TRIGGER:A:LEVEL:D11	1.4000
TRIGGER:A:LEVEL:D12	1.4000
TRIGGER:A:LEVEL:D13	1.4000
TRIGGER:A:LEVEL:D14	1.4000
TRIGGER:A:LEVEL:D15	1.4000
TRIGGER:A:LEVEL:D2	1.4000
TRIGGER:A:LEVEL:D3	1.4000

Table C-1: Default Values (cont.)

Item	Description
TRIGGER:A:LEVEL:D4	1.4000
TRIGGER:A:LEVEL:D5	1.4000
TRIGGER:A:LEVEL:D6	1.4000
TRIGGER:A:LEVEL:D7	1.4000
TRIGGER:A:LEVEL:D8	1.4000
TRIGGER:A:LEVEL:D9	1.4000
TRIGGER:A:LOGIC:CLASS	LOGIC
TRIGGER:A:LOGIC:FUNCTION	AND
TRIGGER:A:LOGIC:INPUT:CH1	X
TRIGGER:A:LOGIC:INPUT:CH2	X
TRIGGER:A:LOGIC:INPUT:CH3	X
TRIGGER:A:LOGIC:INPUT:CH4	X
TRIGGER:A:LOGIC:INPUT:CLOCK:EDGE	RISE
TRIGGER:A:LOGIC:INPUT:CLOCK:SOURCE	NONE
TRIGGER:A:LOGIC:INPUT:D0	X
TRIGGER:A:LOGIC:INPUT:D1	X
TRIGGER:A:LOGIC:INPUT:D10	X
TRIGGER:A:LOGIC:INPUT:D11	X
TRIGGER:A:LOGIC:INPUT:D12	X
TRIGGER:A:LOGIC:INPUT:D13	X
TRIGGER:A:LOGIC:INPUT:D14	X
TRIGGER:A:LOGIC:INPUT:D15	X
TRIGGER:A:LOGIC:INPUT:D2	X
TRIGGER:A:LOGIC:INPUT:D3	X
TRIGGER:A:LOGIC:INPUT:D4	X
TRIGGER:A:LOGIC:INPUT:D5	X
TRIGGER:A:LOGIC:INPUT:D6	X
TRIGGER:A:LOGIC:INPUT:D7	X
TRIGGER:A:LOGIC:INPUT:D8	X
TRIGGER:A:LOGIC:INPUT:D9	X
TRIGGER:A:LOGIC:PATTERN:DELTATIME	8.0000E-9
TRIGGER:A:LOGIC:PATTERN:WHEN	TRUE
TRIGGER:A:LOGIC:PATTERN:WHEN:LESSLIMIT	8.0000E-9
TRIGGER:A:LOGIC:PATTERN:WHEN:MORELIMIT	8.0000E-9
TRIGGER:A:LOGIC:THRESHOLD:CH1	0.0E+0
TRIGGER:A:LOGIC:THRESHOLD:CH2	0.0E+0
TRIGGER:A:LOGIC:THRESHOLD:CH3	0.0E+0

Table C-1: Default Values (cont.)

Item	Description
TRIGGER:A:LOGIC:THRESHOLD:CH4	0.0E+0
TRIGGER:A:LOGIC:THRESHOLD:D0	1.4000
TRIGGER:A:LOGIC:THRESHOLD:D1	1.4000
TRIGGER:A:LOGIC:THRESHOLD:D10	1.4000
TRIGGER:A:LOGIC:THRESHOLD:D11	1.4000
TRIGGER:A:LOGIC:THRESHOLD:D12	1.4000
TRIGGER:A:LOGIC:THRESHOLD:D13	1.4000
TRIGGER:A:LOGIC:THRESHOLD:D14	1.4000
TRIGGER:A:LOGIC:THRESHOLD:D15	1.4000
TRIGGER:A:LOGIC:THRESHOLD:D2	1.4000
TRIGGER:A:LOGIC:THRESHOLD:D3	1.4000
TRIGGER:A:LOGIC:THRESHOLD:D4	1.4000
TRIGGER:A:LOGIC:THRESHOLD:D5	1.4000
TRIGGER:A:LOGIC:THRESHOLD:D6	1.4000
TRIGGER:A:LOGIC:THRESHOLD:D7	1.4000
TRIGGER:A:LOGIC:THRESHOLD:D8	1.4000
TRIGGER:A:LOGIC:THRESHOLD:D9	1.4000
TRIGGER:A:LOWERTHRESHOLD:CH1	0.0E+0
TRIGGER:A:LOWERTHRESHOLD:CH2	0.0E+0
TRIGGER:A:LOWERTHRESHOLD:CH3	0.0E+0
TRIGGER:A:LOWERTHRESHOLD:CH4	0.0E+0
TRIGGER:A:LOWERTHRESHOLD:D0	1.4000
TRIGGER:A:LOWERTHRESHOLD:D1	1.4000
TRIGGER:A:LOWERTHRESHOLD:D10	1.4000
TRIGGER:A:LOWERTHRESHOLD:D11	1.4000
TRIGGER:A:LOWERTHRESHOLD:D12	1.4000
TRIGGER:A:LOWERTHRESHOLD:D13	1.4000
TRIGGER:A:LOWERTHRESHOLD:D14	1.4000
TRIGGER:A:LOWERTHRESHOLD:D15	1.4000
TRIGGER:A:LOWERTHRESHOLD:D2	1.4000
TRIGGER:A:LOWERTHRESHOLD:D3	1.4000
TRIGGER:A:LOWERTHRESHOLD:D4	1.4000
TRIGGER:A:LOWERTHRESHOLD:D5	1.4000
TRIGGER:A:LOWERTHRESHOLD:D6	1.4000
TRIGGER:A:LOWERTHRESHOLD:D7	1.4000
TRIGGER:A:LOWERTHRESHOLD:D8	1.4000
TRIGGER:A:LOWERTHRESHOLD:D9	1.4000

Table C-1: Default Values (cont.)

Item	Description
TRIGGER:A:MODE	AUTO
TRIGGER:A:PULSE:CLASS	WIDTH
TRIGGER:A:PULSEWIDTH:HIGHLIMIT	12.0000E-9
TRIGGER:A:PULSEWIDTH:LOWLIMIT	8.0000E-9
TRIGGER:A:PULSEWIDTH:POLARITY	POSITIVE
TRIGGER:A:PULSEWIDTH:WHEN	LESSTHAN
TRIGGER:A:PULSEWIDTH:WIDTH	8.0000E-9
TRIGGER:A:RUNT:POLARITY	POSITIVE
TRIGGER:A:RUNT:WHEN	OCCURS
TRIGGER:A:RUNT:WIDTH	8.0000E-9
TRIGGER:A:SETHOLD:CLOCK:EDGE	RISE
TRIGGER:A:SETHOLD:CLOCK:SOURCE	CH1
TRIGGER:A:SETHOLD:CLOCK:THRESHOLD	0.0E+0
TRIGGER:A:SETHOLD:DATA:SOURCE	NONE
TRIGGER:A:SETHOLD:DATA:THRESHOLD	9.91E+37
TRIGGER:A:SETHOLD:HOLDTIME	8.0000E-9
TRIGGER:A:SETHOLD:SETTIME	8.0000E-9
TRIGGER:A:SETHOLD:THRESHOLD:CH1	0.0E+0
TRIGGER:A:SETHOLD:THRESHOLD:CH2	0.0E+0
TRIGGER:A:SETHOLD:THRESHOLD:CH3	0.0E+0
TRIGGER:A:SETHOLD:THRESHOLD:CH4	0.0E+0
TRIGGER:A:SETHOLD:THRESHOLD:D0	1.4000
TRIGGER:A:SETHOLD:THRESHOLD:D1	1.4000
TRIGGER:A:SETHOLD:THRESHOLD:D10	1.4000
TRIGGER:A:SETHOLD:THRESHOLD:D11	1.4000
TRIGGER:A:SETHOLD:THRESHOLD:D12	1.4000
TRIGGER:A:SETHOLD:THRESHOLD:D13	1.4000
TRIGGER:A:SETHOLD:THRESHOLD:D14	1.4000
TRIGGER:A:SETHOLD:THRESHOLD:D15	1.4000
TRIGGER:A:SETHOLD:THRESHOLD:D2	1.4000
TRIGGER:A:SETHOLD:THRESHOLD:D3	1.4000
TRIGGER:A:SETHOLD:THRESHOLD:D4	1.4000
TRIGGER:A:SETHOLD:THRESHOLD:D5	1.4000
TRIGGER:A:SETHOLD:THRESHOLD:D6	1.4000
TRIGGER:A:SETHOLD:THRESHOLD:D7	1.4000
TRIGGER:A:SETHOLD:THRESHOLD:D8	1.4000
TRIGGER:A:SETHOLD:THRESHOLD:D9	1.4000

Table C-1: Default Values (cont.)

Item	Description
TRIGGER:A:TIMEOUT:POLARITY	STAYSHIGH
TRIGGER:A:TIMEOUT:TIME	8.0000E-9
TRIGGER:A:TRANSITION:DELTATIME	8.0000E-9
TRIGGER:A:TRANSITION:POLARITY	POSITIVE
TRIGGER:A:TRANSITION:WHEN	SLOWER
TRIGGER:A:TYPE	EDGE
TRIGGER:A:UPPERTHRESHOLD:CH1	800.0000E-3
TRIGGER:A:UPPERTHRESHOLD:CH2	800.0000E-3
TRIGGER:A:UPPERTHRESHOLD:CH3	800.0000E-3
TRIGGER:A:UPPERTHRESHOLD:CH4	800.0000E-3
TRIGGER:B:BY	TIME
TRIGGER:B:EDGE:COUPLING	DC
TRIGGER:B:EDGE:SLOPE	RISE
TRIGGER:B:EDGE:SOURCE	CH1
TRIGGER:B:EVENTS:COUNT	1
TRIGGER:B:LEVEL	0.0E+0
TRIGGER:B:LEVEL:CH1	0.0E+0
TRIGGER:B:LEVEL:CH2	0.0E+0
TRIGGER:B:LEVEL:CH3	0.0E+0
TRIGGER:B:LEVEL:CH4	0.0E+0
TRIGGER:B:LEVEL:D0	1.4000
TRIGGER:B:LEVEL:D1	1.4000
TRIGGER:B:LEVEL:D10	1.4000
TRIGGER:B:LEVEL:D11	1.4000
TRIGGER:B:LEVEL:D12	1.4000
TRIGGER:B:LEVEL:D13	1.4000
TRIGGER:B:LEVEL:D14	1.4000
TRIGGER:B:LEVEL:D15	1.4000
TRIGGER:B:LEVEL:D2	1.4000
TRIGGER:B:LEVEL:D3	1.4000
TRIGGER:B:LEVEL:D4	1.4000
TRIGGER:B:LEVEL:D5	1.4000
TRIGGER:B:LEVEL:D6	1.4000
TRIGGER:B:LEVEL:D7	1.4000
TRIGGER:B:LEVEL:D8	1.4000
TRIGGER:B:LEVEL:D9	1.4000
TRIGGER:B:LOWERTHRESHOLD:CH1	0.0E+0

Table C-1: Default Values (cont.)

Item	Description
TRIGGER:B:LOWERTHRESHOLD:CH2	0.0E+0
TRIGGER:B:LOWERTHRESHOLD:CH3	0.0E+0
TRIGGER:B:LOWERTHRESHOLD:CH4	0.0E+0
TRIGGER:B:LOWERTHRESHOLD:D0	1.4000
TRIGGER:B:LOWERTHRESHOLD:D1	1.4000
TRIGGER:B:LOWERTHRESHOLD:D10	1.4000
TRIGGER:B:LOWERTHRESHOLD:D11	1.4000
TRIGGER:B:LOWERTHRESHOLD:D12	1.4000
TRIGGER:B:LOWERTHRESHOLD:D13	1.4000
TRIGGER:B:LOWERTHRESHOLD:D14	1.4000
TRIGGER:B:LOWERTHRESHOLD:D15	1.4000
TRIGGER:B:LOWERTHRESHOLD:D2	1.4000
TRIGGER:B:LOWERTHRESHOLD:D3	1.4000
TRIGGER:B:LOWERTHRESHOLD:D4	1.4000
TRIGGER:B:LOWERTHRESHOLD:D5	1.4000
TRIGGER:B:LOWERTHRESHOLD:D6	1.4000
TRIGGER:B:LOWERTHRESHOLD:D7	1.4000
TRIGGER:B:LOWERTHRESHOLD:D8	1.4000
TRIGGER:B:LOWERTHRESHOLD:D9	1.4000
TRIGGER:B:STATE	0
TRIGGER:B:TIME	8.0000E-9
TRIGGER:B:TYPE	EDGE
VERBOSE	1
VIDPIC:AUTOCONTRAST	1
VIDPIC:AUTOCONTRAST:UPDATERATE	20
ZOOM:MODE	0
ZOOM:ZOOM1:POSITION	50.0000
ZOOM:ZOOM1:SCALE	4.0000E-6
ZOOM:ZOOM1:STATE	0

Appendix D: Waveform Transfer (WFMOutpre and CURVe Query) Examples

Here are several examples of command sequences using the **WFMOutpre?** and **CURVe?** queries to transfer data from the oscilloscope to a PC, with different data sources, including Analog, Digital waveforms. Each command sequence example is followed by an explanation of the returned **WFMOutpre?** results. The **WFMOutpre?** values act as settings that apply to the **CURVe** query data being transferred.

Here are several examples of command sequences using the **WFMOutpre?** and **CURVe?** queries to transfer data from the oscilloscope to a PC, with different data sources, including Analog, Digital, and RF frequency domain waveforms. Each command sequence example is followed by an explanation of the returned **WFMOutpre?** results. The **WFMOutpre?** values act as settings that apply to the **CURVe** query data being transferred.

NOTE. *In order to guarantee that the waveform data returned from **CURVe?** queries of multiple waveforms are correlated to the same acquisition, you should use single sequence acquisition mode to acquire the waveform data from a single acquisition. Single sequence acquisition mode is enabled using [ACQuire:STOPAfter SEQUENCE](#).*

Example 1: Analog Waveform (Channels 1–4)

Goal: Transfer 10,000 points of analog channel waveform data from the oscilloscope to a PC.

Command	Comment
:DATA:SOURce CH1	
:DATA:START 1	
:DATA:STOP 10000	
:WFMOutpre:ENCdg BINARY	
:WFMOutpre:BYT_Nr 1	
:HEADer 1	

Command	Comment
:WFMOutpre?	Returns the following values. Each value represents the current settings that a CURVe? query will use to format the data that will be transferred from the oscilloscope to a PC or other device (see next table for explanations): :WFMOUTPRE:BYT_NR 1;BIT_NR 8;ENCDG BINARY;BN_FMT RI;BYT_OR MSB;WFID "Ch1, DC coupling, 100.0mV/div, 4.000us/div, 10000 points, Sample mode";NR_PT 10000;PT_FMT Y;PT_ORDER LINEAR;XUNIT "s";XINCR 4.0000E-9;XZERO -20.0000E-6;PT_OFF 0;YUNIT "V";YMULT 4.0000E-3;YOFF 0.0E+0;YZERO 0.0E+0
:CURVe?	Returns 10,000 data points: :CURVe #510000<10,000 binary bytes of waveform data.>

NOTE. You can also use the *WAVFrm?* query, which concatenates the *WFMOutpre?* and *CURVe?* queries.

WFMOutpre? Query results	Explanation
BYT_NR 1	This value specifies the number of bytes per data point in the waveform data. To change this value automatically, use the WFMOutpre:BYT_Nr command. Note that changing this value automatically changes the BIT_NR value accordingly. (This value can also be set using the DATa:WIDTH command.)
BIT_NR 8	This value specifies the number of bits per data point in the waveform data. To change this value, use the WFMOutpre:BIT_Nr command. Note that changing this value automatically changes the BYT_NR value accordingly.
ENCDG BINARY	This value specifies the encoding of the waveform data. To change this value (the other possibility is ASCII), use the WFMOutpre:ENCdg command. (This value can also be set using the DATa:ENCdg command, which provides the ability to set the WFMOutpre:ENCdg, WFMOutpre:BN_Fmt, and WFMOutpre:BYT_Or values using a single command.)
BN_FMT RI	This value specifies the binary format, which in this case is RI (signed integer). To change this value (the other possibility is RP or positive integer), use the WFMOutpre:BN_Fmt command. Note: this field is not applicable for ASCII encoding.
BYT_OR MSB	This value specifies the byte order for the BINARY encoding, which in this case is MSB (most significant byte first). To change this value to LSB, use the WFMOutpre:BYT_Or command. Note: this field is not applicable for ASCII encoding.
WFID "Ch1, DC coupling, 100.0mV/div, 4.000us/div, 10000 points, Sample mode"	This quoted string represents information about the source waveform that would be returned by a WFMOutpre:WFid? query.
NR_PT 10000	This value indicates the number of data points in the waveform record to be transferred using the CURVe? query. (If you would like to determine only this value, use the WFMOutpre:NR_Pt? query.) Note: this value is typically equal to the full record length, but you also have the option to transfer only a portion of the record length by using the DATa:START and DATa:STOP commands.

WFMOutpre? Query results	Explanation
PT_FMT Y	This value indicates the format of the data points in the waveform record. In this case, the value represents YT format. This is query only – the returned values can be Y for YT format or ENV for envelope format (min/max pairs). (If you would like to determine only this value, use the WFMOutpre:PT_Fmt? query.)
PT_ORDER LINEAR	This value is always LINEar.
XUNIT "s"	This value indicates the units of the x-axis of the waveform record. This is determined by the horizontal settings for the waveform source. Typically, this value is "s", representing seconds. When using the math waveform as a source, the value can be "s" or "Hz". This is query only. (If you would like to determine only this value, use the WFMOutpre:XUnit? query.)
XINCR 4.0000E-9	This value indicates the time, in seconds, or frequency, in hertz, between data points in the waveform record. This is query only. (If you would like to determine only this value, use the WFMOutpre:XINcr? query.)
XZERO -20.0000E-6	This value indicates the time, in seconds, or frequency, in hertz, of the first data point in the waveform record. This time or frequency is relative to the time of the trigger, which is always 0. So, this XZEro time or frequency can be negative. This is query only. (If you would like to determine only this value, use the WFMOutpre:XZEro? query.)
PT_OFF 0	This is a query provided only for compatibility with performance oscilloscopes. The returned value is always 0. (If you would like to determine only this value, use the WFMOutpre:PT_Off? query.)
YUNIT "v"	This value indicates the vertical units of data points in the waveform record. This can be any of several string values, depending upon the vertical units of the source waveform – in this case, volts. This is query only. (If you would like to determine only this value, use the WFMOutpre:YUnit? query.)
YMULT 4.0000E-3	This value indicates the multiplying factor to convert the data point values from digitizing levels to the units specified by the <code>WFMOutpre:YUnit</code> command. This is query only. (If you would like to determine only this value, use the WFMOutpre:YMult? query.)
YOFF 0.0E+0	This value indicates the vertical position of the source waveform in digitizing levels. There are 25 digitizing levels per vertical division. This is query only. (If you would like to determine only this value, use the WFMOutpre:YOff? query.)
YZERO 0.0E+0	This value indicates the vertical offset of the source waveform in units specified by the <code>WFMOutpre:YUnit</code> command. This is query only. (If you would like to determine only this value, use the WFMOutpre:YZEro? query.)

Example 2: Digital Waveform

Goal: Transfer 25 points of digital channel waveform data from the oscilloscope to a PC.

Command	Comment
:DATA:SOURce CH1_D5	
:DATA:START 1	
:DATA:STOP 25	
:WFMOutpre:ENCdg ASCii	
:WFMOutpre:BYT_Nr 1	

Command	Comment
:HEAdEr 1	
:VERBoSe 1	
:WFMOupre?	Returns the following values. Each value represents the current settings that a CURVe? query will use to format the data that will be transferred from the oscilloscope to a PC or other device (see next table for explanations): :WFMOUPRE:BYT_Nr 1;BIT_Nr 8;ENCDG ASCII;BN_FMT RI;BYT_OR MSB;WFID "D5, unknown coupling, 100.0us/div, 10000 points, Digitalmode";NR_PT 25;PT_FMT Y;PT_ORDER LINEAR;XUNIT "s";XINCR 100.0000E-9;XZERO -500.0000E-6;PT_OFF 0;YUNIT "State";YMULT 1.0000;YOFF 0.0E+0;YZERO 0.0E+0
:CURVe?	Returns the following values. Each value represents a data point: :CURVe 0,1,1,1,1,1,1,1,1,1,1,1,1,1,1,0,0,0,0,0,0,0,0

NOTE. You can also use the *WAVFrm?* query, which concatenates the *WFMOupre?* and *CURVe?* queries.

WFMOupre? Query results	Explanation
WFMOUPRE:BYT_Nr 1	This value specifies the number of bytes per data point in the waveform data. To change this value automatically, use the WFMOupre:BYT_Nr command. Note that changing this value automatically changes the BIT_Nr value accordingly. (This value can also be set using the DATa:WIDTh command.)
BIT_Nr 8	This value specifies the number of bits per data point in the waveform data. To change this value, use the WFMOupre:BIT_Nr command. Note that changing this value automatically changes the BYT_Nr value accordingly.
ENCDG ASCII	This value specifies the encoding of the waveform data. To change this value (the other possibility is BINARY), use the WFMOupre:ENCdg command. (This value can also be set using the DATa:ENCdg command, which provides the ability to set the WFMOupre:ENCdg , WFMOupre:BN_Fmt , and WFMOupre:BYT_Or values using a single command.)
BN_FMT RI	This value specifies the binary format, which in this case is RI (signed integer). To change this value (the other possibility is RP or positive integer), use the WFMOupre:BN_Fmt command. Note: this field is not applicable for ASCII encoding.
BYT_OR MSB	This value specifies the byte order for the BINARY encoding, which in this case is MSB (most significant byte first, also known as IBM format). To change this value to LSB, use the WFMOupre:BYT_Or command. Note: this field is not applicable for ASCII encoding.
WFID "D5, unknown coupling, 100.0us/div, 10000 points, Digital mode"	This quoted string represents information about the source waveform that would be returned by a WFMOupre:WFid? query. It cannot be changed.

WFMOutpre? Query results	Explanation
NR_PT 25	This value indicates the number of data points in the waveform record to be transferred using the CURVe? query. (If you would like to determine only this value, use the WFMOutpre:NR_Pt? query.) Note: this value is typically equal to the full record length, but you also have the option to transfer only a portion of the record length by using the DATa:STaRt and DATa:STOp commands.
PT_FMT Y	This value indicates the format of the data points in the waveform record. In this case, the value represents YT format. This is query only. (If you would like to determine only this value, use the WFMOutpre:PT_Fmt? query.)
PT_ORDER LINEAR	This value is always LINEAR.
XUNIT "s"	This value indicates the units of the x-axis of the waveform record. This is determined by the horizontal settings for the waveform source. Typically, this value is "s", representing seconds. When using the math waveform as a source, the value can be "s" or "Hz". This is query only. (If you would like to determine only this value, use the WFMOutpre:XUNit? query.)
XINCR 100.0000E-9	This value indicates the time, in seconds, or frequency, in hertz, between data points in the waveform record. This is query only. (If you would like to determine only this value, use the WFMOutpre:XINcr? query.)
XZERO -500.0000E-6	This value indicates the time, in seconds, or frequency, in hertz, of the first data point in the waveform record. This time or frequency is relative to the time of the trigger, which is always 0. So, this XZEro time or frequency can be negative. This is query only. (If you would like to determine only this value, use the WFMOutpre:XZEro? query.)
PT_OFF 0	This is a query provided only for compatibility with other Tektronix oscilloscopes. The returned value is always 0. (If you would like to determine only this value, use the WFMOutpre:PT_Off? query.)
YUNIT "State"	This value indicates the vertical units of data points in the waveform record. This can be any of several string values, depending upon the vertical units of the source waveform – in this case, State. This is query only. (If you would like to determine only this value, use the WFMOutpre:YUNit? query.)
YMULT 1.0000	This value indicates the multiplying factor to convert the data point values from digitizing levels to the units specified by the <code>WFMOUtpre:YUNit</code> command. This is query only. (If you would like to determine only this value, use the WFMOutpre:YMUlt? query.)
YOFF 0.0E+0	This value indicates the vertical position of the source waveform in digitizing levels. There are 25 digitizing levels per vertical division. This is query only. (If you would like to determine only this value, use the WFMOutpre:YOFF? query.)
YZERO 0.0E+0	This value indicates the vertical offset of the source waveform in units specified by the <code>WFMOUtpre:YUNit</code> command. This is query only. (If you would like to determine only this value, use the WFMOutpre:YZEro? query.)

Example 3: Digital with 4 Bytes Per Point and Zoom Off

Goal: Transfer 25 points of Digital data from the oscilloscope to a PC using 4 bytes per point and Zoom off.

Command	Comment
:DATA:SOURce CH1_D1	
:ACQuire:MAGnivu 0	
:DATA:START 1	
:DATA:STOP 25	
:WFMOutpre:ENCdg ASCii	
:WFMOutpre:BYT_Nr 4	
:HEADer 1	
:VERBoSe 1	
:WFMOutpre?	Returns the following values. Each value represents the current settings that a CURVe? query will use to format the data that will be transferred from the oscilloscope to a PC or other device (see next table for explanations): :WFMOUTPRE:BYT_NR 4;BIT_NR 32;ENCDG ASCII;BN_FMT RI;BYT_OR MSB;WFID "Digital, unknown coupling, 100.0us/div, 10000 points, Digital mode";NR_PT 25;PT_FMT Y;PT_ORDER LINEAR;XUNIT "s";XINCR 100.0000E-9;XZERO -500.0000E-6;PT_OFF 0;YUNIT "State";YMULT 1.0000;YOFF 0.0E+0;YZERO 0.0E+0
:CURVe?	Returns the following values. Each value represents a data point: :CURVe FB386,FB366,FB3E6,FB366,FB3E6,FB32E,FB3A2,FB32E,FB3AA,FB366,FB3EA,FB366,FB3E2,FB36,FB3E6,FB366,FB3E6,FB346,FB3C6,FB346,FB3C6,FB34E,FB3C2,FB34E,FB3CA

NOTE. The returned hexadecimal data values are formatted without leading zeroes. For example, a 4-byte digital value of FB386 should be interpreted as 000FB386.

NOTE. You can also use the *WAVFrm?* query, which concatenates the *WFMOutpre?* and *CURVe?* queries.

WFMOutpre? Query results	Explanation
WFMOUTPRE:BYT_NR 4	This value specifies the number of bytes per data point in the waveform data. To change this value automatically, use the WFMOutpre:BYT_Nr command. Note that changing this value automatically changes the BIT_NR value accordingly. (This value can also be set using the DATA:WIDTH command.)
BIT_NR 32	This value specifies the number of bits per data point in the waveform data. To change this value, use the WFMOutpre:BIT_Nr command. Note that changing this value automatically changes the BYT_NR value accordingly.

WFMOutpre? Query results	Explanation
ENCDG ASCII	This value specifies the encoding of the waveform data. To change this value (the other possibility is BINARY), use the WFMOutpre:ENCDg command. (This value can also be set using the DATA:ENCDg command, which provides the ability to set the WFMOutpre:ENCDg , WFMOutpre:BN_Fmt , and WFMOutpre:BYT_Or values using a single command.)
BN_FMT RI	This value specifies the binary format, which in this case is RI (signed integer). To change this value (the other possibility is RP or positive integer), use the WFMOutpre:BN_Fmt command. Note: this field is not applicable for ASCII encoding.
BYT_OR MSB	This value specifies the byte order for the BINARY encoding, which in this case is MSB (most significant byte first, also known as IBM format). To change this value to LSB, use the WFMOutpre:BYT_Or command. Note: this field is not applicable for ASCII encoding.
WFID "Digital, unknown coupling, 100.0us/div, 10000 points, Digital mode"	This quoted string represents information about the source waveform that would be returned by a WFMOutpre:WFid? query. It cannot be changed.
NR_PT 25	This value indicates the number of data points in the waveform record to be transferred using the CURVe? query. (If you would like to determine only this value, use the WFMOutpre:NR_Pt? query.) Note: this value is typically equal to the full record length, but you also have the option to transfer only a portion of the record length by using the DATA:START and DATA:STOP commands.
PT_FMT Y	This value indicates the format of the data points in the waveform record. In this case, the value represents YT format. This is query only – the returned values can be Y for YT format or ENV for envelope format (min/max pairs). (If you would like to determine only this value, use the WFMOutpre:PT_Fmt? query.)
PT_ORDER LINEAR	This value is always LINEar.
XUNIT "s"	This value indicates the units of the x-axis of the waveform record. This is determined by the horizontal settings for the waveform source. For live channels, this value is "s", representing seconds. When using the math waveform as a source, the value can be "s" or "Hz". This is query only. (If you would like to determine only this value, use the WFMOutpre:XUnit? query.)
XINCR 100.0000E-9	This value indicates the time, in seconds, or frequency, in hertz, between data points in the waveform record. This is query only. (If you would like to determine only this value, use the WFMOutpre:XINcr? query.)
XZERO -500.0000E-6	This value indicates the time, in seconds, or frequency, in hertz, of the first data point in the waveform record. This time or frequency is relative to the time of the trigger, which is always 0. So, this XZErO time or frequency can be negative. This is query only. (If you would like to determine only this value, use the WFMOutpre:XZErO? query.)
PT_OFF 0	This is a query provided only for compatibility with performance oscilloscopes. The returned value is always 0. (If you would like to determine only this value, use the WFMOutpre:PT_Off? query.)
YUNIT "State"	This value indicates the vertical units of data points in the waveform record. This can be any of several string values, depending upon the vertical units of the source waveform – in this case, State. This is query only. (If you would like to determine only this value, use the WFMOutpre:YUnit? query.)

WFMOutpre? Query results	Explanation
YMULT 1.0000	This value indicates the multiplying factor to convert the data point values from digitizing levels to the units specified by the WFMOUTpre:YUNIT command. This is query only. (If you would like to determine only this value, use the WFMOutpre:YMult? query.)
YOFF 0.0E+0	This value indicates the vertical position of the source waveform in digitizing levels. There are 25 digitizing levels per vertical division. This is query only. (If you would like to determine only this value, use the WFMOutpre:YOff? query.)
YZERO 0.0E+0	This value indicates the vertical offset of the source waveform in units specified by the WFMOUTpre:YUNIT command. This is query only. (If you would like to determine only this value, use the WFMOutpre:YZero? query.)

Example 4: Digital with 8 Bytes Per Point and Zoom Off

Goal: Transfer 25 points of Digital data from the oscilloscope to a PC using 8 bytes per point and Zoom off.

Command	Comment
:DATA:SOURce CH1_D1	
:ACQuire:MAGnitu 0	
:DATA:START 1	
:DATA:STOP 25	
:WFMOutpre:ENCdg ASCII	
:WFMOutpre:BYT_Nr 8	
:HEADer 1	
:VERBoSe 1	
:WFMOutpre?	Returns the following values. Each value represents the current settings that a CURVe? query will use to format the data that will be transferred from the oscilloscope to a PC or other device (see next table for explanations): :WFMOUTPRE:BYT_NR 8;BIT_NR 64;ENCDG ASCII;BN_FMT RI;BYT_OR MSB;WFID "Digital, unknown coupling, 100.0us/div, 10000 points, Digital mode";NR_PT 25;PT_FMT Y;PT_ORDER LINEAR;XUNIT "s";XINCR 100.0000E-9;XZERO -500.0000E-6;PT_OFF 0;YUNIT "State";YMULT 1.0000;YOFF 0.0E+0;YZERO 0.0E+0
:CURVe?	Returns the following values. Each value represents a data point: :CURVe 80000FB386,E0000FB386,80000FB3E6,80000FB3E6,80000FB3E6, C8000FB3A6,8C000FB3A6, 8C000FB3A6,84000FB3AE,CC000FB3A6, 8C000FB3E6,8C000FB3E6,84000FB3E6,80000FB3E6,80000FB3E6, 80000FB3E6,80000FB3E6,A0000FB3C6,80000FB3C6,80000FB3C6, 80000FB3C6,88000FB3C6,8C000FB3C6, 8C000FB3C6,84000FB3CE

NOTE. The returned hexadecimal data values are formatted without leading zeroes.

NOTE. You can also use the *WAVFrm?* query, which concatenates the *WFMOutpre?* and *CURVe?* queries.

WFMOutpre? Query results	Explanation
WFMOUTPRE:BYT_NR 8	This value specifies the number of bytes per data point in the waveform data. To change this value automatically, use the WFMOutpre:BYT_Nr command. Note that changing this value automatically changes the BIT_NR value accordingly. (This value can also be set using the DATA:WIDTH command.)
BIT_NR 64	This value specifies the number of bits per data point in the waveform data. To change this value, use the WFMOutpre:BIT_Nr command. Note that changing this value automatically changes the BYT_NR value accordingly.
ENCDG ASCII	This value specifies the encoding of the waveform data. To change this value (the other possibility is BINARY), use the WFMOutpre:ENCDg command. (This value can also be set using the DATA:ENCDg command, which provides the ability to set the WFMOutpre:ENCDg , WFMOutpre:BN_Fmt , and WFMOutpre:BYT_Or values using a single command.)
BN_FMT RI	This value specifies the binary format, which in this case is RI (signed integer). To change this value (the other possibility is RP or positive integer), use the WFMOutpre:BN_Fmt command. Note: this field is not applicable for ASCII encoding.
BYT_OR MSB	This value specifies the byte order for the BINARY encoding, which in this case is MSB (most significant byte first, also known as IBM format). To change this value to LSB, use the WFMOutpre:BYT_Or command. Note: this field is not applicable for ASCII encoding.
WFID "Digital, unknown coupling, 100.0us/div, 10000 points, Digital mode"	This quoted string represents information about the source waveform that would be returned by a WFMOutpre:WFID? query. It cannot be changed.
NR_PT 25	This value indicates the number of data points in the waveform record to be transferred using the CURVe? query. (If you would like to determine only this value, use the WFMOutpre:NR_Pt? query.) Note: this value is typically equal to the full record length, but you also have the option to transfer only a portion of the record length by using the DATA:START and DATA:STOP commands.
PT_FMT Y	This value indicates the format of the data points in the waveform record. In this case, the value represents YT format. This is query only – the returned values can be Y for YT format or ENV for envelope format (min/max pairs). (If you would like to determine only this value, use the WFMOutpre:PT_Fmt? query.)
PT_ORDER LINEAR	This value is always LINEar.
XUNIT "s"	This value indicates the units of the x-axis of the waveform record. This is determined by the horizontal settings for the waveform source. Typically, this value is "s", representing seconds. When using the math waveform as a source, the value can be "s" or "Hz". This is query only. (If you would like to determine only this value, use the WFMOutpre:XUNIT? query.)
XINCR 100.0000E-9	This value indicates the time, in seconds, or frequency, in hertz, between data points in the waveform record. This is query only. (If you would like to determine only this value, use the WFMOutpre:XINcr? query.)

WFMOUtpre? Query results	Explanation
XZERO -500.0000E-6	This value indicates the time, in seconds, or frequency, in hertz, of the first data point in the waveform record. This time or frequency is relative to the time of the trigger, which is always 0. So, this XZero time or frequency can be negative. This is query only. (If you would like to determine only this value, use the WFMOUtpre:XZero? query.)
PT_OFF 0	This is a query provided only for compatibility with other Tektronix oscilloscopes. The returned value is always 0. (If you would like to determine only this value, use the WFMOUtpre:PT_Off? query.)
YUNIT "State"	This value indicates the vertical units of data points in the waveform record. This can be any of several string values, depending upon the vertical units of the source waveform – in this case, State. This is query only. (If you would like to determine only this value, use the WFMOUtpre:YUnit? query.)
YMULT 1.0000	This value indicates the multiplying factor to convert the data point values from digitizing levels to the units specified by the <code>WFMOUtpre:YUnit</code> command. This is query only. (If you would like to determine only this value, use the WFMOUtpre:YMult? query.)
YOFF 0.0E+0	This value indicates the vertical position of the source waveform in digitizing levels. There are 25 digitizing levels per vertical division. This is query only. (If you would like to determine only this value, use the WFMOUtpre:YOff? query.)
YZERO 0.0E+0	This value indicates the vertical offset of the source waveform in units specified by the <code>WFMOUtpre:YUnit</code> command. This is query only. (If you would like to determine only this value, use the WFMOUtpre:YZero? query.)

Example 5: Digital with 4 Bytes Per Point and Zoom On

Goal: Transfer 25 points of Digital data from the oscilloscope to a PC using 4 bytes per point and Zoom on.

Command	Comment
:DATA:SOURce CH1_D1	
:ACQuire:MAGnivu 1	
:DATA:START 1	
:DATA:STOP 25	
:WFMOUtpre:ENCdg ASCii	
:WFMOUtpre:BYT_Nr 4	
:HEADer 1	
:VERBose 1	

Command	Comment
:WFMOutpre?	Returns the following values. Each value represents the current settings that a CURVE? query will use to format the data that will be transferred from the oscilloscope to a PC or other device (see next table for explanations): <pre>:WFMOUTPRE:BYT_NR 4;BIT_NR 32;ENCDG ASCII;BN_FMT RI;BYT_OR MSB;WFID "Digital, unknown coupling, 100.0us/div, 10000 points, Digital mode";NR_PT 25;PT_FMT Y;PT_ORDER LINEAR;XUNIT "s";XINCR 1.2121E-9;XZERO -6.0606E-6;PT_OFF 0;YUNIT "State";YMULT 1.0000;YOFF 0.0E+0;YZERO 0.0E+0</pre>
:CURVE?	Returns the following values. Each value represents a data point: <pre>:CURVE FB6E6,FB666,FB6E6,FB666,FB666,FB6E6,FB666,FB6E6,FB666, FB666, FB6E6,FB666,FB6E6,FB6E6,FB666,FB6E6,FB666,FB666, FB6E6,FB666,FB6E6,FB6E6,FB666,FB6E6,FB666</pre>

NOTE. The returned hexadecimal data values are formatted without leading zeroes.

NOTE. You can also use the *WAVFrm?* query, which concatenates the *WFMOutpre?* and *CURVe?* queries.

WFMOutpre? Query results	Explanation
WFMOUTPRE:BYT_NR 4	This value specifies the number of bytes per data point in the waveform data. To change this value automatically, use the WFMOutpre:BYT_Nr command. Note that changing this value automatically changes the BIT_NR value accordingly. (This value can also be set using the DATa:WIDTH command.)
BIT_NR 32	This value specifies the number of bits per data point in the waveform data. To change this value, use the WFMOutpre:BIT_Nr command. Note that changing this value automatically changes the BYT_NR value accordingly.
ENCDG ASCII	This value specifies the encoding of the waveform data. To change this value (the other possibility is BINARY), use the WFMOutpre:ENCDg command. (This value can also be set using the DATa:ENCDg command, which provides the ability to set the WFMOutpre:ENCDg, WFMOutpre:BN_Fmt,: and WFMOutpre:BYT_Or values using a single command.)
BN_FMT RI	This value specifies the binary format, which in this case is RI (signed integer). To change this value (the other possibility is RP or positive integer), use the WFMOutpre:BN_Fmt command. Note: this field is not applicable for ASCII encoding.
BYT_OR MSB	This value specifies the byte order for the BINARY encoding, which in this case is MSB (most significant byte first, also known as IBM format). To change this value to LSB, use the WFMOutpre:BYT_Or command. Note: this field is not applicable for ASCII encoding.

WFMOUtpre? Query results	Explanation
WFID "Digital, unknown coupling, 100.0us/div, 10000 points, Digital mode"	This quoted string represents information about the source waveform that would be returned by a <code>WFMOUtpre:WFid?</code> query. It cannot be changed.
NR_PT 25	This value indicates the number of data points in the waveform record to be transferred using the CURVe? query. (If you would like to determine only this value, use the WFMOUtpre:NR_Pt? query.) Note: this value is typically equal to the full record length, but you also have the option to transfer only a portion of the record length by using the DATA:START and DATA:STOP commands.
PT_FMT Y	This value indicates the format of the data points in the waveform record. In this case, the value represents YT format. This is query only – the returned values can be Y for YT format or ENV for envelope format (min/max pairs). (If you would like to determine only this value, use the WFMOUtpre:PT_Fmt? query.)
PT_ORDER LINEAR	This value is always LINEar.
XUNIT "s"	This value indicates the units of the x-axis of the waveform record. This is determined by the horizontal settings for the waveform source. Typically, this value is "s", representing seconds. When using the math waveform as a source, the value can be "s" or "Hz". This is query only. (If you would like to determine only this value, use the WFMOUtpre:XUnit? query.)
XINCR 1.2121E-9	This value indicates the time, in seconds, or frequency, in hertz, between data points in the waveform record. This is query only. (If you would like to determine only this value, use the WFMOUtpre:XINcr? query.)
XZERO -6.0606E-6	This value indicates the time, in seconds, or frequency, in hertz, of the first data point in the waveform record. This time or frequency is relative to the time of the trigger, which is always 0. So, this XZEro time or frequency can be negative. This is query only. (If you would like to determine only this value, use the WFMOUtpre:XZEro? query.)
PT_OFF 0	This is a query provided only for compatibility with performance oscilloscopes. The returned value is always 0. (If you would like to determine only this value, use the WFMOUtpre:PT_Off? query.)
YUNIT "State"	This value indicates the vertical units of data points in the waveform record. This can be any of several string values, depending upon the vertical units of the source waveform – in this case, State. This is query only. (If you would like to determine only this value, use the WFMOUtpre:YUnit? query.)
YMULT 1.0000	This value indicates the multiplying factor to convert the data point values from digitizing levels to the units specified by the <code>WFMOUtpre:YUnit</code> command. This is query only. (If you would like to determine only this value, use the WFMOUtpre:YMult? query.)
YOFF 0.0E+0	This value indicates the vertical position of the source waveform in digitizing levels. There are 25 digitizing levels per vertical division. This is query only. (If you would like to determine only this value, use the WFMOUtpre:YOff? query.)
YZERO 0.0E+0	This value indicates the vertical offset of the source waveform in units specified by the <code>WFMOUtpre:YUnit</code> command. This is query only. (If you would like to determine only this value, use the WFMOUtpre:YZEro? query.)

Example 6: Digital with 8 Bytes Per Point and MagniVu On

Goal: Transfer 25 points of Digital data from the oscilloscope to a PC using 8 bytes per point and Zoom on.

Command	Comment
:DATA:SOURce CH1_D1	
:ACQuire:MAGnivu 1	
:DATA:START 1	
:DATA:STOP 25	
:WFMOutpre:ENCdg ASCii	
:WFMOutpre:BYT_Nr 8	
:HEAdEr 1	
:VERBoSe 1	
:WFMOutpre?	Returns the following values. Each value represents the current settings that a CURVe? query will use to format the data that will be transferred from the oscilloscope to a PC or other device (see next table for explanations): :WFMOUTPRE:BYT_NR 8;BIT_NR 64;ENCDG ASCII;BN_FMT RI;BYT_OR MSB;WFID "Digital, unknown coupling, 100.0us/div, 10000 points, Digital mode";NR_PT 25;PT_FMT Y;PT_ORDER LINEAR;XUNIT "s";XINCR 1.2121E-9;XZERO -6.0606E-6;PT_OFF 0;YUNIT "State";YMULT 1.0000;YOFF 0.0E+0;YZERO 0.0E+0
:CURVe?	Returns the following values. Each value represents a data point: :CURVe 80000FB787,80000FB787,E8000FB787,8C000FB7E7,8C000FB7E7,84000FB7EF,CC000FB7A7,8C000FB7A7,8C000FB7A7,84000FB7A7,84000FB7A7,80000FB7A7,80000FB7A7,80000FB7A7,80000FB7A7,80000FB7A7,80000FB7A7,80000FB7A7,C8000FB7A7,8C000FB7E7,8C000FB7E7,84000FB7EF,CC000FB7A7,8C000FB7A7,8C000FB7A7

NOTE. The returned hexadecimal data values are formatted without leading zeroes.

NOTE. You can also use the *WAVFrm?* query, which concatenates the *WFMOutpre?* and *CURVe?* queries.

WFMOutpre? Query results	Explanation
WFMOUTPRE:BYT_NR 8	This value specifies the number of bytes per data point in the waveform data. To change this value automatically, use the WFMOutpre:BYT_Nr command. Note that changing this value automatically changes the BIT_NR value accordingly. (This value can also be set using the DATA:WIDTH command.)
BIT_NR 64	This value specifies the number of bits per data point in the waveform data. To change this value, use the WFMOutpre:BIT_Nr command. Note that changing this value automatically changes the BYT_NR value accordingly.
ENCDG ASCII	This value specifies the encoding of the waveform data. To change this value (the other possibility is BINARY), use the WFMOutpre:ENCdg command. (This value can also be set using the DATA:ENCdg command, which provides the ability to set the WFMOutpre:ENCdg, WFMOutpre:BN_Fmt, and WFMOutpre:BYT_Or values using a single command.)
BN_FMT RI	This value specifies the binary format, which in this case is RI (signed integer). To change this value (the other possibility is RP or positive integer), use the WFMOutpre:BN_Fmt command. Note: this field is not applicable for ASCII encoding.
BYT_OR MSB	This value specifies the byte order for the BINARY encoding, which in this case is MSB (most significant byte first, also known as IBM format). To change this value to LSB, use the WFMOutpre:BYT_Or command. Note: this field is not applicable for ASCII encoding.
WFID "Digital, unknown coupling, 100.0us/div, 10000 points, Digital mode"	This quoted string represents information about the source waveform that would be returned by a WFMOutpre:WFID? query. It cannot be changed.
NR_PT 25	This value indicates the number of data points in the waveform record. (If you would like to determine only this value, use the WFMOutpre:NR_Pt? query.) Note: this value is typically equal to the full record length, but you also have the option to transfer only a portion of the record length by using the DATA:START and DATA:STOP commands.
PT_FMT Y	This value indicates the format of the data points in the waveform record. In this case, the value represents YT format. This is query only – the returned values can be Y for YT format or ENV for envelope format (min/max pairs). (If you would like to determine only this value, use the WFMOutpre:PT_Fmt? query.)
PT_ORDER LINEAR	This value is always LINEar.
XUNIT "s"	This value indicates the units of the x-axis of the waveform record. This is determined by the horizontal settings for the waveform source. Typically, this value is "s", representing seconds. When using the math waveform as a source, the value can be "s" or "Hz". This is query only. (If you would like to determine only this value, use the WFMOutpre:XUnit? query.)
XINCR 1.2121E-9	This value indicates the time, in seconds, or frequency, in hertz, between data points in the waveform record. This is query only. (If you would like to determine only this value, use the WFMOutpre:XINcr? query.)
XZERO -6.0606E-6	This value indicates the time, in seconds, or frequency, in hertz, of the first data point in the waveform record. This time or frequency is relative to the time of the trigger, which is always 0. So, this XZEro time or frequency can be negative. This is query only. (If you would like to determine only this value, use the WFMOutpre:XZEro? query.)
PT_OFF 0	This is a query provided only for compatibility with other Tektronix oscilloscopes. The returned value is always 0. (If you would like to determine only this value, use the WFMOutpre:PT_Off? query.)

WFMOutpre? Query results	Explanation
YUNIT "State"	This value indicates the vertical units of data points in the waveform record. This can be any of several string values, depending upon the vertical units of the source waveform – in this case, State. This is query only. (If you would like to determine only this value, use the WFMOutpre:YUnit? query.)
YMULT 1.0000	This value indicates the multiplying factor to convert the data point values from digitizing levels to the units specified by the <code>WFMOutpre:YUnit</code> command. This is query only. (If you would like to determine only this value, use the WFMOutpre:YMult? query.)
YOFF 0.0E+0	This value indicates the vertical position of the source waveform in digitizing levels. There are 25 digitizing levels per vertical division. This is query only. (If you would like to determine only this value, use the WFMOutpre:YOff? query.)
YZERO 0.0E+0	This value indicates the vertical offset of the source waveform in units specified by the <code>WFMOutpre:YUnit</code> command. This is query only. (If you would like to determine only this value, use the WFMOutpre:YZero? query.)

Example 7: RF Frequency Domain Waveform

Goal: Transfer 5 points of RF frequency domain data from the oscilloscope to a PC using 4 bytes per point.

NOTE. A frequency domain waveform used in data transfer from the oscilloscope to a PC or other device may be one of the four RF frequency domain traces or the Spectrum Math waveform.

Command	Comment
:DATA:SOURce RF_NORMa1	
:DATA:START 495	
:DATA:STOP 505	
:WFMOutpre:ENCdg ASCii	
:WFMOutpre:BYT_Nr 4	
:HEADer 1	
:VERBoSe 1	

Command	Comment
:WFMOupre?	Returns the following values. Each value represents the current settings that a CURVe? query will use to format the data that will be transferred from the oscilloscope to a PC or other device (see next table for explanations): :WFMOUPRE:BYT_NR 4;BIT_NR 32;ENCDG ASCII;BN_FMT FP;BYT_OR MSB;WFID "RF_NORMAL, unknown coupling, 10.00w/div, 200.0MHz/div, 1001 points, Sample mode, Center Freq: 2.400GHz, Span: 2.000GHz, Reference Level: 1.000mw";NR_PT 21;PT_FMT Y;PT_ORDER LINEAR;XUNIT "Hz";XINCR 2.0000E+6;XZERO 1.4000E+9;PT_OFF 0;YUNIT "W";YMULT 1.5625E-3;YOFF 0.0E+0;YZERO 0.0E+0;DOMAIN FREQUENCY;WFMTYPE RF_FD;CENTERFREQUENCY 2.4000E+9;SPAN 2.0000E+9;REFLEVEL 1.0000E-3
:CURVe?	Returns the following values. Each value represents a data point: :CURVE 2.43108E-10,1.62648E-10,2.78478E-10,4.15163E-10,7.66223E-10,7.63905E-10,3.68375E-10,3.42668E-10,3.06422E-10,1.81561E-10,2.0223E-10,4.24327E-10,1.95298E-10,2.1304E-10,9.41791E-11,4.56513E-10,4.33067E-10,6.57215E-11,1.65388E-10,9.09116E-10,5.39507E-10

WFMOupre? Query results	Explanation
BYT_NR 4	This value specifies the number of bytes per data point in the waveform data. This value is fixed for RF frequency domain traces.
BIT_NR 32	This value specifies the number of bits per data point in the waveform data. This value is fixed for RF frequency domain traces.
ENCDG ASCII	This value specifies the encoding of the waveform data. To change this value (the other possibility for RF frequency domain traces is FPBINARY), use the WFMOupre:ENCdg command. This value can also be set using the DATA:ENCdg command.
BN_FMT FP	This value specifies the binary format, which in this case is FP (floating point). This value is fixed for RF frequency domain traces. Note: This field is not applicable for ASCII encoding.
BYT_OR MSB	This value specifies the byte order for the BINARY encoding, which in this case is MSB (most significant byte first, also known as IBM format). To change this value to LSB, use the WFMOupre:BYT_Or command. Note: This field is not applicable for ASCII encoding.
WFID "RF_NORMAL, unknown coupling, 10.00W/div, 200.0MHz/div, 1001 points, Sample mode, Center Freq: 2.400GHz, Span: 2.000GHz, Reference Level: 1.000mW"	This quoted string represents information about the source waveform that would be returned by a WFMOupre:WFId? Query. It cannot be changed.
NR_PT 21	This value indicates the number of data points in the waveform record. (If you would like to determine only this value, use the WFMOupre:NR_Pt? query.) Note: This value is typically equal to the full record length of the RF frequency domain trace, but you also have the option to transfer only a portion of the record using the DATs:STARt and DATA:STOP commands.
PT_FMT Y	This value indicates the format of the data points in the waveform record. In this case, the value represents YF (Amplitude vs. Frequency) format. This is query only. (If you would like to determine only this value, use the WFMOupre:PT_Fmt? query.
PT_ORDER LINEAR .	This value is always LINear

XUNIT "Hz"	This value indicates the units of the x-axis of the waveform record. This value is always "Hz" for RF frequency domain traces. This is query only. (If you would like to determine only this value, use the WFMOutpre:XUNit? query.)
XINCR 2.0000E+6	This value indicates the frequency, in hertz, between data points in the waveform record. This is query only. (If you would like to determine only this value, use the WFMOutpre:XINcr? query.)
XZERO 1.4000E+9	This value indicates the frequency, in hertz, of the first data point in the waveform record. This frequency is relative to the time of the trigger, which is always 0. This XZEro frequency can be negative. This is query only. (If you would like to determine only this value, use the WFMOutpre:XZEro? query.)
PT_OFF 0	This is a query provided only for compatibility with other Tektronix oscilloscopes. The returned value is always 0. (If you would like to determine only this value, use the WFMOutpre:PT_Off? query.)
YUNIT "W"	This value indicates the units of data points in the waveform record. This value depends on the vertical units of the source waveform – in this case, watts. This is query only. (If you would like to determine only this value, use the WFMOutpre:YUNit? query.)
YMULT 1.5625E-3	This value indicates the multiplying factor to convert the data point values to the units specified by the:WFMOutpre:YUNit command. This is query only. (If you would like to determine only this value, use the WFMOutpre:YMUlt? query.)
YOFF 0.0E+0	This value indicates the vertical position of the source waveform. This value is unused for RF frequency domain traces and is always 0.
YZERO 0.0E+0	This value indicates the vertical offset of the source waveform. This value is unused for RF frequency domain traces and is always 0.
DOMAIN FREQUENCY	This value indicates the domain in which the source waveform is displayed and stored. For RF time domain traces, the domain is Time and waveform transfer information is treated as integer information. For RF frequency domain traces, the domain is Frequency and waveform transfer information is treated as floating point information. This is query only. (If you would like to determine only this value, use the WFMOutpre:DOMain? query.)
WFMTYPE RF_FD	This value indicates the type of the source waveform. RF_FD indicates an RF frequency domain trace (frequency domain waveform). This is query only. (If you would like to determine only this value, use the WFMOutpre:WFMTYPE? query.)
CENTERFREQUENCY 2.4000E+9	This value indicates the center frequency, in hertz, of the source waveform. This is query only. (If you would like to determine only this value, use the WFMOutpre:CENTERFREQuency? query.)
SPAN 2.0000E+9	This value indicates the frequency span, in hertz, of the source waveform. This is query only. (If you would like to determine only this value, use the WFMOutpre:SPAN? query.)
REFLEVEL 1.0000E-3	This value indicates the reference level, in watts, of the source waveform. This is query only. (If you would like to determine only this value, use the WFMOutpre:REFLEvel? query.)

Appendix E: Search and Trigger Command Sequence Examples

The following are some example command sequences that show a variety of different searches and triggers. The commands in these sequences are not order-dependent.

To use these examples, connect channel 1, channel 2, channel 3 and channel 4 to the probe compensation signal located on the right hand side of the front panel.

The search and trigger command group sections contain more information on general search and trigger concepts. ,

Example 1: Single Threshold Edge Search

Goal: Search the channel 2 waveform and place a mark at each instance where it crosses below a threshold of 1.4 volts.

Command	Comment
*RST;,*OPC?	Resets the oscilloscope and waits for that operation to complete (approximately 1 to 5 seconds depending on the complexity of the previous setup).
display:waveview1:ch2:state 1	Turns the CH2 waveform on.
:AUTOset EXECute>,*OPC?	Autosets the displayed waveform CH2 and waits for the auto setup to complete.
SEARCH:SEARCH1:TRIGger:A:TYPe EDGE	Specifies that this will be an edge search (a mark will be placed when the source waveform passes through a specified threshold level in the specified direction).
SEARCH:SEARCH1:TRIGger:A:EDGE:SOUrce CH2	Specifies the CH2 waveform as the source waveform.
SEARCH:SEARCH1:TRIGger:A:LOWerthreshold:CH2 1.4	Specifies 1.4 volts as the threshold level.
SEARCH:SEARCH1:TRIGger:A:EDGE:SLOpe FALL	Specifies the falling edge as the direction.
SEARCH:SEARCH1:STATE 1	Turns the search on.
SEARCH:SEARCH1:TOTal?	Returns 4, indicating that CH2 fell below the 1.4 volt threshold 4 times.
SEARCH:SEARCH1:LIST?	Returns a list of the 4 marks: CH2 ,11.2411,11.2411,11.2411,-1.5504E-3, 0.0E+0,0.0E+0,0.0E+0; CH2 ,37.0737,37.0737,37.0737, -517.0517E-6,0.0E+0,0.0E+0,0.0E+0; CH2 ,62.9163,62.9163,62.9163, 516.6517E-6,0.0E+0,0.0E+0, 0.0E+0; CH2 ,88.7489,88.7489,88.7489, 1.5500E-3,0.0E+0,0.0E+0,0.0E+0

Example 2: Single Threshold Edge Trigger

Goal: Trigger on the channel 2 waveform when the waveform crosses below a threshold of 1.4 volts.

Command	Comment
*RST	Resets the oscilloscope. Wait for the reset to complete (approximately 3 seconds).
:DISPLAY:WAVEVIEW1:CH2:STATE 1	Turns the CH2 waveform on.
:TRIGger:A:TYPE EDGE	Specifies that this will be an edge trigger (trigger will occur when the source waveform passes through a specified threshold level in the specified direction).
:TRIGger:A:EDGE:SOURce CH2	Specifies the CH2 waveform as the source waveform.
:TRIGger:A:LOWerthreshold:CH2 1.4	Specifies 1.4 volts as the threshold level.
:TRIGger:A:EDGE:SLOpe FALL	Specifies as falling edge as the direction.
:TRIGger:STATE?	Should return TRIGGER, not AUTO.

Example 3: Dual Threshold Runt Search

Goal: Search the channel 3 waveform for negative runt pulses and place a mark at each instance when the waveform drops below an upper threshold of 1.4 volts, but does not cross a lower threshold of -2 volts before re-crossing the upper threshold. The pulse width must be less than 600 microseconds.

Command	Comment
*RST	Resets the oscilloscope. Wait for the reset to complete (approximately 3 seconds).
:DISPLAY:WAVEVIEW1:CH3:STATE 1	Turns the CH3 waveform on.
:AUTOset EXECute	Autosets the displayed waveform CH3. Wait for the autoset to complete (approximately 3 seconds).
SEARCH:SEARCH1:TRIGger:A:TYPE RUNT	Specifies that this will be a runt search (a mark will be placed on a pulse amplitude that crosses one threshold but fails to cross a second threshold before re-crossing the first).
SEARCH:SEARCH1:TRIGger:A:EDGE:SOURce CH3	Specifies to use channel 3 as the source waveform.
SEARCH:SEARCH1:TRIGger:A:LOWerthreshold:CH3 -2	Specifies to use -2 volts as the lower threshold.
SEARCH:SEARCH1:TRIGger:A:UPPerthreshold: CH3 1.4	Specifies to use 1.4 volts as the upper threshold.
SEARCH:SEARCH1:TRIGger:A:RUNT:POLarity NEGative	Specifies to search for when the runt polarity is negative.
SEARCH:SEARCH1:TRIGger:A:RUNT:WIDTH 600E-6	Specifies a pulse width of 600E-6 seconds.
SEARCH:SEARCH1:TRIGger:A:RUNT:WHEn LESSthan	Specifies to search for when the pulse width is less than 600E-6 seconds.
SEARCH:SEARCH1:STATE 1	Turns the search on.

Command	Comment
SEARCH:SEARCH1:TOTal?	Returns 3, indicating that 3 total negative runt pulses were less than 600E-6 seconds wide
SEARCH:SEARCH1:LIST?	Returns a list of 3 marks: CH3 ,11.2311,24.1624,24.1624,1.0335E3,0.0E+0,0.0E+0,0.0E+0; CH3 ,37.0837,50.0150,50.0150,600.0600E9,0.0E+0,0.0E+0,0.0E+0; CH3 ,62.9263,75.8576,75.8576,1.0343E-3,0.0E+0,0.0E+0,0.0E+0

NOTE. You could use a similar command sequence with a transition type search.

Example 4: Single Threshold Logic Search on Three Waveforms

Goal: Search the channel 1, 2 and 3 waveforms and place a mark at each instance when either channel 1 is above 1.4 volts, channel 2 is above 1.5 volts, or channel 3 is above 1.3 volts.

Command	Comment
*RST	Resets the oscilloscope. Wait for the reset to complete (approximately 3 seconds).
:DISPLAY:WAVEVIEW1:CH1:STATE 1	Turns the CH1 waveform on.
:DISPLAY:WAVEVIEW1:CH2:STATE 1	Turns the CH2 waveform on.
:DISPLAY:WAVEVIEW1:CH3:STATE 1	Turns the CH3 waveform on.
:AUTOset EXECute	Autosets the displayed waveforms CH1, CH2 and CH3. Wait for the autoset to complete (approximately 3 seconds).
SEARCH:SEARCH1:TRIGger:A:TYPE LOGic	Specifies that this will be a logic search (a mark will be placed when all channels transition to the specified state).
SEARCH:SEARCH1:TRIGger:A:LOGic:INPut:CH1 HIGH	Specifies the Boolean logic criteria for channel 1; in this case, high.
SEARCH:SEARCH1:TRIGger:A:LOGic:INPut:CH2 HIGH	Specifies the Boolean logic criteria for channel 2; in this case, high.
SEARCH:SEARCH1:TRIGger:A:LOGic:INPut:CH3 HIGH	Specifies the Boolean logic criteria for channel 3; in this case, high.
SEARCH:SEARCH1:TRIGger:A:LOGic:THReshold:CH1 1.4	Specifies to use 1.4 volts as the threshold for CH1.
SEARCH:SEARCH1:TRIGger:A:LOGic:THReshold:CH2 1.5	Specifies to use 1.5 volts as the threshold for CH2.
SEARCH:SEARCH1:TRIGger:A:LOGic:THReshold:CH3 1.3	Specifies to use 1.3 volts as the threshold for CH3.
SEARCH:SEARCH1:TRIGger:A:LOGic:PATtern:WHEn TRUE	Specifies the condition for generating a logic pattern search; in this case, true.
SEARCH:SEARCH1:TRIGger:A:LOGic:FUNCTion OR	Specifies the logic operator for the logic search; in this case, OR.
SEARCH:SEARCH1:STATE 1	Turns the search on.

Command	Comment
SEARCH:SEARCH1:TOTa?	Returns 3 or 4, indicating 3 or 4 times when one of channels 1, 2 or 3 became high. Note: Depending upon the number of transitions displayed, you may get 3 or 4 search marks for this example. If you get 3 search marks, try adjusting the horizontal position until you see 4 search marks.
SEARCH:SEARCH1:LIST?	Returns a list of 3 marks on 3 waveforms: CH1 ,24.1600,24.1600,24.1600,1.0336E3,0.0E+0,0.0E+0,0.0E+0; CH2 ,24.1600,24.1600,24.1600,1.0336E3,0.0E+0,0.0E+0,0.0E+0; CH3 ,24.1600,24.1600,24.1600,1.0336E3,0.0E+0,0.0E+0,0.0E+0; CH1 ,50.0000,50.0000,50.0000,0.0E+0,0.0E+0,0.0E+0,0.0E+0; CH2 ,50.0000,50.0000,50.0000,0.0E+0,0.0E+0,0.0E+0,0.0E+0; CH3 ,50.0000,50.0000,50.0000,0.0E+0,0.0E+0,0.0E+0,0.0E+0; CH1 ,75.8300,75.8300,75.8300,1.0332E3,0.0E+0,0.0E+0,0.0E+0; CH2 ,75.8300,75.8300,75.8300,1.0332E3,0.0E+0,0.0E+0,0.0E+0; CH3 ,75.8300,75.8300,75.8300,1.0332E-3,0.0E+0,0.0E+0,0.0E+0

Glossary

ASCII

Acronym for the American Standard Code for Information Interchange. Controllers transmit commands to the instrument using ASCII character encoding.

Address

A 7-bit code that identifies an instrument on the communication bus. The digitizing instrument must have a unique address for the controller to recognize and transmit commands to it.

Backus-Naur Form (BNF)

A standard notation system for command syntax diagrams. The syntax diagrams in this manual use BNF notation.

Controller

A computer or other device that sends commands to and accepts responses from the digitizing instrument.

EOI

A mnemonic referring to the control line End or Identify. One of the two possible end-of-message terminators.

EOM

A generic acronym referring to the end-of-message terminator. The end-of-message terminator can be either an EOI or the ASCII code for line feed (LF).

Equivalent-Time sampling (ET)

A sampling mode in which the instrument acquires signals over many repetitions of the event. This instrument uses a type of equivalent time sampling called random equivalent time sampling. It utilizes an internal clock that runs asynchronously with respect to the input signal and the signal trigger. The instrument takes samples continuously, independent of the trigger position, and displays them based on the time difference between the sample and the trigger. Although the samples are taken sequentially in time, they are random with respect to the trigger.

Real-Time sampling

A sampling mode where the instrument samples fast enough to completely fill a waveform record from a single trigger event. Use real-time sampling to capture single-shot or transient events.

IEEE

An acronym for the Institute for Electrical and Electronic Engineers.

Serial poll

A device on the bus can request service from the Controller by asserting the SRQ line. When a controller acknowledges the SRQ, it serial polls each device on the bus to determine which device on the bus requested service. Any device requesting service returns a status byte indicating it needs to be serviced and then

unasserts the SRQ line. Devices not requiring service return a status byte that indicates they do not need servicing.

TEKSecure

A Tektronix custom command that initializes both waveform and setup memories. This overwrites any previously stored data.

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